UNITED STATES Coast Pilot®

2

Atlantic Coast: Cape Cod, Massachusetts to Sandy Hook, New Jersey

2026 (55th) Edition

This edition cancels the 54th Edition and includes all previously published corrections.

Weekly updates to this edition are available at:
nauticalcharts.noaa.gov/publications/coast-pilot/index.html

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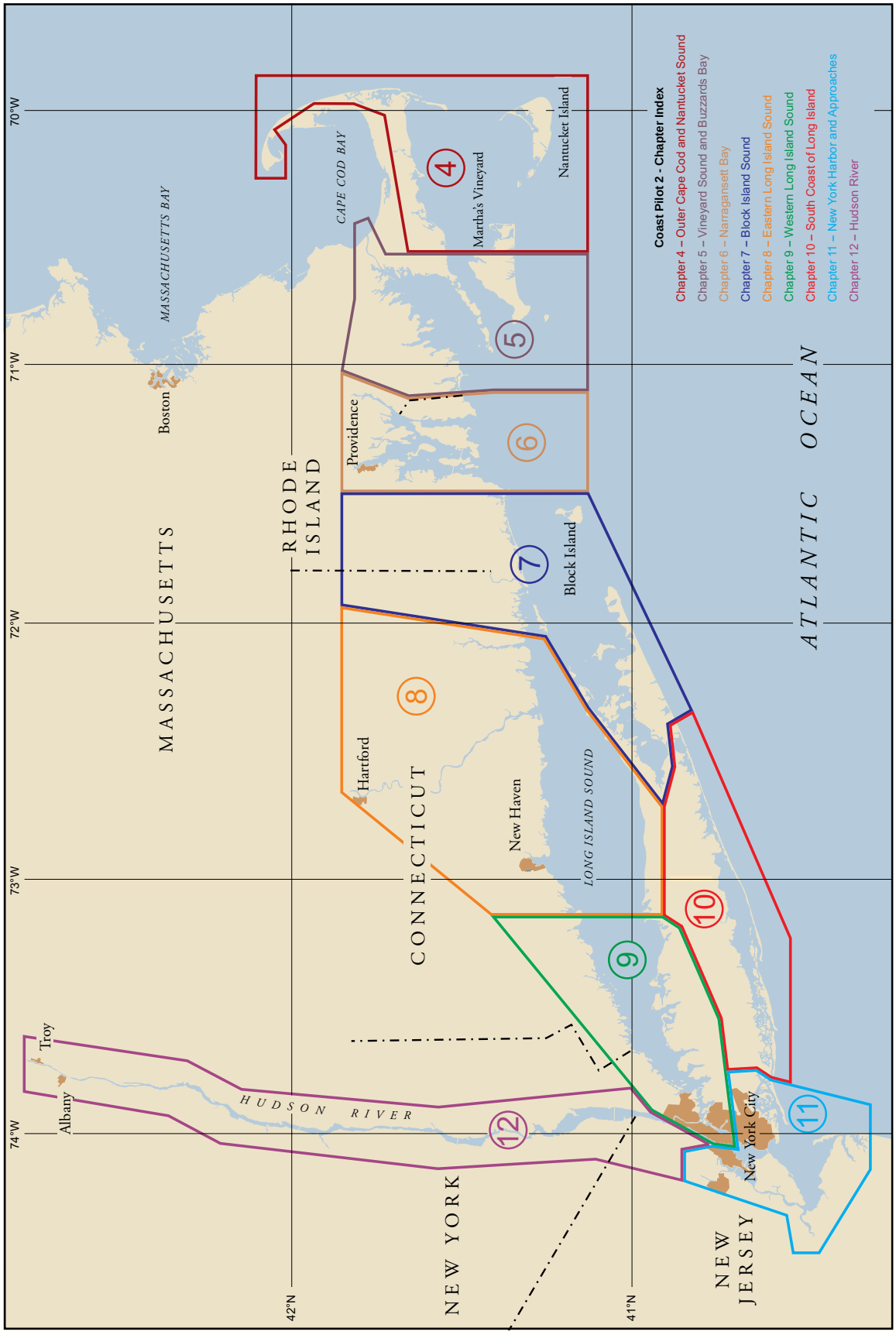

Coast Pilot 2 - Chapter Index
Chapter 4 – Outer Cape Cod and Nantucket Sound
Chapter 5 – Vineyard Sound and Buzzards Bay
Chapter 6 – Narragansett Bay
Chapter 7 – Block Island Sound
Chapter 8 – Eastern Long Island Sound
Chapter 9 – Western Long Island Sound
Chapter 10 – South Coast of Long Island
Chapter 11 – New York Harbor and Approaches
Chapter 12 – Hudson River
70°W
71°W
72°W
73°W
74°W
42°N
41°N
MASSACHUSETTS BAY
CAPE COD BAY
Boston
Nantucket Island
Martha's Vineyard
Providence
RHODE ISLAND
MASSACHUSETTS
Block Island
ATLANTIC OCEAN
Hartford
CONNECTICUT
New Haven
LONG ISLAND SOUND
Troy
Albany
HUDSON RIVER
NEW YORK
New York City
NEW JERSEY
4
5
6
7
8
9
10
11
12

Preface

The United States Coast Pilot is published by the National Ocean Service (NOS), National Oceanic and Atmospheric Administration (NOAA), pursuant to the Act of 6 August 1947 (33 U.S.C. 883a and b), and the Act of 22 October 1968 (44 U.S.C. 1310).

The Coast Pilot supplements the navigational information shown on NOAA nautical charts. The publication is continually updated and maintained from inspections conducted by NOAA survey vessels and field parties, corrections published in Notices to Mariners, information from other Federal agencies, State and local governments, maritime and pilots' associations, port authorities, and concerned mariners.

NOAA's Office of Coast Survey encourages public feedback regarding its suite of nautical charting products and services through **ASSIST**, Coast Survey's stakeholder engagement and feedback tool. This allows customers to submit questions or comments or to report an error with NOAA's nautical charts and products.
Customers can access **ASSIST** at *www.nauticalcharts.noaa.gov/customer-service/assist/*
Those who prefer to communicate by telephone can contact Coast Survey at 1–888–990–6622.

Coast Pilot corrections are no longer published in the NGA Notice to Mariners effective 01 January 2021. Additional information regarding the NGA policy change can be referenced at *msi.nga.mil/NTM* in the Notice to Mariners 52/20 Hydrogram and Marine Information sections.

Coast Pilot Updates

Check for weekly critical updates for this edition at *nauticalcharts.noaa.gov/publications/coast-pilot/index.html*
(See **33 CFR 164.33 Charts and Publications**, chapter 2, for regulations.)

Customers may print the specifically affected paragraphs to revise this book, or download an updated file (PDF) of the entire volume.

A *Weekly Record of Updates* is provided directly preceding the index.

Contents

Preface III

Chapter 1: General Information 1

Chapter 2: Navigation Regulations 37

Chapter 3: Cape Cod To Sandy Hook 163

Chapter 4: Outer Cape Cod and Nantucket Sound 185

Chapter 5: Vineyard Sound and Buzzards Bay 207

Chapter 6: Narragansett Bay 233

Chapter 7: Block Island Sound 253

Chapter 8: Eastern Long Island Sound 275

Chapter 9: Western Long Island Sound. 303

Chapter 10: South Coast of Long Island 335

Chapter 11: New York Harbor and Approaches 347

Chapter 12: Hudson River 375

Navigation Rules 389

Appendix A 415

Weekly Record of Updates. 421

Index 425

General Information

(1)

UNITED STATES COAST PILOT®

(2) The United States Coast Pilot, published by the National Oceanic and Atmospheric Administration (NOAA), is a series of ten nautical books (volumes) that encompasses a wide variety of information important to navigators of U.S. coastal/intracoastal waters and the waters of the Great Lakes. The Coast Pilot is intended to be used as a supplement to NOAA nautical charts. Much of the content cannot be shown graphically on the charts and is not readily available elsewhere. Topics which are covered include environmental factors of weather, climate, ice conditions, tides, water levels, currents, prominent coastal features and landmarks. Specific information on vertical clearances, wharf descriptions, small-craft facilities, hazards, dredged channels and depths are also provided. Navigation services and regulations are also identified including pilotage, towing, anchorages, routes and traffic separation schemes, environmental protection, and other Federal laws.

(3) New editions of each volume are issued annually. Fully updated files are posted weekly on the Internet, and are also available through NOAA Certified Chart Agents at *www.nauticalcharts.noaa.gov*.

(4) **Amendments** to this publication are available at *nauticalcharts.noaa.gov/publications/coast-pilot/index.html.*

(5) <Deleted Paragraph>

(5)

Using the Coast Pilot

(6) **Chapter 1** contains definitions of general and standard terms used throughout the volume, discussions of NOAA charting products and services, descriptions of maritime services by various U.S. Government agencies, Notices to Mariners and other information pertinent to safe navigation.

(7) **Chapter 2** contains selected extracts from the Code of Federal Regulations (CFR) that affect mariners.

(8) **Chapter 3** contains general information that is peculiar to the region covered by a particular Coast Pilot volume. For example, practical information regarding offshore currents and dangers, coastal aids to navigation, prominent landmarks and the general character of the coast and depths helpful in approaching the region.

(9) In **Chapter 4 and the remaining numbered chapters**, the detailed description of the region begins. A map precedes each chapter and outlines the nautical charts used in the area to be discussed. In these chapters, as much as possible, the coastal description is in geographic sequence, north to south on the east coast, east to west on the gulf coast, clockwise around each of the Great Lakes and south to north on the west coast and Alaskan coast. Features are described as they appear on the largest scale chart, with that chart number prominently shown in blue.

(10) **Appendix A** contains contact information regarding the various products, services and agencies detailed throughout the volume.

(11) **Navigation Rules**— preceding Appendix A, contains the International (72 COLREGS) and Inland Navigation Rules, technical Annexes, and associated Federal rules and regulations.

(12) The **Weekly Record of Updates** is intended as a log for critical updates applied to this volume.

(13) The **Index** contains geographic names mentioned throughout a Coast Pilot volume. These names are boldfaced and indexed along with the number of the largest scale chart on which the entire feature appears. Asterisks preceding a chart number in the index of Coast Pilot 5 indicate charts published by the National Geospatial-Intelligence Agency, and in the index of Coast Pilot 6, charts published by the Canadian Hydrographic Service.

(14)

Bearings

(15) Bearings and courses are in degrees true and are measured clockwise from **000°** (north) to **359°**. The bearings of an aid to navigation (e.g., directional light, light sector, range) are given as viewed from the bridge of a vessel toward the light.

(16)

Bridges and Cables

(17) Vertical clearances of bridges and overhead cables are in feet above mean high water unless otherwise stated; clearances in Coast Pilot 6 are in feet above Low Water Datum unless otherwise stated. When the water level is above Low Water Datum, the bridge and overhead cable clearances given in the Coast Pilot and shown on the charts should be reduced accordingly. Clearances of drawbridges are for the closed position, although the open clearances are also given for vertical-lift bridges. Whenever a bridge span over a channel does not open fully to an unlimited clearance position, a minimum clearance for the sections over the channel is given; the same applies to swing and pontoon bridges with openings less than 50 feet horizontally. Clearances given in the Coast Pilot are those approved for nautical charting and are supplied by the U.S. Coast Guard (bridges) and U.S. Army Corps of

Engineers (cables). See charts for horizontal clearances of bridges, as these are generally given in the Coast Pilot only when they are less than 50 feet (15 meters). Tables listing structures across waterways, found in some Coast Pilots, show both horizontal and vertical clearances. Submarine cables are rarely mentioned.

(18)

Cable ferries

(19) Cable ferries are guided by cables fastened to shore and sometimes propelled by a cable rig attached to the shore. Generally, the cables are suspended during crossings and dropped to the bottom when the ferries dock. Where specific operating procedures are known they are mentioned in the text. Since operating procedures vary, mariners are advised to exercise extreme caution and seek local knowledge. **DO NOT ATTEMPT TO PASS A MOVING CABLE FERRY.**

(20)

Courses

(21) These are true and are given in degrees clockwise from **000°** (north) to **359°**. The courses given are the courses to be made good.

(22)

Currents

(23) Stated current velocities are the averages at strength. Velocities are in knots, which are nautical miles per hour. Directions are the true directions to which the currents set (see chapter 3, this book).

(24)

Depths

(25) Depth is the vertical distance from the chart datum to the bottom and is expressed in the same units (feet, meters or fathoms) as those soundings found on the chart. (See Chart Datum, this chapter, for further detail.) The **controlling depth** is the least known depth of a channel. This depth is determined by periodic hydrographic surveys and restricts use of the channel to drafts less than that depth. The **centerline controlling depth** applies only to the channel centerline or close proximity; lesser depths may exist in the remainder of the channel. The **midchannel controlling depth** is the controlling depth of only the middle half of the channel. **Federal project depth** is the original design dredging depth of a channel planned by the U.S. Army Corps of Engineers (USACE) and may be deeper than current conditions. For this reason, project depth must not be confused with controlling depth. **Depths alongside** wharves usually have been reported by owners and/or operators of the waterfront facilities and have not been verified by Government surveys. Since these depths may be subject to change, local authorities should be consulted for the latest controlling depths.

(26) For all maintained channels with controlling depths detailed on charts in tabular form, the Coast Pilot usually states only the project depths. For all other channels which may be depicted on charts with depth legends, notes or soundings, the Coast Pilot will list where to find the most recent information on the latest known surveys. Depths may vary considerably between maintenance dredging.

(27)

Under-keel clearances

(28) It is becoming increasingly evident that economic pressures are causing mariners to navigate through waters of barely adequate depth, with under-keel clearances being finely assessed from the charted depths, predicted tide levels and depths recorded by echo sounders.

(29) It cannot be too strongly emphasized that even charts based on modern surveys may not show all sea-bed obstructions or the shoalest depths, and actual tide levels may be appreciably lower than those predicted.

(30) In many ships an appreciable correction must be applied to shoal soundings recorded by echo sounders due to the horizontal distance between the transducers. This separation correction, which is the amount by which recorded depths therefore exceed true depths, increases with decreasing depths to a maximum equal to half the distance apart of the transducers; at this maximum the transducers are aground. Ships whose transducers are more than 6 feet (1.8 meters) apart should construct a table of true and recorded depths using the Traverse Tables. (Refer to the topic on echo soundings elsewhere in chapter 1.)

(31) Other appreciable corrections, which must be applied to many ships, are for settlement and squat. These corrections depend on the depth of water below the keel, the hull form and the speed of the ship.

(32) Settlement causes the water level around the ship to be lower than would otherwise be the case. It will always cause echo soundings to be less than they would otherwise be. Settlement is appreciable when the depth is less than seven times the draft of the ship and increases as the depth decreases and the speed increases.

(33) Squat denotes a change in trim of a ship underway, relative to her trim when stopped. It usually causes the stern of a vessel to sit deeper in the water. However, it is reported that in the case of mammoth ships, squat causes the bow to sit deeper. Depending on the location of the echo sounding transducers, this may cause the recorded depth to be greater or less than it ought to be. **Caution and common sense are continuing requirements for safe navigation.**

(34)

Distances

(35) These are in nautical miles unless otherwise stated. A nautical mile is one minute of latitude, or approximately 2,000 yards, and is about 1.15 statute miles.

(36) Coast Pilot 6 is in statute miles unless otherwise stated. A statute mile is 5,280 feet or about 0.87 nautical mile.

(37)

Geographic Coordinates

(38) Geographic coordinates listed in the Coast Pilot are referred to North American Datum of 1983 (NAD 83)

unless otherwise noted for certain CFR extracts in chapter 2.

(39)

Heights

(40) These are in feet (meters) above the tidal datum used for that purpose on the charts, usually mean high water. However, the heights of the decks of piers and wharves are given in feet (meters) above the chart datum for depths.

(41) Coast Pilot 6 is in feet (meters) above the chart datum used for that purpose on the charts, usually Low Water Datum.

(42)

Light and Sound Signal Characteristics

(43) These are not described in the Coast Pilot. Also, light sectors and visible ranges are generally not fully described. This information can be found in U.S. Coast Guard Light Lists.

(44)

Obstructions

(45) Wrecks and other obstructions are mentioned only if they are relatively permanent and in or near normal traffic routes.

(46)

Radio Navigational Aids

(47) For detailed information on Radio Navigation Aids see the **United States Coast Guard Light Lists** and the National Geospatial-Intelligence Agency's **Radio Navigational Aids, Publication 117**.

(48)

Ranges

(49) These are not fully described. "**A 339°** Range" means that the rear structure bears **339°** from the front structure. (See United States Coast Guard Light Lists.)

(50)

Reported information

(51) Information received by NOAA from various sources concerning depths, dangers, currents, facilities, and other topics, which has not been verified by Government surveys or inspections, is often included in the Coast Pilot; such **unverified information** is qualified as "reported" and should be regarded with caution.

(52)

Tides

(53) Tidal information, including real-time water levels, tide predictions and tidal current predictions are available at *tidesandcurrents.noaa.gov*.

(54)

Time

(55) Unless otherwise stated, all times are given in local standard time in the 24-hour system. (Noon is 1200, 2:00 p.m. is 1400 and midnight is 0000.)

(56)

Winds

(57) Directions are the true directions from which the winds blow; however, sometimes (rarely) compass points are used. Unless otherwise indicated, speeds are given in knots, which are nautical miles per hour.

(58) <Deleted Paragraph>

(58)

NAUTICAL CHARTS

(59) NOAA produces and maintains a suite of over 1,000 nautical charts that cover the U.S. coastal waters, the Great Lakes and U.S. territories. These charts provide a graphic representation of water depths, the shoreline, prominent topographic and man-made features, aids to navigation and other navigational information useful to the mariner. NOAA's charts are available in a variety of digital formats designed to meet the specific requirements of all mariners. Paper copies may also be obtained through one of NOAA's Print-on-Demand partners.

(59) <59-60 Deleted>

(60)

Paper Print on Demand Nautical Charts

(61) <Deleted Paragraph>

(61) The content of Print-On-Demand (POD) charts is updated weekly by NOAA with the most current U.S. Coast Guard Local Notice to Mariners and other critical safety information. POD charts are printed under the authority of NOAA and shipped through partnerships between NOAA and commercial providers. POD information and a list of participating POD chart agents can be found at *nauticalcharts.noaa.gov/publications/print-agents.html#paper-charts-mobile*.

(62) <Deleted Paragraph>

(62)

Portable Document Format (PDF) Nautical Charts

(63) <Deleted Paragraph>

(63) Almost all of NOAA's nautical charts may be downloaded for free as Portable Document Format (PDF) files at *nauticalcharts.noaa.gov/charts/noaa-raster-charts.html#full-size-nautical-charts*. The PDF nautical charts are exact replicas of the images used to produce POD and Raster Navigational Charts (RNC). As such, they also have all the latest updates based on U.S. Coast Guard Local Notices to Mariners, National Geospatial-Intelligence Agency Notices to Mariners and other critical safety information.

(64) Most PDF charts can be printed at the proper scale from any plotter accommodating a 36-inch paper width. When printed properly, PDF charts and POD charts are very similar, but PDF charts have not yet been approved to meet Federal regulations for paper chart carriage requirements as POD charts have.

(65)

BookletCharts

(66) The NOAA BookletChart™ is a product that can be printed by the users for free. They are made to help recreational boaters locate themselves on the water. BookletCharts are reduced in scale and divided into pages for convenience but otherwise contain all the information of the full-scale nautical charts and are updated weekly. For more information visit *nauticalcharts.noaa.gov/charts/noaa-raster-charts.html#booklet-charts*.

(67)

Raster Navigational Charts (NOAA RNC®)

(68) NOAA Raster Navigational Charts (NOAA RNC®) are geo-referenced digital images of NOAA's entire suite of paper charts. NOAA RNCs are official data that can be used in many types of electronic charting systems (ECS), including Raster Chart Display Systems (RCDS) and some Electronic Chart Display and Information Systems (ECDIS). Current regulations support the use of RNCs as a primary means of navigation when ENCs are not available, but they require an accompanying minimal set of up-to-date paper charts. They can integrate position information from the Global Positioning System (GPS) and other navigational sensors, such as radar and automatic identification systems (AIS) to show a vessel's track, waypoints, and planned routes. NOAA RNCs and their weekly updates are available free of charge at *nauticalcharts.noaa.gov/charts/noaa-raster-charts.html*.

(69)

Electronic Navigational Charts (NOAA ENC®)

(70) NOAA Electronic Navigational Charts (NOAA ENC®) are databases of charted objects and their attributes with standardized content, structure and format. They comply with International Hydrographic Organization (IHO) specifications stated in IHO Publication S-57. They may be used as an alternative to paper charts required on SOLAS class vessels.

(71) ENCs are intended for use in electronic charting systems (ECS) as well as Electronic Chart Display and Information Systems (ECDIS). ECDIS are programmable to show as much or as little data as the user requires. They can integrate position information from the Global Positioning System (GPS) and other navigational sensors, such as radar and automatic identification systems (AIS) to show a vessel's track, waypoints and planned routes. Using this information ECDIS can use ENCs to give warning of impending danger in relation to the vessel's position and movement. NOAA ENCs and their updates are available free of charge at *nauticalcharts.noaa.gov/charts/noaa-enc.html*.

(72)

Nautical Chart—New Editions and Corrections

(73) New editions of paper Print-on-Demand (POD) charts are available on the Monday after NOAA clears a new edition for release. Once the authorized POD chart is available, it meets federal chart carriage requirements, and should be put into service immediately. It should be updated from the *last correction and cleared through* dates shown in the lower left corner of the chart.

(74) The chart date is of vital importance to the navigator. When charted information becomes obsolete, further use of the chart for navigation is dangerous. Natural and artificial changes, many of them critical, are occurring constantly; therefore it is important that navigators use up-to-date charts. Nautical charts and publications are available for purchase from authorized POD agents and their sales outlets.

(75) NOAA's "Nautical Chart Update" website allows mariners to update their nautical charts from one database that includes information from NOAA, NGA U.S. Notice to Mariners, U.S. Coast Guard Local Notices to Mariners and the Canadian Coast Guard Notices to Mariners at: *nauticalcharts.noaa.gov/charts/chart-updates.html*.

(76)

Nautical Chart Numbering System

(77) This chart numbering system, adopted by NOAA and National Geospatial-Intelligence Agency (NGA), provides for a uniform method of identifying charts published by both agencies. Nautical charts published by NGA and by the Canadian Hydrographic Service are identified in the Coast Pilot by an asterisk preceding the chart number.

(78)

Chart Scale

(79) The scale of a chart is the ratio of a given distance on the chart to the actual distance that it represents on the earth. For example, one unit of measurement on a 1:10,000 scale chart is equal to 10,000 of the same unit on the earth's surface. Large scale charts show greater detail of a relatively small area. Small scale charts show less detail but cover a larger area. Certain hydrographic information may be omitted on smaller scale charts. **Mariners should always obtain the largest scale coverage for near shore navigation**.

(80) The scales of nautical charts range from 1:2,500 to about 1:5,000,000. Graphic scales are generally shown on charts with scales of 1:80,000 or larger, and numerical scales are given on smaller scale charts. NOAA charts are classified according to scale as follows:

(81) **Sailing charts**, scales 1:600,000 and smaller, are for use in fixing the mariner's position approaching the coast from the open ocean or for sailing between distant coastwise ports. On such charts the shoreline and topography are generalized and only offshore soundings, principal lights, outer buoys and landmarks visible at considerable distances are shown.

(82) **General charts**, scales 1:150,000 to 1:600,000, are for coastwise navigation outside of outlying reefs and shoals.

(83) **Coast charts**, scales 1:50,000 to 1:150,000, are for inshore navigation leading to bays and harbors of considerable width and for navigating large inland waterways.

(84) **Harbor charts**, scales larger than 1:50,000, are for harbors, anchorage areas and the smaller waterways.

(85) **Special charts**, at various scales, cover the Intracoastal waterway and miscellaneous small-craft areas.

(86)

Chart Projections

(87) The **Mercator projection** used on most nautical charts has straight-line meridians and parallels that intersect at right angles. On any particular chart the distances between meridians are equal throughout, but distances between parallels increase progressively from the equator toward the poles so that a straight line between any two points is a rhumb line. This unique property of the Mercator projection is one of the main reasons why it is preferred by the mariner.

(88) The **Polyconic projection** is used on most U.S. nautical charts of the Great Lakes. On this projection, parallels of latitude appear as non-concentric circles, and meridians appear as curved lines converging toward the pole and concave to the central meridian. The scale is correct along any parallel and along the central meridian of the projection. Along other meridians the scale increases with increased difference of longitude from the central meridian.

(89)

Chart Datum, Tidal Waters

(90) Chart Datum is the particular tidal level to which soundings and depth curves on a nautical chart or bathymetric map are referred. The tidal datum of **Mean Lower Low Water** is used on all NOAA charts, except for charts in the Great Lakes and non-tidal inland waterways. For information on **Chart Datum, Great Lakes System**, see Coast Pilot 6, chapter 3.

(91)

Horizontal Datum

(92) Nautical charts are constructed based on one of a number of horizontal datums which are adopted to best represent individual regions around the world. Note that the terms horizontal datum, horizontal geodetic datum, and horizontal control datum are synonymous.

(93) The exact placement of lines of latitude and longitude on a nautical chart is dependent on the referenced horizontal datum. Charts of the United States are currently referenced primarily to the North American Datum of 1983 (NAD 83), and the World Geodetic System 1984 (WGS 84). WGS 84 is equivalent to the NAD 83 for charting purposes.

(94) NAD 83 and WGS 84 have replaced the North American Datum of 1927 and other regional datums as the primary horizontal datum to which NOAA charts are referenced. Since some geographic positions may still be referenced to the older datums, NOAA has included notes on charts which show the amount to shift those positions in latitude and longitude to fit the chart's NAD 83 or WGS 84 projection.

(95) It should be noted that the physical shift between positions on older datums and NAD 83/WGS 84 was significant. Mariners should always be certain the positions they are plotting on a nautical chart are on the same datum as the chart.

(96)

Chart Accuracy

(97) The value of a nautical chart depends upon the accuracy of the surveys on which it is based. The chart reflects what was found by field surveys and what has been reported to NOAA. It also represents general conditions at the time of surveys or reports and does not necessarily portray present conditions. Significant changes may have taken place since the date of the last survey or report.

(98) Each sounding represents an actual measure of depth and location at the time the survey was made, and each bottom characteristic represents a sampling of the surface layer of the sea bottom at the time of the sampling. Areas where sand and mud prevail, especially the entrances and approaches to bays and rivers exposed to strong tidal current and heavy seas, are subject to continual change.

(99) In coral regions and where rocks and boulders abound, it is always possible that surveys may have failed to find every obstruction. Thus, when navigating such waters, customary routes and channels should be followed, and areas where irregular and sudden changes in depth indicate conditions associated with pinnacle rocks, coral heads, or boulders should be avoided.

(100) Information charted as "reported" should be treated with caution when navigating the area, because the actual conditions have not been verified by government surveys.

(101)

Source Diagrams and Zone of Confidence Diagrams

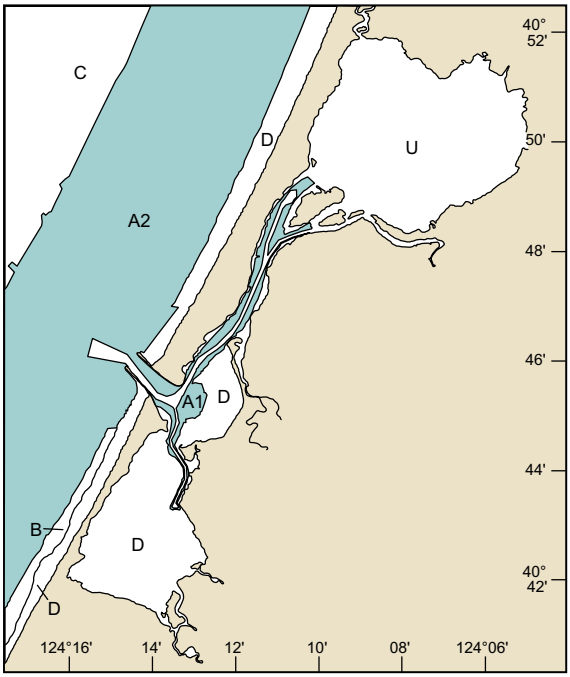
(102) The age and accuracy of hydrographic survey data that support nautical charts can vary. Depth information on nautical charts, paper or digital, is based on data from the latest available hydrographic survey, which in many cases may be quite old. Diagrams are provided on nautical charts to assist mariners in assessing hydrographic survey data and the associated level of risk to navigate in a particular area. There are currently two types of diagrams shown on NOAA paper and raster navigational charts (RNCs) of 1:500,000 scale and larger—**Zone of Confidence (ZOC) Diagrams** and **Source Diagrams**. ZOC information (designated CATZOC) is also found on electronic navigational charts (ENCs). This provides consistency in the display of source data between ENCs and newer paper charts.

(103) Both types of diagrams consist of a graphic representation of the extents of hydrographic surveys within the chart and accompanying table of related survey quality categories. CATZOC information on an ENC, unlike the diagrams on a paper chart or RNC, is displayed over the ENC data using symbols rather than letters. These symbols are displayed on a separate layer, which can be viewed when planning a route, then switched off until needed again at another time.

(106)

Zone of Confidence Diagrams

ZOC CATEGORIES				
ZOC	DATE	POSITION ACCURACY	DEPTH ACCURACY	SEAFLOOR COVERAGE
A1	2008–2016	± 16.4 ft	= 1.6 feet + 1% depth	All significant seafloor features detected
A2	—	± 65.6 ft	= 3.3 feet + 2% depth	All significant seafloor features detected
B	2005	± 164.0 ft	= 3.3 feet + 2% depth	Uncharted features hazardous to surface navigation are not expected but may exist
C	—	± 1640.4 ft	= 6.6 feet + 2% depth	Depth anomalies may be expected
D	—	Worse than ZOC C	Worse than ZOC C	Large depth anomolies may be expected
U	Unassessed – The quality of the bathymetric data has yet to be assessed.			



(104) On **ZOC Diagrams**, the quality of the hydrographic data is assessed according to six categories; five quality categories for assessed data (A1, A2, B, C and D) and a sixth category (U) for data that has not yet been assessed. On the ENC, the categories are shown using a rating system of stars—the higher the quality, the greater the number of stars. Assessment of hydrographic data quality and classification into zones of confidence is based on a combination of: survey date, position accuracy, depth accuracy and sea floor coverage (the survey's ability to detect objects on the seafloor.)

(105) **Source Diagrams** provide the mariner with additional information about the density and adequacy of the sounding data depicted on the chart. The adequacy with which sounding data reflects the configuration of the bottom depends on the following factors: survey technology employed (sounding and navigation equipment), survey specifications in effect (prescribed survey line spacing and sounding interval) and type of bottom (e.g., rocky with existence of submerged pinnacles, flat sandy, coastal deposits subject to frequent episodes of deposition and erosion). Source diagrams will be replaced with ZOC diagrams as new editions are created.

(108)

Chart Symbols, Abbreviations and Terms

(109) The standard symbols and abbreviations approved for use on nautical charts produced by the U.S. Government are described in **U.S. Chart No. 1: Symbols, Abbreviations and Terms used on Paper and Electronic Navigational Charts.** This reference, jointly maintained by the National Geospatial-Intelligence Agency (NGA) and NOAA, is available at *nauticalcharts.noaa.gov/publications/us-chart-1.html.*

(110) The publication **Chart 1: Symbols, Abbreviations and Terms** published by the Canadian Hydrographic Service, is available online at *charts.gc.ca/publications/chart1-carte1/index-eng.asp.*

(111) Some symbols and abbreviations used on foreign charts, including reproductions of foreign charts made by NGA, are different than those used on U.S. charts. It is recommended that mariners who use foreign charts also obtain the symbol sheet or Chart No. 1 produced by the appropriate foreign agency.

(112) Mariners are warned that the buoyage systems, shapes and colors used by other countries often have a different significance than the U.S. system.

(113)

Areas with Blue Tint

(114) A blue tint is shown in water areas on many charts to accentuate shoals and other areas considered dangerous for navigation when using that particular chart. Since the danger curve varies with the intended purpose of a chart a careful inspection should be made to determine the contour depth of the blue tint areas.

(115)

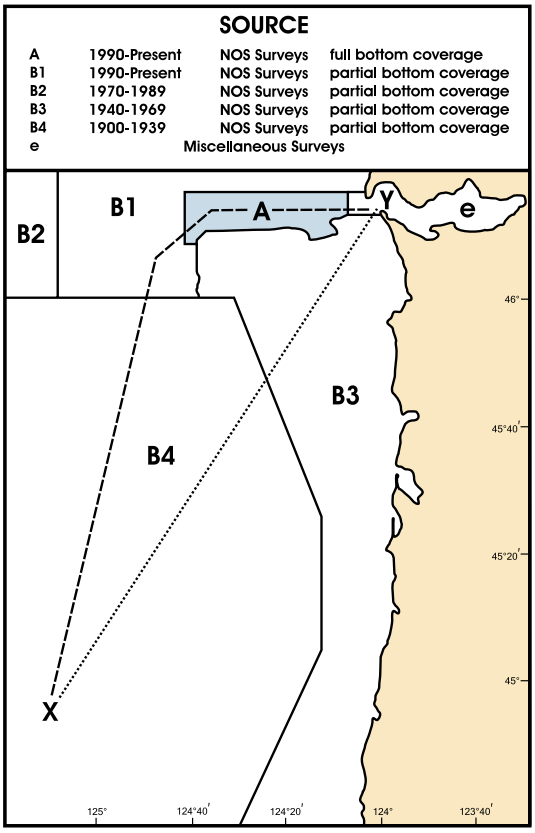
Bridge and Cable Clearances

(116) For bascule bridges whose spans do not open to a full vertical position, unlimited overhead clearance is not

(107)

Source Diagrams

Referring to the accompanying sample Source Diagram to the right and the previous discussion of survey methods over time, transiting from Point X to Point Y, along the track indicated by the dotted line, would have the following information available about the relative quality of the depth information shown on the chart.



Point X lies in an area surveyed by NOAA between 1900-1939. The sounding data in this area would have been collected by leadline. Depths between sounding points can only be inferred, and undetected features might exist between the sounding points in areas of irregular relief — caution should be exercised.

The transit then crosses an area surveyed by NOAA between 1940-1969. The sounding data in this area would have been collected by continuous recording single beam echo sounder. It is possible that features could have been missed between sounding lines, although echo sounders record all depths along a sounding line with varying beam widths.

The transit ends in an area charted from miscellaneous surveys. These surveys may be too numerous to depict or may vary in age, reliability, origin or technology used. No inferences about the fitness of the data can be made in this area from the diagram.

Referring again to the accompanying sample Source Diagram, and the previous discussion of survey methods over time, a mariner could choose to transit from Point X to Point Y, along the track shown with a dashed line.

The transit starts again in an area surveyed by NOAA between 1900-1939. The sounding data in this area would have been collected by leadline. Depths between sounding points can only be inferred, and undetected features might still exist between the sounding points in areas of irregular relief — caution should be exercised.

The transit then crosses an area surveyed by NOAA between 1990–present, with partial bottom coverage. The data is collected in metric units and acquired by continuous recording single beam echo sounder. It is possible that features could have been missed between the sounding lines, although echo sounders record all depths along a sounding line with varying beam widths.

The transit then crosses into an area surveyed by NOAA etween 1990–present, having full bottom coverage. This area of the charted diagram is shaded with a blue screen to draw attention to the fact that full bottom coverage has been achieved. The data in this area would have been collected in metric units and acquired by side scan sonar or multibeam sonar technology. Undetected features in this area, at the time of the survey, would be unlikely.

The transit ends in an area charted from miscellaneous surveys. These surveys may be too numerous to depict or may vary in age, reliability, origin or technology used. No inferences about the fitness of the data can be made in this area from the diagram. By choosing to transit along the track shown by the dashed line, the mariner would elect to take advantage of survey information that is more recent and collected with modern technology.

available for the entire charted horizontal clearance when the bridge is open, due to the inclination of the drawspans over the channel.

(117) Charted in black text, vertical clearances of overhead cables are for the lowest wires at mean high water as authorized and permitted by the U.S. Army Corps of Engineers (USACE). Reported clearances received from sources other than the USACE are labeled as such. When provided, safe vertical clearances are shown in magenta text and indicate the highest points of a ship that can pass under an overhead power cable without risk of electrical discharge from the cable to the ship or without making contact with a bridge. **Vessels with masts, stacks, booms or antennas should allow su f cient clearance under power cables to avoid arcing.**

(118)

Submarine Cables and Submerged Pipelines

(119) **Submarine cables** and **submerged pipelines** cross many waterways used by both large and small vessels, but all of them may not be charted. For inshore areas, they usually are buried beneath the seabed, but for offshore areas they may lie on the ocean floor. Warning signs are often posted to warn mariners of their existence.

(120) The installation of submarine cables or pipelines in U.S. waters or the Continental Shelf of the United States is under the jurisdiction of one or more Federal agencies, depending on the nature of the installation. They are shown on the charts when the necessary information is reported to NOAA and they have been recommended for charting by the responsible agency. The chart symbols for submarine cable and pipeline areas are usually shown for inshore areas, whereas chart symbols for submarine cable and pipeline routes may be shown for offshore areas. Submarine cables and pipelines are not described in the Coast Pilots.

(121) In view of the serious consequences resulting from damage to submarine cables and pipelines, vessel operators should take special care when anchoring, fishing or engaging in underwater operations near areas where these cables or pipelines may exist or have been reported to exist. Mariners are also warned that the areas where cables and pipelines were originally buried may have changed and they may be exposed; extreme caution should be used when operating vessels in depths of water comparable to the vessel's draft.

(122) Certain cables carry high voltage, while many pipelines carry natural gas under high pressure or petroleum products. Electrocution, fire or explosion with injury, loss of life or a serious pollution incident could occur if they are broached.

(123) Vessels fouling a submarine cable or pipeline should attempt to clear without undue strain. Anchors or gear that cannot be cleared should be slipped, but no attempt should be made to cut a cable or a pipeline.

(124)

Artificial Obstructions to Navigation

(125) **Disposal areas** are designated by the U.S. Army Corps of Engineers for depositing dredged material where there is sufficient depth not to cause shoaling or create a danger to surface navigation. The areas are charted without blue tint, and soundings and depth curves are retained.

(126) **Disposal sites** are areas established by Federal regulation (**40 CFR 220** through **228**) in which dumping of dredged and fill material and other nonbuoyant objects is allowed with the issuance of a permit. Dumping of dredged and fill material is supervised by the U.S. Army Corps of Engineers and all other dumping by the Environmental Protection Agency (EPA). (See U.S. Army Corps of Engineers and Environmental Protection Agency, this chapter, and Appendix A for office addresses.)

(127) **Dumping grounds** are also areas that were established by Federal regulation (**33 CFR 205**). However, these regulations have been revoked and the use of the areas discontinued. These areas will continue to be shown on nautical charts until such time as they are no longer considered to be a danger to navigation.

(128) Disposal Sites and Dumping Grounds are rarely mentioned in the Coast Pilot, but are shown on nautical charts. **Mariners are advised to exercise caution in the vicinity of all dumping areas.**

(129) **Spoil areas** are for the purpose of depositing dredged material, usually near and parallel to dredged channels. Spoil areas are usually charted from survey drawings from U.S. Army Corps of Engineers after-dredging surveys, though they may originate from private or other Government agency surveys. On nautical charts, spoil areas are tinted blue, labeled and have all soundings and depth curves omitted from within their boundaries. Spoil areas present a hazard to navigation and even the smallest craft should avoid crossing them.

(130) **Fish havens** are artificial shelters constructed of various materials including rocks, rubble, derelict barges/oil rigs and specially designed precast structures. This material is placed on the sea floor to simulate natural reefs and attract fish. Fish havens are often located near fishing ports or major coastal inlets and are usually considered hazards to shipping. Before such a reef may be built, the U.S Army Corps of Engineers must issue a permit specifying the location and depth over the reef. Constructed of rigid material and projecting above the bottom, they can impede surface navigation and therefore represent an important feature for charting. Fish havens may be periodically altered by the addition of new material, thereby possibly increasing the hazard. They are outlined and labeled on charts and show the minimum authorized depth when known. Fish havens are tinted blue if they have a minimum authorized depth of 11 fathoms or less. If the minimum authorized depth is unknown and they are in depths greater than 11 fathoms, they are considered a danger to navigation. Navigators

should be cautious about passing over fish havens or anchoring in their vicinity.

(131) **Fishtrap areas** are areas established by the U.S. Army Corps of Engineers, or State or local authority, in which traps may be built and maintained according to established regulations. The fish stakes that may exist in these areas are obstructions to navigation and may be dangerous. The limits of fishtrap areas and a cautionary note are usually charted. Navigators should avoid these areas.

(132)

Local Magnetic Disturbances

(133) If measured values of magnetic variation differ from the expected (charted) values by several degrees, a magnetic disturbance note will be printed on the chart. The note will indicate the location and magnitude of the disturbance, but the indicated magnitude should not be considered as the largest possible value that may be encountered. Large disturbances are more frequently detected in the shallow waters near land masses than on the deep sea. Generally, the effect of a local magnetic disturbance diminishes rapidly with distance, but in some locations there are multiple sources of disturbances and the effects may be distributed for many miles.

(134)

Compass Roses

(135) Each compass rose shows the date, magnetic variation and the annual change in variation. Prior to the new edition of a nautical chart, the compass roses are reviewed. Corrections for annual change and other revisions may be made as a result of newer and more accurate information. On some general and sailing charts, the magnetic variation is shown by isogonic lines in addition to the compass roses.

(136)

Echo Soundings

(137) The echo sounder on a ship may indicate small variations from charted soundings; this may be due to the fact that various corrections (instrument corrections, settlement and squat, draft and velocity corrections) are made to echo soundings in surveying which are not normally made in ordinary navigation, or to observational errors in reading the echo sounder. Instrument errors vary between different equipment and must be determined by calibration aboard ship. Most types of echo sounders are factory calibrated for a velocity of sound in water of 800 fathoms per second, but the actual velocity may differ from the calibrated velocity by as much as 5 percent, depending upon the temperature and salinity of the waters in which the vessel is operating; the highest velocities are found in warm, highly saline water and the lowest in icy freshwater. Velocity corrections for these variations are determined and applied to echo soundings during hydrographic surveys. All echo soundings must be corrected for the vessel's draft, unless the draft observation has been set on the echo sounder.

(138) Observational errors include misinterpreting false echoes from schools of fish, seaweed, etc., but the most serious error that commonly occurs is where the depth is greater than the scale range of the instrument; a 400–fathom scale indicates 15 fathoms when the depth is 415 fathoms. Caution in navigation should be exercised when wide variations from charted depths are observed.

(139)

NOTICES TO MARINERS

(140) **Notices to Mariners** are published to advise operators of marine information affecting the safety of navigation. The notices include changes in aids to navigation, depths in channels, bridge and overhead cable clearances, reported dangers and other useful marine information. They should be used routinely for updating the latest editions of nautical charts and related publications.

(141) **Local Notices to Mariners** are issued by each Coast Guard District Commander for the waters under their jurisdiction. (See Appendix A for Coast Guard district(s) covered by this volume.) These notices are usually published weekly and are available at *navcen.uscg.gov*.

(142) **U.S. Notice to Mariners**, published weekly by the National Geospatial-Intelligence Agency, are prepared jointly with NOAA and the Coast Guard. These notices contain selected items from the Local Notices to Mariners and other reported marine information required by oceangoing vessels operating in both foreign and domestic waters. Special items covering a variety of subjects and generally not discussed in the Coast Pilot or shown on nautical charts are published annually in Notice to Mariners No. 1. These items are important to the mariner and should be read for future reference. These notices are available at *msi.nga.mil/NGAPortal/MSI.portal*.

(143) **Broadcast Notices to Mariners** are made by the Coast Guard to report deficiencies and important changes in aids to navigation. (See Navigational Warnings, Information and Weather, this chapter.)

(144) The **Special Notice to Mariners** is an annual publication containing important information for mariners on a variety of subjects which supplements information not usually found on charts and in navigational publications. It includes excerpts from various Federal laws and regulations regarding marine pollution reporting, aids to navigation and Vessel Traffic Service (VTS) procedures. There are tips for trip planning, updates to the Rules of the Road and information on local hazards. Also included are points of contact, phone numbers and email addresses for various subject matter experts to assist the mariner in locating further information.

(145) Vessels operating within the limits of the Coast Guard districts can obtain information affecting NOAA charts and related publications from the Local Notices to Mariners. Small craft using the Intracoastal Waterway and other waterways and small harbors within the United

States that are not normally used by oceangoing vessels will require the Local Notices to Mariners to keep charts and related publications up to date.

(146)

AIDS TO NAVIGATION

(147)

U.S. Aids to Navigation System

(148) The navigable waters of the United States are marked to assist navigation using the U.S. Aids to Navigation System, a system consistent with the International Association of Marine Aids to Navigation and Lighthouse Authorities (IALA) Maritime Buoyage System. The **IALA Maritime Buoyage System** is followed by most of the world's maritime nations and will improve maritime safety by encouraging conformity in buoyage systems worldwide. IALA buoyage is divided into two regions made up of Region A and Region B. All navigable waters of the United States follow IALA Region B, except U.S. possessions west of the International Date Line and south of 10° north latitude, which follow IALA Region A. Lateral aids to navigation in Region A vary from those located within Region B. Nonlateral aids to navigation are the same as those used in Region B. Appropriate nautical charts and publications should be consulted to determine whether the Region A or Region B marking schemes are in effect for a given area.

(149) As standard protocol, the U.S. Coast Guard reported assigned positions of aids to navigation uses the North American Datum of 1983 (NAD 83). Due to the development of new navigational systems and the retirement of old systems, the World Geodetic System 1984 (WGS 84) has become the preferred standard. In 2020, the U.S. Coast Guard Chief of the Office of Navigation Systems (CG-NAV) announced that all geographic coordinates for aids to navigation assigned positions will be reported using WGS 84.

(150)

Reporting Defects in Aids to Navigation

(151) Promptly notify the nearest Coast Guard District Commander if an aid to navigation is observed to be missing, sunk, capsized, out of position, damaged, extinguished or showing improper characteristics.

(152) **Aids to navigation** in United States waters of the Great Lakes and their connecting waters, except for the St. Lawrence River, are maintained by the U.S. Coast Guard. Local jurisdiction for the region is assigned to the Commander, Ninth Coast Guard District. The Lake Champlain region and the Hudson River are under the jurisdiction of the Commander, First Coast Guard District. (See Appendix A for the addresses.)

(153) It is unlawful to establish or maintain any aid similar to those maintained by the U.S. Coast Guard without first obtaining permission from the Coast Guard District Commander. The licensed officer in command of a vessel which collides with any aid must report the fact promptly to the nearest U.S. Coast Guard Sector.

(154)

Lights

(155) **Nominal range** is the term for the range of visibility of lights as defined in the U.S. Coast Guard Light List and shown on nautical charts. It is the maximum distance a light can be seen in clear weather (meteorological visibility of 10 nautical miles). Nominal range is listed for all lighted aids to navigation except range lights, directional lights, and private aids to navigation.

(156) **Luminous range** is the greatest distance a light may be seen given its nominal range and the prevailing meteorological visibility. The Luminous Range Diagram, found in the U.S. Coast Guard Light List, enables the mariner to determine the approximate luminous range of a light when the nominal range and the prevailing meteorological visibility are known. The nominal range and the luminous range do not take into account elevation, observer's height of eye, or the curvature of the earth.

(157) **Geographic range** is the greatest distance the curvature of the earth permits an object, of a given height, to be seen from a particular height of eye without regard to luminous intensity or visibility conditions. To determine the actual geographic range for height of eye, the geographic range must be corrected by a distance corresponding to the height difference. The Geographic Range Table, found in the U.S. Coast Guard Light List, gives the approximate geographic range of visibility for an object which may be seen by an observer at sea level.

(158) The maximum distances at which lights can be seen may at times be increased by abnormal atmospheric refraction and may be greatly decreased by unfavorable weather conditions such as fog, rain, haze or smoke. All except the most powerful lights are easily obscured by such conditions. In some conditions of the atmosphere white lights may have a reddish hue. During weather conditions which tend to reduce visibility, colored lights are more quickly lost to sight than white lights. Navigational lights should be used with caution because of the following conditions that may exist.

(159) A light may be extinguished and the fact not reported to the Coast Guard for correction, or a light may be located in an isolated area where it will take time to correct.

(160) In regions where ice conditions prevail the lantern panes of unattended lights may become covered with ice or snow, which will greatly reduce the visibility and may also cause colored lights to appear white.

(161) Brilliant shore lights used for advertising and other purposes, particularly those in densely populated areas, make it difficult to identify a navigational light.

(162) At short distances flashing lights may show a faint continuous light between flashes.

(163) The distance of an observer from a light cannot be estimated by its apparent intensity. The characteristics of lights in an area should always be checked in order that powerful lights visible in the distance not be mistaken for nearby lights showing similar characteristics at low intensity such as those on lighted buoys.

(164) The apparent characteristic of a complex light may change with the distance of the observer, due to color and intensity variations among the different lights of the group. The characteristic as charted and shown in the Light List may not be recognized until nearer the light.

(165) Motion of a vessel in a heavy sea may cause a light to alternately appear and disappear, and thus give a false characteristic.

(166) Where lights have different colored sectors, be guided by the correct bearing of the light; do not rely on being able to accurately observe the point at which the color changes. On either side of the line of demarcation of colored sectors there is always a small arc of uncertain color.

(167) On some bearings from the light, the range of visibility of the light may be reduced by obstructions. In such cases, the obstructed arc might differ with height of eye and distance. When a light is cut off by adjoining land and the arc of visibility is given, the bearing on which the light disappears may vary with the distance of the vessel from which observed and with the height of eye. When the light is cut off by a sloping hill or point of land, the light may be seen over a wider arc by a ship far off than by one closer.

(168) Arcs of circles drawn on charts around a light are not intended to give information as to the distance at which it can be seen, but solely to indicate, in the case of lights which do not show equally in all directions, the bearings between which the variation of visibility or obscuration of the light occurs.

(169) Lights of equal candlepower but of different colors may be seen at different distances. This fact should be considered not only in predicting the distance at which a light can be seen, but also in identifying it.

(170) Lights should not be passed close aboard, because in many cases riprap mounds are maintained to protect the structure against ice damage and scouring action.

(171) Many prominent towers, tanks, smokestacks, buildings and other similar structures, charted as landmarks, display flashing and/or fixed red aircraft obstruction lights. Lights shown from landmarks are charted only when they have distinctive characteristics to enable the mariner to positively identify the location of the charted structure.

(172)

Articulated Lights

(173) An articulated light is a vertical pipe structure supported by a submerged buoyancy chamber and attached by a universal coupling to a weighted sinker on the seafloor. The light, allowed to move about by the universal coupling, is not as precise as a fixed aid. However, it has a much smaller watch circle than a conventional buoy, because the buoyancy chamber tends to force the pipe back to a vertical position when it heels over under the effects of wind, wave or current.

(174) Articulated lights are primarily designed to mark narrow channels with greater precision than conventional buoys.

(175)

Daybeacons

(176) Daybeacons are unlighted aids affixed to stationary structures. They are marked with dayboards for daytime identification. The dayboards aid navigation by presenting one of several standard shapes and colors which have navigational significance. Dayboards are sometimes referred to as daymarks.

(177) Daybeacons are found on-shore and in shallow water. They are frequently used to mark channel edges.

(178)

Articulated Daybeacons

(179) Articulated daybeacons are similar to articulated lights, described above, except they are unlighted.

(180)

Buoys

(181) The aids to navigation depicted on charts comprise a system consisting of fixed and floating aids with varying degrees of reliability. Therefore, prudent mariners will not rely solely on any single aid to navigation, particularly a floating aid.

(182) The approximate position of a buoy is represented by the dot or circle associated with the buoy symbol. The approximate position is used because of practical limitations in positioning and maintaining buoys and their sinkers in precise geographical locations. These limitations include, but are not limited to, inherent imprecisions in position fixing methods, prevailing atmospheric and sea conditions, the slope of and the material making up the seabed, the fact that buoys are moored to sinkers by varying lengths of chain and the fact that buoy body and/or sinker positions are not under continuous surveillance, but are normally checked only during periodic maintenance visits which often occur more than a year apart. The position of the buoy body can be expected to shift inside and outside of the charting symbol due to the forces of nature. The mariner is also cautioned that buoys are liable to be carried away, shifted, capsized, sunk, etc. Lighted buoys may be extinguished or sound signals may not function as a result of ice, running ice or other natural causes, collisions or other accidents.

(183) For the foregoing reasons, a prudent mariner must not rely completely upon the charted position or operation of floating aids to navigation but will also utilize bearings from fixed objects and aids to navigation on shore. Further, a vessel attempting to pass close aboard always risks collision with a yawing buoy or with the obstruction the buoy marks.

(184) Buoys may not always properly mark shoals or other obstructions due to shifting of the shoals or of the buoys. Buoys marking wrecks or other obstructions are usually placed on the seaward or channelward side and not directly over a wreck. Since buoys may be located some distance

from a wreck they are intended to mark, and since sunken wrecks are not always static, extreme caution should be exercised when operating in the vicinity of such buoys.

(185)

Automatic Identification System (AIS) Aids to Navigation

(186) AIS is an automatic communication and identification system intended to improve the safety of navigation by assisting the efficient operation of a Vessel Traffic Services (VTS), ship reporting, ship-to-ship and ship-to-shore operations. AIS is increasingly being used as an aid to navigation. An AIS-equipped aid to navigation may provide a positive identification of the aid. It may also have the capability to transmit an accurate position and provide additional information such as actual tide height and/or weather information.

(187) The AIS message may represent an aid to navigation that physically exists (physical AIS Aid to Navigation) or the message, transmitted from a remote location, may represent an aid to navigation that does not physically exist (virtual AIS Aid to Navigation). A virtual aid to navigation is a digital information object promulgated by an authorized service provider that can be presented on navigational systems.

(188) Physical AIS aids to navigation are charted with the symbol for the physical aid (such as a buoy or light) with a magenta circle surrounding the symbol and labeled AIS. Virtual aids to navigation are charted with a small central dot with a topmark symbol indicating the purpose of the aid, surrounded by a magenta circle and labeled V-AIS. Temporary AIS aids to navigation and stations remotely transmitting an AIS signal are not charted. See U.S. Chart No. 1, Section S, for additional information and examples.

(190)

Bridge Lights and Clearance Gages

(191) The Coast Guard regulates marine obstruction lights and clearance gages on bridges across navigable waters. Where installed, clearance gages are generally vertical numerical scales, reading from top to bottom, and show the actual vertical clearance between the existing water level and the lowest point of the bridge over the channel; the gages are normally on the right-hand pier or abutment of the bridge, on both the upstream and downstream sides.

(192) Bridge lights are fixed red or green and are privately maintained; they are generally not charted or described in the text of the Coast Pilot. All bridge piers (and their protective fenders) and abutments that are in or adjacent to a navigation channel are marked on all channel sides by red lights. On each channel span of a fixed bridge, there is a range of two green lights marking the center of the channel and a red light marking both edges of the channel, except that when the margins of the channel are confined by bridge piers, the red lights on the span are omitted, since the pier lights then mark the channel edges. For multiplespan fixed bridges, the main-channel span may also be marked by three white lights in a vertical line above the green range lights.

(193) On all types of drawbridges, one or more red lights are shown from the drawspan (higher than the pier lights) when the span is closed; when the span is open, the higher red lights are obscured and one or two green lights are shown from the drawspan, higher than the pier lights. The number and location of the red and green lights depend upon the type of drawbridge.

(194) Bridges and their lighting, construction and maintenance are set forth in **33 CFR 114**, **115**, **116**, and **118** (not carried in this Coast Pilot). Aircraft obstruction lights prescribed by the Federal Aviation Administration may operate at certain bridges.

(195)

Sound Signals

(196) Caution should be exercised in the use of sound signals for navigation purposes. They should be considered solely as warning devices.

(197) Sound travels through the air in a variable manner, even without the effects of wind, and, therefore the hearing of sound signals cannot be implicitly relied upon.

(198) Experience indicates that distances must not be judged only by the intensity of the sound; that occasionally there may be areas close to a sound signal in which it is not heard; and that fog may exist not far from a station, yet not be seen from it, so the signal may not be operating. It is not always possible to start a sound signal immediately when fog is observed.

(199)

Channel Markers

(200) Lights, daybeacons, and buoys along dredged channels do not always mark the bottom edges. Due to local conditions, aids may be located inside or outside the channel limits shown by dashed lines on a chart. The Light List tabulates the offset distances for these aids in many instances.

(201) Aids may be moved, discontinued or replaced by other types to facilitate dredging operations. Mariners should exercise caution when navigating areas where dredges with auxiliary equipment are working.

(202) Temporary changes in aids are not included on the charts.

(203)

Light Lists

(204) The Coast Guard Light Lists are a means for communicating aids to navigation information to the maritime public. They are updated weekly and

available for download on the United States Coast Guard Navigation Center's website at *www.navcen.uscg.gov*. Mariners should refer to these lists for detailed information regarding the characteristics and visibility of lights, and the description of light structures, buoys, sound signals and electronic aids.

(205)

ELECTRONIC POSITIONING SYSTEMS

(206) **Global Positioning System (GPS)**permits land, sea, and airborne users to determine their three-dimensional position, velocity and time 24 hours a day, in all weather, anywhere in the world. The basic system is defined as a constellation of satellites, the navigation payloads which produce the GPS signals, ground stations, data links and associated command and control facilities, that are operated and maintained by the Department of Defense. Please report GPS problems or anomalies at *navcen.uscg.gov* or contact the USCG Navigation Information Service at 703–313–5900.

(207)

LORAN-C

(208) LORAN, an acronym for LOng RAnge Navigation, was an electronic aid to navigation consisting of shore-based radio transmitters. In accordance with the Department of Homeland Security Appropriations Act, the U.S. Coast Guard terminated the transmission of all LORAN-C signals as of August 2010, rendering them unusable and permanently discontinued. For more details, visit *navcen.uscg.gov*. The Coast Guard strongly urges mariners accustomed to using LORAN-C for navigation to shift to a GPS navigation system and become familiar with its operation. NOAA is removing LORAN-C lines of position from all of its charts as new editions are published.

(209)

SEARCH AND RESCUE

(210)

Coast Guard Search and Rescue

(211) The Coast Guard conducts and/or coordinates search and rescue operations for surface vessels or aircraft that are in distress or overdue. Search and rescue vessels and aircraft have special markings, including a wide slash of red-orange and a small slash of blue on the forward portion of the hull or fuselage. Other parts of aircraft, normally painted white, may have other areas painted red to facilitate observation. The cooperation of vessel operators with Coast Guard helicopters, fixed-wing aircraft, and vessels may mean the difference between life and death for some seaman or aviator; such cooperation is greatly facilitated by the prior knowledge on the part of vessel operators of the operational requirements of Coast Guard equipment and personnel, of the international distress signals and procedures and of good seamanship.

(212) <Deleted Paragraph>

(212)

Search and Rescue Great Lakes

(213) The United States Coast Guard has established a toll-free search and rescue telephone number for the Great Lakes. The number is intended for use when the telephone number of the nearest Coast Guard station is unknown or when that station cannot be contacted. The toll-free number should not be used without first attempting to contact the nearest Coast Guard station. In all Great Lakes States the telephone number is 800-321-4400. This number is to be used for public reports of distress incidents, suspicious sightings, pollution or other maritime concerns.

(214)

Radiotelephone Distress Message

(215) Distress calls indicate a vessel or aircraft is threatened by grave and imminent danger and requests immediate assistance. They have absolute priority over all other transmissions. All stations which hear a distress call must immediately cease any transmission capable of interfering with the distress traffic and continue to listen on the frequency used for the emission of the distress call. This call should not be addressed to a particular station, and acknowledgment of receipt should not be given before the distress message which follows it is sent.

(216) Distress calls are made on VHF-FM channel 16 (MAYDAY). For less serious situations than warrant the distress procedure, the radiotelephone urgency signal consisting of three repetitions of the word PAN-PAN (pronounced PAWN-PAWN), or the safety signal SECURITE (pronounced SECURITAY) spoken three times, are used as appropriate. For complete information on emergency radio procedures, see **47 CFR 80** or **Radio Navigational Aids, Pub. 117**.

(217)

Global Maritime Distress and Safety System (GMDSS)

(218) This international system, developed by the International Maritime Organization (IMO), is based on a combination of satellite and terrestrial radio services and has changed international distress communications from being primarily ship-to-ship based to primarily ship-to-shore (Rescue Coordination Center) based. Prior to the GMDSS, the number and types of radio safety equipment required to be carried by vessels depended upon the tonnage. Under GMDSS, the number and type of radio safety equipment vessels are required to carry depend on the areas in which they travel; GMDSS sea areas are defined by governments. All GMDSS-regulated ships must carry a satellite Emergency Position Indicating Radio Beacon (EPIRB), a NAVTEX receiver (if they travel in any areas served by NAVTEX), an Inmarsat-C SafetyNET receiver (if they travel in any areas not served by NAVTEX), a DSC-equipped VHF radiotelephone, two or more VHF handhelds and a search and rescue radar transponder (SART).

(219)

Automated Mutual Assistance Vessel Rescue System (AMVER)

(220) AMVER is a worldwide voluntary ship reporting system operated by the United States Coast Guard to promote safety of life and property at sea. AMVER's mission is to quickly provide search and rescue (SAR) authorities, on demand, accurate information on the positions and characteristics of vessels near a reported distress. Any merchant vessel anywhere on the globe, on a voyage of greater than 24 hours duration, is welcome in the AMVER system and family. International participation is voluntary regardless of the vessel's flag of registry, the nationality of the owner or company or ports of call.

(221) According to U.S. Maritime Administration (MARAD) regulations, U.S. flag merchant vessels of 1,000 gross tons or more operating in foreign commerce and foreign flag vessels of 1,000 gross tons or more for which an Interim War Risk Insurance Binder has been issued under the provisions of Title XII, Merchant Marine Act, 1936, must report and regularly update their voyages and positions to AMVER in accordance with instructions set forth in the AMVER Ship Reporting System Manual. For more information contact AMVER Maritime Relations U.S. Coast Guard, 1 South Street Battery Park Building, New York, NY 10004; Phone: 212–668–7764, Fax: 212-668-7684, Telex: 127594-AMVER NYK, or go to *amver.com*.

(222)

COSPAS-SARSAT

(223) COSPAS: Space System for Search of Distress Vessels - SARSAT: Search and Rescue Satellite-Aided Tracking. COSPAS-SARSAT is an international satellite system designed to provide distress alert and location data to assist search and rescue operations using satellites and ground facilities to detect and locate the signals of distress beacons operating on 406 MHz. For more information on the Cospas-Sarsat System go to *cospas-sarsat.int*.

(224)

Digital Selective Calling (DSC)

(225) The U.S. Coast Guard offers VHF and MF/HF radiotelephone service to mariners as part of the Global Maritime Distress and Safety System. This service, called digital selective calling (DSC), allows mariners to instantly send an automatically formatted distress alert to the Coast Guard or other rescue authority anywhere in the world. Digital selective calling also allows mariners to initiate or receive distress, urgency, safety and routine radiotelephone calls to or from any similarly equipped vessel or shore station, without requiring either party to be near a radio loudspeaker. Each ship or shore station equipped with a DSC terminal has a unique Maritime Mobile Station Identity (MMSI). This is a nine-digit number that specifically identifies a ship, coast station, or group of stations. The DSC system alerts an operator when a distress call is received. It will provide the operator with a pre-formatted message that can include the distressed vessel's nine-digit MMSI, location, nature of distress, desired mode of communication and preferred working frequency.

(226)

Emergency Position Indicating Radiobeacons (EPIRB)

(227) EPIRBs emit a radio signal that can be used to locate mariners in distress. SARSAT satellites can locate the position of a 406 MHz EPIRB which greatly increases a mariner's chances of survival. While orbiting the earth, the satellites continuously monitor EPIRB frequencies. When SARSAT receives an EPIRB signal, it determines the beacon's position that is ultimately relayed to the nearest Coast Guard Rescue Coordination Center where rescue units are dispatched to the scene.

(228) Mariners should ensure that their EPIRB is in working condition and stowed properly at all times to avoid non-distress emissions. Mariners are required to register their 406 MHz EPIRBs for improved search and rescue response and keep the registration current at all times. Registration can be accomplished online at *beaconregistration.noaa.gov*.

(229) <Deleted Paragraph>

(229)

EPIRB Types

Type	Frequency	Description
Cat I	406 MHz	Float-free, automatically activated EPIRB. Detectable by satellite anywhere in the world. Recognized by the Global Maritime and Distress Safety System (GMDSS).
Cat II	406 MHz	Similar to Category I, except is manually activated. Some models are also water activated.

(230) <Deleted Paragraph>

(230)

Medical Advice

(231) Ships at sea with no medical personnel embarked and experiencing a medical emergency onboard can receive medical advice via radiotelex, radiotelephony or Inmarsat. Messages are generally addressed RADIOMEDICAL followed by the name of the coast station to which the message is sent. The priority of the message should depend on the severity of the ailment. In extreme emergency, the urgency signal (PAN-PAN) should precede the address. Messages are sent using distress and safety frequencies.

(232)

Vessel Identification

(233) Coast Guard search and rescue aircraft and surface craft use radar to assist in locating disabled vessels. Wooden and fiberglass vessels are often poor radar targets. Operators of disabled craft that are the object of a search are requested to hoist, as high above the waterline as possible, a radar-reflecting device. If no special radar-reflecting device is aboard, an improvised device can be used. This should consist of metallic objects of irregular shape. The more irregular the shape, the better will be the

(241)

U.S. VHF Channel Information

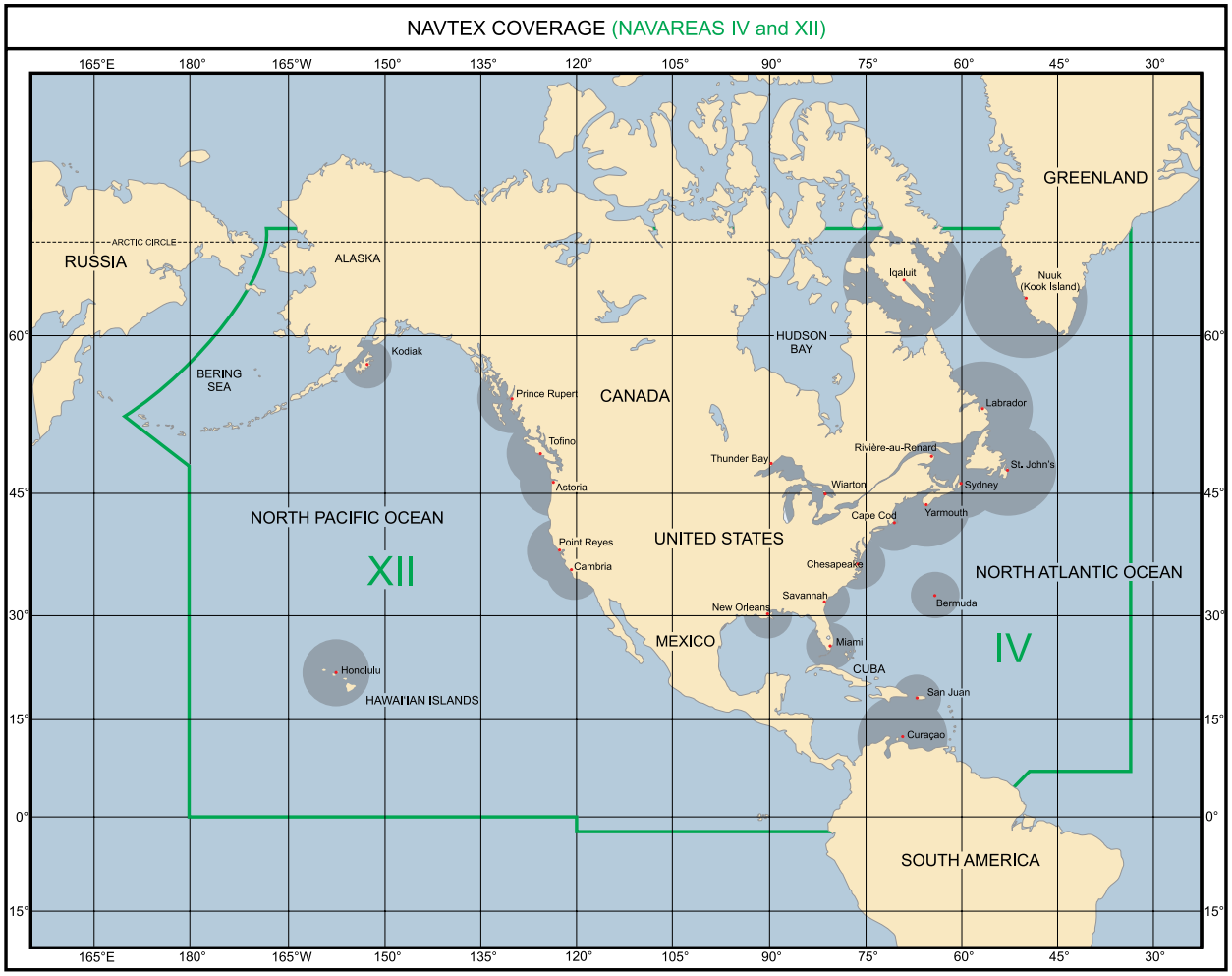
New Channel	Old Channel	Ship Frequency (MHz)		Channel Usage
		Transmit	Receive	
1001	01A	156.050	156.050	Port Operations and Commercial, VTS. Available only in New Orleans / Lower Mississippi area.
1005	05A	156.250	156.250	Port Operations or VTS in the Houston, New Orleans and Seattle areas
06	06	156.300	156.300	Intership Safety
1007	07A	156.350	156.350	Commercial. VDSMS
08	08	156.400	156.400	Commercial (Intership only) VDSMS
09	09	156.450	156.450	Boater Calling; Commercial and Non-commercial. VDSMS
10	10	156.500	156.500	Commercial. VDSMS
11	11	156.550	156.550	Commercial; VTS in selected areas. VDSMS
12	12	156.600	156.600	Port Operations; VTS in selected areas
13	13	156.650	156.650	Intership Navigation Safety (Bridge-to-Bridge). Ships greater than 20m maintain a listening watch on this channel in U.S. waters.
14	14	156.700	156.700	Port Operations; VTS in selected areas
15	15	- -	156.750	Environmental (Receive only) Used by Class C EPIRBs
16	16	156.800	156.800	International Distress, Safety and Calling. Ships required to carry radio, USCG, and most coast stations maintain a listening watch on this channel. *(Refer to: Radio Watchkeeping Regulations).*
17	17	156.850	156.850	State and local government maritime control
1018	18A	156.900	156.900	Commercial. VDSMS
1019	19A	156.950	156.950	Commercial. VDSMS
20	20	157.000	161.600	Port Operations (duplex)
1020	20A	157.000	157.000	Port Operations
1021	21A	157.050	157.050	U.S. Coast Guard only
1022	22A	157.100	157.100	Coast Guard Liaison and Maritime Safety Information Broadcasts. *(Broadcasts announced on Channel 16)*
1023	23A	157.150	157.150	U.S. Coast Guard only
24	24	157.200	161.800	Public Correspondence (Marine Operator). VDSMS
25	25	157.250	161.850	Public Correspondence (Marine Operator). VDSMS
26	26	157.300	161.900	Public Correspondence (Marine Operator). VDSMS
27	27	157.350	161.950	Public Correspondence (Marine Operator). VDSMS
28	28	157.400	162.000	Public Correspondence (Marine Operator). VDSMS
1063	63A	156.175	156.175	Port Operations and Commercial, VTS. Available only in New Orleans / Lower Mississippi area.
1065	65A	156.275	156.275	Port Operations
1066	66A	156.325	156.325	Port Operations
67	67	156.375	156.375	Commercial. Used for Bridge-to-Bridge communications in lower Mississippi River. (Intership only.)
68	68	156.425	156.425	Non-Commercial. VDSMS
69	69	156.475	156.475	Non-Commercial. VDSMS
70	70	156.525	156.525	Digital Selective Calling (voice communications not allowed)
71	71	156.575	156.575	Non-Commercial. VDSMS
72	72	156.625	156.625	Non-Commercial (Intership only). VDSMS
73	73	156.675	156.675	Port Operations
74	74	156.725	156.725	Port Operations
77	77	156.875	156.875	Port Operations (Intership only)
1078	78A	156.925	156.925	Non-Commercial. VDSMS
1079	79A	156.975	156.975	Commercial. Non-commercial in Great Lakes only. VDSMS
1080	80A	157.025	157.025	Commercial. Non-commercial in Great Lakes only. VDSMS
1081	81A	157.075	157.075	U.S. Government only - Environmental protection operations.
1082	82A	157.125	157.125	U.S. Government only
1083	83A	157.175	157.175	U.S. Coast Guard only
84	84	157.225	161.825	Public Correspondence (Marine Operator). VDSMS
85	85	157.275	161.875	Public Correspondence (Marine Operator). VDSMS
86	86	157.325	161.925	Public Correspondence (Marine Operator). VDSMS
87	87	157.375	157.375	Public Correspondence (Marine Operator). VDSMS
88	88A	157.425	157.425	Commercial (Intership only). VDSMS
AIS 1	AIS 1	161.975	161.975	Automatic Identification System (AIS)
AIS 2	AIS 2	162.025	162.025	Automatic Identification System (AIS)

Boaters should normally use channels listed as Non-Commercial. Channel 16 is used for calling other stations or for distress alerting. Channel 13 should be used to contact a ship when there is danger of collision. All ships of length 20m or greater are required to guard VHF channel 13, in addition to VHF channel 16, when operating within U.S. territorial waters. Users may be fined by the FCC for improper use of these channels. Frequencies are in megahertz (MHz). Modulation is 16KF3E or 16KG3E. VDSMS (VHF Digital Small Message Services). Transmissions of short digital messages in accordance with RTCM Standard 12301.1 is allowed. Four-digit VHF maritime channel numbers: US channel 05A is now designated and recognized internationally by the channel 1005, and the U.S. Coast Guard channel 22A is now designated and recognized internationally by the channel "1022". These new channel number designations are being displayed on new models of VHF marine radios.

Further information can be obtained by visiting the following:

U.S. VHF Channel Information - *https://www.navcen.uscg.gov/us-vhf-channel-information*
Radio Watchkeeping Regulations - *https://www.navcen.uscg.gov/radio-watchkeeping-regulations*
International Telecommunications Union (ITU) Radio Regulations Appendix 18 - *https://navcen.uscg.gov/international-vhf-marine-radio-channels-freq*

(247)



radar-reflective quality. For quick identification at night, shine spotlights straight up. If aircraft are involved, once you are identified, turn lights away so as not to blind aircraft crew.

(234)

Float Plan

(235) Small craft operators should prepare a float plan before starting a trip and leave it ashore with a yacht club, marina, friend or relative. It is advisable to regularly use a checking-in procedure by radio or telephone for each point specified in the float plan. A float plan is vital for determining if a boat is overdue and will assist in locating a missing vessel in the event search and rescue operations become necessary.

(236)

NAVIGATIONAL WARNINGS, INFORMATION AND WEATHER

(237) Marine radio warnings and weather are disseminated by many sources and through several types of transmissions. For complete information on radio warnings and weather, see **Radio Navigational Aids, Pub. 117** and the National Weather Service (NWS) publication **Worldwide Marine Radiofacsimile Broadcast Schedules**.

(238) Radio navigational warning broadcasts are designed to provide the mariner with up-to-date marine information vital to safe navigation. There are three types of broadcasts: coastal and local, long range and worldwide.

(239) Coastal and local warnings are generally restricted to ports, harbors and coastal waters and involve items of local interest. Usually, local or short-range warnings are broadcast from a single coastal station, frequently by voice and also radiotelegraph, to assist small craft operators in the area. The information is often quite detailed. Foreign area broadcasts are frequently in English as well as the native language. In the United States, short-range radio navigational warnings are broadcast by the U.S. Coast Guard Districts via NAVTEX and subordinate coastal radio stations.

(240) Long range warnings are intended primarily to assist mariners on the high seas by promulgating navigational safety information concerning port and harbor approaches, coastlines and major ocean areas. Long-range radio navigational warnings are usually broadcast by means of radiotelegraphy and in many instances by radio-teletypewriter. A NAVAREA system of navigational warning areas has been developed providing worldwide coverage using standard format and procedures. The U.S.

participates as Area Coordinator for both NAVAREA IV (Western North Atlantic) and NAVAREA XII (Eastern North Pacific).

(242) The United States also maintains worldwide coverage using the HYDROLANT/HYDROPAC Navigational Warning System outside of NAVAREAs IV and XII.

(243)

NAVTEX

(244) NAVTEX is a standard international method of broadcasting notices to mariners and marine weather forecasts using small, low cost receivers designed to be installed in the pilothouse of a vessel. NAVTEX receivers screen incoming messages, inhibiting those which had been previously received or are of a category not of interest to the user, and print the rest on adding machine-size paper. NAVTEX not only provides marine information previously available only to those knowledgeable in Morse code but also allows any mariner who cannot man a radio full time to receive safety information at any hour. All NAVTEX transmissions are made on 518 kHz. Mariners who do not have NAVTEX receivers but have Simplex Teletype Over Radio (SITOR) radio equipment can also receive these broadcasts by operating it in the Forward Error Correction (FEC) mode and tuning to 518 kHz.

(245) Information broadcast over NAVTEX includes offshore weather forecasts, offshore marine advisory warnings, search and rescue information and navigational information that applies to waters from the line of demarcation (separating Inland Rules from COLREG Rule waters) to 200 miles offshore. Navigational information that affects the safety of navigation of deep draft (15 feet or more) vessels within the U.S. Inland Rules waters will also be included. Gulf Stream location is also included from Miami and Portsmouth. Coastal and high seas weather forecasts are not being broadcast over NAVTEX. The Safety of Life at Sea Convention, as amended in 1988, requires vessels regulated by that convention to carry NAVTEX receivers.

(246) See Appendix A, U.S. NAVTEX Transmitting Stations, for a list of NAVTEX broadcast stations and message content covered by this Coast Pilot.

(248)

Broadcast Notice to Mariners

(249) The U.S. Coast Guard transmits **Urgent Marine Information Broadcast** (UMIB) safety warnings and communicates with mariners on VHF channel 1022 (previously channel 22A). These safety broadcasts contain information such as notices to mariners, storm warnings, distress warnings and other pertinent information that is vital for safe navigation. Following a preliminary call on VHF-FM channel 16 (156.800 MHz), mariners are instructed to shift to VHF-FM channel 1022 (157.100 MHz). A shipboard radio tuned to channel 1022 (U.S. mode) both transmits and listens on 157.100 MHz, and can receive UMIBs transmitted by the U.S Coast Guard. A shipboard radio tuned to channel 22 (international mode) transmits on 157.100 MHz and listens on 161.700 MHz, and cannot receive these safety warnings. In accordance with **33 CFR 26.03(d)**, "The radiotelephone required by (this Bridge-to-bridge radiotelephone regulation) must be capable of transmitting and receiving on VHF FM channel 22A (157.100 MHz)." This regulation applies to foreign ships in U.S. waters as well as to U.S. ships.

(250)

NOAA Weather Radio Broadcasts

(251) NOAA Weather Radio provides continuous broadcasts of the latest weather information directly from (NWS) offices. In addition to general weather information, marine weather is provided by stations along the sea coasts and the Great Lakes. During severe weather, NWS forecasters can interrupt the regular broadcasts and substitute special warning messages. The stations operate 24 hours daily, and messages are repeated every 4 to 6 minutes and are routinely revised every 1 to 3 hours or more frequently if necessary. The broadcasts are made on seven VHF-FM frequencies, 162.40 to 162.55 MHz. The 162.475 MHz frequency is only used in special cases where needed to avoid channel interference. They can usually be heard as far as 40 miles from the antenna site, sometimes more. The effective range depends on many factors, including the height of the broadcast antenna, terrain, quality of the receiver and the type of receiving antenna. As a general rule, listeners close to or perhaps beyond the 40 mile range should have a good quality receiver system to get reliable reception. (See Appendix A for a list of these stations in the area covered by this Coast Pilot.)

(252)

Commercial Maritime Coast Stations and Weather Nets

(253) Commercial maritime coast stations, which provide communications services, broadcast weather information to ships at sea as a public service, or make forecast information available on demand, either free or for a nominal fee. These transmissions are most commonly performed using HF SITOR and Pactor/E-Mail; however, several of these stations also offer services via Inmarsat satellite and other means.

(254) There are also a number of maritime weather *nets* operating on commercial marine VHF, MF and HF, where weather information is exchanged. These *nets* are extremely popular in areas of the world that have a large yachting population and where weather is dynamic, such as in the Caribbean, and typically incorporate volunteers ashore.

(255) Information on commercial maritime coast stations, including schedules and frequencies, is available in the **Radio Navigational Aids, Pub. 117**.

(256)

Standard Abbreviations for Broadcasts

(257) A listing of Standard Abbreviations for Textual Maritime Safety Broadcasts can be found in this chapter.

(258)

Standard Abbreviations Used in Broadcasts

Aids to Navigation

AERO RBN — Aeronautical Radiobeacon
ART DBN — Articulated Daybeacon
ART LT — Articulated Light
DESTR — Destroyed
DISCONTD — Discontinued
ESTAB — Established
ELB — Exposed Location Buoy
FOG SIG — Fog Signal Station
LNB — Large Navigation Buoy
LT — Light
LLNR — Light List Number
LBB — Lighted Bell Buoy
LB — Lighted Buoy
LGB — Lighted Gong Buoy
LHB — Lighted Horn Buoy
LWB — Lighted Whistle Buoy
ODAS — Ocean Data Acquisition System
PRIV MAINTD — Privately Maintained
RACON — Radar Beacon
RA REF — Radar Reflector
TRUB — Temporarily Replaced by Unlighted Buoy
TRLB — Temporarily Replaced by Lighted Buoy
WHIS — Whistle

Light Characteristics

AL — Alternating
CHAR — Characteristic
FL(2+1) — Composite Group-Flashing
OC(2+1) — Composite Group-Occulting
Q — Continuous Quick-Flashing
FFL — Fixed and Flashing
F — Fixed
FL(3) — Group-Flashing
OC(2) — Group-Occulting
IQ — Interrupted Quick-Flashing
ISO — Isophase
MO(A) — Morse Code
OC — Occulting
FL — Single-Flashing

Colors (Color refers to light characteristics of Aids to Navigation only)

B — Black
BU — Blue
G — Green
OR — Orange
R — Red
W — White
Y — Yellow

Organizations

CCGD(#) — Commander, Coast Guard District (#)
CG — Coast Guard
COE — Corps of Engineers
NGA — National Geospatial-Intelligence Agency
NOS — National Ocean Service
NWS — National Weather Service

Vessels

A/C — Aircraft
F/V — Fishing Vessel
LNG — Liquified Natural Gas Carrier
M/V — Motor Vessel*
P/C — Pleasure Craft
R/V — Research Vessel
S/V — Sailing Vessel

* M/V includes: Steam Ship, Container Vessel, Cargo Vessel, etc.

Compass Directions

N — North
S — South
E — East
W — West
NE — Northeast
NW — Northwest
SE — Southeast
SW — Southwest

Various

ANCH — Anchorage
ANCH PROHIB — Anchorage Prohibited
APPROX — Approximate
ATLC — Atlantic
AUTH — Authorized
AVG — Average
BRG — Bearing
BKW — Breakwater
BNM — Broadcast Notice to Mariners
CHAN — Channel
CFR — Code of Federal Regulations
CONT — Continue
DEG — Degrees (temp, geo-position)
DIA — Diameter
ED — Edition
EFF — Effect/Effective
ENTR — Entrance
EXPLOS ANCH — Explosive Anchorage
FM(S) — Fathoms
FT — Foot/Feet
HBR — Harbor
HT — Height
HZ — Hertz
HOR CL — Horizontal Clearance
HR — Hour
COLREGS — International Regulations for Preventing Collisions at Sea
KHZ — Kilohertz
KM — Kilometer
KT(S) — Knot(s)
LAT — Latitude
LNM — Local Notice to Mariners
LONG — Longitude
MAINTD — Maintained
MAX — Maximum
MHZ — Megahertz
MB — Millibar
MM — Millimeter
MIN — Minute (time, geo position)
MOD — Moderate
MT —Mountain, Mount
NM — Nautical Mile(s)
NTM — Notice to Mariners
OBSTR — Obstruction
OCCASION — Occasion/Occasionally
OPAREA — Operating Area
PAC — Pacific
PT(S) — Point(s)
POS — Position
PA —Position Approximate
PRES — Pressure
PRIV — Private/Privately
PROHIB — Prohibited
PUB — Publication
RGE — Range
REP — Reported
RESTR — Restricted
RK — Rock
ST — Saint
SEC — Second (time, geo position)
SIG STA — Signal Station
STA — Station
SM — Statute Mile(s)
S SIG STA — Storm Signal Station
TEMP — Temporary
TSTORM — Thunderstorm
THRU — Through
T — True
UNCOV — Uncovers
UTC — Universal Coordinate Time
UMIB — Urgent Marine Information Broadcast
VEL — Velocity
VERT CL — Vertical Clearance
VIS — Visibility
YD — Yard(s)
WARN — Warning
WX — Weather
WK — Wreck

These abbreviations were jointly approved by the U.S. Coast Guard, National Weather Service, National Geospatial-Intelligence Agency and the Radio Technical Commission for Maritime Services. In addition to appearing in radio broadcasts of the U.S. Coast Guard and National Weather Service, they appear in Notices to Mariners of the U.S. Coast Guard and National Geospatial-Intelligence Agency and in NAVTEX.

(259)

Voluntary Observing Ship Program (VOS)

(260) The Voluntary Observing Ship program is organized for the purpose of obtaining weather and oceanographic observations from moving ships. An international program under World Meteorological Organization auspices, the VOS has over 5000 vessels participating from 23 countries. Any vessel willing to take and transmit observations in marine areas can join the program. Weather observations are essential to meteorologists preparing weather forecasts for coastal, offshore and high seas areas. For more information on the VOS, including a comprehensive observing handbook, visit *vos.noaa.gov*.

(261)

National Institute of Standards and Technology (NIST)

(262) The National Institute of Standards and Technology maintains the standards for time and frequency for most users in the United States. NIST provides a variety of services designed to deliver time and frequency signals to the people who need them. The signals are broadcast via several mediums, including high and low frequency radio, the Internet and telephone lines. Broadcasts of time and frequency signals are made by stations operating in the part of the radio spectrum that is properly known as high frequency (HF) but is commonly called shortwave. Station WWV is located just north of Fort Collins, Colorado, and station WWVH is located on the island of Kaua'i, Hawaii. Both stations broadcast continuous time and frequency signals on 2.5, 5, 10 and 15 MHz; WWV also broadcasts on 20 MHz.

(263) **NIST Time and Frequency Services, Special Publication 432** gives a detailed description of the signals and services offered by NIST, how they work and how you can use them. The publication is available for download at *nist.gov/pml/div688/generalpubs.cfm*.

(264)

CAUTIONARY INFORMATION

(265)

Hurricanes and Tropical Storms

(266) Hurricanes, tropical storms and other major storms may cause considerable damage to marine structures, aids to navigation and moored vessels, resulting in submerged debris in unknown locations. Fixed aids to navigation may have been damaged or destroyed. Buoys may have been moved from charted positions, damaged, sunk, extinguished or otherwise made inoperative. Mariners should not rely upon the position or operation of an aid to navigation. Charted soundings, channel depths and shoreline may not reflect actual conditions following these storms. Wrecks and submerged obstructions may have been displaced from charted locations. Pipelines may have become uncovered or moved. Mariners are urged to exercise extreme caution and are requested to report aids to navigation discrepancies and hazards to navigation to the U.S. Coast Guard.

(267)

Destructive Waves

(268) Unusual sudden changes in water level can be caused by tsunamis or violent storms. These two types of destructive waves have become commonly known as **tidal waves,** a name which is technically incorrect as they are not the result of tide-producing forces.

(269) **Tsunamis** (seismic sea waves) are ocean waves generated by any rapid large-scale disturbance of the sea water. Most tsunamis are generated by earthquakes, but they may also be caused by volcanic eruptions, landslides, undersea slumps or meteor impacts.

(270) The waves radiate outward in all directions from the disturbance and can propagate across entire ocean basins. Tsunami waves are distinguished from ordinary ocean waves by their great length between peaks, often exceeding 100 miles in the deep ocean, and by the long interval of time between these peaks, ranging from five minutes to an hour. The speed at which tsunamis travel depends on the ocean depth. A tsunami can exceed 500 knots in the deep ocean but slows to 20 or 30 knots in the shallow water near land. In less than 24 hours, a tsunami can cross the entire Pacific Ocean.

(271) In the deep ocean, a tsunami is barely noticeable and will only cause a small and slow rising and falling of the sea surface as it passes. Only as it approaches land does a tsunami become a hazard. As the tsunami approaches land and shallow water, the waves slow down and become compressed, causing them to grow in height. In the best of cases, the tsunami comes onshore like a quickly rising tide and causes a gentle flooding of low-lying coastal areas. In the worst of cases, a bore will form.

(272) A bore is a wall of turbulent water that can exceed several yards in height and can rush onshore with great destructive power. Behind the bore is a deep and fast-moving flood that can pick up and sweep away almost anything in its path. Minutes later, the water will drain away as the trough of the tsunami wave arrives, sometimes exposing great patches of the sea floor, then the water will rush in again as before, causing additional damage. This destructive cycle may repeat many times before the hazard finally passes. Sometimes the first noticeable part of the wave is the trough, which causes a recession of the water from shore, and people who have gone out to investigate this unusual exposure of the beach have been engulfed by the oncoming crest. Such an unexplained withdrawal of the sea should be considered as nature's warning of an approaching wave.

(273) Tsunamis do not have a season and do not occur regularly or frequently. Yet they pose a major threat to the coastal populations of the Pacific and other world oceans and seas. Nothing can be done to prevent them, but their adverse impact can be reduced with proper planning. The loss of life and property can be lessened if shipmasters and others acquaint themselves with the behavior of these waves so that intelligent action can be taken when they become imminent.

(274) <Deleted Paragraph>

(274) NOAA oversees the U.S. Tsunami Program with its mission to provide a 24-hour detection and warning system and increase public awareness about the threat of tsunamis. The NOAA National Weather Service operates two tsunami warning centers The West Coast/Alaska Tsunami Warning Center in Palmer, Alaska, and the Richard H. Hagemeyer Pacific Tsunami Warning Center in ‘Ewa Beach, Hawaii: *www.tsunami.gov*. These centers continuously monitor data from seismological and tidal stations, evaluate earthquakes that have the potential to generate tsunamis and disseminate tsunami information and warning bulletins to government authorities and the public.

(275) A tsunami warning is issued when a potential tsunami with significant inundation is imminent or expected. Warnings alert the public that widespread, dangerous coastal flooding accompanied by powerful currents is possible and may continue for several hours after arrival of the initial wave. Warnings also alert emergency management officials to take action for the entire tsunami hazard zone. When a tsunami warning has been issued, use a NOAA Weather Radio or stay tuned to a Coast Guard emergency frequency station or a local radio or television station for updated emergency information.

(276)

Storm Surge

(277) <Deleted Paragraph>

(277) A considerable rise or fall in the level of the sea along a particular coast may result from strong winds and sharp change in barometric pressure. In cases where the water level is raised, higher waves can form with greater depth, and the combination can be destructive to low regions, particularly at high stages of tide. Extreme low levels can result in depths which are considerably less than those shown on nautical charts. This type of wave occurs especially in coastal regions bordering on shallow waters which are subject to tropical storms.

(278) **Seiche** is a stationary vertical wave oscillation with a period varying from a few minutes to an hour or more but somewhat less than the tidal periods. It is usually attributed to external forces such as strong winds, changes in barometric pressure, swells or tsunamis disturbing the equilibrium of the water surface. Seiche is found both in enclosed bodies of water and superimposed upon the tides of the open ocean. When the external forces cause a short-period horizontal oscillation on the water, it is called **surge**.

(279) <Deleted Paragraph>

(279) The combined effect of seiche and surge sometimes makes it difficult to maintain a ship in its position alongside a pier even though the water may appear to be completely undisturbed, and heavy mooring lines have been parted repeatedly under such conditions. Pilots advise taut lines to reduce the effect of the surge.

(280) <Deleted Paragraph>

(280)

Immersion Hypothermia

(281) Immersion hypothermia is the loss of heat when a body is immersed in water. With few exceptions, humans die if their core temperature of approximately 99.7° F drops below 78.6° F. Cardiac arrest is the most common direct cause of death. During prolonged immersion, the main threat to life is cold or cold and drowning combined.

(282)

SURVIVAL TIME VERSUS WATER TEMPERATURE

Water Temperature (°F)	Exhaustion or Unconsciousness	Expected Time of Survival
32	15 minutes	15 to 45 minutes
32 to 41	15-30 minutes	30 to 90 minutes
41 to 50	30-60 minutes	1 to 3 hours
50 to 59	1-2 hours	1 to 6 hours
59 to 68	2-7 hours	2 to 40 hours
68 to 77	3-12 hours	3 hours to indefinite
77 and above	indefinite	indefinite

(283) The length of time that a human survives in water depends on the water temperature and to a lesser extent on the person’s behavior and body type. The table shows approximate human survival time in the sea. Body type can cause deviations, as small people become hypothermic more rapidly than large people. The cooling rate can be slowed by the person’s behavior and insulated gear. The Heat Escape Lessening Posture (HELP) was developed for those in the water alone and the huddle for small groups. Both require a PFD (personal flotation device), or life preserver. HELP involves holding the arms close to the body, keeping the thighs together, and raising the knees to protect the groin area. In the huddle, people face each other and keep their bodies as close together as possible. These positions improve survival time to approximately two times that of a swimmer and one and a half times that of a person in the passive position.

(284) Near-drowning victims in cold water (less than 70° F) are revivable for much longer periods than usual. Keys to a successful revival are immediate cardiopulmonary resuscitation (CPR) and administration of pure oxygen. Total re-warming is not necessary at first. The whole revival process may take hours and require medical help.

(285)

Wind Chill and Frostbite

(286) When the body is warmer than its surroundings, it begins to lose heat. The rate of loss depends on barriers such as clothing and insulation, the speed of air movement

and air temperature. Heat loss increases dramatically in moving air that is colder than skin temperature (91.4° F). Even a light wind increases heat loss, and a strong wind can lower the body temperature if the rate of loss is greater than the body's heat replacement rate.

(287) When skin temperature drops below 50° F, there is a marked constriction of blood vessels, leading to vascular stagnation, oxygen want and cellular damage. The first indication that something is wrong is a painful tingling. Swelling of varying extent follows, provided freezing has not occurred. Excruciating pain may be felt if the skin temperature is lowered rapidly, but freezing of localized portions of the skin may be painless when the rate of change is slow. Possible effects of cold include cold allergy (welts), chilblains, which appear as reddened, warm, itching, swollen patches on the fingers and toes, and trench foot and immersion foot, which present essentially the same picture. Both result from exposure to cold and lack of circulation. Wetness can add to the problem as water and wind soften the tissues and accelerate heat loss.

(288) Frostbite usually begins when the skin temperature falls within the range of 14° to 4° F. Ice crystals form in the tissues and small blood vessels. The rate of heat loss determines the rate of freezing, which is accelerated by wind, wetness, extreme cold and poor blood circulation. Parts of the body susceptible to freezing are those with surfaces large in relation to their volume, such as toes, fingers, ears, nose, chin and cheeks.

(289) Injuries from the cold may, to a large extent, be prevented by maintaining natural warmth through the use of proper footgear and adequate, dry clothing, by avoiding cramped positions and constricting clothing and by active exercise of the hands, legs and feet.

(290)

MARINE POLLUTION

(290) <Deleted Paragraph>

(291)

The Federal Water Pollution Control Act (Clean Water Act)

(291) <Deleted Paragraph>

(292) The Federal Water Pollution Control Act (FWPCA) or Clean Water Act (CWA) was passed to restore and maintain the chemical, physical and biological integrity of the waters within the United States.

(293)

No-Discharge Zones

(293) <Deleted Paragraph>

(294) Section 312 of the FWPCA gives the Environmental Protection Agency (EPA) and States the authority to designate certain areas as No-Discharge Zones (NDZ) for vessel sewage. Freshwater lakes, freshwater reservoirs or other freshwater impoundments whose entrances and exits prohibit traffic by regulated vessels (vessels with installed toilets) are, by regulation, NDZs. Rivers that do not support interstate navigation vessel traffic are also NDZs by regulation. Water bodies that can be designated as NDZs by States and EPA include the Great Lakes and their connecting waterways, freshwater lakes and impoundments accessible through locks and other flowing waters that support interstate navigation by vessels subject to regulation.

(295) Inside NDZ waters, discharge of any sewage, whether treated or untreated, is completely prohibited.

(296) Discharge of sewage in waters not designated as NDZs is regulated by the Marine Sanitation Device Standard (see **40 CFR 140** in chapter 2.)

(297) Additional information concerning the regulations may be obtained from *water.epa.gov*.

(298)

Oil Spill Reporting

(299) Reporting requirements for any oil discharge, noxious liquid substance or harmful substance occurring in waters under U.S. jurisdiction are found in **33 CFR 153**, subpart B (not in this Coast Pilot.) Any person in charge of a vessel or an onshore/offshore facility must, as soon as they have knowledge of any discharge of oil or a hazardous substance, immediately notify the National Response Center (NRC) at 800-424-8802 or NRC@uscg.mil.

(300)

Ocean Dumping

(301) The Marine Protection Research and Sanctuaries Act of 1972, as amended (33 USC 1401 et seq.), regulates the dumping of all material, except fish waste, into ocean waters. Radiological, chemical and biological warfare agents and other high level radioactive wastes are expressly banned from ocean disposal. The USACE issues permits for the disposal of dredged spoils; the EPA is authorized to issue permits for all other dumping activities. Surveillance and enforcement to prevent unlawful transportation of material for dumping or unlawful dumping under the Act has been assigned to the U.S. Coast Guard. The Act provides civil penalties of up to $50,000 and criminal penalties of up to $50,000 and/or one year imprisonment.

(301) <Deleted Paragraph>

(302)

SELECT NAVIGATION RULES

(303)

Improper use of searchlights

(304) No person shall flash or cause to be flashed the rays of a searchlight or other blinding light onto the bridge or into the pilothouse of any vessel underway. The International Code Signal "PG2" may be made by a vessel inconvenienced by the glare of a searchlight in order to apprise the offending vessel of the fact.

(305)

Use of Radar

(306) <Deleted Paragraph>

(306) Navigation Rules, International-Inland, Rule 7, states, in part, that every vessel shall use all available

means appropriate to the prevailing circumstances and conditions to determine if risk of collision exists. If there is any doubt such risk shall be deemed to exist. Proper use shall be made of radar equipment if fitted and operational, including long-range scanning to obtain early warning of risk of collision and radar plotting or equivalent systematic observation of detected objects.

(307) <Deleted Paragraph>

(307) This rule places an additional responsibility on vessels that are equipped and manned to use radar to do so while underway during periods of reduced visibility without in any way relieving commanding officers of the responsibility of carrying out normal precautionary measures.

(308) <Deleted Paragraph>

(308) Navigation Rules, International-Inland, Rules 6, 7, 8, and 19 apply to the use of radar.

(309)

Danger signal

(310) Navigation Rules, International-Inland, Rule 34(d), states that when vessels in sight of one another are approaching each other and from any cause either vessel fails to understand the intentions or actions of the other or is in doubt whether sufficient action is being taken by the other to avoid collision, the vessel in doubt shall immediately indicate such doubt by giving at least five short and rapid blasts on the whistle. Such signal may be supplemented by a light signal of at least five short and rapid flashes.

(311)

Narrow channels

(312) Navigation Rules, International-Inland, Rule 9(b) states that a vessel of less than 20 meters in length or a sailing vessel shall not impede the passage of a vessel that can safely navigate only within a narrow channel or fairway.

(313)

REGULATED WATERS

(314)

Traffic Separation Schemes (Traffic Lanes)

(315) To increase the safety of navigation, particularly

3 Converule 9(b)

variety of circumstances, such as with bays, ports, mouths of rivers, deeply indented coastlines, fringing reefs, and roadsteads. (LOSC arts. 6-15). Consistent with these rules, the U.S. baselines are the mean of the lower low tides as depicted on the largest-scale NOAA nautical charts. The U.S. normal baselines are ambulatory and subject to changes as the coastline accretes and erodes. NOAA's nautical charts depict the baselines from which the seaward limits of the U.S. territorial sea, contiguous zone, and exclusive economic zone are measured as well as the seaward boundary of the Three Nautical Mile Line and the Natural Resources Boundary, as described below.

(325)

Internal Waters

(326) Internal waters are the waters on the landward side of the baselines from which the breadth of the territorial sea is measured. (LOSC art. 8). The United States has full sovereignty over its internal waters as if they were part of its land territory and may exclude foreign flag vessels from its internal waters subject to the right of entry of vessels in distress. The right of innocent passage does not apply in internal waters. Ships and aircraft may not enter or overfly internal waters without permission of the coastal State. Examples of internal waters include rivers, harbors, lagoons, some bays and canals, and lakes, including the Great Lakes.

(327)

Territorial Sea

(328) Each coastal State may claim a territorial sea that extends seaward up to 12 nautical miles (nm) from its baselines. (LOSC arts. 3, 4). The coastal State exercises sovereignty over its territorial sea, the airspace above it, and the seabed and subsoil beneath it. (LOSC art. 2). Foreign flagged ships enjoy the right of innocent passage while transiting the territorial sea subject to laws and regulations adopted by the coastal State that are in conformity with the **Law of the Sea Convention** (LOSC arts. 17-26) and other rules of international law relating to such passage.

(329) In 1988, the United States claimed a 12 nm territorial sea. The extension of the territorial sea of the United States includes the belt of seas around the Commonwealth of Puerto Rico, Guam, American Samoa, the U.S. Virgin Islands, the Commonwealth of the Northern Mariana Islands and any other territory or possession over which the United States exercises sovereignty. (Presidential Proclamation No. 5928 of December 27, 1988, 54 Fed. Reg. 777 (Dec. 27, 1988)). The territorial sea of the United States extends seaward to 12 nm from the baselines, which is determined in accordance with the Law of the Sea Convention except as otherwise established in a maritime boundary treaty of the United States. Vessels of all States navigating through the territorial sea enjoy the right of innocent passage. (LOSC art. 17). However, as a coastal State, the United States may adopt certain laws and regulations relating to innocent passage so long as they are in conformity with the provisions of LOSC and other rules of international law. (LOSC art. 21(1)).

(330)

Contiguous Zone

(331) Each coastal State may claim a contiguous zone adjacent to and beyond its territorial sea that extends seaward up to 24 nm from its baselines. (LOSC art. 33(1) & (2)). In its contiguous zone, a coastal State may exercise the control necessary to prevent the infringement of its customs, fiscal, immigration, or sanitary laws and regulations within its territory or territorial sea, and punish infringement of those laws and regulations committed within its territory or territorial sea. (LOSC art. 33(1)

(324)

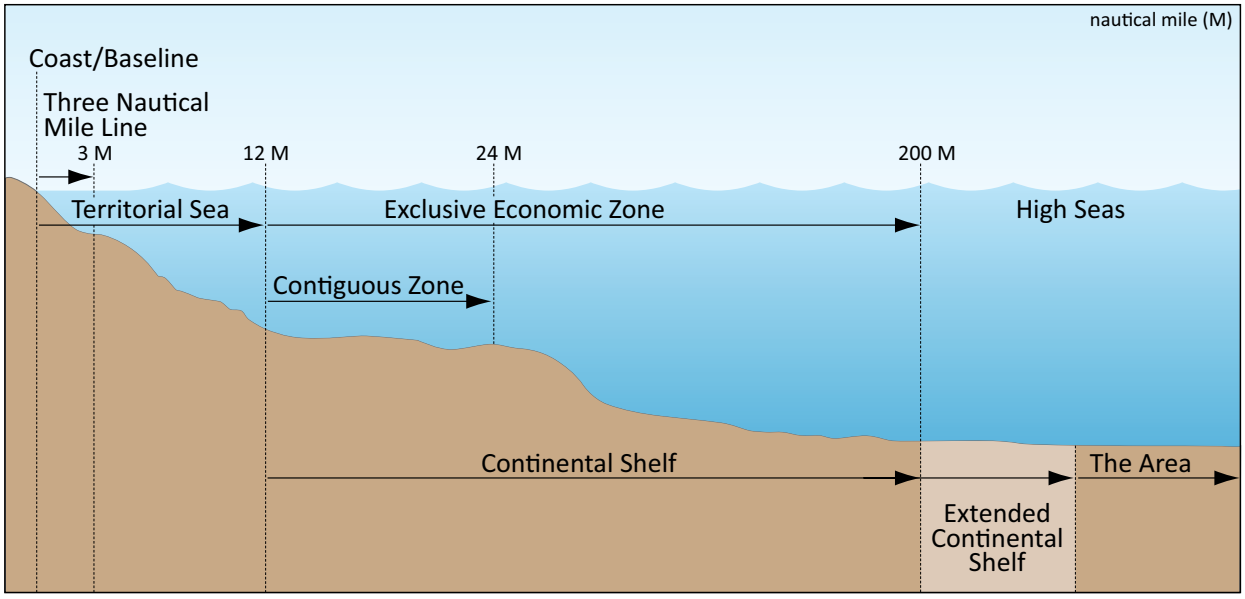


Figure 1: Offshore extent of maritime zones beyond internal waters
https://www.noaa.gov/maritime-zones-and-boundaries

(a) & (b)). In addition, in order to control traffic in archeological and historical objects, a coastal State may presume that the removal of archeological and historical objects (e.g., underwater cultural heritage) found at sea within its contiguous zone without its approval would result in an infringement of its laws and regulations. (LOSC art. 303).

(332) The contiguous zone of the United States includes the waters off of all U.S. coastal states, the Commonwealth of Puerto Rico, Guam, American Samoa, the U.S. Virgin Islands, the Commonwealth of the Northern Mariana Islands and any other territory or possession over which the United States exercises sovereignty. In 1999, the United States claimed a contiguous zone extending from 12 to 24 nm offshore. (Presidential Proclamation No. 7219 of August 2, 1999, Contiguous Zone of the United States, 64 Fed. Reg. 48,701 (Aug. 8, 1999)).

(333)

Exclusive Economic Zone

(334) Each coastal State may claim an exclusive economic zone (EEZ) beyond and adjacent to its territorial sea that extends seaward up to 200 nm from its baselines (or out to a maritime boundary with another coastal State). (LOSC art. 55). Within its EEZ, a coastal State has: (a) sovereign rights for the purpose of exploring, exploiting, conserving, and managing natural resources, whether living or nonliving, of the seabed and subsoil and the superjacent waters and with regard to other activities for the economic exploitation and exploration of the zone, such as the production of energy from the water, currents, and winds; (b) jurisdiction as provided for in the relevant provisions of the LOSC with regard to the establishment and use of artificial islands, installations, and structures, marine scientific research, and the protection and preservation of the marine environment; and (c) other rights and duties provided for in the LOSC. (LOSC art. 56).

(335) The United States claimed a 200 nm EEZ in 1983. The U.S. EEZ extends no more than 200 nm from the territorial sea baselines and is adjacent to the 12 nm territorial sea of the United States, including the Commonwealth of Puerto Rico, Guam, American Samoa, the U.S. Virgin Islands, the Commonwealth of the Northern Mariana Islands and any other territory or possession over which the United States exercises sovereignty. (Presidential Proclamation No. 5030 of March 10, 1983; Exclusive Economic Zone and Maritime Boundaries; Notice of Limits; 60 Fed. Reg. 43,825 (Aug. 23, 1995)). As such, the exclusive economic zone overlaps the 12 nm-24 nm contiguous zone. In December 2023, the U.S. Department of State released the geographic coordinates defining the outer limits of the U.S. extended continental shelf. (U.S. Dep't of State, Announcement of U.S. Extended Continental Shelf Outer Limits (Dec. 19, 2023); see also Continental Shelf and Maritime Boundaries; Notice of Limits; 88 Fed. Reg. 88,470 (Dec. 21, 2023)). In addition, the U.S. Department of State also updated information pertaining to the geographic coordinates defining the outer limits of the U.S. EEZ. (Exclusive Economic Zone and Maritime Boundaries; Notice of Limits; 88 Fed. Reg. 88,477 (Dec. 21, 2023)).

(336) Note: Under certain U.S. fisheries laws, such as the Magnuson-Stevens Fishery Conservation and Management Act, the term EEZ is defined as having an inner boundary that is coterminous with the seaward (or outer) boundary of each of the individual coastal states of the U.S. See 16 U.S.C. § 1802(11). Under the Submerged Lands Act, the seaward boundary of each of the individual coastal states is generally three nautical (or geographic) miles from the coast line. The seaward boundaries of Florida (Gulf of America coast only), Texas, and Puerto Rico extend nine nautical miles from the coast line. In the Great Lakes, each U.S. state's seaward boundary may extend to the international maritime boundary with Canada. See 43 U.S.C. § 1312. Under the Submerged Lands Act, a coastal state's seaward boundary may be fixed by Supreme Court decree. (See below for further information on the Three Nautical Mile Line and the Natural Resources Boundary).

(337)

Continental Shelf

(338) The continental shelf of a coastal State is comprised of the seabed and subsoil of the submarine areas that extend beyond its territorial sea throughout the natural prolongation of its land territory to the outer edge of the continental margin, or to a distance of 200 nm from its baselines where the outer edge of the continental margin does not extend up to that distance. (LOSC art. 76(1)). The extent of the continental shelf can also be limited by a maritime boundary with another coastal State. (LOSC art. 76(10)).

(339) Where the outer edge of a coastal State's continental margin extends beyond 200 nm from its baselines, the outer limits of its continental shelf are determined in accordance with Article 76 paragraphs 2-7 of the **Law of the Sea Convention.** (LOSC art. 76 (2-7)). The portion of a coastal State's continental shelf that lies beyond the 200 nm limit is often called the extended continental shelf (ECS).

(340) A coastal State exercises sovereign rights and exclusive jurisdiction over its continental shelf for the purpose of exploring it and exploiting its natural resources, as well as for other purposes specified in the Law of the Sea Convention. The natural resources of the continental shelf consist of the mineral and other non-living resources of the seabed and subsoil together with living organisms belonging to sedentary species, that is to say, organisms which, at the harvestable stage, either are immobile on or under the seabed or are unable to move except in constant physical contact with the seabed or subsoil. (LOSC art. 77).

(341) In 1945, the United States proclaimed jurisdiction and control over its continental shelf (Presidential Proclamation No. 2667 of Sept. 28, 1945; 10 Fed. Reg.

12,303 (Oct. 2, 1945)). Consistent with international law, the United States exercises its continental shelf rights out to a distance of at least 200 nautical miles from the baselines through several domestic laws. The U.S. Extended Continental Shelf Project, led by the U.S. Department of State, NOAA, and the U.S. Geological Survey, determines the outer limits of the U.S. continental shelf beyond 200 nautical miles (i.e., extended continental shelf). In December 2023, the U.S. Department of State released the geographic coordinates defining the outer limits of the U.S. extended continental shelf. (U.S. Dep't of State, Announcement of U.S.Extended Continental Shelf Outer Limits (Dec. 19, 2023); see also Continental Shelf and Maritime Boundaries; Notice of Limits; 88 Fed. Reg. 88,470 (Dec. 21, 2023)).

(342)

High Seas

(343) The high seas comprise all parts of the sea that are not included in the exclusive economic zone, the territorial sea or the internal waters of a State, or in the archipelagic waters of an archipelagic State. (LOSC art. 86).

(344)

Area

(345) The Area is comprised of the seabed and ocean floor and subsoil thereof beyond the limits of national jurisdiction. (LOSC art. 1(1)). It does not include superjacent waters (i.e., the water column) or the air space above those waters. (LOSC arts. 1(1), 135). No State may claim or exercise sovereignty or sovereign rights over any part of the Area or its resources. (LOSC art. 137(1)). The term "resources" means all solid, liquid or gaseous mineral resources in situ in the Area at or beneath the seabed, including polymetallic nodules. (LOSC art. 133(a)).

(346)

Straits Used for International Navigation

(347) "Straits used for international navigation" are those that are used or are capable of use for international navigation between one area of the high seas or exclusive economic zone ("EEZ") and another area of the high seas or EEZ. (LOSC art. 37). Part III of the Law of the Sea Convention (LOSC arts. 34-45) describes the regime of transit passage through such straits and the rights, jurisdiction, and duties of the States bordering such straits. Transit passage means the exercise in accordance with Part III of the Law of the Sea Convention of the freedom of navigation and overflight solely for the purpose of continuous and expeditious transit of the strait. (LOSC arts. 37, 38). The right of transit passage applies throughout straits used or capable of use for international navigation, including to all normally used approaches to and from such straits. Ships and aircraft in transit passage must comply with the duties outlined in LOSC article 39, which include proceeding without delay and refraining from any activities other than those incident to their normal modes of continuous and expeditious transit. Ships in transit passage may not carry out any research or survey activities without the prior authorization of the States bordering the strait. (LOSC art. 40). States bordering straits used for international navigation may designate sea lanes and prescribe traffic separation schemes for navigation in accordance with Part III where necessary to promote safe passage of ships. (LOSC art. 41). They may also adopt laws and regulations relating to transit passage in respect of certain activities, such as fishing. (LOSC art. 42). States bordering straits used for international navigation may not hamper transit passage. (LOSC art. 44). The transit passage regime does not otherwise affect the legal status of the waters forming an international strait or the exercise of sovereignty or jurisdiction by the bordering States over the waters, air space, seabed, and subsoil of the strait. (LOSC art. 34).

(348)

Three Nautical Mile Line

(349) The Three Nautical Mile Line, as measured from the territorial sea baselines and previously identified as the outer limit of the U.S. territorial sea, is retained on NOAA charts because it continues to be used in certain federal laws.

(350) Note: Since the "coast line," a term used in the Submerged Lands Act (43 USC Section 1301 et seq.), and the baselines are determined using the same criteria under international law, the Three Nautical Mile Line is generally the same as the seaward boundaries of individual U.S. coastal states under the Submerged Lands Act. There are exceptions; therefore, the Three Nautical Mile Line does not necessarily depict the seaward boundaries of all U.S. coastal states in all circumstances under the Submerged Lands Act.

(351)

Natural Resources Boundary

(352) The nine (9) nm Natural Resources Boundary is the seaward boundary of the submerged lands of Puerto Rico, Texas and the Gulf coast of Florida. It coincides with the inner limit of the U.S. outer continental shelf under the Outer Continental Shelf Lands Act (43 U.S.C. Section 1331 et seq.).

(353)

SUPPLEMENTAL INFORMATION

(354)

Notification of Arrival and Vessel Response Plans

(355) A **Notif cation of Arrival (NOA)** must be submitted by all U.S. and foreign vessels bound for or departing from ports or places in the United States. (See **33 CFR 160 – Subpart C**, chapter 2). Additionally, tank vessels and non-tank vessels are required to submit an oil spill response plan. (See **33 CFR 155**– Subparts D and J, not contained in this Coast Pilot.)

(356)

Marine Protected Area (MPA)

(357) Marine Protected Areas (MPAs) are particular places in ocean, coastal and estuarine ecosystems where vital

natural and cultural resources are given greater protection than in surrounding waters. MPAs have been established in the U.S. for more than a century. Currently, there are over 1,700 MPAs in U.S. marine waters and the Great Lakes, with levels of protection ranging from a few "no-take" areas that prohibit all extractive uses to the more common multiple use areas that allow vessel access, anchoring, fishing and non-consumptive activities. MPAs are managed by dozens of Federal, state, tribal and local authorities. For detailed information on MPA locations, types, interactive map, purposes and legal restrictions, visit *marineprotectedareas.noaa.gov.*

(358)

Archaeological Resource Preservation

(359) Under Federal and state laws it is illegal to destroy, deface, collect, transport, sell or trade archaeological, cultural, submerged and historic resources without authorization. Applicable laws include, but are not limited to, the Historic Sites Act, the Archaeological Resource Protection Act, the National Historic Preservation Act the Abandoned Shipwreck Act, and the Sunken Military Craft Act. These laws protect archaeological resources on lands administered by the National Park Service, U.S. Fish and Wildlife Service, Bureau of Land Management, and National Marine Sanctuaries as well as state, private and Native lands.

(360)

DEPARTMENT OF AGRICULTURE

(361)

Animal and Plant Health Inspection Service

(362) The Animal and Plant Health Inspection Service is responsible for protecting the Nation's animal population, food and fiber crops and forests from invasion by foreign pests. They administer agricultural quarantine and restrictive orders issued under authority provided in various acts of Congress. The regulations prohibit or restrict the importation or interstate movement of live animals, meats, animal products, plants, plant products, soil, injurious insects, and associated items that may introduce or spread plant pests and animal diseases which may be new to or not widely distributed within the United States or its territories. Inspectors examine imports at ports of entry as well as the vessel, its stores and crew or passenger baggage.

(363) The Service also provides an inspection and certification service for exporters to assist them in meeting the quarantine requirements of foreign countries. (See **Appendix A** for a list of ports where agricultural inspectors are located and inspections conducted.)

(363) <Deleted Paragraph>

(364)

DEPARTMENT OF COMMERCE

(365)

National Oceanic and Atmospheric Administration (NOAA)

(366) The National Oceanic and Atmospheric Administration (NOAA) conducts research and gathers data about the global oceans, atmosphere, space and sun, and applies this knowledge to improve our understanding and stewardship of the environment.

(367) NOAA provides services to the nation and the public through five major organizations: the National Ocean Service; the National Weather Service; the National Marine Fisheries Service; the National Environmental Satellite, Data and Information Service (NESDIS); and NOAA Research; and numerous special program units. In addition, NOAA research and operational activities are supported by the Nation's seventh uniformed service, the NOAA Corps, a commissioned officer corps of men and women who operate NOAA ships and aircraft and serve in scientific and administrative positions.

(368)

National Ocean Service (NOS)

(369) The National Ocean Service's primary concern is the health and safety of our Nation's coastal and oceanic environment. Within NOS, the **O f ce of Coast Survey** is responsible for producing and maintaining the suite of over 1000 nautical charts and the Coast Pilots that cover the coastal waters of the U.S. and its territories. Nautical charts are published primarily for the use of the mariner but serve the public interest in many other ways. Cartographers in Coast Survey receive and compile information from a variety of government and non-governmental sources for portrayal on nautical charts and the Coast Pilots. In addition, Coast Survey hydrographers, as well as private contractors, conduct new surveys that are used to update these products. The principal facilities of Coast Survey are located at NOAA headquarters in Silver Spring, MD; Norfolk, VA (Marine Operations Center Atlantic); and Seattle, WA (Western Regional Center).

(370) The **Center for Operational Oceanographic Products and Services (CO-OPS)** collects and distributes observations and predictions of water levels and currents to ensure safe, efficient and environmentally sound maritime commerce. Users can find a variety of information, including observed water level and currents data, tide and current predictions, sea level trends and coastal inundation information. Tides and Currents information is available at *tidesandcurrents.noaa.gov.*

(371) **PORTS® (Physical Oceanographic Real-Time System)** is a decision support tool that improves the safety and efficiency of maritime commerce and coastal resource management. Data from PORTS® supports navigation safety, improves the efficiency of U.S. ports and harbors, and ensures the protection of coastal marine resources.

PORTS® collects and disseminates observations of water levels, currents, salinity, bridge air gap and meteorological parameters (e.g., winds, atmospheric pressure, air and water temperatures) that mariners need to navigate safely and allows seaport and terminal facilities to make good decisions. PORTS® data and information is provided via the internet at *tidesandcurrents.noaa.gov/ports_info.html* and, in some areas, via telephone voice response.

(372)

NOAA Tide Predictions and Tidal Current Predictions

(373) NOAA discontinued the annual printed Tide Tables and Tidal Current Tables in 2020, and has transitioned to providing this information digitally online. The online prediction service equals or exceeds the accuracy of the historically printed publications. Tide and tidal current predictions needed for navigation can be generated in real-time for any time period required by the mariner, for as short as one day, or as long as an entire year. All predictions for U.S. waters are available at *tidesandcurrents.noaa.gov*.

(374) U.S. Coast Guard regulations do not consider access to NOAA's online prediction service "while navigating" as meeting carriage requirements. In order to use predictions from these services, the predictions must be generated in advance and either be stored on the user's device as an electronic file (PDF, screen image, data table, etc.), or as a printed page.

(375) **NOAA Tide Predictions** - *tidesandcurrents.noaa.gov/tide_predictions.html* - allows the generation of predicted tides for more than 3000 locations along the U.S. coastline. Tide predictions may be generated as times and heights of high and low tides for all locations, or as interval predictions (hourly, 15-minute, 6-minute, etc.) for many locations. Tide predictions may be generated for past, present, or future dates; with lengths of 1 day to 1 month, or the full calendar year. The Users Guide - *tidesandcurrents.noaa.gov/PageHelp.html* - describes the displays, formats, additional capabilities, and uses of this online service.

(376) **Caution** –When using Tide Predictions, slack water should not be confused with high or low water. For ocean stations there is often little difference between the time of high or low water and the start of flood/ebb currents; but for places in narrow channels, landlocked harbors or on tidal rivers, the time of slack current may differ by several hours from the time of high or low water. The relationship of the times of high or low water to the flood and ebb of the current depends upon a number of factors unique to each location; no simple general rule can be given which applies to every location. For navigation or other activities which depend on slack water, tidal current predictions should be used to provide times of slack water.

(377) **NOAA Tidal Current Predictions** –*https://tidesandcurrents.noaa.gov/* - allows the generation of predicted currents for more than 2500 locations along the U.S. coastline. Tidal current predictions may be generated as times and speeds of flood/ebb currents and times of slack water for all locations; or as interval predictions of speed (hourly, 30-minute, 6-minute) for many locations. Tidal current predictions may be generated for past, present or future dates; with length of 1 day to 2 weeks, or the full calendar year. The Users Guide - *https://tidesandcurrents.noaa.gov/PageHelp.html* - describes the displays, formats, additional capabilities, and uses of this online service.

(378)

National Weather Service (NWS)

(379)

National Data Buoy Center Meteorological Buoys

(380) The National Data Buoy Center (NDBC) deploys moored meteorological buoys that provide weather data directly to the mariner as well as to marine forecasters.

(381) These buoys have a watch circle radius (WCR) of 2,000 to 4,000 yards from assigned position (AP). In addition, any mooring in waters deeper than 1,000 feet will have a floating "loop" or catenary that may be as little as 500 feet below the surface. This catenary could be anywhere within the buoy's WCR. Any underwater activity within this radius may contact the mooring, causing a failure.

(382) To avoid cutting or damaging a mooring, mariners are urged to exercise extreme caution when navigating in the vicinity of meteorological buoys and to remain well clear of the watch circle. If a mooring is accidentally contacted or cut, please notify NDBC at 228-688-2835 or 228-688-2436.

(383) For further information relating to these buoys visit *ndbc.noaa.gov*.

(383) <Deleted Paragraph>

(384)

Marine Weather Forecasts

(385) The NWS provides marine weather forecasts and warnings for the U.S. coastal waters, the Great Lakes, offshore waters and high seas areas. Scheduled marine forecasts are issued four times daily from **National Weather Service O f ces** with local areas of responsibility around the United States, Guam, American Samoa and Puerto Rico. (See Appendix A for NWS Offices located in the area covered by this Coast Pilot.)

(386) Typically, marine forecasts contain information on wind speed and direction, wave heights, visibility, weather and a general synopsis of weather patterns affecting the region. The forecasts are supplemented with special marine warnings and statements, radar summaries, marine observations, small-craft advisories, gale warnings, storm warnings and various categories of tropical cyclone warnings, e.g., tropical depression, tropical storm and hurricane warnings. Specialized products such as coastal flood, seiche, and tsunami warnings, heavy surf advisories, low water statements, ice forecasts and outlooks and lake shore warnings

and statements are issued as necessary. For further information, visit: *https://www.weather.gov/marine/* .

(387) The principal means of disseminating marine weather services and products in coastal areas is **NOAA Weather Radio**. This network of more than 900 transmitters, covering all 50 states, adjacent coastal waters, Puerto Rico, the U.S. Virgin Islands and the U.S. Pacific Territories, is operated by the NWS and provides continuous broadcasts of weather information for the general public. These broadcasts repeat recorded messages every 4 to 6 minutes. Messages are updated periodically, usually every 2-3 hours and amended as required to include the latest information. When severe weather threatens, routine transmissions are interrupted and the broadcast is devoted to emergency warnings. (See Appendix A for NOAA Weather Radio Stations covered by this Coast Pilot.)

(388) In coastal areas, the programming is tailored to the needs of the marine community. Each coastal marine forecast covers a specific area. For example, "Cape Henlopen to Virginia Beach, out 20 miles." The broadcast range is about 40 miles from the transmitting antenna site, depending on terrain and quality of the receiver used. When transmitting antennas are on high ground, the range is somewhat greater, reaching 60 miles or more. Some receivers are equipped with a warning alert device that can be turned on by means of a tone signal controlled by the NWS office concerned. This signal is transmitted for 13 seconds preceding an announcement of a severe weather warning.

(389) Marine weather warnings are displayed to small-craft operators and others within sight of the shore by the flags, pennants and lights of the **Coastal Warning Display** program. These displays are meant to warn the public of approaching storm conditions and visually communicate that citizens should take personal responsibility for individual safety in the face of an approaching storm. Anyone observing the signals displayed by the program is urged to tune to the NWS radio broadcasts for the latest information. (See **National Weather Service Coastal Warning Displays** illustration for additional information.)

(391) NWS marine weather products are also disseminated to marine users through the broadcast facilities of the Coast Guard, Navy and commercial marine radio stations. Details on these broadcasts including times, frequencies and broadcast content are listed on the NWS internet site, **Marine Product Dissemination Information**, *https://www.weather.gov/marine/nws_dissemination* .

(392) Ships of all nations share equally in the effort to report weather observations. These reports enable meteorologists to create a detailed picture of wind, wave and weather patterns over the open waters that no other data source can provide and upon which marine forecasts are based. The effectiveness and reliability of these forecasts and warnings plus other services to the marine community are strongly linked to the observations received from mariners. There is an especially urgent need for ship observations in the coastal waters, and the NWS asks that these be made and transmitted whenever possible. Many storms originate and intensify in coastal areas. There may be a great difference in both wind direction and speed between the open sea, the offshore waters and on the coast itself.

(393) Information on how ships, commercial fishermen, offshore industries and others in the coastal zone may participate in the marine observation program is available from **National Weather Service Port Meteorological O f cers (PMOs)**. PMOs are located in major U.S. port cities where they visit ships in port to assist masters and mates with the weather observation program, provide instruction on the interpretation of weather charts, calibrate barometers and other meteorological instruments and discuss marine weather communications and marine weather requirements affecting the ships' operations. (For further information on the Voluntary Observing Ship Program and PMOs, go to *vos.noaa.gov*.)

(394)

Space Weather Prediction Center (SWPC)

(395) The Space Weather Prediction Center provides real-time monitoring and forecasting of solar and geophysical events that impact satellites, power grids, communications, navigation and many other technological systems. (See Space Weather Prediction Center in Appendix A.)

(396)

National Environmental Satellite, Data, and Information Service (NESDIS)

(397) Among its functions, NESDIS archives, processes and disseminates the non-real-time meteorological and oceanographic data collected by government agencies and private institutions. Marine weather observations are collected from ships at sea on a voluntary basis. About one million observations are received annually at NESDIS's National Climatic Center. They come from vessels representing every maritime nation. These observations, along with land data, are returned to the mariners in the form of climatological summaries and atlases for coastal and ocean areas. They are available in such NOAA publications as the **U.S. Coast Pilot, Mariners Weather Log** and **Local Climatological Data, Annual Summary**. They also appear in the National Geospatial-Intelligence Agency's **Pilot Chart Atlases** and **Sailing Directions Planning Guides**.

(398)

DEPARTMENT OF DEFENSE

(399)

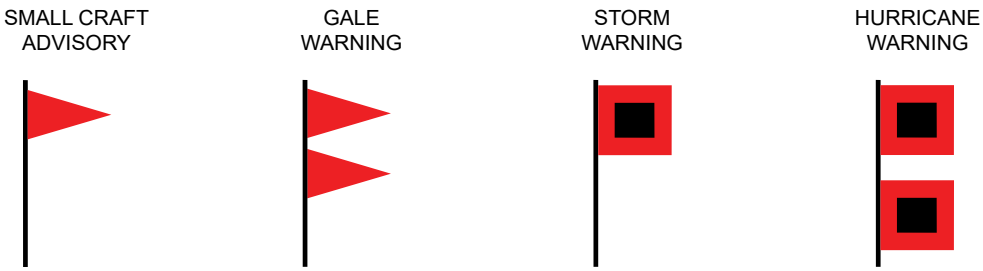
National Geospatial-Intelligence Agency (NGA)

(400) The National Geospatial-Intelligence Agency provides hydrographic, navigational, topographic, and geodetic data, charts, maps and related products and services to the Armed Forces, other Federal Agencies, the Merchant Marine and mariners in general. Publications include Sailing Directions, List of Lights,

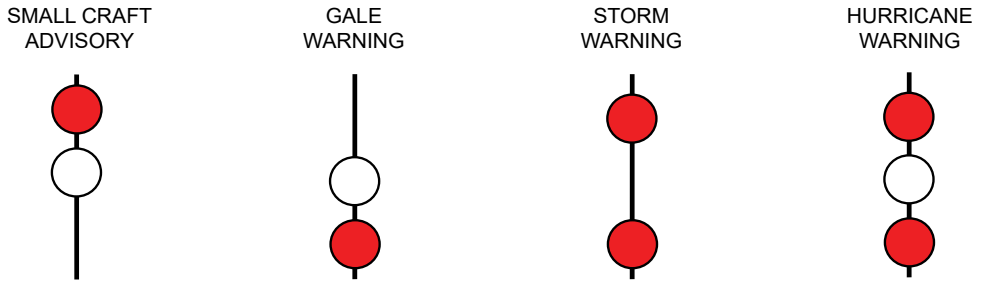
(390)

NATIONAL WEATHER SERVICE COASTAL WARNING DISPLAYS

DAYTIME SIGNALS



NIGHT (LIGHT) SIGNALS



SMALL CRAFT ADVISORY: An advisory issued by coastal and Great Lakes Weather Forecast Offices (WFO) for areas included in the Coastal Waters Forecast or Nearshore Marine Forecast (NSH) products. Thresholds governing the issuance of small craft advisories are specific to geographic areas. A Small Craft Advisory may also be issued when sea or lake ice exists that could be hazardous to small boats. There is no precise definition of a small craft. Any vessel that may be adversely affected by Small Craft Advisory criteria should be considered a small craft. Other considerations include the experience of the vessel operator, and the type, overall size, and sea worthiness of the vessel. There is no legal definition of "small craft". The Small Craft Advisory is an advisory in Coastal Waters and Nearshore forecasts for sustained winds, frequent gusts, or sea/wave conditions, exceeding defined thresholds specific to geographic areas. A Small Craft Advisory may also be issued when sea or lake ice exists that could be hazardous to small boats.

Eastern (ME to SC, Lake Erie, Lake Ontario) – Sustained winds or frequent gusts ranging between 25 and 33 knots (except 20 to 25 knots, lower threshold area dependent, to 33 knots for harbors, bays, etc.) and/or seas or waves 5 to 7 feet and greater, area dependent.

Central (MN to OH) – Sustained winds or frequent gusts (on the Great Lakes) between 22 and 33 knots inclusive, and/or seas or waves greater than 4 feet.

Southern (GA to TX and Caribbean) – Sustained winds of 20 to 33 knots, and/or forecast seas 7 feet or greater that are expected for more than 2 hours.

Western (WA..CA) - Sustained winds of 21 to 33 knots, potentially in combination with wave heights exceeding 10 feet (or wave steepness values exceeding local thresholds).

Alaska (AK) – Sustained winds or frequent gusts of 23 to 33 knots. A small craft advisory for rough seas may be issued for sea/wave conditions deemed locally significant, based on user needs, and should be no lower than 8 feet.

Hawaii (HI), Samoa – Sustained winds 25 knots or greater and seas 10 feet or greater.

Guam and the Northern Mariana Islands – Sustained winds 22 to 33 knots and/or combined seas of 10 feet or more. "Frequent gusts"are typically long duration conditions (greater than 2 hours).

For a list of NWS Weather Offices by Region, refer to the following website: http://www.nws.noaa.gov/organization.php

GALE WARNING: To indicate winds within the range 34 to 47 knots are forecast for the area.

STORM WARNING: To indicate winds 48 knots and above, no matter how high the speed, are forecast for the area. However, if the winds are associated with a tropical cyclone (hurricane), the STORM WARNING indicates that winds within the range 48-63 knots are forecast.

HURRICANE WARNING: Issued only in connection with a tropical cyclone (hurricane) to indicate that winds 64 knots and above are forecast for the area.

NOTE: A "HURRICANE WATCH" is an announcement issued by the National Weather Service via press and television broadcasts whenever a tropical storm or hurricane becomes a threat to a coastal area. The "Hurricane Watch" announcement is not a warning, rather it indicates that the hurricane is near enough that everyone in the area covered by the "Watch" should listen to their radios for subsequent advisories and be ready to take precautionary action in case hurricane warnings are issued.

NOTE: A SPECIAL MARINE WARNING is issued whenever a severe local storm or strong wind of brief duration is imminent and is not covered by existing warnings or advisories. No visual displays will be used in connection with the Special Marine Warning Bulletin; boaters will be able to receive thesespecial warnings by keeping tuned to a NOAA Weather Radio station or to Coast Guard and commercial radio stations that transmit marine weather information.

Distances Between Ports, Radio Navigational Aids, International Code of Signals, American Practical Navigator (Bowditch) and Notice to Mariners. (See NGA Procurement Information in Appendix A.)

(401)

Army Corps of Engineers

(402) The U.S. Army Corps of Engineers has charge of the improvement of the rivers and harbors of the United States and of miscellaneous other civil works, which include the administration of certain Federal laws enacted for the protection and preservation of navigable waters of the United States; the establishment of regulations for the use, administration, and navigation of navigable waters; the establishment of harbor lines; the removal of sunken vessels obstructing or endangering navigation; and the granting of permits for structures or operations in navigable waters and for discharges and deposits of dredged and fill materials in these waters.

(403) **Restricted areas** in most places are defined and regulations governing them are established by the USACE. The regulations are enforced by the authority designated in the regulations, and the areas are shown on the large-scale charts of the National Ocean Service. Copies of the regulations may be obtained at the District offices of the USACE. The regulations also are included in the appropriate Coast Pilot.

(404) Information concerning the various ports, improvements, channel depths, navigable waters and the condition of the Intracoastal Waterways in the areas under their jurisdiction may be obtained direct from the District Engineer Offices. (See Appendix A for addresses.)

(405) The USACE has general supervision of location, construction and manner of maintenance of all **fishtraps,** weirs, pounds or other fishing structures in the navigable waters of the United States. Where state and/or local controls are sufficient to regulate these structures, including that they do not interfere with navigation, the USACE leaves such regulation to the state or local authority. (See **33 CFR 330** (not carried in this Pilot) for applicable Federal regulations.) Construction permits issued by the Engineers specify the lights and signals required for the safety of navigation.

(406) **Fish havens**, artificial reefs constructed to attract fish, can be established in U.S. coastal waters only as authorized by a USACE permit; the permit specifies the location, extent and depth over these mounds of rubble.

(407)

Naval Observatory

(408) The United States Naval Observatory (USNO) provides a wide range of astronomical data and products and serves as the official source of time for the U.S. Department of Defense and a standard of time for the entire United States. The USNO provides earth orientation products such as the latest 24-hour and 48-hour sets of GPS satellite orbits, the latest determinations and predictions for polar motion and information for GPS users. The USNO also maintains a reference for precise time (USNO Master Clock) and monitors the GPS constellation. For extensive information on the USNO products available, visit: *https://www.cnmoc.usff.navy.mil/usno/* or contact by telephone at 202-762-1467.

(409)

DEPARTMENT OF HEALTH AND HUMAN SERVICES

(410)

Food and Drug Administration (FDA)

(411) Under the provisions of the Control of Communicable Diseases Regulations (**21 CFR 1240**) and Interstate Conveyance Sanitation Regulations (**21 CFR 1250**), vessel companies operating in interstate traffic must obtain potable water for drinking and culinary purposes only at watering points found acceptable to the FDA. Water supplies used in watering point operations must also be inspected to determine compliance with applicable Interstate Quarantine Regulations (**42 CFR 72**). These regulations are based on authority contained in the Public Health Service Act (PL 78–410). Penalties for violation of any regulation prescribed under authority of the Act are provided for under Section 368 (42 USC 271) of the Act.

(412)

Public Health Service

(413) The Public Health Service administers foreign quarantine procedures at U.S. ports of entry.

(414) All vessels arriving in the United States are subject to public health inspection. Vessels subject to routine boarding for quarantine inspection are only those which have had on board during the 15 days preceding the date of expected arrival or during the period since departure (whichever period of time is shorter) the occurrence of any death or ill person among passengers or crew (including those who have disembarked or have been removed). The master of a vessel must report such occurrences immediately by radio to the quarantine station at or nearest the port at which the vessel will arrive.

(415) In addition, the master of a vessel carrying 13 or more passengers must report by radio 24 hours before arrival the number of cases (including zero) of diarrhea in passengers and crew recorded in the ship's medical log during the current cruise. All cases that occur after the 24 hour report must also be reported not less than 4 hours before arrival.

(416) *Ill person* means a person who:

(417) 1. Has a temperature of 100°F (or 38°C) or greater, accompanied by a rash, glandular swelling or jaundice, or which has persisted for more than 48 hours; or

(418) 2. Has diarrhea, defined as the occurrence in a 24 hour period of three or more loose stools or of a greater than normal (for the person) amount of loose stools.

(419) Vessels arriving at ports under control of the United States are subject to sanitary inspection to determine whether measures should be applied to prevent the

introduction, transmission or spread of communicable disease.

(420) Specific public health laws, regulations, policies and procedures may be obtained by contacting U.S. Quarantine Stations, U.S. Consulates or the Chief Program Operations, Division of Quarantine, Centers for Disease Control, Atlanta, GA 30333. (See Appendix A for addresses of U.S. Public Health Service Quarantine Stations.)

(421)

DEPARTMENT OF HOMELAND SECURITY

(422)

Citizenship and Immigration Services

(423) The **U.S. Citizenship and Immigration Service** (USCIS) is the federal agency that oversees lawful immigration to the United States. the Service enhances security and improves the efficiency of national immigration services by exclusively focusing on the administration of benefit applications. No person may enter the United States until they have been inspected by an immigration officer. A list of the offices covered by this Coast Pilot is given in Appendix A.

(424)

U.S. Coast Guard

(425) The U.S. Coast Guard has among its duties the enforcement of the laws of the United States on the high seas and in coastal and inland waters of the U.S. and its possessions; enforcement of navigation and neutrality laws and regulations; establishment and enforcement of navigational regulations upon the Inland Waters of the United States, including the establishment of a demarcation line separating the high seas from waters upon which U.S. navigational rules apply; administration of the Oil Pollution Act of 1990, as amended; establishment and administration of vessel anchorages; approval of bridge locations and clearances over navigable waters; administration of the alteration of obstructive bridges; regulation of drawbridge operations; inspection of vessels of the Merchant Marine; admeasurement of vessels; documentation of vessels; preparation and publication of merchant vessel registers; registration of stack insignia; port security; issuance of Merchant Marine licenses and documents; search and rescue operations; investigation of marine casualties and accidents and suspension and revocation proceedings; destruction of derelicts; operation of aids to navigation; maintenance and issuance of Light Lists and Local Notices to Mariners; and operation of ice-breaking facilities.

(426) Issuance of certificates of registry (more commonly referred to as Certificates of Documentation) with endorsements indicating eligibility of vessels that measure at least 5 net tons to engage in various trades for commercial vessels and certain recreational vessels that are numbered either by the Coast Guard or by a state having an approved numbering system (the latter is the most common) and the administration of the various laws pertaining thereto are functions of the Coast Guard and specifically the National Vessel Documentation Center. Owners of vessels may obtain the necessary information from the National Vessel Documentation Center either by mail to the National Vessel Documentation Center, 792 T.J. Jackson Drive, Falling Waters, WV 25419-9502; via toll free number: 800-799-8362; or via online at: *dco.uscg.mil/Our-Organization/Deputy-for-Operations-Policy-and-Capabilities-DCO-D/National-Vessel-Documentation-Center/.*

(427)

U.S. Customs and Border Protection

(428) The U.S. Customs and Border Protection administers certain laws relating to:

(429) – entry and clearance of vessels and permits for certain vessel movements between points in the United States

(430) – prohibitions against coastwise transportation of passengers and merchandise

(431) – salvage

(432) – dredging and towing by foreign vessels

(433) – certain activities of vessels in the fishing trade

(434) – regular and special tonnage taxes on vessels

(435) – landing and delivery of foreign merchandise (including unlading, appraisement, lighterage, drayage, warehousing and shipment in bond)

(436) – collection of customs duties, including duty on imported pleasure boats and yachts and 50% duty on foreign repairs to American vessels engaged in trade

(437) – customs treatment of sea and ship's stores while in port and the baggage of crewmen and passengers

(438) – illegally imported merchandise

(439) – remission of penalties or forfeiture if customs or navigation laws have been violated.

(440) Customs and Border Protection also cooperates with many other Federal agencies in the enforcement of statutes for which they are responsible for. Customs districts and ports of entry are listed in Appendix A.

(441) The Customs and Border Protection office may issue, without charge, a **cruising license**, normally valid for one year, to a yacht of a foreign country that has a reciprocal agreement with the United States. A foreign yacht holding a cruising license is exempt from having to undergo formal entry and clearance procedures such as filing manifests and obtaining permits to proceed as well as from payment of tonnage tax and entry and clearance fees at all but the first port of entry. These vessels must not engage in trade, violate the laws of the United States or visit a vessel not yet inspected by a Customs Agent and does, within 24 hours of arrival at each port or place in the United States, report the fact of arrival to the nearest customhouse. Countries that have reciprocal agreements granting these privileges to U.S. yachts are:

(442)

Countries with U.S. Cruising License Reciprocity	
Argentina	Honduras

Australia	Ireland
Austria	Italy
Bahama Islands	Jamaica
Belguim	Liberia
Bermuda	Marshall Islands
Canada	Netherlands
Denmark	New Zealand
Finland	Norway
France	Sweden
Germany	Switzerland
Great Britain	Turkey
Greece	

(443) Further information concerning cruising licenses may be obtained from the headquarters port for the customs district in which the license is desired or at *cbp.gov*. U.S. yacht owners planning cruises to foreign ports may contact the nearest customs district headquarters as to customs requirements.

(444)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

(445) The U.S. EPA provides coordinated governmental action to ensure the protection of the environment by abating and controlling pollution on a systematic basis. The ocean dumping permit program of the EPA provides that except when authorized by permit, the dumping of any material into the ocean is prohibited by the "Marine Protection, Research, and Sanctuaries Act of 1972, Public Law 92–532," as amended (33 USC 1401 et seq.).

(446) Permits for the **dumping of dredged material** into waters of the United States, including the territorial sea, and into ocean waters are issued by the U.S. Army Corps of Engineers. Permits for the dumping of fill material into waters of the United States, including the territorial sea, are also issued by the U.S. Army Corps of Engineers. Permits for the dumping of other material in the territorial sea and ocean waters are issued by the EPA.

(447) U.S. Army Corps of Engineers regulations relating to the above are contained in **33 CFR 323** and **324**; EPA regulations are in **40 CFR 220** though **228**. (See Disposal Sites, this chapter.)

(448) Persons or organizations who want to file for an application for an ocean dumping permit should write the EPA Regional Office for the region in which the port of departure is located. (See Appendix A for addresses of regional offices and States in the EPA coastal regions.)

(449) The letter should contain the name and address of the applicant, name and address of person or firm, the name and usual location of the conveyance to be used in the transportation and dumping of the material involved, a physical description where appropriate, and the quantity to be dumped and proposed dumping site.

(450) Everyone who writes EPA will be sent information about a final application for a permit as soon as possible. This final application is expected to include questions about the description of the process or activity giving rise to the production of the dumping material, information on past activities of applicant or others with respect to the disposal of the type of material involved, and a description about available alternative means of disposal of the material with explanations about why an alternative is thought by the applicant to be inappropriate.

(450.001)

FEDERAL AVIATION ADMINISTRATION (FAA)

(450.002)

Obstruction Evaluation/Airport Airspace Analysis (OE/AAA)

(450.003) The **Federal Aviation Administration (FAA)** has established comprehensive policies and procedures to evaluate potential hazards to air navigation posed by construction or alteration projects. These assessments are critical to maintaining the safety and efficiency of the National Airspace System (NAS). To support this effort, the FAA's Air Traffic Organization (ATO) developed the Obstruction Evaluation and Airport Airspace Analysis (OE/AAA) System, a web-based platform that enables public submission and FAA processing of proposed construction or alteration notices. The collection and processing of this information are authorized under **49 U.S.C. § 44718** and **14 CFR Part 77**, which pertains to the notice and study of proposed construction or alteration near navigable airspace.

(450.004) Maritime vessels are required to comply with the provisions of **14 CFR Part 77 Subpart B**. In consideration of procedures and for potential conflict between aircraft and maritime vessels, waterways of the United States with vessel height greater than 200 feet (Air Draft) above the waterline, must be assessed by the FAA. Due to airport, airspace, and waterway proximity, lesser heights may also require assessment. Owners, operators, agents, voyage planners, and masters may consult the **Obstruction Evaluation/Airport Airspace Analysis (OE/AAA) Pre-Screening Tool** at *https://oeaaa.faa.gov/oeaaa/oe3a/main/#/home* to assess applicability for lesser (Air Draft) heights.

(451)

FEDERAL COMMUNICATIONS COMMISSION (FCC)

(452) The Federal Communications Commission controls non-government radio communications in the United States, Guam, Puerto Rico and the Virgin Islands. Commission inspectors have authority to board ships to determine whether their radio stations comply with international treaties, Federal laws and Commission regulations. The commission has field offices in the principal U.S. ports. (See Appendix A for addresses.) Information concerning ship radio regulations and service documents may be obtained from the Federal

Communications Commission, Washington, DC 20554, or from any of the field offices.

(452) <Deleted Paragraph>

(454) <Deleted Paragraph>

(456) <Deleted Paragraph>

(453)

Measurements and Equivalencies

nautical mile — 1,852 meters / 6,076.12 feet
statute mile — 5,280 feet / 1,609.3 meters / 1.6093 kilometers
cable — 0.1 nautical mile (CN) / 720 feet (US)
fathom — 6 feet / 1.8288 meters
foot — 0.3048 meter
inch — 2.54 centimeters
pound (avoirdupois) — 453.59 gram
kilometer — 1,000 meters
knot — 1.6877 feet per second / 0.5144 meters per second
miles/hour (statute) — 1.466 feet per second / 0.44704 meters per second
acre — 43,560 square feet / 4,046.82 square meters
gram — 0.0022046 pound (avoirdupois) / 0.035274 ounce
meter — 39.37 inches / 3.281 feet / 1.0936 yards
short ton — 2,000 pounds
long ton — 2,240 pounds
metric ton — 2,204.6 pounds
kilogram — 2.2 pounds
liter — 1.0567 quarts
barrel (petroleum) — 42 gallons (US)

Conversion Factors

Linear

inches — muiltiply by 25.40 — millimeters
inches — multiply by 2.540 — centimeters
centimeters — multiply by 0.032808 — feet
feet — multiply by 30.48 — centimeters
feet — multiply by 0.3048 — meters
feet — multiply by 0.00016458 — nautical miles
yard — multiply by 0.9144 — meters
meters — multiply by 3.2808 — feet
meters — multiply by 1.094 — yards
meters — multiply by 0.0005399 — nautical miles
statute miles — multiply by 0.86897 — nautical miles
statute miles — multiply by 1.6093 — kilometers
statute miles — multiply by 1,609.3 — meters
nautical miles — multiply by 1.151 — statute miles

Area

acres — multiply by 4,046.9 — square meters
acres — multiply by 43,560 — square feet
acres — multiply by 0.404685 — hectare
hectare — multiply by 2.471054 — acres
hectare — multiply by 10,000 — square meters
hectare — multiply by 1.07639×10^5 — square feet
square feet — multiply by 0.0929 — square meters
square feet — multiply by 0.00002296 — acres
square meters — multiply by 10.764 — square feet
square meters — multiply by 0.0002471 — acres

Depths

fathoms — multiply by 1.8288 — meters
feet — multiply by 0.3048 — meters
meters — multiply by 0.54681 — fathoms
meters — multiply by 3.2808 — feet

Rate

feet/second — multiply by 0.5925 — knots
feet/second — multiply by 0.6818 — miles/hour
feet/second — multiply by 30.48 — centimeters/second
statute miles/hour — multiply by 0.8689 — knots
statute miles/hour — multiply by 1.467 — feet/second
statute miles/hour — multiply by 0.447 — meters/second
knots — multiply by 1.151 — miles/hour
knots — multiply by 0.5144 — meters/second
knots — multiply by 1.6878 — feet/second
centimeters/second — multiply by 0.01944 — miles/hour
centimeters/second — multiply by 0.02237 — miles/hour
centimeters/second — multiply by 0.032808 — feet/second

Mass

grams — multiply by 0.035275 — ounces
grams — multiply by 0.002205 — pounds
ounces — multiply by 28.349 — grams
pounds — multiply by 0.45359 — kilograms
short tons — multiply by 2,000 — pounds
short tons — multiply by 0.89286 — long tons
short tons — multiply by 0.9072 — metric tons
long tons — multiply by 2,240 — pounds
long tons — multiply by 1.12 — short tons
long tons — multiply by 1.016 — metric tons
metric tons — multiply by 1,000 — kilograms
metric tons — multiply by 0.9842 — long tons
metric tons — multiply by 1.1023 — short tons
metric tons — multiply by 2,204.6 — pounds

Volume

barrels (petroleum) — multiply by 42 — gallons (US)
barrels (petroleum) — multiply by 158.99 — liters
barrels (liquid, US) — multiply by 31.5 — gallons (US)
barrels (liquid, US) — multiply by 26.229 — gallons (British)
barrels (liquid, US) — multiply by 119.24 — liters
gallons (US) — multiply by 0.02381 — barrels (petroleum)
gallons (US) — multiply by 3.7854 — liters
liters — multiply by 0.26417 — gallons (US)

(454)

Tips for BOATING CLEAN AND GREEN

Practice preventative engine maintenance
Keep your engine well tuned and practice preventative engine maintenance by regularly checking hoses and lines for chaffing or deterioration.

Use oil absorbants
Place and secure an oil absorbent under the engine and in the bilge. Avoid using bilge cleaners as they may get discharged overboard. It is illegal to use soap to disperse fuel and oil spills. Report oil and chemical spills by calling the EPA National Response Center at 800–424–8802.

Spill-proof your fueling practices
Use a spill proof system like a portable oil change pump to change your oil. Use oil absorbents when fueling and changing the oil. Do not top-off your fuel tank; leave it 10 percent empty to allow fuel to expand as it warms.

Reduce greywater discharges
Use shore-side facilities for laundry, showers, and dish washing whenever possible. Use only phosphate-free and biodegradable soaps. The legality of discharging greywater into a marina or within three miles off the coast varies from place to place. In some areas, there are local ordinances and codes that allow harbor patrol to issue citations for any discharge that is not "clean and clear." To avoid any potential fines and to protect the aquatic environment, do not discharge greywater overboard.

WARNING
HAZARDOUS WASTE

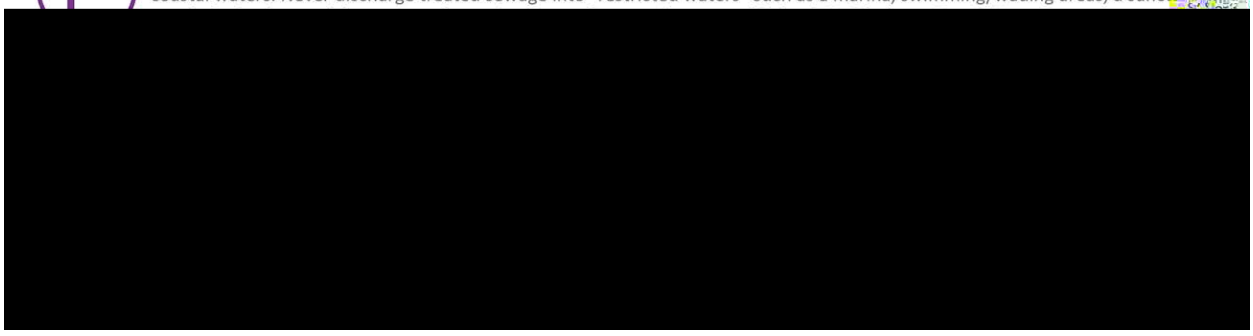
Dispose of hazardous waste properly
Recycle and properly dispose of absorbents, used oil, oil filters, paint, and batteries at your local household hazardous waste collection site.

Minimize boat cleaning and maintenance conducted on the water
- Use more elbow grease.
- Use products that are water-based, biodegradable, phosphate-free, and labeled as less toxic.
- Check out less toxic cleaning alternatives for all types of uses. Visit http://dbw.parks.ca.gov/?page_id=29184.
- Buy only the amount that you need and use products for spot cleaning only.
- Properly handle and store materials. Dispose of hazardous waste legally and safely.

Reduce discharges from bottom paints
- Consider alternative, non-biocide hull coatings.
- Clean the bottom with a soft, non-abrasive sponge.
- Use hull cleaning companies who use green management practices such as monitoring their divers and using non-abrasive scrubbing agents that do not release paint into the water. For more information visit https://www3.epa.gov/npdes/pubs/vgp_hull_husbandry.pdf.

Stow it, don't throw it
Keep your trash on-board. Recycle plastic, glass, metal, and paper. Avoid excess packaging.

Dump at the pump!
It is illegal to discharge untreated sewage anywhere within the three-mile territorial limit including lakes, rivers, reservoirs or coastal waters. Never discharge treated sewage into "restricted waters" such as a marina, swimming/wading areas, a sanctuary



Navigation Regulations

(1) This chapter contains extracts from **Code of Federal Regulations (CFR)** that are of importance to mariners in the area covered by this Coast Pilot. Sections of little value to the mariner are sometimes omitted. Omitted sections are signified by the following [...]

(2) Extracts from the following titles are contained in this chapter.

(3)

Title 33: Navigation and Navigable Waters
- Part 26—Vessel Bridge-to-Bridge Radiotelephone Regulations
- Part 80—COLREGS Demarcation Lines
- Part 81—72 COLREGS: Implementing Rules
- Part 82—72 COLREGS: Interpretive Rules
- Part 88—Annex V: Pilot Rules
- Part 89—Inland Navigation Rules: Implementing Rules
- Part 90—Inland Rules: Interpretive Rules
- Part 110—Anchorage Regulations
- Part 117—Drawbridge Operation Regulations
- Part 157—Rules for the Protection of the Marine Environment Relating to Tank Vessels Carrying Oil in Bulk (in part)
- Part 160—Ports and Waterways Safety-General
- Part 161—Vessel Traffic Management
- Part 162—Inland Waterways Navigation Regulations
- Part 164—Navigation Safety Regulations (in part)
- Part 165—Regulated Navigation Areas and Limited Access Areas
- Part 166—Shipping Safety Fairways
- Part 167—Offshore Traffic Separation Schemes
- Part 169—Ship Reporting Systems
- Part 207—Navigation Regulations
- Part 334—Danger Zones and Restricted Area Regulations

Title 40: Protection of Environment
- Part 140—Marine Sanitation Device Standard

Title 46: Shipping
- Part 15—Manning Requirements

Title 50: Wildlife and Fisheries
- Part 222—General Endangered and Threatened Marine Species
- Part 224—Endangered Marine and Anadromous Species

(4) These regulations can only be amended by the enforcing agency or other authority cited in the regulations. Accordingly, requests for changes to these regulations should be directed to the appropriate agency for action. In those regulations where the enforcing agency is not cited or is unclear, recommendations for changes should be directed to the following Federal agencies for action:

(5) **National Oceanic and Atmospheric Administration**—50 CFR 222, 224 and 226

(6) **United States Coast Guard**—33 CFR 26, 80, 81, 82, 88, 89, 90, 110, 117, 157, 160, 161, 162, 164, 165, 166, 167 and 169; 46 CFR 15

(7) **United States Army Corps of Engineers**—33 CFR 207 and 334

(8) **Environmental Protection Agency**—40 CFR 140

(9)

TITLE 33–NAVIGATION AND NAVIGABLE WATERS

(10)

Part 26–Vessel Bridge-to-Bridge Radiotelephone Regulations

(11)

§26.01 Purpose.

(12) (a) The purpose of this part is to implement the provisions of the Vessel Bridge-to-Bridge Radiotelephone Act. This part–

(13) (1) Requires the use of the vessel bridge-to-bridge radiotelephone;

(14) (2) Provides the Coast Guard's interpretation of the meaning of important terms in the Act;

(15) (3) Prescribes the procedures for applying for an exemption from the Act and the regulations issued under the Act and a listing of exemptions.

(16) (b) Nothing in this part relieves any person from the obligation of complying with the rules of the road and the applicable pilot rules.

(17)

§26.02 Definitions.

(18) For the purpose of this part and interpreting the Act–

(19) *Act* means the "Vessel Bridge-to-Bridge Radiotelephone Act", 33 U.S.C. sections 1201–1208;

(20) *Length* is measured from end to end over the deck excluding sheer;

(21) *Power-driven vessel* means any vessel propelled by machinery; and

(22) *Secretary* means the Secretary of the Department in which the Coast Guard is operating;

(23) *Territorial sea* means all waters as defined in §2.22(a)(1) of this chapter.

(24) *Towing vessel* means any commercial vessel engaged in towing another vessel astern, alongside, or by pushing ahead.

(25) *Vessel Traffic Services (VTS)* means a service implemented under Part 161 of this chapter by the United States Coast Guard designed to improve the safety and efficiency of vessel traffic and to protect the environment. The VTS has the capability to interact with marine traffic

and respond to traffic situations developing in the VTS area.

(26) *Vessel Traffic Service Area* or *VTS Area* means the geographical area encompassing a specific VTS area of service as described in Part 161 of this chapter. This area of service may be subdivided into sectors for the purpose of allocating responsibility to individual Vessel Traffic Centers or to identify different operating requirements.

(27) **Note:** Although regulatory jurisdiction is limited to the navigable waters of the United States, certain vessels will be encouraged or may be required, as a condition of port entry, to report beyond this area to facilitate traffic management within the VTS area.

(28)

§26.03 Radiotelephone required.

(29) (a) Unless an exemption is granted under §26.09 and except as provided in paragraph (a)(4) of this section, this part applies to:

(30) (1) Every power-driven vessel of 20 meters or over in length while navigating;

(31) (2) Every vessel of 100 gross tons and upward carrying one or more passengers for hire while navigating;

(32) (3) Every towing vessel of 26 feet or over in length while navigating; and

(33) (4) Every dredge and floating plant engaged in or near a channel or fairway in operations likely to restrict or affect navigation of other vessels except for an unmanned or intermittently manned floating plant under the control of a dredge.

(34) (b) Every vessel, dredge, or floating plant described in paragraph (a) of this section must have a radiotelephone on board capable of operation from its navigational bridge, or in the case of a dredge, from its main control station, and capable of transmitting and receiving on the frequency or frequencies within the 156-162 Mega-Hertz band using the classes of emissions designated by the Federal Communications Commission for the exchange of navigational information.

(35) (c) The radiotelephone required by paragraph (b) of this section must be carried on board the described vessels, dredges, and floating plants upon the navigable waters of the United States.

(36) (d) The radiotelephone required by paragraph (b) of this section must be capable of transmitting and receiving on VHF-FM channel 1022 (157.1 MHz).

(37) (e) While transiting any of the following waters, each vessel described in paragraph (a) of this section also must have on board a radiotelephone capable of transmitting and receiving on VHF-FM channel 67 (156.375 MHz):

(38) (1) The lower Mississippi River from the territorial sea boundary, and within either the Southwest Pass safety fairway or the South Pass safety fairway specified in 33 CFR 166.200, to mile 242.4 AHP (Above Head of Passes) near Baton Rouge;

(39) (2) The Mississippi River-Gulf Outlet from the territorial sea boundary, and within the Mississippi River-Gulf outlet Safety Fairway specified in 33 CFR 166.200, to that channel's junction with the Inner Harbor Navigation Canal; and

(40) (3) The full length of the Inner Harbor Navigation Canal from its junction with the Mississippi River to that canal's entry to Lake Pontchartrain at the New Seabrook vehicular bridge.

(41) (f) In addition to the radiotelephone required by paragraph (b) of this section, each vessel described in paragraph (a) of this section while transiting any waters within a Vessel Traffic Service Area, must have on board a radiotelephone capable of transmitting and receiving on the VTS designated frequency in Table 161.12(c) (VTS and VMRS Centers, Call Signs/MMSI, Designated Frequencies, and Monitoring Areas).

(42) **Note:** A single VHF-FM radio capable of scanning or sequential monitoring (often referred to as "dual watch" capability) will not meet the requirements for two radios.

(43)

§26.04 Use of the designated frequency.

(44) (a) No person may use the frequency designated by the Federal Communications Commission under section 8 of the Act, 33 U.S.C. 1207(a), to transmit any information other than information necessary for the safe navigation of vessels or necessary tests.

(45) (b) Each person who is required to maintain a listening watch under section 5 of the Act shall, when necessary, transmit and confirm, on the designated frequency, the intentions of his vessel and any other information necessary for the safe navigation of vessels.

(46) (c) Nothing in these regulations may be construed as prohibiting the use of the designated frequency to communicate with shore stations to obtain or furnish information necessary for the safe navigation of vessels.

(47) (d) On the navigable waters of the United States, channel 13 (156.65 MHz) is the designated frequency required to be monitored in accordance with §26.05(a) except that in the area prescribed in §26.03(e), channel 67 (156.375 MHz) is the designated frequency.

(48) (e) On those navigable waters of the United States within a VTS area, the designated VTS frequency is an additional designated frequency required to be monitored in accordance with §26.05.

(49)

§26.05 Use of radiotelephone.

(50) Section 5 of the Act states that the radiotelephone required by this Act is for the exclusive use of the master or person in charge of the vessel, or the person designated by the master or person in charge to pilot or direct the movement of the vessel, who shall maintain a listening watch on the designated frequency. Nothing herein shall be interpreted as precluding the use of portable radiotelephone equipment to satisfy the requirements of this Act.

(51)

§26.06 Maintenance of radiotelephone; failure of radiotelephone.

(52) Section 6 of the Act states–

(53) (a) Whenever radiotelephone capability is required by this Act, a vessel's radiotelephone equipment shall be maintained in effective operating condition. If the radiotelephone equipment carried aboard a vessel ceases to operate, the master shall exercise due diligence to restore it or cause it to be restored to effective operating condition at the earliest practicable time. The failure of a vessel's radiotelephone equipment shall not, in itself, constitute a violation of this Act, nor shall it obligate the master of any vessel to moor or anchor his vessel; however, the loss of radiotelephone capability shall be given consideration in the navigation of the vessel.

(54)

§26.07 Communications.

(55) No person may use the services of, and no person may serve as, a person required to maintain a listening watch under section 5 of the Act, 33 U.S.C. 1204, unless the person can communicate in the English language.

(56)

§26.08 Exemption procedures.

(57) (a) The Commandant has redelegated to the Assistant Commandant for Prevention Policy, U.S. Coast Guard Headquarters, with the reservation that this authority shall not be further redelegated, the authority to grant exemptions from provisions of the Vessel Bridge-to-Bridge Radiotelephone Act and this part.

(58) (b) Any person may petition for an exemption from any provision of the Act or this part;

(59) (c) Each petition must be submitted in writing to Commandant (CG–DCO–D), Attn: Deputy for Operations Policy and Capabilities, U.S. Coast Guard Stop 7318, 2703 Martin Luther King Jr. Avenue SE., Washington, DC 20593–7318, and must state:

(60) (1) The provisions of the Act or this part from which an exemption is requested; and

(61) (2) The reasons why marine navigation will not be adversely affected if the exemption is granted and if the exemption relates to a local communication system how that system would fully comply with the intent of the concept of the Act but would not conform in detail if the exemption is granted.

(62)

§26.09 List of exemptions.

(63) (a) All vessels navigating on those waters governed by the navigation rules for Great Lakes and their connecting and tributary waters (33 U.S.C. 241 *et seq*.) are exempt from the requirements of the Vessel Bridge-to-Bridge Radiotelephone Act and this part until May 6, 1975.

(64) (b) Each vessel navigating on the Great Lakes as defined in the Inland Navigational Rules Act of 1980 (33 U.S.C. 2001 *et seq*.) and to which the Vessel Bridge-to-Bridge Radiotelephone Act (33 U.S.C. 1201–1208) applies is exempt from the requirements in 33 U.S.C. 1203, 1204, and 1205 and the regulations under §§26.03, 26.04, 26.05, 26.06, and 26.07. Each of these vessels and each person to whom 33 U.S.C. 1208(a) applies must comply with Articles VII, X, XI, XII, XIII, XV, and XVI and Technical Regulations 1-9 of "The Agreement Between the United States of America and Canada for Promotion of Safety on the Great Lakes by Means of Radio, 1973."

(65)

Part 80–COLREGS Demarcation Lines

(66)

§80.01 General basis and purpose of demarcation lines.

(67) (a) The regulations in this part establish the lines of demarcation delineating those waters upon which mariners shall comply with the International Regulations for Preventing Collisions at Sea, 1972 (72 COLREGS) and those waters upon which mariners shall comply with the Inland Navigation Rules.

(68) (b) The waters inside of the lines are Inland Rules waters. The waters outside the lines are COLREGS waters.

(69) (c) Geographic coordinates expressed in terms of latitude or longitude, or both, are not intended for plotting on maps or charts whose referenced horizontal datum is the North American Datum of 1983 (NAD 83), unless such geographic coordinates are expressly labeled NAD 83. Geographic coordinates without the NAD 83 reference may be plotted on maps or charts referenced to NAD 83 only after application of the appropriate corrections that are published on the particular map or chart being used.

(70)

§80.135 Hull, Mass. to Race Point, Mass.

(71) (a) Except inside lines described in this section, the 72 COLREGS apply on the harbors, bays, and inlets on the east coast of Massachusetts from the easternmost radio tower at Hull, charted in approximate position latitude 42°16.7'N., longitude 70°52.6'W., to Race Point on Cape Cod.

(72) (b) A line drawn from Canal Breakwater Light 6 south to the shoreline.

(73)

§80.145 Race Point, Mass., to Watch Hill, R.I.

(74) (a) Except inside lines specifically described in this section, the 72 COLREGS shall apply on the sounds, bays, harbors, and inlets along the coast of Cape Cod and the southern coasts of Massachusetts and Rhode Island from Race Point to Watch Hill.

(75) (b) A line drawn from Nobska Point Light to Tarpaulin Cove Light on the southeastern side of Naushon Island; thence from the southernmost tangent of Naushon Island to the easternmost extremity of Nashawena Island; thence from the southwesternmost extremity of Nashawena Island to the easternmost extremity of Cuttyhunk Island; thence from the southwestern tangent of Cuttyhunk Island to the tower on Gooseberry Neck charted in approximate position 41°29.1'N., 71°02.3'W.

(76) (c) A line drawn from Sakonnet Breakwater Light 2 tangent to the southernmost part of Sachuest Point charted in approximate position 41°28.5'N., 71°14.8'W.

(77) (d) An east-west line drawn through Beavertail Light between Brenton Point and the Boston Neck shoreline.

(78)

§80.150 Block Island, R.I.

(79) The 72 COLREGS shall apply on the harbors of Block Island.

(80)

§80.155 Watch Hill, RI to Montauk Point, NY.

(81) (a) A line drawn from 41°18'13.999"N., 071°51'30.300"W. (Watch Hill Light) to East Point on Fishers Island.

(82) (b) A line drawn from Race Point to 41°14'36.509"N., 072°02'49.676"W. (Race Rock Light); thence to 41°12'22.900"N., 072°06'24.700"W. (Little Gull Island Light) thence to East Point on Plum Island.

(83) (c) A line drawn from 41°10'16.704"N., 072°12'21.684"W. (Plum Island Harbor East Dolphin Light) to 41°10'17.262"N., 072°12'23.796"W. (Plum Island Harbor West Dolphin Light).

(84) (d) A line drawn from 41°10'25.745"N., 072°12'42.137"W. (Plum Gut Light) to 41°09'48.393"N., 072°13'25.014"W. (Orient Point Light); thence to Orient Point.

(85) (e) A line drawn from 41°06'35.100"N., 072°18'21.400"W. (Long Beach Bar Light) to Cornelius Point.

(86) (f) A line drawn from 41°04'12.000"N., 072°16'48.000"W. (Coecles Harbor Entrance Light) to Sungic Point.

(87) (g) A line drawn from Nicholl Point to 41°02'25.166"N., 072°15'42.971"W. (Cedar Island Light 3CI).

(88) (h) A line drawn from 41°02'06.060"N., 072°11'19.560"W. (Threemile Harbor West Breakwater Light) to 41°02'05.580"N., 072°11'15.777"W. (Threemile Harbor East Breakwater Light).

(89) (i) A line drawn from 41°04'44.210"N., 071°56'20.308"W. (Montauk West Jetty Light 2) to 41°04'46.095"N., 071°56'14.168"W. (Montauk East Jetty Light 1).

(90)

§80.160 Montauk Point, NY to Atlantic Beach, NY.

(91) (a) A line drawn from the 40°50'17.952"N., 072°28'29.010"W. (Shinnecock Inlet Breakwater Light 2) to 40°50'23.490"N., 072°28'40.122"W. (Shinnecock Inlet Breakwater Light 1).

(92) (b) A line drawn from 40°45'47.763"N., 072°45'11.095"W. (Moriches Inlet Breakwater Light 2) to 40°45'49.692"N., 072°45'21.719"W. (Moriches Inlet Breakwater Light 1).

(93) (c) A line drawn from the westernmost point on Fire Island to the southernmost extremity of the spit of land at the western end of Oak Beach.

(94) (d) A line drawn from 40°34'23.568"N., 073°34'32.364"W. (Jones Inlet Light) 322° true across Jones Inlet to the shoreline.

(95)

Part 81–72 COLREGS: IMPLEMENTING RULES

(96)

§81.1 Definitions.

(97) As used in this part:

(98) *72 COLREGS* refers to the International Regulations for Preventing Collisions at Sea, 1972, done at London, October 20, 1972, as rectified by the Proces-Verbal of December 1, 1973, as amended.

(99) *A vessel of special construction or purpose* means a vessel designed or modified to perform a special function and whose arrangement is thereby made relatively inflexible.

(100) *Interference with the special function of the vessel* occurs when installation or use of lights, shapes, or sound-signaling appliances under 72 COLREGS prevents or significantly hinders the operation in which the vessel is usually engaged.

(101)

§81.3 General.

(102) Vessels of special construction or purpose which cannot fully comply with the light, shape, and sound signal provisions of 72 COLREGS without interfering with their special function may instead meet alternative requirements. The Chief of the Prevention Division in each Coast Guard District Office makes this determination and requires that alternative compliance be as close as possible with the 72 COLREGS. These regulations set out the procedure by which a vessel may be certified for alternative compliance. The information collection and recordkeeping requirements in §§81.5 and 81.18 have been approved by the Office of Management and Budget under OMB control No. 1625-0019.

(103)

Alternative Compliance

(104)

§81.5 Application for a Certificate of Alternative Compliance.

(105) (a) The owner, builder, operator, or agent of a vessel of special construction or purpose who believes the vessel cannot fully comply with the 72 COLREGS light, shape, or sound signal provisions without interference with its special function may apply for a determination that alternative compliance is justified. The application must be in writing, submitted to the Chief of the Prevention Division of the Coast Guard District in which the vessel is being built or operated, and include the following information:

(106) (1) The name, address, and telephone number of the applicant.

(107) (2) The identification of the vessel by its:

(108) (i) Official number;

(109) (ii) Shipyard hull number;

(110) (iii) Hull identification number; or

(111) (iv) State number, if the vessel does not have an official number or hull identification number.

(112) (3) Vessel name and home port, if known.

(113) (4) A description of the vessel's area of operation.

(114) (5) A description of the provision for which the Certificate of Alternative Compliance is sought, including:

(115) (i) The 72 COLREGS Rule or Annex section number for which the Certificate of Alternative Compliance is sought;

(116) (ii) A description of the special function of the vessel that would be interfered with by full compliance with the provision of that Rule or Annex section; and

(117) (iii) A statement of how full compliance would interfere with the special function of the vessel.

(118) (6) A description of the alternative installation that is in closest possible compliance with the applicable 72 COLREGS Rule or Annex section.

(119) (7) A copy of the vessel's plans or an accurate scale drawing that clearly shows:

(120) (i) The required installation of the equipment under the 72 COLREGS,

(121) (ii) The proposed installation of the equipment for which certification is being sought, and

(122) (iii) Any obstructions that may interfere with the equipment when installed in:

(123) (A) The required location; and

(124) (B) The proposed location.

(125) (b) The Coast Guard may request from the applicant additional information concerning the application.

(126)

§81.9 Certificate of Alternative Compliance: Contents.

(127) The Chief of the Prevention Division issues the Certificate of Alternative Compliance to the vessel based on a determination that it cannot comply fully with 72 COLREGS light, shape, and sound signal provisions without interference with its special function. This Certificate includes—

(128) (a) Identification of the vessel as supplied in the application under §81.5(a)(2);

(129) (b) The provision of the 72 COLREGS for which the Certificate authorizes alternative compliance;

(130) (c) A certification that the vessel is unable to comply fully with the 72 COLREGS lights, shape, and sound signal requirements without interference with its special function;

(131) (d) A statement of why full compliance would interfere with the special function of the vessel;

(132) (e) The required alternative installation;

(133) (f) A statement that the required alternative installation is in the closest possible compliance with the 72 COLREGS without interfering with the special function of the vessel;

(134) (g) The date of issuance;

(135) (h) A statement that the Certificate of Alternative Compliance terminates when the vessel ceases to be usually engaged in the operation for which the certificate is issued.

(136)

§81.17 Certificate of Alternative Compliance: Termination.

(137) The Certificate of Alternative Compliance terminates if the information supplied under §81.5(a) or the Certificate issued under §81.9 is no longer applicable to the vessel.

(138)

§81.18 Notice and record of certification of vessels of special construction or purpose.

(139) (a) In accordance with 33 U.S.C. 1605(c), a notice is published in the Federal Register of the following:

(140) (1) Each Certificate of Alternative Compliance issued under §81.9; and

(141) (2) Each Coast Guard vessel determined by the Commandant to be a vessel of special construction or purpose.

(142) (b) Copies of Certificate of Alternative Compliance and documentation concerning Coast Guard vessels are available for inspection at Marine Transportation Systems Directorate, U.S. Coast Guard Headquarters, (CG-5PW), Stop 7509, 2703 Martin Luther King Avenue SE., Washington, DC 20593-7509.

(143) (c) The owner or operator of a vessel issued a Certificate shall ensure that the vessel does not operate unless the Certificate of Alternative Compliance or a certified copy of that Certificate is on board the vessel and available for inspection by Coast Guard personnel.

(144)

Exemptions

(145)

§81.20 Lights and sound signal appliances.

(146) Each vessel under the 72 COLREGS, except the vessels of the Navy, is exempt from the requirements of the 72 COLREGS to the limitation for the period of time stated in Rule 38 (a), (b), (c), (d), (e), (f), and (g) if:

(147) (a) Her keel is laid or is at a corresponding stage of construction before July 15, 1977; and

(148) (b) She meets the International Regulations for Preventing Collisions at Sea, 1960 (77 Stat. 194, 33 U.S.C. 1051-1094).

(149)

Part 82—72 COLREGS: INTERPRETATIVE RULES

(150)

§82.1 Purpose.

(151) This part contains the interpretative rules concerning the 72 COLREGS that are adopted by the Coast Guard for the guidance of the public.

(152)

§82.3 Pushing vessel and vessel being pushed: Composite unit.

(153) Rule 24(b) of the 72 COLREGS states that when a pushing vessel and a vessel being pushed ahead are rigidly connected in a composite unit, they are regarded as a power-driven vessel and must exhibit the lights under Rule 23. A "composite unit" is interpreted to be a pushing vessel that is rigidly connected by mechanical means to a vessel being pushed so they react to sea and swell as one vessel. "Mechanical means" does not include the following:

(154) (a) Lines.

(155) (b) Hawsers.

(156) (c) Wires.

(157) (d) Chains.

(158)

§82.5 Lights for moored vessels.

(159) For the purposes of Rule 30 of the 72 COLREGS, a *vessel at anchor* includes a barge made fast to one or more mooring buoys or other similar device attached to the sea or river floor. Such a barge may be lighted as a vessel at anchor in accordance with Rule 30, or may be lighted on the corners in accordance with 33 CFR 83.30(h) through (l).

(160)

§82.7 Sidelights for unmanned barges.

(161) An unmanned barge being towed may use the exception of COLREGS Rule 24(h). However, this exception only applies to the vertical sector requirements.

(162)

Part 88—ANNEX V: PILOT RULES

(163)

§88.01 Purpose and applicability.

(164) This part applies to all vessels operating on United States inland waters and to United States vessels operating on the Canadian waters of the Great Lakes to the extent there is no conflict with Canadian law.

(165)

§88.03 Definitions.

(166) The terms used in this part have the same meaning as the terms defined in part 83 of this subchapter.

(167)

§88.05 Law enforcement vessels.

(168) (a) Law enforcement vessels may display a flashing blue light when engaged in direct law enforcement or public safety activities. This light must be located so that it does not interfere with the visibility of the vessel's navigation lights.

(169) (b) The blue light described in this section may be displayed by law enforcement vessels of the United States and the States and their political subdivisions.

(170)

§88.07 Public safety activities.

(171) (a) Vessels engaged in government sanctioned public safety activities, and commercial vessels performing similar functions, may display an alternately flashing red and yellow light signal. This identification light signal must be located so that it does not interfere with the visibility of the vessel's navigation lights. The identification light signal may be used only as an identification signal and conveys no special privilege. Vessels using the identification light signal during public safety activities must abide by the Inland Navigation Rules, and must not presume that the light or the exigency gives them precedence or right of way.

(172) (b) Public safety activities include but are not limited to patrolling marine parades, regattas, or special water celebrations; traffic control; salvage; firefighting; medical assistance; assisting disabled vessels; and search and rescue.

(173)

Part 89—INLAND NAVIGATION RULES: IMPLEMENTING RULES

(174)

Subpart A—Certificate of Alternative Compliance

(175)

§89.1 Definitions.

(176) As used in this subpart:

(177) *Inland Rules* refers to the Inland Navigation Rules contained in the Inland Navigational Rules Act of 1980 (Pub. L. 96-591) and the technical annexes established under that act.

(178) *A vessel of special construction or purpose* means a vessel designed or modified to perform a special function and whose arrangement is thereby made relatively inflexible.

(179) *Interference with the special function of the vessel* occurs when installation or use of lights, shapes, or sound-signaling appliances under the Inland Rules prevents or significantly hinders the operation in which the vessel is usually engaged.

(180)

§89.3 General.

(181) Vessels of special construction or purpose which cannot fully comply with the light, shape, and sound signal provisions of the Inland Rules without interfering with their special function may instead meet alternative requirements. The Chief of the Prevention Division in each Coast Guard District Office makes this determination and requires that alternative compliance be as close as possible with the Inland Rules. These regulations set out the procedure by which a vessel may be certified for alternative compliance. The information collection and recordkeeping requirements in §§89.5 and 89.18 have been approved by the Office of Management and Budget under OMB control No. 1625-0019.

(182)

§89.5 Application for a Certificate of Alternative Compliance.

(183) (a) The owner, builder, operator, or agent of a vessel of special construction or purpose who believes the vessel cannot fully comply with the Inland Rules light, shape, or sound signal provisions without interference with its special function may apply for a determination that alternative compliance is justified. The application must be in writing, submitted to the Chief of the Prevention Division of the Coast Guard District in which the vessel is being built or operated, and include the following information:

(184) (1) The name, address, and telephone number of the applicant.

(185) (2) The identification of the vessel by its:

(186) (i) Official number;

(187) (ii) Shipyard hull number;

(188) (iii) Hull identification number; or

(189) (iv) State number, if the vessel does not have an official number or hull identification number.

(190) (3) Vessel name and home port, if known.

(191) (4) A description of the vessel's area of operation.

(192) (5) A description of the provision for which the Certificate of Alternative Compliance is sought, including:

(193) (i) The Inland Rules Rule or Annex section number for which the Certificate of Alternative Compliance is sought;

(194) (ii) A description of the special function of the vessel that would be interfered with by full compliance with the provision of that Rule or Annex section; and

(195) (iii) A statement of how full compliance would interfere with the special function of the vessel.

(196) (6) A description of the alternative installation that is in closest possible compliance with the applicable Inland Navigation Rules Rule or Annex section.

(197) (7) A copy of the vessel's plans or an accurate scale drawing that clearly shows:

(198) (i) The required installation of the equipment under the Inland Rules,

(199) (ii) The proposed installation of the equipment for which certification is being sought, and

(200) (iii) Any obstructions that may interfere with the equipment when installed in:

(201) (A) The required location; and

(202) (B) The proposed location.

(203) (b) The Coast Guard may request from the applicant additional information concerning the application.

(204)

§89.9 Certificate of Alternative Compliance: Contents.

(205) The Chief of the Prevention Division issues the Certificate of Alternative Compliance to the vessel based on a determination that it cannot comply fully with Inland Rules light, shape, and sound signal provisions without interference with its special function. This Certificate includes:

(206) (a) Identification of the vessel as supplied in the application under §89.5(a)(2);

(207) (b) The provision of the Inland Rules for which the Certificate authorizes alternative compliance;

(208) (c) A certification that the vessel is unable to comply fully with the Inland Rules light, shape, and sound signal requirements without interference with its special function;

(209) (d) A statement of why full compliance would interfere with the special function of the vessel;

(210) (e) The required alternative installation;

(211) (f) A statement that the required alternative installation is in the closest possible compliance with the Inland Rules without interfering with the special function of the vessel;

(212) (g) The date of issuance;

(213) (h) A statement that the Certificate of Alternative Compliance terminates when the vessel ceases to be usually engaged in the operation for which the certificate is issued.

(214)

§89.17 Certificate of Alternative Compliance: Termination.

(215) The Certificate of Alternative Compliance terminates if the information supplied under §89.5(a) or the Certificate issued under §89.9 is no longer applicable to the vessel.

(216)

§89.18 Record of certification of vessels of special construction or purpose.

(217) (a) Copies of Certificates of Alternative Compliance and documentation concerning Coast Guard vessels are available for inspection at the offices of the Marine Transportation Systems Directorate, U.S. Coast Guard Headquarters (CG-5PW), Stop 7509, 2703 Martin Luther King Avenue SE., Washington, DC 20593-7509.

(218) (b) The owner or operator of a vessel issued a Certificate shall ensure that the vessel does not operate unless the Certificate of Alternative Compliance or a certified copy of that Certificate is on board the vessel and available for inspection by Coast Guard personnel.

(219)

Subpart B—Waters Upon Which Certain Inland Navigation Rules Apply

(220)

§89.21 Purpose.

(221) Inland Navigation Rules 9(a)(ii), 14(d), and 15(b) apply to the Great Lakes, and along with 24(i), apply on the "Western Rivers" as defined in Rule 3(1), and to additional specifically designated waters. The purpose of this Subpart is to specify those additional waters upon which Inland Navigation Rules 9(a)(ii), 14(d), 15(b), and 24(i) apply.

(222)

§89.23 Definitions.

(223) As used in this subpart:

(224) *Inland Rules* refers to the Inland Navigation Rules contained in the Inland Navigational Rules Act of 1980 (Pub. L. 96-591, 33 U.S.C. 2001 et. seq.) and the technical annexes established under that Act.

(225)

§89.25 Waters upon which Inland Rules 9(a)(ii), 14(d), and 15(b) apply.

(226) Inland Rules 9(a)(ii), 14(d), and 15(b) apply on the Great Lakes, the Western Rivers, and the following specified waters:

(227) (a) Tennessee-Tombigbee Waterway.

(228) (b) Tombigbee River.

(229) (c) Black Warrior River.

(230) (d) Alabama River.

(231) (e) Coosa River.

(232) (f) Mobile River above the Cochrane Bridge at St. Louis Point.

(233) (g) Flint River.

(234) (h) Chattahoochee River.

(235) (i) The Apalachicola River above its confluence with the Jackson River.

(236)

§89.27 Waters upon which Inland Rule 24(j) applies.

(237) (a) Inland Rule 24(j) applies on the Western Rivers and the specified waters listed in §89.25 (a) through (i).

(238) (b) Inland Rule 24(j) applies on the Gulf Intracoastal Waterway from St. Marks, Florida, to the Rio Grande, Texas, including the Morgan City-Port Allen Alternate Route and the Galveston-Freeport Cutoff, except that a power-driven vessel pushing ahead or towing alongside shall exhibit the lights required by Inland Rule 24(c), while transiting within the following areas:

(239) (1) St. Andrews Bay from the Hathaway Fixed Bridge at Mile 284.6 East of Harvey Locks (EHL) to the DuPont Fixed Bridge at Mile 295.4 EHL.

(240) (2) Pensacola Bay, Santa Rosa Sound and Big Lagoon from the Light "10" off of Trout Point at Mile 176.9 EHL to the Pensacola Fixed Bridge at Mile 189.1 EHL.

(241) (3) Mobile Bay and Bon Secour Bay from the Dauphin Island Causeway Fixed Bridge at Mile 127.7 EHL to Little Point Clear at Mile 140 EHL.

(242) (4) Mississippi Sound from Grand Island Waterway Light "1" at Mile 53.8 EHL to Light "40" off the West Point of Dauphin Island at Mile 118.7 EHL.

(243) (5) The Mississippi River at New Orleans, Mississippi River-Gulf Outlet Canal and the Inner Harbor Navigation Canal from the junction of the Harvey Canal and the Algiers Alternate Route at Mile 6.5 West of Harvey Locks (WHL) to the Michoud Canal at Mile 18 EHL.

(244) (6) The Calcasieu River from the Calcasieu Lock at Mile 238.6 WHL to the Ellender Lift Bridge at Mile 243.6 WHL.

(245) (7) The Sabine Neches Canal from mile 262.5 WHL to mile 291.5 WHL.

(246) (8) Bolivar Roads from the Bolivar Assembling Basin at Mile 346 WHL to the Galveston Causeway Bridge at Mile 357.3 WHL.

(247) (9) Freeport Harbor from Surfside Beach Fixed Bridge at Mile 393.8 WHL to the Bryan Beach Pontoon Bridge at Mile 397.6 WHL.

(248) (10) Matagorda Ship Channel area of Matagorda Bay from Range "K" Front Light at Mile 468.7 WHL to the Port O'Connor Jetty at Mile 472.2 WHL.

(249) (11) Corpus Christi Bay from Redfish Bay Day Beacon "55" at Mile 537.4 WHL when in the Gulf Intracoastal Waterway main route or from the north end of Lydia Ann Island Mile 531.1A when in the Gulf Intracoastal Waterway Alternate Route to Corpus Christi Bay LT 76 at Mile 543.7 WHL.

(250) (12) Port Isabel and Brownsville Ship Channel south of the Padre Island Causeway Fixed Bridge at Mile 665.1 WHL.

(251)

Part 90—INLAND RULES: INTERPRETATIVE RULES

(252)

§90.1 Purpose.

(253) This part contains the interpretative rules for the Inland Rules. These interpretative rules are intended as a guide to assist the public and promote compliance with the Inland Rules.

(254)

§90.3 Pushing vessel and vessel being pushed: Composite unit.

(255) Rule 24(b) of the Inland Rules states that when a pushing vessel and a vessel being pushed ahead are rigidly connected in a composite unit, they are regarded as a power-driven vessel and must exhibit the lights prescribed in Rule 23. A "composite unit" is interpreted to be the combination of a pushing vessel and a vessel being push ahead that are rigidly connected by mechanical means so they react to sea and swell as one vessel. Mechanical means does not include lines, wires, hawsers, or chains.

(256)

§90.5 Lights for moored vessels.

(257) A *vessel at anchor* includes a vessel made fast to one or more mooring buoys or other similar device attached to the ocean floor. Such vessels may be lighted as a vessel at anchor in accordance with Rule 30, or may be lighted on the corners in accordance with 33 CFR 88.30(h) through (l).

(258)

§90.7 Sidelights for unmanned barges.

(259) An unmanned barge being towed may use the exception of COLREGS Rule 24(h). However, this exception only applies to the vertical sector requirements for sidelights.

(260)

Part 110–Anchorage Regulations

(261)

§110.1 General.

(262) (a) The areas described in subpart A of this part are designated as special anchorage areas for the purposes of rule 30 (33 CFR 83.30) and rule 35 (33 CFR 83.35) of the Inland Navigation Rules, 33 CFR Chapter I, Subchapter E. Vessels of less than 20 meters in length; and barges, canal boats, scows, or other nondescript craft, are not required to sound signals required by rule 35 of the Inland Navigation Rules. Vessels of less than 20 meters are not required to exhibit anchor lights or shapes required by rule 30 of the Inland Navigation Rules.

(263) (b) The anchorage grounds for vessels described in Subpart B of this part are established, and the rules and regulations in relation thereto adopted, pursuant to the authority contained in section 7 of the act of March 4, 1915, as amended (38 Stat. 1053; 33 U.S.C. 471).

(264) (c) All bearings in the part are referred to true meridian.

(265) (d) Geographic coordinates expressed in terms of latitude or longitude, or both, are not intended for plotting on maps or charts whose referenced horizontal datum is the North American Datum of 1983 (NAD 83), unless such geographic coordinates are expressly labeled NAD 83. Geographic coordinates without the NAD 83 reference may be plotted on maps or charts referenced to NAD 83 only after application of the appropriate corrections that are published on the particular map or chart being used.

(266)

§ 110.1a Anchorages under Ports and Waterways Safety provisions.

(267) (a) The anchorages listed in this section are regulated under **46 U.S.C. Chapter 700**, "Ports and Waterways Safety":

(268) (1) Section 110.155 *Port of New York*.

(269) (2) [Reserved]

(270) (b) [Reserved]

(271)

Subpart A–Special Anchorage Areas

(272)

§110.38 Edgartown Harbor, Mass.

(273) An area in the inner harbor easterly of the project channel and south of Chappaquiddick Point bounded as follows: Beginning at

(274) 41°23'19"N., 70°30'32"W.; thence southeasterly along the shore to

(275) 41°22'52"N., 70°30'12"W.; thence

(276) 287°30', 1,600 feet; thence

(277) 327°30', 700 feet; thence

(278) 359°, 800 feet; thence

(279) 024°15', approximately 900 feet to the point of beginning.

(280)

§110.40 Silver Beach Harbor, North Falmouth, Mass.

(281) All the waters of the harbor northward of the inner end of the entrance channel.

(282)

§110.45 Onset Bay, Mass.

(283) Northerly of a line extending from the northernmost point of Onset Island to the easternmost point of Wickets Island; easterly of a line extending from the easternmost point of Wickets Island to the southwest extremity of Point Independence; southerly of the shoreline; and westerly of the shoreline and of a line bearing due north from the northernmost point of Onset Island.

(284)

§110.45a Mattapoisett Harbor, Mattapoisett, Mass.

(285) (a) Area No. 1 beginning at a point on the shore at

(286) 41°39'23"N., 70°48'50"W.; thence 138.5°T. to

(287) 41°38'45"N., 70°48'02"W.; thence 031°T. to

(288) 41°39'02"N., 70°47'48"W.; thence along the shore to the point of beginning.

(289) (b) Area No. 2 beginning at a point on the shore at

(290) 41°39'24"N., 70°49'02"W.; thence 142.5°T to

(291) 41°38'10"N., 70°47'45"W.; thence 219°T. to

(292) 41°37'54"N., 70°48'02"W.; thence along the shore to the point of beginning.

(293)

§110.46 Newport Harbor, Newport, R.I.

(294) (a) *Area No. 1*. The waters of Brenton Cove south of a line extending from

(295) 41°28'50"N., 71°18'58"W.; to

(296) 41°28'45"N., 71°20'08"W.; thence along the shoreline to the point of beginning.

(297) (b) *Area No. 2*. The waters east of Goat Island beginning at a point bearing 090°, 245 yards from Goat Island Shoal Light; thence

(298) 007°, 505 yards; thence

(299) 054°, 90 yards; thence

(300) 086°, 330 yards; thence

(301) 122°, 90 yards; thence

(302) 179°, 290 yards; thence

(303) 228°, 380 yards; thence

(304) 270°, 250 yards to the point of beginning.

(305) (c) *Area No. 3*. The waters north of Goat Island Causeway Bridge beginning at Newport Harbor Light; thence 023° to the southwest corner of Anchorage E; thence 081° following the southerly boundary of Anchorage E to the shoreline; thence south along the shoreline to the east foot of the Goat Island Causeway Bridge; thence west following Goat Island Causeway Bridge to the shoreline of Goat Island; thence north following the east shore of Goat Island to the point of beginning.

(306)

§110.47 Little Narragansett Bay, Watch Hill, R.I.

(307) All of the navigable waters of Watch Hill Cove southeasterly of a line beginning at the shore end of the United States project groin on the southerly shore of the

cove and running 41°30' true, to the northerly shore of the cove at a point about 200 feet west of the west side of the shore end of Meadow Lane, with the exception of a 100-foot wide channel running from the westerly end of the cove in a southeasterly direction to the Watch Hill Yacht Club pier, thence along in front of the piers on the easterly side of the cove northerly to the shore at the north end of the cove.

(308)

§110.48 Thompson Cove on east side of Pawcatuck River below Westerly, R.I.

(309) Eastward of a line extending from the channelward end of Thompson Dock at the northern end of Thompson Cove 184° to the shore at the southern end of Thompson Cove.

(310)

§110.50 Stonington Harbor, Conn.

(311) (a) *Area No. 1*. Beginning at the southeastern tip of Wamphassuc Point; thence to the northwesterly end of Stonington Inner Breakwater; thence along the breakwater to

(312) 71°54'50.5"W.; thence to

(313) 41°20'25.3"N., 71°54'50.5"W.; thence to a point on the shoreline at

(314) 41°20'32"N., 71°54'54.8"W.; thence along the shoreline to the point of beginning.

(315) (b) *Area No. 2*. Beginning at a point on the shoreline at

(316) 41°19'55.8"N., 71°54'28.9"W.; thence to

(317) 41°19'55.8"N., 71°54'37.1"W.; thence to

(318) 41°20'01.6"N., 71°54'38.8"W.; thence to

(319) 41°20'02.0"N., 71°54'34.3"W.; thence along the shoreline to the point of beginning.

(320) (c) *Area No. 3*. Beginning at a point on the shoreline at

(321) 41°20'29.5"N., 71°54'43.0"W.; thence to

(322) 41°20'25.6"N., 71°54'48.5"W.; thence to

(323) 41°20'10.7"N., 71°54'48.5"W.; thence to the shoreline at

(324) 41°20'10.7"N.; thence along the shoreline to the point of beginning.

(325)

§110.50a Fishers Island Sound, Stonington, Conn.

(326) An area on the east side of Mason Island bounded as follows:

(327) Beginning at the shore line on the easterly side of Mason Island at

(328) 41°20'06"N.; thence due east about 600 feet to

(329) 41°20'06"N., 71°57'37"W.; thence due south about 2,400 feet to

(330) 41°19'42"N., 71°57'37"W.; thence due west about 1,000 feet to the shore line on the easterly side of Mason Island at

(331) 41°19'42"N.; thence along the shore line to the point of beginning.

(332)

§110.50b Mystic Harbor, Groton and Stonington, Conn.

(333) (a) *Area No. 1*. Beginning at Ram Point on the westerly side of Mason Island at

(334) 41°19'44"N., 71°58'42"W.; thence to

(335) 41°19'30"N., 71°58'43"W.; thence to

(336) 41°19'36"N., 71°58'58"W.; thence to

(337) 41°19'45"N., 71°58'56"W.; thence to the point of beginning.

(338) (b) *Area No. 2*. Beginning at a point about 250 feet southerly of Area 1 and on line with the easterly limit of Area 1 at

(339) 41°19'27"N., 71°58'44"W.; thence to

(340) 41°19'19"N., 71°58'45"W.; thence to

(341) 41°19'25"N., 71°58'59"W.; thence to

(342) 41°19'33"N., 71°58'58"W.; thence to the point of beginning.

(343)

§110.50c Mumford Cove, Groton, Conn.

(344) (a) *Area No. 1*. Beginning at a point on the easterly shore of Mumford Cove at

(345) 41°19'36"N., 72°01'06"W.; to

(346) 41°19'30"N., 72°01'04"W.; thence to the shore at

(347) 41°19'31"N., 72°01'00"W.; and thence along the shoreline to the point of beginning.

(348) (b) *Area No. 2*. Beginning at a point on the easterly shore of Mumford Cove at

(349) 41°19'15.0"N., 72°00'54.0"W.; thence to

(350) 41°19'14.5"N., 72°00'59.0"W.; thence to

(351) 41°19'11.0"N., 72°00'58.0"W.; thence to

(352) 41°19'10.0"N., 72°00'54.0"W.; thence to

(353) 41°19'12.5"N., 72°00'52.0"W.; thence to

(354) 41°19'14.0"N., 72°00'55.0"W.; and thence to the point of beginning.

(355)

§110.51 Groton, Conn.

(356) The waters between an unnamed cove and Pine Island. (a) Beginning at a point on the shoreline of Avery Point at

(357) 41°19'01.4"N., 72°03'42.8"W.; thence to a point in the cove at

(358) 41°19'02.5"N., 72°03'36.2"W.; thence southeasterly to a point at

(359) 41°18'56.2"N., 72°03'34.2"W.; thence northeasterly to

(360) 41°19'02.5"N., 72°03'19.2"W.; thence terminating at the tip of Jupiter Point at

(361) 41°19'04.4"N., 72°03'19.7"W. DATUM: NAD 83

(362) (b) Beginning at a point on the shoreline of Pine Island at

(363) 41°18'47.1"N., 72°03'36.8"W.; thence northerly to

(364) 41°18'54.1"N., 72°03'35.4"W.; thence northeasterly to a point at

(365) 41°19'01.2"N., 72°03'19.3"W.; thence terminating at a point at

(366) 41°18'54.0"N., 72°03'17.5"W. DATUM NAD 83

(367)

§110.52 Thames River, New London, Conn.

(368) (a) *Area No. 1*. An area in the westerly part of Greens Harbor bounded as follows: Beginning at a point on the shore 100 yards southeasterly of the southerly side of Thames Street extended; thence 84°, 420 yards; thence 156°, 425 yards; thence 240°, 210 yards to the shore; and thence northwesterly along the shore to the point of beginning.

(369) (b) *Area No. 2*. An area in the westerly part of Greens Harbor bounded as follows: Beginning at a point on the shore 15 yards southeasterly of the southerly side of Converse Place extended; thence 54°, 170 yards; thence 114°30', 550 yards; thence 266°30', 250 yards; thence 234°, 230 yards, to the shore; and thence northwesterly along the shore to the point of beginning.

(370) (c) *Area No. 3*. An area on the westerly side of the Thames River in the vicinity of Jacobs Rock, the location of the U.S. Coast Guard Academy Sailing Center, bounded as follows: Beginning at the point on the shore where the north side of the Jacobs Rock causeway meets the western shoreline; thence northerly along the western shore of the Thames River a distance of 200 yards; thence 090°, 240 yards; thence 180°, 200 yards to the Jacobs Rock causeway; thence westerly along the causeway to the point of beginning.

(371) (d) *Area No. 4*. An area in the western part of the Thames River, north of the highway bridge, bounded as follows: Beginning at a point 125 yards north of the highway bridge at

(372) 41°21'56"N., 72°05'32"W.; thence easterly to

(373) 41°21'56"N., 72°05'27"W.; thence northerly to

(374) 41°22'12"N., 72°05'27"W.; thence westerly to

(375) 41°22'12"N., 72°05'47"W.; thence southeasterly to

(376) 41°22'02"N., 72°05'40"W.; thence downriver along the charted foul grounds to the point of beginning.

(377)

§110.53 Niantic, Conn.

(378) Beginning on the shoreline at

(379) 41°18'25.3"N., 72°12'16.3"W.; thence to

(380) 41°18'23.3"N., 72°12'11.6"W.; thence to

(381) 41°18'50.7"N., 72°11'51.5"W.; thence to the shore at

(382) 41°18'56.5"N., 72°12'05.6"W.; thence along the shoreline to the point of beginning.

(383)

§110.54 Long Island Sound, on west side of entrance to Pataguanset River, Conn.

(384) An area east of Giants Neck (formerly known as Grant Neck) described as follows: Beginning at a point bearing

(385) 114°, 75 feet, from the outer end of the breakwater at the south end of Giants Neck; thence

(386) 90°, 1,050 feet; thence

(387) 22°17'30", 2,140 feet; thence

(388) 283°27'15.5", 240 feet; thence

(389) 220°36'39", 1,252.6 feet; thence

(390) 295°23'16.5", 326.5 feet; thence

(391) 269°02'42.6", 240 feet; thence

(392) 261°46'50.9", 181.9 feet; thence

(393) 226°28'07.7", 275.9 feet; thence

(394) 147°43'27.7", 449.4 feet; thence

(395) 238°01'35.8", 379.6 feet; and thence approximately

(396) 156°31'05.8", 462.11 feet, to the point of beginning.

(397)

§110.55 Connecticut River, Conn.

(398) (a) *West of Calves Island at Old Saybrook*. Beginning at a point bearing

(399) 254°09'16", 153 yards, from Calves Island 20 Light; thence

(400) 157°, 1,037 yards; thence

(401) 175°, 150 yards; thence

(402) 265°, 250 yards; thence

(403) 350°, 660 yards; thence

(404) 337°, 460 yards; and thence approximately

(405) 67°, 135 yards, to the point of beginning.

(406) (a–1) *Area No. 1, at Essex*. Beginning at a point on the shore on the west side of Haydens Point bearing approximately 211°, 270 yards, from Haydens Point Light; thence

(407) 270°, 160 yards; thence

(408) 000°, 140 yards; thence

(409) 300°, 190 yards; thence

(410) 330°, 400 yards; thence

(411) 090°, 60 yards; thence

(412) 150°, 350 yards; thence

(413) 120°, about 434 yards to a point on the shore; thence along the shore to the point of beginning.

(414) (b) *Area No. 2, at Essex*. Beginning at

(415) 41°21'22"N., 72°22'53"W.; thence

(416) 205°30', 375 yards; thence

(417) 194°31', 100 yards; thence

(418) 185°00', 440 yards; thence

(419) 153°30', 80 yards; thence

(420) 121°00', 220 yards; thence

(421) 000°00', approximately 1060 yards to the point of beginning.

(422) (c) *West of Brockway Island at Essex*. That portion of the waters northwest of a line ranging 238° from latitude 41°22'20.7", longitude 72°22'49.8" to the shoreline; southwest of a line connecting a point at latitude 41°22'20.7", longitude 72°22'49.8" and a point at latitude 41°22'28.2", longitude 72°22'56"; and southeast of a line ranging 238° from latitude 41°22'28.2", longitude 72°22'56" to the shoreline.

(423) (d) *Upper Bay*—(1) *Anchorage No. 20-A*. (i) All waters bound by the following points:

(424) 40°42′06.9″N., 074°02′18.0″W.; thence to

(425) 40°42′05.4″N., 074°01′56.9″W.; thence to

(426) 40°41′54.9″N., 074°01′57.7″W.; thence to

(427) 40°41′54.0″N., 074°02′12.0″W.; thence to

(428) 40°41′54.4″N., 074°02′11.7″W.; thence to

(429) 40°41′57.5″N., 074°02′07.5″W.; thence to

(430) 40°42′06.1″N., 074°02′19.1″W.; thence to the point of origin (NAD 83).

(431) (ii) See 33 CFR 110.155(d)(6), (d)(16), and (l).

(432) (2) *Anchorage No. 20-B*. (i) All waters bound by the following points:

(433) 40°41′46.2″N., 074°02′23.0″W.; thence to

(434) 40°41′42.4″N., 074°02′00.5″W.; thence to

(435) 40°41′35.7″N., 074°02′02.7″W.; thence to

(436) 40°41′30.3″N., 074°02′06.3″W.; thence to

(437) 40°41′41.9″N., 074°02′29.2″W.; thence to the point of origin (NAD 83).

(438) (ii) See 33 CFR 110.155(d)(6), (d)(16), and (l).

(439) (3) *Anchorage No. 20-C*. (i) All waters bound by the following points:

(440) 40°41′42.4″N., 074°02′41.5″W.; thence to

(441) 40°41′25.8″N., 074°02′09.2″W.; thence to

(442) 40°41′02.1″N., 074°02′24.7″W.; thence to

(443) 40°41′09.4″N., 074°02′40.0″W.; thence to

(444) 40°41′13.3″N., 074°02′41.5″W.; thence to

(445) 40°41′15.8″N., 074°02′32.6″W.; thence to

(446) 40°41′25.3″N., 074°02′29.1″W.; thence to

(447) 40°41′33.0″N., 074°02′44.5″W.; thence to

(448) 40°41′32.5″N., 074°02′48.8″W.; thence to the point of origin (NAD 83).

(449) (ii) See 33 CFR 110.155(d)(6), (d)(16), and (l).

(450) (e) *Area No. 2, at Lord Island*. Beginning at 41°26′11″N., 72°27′16″W.; thence extending south southeasterly to

(451) 41°26′03″N., 72°27′02″W.; thence extending southeasterly to

(452) 41°25′59″N., 72°26′51″W.; thence extending southwesterly to

(453) 41°25′58″N., 72°26′52″W.; thence extending northwesterly to

(454) 41°26′05″N., 72°27′11″W.; thence extending north northwesterly to

(455) 41°26′10″N., 72°27′20″W.; thence extending easterly to the point of beginning.

(456) (e–1) *Area No. 1 at Chester*. Beginning at a point about 600 feet southeasterly of the entrance of Chester Creek, at

(457) 41°24'23"N., 72°25'41"W.; to

(458) 41°24'05"N., 72°25'41"W.; to

(459) 41°24'05"N., 72°25'32"W.; to

(460) 41°24'23"N., 72°25'32"W.; thence due west about 600 feet to the point of beginning.

(461) (e–2) *Area No. 2 at Chester*. That area south of latitude 41°24'43.9", west of longitude 72°25'35", north of latitude 41°24'33.4", and east of longitude 72°25'40.8".

(462) (f) *Vicinity of Mouse Island Bar below Portland*. On the north side of the river shoreward of lines described as follows:

(463) (1) Beginning at a point bearing 02°, 175 yards, from Mouse Island 73 Light; thence 270°, 480 yards; and thence due north, approximately 230 yards, to the shore.

(464) (2) Beginning at the said point bearing 02°, 175 yards, from Mouse Island 73 Light; thence 70°, 400 yards; and thence 350°, approximately 250 yards, to the shore.

(465) (g) *Area at Portland*. Beginning at a point on the shore, about 700 feet southeasterly from the easterly end of the New York, New Haven and Hartford Railroad Company bridge at

(466) 41°33'55"N., 72°38'43"W.; thence 250° to

(467) 41°33'54"N., 72°38'46"W.; thence 160° to

(468) 41°33'48"N., 72°38'43W."; thence 145° to

(469) 41°33'44"N., 72°38'39"W.; thence 055° to a point on the shore at

(470) 41°33'47"N., 72°38'32"W.; thence along the shore to the point of beginning.

(471)

§110.55a Five Mile River, Norwalk and Darien, Conn.

(472) The water area of the Five Mile River beginning at a point on the southeast shore of Butler Island at 41°03'27.5"N., 73°26'52"W.; thence following the shoreline northerly along the westerly side of Five Mile River to the highway bridge at Route 136 (White Bridge); thence easterly along the southerly side of the highway bridge to the easterly side of Five Mile River; thence following the shoreline southerly along the easterly side of Five Mile River to a point on the southwest shore at Rowayton at 41°03'30"N., 73°26'47"W., thence 242° to the point of beginning, except those areas within the designated project channel as shown by dotted lines on the Five Mile River on Chart No. 12368 (formerly C and GS Chart No. 221) issued by National Oceanic and Atmospheric Administration, U.S. Department of Commerce.

(473)

§ 110.55b Connecticut River, Old Saybrook, Connecticut.

(474) (a) *Special anchorage area A*. All of the waters enclosed by a line beginning at latitude 41°19'54.75"N., longitude 072°21'08.40"W.; thence to latitude 41°19'21.50"N., longitude 072°20'49.65"W.; thence to latitude 41°19'17.80"N., longitude 072°20'49.25"W.; thence to latitude 41°19'17.05"N., longitude 72°20'59"W.; thence to latitude 41°19'25.40"N., longitude 72°21'00.95"W.; thence to latitude 41°19'29.50"N., longitude 72°21'17.60"W.; thence to latitude 41°19'35.40"N., longitude 72°21'22.90"W.; thence to latitude 41°19'52.35"N., longitude 72°21'26.10"W.; thence to the point of beginning.

(475) (b) *Special anchorage area B*. All of the waters enclosed by a line beginning at latitude 41°17'26"N., longitude 072°21'04"W.; thence to latitude 41°17'24.60"N., longitude 072°21'16"W.; thence to latitude 41°17'20"N., longitude 072°21'09"W.; thence to latitude 41°17'16"N., longitude 072°21'05"W.; thence to latitude 41°17'16"N., longitude 072°21'03"W.; thence to latitude 41°17'21.5"N., longitude 072°21'04.5"W.; thence to the point of beginning.

(476) (c) *Special anchorage area C*. All of the waters enclosed by a line beginning at latitude 41°17'27"N., longitude 072°21'35"W.; thence to latitude 41°17'24"N., longitude 072°22'01"W.; thence to latitude 41°17'16"N.,

longitude 072°22'00"W.; thence to latitude 41°17'19"N., longitude 072°21'33"W.; thence to the point of beginning.

(477)

§110.56 Noroton Harbor, Darien, Conn.

(478) (a) Beginning at a point on the southwesterly side of Long Neck Point at

(479) 41°02'10"N., 73°28'44"W.; thence northwesterly to

(480) 41°02'17"N., 73°29'11"W.; thence in a north-northwesterly direction to the southeast side of Pratt Island at

(481) 41°02'28"N., 73°29'17"W.; thence following the shoreline around the easterly and northerly sides of Pratt Island, the westerly and northerly sides of Pratt Cove, and the westerly side of the Darien River to the causeway and dam at Gorham Pond on the north; thence along the downstream side of the causeway and dam to the easterly side of the Darien River, thence along the easterly shoreline to the point of beginning.

(482)

§110.58 Cos Cob Harbor, Greenwich, Conn.

(483) (a) *Area A*. Beginning at the mean low water line about 2,800 feet downstream from the easterly end of the New York, New Haven and Hartford Railroad Bridge at

(484) 41°01'23"N., 73°35'40"W.; thence to

(485) 41°01'23"N., 73°35'42"W.; thence to

(486) 41°01'02"N., 73°35'50"W.; thence to

(487) 41°01'02"N., 73°35'48"W.; thence extending along the mean low water line to the point of beginning.

(488) (b) *Area B*. Beginning at the mean low water line about 700 feet downstream from the westerly end of the New York, New Haven and Hartford Railroad Bridge at

(489) 41°01'42"N., 73°35'47"W.; thence to

(490) 41°01'42"N., 73°35'45"W.; thence to

(491) 41°01'23"N., 73°35'44"W.; thence to

(492) 41°01'04"N., 73°35'52"W.; thence to

(493) 41°01'02"N., 73°35'55"W.; thence to

(494) 41°01'02"N., 73°36'00"W.; thence to

(495) 41°01'05"N., 73°36'00"W.; thence along the mean low water line to the point of beginning.

(496)

§110.59 Eastern Long Island, NY.

(497) (a) *Huntington Harbor*. Beginning on the shoreline at 40°54'19.5"N., 73°26'07.9"W.; thence to 40°54'19.5"N., 73°26'02.4"W.; thence along the eastern shoreline to the Mill Dam Road Bridge; thence along the downstream side of the bridge to the westerly side of Huntington Harbor; thence along the western shoreline to the point of beginning.

(498) (b) *Centerport Harbor*. Beginning at the shoreline at 40°54'00"N., 73°22'55.3"W.; thence to 40°54'03.8"N., 73°22'52.1"W.; thence along the eastern shoreline to the Mill Dam Bridge; thence along the downstream side of the bridge to the westerly side of Centerport Harbor; thence along the western shoreline to the point of beginning.

(499) (c) *Northport Harbor*. Beginning on the shoreline at

(500) 40°54′25″N., 73°22′05″W.; thence to

(501) 40°54′37.5″N., 73°21′32.9″ W.; thence along the eastern shoreline to

(502) 40°53′33.1″N., 73°21′28.2″W.; thence to

(503) 40°53′25.8″N., 73°21′37.7″W.; thence along the shoreline to the point of beginning.

(504) (d) *Cold Spring Harbor*. That portion of the waters of Cold Spring Harbor easterly of a line ranging from the cupola in the extreme inner harbor through Cold Spring Harbor Light; southerly of a line ranging from the southernmost point of an L-shaped pier off Wawepex Grove through the Clock Tower at Laurelton and northerly of a line ranging from the outer end of the Socony Mobil Oil Company's pier at Cold Spring Harbor through the Clock Tower at Laurelton, with the exception of an area within a 300-foot radius of the outer end of the Socony Mobil Oil Company's pier.

(505) (e) *Oyster Bay Harbor, New York*. That portion of Oyster Bay Harbor adjacent to the easterly side of Centre Island, westerly of a line on range with Cold Spring Harbor Light and the Stone House on the end of Plum Point, Centre Island.

(506) (f) *Harbor of Oyster Bay, Oyster Bay, New York*. The water area north of the town of Oyster Bay enclosed by a line beginning on the shoreline at

(507) 40°52'35.5"N., 73°32'17"W.; thence to

(508) 40°52'59.5"N., 73°32'18"W.; thence to

(509) 40°53'00"N., 73°30'53"W.; thence to

(510) 40°52'39"N., 73°30'54"W.; thence to the shoreline at

(511) 40°52'25"N., 73°31'18"W.; thence following the shoreline to the point of beginning.

(512) (g) *Harbor of Oyster Bay, New York, Moses Point to Brickyard Point*. That portion of the waters of the Harbor of Oyster Bay enclosed by a line beginning at Moses Point on Centre Island at

(513) 40°53'11"N., 73°31'14"W.; thence to

(514) 40°53'02"N., 73°31'22"W.; thence to

(515) 40°53'02"N., 73°32'00"W.; thence to Brickyard Point on Centre Island at

(516) 40°53'06"N., 73°32'00"W.; thence following the shoreline to the point of beginning.

(517) (h) *Coecles Harbor at Shelter Island, New York*. That portion of Coecles Harbor bounded on the North by a line drawn between the northernmost point of land at Sungic Point and 41°04'09"N., 72°17'54"W., thence eastward along the shoreline to the point of origin.

(518) (i) *West Neck Harbor at Shelter Island, New York*. That portion of West Neck Harbor bounded on the North by a line drawn between 41°02'48"N., 72°20'27"W. and a point on Shell Beach located at 41°02'29"N., 72°20'59"W.; thence eastward along the shoreline to the point of origin.

(519)

§110.60 Captain of the Port, New York.

(520) (a) *Western Long Island Sound*. (1) *Glen Island*. All waters surrounding Glen Island bound by the following points:

(521) 40°52'53.1"N., 073°46'58.9"W.; thence to

(522) 40°52'46.6"N., 073°47'02.7"W.; thence to
(523) 40°53'01.3"N., 073°47'22.6"W.; thence to a line drawn from
(524) 40°53'24.4"N., 073°46'56.7"W. to
(525) 40°53'20.6"N., 073°46'51.2"W., excluding all waters within 25 feet of the 50-foot channel west and south of Glen Island.
(526) (2) *Echo Bay*. All waters northwest of a line drawn from
(527) 40°54'10.0"N., 073°45'52.9"W. to
(528) 40°54'25.0"N., 073°45'38.4"W.
(529) (3) *Glen Island, East*. All waters east of Glen Island, bound by the following points:
(530) 40°53'01.4"N., 073°46'51.4"W.; thence to
(531) 40°53'03.1"N., 073°46'44.4"W.; thence to
(532) 40°53'06.2"N., 073°46'38.0"W.; thence to
(533) 40°53'15.0"N., 073°46'44.00"W.; thence along the shoreline to the point of origin.
(534) (4) *City Island, Eastern Shore*. All waters bound by the following points:
(535) 40°50'12.0"N., 073°46'57.3"W.; thence to
(536) 40°50'31.9"N., 073°46'18.3"W.; thence to
(537) 40°51'17.0"N., 073°46'49.9"W.; thence to
(538) 40°51'19.8"N., 073°46'51.3"W.; thence to
(539) 40°51'47.0"N., 073°47'02.5"W.; thence to
(540) 40°51'28.5"N., 073°47'31.7"W.; thence to
(541) 40°51'25.1"N., 073°47'29.9"W.; thence along the shoreline to the point of origin, excluding the Cable and Pipeline Area between City and Hart Islands.
(542) (5) *City Island, Western Shore*. All waters bound by the following points:
(543) 40°50'11.6"N., 073°46'58.4"W.; thence to
(544) 40°50'02.5"N., 073°47'23.3"W.; thence to
(545) 40°50'43.7"N., 073°47'56.0"W.; thence to
(546) 40°51'15.9"N., 073°47'36.0"W.; thence to
(547) 40°51'15.9"N., 073°47'28.6"W.; thence along the shoreline to the point of origin.
(548) (6) *Eastchester Bay, Western Shore*. All waters shoreward of a line connecting the following points:
(549) 40°49'31.3"N., 073°48'26.3"W.; thence to
(550) 40°50'56.4"N., 073°48'49.2"W.; thence to
(551) 40°50'55.3"N., 073°48'55.4"W.; thence along the shoreline to the point of origin.
(552) (7) *Eastchester Bay, Locust Point*. All waters west of a line drawn from
(553) 40°48'56.3"N., 073°47'56.2"W. to
(554) 40°48'34.4"N., 073°47'56.2"W.
(555) (8) *Manhasset Bay, Plum Point*. All waters bound by the following points:
(556) 40°50'02.9"N., 073°43'37.3"W.; thence to
(557) 40°49'54.0"N., 073°43'14.9"W.; thence to
(558) 40°50'06.6"N., 073°42'51.0"W.; thence to
(559) 40°50'18.6"N., 073°42'51.0"W.; thence along the shoreline to the point of origin; excluding the seaplane restricted area described in § 162.
(560) (9) *Manhasset Bay, Toms Point*. All waters bound by the following points:
(561) 40°50'20.6"N., 073°42'49.5"W.; thence to
(562) 40°50'05.3"N., 073°42'49.4"W.; thence to
(563) 40°49'58.6"N., 073°42'39.0"W.; thence to
(564) 40°49'48.9"N., 073°42'55.6"W.; thence to
(565) 40°49'49.3"N., 073°42'20.4"W.; thence to
(566) 40°50'02.5"N., 073°42'14.2"W.; thence to
(567) 40°50'11.8"N., 073°42'15.4"W.; thence along the shoreline to the point of origin.
(568) (10) *Manhasset Bay, at Port Washington*. All waters bound by the following points:
(569) 40°49'44.9"N., 073°42'11.3"W.; thence to
(570) 40°49'44.3"N., 073°43'03.2"W.; thence to
(571) 40°49'06.8"N., 073°42'46.6"W.; thence to
(572) 40°49'07.0"N., 073°42'16.2"W.; thence along the shoreline to the point of origin.
(573) (11) *Manhasset Bay, West Shore*. All waters bound by the following points:
(574) 40°49'24.6"N., 073°43'40.2"W.; thence to
(575) 40°49'33.2"N., 073°43'28.3"W.; thence to
(576) 40°49'43.8"N., 073°43'53.5"W.; thence to
(577) 40°49'39.2"N., 073°43'57.9"W.; thence along the shoreline to the point of origin.
(578) (12) *Manhasset Bay, Plandome*. All waters bound by the following points:
(579) 40°48'41.6"N., 073°42'31.7"W.; thence to
(580) 40°48'43.6"N., 073°42'42.5"W.; thence to
(581) 40°48'29.0"N., 073°42'44.4"W.; thence to
(582) 40°48'27.3"N., 073°42'35.6"W.; thence along the shoreline to the point of origin.
(583) (13) *Elm Point*. All waters bound by the following points:
(584) 40°49'01.0"N., 073°45'41.9"W.; thence to
(585) 40°49'04.4"N., 073°45'45.3"W.; thence to
(586) 40°49'13.8"N., 073°45'38.7"W.; thence to
(587) 40°49'18.9"N., 073°45'28.3"W.; thence to
(588) 40°49'08.9"N., 073°45'17.5"W.; thence along the shoreline to the point of origin.
(589) (14) *Little Neck Bay*. All waters east of a line drawn from
(590) 40°47'39.4"N., 73°46'27.1"W.; thence to
(591) 40°48'36.6"N., 073°45'58.5"W.; thence to
(592) 40°48'36.4"N., 073°45'48.4"W.; thence along the shoreline to the point of origin.
(593) (15) *Hempstead Harbor, Mosquito Neck*. All waters bound by the following points:
(594) 40°51'43.0"N., 073°39'37.1"W.; thence to
(595) 40°51'09.4"N., 073°39'32.4"W.; thence to
(596) 40°51'14.6"N., 073°39'08.9"W.; thence to
(597) 40°51'20.0"N., 073°38'56.1"W.; thence along the shoreline and breakwater to the point of origin.
(598) (16) *Hempstead Harbor, Sea Cliff*. All waters bound by the following points:
(599) 40°51'16.7"N., 073°38'51.9"W.; thence to
(600) 40°51'12.9"N., 073°39'07.2"W.; thence to
(601) 40°51'03.6"N., 073°39'31.6"W.; thence to
(602) 40°50'24.7"N., 073°39'26.4"W.; thence to
(603) 40°50'22.0"N., 073°39'10.2"W.; thence along the shoreline to the point of origin.

(604) (b) *East River and Flushing Bay*. (1) *Flushing Bay, College Point North*. All waters bound by the following points:

(605) 40°47'37.5"N., 073°51'13.4"W.; thence to

(606) 40°47'10.3"N., 073°51'34.0"W.; thence to

(607) 40°47'09.1"N., 073°51'32.6"W.; thence along the shoreline to the point of origin.

(608) (2) *Flushing Bay, College Point South*. All waters bound by the following points:

(609) 40°47'01.8"N., 073°51'29.2"W.; thence to

(610) 40°47'01.8"N., 073°51'33.2"W.; thence to

(611) 40°46'31.7"N., 073°51'15.9"W.; thence to

(612) 40°46'46.1"N., 073°50'58.6"W.; thence along the shoreline to the point of origin.

(613) (3) *Flushing Bay, Cape Ruth*. All waters bound by the following points:

(614) 40°46'39.9"N., 073°50'56.1"W.; thence to

(615) 40°46'29.2"N., 073°51'14.3"W.; thence to

(616) 40°46'12.3"N., 073°51'04.3"W.; thence to

(617) 40°46'15.2"N., 073°50'55.2"W.; thence along the shoreline to the point of origin.

(618) (4) *Flushing Bay, Southeast Area*. All waters south of a line drawn from 40°45'41.4"N., 073°50'57.2"W. to 40°45'51.7"N., 073°50'34.2"W.

(619) (5) *Flushing Bay, Southwest Area*. All waters bound by the following points:

(620) 40°45'36.7"N., 073°51'16.3"W.; thence to

(621) 40°45'48.5"N., 073°50'58.4"W.; thence to

(622) 40°45'51.3"N., 073°50'59.2"W.; thence to

(623) 40°45'49.4"N., 073°51'07.5"W.; thence to

(624) 40°45'58.7"N., 073°51'13.4"W.; thence to

(625) 40°46'02.1"N., 073°51'20.1"W.; thence to

(626) 40°45'54.8"N., 073°51'28.7"W.; thence to

(627) 40°45'46.2"N., 073°51'35.3"W.; thence northward along the shoreline and breakwater to the point of origin.

(628) (6) *Flushing Bay, West Area*. All waters bound by the following points:

(629) 40°46'51.1"N., 073°52'07.3"W.; thence to

(630) 40°47'11.2"N., 073°51'47.1"W.; thence to

(631) 40°47'01.9"N., 073°51'39.6"W.; thence to

(632) 40°46'28.3"N., 073°51'20.0"W.; thence to the point of origin.

(633) (7) *Bowery Bay*. All waters bounded by the following points:

(634) 40°46'58.4"N., 073°53'44.1"W.; thence to

(635) 40°47'03.3"N., 073°53'37.4"W.; thence to

(636) 40°47'00.3"N., 073°53'29.3"W.; thence to

(637) 40°46'57.0"N., 073°53'29.8"W.; thence to

(638) 40°46'59.9"N., 073°53'34.2"W.; thence to

(639) 40°46'58.5"N., 073°53'35.8"W.; thence to

(640) 40°46'57.1"N., 073°53'33.8"W.; thence to

(641) 40°46'55.9"N., 073°53'35.2"W.; thence to

(642) 40°46'58.2"N., 073°53'39.0"W.; thence to

(643) 40°46'56.1"N., 073°53'41.4"W.; thence along the shoreline and pier to the point of origin.

(644) (c) *Hudson River*. (1) *Yonkers, Greystone Station*. All waters bound by the following points:

(645) 40°58'19.8"N., 073°53'22.8"W.; thence to

(646) 40°58'21.1"N., 073°53'28.7"W.; thence to

(647) 40°58'42.7"N., 073°53'20.3"W.; thence to

(648) 40°58'41.8"N., 073°53'15.4"W.; thence along the shoreline to the point of origin.

(649) (2) *Yonkers, North Glenwood*. All waters bound by the followings points:

(650) 40°57'26.8"N., 073°53'46.6"W.; thence to

(651) 40°57'27.3"N., 073°53'48.8"W.; thence to

(652) 40°57'55.3"N., 073°53'34.4"W.; thence to

(653) 40°57'53.6"N., 073°53'28.6"W.; thence along the shoreline to the point of origin.

(654) (3) *Nyack*. That portion of the Hudson River bound by the following points:

(655) 41°06'06.8"N., 073°54'55,5"W.; thence to

(656) 41°06'06.8"N., 073°54'18.0"W.; thence to

(657) 41°05'00.0"N., 073°54'18.0"W.; thence to

(658) 41°05'00.0"N., 073°55'02.2"W.; thence along the along the shoreline to the point of origin (NAD 1983), excluding a fairway in the charted cable area that is marked with buoys.

(659) (4) *Manhattan, Fort Washington Point*. All waters bound by the following points:

(660) 40°51'08.1"N., 073°56'36.7"W.; thence to

(661) 40°51'09.4"N., 073°56'40.9"W.; thence to

(662) 40°52'08.3"N., 073°55'56.6"W.; thence along the shoreline to the point of origin.

(663) (5) *Yonkers, Main Street*. All waters bound by the following points:

(664) 40°56'15.4"N., 073°54'11.2"W.; thence to

(665) 40°56'16.7"N., 073°54'20.2"W.; thence to

(666) 40°56'08.9"N., 073°54'22.6"W.; thence to

(667) 40°56'07.9"N., 073°54'16.9"W.; thence to

(668) 40°56'07.0"N., 073°54'17.3"W. to

(669) (6) *Yonkers, JFK Marina*. All waters bound by the following points:

(670) 40°57'28.5"N., 073°53'46.0"W.; thence to

(671) 40°57'30.5"N., 073°53'56.8"W.; thence to

(672) 40°57'07.5"N., 073°54'06.2"W.; thence to

(673) 40°57'06.0"N., 073°53'59.5"W.; thence along the shoreline to the point of origin.

(674) (7) *Hastings-on-Hudson*. All waters bound by the following points:

(675) 40°59'56.0"N., 073°53'05.4"W.; thence to

(676) 40°59'56.3"N., 073°53'09.6"W.; thence to

(677) 41°00'05.1"N., 073°53'09.2"W.; thence to

(678) 41°00'14.7"N., 073°53'06.4"W.; thence to

(679) 41°00'14.5"N., 073°53'00.5"W.; thence along the shoreline to the point of origin.

(680) (8) *Tarrytown*. All waters bound by the following points:

(681) 41°04'21.0"N., 073°52'03.4"W.; thence to

(682) 41°04'21.0"N., 073°52'11.3"W.; thence to

(683) 41°04'13.6"N., 073°52'11.0"W.; thence to

(684) 41°04'13.6"N., 073°52'00.5"W.; thence along the shoreline to the point of origin.

(685) (9) *West Point*. All waters west of a line drawn from 41°23'10.0"N., 073°57'18.1"W. to 41°23'23.5"N., 073°57'11.5"W.

(686) (10) *Haverstraw*. That portion of the Hudson River bound by the following points:
(687) 41°11'25.2"N., 073°57'19.9"W.; thence to
(688) 41°11'34.2"N., 073°57'00.8"W.; thence to
(689) 41°11'41.9"N., 073°57'07.5"W.; thence to
(690) 41°11'31.8"N., 073°57'26.5"W.; thence to
(691) 41°11'30.8"N., 073°57'24.9"W.; thence to the point of origin.
(692) (11) *Cedar Hill*. All waters bounded by the following points:
(693) 42°32'33.1"N., 073°45'33.1"W.; thence to
(694) 42°32'33.1"N., 073°45'28.3"W.; thence to
(695) 42°32'49.2"N., 073°45'26.6"W.; thence to
(696) 42°32'49.3"N., 073°45'31.1"W.; thence along the shoreline to the point of origin.
(697) (12) *79th Street Boat Basin South*. All waters of the Hudson River enclosed by a line beginning at the northwest corner of the 70th Street pier at approximate position
(698) 40°46′47.10″N, 073°59′29.13″W; thence to
(699) 40°47′02.60″N, 073°59′17.88″W; thence to
(700) 40°46′59.73″N, 073°59′13.01″W; thence along the shoreline and pier to the point of beginning.
(701) (d) *New York Harbor*. (1) *Newark Bay, Southeast*. All waters bound by the following points:
(702) 40°39'27.9"N., 074°08'07.1"W.; thence to
(703) 40°39'31.7"N., 074°08'13.4"W.; thence to
(704) 40°39'31.4"N., 074°08'24.6"W.; thence to
(705) 40°39'52.4"N., 074°08'11.7"W.; thence to
(706) 40°39'47.8"N., 074°07'59.4"W.; thence along the shoreline to the point of origin.
(707) (2) *Great Kills Harbor*. All waters northeast of a line connecting the following points:
(708) 40°32'06.4"N., 074°08'24.5"W.; thence to
(709) 40°32'06.9"N., 074°08'25.8"W.; thence to
(710) 40°32'19.0"N., 074°08'21.1"W.; thence to
(711) 40°32'28.1"N., 074°08'24.3"W.; thence to
(712) 40°32'40.3"N., 074°08'08.4"W.; thence to
(713) 40°32'45.2"N., 074°08'11.4"W.; thence along the northern and eastern shoreline to the point of origin.
(714) (3) *Jamaica Bay, Canarsie Beach*. All waters bound by the following points:
(715) 40°37'22.0"N., 073°53'43.5"W.; thence to
(716) 40°37'18.4"N., 073°53'32.9"W.; thence to
(717) 40°37'37.6"N., 073°53'06.5"W.; thence to
(718) 40°37'42.9"N., 073°53'14.4"W.; thence along the shoreline to the point of origin.
(719) (4) *Jamaica Bay, East Broad Channel*. All waters bound by the following points:
(720) 40°35'48.5"N., 073°49'12.5"W.; thence to
(721) 40°35'50.2"N., 073°49'04.7"W.; thence to
(722) 40°36'23.4"N., 073°48'56.3"W.; thence along the shoreline to the point of origin.
(723) (5) *Sheepshead Bay, West*. All waters bound by the following points:
(724) 40°35'00.0"N., 073°56'54.8"W.; thence to
(725) 40°34'58.9"N., 073°57'09.6"W.; thence to
(726) 40°34'56.6"N., 073°57'09.1"W.; thence to
(727) 40°34'57.5"N., 073°56'54.4"W.; thence to the point of origin.
(728) (6) *Sheepshead Bay, North*. All waters bound by the following points:
(729) 40°34'58.5"N., 073°56'00.5"W.; thence to
(730) 40°34'58.6"N., 073°56'26.0"W.; thence to
(731) 40°34'56.6"N., 073°56'26.8"W.; thence to
(732) 40°34'54.8"N., 073°56'24.8"W.; thence to
(733) 40°34'55.4"N., 073°56'10.1"W.; thence to
(734) 40°34'57.9"N., 073°56'00.5"W.; thence to the point of origin.
(735) (7) *Sheepshead Bay, South*. All waters bound by the following points:
(736) 40°34'54.2"N., 073°56'01.8"W.; thence to
(737) 40°34'53.6"N., 073°56'27.2"W.; thence to
(738) 40°34'55.8"N., 073°56'43.6"W.; thence to
(739) 40°34'54.5"N., 073°56'43.6"W.; thence to
(740) 40°34'52.0"N., 073°56'34.0"W.; thence to
(741) 40°34'53.1"N., 073°56'01.6"W.; thence to the point of origin.
(742) (8) *Lower Bay, Point Comfort*. All waters bound by the following points:
(743) 40°27'18.5"N., 074°08'24.5"W.; thence to
(744) 40°27'37.4"N., 074°08'51.8"W.; thence to
(745) 40°27'51.4"N., 074°08'31.9"W.; thence to
(746) 40°27'49.7"N., 074°07'44.9"W.; thence to
(747) 40°27'15.3"N., 074°07'45.7"W.; thence along the shoreline to the point of origin.
(748) (9) *Perth Amboy, NJ*. All waters bound by the following points:
(749) 40°30'26.00"N., 074°15'42.00"W.; thence to
(750) 40°30'24.29"N., 074°15'35.20"W.; thence to
(751) 40°30'02.79"N., 074°15'44.16"W.; thence to
(752) 40°29'35.70"N., 074°16'08.88"W.; thence to
(753) 40°29'31.00"N., 074°16'20.75"W.; thence to
(754) 40°29' 47.26"N., 074°16'49.82"W.; thence to
(755) 40°30'02.00"N., 074°16'41.00"W., thence along the shoreline to the point of origin.
(756) (e) *Datum*. All positions are NAD 1983.
(757)

Subpart B–Anchorage Grounds

(758)

§110.140 Buzzards Bay, Nantucket Sound, and adjacent waters, Mass.

(759) (a) *New Bedford Outer Harbor*–(1) *Anchorage A*. West of Sconticut Neck, and shoreward of a line described as follows: Beginning at a point 100 yards southwest of Fort Phoenix Point; thence 154° along a line which passes 100 yards east of New Bedford Channel Buoys 8, 6, and 4, to a point bearing approximately 130°, 225 yards, from New Bedford Channel Buoy 4; thence 87°, 340 yards; thence 156° along a line approximately one mile to its intersection with a line ranging 87° from the cupola on Clarks Point; thence 87° to Sconticut Neck.
(760) (2) *Anchorage B*. All waters bounded by a line beginning at

(761) 41°36′42.3″N., 70°54′24.9″W.; thence to
(762) 41°36′55.5″N., 70°54′06.6″W.; thence to
(763) 41°36′13.6″N., 70°53′40.2″W.; thence to
(764) 41°36′11.1″N., 70°54′07.6″W.; thence along the shoreline to the beginning point.

(765) (b) *Buzzards Bay near entrance to approach channel to Cape Cod Canal*–(1) *Anchorage C*. West of a line parallel to and 850 feet westward from the centerline of Cleveland Ledge Channel; north of a line bearing 129° from the tower on Bird Island; east of a line bearing 25°30' and passing through Bird Island Reef Bell Buoy 13; and south of a line bearing 270° from Wings Neck Light.

(766) Each vessel must obtain permission to proceed to Anchorage C from the U.S. Army Corps of Engineers Cape Cod Canal Control traffic controller.

(767) (2) *Anchorage D*. Beginning at a point bearing 185°, 1,200 yards, from Hog Island Channel 4 Light; thence 129° to a point bearing 209°, approximately 733 yards, from Wings Neck Light; thence 209° to Southwest Ledge Buoy 10; thence 199° along a line to its intersection with a line bearing 129° from the tower on Bird Island; thence 309° to a point 850 feet easterly, right angle distance, from the centerline of Cleveland Ledge Channel; thence northeasterly along a line parallel to and 850 feet eastward from the centerline of Cleveland Ledge Channel to its intersection with a line bearing 218°30' from the point of beginning; thence 38°30' to the point of beginning.

(768) Each vessel must obtain permission to proceed to Anchorage D from the U.S. Army Corps of Engineers Cape Cod Canal Control traffic controller.

(769) (3) *Anchorage L*. The area of water bounded by lines connecting the following points:
(770) 41°30'11"N., 70°48'10"W.; to
(771) 41°30'46"N., 70°48'45"W.; to
(772) 41°32'24"N., 70°45'50"W.; to
(773) 41°31'48"N., 70°45'15"W. and thence to start.

(774) (4) *Anchorage M–(westside)*. The waters bounded by a line connecting the following points:
(775) 41°35'35"N., 70°44'47"W.; to
(776) 41°36'24"N., 70°45'53"W.; to
(777) 41°35'00"N., 70°47'53"W.; to
(778) 41°34'12"N., 70°46'47"W. and thence to the beginning.

(779) (5) Each vessel that anchors in these anchorages must notify the U.S. Army Corps of Engineers Cape Cod Canal Control traffic controller when it anchors, and provide the vessel's name, length, draft, cargo, and its position.

(780) (6) Each vessel anchored in these anchorages must notify U.S. Army Corps of Engineers Cape Cod Canal Control traffic controller when it weighs anchor.

(781) (7) No vessel may anchor unless it maintains a bridge watch, guards and answers Channel 16 FM, and maintains an accurate position plot.

(782) (8) No vessel may anchor unless it maintains the capability to get underway within 30 minutes; except with prior approval of the Coast Guard Captain of the Port Southeastern New England.

(783) (9) No vessel may anchor in a "dead ship" status (propulsion or control unavailable for normal operations) without the prior approval of the Coast Guard Captain of the Port Southeastern New England.

(784) (10) No vessel may conduct lightering operations within these anchorages.

(785) (c) *Vineyard and Nantucket Sounds*–(1) *Anchorage E***.** South of a line beginning at a point bearing 180° about 3.25 miles from Cuttyhunk Light; thence 65° to a point bearing 180°, 0.625 mile from Nashawena Lighted Whistle Buoy; thence 57°30' passing 600 yards northerly of Middle Ground Lighted Bell Buoy 25A, to a point bearing 145°, 1.25 miles from Nobska Point Light; southwest of a line ranging 113° through West Chop Buoy 25 to East Chop Flats Bell Buoy 23; and west of a line bearing 163° between East Chop Flats Bell Buoy 23 and Lone Rock Buoy 1; and northerly of a line bearing 269° between Lone Rock Buoy 1 and a point on the mainland at Oak Bluffs about 0.30 mile southerly of Oak Bluffs Wharf.

(786) (2) *Anchorage F*. Southeast of the Elizabeth Islands, north of a line ranging 97°30' from Cuttyhunk Light toward Nashawena Lighted Whistle Buoy to a point 0.375 mile from that buoy; northwest of a line bearing 57°30' from the last-named point to a point opposite the entrance to Woods Hole; and southwest of a line from the shore of Nonamesset Island bearing 114° and ranging through West Chop Light and East Chop Light.

(787) (3) *Anchorage G*. South of a line beginning at a point on the mainland at Oak Bluffs about 0.30 mile southerly of Oak Bluffs Wharf bearing 89° to Lone Rock Buoy 1; thence 113° from Lone Rock Buoy 1 to Outer Flats Bell Buoy 17; thence 86° to Cross Rip Lightship; thence 118°30' to Tuckernuck Shoal Bell Buoy 7; thence ranging 149° toward Brant Point Light to the breakwater at Brant Point.

(788) (4) *Anchorage H***.** In the vicinity of Squash Meadow shoal, east of a line ranging 163° through Squash Meadow West End Buoy 21; north of lines parallel to and 0.5 mile northerly from lines joining Lone Rock Buoy 1, Outer Flats Bell Buoy 17, and Cross Rip Lightship; and south of a line ranging 97° from East Chop Light toward Cross Rip Lightship.

(789) (5) *Anchorage I***.** Northerly of a line ranging 109° from Nobska Point Light toward Hedge Fence Lighted Horn and Gong Buoy 16, and of a line ranging 97°30' through Hedge Fence East End Buoy to Halfmoon Shoal Lighted Bell Buoy 12, thence 73° to Handkerchief Shoal Buoy 16, and thence to the westernmost point of Monomoy Island.

(790) (6) *Anchorage J*. East of a line bearing 329°, parallel to and 0.875 mile northeasterly of a line running from Brant Point Light through Tuckernuck Shoal Bell Buoy 7, from Coatue Beach to a point 1.25 miles southeasterly from a line between Halfmoon Shoal Lighted Bell Buoy 12 and Handkerchief Shoal Buoy 16; thence 73°, parallel to and 1.25 miles southeasterly from a line running from Halfmoon Shoal Lighted Bell Buoy 12 through

Handkerchief Shoal Buoy 16, to a point bearing 215° from Stone Horse North End Lighted Bell Buoy 9; thence 35° to Stone Horse North End Lighted Bell Buoy 9; thence 70° to a point bearing 207° from Pollock Rip Lightship; and thence 27° through, and to a point 5.0 miles northeasterly from, Pollock Rip Lightship.

(791) (7) *Anchorage K*. North of a line tangent to the southeasterly edge of Monomoy Point and extending to Bearse Shoal North End Buoy 2A and west of a line bearing 7° from Bearse Shoal North End Buoy 2A to Chatham Bar Buoy 2.

(792) (d) *The regulations*. (1) Floats or buoys for marking anchors or moorings in place will be allowed in all areas. Fixed mooring piles or stakes are prohibited.

(793) (2) Except in cases of great emergency, no vessels shall be anchored in New Bedford Outer Harbor, Buzzards Bay near the entrance to the approach channel to Cape Cod Canal, or Vineyard and Nantucket Sounds, outside of the anchorage areas defined in paragraphs (a) to (c) of this section.

(794) (3) Anchors must not be placed outside the anchorage areas, nor shall any vessel be so anchored that any portion of the hull or rigging will at any time extend outside the boundaries of the anchorage area.

(795) (4) Any vessel anchoring under the circumstances of great emergency outside any anchorage area must be placed near the edge of the channel and in such position as not to interfere with the free navigation of the channel, nor obstruct the approach to any pier nor impede the movement of any boat, and shall move away immediately after the emergency ceases or upon notification by an officer of the Coast Guard.

(796) (5) A vessel upon being notified to move into the anchorage limits or to shift its position in anchorage grounds must get under way at once or signal for a tug, and must change position as directed with reasonable promptness.

(797) (6) Whenever the maritime or commercial interests of the United States so require, any officer of the Coast Guard is hereby empowered to shift the position of any vessel anchored within the anchorage areas, of any vessel anchored outside the anchorage areas, and of any vessel which is so moored or anchored as to impede or obstruct vessel movements in any channel.

(798) (7) Nothing in this section shall be construed as relieving the owner or person in charge of any vessel from the penalties of the law for obstructing navigation or for obstructing or interfering with range lights, or for not complying with the navigation laws in regard to lights, fog signals, or for otherwise violating the law.

(799)

§110.142 Nantucket Harbor, Mass.

(800) (a) *The anchorage grounds*. In the Nantucket Harbor, beginning at a point 210 yards, 090° from Brant Point Light; thence easterly to

(801) 41°17'23.0"N., 70°05'14.5"W.; thence southerly to

(802) 41°17'03.0"N., 70°05'14.5"W.; thence southwesterly to

(803) 41°16'54.0"N., 70°05'23.0"W.; thence northwesterly to

(804) 41°16'55.0"N., 70°05'31.0"W.; thence northeasterly to

(805) 41°17'07.5"N., 70°05'27.0"W.; thence to the point of beginning.

(806) (b) *The regulations*. The anchorage is for the use of commercial and pleasure craft. Temporary floats or buoys for marking anchors or moorings in place will be allowed. Fixed mooring piles or stakes are prohibited. The anchoring of vessels including the placing of anchors and moorings is subject to the supervision and approval of the local harbor master.

(807)

§110.145 Narragansett Bay, R.I.

(808) (a) *East Passage*–(1) *Anchorage A*. East of Conanicut Island, beginning at the easterly extremity of the Dumplings; extending 009° to a point at

(809) 41°29'28.0"N., 71°21'05.5"W.; thence

(810) 356°, 5,350 feet; thence

(811) 024°, 5,700 feet; thence

(812) 012°, 1,100 feet; thence

(813) 311.5°, 2,300 feet; thence

(814) 351°, 5,350 feet; thence

(815) 270°, 3,200 feet to the easterly side of Conanicut Island; thence generally along the easterly side of the island to a point on the easterly side of the island due west of the Dumplings; and thence due east to the point of beginning; excluding the approach to the Jamestown Ferry, a zone 900 feet wide to the southward of a line ranging 103° from a point, 300 feet north of the existing ferry landing toward the spire of Trinity Church, Newport.

(816) (i) That portion of the area to the northward of the approach of the Jamestown Ferry shall be restricted for the anchorage of vessels of the U.S. Navy. In that portion of the area to the southward of the approach of the Jamestown Ferry, the requirements of the Navy shall predominate.

(817) (ii) Temporary floats or buoys for marking anchors or moorings in place shall be allowed in this area. Fixed mooring piles or stakes will not be allowed.

(818) (2) *Anchorage B*. Off the west shore of Aquidneck Island to north of Coggeshall Point, northerly of a line ranging 075° from a point on the easterly end of Gould Island, latitude 41°32'13", longitude 71°20'40.5", toward the shore of Aquidneck Island; east of a line ranging 019° from the easternmost of the Dumplings to latitude 41°36'16", longitude 71°17'48"; thence northeast to latitude 41°36'53", longitude 71°17'07.5"; thence east to latitude 41°36'53", longitude 71°16'40", thence southwesterly to latitude 41°35'54", longitude 71°17'17.5"; thence southeasterly to the shore at the easterly end of the north boundary of the cable area in the vicinity of Coggeshall Point; excluding the cable area in the vicinity of Coggeshall Point.

(819) (i) *Anchorage B–1*. Off the southerly end of Prudence Island beginning at

(820) 41°34'08.9"N., 71°19'25.8"W.; thence

(821) 019°, 1,900 feet; thence

(822) 289°, 1,900 feet; thence

(823) 199°, 1,900 feet; thence

(824) 109°, 1,900 feet to the point of beginning.

(825) (a) In this area the requirements of the Navy shall predominate.

(826) (b) Temporary floats or buoys for marking anchors or moorings in place will be allowed in this area. Fixed mooring piles or stakes will not be allowed.

(827) (ii) [Reserved]

(828) (3) *Anchorage C*.

(829) (i) [Reserved]

(830) (ii) West of Coasters Harbor Island, west of a line bearing 351° from Tracey Ledge Buoy 5 through Seventeen-foot Spot Buoy northeast of Gull Rocks; south of a line bearing 292° from the cupola at the Naval War College; east of a line ranging 19° from the easternmost of the Dumplings toward Dyer Island North Point Shoal Lighted Bell Buoy 12A; and north of latitude 41°30'22" which parallel passes through a point 230 yards north of Rose Island Shoal Northeast End Buoy 8.

(831) (iii) In this area the requirements of the Navy shall predominate.

(832) (iv) Temporary floats or buoys for marking anchors or moorings in place will be allowed in this area. Fixed mooring piles or stakes will not be allowed.

(833) (4) *Anchorage D*. West of Goat Island, an area bounded by the following coordinates:

(834) Northeast Corner: 41°29.484′ N, 071°19.975′ W

(835) Northwest Corner: 41°29.484′ N, 071°20.578′ W

(836) Southwest Corner: 41°29.005′ N, 071°20.578′ W

(837) Southeast Corner: 41°29.005′ N, 071°19.975′ W

(838) (i) In this area the requirements of the Navy shall predominate from May 1 to October 1, subject at all times to such adjustments as may be necessary to accommodate all classes of vessels which may require anchorage room.

(839) (ii) Temporary floats or buoys for marking anchors or moorings in place will be allowed in this area. Fixed mooring piles or stakes will not be allowed.

(840) (iii) Should any part of an anchored vessel extend into the recommended vessel route in the East Passage of Narragansett Bay, a securite call notifying mariners of the vessel's exact position and status shall be made at least hourly on VHF channels 13 and 16.

(841) (iv) As much as practicable vessels anchoring will do so in the following order:

(842) (A) Primary anchoring point: 41°29.25′N., 071°20.15′W.

(843) (B) Secondary anchoring point: 41°29.38′N., 071°20.45′W.

(844) (C) Tertiary anchoring point: 41°29.15′N., 071°20.50′W.

(845) Note to paragraph (a): ''Anchoring point'' is the intended position of the anchor at rest on the bottom of the anchorage. All coordinates referenced use datum: NAD 83.

(846) (5) *Anchorage E*. South of Coasters Harbor Island, east of a line bearing 341° from the outer end of Briggs Wharf to the southwestern shore of Coasters Harbor Island near the War College Building; and north of a line ranging 265° from the flagstaff at Fort Greene toward Rose Island Light.

(847) (i) In this area the requirements of the naval service will predominate from May 1 to October 1, but will at all times be subject to such adjustment as may be necessary to accommodate all classes of vessels that may require anchorage room.

(848) (ii) Temporary floats or buoys for marking anchors or moorings in place will be allowed in this area. Fixed mooring piles or stakes will not be allowed.

(849) (b) *West Passage* (1) *Anchorage H***.** North of a line 1,000 yards long bearing 88° from Bonnet Point; west of a line bearing 3° from the eastern end of the last-described line; and south of a line ranging 302° through a point 200 yards south of the Kearny wharf toward the church spire at South Ferry, Boston Neck.

(850) (i) Temporary floats or buoys for marking anchors or moorings in place will be allowed in this area. Fixed mooring piles or stakes will not be allowed.

(851) (2) *Anchorage I*. North of a line 1,000 yards long bearing 88° from Bonnet Point to the shore at Austin Hollow; east of a line bearing 183° from Dutch Island Light; and south of a line ranging 302° through a point 200 yards south of the Kearny wharf toward the church spire at South Ferry, Boston Neck.

(852) (i) Temporary floats or buoys for marking anchors or moorings in place will be allowed in this area. Fixed mooring piles or stakes will not be allowed.

(853) (3) *Anchorage J***.** At Saunderstown, south of a line ranging 110° from the south side of the ferry wharf toward the cable crossing sign on Dutch Island; west of a line ranging 192° from Plum Beach Shoal Buoy 1 PB toward the east shore of The Bonnet; and north of a line from the shore ranging 108° toward Dutch Island Light and the north end of the wharf at Beaver Head.

(854) (i) Temporary floats or buoys for marking anchors or moorings in place will be allowed in this area. Fixed mooring piles or stakes will not be allowed.

(855) (4) *Anchorage K*. In the central and southern portion of Dutch Island Harbor, north of a line ranging 106° from Beaver Head Point Shoal Buoy 2 toward the Jamestown standpipe; east of a line ranging 14° from Beaver Head Point Shoal Buoy 2 toward the inshore end of the engineer wharf, Dutch Island; southeast of a line ranging 50° from Dutch Island Light toward the windmill north of Jamestown; and south of a line parallel to and 100 yards southwesterly from a line ranging 132° from the engineer wharf, Dutch Island, and the west ferry wharf, Jamestown.

(856) (i) Temporary floats or buoys for marking anchors or moorings in place will be allowed in this area. Fixed mooring piles or stakes will not be allowed.

(857) (5) *Anchorage L*. North of a line ranging 101° from a point on shore 300 yards northerly of the Saunderstown ferry wharf toward the entrance to Round Swamp, Conanicut Island; west of a line bearing 15° parallel to and 1,000 feet westerly from a line joining the western point of Dutch Island and Twenty-three Foot Rock Buoy 4, and a line ranging 6° from Dutch Island Light toward Warwick Light; and south of a line ranging 290° from Sand Point, Conanicut Island, to Wickford Harbor Light, and a line bearing 226° from Wickford Harbor Light to Poplar Point tower.

(858) (i) Temporary floats or buoys for marking anchors or moorings in place will be allowed in this area. Fixed mooring piles or stakes will not be allowed.

(859) (6) *Anchorage M*. East and north of Dutch Island, northeast of a line ranging 316° from the inshore end of the west ferry wharf, Jamestown, toward the north end of Dutch Island to a point bearing 88°, 200 yards, from the engineer wharf, Dutch Island, thence ranging 3° toward the shore of Conanicut Island at Slocum Ledge; north of a line 200 yards off the Dutch Island shore ranging 281° from the entrance to Round Swamp toward a point on shore 300 yards northerly from the Saunderstown ferry wharf; east of a line ranging 15° from the western point of Dutch Island to Twenty-three Foot Rock Buoy 4; and south of a line bearing 77° from Twenty-three Foot Rock Buoy 4 to the shore.

(860) (i) Temporary floats or buoys for marking anchors or moorings in place will be allowed in this area. Fixed mooring piles or stakes will not be allowed.

(861) (7) *Anchorage N*. West of the north end of Conanicut Island, south of a line bearing 262° from Conanicut Island Light; east of a line bearing 8° from Twenty-three Foot Rock Buoy 4; and north of a line ranging 290° from Sand Point toward Wickford Harbor Light.

(862) (i) Temporary floats or buoys for marking anchors or moorings in place will be allowed in this area. Fixed mooring piles or stakes will not be allowed.

(863) (c) *Bristol Harbor*–(1) *Anchorage O*. South of the south line of Franklin Street extended westerly; west of a line bearing 164°30' parallel to and 400 feet westerly from the State harbor line between Franklin and Constitution Streets, and of a line ranging 244° from a point on the north line of Constitution Street extended 400 feet beyond the State harbor line toward Usher Rock Buoy 3; and north of the north line of Union Street extended to the Popasquash Neck Shore.

(864) (i) Temporary floats or buoys for marking anchors or moorings in place will be allowed in this area. Fixed mooring piles or stakes will not be allowed.

(865) (d) *The regulations*. (1) Except in cases of great emergency, no vessel shall be anchored in the entrances to Narragansett Bay, in Newport Harbor, or in Bristol Harbor, outside of the anchorage areas defined in paragraphs (a), (b) and (c) of this section.

(866) (2) Anchors must not be placed outside the anchorage areas, nor shall any vessel be so anchored that any portion of the hull or rigging shall at any time extend outside the boundaries of the anchorage area. However, Anchorage D (paragraph (a)(4) of this section) is exempt from this requirement.

(867) (3) Any vessel anchoring under the circumstances of great emergency outside the anchorage areas must be placed near the edge of the channel and in such position as not to interfere with the free navigation of the channel, nor obstruct the approach to any pier, nor impede the movement of any boat, and shall move away immediately after the emergency ceases, or upon notification by an officer of the Coast Guard.

(868) (4) A vessel upon being notified to move into the anchorage limits or to shift its position on anchorage grounds must get under way at once or signal for a tug, and must change position as directed with reasonable promptness.

(869) (5) Whenever the maritime or commercial interests of the United States so require, any officer of the Coast Guard is hereby empowered to shift the position of any vessel anchored within the anchorage areas, of any vessel anchored outside the anchorage areas, and of any vessel which is so moored or anchored as to impede or obstruct vessel movements in any channel.

(870) (6) Nothing in this section shall be construed as relieving the owner or person in charge of any vessel from the penalties of the law for obstructing navigation or for obstructing or interfering with range lights, or for not complying with the navigation laws in regard to lights, fog signals, or for otherwise violating the law.

(871)

§110.146 Long Island Sound.

(872) (a) *Anchorage grounds*. (1) *Bridgeport Anchorage Ground*. That portion of Long Island Sound enclosed by a line connecting the following points:

(873) 41°04'52"N., 73°14'04"W.; thence to

(874) 41°03'45"N., 73°14'04"W.; thence to

(875) 41°03'45"N., 73°11'39"W.; thence to

(876) 41°02'50"N., 73°12'08"W.; thence to

(877) 41°02'50"N., 73°16'18"W.; thence to

(878) 41°04'52"N., 73°16'18"W.; returning to point of origin.

(879) (2) *New Haven North Anchorage Ground*. That portion of Long Island Sound enclosed by a line connecting the following points:

(880) 41°12'18"N., 72°52'36"W.; thence to

(881) 41°12'18"N., 72°49'36"W.; thence to

(882) 41°10'12"N., 72°48'18"W.; thence to

(883) 41°10'12"N., 72°52'12"W.; thence to

(884) 41°11'06"N., 72°53'06"W.; returning to point of origin.

(885) (3) *New Haven South Anchorage Ground*. That portion of Long Island Sound enclosed by a line connecting the following points:

(886) 41°09'30"N., 72°47'48"W.; thence to

(887) 41°08'36"N., 72°47'24"W.; thence to

(888) 41°08'36"N., 72°51'24"W.; thence to

(889) 41°09'30"N., 72°51'24"W.; returning to point of origin.

(890) (4) *New London Anchorage Ground*. That portion of Long Island Sound enclosed by a line connecting the following points:

(891) 41°14'11"N., 72°15'38" W.; thence to

(892) 41°15'05"N., 72°16'02" W.; thence to

(893) 41°15'39"N., 72°13'21" W.; thence to

(894) 41°14'45"N., 72°12'57" W.; returning to point of origin.

(895) (5) *Northport Anchorage Ground*. That portion of Long Island Sound enclosed by a line connecting the following points:

(896) 40°58'48"N., 73°16'30"W.; thence to

(897) 40°57'42"N., 73°11'42"W.; thence to

(898) 40°56'30"N., 73°13'30"W.; thence to

(899) 40°57'36"N., 73°18'12"W.; returning to point of origin.

(900) (6) *Port Jefferson Anchorage Ground*. That portion of Long Island Sound enclosed by a line connecting the following points:

(901) 41°01'48"N., 73°04'54"W.; thence to

(902) 41°01'48"N., 73°00'00"W.; thence to

(903) 41°00'18"N., 73°00'00"W.; thence to

(904) 41°00'18"N., 73°04'54"W.; returning to point of origin.

(905) (7) *Riverhead Anchorage Ground*. That portion of Long Island Sound enclosed by a line connecting the following points:

(906) 41°03'00"N., 72°42'00"W.; thence to

(907) 41°04'00"N., 72°36'00"W.; thence to

(908) 41°02'00"N., 72°35'24"W.; thence to

(909) 41°01'24" N., 72°41'24"W.; returning to point of origin.

(910) (8) All coordinates referenced use datum: NAD 83.

(911) (b) *General regulations*. (1) These anchorages are designated for general purposes, but are intended primarily for use by commercial vessels of 300 gross tons and greater and all tank vessels including tank barges. Except in emergencies, commercial vessels of 300 gross tons and greater and all tank vessels, including tank barges, anchoring in the Captain of the Port Long Island Sound Zone inside the line of demarcation shall anchor in the anchorage grounds described above.

(912) (2) Prior to anchoring in the anchorage area, all vessels shall notify the Coast Guard Captain of the Port via VHF-FM Channel 16.

(913) (3) In anchorages where lightering and bunkering operations are authorized, the Captain of the Port must be notified at least four hours in advance of a vessel conducting lightering or bunkering operations, as required by 156.118 of this title. In addition, all lightering and bunkering operations must be done in accordance with 156.120 of this title.

(914) (4) Within an anchorage, navigation is prohibited within 500 yards of an anchored vessel that is conducting bunkering or lightering operations. In accordance with the "Regulated Navigation Area: Long Island Sound Marine Inspection and Captain of the Port Zone," 33 CFR 165.153(d)(7), navigation also is prohibited within 100 yards of a vessel engaged in commercial service.

(915) (5) Any vessel conducting lightering or bunkering operations shall display by day a red flag at its mast head or at least 10 feet above the upper deck if the vessel has no mast, and by night the flag must be illuminated by spotlight. These signals shall be in addition to day signals, lights, and whistle signals required by rules 30 (33 U.S.C. 2030) and 35 (33 U.S.C. 2035) of the Inland Navigation Rules when at anchor in a general anchorage area.

(916) (6) Except as otherwise provided, a vessel may not occupy an anchorage for more than 30 days, unless the vessel obtains written permission from the Captain of the Port.

(917) (7) If a request is made for the long-term lay up of a vessel, the Captain of the Port may establish special conditions with which the vessel must comply in order for such a request to be approved.

(918) (8) The Captain of the Port may prescribe specific conditions for vessels anchoring within the anchorage grounds described in this section, pursuant to 33 CFR 109.05. These conditions may include, but are not limited to: The number and location of anchors; scope of chain; readiness of the engineering plant and equipment; use of tugs; and requirements for maintaining communication guards on selected radio frequencies.

(919) (9) No vessel in such condition that it is likely to sink or otherwise become a menace or obstruction to navigation or anchorage of other vessels shall occupy an anchorage, except in cases where unforeseen circumstances create conditions of imminent peril to personnel, and then only for such period as may be authorized by the Captain of the Port.

(920) (10) All vessels anchored within the designated anchorage grounds shall comply with the regulations found in 33 CFR 164.19 and shall maintain a continuous bridge watch by a licensed deck officer proficient in English, monitoring VHF-FM Channel 16. This individual shall confirm that the ship's crew performs frequent checks of the vessel's position to ensure the vessel is not dragging anchor. A second VHF-FM radio monitoring Channel 13 is strongly recommended.

(921) (11) Anchors shall be placed well within the anchorage grounds so that no portion of the hull or rigging will at any time extend outside of the anchorage area.

(922) (12) The Coast Guard Captain of the Port may close the anchorage area and direct vessels to depart the anchorage during periods of adverse weather or at other times as deemed necessary in the interest of port safety and security.

(923) (13) Any vessel anchored in these grounds must be capable of getting underway if ordered by the Captain of the Port and must be able to do so within two (2) hours of notification by the Captain of the Port. If a vessel will not be able to get underway within two (2) hours of notification, permission must be requested from the Captain of the Port to remain in the anchorage. No

vessel shall anchor in a "dead ship" status (propulsion or control unavailable for normal operations) without prior approval of the Captain of the Port.

(924) (14) Fixed moorings, piles or stakes are prohibited.

(925)

§110.147 New London Harbor, Conn.

(926) (a) *The anchorage grounds*–(1) *Anchorage A*. In the Thames River east of Shaw Cove, bounded by lines connecting points which are the following bearings and distances from Monument, Groton (41°21'18"N., 72°04'48"W.): 243°, 1,400 yards; 246°, 925 yards; 217°, 1,380 yards; and 235°, 1,450 yards.

(927) (2) *Anchorage B*. In the Thames River southward of New London, bounded by lines connecting points which are the following bearings and distances from New London Harbor Light (41°18'59"N., 72°05'25"W.): 002°, 2,460 yards; 009°, 2,480 yards; 026°, 1,175 yards; and 008°, 1,075 yards.

(928) (3) *Anchorage C*. In the Thames River southward of New London Harbor, bounded by lines connecting a point bearing 100°, 450 yards from New London Harbor Light, a point bearing 270°, 575 yards from New London Ledge Light (41°18'21"N., 72°04'39"W.), and a point bearing 270°, 1,450 yards from New London Ledge Light.

(929) (4) *Anchorage D*. In Long Island Sound approximately two miles west-southwest of New London Ledge Light, bounded by lines connecting points which are the following bearings and distances from New London Ledge Light; 246°, 2.6 miles; 247°, 2.1 miles; 233°, 2.1 miles; and 235°, 2.6 miles.

(930) (5) *Anchorage E*. The waters at the mouth of New London Harbor one mile southeast of New London Ledge Light beginning at

(931) 41°17'26"N., 72°04'21"W.; thence northeasterly to

(932) 41°17'38"N., 72°03'54"W.; thence southeasterly to

(933) 41°16'50"N., 72°03'16"W.; and thence southwesterly to

(934) 41°16'38"N., 72°03'43"W.; and thence northwesterly to the point of beginning.

(935) (6) *Anchorage F*. The waters off the mouth of New London Harbor two miles southeast of New London Ledge Light beginning at

(936) 41°16'00"N., 72°03'13"W.; thence westerly to

(937) 41°16'00"N., 72°03'38"W.; thence northerly to

(938) 41°16'35"N., 72°03'38"W.; thence easterly to

(939) 41°16'35"N., 72°03'13"W.; and thence southerly to the point of beginning.

(940) (b) *The regulations*–(1) Anchorage A is for barges and small vessels drawing less than 12 feet.

(941) (2) Anchorage F is reserved for the use of naval vessels and, except in cases of emergency, no other vessel may anchor in Anchorage F without permission from the Captain of the Port, New London, CT.

(942) (3) Except in emergencies, vessels shall not anchor in New London Harbor or the approaches thereto outside the anchorages defined in paragraph (a) of this section unless authorized to do so by the Captain of the Port.

(943)

§110.148 Johnsons River at Bridgeport, Conn.

(944) (a) *The anchorage grounds*. In Johnsons River, beginning at

(945) point "A" 41°10'12.3"N., 73°09'50.2"W.; to

(946) point "B" 41°10'12.3"N., 73°09'52.1"W.; to

(947) point "C" 41°10'10.0"N., 73°09'54.9"W.; to

(948) point "D" 41°10'05.0"N., 73°09'56.1"W.; to

(949) point "E" 41°10'04.0"N., 73°09'55.9"W.; to

(950) point "F" 41°10'05.0"N., 73°09'54.5"W.; to

(951) point "G" 41°10'05.8"N., 73°09'54.5"W.; thence to the point of beginning.

(952) (b) *The regulations*. The anchorage is for use by commercial and pleasure craft. Temporary floats or buoys for marking anchors or moorings will be allowed. The anchoring of vessels and placing of temporary anchors or mooring piles are under the jurisdiction of the local harbor master. Fixed mooring piles or stakes will not be allowed.

(953)

§110.149 Narragansett Bay, RI.

(954) (a) *Brenton Point Anchorage Ground*. An area bounded by the following coordinates:

(955) 41°22′37.1″ N, 71°14′40.3″W; thence to

(956) 41°20′42.8″ N, 71°14′40.3″W; thence to

(957) 41°18′24.1″ N, 71°20′32.5″W; thence to

(958) 41°20′22.6″ N, 71°20′32.5″W; thence back to point of origin.

(959) (b) The following regulations apply in the Brenton Point anchorage ground.

(960) (1) Prior to anchoring within the anchorage area, all vessels shall notify the Coast Guard Captain of the Port via VHF–FM Channel 16.

(961) (2) Except as otherwise provided, no vessel may occupy this anchorage ground for a period of time in excess of 96 hours without prior approval of the Captain of the Port.

(962) (3) If a request is made for the longterm lay up of a vessel, the Captain of the Port may establish special conditions with which the vessel must comply in order for such a request to be approved.

(963) (4) No vessel in such condition that it is likely to sink or otherwise become a menace or obstruction to navigation or anchorage of other vessels shall occupy an anchorage except in cases where unforeseen circumstances create conditions of imminent peril to personnel and then only for such period as may be authorized by the Captain of the Port.

(964) (5) Anchors shall be placed well within the anchorage areas so that no portion of the hull or rigging will at any time extend outside of the anchorage area.

(965) (6) The Coast Guard Captain of the Port may close the anchorage area and direct vessels to depart the anchorage during periods of adverse weather or at other

times as deemed necessary in the interest of port safety and security.

(966) (7) Any vessel anchored in these grounds must be capable of getting underway if ordered by the Captain of the Port and must be able to do so within two hours of notification by the Captain of the Port. If a vessel will not be able to get underway within two hours of notification, permission must be requested from the Captain of the Port to remain in the anchorage. No vessel shall anchor in a ''dead ship'' status (propulsion or control unavailable for normal operations) without prior approval of the Captain of the Port.

(967) (8) Brenton Point anchorage ground is a general anchorage area reserved primarily for commercial vessels waiting to enter Narragansett Bay.

(968) (9) Temporary floats or buoys for marking anchors or moorings in place will be allowed in this area. Fixed mooring piles or stakes will not be allowed.

(969) (10) All coordinates referenced use datum: NAD 83.

(970)

§110.150 Block Island Sound N.Y.

(971) (a) *The anchorage ground*. A ¾- by 2-mile rectangular area approximately 3 miles east-northeast of Gardiners Island with the following coordinates:

(972) 41°06'12"N., 72°00'05"W.

(973) 41°07'40"N., 72°01'54"W.

(974) 41°08'12"N., 72°01'10"W.

(975) 41°06'46"N., 71°59'18"W.

(976) (b) *The regulations*. This anchorage ground is for use of U.S. Navy submarines. No vessel or person may approach or remain within 500 yards of a U.S. Navy submarine anchored in this anchorage ground.

(977)

§110.155 Port of New York.

(978) (a) *Long Island Sound*–(1) *Anchorage No. 1*. Southwest of a line between Neptune Island and Glen Island ranging from Aunt Phebe Rock Light and tangent to the north edge of Glen Island; southwest of a line tangent to the northeast edge of Glen Island and Goose Island breakwater; southwest of a line bearing southeasterly from the southwest end of Goose Island breakwater and on range with the south gable of the Casino on the northeast end of Glen Island; west of a line ranging from the east edge of Goose Island breakwater to the west edge of the north end of Hart Island; west of Hart Island; and northwest of a line extending from Hart Island Light to Locust Point; excluding from this area, however, (i) the waters northeast of a line ranging 303° from the southwest end of Hart Island; northwest of a line ranging from the water tank at the north end of Davids Island 207°40' to the northwest end of City Island; and south of latitude 40°52'12"; and (ii) the waters west of Hunter Island; and south of a line ranging from the most southerly end of Glen Island tangent to the most northerly end of Hunter Island.

(979) (i) Boats shall not anchor in this area in buoyed channels.

(980) (ii) Boats shall be so anchored as to leave at all times an open, usable channel, at least 50 feet wide, west and south of Glen Island.

(981) **NOTE:** The special anchorage area in this anchorage is described in §110.60.

(982) (2) [Reserved]

(983) (b) *East River*–(1) *Anchorage No. 6*. On Hammond Flats north of a line bearing 260° from the head of the pier on Throgs Neck at the foot of Pennyfield Avenue to the north tower of Bronx-Whitestone Bridge at Old Ferry Point.

(984) (2) [Reserved]

(985) (3) *Anchorage No. 8*. North of a line bearing 259° between the north tower of the Bronx-Whitestone Bridge at Old Ferry Point and a point at latitude 40°47'57", longitude 73°52'16"; thence east of a line bearing 0° to latitude 40°48'06"; thence southeast of a line parallel to the bulkhead extending northeasterly to latitude 40°48'20"; thence north of a line bearing 296° to shore.

(986) (4) *Anchorage No. 9*. East of a line from College Point Reef Light tangent to the west side of College Point; and south of a line from College Point Reef Light to Whitestone Point.

(987) (5) *Anchorage No. 10*. An area in Flushing Bay, beginning at a point on shore at La Guardia Airport at

(988) 40°46'49"N., 73°52'21"W.; to

(989) 40°47'20"N., 73°51'55"W.; to

(990) 40°47'38"N., 73°51'15"W.; and an area on the west side of Bowery Bay, beginning at

(991) 40°46'58"N., 73°53'46"W.; to

(992) 40°47'03"N., 73°53'39"W.; to

(993) 40°47'00"N., 73°53'31"W.; to

(994) 40°46'55"N., 73°53'32"W.; to

(995) 40°46'49"N., 73°53'39"W.

(996) **NOTE:** Special anchorage areas in this anchorage are described in §110.60.

(997) (6) *Anchorage No. 11*. An area in East River beginning at a point on a pierhead at

(998) 40°47'55.0"N., 73°53'19.5"W.; to

(999) 40°47'40.0"N., 73°51'58.0"W.; to

(1000) 40°47'16.0"N., 73°52'15.0"W.

(1001) (7) [Reserved]

(1002) (8) *Anchorage No. 14*. In Hallets Cove, east of a line from a point on shore 100 feet west of the southerly prolongation of 2nd Street, Astoria, to Gibbs Point.

(1003) (c) *Hudson River* (1) *Anchorage No. 16*. North of a line on a range with the north side of the north pier of the Union Dry Dock and Repair Company Shipyard, Edgewater, New Jersey; west of a line ranging 25° from a point 120 yards east of the east end of said pier to a point (500 yards from the shore and 915 yards from the Fort Lee flagpole) on a line ranging approximately 100°22' from the Fort Lee flagpole toward the square chimney on the Medical Center Building at 168th Street, Manhattan; and south of said line ranging between the Fort Lee flagpole and the square chimney on the Medical Center Building.

(1004) (i) When the use of Anchorage No. 16 is required by naval vessels, the vessels anchored therein shall move when the Captain of the Port directs them.

(1005) (ii) [Reserved]

(1006) (2) *Anchorage No. 17*. All waters of the Hudson River bound by the following points:

(1007) 40°56′26.66″N, 073°55′12.06″W; thence to

(1008) 40°56′22.54″N, 073°54′49.77″W; thence to

(1009) 40°55′56.00″N, 073°54′58.00″W; thence to

(1010) 40°55′54.15″N, 073°54′46.96″W; thence to

(1011) 40°54′18.43″N, 073°55′21.12″W; thence to

(1012) 40°52′27.59″N, 073°56′14.32″W; thence to

(1013) 40°51′34.20″N, 073°56′52.64″W; thence to

(1014) 40°51′20.76″N, 073°57′31.75″W; thence along the shoreline to the point of origin (NAD 83).

(1015) (i) When the use of Anchorage No. 17 is required by naval vessels, the vessels anchored therein shall move when the Captain of the Port directs them.

(1016) (ii) [Reserved]

(1017) (3) *Anchorage No. 18–A*. East of lines bearing 8° from the northwest corner of the crib icebreaker north of the New York Central Railroad Company drawbridge across Spuyten Duyvil Creek (Harlem River) to a point 250 yards offshore and on line with the New York Central Railroad signal bridge at the foot of West 231st Street, extended, at Spuyten Duyvil, Bronx, New York; thence bearing 19° to the channelward face of the Mount St. Vincent Dock at the foot of West 261st Street, Riverdale, Bronx, New York.

(1018) (i) When the use of Anchorage No. 18–A is required by naval vessels, the vessels anchored therein shall move when the Captain of the Port directs them.

(1019) (4) *Anchorage No. 18*. All waters of the Hudson River bound by the following points:

(1020) 40°56′54.0″N, 073°54′40.0″W; thence to

(1021) 40°56′51.0″N, 073°54′24.0″W; thence to

(1022) 40°55′53.0″N, 073°54′40.0″W; thence to

(1023) 40°55′56.0″N, 073°54′58.0″W; thence to the point of origin (NAD 83).

(1024) (i) This anchorage ground is reserved for use by ships only.

(1025) (ii) [Reserved]

(1026) (5) *Anchorages No. 19 East and 19 West*.

(1027) (i) *Anchorage No. 19 East*. All waters of the Hudson River bound by the following points:

(1028) 40°49'42.6"N., 073°57'14.7"W.; thence to

(1029) 40°49'45.9"N., 073°57'22.0"W.; thence to

(1030) 40°49'52.0"N., 073°57'22.0"W.; thence to

(1031) 40°50'08.3"N., 073°57'10.8"W.; thence to

(1032) 40°50'55.4"N., 073°56'59.7"W.; thence to

(1033) 40°51'02.5"N., 073°56'57.4"W.; thence to

(1034) 40°51'00.8"N., 073°56'49.4"W.; thence along the shoreline to the point of origin.

(1035) (ii) *Anchorage No. 19 West*. All waters of the Hudson River bound by the following points:

(1036) 40°46'56.3"N., 073°59'42.2"W.; thence to

(1037) 40°47'36.9"N., 073°59'11.7"W.; thence to

(1038) 40°49'31.3"N., 073°57'43.8"W.; thence to

(1039) 40°49'40.2"N., 073°57'37.6"W.; thence to

(1040) 40°49'52.4"N., 073°57'37.6"W.; thence to

(1041) 40°49'57.7"N., 073°57'47.3"W.; thence to

(1042) 40°49'32.2"N., 073°58'12.9"W.; thence to

(1043) 40°49'00.7"N., 073°58'33.1"W.; thence to

(1044) 40°48'28.7"N., 073°58'53.8"W.; thence to

(1045) 40°47'38.2"N., 073°59'31.2"W.; thence to

(1046) 40°47'02.7"N., 073°59'57.4"W.; thence to the point of origin.

(1047) (iii) The following regulations apply to 33 CFR 110.155(c)(5)(i) and (ii):

(1048) (A) No vessel may conduct lightering operations in these anchorage grounds without permission from the Captain of the Port. When lightering is authorized, the Captain of the Port New York must be notified at least four hours in advance of a vessel conducting lightering operations as required by 156.118 of this title.

(1049) (B) Any vessel conducting lightering or bunkering operations shall display by day a red flag (46 CFR 35.30–1; Pub 102; International Code of Signals signaling instructions) at its mast head or at least 10 feet above the upper deck if the vessel has no mast, and by night the flag must be illuminated by spotlight. These signals shall be in addition to day signals, lights and whistle signals as required by rules 30 (33 USC 2030 and 33 CFR 83.30) and 35 (33 USC 2035 and 33 CFR 83.35) of the Inland Navigation Rules when at anchor in a general anchorage area.

(1050) (C) Within an anchorage, fishing and navigation are prohibited within 500 yards of an anchored vessel displaying a red flag.

(1051) (D) These anchorage grounds are only authorized for use by tugs and/or barges.

(1052) (E) No vessel may occupy this anchorage ground for a period of time in excess of 96 hours without prior approval of the Captain of the Port.

(1053) (F) No vessel may anchor in Anchorage No. 19 East or No. 19 West without permission from the Captain of the Port.

(1054) (G) Each vessel shall report its position within Anchorage No. 19 East or No. 19 West to the Captain of the Port immediately after anchoring.

(1055) (H) All coordinates referenced use datum: NAD 83.

(1056) (6) *Anchorage No. 19–A*. An area located west of Hyde Park enclosed by the coordinates starting at

(1057) 41°48'35"N., 73°57'00"W.; to

(1058) 41°48'35"N., 73°56'44"W.; to

(1059) 41°47'32"N., 73°56'50"W.; to

(1060) 41°47'32"N., 73°57'10"W.; thence back to

(1061) 41°48'35"N., 73°57'00"W. (NAD 1983).

(1062) (i) No vessel may anchor in Anchorage 19–A from December 16 to the last day of February without permission from the Captain of the Port, New York.

(1063) (ii) No vessel less than 20 meters in length may anchor in Anchorage 19–A without prior approval of the Captain of the Port, New York.

(1064) (d) *Upper Bay*–(1) *Anchorage No. 20–A*.

(1065) That area enclosed by coordinates starting at:

(1066) 40°42'02.5"N., 074°02'25.5"W.; thence to
(1067) 40°42'06.5"N., 074°02’19.5"W.; thence to
(1068) 40°42'05.0”N., 074°01'58.4"W.; thence to
(1069) 40°41'54.5"N., 074°01'59.2"W.; thence to
(1070) 40°41'53.0"N., 074°02'23.0"W.; thence to
(1071) (i) See 33 CFR 110.155(d)(6), (d)(16), and (1).
(1072) (2) *Anchorage No. 20–B*.
(1073) That area enclosed by coordinates starting at:
(1074) 40°41'47.0"N., 074°02'31.5"W.; thence to
(1075) 40°41'42.0"N., 074°01'02.0"W.; thence to
(1076) 40°41'35.3"N., 074°02'04.2"W.; thence to
(1077) 40°41'29.9"N., 074°02'07.8"W.; thence to
(1078) 40°41'42.6"N., 074°02'32.7"W.; thence back to
(1079) 40°41′47.0″N., 74°02′31.5″W.
(1080) (i) See 33 CFR 110.155(d)(6), (d)(16), and (1).
(1081) (3) *Anchorage No. 20–C*.
(1082) That area enclosed by coordinates starting at:
(1083) 40°41'42.0"N., 074°02'43.0"W.; thence to
(1084) 40°41'25.4"N., 074°02'10.7"W.; thence to
(1085) 40°41'01.7"N., 074°02'26.2"W.; thence to
(1086) 40°41'09.0"N., 074°02'41.5"W.; thence to
(1087) 40°41'20.0"N., 074°02'59.2"W.; thence back to
(1088) 40°41'42.0"N., 074°02'43.0"W.
(1089) (i) See 33 CFR 110.155(d)(6), (d)(16), and (l).
(1090) (4) *Anchorage No. 20–D*. That area enclosed by coordinates starting at
(1091) 40°41'09.5"N., 74°02'49.5"W.; to
(1092) 40°40'59.2"N., 74°02'27.9"W.; to
(1093) 40°40'44.5"N., 74°02'37.5"W.; to
(1094) 40°40'42.7"N., 74°03'07.6"W.; thence back to
(1095) 40°41′09.5″N., 74°02′49.5″W.
(1096) (i) See 33 CFR 110.155 (d)(6), (d)(16), and (1).
(1097) (5) *Anchorage No. 20–E*. That area enclosed by coordinates starting at
(1098) 40°40'38.2"N., 74°02'59.6"W.; to
(1099) 40°40'39.4"N., 74°02'40.9"W.; to
(1100) 40°40'09.2"N., 74°03'00.7"W.; to
(1101) 40°40'24.4"N., 74°03'24.6"W.; thence back to.
(1102) 40°40′38.2″N., 74°02′59.6″W.
(1103) (i) See 33 CFR 110.155 (d)(6), (d)(16), and (1).
(1104) (6) No vessel may occupy this anchorage for a period of time in excess of 72 hours without the prior approval of the Captain of the Port.
(1105) (7) *Anchorage No. 20–F*. All waters bound by the following points:
(1106) 40°40'12.2"N., 074°03'39.9"W.; thence to
(1107) 40°39'53.9"N., 074°03'09.6"W.; thence to
(1108) 40°39'38.9"N., 074°03'19.5"W.; thence to
(1109) 40°39'53.5"N., 074°03'53.7"W.; thence to the point of origin (NAD 83).
(1110) (i) See 33 CFR 110.155 (d)(9), (d)(16), and (1).
(1111) (ii) [Reserved]
(1112) (8) *Anchorage No. 20–G*. That area enclosed by coordinates starting at
(1113) 40°39'30.1"N., 74°04'08.0"W.; to
(1114) 40°39'32.0"N., 74°03'53.5"W.; to
(1115) 40°39'27.5"N., 74°03'42.5"W.; to
(1116) 40°39'13.0"N., 74°03'51.0"W.; to
(1117) 40°39'09.5"N., 74°04'23.1"W.; thence back to
(1118) 40°39′30.1″N., 74°04′08.0″W.
(1119) (i) See 33 CFR 110.155 (d)(9), (d)(16), and (l).
(1120) (9) This anchorage is designated a naval anchorage. The Captain of the Port may permit commercial vessels to anchor temporarily in this anchorage, ordinarily not more than 24 hours, when the anchorage will not be needed for naval vessels. Upon notification of an anticipated naval arrival, any commercial vessel so anchored must relocate at its own expense.
(1121) (10) *Anchorage No. 21–A*. That area enclosed by coordinates starting at
(1122) 40°40'22.5"N., 74°01'35.2"W.; to
(1123) 40°40'20.5"N., 74°01'27.7"W.; to
(1124) 40°39'48.9"N., 74°01'22.4"W.; to
(1125) 40°38'54.7"N., 74°02'18.9"W.; to
(1126) 40°39'03.0"N., 74°02'26.3"W.; thence back to the beginning.
(1127) (i) See 33 CFR 110.155 (d)(16) and (l).
(1128) (ii) No vessel may occupy this anchorage for a period of time in excess of 96 hours without prior approval of the Captain of the Port.
(1129) (11) *Anchorage No. 21–B*. That area enclosed by coordinates starting at
(1130) 40°40'23.8"N., 74°02'10.9"W.; to
(1131) 40°40'26.2"N., 74°01'49.5"W.; to
(1132) 40°40'22.5"N., 74°01'35.2"W.; to
(1133) 40°39'03.0"N., 74°02'26.3"W.; to
(1134) 40°38'54.7"N., 74°02'18.9"W.; to
(1135) 40°38'43.7"N., 74°02'30.3"W.; to
(1136) 40°39'19.3"N., 74°03'03.3"W.; to
(1137) 40°39'22.3"N., 74°03'02.4"W.; to
(1138) 40°40'18.6"N., 74°02'25.5"W.; thence back to
(1139) 40°40′23.8″N., 74°02′10.9″W.
(1140) (i) See 33 CFR 110.155 (d)(16) and (l).
(1141) (ii) No vessel with a draft of 10 feet (3.048 meters) or less may occupy this anchorage without the prior approval of the Captain of the Port.
(1142) (iii) No vessel may occupy this anchorage for a period of time in excess of 96 hours without prior approval of the Captain of the Port.
(1143) (12) *Anchorage No. 21–C*. That area enclosed by coordinates starting at
(1144) 40°39'19.3"N., 74°03'03.3"W.; to
(1145) 40°38'43.7"N., 74°02'30.3"W.; to
(1146) 40°38'41.6"N., 74°02'32.5"W.; to
(1147) 40°38'03.0"N., 74°02'48.7"W.; to
(1148) 40°38'03.0"N., 74°03'03.5"W.; to
(1149) 40°38'38.4"N., 74°03'15.5"W.; thence back to
(1150) 40°39′19.3″N., 74°03′03.3″W.
(1151) (i) See 33 CFR 110.155 (d)(16) and (l).
(1152) (ii) No vessel with a draft of 33 feet (10.0584 meters) or less may occupy this anchorage without the prior approval of the Captain of the Port.
(1153) (iii) No vessel may occupy this anchorage for a period of time in excess of 96 hours without prior approval of the Captain of the Port.

(1154) (13) *Anchorage No. 23–A*. That area enclosed by coordinates starting at

(1155) 40°38'36.5"N., 74°04'13.5"W.; to

(1156) 40°38'37.0"N., 74°03'49.0"W.; to

(1157) 40°38'23.4"N., 74°03'37.2"W.; to

(1158) 40°37'49.5"N., 74°03'25.7"W.; to

(1159) 40°37'49.8"N., 74°03'50.1"W.; to

(1160) 40°37'50.0"N., 74°03'50.2"W.; to

(1161) 40°37'53.0"N., 74°04'07.0"W.; thence back to

(1162) 40°38'36.5"N., 74°04'13.5"W.

(1163) (i) See 33 CFR 110.155 (d)(16) and (l).

(1164) (ii) No vessel may occupy this anchorage for a period of time in excess of 48 hours without the prior approval of the Captain of the Port.

(1165) (iii) No vessel with a length overall in excess of 670 feet (204.216 meters) may occupy this anchorage without the prior approval of the Captain of the Port.

(1166) (iv) No vessel with a draft of 40 feet (12.192 meters) or more may occupy this anchorage without the prior approval of the Captain of the Port unless it anchors within 5 hours after ebb current begins at the Narrows.

(1167) (v) See 33 CFR 334.85 for information on anchoring near the U.S. Navy restricted area adjacent to this anchorage.

(1168) (14) *Anchorage No. 23–B*. That area enclosed by coordinates starting at

(1169) 40°37'49.8"N., 74°03'50.1"W.; to

(1170) 40°37'49.5"N., 74°03'25.7"W.; to

(1171) 40°37'27.0"N., 74°03'18.1"W.; to

(1172) 40°37'23.0"N., 74°03'59.0"W.; to

(1173) 40°37'30.0"N., 74°04'04.0"W.; to

(1174) 40°37'37.5"N., 74°03'46.0"W.; thence back to

(1175) 40°37'49.8"N., 74°03'50.1"W.

(1176) (i) See 33 CFR 110.155 (d)(13)(ii) and (iv), (d)(16), and (l).

(1177) (ii) No vessel with a length overall of 670 feet (204.216 meters) or less may occupy this anchorage without the prior approval of the Captain of the Port.

(1178) (iii) See 33 CFR 334.85 for information on anchoring near the U.S. Navy restricted area adjacent to this anchorage.

(1179) (15) *Anchorage No. 24*.That area enclosed by coordinates starting at

(1180) 40°37'23.0"N., 74°03'59.0"W.; to

(1181) 40°37'27.0"N., 74°03'18.1"W.; to

(1182) 40°36'40.1"N., 74°03'02.2"W.; to

(1183) 40°36'25.5"N., 74°02'56.4"W.; to

(1184) 40°36'21.0"N., 74°03'11.0"W.; to

(1185) 40°36'25.0"N., 74°03'17.5"W.; thence back to

(1186) 40°37′23.0″N., 74°03′59.0″W.

(1187) (i) See 33 CFR 110.155 (d)(13) (ii) and (iv), (d)(16), and (l).

(1188) (ii) No vessel with a length overall of less than 800 feet (243.84 meters), or with a draft of less than 40 feet (12.192 meters) may occupy this anchorage without the prior approval of the Captain of the Port.

(1189) (16) Any vessel anchored in or intending to anchor in Federal Anchorage 20–A through 20–G, 21–A through 21–C, 23–A and 23–B, 24 or 25 must comply with the following requirements:

(1190) (i) No vessel may anchor unless it notifies the Captain of the Port when it anchors, of the vessel's name, length, draft, and its position in the anchorage.

(1191) (ii) Each vessel anchored must notify the Captain of the Port when it weighs anchor.

(1192) (iii) No vessel may conduct lightering operations unless it notifies the Captain of the Port before it begins lightering operations.

(1193) (iv) Each vessel lightering must notify the Captain of the Port at the termination of lightering.

(1194) (v) No vessel may anchor unless it maintains a bridge watch, guards and answers Channel 16 FM, and maintains an accurate position plot.

(1195) (vi) If any vessel is so close to another that a collision is probable, each vessel must communicate with the other vessel and the Captain of the Port on Channel 16 FM and shall act to eliminate the close proximity situation.

(1196) (vii) No vessel may anchor unless it maintains the capability to get underway within 30 minutes except with prior approval of the Captain of the Port.

(1197) (viii) No vessel may anchor in a "dead ship" status (propulsion or control unavailable for normal operations) without the prior approval of the Captain of the Port.

(1198) (ix) Each vessel in a "dead ship" status must engage an adequate number of tugs alongside during tide changes. A tug alongside may assume the Channel 16 FM radio guard for the vessel after it notifies the Captain of the Port.

(1199) (x) No vessel may lighter in a "dead ship" status without prior approval from the Captain of the Port.

(1200) (e) *Lower Bay*–(1) *Anchorage No. 25*.That area enclosed by coordinates starting at

(1201) 40°35'58.2"N., 74°02'18.4"W.; to

(1202) 40°36'12.0"N., 74°01'29.0"W.; to

(1203) 40°36'03.0"N., 74°00'52.5"W.; to

(1204) 40°34'57.5"N., 74°00'25.0"W.; to

(1205) 40°34'40.0"N., 74°01'03.0"W.; to

(1206) 40°34'53.0"N., 74°01'56.1"W.; to

(1207) 40°35'23.9"N., 74°02'04.8"W.; thence back to

(1208) 40°35′58.2″N., 74°02′18.4″W.

(1209) (i) See 33 CFR 110.155 (d)(16) and (l).

(1210) (ii) When the use of this anchorage is required by naval vessels, any commercial vessels anchored therein must move when directed by the Captain of the Port.

(1211) (iii) No vessel may occupy this anchorage for a period of time in excess of 96 hours without prior approval of the Captain of the Port.

(1212) (f) *Lower Bay, Raritan Bay, Sandy Hook Bay, and Atlantic Ocean.* (1) *Anchorage No. 26*. In Raritan and Sandy Hook Bays all waters bound by the following points: 40°30′06.74"N., 074°10'04.96"W.; thence to 40°28′59.44"N., 074°05′00.00"W.; thence to 40°28′44.94"N., 074°05′00.00"W.; thence to 40°29′05.02"N., 074°07′30.56"W.; thence to 40°29′17.49"N., 074°10′16.50"W.; thence to the point of origin (NAD 83).

(1213) (2) *Anchorage No. 27*. In the Atlantic Ocean all waters bound by the following points: 40°28′49.27"N., 074°00′12.13"W.; thence to 40°28′52.12"N., 074°00′00.56"W.; thence to 40°28′40.88"N., 073°58′51.95"W.; thence to 40°25′57.91"N., 073°54′55.56"W.; thence to 40°23'45.55"N., 073°54′54.89"W.; thence to 40°23′45.38"N., 073°58′32.10"W.; thence along the shoreline to the point of origin (NAD 83).

(1214) (3) *Anchorage No. 28*. In Lower Bay all waters bound by the following points: 40°30′02.30"N., 074°08′52.69"W.; thence to 40°29′10.10"N., 074°04′59.65"W.; thence to 40°29′09.99"N., 074°02′57.75"W.; thence to 40°31′52.89"N., 074°02′39.89"W.; thence to 40°31′59.72"N., 074°03′25.13"W.; thence to 40°31′28.57"N., 074°03′40.70"W.; thence to 40°30′26.24"N., 074°05′11.46"W.; thence to 40°30′19.01"N., 074°06′21.37"W.; thence to 40°30′21.53"N., 074°08′46.19"W.; thence to the point of origin (NAD 83).

(1215) (g) [Reserved]

(1216) (h) [Reserved]

(1217) (i) *Arthur Kill*–(1) *Anchorage No. 41*. The passage between Pralls Island and Staten Island included between a line running 29° from the extreme northwest point of Pralls Island to a point on Staten Island and a line from the southern point of Pralls Island to the north side of the mouth of Neck Creek at Travis, Staten Island.

(1218) (2) *Anchorage No. 42*. East of lines ranging from the head of the Tottenville Shipyard Company pier at Tottenville, Staten Island, to the first pier of the Outerbridge Crossing west from the Staten Island shore, thence to Arthur Kill Light 10, thence to Arthur Kill Light 14, and thence to Arthur Kill Lighted Buoy 16; and south of a line from thence to Smoking Point.

(1219) (j) *Raritan Bay*–(1) *Anchorage No. 44*. An area in Raritan Bay located at the junction of Arthur Kill and Raritan River, beginning at

(1220) 40°03'07"N., 74°15'30"W.; to

(1221) 40°30'01"N., 74°15'30"W.; to

(1222) 40°29'27"N., 74°15'06"W.; to

(1223) 40°29'24"N., 74°15'01"W.; to

(1224) 40°29'15"N., 74°14'55"W.; to

(1225) 40°29'14"N., 74°15'25"W.; to

(1226) 40°29'48"N., 74°15'48"W.; thence to the point of beginning.

(1227) (i) The anchorage is restricted to deepdraft vessels except that barges may moor in that portion of the anchorage southerly of latitude 40°29'22".

(1228) (ii) No vessel shall occupy the deepdraft portion of the anchorage for a longer period than 48 hours without a permit from the Captain of the Port.

(1229) (2) *Anchorage No. 45*. West of the Raritan Bay Channel leading into Arthur Kill; north of the Raritan River Channel leading into Raritan River; and east of the Cutoff Channel between Raritan River and Arthur Kill, except that part of the said area occupied by Anchorage No. 44.

(1230) (i) Vessels must not anchor in the channel to Keyport Harbor west of lines ranging from Keyport Channel Buoy 1 to Keyport Channel Buoy 9, thence through Keyport Channel Buoys 11 and 13 to the northeast corner of the easterly steamboat wharf; and east of a line extending from a point 400 yards west of Keyport Channel Buoy 1 tangent to the west shore at the mouth of Matawan Creek.

(1231) (ii) [Reserved]

(1232) (k) [Reserved]

(1233) (l) *General regulations*. (1) No vessel in excess of 800 feet (243.84 meters) in length overall or 40 feet (12.192 meters) in draft may anchor unless it notifies the Captain of the Port at least 48 hours prior to entering Ambrose Channel.

(1234) (2) Except in cases of great emergency, no vessel shall be anchored in the navigable waters of the Port of New York outside of the anchorage areas established in this section, nor cast anchor within a cable or pipe line area shown on a Government chart, nor be moored, anchored, or tied up to any pier, wharf, or vessel in such manner as to obstruct or endanger the passage of any vessel in transit by, or to or from, adjacent wharves, piers, or slips.

(1235) (3) No vessel shall occupy for a longer period than 30 days, unless a permit is obtained from the Captain of the Port for that purpose, any anchorage for which the time of occupancy is not otherwise prescribed in this section. No vessel in a condition such that it is likely to sink or otherwise become a menace or obstruction to navigation or anchorage of other vessels shall occupy an anchorage except in an emergency, and then only for such period as may be permitted by the Captain of the Port.

(1236) (4) Whenever, in the opinion of the Captain of the Port, such action may be necessary, that officer may require any or all vessels in any designated anchorage area to moor with two or more anchors.

(1237) (5) Every vessel whose crew may be reduced to such number that it will not have sufficient men on board to weigh anchor at any time shall be anchored with two anchors, with mooring swivel put on before the crew shall be reduced or released, unless the Captain of the Port shall waive the requirement of a mooring swivel.

(1238) (6) Anchors of all vessels must be placed well within the anchorage areas, so that no portion of the hull or rigging shall at any time extend outside the boundaries of the anchorage area.

(1239) (7) Any vessel anchoring under circumstances of great emergency outside of the anchorage areas must be placed near the edge of the channel and in such position as not to interfere with the free navigation of the channel nor obstruct the approach to any pier nor impede the movement of any boat, and shall move away immediately after the emergency ceases, or upon notification by the Captain of the Port.

(1240) (8) Operations near commercial mooring buoys permitted by the District Engineer, U.S. Army Corps of Engineers.

(1241) (i) No vessel shall continuously occupy a mooring when a vessel in regular traffic requires the berth or when navigation would be menaced or inconvenienced thereby.

(1242) (ii) No vessel shall moor or anchor in any anchorage in such a manner as to interfere with the use of a duly authorized mooring buoy. Nor shall any vessel moored to a buoy authorized by the District Engineer, U.S. Army Corps of Engineers be moored such that any portion of that vessel comes within 50 feet of a marked or dredged channel.

(1243) (iii) No vessel shall be operated within the limits of an anchorage at speed exceeding 6 knots when in the vicinity of a moored vessel.

(1244) (iv) In an emergency the Captain of the Port may shift the position of any unattended vessel moored in or near any anchorage.

(1245) (9) Barge dispensing stations and stake boats may be anchored in such places as the Captain of the Port may designate.

(1246) (10) Upon approval of the District Engineer, Corps of Engineers, the Captain of the Port may permit wrecking plant or other vessels legally engaged in recovering sunken property, or in laying or repairing pipe lines or cables legally established, or plant engaged in dredging operations, to anchor within channels of the Port of New York. Permit issued by the Captain of the Port is not necessary for plant engaged upon works of river and harbor improvement under the supervision of the District Engineer, but the District Engineer will notify the Captain of the Port in advance of all such proposed work.

(1247) (11) Whenever the maritime or commercial interests of the United States so require, the Captain of the Port is hereby empowered to shift the position of any vessel anchored within the anchorage areas, of any vessel anchored outside the anchorage areas, of any vessel which is so moored or anchored as to impede or obstruct vessel movements in any channel or obstruct or interfere with range lights and of any vessel which, lying at the exterior end of a pier or alongside an open bulkhead, obstructs or endangers the passage of vessels in transit by, or to or from, adjacent wharf property or impedes the movements of vessels entering or leaving adjacent slips.

(1248) (12) A vessel upon being notified to move into the anchorage limits or to shift its position on anchorage grounds, shall get under way at once or signal for a tug, and shall change position as directed, with reasonable promptness.

(1249) (13) Nothing in this section shall be construed as relieving any vessel or the owner or person in charge of any vessel from the penalties of law for obstructing navigation or for obstructing or interfering with range lights, or for not complying with the navigation laws in regard to lights, fog signals, or for otherwise violating law.

(1250) (14) Any vessel prohibited by these rules from anchoring in a specific anchorage because of the vessel's length or draft may anchor in the anchorage with permission from the Captain of the Port.

(1251) (m) *Anchorages for vessels carrying explosives–*

(1252) (1) [Reserved]

(1253) (2) *Anchorage No. 49–F (emergency naval anchorage).* That portion of Sandy Hook Bay bounded by a line bearing 170°, 3,800 yards, from a point bearing 281°30', 2,050 yards from Sandy Hook Light; thence 260°, 500 yards; thence 350°, 3,800 yards; thence 080°, 500 yards, to the point of beginning.

(1254) (i) This anchorage is to be used for the anchorage of naval vessels during emergencies only.

(1255) (ii) No pleasure or commercial craft shall navigate or moor within this area at any time when naval vessels which are moored in the area display a red flag by day or a red light by night.

(1256) (3) *Anchorage No. 49–G (naval anchorage).* That portion of Sandy Hook Bay bounded by a line bearing 208°, 1,350 yards, from a point bearing 292°30', 3,600 yards, from Sandy Hook Light; thence 298°, 620 yards; thence 002°, 1,250 yards; thence 107°, 1,150 yards, to the point of beginning.

(1257) (i) No pleasure or commercial craft shall navigate or moor within this area at any time when vessels which are moored in the area display a red flag by day or red light by night.

(1258) (n) *Regulations for explosive anchorages*. (1) Anchorages Nos. 49–F, and 49–G are reserved for vessels carrying explosives. All vessels carrying explosives shall be within these areas when anchored, except as provided in paragraph (n)(6) of this section.

(1259) (2) A written permit shall be obtained from the Captain of the Port before vessels carrying explosives, or on which explosives are to be loaded, may proceed to the anchorages provided for them; and no vessel shall occupy a berth in such anchorage except by authority of such permit, which permit may be revoked at any time.

(1260) (3) Vessels used in connection with loading or unloading explosives on vessels in anchorage areas, including tugs and stevedore boats, shall carry a written permit from the Captain of the Port. The Captain of the Port may, in his discretion, require every person having business on board vessels which are being loaded with explosives, other than members of the crew, to have a pass from the Captain of the Port in such form as he shall prescribe. Such permit or pass shall be shown whenever required by him or by his authorized agents.

(1261) (4) Whenever any vessel not fitted with mechanical power anchors in the explosives anchorages while carrying explosives, the Captain of the Port may require the attendance of a tug upon such vessel when in his judgment such action is necessary.

(1262) (5) Vessels carrying explosives shall comply with the general regulations in paragraph (1) of this section when applicable.

(1263) (6) The District Engineer, Corps of Engineers, may authorize, in writing, a vessel carrying explosives for use on river and harbor works or on other work under federal permit issued by the District Engineer to anchor in or near the vicinity of such work without a permit from the

Captain of the Port. The District Engineer will prescribe the quantities of such explosives allowed on such vessel and the conditions under which they are to be stored and handled, and will furnish the Captain of the Port with a copy of such safety instructions together with a copy of his written authorization.

(1264) (7) Every vessel loading, unloading, transporting, or containing explosives shall display by day a red flag at least 16 square feet in area at its masthead, or at least 10 feet above the upper deck if the vessel has no mast, and shall display by night a red light in the same position specified for the flag.

(1265) (8) When local regulations of any place require previous local authority for the transfer of explosives or fireworks between vessels or between a vessel and a wharf or other place ashore, the Captain of the Port will permit the removal from the anchorage of such vessel containing explosives to any place covered by such local regulations only when he is satisfied that the required local authority has been granted.

(1266) **NOTE:** The anchorage in this section are regulated under Title I, Ports and Waterways Safety Act of 1972 as stated in §110.1a(a) of this Part. The penalties for violating regulations under this Act are stated in §110.1a (b) of this Part.

(1267)

§110.156 Randall Bay, Freeport, Long Island, N.Y.

(1268) (a) *The anchorage grounds*. Southward of a line 312 feet south of and parallel to the south side of Casino Street; eastward of a line 215 feet east of and parallel to the east side of West Side Avenue, said line extending southerly to a point 233 feet north of the prolonged north side of Clinton Street; northeastward of a line from the last-mentioned point to a point 243 feet southerly of the prolonged south side of Clinton Street and 210 feet east of the east side of Prospect Street; eastward of a line 210 feet east of and parallel to the east side of Prospect Street; northward of a line 25 feet north of and parallel to the prolonged north side of Suffolk Street; westward of a line 210 feet west of and parallel to the west side of South Long Beach Avenue, said line extending northerly to a point 222 feet south of the prolonged south side of Queens Street; southwestward of a line from the last-mentioned point to a point 74 feet northerly of the prolonged north side of Queens Street and 120 feet west of the west side of Roosevelt Avenue; and westward of a line 120 feet west of and parallel to the west side of Roosevelt Avenue.

(1269) (b) *The regulations*. (1) When applied for, a berth in this anchorage, if available, may be assigned to any vessel by the Captain of the Port of Long Island Sound.

(1270) (2) The Captain of the Port is authorized to issue permits for maintaining mooring buoys within the anchorage. The method of anchoring these buoys shall be as prescribed by the Captain of the Port.

(1271) (3) No vessel shall anchor in the anchorage in such manner as to interfere with the use of a duly authorized mooring buoy.

(1272) (4) No vessel shall be navigated within the anchorage at a speed exceeding six knots.

(1273) (5) In case of emergencies, the Captain of the Port is authorized to shift the position of any unattended vessel moored in or near the anchorage.

(1274)

Part 117–Drawbridge Operation Regulations

(1275)

Subpart A–General Requirements

(1276)

§117.1 Purpose.

(1277) (a) This part prescribes the general and special drawbridge operating regulations that apply to the drawbridges across the navigable waters of the United States and its territories. The authority to regulate drawbridges across the navigable waters of the United States is vested in the Secretary of Homeland Security.

(1278) (b) Subpart A contains the general operation requirements that apply to all drawbridges.

(1279) (c) Subpart B contains specific requirements for operation of individual drawbridges. These requirements are in addition to or vary from the general requirements in Subpart A. Specific sections in subpart B that vary from a general requirement in Subpart A supersede the general requirement. All other general requirements in Subpart A, that are not at variance, apply to the drawbridges and removable span bridges listed in Subpart B.

(1280)

§117.4 Definitions.

(1281) The following definitions apply to this part:

(1282) *Appurtenance* means an attachment or accessory extending beyond the hull or superstructure that is not an integral part of the vessel and is not needed for a vessel's piloting, propelling, controlling, or collision avoidance capabilities.

(1283) *Automated drawbridge* means a drawbridge that is operated by an automated mechanism, not a drawtender. An automated drawbridge is normally kept in the open to navigation position and closes when the mechanism is activated.

(1284) *Deviation* means a District Commander's action authorizing a drawbridge owner to temporarily not comply with the drawbridge opening requirements in this part.

(1285) *Drawbridge* means a bridge with an operational span that is intended to be opened for the passage of waterway traffic.

(1286) *Drawspan* means the operational span of a drawbridge.

(1287) *Lowerable* means a non-structural vessel appurtenance that is or can be made flexible, hinged, collapsible, or telescopic so that it can be mechanically or manually lowered.

(1288) *Nonstructural* means that the item is not rigidly fixed to the vessel and can be relocated or altered.

(1289) *Not essential to navigation* means that a nonstructural vessel appurtenance, when in the lowered position, would not adversely affect the vessel's piloting, propulsion, control, or collision-avoidance capabilities.

(1290) *Public vessel* means a vessel that is owned and operated by the United States Government and is not engaged in commercial service, as defined in 46 U.S.C. 2101.

(1291) *Remotely operated drawbridge* means a drawbridge that is operated by remote control from a location away from the drawbridge.

(1292) *Removable span bridge* means a bridge that requires the complete removal of a span by means other than machinery installed on the bridge to open the bridge to navigation.

(1293) *Untended* means that there is no drawtender at the drawbridge.

(1294)

§117.5 When the drawbridge must open.

(1295) Except as otherwise authorized or required by this part, drawbridges must open promptly and fully for the passage of vessels when a request or signal to open is given in accordance with this subpart.

(1296)

§117.7 General requirements of drawbridge owners.

(1297) Except for drawbridges that have been authorized, before January 3, 2007, to remain closed to navigation or as otherwise specified in subpart B, drawbridge owners must:

(1298) (a) Provide the necessary drawtender(s) for the safe and prompt opening of the drawbridge.

(1299) (b) Maintain the working machinery of the drawbridge in good operating condition.

(1300) (c) Cycle the drawspan(s) periodically to ensure operation of the drawbridge.

(1301) (d) Ensure that the drawbridge operates in accordance with the requirements of this part.

(1302) (e) Any drawbridge allowed to remain closed to navigation prior to January 3, 2007, when necessary, must be returned to operable condition within the designated time set forth by the District Commander and will become subject to the requirements of this part.

(1303)

§117.8 Permanent changes to drawbridge operation.

(1304) (a) Anyone may submit a written request to the District Commander for a permanent change to a drawbridge operating requirement. The request must include documentation supporting or justifying the requested change.

(1305) (b) If after evaluating the request, the District Commander determines that the requested change is not needed, he or she will respond to the request in writing and provide the reasons for denial of the requested change.

(1306) (c) If the District Commander decides that a change may be needed, he or she will begin a rulemaking to implement the change.

(1307)

§117.9 Delaying opening of a draw.

(1308) No person shall unreasonably delay the opening of a draw after the signals required by §117.15 have been given.

(1309) **NOTE:** Trains are usually controlled by the block method. That is, the track is divided into blocks or segments of a mile or more in length. When a train is in a block with a drawbridge, the draw may not be able to open until the train has passed out of the block and the yardmaster or other manager has "unlocked" the drawbridge controls. The maximum time permitted for delay is defined in Subpart B for each affected bridge. Land and water traffic should pass over or through the draw as soon as possible in order to prevent unnecessary delays in the opening and closure of the draw.

(1310)

§117.11 Unnecessary opening of the draw.

(1311) No vessel owner or operator shall–

(1312) (a) Signal a drawbridge to open if the vertical clearance is sufficient to allow the vessel, after all lowerable nonstructural vessel appurtenances that are not essential to navigation have been lowered, to safely pass under the drawbridge in the closed position; or

(1313) (b) Signal a drawbridge to open for any purpose other than to pass through the drawbridge opening.

(1314)

§117.15 Signals.

(1315) (a) *General*. (1) The operator of each vessel requesting a drawbridge to open shall signal the drawtender and the drawtender shall acknowledge that signal. The signal shall be repeated until acknowledged in some manner by the drawtender before proceeding.

(1316) (2) The signals used to request the opening of the draw and to acknowledge that request shall be sound signals, visual signals, or radiotelephone communications described in this subpart.

(1317) (3) Any of the means of signaling described in this subpart sufficient to alert the party being signaled may be used.

(1318) (b) *Sound signals*. (1) Sound signals shall be made by whistle, horn, megaphone, hailer, or other device capable of producing the described signals loud enough to be heard by the drawtender.

(1319) (2) As used in this section, "prolonged blast" means a blast of four to six seconds duration and "short blast" means a blast of approximately one second duration.

(1320) (3) The sound signal to request the opening of a draw is one prolonged blast followed by one short blast sounded not more than three seconds after the prolonged blast. For vessels required to be passed through a draw during a scheduled closure period, the sound signal to request the opening of the draw during that period is five short blasts sounded in rapid succession.

(1321) (4) When the draw can be opened immediately, the sound signal to acknowledge a request to open the draw is one prolonged blast followed by one short blast sounded not more than 30 seconds after the requesting signal.

(1322) (5) When the draw cannot be opened immediately, or is open and shall be closed promptly, the sound signal to acknowledge a request to open the draw is five short blasts sounded in rapid succession not more than 30 seconds after the vessel's opening signal. The signal shall be repeated until acknowledged in some manner by the requesting vessel.

(1323) (c) *Visual signals*. (1) The visual signal to request the opening of a draw is–

(1324) (i) A white flag raised and lowered vertically; or

(1325) (ii) A white, amber, or green light raised and lowered vertically.

(1326) (2) When the draw can be opened immediately, the visual signal to acknowledge a request to open the draw, given not more than 30 seconds after the vessel's opening signal, is–

(1327) (i) A white flag raised and lowered vertically;

(1328) (ii) A white, amber, or green light raised and lowered vertically; or

(1329) (iii) A fixed or flashing white, amber, or green light or lights.

(1330) (3) When the draw cannot be opened immediately, or is open and must be closed promptly, the visual signal to acknowledge a request to open the draw is–

(1331) (i) A red flag or red light swung back and forth horizontally in full sight of the vessel given not more than 30 seconds after the vessel's opening signal; or

(1332) (ii) A fixed or flashing red light or lights given not more than 30 seconds after the vessel's opening signal.

(1333) (4) The acknowledging signal when the draw cannot open immediately or is open and must be closed promptly shall be repeated until acknowledged in some manner by the requesting vessel.

(1334) (d) *Radiotelephone communications*. (1) Radiotelephones may be used to communicate the same information provided by sound and visual signals.

(1335) (2) The vessel and the drawtender shall monitor the frequency used until the vessel has cleared the draw.

(1336) (3) When radiotelephone contact cannot be initiated or maintained, sound or visual signals under this section shall be used.

(1337)

§117.17 Signaling for contiguous drawbridges.

(1338) When a vessel must pass two or more drawbridges close together, the opening signal is given for the first bridge. After acknowledgment from the first bridge that it will promptly open, the opening signal is given for the second bridge, and so on until all bridges that the vessel must pass have been given the opening signal and have acknowledged that they will open promptly.

(1339)

§117.19 Signaling when two or more vessels are approaching a drawbridge.

(1340) When two or more vessels are approaching the same drawbridge at the same time, or nearly the same time, whether from the same or opposite directions, each vessel shall signal independently for the opening of the draw and the drawtender shall reply in turn to the signal of each vessel. The drawtender need not reply to signals by vessels accumulated at the bridge for passage during a scheduled open period.

(1341)

§117.21 Signaling for an opened drawbridge.

(1342) When a vessel approaches a drawbridge with the draw in the open position, the vessel shall give the opening signal. If no acknowledgment is received within 30 seconds, the vessel may proceed, with caution, through the open draw.

(1343)

§117.23 Installation of radiotelephones.

(1344) (a) When the District Commander deems it necessary for reasons of safety of navigation, the District Commander may require the installation and operation of a radiotelephone on or near a drawbridge.

(1345) (b) The District Commander gives written notice of the proposed requirement to the bridge owner.

(1346) (c) All comments the owner wishes to submit shall be submitted to the District Commander within 30 days of receipt of the notice under paragraph (b) of this section.

(1347) (d) If, upon consideration of the comments received, the District Commander determines that a radiotelephone is necessary, the District Commander notifies the bridge owner that a radiotelephone shall be installed and gives a reasonable time, not to exceed six months, to install the radiotelephone and commence operation.

(1348)

§117.24 Radiotelephone installation identification.

(1349) (a) The Coast Guard authorizes, and the District Commander may require the installation of a sign on drawbridges, on the upstream and downstream sides, indicating that the bridge is equipped with and operates a VHF radiotelephone in accordance with §117.23.

(1350) (b) The sign shall give notice of the radiotelephone and its calling and working channels–

(1351) (1) In plain language; or

(1352) (2) By a sign consisting of the outline of a telephone handset with the long axis placed horizontally and a vertical three-legged lightning slash superimposed over the handset. The slash shall be as long vertically as the handset is wide horizontally and normally not less than 27 inches and no more than 36 inches long. The preferred calling channel should be shown in the lower left quadrant and the preferred working channel should be shown in the lower right quadrant.

(1353) **Note**: It is recommended that the radiotelephone sign be similar in design to the Service Signs established by the Federal Highway Administration (FHWA) in U.S.

Road Symbol Signs using Reflective Blue and Reflective White colors. Color and design information is available from the District Commander of the Coast Guard District in which the bridge is located.

(1354)

§117.31 Drawbridge operations for emergency vehicles and emergency vessels.

(1355) (a) Upon receiving notification that an emergency vehicle is responding to an emergency situation, a drawtender must make all reasonable efforts to have the drawspan closed at the time the emergency vehicle arrives.

(1356) (b) When a drawtender receives notice, or a proper signal as provided in §117.15 of this part, the drawtender shall take all reasonable measures to have the draw opened, regardless of the operating schedule of the draw, for passage of the following, provided this opening does not conflict with local emergency management procedures which have been approved by the cognizant Coast Guard Captain of the Port:

(1357) (1) Federal, State, and local government vessels used for public safety;

(1358) (2) Vessels in distress where a delay would endanger life or property;

(1359) (3) Commercial vessels engaged in rescue or emergency salvage operations; and

(1360) (4) Vessels seeking shelter from severe weather.

(1361)

§117.33 Closure of draw for natural disasters or civil disorders.

(1362) Drawbridges need not open for the passage of vessels during periods of natural disasters or civil disorders declared by the appropriate authorities unless otherwise provided for in Subpart B or directed to do so by the District Commander.

(1363)

§117.35 Temporary change to a drawbridge operating schedule.

(1364) (a) For any temporary change to the operating schedule of a drawbridge, lasting less than or equal to 180 days, the District Commander may issue a deviation approval letter to the bridge owner and publish a "Notice of temporary deviation from regulations" in the **Federal Register**.

(1365) (b) If the time period for a temporary change to the operating schedule of a drawbridge will be greater then 180 days, the District Commander will follow appropriate rulemaking procedures and publish a temporary rule in the **Federal Register** prior to the start of the action.

(1366) (c) *Request for change*. (1) To temporarily change the drawbridge-operating requirements the bridge owner must submit a written request to the District Commander for approval of the change.

(1367) (2) The request must describe the reason for the deviation and the dates and times scheduled for the start and end of the change.

(1368) (3) Requests should be submitted as early as possible, preferably 90 days before the start of the action. District Commanders have discretion to accept requests submitted less than 90 days before a needed change if those requests can be processed before the date of the needed change.

(1369) (d) *Determination*. The District Commander's determination to allow the schedule change is normally forwarded to the bridge owner within ten working days after receipt of the request. If the request is denied, the reasons for the denial will be set out in the District Commander's decision letter.

(1370) (e) The drawbridge must return to its regular operating schedule immediately at the end of the designated time period.

(1371) (f) If the authorized deviation period for an event is broken into separate time periods on the same day or on consecutive days, the drawbridge must provide openings for navigation between authorized schedule changes.

(1372) (g) The District Commander will also announce the change to the operating schedule in the Local Notice to Mariners and other appropriate local media.

(1373)

§117.36 Closure of drawbridge for emergency repair.

(1374) (a) When a drawbridge unexpectedly becomes inoperable, or should be immediately rendered inoperable because of mechanical failure or structural defect, the drawbridge owner must notify the District Commander of the closure without delay and give the reason for the emergency closure of the drawbridge and an estimated time when the drawbridge will be returned to operating condition.

(1375) (b) The District Commander will notify mariners about the drawbridge status through Broadcast Notices to Mariners, Local Notice to Mariners and any other appropriate local media.

(1376) (c) Repair work under this section must be performed with all due speed in order to return the drawbridge to operation as soon as possible.

(1377)

§117.39 Authorized closure of drawbridge due to infrequent requests for openings.

(1378) (a) When there have been no requests for drawbridge openings for at least two years, a bridge owner may

telephone. To enable mariners to request bridge openings, the owner shall maintain and monitor a telephone at the bridge and provide a means for mariners to secure their boats upstream and downstream of the bridge in order to use this telephone.

(1418)

§117.205 Connecticut River.

(1419) (a) The owners of the AMTRAK Old Saybrook-Old Lyme Bridge, mile 3.4, the Route 82 Bridge, mile 16.8, and the CONRAIL Middletown-Portland Bridge, mile 32.0, shall provide, and keep in good legible condition, clearance gauges with figures not less than twelve (12) inches which designed, installed and maintained according to the provisions of §118.160 of this chapter.

(1420) (b) The draws of the AMTRAK Old Saybrook-Old Lyme Bridge, mile 3.4, and the CONRAIL Middletown-Portland Bridge, mile 32.0, shall be opened as soon as practicable for all non-commercial vessels that cannot pass under the closed draws, but in no case shall the delay be more than 20 minutes from the time the opening was requested.

(1421) (c) The draw of the Route 82 Bridge, mile 16.8, at East Haddam, shall operate as follows:

(1422) (1) *From May 1 through October 31*: The draw shall open on signal for commercial vessels. For recreational vessels, the draw shall open on signal, except that from 6 a.m. to 8 p.m., the draw need open for recreational vessels on the hour only.

(1423) (2) *From November 1 through April 30*: The draw shall open on signal for all vessels, except that from 8 p.m. to 4 a.m., the draw shall open on signal if at least six-hours notice is given by calling the number posted at the bridge.

(1424)

§117.207 Housatonic River.

(1425) (a) The draw of the US 1 Bridge, mile 3.5, at Stratford, shall open on signal; except that, from 7 a.m. to 9 a.m., Monday through Friday, and 4 p.m. to 5:45 p.m. daily, the draw need not open for the passage of vessels. From December 1 through March 31, from 8 p.m. to 4 a.m., the draw shall open on signal if at least six-hours notice is given by calling the number posted at the bridge.

(1426) (b) The draw of the Metro-North (Devon) bridge, mile 3.9 at Stratford, shall operate as follows:

(1427) (1) The draw shall open on signal between 4:30 a.m. and 9 p.m. after at least a two-hour advance notice is given; except that, from 5:45 a.m. through 9:45 a.m. and from 4 p.m. through 8 p.m., Monday through Friday excluding holidays, the draw need not open for the passage of vessel traffic unless an emergency exists.

(1428) (2) From 9 p.m. through 4:30 a.m. the draw shall open on signal after at least a four-hour advance notice is given.

(1429) (3) A delay in opening the draw not to exceed 10 minutes may occur when a train scheduled to cross the bridge without stopping has entered the drawbridge lock.

(1430) (4) Requests for bridge openings may be made by calling the bridge via marine radio VHF FM Channel 13 or the telephone number posted at the bridge.

(1431)

§117.209 Mianus River.

(1432) The draw of the Metro-North (Cos Cob) bridge, mile 1.0 at Greenwich, will operate as follows:

(1433) (a) The draw will open on signal between 4:30 a.m. and 9 p.m. after at least a two-hour advance notice is given; except that, from 5:45 a.m. through 9:45 a.m. and from 4 p.m. through 8 p.m., Monday through Friday excluding holidays, the draw need not open for the passage of vessel traffic unless an emergency exists.

(1434) (b) From 9 p.m. through 4:30 a.m. the draw will open on signal after at least a four-hour advance notice is given.

(1435) (c) A delay in opening the draw not to exceed 10 minutes may occur when a train scheduled to cross the bridge without stopping has entered the drawbridge lock.

(1436) (d) Requests for bridge openings may be made by calling the bridge via marine radio VHF FM Channel 13 or the telephone number posted at the bridge.

(1437)

§117.211 Mystic River.

(1438) (a) The draw of the Amtrak railroad bridge, mile 2.4 at Mystic, shall operate as follows:

(1439) (1) From April 1 to October 31, the draw shall open on signal.

(1440) (2) From November 1 to March 31, the draw shall open on signal from 5 a.m. to 9 p.m. From 9 p.m. to 5 a.m., the draw shall open on signal if at least eight hours notice is given.

(1441) (3) Commercial vessels shall be passed immediately at any time; however, the opening may be delayed up to eight minutes to allow trains, which have entered the drawbridge block and are scheduled to cross the bridge without stopping, to clear the block.

(1442) (4) All other vessels shall be passed as soon as practicable but no later than 20 minutes after the signal to open is given.

(1443) (b) The draw of the U.S. 1 Bridge, mile 2.8, at Mystic, shall open on signal except:

(1444) (1) From May 1 through October 31, from 7:40 a.m. to 6:40 p.m., the draw need only open hourly at twenty minutes before the hour.

(1445) (2) From November 1 through April 30, from 8 p.m. to 4 a.m., the draw shall open on signal if at least six-hours notice is given by calling the number posted at the bridge.

(1446)

§117.213 New Haven Harbor, Quinnipiac and Mill Rivers.

(1447) The draws of the Tomlinson Bridge, mile 0.0, the Ferry Street Bridge, mile 0.7, and the Grand Avenue Bridge, mile 1.3, across the Quinnipiac River, and the Chapel Street Bridge, mile 0.4, across the Mill River, shall operate as follows:

(1448) (a) The draw of the Tomlinson Bridge at mile 0.0, across the Quinnipiac River shall open on signal; except

that, from 7:30 a.m. to 8:30 a.m., noon to 12:15 p.m., 12:45 p.m. to 1 p.m., and 4:45 p.m. to 5:45 p.m., Monday through Friday, except Federal holidays, the draw need not open for the passage of vessel traffic.

(1449) (b) The draw of the Ferry Street Bridge at mile 0.7, across Quinnipiac River, shall open on signal; except that, from 7:30 a.m. to 8:30 a.m. and 4:45 p.m. to 5:45 p.m., Monday through Friday, except Federal holidays, the draws need not open for the passage of vessel traffic. From 9 p.m. to 5 a.m. the draw shall open on signal if at least a one-hour advance notice is given by calling the number posted at the bridge.

(1450) (c) The draw of the Grand Avenue Bridge at mile 1.3, across the Quinnipiac River shall open on signal; except that, from 7:30 a.m. to 8:30 a.m. and 4:45 p.m. to 5:45 p.m., Monday through Friday, except Federal holidays, the draw need not open for the passage of vessel traffic. From 9 p.m. to 5 a.m. the draw shall open on signal if at least a one-hour advance notice is given by calling the number posted at the bridge.

(1451) (d) The draw of the Chapel Street Bridge at mile 0.4, across the Mill River shall open on signal; except that, from 7:30 a.m. to 8:30 a.m. and 4:45 p.m. to 5:45 p.m., Monday through Friday, except Federal holidays, the draw need not open for the passage of vessel traffic. From 9 p.m. to 5 a.m. the draw shall open on signal after at least a one-hour advance notice is given by calling the number posted at the bridge.

(1452)

§117.215 Niantic River.

(1453) (a) The draw of the Amtrak Bridge, mile 0.0, at Niantic, shall open on signal at all times. When a train scheduled to cross the bridge without stopping has entered the drawbridge block, a delay in opening the draw may occur until the train has cleared the block. The delay should not exceed 10 minutes.

(1454) (b) The draw of the S156 Bridge, mile 0.1, at Niantic, shall open on signal; except that, from 7 a.m. to 8 a.m., and 4 p.m. to 5 p.m., Monday through Friday, except holidays, the draw shall open only for the passage of commercial vessels. From November 1 through April 30, from 8 p.m. to 4 a.m., the draw shall open on signal if at least six hours notice is given by calling the number posted at the bridge.

(1455)

§117.217 Norwalk River.

(1456) (a) The draw of the Washington Street S136 Bridge, mile 0.0, at Norwalk, shall operate as follows:

(1457) (1) The draw shall open on signal; except that, from 7 a.m. to 8:45 a.m., 11:45 a.m. to 1:15 p.m. and 4 p.m. to 6 p.m., Monday through Friday, except holidays, the draw need not be opened for the passage of vessels that draw less than 14 feet of water.

(1458) (2) The draw need not open for the passage of vessel traffic, from 10 a.m. to 12 p.m., on the first Saturday in December, to facilitate the running of the annual Norwalk River Fun Run. Should inclement weather force the postponement of the race the above bridge closure shall be implemented the next day, the first Sunday after the first Saturday in December, from 10 a.m. to 12 p.m.

(1459) (3) The bridge opening signal is three short blasts. Vessels drawing 14 feet of water or more shall add one prolonged blast after the three short blasts.

(1460) (b) The draw of the Metro-North ''WALK'' Bridge, mile 0.1, at Norwalk, shall operate as follows:

(1461) (1) The draw shall open on signal between 4:30 a.m. and 9 p.m. after at least a two hour advance notice is given; except that, from 5:45 a.m. through 9:45 a.m. and from 4 p.m. through 8 p.m., Monday through Friday excluding holidays, the draw need not open for the passage of vessel traffic unless an emergency exists.

(1462) (2) From 9 p.m. through 4:30 a.m. the draw shall open on signal after at least a four hour advance notice is given.

(1463) (3) A delay in opening the draw not to exceed 10 minutes may occur when a train scheduled to cross the bridge without stopping has entered the drawbridge lock.

(1464) (4) Requests for bridge openings may be made by calling the bridge via marine radio VHF-FM Channel 13 or the telephone number posted at the bridge.

(1465)

§117.219 Pequonnock River.

(1466) (a) The draw of the Stratford Avenue Bridge at mile 0.1, at Bridgeport, shall open on signal; except that, from 6:45 a.m. to 7:15 a.m., 7:45 a.m. to 8:15 a.m., 11:45 a.m. to 1:15 p.m., and 4:30 p.m. to 6:10 p.m., the draw need not open for the passage of vessel traffic. From December 1 through March 31, from 8 p.m. to 4 a.m., the draw shall open on signal if at least a six-hour notice is given by calling the number posted at the bridge.

(1467) (b) The draw of the Metro-North Peck Bridge at mile 0.3, at Bridgeport shall operate as follows:

(1468) (1) The draw shall open on signal between 5:45 a.m. to 9 p.m. if at least four hours advance notice is given; except that, from 5:45 a.m. to 9:45 a.m., and 4 p.m. to 8 p.m., Monday through Friday excluding holidays, the draw need not open for the passage of vessel traffic unless an emergency exists.

(1469) (2) From 9 p.m. to 5:45 a.m., the draw shall open on signal if at least an eight hour notice is given.

(1470) (3) A delay in opening the draw not to exceed 15 minutes may occur when a train scheduled to cross the bridge without stopping has entered the drawbridge block.

(1471) (4) Requests for bridge openings may be made by calling the telephone number posted at the bridge.

(1472)

§117.221 Saugatuck River.

(1473) (a) Public vessels of the United States must be passed through as soon as possible.

(1474) (b) The draw of the Metro-North ''SAGA'' bridge, mile 1.1 at Saugatuck, shall operate as follows:

(1475) (1) The draw shall open on signal between 4:30 a.m. and 9 p.m. after at least a two-hour advance notice is given;

except that, from 5:45 a.m. through 9:45 a.m. and from 4 p.m. through 8 p.m., Monday through Friday excluding holidays, the draw need not open for the passage of vessel traffic unless an emergency exists.

(1476) (2) From 9 p.m. through 4:30 a.m. the draw shall open on signal after at least a four-hour advance notice is given.

(1477) (3) A delay in opening the draw not to exceed 10 minutes may occur when a train scheduled to cross the bridge without stopping has entered the drawbridge lock.

(1478) (4) Requests for bridge openings may be made by calling the bridge via marine radio VHF FM Channel 13 or the telephone number posted at the bridge.

(1479) (c) The draw of the Route 136 Bridge, mile 1.3 at Saugatuck shall operate as follows:

(1480) (1) Year-round, need not open weekdays, except federal holidays, from 7 a.m. to 8:30 a.m. and 5:30 p.m. to 7:30 p.m.

(1481) (2) From April 15-October 31, open on signal if at least two hours notice is given, except as provided in paragraph (c)(1) of this section.

(1482) (3) From November 1-April 14, open on signal:

(1483) (i) From 8:30 a.m. to 3 p.m. if at least eight hours notice is given:

(1484) (ii) From 3 p.m. to 8:30 a.m., if at least 24 hours notice is given, except as provided in paragraph (c)(1) of this section.

(1485)

§117.223 Shaw Cove.

(1486) The draw of the Amtrak bridge, mile 0.0 at New London, shall open on signal from December 1 through March 31 from 8 a.m. to 5 p.m. Monday through Friday. From December 1 through March 31 from 5 p.m. to 8 a.m. and on Saturdays and Sundays, the draw shall open on signal if at least eight hours notice is given. From April 1 through November 30 from 5 a.m. to 10 p.m., the draw shall open on signal; and, from 10 p.m. to 5 a.m., the draw shall open on signal if at least one hour notice is given. A delay of up to 10 minutes may be expected if a train is approaching so closely that it may not be safely stopped. When a vessel is in an emergency that may endanger life or property, the draw shall open as soon as possible.

(1487)

§117.224 Thames River.

(1488) The draw of the Amtrak Bridge, mile 3.0, at New London, shall operate as follows:

(1489) (a) The draw shall open on signal to 75 feet above mean high water for all vessel traffic unless a full bridge opening to 135.3 feet above mean high water is requested.

(1490) (b) The 75 foot opening will be signified by a range light display with one solid green light and one flashing green light and the full 135.3 foot opening will be signified with two solid green range lights.

(1491) (c) The draw shall open on signal for public vessels of the United States and commercial vessels; except that, when a train scheduled to cross the bridge without stopping has passed the Midway, Groton, or New London stations and is in motion toward the bridge, the lift span shall not be opened until the train has crossed the bridge.

(1492) (d) The draw shall open on signal as soon as practicable for all other vessel traffic but no later than 20 minutes after the signal to open is given.

(1493)

§117.225 Yellow Mill Channel.

(1494) The drawspan of the Stratford Avenue Bridge, mile 0.3 at Bridgeport, must open on signal if at least 24-hours notice is given. Public vessels of the United States must pass through as soon as possible.

(1495) **MASSACHUSETTS**

(1496)

§117.585 Acushnet River.

(1497) (a) The New Bedford-Fairhaven RT-6 Bridge, mile 0.0, will open promptly, provided proper signal is given, on the following schedule:

(1498) (1) On the hour between 6:00 a.m. and 10:00 a.m. inclusive.

(1499) (2) At a quarter past the hour between 11:15 a.m. and 6:15 p.m. inclusive.

(1500) (3) At all other times on call.

(1501) (b) The draw will be opened at any time for vessels whose draft exceeds 15 feet, for vessels owned or operated by the U.S. Government, the State of Massachusetts, or by local authorities.

(1502) (c) Each opening of the draw, from the time vehicular traffic flow is stopped until the flow resumes, shall not exceed 15 minutes except for vessels whose draft exceeds 15 feet or in extraordinary circumstances.

(1503) (d) From 6 p.m. on December 24 to midnight on December 25 and from 6 p.m. on December 31 to midnight on January 1, the draw shall open on signal if at least a two-hour notice is given by calling the number posted at the bridge.

(1504)

§117.587 Apponagansett River.

(1505) (a) The draw of the Padanaram Bridge, mile 1.0, shall open on signal from 1 May through 31 October, between 6 a.m. and 9 p.m., daily, as follows:

(1506) (1) The bridge shall open on signal, twice an hour, on the hour and half hour between 6 a.m. and 9 a.m. and between 8 p.m. and 9 p.m.

(1507) (2) The bridge shall open on signal, once an hour, on the hour between 9 a.m. and 8 p.m.

(1508) (b) At all other times the bridge shall open if at least four (4) hours advance notice is given.

(1509) (c) The owners of this bridge shall provide and maintain mooring facilities for vessels to make fast while waiting for the bridge to open.

(1510) (d) The owners of this bridge shall provide and keep in good legible condition, clearance gauges for each draw with figures not less than twelve (12) inches high designed, installed and maintained according to the provisions of section 118.160 of this chapter.

(1511)

§117.589 Cape Cod Canal.

(1512) The draw of the Conrail railroad bridge, mile 0.7 at Bourne, shall operate as follows:

(1513) (a) The draw is normally in the fully open position except for the passage of trains or for maintenance. No signal is required if the draw is in the fully open position.

(1514) (b) If the draw is not in the fully open position, the opening signal is one prolonged and one short blast.

(1515) (c) Signals to be sounded from the bridge are–

(1516) (1) Immediately preceding the opening of the draw, one prolonged blast;

(1517) (2) Immediately preceding the closing of the draw, two prolonged blasts;

(1518) (3) When a vessel has sounded the opening signal and the draw cannot be opened immediately, five short blasts in a rapid succession; and

(1519) (4) When the draw is closed and visibility is reduced in foggy weather, five short blasts in rapid succession every two minutes.

(1520)

§117.598 Eel Pond Channel.

(1521) The following requirements apply to the draw of Eel Pond (Water Street) drawbridge at mile 0.0 at Falmouth, Massachusetts.

(1522) (a) The draw shall open at all times as soon as possible for public vessels of the United States, State or local vessels used for public safety, and vessels in distress. The opening signal for these vessels shall be four or more short blast of a whistle, horn, or radio request.

(1523) (b) The owners of this bridge shall provide and keep in good legible condition clearance gauges for each draw with figures not less than 12 inches high designed, installed and maintained according to the provisions of section 118.160 of these regulations.

(1524) (c) The draw shall operate as follows:

(1525) (1) On signal from October 15 through May 14, from 8 a.m. to 5 p.m. except as provided in paragraph (c)(3)(i) of this section.

(1526) (2) Need open on signal only on the hour and half hour as follows:

(1527) (i) From May 15 through June 14 and from September 16 through October 14, from 7 a.m. to 7 p.m.

(1528) (ii) From June 15 through September 15, from 6 a.m. to 9 p.m.

(1529) (3) The draw shall open on signal if at least 8 hours advance notice is given:

(1530) (i) At all times on Christmas, New Years, Easter and all Sundays in January and February.

(1531) (ii) At all other times not stipulated in paragraphs (c)(1) and (c)(2) of this section.

(1532)

§117.607 Mitchell River.

(1533) The Chatham Highway Bridge, at mile 0.2, at Chatham, Massachusetts, shall operate as follows:

(1534) (a) From May 1 through October 31, the draw shall open on signal from 8 a.m. to 5 p.m., if at least one-hour notice is given and from 5 p.m. to 8 a.m. the draw shall open on signal if at least 12-hours notice is given by calling the Chatham Harbormasters Department.

(1535) (b) From November 1 through April 30, the draw shall open on signal if at least a 24-hours advance notice is given by calling the Chatham Harbormasters Department.

(1536)

§117.619 Taunton River.

(1537) (a) The Brightman Street (Route-6) Bridge at mile 1.8, between Fall River and Somerset, shall operate as follows:

(1538) (b) The draw shall open on signal between 5 a.m. and 9 p.m., daily. From 9 p.m. through 5 a.m. the draw shall open on signal after at least a one-hour advance notice is given by calling the number posted at the bridge.

(1539) (c) From June 1 through August 31, the draw need not open for the passage of pleasure craft from 7 a.m. to 9:30 a.m., and from 4 p.m. to 6:30 p.m., Monday through Friday, except holidays. The draw shall open for commercial vessel at all times.

(1540) (d) From 6 p.m. on December 24 to midnight on December 25, and from 6 p.m. on December 31 to midnight on January 1, the draw shall open on signal if at least a two-hour advance notice is given by calling the number posted at the bridge.

(1541) (e) The owner of the bridge shall provide and keep in good legible condition clearance gauges located on both upstream and downstream sides of the draw with figures not less than twelve inches in height, designed, installed and maintained according to the provisions of §118.160 of this chapter.

(1542) (f) The draw of the Veterans Memorial Bridge, mile 2.1, across the Taunton River between Fall River and Somerset, shall operate as follows:

(1543) (1) From 7 a.m. through 3 p.m. the draw shall open on signal.

(1544) (2) From 3 p.m. through 7 a.m. the draw shall open on signal provided a two hour advance notice is given by calling the number posted at the bridge.

(1545)

§117.620 Westport River-East Branch.

(1546) The Westport Point Bridge, mile 1.2 at Westport, shall operate as follows:

(1547) (a) Public vessels of the United States must be passed as soon as possible.

(1548) (b) The owners of this bridge shall provide and keep in good legible condition clearance gauges for each draw with figures not less than 12 inches high designed, installed, and maintained, according to the provisions of §118.160 of this chapter.

(1549) (c) That the drawspan for the Westport Point Drawbridge, mile 1.2 at Westport, must open on signal if at least 24 hours notice is given.

(1550)

§117.622 Weymouth Fore River

(1551) The draw of the Quincy Weymouth SR3A bridge, mile 3.5 between Quincy Point and North Weymouth, Massachusetts, shall open on signal, except that:

(1552) (a) From 6:30 a.m. to 9 a.m. and from 4:30 p.m. to 6:30 p.m., Monday through Friday, except holidays observed in the locality, the draw need not be opened.

(1553) (b) The draw shall open on signal at all times for self-propelled vessels greater than 10,000 gross tons.

(1554) (c) From noon to 6 p.m. on Thanksgiving Day, from 6 p.m. on December 24 to midnight on December 25, and from 6 p.m. on December 31 to midnight on January 1, the draw shall open on signal after at least a two-hour advance notice is given by calling the number posted at the bridge.

(1555) **NEW JERSEY**

(1556)

§117.702 Arthur Kill

(1557) (a) The draw of the Arthur Kill (AK) Railroad Bridge shall be maintained in the full open position for navigation at all times, except during periods when it is closed for the passage of rail traffic.

(1558) (b) The bridge owner/operator shall maintain a dedicated telephone hot line for vessel operators to call the bridge in advance to coordinate anticipated bridge closures. The telephone hot line number shall be posted on signs at the bridge clearly visible from both the up and downstream sides of the bridge.

(1559) (c) Tide constrained deep draft vessels shall notify the bridge operator, daily, of their expected times of vessel transits through the bridge, by calling the designated telephone hot line.

(1560) (d) The bridge shall not be closed for the passage of rail traffic during any predicted high tide period if a tide constrained deep draft vessel has provided the bridge operator with an advance notice of their intent to transit through the bridge. For the purposes of this regulation, the predicted high tide period shall be considered to be from two hours before each predicted high tide to a half-hour after each predicted high tide taken at the Battery, New York.

(1561) (e) The bridge operator shall issue a manual broadcast notice to mariners of the intent to close the bridge for a period of up to 30 minutes for the passage of rail traffic, on VHF-FM channels 13 and 16 (minimum range of 15 miles) 90 minutes before and again at 75 minutes before each bridge closure.

(1562) (f) Beginning at 60 minutes prior to each bridge closure, automated or manual broadcast notice to mariners must be repeated at 15 minute intervals and again at 10 and 5 minutes prior to each bridge closure and once again as the bridge begins to close, at which point the appropriate sound signal will be given.

(1563) (g) Two 15 minute bridge closures may be provided each day for the passage of multiple rail traffic movements across the bridge. Each 15 minute bridge closure shall be separated by at least a 30 minute period when the bridge is returned to and remains in the full open position. Notification of the two 15 minute closures shall follow the same procedures outlined in paragraphs (e) and (f) above.

(1564) (h) A vessel operator may request up to a 30 minute delay for any bridge closure in order to allow vessel traffic to meet tide or current requirements; however, the request to delay the bridge closure must be made within 30 minutes following the initial broadcast for the bridge closure. Requests received after the initial 30 minute broadcast will not be granted.

(1565) (i) In the event of a bridge operational failure, the bridge operator shall immediately notify the Coast Guard Captain of the Port New York. The bridge owner/operator must provide and dispatch a bridge repair crew to be on scene at the bridge no later than 45 minutes after the bridge fails to operate. A repair crew must remain on scene during the operational failure until the bridge has been fully restored to normal operations or until the bridge is raised and locked in the fully open position.

(1566) (j) When the bridge is not tended locally it must be operated from a remote location. A sufficient number of closed circuit TV cameras, approved by the Coast Guard, shall be operated and maintained at the bridge site to enable the remotely located bridge tender to have full view of both river traffic and the bridge.

(1567) (k) VHF-FM channels 13 and 16 shall be maintained and monitored to facilitate communication in both the remote and local control locations. The bridge shall also be equipped with directional microphones and horns to receive and deliver signals to vessels.

(1568) (l) Whenever the remote control system equipment is disabled or fails to operate for any reason, the bridge operator shall immediately notify the Captain of the Port New York. The bridge shall be physically tended and operated by local control as soon as possible, but no more than 45 minutes after malfunction or disability of the remote system.

(1569) (m) Mechanical bypass and override capability of the remote operation system shall be provided and maintained at all times.

(1570)

§117.709 Cheesequake Creek.

(1571) (a) The draw of the S35 Bridge, at mile 0.0, at Morgan, South Amboy, New Jersey, shall operate as follows:

(1572) (1) From April 1 through November 30 from 7 a.m. to 8 p.m., the draw need only open on the hour. From 8 p.m. to 11 p.m. the draw shall open on signal. From 11 p.m. to 7 a.m. the draw shall open after at least a two hour advance notice is given by calling the number posted at the bridge.

(1573) (2) From December 1 through March 31, the draw shall open on signal after at least a two hour advance notice is given by calling the number posted at the bridge.

(1574) (b) The draw of the New Jersey Transit Rail Operations railroad bridge, mile 0.2, operates as follows:

(1575) (1) The draw shall open on signal; except that, at least four hours notice is required–

(1576) (i) From January 1 through March 31 from 6 p.m. to 6 a.m.;

(1577) (ii) From April 1 through April 30 and November 1 through November 30 from 10 p.m. to 6 a.m. Monday through Thursday, and midnight Sunday through 6 a.m. Monday; and

(1578) (iii) From December 1 through December 31 from 10 p.m. to 6 a.m.

(1579) (2) The owners of the bridge shall provide and keep in good legible condition two board gages painted white with black figures not less than eight inches high to indicate the vertical clearance under the closed draw at all stages of the tide. The gages shall be so placed on the bridge that they are plainly visible to operators of vessels approaching the bridge either up or downstream.

(1580)

§117.718 Elizabeth River.

(1581) The draw of the South Front Street bridge, mile 0.0 at Elizabeth, shall open on signal; except that, from 12 midnight to 7 a.m., the draw shall open on signal if at least three hours notice is given.

(1582)

§117.723 Hackensack River.

(1583) (a) The following requirements apply to all bridges across the Hackensack River:

(1584) (1) The owners of each bridge shall provide and keep in good legible condition clearance gauges for each draw, with figures not less than 18 inches high for bridges below the turning basin at mile 4.0, and 12 inches high for bridges above mile 4.0. The gauges shall be designed, installed and maintained according to the provisions of §118.160 of this chapter.

(1585) (2) Train and locomotives shall be controlled so that any delay in opening the draw shall not exceed 10 minutes. However, if a train moving toward the bridge has crossed the home signal for the bridge before the signal requesting the opening of the bridge is given, the train may continue across the bridge and must clear the bridge interlocks before stopping or reversing.

(1586) (3) New Jersey Transit Rail Operations' (NJTRO) roving crews shall consist of two qualified operators on each shift, each having a vehicle which is equipped with marine and railroad radios, a cellular telephone, and emergency bridge repair and maintenance tools. This crew shall be split with one drawtender stationed at Upper Hack and the other drawtender at the NJTRO HX drawbridge. Adequate security measures shall be provided to prevent vandalism to the bridge operating controls and mechanisms to ensure prompt openings of NJTRO bridges.

(1587) (4) Except as provided in paragraphs (b) through (j) of this section, the draws shall open on signal.

(1588) (b) The draw of the PATH Bridge, mile 3.0, at Jersey City, shall open on signal provided at least a two-hour advance notice is provided by calling the number posted at the bridge. The draw need not open for the passage of vessel traffic Monday through Friday, except Federal holidays, from 6 a.m. to 10 a.m. and from 4 p.m. to 8 p.m. Additional bridge openings shall be provided for commercial vessels from 6 a.m. to 7:20 a.m.; 9:20 a.m. to 10 a.m.; 4 p.m. to 4:30 p.m. and from 6:50 p.m. to 8 p.m. provided at least a two-hour advance notice is given by calling the number posted at the bridge.

(1589) (c) The draw of the Hack-Freight Railroad Bridge at mile 3.1, shall open on signal at all times, except as provided in paragraph (a)(2) of this section. The bridge shall be operated from a remote location at all times, except when it is tended locally. Sufficient closed circuit television cameras, approved by the Coast Guard, shall be operated and maintained at the bridge site to enable the remotely located bridge tender to have full view of both river traffic and the bridge.

(1590) (1) Radiotelephone Channel 13/16 VHF–FM shall be maintained and utilized to facilitate communication in both remote and local control locations. The bridge shall also be equipped with directional microphones and horns to receive and deliver signals to vessels.

(1591) (2) Whenever the remote control system equipment is partially disabled or fails for any reason, the bridge shall be physically tended and operated by local control as soon as possible, but no more than 45 minutes after malfunction or disability of the remote system. Mechanical bypass and override capability of the remote system shall be provided and maintained.

(1592) (d) Except as provided in paragraph (a)(2) of this section, the draw of the NJTRO Lower Hack Bridge, mile 3.4, at Jersey City shall open on signal if at least a one-hour advance notice is given to the drawtender at the Upper Hack bridge, mile 6.9, at Secaucus, New Jersey by calling the number posted at the bridge. In the event the NJTRO HX draw tender is at the Newark/Harrison (Morristown Line) Bridge, mile 5.8, on the Passaic River, up to an additional half hour delay is permitted.

(1593) (e) The draw of the Amtrak Portal Bridge, mile 5.0, at Little Snake Hill, New Jersey, shall only open to 55 feet horizontal clearance in the east channel and the west channel will be closed to all navigation. The draw need not open for the passage of vessel traffic from 5 a.m. to 10 a.m. and from 3 p.m. to 8 p.m. Additional bridge openings shall be provided for tide restricted commercial vessels between 7 a.m. and 8 a.m. and between 5 p.m. and 6 p.m., if at least a two-hour advance notice is given by calling the number posted at the bridge. At all other times the bridge shall open on signal if at least two-hour advance notice is given.

(1594) (f) Except as provided in paragraph (a)(2) of this section, the draw of the NJTRO Upper Hack Bridge, mile 6.9 at Secaucus, N.J. shall open on signal unless the drawtender is at the NJTRO HX Bridge, mile 7.7 at

Secaucus, N.J. over the Hackensack River, then up to a half hour delay is permitted.

(1595) (g) Except as provided in paragraph (a)(2) of this section, the draw of the NJTRO HX Bridge at mile 7.7, shall open on signal if at least a half hour notice is given to the drawtender at the Upper Hack Bridge.

(1596) (h) The draw of the Harold J. Dillard Memorial (Court Street) Bridge, mile 16.2, Hackensack, shall open on signal if at least four hours notice is given.

(1597) (i) The draw of the New York Susquehanna and Western Railroad bridge, mile 16.3, and the Midtown bridge, mile 16.5, both at Hackensack, need not be opened for the passage of vessels, however, the draws shall be restored to operable condition within 12 months after notification by the District Commander to do so.

(1598) (j) The draw of the Route 1 & 9 Bridge, mile 1.8, at Jersey City, shall open on signal; except that, from 11 p.m. to 7 a.m., the draw shall open on signal if at least two hours advance notice is given by calling the number posted at the bridge.

(1599) (k) The draw of the Route 7 Bridge, mile 3.1, at Jersey City, shall open on signal if at least four hours advance notice is given by calling the number posted at the bridge.

(1600)

§117.734 Navesink River (Swimming River).

(1601) The Oceanic Bridge, mile 4.5, shall open on signal; except that, from December 1 through March 31, the draw shall open on signal, if at least a twenty-four hour notice is given by calling the number posted at the bridge. The owner of this bridge shall provide and keep in good legible condition clearance gauges with figures not less than eight inches high, designed, installed and maintained according to the provisions of §118.160 of this chapter.

(1602)

§117.735 Newark Bay.

(1603) The following requirements apply to all bridges across this waterway:

(1604) (a) Public vessels of the United States, state or local vessels used for public service, and vessels in distress shall be passed through the draw without delay. The opening signal from these vessels is four or more short blasts of a whistle or horn or a radio request.

(1605) (b) The owners of these bridges shall provide and keep in good legible condition two board gages painted white with black figures not less than 12 inches high to indicate the vertical clearance under the closed draw at all stages of the tide. The gages shall be so placed on the bridge that they are plainly visible to operators of vessels approaching the bridge either up or downstream.

(1606) (c) Trains and locomotives shall be controlled so that any delay in opening the draw span shall not exceed five minutes. However, if a train moving toward the bridge has crossed the home signal for the bridge before the signal requesting opening of the bridge is given, the train may continue across the bridge and must clear the bridge interlocks before stopping.

(1607)

§117.736 Oceanport Creek.

(1608) The drawspan for the New Jersey Transit Rail Operations Drawbridge, mile 8.4 near Oceanport, must open on signal from May 15 through September 15 between 5 a.m. and 9 p.m.; except that, the drawspan need not open 6 a.m. to 7:45 a.m. and 5:30 p.m. to 7:30 p.m. on weekdays, excluding all federal holidays except for Martin Luther King Day. The drawspan must open on signal upon four hours notice from May 15 through September 15 between 9 p.m. and 5 a.m., and from September 16 through May 14; except that, the drawspan need not be opened from 6 a.m. to 7:45 a.m. and 5:30 p.m. to 7:30 p.m. on weekdays, excluding all federal holidays except for Martin Luther King Day. Public vessels of the United States must be passed as soon as possible at anytime.

(1609)

§117.738 [Removed].

(1610)

§117.739 Passaic River.

(1611) (a) The following requirements apply to all bridges in this section across the Passaic River:

(1612) (1) The owners of these bridges shall provide, and keep in good legible condition, clearance gauges with figures not less than twelve (12) inches high designed, installed and maintained according to the provisions of §118.160 of this chapter.

(1613) (2) New Jersey Transit Rail Operations' (NJTRO) roving crews shall consist of an adequate number of operators to ensure NJTRO bridges are operated according to the requirements of this section.

(1614) (b) The draw of the Routes 1 & 9 (Lincoln Highway) Bridge, mile 1.8, at Newark, shall open on signal if at least four hours notice is given.

(1615) (c) The draw of CONRAIL's Point-No-Point Railroad Bridge, mile 2.6, at Newark, shall open on signal if at least four hours notice is given to the CONRAIL Movement Desk. After the signal to open is given, the opening may be delayed no more than ten minutes.

(1616) (d) The draw of the Jackson Street Bridge, mile 4.6, shall open on signal if at least four hours notice is given by calling the number posted at the bridge.

(1617) (e) The draw of the Bridge Street Bridge, mile 5.6, shall open on signal if at least four hours notice is given by calling the number posted at the bridge.

(1618) (f) The draw of the NJTRO Newark-Harrison (Morristown Line) Bridge, mile 5.8, at Harrison, New Jersey shall open on signal if at least one hour advance notice is given to the drawtender at Upper Hack Bridge mile 6.9, across the Hackensack River at Secaucus, N.J. In the event the HX drawtender is at the Lower Hack Bridge, mile 3.4 on the Hackensack River, at Jersey City then up to an additional half hour delay in opening is permitted. After the signal to open is given, the opening may be delayed no more than ten minutes. From 7:15 a.m. to 9 a.m. and from 4:30 p.m. to 6:50 p.m., Monday

through Friday except federal holidays, the draw need not open.

(1619) (g) The Route 280 Bridge, mile 5.8, at Harrison, New Jersey, shall open on signal if at least 24 hours notice is given by calling the number posted at the bridge.

(1620) (h) The draw of the Clay Street Bridge, mile 6.0, shall open on signal if at least four hours notice is given by calling the number posted at the bridge.

(1621) (i) The draw of the NJTRO (West Arlington) Bridge, mile 8.0, at Kearney, shall open on signal from 7 a.m. to 11 p.m. if at least eight hours notice is given. After the signal to open is given, the opening may be delayed no more than ten minutes. From 11 p.m. to 7 a.m., the draw need not be opened.

(1622) (j) The draw of the Route 7 (Rutgers Street) Bridge, mile 8.9, at Belleville, shall open on signal if at least four hours notice is given.

(1623) (k) The draw of the Avondale Bridge, mile 10.7, at Lyndhurst, shall open on signal if at least four hours notice is given.

(1624) (l) The draw of the NJTRO Bridge, mile 11.7, shall open on signal after at least a 24 hour notice is given by calling the number posted at the bridge.

(1625) (m) West Eighth Street Bridge, mile 15.3, at Garfield need not open for the passage of vessels.

(1626)

§117.743 Rahway River.

(1627) The draw of the Conrail Bridge, mile 2.0, across the Rahway River, at Linden, New Jersey, shall operate as follows:

(1628) (a) The draw shall remain in the full open position at all times, and shall only be closed for the passage of rail traffic or the performance of maintenance authorized in accordance with subpart A of this part.

(1629) (b) The draw shall be remotely operated by a bridge/train dispatcher located at the Conrail Dispatch Office at Mount Laurel, New Jersey.

(1630) (c) A marine traffic light system shall be maintained at the bridge and display flashing green lights to indicate that vessels may pass through the bridge, and flashing red lights anytime the bridge is not in the full open position.

(1631) (d) An infrared sensor system shall be maintained at the bridge to determine that no conflict with vessel traffic exists while the bridge is closing.

(1632) (e) Before the bridge may be closed from the remote location, an on-site train crewmember shall observe the waterway for any vessel traffic. All approaching vessels shall be allowed to pass before the bridge may close. The on-scene train crewmember shall then communicate with the bridge/train dispatcher at the Conrail Dispatch Office, at Mount Laurel, either by radio or telephone, to request the bridge be closed.

(1633) (f) While the bridge is moving from the full open to full closed position, the bridge/train dispatcher shall maintain constant surveillance of the navigational channel at the bridge using the infrared sensor system.

(1634) (g) If the infrared sensors detect a vessel or other obstruction approaching or under the bridge before the draw is fully lowered and locked, the closing sequence shall be stopped, automatically, and the draw shall be raised to its full open position until the channel is clear.

(1635) (h) During the downward bridge closing movement, the marine traffic light system located at the bridge will change from flashing green to flashing red, the public address system shall announce that the bridge shall be closing, and the horn shall sound two times, pause 10 seconds, then repeat two horn blasts until the bridge is seated and fully locked down.

(1636) (i) When all rail traffic has cleared the bridge, the bridge/train dispatcher shall sound the horn five-times to signal that the draw is about to open.

(1637) (j) In the event of a failure, or obstruction to the infrared sensor system, the bridge shall immediately be returned to the full open position until the problem is corrected.

(1638) (k) In the event of a loss of communication between the on-site personnel and the bridge/train dispatcher, the bridge shall immediately be returned to the full open position until the problem is corrected.

(1639) (l) Should the draw become inoperable from the remote site while the bridge is in the closed position, a bridge tender, maintenance personnel, or engineer shall be deployed to be on scene within one hour from the time the draw becomes inoperable until the bridge can be returned to the full open position.

(1640) (m) Trains shall be controlled so that any delay in opening of the draw shall not exceed ten minutes after a train has crossed the bridge; except, as provided in 33 CFR 117.31(b). However, if a train moving toward the bridge has crossed the home signal for the bridge, the train may continue across the bridge and must clear the bridge interlocks before stopping.

(1641)

§117.747 Raritan River

(1642) (a) The draw of New Jersey Transit Rail Operations Railroad Bridge at mile 0.5 shall open on signal; except that, from 6 a.m. to 9:30 a.m. and 4:30 p.m. to 7:30 p.m., Monday through Friday, except holidays, the bridge need not open.

(1643) (b) The bridge owner shall provide and keep in good legible condition two clearance gauges with figures not less than 12 inches high designed, installed and maintained according to the provisions of §118.160 of this chapter.

(1644) (c) Trains and locomotives shall be controlled so that any delay in opening the draw span shall not exceed ten minutes. However, if a train moving toward the bridge has crossed the home signal for the bridge before the signal requesting opening of the bridge is given, the train may continue across the bridge and must clear the bridge interlocks before the bridge may be opened.

(1645)

§117.755 Shrewsbury River.

(1646) The draw of the Sea Bright Bridge highway bridge at mile 4.0, across the Shrewsbury River at Sea Bright, New Jersey, shall operate as follows:

(1647) (a) The draw shall open on signal at all times; except that, from the Friday before Memorial Day through Labor Day, on Friday, Saturday, Sunday and holidays, between 9 a.m. and 7.pm., the draw need only open on the hour.

(1648) (b) The draw need not be opened at any time for a sail boat unless it is operating under auxiliary power or is being towed by powered vessel.

(1649) (c) The owners of the bridge shall keep in good legible condition two clearance gages with figures not less than eight inches high, designed, installed, and maintained according to the provisions of §118.160 of this chapter.

(1650)

§117.756 South River.

(1651) The draw of the Conrail bridge, mile 2.8 at South River shall open on weekdays (exclusive of holidays) from December 1 through the last day of February if at least four hours notice is given. From March 1 through November 30, and December 1 through the last day of February on weekends and holidays the draw shall be maintained open to navigation except for closure to accommodate passage of a train. The draw shall be opened as soon as possible at all times for passage of a public vessel of the United States.

(1652) **NEW YORK**

(1653)

§117.771 Bronx River.

(1654) (a) The draw of the Conrail Bridge, mile 1.6 at the Bronx, New York, need not be opened for the passage of vessels.

(1655) (b) The owners of the Conrail Bridge, mile 1.6, at the Bronx, New York, shall provide and keep in good legible condition two clearance gauges designed, installed and maintained in accordance with the provisions of §118.160 of this chapter.

(1656)

§117.779 [Removed].

(1657)

§117.781 East River.

(1658) The following requirements apply to the Roosevelt Island bridge, mile 6.4 at New York City, as follows:

(1659) (a) Public vessels of the United States Government, state or local vessels used for public safety, and vessels in distress shall be passed through the draws of each bridge as soon as possible without delay at anytime. The opening signal from these vessels shall be four or more short blasts of a whistle, horn or radio request.

(1660) (b) The owners of each bridge shall provide and keep in good legible condition clearance gauges for each draw with figures not less than 12 inches high designed, installed and maintained according to the provisions of §118.160 of these regulations.

(1661) (c) The draw of the Roosevelt Island bridge shall open on signal if at least two hour advance notice is given to the drawtender at the Grand Street/Avenue bridge, mile 3.1 across Newtown Creek (East Branch), the New York Department of Transportation (NYCDOT) Radio Hotline or NYCDOT Bridge Operations Office. In the event the drawtender is at Borden Avenue or Hunters Point Avenue bridges mile 1.2 and 1.4, respectively, across Dutch Kills, up to an additional half hour delay may be required.

(1662)

§117.787 Gowanus Canal.

(1663) The draws of the Ninth Street Bridge, mile 1.4, the Third Street Bridge, mile 1.8, the Carroll Street Bridge, mile 2.0, and the Union Street Bridge, mile 2.1, at Brooklyn, shall open on signal, if at least a two-hour advance notice is given to the New York City Department of Transportation (NYCDOT), Radio Hotline, or the NYCDOT Bridge Operations Office.

(1664)

§117.789 Harlem River.

(1665) (a) The draws of all railroad bridges across the Harlem River may remain in the closed position from the time a train scheduled to cross the bridge is within five minutes from the bridge, and until that train has fully crossed the bridge. The maximum time permitted for delay shall not exceed ten (10) minutes. Land and water traffic should pass over or through the draw as soon as possible to prevent unnecessary delays in the opening and closure of the draw.

(1666) (b)(1) The draws of the bridges at 103 Street, mile 0.0, 125 Street (Triborough), mile 1.3, Willis Avenue, mile 1.5, Third Avenue, mile 1.9, Madison Avenue, mile 2.3, 145 Street, mile 2.8 Macombs Dam, mile 3.2, 207 Street, mile 6.0, and the Broadway Bridge, mile 6.8 shall open on signal if at least a four-hour advance notice is given to New York City Highway Radio (Hotline) Room and the Triborough Bridge and Tunnel Authority (TBTA) for the 125 Street (Triborough), mile 1.3. The draws of the above bridges, except the Broadway Bridge, need not open for the passage of vessel traffic from 6 a.m. to 9 a.m. and 5 p.m. to 7 p.m., Monday through Friday, except federal holidays. The draw of the Broadway Bridge need not open for the passage of vessel traffic from 7 a.m. to 10 a.m. and 4 p.m. to 7 p.m., Monday through Friday, except federal holidays.

(1667) (2) The draws of the Willis Avenue Bridge, mile 1.5, Third Avenue Bridge, mile 1.9, and the Madsion Avenue Bridge, mile 2.3, need not open for the passage of vessel traffic at various times between 8 a.m. and 5 p.m. on the first Sunday in May and November. The exact time and date of each bridge closure will be published in the Local Notice to Mariners several weeks prior to each closure.

(1668) (c) The draw of the Metro North (Park Avenue) Bridge, mile 2.1, shall open on signal, except, as provided in paragraph (a) of this section, if a least a four-hour

advance notice is given. The draw need not open for the passage of vessel traffic from 5 a.m. to 10 a.m. and 4 p.m. to 8 p.m., Monday through Friday, except Federal holidays.

(1669) (d) The draw of the Spuyten Duyvil railroad bridge, mile 7.9, shall open on signal at all times, except as provided in paragraph (a) of this section.

(1670)

§117.791 Hudson River.

(1671) (a) The draws of the bridges listed in this section shall open as soon as possible at any time for the passage of the following vessels:

(1672) (1) Downbound vessels during a freshet of a height exceeding an elevation determined by the District Commander.

(1673) (2) Public vessels of the United States.

(1674) (3) Vessels of 500 tons or more.

(1675) (4) Tugs with a tow on a hawser.

(1676) (b) The draws of the bridges listed in this section shall not remain open for more than 15 minutes and may remain closed for up to 10 minutes to allow accumulated land traffic to pass.

(1677) (c) The draw of the Livingston Ave. (Amtrak) Bridge, mile 146.2 between Albany and Rensselaer, shall open on signal; except that, from April 1 through December 15, from 11:00 p.m. to 7:00 a.m., the draw shall open on signal if at least 4 hours notice is given and from December 16 through March 31, the draw shall open on signal if at least 24 hours notice is given.

(1678) (d) The draw of the Troy-Menands Bridge, mile 150.2 between Troy and Menands, need not be opened for the passage of vessels.

(1679) (e) The draw of the Troy-Green Bridge, mile 152.7, between Troy and Green Island, operates as follows:

(1680) (1) From April 1 through December 15 the draw shall open on signal if at least a twenty four hour advance notice is given by calling the number posted at the bridge.

(1681) (2) From December 16 through March 31, the draw need not open for the passage of vessel traffic.

(1682)

§117.793 Hutchinson River (Eastchester Creek).

(1683) (a) The following requirements apply to all bridges across Hutchinson River (Eastchester Creek);

(1684) (1) The owners of each bridge shall provide and keep in good legible condition clearance gauges for each draw with figures not less than 12 inches high designed, installed and maintained according to the provision of §118.160 of this chapter.

(1685) (2) Trains and locomotives shall be controlled so that any delay in opening the draw shall not exceed ten minutes except as provided in §117.31(b). However, if a train moving toward the bridge has crossed the home signal for the bridge before the signal requesting opening of the bridge is given, the train may continue across the bridge and must clear the bridge interlocks before stopping.

(1686) (3) Except as provided in paragraphs (b) and (c) of this section each draw shall open on signal.

(1687) (b) The draw of the Hutchinson River Parkway Bridge, mile 0.9, at the Bronx, New York shall open on signal if at least a two-hour notice is given to the New York City Department of Transportation (NYCDOT) Radio Hotline, or the NYCDOT Bridge Operations Office.

(1688) (c) The draw of the South Fulton Avenue Bridge, mile 2.9, shall open on signal from three hours before to three hours after the predicted high tide. For the purposes of this section, predicted high tide occurs four hours after predicted high water for New York (Battery), as given in the tide tables published by private entities using data provided by the National Ocean Service.

(1689) (1) At all other times, the bridge shall open on signal if at least four hours advance notice is given to the Westchester County Road Maintenance Division during normal work hours or to the County's Parkway Police at all other times.

(1690) (2) The bridge tender shall honor requests for opening within six hours after predicated high water if such request is given to the bridge tender while he or she is on station (three hours before to three hours after predicted high tide).

(1691)

§117.795 Jamaica Bay and Connecting Waterways.

(1692) (a) The draw of the Marine Parkway bridge, mile 3.0 over Rockaway Inlet, shall open on signal Monday through Friday from 8 a.m., to 4 p.m. At all other times, the draw shall open on signal if at least eight hours notice is given: however, the draw shall open on signal if at least a one hour notice is given for the passage of U.S. Navy or National Oceanic and Atmospheric Administration vessels.

(1693) (b) The draw of the Beach Channel railroad bridge shall open on signal; except that, the draw need not open for the passage of vessel traffic, 6:45 a.m. to 8:20 a.m. and 5 p.m. to 6:45 p.m., Monday through Friday, except Federal holidays.

(1694)

§117.799 Long Island, New York Inland Waterway from East Rockaway Inlet to Shinnecock Canal.

(1695) (a) At all times, public vessels of the United States must be passed through the drawspan of each drawbridge listed in this section as soon as possible.

(1696) (b) The draw of each bridge listed in this section need not be opened for sailing vessels, unless the vessels are under machinery power or under tow, if an opening would unduly delay other vessel or vehicular traffic.

(1697) (c) The owners of the bridges listed in this section shall provide and keep in good legible condition two board gages painted white with black figures not less than eight inches high to indicate the vertical clearance under the closed draw at all stages of the tide. The gages shall be so placed on the bridges that they are plainly visible

to operators of vessels approaching the bridges either up or downstream.

(1698) (d) The draws of the West Bay Bridge, mile 0.1, across Quantuck Canal, Beach Lane Bridge, mile 1.1, across Quantuck Canal, Quoque Bridge, mile 1.1, across Quoque Canal, and the Smith Point Bridge, mile 6.1, across Narrow Bay, shall open on signal, from October 1 through April 30, from 8 a.m. to 4 p.m., and from May 1 through September 30, from 6 a.m. to 10 p.m. At all other times during these periods, the draws shall open as soon as possible but not more than one hour after a request to open is received.

(1699) (e) The draw of the Atlantic Beach Bridge across Reynolds Channel, mile 0.4, shall operate as follows:

(1700) (1) From October 1 through May 14 the draw shall open on signal from 8 a.m. to midnight.

(1701) (2) From midnight to 8 a.m. year-round, the draw shall open on signal if at least eight (8) hours of notice is given by calling the Bridge Tower at 516–239–1821.

(1702) (3) From May 15 through September 30, the bridge will open on signal except from 4 p.m. to 7 p.m. on weekdays, and from 11 a.m. to 9 p.m. on Saturdays, Sundays, Memorial Day, Independence Day, and Labor Day, when the bridge will open on the hour and half-hour.

(1703) (f) The draw of the Loop Parkway Bridge across Long Creek, mile 0.7, shall open for commercial vessels engaged in commerce, the draw shall open Monday through Friday from 6:20 a.m. to 9:50 a.m. and 3:20 p.m. to 7:20 p.m. on signal at 20 and 50 minutes after the hour, and on signal at all other times. For all other vessels, the draw shall open on Monday thru Friday from 6:20 a.m. to 7:20 p.m. on signal at 20 and 50 minutes after the hour, and the draw shall open on Saturday, Sunday and Federal Holidays from 7:20 a.m. to 8:20 p.m. on signal at 20 and 50 minutes after the hour, andon signal at all other times.

(1704) (g) The draw of the Long Beach Bridge across Reynolds Channel, mile 4.7, shall open on signal; except that:

(1705) (1) From midnight to 8 a.m. year-round, the draw shall open on signal if at least four hours notice is given; and

(1706) (2) From 3 p.m. to 8 p.m. on Saturdays, Sundays, and holidays from May 15 through September 30, the draw need be opened only on the hour and the half hour.

(1707) (3) From 10 p.m. to midnight on July 3 each year the draw need not open for the passage of vessel traffic.

(1708) (h) The draw of the Meadowbrook State Parkway Bridge, mile 12.8, across Sloop Channel, shall open on signal if at least one-half hour notice is given to the New York State Department of Transportation, as follows:

(1709) (1) Every other hour on the even hour.

(1710) (2) From April 1 through October 31, on Saturday, Sundays, and Federal holidays, every three hours beginning at 1:30 a.m. Notice may be given from the telephone located at the moorings on each side of the bridge or by marine radio.

(1711) (3) From 9 p.m. to midnight, on the Fourth of July, the Meadowbrook State Parkway Bridge need not open for the passage of vessel traffic.

(1712) (i) The draw of the Captree State Parkway bridge at mile 30.7, across State Boat Channel, at Captree Island, shall open on signal if at least one half hour advance notice is given by calling the number posted at the bridge as follows:

(1713) (1) Every other hour on the even hour.

(1714) (2) From April 1 through October 31, on Saturday, Sunday, and Federal holidays every three hours beginning at 3 a.m.

(1715)

§117.800 Mill Neck Creek.

(1716) The draw of the Bayville Bridge, mile 0.1, at Oyster Bay, New York, shall open on signal between 7 a.m. and 11 p.m., from May 1 through October 31, and between 7 a.m. and 5 p.m., Monday through Friday, from November 1 through April 30. At all other times the draw shall open on signal provided at least a two-hour advance notice is given by calling the number posted at the bridge.

(1717)

§117.801 Newtown Creek, Dutch Kills, English Kills and their tributaries.

(1718) (a) The following requirements apply to all bridges across Newtown Creek, Dutch Kills, English Kills, and their tributaries:

(1719) (1) The owners of all bridges across Newtown Creek, Dutch Kills, English Kills and their tributaries listed under this section, shall provide and keep in good legible condition two clearance gauges with figures not less than 12 inches high designed, installed and maintained according to the provisions of §118.160 of this chapter.

(1720) (2) Trains and locomotives shall be controlled so that any delay in opening the draw shall not exceed five minutes. If a train moving toward the bridge has crossed the home signal for the bridge before the request to open the bridge is given, that train may continue across the bridge, but must clear the interlock before stopping.

(1721) (b) The draws of the Long Island Railroad bridges, at mile 1.1, across Dutch Kills at Queens, shall open on signal if at least six-hours advance notice is given to the Long Island Railroad Movement Bureau, except as provided in paragraph (a)(2) of this section.

(1722) (c) The draw of the Borden Avenue Bridge, mile 1.2, across Dutch Kills at Queens, shall open on signal if at least a two-hour advance notice is given to the New York City Department of Transportation (NYCDOT) Radio Hotline or NYCDOT Bridge Operations Office.

(1723) (d) The draw of the Hunters Point Avenue Bridge, mile 1.4, across Dutch Kills at Queens, shall open on signal if at least a two-hour advance notice is given to the New York City Department of Transportation (NYCDOT) Radio Hotline or the NYCDOT Bridge Operations Office.

(1724) (e) The draw of the Metropolitan Avenue Bridge, mile 3.4, across English Kills at New York City, shall open on signal if at least a two-hour advance notice is given to the New York City Department of Transportation (NYCDOT) Radio Hotline or the NYCDOT Bridge Operations Office.

(1725) (f) The draw of the Grand Street/Avenue Bridge, mile 3.1, across Newtown Creek (East Branch) between Brooklyn and Queens, shall open on signal if at least a two-hour advance notice is given to the New York City Department of Transportation (NYCDOT) Radio Hotline or the NYCDOT Bridge Operations Office.

(1726) (g)(1) The draw of the Pulaski Bridge, mile 0.6, and the Greenpoint Avenue Bridge, mile 1.3, shall open on signal if at least a two hour advance notice is given to the New York City Department of Transportation Radio (Hotline) Room.

(1727) (2) The Pulaski Bridge, mile 0.6, need not open for vessel traffic at various times between 8 a.m. and 5 p.m. on the first Sunday in both May and November. The exact time and date of the bridge closure will be published in the Local Notice to Mariners several weeks prior to the first Sunday of both May and November.

(1728)

§117.805 [Removed].

(1729)

§117.813 [Removed and Reserved].

(1730)

§117.815 Westchester Creek.

(1731) The draw of the Bruckner Boulevard/Unionport Bridge, mile 1.7, at the Bronx, New York, shall open on signal if at least a two-hour advance notice is given to the New York City Department of Transportation (NYCDOT) radio hotline, or the NYCDOT Bridge Operations Office. The draw need not be opened for vessel traffic from 7 a.m. to 9 a.m. and 4 p.m. to 6 p.m., Monday through Friday. The owner of the bridge shall provide clearance gauges according to the provisions of §118.160 of this chapter.

(1732)

Part 157–Rules for the Protection of the Marine Environment relating to Tank Vessels Carrying Oil in Bulk.

(1733)

Subpart A–General

(1734)

§157.01 Applicability.

(1735) (a) Unless otherwise indicated, this part applies to each vessel that carries oil in bulk as cargo and that is:

(1736) (1) Documented under the laws of the United States (a U.S. vessel); or

(1737) (2) Any other vessel that enters or operates in the navigable waters of the United States, or that operates, conducts lightering under 46 U.S.C. 3715, or receives cargo from or transfers cargo to a deepwater port under 33 U.S.C. 1501 *et seq*., in the United States Exclusive Economic Zone, as defined in 33 U.S.C. 2701(8).

(1738) (b) This part does not apply to a vessel exempted under 46 U.S.C. 2109 or 46 U.S.C. 3702.

(1739)

§157.02 Incorporation by reference: Where can I get a copy of the publications mentioned in this part?

(1740) (a) Certain material is incorporated by reference into this part with the approval of the Director of the Federal Register under 5 U.S.C. 552(a) and 1 CFR part 51. To enforce any edition other than that specified in this section, the Coast Guard must publish notice of change in the **Federal Register** and the material must be available to the public. All approved material is available for inspection at the National Archives and Records Administration (NARA). For information on the availability of this material at NARA, call 202–741–6030 or go to *http://www.archives.gov/federal_register/code_of_federal_regulations/ibr_locations.html*. Also, it is available for inspection at the Coast Guard Headquarters. Contact Commandant (CG–ENG), Attn: Office of Design and Engineering Standards, U.S. Coast Guard Stop 7509, 2703 Martin Luther King Jr. Avenue SE., Washington, DC 20593–7509; telephone 202–372–1375. The material is also available from the sources indicated in this section.

(1741) *(b) International Maritime Organization (IMO)*—4 Albert Embankment, London SE1 7SR, United Kingdom.

(1742) (1) IMCO Assembly Resolution A.393(X), adopted on 14 November 1977, Recommendation on International Performance and Test Specifications For Oily Water Separating Equipment and Oil Content Meters ("A.393(x)"), incorporation by reference approved for §157.12.

(1743) (2) IMO Assembly Resolution A.496(XII), Adopted on 19 November 1981, Agenda Item 11, Guidelines and Specifications for Oil Discharge Monitoring and Control Systems for Oil Tankers ("A.496(XII)"), incorporation by reference approved for §157.12.

(1744) (3) IMO Assembly Resolution A.586(14), Adopted on 20 November 1985, Agenda item 12, Revised Guidelines and Specifications for Oil Discharge Monitoring and Control Systems for Oil Tankers ("A.586(14)"), incorporation by reference approved for §157.12.

(1745) (4) IMO Marine Environment Protection Committee Resolution MEPC.13 (19), adopted on 9 December 1983, Guidelines for Plan Approval and Installation Survey of Oil Discharge Monitoring and Control Systems for Oil Tankers and Environmental Testing of Control Sections Thereof ("MEPC.13(19)"), incorporation by reference approved for §157.12.

(1746) (5) IMO Marine Environment Protection Committee Resolution MEPC.108(49), Adopted on 18 July 2003, Revised Guidelines and Specifications for Oil Discharge Monitoring and Control Systems for Oil Tankers

("MEPC.108(49)"), incorporation by reference approved for §157.12.

(1747) (6) IMO Assembly Resolution A.601(15), Provision and Display of Manoeuvring Information on Board Ships, Annex sections 1.1, 2.3, 3.1, and 3.2 with appendices, adopted on 19 November 1987 ("A.601(15)"), incorporation by reference approved for §157.450.

(1748) (7) IMO Assembly Resolution A.744(18), Guidelines on the Enhanced Programme of Inspections During Surveys of Bulk Carriers and Oil Tankers, Annex B sections 1.1.3-1.1.4, 1.2-1.3, 2.1, 2.3-2.6, 3-8, and Annexes 1-10 with appendices, adopted 4 November 1993 ("A.744(18)"), incorporation by reference approved for §157.430.

(1749) (8) IMO Assembly Resolution A.751(18), Interim Standards for Ship Manoeuvrability, Annex sections 1.2, 2.3-2.4, 3-4.2, and 5, adopted 4 November 1993 with Explanatory Notes in MSC/Circ. 644 dated 6 June 1994 ("A.751(18)"), incorporation by reference approved for §157.445.

(1750) (9) MARPOL Consolidated Edition 2011, Annex I, Regulations for the prevention of pollution by oil, Chapter 4—Requirements for the cargo area of oil tankers, Part A—Construction, Regulation 22, "Pump-room bottom protection," (Annex I, Regulation 22) incorporation by reference approved for §157.14.

(1751) (10) MARPOL Consolidated Edition 2011, Annex I, Regulations for the prevention of pollution by oil, Chapter 4—Requirements for the cargo area of oil tankers, Part A—Construction, Regulation 23, "Accidental oil outflow performance," (Annex I, Regulation 23) incorporation by reference approved for §157.20.

(1752) (c) *Oil Companies International Marine Forum (OCIMF)* 27 Queen Anne's Gate, London, SW1H 9BU, England].

(1753) (1) International Safety Guide for Oil Tankers and Terminals, Fourth Edition, Chapters 6, 7, and 10, 1996, incorporation by reference approved for §157.435.

(1754) (2) [Reserved]

(1755)

§157.03 Definitions.

(1756) Except as otherwise stated in a subpart:

(1757) *Amidships* means the middle of the length.

(1758) *Animal fat* means a non-petroleum oil, fat, or grease derived from animals and not specifically identified elsewhere in this part.

(1759) *Ballast voyage* means the voyage that a tank vessel engages in after it leaves the port of final cargo discharge.

(1760) *Breadth* or *B* means the maximum molded breadth of a vessel in meters.

(1761) *Cargo tank length* means the length from the forward bulkhead of the forwardmost cargo tanks, to the after bulkhead of the aftermost cargo tanks.

(1762) *Center tank* means any tank inboard of a longitudinal bulkhead.

(1763) *Clean ballast* means ballast which:

(1764) (1) If discharged from a vessel that is stationary into clean, calm water on a clear day, would not–

(1765) (i) Produce visible traces of oil on the surface of the water or on adjoining shore lines; or

(1766) (ii) Cause a sludge or emulsion to be deposited beneath the surface of the water or upon adjoining shore lines; or

(1767) (2) If verified by an approved oil discharge monitoring and control system, has an oil content that does not exceed 15 p.m.

(1768) *Combination carrier* means a vessel designed to carry oil or solid cargoes in bulk.

(1769) *Crude oil* means any liquid hydrocarbon mixture occurring naturally in the earth, whether or not treated to render it suitable for transportation, and includes crude oil from which certain distillate fractions may have been removed, and crude oil to which certain distillate fractions may have been added.

(1770) *Deadweight* or *DWT* means the difference in metric tons between the lightweight displacement and the total displacement of a vessel measured in water of specific gravity 1.025 at the load waterline corresponding to the assigned summer freeboard.

(1771) *Dedicated clean ballast tank* means a cargo tank that is allocated solely for the carriage of clean ballast.

(1772) *Domestic trade* means trade between ports or places within the United States, its territories and possessions, either directly or via a foreign port including trade on the navigable rivers, lakes, and inland waters.

(1773) *Double bottom* means watertight protective spaces that do not carry any oil and which separate the bottom of tanks that hold any oil within the cargo tank length from the outer skin of the vessel.

(1774) *Double hull* means watertight protective spaces that do not carry any oil and which separate the sides, bottom, forward end, and aft end of tanks that hold any oil within the cargo tank length from the outer skin of the vessel as prescribed in §157.10d.

(1775) *Doubles sides* means watertight protective spaces that do not carry any oil and which separate the sides of tanks that hold any oil within the cargo tank length from the outer skin of the vessel.

(1776) *Existing vessel* means any vessel that is not a new vessel.

(1777) *Fleeting or assist towing vessel* means any commercial vessel engaged in towing astern, alongside, or pushing ahead, used solely within a limited geographic area, such as a particular barge fleeting area or commercial facility, and used solely for restricted service, such as making up or breaking up larger tows.

(1778) *Foreign trade* means any trade that is not domestic trade.

(1779) *From the nearest land* means from the baseline from which the territorial sea of the United States is established in accordance with international law.

(1780) *Fuel oil* means any oil used as fuel for machinery in the vessel in which it is carried.

(1781) *Inland vessel* means a vessel that is not oceangoing and that does not operate on the Great Lakes.

(1782) *Instantaneous rate of discharge of oil content* means the rate of discharge of oil in liters per hour at any instant, divided by the speed of the vessel in knots at the same instant.

(1783) *Integrated tug barge* means a tug and a tank barge with a mechanical system that allows the connection of the propulsion unit (the tug) to the stern of the cargo carrying unit (the tank barge) so that the two vessels function as a single self-propelled vessel.

(1784) *Large primary structural member* includes any of the following:

(1785) (1) Web frames.

(1786) (2) Girders.

(1787) (3) Webs.

(1788) (4) Main brackets.

(1789) (5) Transverses.

(1790) (6) Stringers.

(1791) (7) Struts in transverse web frames when there are 3 or more struts and the depth of each is more than 1/15 of the total depth of the tank.

(1792) *Length* or *L* means the distance in meters from the fore side of the stem to the axis of the rudder stock on a waterline at 85 percent of the least molded depth measured from the molded baseline, or 96 percent of the total length on that waterline, whichever is greater. In vessels designed with drag, the waterline is measured parallel to the designed waterline.

(1793) *Lightweight* means the displacement of a vessel in metric tons without cargo, fuel oil, lubricating oil, ballast water, fresh water, and feedwater in tanks, consumable stores, and any persons and their effects.

(1794) *Major conversion* means a conversion of an existing vessel that:

(1795) (1) Substantially alters the dimensions or carrying capacity of the vessel, except a conversion that includes only the installation of segregated ballast tanks, dedicated clean ballast tanks, a crude oil washing system, double sides, a double bottom, or a double hull;

(1796) (2) Changes the type of vessel;

(1797) (3) Substantially prolongs the vessel's service life; or

(1798) (4) Otherwise so changes the vessel that it is essentially a new vessel, as determined by the Commandant (CG–CVC).

(1799) *MARPOL 73/78* means the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocol of 1978 relating to that Convention. A copy of MARPOL 73/78 is available from the International Maritime Organization, 4 Albert Embankment, London, SE1, 7SR, England.

(1800) *New vessel* means:

(1801) (1) A U.S. vessel in domestic trade that:

(1802) (i) Is constructed under a contract awarded after December 31, 1974;

(1803) (ii) In the absence of a building contract, has the keel laid or is at a similar stage of construction after June 30, 1975;

(1804) (iii) Is delivered after December 31, 1977; or

(1805) (iv) Has undergone a major conversion for which:

(1806) (A) The contract is awarded after December 31, 1974;

(1807) (B) In the absence of a contract, conversion is begun after June 30, 1975; or

(1808) (C) Conversion is completed after December 31, 1977; and

(1809) (2) A foreign vessel or a U.S. vessel in foreign trade that:

(1810) (i) Is constructed under a contract awarded after December 31, 1975;

(1811) (ii) In the absence of a building contract, has the keel laid or is at a similar stage of construction after June 30, 1976;

(1812) (iii) Is delivered after December 31, 1979; or

(1813) (iv) Has undergone a major conversion for which:

(1814) (A) The contract is awarded after December 31, 1975;

(1815) (B) In the absence of a contract, conversion is begun after June 30, 1976; or

(1816) (C) Conversion is completed after December 31, 1979.

(1817) *Non-petroleum oil* means oil of any kind that is not petroleum-based. It includes, but is not limited to, animal fat and vegetable oil.

(1818) *Oceangoing* has the same meaning as defined in §151.05 of this chapter.

(1819) *Officer in charge of a navigational watch* means any officer employed or engaged to be responsible for navigating or maneuvering the vessel and for maintaining a continuous vigilant watch during his or her periods of duty and following guidance set out by the master, international or national regulations, and company policies.

(1820) *Oil* means oil of any kind or in any form including, but not limited to, petroleum, fuel oil, sludge, oil refuse, and oil mixed with wastes other than dredged spoil. This includes liquid hydrocarbons as well as animal and vegetable oils.

(1821) *Oil cargo residue* means any residue of oil cargo whether in solid, semi-solid, emulsified, or liquid form from cargo tanks and cargo pump room bilges, including but not limited to, drainages, leakages, exhausted oil, muck, clingage, sludge, bottoms, paraffin (wax), and any constituent component of oil. The term "oil cargo residue" is also known as "cargo oil residue."

(1822) *Oil residue* means–

(1823) (1) Oil cargo residue; and

(1824) (2) Other residue of oil whether in solid, semi-solid, emulsified, or liquid form resulting from drainages, leakages, exhausted oil and other similar occurrences from machinery spaces.

(1825) *Oil spill response vessel* means a vessel that is exclusively dedicated to operations to prevent or mitigate

environmental damage due to an actual or impending accidental oil spill. This includes a vessel that performs routine service as an escort for a tank vessel, but excludes a vessel that engages in any other commercial activity, such as the carriage of any type of cargo.

(1826) *Oil tanker* means a vessel that is constructed or adapted primarily to carry crude oil or products in bulk as cargo. This includes a tank barge, a tankship, and a combination carrier, as well as a vessel that is constructed or adapted primarily to carry noxious liquid substances in bulk as cargo and which also carries crude oil or products in bulk as cargo.

(1827) Oily mixture means a mixture, in any form, with any oil content. "Oily mixture" includes, but is not limited to—

(1828) (1) Slops from bilges;

(1829) (2) Slops from oil cargoes (such as cargo tank washings, oily waste, and oily refuse);

(1830) (3) Oil residue; and

(1831) (4) Oily ballast water from cargo or fuel oil tanks, including any oil cargo residue.

(1832) *Other non-petroleum oil* means an oil of any kind that is not petroleum oil, an animal fat, or a vegetable oil.

(1833) *Permeability of a space* means the ratio of volume within a space that is assumed to be occupied by water to the total volume of that space.

(1834) *Petroleum oil* means petroleum in any form, including but not limited to, crude oil, fuel oil, sludge, oil residue, and refined products.

(1835) *Primary towing vessel* means any vessel engaged in towing astern, alongside, or pushing ahead and includes the tug in an integrated tug barge. It does not include fleeting or assist towing vessels.

(1836) *Product* means any liquid hydrocarbon mixture in any form, except crude oil, petrochemicals, and liquefied gases.

(1837) *Segregated ballast* means the ballast water introduced into a tank that is completely separated from the cargo oil and fuel oil system and that is permanently allocated to the carriage of ballast.

(1838) *Slop tank* means a tank specifically designated for the collection of cargo drainings, washings, and other oily mixtures.

(1839) *Tank* means an enclosed space that is formed by the permanent structure of a vessel, and designed for the carriage of liquid in bulk.

(1840) *Tank barge* means a tank vessel not equipped with a means of self-propulsion.

(1841) *Tank vessel* means a vessel that is constructed or adapted primarily to carry, or that carries, oil or hazardous material in bulk as cargo or cargo residue, and that–

(1842) (1) Is a vessel of the United States;

(1843) (2) Operates on the navigable waters of the United States; or

(1844) (3) Transfers oil or hazardous material in a port or place subject to the jurisdiction of the United States. This does not include an offshore supply vessel, or a fishing vessel or fish tender vessel of not more than 750 gross tons when engaged only in the fishing industry.

(1845) *Tankship* means a tank vessel propelled by mechanical power or sail.

(1846) *Vegetable oil* means a non-petroleum oil or fat not specifically identified elsewhere in this part that is derived from plant seeds, nuts, kernels, or fruits.

(1847) *Wing tank* means a tank that is located adjacent to the side shell plating.

(1848)

§157.04 Authorization of classification societies.

(1849) (a) The Coast Guard may authorize any classification society (CS) to perform certain plan reviews, certifications, and inspections required by this part on vessels classed by that CS, except that only U.S. classification societies may be authorized to perform those plan reviews, inspections, and certifications for U.S. vessels.

(1850) (b) If a CS desires authorization to perform the plan reviews, certifications, and inspections required under this part, it must submit to the Commandant (CG–CVC), Attn: Office of Commercial Vessel Compliance, U.S. Coast Guard Stop 7501, 2703 Martin Luther King Jr. Avenue SE., Washington, DC 20593–7501, evidence from the governments concerned showing that they have authorized the CS to inspect and certify vessels on their behalf under the MARPOL 73/78.

(1851) (c) The Coast Guard notifies the CS in writing whether or not it is accepted as an authorized CS. If authorization is refused, reasons for the refusal are included.

(1852) (d) Acceptance as an authorized CS terminates unless the following are met:

(1853) (1) The authorized CS must have each Coast Guard regulation that is applicable to foreign vessels on the navigable waters of the United States.

(1854) (2) Each issue concerning equivalents to the regulations in this part must be referred to the Coast Guard for determination.

(1855) (3) Copies of any plans, calculations, records of inspections, or other documents relating to any plan review, inspection, or certification performed to meet this part must be made available to the Coast Guard.

(1856) (4) Each document certified under §§157.116(a)(2), 157.118(b)(1)(ii), and 157.216(b)(1)(ii) must be marked with the name or seal of the authorized CS.

(1857) (5) A copy of the final documentation that is issued to each vessel that is certified under this part must be referred to the Commandant (CG-CVC), Attn: Office of Commercial Vessel Compliance, U.S. Coast Guard Stop 7501, 2703 Martin Luther King Jr. Avenue SE., Washington, DC 20593–7501.

(1858)

Subpart B–Design, Equipment, and Installation

(1859)

§157.08 Applicability of Subpart B.

(1860) **NOTE:** An "oil tanker" as defined in §157.03 includes barges as well as self-propelled vessels.

(1861) (a) Sections 157.10d and 157.11(g) apply to each vessel to which this part applies.

(1862) (b) Sections 157.11 (a) through (f), 157.12, 157.15, 157.19(b)(3), 157.33, and 157.37 apply to each vessel to which this part applies that carries 200 cubic meters or more of crude oil or products in bulk as cargo, as well as to each oceangoing oil tanker to which this part applies of 150 gross tons or more. These sections do not apply to a foreign vessel which remains beyond the navigable waters of the United States and does not transfer oil cargo at a port or place subject to the jurisdiction of the United States.

(1863) (c) Section 157.21 applies to each oil tanker to which this part applies of 150 gross tons or more that is oceangoing or that operates on the Great Lakes. This section does not apply to a foreign vessel which remains beyond the navigable waters of the United States and does not transfer oil cargo at a port or place subject to the jurisdiction of the United States.

(1864) (d) Sections in subpart B of 33 CFR part 157 that are not specified in paragraphs (a) through (c) of this section apply to each oceangoing oil tanker to which this part applies of 150 gross tons or more, unless otherwise indicated in paragraphs (e) through (m) of this section. These sections do not apply to a foreign vessel which remains beyond the navigable waters of the United States and does not transfer oil cargo at a port or place subject to the jurisdiction of the United States.

(1865) (e) Sections 157.11 (a) through (f), 157.12, and 157.15 do not apply to a vessel, except an oil tanker, that carries less than 1,000 cubic meters of crude oil or products in bulk as cargo and which retains oil mixtures on board and discharges them to a reception facility.

(1866) (f) Sections 157.11 (a) through (f), 157.12, 157.13, and 157.15 do not apply to a tank vessel that carries only asphalt, carbon black feedstock, or other products with similar physical properties, such as specific gravity and cohesive and adhesive characteristics, that inhibit effective product/water separation and monitoring.

(1867) (g) Sections 157.11 (a) through (f), 157.12, 157.13, 157.15, and 157.23 do not apply to a tank barge that cannot ballast cargo tanks or wash cargo tanks while underway.

(1868) (h) Sections 157.19 and 157.21 do not apply to a tank barge that is certificated by the Coast Guard for limited short protected coastwise routes if the barge is otherwise constructed and certificated for service exclusively on inland routes.

(1869) (i) Section 157.09(d) does not apply to any:

(1870) (1) U.S. vessel in domestic trade that is constructed under a contract awarded before January 8, 1976;

(1871) (2) U.S. vessel in foreign trade that is constructed under a contract awarded before April 1, 1977; or

(1872) (3) Foreign vessel that is constructed under a contract awarded before April 1, 1977.

(1873) (j) Sections 157.09 and 157.10a do not apply to a new vessel that:

(1874) (1) Is constructed under a building contract awarded after June 1, 1979;

(1875) (2) In the absence of a building contract, has the keel laid or is at a similar stage of construction after January 1, 1980;

(1876) (3) Is delivered after June 1, 1982; or

(1877) (4) Has undergone a major conversion for which:

(1878) (i) The contract is awarded after June 1, 1979;

(1879) (ii) In the absence of a contract, conversion is begun after January 1, 1980; or

(1880) (iii) Conversion is completed after June 1, 1982.

(1881) (k) Sections 157.09(b)(3), 157.10(c)(3), 157.10a(d)(3), and 157.10b(b)(3) do not apply to tank barges.

(1882) (1) Section 157.10b does not apply to tank barges if they do not carry ballast while they are engaged in trade involving the transfer of crude oil from an offshore oil exploitation or production facility on the Outer Continental Shelf of the United States.

(1883) (m) Section 157.12 does not apply to a U.S. vessel that:

(1884) (1) Is granted an exemption under Subpart F of this part; or

(1885) (2) Is engaged solely in voyages that are:

(1886) (i) Between ports or places within the United States, its territories or possessions;

(1887) (ii) Of less than 72 hours in length; and

(1888) (iii) At all times within 50 nautical miles of the nearest land.

(1889) (n) Section 157.10d does not apply to:

(1890) (1) A vessel that operates exclusively beyond the navigable waters of the United States and the United States Exclusive Economic Zone, as defined in 33 U.S.C. 2701(8);

(1891) (2) An oil spill response vessel;

(1892) (3) Before January 1, 2015–

(1893) (i) A vessel unloading oil in bulk as cargo at a deepwater port licensed under the Deepwater Port Act of 1974 (33 U.S.C. 1501 *et seq*.); or

(1894) (ii) A delivering vessel that is offloading oil in bulk as cargo in lightering activities–

(1895) (A) Within a lightering zone established under 46 U.S.C. 3715(b)(5); and

(1896) (B) More than 60 miles from the territorial sea base line, as defined in 33 CFR 2.20.

(1897) (4) A vessel documented under 46 U.S.C., Chapter 121, that was equipped with a double hull before August 12, 1992;

(1898) (5) A barge of less than 1,500 gross tons as measured under 46 U.S.C., Chapter 145, carrying refined petroleum in bulk as cargo in or adjacent to waters of the Bering

Sea, Chukchi Sea, and Arctic Ocean and waters tributary thereto and in the waters of the Aleutian Islands and the Alaskan Peninsula west of 155 degrees west longitude; or

(1899) (6) A vessel in the National Defense Reserve Fleet pursuant to 50 App. U.S.C. 1744.

(1900) (o) Section 157.11(h) applies to every oil tanker delivered on or after January 1, 2010, meaning an oil tanker—

(1901) (1) For which the building contract is placed on or after January 1, 2007;

(1902) (2) In the absence of a building contract, the keel of which is laid or which is at a similar stage of construction on or after July 1, 2007;

(1903) (3) The delivery of which is on or after January 1, 2010; or

(1904) (4) That has undergone a major conversion—

(1905) (i) For which the contract is placed on or after January 1, 2007;

(1906) (ii) In the absence of a contract, the construction work of which is begun on or after July 1, 2007; or

(1907) (iii) That is completed on or after January 1, 2010.

(1908)

§157.10d Double hulls on tank vessels.

(1909) (a) With the exceptions stated in §157.08(n), this section applies to a tank vessel–

(1910) (1) For which the building contract is awarded after June 30, 1990; or

(1911) (2) That is delivered after December 31, 1993;

(1912) (3) That undergoes a major conversion for which;

(1913) (i) The contract is awarded after June 30, 1990; or

(1914) (ii) Conversion is completed after December 31, 1993; or

(1915) (4) That is otherwise required to have a double hull by 46 U.S.C. 3703a(c).

(1916) **NOTE:** The double hull compliance dates of 46 U.S.C. 3703a(c) are set out in appendix G to this part. To determine a tank vessel's double hull compliance date under OPA 90, use the vessel's hull configuration (*i.e.*, single hull; single hull with double sides; or single hull with double bottom) on August 18, 1990.

(1917) (b) Each vessel to which this section applies must be fitted with:

(1918) (1) A double hull in accordance with this section; and

(1919) (2) If §157.10 applies, segregated ballast tanks and a crude oil washing system in accordance with that section.

(1920) (c) Except on a vessel to which §157.10d(d) applies, tanks within the cargo tank length that carry any oil must be protected by double sides and a double bottom as follows:

(1921) (1) Double sides must extend for the full depth of the vessel's side or from the uppermost deck, disregarding a rounded gunwale where fitted, to the top of the double bottom. At any cross section, the molded width of the double side, measured at right angles to the side shell plating, from the side of tanks containing oil to the side shell plating, must not be less than the distance w as shown in Figure 157.10d(c) and specified as follows:

(1922) (i) For a vessel of 5,000 DWT and above: w=[0.5+(DWT/20,000)] meters; or, w=2.0 meters (79 in.), whichever is less, but in no case less than 1.0 meter (39 in.).

(1923) (ii) For a vessel of less than 5,000 DWT: w=[0.4+(2.4)(DWT/20,000)] meters, but in no case less than 0.76 meter (30 in.).

(1924) (iii) For a vessel to which Paragraph (a)(4) of this section applies: w=0.76 meter (30 in.), provided that the double side was fitted under a construction or conversion contract awarded prior to June 30, 1990.

(1926) (2) At any cross section, the molded depth of the double bottom, measured at right angles to the bottom shell plating, from the bottom of tanks containing oil to the bottom shell plating, must not be less than the distance h as shown in Figure 157.10d(c) and specified as follows:

(1927) (i) For a vessel of 5,000 DWT and above: h=B/15;

(1925)

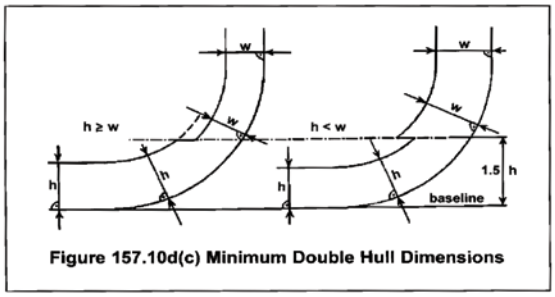


Figure 157.10d(c) Minimum Double Hull Dimensions

or, h=2.0 meters (79 in.), whichever is less, but in no case less than 1.0 meter (39 in.).

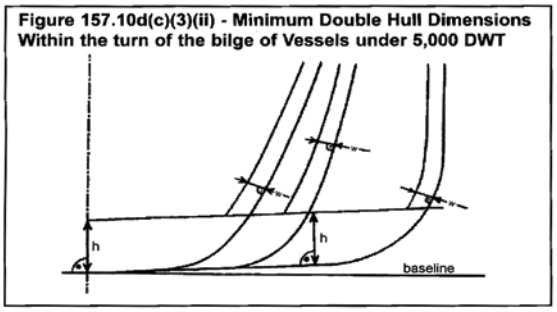
(1928) (ii) For a vessel of less than 5,000 DWT: h=B/15, but in no case less than 0.76 meter (30 in.).

(1929) (iii) For a vessel to which Paragraph (a)(4) of this section applies: h=B/15; or, h=2.0 meters (79 in.), whichever is the lesser, but in no case less than 0.76 meter (30 in.), provided that the double bottom was fitted under a construction or conversion contract awarded prior to June 30, 1990.

(1930) (3) For a vessel built under a contract awarded after September 11, 1992, within the turn of the bilge or at cross sections where the turn of the bilge is not clearly defined, tanks containing oil must be located inboard of the outer shell–

(1931) (i) For a vessel of 5,000 DWT and above: At levels up to 1.5 h above the base line, not less than the distance h,

(1933)



as shown in Figure 157.10d(c) and specified in Paragraph (c)(2) of this section. At levels greater than 1.5 *h* above the base line, not less than the distance *w*, as shown in Figure 157.10d(c) and specified in Paragraph (c)(1) of this section.

(1932) (ii) For a vessel of less than 5,000 DWT: Not less than the distance *h* above the line of the mid-ship flat bottom, as shown in Figure 157.10d(c)(3)(ii) and specified in Paragraph (c)(2) of this section. At levels greater than *h* above the line of the mid-ship flat bottom, not less than the distance *w*, as shown in Figure 157.10d(c)(3)(ii) and specified in Paragraph (c)(1) of this section.

(1934) (4) For a vessel to which §157.10(b) applies that is built under a contract awarded after September 11, 1992.

(1935) (i) The aggregate volume of the double sides, double bottom, forepeak tanks, and afterpeak tanks must not be less than the capacity of segregated ballast tanks required under §157.10(b). Segregated ballast tanks that may be provided in addition to those required under §157.10(b) may be located anywhere within the vessel.

(1936) (ii) Double side and double bottom tanks used to meet the requirements of §157.10(b) must be located as uniformly as practicable along the cargo tank length. Large inboard extensions of individual double side and double bottom tanks, which result in a reduction of overall side or bottom protection, must be avoided.

(1937) (d) A vessel of less than 10,000 DWT that is constructed and certificated for service exclusively on inland or limited short protected coastwise routes must be fitted with double sides and a double bottom as follows:

(1938) (1) A minimum of 61 cm. (2 ft.) from the inboard side of the side shell plate, extending the full depth of the side or from the main deck to the top of the double bottom, measured at right angles to the side shell; and

(1939) (2) A minimum of 61 cm. (2 ft.) from the top of the bottom shell plating, along the full breadth of the vessel's bottom, measured at right angles to the bottom shell.

(1940) (3) For a vessel to which Paragraph (a)(4) of this section applies, the width of the double sides and the depth of the double bottom may be 38 cm. (15 in.), in lieu of the dimensions specified in paragraphs (d)(1) and (d)(2) of this section, provided that the double side and double bottom tanks were fitted under a construction or conversion contract awarded prior to June 30, 1990.

(1941) (4) For a vessel built under a contract awarded after September 11, 1992, a minimum 46 cm. (18 in.) clearance for passage between framing must be maintained throughout the double sides and double bottom.

(1942) (e) Except as provided in Paragraph (e)(3) of this section, a vessel must not carry any oil in any tank extending forward of:

(1943) (1) The collision bulkhead; or

(1944) (2) In the absence of a collision bulk-head, the transverse plane perpendicular to the centerline through a point located:

(1945) (i) The lesser of 10 meters (32.8 ft.) or 5 percent of the vessel length, but in no case less than 1 meter (39 in.), aft of the forward perpendicular;

(1946) (ii) On a vessel of less than 10,000 DWT tons that is constructed and certificated for service exclusively on inland or limited short protected coastwise routes, the lesser of 7.62 meters (25 ft.) or 5 percent of the vessel length, but in no case less than 61 cm. (2 ft.), aft of the headlog or stem at the freeboard deck; or

(1947) (iii) On each vessel which operates exclusively as a box or trail barge, 61 cm. (2 ft.) aft of the headlog.

(1948) (3) This Paragraph does not apply to independent fuel oil tanks that must be located on or above the main deck within the areas described in paragraphs (e)(1) and (e)(2) of this section to serve adjacent deck equipment that cannot be located further aft. Such tanks must be as small and as far aft as is practicable.

(1949) (f) On each vessel, the cargo tank length must not extend aft to any point closer to the stern than the distance equal to the required width of the double side, as prescribed in §157.10d(c)(1) or §157.10d(d)(1).

(1950)

Subpart G–Interim Measures for Certain Tank Vessels Without Double Hulls Carrying Petroleum Oils

(1951)

§157.400 Purpose and applicability.

(1952) (a) The purpose of this subpart is to establish mandatory safety and operational requirements to reduce environmental damage resulting from petroleum oil spills.

(1953) (b) This subpart applies to each tank vessel specified in §157.01 of this part that–

(1954) (1) Is 5,000 gross tons or more;

(1955) (2) Carries petroleum oil in bulk as cargo or oil cargo residue; and

(1956) (3) Is not equipped with a double hull meeting §157.10d of this part, or an equivalent to the requirements of §157.10d, but required to be equipped with a double hull at a date set forth in 46 U.S.C. 3703a (b)(3) and (c)(3).

(1957)

§157.445 Maneuvering performance capability.

(1958) (a) A tankship owner or operator shall ensure that maneuvering tests in accordance with IMO Resolution A.751(18), sections 1.2, 2.3-2.4, 3-4.2, and 5 (with Explanatory Notes in MSC/Circ. 644) have been conducted by July 29, 1997. Completion of maneuvering performance tests must be shown by–

(1959) (1) For a foreign flag tankship, a letter from the flag administration or an authorized classification society, as described in §157.04 of this part, stating the requirements in paragraph (a) of this section have been met; or

(1960) (2) For a U.S. flag tankship, results from the vessel owner confirming the completion of the tests or a letter

from an authorized classification society, as described in §157.04 of this part, stating the requirements in paragraph (a) of this section have been met.

(1961) (b) If a tankship undergoes a major conversion or alteration affecting the control systems, control surfaces, propulsion system, or other areas which may be expected to alter maneuvering performance, the tankship owner or operator shall ensure that new maneuvering tests are conducted as required by paragraph (a) of this section.

(1962) (c) If a tankship is one of a class of vessels with identical propulsion, steering, hydrodynamic, and other relevant design characteristics, maneuvering performance test results for any tankship in the class may be used to satisfy the requirements of paragraph (a) of this section.

(1963) (d) The tankship owner or operator shall ensure that the performance test results, recorded in the format of Appendix 6 of the Explanatory Notes in MSC/Circ. 644., are prominently displayed in the wheelhouse.

(1964) (e) Prior to entering the port or place of destination and prior to getting underway, the tankship master shall discuss the results of the performance tests with the pilot while reviewing the anticipated transit and the possible impact of the tankship's maneuvering capability on the transit.

(1965)

Part 160–Ports and Waterways Safety-General

(1966)

Subpart A–General:

(1967)

§160.1 Purpose.

(1968) Part 160 contains regulations implementing the Ports and Waterways Safety Act (33 U.S.C. 1221) and related statutes.

(1969)

§160.3 Definitions.

(1970) For the purposes of this subchapter:

(1971) *Bulk* means material in any quantity that is shipped, stored, or handled without the benefit of package, label, mark or count and carried in integral or fixed independent tanks.

(1972) *Captain of the Port* means the Coast Guard officer designated by the Commandant to command a Captain of the Port Zone as described in part 3 of this chapter.

(1973) *Commandant* means the Commandant of the United States Coast Guard.

(1974) *Deviation* means any departure from any rule in this subchapter.

(1975) *Director, Vessel Traffic Services* means the Coast Guard officer designated by the Commandant to command a Vessel Traffic Service (VTS) as described in part 161 of this chapter.

(1976) *District Commander* means the Coast Guard officer designated by the Commandant to command a Coast Guard District as described in part 3 of this chapter.

(1977) *ETA* means estimated time of arrival.

(1978) *Length of Tow* means, when towing with a hawser, the length in feet from the stern of the towing vessel to the stern of the last vessel in tow. When pushing ahead or towing alongside, length of tow means the tandem length in feet of the vessels in tow excluding the length of the towing vessel.

(1979) *Person* means an individual, firm, corporation, association, partnership, or governmental entity.

(1980) *State* means each of the several States of the United States, the District of Columbia, the Commonwealth of Puerto Rico, Guam, American Samoa, the United States Virgin Islands, the Trust Territories of the Pacific Islands, the Commonwealth of the Northern Marianas Islands, and any other commonwealth, territory, or possession of the United States.

(1981) *Tanker* means a self-propelled tank vessel constructed or adapted primarily to carry oil or hazardous materials in bulk in the cargo spaces.

(1982) *Tank Vessel* means a vessel that is constructed or adapted to carry, or that carries, oil or hazardous material in bulk as cargo or cargo residue.

(1983) *Vehicle* means every type of conveyance capable of being used as a means of transportation on land.

(1984) *Vessel* means every description of watercraft or other artificial contrivance used, or capable of being used, as a means of transportation on water.

(1985) *Vessel Traffic Services (VTS)* means a service implemented under part 161 of this chapter by the United States Coast Guard designed to improve the safety and efficiency of vessel traffic and to protect the environment. The VTS has the capability to interact with marine traffic and respond to traffic situations developing in the VTS area.

(1986) *Vessel Traffic Service Area* or *VTS Area* means the geographical area encompassing a specific VTS area of service as described in part 161 of this chapter. This area of service may be subdivided into sectors for the purpose of allocating responsibility to individual Vessel Traffic Centers or to identify different operating requirements.

(1987) **Note:** Although regulatory jurisdiction is limited to the navigable waters of the United States, certain vessels will be encouraged or may be required, as a condition of port entry, to report beyond this area to facilitate traffic management within the VTS area.

(1988) *VTS Special Area* means a waterway within a VTS area in which special operating requirements apply.

(1989)

§160.5 Delegations.

(1990) (a) District Commanders and Captains of the Ports are delegated the authority to establish safety zones.

(1991) (b) Under the provisions of 33 CFR 6.04-1 and 6.04-6, District Commanders and Captains of the Ports have been delegated authority to establish security zones.

(1992) (c) Under the provisions of 33 CFR 1.05-1, District Commanders have been delegated authority to establish regulated navigation areas.

(1993) (d) Subject to the supervision of the cognizant Captain of the Port and District Commander, Diretors, Vessel Traffic Services are delegated authority under 33 CFR 1.01-30 to discharge the duties of the Captain of the Port that involve directing the operation, movement and anchorage of vessels within a Vessel Traffic Service area including management of vessel traffic within anchorages, regulated navigation areas and safety zones, and to enforce Vessel Traffic Service and ports and waterways safety regulations. This authority may be exercised by Vessel Traffic Center personnel. The Vessel Traffic Center may, within the Vessel Traffic Service area, provide information, make recommendations, or to a vessel required under Part 161 of this chapter to participate in a Vessel Traffic Service, issue an order, including an order to operate or anchor as directed; require the vessel to comply with orders issued; specify times of entry, movement or departure; restrict operations as necessary for safe operation under the circumstances; or take other action necessary for control of the vessel and the safety of the port or of the marine environment.

(1994)

§160.7 Appeals.

(1995) (a) Any person directly affected by a safety zone or an order or direction issued under this subchapter may request reconsideration by the official who issued it or in whose name it was issued. This request may be made orally or in writing, and the decision of the official receiving the request may be rendered orally or in writing.

(1996) (b) Any person directly affected by the establishment of a safety zone or by an order or direction issued by, or on behalf of, a Captain of the Port may appeal to the District Commander through the Captain of the Port. The appeal must be in writing, except as allowed under paragraph (e) of this section, and shall contain complete supporting documentation and evidence which the appellant wishes to have considered. Upon receipt of the appeal, the District Commander may direct a representative to gather and submit documentation or other evidence which would be necessary or helpful to a resolution of the appeal. A copy of this documentation and evidence is made available to the appellant. The appellant is afforded five working days from the date of receipt to submit rebuttal materials. Following submission of all materials, the District Commander issues a ruling, in writing, on the appeal. Prior to issuing the ruling, the District Commander may, as a matter of discretion, allow oral presentation on the issues.

(1997) (c) Any person directly affected by the establishment of a safety zone or by an order or direction issued by, or on behalf of, a District Commander, or who receives an unfavorable ruling on an appeal taken under paragraph (b) of this section may appeal to the Area Commander through the District Commander. The appeal must be in writing, except as allowed under paragraph (e) of this section, and shall contain complete supporting documentation and evidence which the appellant wishes to have considered. Upon receipt of the appeal, the Area Commander may direct a representative to gather and submit documentation or other evidence which would be necessary or helpful to a resolution of the appeal. A copy of this documentation and evidence is made available to the appellant. The appellant is afforded five working days from the date of receipt to submit rebuttal materials. Following submission of all materials, the Area Commander issues a ruling, in writing, on the appeal. Prior to issuing the ruling, the Area Commander may, as a matter of discretion, allow oral presentation on the issues.

(1998) (d) Any person who receives an unfavorable ruling on an appeal taken under paragraph (c) of this section, may appeal to the Commandant (CG–5P), Attn: Assistant Commandant for Prevention, U.S. Coast Guard Stop 7501, 2703 Martin Luther King Jr. Avenue SE., Washington, DC 20593–7501. The appeal must be in writing, except as allowed under paragraph (e) of this section. The Area Commander forwards the appeal, all the documents and evidence which formed the record upon which the order or direction was issued or the ruling under paragraph (c) of this section was made, and any comments which might be relevant, to the Assistant Commandant for Prevention. A copy of this documentation and evidence is made available to the appellant. The appellant is afforded 5 working days from the date of receipt to submit rebuttal materials to the Assistant Commandant for Prevention. The decision of the Assistant Commandant for Prevention is based upon the materials submitted, without oral argument or presentation. The decision of the Assistant Commandant for Prevention is issued in writing and constitutes final agency action.

(1999) (e) If the delay in presenting a written appeal would have significant adverse impact on the appellant, the appeal under paragraphs (b) and (c) of this section may initially be presented orally. If an initial presentation of the appeal is made orally, the appellant must submit the appeal in writing within five days of the oral presentation to the Coast Guard official to whom the presentation was made. The written appeal must contain, at a minimum, the basis for the appeal and a summary of the material presented orally. If requested, the official to whom the appeal is directed may stay the effect of the action while the ruling is being appealed.

(2000)

Subpart B–Control of Vessel and Facility Operations

(2001)

§160.101 Purpose.

(2002) This subpart describes the authority exercised by District Commanders and Captains of the Ports to insure the safety of vessels and waterfront facilities, and the protection of the navigable waters and the resources therein. The controls described in this subpart are directed to specific situations and hazards.

(2003)

§160.103 Applicability.

(2004) (a) This subpart applies to any–

(2005) (1) Vessel on the navigable waters of the United States, except as provided in paragraphs (b) and (c) of this section;

(2006) (2) Bridge or other structure on or in the navigable waters of the United States; and

(2007) (3) Land structure or shore area immediately adjacent to the navigable waters of the United States.

(2008) (b) This subpart does not apply to any vessel on the Saint Lawrence Seaway.

(2009) (c) Except pursuant to international treaty, convention, or agreement, to which the United States is a party, this subpart does not apply to any foreign vessel that is not destined for, or departing from, a port or place subject to the jurisdiction of the United States and that is in–

(2010) (1) Innocent passage through the territorial sea of the United States;

(2011) (2) Transit through the navigable waters of the United States which form a part of an international strait.

(2012)

§160.105 Compliance with orders.

(2013) Each person who has notice of the terms of an order issued under this subpart must comply with that order.

(2014)

§160.107 Denial of entry.

(2015) Each District Commander or Captain of the Port, subject to recognized principles of international law, may deny entry into the navigable waters of the United States or to any port or place under the jurisdiction of the United States, and within the district or zone of that District Commander or Captain of the Port, to any vessel not in compliance with the provisions of the Port and Tanker Safety Act (33 U.S.C. 1221-1232) or the regulations issued thereunder.

(2016)

§160.109 Waterfront facility safety.

(2017) (a) To prevent damage to, or destruction of, any bridge or other structure on or in the navigable waters of the United States, or any land structure or shore area immediately adjacent to those waters, and to protect the navigable waters and the resources therein from harm resulting from vessel or structure damage, destruction, or loss, each District Commander or Captain of the Port may–

(2018) (1) Direct the handling, loading, unloading, storage, and movement (including the emergency removal, control, and disposition) of explosives or other dangerous articles and substances, including oil or hazardous material as those terms are defined in 46 U.S.C. 2101 on any structure on or in the navigable waters of the United States, or any land structure or shore area immediately adjacent to those waters; and

(2019) (2) Conduct examinations to assure compliance with the safety equipment requirements for structures.

(2020)

§160.111 Special orders applying to vessel operations.

(2021) Each District Commander or Captain of the Port may order a vessel to operate or anchor in the manner directed when–

(2022) (a) The District Commander or Captain of the Port has reasonable cause to believe that the vessel is not in compliance with any regulation, law or treaty;

(2023) (b) The District Commander or Captain of the Port determines that the vessel does not satisfy the conditions for vessel operation and cargo transfers specified in §160.113; or

(2024) (c) The District Commander or Captain of the Port has determined that such order is justified in the interest of safety by reason of weather, visibility, sea conditions, temporary port congestion, other temporary hazardous circumstances, or the condition of the vessel.

(2025)

§160.113 Prohibition of vessel operation and cargo transfers.

(2026) (a) Each District Commander or Captain of the Port may prohibit any vessel subject to the provisions of chapter 37 of Title 46, U.S. Code, from operating in the navigable waters of the United States, or from transferring cargo or residue in any port or place under the jurisdiction of the United States, and within the district or zone of that District Commander or Captain of the Port, if the District Commander or the Captain of the Port determines that the vessel's history of accidents, pollution incidents, or serious repair problems creates reason to believe that the vessel may be unsafe or pose a threat to the marine environment.

(2027) (b) The authority to issue orders prohibiting operation of the vessels or transfer of cargo or residue under paragraph (a) of this section also applies if the vessel:

(2028) (1) Fails to comply with any applicable regulation;

(2029) (2) Discharges oil or hazardous material in violation of any law or treaty of the United States;

(2030) (3) Does not comply with applicable vessel traffic service requirements;

(2031) (4) While underway, does not have at least one deck officer on the navigation bridge who is capable of communicating in the English language.

(2032) (c) When a vessel has been prohibited from operating in the navigable waters of the United States under paragraphs (a) or (b) of this section, the District Commander or Captain of the Port may allow provisional entry into the navigable waters of the United States, or into any port or place under the jurisdiction of the United States and within the district or zone of that District Commander or Captain of the Port, if the owner or operator of such vessel proves to the satisfaction of the District Commander or Captain of the Port, that the vessel is not unsafe or does not pose a threat to the marine

environment, and that such entry is necessary for the safety of the vessel or the persons on board.

(2033) (d) A vessel which has been prohibited from operating in the navigable waters of the United States, or from transferring cargo or residue in a port or place under the jurisdiction of the United States under the provisions of paragraph (a) or (b)(1), (2) or (3) of this section, may be allowed provisional entry if the owner or operator proves, to the satisfaction of the District Commander or Captain of the Port that has jurisdiction, that the vessel is no longer unsafe or a threat to the environment, and that the condition which gave rise to the prohibition no longer exists.

(2034)

§160.115 Withholding of clearance.

(2035) (a) Each District Commander or Captain of the Port may request the Secretary of the Treasury, or the authorized representative thereof, to withhold or revoke the clearance required by 46 U.S.C. App. 91 of any vessel, the owner or operator of which is subject to any penalties under 33 U.S.C. 1232.

(2036)

Subpart C–Notification of Arrival, Hazardous Conditions, and Certain Dangerous Cargoes

(2037)

§160.201 General.

(2038) This subpart contains requirements and procedures for submitting a notice of arrival (NOA), and a notice of hazardous condition. The sections in this subpart describe:

(2039) (a) Applicability and exemptions from requirements in this subpart;

(2040) (b) Required information in an NOA;

(2041) (c) Required updates to an NOA;

(2042) (d) Methods and times for submission of an NOA, and updates to an NOA;

(2043) (e) How to obtain a waiver; and

(2044) (f) Requirements for submission of the notice of hazardous condition.

(2045) **Note to §160.201.** For notice-of-arrival requirements for the U.S. Outer Continental Shelf, see 33 CFR part 146.

(2046)

§160.202 Definitions.

(2047) Terms in this subpart that are not defined in this section or in §160.3 have the same meaning as those terms in 46 U.S.C. 2101. As used in this subpart—

(2048) *Agent* means any person, partnership, firm, company or corporation engaged by the owner or charterer of a vessel to act in their behalf in matters concerning the vessel.

(2049) *Barge* means a non-self propelled vessel engaged in commerce.

(2050) *Boundary waters* mean the waters from main shore to main shore of the lakes and rivers and connecting waterways, or the portions thereof, along which the international boundary between the United States and the Dominion of Canada passes, including all bays, arms, and inlets thereof, but not including tributary waters which in their natural channels would flow into such lakes, rivers, and waterways, or waters flowing from such lakes, rivers, and waterways, or the waters of rivers flowing across the boundary.

(2051) *Carried in bulk* means a commodity that is loaded or carried on board a vessel without containers or labels and received and handled without mark or count.

(2052) *Certain dangerous cargo (CDC)* includes any of the following:

(2053) (1) Division 1.1 or 1.2 explosives as defined in 49 CFR 173.50.

(2054) (2) Division 1.5D blasting agents for which a permit is required under 49 CFR 176.415 or, for which a permit is required as a condition of a Research and Special Programs Administration exemption.

(2055) (3) Division 2.3 "poisonous gas", as listed in 49 CFR 172.101 that is also a "material poisonous by inhalation" as defined in 49 CFR 171.8, and that is in a quantity in excess of 1 metric ton per vessel.

(2056) (4) Division 5.1 oxidizing materials for which a permit is required under 49 CFR 176.415 or for which a permit is required as a condition of a Research and Special Programs Administration exemption.

(2057) (5) A liquid material that has a primary or subsidiary classification of Division 6.1 "poisonous material" as listed in 49 CFR 172.101 that is also a "material poisonous by inhalation," as defined in 49 CFR 171.8 and that is in a bulk packaging, or that is in a quantity in excess of 20 metric tons per vessel when not in a bulk packaging.

(2058) (6) Class 7, "highway route controlled quantity" radioactive material or "fissile material, controlled shipment," as defined in 49 CFR 173.403.

(2059) (7) All bulk liquefied gas cargo carried under 46 CFR 151.50–31 or listed in 46 CFR 154.7 that is flammable and/or toxic and that is not carried as certain dangerous cargo residue (CDC residue).

(2060) (8) The following bulk liquids except when carried as CDC residue:

(2061) (i) Acetone cyanohydrin;

(2062) (ii) Allyl alcohol;

(2063) (iii) Chlorosulfonic acid;

(2064) (iv) Crotonaldehyde;

(2065) (v) Ethylene chlorohydrin;

(2066) (vi) Ethylene dibromide;

(2067) (vii) Methacrylonitrile;

(2068) (viii) Oleum (fuming sulfuric acid); and

(2069) (ix) Propylene oxide, alone or mixed with ethylene oxide.

(2070) (9) The following bulk solids:

(2071) (i) Ammonium nitrate listed as a Division 5.1 (oxidizing) material in 49 CFR 172.101 except when carried as CDC residue; and

(2072) (ii) Ammonium nitrate based fertilizer listed as a Division 5.1 (oxidizing) material in 49 CFR 172.101 except when carried as CDC residue.

(2073) *Certain dangerous cargo residue (CDC residue)* includes any of the following:

(2074) (1) Ammonium nitrate in bulk or ammonium nitrate based fertilizer in bulk remaining after all saleable cargo is discharged, not exceeding 1,000 pounds in total and not individually accumulated in quantities exceeding two cubic feet.

(2075) (2) For bulk liquids and liquefied gases, the cargo that remains onboard in a cargo system after discharge that is not accessible through normal transfer procedures, with the exception of the following bulk liquefied gas cargoes carried under 46 CFR 151.50–31 or listed in 46 CFR 154.7:

(2076) (i) Ammonia, anhydrous;

(2077) (ii) Chlorine;

(2078) (iii) Ethane;

(2079) (iv) Ethylene oxide;

(2080) (v) Methane (LNG);

(2081) (vi) Methyl bromide;

(2082) (vii) Sulfur dioxide; and

(2083) (viii) Vinyl chloride.

(2084) *Charterer* means the person or organization that contracts for the majority of the carrying capacity of a ship for the transportation of cargo to a stated port for a specified period. This includes "time charterers" and "voyage charterers."

(2085) *Crewmember* means all persons carried on board the vessel to provide navigation and maintenance of the vessel, its machinery, systems, and arrangements essential for propulsion and safe navigation or to provide services for other persons on board.

(2086) *Embark* means when a crewmember or a person in addition to the crew joins the vessel.

(2087) *Ferry schedule* means a published document that:

(2088) (1) Identifies locations a ferry travels to and from;

(2089) (2) Lists the times of departures and arrivals; and

(2090) (3) Identifies the portion of the year in which the ferry maintains this schedule.

(2091) *Foreign vessel* means a vessel of foreign registry or operated under the authority of a country except the United States.

(2092) *Great Lakes* means Lakes Superior, Michigan, Huron, Erie, and Ontario, their connecting and tributary waters, the Saint Lawrence River as far as Saint Regis, and adjacent port areas.

(2093) *Gross tons* means the tonnage determined by the tonnage authorities of a vessel's flag state in accordance with the national tonnage rules in force before the entry into force of the International Convention on Tonnage Measurement of Ships, 1969 ("Convention"). For a vessel measured only under Annex I of the Convention, gross tons means that tonnage. For a vessel measured under both systems, the higher gross tonnage is the tonnage used for the purposes of the 300-gross-ton threshold.

(2094) *Hazardous condition* means any condition that may adversely affect the safety of any vessel, bridge, structure, or shore area or the environmental quality of any port, harbor, or navigable waterway of the United States. It may, but need not, involve collision, allision, fire, explosion, grounding, leaking, damage, injury or illness of a person aboard, or manning-shortage.

(2095) *Nationality* means the state (nation) in which a person is a citizen or to which a person owes permanent allegiance.

(2096) *Operating exclusively within a single Captain of the Port zone* refers to vessel movements within the boundaries of a single COTP zone, e.g., from one dock to another, one berth to another, one anchorage to another, or any combination of such transits. Once a vessel has arrived in a port in a COPT zone, it would not be considered as departing from a port or place simply because of its movements within that specific port.

(2097) *Operator* means any person including, but not limited to, an owner, a charterer, or another contractor who conducts, or is responsible for, the operation of a vessel.

(2098) *Persons in addition to crewmembers* mean any person onboard the vessel, including passengers, who are not included on the list of crewmembers.

(2099) *Port or place of departure* means any port or place in which a vessel is anchored or moored.

(2100) *Port or place of destination* means any port or place in which a vessel is bound to anchor or moor.

(2101) *Public vessel* means a vessel that is owned or demise-(bareboat) chartered by the government of the United States, by a State or local government, or by the government of a foreign country and that is not engaged in commercial service.

(2102) *Time charterer* means the party who hires a vessel for a specific amount of time. The owner and his crew manage the vessel, but the charterer selects the ports of destination.

(2103) *Voyage charterer* means the party who hires a vessel for a single voyage. The owner and his crew manage the vessel, but the charterer selects the ports of destination.

(2104)

§160.203 Applicability.

(2105) (a) This subpart applies to the following vessels that are bound for or departing from ports or places within the navigable waters of the United States, as defined in 33 CFR 2.36(a), which includes internal waters and the territorial seas of the United States, and any deepwater port as defined in 33 CFR 148.5:

(2106) (1) U.S. vessels in commercial service, and

(2107) (2) All foreign vessels.

(2108) (b) Unless otherwise specified in this subpart, the owner, agent, master, operator, or person in charge of a vessel regulated by this subpart is responsible for compliance with the requirements in this subpart.

(2109) (c) Towing vessels controlling a barge or barges required to submit an NOA under this subpart must submit only one NOA containing the information required for the towing vessel and each barge under its control.

(2110)

§160.204 Exemptions and exceptions.

(2111) (a) Except for reporting notice of hazardous conditions, the following vessels are exempt from requirements in this subpart:

(2112) (1) A passenger or offshore supply vessel when employed in the exploration for or in the removal of oil, gas, or mineral resources on the continental shelf.

(2113) (2) An oil spill response vessel (OSRV) when engaged in actual spill response operations or during spill response exercises.

(2114) (3) After December 31, 2015, a vessel required by 33 CFR 165.830 or 165.921 to report its movements, its cargo, or the cargo in barges it is towing.

(2115) (4) A United States or Canadian vessel engaged in the salving operations of any property wrecked, or rendering aid and assistance to any vessels wrecked, disabled, or in distress, in waters specified in Article II of the 1908 Treaty of Extradition, Wrecking and Salvage (35 Stat. 2035; Treaty Series 502).

(2116) (5) The following vessels neither carrying certain dangerous cargo nor controlling another vessel carrying certain dangerous cargo:

(2117) (i) A foreign vessel 300 gross tons or less not engaged in commercial service.

(2118) (ii) A vessel operating exclusively within a single Captain of the Port zone. Captain of the Port zones are defined in 33 CFR part 3.

(2119) (iii) A U.S. towing vessel and a U.S. barge operating solely between ports or places of the contiguous 48 states, Alaska, and the District of Columbia.

(2120) (iv) A public vessel.

(2121) (v) Except for a tank vessel, a U.S. vessel operating solely between ports or places of the United States on the Great Lakes.

(2122) (vi) A U.S. vessel 300 gross tons or less, engaged in commercial service not coming from a foreign port or place.

(2123) (vii) Each ferry on a fixed route that is described in an accurate schedule that is submitted by the ferry operator, along with information in paragraphs (a)(5)(vii)(A) through (J) of this section, to the Captain of the Port for each port or place of destination listed in the schedule at least 24 hours in advance of the first date and time of arrival listed on the schedule. At least 24 hours before the first date and time of arrival listed on the ferry schedule, each ferry operator who submits a schedule under paragraph (a)(5)(vii) of this section must also provide the following information to the Captain of the Port for each port or place of destination listed in the schedule for the ferry, and if the schedule or the following submitted information changes, the ferry operator must submit an updated schedule at least 24 hours in advance of the first date and time of arrival listed on the new schedule and updates on the following items whenever the submitted information is no longer accurate:

(2124) (A) Name of the vessel;

(2125) (B) Country of registry of the vessel;

(2126) (C) Call sign of the vessel;

(2127) (D) International Maritime Organization (IMO) international number or, if the vessel does not have an assigned IMO international number, the official number of the vessel;

(2128) (E) Name of the registered owner of the vessel;

(2129) (F) Name of the operator of the vessel;

(2130) (G) Name of the vessel's classification society or recognized organization, if applicable;

(2131) (H) Each port or place of destination;

(2132) (I) Estimated dates and times of arrivals at and departures from these ports or places; and

(2133) (J) Name and telephone number of a 24-hour point of contact.

(2134) (b) A vessel less than 500 gross tons is not required to submit the International Safety Management (ISM) Code Notice (Entry 7 in Table 160.206 of §160.206).

(2135) (c) A U.S. vessel is not required to submit the International Ship and Port Facility Security (ISPS) Code Notice information (Entry 8 in Table 160.206 of §160.206).

(2136)

§160.205 Notices of arrival.

(2137) The owner, agent, Master, operator, or person in charge of a vessel must submit notices of arrival consistent with the requirements in this subpart.

(2138)

§160.206 Information required in an NOA.

(2139) (a) *Information required*. With the exceptions noted in paragraph (b) of this section, each NOA must contain all of the information items specified in Table 160.206. Vessel owners and operators should protect any personal information they gather in preparing notices for transmittal to the National Vessel Movement Center (NVMC) to prevent unauthorized disclosure of that information.

(2140) (b) *Exceptions*. If a crewmember or person on board other than a crewmember is not required to carry a passport for travel, then passport information required in Table 160.206 by items (4)(iv) and (5)(iv) need not be provided for that person.

(2142)

§160.208 Updates to a submitted NOA.

(2143) (a) Unless otherwise specified in this section, whenever events cause NOA information submitted for a vessel to become inaccurate, or the submitter to realize that data submitted was inaccurate, the owner, agent, Master, operator, or person in charge of that vessel must submit an update within the times required in §160.212.

(2144) (b) Changes in the following information need not be reported:

(2145) (1) Changes in arrival or departure times that are less than six (6) hours;

(2146) (2) Changes in vessel location or position of the vessel at the time of reporting (entry (2)(vi) to Table 160.206); and

(2141)

Table 160.206 – NOA Information Items		
Required Information	**Vessels neither carrying CDC nor controlling another vessel carrying CDC**	**Vessels carrying CDC or controlling another vessel carrying CDC**
(1) Vessel Information		
(i) Name	X	X
(ii) Name of the registered owner	X	X
(iii) Country of registry	X	X
(iv) Call sign	X	X
(v) International Maritime Organization (IMO) international number or, if vessel does not have an assigned IMO international number, substitute with official number	X	X
(vi) Name of the operator	X	X
(vii) Name of the charterer	X	X
(viii) Name of classification society or recognized organization	X	X
(ix) Maritime Mobile Service Identity (MMSI) number, if applicable	X	X
(x) Whether the vessel is 300 gross tons or less (yes or no)	X	X
(xi) USCG Vessel Response Plan Control Number, if applicable	X	X
(2) Voyage Information		
(i) Names of last five foreign ports or places visited	X	X
(ii) Dates of arrival and departure for last five foreign ports or places visited	X	X
(iii) For the port or place of the United States to be visited, list the names of the receiving facility, the port or place, the city, and the state	X	X
(iv) For the port or place in the United States to be visited, the estimated date and time of arrival	X	X
(v) For the port or place in the United States to be visited, the estimated date and time of departure	X	X
(vi) The location (port or place and country) or position (latitude and longitude or waterway and mile marker) of the vessel at the time of reporting	X	X
(vii) The name and telephone number of a 24-hour point of contact	X	X
(viii) Whether the vessel's voyage time is less than 24 hours (yes or no)	X	X
(ix) Last port or place of departure	X	X
(x) Dates of arrival and departure for last port or place of departure	X	X
(3) Cargo Information		
(i) A general description of cargo, other than CDC, on board the vessel (e.g. grain, container, oil, etc.)	X	X
(ii) Name of each CDC carried, including cargo UN number, if applicable	–	X
(iii) Amount of each CDC carried	–	X
(4) Information for each Crewmember On Board		
(i) Full name	X	X
(ii) Date of birth	X	X
(iii) Nationality	X	X
(iv) Passport* or mariners document number (type of identification and number)	X	X
(v) Position or duties on the vessel	X	X
(vi) Where the crewmembers embarked (list port or place and country)	X	X
(5) Information for each Person On Board in Addition to Crew		
(i) Full name	X	X
(ii) Date of birth	X	X
(iii) Nationality	X	X
(iv) Passport number*	X	X
(v) Where the person embarked (list port or place and country)	X	X
(6) Operational condition of equipment required by 33 CFR part 164 of this chapter (see note to table)	X	X
(7) International Safety Management (ISM) Code Notice		
(i) The date of expiration for the company's Document of Compliance certificate that covers the vessel	X	X
(ii) The date of expiration for the vessel's Safety Management Certificate	X	X
(iii) The name of the Flag Administration, or the recognized organization(s) representing the vessel Flag Administration, that issued those certificates	X	X
(8) International Ship and Port Facility Code (ISPS) Notice		
(i) The date of issuance for the vessel's International Ship Security Certificate (ISSC), if any	X	X
(ii) Whether the ISSC, if any, is an initial Interim ISSC, subsequent and consecutive Interim ISSC, or final ISSC	X	X
(iii) Declaration that the approved ship security plan, if any, is being implemented	X	X
(iv) If a subsequent and consecutive Interim ISSC, the reasons therefore	X	X
(v) The name and 24-hour contact information for the Company Security Officer	X	X
(vi) The name of the Flag Administration, or the recognized security organization(s) representing the vessel Flag Administration that issued the ISSC	X	X
Note to Table 160.206. For items with an asterisk (*), see paragraph (b) of this section. Submitting a response for item 6 indicating that navigation equipment is not operating properly does not serve as notice to the District Commander, Captain of the Port, or Vessel Traffic Center, under 33 CFR 164.53.		

(2170) (b) Any hazardous conditions as defined in §160.202; and

(2171) (c) If the vessel is carrying certain dangerous cargo or controlling a vessel carrying certain dangerous cargo, the amount and name of each CDC carried, including cargo UN number if applicable.

(2172)

§160.216 Notice of hazardous conditions.

(2173) (a) Whenever there is a hazardous condition either on board a vessel or caused by a vessel or its operation, the owner, agent, master, operator, or person in charge must immediately notify the nearest Coast Guard Sector Office or Group Office, and in addition submit any report required by 46 CFR 4.05-10.

(2174) (b) When the hazardous condition involves cargo loss or jettisoning as described in 33 CFR 97.115, the notification required by paragraph (a) of this section must include—

(2175) (1) What was lost, including a description of cargo, substances involved, and types of packages;

(2176) (2) How many were lost, including the number of packages and quantity of substances they represent;

(2177) (3) When the incident occurred, including the time of the incident or period of time over which the incident occurred;

(2178) (4) Where the incident occurred, including the exact or estimated location of the incident, the route the ship was taking, and the weather (wind and sea) conditions at the time or approximate time of the incident; and

(2179) (5) How the incident occurred, including the circumstances of the incident, the type of securing equipment that was used, and any other material failures that may have contributed to the incident.

(2180)

Part 161–Vessel Traffic Management

(2181)

Subpart A–Vessel Traffic Services

(2182)

General Rules

(2183)

§161.1 Purpose and Intent.

(2184) (a) The purpose of this part is to promulgate regulations implementing and enforcing certain sections of the Ports and Waterways Safety Act (PWSA) setting up a national system of Vessel Traffic Services that will enhance navigation, vessel safety, and marine environmental protection and promote safe vessel movement by reducing the potential for collisions, rammings, and groundings, and the loss of lives and property associated with these incidents within VTS areas established hereunder.

(2185) (b) Vessel Traffic Services provide the mariner with information related to the safe navigation of a waterway. This information, coupled with the mariner's compliance with the provisions set forth in this part, enhances the safe routing of vessels through congested waterways or waterways of particular hazard. Under certain circumstances, a VTS may issue directions to control the movement of vessels in order to minimize the risk of collision between vessels, or damage to property or the environment.

(2186) (c) The owner, operator, charterer, master, or person directing the movement of a vessel remains at all times responsible for the manner in which the vessel is operated and maneuvered, and is responsible for the safe navigation of the vessel under all circumstances. Compliance with these rules or with a direction of the VTS is at all times contingent upon the exigencies of safe navigation.

(2187) (d) Nothing in this part is intended to relieve any vessel, owner, operator, charterer, master, or person directing the movement of a vessel from the consequences of any neglect to comply with this part or any other applicable law or regulations (e.g., the International Regulations for Prevention of Collisions at Sea, 1972 (72 COLREGS) or the Inland Navigation Rules) or of the neglect of any precaution which may be required by the ordinary practice of seamen, or by the special circumstances of the case.

(2188)

§161.2 Definitions.

(2189) For the purposes of this part:

(2190) *Center* means a Vessel Traffic Center or Vessel Movement Center.

(2191) *Cooperative Vessel Traffic Services (CVTS)* means the system of vessel traffic management established and jointly operated by the United States and Canada within adjoining waters. In addition, CVTS facilitates traffic movement and anchorages, avoids jurisdictional disputes, and renders assistance in emergencies in adjoining United States and Canadian waters.

(2192) *Hazardous Vessel Operating Condition* means any condition related to a vessel's ability to safely navigate or maneuver, and includes, but is not limited to:

(2193) (1) The absence or malfunction of vessel operating equipment, such as propulsion machinery, steering gear, radar system, gyrocompass, depth sounding device, automatic radar plotting aid (ARPA), radiotelephone, Automatic Identification System equipment, navigational lighting, sound signaling devices or similar equipment.

(2194) (2) Any condition on board the vessel likely to impair navigation, such as lack of current nautical charts and publications, personnel shortage, or similar condition.

(2195) (3) Vessel characteristics that affect or restrict maneuverability, such as cargo or tow arrangement, trim, loaded condition, underkeel or overhead clearance, speed capabilities, power availability, or similar characteristics, which may affect the positive control or safe handling of the vessel or the tow.

(2196) *Navigable waters* means all navigable waters of the United States including the territorial sea of the United States, extending to 12 nautical miles from United States

baselines, as described in Presidential Proclamation No. 5928 of December 27, 1988.

(2197) *Precautionary Area* means a routing measure comprising an area within defined limits where vessels must navigate with particular caution and within which the direction of traffic may be recommended.

(2198) *Published* means available in a widely-distributed and publicly available medium (e.g., VTS User's Manual, ferry schedule, Notice to Mariners).

(2199) *Towing Vessel* means any commercial vessel engaged in towing another vessel astern, alongside, or by pushing ahead.

(2200) *Vessel Movement Center (VMC)* means the shore-based facility that operates the vessel tracking system for a Vessel Movement Reporting System (VMRS) area or zone within such an area. The VMC does not necessarily have the capability or qualified personnel to interact with marine traffic, nor does it necessarily respond to traffic situations developing in the area, as does a Vessel Traffic Service (VTS).

(2201) *Vessel Movement Reporting System (VMRS)* means a mandatory reporting system used to monitor and track vessel movements. This is accomplished by a vessel providing information under established procedures as set forth in this part in the areas defined in Table 161.12(c) (VTS and VMRS Centers, Call Signs/MMSI, Designated Frequencies, and Monitoring Areas).

(2202) *Vessel Movement Reporting System (VMRS) User* means a vessel, or an owner, operator, charterer, Master, or person directing the movement of a vessel that is required to participate in a VMRS.

(2203) *Vessel Traffic Center (VTC)* means the shore-based facility that operates the vessel traffic service for the Vessel Traffic Service area or zone within such an area.

(2204) *Vessel Traffic Services (VTS)* means a service implemented by the United States Coast Guard designed to improve the safety and efficiency of vessel traffic and to protect the environment. The VTS has the capability to interact with marine traffic and respond to traffic situations developing in the VTS area.

(2205) *Vessel Traffic Service Area* or *VTS Area* means the geographical area encompassing a specific VTS area of service. This area of service may be subdivided into zones for the purpose of allocating responsibility to individual Vessel Traffic Centers or to identify different operating requirements.

(2206) **Note:** Although regulatory jurisdiction is limited to the navigable waters of the United States, certain vessels will be encouraged or may be required, as a condition of port entry, to report beyond this area to facilitate traffic management within the VTS area.

(2207) *VTS Special Area* means a waterway within a VTS area in which special operating requirements apply.

(2208) *VTS User* means a vessel or an owner, operator, charterer, Master, or person directing the movement of a vessel within a VTS area that is:

(2209) (1) Subject to the Vessel Bridge-to-Bridge Radiotelephone Act;

(2210) (2) Required to participate in a VMRS; or

(2211) (3) Equipped with a required Coast Guard type-approved Automatic Identification System (AIS).

(2212) *VTS User's Manual* means the manual established and distributed by the VTS to provide the mariner with a description of the services offered and rules in force for that VTS. Additionally, the manual may include chartlets showing the area and zone boundaries, general navigational information about the area, and procedures, radio frequencies, reporting provisions and other information which may assist the mariner while in the VTS area.

(2213)

§161.3 Applicability.

(2214) The provisions of this subpart shall apply to each VTS User and may also apply to any vessel while underway or at anchor on the navigable waters of the United States within a VTS area, to the extent the VTS considers necessary.

(2215)

§161.4 Requirement to carry the rules.

(2216) Each VTS User shall carry on board and maintain for ready reference a copy of these rules.

(2217) **Note:** These rules are contained in the applicable U.S. Coast Pilot, the VTS User's Manual which may be obtained by contacting the appropriate VTS or downloaded from the Coast Guard Navigation Center website (https://www.navcen.uscg.gov).

(2218)

§161.5 Deviations from the rules.

(2219) (a) Requests to deviate from any provision in this part, either for an extended period of time or if anticipated before the start of a transit, must be submitted in writing to the appropriate District Commander. Upon receipt of the written request, the District Commander may authorize a deviation if it is determined that such a deviation provides a level of safety equivalent to that provided by the required measure or is a maneuver considered necessary for safe navigation under the circumstances. An application for an authorized deviation must state the need and fully describe the proposed alternative to the required measure.

(2220) (b) Requests to deviate from any provision in this part due to circumstances that develop during a transit or immediately preceding a transit may be made to the appropriate VTC. Requests to deviate must be made as far in advance as practicable. Upon receipt of the request, the VTC may authorize a deviation if it is determined that, based on vessel handling characteristics, traffic density, radar contacts, environmental conditions and other relevant information, such a deviation provides a level of safety equivalent to that provided by the required measure or is a maneuver considered necessary for safe navigation under the circumstances.

(2221)

§161.6 Preemption.

(2222) The regulations in this part have preemptive impact over State laws or regulations on the same subject matter.

The Coast Guard has determined, after considering the factors developed by the Supreme Court in *U.S. v. Locke,* 529 U.S. 89 (2000), that by enacting Chapter 25 of the Ports and Waterways Safety Act (33 U.S.C. 1221 *et seq*.), Congress intended that Coast Guard regulations preempt State laws or regulations regarding vessel traffic services in United States ports and waterways.

(2223)

Services, VTS Measures, and Operating Requirements

(2224)

§161.10 Services.

(2225) To enhance navigation and vessel safety, and to protect the marine environment, a VTS may issue advisories, or respond to vessel requests for information, on reported conditions within the VTS area, such as:

(2226) (a) Hazardous conditions or circumstances;

(2227) (b) Vessel congestion;

(2228) (c) Traffic density;

(2229) (d) Environmental conditions;

(2230) (e) Aids to navigation status;

(2231) (f) Anticipated vessel encounters;

(2232) (g) Another vessel's name, type, position, hazardous vessel operating conditions, if applicable, and intended navigation movements, as reported;

(2233) (h) Temporary measures in effect;

(2234) (i) A description of local harbor operations and conditions, such as ferry routes, dredging, and so forth;

(2235) (j) Anchorage availability; or

(2236) (k) Other information or special circumstances.

(2237)

§161.11 VTS measures.

(2238) (a) A VTS may issue measures or directions to enhance navigation and vessel safety and to protect the marine environment, such as, but not limited to:

(2239) (1) Designating temporary reporting points and procedures;

(2240) (2) Imposing vessel operating requirements; or

(2241) (3) Establishing vessel traffic routing schemes.

(2242) (b) During conditions of vessel congestion, restricted visibility, adverse weather, or other hazardous circumstances, a VTS may control, supervise, or otherwise manage traffic, by specifying times of entry, movement, or departure to, from, or within a VTS area.

(2243)

§161.12 Vessel operating requirements.

(2244) (a) Subject to the exigencies of safe navigation, a VTS User shall comply with all measures established or directions issued by a VTS.

(2245) (b) If, in a specific circumstance, a VTS User is unable to safely comply with a measure or direction issued by the VTS, the VTS User may deviate only to the extent necessary to avoid endangering persons, property or the environment. The deviation shall be reported to the VTS as soon as is practicable.

(2246) (c) When not exchanging voice communications, a VTS User must maintain a listening watch as required by §26.04(e) of this chapter on the VTS frequency designated in Table 161.12(c) (VTS and VMRS Centers, Call Signs/MMSI, Designated Frequencies, and Monitoring Areas). In addition, the VTS User must respond promptly when hailed and communicated in the English language.

(2247) **Note to §161.12(c):** As stated in 47 CFR 80.148(b), a very high frequency watch on Channel 16 (156.800 MHz) is not required on vessels subject to the Vessel Bridge-to-Bridge Radiotelephone Act and participating in a Vessel Traffic Service (VTS) system when the watch is maintained on both the vessel bridge-to-bridge frequency and a designated VTS frequency.

(2250) (d) As soon as practicable a VTS User shall notify the VTS of any of the following:

(2251) (1) A marine casualty as defined in 46 CFR 4.05-1;

(2252) (2) Involvement in the ramming of a fixed or floating object;

(2253) (3) A pollution incident as defined in §151.15 of this chapter;

(2254) (4) A defect or discrepancy in an aid to navigation;

(2255) (5) A hazardous condition as defined in §160.202 of this chapter;

(2256) (6) Improper operation of vessel equipment required by Part 164 of this chapter;

(2257) (7) A situation involving hazardous materials for which a report is required by 49 CFR 176.48; and

(2258) (8) A hazardous vessel operating condition as defined in §161.2.

(2259)

§161.13 VTS Special Area Operating Requirements.

(2260) The following operating requirements apply within a VTS Special Area:

(2261) (a) A VTS User shall, if towing astern, do so with as short a hawser as safety and good seamanship permits.

(2262) (b) A VMRS User shall:

(2263) (1) Not enter or get underway in the area without prior approval of the VTS;

(2264) (2) Not enter a VTS Special Area if a hazardous vessel operating condition or circumstance exists;

(2265) (3) Not meet, cross, or overtake any other VMRS User in the area without prior approval of the VTS; and

(2266) (4) Before meeting, crossing, or overtaking any other VMRS User in the area, communicate on the designated vessel bridge-to-bridge radiotelephone frequency, intended navigation movements, and any other information necessary in order to make safe passing arrangements. This requirement does not relieve a vessel of any duty prescribed by the International Regulations for Prevention of Collisions at Sea, 1972 (72 COLREGS) or the Inland Navigation Rules.

(2248)

TABLE to §161.12(c)-VTS and VMRS Centers, Call Signs/MMSI, Designated Frequencies, and Monitoring Areas

Center MMSI[1] Call Sign	Designated frequency (Channel designation)—purpose[2]	Monitoring Area [3, 4]
Berwick Bay 003669950 *Berwick Traffic*	156.550 MHz (Ch. 11)	The waters south of 29°45'N, west of 91°10'W, north of 29°37'N, and east of 91°18'W.
Buzzards Bay *Buzzards Bay Control*[5]	156.600 MHz (Ch. 12)	The waters east and north of a line drawn from the southern tangent of Sakonnet Point, Rhode Island, in approximate position latitude 41°27.20′ N., longitude 71°11.70′ W., to the Buzzards Bay Entrance Light in approximate position latitude 41°23.8′ N., longitude 71°02.00′ W., and then to the southwestern tangent of Cuttyhunk Island, Massachusetts, at approximate position latitude 41°24.60′ N., longitude 70°57.00′ W., and including all of the Cape Cod Canal to its eastern entrance, except that the area of New Bedford harbor within the confines (north of) the hurricane barrier, and the passages through the Elizabeth Islands, is not considered to be ''Buzzards Bay''.
Houston-Galveston 003669954		The navigable waters north of 29°00.00′ N., west of 94°20.00′ W., south of 29°49.00′ N., and east of 95°20.00′ W.
Houston Traffic	156.550 MHz (Ch. 11) 156.250 MHz (Ch. 5A) —For sailing plans only	The navigable waters north of a line extending due west from the southern most end of Exxon Dock #1 (20°43.37′ N, 95°01.27′ W.)
Houston Traffic	156.600 MHz (Ch. 12) 156.250 MHz (Ch. 5A) —For sailing plans only	The navigable waters south of a line extending due west from the southern most end of Exxon Dock #1 (29°43.37′ N, 95°01.27′ W.).
Los Angeles/Long Beach 03660465 *San Pedro Traffic*	156.700 MHz (Ch. 14)	*Vessel Movement Reporting System Area*: The navigable waters within a 25 nautical mile radius of Point Fermin Light (33°42.30′ N, 118°17.60′ W.).
Louisville 003669732 *Louisville Traffic*	156.650 MHz (Ch. 13)	The waters of the Ohio River between McAlpine Locks (Mile 606) and Twelve Mile Island (Mile 593), only when the McAlpine upper pool gauge is at approximately 13.0 feet or above.
Lower Mississippi River 0036699952 *New Orleans Traffic*	156.550 MHz (Ch. 11)	The navigable waters of the Lower Mississippi River below 29°55.30′ N, 89°56.60′ W (Saxonholm Light) at 86.0 miles Above Head of Passes (AHP), extending down river to Southwest Pass, and, within a 12 nautical mile radius around 28°54.30′ N, 89°25.70′ W (Southwest Pass Entrance Light) at 20.1 miles Below Head of Passes.
New Orleans Traffic	156.600 MHz (Ch. 12)	The navigable waters of the Lower Mississippi River bounded on the north by a line drawn perpendicular on the river at 29°55.50′ N., 90°12.77′ W. (Upper Twelve Mile Point) at 109.0 miles AHP and on the south by a line drawn perpendicularly at 29°55.30′ N., 89°55.60′ W. (Saxonholm Light) at 86.0 miles AHP.
New Orleans Traffic	156.250 MHz (Ch. 05A)	The navigable waters of the Lower Mississippi River below 30°38.70′ N., 91°17.50′ W. (Port Hudson Light) at 254.5 miles AHP bounded on the south by a line drawn perpendicular on the river at 29°55.50′ N., 90°12.77′ W. (Upper Twelve Mile Point) at 109.0 miles AHP.
New York 003669951 *New York Traffic*	156.550 MHz (Ch. 11) —For sailing plans only 156.600 MHz (Ch. 12) —For vessels at anchor	The area consists of the navigable waters of the Lower New York Bay bounded on the east by a line drawn from Norton Point to Breezy Point; on the south by a line connecting the entrance buoys at the Ambrose Channel, Swash Channel, and Sandy Hook Channel to Sandy Hook Point; and on the southeast including the waters of Sandy Hook Bay south to a line drawn at latitude 40°25.00′ N.; then west in the Raritan Bay to the Raritan River Railroad Bridge, then north into waters of the Arthur Kill and Newark Bay to the Lehigh Valley Draw Bridge at latitude 40°41.90′ N.; and then east including the waters of the Kill Van Kull and the Upper New York Bay north to a line drawn east-west from the Holland Tunnel ventilator shaft at latitude 40°43.70′ N., longitude 74°01.60′ W., in the Hudson River; and then continuing east including the waters of the East River to the Throgs Neck Bridge, excluding the Harlem River.
New York Traffic	156.700 MHz (Ch. 14)	The navigable waters of the Lower New York Bay west of a line drawn from Norton Point to Breezy Point; and north of a line connecting the entrance buoys of Ambrose Channel, Swash Channel, and Sandy Hook Channel, to Sandy Hook Point; on the southeast including the waters of the Sandy Hook Bay south to a line drawn at latitude 40°25.00′ N.; then west into the waters of Raritan Bay East Reach to a line drawn from Great Kills Light south through Raritan Bay East Reach LGB #14 to Comfort PT, NJ; then north including the waters of the Upper New York Bay south of 40°42.40′ N. (Brooklyn Bridge) and 40°43.70′ N. (Holland Tunnel Ventilator Shaft); west through the KVK into the Arthur Kill north of 40°38.25′ N. (Arthur Kill Railroad Bridge); then north into the waters of the Newark Bay, south of 40°41.95′ N. (Lehigh Valley Draw Bridge).
New York Traffic	156.600 MHz (Ch. 12)	The navigable waters of the Raritan Bay south to a line drawn at latitude 40°26.00′ N.; then west of a line drawn from Great Kills Light south through the Raritan Bay East Reach LGB #14 to Point Comfort, NJ; then west to the Raritan River Railroad Bridge; and north including the waters of the Arthur Kill to 40°28.25′ N. (Arthur Kill Railroad Bridge); including the waters of the East River north of 40°42.40′ N. (Brooklyn Bridge) to the Throgs Neck Bridge, excluding the Harlem River.
Port Arthur 003669955 *Port Arthur Traffic*	156.050 MHz (Ch. 01A)	The navigable waters of the Sabine-Neches Canal south of 29°52.70′ N.; Port Arthur Canal; Sabine Pass Channel; Sabine Bank Channel; Sabine Outer Bar Channel; the offshore safety fairway; and the ICW from High Island to its intersection with the Sabine-Neches Canal.
Port Arthur Traffic	156.275 MHz (Ch. 65A)	The navigable waters of the Neches River; Sabine River; and Sabine-Neches Waterway north of 29°52.70′ N.; and the ICW from its intersection with the Sabine River to MM 260.
Port Arthur Traffic	156.675 MHz (Ch. 73)[6]	The navigable waters of the Calcasieu Channel; Calcasieu River Channel; and the ICW from MM 260 to MM 191.
Prince William Sound 003669958 *Valdez Traffic*	156.650 MHz (Ch. 13)	The navigable waters south of 61°05.00′ N., east of 147°20.00′ W., north of 60°00.00′ N., and west of 146°30.00′ W.; and, all navigable waters in Port Valdez.
Puget Sound[7] *Seattle Traffic* 003669957	156.700 MHz (Ch. 14)	The waters of Puget Sound, Hood Canal and adjacent waters south of a line connecting Nodule Point and Bush Point in Admiralty Inlet and south of a line drawn due east from the southernmost tip of Possession Point on Whidbey Island to the shoreline.
Seattle Traffic 003669957	156.250 MHz (Ch. 5A)	The waters of the Salish Sea east of 124°40.00′ W. excluding the waters in the central portion of the Salish Sea north and east of Race Rocks; the navigable waters of the Strait of Georgia east of 122°52.00′ W.; the San Juan Island Archipelago, Rosario Strait, Bellingham Bay; Admiralty Inlet north of a line connecting Nodule Point and Bush Point and all waters east of Whidbey Island north of a line drawn due east from the southernmost tip of Possession Point on Whidbey Island to the shoreline.
Tofino Traffic 003160012	156.725 MHz (Ch. 74)	The waters west of 124°40.00′ W. within 50 nautical miles of the coast of Vancouver Island including the waters north of 48°00.00′ N., and east of 127°00.00′ W.
Victoria Traffic 003160010	156.550 MHz (Ch. 11)	The waters of the Strait of Georgia west of 122°52.00′ W., the navigable waters of the central Salish Sea north and east of Race Rocks, including the Gulf Island Archipelago, Boundary Pass and Haro Strait.
San Francisco 003669956 *San Francisco Traffic*	156.700 MHz (Ch. 14)	The navigable waters of the San Francisco Offshore Precautionary Area, the navigable waters shoreward of the San Francisco Offshore Precautionary Area east of 122°42.00′ W. and north of 37°40.00′ N. extending eastward through the Golden Gate, and the navigable waters of San Francisco Bay and as far east as the port of Stockton on the San Joaquin River, as far north as the port of Sacramento on the Sacramento River.
San Francisco Traffic	156.600 MHz (Ch. 12)	The navigable waters within a 38 nautical mile radius of Mount Tamalpais (37°55.80′ N., 122°34.60′ W.) west of 122°42.00′ W. and south of 37°40.00′ N. and excluding the San Francisco Offshore Precautionary Area.
St. Marys River 003669953 *Soo Traffic*	156.600 MHz (Ch. 12)	The waters of the St. Marys River and lower Whitefish Bay from 45°57.00′ N. (De Tour Reef Light) to the south, to 46°38.70′ N. (Ile Parisienne Light) to the north, except the waters of the St. Marys Falls Canal and to the east along a line from La Pointe to Sims Point, within Potagannissing Bay and Worsley Bay.

Notes:

[1] Maritime Mobile Service Identifier (MMSI) is a unique nine-digit number assigned that identifies ship stations, ship earth stations, coast stations, coast earth stations, and group calls for use by a digital selective calling (DSC) radio, an INMARSAT ship earth station or AIS. AIS requirements are set forth in §§161.21 and 164.46 of this subchapter. The requirements set forth in §161.21 of this subchapter apply in those areas denoted with an MMSI number, except for Louisville and Los Angeles/Long Beach.

[2] In the event of a communication failure, difficulties or other safety factors, the Center may direct or permit a user to monitor and report on any other designated monitoring frequency or the bridge-to-bridge navigational frequency, 156.650 MHz (Channel 13) or 156.375 MHz (Ch. 67), to the extent that doing so provides a level of safety beyond that provided by other means. The bridge-to-bridge navigational frequency, 156.650 MHz (Ch. 13), is used in certain monitoring areas where the level of reporting does not warrant a designated frequency.

[3] All geographic coordinates (latitude and longitude) are expressed in North American Datum of 1983 (NAD 83).

[4] Some monitoring areas extend beyond navigable waters. Although not required, users are strongly encouraged to maintain a listening watch on the designated monitoring frequency in these areas. Otherwise, they are required to maintain watch as stated in 47 CFR 80.148.

[5] In addition to the vessels denoted in Section 161.16 of this chapter, requirements set forth in subpart B of 33 CFR part 161 also apply to any vessel transiting VMRS Buzzards Bay required to carry a bridge-to-bridge radiotelephone by part 26 of this chapter.

[6] Until otherwise directed, full VTS services will not be available in the Calcasieu Channel, Calcasieu River Channel, and the ICW from MM 260 to MM 191. Vessels may contact Port Arthur Traffic on the designated VTS frequency to request advisories, but are not required to monitor the VTS frequency in this zone.

[7] A Cooperative Vessel Traffic Service was established by the United States and Canada within adjoining waters. The appropriate Center administers the rules issued by both nations; however, enforces only its own set of rules within its jurisdiction. Note, the bridge-to-bridge navigational frequency, 156.650 MHz (Ch. 13), is not so designated in Canadian waters, therefore users are encouraged and permitted to make passing arrangements on the designated monitoring frequencies.

(2249)

TABLE to §161.12(c)-VTS and VMRS Centers, Call Signs/MMSI, Designated Frequencies, and Monitoring Areas

Center MMSI[1] Call Sign	Designated frequency (Channel designation)—purpose[2]	Monitoring Area [3, 4]
Port Arthur 003669955 *Port Arthur Traffic*	156.050 MHz (Ch. 01A)	The navigable waters of the Sabine-Neches Canal south of 29°52.70′ N.; Port Arthur Canal; Sabine Pass Channel; Sabine Bank Channel; Sabine Outer Bar Channel; the offshore safety fairway; and the ICW from High Island to its intersection with the Sabine-Neches Canal.
Port Arthur Traffic	156.275 MHz (Ch. 65A)	The navigable waters of the Neches River; Sabine River; and Sabine-Neches Waterway north of 29°52.70′ N.; and the ICW from its intersection with the Sabine River to MM 260.
Port Arthur Traffic	156.675 MHz (Ch. 73)[6]	The navigable waters of the Calcasieu Channel; Calcasieu River Channel; and the ICW from MM 260 to MM 191.
Prince William Sound 003669958 *Valdez Traffic*	156.650 MHz (Ch. 13)	The navigable waters south of 61°05.00′ N., east of 147°20.00′ W., north of 60°00.00′ N., and west of 146°30.00′ W.; and, all navigable waters in Port Valdez.
Puget Sound[7] *Seattle Traffic* 003669957	156.700 MHz (Ch. 14)	The waters of Puget Sound, Hood Canal and adjacent waters south of a line connecting Nodule Point and Bush Point in Admiralty Inlet and south of a line drawn due east from the southernmost tip of Possession Point on Whidbey Island to the shoreline.
Seattle Traffic 003669957	156.250 MHz (Ch. 5A)	The waters of the Salish Sea east of 124°40.00′ W. excluding the waters in the central portion of the Salish Sea north and east of Race Rocks; the navigable waters of the Strait of Georgia east of 122°52.00′ W.; the San Juan Island Archipelago, Rosario Strait, Bellingham Bay; Admiralty Inlet north of a line connecting Nodule Point and Bush Point and all waters east of Whidbey Island north of a line drawn due east from the southernmost tip of Possession Point on Whidbey Island to the shoreline.
Tofino Traffic 003160012	156.725 MHz (Ch. 74)	The waters west of 124°40.00′ W. within 50 nautical miles of the coast of Vancouver Island including the waters north of 48°00.00′ N., and east of 127°00.00′ W.
Victoria Traffic 003160010	156.550 MHz (Ch. 11)	The waters of the Strait of Georgia west of 122°52.00′ W., the navigable waters of the central Salish Sea north and east of Race Rocks, including the Gulf Island Archipelago, Boundary Pass and Haro Strait.
San Francisco 003669956 *San Francisco Traffic*	156.700 MHz (Ch. 14)	The navigable waters of the San Francisco Offshore Precautionary Area, the navigable waters shoreward of the San Francisco Offshore Precautionary Area east of 122°42.00′ W. and north of 37°40.00′ N. extending eastward through the Golden Gate, and the navigable waters of San Francisco Bay and as far east as the port of Stockton on the San Joaquin River, as far north as the port of Sacramento on the Sacramento River.
San Francisco Traffic	156.600 MHz (Ch. 12)	The navigable waters within a 38 nautical mile radius of Mount Tamalpais (37°55.80′ N., 122°34.60′ W.) west of 122°42.00′ W. and south of 37°40.00′ N. and excluding the San Francisco Offshore Precautionary Area.
St. Marys River 003669953 *Soo Traffic*	156.600 MHz (Ch. 12)	The waters of the St. Marys River and lower Whitefish Bay from 45°57.00′ N. (De Tour Reef Light) to the south, to 46°38.70′ N. (Ile Parisienne Light) to the north, except the waters of the St. Marys Falls Canal and to the east along a line from La Pointe to Sims Point, within Potagannissing Bay and Worsley Bay.

Notes:

[1] Maritime Mobile Service Identifier (MMSI) is a unique nine-digit number assigned that identifies ship stations, ship earth stations, coast stations, coast earth stations, and group calls for use by a digital selective calling (DSC) radio, an INMARSAT ship earth station or AIS. AIS requirements are set forth in §§161.21 and 164.46 of this subchapter. The requirements set forth in §161.21 of this subchapter apply in those areas denoted with an MMSI number, except for Louisville and Los Angeles/Long Beach.

[2] In the event of a communication failure, difficulties or other safety factors, the Center may direct or permit a user to monitor and report on any other designated monitoring frequency or the bridge-to-bridge navigational frequency, 156.650 MHz (Channel 13) or 156.375 MHz (Ch. 67), to the extent that doing so provides a level of safety beyond that provided by other means. The bridge-to-bridge navigational frequency, 156.650 MHz (Ch. 13), is used in certain monitoring areas where the level of reporting does not warrant a designated frequency.

[3] All geographic coordinates (latitude and longitude) are expressed in North American Datum of 1983 (NAD 83).

[4] Some monitoring areas extend beyond navigable waters. Although not required, users are strongly encouraged to maintain a listening watch on the designated monitoring frequency in these areas. Otherwise, they are required to maintain watch as stated in 47 CFR 80.148.

[5] In addition to the vessels denoted in Section 161.16 of this chapter, requirements set forth in subpart B of 33 CFR part 161 also apply to any vessel transiting VMRS Buzzards Bay required to carry a bridge-to-bridge radiotelephone by part 26 of this chapter.

[6] Until otherwise directed, full VTS services will not be available in the Calcasieu Channel, Calcasieu River Channel, and the ICW from MM 260 to MM 191. Vessels may contact Port Arthur Traffic on the designated VTS frequency to request advisories, but are not required to monitor the VTS frequency in this zone.

[7] A Cooperative Vessel Traffic Service was established by the United States and Canada within adjoining waters. The appropriate Center administers the rules issued by both nations; however, enforces only its own set of rules within its jurisdiction. Note, the bridge-to-bridge navigational frequency, 156.650 MHz (Ch. 13), is not so designated in Canadian waters, therefore users are encouraged and permitted to make passing arrangements on the designated monitoring frequencies.

(2281)

TABLE 161.18(a) – The IMO Standard Ship Reporting System

A	ALPHA	Ship	Name, call sign or ship station identity, and flag.
B	BRAVO	Dates and time of events	A 6 digit group giving day of month (first two digits), hours and minutes (last four digits). If other than UTC state time zone used.
C	CHARLIE	Position	A 4 digit group giving latitude in degrees and minutes suffixed with N (north) or S (south) and a 5 digit group giving longitude in degrees and minutes suffixed with E (east) or W (west); or
D	DELTA	Position	True bearing (first 3 digits) and distance (state distance) in nautical miles from a clearly identified landmark (state landmark).
E	ECHO	True course	A 3 digit group.
F	FOXTROT	Speed in knots and tenths of knots	A 3 digit group.
G	GOLF	Port of Departure	Name of last port of call.
H	HOTEL	Date, time and point of entry system	Entry time expressed as in (B) and into the entry position expressed as in (C) or (D).
I	INDIA	Destination and expected time of arrival	Name of port and date time group expressed as in (B).
J	JULIET	Pilot	State whether a deep sea or local pilot is on board.
K	KILO	Date, time and point of exit from system	Exit time expressed as in (B) and exit position expressed as in (C) or (D).
L	LIMA	Route information	Intended track.
M	MIKE	Radio	State in full names of communications stations/frequencies guarded.
N	NOVEMBER	Time of next report	Date time group expressed as in (B).
O	OSCAR	Maximum present static draught in meters	4 digit group giving meters and centimeters.
P	PAPA	Cargo on board	Cargo and brief details of any dangerous cargoes as well as harmful substances and gases that could endanger persons or the environment.
Q	QUEBEC	Defects, damage, deficiencies or limitations	Brief detail of defects, damage, deficiencies or other limitations.
R	ROMEO	Description of pollution or dangerous goods lost	Brief details of type of pollution (oil, chemicals, etc.) or dangerous goods lost overboard; position expressed as in (C) or (D).
S	SIERRA	Weather conditions	Brief details of weather and sea conditions prevailing.
T	TANGO	Ship's representative and/or owner	Details of name and particulars of ship's representative and/or owner for provision of information.
U	UNIFORM	Ship size and type	Details of length, breadth, tonnage, and type, etc., as required.
V	VICTOR	Medical personnel	Doctor, physician's assistant, nurse, no medic.
W	WHISKEY	Total number of persons on board	State number.
X	XRAY	Miscellaneous	Any other information as appropriate. [i.e., a detailed description of a planned operation, which may include: its duration; effective area; any restrictions to navigation; notification procedures for approaching vessels; in addition, for a towing operation: configuration, length of the tow, available horsepower, etc.; for a dredge or floating plant: configuration of pipeline, mooring configuration, number of assist vessels, etc.].

(2267)

Subpart B–Vessel Movement Reporting System

(2268)

§161.15 Purpose and Intent.

(2269) (a) A Vessel Movement Reporting System (VMRS) is a system used to monitor and track vessel movements within a VTS or VMRS area. This is accomplished by requiring that vessels provide information under established procedures as set forth in this part, or as directed by the Center.

(2270) (b) To avoid imposing an undue reporting burden or unduly congesting radiotelephone frequencies, reports shall be limited to information which is essential to achieve the objectives of the VMRS. These reports are consolidated into three reports (sailing plan, position, and final).

(2271)

§161.16 Applicability.

(2272) Unless otherwise stated, the provisions of this subpart apply to the following vessels and VMRS Users:

(2273) (a) Every power-driven vessel of 40 meters (approximately 131 feet) or more in length, while navigating;

(2274) (b) Every towing vessel of 8 meters (approximately 26 feet) or more in length, while navigating; or

(2275) (c) Every vessel certificated to carry 50 or more passengers for hire, when engaged in trade.

(2276)

§161.17 [Removed and Reserved]

(2277)

§161.18 Reporting requirements.

(2278) (a) A Center may: (1) Direct a vessel to provide any of the information set forth in Table 161.18(a) (IMO Standard Ship Reporting System);

(2279) (2) Establish other means of reporting for those vessels unable to report on the designated frequency; or

(2280) (3) Require reports from a vessel in sufficient time to allow advance vessel traffic planning.

(2282) (b) All reports required by this part shall be made as soon as is practicable on the frequency designated in Table 161.12(c) (VTS and VMRS Centers, Call Signs/MMSI, Designated Frequencies, and Monitoring Areas).

(2283) (c) When not exchanging communications, a VMRS User must maintain a listening watch as described in §26.04(e) of this chapter on the frequency designated in Table 161.12(c) (VTS and VMRS Centers, Call Signs/MMSI, Designated Frequencies, and Monitoring Areas). In addition, the VMRS User must respond promptly when hailed and communicate in the English language.

(2284) **Note:** As stated in 47 CFR 80.148(b), a VHF watch on Channel 16 (156.800 MHz) is not required on vessels subject to the Vessel Bridge-to-Bridge Radiotelephone Act and participating in a Vessel Traffic Service (VTS) system when the watch is maintained on both the vessel bridge-to-bridge frequency and a designated VTS frequency.

(2285) (d) A vessel must report:

(2286) (1) Any significant deviation from its Sailing Plan, as defined in §161.19, or from previously reported information; or

(2287) (2) Any intention to deviate from a VTS issued measure or vessel traffic routing system.

(2288) (e) When reports required by this part include time information, such information shall be given using the local time zone in effect and the 24-hour military clock system.

(2289)

§161.19 Sailing Plan (SP).

(2290) Unless otherwise stated, at least 15 minutes before navigating a VTS area, a vessel must report the:

(2291) (a) Vessel name and type;

(2292) (b) Position;

(2293) (c) Destination and ETA;

(2294) (d) Intended route;

(2295) (e) Time and point of entry; and

(2296) (f) Dangerous cargo on board or in its tow, as defined in §160.202 of this chapter.

(2297)

§161.20 Position Report (PR).

(2298) A vessel must report its name and position:

(2299) (a) Upon point of entry into a VMRS area;

(2300) (b) At designated points as set forth in Subpart C; or

(2301) (c) When directed by the Center.

(2302)

§161.21 Automated reporting.

(2303) (a) Unless otherwise directed, vessels equipped with an Automatic Identification System (AIS) are required to make continuous, all stations, AIS broadcasts, in lieu of voice Position Reports, to those Centers denoted in Table 161.12(c) of this part.

(2304) (b) Should an AIS become non-operational, while or prior to navigating a VMRS area, it should be restored to operating condition as soon as possible, and, until restored a vessel must:

(2305) (1) Notify the Center;

(2306) (2) Make voice radio Position Reports at designated reporting points as required by §161.20(b) of this part; and

(2307) (3) Make any other reports as directed by the Center.

(2308)

§161.22 Final Report (FR).

(2309) A vessel must report its name and position:

(2310) (a) On arrival at its destination; or

(2311) (b) When leaving a VTS area.

(2312)

§161.23 Reporting exemptions.

(2313) (a) Unless otherwise directed, the following vessels are exempted from providing Position and Final Reports due to the nature of their operation:

(2314) (1) Vessels on a published schedule and route;

(2315) (2) Vessels operating within an area of a radius of three nautical miles or less; or

(2316) (3) Vessels escorting another vessel or assisting another vessel in maneuvering procedures.

(2317) (b) A vessel described in paragraph (a) of this section must:

(2318) (1) Provide a Sailing Plan at least 5 minutes but not more than 15 minutes before navigating within the VMRS area; and

(2319) (2) If it departs from its promulgated schedule by more than 15 minutes or changes its limited operating area, make the established VMRS reports, or report as directed.

(2320)

Subpart C–Vessel Traffic Service and Vessel Movement Reporting System Areas and Reporting Points

(2321) **Note:** All geographic coordinates contained in part 161 (latitude and longitude) are expressed in North American Datum of 1983 (NAD 83).

(2322)

§161.25 Vessel Traffic Service New York.

(2323) The area consists of the navigable waters of the Lower New York Harbor bounded on the east by a line drawn from Norton Point to Breezy Point; on the south by a line connecting the entrance buoys at the Ambrose Channel, Swash Channel, and Sandy Hook Channel to Sandy Hook Point; and on the southeast including the waters of Sandy Hook Bay south to a line drawn at 40°25'N.; then west into waters of the Raritan Bay to the Raritan River Rail Road Bridge; and then north including the waters of the Arthur Kill and Newark Bay to the Lehigh Valley Draw Bridge at 40°41.9'N.; and then east including the waters of the Kill Van Kull and Upper New York Bay north to a line drawn east-west from the Holland Tunnel Ventilator Shaft at 40°43.7'N., 74°01.6'W. in the Hudson River; and then continuing east including the waters of the East

River to the Throgs Neck Bridge, excluding the Harlem River.

(2324) **Note:** Although mandatory participation in VTSNY is limited to the area within the navigable waters of the United States, VTSNY will provide services beyond those waters. Prospective users are encouraged to report beyond the area of required participation in order to facilitate advance vessel traffic management in the VTS area and to receive VTSNY advisories and/or assistance.

(2325)

Part 162–Inland Waterways Navigation Regulations

(2326)

§162.1 General.

(2327) Geographic coordinates expressed in terms of latitude or longitude, or both, are not intended for plotting on maps or charts whose referenced horizontal datum is the North American Datum of 1983 (NAD 83), unless such geographic coordinates are expressly labeled NAD 83. Geographic coordinates without the NAD 83 reference may be plotted on maps or charts referenced to NAD 83 only after application of the appropriate corrections that are published on the particular map or chart being used.

(2328)

§162.5 Definitions.

(2329) The following definition applies to this part:

(2330) *Merchant mariner credential* or *MMC* means the credential issued by the Coast Guard under 46 CFR part 10. It combines the individual merchant mariner's document, license, and certificate of registry enumerated in 46 U.S.C. subtitle II part E as well as the STCW endorsement into a single credential that serves as the mariner's qualification document, certificate of identification, and certificate of service.

(2331)

§162.15 Manhasset Bay, N.Y.; seaplane restricted area.

(2332) (a) *The restricted area*. An area in Manhasset Bay between the shore at Manorhaven on the north and the southerly limit line of the special anchorage area in Manhasset Bay, west area at Manorhaven (described in 33 CFR 110.60), on the south; its axis being a line bearing 166°50' true from latitude 40°50'17.337", longitude 73°43'03.877", which point is on the south side of Orchard Beach Boulevard at Manorhaven; and being 100 feet wide for a distance of 380 feet in a southerly direction from the south side of Orchard Beach Boulevard, and thence flaring to a width of 300 feet at the southerly limit line.

(2333) (b) *The regulations*. (1) Vessels shall not anchor or moor within the restricted area.

(2334) (2) All vessels traversing the area shall pass directly through without unnecessary delay, and shall give seaplanes the right-of-way at all times.

(2335)

§162.20 Flushing Bay near La Guardia Airport, Flushing, N.Y.; restricted area.

(2336) (a) *The area*. An area in the main channel in Flushing Bay extending for a distance of 300 feet on either side of the extended center line of Runway No. 13–31 at La Guardia Airport.

(2337) (b) *The regulations*. (1) All vessels traversing the area shall pass directly through without unnecessary delay.

(2338) (2) No vessels having a height of more than 35 feet with reference to the plane of mean high water shall enter or pass through the area whenever visibility is less than one mile.

(2339)

Part 164–Navigation Safety Regulations (in part). For a complete description of this part see 33 CFR 164.

(2340)

§164.01 Applicability.

(2341) (a) This part (except as specifically limited by this section) applies to each self-propelled vessel of 1600 or more gross tons (except as provided in paragraphs (c) and (d) of this section, or for foreign vessels described in §164.02) when it is operating in the navigable waters of the United States except the St. Lawrence Seaway.

(2342) (b) Sections 164.70 through 164.82 of this part apply to each towing vessel of 12 meters (39.4 feet) or more in length operating in the navigable waters of the United States other than the St. Lawrence Seaway; except that a towing vessel is exempt from the requirements of §164.72 if it is–

(2343) (1) Used solely within a limited geographic area, such as a fleeting-area for barges or a commercial facility, and used solely for restricted service, such as making up or breaking up larger tows;

(2344) (2) Used solely for assistance towing as defined by 46 CFR 10.103;

(2345) (3) Used solely for pollution response; or

(2346) (4) Any other vessel exempted by the Captain of the Port (COTP). The COTP, upon written request, may, in writing, exempt a vessel from §164.72 for a specified route if he or she decides that exempting it would not allow its unsafe navigation under anticipated conditions.

(2347) (c) Provisions of §§164.11(a)(2) and (c), 164.30, 164.33, and 164.46 do not apply to warships or other vessels owned, leased, or operated by the United States Government and used only in government noncommercial service when these vessels are equipped with electronic navigation systems that have met the applicable agency regulations regarding navigation safety.

(2348) (d) Provisions of §164.46 apply to some self-propelled vessels of less than 1600 gross tonnage.

(2349)

§164.02 Applicability exception for foreign vessels.

(2350) (a) Except for §164.46(c), none of the requirements of this part apply to foreign vessels that:

(2351) (1) Are not destined for, or departing from, a port or place subject to the jurisdiction of the United States; and

(2352) (2) Are in:

(2353) (i) Innocent passage through the territorial sea of the United States; or

(2354) (ii) Transit through navigable waters of the United States which form a part of an international strait.

(2355)

§164.03 Incorporation by reference.

(2356) (a) Certain material is incorporated by reference into this part with the approval of the Director of the Federal Register under 5 U.S.C. 552(a) and 1 CFR part 51. To enforce any edition other than that specified in this section, the Coast Guard must publish notice of the change in the Federal Register and the material must be available to the public. All approved material is available for inspection at the National Archives and Records Administration (NARA). For more information on the availability of this material at NARA, call 202–741–6030, or go to: *www.archives.gov/federal-register/cfr/ibr-locations.html*. Also, it is available for inspection at the Commandant (CG-NAV), U.S. Coast Guard Stop 7418, Attn: Office of Navigation Systems, 2703 Martin Luther King Jr. Ave. SE., Washington, DC 20593-7418, telephone 202–372–1565, and is available from the sources listed below.

(2357) (b) American Petroleum Institute (API), 1220 L Street NW., Washington, DC 20005-4070, 202–682–8000, *www.api.org*:

(2358) (1) API Specification 9A, Specification for Wire Rope, Section 3, Properties and Tests for Wire and Wire Rope, May 28, 1984, IBR approved for §164.74.

(2359) (2) [Reserved]

(2360) (c) ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959, 610–832–9585, *www.astm.org*:

(2361) (1) ASTM D4268-93, Standard Test Method for Testing Fiber Rope, IBR approved for §164.74.

(2362) (2) [Reserved]

(2363) (d) Cordage Institute, 350 Lincoln Street, Hingham, MA 02043.

(2364) (1) CIA-3, Standard Test Methods for Fiber Rope Including Standard Terminations, Revised, June 1980, IBR approved for §164.74.

(2365) (2) [Reserved]

(2366) (e) International Maritime Organization (IMO), 4 Albert Embankment, London SE1 7SR, United Kingdom, *www.imo.org*:

(2367) (1) IMO Resolution A342(IX), Recommendation on Performance Standards for Automatic Pilots, November 12, 1975, IBR approved for §164.13.

(2368) (2) IMO Resolution A.917(22), Guidelines for the Onboard Operational Use of Shipborne Automatic Identification System (AIS), January 25, 2002, IBR approved for §164.46.

(2369) (3) SN/Circ.227, Guidelines for the Installation of a Shipborne Automatic Identification System (AIS), January 6, 2003, IBR approved for §164.46.

(2370) (4) SN/Circ.244, Guidance on the Use of the UN/LOCODE in the Destination Field in AIS Messages, December 15, 2004, IBR approved for §164.46.

(2371) (5) SN/Circ.245, Amendments to the Guidelines for the Installation of a Shipborne Automatic Identification System (AIS)(SN/Circ.227), December 15, 2004, IBR approved for §164.46.

(2372) (6) SOLAS, International Convention for the Safety of Life at Sea, 1974, and 1988 Protocol relating thereto, 2000 Amendments, effective January and July 2002, (SOLAS 2000 Amendments), IBR approved for §164.46.

(2373) (7) Conference resolution 1, Adoption of amendments to the Annex to the International Convention for the Safety of Life at Sea, 1974, and amendments to Chapter V of SOLAS 1974, adopted on December 12, 2002, IBR approved for §164.46.

(2374) (8) SN.1/Circ.289, Guidance on the Use of AIS Application-Specific Messages, June 2, 2010, IBR approved for §164.46.

(2375) (f) National Marine Electronics Association (NMEA), 7 Riggs Avenue, Severna Park, MD 21146, 800–808–6632, *www.nmea.org*:

(2376) (1) NMEA 0400, Installation Standard for Marine Electronic Equipment used on Moderate-Sized Vessels, Version 3.10, February 2012, IBR approved for §164.46.

(2377) (2) [Reserved]

(2378) (g) Radio Technical Commission for Maritime Services (RTCM), 1611 N. Kent St., Suite 605, Arlington, VA 22209, 703-527-2000, *www.rtcm.org*:

(2379) (1) RTCM Paper 12-78/DO-100, Minimum Performance Standards, Loran C Receiving Equipment, 1977, IBR approved for §164.41.

(2380) (2) RTCM Paper 71-95/SC112-STD, RTCM Recommended Standards for Marine Radar Equipment Installed on Ships of Less Than 300 Tons Gross Tonnage, Version 1.1, October 10, 1995, IBR approved for §164.72.

(2381) (3) RTCM Paper 191-93/SC112-X, RTCM Recommended Standards for Maritime Radar Equipment Installed on Ships of 300 Tons Gross Tonnage and Upwards, Version 1.2, December 20, 1993, IBR approved for **§164.72**.

(2382) (h) International Electrotechnical Commission (IEC), 3, rue de Varembe, Geneva, Switzerland, +41 22 919 02 11, *http://www.iec.ch/*. Email: info@iec.ch.

(2383) (1) IEC 62065 (IEC 62065 2002–03), Maritime navigation and radiocommunication equipment and systems—Track control systems— Operational and performance requirements, methods of testing and required test results, First Edition, dated 2002, IBR approved for **§ 164.13(d)**.

(2384) (2) IEC 62065 (IEC 62065 2014–02), Maritime navigation and radiocommunication equipment and systems—Track control systems— Operational and

performance requirements, methods of testing and required test results, Edition 2.0, dated 2014, IBR approved for **§ 164.13(d)**.

(2385)

§164.11 Navigation under way: General.

(2386) The owner, master, or person in charge of each vessel underway shall ensure that:

(2387) (a) The wheelhouse is constantly manned by persons who–

(2388) (1) Direct and control the movement of the vessel; and

(2389) (2) Fix the vessel's position;

(2390) (b) Each person performing a duty described in paragraph (a) of this section is competent to perform that duty;

(2391) (c) The position of the vessel at each fix is plotted on a chart of the area and the person directing the movement of the vessel is informed of the vessel's position;

(2392) (d) Electronic and other navigational equipment, external fixed aids to navigation, geographic reference points, and hydrographic contours are used when fixing the vessel's position;

(2393) (e) Buoys alone are not used to fix the vessel's position;

(2394) **Note:** Buoys are aids to navigation placed in approximate positions to alert the mariner to hazards to navigation or to indicate the orientation of a channel. Buoys may not maintain an exact position because strong or varying currents, heavy seas, ice, and collisions with vessels can move or sink them or set them adrift. Although buoys may corroborate a position fixed by other means, buoys cannot be used to fix a position: however, if no other aids are available, buoys alone may be used to establish an estimated position.

(2395) (f) The danger of each closing visual or each closing radar contact is evaluated and the person directing the movement of the vessel knows the evaluation;

(2396) (g) Rudder orders are executed as given;

(2397) (h) Engine speed and direction orders are executed as given;

(2398) (i) Magnetic variation and deviation and gyrocompass errors are known and correctly applied by the person directing the movement of the vessel;

(2399) (j) A person whom he has determined is competent to steer the vessel is in the wheelhouse at all times (See also 46 U.S.C. 8702(d), which requires an able seaman at the wheel on U.S. vessels of 100 gross tons or more in narrow or crowded waters during low visibility.);

(2400) (k) If a pilot other than a member of the vessel's crew is employed, the pilot is informed of the draft, maneuvering characteristics, and peculiarities of the vessel and of any abnormal circumstances on the vessel that may affect its safe navigation.

(2401) (1) Current velocity and direction for the area to be transited are known by the person directing the movement of the vessel;

(2402) (m) Predicted set and drift are known by the person directing movement of the vessel;

(2403) (n) Tidal state for the area to be transited is known by the person directing movement of the vessel;

(2404) (o) The vessel's anchors are ready for letting go;

(2405) (p) The person directing the movement of the vessel sets the vessel's speed with consideration for–

(2406) (1) The prevailing visibility and weather conditions;

(2407) (2) The proximity of the vessel to fixed shore and marine structures;

(2408) (3) The tendency of the vessel underway to squat and suffer impairment of maneuverability when there is small underkeel clearance;

(2409) (4) The comparative proportions of the vessel and the channel;

(2410) (5) The density of marine traffic;

(2411) (6) The damage that might be caused by the vessel's wake;

(2412) (7) The strength and direction of the current; and

(2413) (8) Any local vessel speed limit;

(2414) (q) The tests required by §164.25 are made and recorded in the vessel's log; and

(2415) (r) The equipment required by this part is maintained in operable condition.

(2416) (s) Upon entering U.S. waters, the steering wheel or lever on the navigating bridge is operated to determine if the steering equipment is operating properly under manual control, unless the vessel has been steered under manual control from the navigating bridge within the preceding 2 hours, except when operating on the Great Lakes and their connecting and tributary waters.

(2417) (t) At least two of the steering-gear power units on the vessel are in operation when such units are capable of simultaneous operation, except when the vessel is sailing on the Great Lakes and their connecting and tributary waters, and except as required by paragraph (u) of this section.

(2418) (u) On each passenger vessel meeting the requirements of the International Convention for the Safety of Life at Sea, 1960 (SOLAS 60) and on each cargo vessel meeting the requirements of SOLAS 74 as amended in 1981, the number of steering-gear power units necessary to move the rudder from 35 on either side to 30 on the other in not more than 28 seconds must be in simultaneous operation.

(2419)

§164.13 Navigation underway: tankers.

(2420) (a) As used in this section, "tanker" means a self-propelled tank vessel, including integrated tug barge combinations, constructed or adapted primarily to carry oil or hazardous material in bulk in the cargo spaces and inspected and certificated as a tanker.

(2421) (b) Each tanker must have an engineering watch capable of monitoring the propulsion system, communicating with the bridge, and implementing manual control measures immediately when necessary. The watch must be physically present in the machinery spaces or in the main control space and must consist of at

least an engineer with an appropriately endorsed license or merchant mariner credential.

(2422) (c) Each tanker must navigate with at least two deck officers with an appropriately endorsed license or merchant mariner credential on watch on the bridge, one of whom may be a pilot. In waters where a pilot is required, the second officer, must be an individual holding an appropriately endorsed license or merchant mariner credential and assigned to the vessel as master, mate, or officer in charge of a navigational watch, who is separate and distinct from the pilot.

(2423) (d) This paragraph (d) has preemptive effect over State or local regulation within the same field. A tanker may navigate using a heading or track control system only if:

(2424) (1) The tanker is at least one-half nautical mile (1,012 yards) beyond the territorial sea baseline, as defined in **33 CFR 2.20**;

(2425) (i) Not within waters specified in **33 CFR part 110** (anchorages), or; (ii) Not within waters specified as precautionary areas in **33 CFR part 167**, and;

(2426) (2) There is a person, competent to steer the vessel, present to assume manual control of the steering station at all times including, but not limited to, the conditions listed in **46 CFR 35.20–45(a)** through **(c)**; and

(2427) (3) The system meets the heading or track control specifications of either IEC 62065 (2002–03) or IEC 62065 (2014–02) (incorporated by reference, see **§ 164.03**).

(2428)

§164.15 Navigation bridge visibility.

(2429) (a) The arrangement of cargo, cargo gear, and trim of all vessels entering or departing from U.S. ports must be such that the field of vision from the navigation bridge conforms as closely as possible to the following requirements:

(2430) (1) From the conning position, the view of the sea surface must not be obscured by more than the lesser of two ship lengths or 500 meters (1,640 feet) from dead ahead to 10 degrees on either side of the vessel. Within this arc of visibility any blind sector caused by cargo, cargo gear, or other permanent obstruction must not exceed 5 degrees.

(2431) (2) From the conning position, the horizontal field of vision must extend over an arc from at least 22.5 degrees abaft the beam on one side of the vessel, through dead ahead, to at least 22.5 degrees abaft the beam on the other side of the vessel. Blind sectors forward of the beam caused by cargo, cargo gear, or other permanent obstruction must not exceed 10 degrees each, nor total more than 20 degrees, including any blind sector within the arc of visibility described in paragraph (a)(1) of this section.

(2432) (3) From each bridge wing, the field of vision must extend over an arc from at least 45 degrees on the opposite bow, through dead ahead, to at least dead astern.

(2433) (4) From the main steering position, the field of vision must extend over an arc from dead ahead to at least 60 degrees on either side of the vessel.

(2434) (b) A clear view must be provided through at least two front windows at all times regardless of weather conditions.

(2435)

§164.19 Requirements for vessels at anchor.

(2436) The master or person in charge of each vessel that is anchored shall ensure that–

(2437) (a) A proper anchor watch is maintained;

(2438) (b) Procedures are followed to detect a dragging anchor; and

(2439) (c) Whenever weather, tide, or current conditions are likely to cause the vessel's anchor to drag, action is taken to ensure the safety of the vessel, structures, and other vessels, such as being ready to veer chain, let go a second anchor, or get underway using the vessel's own propulsion or tug assistance.

(2440)

§164.25 Tests before entering or getting underway.

(2441) (a) Except as provided in paragraphs (b) and (c) of this section no person may cause a vessel to enter into or get underway on the navigable waters of the United States unless no more than 12 hours before entering or getting underway, the following equipment has been tested:

(2442) (1) Primary and secondary steering gear. The test procedure includes a visual inspection of the steering gear and its connecting linkage, and where applicable, the operation of the following:

(2443) (i) Each remote steering gear control system.

(2444) (ii) Each steering position located on the navigating bridge.

(2445) (iii) The main steering gear from the alternative power supply, if installed.

(2446) (iv) Each rudder angle indicator in relation to the actual position of the rudder.

(2447) (v) Each remote steering gear control system power failure alarm.

(2448) (vi) Each remote steering gear power unit failure alarm.

(2449) (vii) The full movement of the rudder to the required capabilities of the steering gear.

(2450) (2) All internal vessel control communications and vessel control alarms.

(2451) (3) Standby or emergency generator, for as long as necessary to show proper functioning, including steady state temperature and pressure readings.

(2452) (4) Storage batteries for emergency lighting and power systems in vessel control and propulsion machinery spaces.

(2453) (5) Main propulsion machinery, ahead and astern.

(2454) (b) Vessels navigating on the Great Lakes and their connecting and tributary waters, having once completed the test requirements of this sub-part, are considered to remain in compliance until arriving at the next port of call on the Great Lakes.

(2455) (c) Vessels entering the Great Lakes from the St. Lawrence Seaway are considered to be in compliance with this subpart if the required tests are conducted preparatory to or during the passage of the St. Lawrence Seaway or within one hour of passing Wolfe Island.

(2456) (d) No vessel may enter, or be operated on the navigable waters of the United States unless the emergency steering drill described below has been conducted within 48 hours prior to entry and logged in the vessel logbook, unless the drill is conducted and logged on a regular basis at least once every three months. This drill must include at a minimum the following:

(2457) (1) Operation of the main steering gear from within the steering gear compartment.

(2458) (2) Operation of the means of communications between the navigating bridge and the steering compartment.

(2459) (3) Operation of the alternative power supply for the steering gear if the vessel is so equipped.

(2460)

§164.30 Charts, publications, and equipment: General.

(2461) No person may operate or cause the operation of a vessel unless the vessel has the marine charts, publications, and equipment as required by §§164.33 through 164.41 of this part.

(2462)

§164.33 Charts and publications.

(2463) (a) Each vessel must have the following:

(2464) (1) Marine charts of the area to be transited, published by the National Ocean Service, U.S. Army Corps of Engineers, or a river authority that–

(2465) (i) Are of a large enough scale and have enough detail to make safe navigation of the area possible; and

(2466) (ii) Are currently corrected.

(2467) (2) For the area to be transited, a currently corrected copy of, or applicable currently corrected extract from, each of the following publications:

(2468) (i) U.S. Coast Pilot.

(2469) (ii) Coast Guard Light List.

(2470) (3) For the area to be transited, the current edition of, or applicable current extract from:

(2471) (i) Tide tables published by private entities using data provided by the National Ocean Service.

(2472) (ii) Tidal current tables published by private entities using data provided by the National Ocean Service, or river current publication issued by a river authority.

(2473) (b) As an alternative to the requirements for paragraph (a) of this section, a marine chart or publication, or applicable extract, published by a foreign government may be substituted for a U.S. chart and publication required by this section. The chart must be of large enough scale and have enough detail to make safe navigation of the area possible, and must be currently corrected. The publication, or applicable extract, must singly or in combination contain similar information to the U.S. Government publication to make safe navigation of the area possible. The publication, or applicable extract must be currently corrected, with the exceptions of tide and tidal current tables, which must be the current editions.

(2474) (c) As used in this section, "currently corrected" means corrected with changes contained in all Notices to Mariners published by the National Geospatial-Intelligence Agency, or an equivalent foreign government publication, reasonably available to the vessel, and that is applicable to the vessel's transit.

(2475)

§164.35 Equipment: All vessels.

(2476) Each vessel must have the following:

(2477) (a) A marine radar system for surface navigation.

(2478) (b) An illuminated magnetic steering compass, mounted in a binnacle, that can be read at the vessel's main steering stand.

(2479) (c) A current magnetic compass deviation table or graph or compass comparison record for the steering compass, in the wheelhouse.

(2480) (d) A gyrocompass.

(2481) (e) An illuminated repeater for the gyrocompass required by paragraph (d) of this section that is at the main steering stand, unless that gyrocompass is illuminated and is at the main steering stand.

(2482) (f) An illuminated rudder angle indicator in the wheelhouse.

(2483) (g) The following maneuvering information prominently displayed on a fact sheet in the wheelhouse:

(2484) (1) A turning circle diagram to port and starboard that shows the time and distance and advance and transfer required to alter course 90 degrees with maximum rudder angle and constant power settings, for either full and half speeds, or for full and slow speeds. For vessels whose turning circles are essentially the same for both directions, a diagram showing a turning circle in one direction, with a note on the diagram stating that turns to port and starboard are essentially the same, may be substituted.

(2485) (2) The time and distance to stop the vessel from either full and half speeds, or from full and slow speeds, while maintaining approximately the initial heading with minimum application of the rudder.

(2486) (3) For each vessel with a fixed propeller, a table of shaft revolutions per minute for a representative range of speeds.

(2487) (4) For each vessel with a controllable pitch propeller, a table of control settings for a representative range of speeds.

(2488) (5) For each vessel that is fitted with an auxiliary device to assist in maneuvering, such as a bow thruster, a table of vessel speeds at which the auxiliary device is effective in maneuvering the vessel.

(2489) (6) The maneuvering information for the normal load and normal ballast condition for–

(2490) (i) Calm weather—wind 10 knots or less, calm sea;

(2491) (ii) No current;

(2492) (iii) Deep water conditions—water depth twice the vessel's draft or greater; and

(2493) (iv) Clean hull.

(2494) (7) At the bottom of the fact sheet, the following statement:

(2495)

WARNING

The response of the (name of the vessel) may be different from that listed above if any of the following conditions, upon which the maneuvering information is based, are varied:
(1) Calm weather—wind 10 knots or less, calm sea;
(2) No current;
(3) Water depth twice the vessel's draft or greater;
(4) Clean hull; and
(5) Intermediate drafts or unusual trim.

(2496) (h) An echo depth sounding device.

(2497) (i) A device that can continuously record the depth readings of the vessel's echo depth sounding device, except when operating on the Great Lakes and their connecting and tributary waters.

(2498) (j) Equipment on the bridge for plotting relative motion.

(2499) (k) Simple operating instructions with a block diagram, showing the change-over procedures for remote steering gear control systems and steering gear power units, permanently displayed on the navigating bridge and in the steering gear compartment.

(2500) (1) An indicator readable from the centerline conning position showing the rate of revolution of each propeller, except when operating on the Great Lakes and their connecting and tributary waters.

(2501) (m) If fitted with controllable pitch propellers, an indicator readable from the centerline conning position showing the pitch and operational mode of such propellers, except when operating on the Great Lakes and their connecting and tributary waters.

(2502) (n) If fitted with lateral thrust propellers, an indicator readable from the centerline conning position showing the direction and amount of thrust of such propellers, except when operating on the Great Lakes and their connecting and tributary waters.

(2503) (o) A telephone or other means of communication for relaying headings to the emergency steering station. Also, each vessel of 500 gross tons and over and constructed on or after June 9, 1995 must be provided with arrangements for supplying visual compass-readings to the emergency steering station.

(2504)

§164.37 Equipment: Vessels of 10,000 gross tons or more.

(2505) (a) Each vessel of 10,000 gross tons or more must have, in addition to the radar system under §164.35(a), a second marine radar system that operates independently of the first.

(2506) **Note:** Independent operation means two completely separate systems, from separate branch power supply circuits or distribution panels to antennas, so that failure of any component of one system will not render the other system inoperative.

(2507) (b) On each tanker of 10,000 gross tons or more that is subject to 46 U.S.C. 3708, the dual radar system required by this part must have a short range capability and a long range capability and each radar must have true north features consisting of a display that is stabilized in azimuth.

(2508)

§164.38 Automatic radar plotting aids (ARPA). (See 33 CFR 164.)

(2509)

§164.39 Steering gear: Foreign tankers.

(2510) (a) This section applies to each foreign tanker of 10,000 gross tons or more, except a public vessel, that–

(2511) (1) Transfers oil at a port or place subject to the jurisdiction of the United States; or

(2512) (2) Otherwise enters or operates in the navigable waters of the United States, except a vessel described by §164.02 of this part.

(2513) (b) *Definitions*. The terms used in this section are as follows:

(2514) *Constructed* means the same as in Chapter II-1, Regulations 1.1.2 and 1.1.3.1, of SOLAS 74.

(2515) *Existing tanker* means a tanker–

(2516) (1) For which the building contract is placed on or after June 1, 1979;

(2517) (2) In the absence of a building contract, the keel of which is laid or which is at a similar stage of construction on or after January 1, 1980;

(2518) (3) The delivery of which occurs on or after June 1, 1982; or

(2519) (4) That has undergone a major conversion contracted for on or after June 1, 1979; or construction of which was begun on or after January 1, 1980, or completed on or after June 1, 1982.

(2520) *Public vessel, oil, hazardous materials,* and *foreign vessel* mean the same as in 46 U.S.C. 2101.

(2521) *SOLAS 74* means the International Convention for the Safety of Life at Sea, 1974, as amended.

(2522) *Tanker* means a self-propelled vessel defined as a tanker by 46 U.S.C. 2101(38) or as a tank vessel by 46 U.S.C. 2101(39).

(2523) (c) Each tanker constructed on or after September 1, 1984, must meet the applicable requirements of Chapter II-1, Regulations 29 and 30, of SOLAS 74.

(2524) (d) Each tanker constructed before September 1, 1984, must meet the requirements of Chapter II-1, Regulation 29.19, of SOLAS 74.

(2525) (e) Each tanker of 40,000 gross tons or more, constructed before September 1, 1984, that does not meet the single-failure criterion of Chapter II-1, Regulation 29.16, of SOLAS 74, must meet the requirements of Chapter II-1, Regulation 29.20, of SOLAS 74.

(2526) (f) Each tanker constructed before September 1, 1984, must meet the applicable requirements of Chapter II-1, Regulations 29.14 and 29.15, of SOLAS 74.

(2527)

§164.40 Devices to indicate speed and distance.

(2528) (a) Each vessel required to be fitted with an Automatic Radar Plotting Aid (ARPA) under §164.38 of this part

must be fitted with a device to indicate speed and distance of the vessel either through the water, or over the ground.

(2529) (b) The device must meet the following specifications:

(2530) (1) The display must be easily readable on the bridge by day or night.

(2531) (2) Errors in the indicated speed, when the vessel is operating free from shallow water effect, and from the effects of wind, current, and tide, should not exceed 5 percent of the speed of the vessel, or 0.5 knot, whichever is greater.

(2532) (3) Errors in the indicated distance run, when the vessel is operating free from shallow water effect, and from the effects of wind, current, and tide, should not exceed 5 percent of the distance run of the vessel in one hour or 0.5 nautical mile in each hour, whichever is greater.

(2533)

§164.41 Electronic position fixing devices.

(2534) (a) Each vessel calling at a port in the continental United States, including Alaska south of Cape Prince of Wales, except each vessel owned or bareboat chartered and operated by the United States, or by a state or its political subdivision, or by a foreign nation, and not engaged in commerce, must have a satellite navigation receiver with—

(2535) (1) Automatic acquisition of satellite signals after initial operator settings have been entered; and

(2536) (2) Position updates derived from satellite information during each usable satellite pass.

(2537) (b) A system that is found by the Commandant to meet the intent of the statements of availability, coverage, and accuracy for the U.S. Coastal Confluence Zone (CCZ) contained in the U.S. "Federal Radionavigation Plan" (Report No. DOD-NO 4650.4-P, I or No. DOT-TSC-RSPA-80-16, I). A person desiring a finding by the Commandant under this subparagraph must submit a written application describing the device to the Commandant (CG-DCO-D), Attn: Deputy for Operations Policy and Capabilities, U.S. Coast Guard Stop 7318, 2703 Martin Luther King Jr. Avenue SE., Washington, DC 20593-7318. After reviewing the application, the Commandant may request additional information to establish whether or not the device meets the intent of the Federal Radionavigation Plan. Note: The Federal Radionavigation Plan is available from the National Technical Information Service, Springfield, Va. 22161, with the following Government Accession Numbers:

(2538) Vol 1, ADA 116468

(2539) Vol 2, ADA 116469

(2540) Vol 3, ADA 116470

(2541) Vol 4, ADA 116471

(2542)

§164.42 Rate of turn indicator.

(2543) Each vessel of 100,000 gross tons or more constructed on or after September 1, 1984, shall be fitted with a rate of turn indicator.

(2544)

§164.43 [Removed]

(2545)

§164.46 Automatic Identification System.

(2546) (a) *Definitions*. As used in this section—

(2547) *Automatic Identification Systems* or *AIS* means a maritime navigation safety communications system standardized by the International Telecommunication Union (ITU), adopted by the International Maritime Organization (IMO), that—

(2548) (1) Provides vessel information, including the vessel's identity, type, position, course, speed, navigational status and other safety-related information automatically to appropriately equipped shore stations, other ships, and aircraft;

(2549) (2) Receives automatically such information from similarly fitted ships, monitors and tracks ships; and

(2550) (3) Exchanges data with shore-based facilities.

(2551) *Gross tonnage* means tonnage as defined under the International Convention on Tonnage Measurement of Ships, 1969.

(2552) *International voyage* means a voyage from a country to which the present International Convention for the Safety of Life at Sea applies to a port outside such country, or conversely.

(2553) *Properly installed, operational* means an Automatic Identification System (AIS) that is installed and operated using the guidelines set forth by the International Maritime Organization (IMO) Resolution A.917(22) and Safety of Navigation Circulars (SN/Circ.) 227, 244, 245, and SN.1/Circ.289; or National Marine Electronics Association (NMEA) Installation Standard 0400-3.10 in lieu of SN/Circ.227 and 245 (incorporated by reference, see §164.03).

(2554) (b) *AIS carriage*—(1) *AIS Class A device*. The following vessels must have on board a properly installed, operational Coast Guard type-approved AIS Class A device:

(2555) (i) A self-propelled vessel of 65 feet or more in length, engaged in commercial service.

(2556) (ii) A towing vessel of 26 feet or more in length and more than 600 horsepower, engaged in commercial service.

(2557) (iii) A self-propelled vessel that is certificated to carry more than 150 passengers.

(2558) (iv) A self-propelled vessel engaged in dredging operations in or near a commercial channel or shipping fairway in a manner likely to restrict or affect navigation of other vessels.

(2559) (v) A self-propelled vessel engaged in the movement of—

(2560) (A) Certain dangerous cargo as defined in subpart C of part 160 of this chapter, or

(2561) (B) Flammable or combustible liquid cargo in bulk that is listed in 46 CFR 30.25-1, Table 30.25-1.

(2562) (2) *AIS Class B device*. Use of a Coast Guard type-approved AIS Class B device in lieu of an AIS Class A

device is permissible on the following vessels if they are not subject to pilotage by other than the vessel Master or crew:

(2563) (i) Fishing industry vessels;

(2564) (ii) Vessels identified in paragraph (b)(1)(i) of this section that are certificated to carry less than 150 passengers and that—

(2565) (A) Do not operate in a Vessel Traffic Service (VTS) or Vessel Movement Reporting System (VMRS) area defined in Table 161.12(c) of §161.12 of this chapter, and

(2566) (B) Do not operate at speeds in excess of 14 knots; and

(2567) (iii) Vessels identified in paragraph (b)(1)(iv) of this section engaged in dredging operations.

(2568) **Note to paragraph (b):** Under 33 U.S.C. 1223(b)(3) and 33 CFR 160.111, a Coast Guard Captain of the Port (COTP) may restrict the operation of a vessel if he or she determines that by reason of weather, visibility, sea conditions, port congestion, other hazardous circumstances, or the condition of such vessel, the restriction is justified in the interest of safety. In certain circumstances, if a COTP is concerned that the operation of a vessel not subject to §164.46 would be unsafe, the COTP may determine that voluntary installation of AIS by the operator would mitigate that concern. Fishing industry vessels include fishing vessels, fish processing vessels, and fish tender vessels as defined in 46 U.S.C. 2101.

(2569) (c) *SOLAS provisions*. The following self-propelled vessels must comply with International Convention for Safety of Life at Sea (SOLAS), as amended, Chapter V, regulation 19.2.1.6 (Positioning System), 19.2.4 (AIS Class A), and 19.2.3.5 (Transmitting Heading Device) or 19.2.5.1 (Gyro Compass) as applicable (Incorporated by reference, see §164.03):

(2570) (1) A vessel of 300 gross tonnage or more, on an international voyage.

(2571) (2) A vessel of 150 gross tonnage or more, when carrying more than 12 passengers on an international voyage.

(2572) (d) *Operations*. The requirements in this paragraph are applicable to any vessel equipped with AIS.

(2573) (1) Use of AIS does not relieve the vessel of the requirements to sound whistle signals or display lights or shapes in accordance with the International Regulations for Preventing Collisions at Sea, 1972 (72 COLREGS), 28 U.S.T. 3459, T.I.A.S. 8587, or Inland Navigation Rules, 33 CFR part 83; nor of the radio requirements of the Vessel Bridge-to-Bridge Radiotelephone Act, 33 U.S.C. 1201-1208, part 26 of this chapter, and 47 CFR part 80.

(2574) (2) AIS must be maintained in effective operating condition, which includes—

(2575) (i) The ability to reinitialize the AIS, which requires access to and knowledge of the AIS power source and password;

(2576) (ii) The ability to access AIS information from the primary conning position of the vessel;

(2577) (iii) The accurate broadcast of a properly assigned Maritime Mobile Service Identity (MMSI) number;

(2578) (iv) The accurate input and upkeep of all AIS data fields and system updates; and

(2579) (v) For those vessels denoted in paragraph (b) of this section, the continual operation of AIS and its associated devices (*e.g.*, positioning system, gyro, converters, displays) at all times while the vessel is underway or at anchor, and, if moored, at least 15 minutes prior to getting underway; except when its operation would compromise the safety or security of the vessel or a security incident is imminent. The AIS should be returned to continuous operation as soon as the compromise has been mitigated or the security incident has passed. The time and reason for the silent period should be recorded in the ship's official log and reported to the nearest Captain of the Port or Vessel Traffic Center (VTC).

(2580) (3) AIS safety-related text messaging must be conducted in English and solely to exchange or communicate pertinent navigation safety information (analogous to a SECURITE broadcast). Although not prohibited, AIS text messaging should not be relied upon as the primary means for broadcasting distress (MAYDAY) or urgent (PAN PAN) communications. (47 CFR 80.1109, Distress, urgency, and safety communications).

(2581) (4) AIS application-specific messaging (ASM) is permissible, but is limited to applications adopted by the International Maritime Organization (such as IMO SN.1/Circ.289) or those denoted in the International Association of Marine Aids to Navigation and Lighthouse Authorities' (IALA) ASM Collection for use in the United States or Canada, and to no more than one ASM per minute.

(2582) **Note 1 to §164.46(d):** The Coast Guard has developed the "USCG AIS Encoding Guidance" to help ensure consistent and accurate data encoding (input) by AIS users. This Guide is available at our "AIS Frequently Asked Questions" (FAQ #2) World Wide Web page at *www.navcen.uscg.gov/ais-frequently-asked-questions#2*. Although of great benefit, the interfacing or installation of other external devices or displays (*e.g.*, transmitting heading device, gyro, rate of turn indicator, electronic charting systems, and radar), is not currently required except as denoted in §164.46(c). Most application-specific messages require interfacing to an external system that is capable of their portrayal, such as equipment certified to meet Radio Technical Commission for Maritime Services (RTCM) electronic chart system (ECS) standard 10900 series.

(2583) (e) *Watchkeeping*. AIS is primarily intended for use by the Master or person in charge of the vessel, or by the person designated by the Master or person in charge to pilot or direct the movement of the vessel, who must maintain a periodic watch for AIS information.

(2584) (f) *Portable AIS*. The use of a portable AIS is permissible only to the extent that electromagnetic interference does not affect the proper function of existing navigation and communication equipment on board and

such that only one AIS device may be transmitting on board a vessel at any one time.

(2585) (g) *AIS Pilot Plug*. The AIS Pilot Plug on any vessel subject to pilotage by other than the vessel Master or crew must be readily available and easily accessible from the primary conning position of the vessel and permanently affixed (not an extension cord) and adjacent (within 3 feet) to a 120-volt 50/60 Hz AC power receptacle (NEMA 5-15).

(2586) (h) *Exceptions*. The following vessels may seek up to a 5-year deviation from the AIS requirements of this section by requesting a deviation under §164.55.

(2587) (1) Vessels that operate solely within a very confined area (*e.g.*, less than a 1 nautical-mile radius, shipyard, or barge fleeting facility);

(2588) (2) Vessels that conduct only short voyages (less than 1 nautical mile) on a fixed schedule (*e.g.*, a bank-to-bank river ferry service or a tender vessel);

(2589) (3) Vessels that are not likely to encounter other AIS-equipped vessels;

(2590) (4) Vessels whose design or construction makes it impracticable to operate an AIS device (*e.g.*, those that lack electrical power, have an exposed or open cabin, or are submersible); or

(2591) (5) Vessels denoted in paragraph (b)(2) that seek a deviation from requirements in paragraphs (d)(2)(ii) and (e) of this section because their AIS Class B device lacks a display.

(2592) (i) *Prohibition*. Except for maritime support stations (see 47 CFR 80.5) licensed by the Federal Communications Commission (FCC), broadcasts from AIS Class A or B devices on aircraft, non-self propelled vessels or from land are prohibited.

(2593) (j) *Implementation date*. Those vessels identified in paragraphs (b) and (c) of this section that were not previously subject to AIS carriage must install AIS no later than March 1, 2016.

(2594)

§164.51 Deviations from rules: Emergency.

(2595) Except for the requirements of §164.53(b), in an emergency, any person may deviate from any rule in this part to the extent necessary to avoid endangering persons, property, or the environment.

(2596)

§164.53 Deviations from rules and reporting: Non-operating equipment.

(2597) (a) If during a voyage any equipment required by this part stops operating properly, the person directing the movement of the vessel may continue to the next port of call, subject to the directions of the District Commander or the Captain of the Port, as provided by 33 CFR 160.

(2598) (b) If the vessel's automatic identification system (AIS), radar, radio navigation receivers, gyrocompass, echo depth sounding device, or primary steering gear stops operating properly, the person directing the movement of the vessel must report or cause to be reported that it is not operating properly to the nearest Captain of the Port, District Commander, or, if participating in a Vessel Traffic Service, to the Vessel Traffic Center, as soon as possible.

(2599)

§164.55 Deviations from rules: Continuing operation or period of time.

(2600) The Captain of the Port, upon written application, may authorize a deviation from any rule in this part if he determines that the deviation does not impair the safe navigation of the vessel under anticipated conditions and will not result in a violation of the rules for preventing collisions at sea. The authorization may be issued for vessels operating in the waters under the jurisdiction of the Captain of the Port for any continuing operation or period of time the Captain of the Port specifies.

(2601)

§164.61 Marine casualty reporting and record retention.

(2602) When a vessel is involved in a marine casualty as defined in 46 CFR 4.03-1, the master or person in charge of the vessel shall:

(2603) (a) Ensure compliance with 46 CFR 4.05, "Notice of Marine Casualty and Voyage Records," and

(2604) (b) Ensure that the voyage records required by 46 CFR 4.05-15 are retained for:

(2605) (1) 30 days after the casualty if the vessel remains in the navigable waters of the United States; or

(2606) (2) 30 days after the return of the vessel to a United States port if the vessel departs the navigable waters of the United States within 30 days after the marine casualty.

(2607)

§164.70 Definitions.

(2608) For purposes of §§164.72 through 164.82, the term–

(2609) *Current edition* means the most recent published version of a publication, chart, or map required by §164.72.

(2610) *Currently corrected edition* means a current or previous edition of a publication required by §164.72, corrected with changes that come from Notice to Mariners (NTMs) or Notices to Navigation reasonably available and that apply to the vessel's transit. Hand-annotated river maps from U.S. Army Corps of Engineers (USACE) are currently corrected editions if issued within the previous 5 years.

(2611) *Great Lakes* means the Great Lakes and their connecting and tributary waters including the Calumet River as far as the Thomas J. O'Brien Lock and Controlling Works (between miles 326 and 327), the Chicago River as far as the east side of the Ashland Avenue Bridge (between miles 321 and 322), and the Saint Lawrence River as far east as the lower exit of Saint Lambert Lock.

(2612) *Merchant mariner credential* or *MMC* means the credential issued by the Coast Guard under 46 CFR part 10. It combines the individual merchant mariner's document, license, and certificate of registry enumerated in 46 U.S.C. subtitle II part E as well as the STCW endorsement into a single credential that serves as the mariner's qualification

document, certificate of identification, and certificate of service.

(2613) *Swing-meter* means an electronic or electric device that indicates the rate of turn of the vessel on board which it is installed.

(2614) *Towing vessel* means a commercial vessel engaged in or intending to engage in pulling, pushing or hauling alongside, or any combination of pulling, pushing, or hauling alongside.

(2615) *Western Rivers* means the Mississippi River, its tributaries, South Pass, and Southwest Pass, to the navigational-demarcation lines dividing the high seas from harbors, rivers, and other inland waters of the United States, and the Port Allen-Morgan City Alternative Route, and that part of the Atchafalaya River above its junction with the Port Allen-Morgan City Alternative Route including the Old River and the Red River and those waters specified by §§89.25 and 89.27 of this chapter, and such other, similar waters as are designated by the COTP.

(2616)

§164.72 Navigational-safety equipment, charts or maps, and publications required on towing vessels.

(2617) (a) Except as provided by §164.01(b), each towing vessel must be equipped with the following navigational-safety equipment:

(2618) (1) *Marine radar*. By August 2, 1997, a marine radar that meets the following applicable requirements:

(2619) (i) For a vessel of less than 300 tons gross tonnage that engages in towing on navigable waters of the U.S., including Western Rivers, the radar must meet–

(2620) (A) The requirements of the Federal Communications Commission (FCC) specified by 47 CFR part 80; and

(2621) (B) RTCM Standard for Marine Radar Equipment Installed on Ships of Less Than 300 Tons Gross Tonnage, RTCM Paper-71-95/SC112-STD, Version 1.1, display Category II and stabilization Category Bravo.

(2622) (ii) For a vessel of less than 300 tons gross tonnage that engages in towing seaward of navigable waters of the U.S. or more than three nautical miles from shore on the Great Lakes, the radar must meet–

(2623) (A) The requirements of the FCC specified by 47 CFR part 80; and

(2624) (B) RTCM Standard for Marine Radar Equipment Installed on Ships of Less Than 300 Tons Gross Tonnage, RTCM Paper 71-95/SC112-STD, Version 1.1, display Category I and stabilization Category Alpha.

(2625) (iii) For a vessel of 300 tons gross tonnage or more that engages in towing on navigable waters of the U.S., including Western rivers, the radar must meet–

(2626) (A) The requirements of the Federal Communications Commission (FCC) specified by 47 CFR part 80; and

(2627) (B) RTCM Recommended Standards for Marine Radar Equipment Installed on Ships of 300 Tons Gross Tonnage and Upwards, RTCM Paper 191-93/SC112-X, Version 1.2 except the requirements for azimuth stabilization in paragraph 3.10.

(2628) (iv) For a vessel of 300 tons gross tonnage or more that engages in towing seaward of navigable waters of the U.S. or more than three nautical miles from shore on the Great Lakes, the radar must meet–

(2629) (A) The requirements of the FCC specified by 47 CFR part 80; and

(2630) (B) RTCM Recommended Standards for Marine Radar Equipment Installed on Ships of 300 Tons Gross Tonnage and Upwards, RTCM Paper 191-93/SC112-X, Version 1.2.

(2631) (v) A towing vessel with an existing radar must meet the applicable requirements of paragraphs (a)(1)(i) through (iv) of this section by August 2, 1998; except that a towing vessel with an existing radar must meet the display and stabilization requirements of paragraph (a)(1)(ii)(B) of this section by August 2, 2001.

(2632) (2) *Searchlight*. A searchlight, directable from the vessel's main steering station and capable of illuminating objects at a distance of at least two times the length of the tow.

(2633) (3) *VHF-FM Radio*. An installation or multiple installations of VHF-FM radios as prescribed by part 26 of this chapter and 47 CFR part 80, to maintain a continuous listening watch on the designated calling channel, VHF-FM Channel 13 (except on portions of the Lower Mississippi River, where VHF-FM Channel 67 is the designated calling channel), and to separately monitor the International Distress and Calling Channel, VHF-FM Channel 16, except when transmitting or receiving traffic on other VHF-FM channels or when participating in a Vessel Traffic Service (VTS) or monitoring a channel of a VTS. (Each U.S. towing vessel of 26 feet (about 8 meters) or more in length, except a public vessel, must hold a ship-radio-station license for radio transmitters (including radar and EPIRBs), and each operator must hold a restricted operator's license or higher. To get an application for either license, call (800) 418-FORM or (202) 418-FORM, or write to the FCC; Wireless Bureau, Licensing Division; 1270 Fairfield Road; Gettysburg, PA 17325-7245.)

(2634) (4) *Magnetic Compass*. Either–

(2635) (i) An illuminated swing-meter or an illuminated card-type magnetic steering compass readable from the vessel's main steering station, if the vessel engages in towing exclusively on Western Rivers; or

(2636) (ii) An illuminated card-type magnetic steering compass readable from the vessel's main steering station.

(2637) (5) *Echo Depth-Sounding Device*. By August 2, 2001, an echo depth-sounding device readable from the vessel's main steering station, unless the vessel engages in towing exclusively on Western Rivers.

(2638) (6) *Electronic Position-Fixing Device*. An electronic position-fixing device, a satellite navigational system such as the Global Positioning System (GPS) as required by §164.41, if the vessel engages in towing seaward of navigable waters of the U.S. or more than three nautical miles from shore on the Great Lakes.

(2658)

TABLE 164.72 – Equipment, Charts or Maps, and Publications of Towing Vessels for 12 Meters or More in Length

	Western Rivers	U.S. Navigable Waters (other than Western Rivers)	Waters seaward of Navigable Waters and 3 NM or more from shore on the Great Lakes
Marine Radar: Towing Vessels of less than 300 GT	RTCM Paper 71-95/SC112-STD Version 1.1 Display Category II[1] Stabilization Category BRAVO	RTCM Paper 71-95/SC112-STD Version 1.1 Display Category II[1] Stabilization Category BRAVO	RTCM Paper 71-95/SC112-STD Version 1.1 Display Category I[2] Stabilization Category ALPHA
Towing Vessels of 300 GT or more	RTCM Paper 191-93/SC112-X Version 1.2 (except the Azimuth stabilization requirement in paragraph 3.10)[1]	RTCM Paper 191-93/SC112-X Version 1.2 (except the Azimuth stabilization requirement in paragraph 3.10)[1]	RTCM Paper 191-93/SC112-X Version 1.2[1]
Searchlight	X	X	X
VHF-FM Radio	X	X	X
Magnetic Compass	X[3]	X	X
Swing Meter	X[3]		
Echo Depth-sounding Device		X	X
Electronic Position Fixing Device			X
Charts or Maps	(1) Large enough scale (2) Current edition or currently corrected edition	(1) Large enough scale (2) Current edition or currently corrected edition	(1) Large enough scale (2) Currently corrected edition
General Publications	(1) U.S. Coast Guard Light List (2) Notices to Navigation or Local Notices to Mariners (3) River-current Tables	(1) U.S. Coast Guard Light List (2) Local Notices to Mariners (3) Tidal-current Tables (4) Tide Tables (5) U.S. Coast Pilot	(1) U.S. Coast Guard Light List (2) Local Notices to Mariners (3) Tidal-current Tables (4) Tide Tables (5) U.S. Coast Pilot

Notes:
[1] Towing vessels with existing radar must meet this requirement by August 2, 1998.
[2] Towing vessels with existing radar must meet this requirement by August 2, 1998 but do not need to meet the display and stabilization requirements until August 2, 2001.
[3] A towing vessel may carry either a swing-meter or a magnetic compass.

(2639) (b) Each towing vessel must carry on board and maintain the following:

(2640) (1) *Charts or maps*. Marine charts or maps of the areas to be transited, published by the National Ocean Service (NOS), the ACOE, or a river authority that satisfy the following requirements:

(2641) (i) The charts or maps must be of a large enough scale and have enough detail to make safe navigation of the areas possible.

(2642) (ii) The charts or maps must be either–

(2643) (A) Current editions or currently corrected editions, if the vessel engages in towing exclusively on navigable waters of the U.S., including Western Rivers; or

(2644) (B) Currently corrected editions, if the vessel engages in towing seaward of navigable waters of the U.S. or more than three nautical miles from shore on the Great Lakes.

(2645) (iii) The charts or maps may be, instead of charts or maps required by paragraphs (b)(1) (i) and (ii) of this section, currently corrected marine charts or maps, or applicable extracts, published by a foreign government. These charts or maps, or applicable extracts, must contain information similar to that on the charts or maps required by paragraphs (b)(1) (i) and (ii) of this section, be of large enough scale, and have enough detail to make safe navigation of the areas possible, and must be currently corrected.

(2646) (2) *General publications*. A currently corrected edition of, or an applicable currently corrected extract from, each of the following publications for the area to be transited:

(2647) (i) If the vessel is engaged in towing exclusively on Western Rivers–

(2648) (A) U.S. Coast Guard Light List;

(2649) (B) Applicable Notices to Navigation published by the ACOE, or Local Notices to Mariners (LNMs) published by the Coast Guard, for the area to be transited, when available; and

(2650) (C) River-current tables published by a river authority, if available.

(2651) (ii) If the vessel is engaged other than in towing exclusively on Western Rivers–

(2652) (A) Coast Guard Light List;

(2653) (B) Notices to Mariners published by the National Geospatial-Intelligence Agency, or LNMs published by the Coast Guard;

(2654) (C) Tidal-Current tables published by private entities using data provided by the NOS, or river-current tables published by the ACOE or a river authority:

(2655) (D) Tide tables published by private entities using data provided by the NOS; and

(2656) (E) U.S. Coast Pilot.

(2657) (c) Table 164.72, following, summarizes the navigational-safety equipment, charts or maps, and

publications required for towing vessels of 12 meters or more in length engaged in towing:

(2659)

§164.74 Towline and terminal gear for towing astern.

(2660) (a) *Towline*. The owner, master, or operator of each vessel towing astern shall ensure that the strength of each towline is adequate for its intended service, considering at least the following factors:

(2661) (1) The size and material of each towline must be–

(2662) (i) Appropriate for the horsepower or bollard pull of the vessel;

(2663) (ii) Appropriate for the static loads and dynamic loads expected during the intended service;

(2664) (iii) Appropriate for the sea conditions expected during the intended service;

(2665) (iv) Appropriate for exposure to the marine environment and to any chemicals used or carried on board the vessel;

(2666) (v) Appropriate for the temperatures of normal stowage and service on board the vessel;

(2667) (vi) Compatible with associated navigational-safety equipment; and

(2668) (vii) Appropriate for the likelihood of mechanical damage.

(2669) (2) Each towline as rigged must be–

(2670) (i) Free of knots;

(2671) (ii) Spliced with a thimble, or have a poured socket at its end; and

(2672) (iii) Free of wire clips except for temporary repair, for which the towline must have a thimble and either five wire clips or as many wire clips as the manufacturer specifies for the nominal diameter and construction of the towline, whichever is more.

(2673) (3) The condition of each towline must be monitored through the–

(2674) (i) Keeping on board the towing vessel or in company files of a record of the towline's initial minimum breaking strength as determined by the manufacturer, by a classification ("class") society authorized in §157.04 of this chapter, or by a tensile test that meets API Specification 9A, Specification for Wire Rope, Section 3; ASTM D 4268 (incorporated by reference, see §164.03), Standard Test Method for Testing Fiber Ropes; or Cordage Institute CIA 3, Standard Test Methods for Fiber Rope Including Standard Terminations;

(2675) (ii) If the towline is purchased from another owner, master, or operator of a vessel with the intent to use it as a towline or if it is retested for any reason, keeping on board the towing vessel or in company files of a record of each retest of the towline's minimum breaking strength as determined by a class society authorized in §157.04 of this chapter or by a tensile test that meets API Specification 9A, Section 3; ASTM D 4268; (incorporated by reference, see §164.03) or Cordage Institute CIA 3, Standard Test Methods;

(2676) (iii) Conducting visual inspections of the towline in accordance with the manufacturer's recommendations, or at least monthly, and whenever the serviceability of the towline is in doubt (the inspections being conducted by the owner, master, or operator, or by a person on whom the owner, master, or operator confers the responsibility to take corrective measures appropriate for the use of the towline);

(2677) (iv) Evaluating the serviceability of the whole towline or any part of the towline, and removing the whole or part from service either as recommended by the manufacturer or a class society authorized in §157.04 of this chapter or in accordance with a replacement schedule developed by the owner, master, or operator that accounts for at least the–

(2678) (A) Nautical miles on, or time in service of, the towline;

(2679) (B) Operating conditions experienced by the towline;

(2680) (C) History of loading of the towline;

(2681) (D) Surface condition, including corrosion and discoloration, of the towline;

(2682) (E) Amount of visible damage to the towline;

(2683) (F) Amount of material deterioration indicated by measurements of diameter and, if applicable, measurements of lay extension of the towline; and

(2684) (G) Point at which a tensile test proves the minimum breaking strength of the towline inadequate by the standards of paragraph (a)(1) of this section, if necessary; and

(2685) (v) Keeping on board the towing vessel or in company files of a record of the material condition of the towline when inspected under paragraphs (a)(3)(iii) and (iv) of this section. Once this record lapses for three months or more, except when a vessel is laid up or out of service or has not deployed its towline, the owner, master, or operator shall retest the towline or remove it from service.

(2686) (b) *Terminal gear*. The owner, master, or operator of each vessel towing astern shall ensure that the gear used to control, protect, and connect each towline meets the following criteria:

(2687) (1) The material and size of the terminal gear are appropriate for the strength and anticipated loading of the towline and for the environment;

(2688) (2) Each connection is secured by at least one nut with at least one cotter pin or other means of preventing its failure;

(2689) (3) The lead of the towline is appropriate to prevent sharp bends in the towline from fairlead blocks, chocks, or tackle;

(2690) (4) There is provided a method, whether mechanical or non-mechanical, that does not endanger operating personnel but that easily releases the towline;

(2691) (5) The towline is protected from abrasion or chafing by chafing gear, lagging, or other means;

(2692) (6) Except on board a vessel towing in ice on Western Rivers or one using a towline of synthetic or natural fiber,

there is fitted a winch that evenly spools and tightly winds the towline; and

(2693) (7) If a winch is fitted, there is attached to the main drum a brake that has holding power appropriate for the horsepower or bollard pull of the vessel and can be operated without power to the winch.

(2694)

§164.76 Towline and terminal gear for towing alongside and pushing ahead.

(2695) The owner, master, or operator of each vessel towing alongside or pushing ahead shall ensure the face wires, spring lines, and push gear used–

(2696) (a) Are appropriate for the vessel's horsepower;

(2697) (b) Are appropriate for the arrangement of the tow;

(2698) (c) Are frequently inspected; and

(2699) (d) Remain serviceable.

(2700)

§164.78 Navigation under way: Towing vessels.

(2701) (a) The owner, master, or operator of each vessel towing shall ensure that each person directing and controlling the movement of the vessel–

(2702) (1) Understands the arrangement of the tow and the effects of maneuvering on the vessel towing and on the vessel, barge, or object being towed;

(2703) (2) Can fix the position of the vessel using installed navigational equipment, aids to navigation, geographic reference-points, and hydrographic contours;

(2704) (3) Does not fix the position of the vessel using buoys alone (Buoys are aids to navigation placed in approximate positions either to alert mariners to hazards to navigation or to indicate the orientation of a channel. They may not maintain exact charted positions, because strong or varying currents, heavy seas, ice and collisions with vessels can move or sink them or set them adrift. Although they may corroborate a position fixed by other means, they cannot fix a position; however, if no other aids are available, buoys alone may establish an estimated position.);

(2705) (4) Evaluates the danger of each closing visual or radar contact;

(2706) (5) Knows and applies the variation and deviation, where a magnetic compass is fitted and where charts or maps have enough detail to enable this type of correction;

(2707) (6) Knows the speed and direction of the current, and the set, drift, and tidal state for the area to be transited;

(2708) (7) Proceeds at a safe speed taking into account the weather, visibility, density of traffic, draft of tow, possibility of wake damage, speed and direction of the current, and local speed-limits; and

(2709) (8) Monitors the voyage plan required by §164.80.

(2710) (b) The owner, master, or operator of each vessel towing shall ensure that the tests and inspections required by §164.80 are conducted and that the results are entered in the log or other record carried on board.

(2711)

§164.80 Tests, inspections, and voyage planning.

(2712) (a) The owner, master, or operator of each towing vessel of less than 1,600 GT shall ensure that the following tests and inspections of gear occur before the vessel embarks on a voyage of more than 24 hours or when each new master or operator assumes command:

(2713) (1) *Steering-systems*. A test of the steering-gear-control system; a test of the main steering gear from the alternative power supply, if installed; a verification of the rudder-angle indicator relative to the actual position of the rudder; and a visual inspection of the steering gear and its linkage.

(2714) (2) *Navigational equipment*. A test of all installed navigational equipment.

(2715) (3) *Communications*. Operation of all internal vessel control communications and vessel-control alarms, if installed.

(2716) (4) *Lights*. Operation of all navigational lights and all searchlights.

(2717) (5) *Terminal gear*. Visual inspection of tackle; of connections of bridle and towing pendant, if applicable; of chafing gear; and the winch brake, if installed.

(2718) (6) *Propulsion systems*. Visual inspection of the spaces for main propulsion machinery, of machinery, and of devices for monitoring machinery.

(2719) (b) The owner, master, or operator of each towing vessel of 1,600 GT or more shall ensure that the following tests of equipment occur at the frequency required by §164.25 and that the following inspections of gear occur before the vessel embarks on a voyage of more than 24 hours or when each new master or operator assumes command:

(2720) (1) *Navigational equipment*. Tests of onboard equipment as required by §164.25.

(2721) (2) *Terminal gear*. Visual inspection of tackle; of connections of bridle and towing pendant, if applicable; of chafing gear; and of the winch brake, if installed.

(2722) (c)(1) The voyage-planning requirements outlined in this section do not apply to you if your towing vessel is–

(2723) (i) Used solely for any of the following services or any combination of these services–

(2724) (A) Within a limited geographic area, such as fleeting-area for barges or a commercial facility, and used for restricted service, such as making up or breaking up larger tows;

(2725) (B) For harbor-assist;

(2726) (C) For assistance towing as defined by 46 CFR 10.103;

(2727) (D) For response to emergency or pollution;

(2728) (ii) A public vessel that is both owned, or demise chartered, and operated by the United States Government or by a government of a foreign country; and that is not engaged in commercial service;

(2729) (iii) A foreign vessel engaged in innocent passage; or

(2730) (iv) Exempted by the Captain of the Port (COTP).

(2731) (2) If you think your towing vessel should be exempt from these voyage planning requirements for a specified route, you should submit a written request to the appropriate COTP. The COTP will provide you with a written response granting or denying your request.

(2732) (3) If any part of a towing vessel's intended voyage is seaward of the baseline (*i.e.,* the shoreward boundary) of the territorial sea of the U.S., then the owner, master, or operator of the vessel, employed to tow a barge or barges, must ensure that the voyage with the barge or barges is planned, taking into account all pertinent information before the vessel embarks on the voyage. The master must check the planned route for proximity to hazards before the voyage begins. During a voyage, if a decision is made to deviate substantially from the planned route, then the master or mate must plan the new route before deviating from the planned route. The voyage plan must follow company policy and consider the following (related requirements noted in parentheses):

(2733) (i) Applicable information from nautical charts and publications (also see paragraph (b) of section 164.72), including Coast Pilot, Coast Guard Light List, and Coast Guard Local Notice to Mariners for the port of departure, all ports of call, and the destination;

(2734) (ii) Current and forecast weather, including visibility, wind, and sea state for the port of departure, all ports of call, and the destination (also see paragraphs (a)(7) of section 164.78 and (b) of section 164.82);

(2735) (iii) Data on tides and currents for the port of departure, all ports of call, and the destination, and the river stages and forecast, if appropriate;

(2736) (iv) Forward and after drafts of the barge or barges and under-keel and vertical clearances (air-gaps) for all bridges, ports, and berthing areas;

(2737) (v) Pre-departure checklists;

(2738) (vi) Calculated speed and estimated time of arrival at proposed waypoints;

(2739) (vii) Communication contacts at any Vessel Traffic Services, bridges, and facilities, and any port-specific requirements for VHF radio;

(2740) (viii) Any master's or operator's standing orders detailing closest points of approach, special conditions, and critical maneuvers; and

(2741) (ix) Whether the towing vessel has sufficient power to control the tow under all foreseeable circumstances.

(2742)

§164.82 Maintenance, failure, and reporting.

(2743) (a) *Maintenance*. The owner, master, or operator of each towing vessel shall maintain operative the navigational-safety equipment required by §164.72.

(2744) (b) *Failure*. If any of the navigational-safety equipment required by §164.72 fails during a voyage, the owner, master, or operator of the towing vessel shall exercise due diligence to repair it at the earliest practicable time. He or she shall enter its failure in the log or other record carried on board. The failure of equipment, in itself, does not constitute a violation of this rule; nor does it constitute unseaworthiness; nor does it obligate an owner, master, or operator to moor or anchor the vessel. However, the owner, master, or operator shall consider the state of the equipment—along with such factors as weather, visibility, traffic, and the dictates of good seamanship—in deciding whether it is safe for the vessel to proceed.

(2745) (c) *Reporting*. The owner, master, or operator of each towing vessel whose equipment is inoperative or otherwise impaired while the vessel is operating within a Vessel Traffic Service (VTS) Area shall report the fact as required by 33 CFR 161.124. (33 CFR 161.124 requires that each user of a VTS report to the Vessel Traffic Center as soon as practicable:

(2746) (1) Any absence or malfunction of vessel-operating equipment for navigational safety, such as propulsion machinery, steering gear, radar, gyrocompass, echo depth-sounding or other sounding device, automatic dependent surveillance equipment, or navigational lighting;

(2747) (2) Any condition on board the vessel likely to impair navigation, such as shortage of personnel or lack of current nautical charts or maps, or publications; and

(2748) (3) Any characteristics of the vessel that affect or restrict the maneuverability of the vessel, such as arrangement of cargo, trim, loaded condition, under-keel clearance, and speed.)

(2749) (d) *Deviation and authorization*. The owner, master, or operator of each towing vessel unable to repair within 96 hours an inoperative marine radar required by §164.72(a) shall so notify the Captain of the Port (COTP) and shall seek from the COTP both a deviation from the requirements of this section and an authorization for continued operation in the area to be transited. Failure of redundant navigational-safety equipment, including but not limited to failure of one of two installed radars, where each satisfies §164.72(a), does not necessitate either a deviation or an authorization.

(2750) (1) The initial notice and request for a deviation and an authorization may be spoken, but the request must also be written. The written request must explain why immediate repair is impracticable, and state when and by whom the repair will be made.

(2751) (2) The COTP, upon receiving even a spoken request, may grant a deviation and an authorization from any of the provisions of §§164.70 through 164.82 for a specified time if he or she decides that they would not impair the safe navigation of the vessel under anticipated conditions.

(2752)

Part 165–Regulated Navigation Areas and Limited Access Areas

(2753)

Subpart A–General

(2754)

§165.1 Purpose of part.

(2755) The purpose of this part is to:

(2756) (a) Prescribe procedures for establishing different types of limited or controlled access areas and regulated navigation areas;

(2757) (b) Prescribe general regulations for different types of limited or controlled access areas and regulated navigation areas;

(2758) (c) Prescribe specific requirements for established areas; and

(2759) (d) List specific areas and their boundaries.

(2760)

§165.3 Definitions.

(2761) The following definitions apply to this part:

(2762) *Credential* means any or all of the following:

(2763) (1) Merchant mariner's document.

(2764) (2) Merchant mariner's license.

(2765) (3) STCW endorsement.

(2766) (4) Certificate of registry.

(2767) (5) Merchant mariner credential.

(2768) *Merchant mariner credential or MMC* means the credential issued by the Coast Guard under 46 CFR part 10. It combines the individual merchant mariner's document, license, and certificate of registry enumerated in 46 U.S.C. subtitle II part E as well as the STCW endorsement into a single credential that serves the mariner's qualification document, certificate of identification, and certificate of service.

(2769)

§165.5 Establishment procedures.

(2770) (a) A safety zone, security zone, or regulated navigation area may be established on the initiative of any authorized Coast Guard official.

(2771) (b) Any person may request that a safety zone, security zone, or regulated navigation area be established. Except as provided in paragraph (c) of this section, each request must be submitted in writing to either the Captain of the Port or District Commander having jurisdiction over the location as described in 33 CFR part 3 of this chapter, and include the following:

(2772) (1) The name of the person submitting the request;

(2773) (2) The location and boundaries of the safety zone, security zone, or regulated navigation area;

(2774) (3) The date, time, and duration that the safety zone, security zone, or regulated navigation area should be established;

(2775) (4) A description of the activities planned for the safety zone, security zone, or regulated navigation area;

(2776) (5) The nature of the restrictions or conditions desired; and

(2777) (6) The reason why the safety zone, security zone, or regulated navigation area is necessary.

(2778) (Requests for safety zones, security zones, and regulated navigation areas are approved by the Office of Management and Budget under control number 1625-0020.

(2779) (c) Safety Zones and Security Zones. If, for good cause, the request for a safety zone or security zone is made less than 5 working days before the zone is to be established, the request may be made orally, but it must be followed by a written request within 24 hours.

(2780)

§165.7 Notification.

(2781) (a) The establishment of these limited access areas and regulated navigation areas is considered rulemaking. The procedures used to notify persons of the establishment of these areas vary depending upon the circumstances and emergency conditions. Notification may be made by marine broadcasts, local notice to mariners, local news media, distribution in leaflet form, and on-scene oral notice, as well as publication in the **Federal Register**.

(2782) (b) Notification normally contains the physical boundaries of the area, the reasons for the rule, its estimated duration, and the method of obtaining authorization to enter the area, if applicable, and special navigational rules, if applicable.

(2783)

§165.8 Geographic coordinates.

(2784) Geographic coordinates expressed in terms of latitude or longitude, or both, are not intended for plotting on maps or charts whose referenced horizontal datum is the North American Datum of 1983 (NAD 83), unless such geographic coordinates are expressly labeled NAD 83. Geographic coordinates without the NAD 83 reference may be plotted on maps or charts referenced to NAD 83 only after application of the appropriate corrections that are published on the particular map or chart being used.

(2785)

§165.9 Geographic application of limited and controlled access areas and regulated navigation areas.

(2786) (a) *General*. The geographic application of the limited and controlled access areas and regulated navigation areas in this part are determined based on the statutory authority under which each is created.

(2787) (b) *Safety zones and regulated navigation areas*. These zones and areas are created under the authority of the Ports and Waterways Safety Act, 46 U.S.C. 70001–70041. Safety zones established under 46 U.S.C. 70116 and regulated navigation areas may be established in waters subject to the jurisdiction of the United States as defined in §2.38 of this chapter, including the territorial sea to a seaward limit of 12 nautical miles from the baseline.

(2788) (c) *Security zones*. These zones have two sources of authority–46 U.S.C. Chapter 700, and the Act of June

15, 1917, as amended by both the Magnuson Act of August 9, 1950 (“Magnuson Act”), 46 U.S.C. 70051–54, and sec. 104 the Maritime Transportation Security Act of 2002 (Pub. L. 107-295, 116 Stat. 2064). Security zones established under either 46 U.S.C. 70116 or 46 U.S.C. 70051 may be established in waters subject to the jurisdiction of the United States as defined in §2.38 of this chapter, including the territorial sea to a seaward limit of 12 nautical miles from the baseline.

(2789) (d) *Naval vessel protection zones*. These zones are issued under the authority of 14 U.S.C. 91 and 633 and may be established in waters subject to the jurisdiction of the United States as defined in §2.38 of this chapter, including the territorial sea to a seaward limit of 12 nautical miles from the baseline.

(2790)

Subpart B–Regulated Navigation Areas

(2791)

§165.10 Regulated navigation areas.

(2792) A regulated navigation area is a water area within a defined boundary for which regulations for vessels navigating within the area have been established under this part.

(2793)

§165.11 Vessel operating requirements (regulations).

(2794) Each District Commander may control vessel traffic in an area which is determined to have hazardous conditions, by issuing regulations:

(2795) (a) Specifying times of vessel entry, movement, or departure to, from, within, or through ports, harbors, or other waters;

(2796) (b) Establishing vessel size, speed, draft limitations, and operating conditions; and

(2797) (c) Restricting vessel operation, in a hazardous area or under hazardous conditions, to vessels which have particular operating characteristics or capabilities which are considered necessary for safe operation under the circumstances.

(2798)

§165.13 General regulations.

(2799) (a) The master of a vessel in a regulated navigation area shall operate the vessel in accordance with the regulations contained in Subpart F.

(2800) (b) No person may cause or authorize the operation of a vessel in a regulated navigation area contrary to the regulations in this Part.

(2801)

Subpart C–Safety Zones

(2802)

§165.20 Safety zones.

(2803) A Safety Zone is a water area, shore area, or water and shore area to which, for safety or environmental purposes, access is limited to authorized persons, vehicles, or vessels. It may be stationary and described by fixed limits or it may be described as a zone around a vessel in motion.

(2804)

§165.23 General regulations.

(2805) Unless otherwise provided in this part:

(2806) (a) No person may enter a safety zone unless authorized by the COTP or the District Commander;

(2807) (b) No person may bring or cause to be brought into a safety zone any vehicle, vessel, or object unless authorized by the COTP or the District Commander;

(2808) (c) No person may remain in a safety zone or allow any vehicle, vessel, or object to remain in a safety zone unless authorized by the COTP or the District Commander; and

(2809) (d) Each person in a safety zone who has notice of a lawful order or direction shall obey the order or direction of the COTP or District Commander issued to carry out the purposes of this subpart.

(2810)

Subpart D–Security Zones

(2811)

§165.30 Security zones.

(2812) (a) A security zone is an area of land, water, or land and water which is so designated by the Captain of the Port or District Commander for such time as is necessary to prevent damage or injury to any vessel or waterfront facility, to safeguard ports, harbors, territories, or waters of the United States or to secure the observance of the rights and obligations of the United States.

(2813) (b) The purpose of a security zone is to safeguard from destruction, loss, or injury from sabotage or other subversive acts, accidents, or other causes of a similar nature:

(2814) (1) Vessels,

(2815) (2) Harbors,

(2816) (3) Ports, and

(2817) (4) Waterfront facilities:

(2818) in the United States and all territory and water, continental or insular, that is subject to the jurisdiction of the United States.

(2819)

§165.33 General regulations.

(2820) Unless otherwise provided in the special regulations in Subpart F of this part:

(2821) (a) No person or vessel may enter or remain in a security zone without the permission of the Captain of the Port;

(2822) (b) Each person and vessel in a security zone shall obey any direction or order of the Captain of the Port;

(2823) (c) The Captain of the Port may take possession and control of any vessel in the security zone;

(2824) (d) The Captain of the Port may remove any person, vessel, article, or thing from a security zone;

(2825) (e) No person may board, or take or place any article or thing on board, any vessel in a security zone without the permission of the Captain of the Port; and

(2826) (f) No person may take or place any article or thing upon any waterfront facility in a security zone without the permission of the Captain of the Port.

(2827)

Subpart E–Restricted Waterfront Areas

(2828)

§165.40 Restricted Waterfront Areas.

(2829) The Commandant, may direct the COTP to prevent access to waterfront facilities, and port and harbor areas, including vessels and harbor craft therein. This section may apply to persons who do not possess the credentials outlined in 33 CFR 125.09 when certain shipping activities are conducted that are outlined in 33 CFR 125.15.

(2830)

Subpart F–Specific Regulated Navigation Areas and Limited Access Areas

(2831)

§165.100 Regulated Navigation Area: Navigable waters within the First Coast Guard District.

(2832) (a) *Regulated navigation area*. All navigable waters of the United States, as that term is used in 33 CFR 2.36, within the geographic boundaries of the First Coast Guard District, as defined in 33 CFR 3.05-1(b).

(2833) (b) *Definitions*. Terms used in this section have the same meaning as those found in 33 CFR 157.03. Single-hull identifies any tank barge that is not a double-hull tank barge.

(2834) (c) *Applicability*. This section applies to primary towing vessels engaged in towing tank barges carrying petroleum oil in bulk as cargo in the regulated navigation area, or as authorized by the District commander.

(2835) (d) *Regulations*–(1) *Positive control for barges*. (i) Except as provided in paragraph (d)(1)(iii) and paragraph 5 of this section, each single-hull tank barge, unless being towed by a primary towing vessel with twin-screw propulsion and with a separate system for power to each screw, must be accompanied by an escort or assist tug of sufficient capability to promptly push or tow the tank barge away from danger of grounding or collision in the event of–

(2836) (A) A propulsion failure;

(2837) (B) A parted towing line;

(2838) (C) A loss of tow;

(2839) (D) A fire;

(2840) (E) Grounding;

(2841) (F) A loss of steering; or

(2842) (G) Any other time a vessel may be operating in a Hazardous Vessel Operating Condition as defined in §161.2 of this Chapter.

(2843) (ii) Double-hull tank barges are exempt from paragraph (d)(1)(i) of this section.

(2844) (iii) The cognizant Captain of the Port (COTP), upon written application, may authorize an exemption from the requirements of paragraph (d)(1)(i) of this section for–

(2845) (A) Any tank barge with a capacity of less than 25,000 barrels, operating in an area with limited depth or width such as a creek or small river; or

(2846) (B) Any tank barge operating on any water within the COTP Zone, if the operator demonstrates to the satisfaction of the COTP that the barge employs an equivalent level of safety to that provided by the positive control provisions of this section. Each request for an exemption under this paragraph must be submitted in writing to the cognizant COTP no later than 7 days before the intended transit.

(2847) (iv) The operator of a towing vessel engaged in towing any tank barge must immediately call for an escort or assist tug to render assistance in the event of any of the occurrences identified in paragraph (d)(1)(i) of this section.

(2848) (2) *Enhanced communications*. Each vessel engaged in towing a tank barge must communicate by radio on marine band or Very High Frequency (VHF) channel 13 or 16, and issue security calls on marine band or VHF channel 13 or 16, upon approach to the following places:

(2849) (i) Execution Rocks Light (USCG Light List No. [LLNR] 21440).

(2850) (ii) Matinecock Point Shoal Lighted Gong Buoy 21 (LLNR 21420).

(2851) (iii) 32A Buoy (LLNR 21380).

(2852) (iv) Cable and Anchor Reef Lighted Bell Buoy 28C (LLNR 21330).

(2853) (v) Stratford Shoal (Middle Ground) Light (LLNR 21260).

(2854) (vi) Old Field Point Light (LLNR 21275).

(2855) (vii) Approach to Stratford Point from the south (NOAA Chart 12370).

(2856) (viii) Falkner Island Light (LLNR 21170).

(2857) (ix) TE Buoy (LLNR 21160).

(2858) (x) PI Buoy (LLNR 21080).

(2859) (xi) Race Rock Light (LLNR 19815).

(2860) (xii) Valiant Rock Lighted Whistle Buoy 11 (LLNR 19825).

(2861) (xiii) Approach to Point Judith in vicinity of Block Island ferry route.

(2862) (xiv) Buzzards Bay Entrance Light (LLNR 630).

(2863) (xv) Buzzards Bay Midchannel Lighted Buoy BB (LLNR 16055).

(2864) (xvi) Cleveland East Ledge Light (LLNR 16080).

(2865) (xvii) Hog Island Channel Lighted Buoys 1 (LLNR 16130) and 2 (LLNR 16135).

(2866) (xviii) Approach to the Bourne Bridge.

(2867) (xix) Approach to the Sagamore Bridge.

(2868) (xx) Approach to the eastern entrance of Cape Cod Canal.

(2869) (3) *Voyage planning*. (i) Each owner or operator of a towing vessel employed to tow a tank barge shall prepare a written voyage plan for each transit of the tank barge.

(2870) (ii) The watch officer is authorized to make modifications to the plan and validate it as necessary.

(2871) (iii) Except as provided in paragraph (d)(3)(iv) of this section, each voyage plan must contain:

(2872) (A) A description of the type, volume, and grade of cargo.

(2873) (B) Applicable information from nautical charts and publications, including Coast Pilot, Coast Guard Light List, and Coast Guard Local Notice to Mariners, for the destination(s).

(2874) (C) Current and forecasted weather, including visibility, wind, and sea state for the destination(s).

(2875) (D) Data on tides and tidal currents for the destination(s).

(2876) (E) Forward and after drafts of the tank barge, and under-keel and vertical clearances for each port and berthing area.

(2877) (F) Pre-departure checklists.

(2878) (G) Calculated speed and estimated times of arrival at proposed waypoints.

(2879) (H) Communication contacts at Vessel Traffic Service (VTS) (if applicable), bridges, and facilities, and port-specific requirements for VHF radio.

(2880) (I) The master's standing orders detailing closest points of approach, special conditions, and critical maneuvers.

(2881) (iv) Each owner or operator of a tank barge on an intra-port transit of not more than four hours may prepare a voyage plan that contains:

(2882) (A) The information described in paragraphs (d)(3)(iii)(D) and (E) of this section.

(2883) (B) Current weather conditions including visibility, wind, and sea state. This information may be entered in either the voyage plan or towing vessel's log book.

(2884) (C) The channels of VHF radio to monitor.

(2885) (D) Other considerations such as availability of pilot, assist tug, berth, and line-handlers, depth of berth at mean low water, danger areas, and security calls.

(2886) (4) *Navigation restriction areas*. Unless authorized by the cognizant COTP, no tank barge may operate in–

(2887) (i) The waters of Cape Cod Bay south of 42°05'N., and east of 70°25'W.; or

(2888) (ii) The waters of Fishers Island Sound east of 72°02'W., and west of 71°55'W.

(2889) (5) *Special Buzzards Bay regulations*. (i) For the purposes of this section, "Buzzards Bay" is the body of water east and north of a line drawn from the southern tangent of Sakonnet Point, Rhode Island, in approximate position 41°27.2'N., 71°11.7'W., to the Buzzards Bay Entrance Light in approximate position 41°23.48'N., 71°02.5'W., and then to the southwestern tangent of Cuttyhunk Island, Massachusetts, at approximate position 41°24.6'N., 70°57.0'W., and including all of the Cape Cod Canal to its eastern entrance, except that the area of New Bedford harbor within the confines (north) of the hurricane barrier, and the passages through the Elizabeth Islands, is not considered to be "Buzzards Bay".

(2890) (ii) *Additional positive control for barges*. Except as provided in paragraph (d)(1)(iii) of this section, each single hull tank barge transiting buzzards Bay and carrying 5,000 or more barrels of oil or other hazardous material must, in addition to its primary tug, be accompanied by an escort tug of sufficient capability to promptly push or tow the tank barge away from danger of grounding or collision in the event of—

(2891) (A) A propulsion failure;

(2892) (B) A parted tow line;

(2893) (C) A loss of tow;

(2894) (D) A fire;

(2895) (E) Grounding;

(2896) (F) A loss of steering; or

(2897) (G) Any other time a vessel may be operating in a Hazardous Vessel Operating Condition as defined in §161.2 of this subchapter.

(2898) (iii) *Federal pilotage*. Each single hull tank barge transiting Buzzards Bay and carrying 5,000 or more barrels of oil or other hazardous material must be under the direction and control of a pilot, who is not a member of the crew, operating under a valid, appropriately endorsed, Federal first class pilot's license issued by the Coast Guard ("federally licensed pilot"). Pilots are required to embark, direct, and control from the primary tug during transits of Buzzards Bay.

(2899) (iv) In addition to the vessels denoted in §161.16 of this chapter, requirement set forth in subpart B of 33 CFR part 161 also apply to any vessel transiting VMRS Buzzards Bay required to carry a bridge-to-bridge radiotelephone by Part 26 of this chapter.

(2900) (A) A WMRS Buzzards Bay user must:

(2901) (1) Not enter or get underway in the area without first notifying the VMRS Center;

(2902) (2) Not enter VMRS Buzzards Bay if a Hazardous Vessel Operating Condition or circumstance per §161.2 of this subchapter exists;

(2903) (3) If towing astern, do so with as short a hawser as safety and good seamanship permits;

(2904) (4) Not meet, cross or overtake any other VMRS user in the area without first notifying the VMRS center;

(2905) (5) Before meeting, crossing, or overtaking any other VMRS user in the area, communicate on the designated vessel bridge-to-bridge radiotelephone frequency, intended navigation movements, and any other information necessary in order to make safe passing arrangements. This requirement does not relieve a vessel of any duty prescribed by the Navigation Rules (COLREGS and their associated Annexes and Inland Navigation Rules (33 CFR subchapter E)).

(2906) (B) [Reserved]

(2907) (e) In addition to the authority for this part 165, this section is also authorized under authority of section 311, Pub. L. 105-383.

(2908)

§165.115 Safety and Security Zones; Pilgrim Nuclear Power Plant, Plymouth, Massachusetts.

(2909) (a) *Location*. All waters of Cape Cod Bay and land adjacent to those waters enclosed by a line beginning at position

(2910) 41°56'59.3"N., 70°34'58.5"W.; thence to

(2911) 41°57'12.2"N., 70°34'41.9"W.; thence to

(2912) 41°56'42.3"N., 70°34'00.1"W.; thence to

(2913) 41°56'29.5"N., 70°34'14.5"W.

(2914) (b) *Regulations*. (1) In accordance with the general regulations in §§165.23 and 165.33 of this part, entry into or movement within these zones is prohibited unless authorized by the Captain of the Port Boston.

(2915) (2) All vessel operators shall comply with the instructions of the COTP or the designated on-scene U.S. Coast Guard patrol personnel. On-scene Coast Guard patrol personnel include commissioned, warrant, and petty officers of the Coast Guard on board Coast Guard, Coast Guard Auxiliary, local, state, and federal law enforcement vessels.

(2916) (3) No person may enter the waters or land area within the boundaries of the safety and security zones unless previously authorized by the Captain of the Port, Boston or his authorized patrol representative.

(2917)

§165.121 Safety and Security Zones: High Interest Vessels, Narragansett Bay, Rhode Island.

(2918) (a) *Location*. (1) All waters of Rhode Island Sound within a ½ mile radius of any high interest vessel while the vessel is anchored within ½ mile of the point 41°25'N., 71°23'W. in the Narragansett Bay Precautionary Area.

(2919) (2) All waters of Rhode Island Sound, Narragansett Bay, the Providence and Taunton Rivers 2 miles ahead and 1 mile astern, and extending 1000 yards on either side of any high interest vessel transiting Narragansett Bay, or the Providence and Taunton Rivers.

(2920) (3) All waters and land within a 1000-yard radius of any high interest vessel moored at a waterfront facility in the Providence Captain of the Port zone.

(2921) (b) *High interest vessels defined*. For purposes of this section, high interest vessels operating in the Providence Captain of the Port zone include the following: barges or ships carrying liquefied petroleum gas (LPG), liquefied natural gas (LNG), chlorine, anhydrous ammonia, or any other cargo deemed to be high interest by the Captain of the Port, Providence.

(2922) (c) *Regulations*. (1) Entry into or movement within these zones, including below the surface of the water, during times in which high interest vessels are present and the zones are enforced is prohibited unless authorized by the COTP Providence or authorized representative.

(2923) (2) The general regulations covering safety and security zones in §§165.23 and 165.33, respectively, of this part apply.

(2924) (3) All persons and vessels shall comply with the instructions of the COTP, and the designated on-scene U.S. Coast Guard personnel. On-scene Coast Guard patrol personnel include commissioned, warrant, and petty officers of the Coast Guard on board Coast Guard, Coast Guard Auxiliary, local, state, and federal law enforcement vessels.

(2925)

§165.122 Regulated Navigation Area: Navigable waters within Narragansett Bay and the Providence River, Rhode Island.

(2926) (a) *Description of the regulated navigation area (RNA)*. The Regulated Navigation Area (RNA) encompasses all of the navigable waters of Narragansett Bay north of the COLREGS demarcation line and west of the Mt. Hope Bridge, and all of the navigable waters of the Providence River from Conimicut Point to the Providence hurricane barrier.

(2927) (b) *Regulations*. (1) All commercial vessels must:

(2928) (i) Maintain a minimum 10% of the vessel's draft as an under-keel clearance when not assisted by tugs, or when not moored at an assigned berth. Under-keel clearance is the minimum clearance available between the deepest point on the vessel and the bottom of the waterway, in calm water.

(2929) (ii) Have at least one mile of visibility to transit the Providence River between 41°43'01.4"N.; 71°20'41.7"W. (Conimicut Light (LLNR 18305)) and 41°47'38.8"N.; 71°22'46.7"W. (Channel Light 42 (LLNR 18580)).

(2930) (2) Vessels over 65 feet in length inbound for berths in the Providence River are required to make Safety Signal (SECURITE) calls on both VHF channels 13 and 16 at the following geographic locations:

(2931) (i) Pilot Boarding Area;

(2932) (ii) Abeam of Castle Hill;

(2933) (iii) Abeam of Sandy Point;

(2934) (iv) Abeam of 41°43'01.4"N.; 71°20'41.7"W. (Conimicut Light (LLNR 18305));

(2935) (v) Abeam of Sabin Point; and

(2936) (vi) Upon mooring.

(2937) (3) Vessels over 65 feet in length inbound for berths in Mount Hope Bay or in the Taunton River are required to make SECURITE calls on both VHF channels 13 and 16 at the following geographic locations:

(2938) (i) Pilot Boarding Area;

(2939) (ii) Abeam of Castle Hill;

(2940) (iii) Abeam of Sandy Point; and

(2941) (iv) At position 41°39'32.4"N.; 71°14'02.6"W. (Mount Hope Bay Junction Lighted Gong Buoy "MH" (LLNR 18790)).

(2942) (4) Vessels over 65 feet in length outbound for sea down the Providence River Channel shall make SECURITE calls on VHF channels 13 and 16 at the following geographic locations:

(2943) (i) One-half hour prior to departure from the berth;

(2944) (ii) At departure from the berth;

(2945) (iii) Abeam of Sabin Point;

(2946) (iv) Abeam of Gaspee Point; and

(2947) (v) Abeam of position 41°43'01.4"N.; 71°20'41.7"W. (Conimicut Light (LLNR 18305)).

(2948) (5) Vessels over 65 feet in length outbound for sea down from Mount Hope Bay through Narragansett Bay are required to make SECURITE calls on VHF channels 13 and 16 at the following geographic locations:

(2949) (i) One-half hour prior to departure from the berth;

(2950) (ii) At departure from the berth; and

(2951) (iii) At position 41°39'32.4"N.; 71°14'02.6"W. (Mount Hope Bay Junction Lighted Gong Buoy "MH" (LLNR 18790)).

(2952) (6) Vessels 65 feet and under in length, and all recreational vessels, when meeting deep draft commercial vessel traffic in all locations within this RNA shall keep out of the way of the oncoming deep draft commercial vessel. Nothing in this regulation, however, relieves a vessel of any duty prescribed in the Inland Navigation Rules (33 CFR subchapter E).

(2953) (7) The Captain of the Port (COTP) Southeastern New England may authorize a deviation from these regulations. Parties wishing to request a deviation must do so in advance by contacting the COTP Southeastern New England, at 508-457-3211, or via VHF channel 13 (156.7 MHz), or VHF channel 16 (156.8 MHz). Any person or vessel receiving permission from the COTP to deviate from these regulations must comply with any specific instructions provided by the COTP.

(2954) (c) *Enforcement*. Violations of this RNA should be reported to the COTP Southeastern New England at 508-457-3211. Persons found in violation of these regulations may be subject to civil or criminal penalties as provided for in 46 U.S.C. 70036.

(2955)

§165.123 Cruise Ships, Sector Southeastern New England Captain of the Port (COTP) Zone.

(2956) (a) *Location*. The following areas are security zones: All navigable waters within the Southeastern New England Captain of the Port (COTP) Zone, extending from the surface to the sea floor:

(2957) (1) Within a 200-yard radius of any cruise ship that is underway and is under escort of U.S. Coast Guard law enforcement personnel or designated representative, or

(2958) (2) Within a 100-yard radius of any cruise ship that is anchored, at any berth or moored.

(2959) (b) *Definitions*. For the purposes of this section—

(2960) *Cruise ship* means a passenger vessel as defined in 46 U.S.C. 2101(22), that is authorized to carry more than 400 passengers and is 200 or more feet in length. A cruise ship under this section will also include ferries as defined in 46 CFR 2.10-25 that are authorized to carry more than 400 passengers and are 200 feet or more in length.

(2961) *Designated representative* means any Coast Guard commissioned, warrant, or petty officer who has been designated by the COTP to act on the COTP's behalf. The designated representative may be on a Coast Guard vessel, or onboard Federal, state, or a local agency vessel that is authorized to act in support of the Coast Guard.

(2962) *Southeastern New England COTP Zone* is as defined in 33 CFR 3.05–20.

(2963) (c) *Enforcement*. The security zones described in this section will be activated and enforced upon entry of any cruise ship into the navigable waters of the United States (see 33 CFR 2.36(a) to include the 12 NM territorial sea) in the Southeastern New England COTP zone. This zone will remain activated at all times while a cruise ship is within the navigable waters of the United States in the Sector Southeastern New England COTP Zone. In addition, the Coast Guard may broadcast the area designated as a security zone for the duration of the enforcement period via Broadcast Notice to Mariners.

(2964) (d) *Regulations*. (1) In accordance with the general regulations in 33 CFR part 165, subpart D, no person or vessel may enter or move within the security zones created by this section unless granted permission to do so by the COTP Southeastern New England or the designated representative.

(2965) (2) All persons and vessels granted permission to enter a security zone must comply with the instructions of the COTP or the designated representative. Emergency response vessels are authorized to move within the zone, but must abide by the restrictions imposed by the COTP or the designated representative.

(2966) (3) No person may swim upon or below the surface of the water within the boundaries of these security zones unless previously authorized by the COTP or his designated representative.

(2967) (4) Upon being hailed by a U.S. Coast Guard vessel or the designated representative, by siren, radio, flashing light or other means, the operator of the vessel shall proceed as directed.

(2968) (5) Vessel operators desiring to enter or operate within the security zone shall contact the COTP or the designated representative via VHF channel 16 or 508–457–3211 (Sector Southeastern New England command center) to obtain permission to do so.

(2969)

§165.125 Regulated Navigation Area; EPA Superfund Site, New Bedford Harbor, Massachusetts.

(2970) (a) *Location*. The regulated navigation area encompasses all waters bounded by a line beginning at

(2971) 41°37′22.5″N., 70°54′34.1″W.; thence to

(2972) 41°37′14.4″N., 70°54′19.6″W.; thence to

(2973) 41°36′58.5″N., 70°54′08.1″W.; thence to

(2974) 41°36′45.0″N., 70°54′26.9″W.; thence along the shoreline and south side of the hurricane barrier to the beginning point.

(2975) (b) *Regulations*. (1) All vessels and persons are prohibited from activities that would disturb the seabed within the regulated navigation area, including but not limited to anchoring, dragging, trawling, and spudding. Vessels may otherwise transit or navigate within this area without reservation.

(2976) (2) The prohibition described in paragraph (b)(1) of this section shall not apply to vessels or persons engaged

in activities associated with remediation efforts in the New Bedford Harbor Superfund Site, provided that the Coast Guard Captain of the Port (COTP) Southeastern New England, is given advance notice of those activities by the U.S. Environmental Protection Agency (EPA).

(2977) (c) *Waivers*. The Captain of the Port (COTP) Southeastern New England may, in consultation with the U.S. EPA, authorize a waiver from this section if he or she determines that the proposed activity can be performed without undue risk to environmental remediation efforts. Requests for waivers should be submitted in writing to Commander, U.S. Coast Guard Sector Southeastern New England, 1 Little Harbor Road, Woods Hole, MA, 02543, with a copy to the U.S. Environmental Protection Agency, Region 1, New Bedford Harbor Remedial Project Manager, 5 Post Office Square, Suite 100 (OSRR07), Boston, MA 02109, to facilitate review by the EPA and U.S. Coast Guard.

(2978)

§165.130 [Removed]

(2979)

§165.150 New Haven Harbor, Quinnipiac River, Mill River.

(2980) (a) *Boundaries*. The following is a regulated navigation area: The waters surrounding the Tomlinson Bridge and Pearl Harbor Memorial Bridge (I–95 Bridge) located within a line extending from a point A at 41°17′50.35″N, 072°54′34.37″W (the southeast corner of the Magellan Pink Tanks Terminal dock) thence along a line 126°T to point B at 41°17′42.35″N, 072°54′19.37″W (the southwest corner of the Gulf facility) thence north along the shoreline to point C at 41°17′57.35″N, 072°54′04.37″W (the northwest corner of the R & H Terminal dock) thence along a line 303°T to point D at 41°18′05.35″N, 072°54′21.37″W (the west bank of the mouth of the Mill River) thence south along the shoreline to point of origin. All coordinates are North American Datum 1983.

(2981) (b) *Regulations*. (1) No person may operate a vessel or tow a barge in this Regulated Navigation Area in violation of these regulations.

(2982) (2) Applicability. The regulations apply to barges with a freeboard greater than ten feet and to any vessel towing or pushing these barges on outbound transits of the Tomlinson Bridge.

(2983) (3) Regulated barges may not transit the bridge–

(2984) (i) During the period from one hour to five hours after high water slack,

(2985) (ii) When the wind speed at the bridge is greater than twenty knots, and

(2986) (iii) With the barge being towed on a hawser, stern first.

(2987) (4) Regulated barges with a beam greater than fifty feet must be pushed ahead through the bridge.

(2988) (5) If the tug operator does not have a clear view over the barge when pushing ahead, the operator shall post a lookout on the barge with a means of communication with the operator.

(2989) (6) Regulated barges departing the Mill River may transit the bridge only between sunrise and sunset. Barges must be pushed ahead of the tug, bow first, with a second tug standing by to assist at the bow.

(2990) (7) Nothing in this section is intended to relieve any person from complying with:

(2991) (i) Applicable Navigation and Pilot Rules for Inland Waters;

(2992) (ii) Any other laws or regulations;

(2993) (iii) Any order or direction of the Captain of the Port.

(2994) (8) The Captain of the Port Sector Long Island Sound (COTP) may issue an authorization to deviate from any regulation in paragraph (b) of this section if the COTP determines that an alternate operation can be done safely.

(2995) (9) The COTP may temporarily close the RNA for any situation the COTP determines would create an imminent hazard to waterway users in the RNA. Entry into the RNA during temporary closure is prohibited unless authorized by the COTP or the COTP's designated representative. The COTP or designated representative may order the removal of any vessel or equipment within the RNA. To assure wide advance notice of each closure among affected mariners, the COTP may use means including, but not limited to, Broadcast Notice to Mariners and Local Notice to Mariners. The COTP will announce the dates and times of the closure and whether exceptions will be authorized for emergency or other specific vessel traffic.

(2996)

§165.153 Regulated Navigation Area: Long Island Sound Marine Inspection and Captain of the Port Zone.

(2997) (a) *Regulated Navigation Area location*. All waters of the Long Island Sound Marine Inspection and Captain of the Port (COTP) Zone, as delineated in 33 CFR 3.05-35, extending seaward 12 nautical miles from the territorial sea baseline, are established as a regulated navigation area (RNA).

(2998) (b) *Applicability*. This section applies to all vessels operating within the RNA excluding public vessels.

(2999) (c) *Definitions*. The following definitions apply to this section:

(3000) *Commercial service* means any type of trade or business involving the transportation of goods or individuals, except service performed by a combatant vessel.

(3001) *Ferry* means a vessel that:

(3002) (1) Operates in other than ocean or coastwise service;

(3003) (2) Has provisions only for deck passengers or vehicles, or both;

(3004) (3) Operates on a short run on a frequent schedule between two points over the most direct water route; and

(3005) (4) Offers a public service of a type normally attributed to a bridge or tunnel.

(3006) *Public vessels* means vessels owned or bareboat chartered and operated by the United States, or by a State or political subdivision thereof, or by a foreign nation, except when such vessel is engaged in commercial service.

(3007) *Territorial sea baseline* means the line defining the shoreward extent of the territorial sea of the United States drawn according to the principles, as recognized by the United States, of the Convention on the Territorial Sea and the Contiguous Zone, 15 U.S.T. 1606, and the 1982 United Nations Convention on the Law of the Sea (UNCLOS), 21 I.L.M. 1261. Normally, the territorial sea baseline is the mean low water line along the coast of the United States.

(3008) (d) *Regulations*. (1) Speed restrictions in the vicinity of Naval Submarine Base New London and Lower Thames River. Unless authorized by the Captain of the Port (COTP), vessels of 300 gross tons or more may not proceed at a speed in excess of eight knots in the Thames River from New London Harbor channel buoys 7 and 8 (Light List numbers 21875 and 21880 respectively) north through the upper limit of the Naval Submarine Base New London Restricted Area, as that area is specified in 33 CFR 334.75(a). The U.S. Navy and other Federal, State and municipal agencies may assist the U.S. Coast Guard in the enforcement of this rule.

(3009) (2) *Enhanced communications*. Vessels of 300 gross tons or more and all vessels engaged in towing barges must issue securité calls on marine band or Very High Frequency (VHF) radio channel 16 upon approach to the following locations:

(3010) (i) Inbound approach to Cerberus Shoal; and

(3011) (ii) Outbound approach to Race Rock Light (USCG Light List No. 19815).

(3012) (3) All vessels operating within the RNA that are bound for a port or place located in the United States or that must transit the internal waters of the United States, must be inspected to the satisfaction of the U.S. Coast Guard, before entering waters within three nautical miles from the territorial sea baseline. Vessels awaiting inspection will be required to anchor in the manner directed by the COTP. This section does not apply to vessels operating exclusively within the Long Island Sound Marine Inspection and COTP Zone, vessels on single voyage which depart from and return to the same port or place within the RNA, all towing vessels engaged in coastwise trade, vessels in innocent passage not bound for a port or place subject to the jurisdiction of the United States, and all vessels not engaged in commercial service whose last port of call was in the United States. Vessels requiring inspection by the COTP may contact the COTP via marine band or Very High Frequency (VHF) channel 16, telephone at (203) 468-4401, facsimile at (203) 468-4418, or letter, addressed to Captain of the Port, Long Island Sound, 120 Woodward Ave., New Haven, CT 06512.

(3013) (4) All vessels operating within the RNA that are bound for a port or place located in the United States or that must transit the internal waters of the United States, must obtain authorization from the Captain of the Port (COTP) before entering waters within three nautical miles from the territorial sea baseline. Vessels awaiting COTP authorization to enter waters within three nautical miles from the territorial sea baseline will be required to anchor in the manner directed by the COTP. This section does not apply to vessels operating exclusively within the Long Island Sound Marine Inspection and COTP Zone, vessels on a single voyage which depart from and return to the same port or place within the RNA, all towing vessels engaged in coastwise trade, vessels in innocent passage not bound for a port or place subject to the jurisdiction of the United States, and all vessels not engaged in commercial service whose last port of call was in the United States. Vessels may request authorization from the COTP by contacting the COTP via marine band or Very High Frequency (VHF) channel 16, telephone at 203-468-4401, facsimile at (203) 468-4418, or letter addressed to Captain of the Port, Long Island Sound, 120 Woodward Ave., New Haven, CT 06512.

(3014) (5) Vessels over 1,600 gross tons operating in the RNA within three nautical miles from the territorial sea baseline that are bound for a port or place located in the United States or that must transit the internal waters of the United States must receive authorization from the COTP prior to transiting or any intentional vessel movements, including, but not limited to, shifting berths, departing anchorage, or getting underway from a mooring. This section does not apply to vessels in innocent passage not bound for a port or place subject to the jurisdiction of the United States.

(3015) (6) *Ferry vessels*. Vessels of 300 gross tons or more are prohibited from entering all waters within a 1200-yard radius of any ferry vessel transiting in any portion of the Long Island Sound Marine Inspection and COTP Zone without first obtaining the express prior authorization of the ferry vessel operator, master, COTP, or the designated COTP on-scene patrol.

(3016) (7) *Vessels engaged in commercial service*. No vessel may enter within a 100-yard radius of any vessel engaged in commercial service while that vessel is transiting, moored, or berthed in any portion of the Long Island Sound Marine Inspection and COTP zone without the express prior authorization of the vessel's operator, master, COTP, or the designated COTP on-scene representative.

(3017) (8) *Bridge foundations*. Any vessel operating beneath a bridge must make a direct, immediate and expeditious passage beneath the bridge while remaining within the navigable channel. No vessel may stop, moor, anchor or loiter beneath a bridge at any time. No vessel may approach within a 25-yard radius of any bridge foundation, support, stanchion, pier or abutment except as required for the direct, immediate and expeditious transit beneath a bridge.

(3018) (9) This section does not relieve any vessel from compliance with applicable Navigation Rules

(COLREGS and their associated Annexes and Inland Navigation Rules (33 CFR subchapter E)).

(3019)

§165.154 Safety and Security Zones; Captain of the Port Long Island Sound Zone Safety and Security Zones.

(3020) The following areas are designated safety and security zones:

(3021) (a) *Security zones.*

(3022) (1) Dominion Millstone Nuclear Power Plant, Waterford, CT.

(3023) (i) All navigable waters of Long Island Sound, from surface to bottom, North and Northeast of a line running from Bay Point, at approximate position 41°18'34.20"N, 072°10'24.60" W, to Millstone Point at approximate position 41°18'15.00"N, 072°9'57.60" W (NAD 83).

(3024) (ii) All navigable waters of Long Island Sound, from surface to bottom, West of a line starting at 41°18'42"N, 072°09'39"W, running south to the Eastern most point of Fox Island at approximate position 41°18'24.11"N, 072°09'39.73"W (NAD 83).

(3025) (2) Electric Boat Shipyard, Groton, CT.

(3026) (i) *Location*. All navigable waters of the Thames River, from surface to bottom, West of the Electric Boat Corporation Shipyard enclosed by a line beginning at a point on the shoreline at

(3027) 41°20'16"N, 72°04'47"W; then running West to

(3028) 41°20'16.2"N, 72°04'58"W; then running North to

(3029) 41°20′28.7″N, 72°05′01.7″W; then North-Northwest to

(3030) 41°20′53.3″N, 72°05′04.8″W; then North-Northeast to

(3031) 41°21′02.9″N, 72°05′04.9″W; then running to shoreline at

(3032) 41°21′02.9″N, 72°04′58.2″W (NAD 83).

(3033) (ii) *Application*. Paragraphs (a),(e), (f) of § 165.33 do not apply to public vessels or to vessels owned by, under hire to, or performing work for the Electric Boat Division when operating in the security zone.

(3034) (3) Naval Submarine Base New London, Groton, CT. All navigable waters of the Thames River, from surface to bottom, West of Naval Submarine Base New London, Groton, CT, enclosed by a line beginning at a point on the shoreline at

(3035) 41°23′07.9″N, 72°05′13.7″W; then to

(3036) 41°23′07.9″N, 72°05′16.9″ W; then to

(3037) 41°23′26.4″N, 72°05′30.7″W; then to

(3038) 41°23′42.9″N, 72°05′40.1″W; then to

(3039) 41°23′46.7″N, 72°05′42.3″W; then to

(3040) 41°23′53.9″N, 72°05′44.5″W; then to

(3041) 41°24′08.7″N, 72°05′44.5″W; then to

(3042) 41°24′16.2″N, 72°05′43.4″W; then to a point on the shoreline

(3043) 41°24′16.2″N, 72°05′36.4″W; then along the shoreline to the point of beginning (NAD 83).

(3044) (4) U.S. Coast Guard Academy, New London, CT.

(3045) (i) *Location*. All navigable waters of the Thames River, from surface to bottom, in a 500-yard radius from Jacobs Rock, approximate position 41°22'22"N, 072°05'40"W (NAD 83).

(3046) (ii) *Enforcement period*. This rule will be enforced during visits by highranking officials and times of heighted security.

(3047) (iii) *Notification*. The Captain of the Port will notify the maritime community of periods during which this security zone will be enforced by all appropriate means such as Local Notice to Mariners, Marine Safety Information Radio Broadcasts or on scene notice.

(3048) (5) U.S. Coast Guard Vessels, Long Island Sound COTP Zone. All navigable waters within a 100-yard radius of any anchored U.S. Coast Guard vessel. For the purposes of this section, U.S. Coast Guard vessels includes any commissioned vessel or small boat in the service of the regular U.S. Coast Guard and does not include Coast Guard Auxiliary vessels.

(3049) (b) *Safety zones*. (1) Coast Guard Station Fire Island, Long Island, NY. All waters of Fire Island Inlet from the shore out to a line beginning at a point on shore at

(3050) 40°37'31.4"N, 073°15'41.1"W; then North to

(3051) 40°37'35.6"N, 073°15'43.1"W; then East to

(3052) 40°37'36.7"N, 073°15'39.8"W; then East to

(3053) 40°37'37.8"N, 073°15'36.6"W; then East to

(3054) 40°37'41.1"N, 073°15'33.5"W; then Southeast to

(3055) 40°37'39.7"N, 073°15'27.0"W; then Southeast to

(3056) 40°37'37.5"N, 073°15'22.1"W; then Southeast to

(3057) 40°37'37.6"N, 073°15'19.1"W; then Southeast to point on shore at

(3058) 40°37'33.9"N, 073°15'20.8"W (NAD 83).

(3059) (2) [Reserved]

(3060) (c) *Regulations*. (1) The general regulations contained in §165.23 and §165.33 of this part apply. Entering into, remaining within or cause an article or thing to enter into or remain within these safety and security zones is prohibited unless authorized by the Captain of the Port or a designated representative.

(3061) (2) These safety and security zones are closed to all vessel traffic, except as may be permitted by the Captain of the Port (COTP) or a designated representative. Vessel operators given permission to enter or operate in the security zones must comply with all directions given to them by the COTP or the designated representative.

(3062) (3) The ''designated representative'' is any Coast Guard commissioned, warrant or petty officer who has been designated by the Captain of the Port to act on his/her behalf. The on-scene representative may be on a Coast Guard vessel, a state or local law enforcement vessel, or other designated craft, or may be on shore and will communicate with vessels via VHF–FM radio or loudhailer. In addition, members of the Coast Guard Auxiliary may be present to inform vessel operators of this regulation.

(3063) (4) Vessel operators desiring to enter or operate within the security zones shall request permission to do so

by contacting the Captain of the Port Sector Long Island Sound at 203–468–4401, or via VHF Channel 16.

(3064)

§165.156 Regulated Navigation Area, East Rockaway Inlet to Atlantic Beach Bridge, Nassau County, Long Island, New York.

(3065) (a) *Location*. The following area is a Regulated Navigation Area; All waters of East Rockaway Inlet in an area bounded by lines drawn from position 40°34'56"N., 073°45'19"W., (approximate position of East Rockaway Inlet Breakwater Light, LLNR 31500) running north to a point of land on the northwest side of the inlet at position 40°35'28"N., 073°46'12"W., thence easterly along the shore to the east side of the Atlantic Beach Bridge, State Route 878, over East Rockaway Inlet, thence across the bridge to the south side of East Rockaway Inlet, thence westerly along the shore and across the water to the beginning.

(3066) (b) *Regulations*. (1) The general regulations contained in 33 CFR 165.10, 165.11 and 165.13 apply.

(3067) (2) In accordance with the general regulations, the following regulations apply to vessels carrying petroleum products in excess of 250 barrels:

(3068) (i) The vessel must have plans in place to maintain a minimum of two feet under keel clearance at all times.

(3069) (ii) A vessel requiring a nighttime transit through East Rockaway Inlet may only do so only after receiving approval from the Captain of the Port Long Island Sound.

(3070) (iii) Vessels are prohibited from transiting East Rockaway Inlet if a small craft advisory or greater has been issued for the area unless specific approval is received from the Captain of the Port Long Island Sound.

(3071) (iv) In an emergency, any vessel may deviate from the regulations in this section to the extent necessary to avoid endangering the safety of persons, the environment, and or property. If deviation from the regulations is necessary, the master or their designee shall inform the Coast Guard as soon as it is practicable to do so.

(3072) (c) *Waivers*. (1) The Captain of the Port Long Island Sound may, upon request, waive any regulation in this section.

(3073) (2) An application for a waiver must state the need for the waiver and describe the proposed vessel operations through the Regulated Navigation Area.

(3074)

§165.164 Security Zones; Dignitary Arrival/Departure and United Nations Meetings, New York, NY.

(3075) (a) *Location*. The following areas are security zones:

(3076) (1) *Wall Street Heliport*. All waters of the East River within the following boundaries: East of a line drawn between approximate position 40°42′01″N, 074°00′39″W (east of The Battery) to 40°41′36″N, 074°00′52″W (point north of Governors Island) and north of a line drawn from the point north of Governors Island to the southwest corner of Pier 7 North, Brooklyn; and south of a line drawn between 40°42′14.8″N, 074°00′20.3″W (Wall Street, Manhattan), and the northwest corner of Pier 2 North, Brooklyn (NAD 1983).

(3077) (2) *Randalls and Wards Islands*: All waters of the East River between the Hell Gate Rail Road Bridge (mile 8.2), and a line drawn from a point at approximate position 40°47′27.12″N, 073°54′35.14″W (Lawrence Point, Queens) to a point at approximate position 40°47′52.55″N, 073°54′35.25″W (Port Morris Stacks), and all waters of the Bronx Kill southeast of the Bronx Kill Rail Road Bridge (mile 0.6) (NAD 1983).

(3078) (3) *Marine Air Terminal, LaGuardia Airport Security Zone*. All waters of Bowery Bay, Queens, New York, inside of a line drawn from the start of the Rikers Island Bridge in Queens at approximate position 40°46′37″ N, 073°53′30″ W to the intersecting point on the southern side of Rikers Island at approximate position 40°47′12″ N, 073°53′06″ W, then a line drawn east to the western end of LaGuardia Airport at approximate position 40°47′00″ N, 073°52′44″ W, then a line drawn south following the shoreline back to the point of origin at 40°46′37″ N, 073°53′30″ W (NAD 1983).

(3079) (4) *United Nations Manhattan Shoreline*. All waters of the East River bound by the following points: 40°44′37″N, 073°58′16.5″ W (the base of East 35th Street, Manhattan), then east to 40°44′34.5″N, 073°58′10.5″W (about 180 yards offshore of Manhattan), then northeasterly to 40°45′29″N, 073°57′26.5″W (about 125 yards offshore of Manhattan at the Queensboro Bridge), then northwesterly to 40°45′31″N, 073°57′30.5″W (Manhattan shoreline at the Queensboro Bridge), then southerly along the shoreline to the starting point at 40°44′37″N, 073°58′16.5″W (NAD 1983).

(3080) (5) *United Nations West Channel Closure*. All waters of the East River north of a line drawn from approximate position 40°44′37″N, 073°58′16.5″W (the base of East 35th Street, Manhattan), to approximate position 40°44′31.04″N, 073°58′03.10″W (approximately 400 yards east of the Manhattan shoreline), all waters west of a line drawn from approximate position 40°44′31.04″N, 073°58′03.10″W (approximately 400 yards east of the Manhattan shoreline), to the southern tip of Roosevelt Island at approximate position 40°44′57.96″N, 073°57′41.57″W, then along the western shoreline of Roosevelt Island to the Queensboro Bridge, and all waters south of the Queensboro Bridge (NAD 1983).

(3081) (6) *United Nations Full River Closure*. All waters of the East River north of a line drawn from approximate position 40°44′37″N, 073°58′16.5″W (the base of East 35th Street, Manhattan), to approximate position 40°44′23″N, 073°57′44.5″W (Hunters Point, Long Island City), and south of the Queensboro Bridge (NAD 1983).

(3082) (b) *Definitions*.

(3083) As used in this section—

(3084) *Designated representative* means any Coast Guard commissioned, warrant, or petty officer who has been designated by the COTP to act on the COTP's behalf. A designated representative may be on a Coast Guard

vessel, or onboard a federal, state, or local agency vessel that is authorized to act in support of the Coast Guard.

(3085) *Dignitary* means the President or Vice President of the United States, or visiting heads of foreign states or governments.

(3086) (c) *Regulations*. In accordance with the general regulations in 33 CFR 165.33, no person or vessel may enter or move within a security zone created by this section while that security zone is being enforced unless granted permission to do so by the Coast Guard's First District Commander, the COTP, or a designated representative. Vessel operators and persons given permission to enter or operate in a security zone must comply with all directions given to them by the COTP, or a designated representative. Upon being hailed by a U.S. Coast Guard or other duly authorized law enforcement vessel (e.g. New York City police) by siren, radio, flashing lights, or other means, the operator of a vessel must proceed as directed, and follow any instructions to anchor or moor up to a waterfront facility.

(3087) (d) *Enforcement periods*. The security zone described in paragraph (a)(4) of this section is subject to enforcement at all times. All other security zones established by this section will only be enforced when necessary to protect dignitaries as determined by the COTP.

(3088) (e) *Notification*. Because the security zone described in paragraph (a)(4) of this section is subject to enforcement at all times, the Coast Guard will not necessarily take any action to further notify the public about the enforcement of that zone. As for the enforcement periods for the other security zones contained herein, the Coast Guard will rely on the methods described in 33 CFR 165.7 to notify the public of the time and duration of any enforcement period. The COTP may also notify the public about enforcement of these security zones via *http://homeport.uscg.mil/newyork*.

(3089) (f) *Contact information*. Vessel operators desiring to enter or operate within a security zone shall telephone the COTP at 718–354–4356 or a designated representative via VHF channel 16 to obtain permission to do so.

(3090)

§165.165 Regulated Navigation Area; Hudson River South of the Troy Locks, NY.

(3091) (a) *Regulated navigation area*. All navigable waters of the Hudson River south of the Troy Locks.

(3092) (b) *Definitions*. The following definitions apply to this section:

(3093) (1) *Designated representative* means any Coast Guard commissioned, warrant, or petty officer, or a Federal, State, or local law enforcement officer designated by or assisting the Captain of the Port (COTP) New York.

(3094) (2) *Horsepower (HP)* means the total maximum continuous shaft horsepower of all the vessel's main propulsion machinery.

(3095) (c) *Applicability*. This section applies to tugs with less than 3,000 horsepower when engaged in towing operations.

(3096) (d) *Regulations*. (1) Except as provided in paragraph (c)(3) of this section, vessels less than 3,000 horsepower while engaged in towing operations are not authorized to transit that portion of the Hudson River south of the Troy Locks when ice thickness on average is eight inches or greater.

(3097) (2) All Coast Guard assets enforcing this Regulated Navigation Area can be contacted on VHF marine band radio, channel 13 or 16. The COTP can be contacted at (718) 354-4356, and the public may contact the COTP to suggest changes or improvements in the terms of this Regulated Navigation Area.

(3098) (3) All persons desiring to transit through a portion of the regulated area that has operating restrictions in effect must contact the COTP at telephone number (718) 354-4356 or on VHF channel 13 or 16 to seek permission prior to transiting the affected regulated area.

(3099) (4) The COTP will notify the public of any changes in the status of this Regulated Navigation Area by Marine Safety Information Broadcast on VHF-FM marine band radio, channel 1022 (157.1 MHz).

(3100)

§165.169 Safety and Security Zones: New York Marine Inspection Zone and Captain of the Port Zone.

(3101) (a) *Safety and security zones*. The following waters within the New York Marine Inspection Zone and Captain of the Port Zone are safety and security zones:

(3102) (1) *Indian Point Nuclear Power Station (IPNPS).* All waters of the Hudson River within 300-yard radius of the IPNPS pier in approximate position 41°16'12.4"N., 073°57'16.2"W. (NAD 83).

(3103) (2) *U.S. Coast Guard Cutters and Shore Facilities*. All waters within 100 yards of: Each moored, or anchored, Coast Guard Cutter; Coast Guard Station New York; Staten Island, NY; Coast Guard Station Sandy Hook, NJ; Coast Guard Station Kings Point, NY; and Coast Guard Aids to Navigation Team New York, Bayonne, NJ.

(3104) (3) *Part 105 Facilities*. (i) *Definition*. For the purposes of this section, *Part 105 Facility* means any facility subject to the regulations contained in 33 CFR part 105, including those designated as "Public Access Facilities" as defined in 33 CFR 101.105. For public identification purposes, all of these facilities are required to have signs posted along the shoreline, facing the water, indicating that there is a 25 yard waterfront security zone surrounding the facilities.

(3105) (ii) *Location*. All waters within 25 yards of each Part 105 Facility. When a barge, ferry, or other commercial vessel is conducting transfer operations at a Part 105 Facility, the 25-yard zone is measured from the outboard side of the commercial vessel.

(3106) (iii) *Regulations*. (A) Vessels not actively engaged in passenger, cargo, provision, facility maintenance, or inspection, bunker transfer operations, or docking or undocking operations, authorized in advance by the Facility Security Plan, Facility Security Officer or designated representative, must not enter within any part

of a zone described in paragraph (a)(3) of this section without the express permission of the Cost Guard Captain of the Port, a designated representative or designated on-scene patrol personnel.

(3107) (B) Persons seeking Captain of the Port permission to enter within a particular zone for official business other than authorized passenger, cargo, provision, facility maintenance or inspection, bunker transfer operations or authorized docking or undocking operations may request such authorization by contacting: Commander Coast Guard Sector New York, via the Sector Command Center (SCC), at: 212 Coast Guard Drive, Staten Island, NY 10305 or via fax to (718) 354-4125 or by contacting the Sector Command Center Duty Officer by phone at: (718) 354-4353.

(3108) (C) Vessels may transit through any portion of the zone that extends into the navigable channel for the sole purpose of direct and expeditious transit so long as they as they remain within the navigable channel, maintain the maximum safe distance from the Part 105 Facility, and do not stop or loiter within the zone.

(3109) (4) *Liberty and Ellis Islands*—(i) Location. All waters within 150 yards of Liberty Island and Ellis Island, and the Ellis Island Bridge.

(3110) (ii) *Ellis Island Bridge*. In addition to any person or vessel authorized pursuant to paragraph (b) of this section, vessels may transit underneath the Ellis Island Bridge subject to the following conditions:

(3111) (A) Dates/Times: On weekends only, to include Federally Observed Holidays on a Friday or Monday, from Memorial Day Weekend through October 1 each year, between one hour after sunrise and one hour before sunset.

(3112) (B) Vessel types: Human powered vessels equal to or less than twenty five feet. Human powered vessels must be able to safely navigate under the bridge.

(3113) (C) Notification: Human powered vessels desiring to transit shall contact the United States Park Police Command Center at 212–363–3260 or VHF CH 13 regarding intentions of passage prior to entering the safety and security zone and transiting under the Ellis Island Bridge.

(3114) (D) Route: Transits through the safety and security zone and under the bridge shall occur only at the designated route marked with lights and signage.

(3115) (E) Passage: Vessels transiting under the Ellis Island Bridge shall make expeditious passage and not stop or loiter within the safety and security zone.

(3116) (iii) *Enforcement period*. The safety and security zone described in this subsection is effective at all times. Although certain vessels have permission to enter the safety and security zone to transit under the Ellis Island Bridge subject to the conditions outlined in paragraphs (a)(4)(ii)(A)–(E) of this section, the safety and security zone is in effect permanently and can be enforced at any time. When deemed necessary the COTP may rescind the permission granted in paragraphs (a)(4)(ii)(A)–(E) of this section for any period of time.

(3117) (5) *Bridge Piers and Abutments, Overhead Power Cable Towers, Piers and Tunnel Ventilators*. All waters within 25 yards of any bridge pier or abutment, overhead power cable tower, pier or tunnel ventilators south of the Troy, NY Locks. Vessels may transit through any portion of the zone that extends into the navigable channel for the sole purpose of direct and expeditious transit through the zone so long as they remain within the navigable channel, maintain the maximum safe distance from the waterfront facility and do not stop or loiter within the zone.

(3118) (6) *New York City Passenger Ship Terminal, Hudson River, NY*. (i) *Location*. All navigable waters of the Hudson River bound by the following points: From the point 40°46'09"N., 073°59'48.7"W. on the seawall midway between Pier 92 and 94, thence northwest to approximate position 40°46'14"N., 074°00'00.9"W., approximately 125 yards northwest of Pier 92, thence southwest to approximate position 40°45'56.7"N., 074°00'15.3"W., approximately 150 yards west of Pier 86, thence east to the seawall between Pier 84 and Pier 86 at the approximate position 40°45'49.6"N., 073°59'58.1"W. (NAD 1983), thence northeast along the shoreline to the point of origin.

(3119) (ii) *Regulations*. Vessels not actively engaged in passenger, cargo, provision, facility maintenance or inspection, bunker transfer operations, or docking or undocking operations, authorized in advance by the Facility Security Plan, Facility Security Officer or designated representative, must not enter within any part of a zone described in paragraph (a)(6) of this section without express permission of the Coast Guard Captain of the Port, a designated representative or designated on-scene patrol personnel. Persons seeking Captain of the Port permission to enter within the zone described in paragraph (a)(6) of this section for official business other than authorized passenger, cargo, provision, facility maintenance or inspection, bunker transfer operations or authorized docking or undocking operations may request such authorization by contacting: Commander Coast Guard Sector New York, via the Sector Command Center (SCC), at: 212 Coast Guard Drive, Staten Island, NY 10305, or via fax to (718) 354-4125 or by contacting the Sector Command Center Duty Officer by phone at (718) 354-4353.

(3120) (7) *La Guardia Airport, Bowery and Flushing Bays, Queens, NY*. (i) *Location: 200-Yard Zone*. All waters of Bowery and Flushing Bays within approximately 200 yards of La Guardia Airport bound by the following points: Onshore at Steinway, Queens in approximate position

(3121) 40°46'32.1"N., 073°53'22.4"W., thence to

(3122) 40°46'52.8"N., 073°53'09.3"W., thence to

(3123) 40°46'54.8"N., 073°52'54.2"W., thence to

(3124) 40°46'59.3"N., 073°52'51.3"W., thence to

(3125) 40°47'11.8"N., 073°53'17.3"W., thence to

(3126) 40°47'13.0"N., 073°53'16.1"W. on Rikers Island, thence easterly along the Rikers Island shoreline to approximate position

(3127) 40°47'12.9"N., 073°52'17.9"W., thence to
(3128) 40°47'16.7"N., 073°52'09.2"W., thence to
(3129) 40°47'36.1"N., 073°51'52.5"W., thence to
(3130) 40°47'35.1"N., 073°51'50.5"W., thence to
(3131) 40°47'15.9"N., 073°52'06.4"W., thence to
(3132) 40°47'14.5"N., 073°52'03.1"W., thence to
(3133) 40°47'10.6"N., 073°52'06.7"W., thence to
(3134) 40°47'01.9"N., 073°52'02.4"W., thence to
(3135) 40°46'50.4"N., 073°52'08.1"W., thence to
(3136) 40°46'26.8"N., 073°51'18.5"W., thence to
(3137) 40°45'57.2"N., 073°51'01.8"W., thence to
(3138) 40°45'51.2"N., 073°50'59.6"W., thence to
(3139) 40°45'49.5"N., 073°51'07.2"W., thence to
(3140) 40°45'58.8"N., 073°51'13.2"W., thence to
(3141) 40°46'02.3"N., 073°51'20.1"W., thence to
(3142) 40°45'48.4"N., 073°51'37.0"W., (NAD 1983) thence along the shoreline to the point of origin.

(3143) (ii) *Location: 100-Yard Zone*. All waters of Bowery and Flushing Bays within approximately 100 yards of La Guardia Airport bound by the following points: Onshore at Steinway, Queens in approximate position
(3144) 40°46'32.1"N., 073°53'22.4"W., thence to
(3145) 40°46'50.6"N., 073°53'07.3"W., thence to
(3146) 40°46'53.0"N., 073°52'50.9"W., thence to
(3147) 40°46'57.6"N., 073°52'47.9"W., thence to
(3148) 40°47'11.8"N., 073°53'17.3"W., thence to
(3149) 40°47'13.0"N., 073°53'16.1"W., on Rikers Island, thence easterly along the Rikers Island shoreline to approximate position
(3150) 40°47'12.9"N., 073°52'17.9"W., thence to
(3151) 40°47'16.7"N., 073°52'09.2"W., thence to
(3152) 40°47'36.1"N., 073°51'52.5"W., thence to
(3153) 40°47'35.1"N., 073°51'50.5"W., thence to
(3154) 40°47'15.9"N., 073°52'06.4 15.9" 5 % 9(9(9(M(7'1 .9"N., 073°52'06.4 0° approximate position

of this section will be communicated by the Captain of the Port to the public by marine broadcast, local notice to mariners, or notice posted at *http://www.homeport.uscg.mil/newyork*.

(3206) (9) *NYPD Ammunition Depot, Rodman Neck, Eastchester Bay, NY*. (i) *Location: 150-Yard Zone*. All waters of Eastchester Bay within approximately 150 yards of Rodman Neck bound by the following points: Onshore in approximate position

(3207) 40°51'30.4"N., 073°48'14.9"W., thence to

(3208) 40°51'29.9"N., 073°48'20.7"W., thence to

(3209) 40°51'16.9"N., 073°48'22.5"W., thence to

(3210) 40°51'07.5"N., 073°48'18.7"W., thence to

(3211) 40°50'54.2"N., 073°48'11.1"W., thence to

(3212) 40°50'48.5"N., 073°48'04.6"W., thence to

(3213) 40°50'49.2"N., 073°47'56.5"W., thence to

(3214) 40°51'03.6"N., 073°47'47.3"W., thence to

(3215) 40°51'15.7"N., 073°47'46.8"W., thence to

(3216) 40°51'23.5"N., 073°47'41.9"W., (NAD 1983) thence southwesterly along the shoreline to the point of origin.

(3217) (ii) *Enforcement period*. The zones described in paragraph (a)(9) of this section will be effective at all times. When port security conditions permit, the Captain of the Port will allow vessels to operate within that portion of the waters described in paragraph (a)(9)(i) of this section that lies outside of the waters described in paragraph (a)(9)(ii) of this section. Authorization to enter the waters that lie between the outer boundaries of the zones described in paragraphs (a)(9)(i) and (a)(9)(ii) of this section will be communicated by the Captain of the Port to the public by marine broadcast, local notice to mariners, or notice posted at *http://www.homeport.uscg.mil/newyork*.

(3218) (10) *Port Newark/Port Elizabeth, Newark Bay, NJ*. All waters of Newark Bay bound by the following points:

(3219) 40°41'49.9"N., 074°07'32.2"W., thence to

(3220) 40°41'46.5"N., 074°07'20.4"W., thence to

(3221) 40°41'10.7"N., 074°07'45.9"W., thence to

(3222) 40°40'54.3"N., 074°07'55.7"W., thence to

(3223) 40°40'36.2"N., 074°08'03.8"W., thence to

(3224) 40°40'29.1"N., 074°08'06.3"W., thence to

(3225) 40°40'21.9"N., 074°08'10.0"W., thence to

(3226) 40°39'27.9"N., 074°08'43.6"W., thence to

(3227) 40°39'21.5"N., 074°08'50.1"W., thence to

(3228) 40°39'21.5"N., 074°09'54.3"W., (NAD 1983) thence northerly along the shoreline to the point of origin.

(3229) (11) *Global Marine Terminal, Upper New York Bay*. All waters of Upper New York Bay between the Global Marine and Military Ocean Terminals, west of the New Jersey Pierhead Channel.

(3230) (12) *Approaches to New York, Atlantic Ocean*. (i) *Location*: All waters of the Atlantic Ocean between Ambrose and Hudson Canyon Traffic Lane and the Barnegat to Ambrose Traffic lane bound by the following points:

(3231) 40°21'29.9"N., 073°44'41.0"W., thence to

(3232) 40°21'04.5"N., 073°45'31.4"W., thence to

(3233) 40°15'28.3"N., 073°44'13.8"W., thence to

(3234) 40°15'35.4"N., 073°43'29.8"W., thence to

(3235) 40°19'21.2"N., 073°42'53.0"W., (NAD 1983) thence to point of origin.

(3236) (ii) *Enforcement period*. Enforcement periods for the zone in paragraph (a)(12) of this section will be announced through marine information broadcast or other appropriate method of communication and the zone is activated whenever a vessel is anchored in the area described in paragraph (a)(12)(i) or a Coast Guard patrol vessel is on-scene.

(3237) (iii) *Regulations*. (A) The area described in paragraph (a)(12) of this section is not a Federal Anchorage Ground. Only vessels directed by the Captain of the Port or his or her designated representative to enter this zone are authorized to anchor here.

(3238) (B) Vessels do not need permission from the Captain of the Port to transit the area described in paragraph (a)(12) of this section during periods when that security zone is not being enforced.

(3239) (13) *Liquefied Hazardous Gas (LHG) Vessels*. (i) *Definitions*. For the purposes of this section, *LHG Vessel* means any vessel constructed to carry, in bulk, any of the flammable or toxic products listed in 33 CFR 127.005, Table 127.005.

(3240) (ii) *Location*. All waters within a 200-yard radius of any LHG Vessel that is underway and all waters within a 100-yard radius of any LHG vessel that is moored or at anchor.

(3241) (iii) *Enforcement period*. The zone described in paragraph (a)(13) of this section will be activated upon entry of a LHG Vessel into the navigable waters of the United States (see 33 CFR 2.36(a) to include the 12 NM territorial sea) in the New York Captain of the Port Zone (33 CFR 3.05-30). The LHG Vessel will be identifiable by the requirement to fly the Bravo flag (red international signal flag under Pub. 102, International Code of Signals) from the outermost halyard (above the pilot house) where it can most easily be seen. In addition to visual identification of the LHG Vessel, the Captain of the Port will notify the maritime community of periods during which this zone will be enforced by methods in accordance with 33 CFR 165.7.

(3242) (14) *Cruise ships*. (i) *Definition*. For the purposes of this section, *cruise ship* means a passenger vessel as defined in 46 U.S.C. 2101 (22) that is authorized to carry more than 400 passengers and is 200 or more feet in length. A cruise ship under this section will also include ferries as defined in 46 CFR 2.10-25 that are authorized to carry more than 400 passengers and are 200 feet or more in length.

(3243) (ii) *Location*. All waters within a 100-yard radius of any cruise ship whether underway, anchored, or at berth.

(3244) (iii) *Enforcement period*. The zone described in paragraph (a)(14) of this section will be activated upon entry of any cruise ship into the navigable waters of the United States (See 33 CFR 2.36(a) to include the 12 NM territorial sea) in the New York Captain of the Port Zone (33 CFR 3.05-30). This zone will remain activated at all

times while the cruise ship is within the navigable waters of the United States in the New York Captain of the Port Zone.

(3245) (15) *Designated Vessels*. (i) *Definition*. For the purposes of this section, *Designated Vessels* are vessels carrying government officials, dignitaries, or other passengers requiring protection by the U.S. Secret Service, or other Federal, State or local law enforcement agency; barges or ships carrying petroleum products, chemicals, or other hazardous cargo; and passenger vessels (as defined in 46 U.S.C. 2101(22), that are authorized to carry more than 400 passengers and are less than 200 feet in length.

(3246) (ii) *Location*. All waters within a 100-yard radius of any Designated Vessel.

(3247) (iii) *Enforcement period*. The zone described in paragraph (a)(15) of this section will be activated upon entry of any Designated Vessel into the navigable waters of the United States (see 33 CFR 2.36(a) to include the 12 NM territorial sea) in the New York Captain of the Port Zone (33 CFR 3.05-30). This zone will remain activated at all times while the Designated Vessel is within the navigable waters of the United States in the New York Captain of the Port Zone. The Designated Vessels, including ships and barges carrying petroleum products, chemicals, and other hazardous cargo will be recognized by the requirement to fly the Bravo flag (red international signal flag under Pub 102, International Code of Signals) from the outermost halyard (above the pilot house) where it can be most easily seen. Vessels that are constrained by draft from leaving the channel may transit through the zone for the sole purpose of direct and expeditious transit so long as they remain within the navigable channel, maintain the maximum possible safe distance from the Designated Vessel, and do not stop or loiter within the zone. Designated Vessels carrying government officials, dignitaries, or other passengers requiring protection, and passenger vessels authorized to carry more than 400 passengers and are less than 200 feet in length will be recognizable by their being escorted by a federal, state or local law enforcement or security vessel. The law enforcement or security vessel will be identifiable by flashing light, siren, flags, markings and/or through other means that clearly identify the vessel as engaged in law enforcement or security operations.

(3248) (16) *134th Street Pipeline Metering and Regulating Station*. (i) *Location*. All waters of the Hudson River within 25 yards of the 134th Street Pipeline Metering and Regulating Station.

(3249) (ii) *Regulations*. (A) Vessels not actively engaged in facility maintenance or inspection operations authorized in advance by the Pipeline Security Officer or designated representative, or authorized docking or undocking operations, must not enter within any part of a zone described in paragraph (a)(16) of this section without the express permission of the Coast Guard Captain of the Port, a designated representative or designated on-scene patrol personnel.

(3250) (B) Persons seeking Captain of the Port permission to enter within a particular zone for official business other than authorized passenger, cargo, provision, facility maintenance or inspection, bunker transfer operations or authorized docking or undocking operations may request such authorization by contacting: Commander Coast Guard Sector New York, via the Sector Command Center (SCC), at: 212 Coast Guard Drive, Staten Island, NY 10305, or via fax to (718) 354-4125 or by contacting the Sector Command Center Duty Officer by phone at: (718) 354-4353.

(3251) (17) *Waterfront heliports*. (i) *Location*. All waters on the East River within 25 yards of the East 34th Street and Wall Street Heliports, and all waters of the Hudson River within 25 yards of the West 30th Street Heliport and the Jersey City/Newport Helistop, areas of land or water under and in immediate proximity to them; buildings on such structures or contiguous to them; and equipment and materials on such structures and in such buildings. When a barge, ferry, or other commercial vessel is conducting transfer operations at a waterfront heliport, the 25-yard zone is measured from the outboard side of the commercial vessel.

(3252) (ii) *Regulations*. (A) Vessels not actively engaged in passenger, cargo, provision, facility maintenance or inspection, bunker transfer operations, or docking and undocking operations, authorized in advance by the Facility Security Plan, Facility Security Officer or designated representative, must not enter within any part of a zone described in paragraph (a)(17) of this section without the express permission of the Coast Guard Captain of the Port, a designated representative, or designated on-scene patrol personnel.

(3253) (B) Persons seeking Captain of the Port permission to enter within a particular zone for official business other than authorized passenger, cargo, provision, facility maintenance or inspection, bunker transfer operations or authorized docking and undocking operations may request such authorization by contacting: Commander Coast Guard Sector New York, via the Sector Command Center (SCC), at: 212 Coast Guard Drive, Staten Island, NY 10305, or via fax to (718) 354-4125 or by contacting the Sector Command Center Duty Officer by phone at: (718) 354-4353.

(3254) (C) Vessels entering or departing the marina north of the Newport Helistop are authorized to transit through the safety/security zone around the Newport Helistop during their transit, provided that helicopters are not taking off or landing. No loitering or unnecessary delay is authorized during these transits.

(3255) (b) *Regulations*. (1) Entry into or remaining in a safety or security zone in prohibited unless authorized by the Coast Guard Captain of the Port, New York.

(3256) (2) Persons desiring to transit the area of a safety or security zone may contact the Captain of the Port at 718-354-4088 or on VHF channel 14 (156.7 MHz) or VHF channel 16 (156.8 MHz) to seek permission to transit the area. If permission is granted, all persons and vessels

must comply with the instructions of the Captain of the Port or his or her designated representative.

(3257)

§165.170 [Removed]

(3258)

§165.172 Safety Zone; Underwater Hazard, Gravesend Bay, Brooklyn, NY.

(3259) (a) *Location*. The following area is a safety zone: All navigable waters of Gravesend Bay within a 110-yard radius of a point in position 40°36'30"N., 74°02'14"W. (NAD 83), approximately 70-yards southeast of the Verrazano Bridge Brooklyn tower.

(3260) (b) *Regulations*. (1) The general regulations contained in 33 CFR 165.23 apply.

(3261) (2) Entering into, transiting through, diving, dredging, dumping, fishing, trawling, conducting salvage operations, remaining within or anchoring within this safety zone is prohibited unless authorized by the Captain of the Port (COTP) New York or the designated on-scene representative.

(3262) (3) The "designated on-scene representative" is any Coast Guard commissioned, warrent, or petty officer who has been designated by the COTP New York.

(3263) (4) Vessel operators desiring to enter or operate within the safety zone may contact the COTP New York or the designated representative at the Coast Guard Sector New York Command Center via VHF Channel 16 or by phone at (718) 354-4353 to request permission.

(3264) (5) Vessel operators given permission to enter or operate in the safety zone must comply with all directions given to them by the COTP New York or the on-scene representative.

(3265)

Subpart G-Protection of Naval Vessels

(3266)

§165.2010 Purpose.

(3267) This subpart establishes the geographic parameters of naval vessel protection zones surrounding U.S. naval vessels in the navigable waters of the United States. This subpart also establishes when the U.S. Navy will take enforcement action in accordance with the statutory guideline of 14 U.S.C. 91. Nothing in the rules and regulations contained in this subpart shall relieve any vessel, including U.S. naval vessels, from the observance of the Navigation Rules. The rules and regulations contained in this subpart supplement, but do not replace or supercede, any other regulation pertaining to the safety or security of U.S. naval vessels.

(3268)

§165.2015 Definitions.

(3269) The following definitions apply to this subpart:

(3270) *Atlantic Area* means that area described in 33 CFR 3.04–1 Atlantic Area.

(3271) *Large U.S. naval vessel* means any U.S. naval vessel greater than 100 feet in length overall.

(3272) *Naval defensive sea area* means those areas described in 32 CFR part 761.

(3273) *Naval vessel protection zone* is a 500-yard regulated area of water surrounding large U.S. naval vessels that is necessary to provide for the safety or security of these U.S. naval vessels.

(3274) *Navigable waters of the United States* means those waters defined as such in 33 CFR part 2.

(3275) *Navigation rules* means the Navigation Rules, International-Inland.

(3276) *Official patrol* means those personnel designated and supervised by a senior naval officer present in command and tasked to monitor a naval vessel protection zone, permit entry into the zone, give legally enforceable orders to persons or vessels within the zone, and take other actions authorized by the U.S. Navy.

(3277) *Pacific Area* means that area described in 33 CFR 3.04–3 Pacific Area.

(3278) *Restricted area* means those areas established by the Army Corps of Engineers and set out in 33 CFR part 334.

(3279) *Senior naval officer present in command* is, unless otherwise designated by competent authority, the senior line officer of the U.S. Navy on active duty, eligible for command at sea, who is present and in command of any part of the Department of Navy in the area.

(3280) *U.S. naval vessel* means any vessel owned, operated, chartered, or leased by the U.S. Navy; any pre-commissioned vessel under construction for the U.S. Navy, once launched into the water; and any vessel under the operational control of the U.S. Navy or a Combatant Command.

(3281) *Vessel* means every description of watercraft or other artificial contrivance used, or capable of being used, as a means of transportation on water, except U.S. Coast Guard or U.S. naval vessels.

(3282)

§165.2020 Enforcement authority.

(3283) (a) *Coast Guard*. Any Coast Guard commissioned, warrant or petty officer may enforce the rules and regulations contained in this subpart.

(3284) (b) *Senior naval officer present in command*. In the navigable waters of the United States, when immediate action is required and representatives of the Coast Guard are not present or not present in sufficient force to exercise effective control in the vicinity of large U.S. naval vessels, the senior naval officer present in command is responsible for the enforcement of the rules and regulations contained in this subpart to ensure the safety and security of all large naval vessels present. In meeting this responsibility, the senior naval officer present in command may directly assist any Coast Guard enforcement personnel who are present.

(3285)

§165.2025 Atlantic Area.

(3286) (a) This section applies to any vessel or person in the navigable waters of the United States within the boundaries of the U.S. Coast Guard Atlantic Area, which

includes the First, Fifth, Seventh, Eighth and Ninth U.S. Coast Guard Districts.

(3287) **Note to §165.2025 paragraph** (a): The boundaries of the U.S. Coast Guard Atlantic Area and the First, Fifth, Seventh, Eighth and Ninth U.S. Coast Guard Districts are set out in 33 CFR part 3.

(3288) (b) A naval vessel protection zone exists around U.S. naval vessels greater than 100 feet in length overall at all times in the navigable waters of the United States, whether the large U.S. naval vessel is underway, anchored, moored, or within a floating dry dock, except when the large naval vessel is moored or anchored within a restricted area or within a naval defensive sea area.

(3289) (c) The Navigation Rules shall apply at all times within a naval vessel protection zone.

(3290) (d) When within a naval vessel protection zone, all vessels shall operate at the minimum speed necessary to maintain a safe course, unless required to maintain speed by the Navigation Rules, and shall proceed as directed by the Coast Guard, the senior naval officer present in command, or the official patrol. When within a naval vessel protection zone, no vessel or person is allowed within 100 yards of a large U.S. naval vessel unless authorized by the Coast Guard, the senior naval officer present in command, or official patrol.

(3291) (e) To request authorization to operate within 100 yards of a large U.S. naval vessel, contact the Coast Guard, the senior naval officer present in command, or the official patrol on VHF-FM channel 16.

(3292) (f) When conditions permit, the Coast Guard, senior naval officer present in command, or the official patrol should:

(3293) (1) Give advance notice on VHF-FM channel 16 of all large U.S. naval vessel movements;

(3294) (2) Permit vessels constrained by their navigational draft or restricted in their ability to maneuver to pass within 100 yards of a large U.S. naval vessel in order to ensure a safe passage in accordance with the Navigation Rules; and

(3295) (3) Permit commercial vessels anchored in a designated anchorage area to remain at anchor when within 100 yards of passing large U.S. naval vessels; and

(3296) (4) Permit vessels that must transit via a navigable channel or waterway to pass within 100 yards of a moored or anchored large U.S. naval vessel with minimal delay consistent with security.

(3297) **Note to §165.2025 paragraph** (f): The listed actions are discretionary and do not create any additional right to appeal or otherwise dispute a decision of the Coast Guard, the senior naval officer present in command, or the official patrol.

(3298)

Part 166–Shipping Safety Fairways

(3299)

Subpart A–General

(3300)

§166.100 Purpose.

(3301) The purpose of these regulations is to establish and designate shipping safety fairways and fairway anchorages to provide unobstructed approaches for vessels using U.S. ports.

(3302)

§166.103 Geographic Coordinates.

(3303) Geographic coordinates expressed in terms of latitude or longitude, or both, are not intended for plotting on maps or charts whose referenced horizontal datum is the North American Datum of 1983 (NAD 83), unless such geographic coordinates are expressly labeled NAD 83. Geographic coordinates without the NAD 83 reference may be plotted on maps or charts referenced to NAD 83 only after application of the appropriate corrections that are published on the particular map or chart being used.

(3304)

§166.105 Definitions.

(3305) (a) *Shipping safety fairway* or *fairway* means a lane or corridor in which no artificial island or fixed structure, whether temporary or permanent, will be permitted. Temporary underwater obstacles may be permitted under certain conditions described for specific areas in Subpart B. Aids to navigation approved by the U.S. Coast Guard may be established in a fairway.

(3306) (b) *Fairway anchorage* means an anchorage area contiguous to and associated with a fairway, in which fixed structures may be permitted within certain spacing limitations, as described for specific areas in Subpart B.

(3307)

§166.110 Modification of areas.

(3308) Fairways and fairway anchorages are subject to modification in accordance with 33 U.S.C. 1223(c); 92 Stat. 1473.

(3309)

§166.500 Areas along the Atlantic Coast.

(3310) (a) *Purpose*. Fairways, as described in this section are established to control the erection of structures therein to provide safe vessel routes along the Atlantic Coast.

(3311) (b) *Designated areas*.–(1) *Off New York Shipping Safety Fairway*. (i) Ambrose to Nantucket Safety Fairway. The area enclosed by rhumb lines, [North American Datum of 1927 (NAD-27)], joining points at:

(3312) 40°32'20"N., 73°04'57"W.

(3313) 40°30'58"N., 72°58'25"W.

(3314) 40°34'07"N., 70°19'23"W.

(3315) 40°35'37"N., 70°14'09"W.

(3316) 40°30'37"N., 70°14'00"W.

(3317) 40°32'07"N., 70°19'19"W.

(3318) 40°28'58"N., 72°58'25"W.
(3319) 40°27'20"N., 73°04'57"W.
(3320) (ii) *Nantucket to Ambrose Safety Fairway*. The area enclosed by rhumb lines, NAD-27, joining points at:
(3321) 40°24'20"N., 73°04'58"W.
(3322) 40°22'58'N., 72°58'26"W.
(3323) 40°26'07"N., 70°19'09"W.
(3324) 40°27'37"N., 70°13'46"W.
(3325) 40°22'37"N., 70°13'36"W.
(3326) 40°24'07"N., 70°19'05"W.
(3327) 40°20'58"N., 72°58'26"W.
(3328) 40°19'20"N., 73°04'58"W.

(3329)

Part 167–Offshore Traffic Separation Schemes

(3330)

Subpart A-General

(3331)

§167.1 Purpose.

(3332) The purpose of the regulations in this part is to establish and designate traffic separation schemes and precautionary areas to provide access routes for vessels proceeding to and from U.S. ports.

(3333)

§167.3 Geographic coordinates.

(3334) Geographic coordinates are defined using North American 1927 Datum (NAD 27) unless indicated otherwise.

(3335)

§167.5 Definitions.

(3336) (a) *Area to be avoided* means a routing measure comprising an area within defined limits in which either navigation is particularly hazardous or it is exceptionally important to avoid casualties and which should be avoided by all ships or certain classes of ships.
(3337) (b) *Traffic separation scheme (TSS)* means a designated routing measure which is aimed at the separation of opposing streams of traffic by appropriate means and by the establishment of traffic lanes.
(3338) (c) *Traffic lane* means an area within defined limits in which one-way traffic is established. Natural obstacles, including those forming separation zones, may constitute a boundary.
(3339) (d) *Separation zone or line* means a zone or line separating the traffic lanes in which ships are proceeding in opposite or nearly opposite directions; or separating a traffic lane from the adjacent sea area; or separating traffic lanes designated for particular classes of ships proceeding in the same direction.
(3340) (e) *Precautionary area* means a routing measure comprising an area within defined limits where ships must navigate with particular caution and within which the direction of traffic flow may be recommended.
(3341) (f) *Deep-water route* means an internationally recognized routing measure primarily intended for use by ships that, because of their draft in relation to the available depth of water in the area concerned, require the use of such a route.
(3342) (g) *Two-way route* means a route within defined limits inside which two-way traffic is established, aimed at providing safe passage of ships through waters where navigation is difficult or dangerous.

(3343)

§167.10 Operating rules.

(3344) The operator of a vessel in a TSS shall comply with Rule 10 of the International Regulations for Preventing Collisions at Sea, 1972, as amended.

(3345)

§167.15 Modification of schemes.

(3346) (a) A traffic separation scheme or precautionary area described in this part may be permanently amended in accordance with 33 U.S.C. 1223 (92 Stat. 1473), and with international agreements.
(3347) (b) A traffic separation scheme or precautionary area in this part may be temporarily adjusted by the Commandant of the Coast Guard in an emergency, or to accommodate operations which would create an undue hazard for vessels using the scheme or which would contravene Rule 10 of the International Regulations for Preventing Collisions at Sea, 1972. Adjustment may be in the form of a temporary traffic lane shift, a temporary suspension of a section of the scheme, a temporary precautionary area overlaying a lane, or other appropriate measure. Adjustments will only be made where, in the judgment of the Coast Guard, there is no reasonable alternative means of conducting an operation and navigation safety will not be jeopardized by the adjustment. Notice of adjustments will be made in the appropriate Notice to Mariners and in the Federal Register. Requests by members of the public for temporary adjustments to traffic separation schemes must be submitted 150 days prior to the time the adjustment is desired. Such Requests, describing the interference that would otherwise occur to a TSS, should be submitted to the District Commander of the Coast Guard District in which the TSS is located.

(3348)

Subpart B–Description of Traffic Separation Schemes and Precautionary Areas

(3349)

§167.100 In the approaches to Narragansett Bay, RI, and Buzzards Bay, MA: General.

(3350) The traffic separation scheme in the approaches to Narragansett Bay, RI, and Buzzards Bay, MA, consists of four parts: Two precautionary areas and two approaches—a Narragansett approach and a Buzzards Bay approach. The specific areas in the approaches to Narragansett Bay, RI, and Buzzards Bay, MA, are described in §§167.101 through 167.103. The geographic coordinates in §§167.101 through 167.103 are defined

using North American Datum 1983 (NAD 83), which is equivalent to WGS 1984 datum.

(3351)

§167.101 In the approaches to Narragansett Bay, RI, and Buzzards Bay, MA: Precautionary areas.

(3352) (a) A precautionary area is established with a radius of 5.4 miles and is centered upon geographical position 41°06.00′N., 71°23.30′W.

(3353) (b) A precautionary area is established with a radius of 3.55 miles and is centered upon geographical position 41°25.60′N., 71°23.30′W.

(3354)

§167.102 In the approaches to Narragansett Bay, RI, and Buzzards Bay, MA: Narragansett Bay approach.

(3355) (a) A separation zone 2 miles wide is established and is centered upon the following geographical positions:

(3356) 41°22.70′N., 71°23.30′W.

(3357) 41°11.10′N., 71°23.30′W.

(3358) (b) A traffic lane 1 mile wide is established on each side of the separation zone.

(3359)

§167.103 In the approaches to Narragansett Bay, RI, and Buzzards Bay, MA: Buzzards Bay approach.

(3360) (a) A separation zone 1 mile wide is established and is centered upon the following geographical positions:

(3361) 41°10.20′N., 71°19.10′W.

(3362) 41°21.80′N., 71°07.10′W.

(3363) (b) A traffic lane 1 mile wide is established on each side of the separation zone.

(3364) **Note to §167.103:** A restricted area, 2 miles wide, extending from the southern limit of the Narragansett Bay approach traffic separation zone to latitude 41°24.70’N., has been established. The restricted area will only be closed to ship traffic by the Naval Underwater System Center during periods of daylight and optimum weather conditions for torpedo range usage. The closing of the restricted area will be indicated by the activation of a white strobe light mounted on Brenton Reef Light and controlled by a naval ship supporting the torpedo range activities. There would be no ship restrictions expected during inclement weather or when the torpedo range is not in use.

(3365)

§167.150 Off New York Traffic Separation Scheme: General.

(3366) The specific areas in the Off New York Traffic Separation Scheme and Precautionary Areas are described in §§167.151, 167.152, 167.153, 167.154, and 167.155 of this chapter.

(3367)

§167.151 Off New York: Precautionary areas.

(3368) (a) A circular precautionary area with a radius of 7 miles is established centered upon 40°27.50'N., 73°49.90'W.

(3369) (b) A precautionary area is established between the traffic separation scheme “Eastern Approach, off Nantucket” and the traffic separation schemes “In the Approach to Boston, Massachusetts.” (1) The precautionary area is bounded to the east by a circle of radius 15.5 miles, centered upon geographic position:

(3370) 40°35.00'N., 69°00.00'W., and is intersected by the traffic separation schemes “In the Approach to Boston, Massachusetts” and “Off New York” at the following geographic positions:

(3371) 40°50.33'N., 68°57.00'W.

(3372) 40°23.75'N., 69°14.63'W.

(3373) (2) The precautionary area is bounded to the west by a line connecting the two traffic separation schemes between the following geographic positions:

(3374) 40°36.75'N., 69°15.16'W.

(3375) 40°48.00'N., 69°03.33'W.

(3376)

§167.152 Off New York: Eastern approach, off Nantucket.

(3377) (a) A separation zone is established bounded by a line connecting the following geographic positions:

(3378) 40°28.75'N., 69°14.83'W.

(3379) 40°27.62'N., 70°13.77'W.

(3380) 40°30.62'N., 70°14.00'W.

(3381) 40°31.75'N., 69°14.97'W.

(3382) (b) A traffic lane for westbound traffic is established between the separation zone and a line connecting the following geographic positions:

(3383) 40°36.75'N., 69°15.17'W.

(3384) 40°35.62'N., 70°14.15'W.

(3385) (c) A traffic lane for eastbound traffic is established between the separation zone and a line connecting the following geographic positions:

(3386) 40°22.62'N., 70°13.60'W.

(3387) 40°23.75'N., 73°14.63'W.

(3388)

§167.153 Off New York: Eastern approach.

(3389) (a) A separation zone is established bounded by a line connecting the following geographic positions:

(3390) 40°24.33'N., 73°04.97'W.

(3391) 40°24.20'N., 73°11.50'W.

(3392) 40°26.00'N., 73°40.93'W.

(3393) 40°27.00'N., 73°40.75'W.

(3394) 40°27.20'N., 73°11.50'W.

(3395) 40°27.33'N., 73°04.95'W.

(3396) (b) A traffic lane for westbound traffic is established between the separation zone and a line connecting the following geographic positions:

(3397) 40°32.33'N., 73°04.95'W.

(3398) 40°32.20'N., 73°11.50'W.

(3399) 40°28.00'N., 73°40.73'W.

(3400) (c) A traffic lane for eastbound traffic is established between the separation zone and a line connecting the following geographic positions:

(3401) 40°25.05'N., 73°41.32'W.

(3402) 40°19.20'N., 73°11.50'W.

(3403) 40°19.33'N., 73°04.97'W.

(3404)

§167.154 Off New York: Southeastern approach.

(3405) (a) A separation zone is established bounded by a line connecting the following geographic positions:

(3406) 40°03.10'N., 73°17.93'W.

(3407) 40°06.50'N., 73°22.73'W.

(3408) 40°22.45'N., 73°43.55'W.

(3409) 40°23.20'N., 73°42.70'W.

(3410) 40°08.72'N., 73°20.10'W.

(3411) 40°05.32'N., 73°15.28'W.

(3412) (b) A traffic lane for northwest-bound traffic is established between the separation zone and a line connecting the following geographic positions:

(3413) 40°08.98'N., 73°10.87'W.

(3414) 40°12.42'N., 73°15.67'W.

(3415) 40°24.02'N., 73°41.97'W.

(3416) (c) A traffic lane for southeast-bound traffic is established between the separation zone and a line connecting the following geographic positions:

(3417) 40°21.82'N., 73°44.55'W.

(3418) 40°02.80'N., 73°27.15'W.

(3419) 39°59.43'N., 73°22.35'W.

(3420)

§167.155 Off New York: Southern approach.

(3421) (a) A separation zone is established bounded by a line connecting the following geographic positions:

(3422) 39°45.70'N., 73°48.00'W.

(3423) 40°20.63'N., 73°48.33'W.

(3424) 40°20.87'N., 73°47.07'W.

(3425) 39°45.70'N., 73°44.00'W.

(3426) (b) A traffic lane for northbound traffic is established between the separation zone and a line connecting the following geographic positions:

(3427) 39°45.70'N., 73°37.70'W.

(3428) 40°21.25'N., 73°45.85'W.

(3429) (c) A traffic lane for southbound traffic is established between the separation zone and a line connecting the following geographic positions:

(3430) 40°20.53'N., 73°49.65'W.

(3431) 39°45.70'N., 73°54.40'W.

(3432)

Part 169–Ship Reporting Systems

(3433)

Subpart A–General

(3434)

§169.1 What is the purpose of this part?

(3435) This subpart prescribes the requirements for mandatory ship reporting systems. Ship reporting systems are used to provide, gather, or exchange information through radio reports. The information is used to provide data for many purposes including, but not limited to: navigation safety, maritime security and domain awareness, environmental protection, vessel traffic services, search and rescue, weather forecasting and prevention of marine pollution.

(3436) **Note to §169.1**: For ship reporting system requirements not established by the Coast Guard, see 50 CFR Part 404.

(3437)

§169.5 How are terms used in this part defined?

(3438) As used in this part—

(3439) *Administration* means the Government of the State whose flag the ship is entitled to fly.

(3440) *Cargo ship* means any ship which is not a passenger ship.

(3441) *Flag Administration* means the Government of a State whose flag the ship is entitled to fly.

(3442) *Gross tonnage* means tonnage as defined under the International Convention on Tonnage Measurement of Ships, 1969 (Incorporated by reference, see §169.15).

(3443) *Gross tons* means vessel tonnage measured in accordance with the method utilized by the flag state administration of that vessel.

(3444) *High speed craft* means a craft that is operable on or above the water and is capable of a maximum speed equal to or exceeding V=3.7xdispl.1667, where "V" is the maximum speed and "displ" is the vessel displacement corresponding to the design waterline in cubic meters.

(3445) *High speed passenger craft* means a high speed craft carrying more than 12 passengers.

(3446) *International voyage* means a voyage from a country to which the present International Convention for the Safety of Life at Sea (SOLAS), 1974 applies to a port outside such country, or conversely. For U.S. ships, such voyages will be considered to originate at a port in the United States, regardless of when the voyage actually began. Such voyages for U.S. ships will continue until the ship returns to the United States from its last foreign port.

(3447) *Long range identification and tracking (LRIT) information* or *position report* means report containing the following information:

(3448) (1) The identity of the ship;

(3449) (2) The position of the ship (latitude and longitude); and

(3450) (3) The date and time of the position provided.

(3451) *LRIT Data Center* means a center established by a SOLAS Contracting Government or a group of Contracting Governments, or in the case of International Data Center, by IMO, to request, receive, process, and archive LRIT information. An LRIT Data Center may be National, Regional, Co-operative or International.

(3452) *Mandatory ship reporting system* means a ship reporting system that requires the participation of specified vessels or classes of vessels, and that is established by a government or governments after adoption of a proposed system by the International Maritime Organization (IMO) as complying with all requirements of regulation V/8-1 of the International Convention for the Safety of Life at

Sea, 1974, as amended (SOLAS), except paragraph (e) thereof.

(3453) *Mobile offshore drilling unit* means a self-propelled vessel capable of engaging in drilling operations for the exploration or exploitation of subsea resources.

(3454) *Passenger ship* means a ship that carries more than 12 passengers.

(3455) *Self-propelled ships* means ships propelled by mechanical means.

(3456) *Shore-based authority* means the government appointed office or offices that will receive the reports made by ships entering each of the mandatary ship reporting systems. The office or offices will be responsible for the management and coordination of the system, interaction with participating ships, and the safe and effective operation of the system. Such an authority may or may not be an authority in charge of a vessel traffic service.

(3457) *United States* means the States of the United States, the District of Columbia, Guam, Puerto Rico, the Virgin Islands, American Samoa, the Northern Mariana Islands, and any other territory or possession of the United States.

(3458)

§169.10 What geographic coordinates are used?

(3459) Geographic coordinates expressed in terms of latitude or longitude, or both, are not intended for plotting on maps or charts where the referenced horizontal datum is the North American Datum of 1983 (NAD 83), unless such geographic coordinates are expressly labeled NAD 83. Geographic coordinates without the NAD 83 reference may be plotted on maps or charts referenced to NAD 83 only after application of the appropriate corrections that are published on the particular map or chart being used.

(3460)

§169.15 Incorporation by reference: Where can I get a copy of the publications mentioned in this part?

(3461) (a) Certain material is incorporated by reference into this part with the approval of the Director of the Federal Register under 5 U.S.C. 552(a) and 1 CFR part 51. To enforce any edition other than that specified in this section, the Coast Guard must publish notice of change in the **Federal Register** and the material must be available to the public. All approved material is available for inspection at the National Archives and Records Administration (NARA). For information on the availability of this material at NARA, call 202–741–6030 or go to *http://www.archives.gov/federal_register/code_of_federal_regulations/ibr_locations.html*. Also, it is available for inspection at Coast Guard Headquarters. Contact Commandant (CG-NAV), Attn: Office of Navigation Systems, 2703 Martin Luther King Jr. Avenue SE., Stop 7418, Washington, DC 20593-7418, and is available from the sources indicated in this section.

(3462) (b) *International Electrotechnical Commission (IEC) Bureau Central de la Commission Electrotechnique Internationale,* 3 rue de Varembé, P.O. Box 131, 1211 Geneva 20, Switzerland.

(3463) (1) IEC 60945, Fourth edition 2002-08, Maritime navigation and radiocommunication equipment and systems-General requirements-Methods of testing and required test results, incorporation by reference approved for §169.215.

(3464) (2) [Reserved]

(3465) (c) *International Maritime Organization (IMO),* 4 Albert Embankment, London SE1 7SR, U.K.

(3466) (1) IMO Resolution MSC.202(81), adopted on May 19, 2006, Adoption of Amendments to the International Convention for the Safety of Life at Sea, 1974, as Amended, incorporation by reference approved for §160.240.

(3467) (2) IMO Resolution MSC. 210(81), adopted on May 19, 2006, Performance Standards and Functional Requirements for the Long-Range Identification and Tracking of Ships, incorporation by reference approved for §§169.215 and 169.240.

(3468) (3) IMO Resolution MSC.254(83), adopted on October 12, 2007, Adoption of Amendments to the Performance Standards and Functional Requirements for the Long-Range Identification and Tracking of Ships, incorporation by reference approved for §§169.215 and 169.240.

(3469) (4) IMO Resolution A.694(17), adopted on November 6, 1991, General Requirements for Shipborne Radio Equipment Forming Part of the Global Maritime Distress and Safety System (GMDSS) and for Electronic Navigational Aids, incorporation by reference approved for §165.215.

(3470) (5) International Convention on Tonnage Measurement of Ships, 1969, incorporation by reference approved for §169.5.

(3471)

Subpart B–Establishment of Two Mandatory Ship Reporting Systems for the Protection of North Atlantic Right Whales

(3472)

§169.100 What mandatory ship reporting systems are established by this subpart?

(3473) This subpart prescribes requirements for the establishment and maintenance of two mandatory ship reporting systems for the protection of the endangered northern right whale (also known as the North Atlantic right whale). These two systems are designated for certain areas of the East Coast of the United States. One system is located in the northeast and is identified as WHALESNORTH. The other system is located in the southeast and is identified as WHALESSOUTH.

(3474) **Note:** 50 CFR 224.103(c) contains requirements and procedures concerning North Atlantic right whale approach limitations and avoidance procedures.

(3475)

§169.102 Who is the shore-based authority?

(3476) The U.S. Coast Guard is the shore-based authority for these mandatory ship reporting systems.

(3477)

§169.105 Where is the northeastern reporting system located?

(3478) Geographical boundaries of the northeastern area include the waters of Cape Cod Bay, Massachusetts Bay, and the Great South Channel east and southeast of Massachusetts. The coordinates (NAD 83) of the area are as follows: from a point on Cape Ann, Massachusetts at 42°39'N, 70°37'W; then northeast to 42°45'N, 70°13'W; then southeast to 42°10'N, 68°31'W; then south to 41°00'N, 68°31'W; then west to 41°00'N, 69°17'W; then northwest to 42°05'N, 70°02'W; then west to 42°04'N, 70°10'W; and then along the Massachusetts shoreline of Cape Cod Bay and Massachusetts Bay back to the point on Cape Ann at 42°39'N, 70°37'W.

(3479)

§169.110 When is the northeastern reporting system in effect?

(3480) The mandatory ship reporting system in the northeastern United States operates year-round.

(3481)

§169.115 Where is the southeastern reporting system located?

(3482) Geographical boundaries of the southeastern area include coastal waters within about 25 nautical miles (45 kilometers) along a 90-nautical mile (170-kilometer) stretch of the Atlantic seaboard in Florida and Georgia. The area coordinates (NAD 83) extends from the shoreline east to longitude 80°51.6'W with the southern and northern boundaries at latitude 30°00'N and 31°27'N., respectively.

(3483)

§169.120 When is the southeastern reporting system in effect?

(3484) The mandatory ship reporting system in the southeastern United States operates during the period beginning on November 15 each year through April 16 of the following year.

(3485)

§169.125 What classes of ships are required to make reports?

(3486) Each self-propelled ship of 300 gross tons or greater must participate in the reporting systems, except government ships exempted from reporting by regulation V/8–1(c) of SOLAS. However, exempt ships are encouraged to participate in the reporting systems.

(3487)

§169.130 When are ships required to make reports?

(3488) Participating ships must report to the shore-based authority upon entering the area covered by a reporting system. Additional reports are not necessary for movements made within a system or for ships exiting a system.

(3489)

§169.135 How must the reports be made?

(3490) (a) A ship equipped with INMARSAT C must report in IM0 standard format as provided in §169.140 in table 169.140.

(3491) (b) A ship not equipped with INMARSAT C must report to the Coast Guard using other means, listed below in order of precedence–

(3492) (1) Narrow band direct printing (SITOR),

(3493) (2) HF voice communication, or

(3494) (3) MF or VHF voice communications.

(3495) (c) SITOR or HF reports made directly to the Coast Guard's Communications Area Master Station Atlantic (CAMSLANT) in Chesapeake, VA, or MF or VHF reports made to Coast Guard activities or groups, should only be made by ships not equipped with INMARSAT C. Ships in this category must provide all the required information to the Coast Guard watchstander.

(3496)

§169.140 What information must be included in the report?

(3497) Each ship report made to the shore-based authority must follow the standard reporting and format requirements listed in this section in table 169.140. Current email addresses and telex numbers are published annually in the U.S. Coast Pilot.

(3498)

TABLE 169.140 – Requirements for Ship Reports

Telegraphy	Function	Information required
Name of system	System identifier	Ship reporting system WHALESNORTH or WHALESSOUTH
M	INMARSAT Number	Vessel INMARSAT number
A	Ship	The name, call sign or ship station identity, IMO number, and flag of the vessel.
B	Date and time of event	A 6-digit group giving day of month (first two digits), hours and minutes (last four digits)
E	True course	A 3-digit group indicating true course.
F	Speed in knots and tenths of knots	A 3-digit group.
H	Date, time and point of entry into system	Entry time expressed as in (B) and entry position expressed as-(1) a 4-digit group giving latitude in degrees and minutes suffixed with N (north) or S (south) and a 5-digit group giving longitude in degrees and minutes suffixed with E (east) or W (west); or (2) True bearing (first 3 digits) and distance (state distance) in nautical miles from a clearly identified landmark (state landmark)

TABLE 169.140 – Requirements for Ship Reports		
Telegraphy	**Function**	**Information required**
I	Destination and expected time of arrival	Name of port and date time group expressed as in (B)
L	Route information	Intended track.

(3499)

Subpart C–Transmission of Long Range Identification and Tracking Information

(3500)

§169.200 What is the purpose of this subpart?

(3501) This subpart implements Regulation 19-1 of SOLAS Chapter V (SOLAS V/19-1) and requires certain ships engaged on an international voyage to transmit vessel identification and position information electronically. This requirement enables the Coast Guard to obtain long range identification and tracking (LRIT) information and thus heightens our overall maritime domain awareness, enhances our search and rescue operations, and increases our ability to detect anomalies and deter transportation security incidents.

(3502)

§169.205 What types of ships are required to transmit LRIT information (position reports)?

(3503) The following ships, while engaged on an international voyage, are required to transmit position reports:

(3504) (a) A passenger ship, including high speed passenger craft.

(3505) (b) A cargo ship, including high speed craft, of 300 gross tonnage or more.

(3506) (c) A mobile offshore drilling unit while underway and not engaged in drilling operations.

(3507)

§169.210 Where during its international voyage must a ship transmit position reports?

(3508) The requirements for the transmission of position reports, imposed by the United States, vary depending on the relationship of the United States to a ship identified in §169.205.

(3509) (a) *Flag State relationship*. A U.S. flag ship engaged on an international voyage must transmit position reports wherever they are located.

(3510) (b) *Port State relationship*. A foreign flag ship engaged on an international voyage must transmit position reports after the ship has announced its intention to enter a U.S. port or place under requirements in 33 CFR part 160, subpart C.

(3511) (c) *Coastal State relationship*. A foreign flag ship engaged on an international voyage must transmit position reports when the ship is within 1,000 nautical miles of the baseline of the United States, unless their Flag Administration, under authority of SOLAS V/19-1.9.1, has directed them not to do so.

(3512)

§169.215 How must a ship transmit position reports?

(3513) A ship must transmit position reports using Long Range Identification and Tracking (LRIT) equipment that has been type-approved by their Administration. To be type-approved by the Coast Guard, LRIT equipment must meet the requirements of IMO Resolutions A.694(17), MSC.210(81), and MSC.254(83), and IEC standard IEC 60945 (Incorporated by reference, see §169.15).

(3514)

§169.220 When must a ship be fitted with LRIT equipment?

(3515) A ship identified in §169.205 must be equipped with LRIT equipment—

(3516) (a) Before getting underway, if the ship is constructed on or after December 31, 2008.

(3517) (b) By the first survey of the radio installation after December 31, 2008, if the ship is—

(3518) (1) Constructed before December 31, 2008, and

(3519) (2) Operates within—

(3520) (i) One hundred (100) nautical miles of the United States baseline, or

(3521) (ii) Range of an Inmarsat geostationary satellite, or other Application Service Provider recognized by the Administration, with continuous alerting is available.

(3522) (c) By the first survey of the radio installation after July 1, 2009, if the ship is—

(3523) (1) Constructed before December 31, 2008, and

(3524) (2) Operates within the area or range specified in paragraph (b)(2) of this section as well as outside the range of an Inmarsat geostationary satellite with which continuous alerting is available. While operating in the area or range specified in paragraph (b)(2) of this section, however, a ship must install LRIT equipment by the first survey of the radio installation after December 31, 2008.

(3525)

§169.225 Which Application Service Providers may a ship use?

(3526) A ship may use an Application Service Provider (ASP) recognized by its administration. Some Communication Service Providers may also serve as an ASP.

(3527)

§169.230 How often must a ship transmit position reports?

(3528) A ship's LRIT equipment must transmit position reports at 6-hour intervals unless a more frequent interval is requested remotely by an LRIT Data Center.

(3529)

§169.235 What exemptions are there from reporting?

(3530) A ship is exempt from this subpart if it is—

(3531) (a) Fitted with an operating automatic identification system (AIS), under 33 CFR 164.46, and operates only within 20 nautical miles of the United States baseline,

(3532) (b) A warship, naval auxiliaries or other ship owned or operated by a SOLAS Contracting Government and used only on Government non-commercial service, or

(3533) (c) A ship solely navigating the Great Lakes of North America and their connecting and tributary waters as far east as the lower exit of the St. Lambert Lock at Montreal in the Province of Quebec, Canada.

(3534)

§169.240 When may LRIT equipment be switched off?

(3535) A ship engaged on an international voyage may switch off its LRIT equipment only when it is permitted by its Flag Administration, in circumstances detailed in SOLAS V/19-1.7, or in paragraph 4.4.1, of resolution MSC.210(81), as amended by resolution MSC.254(83) (Incorporated by reference, see §169.15).

(3536)

§169.245 What must a ship master do if LRIT equipment is switched off or fails to operate?

(3537) (a) If a ship's LRIT equipment is switched off or fails to operate, the ship's master must inform his or her Flag Administration without undue delay.

(3538) (b) The master must also make an entry in the ship's logbook that states—

(3539) (1) His or her reason for switching the LRIT equipment off, or an entry that the equipment has failed to operate, and

(3540) (2) The period during which the LRIT equipment was switched off or non-operational.

(3541) **Note to §169.245**: for U.S. vessels, the U.S. Coast Guard serves as the Flag Administration for purposes of this section. All LRIT notifications for the U.S. Flag administration, in addition to requests or questions about LRIT, should be communicated to the U.S. Coast Guard by e-mail addressed to *LRIT@uscg.mil*.

(3542)

Part 207–Navigation Regulations

(3543)

§207.20 Cape Cod Canal, Mass.; use, administration, and navigation.

(3544) (a) *Limit of canal*. The canal, including approaches, extends from the Canal Station Minus 100 in Cape Cod Bay, approximately one and six-tenths (1.6) statute miles seaward of the Canal Breakwater Light, through dredged channels and land cuts to Cleveland Ledge Light in Buzzards Bay approximately four (4) statute miles southwest of Wings Neck.

(3545) (b) *Supervision*. (1) The movement of ships, boats and craft of every description through the canal and the operation and maintenance of the waterway and all property of the United States pertaining thereto shall be under the supervision of the Division Engineer, U.S. Army Engineer Division, New England, Corps of Engineers, Waltham, Massachusetts, or the authorized representative of the division engineer, the Engineer-In-Charge of the Cape Cod Canal. The division engineer or the Engineer-In-Charge from time to time will prescribe rules governing the dimensions of vessels which may transit the waterway, and other special conditions and requirements which will govern the movement of vessels using the waterway.

(3546) (2) The Engineer-In-Charge, through the marine traffic controller on duty, will enforce these regulations and monitor traffic through the canal. The marine traffic controller on duty is the individual responsible for interpretation of these regulations with respect to vessels transiting the canal. Vessels transiting the canal must obey the orders of the marine traffic controller.

(3547) (3) The government has tugs stationed at the West Boat Basin for emergency use on an on-call basis. A patrol vessel is manned and operational 24-hours a day.

(3548) (c) *Communications*. There is a marine traffic controller on duty 24 hours a day, seven days a week, in the traffic control center located at the Canal Administrative Office. The primary method of communications between the canal and vessels transiting will be by VHF-FM Marine radio. The traffic controller can also be contacted by telephone.

(3549) (1) For radio communications, call the traffic controller on channel 16 to establish contact. The transmissions will then be switched to channel 12 or 14 as the working channel to pass information. Channel 13 is also available at the canal office; however, the use of channel 13 should be limited to emergency situations or whenever vessels do not have one of the other channels. All four channels are monitored continuously by the traffic controller. Radio discipline will be adhered to in accordance with FCC rules and regulations.

(3550) (2) For telephone communications with the traffic controller, call (617) 759-4431.

(3551) (3) Vessels shall maintain a radio guard on Marine VHF-FM channel 13 during the entire passage through the canal.

(3552) (4) All radio communications in the vicinity of the canal are tape recorded for future reference.

(3553) (d) *Vessels allowed passage*. The canal is open for passage to all adequately powered vessels properly equipped and seaworthy, of sizes consistent with safe navigation as governed by the controlling depths and widths of the channel and the vertical and horizontal clearances of the bridges over the waterway. The granting of permission for any vessel to proceed through the waterway shall not relieve the owners, agents and operators of full responsibility for its safe passage. No vessel having a greater draft forward than aft will be allowed to transit the canal. Craft of low power and wind driven are required to have and use auxiliary power during passage throughout the canal as defined in paragraph (a) of this section. Low powered vessels will be required to await slack water or favorable current for canal transit.

(3554) (e) *Tows*. (1) Tows shall be made-up outside the canal entrances. All vessels engaged in towing other vessels not equipped with a rudder shall use two lines or a bridle and one tow line. If the vessel in tow is equipped

with a rudder or a ship shaped bow, one tow line may be used. All tow lines of hawsers must be hauled as short as practicable for safe handling of the tows. No towboat will be allowed to enter the waterway with more than two barges in tow unless prior approval is granted by the Engineer-In-Charge; requests must be submitted 12 hours in advance of the passage.

(3555) (2) The maximum length of pontoon rafts using the canal will be limited to 600 feet, and the maximum width to 100 feet. Pontoon rafts exceeding 200 feet in length will be required to have an additional tug on the stern to insure that the tow is kept in line. The tugs used must have sufficient power to handle the raft safely.

(3556) (3) Dead ships are required to transit the canal during daylight hours and must be provided with the number of tugs sufficient to afford safe passage through the canal. (A dead ship will not be allowed to enter the canal unless prior approval is granted by the Engineer-In-Charge; requests must be submitted 12 hours in advance of the passage).

(3557) (f) *Dangerous cargoes*. The master or pilot of any vessel or tow carrying dangerous cargoes must notify the Marine Traffic Controller prior to entering the canal. Dangerous cargoes are defined as those items listed in 33 CFR 126.10 when carried in bulk *(i.e*., quantities exceeding 110 U.S. gallons in one tank) plus Class A explosives (commercial or military) as listed in 49 CFR 173.53 (commercial) and 46 CFR 146.29–100 (military), liquified natural gas and liquified petroleum gas. Transportation of dangerous cargoes through the canal shall be in strict accordance with existing regulations prescribed by law. In addition, vessels carrying dangerous cargoes shall comply with the following requirements.

(3558) (1) They must have sufficient horsepower to overcome tidal currents or they will be required to wait for favorable current conditions.

(3559) (2) Transits will be during daylight hours.

(3560) (3) No transit will be permitted when visibility conditions are unstable or less than 2 miles at the approaches and throughout the entire length of the canal.

(3561) (4) Transits must await a clear canal for passage.

(3562) (g) *Obtaining clearance*. (1) Vessels under 65 feet in length may enter the canal without obtaining clearance. All craft are required to make a complete passage through the canal except excursion craft which may operate and change direction within the canal in accordance with procedures coordinated with the marine traffic controller on duty. When the railroad bridge span is in the closed (down) position, all vessels are directed not to proceed beyond the points designated by the stop signs posted east and west of the railroad bridge. Vessels proceeding with a fair tide (with the current) should turn and stem the current at the designated stop points until the railroad bridge is in the raised (open) position.

(3563) (2) Vessels 65 feet in length and over shall not enter the canal until clearance has been obtained from the marine traffic controller by radio. See paragraph (c) "Communications" for procedures. If a vessel, granted prior clearance, is delayed or stops at the mooring basins, state pier, or the Sandwich bulkhead, a second clearance must be obtained prior to continuing passage through the canal.

(3564) (3) Vessels will be given clearance in the order of arrival, except when conditions warrant one-way traffic, or for any reason an order of priority is necessary, clearance will be granted in the following order.

(3565) (i) First—To vessels owned or operated by the United States, including contractors' equipment employed on canal maintenance or improvement work.

(3566) (ii) Second—To passenger vessels.

(3567) (iii) Third—To tankers and barges docking and undocking at the Canal Electric Terminal.

(3568) (iv) Fourth—To merchant vessels, towboats, commercial fishing vessels, pleasure boats and miscellaneous craft.

(3569) (4) Procedures in adverse weather: Vessels carrying flammable or combustible cargoes as defined in 46 CFR 30.25 will be restricted from passage through the canal when visibility is less than ½ mile. Other vessels may transit the canal in thick weather by use of radar with the understanding that the United States Government will assume no responsibility: *And provided,* That clearance has been obtained from the marine traffic controller.

(3570) (h) *Traffic lights*. There are three sets of traffic lights showing red, green, and yellow that are operated on a continuous basis at the canal. The traffic lights apply to all vessels 65 feet in length and over. The traffic lights are a secondary system that is operated in support of the radio communications system. The traffic lights are located at the easterly canal entrance, Sandwich, and at the westerly entrance to Hog Island Channel at Wings Neck. A third traffic light is located at the Canal Electric Terminal basin on the south side of the canal in Sandwich, and applies only to vessels arriving and departing that terminal.

(3571) (1) *Westbound traffic.* When the green light is on at the eastern (Cape Cod Bay) entrance, vessels may proceed westward through the canal. When the red light is on, any type of vessel 65 feet in length and over must stop clear of the Cape Cod Bay entrance channel. When the yellow light is on, vessels 65 feet in length and over and drawing less than 25 feet may proceed as far as the East Mooring Basin where they must stop. Prior to continuing passage through the canal, clearance must be obtained from the marine traffic controller.

(3572) (2) *Eastbound traffic.* When the green light is on at Wings Neck, vessels may proceed eastward through the canal. When the red light is on, vessels 65 feet and over in length and drawing less than 25 feet must keep southerly of Hog Island Channel Entrance Buoys Nos. 1 and 2 and utilize the general anchorage areas adjacent to the improved channel. Vessel traffic drawing 25 feet and over are directed not to enter the canal channel at the Cleveland Ledge Light entrance and shall lay to or anchor in the vicinity of Buzzards Bay Buoy No. 11 (FLW & Bell) until clearance is granted by the canal marine traffic controller or a green traffic light at Wings Neck

is displayed. When the yellow light is on, vessels may proceed through Hog Island Channel as far as the West Mooring Basin where they must stop. Prior to continuing passage through the canal, clearance must be obtained from the marine traffic controller.

(3573) (i) *Railroad Bridge Signals*. The following signals at the Buzzards Bay Railroad Bridge will be given strict attention.

(3574) (1) The vertical lift span on the railroad bridge is normally kept in the raised (open) position except when it is lowered for the passage of trains, or for maintenance purposes. Immediately preceding the lowering of the span, the operator will sound two long blasts of an air horn. Immediately preceding the raising of the span, the operator will sound one long blast of an air horn. When a vessel or craft of any type is approaching the bridge with the span in the down (closed) position and the span cannot be raised immediately, the operator of the bridge will so indicate by sounding danger signals of four short blasts in quick succession.

(3575) (2) When the lift span is in the down (closed) position in foggy weather or when visibility is obscured by vapor, there will be four short blasts sounded from the bridge every two minutes.

(3576) (j) *Speed*. All vessels are directed to pass mooring and boat basin facilities, the state pier, and all floating plant engaged in maintenance operations of the waterway at a minimum speed consistent with safe navigation. In order to coordinate scheduled rail traffic with the passage of vessels, to minimize erosion of the canal banks and dikes from excessive wave wash and suction, and for the safety of vessels using the canal, the following speed regulations must be observed by vessels of all types, including pleasure craft. The minimum running time for the land cut between the East Mooring Basin (Station 35) and the Administration Office in Buzzards Bay (Station 388) is prescribed as follows:

(3577) Head Tide–60 Minutes

(3578) Fair Tide–30 Minutes

(3579) Slack Tide–45 Minutes

(3580) The minimum running time between the Administration Office (Station 388) and Hog Island Channel westerly entrance Buoy No. 1 (Station 661) is prescribed as follows:

(3581) Head Tide–46 Minutes

(3582) Fair Tide–23 Minutes

(3583) Slack Tide–35 Minutes

(3584) The running time at slack water will apply to any vessel which enters that portion of the canal between stations 35 and 661, within the period of one-half hour before or after the predicted time of slack water as given in the National Ocean Service publication "Current Tables, Atlantic Coast, North America." The minimum running time during a head tide or a fair tide shall apply to any vessel which enters that portion of the canal between Station 35 and 661 at any time other than designated above for time requirements at slack tide. Vessels of any kind unable to make a through transit of the land cut portion of the canal against a head current of 6.0 knots within a maximum time limit of 2 hours 30 minutes shall be required to obtain the assistance of a helper tug at the vessel owner's expense or await favorable tide conditions prior to receiving clearance from the marine traffic controller. In the event vessels within the confines of the canal fail to perform and are unable to make sufficient headway against the currents, the marine traffic controller may activate a helper tug in accordance with paragraph (k) of this section.

(3585) (k) *Management of vessels*. (1) Vessels within the limits of the canal shall comply with applicable navigation rules.

(3586) (2) Vessels within the limits of the canal shall comply with the applicable requirements for the use of pilots established by the Coast Guard, including but not limited to those contained in 46 CFR 157.20–40. Vessels will not be granted clearance to enter the canal until the marine traffic controller has been notified of the name of the pilot who will be handling the vessel.

(3587) (3) The master of a vessel will be responsible for notifying the marine traffic controller as soon as an emergency situation appears to be developing. When in the opinion of the marine traffic controller an emergency exists, he/she can require the master to accept the assistance of a helper vessel. Whether or not assistance is provided by a government vessel or by a private firm under contract to the government, the government reserves the right to seek compensation from the vessel owners for all costs incurred.

(3588) (4) Right of Way: All vessels proceeding with the current shall have the right of way over those proceeding against the current. All craft up to 65 feet in length shall be operated so as not to interfere with the navigation of vessels of greater length.

(3589) (5) Passing of vessels: The passing of one vessel by another when proceeding in the same direction is prohibited except when a leading low powered ship is unable to make sufficient headway. However, extreme caution must be observed to avoid collision, and consideration must be given to the size of the ship to be overtaken, velocity of current and wind, and atmospheric conditions. Masters of vessels involved shall inform the marine traffic controller on duty of developing situations to facilitate coordination of vessel movement. Meeting or passing of vessels at the easterly end of the canal between Station Minus 40 and Station 60 will not be permitted, except in cases of extreme emergency, in order to allow vessels to utilize the center line range to minimize the effects of hazardous eddies and currents. Due to bank suction and tidal set, meeting and passing of vessels at the following location will be avoided:

(3590) (i) Sagamore Bridge.

(3591) (ii) Bourne Bridge.

(3592) (iii) Railroad Bridge.

(3593) (iv) Mass. Maritime Academy.

(3594) (6) Unnecessary delay in canal: Vessels and other type crafts must not obstruct navigation by unnecessarily

idling at low speed when entering or passing through the canal.

(3595) (7) Stopping in the waterway: Anchoring in the Cape Cod Canal Channel is prohibited except in emergencies. For the safety of canal operations it is mandatory that the masters of all vessels anchoring in or adjacent to the canal channel (Cape Cod Bay to Cleveland Ledge Light) for any reason, immediately notify the marine traffic controller.

(3596) (8) Utilization of mooring and boat basins and the Sandwich Bulkhead: Vessels mooring or anchoring in the mooring or boat basins at the Sandwich bulkhead must do so in a manner not to obstruct or impede vessel movements to and from facilities. These facilities are of limited capacity and permission to occupy them for periods exceeding 24 hours must be obtained in advance from the marine traffic controller. Mooring in the West Boat Basin at Buzzards Bay, near the railroad bridge, is not permitted except in an emergency. Fishing boats, yachts, cabin cruisers and other craft utilizing the East Boat Basin on the south side of the canal at Sandwich, Massachusetts are not permitted to tie up at the Corps of Engineers landing float or anchor in a manner to prevent canal floating plant from having ready access to the float. All vessels or barges left unattended must be securely tied with adequate lines or cables. The United States assumes no liability for damages which may be sustained by any craft using the bulkhead at Sandwich or the canal mooring or boat basin facilities. Vessels shall not be left unattended along the face of the government bulkhead. A responsible person with authority to authorize and/or accomplish vessel movement must remain onboard at all times.

(3597) (l) *Grounded, wrecked or damaged vessels*. In the event a vessel is grounded, or so damaged by accident as to render it likely to become an obstruction and/or hazard to navigation in the waterway, the division engineer or the division engineer's authorized representative shall supervise and direct all operations that may be necessary to remove the vessel to a safe locality.

(3598) (m) [Reserved]

(3599) (n) *Deposit of refuse*. No oil or other allied liquids, ashes, or materials of any kind shall be thrown, pumped or swept into the canal or its approaches from any vessel or craft using the waterway, nor shall any refuse be deposited on canal grounds, marine structures, or facilities.

(3600) (o) *Trespass to property*. Subject to the provisions of paragraph (q) of this section trespass upon the canal property is prohibited.

(3601) (p) *Bridges over the canal*. The government owns, operates and maintains all bridges across the canal which include one railroad bridge and two highway bridges. The division engineer or his/her authorized representative may establish rules and regulations governing the use of these bridges.

(3602) (q) *Recreational use of canal*–(1) *Policy*. (i) It is the policy of the Secretary of the Army acting through the Chief of Engineers to provide the public with safe and healthful recreational opportunities within all water resource development projects administered by the Chief of Engineers, including the canal and government lands part thereof. Any recreational use of the canal and those lands shall be at the users own risk.

(3603) (ii) All water resource development projects open for recreational use shall be available to the public without regard to sex, race, creed, color or national origin. No lessee, licensee, or concessionaire providing a service to the public shall discriminate against any person or persons because of sex, race, creed, color or national origin in the conduct of operations under the lease, license or concession contract.

(3604) (2) *Motor vehicles*. Operation of motor vehicles, motorcycles, minibikes, mopeds, motorbikes, snowmobiles, and all types of off-road vehicles is prohibited on government lands and service roads except in areas specifically designated for such operation.

(3605) (3) *Swimming*. Swimming, skin diving, snorkeling, and scuba diving in the canal between the east entrance in Cape Cod Bay and the west entrance at Cleveland Ledge Light are prohibited. Diving operations may be authorized by the Engineer-In-Charge in conjunction with operation and maintenance of the canal.

(3606) (4) *Camping*. Overnight tenting or camping on government land is prohibited except in areas designated by the division engineer. Bourne Scenic Park and Scusset Beach State Reservation are designated camping areas. Persons asleep during hours of darkness in or out of vehicles shall be considered as campers.

(3607) (5) *Fishing*. Persons may fish with rod and line from the banks of the canal on Federally owned property except areas designated by the division engineer. Fishing and lobstering by boat in the Cape Cod Canal between the east entrance in Cape Cod Bay and the west entrance at Cleveland Ledge Light are prohibited. Fishing by boat is permitted in the area west of the State Pier in Buzzards Bay, provided that all craft stay out of the channel defined by United States Coast Guard buoys and beacons. Fish and game laws of the United States and the Commonwealth of Massachusetts will be enforced.

(3608) (6) *Hunting*. Hunting is permitted in accordance with game laws of the United States and the Commonwealth of Massachusetts.

(3609) (7) *Fires*. No open fires will be allowed at any time except by special permission and then shall be continuously overseen and in compliance with state or town laws.

(3610) (8) *Control of animals and pets*. (i) No person shall bring or have horses in camping, picnic, swimming beaches or developed recreation areas.

(3611) (ii) No person shall bring dogs (except seeing eye dogs), cats, or other pets into developed recreation areas unless penned, caged, or on a leash no longer than six feet or otherwise under physical restrictive controls at all times.

(3612) (9) *Restrictions*. (i) The division engineer may establish a reasonable schedule of visiting hours for

all or portions of the project area and close or restrict the public use of all or any portion of the project by the posting of appropriate signs indicating the extent and scope of closure. All persons shall observe such posted restrictions.

(3613) (ii) The operation or use of any audio or other noise producing device including, but not limited to, communications media and vehicles in such a manner as to unreasonably annoy, endanger persons or affect vessel traffic through the canal is prohibited.

(3614) (10) *Explosives, firearms, other weapons and fireworks*. (i) The possession of loaded firearms, ammunition, projectile firing devices, bows and arrows, crossbows, and explosives of any kind is prohibited unless in the possession of a law enforcement officer or Government employee on official duty or used for hunting during the hunting season as permitted under paragraph (q)(6) of this section, or unless written permission has been received from the division engineer.

(3615) (ii) The possession or use of fireworks is prohibited unless written permission has been received from the division engineer.

(3616) (11) *Public property*. Destruction, injury, defacement or removal of public property including natural formations, historical and archeological features and vegetative growth is prohibited without written permission of the division engineer.

(3617) (12) *Abandonment of personal property*. (i) Abandonment of personal property is prohibited. Personal property shall not be left unattended upon the lands or waters of the project except in accordance with this regulation. After a period of 24 hours, abandoned or unattended personal property shall be impounded and stored at a storage point designated by the division engineer. The division engineer shall assess a reasonable impoundment fee, which shall be paid before the impounded property is returned to its owners.

(3618) (ii) The division engineer shall, by public or private sales or otherwise, dispose of all lost, abandoned, or unclaimed personal property that comes into his/her custody or control. However, efforts should be made to find the owner, the owner's heirs or next of kin, or legal representatives. If the owner, heirs or next of kin, or legal representative is determined but not found, the property may not be disposed of until the expiration of 120 days after the date when notice, giving the time and place of the intended sale or other disposition, has been sent by certified or registered mail to that person at last known address. When diligent effort to determine the owner, owner's heirs or next of kin, or legal representative is unsuccessful, the property may be disposed of without delay, except that if it has a fair market value of $25 or more the property generally may not be disposed of until three months after the date it is received at the Cape Cod Canal Administrative Office. The net proceeds from the sale of property shall be placed into the Treasury of the United States as miscellaneous receipts.

(3619) (13) *Lost and found articles*. All abandoned/lost articles shall be deposited by the finder at the Canal Administration office or with Canal ranger. The finder shall leave his/her name, address, and phone number. All lost articles shall be disposed of in accordance with procedures set forth in paragraph (q)(12) of this section.

(3620) (14) *Advertisement*. Advertising by the use of billboards, signs, markers, audio devices or any other means whatever is prohibited unless written permission has been received from the division engineer.

(3621) (15) *Commercial activities*. The engaging in or solicitation of business without the written permission of the division engineer is prohibited.

(3622) (16) *Unauthorized structures*. The construction or placing of any structure of any kind under, upon or over the project lands or waters is prohibited unless a permit has been issued by the division engineer. Structures not under permit are subject to summary removal by the division engineer.

(3623) (17) *Special events*. Prior approval must be obtained from the Engineer-In-Charge for special events, recreational programs and group activities. The public shall not be charged any fee by the sponsor of such event unless the division engineer has approved in writing the proposed schedule of fees.

(3624) (18) *Interference with government employees*. Interference with any government employee in the conduct of official duties pertaining to the administration of these regulations is prohibited.

(3625)

§207.50 Hudson River Lock at Troy, N.Y.; navigation.

(3626) (a) *Authority of lockmaster*. The lockmaster shall be charged with the immediate control and management of the lock, and of the area set aside as the lock area, including the lock approach channels. He shall see that all laws, rules and regulations for the use of the lock and lock area are duly complied with, to which end he is authorized to give all necessary orders and directions in accordance therewith, both to employees of the Government and to any and every person within the limits of the lock or lock area, whether navigating the lock or not. No one shall cause any movement of any vessel, boat, or other floating thing in the lock or approaches except by or under the direction of the lockmaster or his assistants.

(3627) (b) *Signals*. Steamboats or tows desiring lockage in either direction shall give notice to the lock tenders, when not more than three-fourths mile from the lock, by one long blast of (10 seconds' duration), followed by one short blast (of three seconds' duration), on a whistle or horn. When the lock is ready for entrance a green light will be shown from the river wall. An amber light will indicate that the lock is being made ready for entrance. A red light will indicate that the approaching vessel must wait. Whenever local conditions make it advisable the visual signals will be supplemented by sound signals as follows:

(3628) (1) One long blast of a horn to indicate that the vessel must wait.

(3629) (2) One short blast of a horn to indicate that the lock is being made ready for entrance.

(3630) (3) Two short blasts of a horn to indicate permission to enter the lock.

(3631) (4) Four short and rapid blasts to attract attention, indicate caution, and signal danger.

(3632) (c) *Draft of boats*. Deep-draft boats must clear the miter sills by at least 3 inches. Boats drawing too much water will not be allowed to lighter cargo in the entrances.

(3633) (d) *Precedence at the lock*. The vessel arriving first at the lock shall be first to lock through; but precedence shall be given to vessels belonging to the United States and to commercial vessels in the order named. Arrival posts or markers may be established ashore above or below the lock. Vessels arriving at or opposite such posts or markers will be considered as having arrived at the lock within the meaning of this paragraph. If the traffic is crowded in both directions; up and down lockages will usually be made alternately, but the lock tender may permit two or more lockages to be made at one time in the same direction when this will not cause unreasonable delay. In case two or more boats or tows are to enter for the same lockage, they shall enter as directed by the lock tender. No boat shall run ahead of another while in the lock. The boat that enters first shall leave first.

(3634) (e) *Lockage of pleasure boats*. The lockage of pleasure boats, house boats or like craft shall be expedited by locking them through with commercial craft (other than barges carrying gasoline or highly hazardous materials) in order to utilize the capacity of the lock to its maximum. Lockage of pleasure craft may be made with commercial craft carrying petroleum products other than gasoline provided a clear distance of at least 100 feet between such vessels can be maintained in the lock. If, after the arrival of such craft, no separate or combined lockage can be accomplished within a reasonable time, not to exceed the time required for three other lockages, then separate lockage shall be made.

(3635) (f) *Stations while waiting*. Boats waiting their turn to enter the lock must lie at a sufficient distance from the lock and in such a position as to leave sufficient room for the passage of boats leaving the lock.

(3636) (g) *Unnecessary delay*. (1) Boats must not cause delay in entering or leaving the lock. Masters and pilots will be held to a strict accountability in this matter, and those with tows must provide enough men to move barges promptly. Boats failing to enter the lock with reasonable promptness after being signaled will lose their turn.

(3637) (2) Tugboats arriving with their tows in a condition which will delay locking shall lose their turn if so ordered by the lock tender. Leaking boats may be excluded until put in shape to be passed through safely.

(3638) (h) *Mooring*. Boats in the lock or waiting in the entrance shall be moored where directed by the lock tender, by bow, stern, and spring lines, to the snubbing posts or line hooks. Tying boats to the lock ladders is strictly prohibited.

(3639) (i) *Protection of lock gates*. Boats will not be permitted to enter or leave the lock until the lock gates are at rest in the gate recesses and the lock tender has directed the boat to start.

(3640) (j) *Damage to walls, etc*. All craft passing through the lock must be free from projections or sharp corners which might scar the walls or injure other parts. Steamboats must be provided with suitable fenders, etc. One man shall be kept at the head of every tow till it has cleared the lock and guide walls, and shall use the fender to prevent scarring the walls.

(3641) (k) *Handling machinery*. None but employees of the United States will be allowed to move any valve, gate, or other machinery belonging to the lock.

(3642) (l) *Refuse in lock*. Throwing ashes, refuse, or other obstruction in the entrances or in the lock, or on the walls thereof, and passing coal from flats or barges to a steamboat while in the lock is prohibited.

(3643) (m) [Reserved]

(3644) (n) *Trespass on United States property*. Trespass on United States property, or willful injury to the banks, masonry, fences, trees, houses, machinery, or other property of the United States at or near the lock is strictly prohibited.

(3645) (o) *Penalties*. In addition to the penalties prescribed by law, boats which fail to comply with the regulations in this section will thereafter be refused lockage until assurances have been received, satisfactory to the District Engineer, Corps of Engineers, New York, New York, that the regulations will be complied with.

(3646)

§207.60 Federal Dam, Hudson River, Troy, N.Y.; pool level.

(3647) (a) Whenever the elevation of the pool created by the Federal dam at Troy, N.Y., shall fall to a point level with the crest of the main spillway, the elevation of which is + 14.33 feet mean sea level, the operation of the power plant shall cease and further operation thereof shall be suspended until such time as the water level rises to or above + 14.43 feet mean sea level.

(3648) (b) Flashboards may be maintained on the section of the spillway of the dam having an elevation of + 14.33 feet mean sea level in order to increase the elevation of this section to an elevation equal to that of the auxiliary spillway, or + 16.33 feet mean sea level: Provided, That the flashboards are so erected as to drop automatically when the pool level rises to an elevation of + 18.5 feet mean sea level, and conform in other respects to the plans attached thereto.

(3649) (c) The tide staff to be used in determining the elevation of the pool shall be the ceramic tide staff now located on the westerly face of the east lock wall north of the northerly gates, the zero of which is set 2 feet below mean sea level.

(3650) (d) The regulations of the pool level and the maintenance of flashboards shall be subject to the supervision and approval of the District Engineer, New York City.

(3651)

§207.800 Collection of navigation statistics.

(3652) (a) *Definitions*. For the purpose of this regulation the following terms are defined:

(3653) (1) *Navigable waters of the United States* means those waters of the United States that are subject to the ebb and flow of the tide shoreward to the mean high water mark, and/or are presently used, or have been used in the past, or may be susceptible to use to transport interstate or foreign commerce. (See 33 CFR part 329 for a more complete definition of this term.)

(3654) (2) *Offenses and Violations* mean:

(3655) (i) Failure to submit a required report.

(3656) (ii) Failure to provide a timely, accurate, and complete report.

(3657) (iii) Failure to submit monthly listings of idle vessels or vessels in transit.

(3658) (iv) Failure to submit a report required by the lockmaster or canal operator.

(3659) (3) *Leased* or *chartered vessel* means a vessel that is leased or chartered when the owner relinquishes control of the vessel through a contractual agreement with a second party for a specified period of time and/or for a specified remuneration from the lessee. Commercial movements on an affreightment basis are not considered a lease or charter of a particular vessel.

(3660) (4) *Person* or *entity* means an individual, corporation, partnership, or company.

(3661) (5) *Timely* means vessel and commodity movement data must be received by the Waterborne Commerce Statistics Center within 30 days after the close of the month in which the vessel movement or nonmovement takes place.

(3662) (6) *Commercial vessel* means a vessel used in transporting by water, either merchandise or passengers for compensation or hire, or in the course of business of the owner, lessee, or operator of the vessel.

(3663) (7) *Reporting situation* means a vessel movement by an operator that is required to be reported. Typical examples are listed in the instructions on the various ENG Forms. Five typical movements that are required to be reported by vessel operating companies include the following examples: Company A is the barge owner, and the barge transports corn from Minneapolis, MN to New Orleans, LA, with fleeting at Cairo, IL.

(3664) (i) *Lease/Charter:* If Company A leases or charters the barge to Company B, then Company B is responsible for reporting the movements of the barge until the lease/charter expires.

(3665) (ii) *Interline movement:* A barge is towed from Minneapolis to Cairo by Company A, and from Cairo to New Orleans by Company B. Since Company A is the barge owner, and the barge is not leased. Company A reports the entire movement of the barge with an origin of Minneapolis and a destination of New Orleans.

(3666) (iii) *Vessel swap/trade:* Company A swaps barge with Company B to allow Company B to meet a delivery commitment to New Orleans. Since Company A has not leased/chartered the barge, Company A is responsible for filing the report. Company B is responsible for filing the report on the barge which is traded to Company A. The swap or trade will not affect the primary responsibility for reporting the individual vessel movements.

(3667) (iv) *Re-Consignment:* Barge is reconsigned to Mobile, AL. Company A reports the movements as originating in Minneapolis and terminating in Mobile. The point from which barge is reconsigned is not reported, only points of loading and unloading.

(3668) (v) *Fleeting:* Barge is deposited at a New Orleans fleeting area by Company A and towed by Company B from fleeting area to New Orleans area dock for unloading. Company A, as barge owner, reports entire movements from Minneapolis to the dock in New Orleans. Company B does not report any barge movement.

(3669) (b) Implementation of the waterborne commerce statistics provisions of the River and Harbor Act of 1922, as amended by the Water Resources Development Act of 1986 (Pub. L. 99-662), mandates the following.

(3670) (1) *Filing requirements*. Except as provided in paragraph (b)(2) of this section, the person or entity receiving remuneration for the movement of vessels or for the transportation of goods or passengers on the navigable waters is responsible for assuring that the activity report of commercial vessels is timely filed.

(3671) (i) For vessels under lease/charter agreements, the lessee or charterer of any commercial vessel engaged in commercial transportation will be responsible for the filing of said reports until the lease/charter expires.

(3672) (ii) The vessel owner, or his designated agent, is always the responsible party for ensuring that all commercial activity of the vessel is timely reported.

(3673) (2) The following Vessel Information Reports are to be filed with the Army Corps of Engineers, at the address specified on the ENG Form, and are to include:

(3674) (i) *Monthly reports*. These reports shall be made on ENG Forms furnished upon written request of the vessel operating companies to the Army Corps of Engineers. The forms are available at the following address: U.S. Army Corps of Engineers, Waterborne Commerce Statistics Center, Post Office Box 61280, New Orleans, LA 70161-1280.

(3675) (A) All movements of domestic waterborne commercial vessels shall be reported, including but not limited to: Dry cargo ship and tanker moves, loaded and empty barge moves, towboat moves, with or without barges in tow, fishing vessels, movements of crew boats and supply boats to offshore locations, tugboat moves and movements of newly constructed vessels from the shipyard to the point of delivery.

(3676) (B) Vessels idle during the month must also be reported.

(3677) (C) Notwithstanding the above requirements, the following waterborne vessel movements need not be reported:

(3678) *(1)* Movements of recreational vessels.

(3679) *(2)* Movements of fire, police, and patrol vessels.

(3680) *(3)* Movements of vessels exclusively engaged in construction (e.g., piledrivers and crane barges). **Note:** however, that movements of supplies, materials, and crews to or from the construction site must be timely reported.

(3681) *(4)* Movements of dredges to or from the dredging site. However, vessel movements of dredged material from the dredging site to the disposal site must be reported.

(3682) *(5)* Specific movements granted exemption in writing by the Waterborne Commerce Statistics Center.

(3683) (D) ENG Forms 3925 and 3925b shall be completed and filed by vessel operating companies each month for all voyages or vessel movements completed during the month. Vessels that did not complete a move during the month shall be reported as idle or in transit.

(3684) (E) The vessel operating company may request a waiver from the Army Corps of Engineers, and upon written approval by the Waterborne Commerce Center, the company may be allowed to provide the requisite information of the above paragraph (D), on computer printouts, magnetic tape, diskettes, or alternate medium approved by the Center.

(3685) (F) Harbor Maintenance Tax information is required on ENG Form 3925 for cargo movements into or out of ports that are subject to the provisions of section 1402 of the Water Resources Development Act of 1986 (Pub. L. 99-662).

(3686) *(1)* The name of the shipper of the commodity, and the shipper's Internal Revenue Service number or Social Security number, must be reported on the form.

(3687) *(2)* If a specific exemption applies to the shipper, the shipper should list the appropriate exemption code. The specific exemption codes are listed in the directions for ENG Form 3925.

(3688) *(3)* Refer to 19 CFR part 24 for detailed information on exemptions and ports subject to the Harbor Maintenance Tax.

(3689) (ii) *Annual reports*. Annually an inventory of vessels available for commercial carriage of domestic commerce and vessel characteristics must be filed on ENG Forms 3931 and 3932.

(3690) (iii) *Transaction reports*. The sale, charter, or lease of vessels to other companies must also be reported to assure that proper decisions are made regarding each company's duty for reporting vessel movements during the year. In the absence of notification of the transaction, the former company of record remains responsible until proper notice is received by the Corps.

(3691) (iv) *Reports to lockmasters and canal operators*. Masters of self-propelled non-recreational vessels which pass through locks and canals operated by the Army Corps of Engineers will provide the data specified on ENG Forms 3102b, 3102c, and/or 3102d to the lockmaster, canal operator, or his designated representative in the manner and detail dictated.

(3692) (c) *Penalties for noncompliance*. The following penalties for noncompliance can be assessed for offenses and violations.

(3693) (1) *Criminal penalties*. Every person or persons violating the provisions of this regulation shall, for each and every offense, be liable to a fine of not more than $5,000, or imprisonment not exceeding two months, to be enforced in any district court in the United States within whose territorial jurisdiction such offense may have been committed.

(3694) (2) In addition, any person or entity that fails to provide timely, accurate, and complete statements or reports required to be submitted by the regulation in this section may also be assessed a civil penalty of up to $6,270 per violation under 33 U.S.C. 555, as amended.

(3695) (3) *Denial of passage*. In addition to these fines, penalties, and imprisonments, the lockmaster or canal operator can refuse to allow vessel passage.

(3696) (d) *Enforcement policy*. Every means at the disposal of the Army Corps of Engineers will be utilized to monitor and enforce these regulations.

(3697) (1) To identify vessel operating companies that should be reporting waterborne commerce data, The Corps will make use of, but is not limited to, the following sources.

(3698) (i) Data on purchase and sale of vessels.

(3699) (ii) U.S. Coast Guard vessel documentation and reports.

(3700) (iii) Data collected at Locks, Canals, and other facilities operated by the Corps.

(3701) (iv) Data provided by terminals on ENG Form 3926.

(3702) (v) Data provided by the other Federal agencies including the Internal Revenue Service, Customs Service, Maritime Administration, Department of Transportation, and Department of Commerce.

(3703) (vi) Data provided by ports, local facilities, and State or local governments.

(3704) (vii) Data from trade journals and publications.

(3705) (viii) Site visits and inspections.

(3706) (2) *Notice of violation*. Once a reporting violation is determined to have occurred, the Chief of the Waterborne Commerce Statistics Center will notify the responsible party and allow 30 days for the reports to be filed after the fact. If the reports are not filed within this 30-day notice period, then appropriate civil or criminal actions will be undertaken by the Army Corps of Engineers, including the proposal of civil or criminal penalties for noncompliance. Typical cases for criminal or civil action include, but are not limited to, those violations which are willful, repeated, or have a substantial impact in the opinion of the Chief of the Waterborne Commerce Statistics Center.

(3707) (3) *Administrative assessment of civil penalties*. Civil penalties may be assessed in the following manner.

(3708) (i) *Authorization*. If the Chief of the Waterborne Commerce Statistics Center finds that a person or entity has failed to comply with any of the provisions specified herein, he is authorized to assess a civil penalty in accordance with the Class I penalty provisions of 33 CFR part 326. Provided, however, that the procedures in 33 CFR part 326 specifically implementing the Clean Water Act (33 U.S.C. 1319(g)(4)), public notice, comment period, and state coordination, shall not apply.

(3709) (ii) *Initiation*. The Chief of the Waterborne Commerce Statistics Center will prepare and process a proposed civil penalty order which shall state the amount of the penalty to be assessed, describe by reasonable specificity the nature of the violation, and indicate the applicable provisions of 33 CFR part 326.

(3710) (iii) *Hearing requests*. Recipients of a proposed civil penalty order may file a written request for a hearing or other proceeding. This request shall be as specified in 33 CFR part 326 and shall be addressed to the Director of the Water Resources Support Center, Casey Building, Fort Belvoir, VA 22060-5586, who will provide the requesting person or entity with a reasonable opportunity to present evidence regarding the issuance, modification, or revocation of the proposed order. Thereafter, the Director of the Water Resources Center shall issue a final order.

(3711) (4) *Additional remedies*. Appropriate cases may also be referred to the local U.S. Attorney for prosecution, penalty collection, injunctive, and other relief by the Chief of the Waterborne Commerce Statistics Center.

(3712)

Part 334–Danger Zones and Restricted Area Regulations

(3713)

§334.1 Purpose.

(3714) The purpose of this part is to:

(3715) (a) Prescribe procedures for establishing, amending and disestablishing danger zones and restricted areas;

(3716) (b) List the specific danger zones and restricted areas and their boundaries; and

(3717) (c) Prescribe specific requirements, access limitations and controlled activities within the danger zones and restricted areas.

(3718)

§334.2 Definitions.

(3719) (a) *Danger zone*. A defined water area (or areas) used for target practice, bombing, rocket firing or other especially hazardous operations, normally for the armed forces. The danger zones may be closed to the public on a full-time or intermittent basis, as stated in the regulations.

(3720) (b) *Restricted area*. A defined water area for the purpose of prohibiting or limiting public access to the area. Restricted areas generally provide security for Government property and/or protection to the public from the risks of damage or injury arising from the Government's use of that area.

(3721)

§334.3 Special policies.

(3722) (a) *General*. The general regulatory policies stated in 33 CFR part 320 will be followed as appropriate. In addition, danger zone and restricted area regulations shall provide for public access to the area to the maximum extent practicable.

(3723) (b) *Food fishing industry*. The authority to prescribe danger zone and restricted area regulations must be exercised so as not to unreasonably interfere with or restrict the food fishing industry. Whenever the proposed establishment of a danger zone or restricted area may affect fishing operations, the District Engineer will consult with the Regional Director, U.S. Fish and Wildlife Service, Department of the Interior and the Regional Director, National Marine Fisheries Service, National Oceanic & Atmospheric Administration (NOAA).

(3724) (c) *Temporary, occasional or intermittent use*. If the use of the water area is desired for a short period of time, not to exceed thirty days in duration, and that planned operations can be conducted safely without imposing unreasonable restrictions on navigation, and without promulgating restricted area regulations in accordance with the regulations in this section, applicants may be informed that formal regulations are not required. Activities of this type shall not reoccur more often than biennially (every other year), unless danger zone/restricted area rules are promulgated under this Part. Proper notices for mariners requesting that vessels avoid the area will be issued by the Agency requesting such use of the water area, or if appropriate, by the District Engineer, to all known interested persons. Copies will also be sent to appropriate State agencies, the Commandant, U.S. Coast Guard, Washington, DC 20590, and Director, National Geospatial-Intelligence Agency, Hydrographic Center, Washington, DC 20390, ATTN: Code NS 12. Notification to all parties and Agencies shall be made at least two weeks prior to the planned event, or earlier, if required for distribution of Local Notice to Mariners by the Coast Guard.

(3725)

§334.4 Establishment and amendment procedures.

(3726) (a) *Application*. Any request for the establishment, amendment or revocation of a danger zone or restricted area must contain sufficient information for the District Engineer to issue a public notice, and as a minimum must contain the following:

(3727) (1) Name, address and telephone number of requestor including the identity of the command and DoD facility and the identity of a point of contact with phone number.

(3728) (2) Name of waterway and if a small tributary, the name of a larger connecting waterbody.

(3729) (3) Name of closest city or town, county/parish and state.

(3730) (4) Location of proposed or existing danger zone or restricted area with a map showing the location, if possible.

(3731) (5) A brief statement of the need for the area, its intended use and detailed description of the times, dates and extent of restriction.

(3732) (b) *Public notice*. (1) The Corps will normally publish public notices and **Federal Register** documents concurrently. Upon receipt of a request for the establishment, amendment or revocation of a danger zone or restricted area, the District Engineer should forward a copy of the request with his/her recommendation, a copy of the draft public notice and a draft **Federal Register** document to the Office of the Chief of Engineers, ATTN: CECW-OR. The Chief of Engineers will publish the proposal in the **Federal Register** concurrent with the public notice issued by the District Engineer.

(3733) (2) *Content*. The public notice and **Federal Register** documents must include sufficient information to give a clear understanding of the proposed action and should include the following items of information:

(3734) (i) Applicable statutory authority or authorities; (40 Stat. 266; 33 U.S.C. 1) and (40 Stat. 892; 33 U.S.C. 3).

(3735) (ii) A reasonable comment period. The public notice should fix a limiting date within which comments will be received, normally a period not less than 30 days after publication of the notice.

(3736) (iii) The address of the District Engineer as the recipient of any comments received.

(3737) (iv) The identity of the applicant/proponent;

(3738) (v) The name or title, address and telephone number of the Corps employee from whom additional information concerning the proposal may be obtained;

(3739) (vi) The location of the proposed activity accompanied by a map of sufficient detail to show the boundaries of the area(s) and its relationship to the surrounding area.

(3740) (3) *Distribution*. Public notice will be distributed in accordance with 33 CFR 325.3(d)(1). In addition to this general distribution, public notices will be sent to the following Agencies:

(3741) (i) The Federal Aviation Administration (FAA) where the use of airspace is involved.

(3742) (ii) The Commander, Service Force, U.S. Atlantic Fleet, if a proposed action involves a danger zone off the U.S. Atlantic coast.

(3743) (iii) Proposed danger zones on the U.S. Pacific coast must be coordinated with the applicable commands as follows:

(3744) Alaska, Oregon and Washington:

(3745) Commander, Naval Base, Seattle

(3746) California:

(3747) Commander, Naval Base, San Diego

(3748) Hawaii and Trust Territories:

(3749) Commander, Naval Base, Pearl Harbor

(3750) (c) *Public hearing*. The District Engineer may conduct a public hearing in accordance with 33 CFR part 327.

(3751) (d) *Environmental documentation*. The District Engineer shall prepare environmental documentation in accordance with appendix B to 33 CFR part 325.

(3752) (e) *District Engineer's recommendation*. After closure of the comment period, and upon completion of the District Engineer's review he/she shall forward the case through channels to the Office of the Chief of Engineers, ATTN: CECW-OR with a recommendation of whether or not the danger zone or restricted area regulation should be promulgated. The District Engineer shall include a copy of environmental documentation prepared in accordance with appendix B to 33 CFR part 325, the record of any public hearings, if held, a summary of any comments received and a response thereto, and a draft of the regulation as it is to appear in the **Federal Register.**

(3753) (f) *Final decision*. The Chief of Engineers will notify the District Engineer of the final decision to either approve or disapprove the regulations. The District Engineer will notify the applicant/proponent and publish a public notice of the final decision. Concurrent with issuance of the public notice the Office of the Chief of Engineers will publish the final decision in the **Federal Register** and either withdraw the proposed regulation or issue the final regulation as appropriate. The final rule shall become effective no sooner than 30 days after publication in the **Federal Register** unless the Chief of Engineers finds that sufficient cause exists and publishes that rationale with the regulations.

(3754)

§334.5 Disestablishment of a danger zone.

(3755) (a) Upon receipt of a request from any agency for the disestablishment of a danger zone, the District Engineer shall notify that agency of its responsibility for returning the area to a condition suitable for use by the public. The agency must either certify that it has not used the area for a purpose that requires cleanup or that it has removed all hazardous materials and munitions, before the Corps will disestablish the area. The agency will remain responsible for the enforcement of the danger zone regulations to prevent unauthorized entry into the area until the area is deemed safe for use by the public and the area is disestablished by the Corps.

(3756) (b) Upon receipt of the certification required in paragraph (a) of this section, the District shall forward the request for disestablishment of the danger zone through channels to CECW-OR, with its recommendations. Notice of proposed rulemaking and public procedures as outlined in §334.4 are not normally required before publication of the final rule revoking a restricted area or danger zone regulation. The disestablishment/revocation of the danger zone or restricted area regulation removes a restriction on a waterway.

(3757)

§334.6 Datum.

(3758) (a) Geographic coordinates expressed in terms of latitude or longitude, or both, are not intended for plotting on maps or charts whose reference horizontal datum is the North American Datum of 1983 (NAD 83), unless such geographic coordinates are expressly labeled NAD 83.

Geographic coordinates without the NAD 83 reference may be plotted on maps or charts referenced to NAD 83 only after application of the appropriate corrections that are published on the particular map or chart being used.

(3759) (b) For further information on NAD 83 and National Service nautical charts please contact; Director, Coast Survey (N/CG2), National Ocean Service, NOAA, 1315 East-West Highway, Station 6147, Silver Spring, MD 20910-3282.

(3760)

§334.60 Cape Cod Bay south of Wellfleet Harbor, Mass.; naval aircraft bombing target area.

(3761) (a) *The danger zone*. A circular area with a radius of 1,000 yards having its center on the aircraft bombing target hulk James Longstreet in Cape Cod Bay at latitude 41°49'46", longitude 70°02'54".

(3762) (b) *The regulations*. (1) No vessel shall enter or remain in the danger zone at any time, except as authorized by the enforcing agency.

(3763) (2) This section shall be enforced by the Commandant, First Naval District, and such agencies as he may designate.

(3764)

§334.70 Buzzards Bay, and adjacent waters, Mass.; danger zones for naval operations.

(3765) (a) *Atlantic Ocean in vicinity of No Mans Land*–(1) *The area*. The waters surrounding No Mans Land within an area bounded as follows: Beginning at

(3766) 41°12'30", 70°50'30"; thence northwesterly to

(3767) 41°15'30", 70°51'30"; thence northeasterly to

(3768) 41°17'30", 70°50'30"; thence southeasterly to

(3769) 41°16'00", 70°47'30"; thence south to

(3770) 41°12'30", 70°47'30"; thence westerly to the point of beginning.

(3771) (2) *The regulations*. No vessel or person shall at any time enter or remain within a rectangular portion of the area bounded on the north by latitude 41°16'00", on the east by longitude 70°47'30", on the south by latitude 41°12'30", and on the west by longitude 70°50'30", or within the remainder of the area between November 1, and April 30, inclusive, except by permission of the enforcing agency.

(3772) (3) The regulations in this paragraph shall be enforced by the Commandant, First Naval District, and such agencies as he may designate.

(3773)

§334.75 Thames River, Naval Submarine Base New London, Restricted Area.

(3774) (a) *The area*: The open waters of the Thames River approximately 5 nautical miles upriver from its mouth along the boundary between Groton and Waterford, Connecticut, within an area bounded as follows:

(3775) From a point on the eastern shore at

(3776) 41°24'14.4"N., 72°05'38.0"W., then northerly along the coast to

(3777) 41°24'20.0"N., 72°05'37.9"W., then westerly across the river to a point on the western shore at

(3778) 41°24'20.0"N., 72°05'55.5"W., then southerly along the coast to a point on the western shore at

(3779) 41°24'05.0"N., 72°05'55.7"W., then easterly to the western edge of the dredged channel to a point located at

(3780) 41°24'04.1"N., 72°05'51.2"W., then southerly along the western edge of the dredged channel to a point at

(3781) 41°24'00"N., 72°05'52.6"W., then southerly along the western edge of the dredged channel to a point located at

(3782) 41°23'57.1"N., 72°05'52.5"W., then southerly to buoy "11" located at a point at

(3783) 41°23'45.6"N., 72°05'53.7"W., then southerly to buoy "B" on the northeastern shore of Mamacoke Hill to a point at

(3784) 41°23'33.8"N., 72°05'53.7"W., then southerly along the shore to buoy "A" at

(3785) 41°23'25.0"N., 72°05'45.4"W., then southeasterly to buoy "9" at a point located at

(3786) 41°23'15.0"N., 72°05'35.0"W., then easterly to a point on the eastern shore at

(3787) 41°23'15.0"N., 72°05'17.9"W., then northerly along the shore to a point on the eastern shore at

(3788) 41°23'15.8"N., 72°05'17.9"W., then along the following points:

(3789) 41°23'15.8"N., 72°05'22.0"W.

(3790) 41°23'25.9"N., 72°05'29.9"W.

(3791) 41°23'33.8"N., 72°05'34.7"W.

(3792) 41°23'37.0"N., 72°05'38.0"W.

(3793) 41°23'41.0"N., 72°05'40.3"W.

(3794) 41°23'47.2"N., 72°05'42.3"W.

(3795) 41°23'53.8"N., 72°05'43.7"W.

(3796) 41°23'59.8"N., 72°05'43.0"W.

(3797) 41°24'12.4"N., 72°05'43.2"W. Then to the point of beginning on the eastern shore.

(3798) (b) *The regulations*.

(3799) (1) Vessels and other watercraft within the designated navigation channel may proceed through the restricted area at normal operating speeds without stopping. Vessels and watercraft may also utilize the water area within the restricted area located between the western edge of the designated channel and the western shore for fishing, anchoring and other recreational uses. However, all persons, vessels and watercraft, except U.S. military personnel and vessels must leave the restricted area when notified by personnel of the New London Submarine Base that such use will interfere with submarine maneuvering, operations or security.

(3800) (2) Commercial fishermen and shell fishermen may fish within the restricted area provided their vessels display registration numbers issued by the Naval Submarine Base, New London, Connecticut. The registration numbers may be obtained by contacting the Commanding Officer, Naval Submarine Base New London. All commercial fishermen and shell fishermen must also leave the restricted area when notified by personnel of the New London Submarine Base that such use will interfere with submarine maneuvering, operations or security.

(3801) (3) Vessels which are owned, operated or sponsored by local, state municipalities or academic institutions preparing for or participating in a water sport or water related recreational event sponsored by those local or state municipalities or academic institutions, or private or commercial vessels engaged in observing the conduct of the above event shall be exempt from the restrictions above, providing:

(3802) (i) The Commanding Officer, Naval Submarine Base New London, and the Coast Guard Captain of the Port are advised in writing at least 48 hours in advance of the event, or

(3803) (ii) The event was publicized in such a manner that the local public in general had a reasonable opportunity to learn of the event 48 hours in advance.

(3804) (4) The regulations in this section shall be enforced by the Commander, U.S. Naval Submarine Base New London, Connecticut, and such agencies as he/she may designate.

(3805)

§334.80 Narragansett Bay, R.I.; restricted area.

(3806) (a) Beginning at a point on the east shore of Conanicut Island at

(3807) 41°33'15"; thence southeasterly to

(3808) 41°32'44", 71°21'17"; thence southerly to

(3809) 41°32'09", 71°21'17"; thence southeasterly to

(3810) 41°31'50", 71°21'10"; thence southeasterly to

(3811) 41°31'26", 71°20'33"; thence easterly to

(3812) 41°31'27", 71°20'06"; thence northerly to a point on the southwesterly shore of Prudence Island at

(3813) 41°35'00"; thence northerly along the southwesterly shore of Prudence Island to a point at

(3814) 41°35'43", 71°20'15.5"; thence northwesterly to

(3815) 41°37'21", 71°20'48"; thence west to

(3816) 41°37'21", 71°21'48"; and thence south to

(3817) 41°33'54", 71°21'48".

(3818) (b) *The regulations.* (1) No person or vessel shall at any time, under any circumstances, anchor or fish or tow a drag of any kind in the prohibited area because of the extensive cable system located therein.

(3819) (2) Orders and instructions issued by patrol craft or other authorized representatives of the enforcing agency shall be carried out promptly by persons or vessels in or in the vicinity of the prohibited area.

(3820) (3) The regulations in this section shall be enforced by the Commander U.S. Naval Base, Newport, R.I., and such agencies as he may designate.

(3821)

§334.81 Narragansett Bay, East Passage, Coddington Cove, Naval Station Newport, Naval Restricted Area, Newport, Rhode Island.

(3822) (a) *The area*. All of the navigable waters of Coddington Cove east of a line that connects Coddington Point at 41°31'24.0"N., 71°19'24.0"W.; with the outer end of the Coddington Cove breakwater on the north side of the cove at 41°31'55.7"N., 71°19'28.2"W.

(3823) (b) *The regulation*. All persons, swimmers, vessels and other craft, except those vessels under the supervision or contract to local military or Naval authority, vessels of the United States Coast Guard, and local or state law enforcement vessels, are prohibited from entering the restricted area without specific permission from the Commanding Officer, Naval Station Newport, USN, Newport, Rhode Island or his/her authorized representative.

(3824) (c) *Enforcement*. The regulation in this section, promulgated by the United States Army Corps of Engineers, shall be enforced by the United States Navy, Commanding Officer Naval Station Newport, and/or such agencies or persons as he/she may designate.

(3825)

§334.82 Narragansett Bay, East Passage, Coasters Harbor Island, Naval Station Newport, Newport, Rhode Island, Restricted Area.

(3826) (a) *The area*. The waters within a "C-shaped" area adjacent to and surrounding Coasters Harbor Island beginning at Coddington Point at

(3827) 41°31'24.0"N., 71°19'24.0"W.; thence to

(3828) 41°31'21.5"N., 71°19'45.0"W.; thence to

(3829) 41°31'04.2"N., 71°19'52.8"W.; thence to

(3830) 41°30'27.3"N., 71°19'52.8"W.; thence to

(3831) 41°30'13.8"N., 71°19'42.0"W.; thence to

(3832) 41°30'10.2"N., 71°19'32.6"W.; thence to

(3833) 41°30'10.2"N., 71°19'20.0"W.; thence northerly along the mainland shoreline to the point of origin.

(3834) (b) *The regulation*. All persons, swimmers, vessels and other craft, except those vessels under the supervision or contract to local military or Naval authority, vessels of the United States Coast Guard, and Federal, local or State law enforcement vessels, are prohibited from entering the restricted areas without permission from the Commanding Officer Naval Station Newport, USN, Newport, Rhode Island or his/her authorized representative.

(3835) (c) *Enforcement*. (1) The regulation in this section, promulgated by the United States Army Corps of Engineers, shall be enforced by the United States Navy, Commanding Officer Naval Station Newport, Newport, Rhode Island and/or other persons or agencies as he/she may designate.

(3836)

§334.85 New York Harbor, adjacent to the Stapleton Naval Station, Staten Island, New York; restricted area.

(3837) (a) *The area*. The waters of New York Harbor beginning at a point on shore at

(3838) 40°38'02"N., 074°04'24"W.; thence easterly to

(3839) 40°38'02.5"N., 074°04'09"W.; thence southerly to

(3840) 40°37'53"N., 074°04'07"W.; thence east-southeasterly to

(3841) 40°37'50"N., 074°03'50.2"W.; thence south-southeasterly to

(3842) 40°37'37.5"N., 074°03'46"W.; thence southwesterly to the shore line at

(3843) 40°37'24.5"N., 074°04'18"W.; thence northerly along the shore line to the point of origin.

(3844) (b) *The regulations*. (1) The portion of the restricted area extending from the shore out to a line 600 feet east of the U.S. Pierhead Line is closed to all persons and vessels except those vessels owned by, under hire to or performing work for Naval Station New York, Staten Island, New York.

(3845) (2) The portion of the restricted area beginning 600 feet seaward of the U.S. Pierhead Line is open to transiting vessels only. Vessels shall proceed across the area by the most direct route and without unnecessary delay. For vessels under sail, necessary tacking shall constitute a direct route.

(3846) (3) Commercial vessels at anchor will be permitted to swing into the seaward portion of the restricted area while at anchor and during the tide changes.

(3847) (c) *Enforcement*. The regulations in this section shall be enforced by the Commanding Officer, Naval Station New York, and such agencies as he/she shall designate.

(3848)

§334.102 Sandy Hook Bay, Naval Weapons Station EARLE, Piers and Terminal Channel, Restricted Area, Middletown, New Jersey.

(3849) (a) *The area*. All of the navigable waters within the area bounded by these coordinates:

(3850) 40°25'55.6"N., 074°04'31.4"W.; thence to

(3851) 40°26'54.0"N., 074°03'53.0"W.; thence to

(3852) 40°26'58.0"N., 074°04'03.0"W.; thence to

(3853) 40°27'56.0"N., 074°03'24.0"W.; thence to

(3854) 40°27'41.7"N., 074°02'45.0"W.; thence to

(3855) 40°28'23.5"N., 074°02'16.6"W.; thence to

(3856) 40°28'21.2"N., 074°01'56.0"W.; thence to

(3857) 40°28'07.9"N., 074°02'18.6"W.; thence to

(3858) 40°27'39.3"N., 074°02'38.3"W.; thence to

(3859) 40°27'28.5"N., 074°02'10.4"W.; thence to

(3860) 40°26'29.5"N., 074°02'51.2"W.; thence to

(3861) 40°26'31.4"N., 074°02'55.4"W.; thence to

(3862) 40°25'27.1"N., 074°03'39.7"W.; and thence along the shoreline to the point of origin (NAD 83).

(3863) The Department of the Navy plans to install buoys along these coordinates to outline the Restricted Area.

(3864) (b) *The regulation*. (1) Except as set forth in subparagraph (b)(2), no persons, unauthorized vessels or other unauthorized craft may enter the restricted area at any time;

(3865) (2) Vessels are authorized to cross the Terminal Channel provided that there are no naval vessels then transiting the channel bounded by:

(3866) 40°27'41.7"N., 074°02'45.0"W.; thence to

(3867) 40°28'23.5"N., 074°02'16.6"W.; thence to

(3868) 40°28'21.2"N., 074°01'56.0"W.; thence to

(3869) 40°28'07.9"N., 074°02'18.6"W.; thence to

(3870) 40°27'39.3"N., 074°02'38.3"W.; and

(3871) (3) No person may swim in the Restricted Area.

(3872) (c) *Enforcement*. The regulation in this section, promulgated by the U.S. Army Corps of Engineers, shall be enforced by the Commanding Officer, Naval Weapons Station Earle, and/or other persons or agencies as he/she may designate.

(3873)

TITLE 40–PROTECTION OF ENVIRONMENT

(3874)

Part 140–Marine Sanitation Device Standard

(3875)

§140.1 Definitions.

(3876) For the purpose of these standards the following definitions shall apply:

(3877) (a) *Sewage* means human body wastes and the wastes from toilets and other receptacles intended to receive or retain body wastes;

(3878) (b) *Discharge* includes, but is not limited to, any spilling, leaking, pumping, pouring, emitting, emptying, or dumping;

(3879) (c) *Marine sanitation device* includes any equipment for installation onboard a vessel and which is designed to receive, retain, treat, or discharge sewage and any process to treat such sewage;

(3880) (d) *Vessel* includes every description of watercraft or other artificial contrivance used, or capable of being used, as a means of transportation on waters of the United States;

(3881) (e) *New vessel* refers to any vessel on which construction was initiated on or after January 30, 1975;

(3882) (f) *Existing vessel* refers to any vessel on which construction was initiated before January 30, 1975;

(3883) (g) *Fecal coliform bacteria* are those organisms associated with the intestines of warm-blooded animals that are commonly used to indicate the presence of fecal material and the potential presence of organisms capable of causing human disease.

(3884)

§140.2 Scope of standard.

(3885) The standard adopted herein applies only to vessels on which a marine sanitation device has been installed. The standard does not require the installation of a marine sanitation device on any vessel that is not so equipped. The standard applies to vessels owned and operated by the United States unless the Secretary of Defense finds that compliance would not be in the interest of national security.

(3886)

§140.3 Standard.

(3887) (a) (1) In freshwater lakes, freshwater reservoirs or other freshwater impoundments whose inlets or outlets are such as to prevent the ingress or egress by vessel traffic subject to this regulation, or in rivers not capable of navigation by interstate vessel traffic subject to this regulation, marine sanitation devices certified by the U.S. Coast Guard (see 33 CFR part 159, published in 40 FR 4622, January 30, 1975), installed on all vessels shall be

designed and operated to prevent the overboard discharge of sewage, treated or untreated, or of any waste derived from sewage. This shall not be construed to prohibit the carriage of Coast Guard-certified flow-through treatment devices which have been secured so as to prevent such discharges.

(3888) (2) In all other waters, Coast Guard-certified marine sanitation devices installed on all vessels shall be designed and operated to either retain, dispose of, or discharge sewage. If the device has a discharge, subject to paragraph (d) of this section, the effluent shall not have a fecal coliform bacterial count of greater than 1,000 per 100 milliliters nor visible floating solids. Waters where a Coast Guard-certified marine sanitation device permitting discharge is allowed include coastal waters and estuaries, the Great Lakes and inter-connected waterways, fresh-water lakes and impoundments accessible through locks, and other flowing waters that are navigable interstate by vessels subject to this regulation.

(3889) (b) This standard shall become effective on January 30, 1977 for new vessels and on January 30, 1980 for existing vessels (or, in the case of vessels owned and operated by the Department of Defense, two years and five years, for new and existing vessels, respectively, after promulgation of implementing regulations by the Secretary of Defense under section 312(d) of the Act).

(3890) (c) Any vessel which is equipped as of the date of promulgation of this regulation with a Coast Guard-certified flow-through marine sanitation device meeting the requirements of paragraph (a)(2) of this section, shall not be required to comply with the provisions designed to prevent the overboard discharge of sewage, treated or untreated, in paragraph (a)(1) of this section, for the operable life of that device.

(3891) (d) After January 30, 1980, subject to paragraphs (e) and (f) of this section, marine sanitation devices on all vessels on waters that are not subject to a prohibition of the overboard discharge of sewage, treated or untreated, as specified in paragraph (a)(1) of this section, shall be designed and operated to either retain, dispose of, or discharge sewage, and shall be certified by the U.S. Coast Guard. If the device has a discharge, the effluent shall not have a fecal coliform bacterial count of greater than 200 per 100 milliliters, nor suspended solids greater than 150 mg/1.

(3892) (e) Any existing vessel on waters not subject to a prohibition of the overboard discharge of sewage in paragraph (a)(1) of this section, and which is equipped with a certified device on or before January 30, 1978, shall not be required to comply with paragraph (d) of this section, for the operable life of that device.

(3893) (f) Any new vessel on waters not subject to the prohibition of the overboard discharge of sewage in paragraph (a)(1) of this section, and on which construction is initiated before January 31, 1980, which is equipped with a marine sanitation device before January 31, 1980, certified under paragraph (a)(2) of this section, shall not be required to comply with paragraph (d) of this section, for the operable life of that device.

(3894) (g) The degrees of treatment described in paragraphs (a) and (d) of this section are ‘‘appropriate standards’’ for purposes of Coast Guard and Department of Defense certification pursuant to section 312(g)(2) of the Act.

(3895)

§140.4 Complete prohibition.

(3896) (a) Prohibition pursuant to CWA section 312(f)(3): a State may completely prohibit the discharge from all vessels of any sewage, whether treated or not, into some or all of the waters within such State by making a written application to the Administrator, Environmental Protection Agency, and by receiving the Administrator's affirmative determination pursuant to section 312(f)(3) of the Act. [...]

(3897) (b) Prohibition pursuant to CWA section 312(f)(4)(A): a State may make a written application to the Administrator, Environmental Protection Agency, under section 312(f)(4)(A) of the Act, for the issuance of a regulation completely prohibiting discharge from a vessel of any sewage, whether treated or not, into particular waters of the United States or specified portions thereof, which waters are located within the boundaries of such State. Such application shall specify with particularly the waters, or portions thereof, for which a complete prohibition is desired. The application shall include identification of water recreational areas, drinking water intakes, aquatic sanctuaries, identifiable fish-spawning and nursery areas, and areas of intensive boating activities. If, on the basis of the State's application and any other information available to him, the Administrator is unable to make a finding that the waters listed in the application require a complete prohibition of any discharge in the waters or portions thereof covered by the application, he shall state the reasons why he cannot make such a finding, and shall deny the application. If the Administrator makes a finding that the waters listed in the application require a complete prohibition of any discharge in all or any part of the waters or portions thereof covered by the State’s application, he shall publish notice of such findings together with a notice of proposed rule making, and then shall proceed in accordance with 5 U.S.C. 553. If the Administrator’s finding is that applicable water quality standards require a complete prohibition covering a more restricted or more expanded area than that applied for by the State, he shall state the reasons why his finding differs in scope from that requested in the State’s application. [...]

(3898) (c)(1) *Prohibition pursuant to CWA* section 312(f)(4)(B): A State may make written application to the Administrator of the Environmental Protection Agency under section 312(f)(4)(B) of the Act for the issuance of a regulation establishing a drinking water intake no discharge zone which completely prohibits discharge from a vessel of any sewage, whether treated or untreated,

into that zone in particular waters, or portions thereof, within such State. Such application shall:

(3899) (i) Identify and describe exactly and in detail the location of the drinking water supply intake(s) and the community served by the intake(s), including average and maximum expected amounts of inflow;

(3900) (ii) Specify and describe exactly and in detail, the waters, or portions thereof, for which a complete prohibition is desired, and where appropriate, average, maximum and low flows in million gallons per day (MGD) or the metric equivalent;

(3901) (iii) Include a map, either a USGS topographic quadrant map or a NOAA nautical chart, as applicable, clearly marking by latitude and longitude the waters or portions thereof to be designated a drinking water intake zone; and

(3902) (iv) Include a statement of basis justifying the size of the requested drinking water intake zone, for example, identifying areas of intensive boating activities.

(3903) (2) If the Administrator finds that a complete prohibition is appropriate under this paragraph, he or she shall publish notice of such finding together with a notice of proposed rulemaking, and then shall proceed in accordance with 5 U.S.C. 553. If the Administrator's finding is that a complete prohibition covering a more restricted or more expanded area than that applied for by the State is appropriate, he or she shall also include a statement of the reasons why the finding differs in scope from that requested in the State's application.

(3904) (3) If the Administrator finds that a complete prohibition is inappropriate under this paragraph, he or she shall deny the application and state the reasons for such denial.

(3905) (4) For the following waters the discharge from a vessel of any sewage, whether treated or not, is completely prohibited pursuant to CWA section 312(f)(4)(B):

(3906) (i) Two portions of the Hudson River in New York State, the first is bounded by an east-west line through the most northern confluence of the Mohawk River which will be designated by the Troy-Waterford Bridge (126th Street Bridge) on the south and Lock 2 on the north, and the second of which is bounded on the north by the southern end of Houghtaling Island and on the south by a line between the Village of Roseton on the western shore and Low Point on the eastern shore in the vicinity of Chelsea, as described in Items 2 and 3 of 6 NYCRR Part 858.4.

(3907) (ii) [Reserved]

(3908)

§140.5 Analytical procedures.

(3909) In determining the composition and quality of effluent discharge from marine sanitation devices, the procedures contained in 40 CFR part 136, "Guidelines Establishing Test Procedures for the Analysis of Pollutants," or subsequent revisions or amendments thereto, shall be employed.

(3910)

TITLE 46–SHIPPING

(3911)

Part 15–Manning Requirements (in part)

(3912)

Subpart I–Vessels in Foreign Trade

(3913)

§15.1001 General.

(3914) Self-propelled vessels engaged in foreign commerce are required to use a pilot holding a valid MMC or license with appropriate endorsement as a first-class pilot when operating in the navigable waters of the United States specified in this subpart.

(3915)

§15.1030 New York and New Jersey.

(3916) The following U.S. navigable waters located within the States of New York and New Jersey when the vessel is making an intra-port transit, to include, but not limited to, a movement from a dock to a dock, from a dock to an anchorage, from an anchorage to a dock, or from an anchorage to an anchorage, within the following listed operating areas:

(3917) (a) East River from Execution Rocks to New York Harbor, Upper Bay;

(3918) (b) Hudson River from Yonkers, New York to New York Harbor, Upper Bay;

(3919) (c) Raritan River from Grossman Dock/Arsenal to New York Harbor, Lower Bay;

(3920) (d) Arthur Kill Channel;

(3921) (e) Kill Van Kull Channel;

(3922) (f) Newark Bay;

(3923) (g) Passaic River from Point No Point to Newark Bay;

(3924) (h) Hackensack River from the turning basin to Newark Bay; and

(3925) (i) New York Harbor, Upper and Lower Bay.

(3926) **Note to §15.1030:**

(3927) "Intra-port transit" as used in this section includes the movement of a foreign-trade vessel inbound from sea from the point where a State-licensed pilot ceases providing pilotage to another point within the identified areas (*i.e.,* a dock or anchorage). Likewise, intra-port transit also includes the movement of a foreign-trade vessel outbound to sea from a point within the identified areas (*i.e.,* a dock or anchorage) to the point where a State licensed pilot begins providing pilotage.

(3928)

§15.1040 Massachusetts.

(3929) The following U.S. navigable waters located within the State of Massachusetts when the vessel is in transit, but not bound to or departing from a port within the following listed operating areas:

(3930) (a) Cape Cod Bay south of 41°48'54"N.;

(3931) (b) The Cape Cod Canal; and

(3932) (c) Buzzards Bay east of a line extending from the southernmost point of Wilbur Point (41°34'55"N., 70°51'15"W.) to the easternmost point of Pasque Island (41°26'55"N., 70°50'30"W.).

(3933)

TITLE 50–WILDLIFE AND FISHERIES

(3934)

Part 222–General Endangered and Threatened Marine Species

(3935)

Subpart A–Introduction and General Provisions

(3936)

§222.101 Purpose and scope of regulations.

(3937) (a) The regulations of parts 222, 223, and 224 of this chapter implement the Endangered Species Act (Act), and govern the taking, possession, transportation, sale, purchase, barter, exportation, importation of, and other requirements pertaining to wildlife and plants under the jurisdiction of the Secretary of Commerce and determined to be threatened or endangered pursuant to section 4(a) of the Act. These regulations are implemented by the National Marine Fisheries Service, National Oceanic and Atmospheric Administration, U.S. Department of Commerce, This part pertains to general provisions and definitions. Specifically, parts 223 and 224 pertain to provisions to threatened species and endangered species, respectively. Part 226 enumerates designated critical habitat for endangered and threatened species. Certain of the endangered and threatened marine species enumerated in §§ 224.102 and 223.102 are included in Appendix I or II to the Convention on International Trade of Endangered Species of Wild Fauna and Flora. The importation, exportation, and re-exportation of such species are subject to additional regulations set forth at 50 CFR part 23, chapter I.

(3938) (b) For rules and procedures relating to species determined to be threatened or endangered under the jurisdiction of the Secretary of the Interior, see 50 CFR parts 10 through 17. For rules and procedures relating to the general implementation of the Act jointly by the Departments of the Interior and Commerce and for certain species under the joint jurisdiction of both the Secretaries of the Interior and Commerce, see 50 CFR Chapter IV. Marine mammals listed as endangered or threatened and subject to these regulations may also be subject to additional requirements pursuant to the Marine Mammal Protection Act (for regulations implementing that act, see 50 CFR part 216).

(3939) (c) No statute or regulation of any state shall be construed to relieve a person from the restrictions, conditions, and requirements contained in parts 222, 223, and 224 of this chapter. In addition, nothing in parts 222, 223, and 224 of this chapter, including any permit issued pursuant thereto, shall be construed to relieve a person from any other requirements imposed by a statute or regulation of any state or of the United States, including any applicable health, quarantine, agricultural, or customs laws or regulations, or any other National Marine Fisheries Service enforced statutes or regulations.

(3940)

Part 224–Endangered Marine and Anadromous Species

(3941)

§224.103 Special prohibitions for endangered marine mammals.

(3942) (c) *Approaching right whales*. (1) *Prohibitions*. Except as provided under paragraph (c)(3) of this section, it is unlawful for any person subject to the jurisdiction of the United States to commit, attempt to commit, to solicit another to commit, or cause to be committed any of the following acts:

(3943) (i) Approach (including by interception) within 500 yards (460 m) of a right whale by vessel, aircraft, or any other means;

(3944) (ii) Fail to undertake required right whale avoidance measures specified under paragraph (c)(2) of this section.

(3945) (2) *Right whale avoidance measures*. Except as provided under paragraph (c)(3) of this section, the following avoidance measures must be taken if within 500 yards (460 m) of a right whale:

(3946) (i) If underway, a vessel must steer a course away from the right whale and immediately leave the area at slow safe speed.

(3947) (ii) An aircraft must take a course away from the right whale and immediately leave the area at a constant airspeed.

(3948) (3) *Exceptions*. The following exceptions apply to this section, but any person who claims the applicability of an exception has the burden of proving that the exception applies:

(3949) (i) Paragraphs (c)(1) and (c)(2) of this section do not apply if a right whale approach is authorized by the National Marine Fisheries Service through a permit issued under part 222, subpart C, of this chapter (General Permit Procedures) or through a similar authorization.

(3950) (ii) Paragraphs (c)(1) and (c)(2) of this section do not apply where compliance would create an imminent and serious threat to a person, vessel, or aircraft.

(3951) (iii) Paragraphs (c)(1) and (c)(2)of this section do not apply when approaching to investigate a right whale entanglement or injury, or to assist in the disentanglement or rescue of a right whale, provided that permission is received from the National Marine Fisheries Service or designee prior to the approach.

(3952) (iv) Paragraphs (c)(1) and (c)(2) of this section do not apply to an aircraft unless the aircraft is conducting whale watch activities.

(3953) (v) Paragraph (c)(2) of this section does not apply to the extent that a vessel is restricted in her ability to maneuver and, because of the restriction, cannot comply with paragraph (c)(2) of this section.

(3954)

§224.105 Speed restrictions to protect North Atlantic Right Whales.

(3955) (a) The following restrictions apply to: All vessels greater than or equal to 65 ft (19.8 m) in overall length and subject to the jurisdiction of the United States, and all other vessels greater than or equal to 65 ft (19.8 m) in overall length entering or departing a port or place subject to the jurisdiction of the United States. These restrictions shall not apply to U.S. vessels owned or operated by, or under contract to, the Federal Government. This exemption extends to foreign sovereign vessels when they are engaging in joint exercises with the U.S. Department of the Navy. In addition, these restrictions do not apply to law enforcement vessels of a State, or political subdivision thereof, when engaged in law enforcement or search and rescue duties.

(3956) (1) *Southeast U.S.* (south of St. Augustine, FL to north of Brunswick, GA): Vessels shall travel at a speed of 10 knots or less over ground during the period of November 15 to April 15 each year in the area bounded by the following: Beginning at 31°27'00.0"N., 80°51'36.0"W.; thence west to charted mean high water line then south along charted mean high water line and inshore limits of COLREGS limit to a latitude of 29°45'00.0"N., thence east to 29°45'00.0"N., 80°51'36.0"W.; thence back to starting point. (Fig. 1).

(3957) (2) *Mid-Atlantic U.S.* (from north of Brunswick, Georgia to Rhode Island): Vessels shall travel 10 knots or less over ground in the period November 1 to April 30 each year:

(3958) (i) In the area bounded by the following: 33°56'42.0"N., 77°31'30.0"W.; thence along a NW bearing of **313.26°** True to charted mean high water line then south along mean high water line and inshore limits of COLREGS limit to a latitude of 31°27'00.0"N.; thence east to

(3959) 31°27'00.0"N., 80°51'36.0"W.; thence to

(3960) 31°50'00.0"N., 80°33'12.0"W.; thence to

(3961) 32°59'06.0"N., 78°50'18.0"W.; thence to

(3962) 33°28'24.0"N., 78°32'30.0"W.; thence to

(3963) 33°36'30.0"N., 77°47'06.0"W.; thence back to starting point;

(3964) (ii) Within a 20-nm (37 km) radius (as measured seaward from COLREGS delineated coast lines and the center point of the port entrance) (Fig. 2) at the

(3965) (A) Ports of New York/New Jersey:

(3966) 40°29'42.2"N., 73°55'57.6"W.;

(3967) (B) Delaware Bay (Ports of Philadelphia and Wilmington):

(3968) 38°52'27.4"N., 75°01'32.1"W.;

(3969) (C) Entrance to the Chesapeake Bay (Ports of Hampton Roads and Baltimore):

(3970) 37°00'36.9"N., 75°57'50.5"W.; and

(3971) (D) Ports of Morehead City and Beaufort, NC: 34°41'32.0"N., 76°40'08.3"W.; and

(3972) (iii) In Block Island Sound, in the area bounded by the following coordinates: Beginning at

(3973) 40°51'53.7"N., 70°36'44.9"W.; thence to

(3974) 41°20'14.1"N., 70°49'44.1"W.; thence to

(3975) 41°04'16.7"N., 71°51'21.0"W.; thence to

(3976) 40°35'56.5"N., 71°38'25.1"W.; thence back to starting point. (Fig. 2).

(3977) (3) Northeast U.S. (north of Rhode Island):

(3978) (i) In Cape Cod Bay, MA: Vessels shall travel at a speed of 10 knots or less over ground during the period of January 1 to May 15 in Cape Cod Bay, in an area beginning at 42°04'56.5"N., 70°12'00.0"W.; thence north to 42°12'00.0"N., 70°12'00.0"W.; thence due west to charted mean high water line; thence along charted mean high water within Cape Cod Bay back to beginning point. (Fig. 3).

(3979) (ii) Off Race Point: Vessels shall travel at a speed of 10 knots or less over ground during the period of March 1 to April 30 each year in waters bounded by straight lines connecting the following points in the order stated (Fig. 3):

(3980) 42°30'00.0"N., 69°45'00.0"W.; thence to

(3981) 42°30'00.0"N., 70°30'00.0"W.; thence to

(3982) 42°12'00.0"N., 70°30'00.0"W.; thence to

(3983) 42°12'00.0"N., 70°12'00.0"W.; thence to

(3984) 42°04'56.5"N., 70°12'00.0"W.; thence along charted mean high water line and inshore limits of COLREGS limit to a latitude of 41°40'00.0"N., thence due east to 41°41'00.0"N., 69°45'00.0"W.; thence back to starting point.

(3985) (iii) Great South Channel: Vessels shall travel at a speed of 10 knots or less over ground during the period of April 1 to July 31 each year in all waters bounded by straight lines connecting the following points in the order stated (Fig. 3):

(3986) 42°30'00.0"N., 69°45'00.0"W.

(3987) 41°40'00.0"N., 69°45'00.0"W.

(3988) 41°00'00.0"N., 69°05'00.0"W.

(3989) 42°09'00.0"N., 67°08'24.0"W.

(3990) 42°30'00.0"N., 67°27'00.0"W.

(3991) 42°30'00.0"N., 69°45'00.0"W.

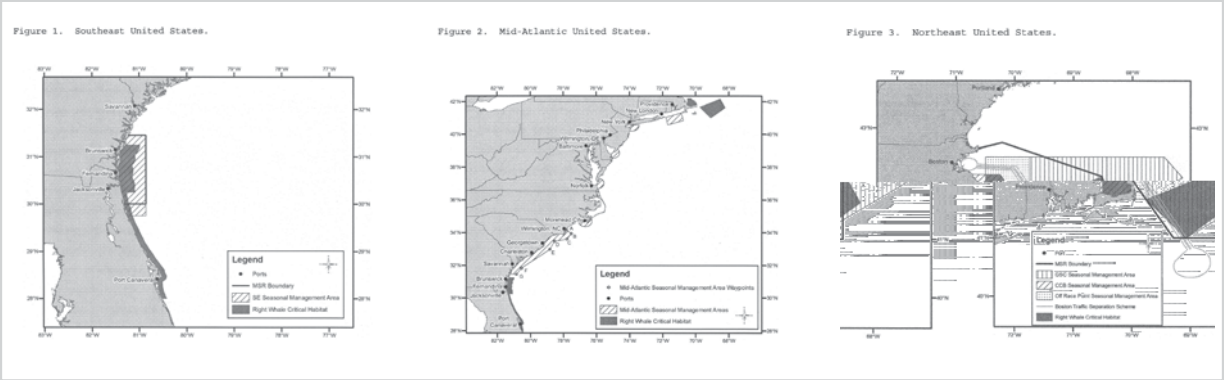
(3992) (b) Except as noted in paragraph (c) of this section, it is unlawful under this section:

(3993) (1) For any vessel subject to the jurisdiction of the United States to violate any speed restriction established in paragraph (a) of this section; or

(3994) (2) For any vessel entering or departing a port or place under the jurisdiction of the United States to violate any speed restriction established in paragraph (a) of this section.

(3995) (c) A vessel may operate at a speed necessary to maintain safe maneuvering speed instead of the required ten knots only if justified because the vessel is in an area where oceanographic, hydrographic and/or meteorological conditions severely restrict the

(3997)



maneuverability of the vessel and the need to operate at such speed is confirmed by the pilot on board or, when a vessel is not carrying a pilot, the master of the vessel. If a deviation from the ten-knot speed limit is necessary, the reasons for the deviation, the speed at which the vessel is operated, the latitude and longitude of the area, and the time and duration of such deviation shall be entered into the logbook of the vessel. The master of the vessel shall attest to the accuracy of the logbook entry by signing and dating it.

(3996) (d) No later than January 1, 2019, the National Marine Fisheries Service will publish and seek comment on a report evaluating the conservation value and economic and navigational safety impacts of this section, including any recommendations to minimize burden of such impacts.

(3998)

Part 226–Designated Critical Habitat

(3999)

§226.101 Purpose and scope.

(4000) The regulations contained in this part identify those habitats designated by the Secretary of Commerce as critical, under section 4 of the Act, for endangered and threatened species under the jurisdiction of the Secretary of Commerce. Those species are enumerated at §223.102 of this chapter if threatened and at §224.101 of this chapter if endangered. For regulations pertaining to the designation of critical habitat, see part 424 of this title; for regulations pertaining to prohibitions against the adverse modification or destruction of critical habitat, see part 402 of this title. Additional information regarding designated critical habitats that is not provided in this section may be obtained upon request to the Office of Protected Resources (*see* §222.102, definition of "Office of Protected Resources").

(4001)

§ 226.203 Critical habitat for North Atlantic right whales (*Eubalaena glacialis*).

(4002) Critical habitat is designated for North Atlantic right whales as described in this section. The textual descriptions in paragraph (b) of this section are the definitive source for determining the critical habitat boundaries. The maps of the critical habitat units provided in paragraph (c) of this section are for illustrative purposes only.

(4003) (a) Physical and biological features essential to the conservation of endangered North Atlantic right whales.

(4004) (1) *Unit 1*. The physical and biological features essential to the conservation of the North Atlantic right whale, which provide foraging area functions in Unit 1 are: The physical oceanographic conditions and structures of the Gulf of Maine and Georges Bank region that combine to distribute and aggregate *C.finmarchicus* for right whale foraging, namely prevailing currents and circulation patterns, bathymetric features (basins, banks, and channels), oceanic fronts, density gradients, and temperature regimes; low flow velocities in Jordan, Wilkinson, and Georges Basins that allow diapausing *C.finmarchicus* to aggregate passively below the convective layer so that the copepods are retained in the basins; late stage *C.finmarchicus* in dense aggregations in the Gulf of Maine and Georges Bank region; and diapausing *C.finmarchicus* in aggregations in the Gulf of Maine and Georges Bank region.

(4005) (2) *Unit 2*. The physical features essential to the conservation of the North Atlantic right whale, which provide calving area functions in Unit 2, are:

(4006) (i) Sea surface conditions associated with Force 4 or less on the Beaufort Scale,

(4007) (ii) Sea surface temperatures of 7°C to 17°C, and

(4008) (iii) Water depths of 6 to 28 meters, where these features simultaneously co-occur over contiguous areas of at least 231 nmi2 of ocean waters during the months of November through April. When these features are available, they are selected by right whale cows and calves in dynamic combinations that are suitable for calving, nursing, and rearing, and which vary, within the ranges specified, depending on factors such as weather and age of the calves.

(4009) (b) *Critical habitat boundaries*. Critical habitat includes two areas (Units) located in the Gulf of Maine and Georges Bank Region (Unit 1) and off the coast of North Carolina, South Carolina, Georgia and Florida (Unit 2).

(4010) (1) *Unit 1*. The specific area on which are found the physical and biological features essential to the conservation of the North Atlantic right whale include all waters, seaward of the boundary delineated by the line connecting the geographic coordinates and landmarks identified herein:

(4011) (i) The southern tip of Nauset Beach (Cape Cod) (41°38.39′N., 69°57.32′W.).

(4012) (ii) From this point, southwesterly to 41°37.19′N., 69°59.11′W.

(4013) (iii) From this point, southward along the eastern shore of South Monomoy Island to 41°32.76′N., 69°59.73′W.

(4014) (iv) From this point, southeasterly to 40°50′N., 69°12′W.

(4015) (v) From this point, east to 40°50′N., 68°50′W.

(4016) (vi) From this point, northeasterly to 42°00′N., 67°55′W.

(4017) (vii) From this point, east to 42°00′N., 67°30′W.

(4018) (viii) From this point, northeast to the intersection of the U.S.-Canada maritime boundary and 42°10′N.

(4019) (ix) From this point, following the U.S.-Canada maritime boundary north to the intersection of 44°49.727′N., 66°57.952′W.; From this point, moving southwest along the coast of Maine, the specific area is located seaward of the line connecting the following points:

(4020)

Latitude	Longitude
44°49.727'N.	66°57.952'W.
44°49.67'N.	66°57.77'W.
44°48.64'N.	66°56.43'W.
44°47.36'N.	66°59.25'W.
44°45.51'N.	67°02.87'W.
44°37.07'N.	67°09.75'W.
44°27.77'N.	67°32.86'W.
44°25.74'N.	67°38.39'W.
44°21.66'N.	67°51.78'W.
44°19.08'N.	68°02.05'W.
44°13.55'N.	68°10.71'W.
44°08.36'N.	68°14.75'W.
43°59.36'N.	68°37.95'W.
43°59.83'N.	68°50.06'W.
43°56.72'N.	69°04.89'W.
43°50.28'N.	69°18.86'W.
43°48.96'N.	69°31.15'W.
43°43.64'N.	69°37.58'W.
43°41.44'N.	69°45.27'W.
43°36.04'N.	70°03.98'W.
43°31.94'N.	70°08.68'W.
43°27.63'N.	70°17.48'W.
43°20.23'N.	70°23.64'W.
43°04.06'N.	70°36.70'W.
43°02.93'N.	70°41.47'W.
43°02.55'N.	70°43.33'W.

(4021) (x) From this point (43°2.93′N., 70°41.47′W.) on the coast of New Hampshire south of Portsmouth, the boundary of the specific area follows the coastline southward along the coasts of New Hampshire and Massachusetts along Cape Cod to Provincetown southward along the eastern edge of Cape Cod to the southern tip of Nauset Beach (Cape Cod) (41°38.39′N., 69°57.32′W.) with the exception of the area landward of the lines drawn by connecting the following points:

(4022)

Latitude	Longitude		
42°59.986'N.	70°44.654'W.	to	Rye Harbor
42°59.956'N.	70°44.737'W.		Rye Harbor
42°53.691'N.	70°48.516'W.	to	Hampton Harbor
42°53.519'N.	70°48.748'W.		Hampton Harbor
42°49.136'N.	70°48.242'W.	to	Newburyport Harbor
42°48.964'N.	70°48.282'W.		Newburyport Harbor
42°42.145'N.	70°46.995'W.	to	Plum Island Sound
42°41.523'N.	70°47.356'W.		Plum Island Sound
42°40.266'N.	70°43.838'W.	to	Essex Bay
42°39.778'N.	70°43.142'W.		Essex Bay
42°39.645'N.	70°36.715'W.	to	Rockport Harbor
42°39.613'N.	70°36.60'W.		Rockport Harbor
42°20.665'N.	70°57.205'W.	to	Boston Harbor
42°20.009'N.	70°55.803'W.		Boston Harbor
42°19.548'N.	70°55.436'W.	to	Boston Harbor
42°18.599'N.	70°52.961'W.		Boston Harbor
42°15.203'N.	70°46.324'W.	to	Cohasset Harbor
42°15.214'N.	70°47.352'W.		Cohasset Harbor
42°12.09'N.	70°42.98'W.	to	Scituate Harbor
42°12.211'N.	70°43.002'W.		Scituate Harbor
42°09.724'N.	70°42.378'W.	to	New Inlet
42°10.085'N.	70°42.875'W.		New Inlet
42°04.64'N.	70°38.587'W.	to	Green Harbor
42°04.583'N.	70°38.631'W.		Green Harbor
41°59.686'N.	70°37.948'W.	to	Duxbury Bay/ Plymouth Harbor
41°58.75'N.	70°39.052'W.		Duxbury Bay/ Plymouth Harbor
41°50.395'N.	70°31.943'W.	to	Ellisville Harbor
41°50.369'N.	70°32.145'W.		Ellisville Harbor
41°45.87'N.	70°28.62'W.	to	Sandwich Harbor
41°45.75'N.	70°28.40'W.		Sandwich Harbor
41°44.93'N.	70°25.74'W.	to	Scorton Harbor
41°44.90'N.	70°25.60'W.		Scorton Harbor
41°44.00'N.	70°17.50'W.	to	Barnstable Harbor
41°44.00'N.	70°13.90'W.		Barnstable Harbor
41°45.53'N.	70°09.387'W.	to	Sesuit Harbor
41°45.523'N.	70°09.307'W.		Sesuit Harbor
41°45.546'N.	70°07.39'W.	to	Quivett Creek

(4036)

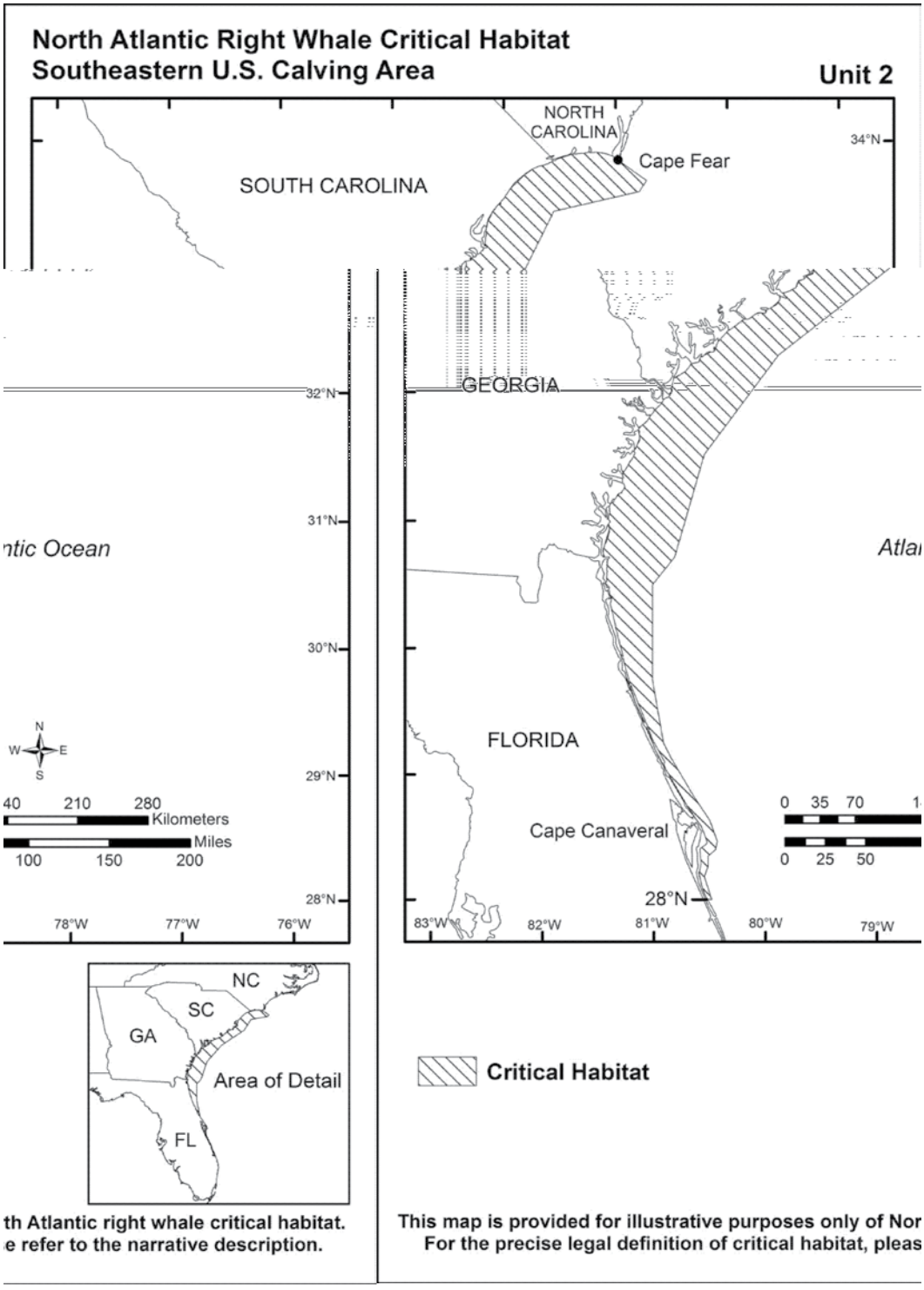
North Atlantic Right Whale Critical Habitat
Northeastern U.S. Foraging Area **Unit 1**

Critical Habitat

200m Depth Contour

This map is provided for illustrative purposes only of North Atlantic right whale critical habitat. For the precise legal definition of critical habitat, please refer to the narrative description.

(4037)


North Atlantic Right Whale Critical Habitat
Southeastern U.S. Calving Area
Unit 2
NORTH CAROLINA
SOUTH CAROLINA
Cape Fear
34°N
GEORGIA
32°N
31°N
30°N
29°N
28°N
FLORIDA
Cape Canaveral
78°W
77°W
76°W
83°W
82°W
81°W
80°W
79°W
N
W
E
S
40 210 280
Kilometers
Miles
100 150 200
0 35 70
0 25 50
NC
SC
GA
FL
Area of Detail
Critical Habitat
th Atlantic right whale critical habitat.
e refer to the narrative description.
This map is provided for illustrative purposes only of Nor
For the precise legal definition of critical habitat, pleas

Latitude	Longitude		
41°45.551'N.	70°07.32'W.		Quivett Creek
41°47.269'N.	70°01.411'W.	to	Namskaket Creek
41°47.418'N.	70°01.306'W.		Namskaket Creek
41°47.961'N.	70°0.561'W.	to	Rock Harbor Creek
41°48.07'N.	70°0.514'W.		Rock Harbor Creek
41°48.432'N.	70°0.286'W.	to	Boat Meadow River
41°48.483'N.	70°0.216'W.		Boat Meadow River
41°48.777'N.	70°0.317'W.	to	Herring River
41°48.983'N.	70°0.196'W.		Herring River
41°55.501'N.	70°03.51'W.	to	Herring River, inside Wellfleet Harbor
41°55.322'N.	70°03.191'W.		Herring River, inside Wellfleet Harbor
41°53.922'N.	70°01.333'W.	to	Blackfish Creek/ Loagy Bay
41°54.497'N.	70°01.182'W.		Blackfish Creek/ Loagy Bay
41°55.503'N.	70°02.07'W.	to	Duck Creek
41°55.753'N.	70°02.281'W.		Duck Creek
41°59.481'N.	70°04.779'W.	to	Pamet River
41°59.563'N.	70°04.718'W.		Pamet River
41°03.601'N.	70°14.269'W.	to	Hatches Harbor
41°03.601'N.	70°14.416'W.		Hatches Harbor
41°48.708'N.	69°56.319'W.	to	Nauset Harbor
41°48.554'N.	69°56.238'W.		Nauset Harbor
41°40.685'N.	69°56.781'W.	to	Chatham Harbor
41°40.884'N.	69°56.28'W.		Chatham Harbor

(4023) (xi) In addition, the specific area does not include waters landward of the 72 COLREGS lines (33 CFR part 80) described below.

(4024) (A) *Portland Head, ME to Cape Ann, MA*.

(4025) (1) A line drawn from the northernmost extremity of Farm Point to Annisquam Harbor Light.

(4026) (2) [Reserved]

(4027) (B) *Cape Ann MA to Marblehead Neck, MA*.

(4028) (1) A line drawn from Gloucester Harbor Breakwater Light to the twin towers charted at latitude 42°35.1′N., longitude 70°41.6′W.

(4029) (2) A line drawn from the westernmost extremity of Gales Point to the easternmost extremity of House Island; thence to Bakers Island Light; thence to Marblehead Light.

(4030) (C) *Hull, MA to Race Point, MA*.

(4031) (1) A line drawn from Canal Breakwater Light 4 south to the shoreline.

(4032) (2) [Reserved]

(4033) (2) *Unit 2*. Unit 2 includes marine waters from Cape Fear, North Carolina, southward to 28°N latitude (approximately 31 miles south of Cape Canaveral, Florida) within the area bounded on the west by the shoreline and the 72 COLREGS lines, and on the east by rhumb lines connecting the following points in the order stated from north to south.

(4034)

Latitude	Longitude
33°51'N.	at shoreline
33°42'N.	77°43'W.
33°37'N.	77°47'W.
33°28'N.	78°33'W.
32°59'N.	78°50'W.
32°17'N.	79°53'W.
31°31'N.	80°33'W.
30°43'N.	80°49'W.
30°30'N.	81°01'W.
29°45'N.	81°01'W.
29°15'N.	80°55'W.
29°08'N.	80°51'W.
28°50'N.	80°39'W.
28°38'N.	80°30'W.
28°28'N.	80°26'W.
28°24'N.	80°27'W.
28°21'N.	80°31'W.
28°16'N.	80°31'W.
28°11'N.	80°33'W.
28°00'N.	80°29'W.
28°00'N.	at shoreline

(4035) (c) Overview maps of the designated critical habitat for the North Atlantic right whale follow.

Cape Cod To Sandy Hook

(1) The Atlantic coast from Cape Cod to Sandy Hook embraces part of the coast of Massachusetts and all of the coasts of Rhode Island, Connecticut and New York. To the mariner this area presents problems of unusual difficulty because of the off-lying shoals, strong and variable currents, large amounts of fog and turbulence of wind and sea in the great storms that so frequently sweep it. Additionally, the mariner is faced with the great volume of waterborne traffic that moves through the area to and from the Port of New York.

(2)

Prominent features

(3) The principal geographic features include Georges Bank, Nantucket and Vineyard Sounds, Buzzards Bay, Narragansett Bay, Long Island Sound and tributaries and New York Harbor and tributaries including the Hudson River.

(4) Cape Cod, a long peninsula jutting eastward from the mainland of Massachusetts, may be likened to an arm bent upward at the elbow. It was originally formed by the last great glacier and has been refashioned by the seas and wind. The outer end of The Cape, as it is called by eastern New Englanders, is a barren region of sand dunes with long yellow beaches, while much of the remainder of the forearm is bleak grassy country. The southern side of the delta-like plain of Cape Cod has been cut along high bluffs by the surf and waves. This section of the coast is covered with growth of pitch pine and scrub oak.

(5) Nantucket, Martha's Vineyard, the Elizabeth Islands and numerous smaller islands were also formed by the glacier. The plains of Martha's Vineyard and Nantucket are broad grassy heaths. The Elizabeth Islands are hilly and partly wooded, and generally the shores are low bluffs.

(6) The western shore of Buzzards Bay is of moderate height, very gently sloping, cleared and cultivated with occasional groves of trees. Several towns and the city of New Bedford are visible along the shores.

(7) Between Buzzards and Narragansett Bays the coast is a mass of sand dunes with steep faces forming a line along the shore. Several headlands along this stretch of coast have fine sand beaches between them.

(8) The boundary line between Massachusetts and Rhode Island strikes the coast just westward of Quicksand Point.

(9) Among the islands in Narragansett Bay are Rhode (Aquidneck) Island, Conanicut and Prudence. These rather large islands are gently sloping, undulating and covered with cultivated fields and orchards and occasional groves of trees.

(10) Westerly from Point Judith to Napatree Point is a continuous line of beaches behind which are many saltponds. These ponds have been formed by the sea breaking through the outer sand barrier and then depositing sand to close the opening. The shore near the water is low, grassy and nearly level but gradually rises with a series of gentle curves to higher wooded lands some distance back.

(11) Block Island is another formation of the glacier. A prominent feature of the island is the entire absence of trees. The surface when viewed from eastward has a grassy undulating appearance, and the hills in many places show steep sandy faces. Near the shoreline the land is low but rapidly rises toward the center of the island to steep hills covered only with grass and dotted occasionally with houses.

(12) The boundary line between Rhode Island and Connecticut follows the Pawcatuck River to above the head of navigation.

(13) The coastline of Connecticut is rockbound and rugged, with numerous sandy beaches and occasional salt meadows or marshland. The surface is mildly rolling near the shore. The depression of small valleys along the shore has created a number of good harbors. The shoreline has been well developed commercially and residentially. It is lined with seaside resorts, state parks and bathing beaches.

(14) The boundary line between Connecticut and New York follows the Byram River for slightly over 1 mile.

(15) Long Island, originally formed by the glacier and thrusting about 105 miles eastward from New York Bay to a point abreast of New London, faces the New England coast across Long Island Sound on the north. Its eastern end is split by Peconic Bay and the 35- and 25-mile peninsulas thus formed are the north and south flukes. The island is almost a plain. On the north coast, bluffs rise to a height of 200 feet. South of these, extending well into the island's midsection, run several chains of hills. The south shore is a barrier beach from about 30 miles west of the eastern extremity to the western end, which has been developed into a series of bathing resorts.

(16)

Disposal sites and dumping grounds

(17) These areas are rarely mentioned in the Coast Pilot but are shown on the nautical charts. (See Disposal Sites and Dumping Grounds, chapter 1, and charts for limits.)

(18)

Aids to navigation

(19) Lights and buoys are the principal guides that mark the approaches to the important harbors. Many of the light stations have sound signals, particularly those in the vicinity of the larger ports.

(20) **Radar** is an important aid in most of this area but should not be relied upon for ranges to the beach in areas such as the south coast of Long Island that offer a relatively low relief. Many of the coastal buoys are equipped with radar reflectors. Radar is of particular importance in detecting other traffic and in the prevention of collisions during periods of low visibility, which are common in this area—refer to the U.S. Coast Guard's Light List for a complete description of navigational aids.

(21)

COLREGS Demarcation Lines

(22) Lines have been established to delineate those waters upon which mariners must comply with the Inland Navigational Rules Act of 1980 (Inland Rules). The waters inside of the lines are **Inland Rules Waters,** and the waters outside of the lines are **COLREGS Waters.** (See **33 CFR 80,** chapter 2, for specific lines of demarcation.)

(23)

Ports and waterways safety

(24) Refer to **33 CFR 160**, chapter 2, for regulations governing vessel operations and requirements for notification of arrivals, hazardous conditions and certain dangerous cargoes to the Captain of the Port.

(25) **Regulated Navigation Areas** have been established within the navigable waters of the First Coast Guard District to increase operational safety for towing vessels and tank barges. (See **33 CFR 165.100**, chapter 2, for limits and regulations.)

(26)

Harbor entrances

(27) The entrances to most of the harbors have dredged channels marked with navigational aids and are easy of access. In some cases jetties and breakwaters extend offshore from the entrances. The entrances to the inlets along the south shore of Long Island are subject to frequent change due to the shifting sand bars.

(28) **Tra f c Separation Schemes (Tra f c Lanes)** have been established in the approaches to Buzzards Bay, Narragansett Bay and New York Harbor. (See chapters 5, 6 and 11, respectively, for details.)

(29) **Vessel Tra f c Service, New York**, operated by the U.S. Coast Guard, serves New York Harbor. (See **33 CFR 161.1** through **161.25**, chapter 2, for regulations.)

(30)

Channels

(31) Federal project depth is a design dredging depth authorized for a channel constructed by the U.S. Army Corps of Engineers (USACE) or their authorized contractor; the project depth may or may not be the goal of maintenance dredging after completion of the original channel, and for this reason, project depth must not be confused with controlling depth. Controlling depth in a channel is its least depth; it restricts use of the channel to drafts less than that depth.

(32) Channel information and minimum depths in deepwater and secondary channels that are regularly maintined by the USACE are available on NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A. Owing to constant shoaling in places, depths may vary considerably between maintenance dredgings. In the case of other channels, the controlling depths printed in the Coast Pilot are from the latest available reports which may, however, be several years old.

(33)

Anchorages

(34) There are numerous anchorages in Nantucket and Vineyard Sounds, Buzzards, Narragansett and Gardiners Bays and Long Island Sound, where vessels with good ground tackle can ride out any gale. Between Cape Cod and Sandy Hook, the more important harbors, either commercially or as harbors of refuge, are New Bedford, Newport, Providence, New London, New Haven and Bridgeport on the mainland, Greenport and Port Jefferson on Long Island, City Island, New York and vast New York Harbor. (See **33 CFR 110**, chapter 2, for limits and regulations.)

(35)

Marine Protected Areas

(36) The chapters that follow may contain references to Marine Protected Areas (MPAs) occurring in navigable coastal waters of the Northeast Atlantic coast. This critical environmental information is intended to inform readers about the location, purpose and legal restrictions of coastal MPAs, with an emphasis on activities of interest to the maritime community. For detailed information on MPAs, visit *marineprotectedareas.noaa.gov*. Some of the major MPAs are listed below.

(37) **Northern Inshore Lobster Waters** includes the State waters of RI, MA, NH and ME.

(38) **Northern Nearshore Lobster Waters** includes the federal waters of RI, MA, NH and ME.

(39) **O f shore Lobster Waters**, about 60 miles offshore to the Exclusive Economic Zone (EEZ) edge, extend from the U.S. Canadian border to Cape Hatteras, NC.

(40) **Southern Nearshore Lobster Waters and Mid-Atlantic Coastal Waters Area** includes the state and federal waters along the continental shelf from Long Island to Cape Hatteras, NC.

(41)

Dangers

(42) The most important dangers confronting the navigator when approaching the area are the great banks and shoals in the eastern approach. The remainder of the isolated dangers throughout the area and in the approaches to the harbors are for the most part well marked and charted.

(43)

Pipelaying barges

(44) With the increased number of pipeline laying operations, operators of all types of vessels should be aware of the dangers of passing close aboard, close ahead or close astern of a jetbarge or pipelaying barge. Pipelaying barges and jetbarges usually move at 0.5 knot or less and have anchors that extend out about 3,500 to 5,000 feet in all directions and that may be marked by lighted anchor buoys. The exposed pipeline behind the pipelaying barge and the area in the vicinity of anchors are hazardous to navigation and should be avoided. The pipeline and anchor cables also represent a submerged hazard to navigation. It is suggested, if safe navigation permits, for all types of vessels to pass well ahead of the pipelaying barge or well astern of the jetbarge. The pipelaying barge, jetbarge and attending vessels may be contacted on VHF-FM channel 16 (156.80 MHz) for passage instructions.

(45)

North Atlantic right whales

(46) The North Atlantic right whale is one of the world's most endangered large whale species. North Atlantic right whales are found primarily in continental shelf waters between Florida and Nova Scotia. They migrate annually along the east coast between the feeding grounds off New England and Canada and the calving grounds off Florida, Georgia and South Carolina. Because right whales mate, rest, feed and nurse their young at the surface, and often do not move out of the way of oncoming ships, they are highly vulnerable to being struck. Pregnant females and females with nursing calves appear to be particularly vulnerable to collisions with ships. Ship strikes and fishing gear entanglements are the two known sources of human-related mortality. Intentionally approaching within 500 yards of right whales is prohibited and is a violation of federal law. (See **50 CFR 224.103**, chapter 2 for limits, regulations and exceptions.)

(47) **Description of North Atlantic right whale:** Right whales are large baleen whales. Adults are generally 45 to 55 feet in length and can weigh up to 70 tons. The body is mostly black, but irregularly shaped white patches may be present on the ventral surface. The best field identification marks are a broad back with no dorsal fin, irregular bumpy white patches (callosities) on the head and a distinctive two-column V-shaped blow when viewed from directly behind or in front of the whale. The whales have broad, paddle-shaped flippers and a broad, deeply notched tail. (See following diagrams and photographs.) Right whales are slow moving and seldom travel faster than 5 or 6 knots. They can stay submerged for 10 to 20 minutes and may appear suddenly when surfacing to breathe. They are often seen alone or in small groups. At times, right whales form large courtship groups of 20 to 30 animals.

(49) **Seasonal occurrence of North Atlantic right whales—**During seasons and in areas where right whales may occur, vessel operators should maintain a sharp lookout for whales and reduce speeds when consistent with safe navigation. In any given year oceanographic variability may affect the seasonal distribution of right whales. In 1986, right whales were frequently sighted within the Stellwagen Bank National Marine Sanctuary throughout the summer, and in the early spring of 1998 a large number of right whales were documented near the Narragansett/Buzzards Bay Traffic Separation Scheme. Two areas in U.S. waters have been designated as critical habitats for North Atlantic right whales: the northeastern foraging area and southeastern calving area. (See **50 CFR 226.203**, chapter 2 for limits, regulations and exceptions).

(50)

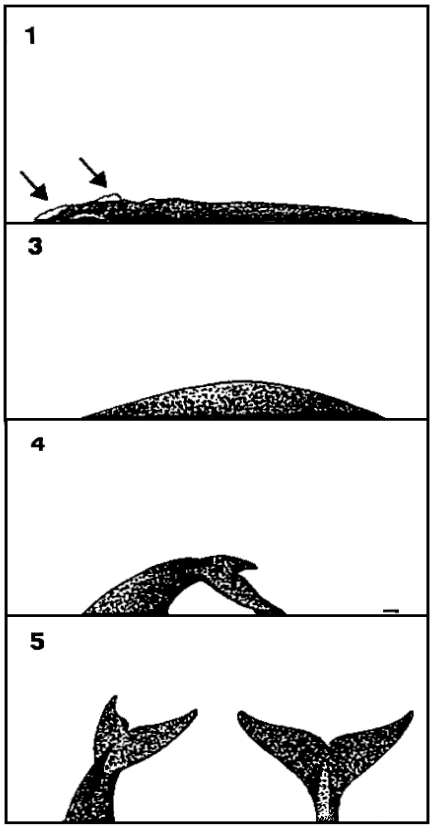
Seasonal occurrence of North Atlantic right whales

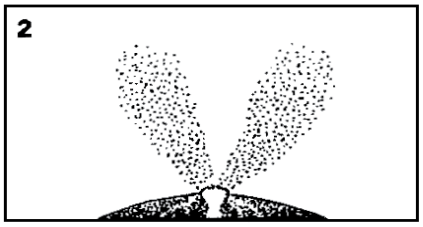
Location	Season	Comments
Central Gulf of Maine (Jordan Basin, Cashes Ledge)	April to June October to December	
Cape Cod Bay	December to May	
Great South Channel, northern edge of Georges Bank	March to July	
Bay of Fundy, Scotian Shelf (Browns Bank, Roseway Basin)	July to October	Most of the population can be found in this area during this time.
Jeffreys Ledge	October to December	Whales are frequently sighted in this area.
Stellwagen Bank National Marine Sanctuary	Year-round	Peak sightings occur in the early spring with infrequent sightings in the summer.
New York to North Carolina	November to April	The migration corridor between right whale habitats is within 30 miles of the Atlantic coast.
South Carolina, Georgia and Florida calving area	November to April	Calving right whale sightings have occurred as far north as Cape Fear, NC and as far south as Miami, FL. Sightings in Gulf waters are rare.

(51) **Mandatory Speed Restrictions:** Vessels 65 feet or greater in length overall (LOA) are subject to mandatory speed restrictions of 10 knots or less in seasonal management areas (SMAs) along the U.S. East Coast during times when right whales are likely to be present (See following maps for locations of SMAs). The Northeastern SMA speed restrictions are in place from January 1 through May 15 in Cape Cod Bay, from March 1 through April 30 off Race Point and from April 1 through July 31 in the Great South Channel. Speed restrictions in the mid-Atlantic U.S. SMAs are in place from November 1 to April 30 and include Block Island Sound, entry into the Ports of New York/New Jersey, Delaware Bay,

(48)

North Atlantic right whale





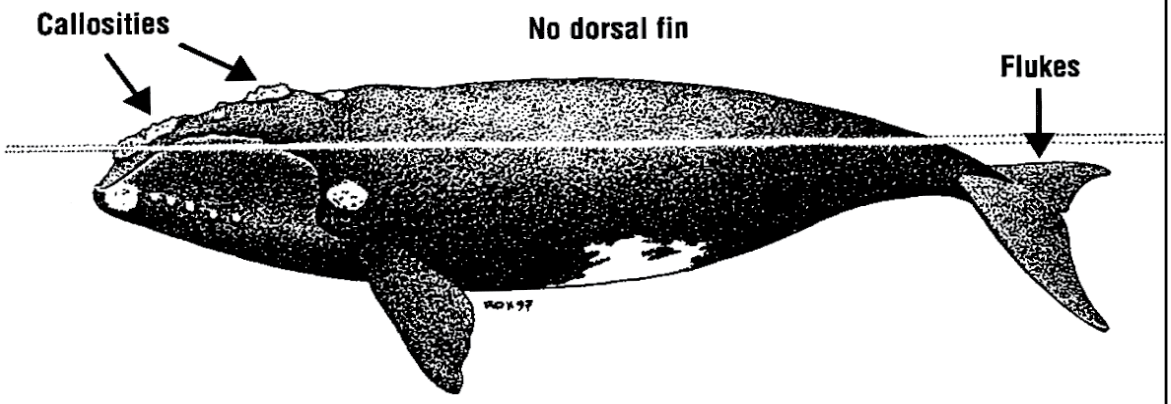
1) Whitish patches of raised and roughened skin (called callosities) on top of the head (see arrows)

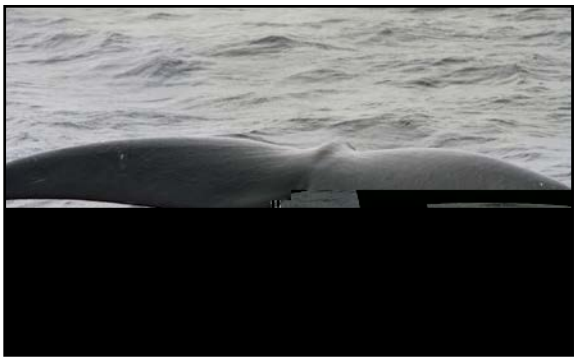
2) V-shaped blow easily visible from in front or behind the whale

3) No dorsal fin on the back

4) Tail flukes often lifted vertically when the animal dives

5) All black tail on the top and underside





Entrance to Chesapeake Bay and the Ports of Morehead City and Beaufort, NC, and within a continuous boundary approximately 20 nautical miles from shore around the major ports of Wilmington, NC, Charleston, SC, and Savannah, GA. Speed restrictions are in place in the Southeastern U.S. SMA from November 15 to April 15; this area extends from shore approximately 30 nautical miles eastward and contains the major ports of Brunswick, GA, Fernandina Beach, FL, and Jacksonville, FL. (See **50 CFR 224.105**, chapter 2 for regulations, limitations and exceptions and complete description of the SMAs.) Boundaries of the SMAs are shown on NOAA Electronic Navigational Charts. NOAA Fisheries may also establish voluntary Dynamic Management Areas (DMAs) when right whales are present in areas and times not covered by the SMAs. Information about established DMAs will be announced over NOAA's customary maritime communication media. Mariners are encouraged to avoid or reduce speeds to 10 knots or less while transiting through DMAs.

(52)

Area to be avoided

(53) In order to significantly reduce the risk of ship strikes to the North Atlantic right whale, an area to be avoided was established in the Great South Channel, east of the Boston Harbor traffic lanes. Ships of 300 gross tons and above should avoid the area bounded by lines connecting the following geographical positions:

(54) 41°44'08"N., 69°34'50"W.;

(55) 42°10'00"N., 68°31'00"W.;

(56) 41°24'53"N., 68°31'00"W.; and

(57) 40°50'28"N., 68°58'40"W. between the period of April 1 through July 31.

(58) **Early Warning and Sighting Advisory Systems:** As weather and conditions permit, dedicated seasonal programs of aerial and vessel surveys are conducted in the Northeast and Southeast U.S. to provide whale sighting information to mariners. Surveys typically occur in the following locations at the specified times: a) Cape Cod Bay, the Gulf of Maine, the Great South Channel and Rhode Island, Block Island, and Long Island Sounds from January through July; b) South Carolina/North Carolina border south to Crescent Beach, FL, from December through March. Survey planes occasionally use VHF-FM channel 16 to contact ships directly if whales have been spotted in close proximity to that vessel. However, many right whales go undetected by surveys. Seasonal right whale advisories and sighting reports are broadcast periodically for these and surrounding areas by Coast Guard Broadcast Notice to Mariners, NAVTEX, NOAA Weather Radio, Cape Cod Canal Vessel Traffic Control and the Bay of Fundy Vessel Traffic Control and are included in the return message from the Right Whale Mandatory Ship Reporting (MSR) systems. General sighting information may be obtained by sending an e-mail to ne.rw.sightings@noaa.gov (Northeast) or se.rw.sightings@noaa.gov (Southeast).

(59) **Precautions when transiting right whale habitat and areas of recently reported right whale sightings:** NOAA recommends the following precautionary measures be taken to avoid adverse interactions with North Atlantic right whales:

(60) Before entering right whale habitat (See "Seasonal Occurrence" table), check Coast Guard Broadcast Notices to Mariners, NAVTEX, NOAA Weather Radio, Mandatory Ship Reporting (MSR) system, Cape Cod Canal Vessel Traffic Control or the Bay of Fundy Vessel Traffic Control as well as other sources for recent right whale sighting reports. Local ship pilots also have information on whale sightings and safe local operating procedures.

(61) Review right whale identification materials and maintain a sharp watch with lookouts familiar with spotting whales. Although right whales are large, their dark color and lack of a dorsal fin can make them difficult to spot.

(62) Avoid transiting through the right whale habitats and areas where right whales have recently been sighted. If transiting between ports within critical habitats, minimize transit distance. Route around observed or recently reported right whales and anticipate delays due to prudent seamanship in response to whale sightings. Avoid transits at night or during periods of low visibility.

(63) If a right whale is sighted from the ship or reported along the intended track of the ship, mariners should exercise caution, post a lookout and reduce speed to 10 knots when consistent with safe navigation. If a right whale is sighted, a vessel must steer a course away from the right whale and immediately leave the area at slow safe speed. Do not assume right whales will move out of the way of an approaching vessel. Mariners should keep in mind that it is illegal to approach a right whale closer than 500 yards. (See **50 CFR 224.103**, chapter 2 for limits, regulations and exceptions.)

(64) Any whale accidentally struck, dead whale carcass and sighting of an injured or entangled whale should be reported immediately to the Coast Guard or NOAA National Marine Fisheries Service noting the precise location, date and time of the accident or sighting. Call 866–755–6622 for reports to NOAA for the area from Virginia to Maine or 877–942–5343 (877–WHALE–HELP) for the area from North Carolina to Florida. In the event of a strike or sighting of a dead, injured or entangled whale, the following information should be provided:

(65) location, date and time of the accident or sighting of a carcass or an entangled whale,

(66) speed and course of the vessel,

(67) vessel specifications such as size and propulsion,

(68) water depth,

(69) environmental conditions such as visibility, wind speed and direction,

(70) description of the impact,

(71) fate of the animal, and

(72) species and size, if known.

(73) **Recommended Two-Way Routes to Avoid Whales:** To reduce the possibility of vessel strikes with right whales, Two-Way Routes were developed for vessels entering and transiting through Cape Cod Bay and arriving and departing the ports of Brunswick, GA, Fernandina Beach, FL, and Jacksonville, FL. The routes were developed from an analysis of historical right whale sightings and are designed to reduce the likelihood of adverse interactions between large vessels and right whales. The routes are found on the latest NOAA Nautical Charts. In 2007, the northern leg of the Boston Traffic Separation Scheme (TSS) was shifted to direct ship traffic away from an area of high whale density. Use of the modified TSS is expected to considerably reduce the risk of striking a whale.

(74) **Mandatory Ship Reporting Systems (MSR) WHALESNORTH and WHALESSOUTH:** Mandatory Ship Reporting (MSR) systems require all vessels, 300 gross tons or greater, to report to the U.S. Coast Guard upon entering two designated reporting areas off the east coast of the United States. (See **33 CFR 169**, chapter 2, for limits and regulations.) Sovereign immune vessels are exempt from the requirement to report but are encouraged to participate.

(75) The two reporting systems will operate independently of each other. The system in the northeastern United States will operate year round and the system in the southeastern United States will operate each year from November 15 through April 15. Reporting ships are only required to make reports when entering a reporting area during a single voyage (that is, a voyage in which a ship is in the area). Ships are not required to report when leaving a port in the reporting area nor when exiting the system.

(76) Mariners should check all MSR messages carefully before transmittal to ensure the message includes the correct address and format. Additional greeting or comments in the message will preclude message receipt by the MSR system. Failure to receive a timely return message from the MSR system that provides locations of recent right whale sightings and precautionary guidance should be reported to the local Coast Guard Sector Office.

(77) **Northeastern reporting system/Southeastern reporting system** (See **33 CFR 169.105** and **169.115**, chapter 2, for limits.)

(78) Vessels shall make reports in accordance with the format in IMO Resolution A.858 (20) in accordance with the International Convention for the Safety of Life at Sea 1974 (SOLAS 74). (See **33 CFR 169.135** and **169.140**, chapter 2, for additional information.) Vessels should report via INMARSAT C or via alternate satellite communications to one of the following addresses:

(79) Email: RightWhale.MSR@noaa.gov or

(80) Telex: 48156090.

(81) Vessels not equipped with INMARSAT C or Telex should submit reports to the U.S. Coast Guard's Communication Area Master Station Atlantic (CAMSLANT) via HF voice frequencies on 4125 kHz, 6215 kHz, 8291 kHz, 12290 kHz and 16420 kHz or by calling 1–800–742–8519x0. Vessels equipped only with VHF-FM voice communications should submit reports to the nearest U.S. Coast Guard Sector.

(82)

Example Report
WHALESNORTH
WHALESNORTH//
M/487654321//
A/CALYPSO/NRUS//
B/031401Z APR//
E/345//
F/10.0//
H/031410Z APR/4104N/06918W//
I/BOSTON/032345Z APR//
L/WP/4104N/06918W/15.5//
L/WP/4210N/06952W/15.5//
L/WP/4230N/07006W/15.5//
WHALESSOUTH
WHALESSOUTH//
M/412345678//
A/BEAGLE/NVES//
B/270810Z MAR//
E/250//
F/10.0//
H/270810Z MAR/3030N/08052W//
I/MAYPORT/271215Z MAR//
L/RL/17.0//

(83)

Georges Bank

(84) **Georges Bank** is an extensive bank with depths of less than 50 fathoms, extending for over 150 miles northeastward from the offshore end of Nantucket Shoals.

(85) In heavy weather the danger area may be considered to be the oval-shaped top of the bank that is about 80 miles long in a northeast and southwest direction and that has a maximum width of about 50 miles. The bottom within this area is extremely broken and irregular, with a great number of ridges and shoal spots having depths of less than 10 fathoms. Between these shoals are channels of varying widths in which depths of about 20 fathoms may be found. All of this area lies within the 30-fathom curve and so much of it has depths of less than 20 fathoms that it may practically all be considered to lie within a generalized 20-fathom curve.

(86) On the southeast side of the bank, outside the 20-fathom curve, the water deepens gradually and with such regularity that soundings would be of considerable value in approaching the bank. On the northwest side the water deepens more rapidly.

(87) The bottom is generally of sand, sometimes with shell, and in places pebbles. Bottom samples as obtained during surveys are shown in a great many places on the charts.

(88) The two principal dangers on Georges Bank are Georges Shoal and Cultivator Shoal, which are near the center of the danger area. Around these shoals the sea breaks in depths of 10 fathoms during heavy weather, and the locality should be avoided by deep-draft vessels.

(89) **Georges Shoal** is a ridge about 13 miles long on which are several shallow depths of 1½ to 3 fathoms.

(90) **Cultivator Shoal**, about 20 miles westward of Georges Shoal, is a ridge nearly 15 miles long, on which depths of 3 to 10 fathoms are found. The 3-fathom spot is near the north end of the shoal. In 1980, a submerged obstruction was reported about 8.7 miles northwest of the 3-fathom spot in about 41°43'N., 68°23'W.; vessels engaged in bottom operations are advised to exercise caution in the area.

(91) The entire area within the 20-fathom curve has an extremely broken bottom. There are numerous ridges and shoal spots on which depths dangerous to navigation, particularly in heavy weather, may be found. These shoal spots generally have steep sides, and very little or no indication of their existence is given by soundings. Tide rips and swirls, as well as overfalls, are common in the vicinity of these spots but are not always visible. They show best with a smooth sea and with the current flowing in certain directions. These disturbances are not usually over the shoalest depths but are commonly alongside them. Small, detached overfalls may be seen in 20 fathoms of water. The tidal currents are rotary with no period of slack water. The velocity at strength is about 2 knots, and the velocity of the minimum current that occurs about midway between the times of strength is about 1 knot. The hourly velocities and directions of the tidal current are shown by means of current roses on National Ocean Service charts.

(92) A navigator must bear in mind while in an area of this character that it is impossible for the surveyor, without a vast expenditure of time, to determine and locate all of the shoalest spots on the many dangerous shoals found. Sudden shoaling on such a bank must be considered an indication of possibly dangerous water. This bank has not been wire dragged.

(93) **Nantucket Shoals** is the general name of the numerous different broken shoals that lie southeastward of Nantucket Island and make this one of the most dangerous parts of the coast of the United States for the navigator. These shoals extend 23 miles eastward and 40 miles southeastward from Nantucket Island. They are shifting in nature, and the depths vary from 3 to 4 feet on some to 4 and 5 fathoms on others, while slues with depths of 10 fathoms or more lead between those farthest offshore. The easterly edge of the shoals has depths of 3 and 4 fathoms in places.

(94)

Area to be avoided

(95) Because of the great danger of stranding and for reasons of environmental protection, the International Maritime Organization (IMO) has established an area to be avoided in the area of Nantucket Shoals. All vessels carrying cargoes of oil or hazardous materials and all other vessels of more than 1,000 gross tons should avoid the area bounded by the following points:

(96) 41°16.5'N., 70°12.5'W.;

(97) 40°43.2'N., 70°00.5'W.;

(98) 40°44.5'N., 69°19.0'W.;

(99) 41°04.5'N., 69°19.0'W.;

(100) 41°23.5'N., 69°31.5'W.; and

(101) 41°23.4'N., 70°02.8'W.

(102) The currents in the area are strong and erratic, reaching a velocity of 3 to 5 knots around the edges of the shoals. They are made erratic by the obstruction of the shoals, in some cases being deflected to such an extent as to cause the direction to change 180° from one side of the shoal to the other.

(103) The tidal current over the shoals is rotary, turning clockwise. Observations in the area indicate an average velocity at strength of about 2.5 knots, but this probably varies appreciably from place to place. Similarly the direction of the current at strength probably depends on the orientation of channels between shoal areas.

(104) Since the current is rotary, there is no true slack. Observations in the area show an average minimum of about 0.5 knot.

(105) The tidal current south of Asia Rip is rotary, turning clockwise. The average velocity at strength is 0.8 knot; the average minimum is 0.6 knot.

(106) Hourly average velocities and directions for Davis Bank and the area south of Asia Rip, referred to predicted times of maximum flood at Pollock Rip Channel, are furnished in the Tidal Current Tables. However the tidal currents are appreciably influenced by winds.

(107) Nantucket Shoals should be entirely avoided by deep-draft vessels when possible and by light-draft vessels without local knowledge, on account of the treacherous currents. There are, however, channels through these various shoals that can be negotiated with local knowledge and caution. In calm weather at slack water these shoals are sometimes difficult to see, and a vessel is liable to be taken into shoaler water than was intended.

(108) Calm, clear days are few; when the sea is calm it is usually foggy, and when clear, it is usually rough. Also to be expected is a considerable amount of hazy weather, which limits visibility.

(109) Should it become necessary to anchor in this area, open sea anchorage may be had anywhere that depths permit. Due consideration should be given to the close proximity of shoals and possibility of dragging due to the winds and currents. Generally it has been found best to avoid the deeper channels and, when rougher water is experienced, to anchor in the lee of a shoal, which would

tend to knock down the heavier swells. A scope of five to one or greater should always be used.

(110)

North Atlantic right whales

(111) Endangered North Atlantic right whales may occur along the northern edge of Georges Bank (peak season: March through July). (See North Atlantic right whales, indexed as such, in this chapter for more information on right whales and recommended measures to avoid collisions.)

(112) All vessels 65 feet or greater in length overall (LOA) and subject to the jurisdiction of the United States are restricted to speeds of 10 knots or less in the Great South Channel Seasonal Management Area between April 1 and July 31. The area is defined as the waters bounded by:

(113) 42°30'N., 69°45'W.;

(114) 42°30'N., 67°27'W.;

(115) 42°09'N., 67°08.4'W.;

(116) 41°00'N., 69°05'W.;

(117) 41°40'N., 69°45'W.; thence back to starting point. (See **50 CFR 224.105**, chapter 2, for regulations, limitations and exceptions.)

(118) **Georges Bank Closure Areas**, Marine Protected Areas (MPAs), extend south and east of Cape Cod to the boundary with the Exclusive Economic Zone (EEZ). **Nantucket Lightship Closed Areas** include waters south and east of Nantucket Island extending to the west edge of Georges Bank.

(119) Nantucket Shoals is made up of the following parts:

(120) **Phelps Bank**, the southeasternmost part of the Nantucket Shoals, is about 6.5 miles long and 2.5 miles wide. A lighted whistle buoy, marking the entrance to the Boston Harbor Traffic Separation Scheme, is about 12 miles eastward of Phelps Bank.

(121) **Asia Rip**, the shoalest point of the bank with 5¾ fathoms, is at the southern end. The wreck of the SS OREGON, covered 3¼ fathoms, is at 40°45'N., 69°19'W., 3 miles south-southeastward of Asia Rip.

(122) **Middle Rip**, with a least-found depth of 4 fathoms and lying north-northwest of Phelps Bank, is about 13.5 miles long and 4.5 miles wide. This shoal consists of two large parts with depths of 4 fathoms on the east and 6 fathoms on the west, separated by a channel with a depth of 7 fathoms and four outlying shoals of 8 to 10 fathoms.

(123) **Fishing Rip**, bow-shaped, with depths of 3 to 10 fathoms, is about 26 miles long north and south and 6.5 miles wide at its widest point. The north point is 20 miles 073° and the south point is 27.5 miles 136°, respectively, from Sankaty Head Light. A large wreck area is near the southern part of Fishing Rip. A wreck and a submerged obstruction are also near the southern portion of the rip in about 41°00.0'N., 69°27.0'W. and 41°01.0'N., 69°29.7'W., respectively.

(124) The unmarked channel westward of Fishing Rip is obstructed by three shoals in the northern section that have least-found depths of 7½, 4½ and 10 fathoms. In the southern part of this channel are four shoals with depths of 8 to 10 fathoms.

(125) **Davis Bank**, the innermost of the outer Nantucket Shoals, is bow shaped and has depths of 2¾ to 10 fathoms of water over it. The bank is about 30 miles long north and south and has a greatest width of 4 miles. The wreck of the vessel PROGRESS is off the inner edge of the bank about 13 miles north-northeastward of the southern end of the bank.

(126) The channel westward of Davis Bank is marked on its west side by lighted and unlighted buoys. A racon is at the northernmost lighted buoy. The use of this channel should be restricted to clear weather due to the strong currents encountered throughout this area.

(127) The inner Nantucket Shoals all lie within the 10-fathom curve. The area is very foul. Only a few of the shoals are described. **Davis South Shoal**, about 20 miles south-southeast of Sankaty Head, consists of two spots of 2¾ and 2½ fathoms about 1.5 miles apart.

(128) **Old South Shoal**, consisting of two spots of 2½ fathoms with a 2-fathom spot and foul ground between them, is about 13.5 miles southeast of Sankaty Head. This shoal is unmarked.

(129) **Great Rip**, about 13 miles east-southeast of Sankaty Head, has depths of 1 to 2¾ fathoms. This shoal is about 7 miles long north and south and 1 to 2 miles wide. About 1.5 miles westward of Great Rip and separated from it by depths of 14 to 19 fathoms is an unnamed and unmarked shoal of 1½ to 2½ fathoms. Breakers are usually observed on the shoal.

(130) **Rose and Crown** is a boot-shaped shoal with its southern end about 10.5 miles east of Sankaty Head. The shoal extends about 5 miles northward and then 3 miles westward. Depths of 1¼ and 1½ fathoms are found in the leg of the boot, a depth of ½ fathom forms the heel and a depth of 1¼ fathoms is found in the toe. Northward of the toe of Rose and Crown is a shoal with foul ground and spots of 1½ and 2½ fathoms. Rose and Crown breaks heavily.

(131) **Bass Rip**, about 2.5 miles eastward of Sankaty Head, is about 3.5 miles long north and south. A depth of ½ fathom is 3 miles 115° from the light. The northern end of the shoal has a depth of 2 fathoms. **Old Man Shoal** extends 4.5 miles southwestward from a point 1.5 miles off the southeastern end of Nantucket Island. Depths of 1¼ to 2¾ fathoms are found on this shoal.

(132) **McBlair Shoal**, the northernmost of the Nantucket Shoals and marked on its northern side by lighted buoys, forms part of the southern side of Great Round Shoal Channel. Depths on this shoal vary from 2¼ to 3½ fathoms.

(133) **Great South Channel** is the passage between the easternmost of the Nantucket Shoals and the westernmost shoal spots of Georges Bank. The approximate center of the channel extends from 40°36'N., 68°55'W. to 41°38'N., 68°55'W. The channel is about 27 miles wide and has depths of 19 fathoms and greater throughout, with lesser depths along the eastern and western edges. The Great

South Channel is a feeding area for endangered North Atlantic right whales in spring and summer (peak season: March through July, although right whales have been seen in the area year round).

(134) Great South Channel lies within the federally designated critical habitat for North Atlantic right whales. In some years, more than a third of the remaining population of North Atlantic right whales can be found in the Great South Channel at any one time. It is illegal to approach closer than 500 yards of any right whale. (See **50 CFR 224.103(c)**, chapter 2, for limits and regulations.) It is recommended that all large vessels (over 100 gross tons) avoid operating in the critical habitat during the peak period of right whale occurrence (March through July). When the area cannot be avoided, precautionary measures should be taken to reduce the risk of ship strikes. (See North Atlantic Right Whales, indexed as such, in chapter 3 for more information on right whales and recommended measures to avoid collisions with whales.)

(135) All vessels 65 feet or greater in length overall (LOA) and subject to the jurisdiction of the United States are restricted to speeds of 10 knots or less in the Great South Channel Seasonal Management Area between April 1 and July 31. The area is defined as the waters bounded by:

(136) 42°30'N., 69°45'W.

(137) 42°30'N., 67°27'W.

(138) 42°09'N., 67°08.4'W.

(139) 41°00'N., 69°05'W.

(140) 41°40'N., 69°45'W.; thence back to starting point. (See **50 CFR 224.105** in chapter 2 for regulations, limitations and exceptions.)

(141) Vessels transiting within the Great South Channel may transit into the WHALESNORTH Mandatory Ship Reporting Area. Each self-propelled ship of 300 gross tons or greater entering WHALESNORTH must participate in the Mandatory Ship Reporting System (See **33 CFR 169**, chapter 2, for limits and regulations, and chapter 3 for sample reports). Sovereign immune vessels are exempt from the requirement to report but are encouraged to participate.

(142) **Restricted Lobster and Gillnet** Marine Protected Areas (MPAs) are in Great South Channel.

(143) **Submarine canyons** are indentations in the edge of the **Continental Shelf**, which is bounded on its seaward side by the 100-fathom curve. They may be traced from depths of 1,000 fathoms or more to the shoaler areas of the Continental Shelf. The navigator who has available some means of echo sounding should have in mind the various canyons found in this locality. The soundings in crossing them are very characteristic in each case, and such soundings may be used to determine the vessel's position with considerable accuracy.

(144) The names of some of the most important submarine canyons are shown on the charts. The longitude following the name is approximate and only given to assist in locating the feature on the chart. **Corsair Canyon**, 66°10'W., on the eastern side of Georges Bank, has a northwesterly trend. On the southern side and toward the western end of Georges Bank, having a northerly trend, are **Lydonia Canyon**, 67°40'W.; **Gilbert Canyon**, 67°50'W.; **Oceanographer Canyon**, 68°05'W.; and **Welker Canyon**, 68°30'W. Southeastward and southward of Nantucket Shoals, having a northerly trend, are **Hydrographer Canyon**, 69°00'W.; **Veatch Canyon**, 69°35'W.; and **Atlantis Canyon**, 70°15'W., **Block Canyon**, 71°20'W., is south-southeasterly of Block Island Sound and has a north-northwesterly trend. **Hudson Canyon**, 72°20'W., extends northwestward to the mouth of the Hudson River. The inshore section of this canyon is called **Mud Gorge**.

(145) Many vessels have been wrecked along this coast as a result of collision, foundering and other causes. Most of the offshore wrecks have been located and surveyed to determine the least depth over the highest projecting part. Dangerous wrecks for the most part are marked by buoys of various colors and shapes and often show a quick-flashing or an interrupted quick-flashing light.

(146) Many vessels have grounded in fog on the south side of Long Island and on Block Island. Probably many of these wrecks could have been avoided if frequent soundings had been taken in approaching the coast. Vessels equipped to do so should make good use of the electronic aids to navigation systems along the coast to check their position frequently.

(147) The coastal waters contain numerous **lobster pots**. Small painted wooden buoys of various designs and colors, secured by small lines, float on the surface; in some cases a second buoy, usually an unpainted wooden stick or bottle and difficult to see, is attached to the lobster pot. These buoys extend from shore out to, and in many cases across, the sailing routes. Small yachts and motor boats are cautioned against fouling them, which is liable to result in a sprung shaft or lost propeller. Fishtraps and fish havens are discussed in chapter 1.

(148) **Fishweirs** are numerous along the outside coast and inside waters. The stakes often become broken off and form a hazard to navigation, especially at night. The areas within which fishweirs are permitted have been established under federal authority and are shown on the ENC's and large scale charts. The exact locations of the weirs within the designated areas are not shown. Strangers should proceed with caution when crossing areas of possible fishweirs and should avoid crossing such areas at night.

(149) **Danger zones** have been established within the area of this Coast Pilot. (See **33 CFR 334**, chapter 2, for limits and regulations.)

(150)

Drawbridges

(151) The general regulations that apply to all drawbridges are given in **33 CFR 117.1** through **117.49**, chapter 2, and the specific regulations that apply only to certain drawbridges are given in **33 CFR 117**, **Subpart B**, chapter 2. Where these regulations apply, references to

them are made in the Coast Pilot under the name of the bridge or the waterway over which the bridge crosses.

(152) The drawbridge opening signals (see **33 CFR 117.15**, chapter 2) have been standardized for most drawbridges within the United States. The opening signals for those few bridges that are nonstandard are given in the specific drawbridge regulations. The specific regulations also address matters such as restricted operating hours and required advance notice for openings.

(153) The mariner should be acquainted with the general and specific regulations for drawbridges over waterways to be transited.

(154)

Routes

(155) Approaching this section of the coast is dangerous for all vessels because of the off-lying banks and shoals, the strong and variable currents, frequency of fog and the broken nature of the bottom. Soundings alone are of little value in establishing the position of a vessel, but the depth should be checked frequently to ensure that the vessel clears all dangers.

(156) In thick weather especially, the greatest caution is necessary, and vessels equipped to do so should make good and timely use of the electronic aids to navigation systems to check their position frequently. The depth should never be shoaled to less than 15 fathoms without an accurate fix having been obtained, and it is advisable to remain offshore in depths of 20 fathoms or more.

(157) The part of Georges Bank lying between latitude 41°05'N., and 42°00'N., and longitude 67°17'W., and 68°35'W. should be avoided. In heavy weather the sea breaks on the spots with 10 fathoms or less, and strong tide rips are encountered. The tide rips do not always indicate shoal water.

(158) Vessels passing southward of the dangerous part of Georges Bank should keep in 30 fathoms or more. Approaching this part of the bank from eastward or southward, the water shoals gradually. Approaching from the westward, the depths are irregular and the water shoals abruptly in places of 20 fathoms or less. On the north side of Georges Bank between longitudes 66°00'W., and 68°00'W., the 100-fathom and 50-fathom curves are only a few miles apart, and when approaching the dangerous part of the bank from northward 50 fathoms may be taken as a good depth to avoid the shoals.

(159) Vessels equipped with echo sounding devices and following the 100-fathom curve along the south side of Georges Bank can frequently verify their position when crossing the several submarine gorges or canyons.

(160) Approaching New York from the area south of Asia Rip, a slight allowance should be made for a southwesterly set of the current. Should the wind be easterly, it is customary to allow, in order to make a course good, a set of the current with it of at least 0.5 knot.

(161) The **North Atlantic Lane Routes** are described in **NV PUB. 106, Atlas of Pilot Charts, North Atlantic Ocean (including the Gulf of America)**, published by the National Geospatial-Intelligence Agency, Washington, DC.

(162) Deep-draft vessels coming from Cape Hatteras, Chesapeake Bay, Delaware Bay or New York usually enter the precautionary area southeast of Asia Rip, then head north through Great South Channel to Cape Cod or the Gulf of Maine.

(163) Vessels of medium draft coming from the southward, or southbound from Boston or ports farther east, may use Cape Cod Canal or Vineyard and Nantucket Sounds via Pollock Rip Channel. Great Round Shoal Channel is also available, but seldom used, as an entrance to or exit from Nantucket Sound. The controlling depth for these passages is from 27 to 32 feet. They avoid Nantucket Shoals and are used by coasting vessels. Small vessels and pleasure craft usually pass through Long Island Sound when proceeding coastwise.

(164)

Current

(165) The Tidal Current Tables at *tidesandcurrents.noaa.gov/currents13* should be consulted for specific information about times, directions and velocities of the current at the numerous locations throughout the area. It must be borne in mind that the current to which a vessel is subjected at any time is the combination of tidal current, wind current and other currents such as those due to drainage or oceanic circulation.

(166) Away from the immediate vicinity of the shore, the tidal currents are generally rotary. They shift direction, usually clockwise, at an average rate of about 30° an hour. They attain velocities of 1 to 2.4 knots or more throughout the Nantucket Shoals-Georges Bank area, the larger velocities occurring generally over the shoaler parts of the area. Between Nantucket Island and Sandy Hook their velocities generally do not exceed 0.5 knot except in the vicinities of the entrances to the larger bays and inland waterways, where the velocities increase as the entrances are approached. For considerable distances from the entrances, strengths of flood and ebb set, respectively, toward and away from those entrances, and minimums of velocity, corresponding to the slacks of reversing currents, set at right angles to the directions of the flood and ebb strengths.

(167) Offshore and away from the influence of the tidal flow into and out of the Gulf of Maine and the larger bays, the tidal current maintains an approximate uniform velocity. Shifting its direction continuously to the right, it sets in all directions of the compass during each tidal cycle of 12.4 hours.

(168) In the offshore area between Cape Cod and Sandy Hook there is a resultant southward drift that is stronger in winter than in summer and has an average velocity less than 0.1 knot.

(169)

Wind current

(170) Wind currents are very complicated. Their velocities and directions depend upon a number of factors such as

(176)

Mean Surface Water Temperatures (°C) and Densities

	Years		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Mean
Cape Cod Canal (east entrance)	16	Temp	1.1	0.4	1.8	5.3	9.4	12.8	14.7	16.2	15.9	13.3	9.4	4.7	8.8
		Density	23.5	23.5	23.4	23.3	23.1	23.2	23.4	23.4	23.4	23.5	23.4	23.5	23.4
Cape Cod Canal (west entrance)	15	Temp	0.8	0.4	2.2	6.1	10.4	15.0	17.6	18.7	17.4	13.9	9.6	4.3	9.7
		Density	23.2	23.2	23.1	22.9	22.9	23.0	23.2	23.3	23.3	23.3	23.2	23.2	23.2
Woods Hole, MA	27	Temp	1.2	0.5	2.4	6.8	11.8	17.0	20.9	21.6	19.7	15.6	10.5	4.9	11.1
		Density	23.2	23.2	23.1	23.2	23.3	23.5	23.6	23.5	23.6	23.6	23.5	23.3	23.4
Newport, RI	16	Temp	2.3	1.6	3.1	6.3	10.9	16.3	19.8	20.6	18.6	15.3	11.3	5.8	11.0
		Density	22.7	22.5	22.5	22.3	22.8	23.1	23.5	23.4	23.5	23.4	23.1	22.9	23.0
New London, CT	24	Temp	2.3	2.4	4.3	9.3	14.3	19.2	22.1	22.5	20.3	16.2	10.9	5.5	12.5
		Density	13.6	12.4	10.1	9.2	10.4	13.9	17.4	18.7	19.2	18.7	16.2	14.1	14.5
Bridgeport, CT	7	Temp	3.9	3.6	5.8	10.1	15.7	21.7	24.8	26.3	24.4	19.6	13.9	8.0	14.8
		Density	19.5	18.4	18.1	17.7	17.6	18.0	19.1	19.8	19.9	20.1	20.0	19.7	19.0
Plum Island, NY (Long Island Sound)	10	Temp	2.9	1.3	2.4	5.6	9.7	14.6	18.8	20.3	19.3	15.8	11.5	6.4	10.7
		Density	21.6	21.6	21.4	20.9	20.8	21.3	21.8	22.2	24.7	22.4	22.0	21.8	21.9
Montauk, NY (Fort Pond Bay)	23	Temp	2.2	1.5	3.1	6.6	10.8	16.1	20.1	21.1	19.6	15.9	11.0	5.7	11.1
		Density	22.5	22.4	22.3	21.7	21.8	22.2	22.6	22.8	23.1	23.1	22.9	22.6	22.5
Willets Point, NY (East River)	39	Temp	1.9	1.0	2.8	7.0	12.2	17.1	20.3	22.1	21.2	16.8	11.1	5.1	11.6
		Density	18.8	18.5	18.3	17.7	17.7	18.1	18.6	19.1	19.2	19.0	19.0	18.8	18.6
New York, NY (The Battery)	44	Temp	2.8	2.0	3.7	7.6	12.9	18.3	21.9	22.9	21.4	16.8	11.4	6.0	12.9
		Density	15.5	15.4	13.6	12.1	13.7	15.7	17.0	17.5	17.5	17.3	16.2	15.3	15.8
Bear Mountain, NY (Hudson River)	5	Temp	0.7	0.4	1.8	7.6	14.1	20.6	24.2	25.2	23.8	17.8	10.8	3.8	12.6
		Density	-0.6	-0.8	-0.7	-0.8	-0.9	-0.8	-0.4	-0.2	0.2	0.0	-0.5	-0.5	-0.5
New York, NY (Fort Hamilton)	12	Temp	2.1	1.2	2.4	6.5	11.8	17.0	20.8	21.8	19.8	15.3	9.8	4.9	11.1
		Density	16.6	17.3	14.8	12.5	15.2	17.0	18.1	19.0	18.7	17.9	16.8	16.5	16.7
Sandy Hook, NJ	33	Temp	1.4	1.4	4.1	9.1	14.6	20.2	23.4	23.5	20.8	15.1	9.4	3.8	12.2
		Density	17.0	16.9	15.9	14.9	16.2	17.7	19.0	19.1	19.2	19.1	18.4	17.5	17.6

Temperature (Celsius)
F (Fahrenheit) = 1.8C (Celsius) + 32

Density as used in this table is the specific gravity of the sea water or the ratio between the weight of a sea-water sample and the weight of an equal volume of distilled water at 15°C (59°F).

velocity, direction and duration of the wind, the proximity of the coast and the direction of the coastline. Generally in the Northern Hemisphere the wind-driven current sets somewhat to the right of the wind, but in coastal waters there are many exceptions to this general rule, the current often setting to the left of the wind, due to the tendency of the current to follow the direction of the coastline or to other local conditions.

(171) The velocity of the current relative to that of the wind also varies with the location. It follows, therefore, that local wind current information is desirable. Such information based upon extensive current and wind observations at a number of stations is available from the Tidal Current prediction service at *tidesandcurrents.noaa.gov*. Links to a user guide for this service can be found in chapter 1 of this book.

(172) The largest current velocities likely to occur during storms at a number of locations offshore and in the sounds are given as follows: 1.5 miles east of Broken Part of Pollock Rip, 2.5 knots; Stone Horse Shoal, 4 knots; 1 mile east of Great Round Shoal Channel Lighted Buoy 2, Nantucket Entrance, 2.5 knots; 13 miles southeast of Asia Rip, 2.5 knots; Cross Rip Shoal, 2.5 knots; Hedge Fence Lighted Gong Buoy 22, Nantucket Sound, 2.5 knots; 1.5 miles southeast of Buzzards Bay Entrance Light, 2 knots; Brenton Reef, 1.5 knots; 0.5 mile south of Bartlett Reef, Long Island Sound, 2.5 knots; 3 miles southward of Cornfield Point, 4 knots; 6.25 miles northwest of Cholera Bank, 1.5 knots.

(173)

Weather, Cape Cod to Sandy Hook

(174) From Georges Bank and the shoals of Nantucket to New York Harbor, fog, currents, winds and waves are constant threats to safe navigation. The following text describes the weather problems that face the mariner when navigating these waters. This section presents an overall, seasonal picture of the weather that can be expected in the offshore waters along the coast of the mid-Atlantic region from Cape Cod, MA, to Sandy Hook, NJ. Detailed information, particularly concerning navigational weather hazards, can be found in the weather articles in the following chapters.

(175) All weather articles in this volume are the product of the National Oceanographic Data Center (NODC) and

(178)

METEOROLOGICAL TABLE – COASTAL AREA OFF CHATHAM, MA (Between 41°N to 42°N and 66°W to 70°W)

WEATHER ELEMENTS	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	ANNUAL
Wind > 33 knots [1]	7.5	7.2	5.3	2.6	0.8	0.3	0.1	0.2	0.7	1.6	4.1	7.1	2.5
Wave Height > 9 feet [1]	11.8	11.7	9.6	6.5	2.7	0.9	0.3	1.1	2.6	5.6	10.1	11.1	5.0
Visibility < 2 nautical miles [1]	6.8	9.0	10.5	13.8	24.8	36.0	41.3	30.4	14.9	7.0	5.4	4.2	18.9
Precipitation [1]	17.7	17.4	10.9	9.3	6.9	4.5	3.9	4.2	5.3	6.6	9.6	14.3	7.8
Temperature > 69° F	0.0	0.1	0.0	0.0	0.2	0.8	6.2	9.7	4.0	0.6	0.0	0.1	2.5
Mean Temperature (°F)	38.1	37.9	39.5	43.4	49.3	56.0	62.5	64.6	61.5	55.9	49.4	42.1	52.8
Temperature < 33° F [1]	22.2	24.7	10.2	0.8	0.2	0.0	0.0	0.0	0.0	0.0	0.4	9.8	3.9
Mean RH (%)	83	82	82	84	88	89	90	89	84	81	80	80	85
Overcast or Obscured [1]	48.7	47.6	42.7	40.9	43.4	43.6	43.8	36.8	29.9	27.0	37.9	46.3	39.0
Mean Cloud Cover (8^{ths})	6.2	6.0	5.4	5.0	5.2	5.4	5.4	5.0	4.4	4.4	5.4	6.1	5.2
Mean SLP (mbs)	1015	1014	1014	1015	1015	1016	1016	1017	1019	1018	1017	1016	1016
Ext. Max. SLP (mbs)	1043	1048	1055	1056	1051	1039	1042	1043	1048	1047	1042	1047	1056
Ext. Min. SLP (mbs)	962	957	964	964	973	981	990	988	985	970	975	964	957
Prevailing Wind Direction	NW	NW	NW	SW	SW	SW	SW	SW	SW	W	W	NW	SW
Thunder and Lightning [1]	0.3	0.3	0.4	0.3	0.3	0.5	0.7	0.6	0.3	0.4	0.3	0.1	0.4

METEOROLOGICAL TABLE – COASTAL AREA SOUTH OF NANTUCKET, MA (Between 40°N to 41°N and 66°W to 70°W)

WEATHER ELEMENTS	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	ANNUAL
Wind > 33 knots [1]	8.1	7.2	5.7	2.8	0.6	0.3	0.2	0.4	1.2	2.9	4.9	7.4	3.2
Wave Height > 9 feet [1]	14.5	13.5	10.6	7.7	3.2	1.4	0.6	1.1	2.5	5.2	8.8	11.5	6.3
Visibility < 2 nautical miles [1]	6.6	8.8	9.9	13.6	23.5	27.5	27.8	17.2	8.9	5.0	4.7	4.2	14.3
Precipitation [1]	15.6	13.9	9.8	9.1	6.3	4.7	4.3	4.9	5.4	7.0	9.7	13.4	8.3
Temperature > 69° F	0.1	0.0	0.1	0.1	0.6	5.9	27.3	39.5	19.3	3.9	0.4	0.1	8.3
Mean Temperature (°F)	39.9	39.8	41.1	45.1	51.6	59.9	66.9	68.7	65.3	58.8	51.8	44.1	53.1
Temperature < 33° F [1]	17.2	19.2	7.8	1.2	0.2	0.0	0.1	0.1	0.0	0.0	0.3	6.3	4.0
Mean RH (%)	83	83	83	85	87	88	89	87	84	81	81	82	85
Overcast or Obscured [1]	45.5	42.4	37.8	35.7	38.6	36.9	34.9	27.6	25.8	26.1	35.2	42.2	35.6
Mean Cloud Cover (8^{ths})	6.0	5.7	5.2	4.7	5.0	4.9	4.9	4.5	4.3	4.5	5.4	5.9	5.1
Mean SLP (mbs)	1015	1015	1015	1015	1016	1016	1016	1017	1018	1018	1016	1016	1016
Ext. Max. SLP (mbs)	1044	1049	1054	1049	1047	1040	1037	1040	1050	1051	1047	1056	1056
Ext. Min. SLP (mbs)	962	965	967	961	976	983	987	987	986	972	972	963	961
Prevailing Wind Direction	NW	NW	NW	W	SW	SW	SW	SW	NE	W	NW	NW	W
Thunder and Lightning [1]	0.5	0.4	0.3	0.3	0.5	0.6	0.7	0.9	0.8	0.6	0.5	0.3	0.5

[1] Percentage Frequency

the National Climatic Data Center (NCDC). The meteorological and climatological tables are the product of the NCDC. Both centers are entities of the National Environmental Satellite, Data, and Information Service (NESDIS) of the National Oceanic and Atmospheric Administration (NOAA). If further information is needed in relation to the content of the weather articles, meteorological tables or climatological tables, contact the National Climatic Data Center, Attn: Customer Service Division, Federal Building, 151 Patton Avenue, Room 120, Asheville, NC 28801-5001. You may also contact the CSD at 828–271–4994 or fax your request to 828–271–4876.

(177) Climatological tables and meteorological tables for coastal locations relevant to discussions within this volume are located in this chapter (chapter 3) and in following chapters within the appropriate port text description. The climatological tables are a special extraction from the International Station Meteorological Climate Summary (ISMCS). The ISMCS is a CD-ROM jointly produced by the National Climatic Data Center (NCDC), Fleet Numerical Meteorology and Oceanography Detachment-Asheville, and the U.S. Air Force Environmental Technical Applications Center, Operating Location–A. The meteorological tables for the ocean areas are compiled from observations made by ships in passage and extracted from the National Climatic Data Center's Tape Deck-1129, Surface Marine Observations. Listed in Appendix A are National Weather Service offices and radio stations that transmit weather information.

(179)

Extratropical cyclones

(180) One of the biggest problems in these waters is the winter storm; the most powerful of these is the "Nor'easter." It generates rough seas, strong winds and high tides that threaten safety at sea and cause damage in

(183)

METEOROLOGICAL TABLE – COASTAL AREA SOUTH OF MARTHA'S VINEYARD, MA (Between 40°N to 42°N and 70°W to 72°W)

WEATHER ELEMENTS	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	ANNUAL
Wind > 33 knots [1]	5.9	5.0	3.8	1.9	0.9	0.3	0.1	0.3	1.2	2.3	4.0	5.8	2.4
Wave Height > 9 feet [1]	10.8	12.4	8.5	5.6	2.1	1.0	0.5	0.8	2.6	3.3	5.4	6.6	4.9
Visibility < 2 nautical miles [1]	6.6	7.2	9.9	11.6	17.2	19.0	19.6	10.9	7.5	4.6	4.2	4.9	10.7
Precipitation [1]	12.4	12.2	9.4	8.6	7.2	4.5	4.0	4.8	6.2	5.8	9.3	12.3	7.8
Temperature > 69° F	0.1	0.1	0.0	0.0	0.4	6.3	42.6	51.6	18.5	1.8	0.2	0.2	11.1
Mean Temperature (°F)	37.5	36.3	40.0	45.2	52.5	62.1	69.5	70.5	65.6	58.1	49.6	41.9	53.3
Temperature < 33° F [1]	27.0	29.8	11.1	0.6	0.1	0.0	0.0	0.0	0.0	0.1	1.7	12.5	6.1
Mean RH (%)	80	80	81	82	84	86	85	84	82	78	78	77	82
Overcast or Obscured [1]	40.1	37.9	36.0	35.4	34.9	31.2	31.6	25.9	27.8	24.4	32.1	37.3	32.7
Mean Cloud Cover (8ths)	5.4	5.2	4.8	4.6	4.8	4.6	4.8	4.3	4.4	4.2	5.0	5.4	4.8
Mean SLP (mbs)	1016	1017	1016	1016	1016	1015	1016	1017	1018	1018	1017	1018	1016
Ext. Max. SLP (mbs)	1050	1049	1050	1040	1040	1038	1042	1046	1046	1044	1046	1050	1050
Ext. Min. SLP (mbs)	962	958	975	975	976	985	987	987	982	982	973	981	958
Prevailing Wind Direction	NW	NW	W	W	SW	SW	SW	SW	SW	W	W	NW	SW
Thunder and Lightning [1]	0.2	0.2	0.2	0.3	0.5	0.6	1.1	0.8	0.4	0.4	0.2	0.2	0.5

METEOROLOGICAL TABLE – COASTAL AREA AROUND LONG ISLAND, NY (Between 40°N to 42°N and 72°W to 75°W)

WEATHER ELEMENTS	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	ANNUAL
Wind > 33 knots [1]	4.0	3.4	2.9	1.5	0.7	0.2	0.1	0.3	0.5	1.8	2.7	4.5	1.8
Wave Height > 9 feet [1]	4.1	5.2	3.8	2.7	0.7	0.5	0.2	0.5	1.0	2.0	2.7	3.5	2.2
Visibility < 2 nautical miles [1]	7.1	7.7	10.2	9.3	15.4	12.3	9.8	5.9	5.1	5.5	4.6	5.3	8.2
Precipitation [1]	11.3	11.5	9.4	8.7	8.0	4.9	5.6	4.9	6.1	6.1	9.1	10.6	8.0
Temperature > 69° F	0.1	0.1	0.0	0.2	2.0	16.8	63.1	67.9	28.6	2.4	0.1	0.1	15.6
Mean Temperature (°F)	36.2	35.6	40.5	46.9	55.5	65.1	72.0	72.2	67.3	58.5	49.5	40.4	53.7
Temperature < 33° F [1]	31.6	32.3	10.8	0.7	0.0	0.0	0.0	0.0	0.0	0.0	2.1	17.4	7.5
Mean RH (%)	82	82	80	79	84	83	83	80	80	79	77	79	81
Overcast or Obscured [1]	33.4	32.7	32.5	31.4	32.2	27.9	26.4	23.7	24.6	22.4	27.9	31.1	28.8
Mean Cloud Cover (8ths)	4.9	4.7	4.6	4.6	4.7	4.6	4.5	4.2	4.2	4.0	4.6	4.8	4.5
Mean SLP (mbs)	1017	1017	1015	1015	1016	1015	1016	1016	1018	1018	1017	1018	1016
Ext. Max. SLP (mbs)	1050	1050	1050	1045	1042	1040	1040	1037	1037	1044	1045	1046	1050
Ext. Min. SLP (mbs)	959	961	961	977	989	981	988	986	978	970	966	981	959
Prevailing Wind Direction	NW	NW	NW	W	SW	SW	SW	SW	SW	W	NW	NW	SW
Thunder and Lightning [1]	0.2	0.1	0.1	0.3	0.8	1.0	1.3	0.9	0.9	0.3	0.6	0.1	0.5

[1] Percentage Frequency

port. These storms do not often come without warning. Approaching from the U.S. mainland or from the seas to the south they are usually well forecasted. Difficulty arises when they develop or deepen explosively off the mid-Atlantic coast. Sometimes called "Hatteras Storms," these lows can grow from small, weak frontal waves to full-blown systems in less than 24 hours. Not only can their circulation expand to cover most of the western North Atlantic but they often accelerate rapidly northeastward. In the exposed waters these storms can generate 40-foot (12 m) waves and hurricane-force winds. Each year more than 40 extratropical systems move across or close to this coast. They average about two to four per month, but as many as ten can affect the region in a single month. Most systems are weak but a few generate gales and rough seas for hundreds of miles, particularly from September through April.

(181) The major winter storm track runs in a line approximately from Cape Hatteras to Cape Cod. Most of the storms that follow this track intensify; the center of intensification is off Delaware Bay. In addition to the forecast, certain atmospheric changes indicate a storm is approaching. The most dependable early indicator is falling pressure. A definite weather change is likely if you observe pressure falls exceeding 2 mb every 3 hours; a drop of 5 mb/3 hours indicates a strong change while 10 mb/3 hours warns of an impending extreme event.

(182) As a storm approaches, winds strengthen, clouds thicken and lower and precipitation begins. Early in the storm's life wind waves can become steep very quickly, making it difficult to reach port especially when you have to navigate an inlet where breaking waves are treacherous. In deeper waters, waves can build to over 20 feet. During winter the possibility of superstructure icing calls for an

early course of action based upon the latest forecast and a knowledge of your vessel.

(184)

Cold fronts

(185) This weather hazard usually approaches from the west through north. Ahead of the front, winds are usually squally and often blow out of the south through southwest. Cirrus clouds give way to Altocumulus or Altostratus and Nimbostratus, then Cumulonimbus. Pressure falls moderately and showers, and perhaps thunderstorms, occur. Seas become choppy. With the frontal passage winds shift rapidly to the west and northwest. Strong gusts and squalls continue. Clearing usually occurs a short distance behind the front as the cold air moves in. Cold fronts can move through the area quite rapidly. Their speed varies from about 10 to 20 knots in summer up to 40 knots in winter. From spring through fall these fronts are often preceded by dense fog.

(186) During the spring and summer when the air ahead of the cold front may be very unstable, a line of thunderstorms, known as a squall line, may develop. These instability lines can form 50 to 300 miles ahead of a fast moving front. They may even contain tornados or waterspouts. These storms can inflict considerable damage on fishing vessels and small craft.

(187)

Tropical cyclones

(188) A tropical cyclone is a warm core, low-pressure system that develops over tropical oceans. It exhibits a rotary, counterclockwise circulation in the Northern Hemisphere around a center or "eye." In small tropical cyclones the diameter of the area of destructive winds may not exceed 25 miles while in the greatest storms the diameter may reach 500 miles. At the center is a comparatively calm, sometimes clear, area known as the eye. The diameter of the eye can vary from about 5 to 25 miles. Winds are usually strongest near the center. They can reach 175 knots or more in an intense hurricane. In the North Atlantic Region (West Indies, Caribbean Sea, Gulf of America and waters off the U.S. East Coast) the following terminology is used in tropical cyclone warnings issued by the National Hurricane Center (National Weather Service):

(189) (1) **Tropical depression**–An organized system of clouds and thunderstorms with a defined circulation and maximum sustained winds of 38 miles per hour (33 knots) or less.

(190) (2) **Tropical storm**–An organized system of strong thunderstorms with a defined circulation and maximum sustained winds between 39 and 73 miles per hour (34 to 63 knots).

(191) (3) **Hurricane**–An intense tropical weather system with a well-defined circulation and a maximum sustained wind speed of 74 miles per hour (64 knots) or greater.

(192) While the following term is not normally used in tropical cyclone advisories it may appear in related products.

(193) (4) **Tropical wave**–A minor tropical disturbance in the easterly trade winds, which could develop into a tropical depression but lacks evidence of a closed circulation; also known as easterly wave.

(194) Along the coast, greater damage may be inflicted by water than by wind. Prolonged winds blowing toward shore can increase water levels from about 3 to 10 feet (1 to 3 m) above normal. This storm tide may begin when the tropical cyclone center is 500 miles or more away. It gradually increases until the winds change direction. On top of this the low pressure in the storm's center can create a ridge or wall of water known as a surge. This will move in the direction of the storm's movement and can be disastrous. The effect may be similar to that of a tsunami (seismic sea wave) caused by earthquakes in the ocean floor. Storm surges can push these tides to 20 feet (6.1 m) or more above normal. About 3 to 4 feet (1 to 1.2 m) of this is due to the decrease of atmospheric pressure and the rest to the strong winds. Additional water damage results from the pounding of sea and swell. Torrential rains, generated by tropical cyclones, can cause both flash floods and river floods from inland rains.

(195)

Tropical cyclone climatology

(196) In an average season, nine or ten tropical cyclones develop and five of these reach hurricane strength; about two hurricanes reach the U.S. While they may develop in any month, June through November is generally considered the tropical cyclone season, with a peak in August, September and October. Early and pre-season storms, from May through mid-July, are most likely to originate in the western Caribbean Sea and Gulf waters. From mid-July through late September this development is spread through the main basin of the tropical Atlantic and a much more persistent westerly movement is noticeable. From late September through November, activity gradually confines itself to the Caribbean and Gulf region. A northerly movement, similar to early season storms, becomes more apparent. However, because of the large reservoir of heat available at the end of the season, these storms are often more intense than their early season counterparts.

(197) The most common path is curved, the storms first moving in a general westward direction, turning later to the northwestward and finally toward the northeast. A considerable number, however, remain in low latitudes and do not turn appreciably toward the north. Freak movements are not uncommon, and there have been storms that described loops, hairpin-curved paths and other irregular patterns. Movement toward the southeast is rare, and, in any case, of short duration. The expanse of the Gulf and Caribbean waters, including bordering coastal regions, as well as the Atlantic Coast are subject to these storms during the hurricane season.

(198) The average speed of movement of tropical cyclones is about 10 to 15 knots. This speed, however, varies considerably according to the storm's location,

development and the associated surface and upper air patterns. The highest rates of speed usually occur in the middle and higher latitudes and range up to 40 to 50 knots. Storms are slowest during recurvature or when looping. They can also become stationary in the absence of steering currents.

(199)

Hurricane warnings and forecasts

(200) The civilian hurricane warning service for the North Atlantic is provided by the **National Hurricane Center / Tropical Prediction Center**, Miami, Florida. It collates ship, aircraft, radar and satellite data to produce and issue tropical cyclone warnings and forecasts for the North Atlantic Ocean, including the Caribbean Sea and Gulf of America as well as the Eastern North Pacific Ocean. Its principal product is the Tropical Cyclone Advisory message especially tailored for Marine, Aviation, Military and public interests. They are issued every 6 hours with intermediate bulletins provided when needed.

(201) For tropical storms and hurricanes threatening to cross the coast of the U.S., coastal warnings are issued to the public by the National Hurricane Center through local Hurricane Warning Offices in order that defense against damage, and perhaps evacuation, can be implanted. Two levels of warnings are employed. The "Hurricane Watch" is a preliminary alert that a hurricane may threaten a specified portion of the coast. It is issued approximately 36 hours before landfall could occur. The second level is the "Hurricane Warning," which indicates that hurricane conditions are expected within 24 hours in advance of landfall. It is aimed at providing the best compromise between timeliness and accuracy for civil defense purposes so that its warning may be too late to allow ocean-going vessels to get underway and complete a successful evasion in open water. To compensate for this, the Marine Advisory contains additional guidance in the form of probabilities of hurricane strikes, for coastal locations and even offshore coordinates, and storm position forecasts for up to 72 hours in advance.

(202)

Hurricane havens

(203) This section is condensed from the **Hurricane Havens Handbook for the North Atlantic Ocean** published by the Marine Meteorology Division, Naval Research Laboratory, Monterey, CA 93943, and available on the internet at *nrlmry.navy.mil/pubs.htm*. While this study concentrates on New York, NY, New London, CT, and Newport, RI, the climatology and principles of navigation can be applied to the entire region; the navigation information can be applied to winter storms as well. For practical purposes any tropical cyclone that approaches within 180 miles is considered a "threat." Data is also incorporated from the Global Tropical/Extratropical Cyclone Climatic Atlas CD-ROM jointly produced by the National Climatic Data Center and the Fleet Numerical Meteorology and Oceanography Detachment-Asheville.

(204) The classical doctrine held by most mariners is that ocean-going ships should leave ports that are threatened by a hurricane. Despite this natural caution, ships continue to be damaged in port or after leaving port, as a result of tropical cyclone encounters. This often stems from the difficulty in forecasting tropical cyclone movement, although these forecasts have improved significantly in the past two decades. In addition to evaluating the forecast it is necessary to assess the risks of remaining in port or putting to sea according to the circumstances of the threat, the facilities of the port and the capabilities of the vessel and crew. For an evaluation as to a course of action, several factors are important. The risk of a particular port experiencing a hurricane is often dependent on seasonal and geographic influences. Forecasts of hurricane movements are more reliable in some areas, particularly the lower latitudes. In the mid-latitudes where storms are often recurving, the difficulty increases. It is important to know the sheltering capabilities of the port that is being considered and the speed of advance of tropical cyclones in the latitudes that you may be sailing. When the tropical cyclone speeds approach or exceed vessel speed, options become limited.

(205) Of the 117 tropical cyclones that threatened New York from 1842 to 1995, 100 occurred from August through October with the main threat in September. The hurricane (winds >64 knots) threat has a peak in August and September; 81 of the 117 hurricanes occurred in those months. Tropical cyclones usually move in from the south or southwest. During this same period New Haven was threatened by 108 tropical cyclones, 91 of which occurred from August through October. Hurricanes are most likely during August and September when 75 out of the total of 108 occurred. The direction of approach is most likely from the south or southwest. Because of the natural protection offered by the shape of the coast from Cape Cod to Cape Hatteras, most recurving storms either make landfall south of Hatteras or pass New England well offshore to the southeast. The majority of storms pass well to the southeast of New England, following the Gulf Stream. Occasionally storms accelerate on a more northerly track similar to the disastrous hurricane of 1938, which advanced rapidly up the east coast, offshore near Hatteras, across central Long Island, into Connecticut and finally through Vermont. This hurricane's forward speed reached 52 knots, an advance that would be difficult to prepare for, even with today's sophisticated warning methods. It is the exceptionally fast-moving storm that poses the greatest threat. For example, based on climatology, a September storm located off Miami would reach New York in about 3 or 4 days. However, the 1938 hurricane traveled this distance in about 30 hours. Tropical cyclones tend to accelerate as they move north of about 30°N. Forward speeds range from 25 to 30 knots for those crossing the New York–New England coast compared to 20 to 25 knots for those passing offshore to the southeast.

(206) Since wind records were available in the New York Harbor area, sustained winds have reached hurricane force (64 knots) only once. The September 1944 hurricane produced 64-knot winds at Central Park and 70-knot winds at La Guardia. Other hurricanes that have caused considerable damage were storms in September 1821, September 1938, August 1954 (Carol) and September 1960 (Donna). During a recent 44-year period along the Connecticut-Rhode Island coast, three hurricanes produced winds that have been estimated to have reached at least minimal hurricane strength. The 1944 hurricane, Carol and the 1938 storm were the three. The 1938 storm was the worst as winds in the New London area were estimated at 78 to 87 knots.

(207) In addition to strong winds, the hurricane brings rough seas, heavy rains and storm surges. New York's Lower Bay is subject to wave action due to an open quadrant, east through south, to the Atlantic. The size and depth of the bay also provide sufficient fetch for a strong wind to generate destructive waves. Deep ocean swells approaching from the open quadrant would be reduced by shoals at the entrance to Lower Bay, between Sandy Hook and Rockaway Point. Upper Bay, Newark Bay, lower Hudson River and East River are subject to limited wave action. Long Island Sound is a deep water sound with a generous fetch in an east-west direction. New London Harbor is well protected from wave action. Although a west wind can produce large seas in the Sound they are greatly reduced on entering the harbor channel. Within Narragansett Bay wave action is severely limited by short fetch for most wind directions. Wave action generated within the Bay will create minimal problems for ships at anchor if the scope of chain employed is set to give the best riding conditions.

(208) Storm tides can produce a high water level, which in addition to inundating coastal areas, may allow wind waves to cause destruction in areas normally unaffected by waves. Combined storm surge and tide have produced water levels of over 10 feet (3 m) above mean low water in the New York Harbor area and levels greater than 15 feet (4.6 m) above mean low water in western Long Island Sound. New London is one of the few east coast ports to have experienced a major storm surge in this century. The storm surge of September 21, 1938 hit New London as an apparent tidal bore (wall of water) causing considerable destruction. This surge was slightly greater than that expected once in a hundred years and was likely due to the fast moving nature of this hurricane. At Newport storm tides were measured at 10.8 feet (3.3 m) above mean sea level during the 1938 hurricane. The top winter extratropical storm produced a 6.0-ft (1.8 m) surge on the 30th of November, 1963.

(209) In summary, New York Harbor is recommended as a hurricane haven. It is a large national harbor with many excellent berthing facilities and good deep-water anchorages. Natural topographic features and numerous man-made structures offer good wind protection. The bathymetry and orientation of the harbor relative to the normal path of hurricanes tend to mitigate the wind wave and ocean swell danger although storm surge is a sufficient threat. The main New London harbor is not a haven for most vessels during a hurricane although the inner harbor is considered safe for most ships. The surrounding topography provides some protection from east through southeast winds for the eastern shore of the main and inner harbor, however the lower western shore of the main harbor is very exposed to southeast through south winds. The entire harbor is subject to the possibility of major storm surge flooding. The port of Newport is located inside Narragansett Bay, which has deep water anchorages within its confines. Although these anchorages are not well sheltered from winds, they have proven hurricane haven properties for ships able to steam at anchor.

(210) Flooding associated with hurricane-induced high tides is the principal threat to small craft in the area. They should be hoisted and secured ashore above projected flood levels whenever possible. Best protection is inside some type of storage building to prevent possible damage by flying objects or to prevent the possibility of broken tie-downs in high winds. Local knowledge is the best guide to weathering a storm in small harbors.

(211)

Waves

(212) In late March of 1984 a 968-mb low off the New Jersey coast generated a 33-foot (10.1 m) wave at Buoy 44005 (42.7°N., 68.3°W) while Buoy 41002 (40.1°N., 73.0°W) measured a 47-foot (14.3 m) wave during Gloria in September 1985. Systems similar to these are partly responsible for the rough seas encountered along this coast from September through April. The buoy closest to the area, 44003, (40.8°N., 68.5°W), in 10 years of operation has measured a 29-foot (8.8 m) wave in February and 25-foot (7.6 m) waves from October through April. It has been estimated that over the open waters along this coast maximum significant waves should reach 30 feet (9 m). The table below (extracted from Marine Weather of Western Washington. Kenneth E. Lilly, Jr., Commander, NOAA, Starpath School of Navigation, 1983) shows the relationship between significant and other wave heights.

(213) This table can be used to project a range of wave heights that might be expected in deep water. If significant wave heights of 10 feet (3 m) are forecast then the most frequently observed waves should be 5- to 6-foot (1.7 to 1.8 m) range while one wave in 100 should reach 17 feet.

(214) A giant or rogue wave might reach 25 feet (7.6 m) in these circumstances. These rogue or "killer" waves occur when the large number of different waves that make up a sea occasionally reinforce each other. This action creates a wave that is much steeper and higher than the surrounding waves. These rogue waves often occur in a stormy sea and are described by mariners who have experienced them, as coming out of nowhere and disappearing just as quickly. If significant wave heights

are observed at 20 feet (6.1 m) then a rogue wave could reach 50 feet (15.2 m) if the water depth could support it.

(215)

Wave Heights from Significant Wave Heights (SWH)	
Most frequent wave heights	0.5 x SWH
Average wave heights	0.6 x SWH
Significant wave height (average height of highest 33%)	1.0 x SWH
Height of highest 10% of the waves	1.3 x SWH
One wave in 1,175 waves	1.9 x SWH
One wave in 300,000 waves	2.5 x SWH

(216) Rough sea conditions are usually generated by gales out of the northwest through northeast. Waves greater than 8 feet (2.4 m) occur about 10 to 15 percent of the time in winter. From fall through spring, wave heights of more than 7 feet (2.1 m) frequently last one day or more; in midwinter they often last 2 days or more. In addition to coastal storms, cold fronts with rapidly shifting winds can create dangerous seas.

(217) Steep waves are often more dangerous than high waves with a gentle slope. Waves appear menacing when the ratio of wave height to length reaches about 1/18. They begin to break when this ratio is about 1/10. Steepest waves develop when strong winds first begin to blow or early in a storm's life. The ship no longer rides easily but is slammed. Steep waves are particularly dangerous to small craft. When wave heights are greater than 5 feet, periods of less than 6 seconds can create problems for boats under 100 feet in length. Waves of 10 feet or more with periods of 6 to 10 seconds can affect comfort in 100- to 200-foot (30.5 to 61 m) vessels. When wind waves reach 20 feet they become hazardous to vessels under 200 feet in length and provide a rough ride for larger ships. Waves moving into shallow water become steeper and break when the depth is about 1.3 times the wave height. Areas such as Nantucket Shoal and Georges Shoals are dangerous in heavy weather. Wave steepness is also increased by tidal currents, particularly when they oppose the wind.

(218) Swells can create problems for larger vessels. About one-half of the waves of 10 feet (3 m) or more, in these waters, are swells from distant storms. They are uncomfortable to ships that roll or pitch in sympathy. Swells with 500- to 1,000-foot (152 to 305 m) wave lengths affect ships of these lengths. When steaming into such swells a resonance is set up until the bow digs into the waves. The resulting pitch will cause more of a power loss than a roll caused by a sea. Swells with wave lengths that range from about three-fourths to twice the ship's length can have this effect. Pitching is heaviest when the ship's speed produces synchronism between the period of encounter and the ship's natural pitching period—this often occurs at or near normal ship speeds.

(219) When in running before a following sea, the greatest danger arises when speed is equal to that of the waves or when the waves overtake the ship so slowly that an almost static situation is created with the vessel lying on the wave crest. In this latter case stability is so reduced that a small vessel could capsize. Waves on the quarter or astern can also result in very poor steering quality. As seas move along the vessel from aft to forward the rudder is less effective and the boat may be slewed across the face of a sea filling the decks with water as she broaches. She could lose her stability and capsize, particularly if the boat is trimmed by the head.

(220)

Winds

(221) Migratory weather systems cause winds that frequently change in strength and direction. In general winds are generally westerly but often take on a northerly component in winter and a southerly one in summer. Strongest winds are generated by lows and cold fronts in fall and winter and by fronts and thunderstorms during spring and summer. Extreme winds are usually associated with a hurricane or severe northeaster and could reach 125 knots. Sustained winds of 100 knots should occur about every 50 years on the average; gusts are usually about 30 percent higher.

(222) In the open seas, away from the influence of land, winds are stronger and less complex. From December through March they are mainly out of the west through north with gales occurring about 6 to 12 percent of the time. Windspeeds, in general, increase with distance from the coast. If winds persist for a long time over a long fetch they will generate rough seas. Winter windspeeds of 20 knots or more persist for more than 12 hours about 50 percent of the time; however these winds often shift and a new fetch is established. Summer winds are usually out of the south through southwest and gales are infrequent. During the spring and fall winds are more variable.

(223) Coastal winds are complex since they are influenced by the topography. Over land speeds are reduced. However, channels and headlands can redirect the wind and even increase the speed by funneling the wind. In general you will find southerly components in summer and northerly ones in winter. In sheltered waters like Buzzard Bay, Narragansett Bay and the harbors of Long Island Sound there are a large percentage of calms, particularly during the morning hours. When the existing circulation is weak and there is a difference between land and water temperature, a land-sea breeze circulation may be set up. As the land heats faster than the water, a sea breeze is established during the day; this onshore flow may reach 15 knots or more. At night the land cools more rapidly often resulting in a weak breeze off the land. In many locations the sea breeze serves to reinforce the prevailing summer wind.

(224)

Visibilities

(225) Fog, precipitation, smoke and haze all reduce visibilities. Fog is the most restrictive and persistent. It forms when warm, moist air moves across colder water, when very cold air moves over warmer water or when

moist air is cooled to near its dew point by radiation or rainfall. These conditions can be triggered by a number of weather situations.

(226) Prior to the arrival of a cold front there is often a warm, southerly flow of air across cool Gulf waters resulting in dense fog. Warm or stationary fronts can also bring fog while rainfall from lows and fronts can create an evaporation fog. Along the coast radiation fog is common on clear, calm nights although it usually burns off during the morning hours. In the spring, coastal fog may occur near the mouths of rivers and streams that are fed by cold snowmelt.

(227) Sea temperatures increase, in general, from north to south, but the variation is usually only a few degrees over open water. Close to the coast, water temperatures are usually warmer in summer and colder in winter than offshore readings. Water temperatures in summer range from about 66 °F to 74 °F, while in winter the range is from about 34 °F to 37 °F.

(228) Advection fog is most common in late spring and early summer when south and southwest winds bring warm humid air over the still-cold Labrador Current. Near Georges Bank visibilities fall to less than 1 mile up to 30 percent of the time. While these frequencies drop to the southwest, fog remains a problem in this season.

(229) The areas along the coast, at the heads of bays within the rivers, may be comparatively clear while fog is very thick outside. The frequency of fog over land and water is usually in opposition. Land fog is often most frequent in fall and winter compared to the spring and summer maximum of sea fog. Consequently figures for poor visibility at inland or sheltered harbors are no guide to conditions at sea or in the approaches.

(230)

Superstructure icing

(231) Heavy winter weather can cause ice to collect on ships sailing these waters. At its worst superstructure icing can sink a vessel. When air temperature drops below the freezing point of sea water (About 28.6 °F) strong winds and rough seas will cause large amounts of sea spray to freeze to the superstructure and those parts of the hull that escape a frequent washing by the sea. Ice amounts increase rapidly with falling air and sea temperatures as well as increasing windspeeds. The most dangerous conditions exist when gales last for several days in temperatures of 28 °F or lower. The ice buildup on a trawler can exceed 5 tons per hour.

(232) A moderate rate of ice accumulation usually occurs when air temperatures are equal to or less than 28 °F with winds of 13 knots or more. When air temperatures drop to 16 °F or below and winds reach 30 knots or greater, ice collects more rapidly. On a 300- to 500-ton vessel it would accumulate at more than 4 tons per hour and is called severe. December, January and February are the worst months. The potential for moderate icing exists about 5 to 10 percent of the time.

(233) In addition to sea spray, ice is also caused by freezing rain or drizzle and fog in freezing conditions. While these two causes could create enough weight on the rigging to cause it to fall, this is minor in comparison with the freezing spray hazard. Icing on the superstructure elevates the center of gravity, decreasing the metacentric height. It increases the sail area and heeling moment due to wind action. Its non-uniform distribution changes the trim. It can hamper steerability and lower ship speed. Icing also creates hazardous deck conditions.

(234) If you can't avoid the weather conditions that cause icing, experience and research have helped develop some guidelines. The first two courses of action when encountering potential icing conditions are to seek shelter from the sea and to steer towards warmer water. Once icing has begun it is prudent to slow down enough so that little or no spray is taken aboard. It is also important to keep ice from building up by whatever means are available. This includes crewmen using tools or baseball bats to remove ice from the deck and superstructure.

(235) Any effort to control the rate of accumulation will buy time. In general heaving to with the bow into the wind and sea as much as possible and varying the course slightly to ensure a minimum symmetrical build up is a good rule. However, experiments have shown that on a trawler with its stern to the wind, loss of stability is only about one-half of that in the ahead condition. When the wind is 30 degrees off the bow the loss of stability is 50 percent greater than in the ahead condition. Also ice accumulates more rapidly on the windward side causing a heeling into the wind. This listing is partially offset by the action of the wind so that a shift to a reciprocal course after icing has built up could be disastrous. When ice builds up significantly it is important to remember that the removal of one ton of ice 50 feet from the vessel's center of gravity is as effective as removing 10 tons of ice 5 feet above the center of gravity.

(236)

Optical phenomena

(237) Optical phenomena range from electromagnetic displays to intricate geometrical patterns. The aurora and Saint Elmo's fire are electromagnetic displays. Halos, coronas, parhelia, sun pillars and related effects are optical phenomena associated with the refraction and diffraction of light through suspended cloud particles; mirages, looming and twilight phenomena such as the "green flash" are associated with refraction of light through air of varying density. Occasionally, sunlight is refracted simultaneously by cloud suspensions and by dense layers of air producing complex symmetric patterns of light around the sun. A mirage is caused by refraction of light rays in a layer of air whose density increases or decreases rapidly, near the surface. A marked decrease in density with increasing altitude causes looming, towering and superior mirages. Looming occurs when objects appear to rise above their true elevation. Objects below the horizon may actually be brought into view. This

apparent effect often leads to a serious underestimation of horizontal distances. Unimpressive landmarks, and distant ships may acquire startling characteristics through apparent vertical stretching; this phenomenon is known as towering. A superior mirage is so named because of the appearance of an image above the actual object. Ships have been seen with an inverted image above and an upright image floating above that.

(238) Inferior mirages result from the upward bending of light rays in an unstable air mass. This phenomenon is observed locally whenever a superheated land mass or a wide expanse of open water is overrun by cold air. Sinking below the horizon of relatively close objects, may result in an overestimation of horizontal distances. Occasionally, a complicated vertical temperature distribution may transform hilly coastlines into impressive walls of lofty pinnacles. This phenomenon is known as Fata Morgana. On clear days, just as the upper rim of the sun disappears below the horizon, green light is sometimes refracted from the solar spectrum. This brief phenomenon is called the green flash.

(239) Floating ice crystals (cirriform clouds, light snow flakes, ice fog or drifting snow) may cause the refraction of light into a variety of faintly colored arcs and halos. This phenomenon, which may be recognized from the fact that the red band is closest to the light source, includes halos, arcs that open toward or away from the sun, mock images and various geometrical figures that may be located in various parts of the sky with references to the sun.

(240) Fogbows, resulting from refraction through suspended water particles, are seen in the region of the sky directly opposite from the sun, or the antisolar point. These bows, although occasionally brilliantly colored, are normally seen as broad white bands with faintly colored borders. Rainbows are also observed.

(241) When atmospheric particles are about equal in size to the wavelength of light, diffraction is likely to occur. Diffractional phenomena frequently show properties similar to those of refraction except for the reversal in the spectrum colors, violet now being closest to the source of light. The Brocken bow, or glory, appears on clouds or fog banks as a colored ring around the projected shadow of the observers head. The solar and lunar coronas, which are observed only through high clouds, resemble the halo except that they may assume increasingly larger diameters as the size of the particles decrease. When the light from the sun or the moon is diffracted by cirrus or cirrostratus, iridescence may sharply delineate the outline of clouds in brilliant green, blue, pink, orange or purple.

(242) Refraction of sunlight takes place whenever the intervening particles are larger than the wavelength. Thus, sunlight that is reflected from ice crystals is transformed into sun pillars and parhelic circles. When both phenomena occur in combination they form the remarkable sun cross. Paricelenci circles are observed with moonlight.

(243) The **auroral borealis** (northern lights) and **St. Elmo's fre** are two types of electrical phenomena sometimes observed in this region. The zone of maximum auroral frequency extends along the periphery of a 20- to 25- degree circle whose center is at the magnetic pole. Auroras are generally associated with moonless nights. An artificial maximum exists in winter because of the longer hours of darkness. No conclusive evidence is available to show that a seasonal variation in the frequency of auroras exists. However, periods of intense sunspot activity are reflected in a maximum occurrence of this electrical phenomenon.

(244) Generally auroras may be classified as having either a ray structure (rays, streams, draperies, corona) or a nebulous appearance (homogeneous quiet arc, homogeneous band, pulsating arcs, pulsating surfaces, diffuse luminous surfaces and feeble glow). Flaming auroras, which fall in neither category, may be added to this list. Moreover, auroras may remain uniformly red, green or purple or assume a rapid succession of these colors. Brilliant shifting auroras are invariably accompanied by magnetic storms and electrical interference with communications.

(245) St. Elmo's fire is occasionally observed in this area, but because of its faintness it is most commonly observed during the night hours and on dark overcast days. These eerie flickers of bluish light are usually caused by the unusual electrification of the snow-filled air, which is most likely when the wind is strong. St. Elmo's fire is restricted to the tips of such objects as ship masts, wind vanes and airplane wings.

(246)

Dew point

(247) The temperature at which condensation to water droplets occurs is called the dew point. If this dew point is above freezing, condensation will be in the form of water. When the dew point reaches freezing, ice crystals will be deposited on cold surfaces. Knowledge of the dew point along with cargo temperature and moisture content is vital for hold ventilation decisions. It is also a parameter used in forecasting fog formation.

(248)

Cargo care

(249) When free air has a dew point temperature higher than the temperature of the surface with which it comes in contact, the air is often cooled sufficiently below its dew point to release moisture. When this happens on board ship, condensation will take place on relatively cold cargo or on the ship's structure within the hold where it later drips onto the cargo. Thus, if cargo is stowed in a cool climate and the vessel sails into warmer waters, ventilation of the hold with outside air will likely lead to sweat damage in any cargo sensitive to moisture. Under such conditions external ventilation should, as a rule, be closed off entirely, unless the cargo generates internal heat, that hazard being greater than sweat damage. In the opposite case, when a vessel is loaded during a warm

period, and moves into cooler weather, vulnerable cargo should be ventilated.

(250) A safe rule for ventilation directed toward moisture control may be stated as follows: Whenever accurate measurements show the outside air has a dew point below the dew point of the air surrounding the cargo to be protected, such outside air is capable of removing moisture from the hold and the ventilation process can be safely started. Whenever the reverse is true, and the outside dew point is higher than the dew point temperature around the cargo, then ventilation will increase the moisture content of the hold and may readily result in sweating within the ship. The above does not take into account possible fumes or gases in the compartment. In such cases discretion must be used.

(251)

Ice

(252) (Refer to discussion under ports affected.)

(253) During some winter months or when threatened by icing conditions, lighted buoys may be removed from station or replaced by unlighted buoys; unlighted buoys, daybeacons and lights on marine sites also may be removed. (See Light List.)

(254) The **International Ice Patrol (IIP)** was formed in 1914 to patrol the Grand Banks of Newfoundland and detect icebergs and warn mariners of their location. Under the 1974 Safety of Life at Sea (SOLAS) Convention, 17 member-nations agree to share the $5 million annual cost of operating the patrol. The U.S. Coast Guard conducts the patrol and maintains IIP records.

(255) The IIP is coordinated from its operations center at New London, Connecticut. Its staff numbers 16, including Coast Guard and civil service specialists. The ice season typically runs from February through July but can last longer. Flying out of St. John's, Newfoundland, USCG aircraft cover the ice danger area, a piece of water twice the size of the State of Texas. Its southern boundary is the latitude of New York City and it reaches halfway across the Atlantic with Newfoundland on the northwest and Greenland and Iceland on its north and northeast. A normal flight lasts seven hours and can cover 35,000 square miles.

(256) Once sighted, a berg's location, size and shape are entered into a computer drift model, used until the berg is re-sighted or melts. The IIP attempts to locate and track all icebergs south of the 50th parallel, and particularly those south of 48°N., which may be hazardous to navigation near the Grand Banks. When sighting data is entered into the drift program, predicted positions of bergs are calculated for 1200 UTC.

(257) All shipping is requested to assist the IIP by reporting all sightings of ice at once to the IIP through any U.S. or Canadian Coast Guard communications station. Ice sightings reports should include: precise position, size and shape of berg, sea surface temperature, and concentration and thickness of sea ice. Reports can be sent to COMINTICEPAT NEW LONDON CT through INMARSAT-C: Code 42. There is no charge for iceberg reports made using Code 42.

(258) The IIP Operations Center can be reached by telephone at 860–271–2626 or 877–423–7287 or via the Coast Guard Atlantic Area Operations Center at 757–398–6700.

(259) A radio facsimile chart of the area depicting 1200Z ice distribution is broadcast three times daily. A list of the radio stations broadcasting IIP Bulletins and frequencies and times of broadcasts is published annually in Local Notices to Mariners of the First Coast Guard District and in Radio Navigational Aids, Pub. 117, issued by the National Geospatial-Intelligence Agency.

(260) The IIP seeks comment on its services to mariners, particularly on the effectiveness of the times and frequencies of radio transmissions. Mariners are requested to mail facsimile charts received at sea to:

(261) International Ice Patrol, 1 Chelsea Street, New London, CT 06320. The frequency used, time of receipt and vessel position at time of receipt should be indicated. Additional customer comments can be directed to IIP Customer Service at 877–423–7287.

(262)

Sizes and Types of Icebergs

Size	Height		Length	
	(feet)	(meters)	(feet)	(meters)
Growler	0-3	0-1	0-19	0-5
Small (S)	4-50	1-15	20-200	6-60
Medium (M)	51-150	16-45	201-400	61-122
Large (L)	151+	46+	401+	123+

Shape	Description
Blocky	Steep sides with flat top. Very solid. Length-height ratio less than 5:1.
Tilted Blocky	Blocky iceberg which has tilted to present a triangular shape from the
Drydock	Eroded such that a large U-shaped slot is formed with twin columns. Slot extends into or near waterline.
Pinnacled	Large central spiral or pyramid.
Dome	Large round smooth top. Solid-type iceberg.
Tabular	Flat-topped iceberg with length-height ratio greater than 5:1.

(263)

Principal ports

(264) The principal deep-draft commercial ports within the area of this Coast Pilot are New Bedford and Fall River, MA; Tiverton and Providence, RI; New London and Bridgeport, CT; New York, Albany and Port Jefferson, NY; and Elizabeth and Newark, NJ.

(265) Other deep-draft facilities are located on Cape Cod Canal; Narragansett Bay; off Northville and Northport,

NY, on Long Island Sound; and on the Hudson River between New York City and Albany, NY.

(266)

Pilotage

(267) Pilotage, with few minor exceptions, is compulsory for all foreign vessels and U.S. vessels under register entering and departing the Port of New York and New Jersey and other ports within the area of this Coast Pilot and for all such vessels transiting Block Island Sound, Narragansett Bay, and Long Island Sound. (See **33 CFR 207.20**, chapter 2, for Pilotage Regulations on the Cape Cod Canal.)

(268) Pilotage is optional for coastwise vessels that have on board a pilot properly licensed by the Federal Government for the waters which the vessel travels.

(269) Arrangements for pilots should be made by the ships' agents at least 24 hours in advance at all of the ports. New York is the only port at which the pilot boat remains on station. Detailed information on pilotage procedures is given in the text for the ports concerned.

(270)

Towage

(271) Tugs are available at all major ports; they can usually be obtained for the smaller ports on advance notice if none are available locally. Arrangements for tugs should be made in advance through ships' agents or the pilots. (See the text for the ports concerned as to the availability of tugs.)

(272)

Vessel arrival inspections

(273) Quarantine, customs, immigration and agricultural quarantine officials are stationed in most major U.S. ports. (See Appendix A for addresses.) Vessels subject to such inspections generally make arrangements in advance through ships' agents. Unless otherwise directed, officials usually board vessels at their berths.

(274) **Harbormasters**, where appointed, are mentioned in the text. They usually have charge of the anchorage and berthage of vessels.

(275)

Supplies

(276) General supplies, including fuel oil, diesel oil and fuel, gasoline, water and marine supplies are available at the principal ports. Similar items but in more limited quantities can be obtained at many places mentioned under descriptions of the different ports.

(277)

Repairs-salvage-wrecking

(278) Complete facilities for large vessels are available in New York Harbor. The extent and types of facilities at other places are shown in the text under the description of the ports.

(279)

Small-craft facilities

(280) There are numerous places where fuel, supplies, repairs, slips for dockage and launching ramps are available for small craft. For the various towns and isolated places, the Coast Pilot includes generalized information about marine facilities.

(281) **A vessel of less than 65.6 feet (20 meters) in length or a sailing vessel shall not impede the passage of a vessel that can safely navigate only within a narrow channel or fairway. (Navigation Rules, International-Inland Rule 9(b).)**

(282)

Standard time

(283) The area covered by this Coast Pilot uses eastern standard time (e.s.t.), which is 5 hours slow of Coordinated Universal Time (UTC). For example, when it is 1000 UTC, it is 0500 at New York City.

(284)

Daylight saving time

(285) Throughout the area of this Coast Pilot, clocks are advanced 1 hour on the second Sunday of March and are set back to standard time on the first Sunday of November.

(286)

Legal public holidays

(287) New Year's Day, January 1; Martin Luther King, Jr.'s Birthday, third Monday in January; Washington's Birthday, third Monday in February; Memorial Day, last Monday in May; Independence Day, July 4; Labor Day, first Monday in September; Columbus Day, second Monday in October; Veterans Day, November 11; Thanksgiving Day, fourth Thursday in November; and Christmas Day, December 25. The national holidays are observed by employees of the Federal Government and the District of Columbia and may not be observed by all the states in every case.

(288) In addition, the following holidays are also observed in the states covered by this Coast Pilot:

(289) Lincoln's Birthday, February 12: CT, NJ and NY.

(290) Evacuation Day, March 17: MA, Boston and Suffolk County only.

(291) Good Friday: CT and NJ.

(292) Patriots Day, third Monday in April: MA.

(293) Rhode Island Independence Day, May 4: RI.

(294) Bunker Hill Day, June 17: MA, Boston and Suffolk County only.

(295) Victory Day, second Monday in August: RI.

(296) General Election Day, first Tuesday after the first Monday in November: NJ, NY and RI.

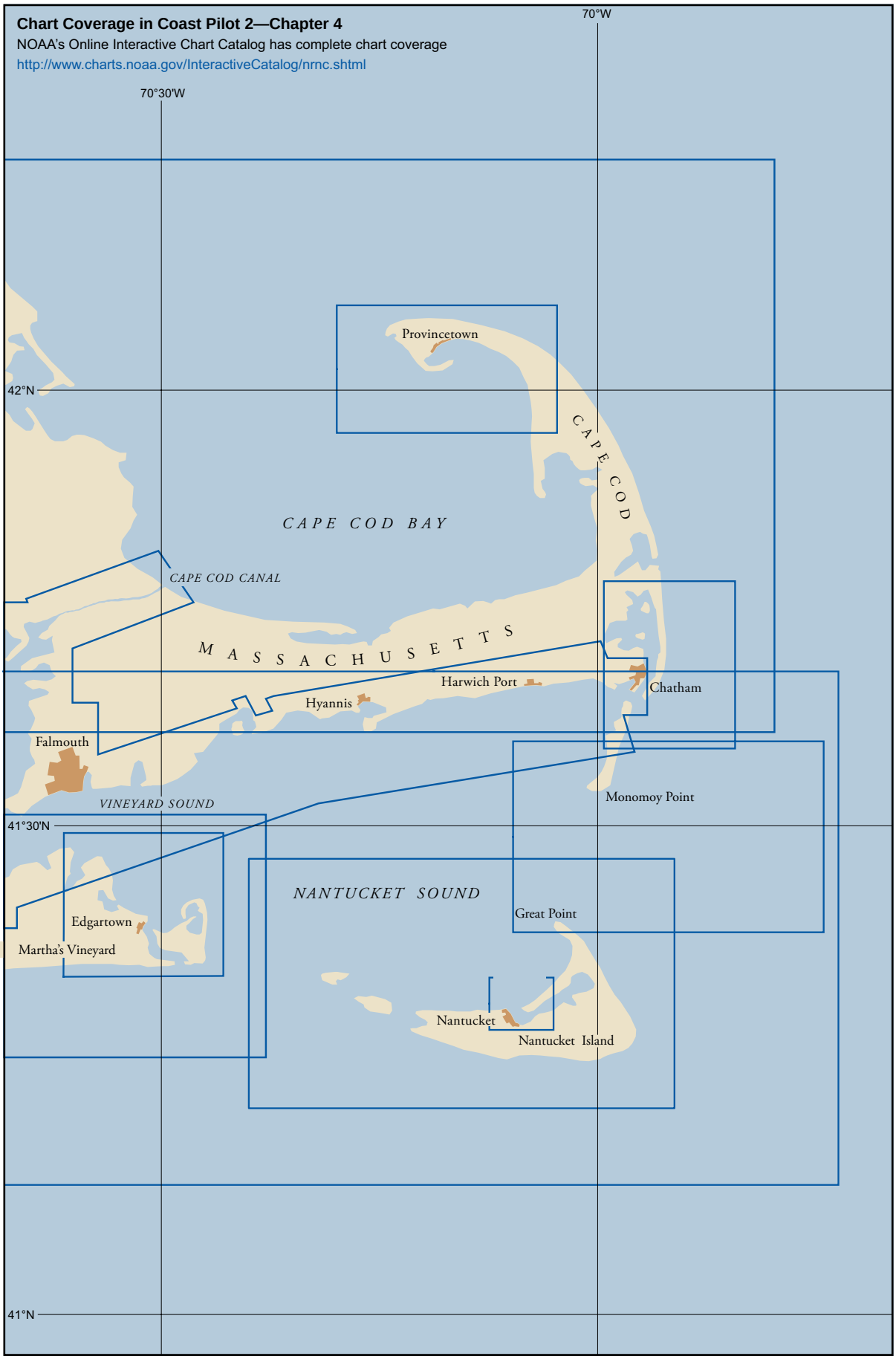


Chart Coverage in Coast Pilot 2—Chapter 4
NOAA's Online Interactive Chart Catalog has complete chart coverage
http://www.charts.noaa.gov/InteractiveCatalog/nrnc.shtml
70°W
70°30'W
42°N
41°30'N
41°N
Provincetown
CAPE COD
CAPE COD BAY
CAPE COD CANAL
MASSACHUSETTS
Harwich Port
Chatham
Hyannis
Falmouth
Monomoy Point
VINEYARD SOUND
NANTUCKET SOUND
Great Point
Edgartown
Martha's Vineyard
Nantucket
Nantucket Island

Outer Cape Cod and Nantucket Sound

(1) This chapter describes the outer shore of Cape Cod and Nantucket Sound including Nantucket Island and the southern and eastern shores of Martha's Vineyard. Also described are Nantucket Harbor, Edgartown Harbor and the other numerous fishing and yachting centers along the southern shore of Cape Cod bordering Nantucket Sound.

(2)

COLREGS Demarcation Lines

(3) The lines established for this part of the coast are described in **33 CFR 80.135** and **80.145**, chapter 2.

(4)

No-Discharge Zone

(5) The State of Massachusetts, with the approval of the Environmental Protection Agency, has established a No-Discharge Zone (NDZ) in all coastal waters of Massachusetts covered by this chapter except a small area from Woods Hole to Vineyard Haven, extending about 3 miles offshore.

(6) Within the NDZ, discharge of sewage, whether treated or untreated, from all vessels is prohibited. Outside the NDZ, discharge of sewage is regulated by 40 CFR 140 (see chapter 2).

(7)

Cape Cod

(8) **Cape Cod** is a long peninsula forming the easterly extremity of Massachusetts. It makes out from the mainland in an easterly direction for 31 miles, then extends northward for over 20 miles. This cape forms the southern and eastern shores of Cape Cod Bay, the northern shore of Nantucket Sound and the eastern shore of Buzzards Bay. The northern trend of Cape Cod, forming what is sometimes called the **Hook of the Cape**, is known as the Lower Cape. This section is well settled and composed almost entirely of sandy lands, with high bare sand dunes and low nearly level plains. The portion of Cape Cod between Chatham and Cape Cod Canal is known as the Upper Cape. This region is wooded and is well settled by numerous towns and villages.

(9)

Current

(10) The tidal current velocities between Race Point and Highland Light are very strong but diminish to less than 1 knot between Highland Light and Chatham Light. Strengths of flood and ebb set northward and southward, respectively, along the coast. The time of current changes rapidly, the strength of flood or ebb occurring about 2 hours later off Nauset Beach Light than off Chatham Light.

(11)

North Atlantic right whales

(12) Federally designated critical habitat for the endangered North Atlantic right whale lies within Cape Cod Bay (See **50 CFR 226.101** and **226.203**, chapter 2, for habitat boundary). It is illegal to approach closer than 500 yards of any right whale. (See **50 CFR 224.103(c)**, chapter 2, for limits and regulation.) **Recommended Two-Way Whale Avoidance Routes** and a **Recommended Two-Way Whale Avoidance Track** have been charted within Cape Cod Bay to reduce interactions between vessels and right whales. When right whales are present in Cape Cod Bay (peak season: December through May), NOAA recommends that mariners use these routes and take the precautionary measures recommended in chapter 3 to reduce the risk of ship strikes. (See North Atlantic right whales, indexed as such, in chapter 3 for more information on right whales and recommended measures to avoid collisions with whales.) **Caution:** Full bottom coverage surveys have not been conducted within the entire route, so uncharted dangers may exist.

(13) All vessels 65 feet or greater in length overall (LOA) and subject to the jurisdiction of the United States are restricted to speeds of 10 knots or less in the Cape Cod Bay Seasonal Management Area between January 1 and May 15. The area is defined as all waters of Cape Cod Bay with a northern boundary of 42°04'56.5"N., 70°12'W., to 42°12'W., thence due west back to shore. The restriction applies to a Seasonal Management Area off Race Point between March 1 and April 30. The area is defined as the waters bounded by:

(14) 42°04'56.5"N., 70°12'W.;

(15) 42°12'N., 70°12'W.;

(16) 42°12'N., 70°30'W.;

(17) 42°30'N., 70°30'W.;

(18) 42°30'N., 69°45'W.;

(19) 41°41'N., 69°45'W.; thence due west to the shore. (See **50 CFR 224.105** in chapter 2 for regulations, limitations and exceptions.)

(20) The waters of Cape Cod Bay lie within the WHALESNORTH Mandatory Ship Reporting Area. Each self-propelled ship of 300 gross tons or greater entering WHALESNORTH must participate in the Mandatory Ship Reporting System (See **33 CFR 169**, chapter 2, for limits and regulations, and chapter 3 for sample reports). Sovereign immune vessels are exempt

from the requirement to report but are encouraged to participate.

(21)

Area to be avoided

(22) In order to significantly reduce the risk of ship strikes to the North Atlantic right whale, an area to be avoided was established in the Great South Channel, east of the Boston Harbor traffic lanes. Ships of 300 gross tons and above should avoid the area bounded by lines connecting the following geographical positions:

(23) 41°44'08"N., 69°34'50"W.;

(24) 42°10'00"N., 68°31'00"W.;

(25) 41°24'53"N., 68°31'00"W.; and

(26) 40°50'28"N., 68°58'40"W. between the period of April 1 through July 31.

(27) In 2023, an unexploded ordnance hazard area, formerly a firing range active from 1952 to 1961, was reported off the coast of Cape Cod extending about 15 miles E of Nauset Beach in a fanned area. Mariners are cautioned against anchoring, dredging or trawling in this area.

(28) **Provincetown Harbor**, formed by a turn in the northern end of the hook of Cape Cod, has a diameter of about 2 miles. It is one of the best harbors on the Atlantic Coast, having a sizable anchorage area with excellent holding ground. Coasters and fishermen find protection here in gales from any direction.

(29) The historical town of **Provincetown**, on the northwestern side of the harbor, is at the site of the first landing of the MAYFLOWER in the new world. Supplies and hull repair facilities are available in Provincetown. Engine repairs are available by arrangement.

(30) The approach and entrance to the harbor are free of dangers and are marked by three lights and by Pilgrim Monument, a slim stone structure 348 feet above the water; two standpipes are about 0.2 mile westward and another is 1.5 miles northeastward of the monument. A 2,500-foot stone breakwater is about 300 yards southeastward of the end of the town pier (MacMillan Wharf). The east and west ends of the breakwater are each marked by a light. Strangers should exercise caution when operating in the area. Numerous fishing vessels work out of Provincetown during the year. During the summer, floats are set out that are capable of mooring vessels up to 40 feet. Larger vessels must tie up at permanent piers. Anchorage inside the breakwater is reported to be fair to poor, mud bottom with much debris. The marina close southwest of MacMillan Wharf maintains 40 mooring buoys on the west side of the harbor.

(31) All moorings and berthing in the harbor are under the control of the **harbormaster,** who has an office at the end of the town pier. The harbormaster monitors VHF-FM channel 16, 24 hours a day.

(32) The Captain of the Port, Boston, has established a fairway 100 yards wide extending from 42°02'00"N., 70°09'35"W. to 42°02'43"N., 70°11'01"W., and in the area extending about 100 yards around the piers in Provincetown Harbor. Anchorage is prohibited in this fairway.

(33) **Provincetown Coast Guard Station** is about 0.4 mile southwest of the town pier.

(34) The finger pier northeastward of the Coast Guard pier is in ruins. Mariners should exercise caution while navigating in the area.

(35) **Cape Cod Canal is described in chapter 5. Complete information about the harbors and ports in Cape Cod Bay is contained in United States Coast Pilot 1, Atlantic Coast, Eastport to Cape Cod.**

(36) Between **Wood End** (42°01'12"N., 70°11'19"W.) and **Race Point**, the westernmost point of Cape Cod, shoals that rise abruptly from deep water extend a maximum of about 0.6 mile from shore. **Race Point Light** (42°03'44"N., 70°14'35"W.), 41 feet above the water, is shown from a white tower on the northwest point of Cape Cod.

(37) **Peaked Hill Bar** includes shoals with a least depth of 11 feet about 3.5 miles northeast of Race Point Light.

(38) The bar is about 0.6 mile offshore and extends for about 4 miles paralleling the coastline. This area should be given a berth of at least 2 miles. Keeping in a depth of 20 fathoms will ensure passing 2.5 to 3 miles off the eastern side of Cape Cod.

(39) From Race Point, the shore of Cape Cod curves northeastward, eastward and then southeastward for about 9 miles to the **Highlands** and is composed of bare sand dunes of various heights. The sand dunes begin to be covered with a brownish-looking growth of grass, and the land is higher as the Highlands are approached. The water shoals somewhat abruptly within 0.5 mile of the shore and care must be taken not to go inside the 10-fathom curve.

(40) **Highland Light** (42°02'22"N., 70°03'39"W.), 170 feet above the water, is shown from a 66-foot white tower with covered way to the dwelling on a high bluff of the Highlands.

(41) Prominent objects include a stone crenellated tower, a red brick stack and three spherical radar domes on the summit of a ridge, 0.5 mile south of Highland Light.

(42) The shore southward from the Highlands for 12 miles to Nauset Beach Light has a slight curve. The terrain continues hilly with narrow valleys at intervals. From Highland Light to near the Chatham entrance the water continues to shoal abruptly within 0.5 mile of the shore.

(43) Two spires at **Truro**, 2.5 miles south of Highland Light, are prominent. A tank stands out near the shore about 9 miles south-southeastward of Highland Light in **South Well f eet**.

(44) **Nauset Beach Light** (41°51'36"N., 69°57'12"W.), 120 feet above the water, is shown from a 48-foot conical tower, the upper part red and the lower part white, on the beach at **Eastham**. The buildings of a former Coast Guard station, about 1 mile south of the light, are conspicuous.

(45) The coast from Nauset Beach Light turns gradually southward to Chatham, a distance of 11 miles, and

becomes lower and less steep. The terrain slopes gently back from the beach and is covered with a scanty growth of grass.

(46) **Nauset Harbor**, 3.3 miles south of Nauset Beach Light, is used by small local craft. The area offshore of the harbor is a chain of shifting sandbars. Breakers are present in this vicinity at all stages of the tide and even during the calmest weather. The area is extremely dangerous for any vessel larger than a runabout or for anyone without local knowledge; strangers should never attempt to enter.

(47) **Nauset Beach**, on the barrier island east of **North Chatham**, is continually changing due to the effects of erosion and shifting sands. Waves wash across the beach into the sound in several places during periods of high water. There is a break in the beach centered in approximately 41°42'16"N., 69°55'47"W. The break is unmarked, extremely dangerous, and should be avoided by those without local knowledge.

(48)

Chatham to The River

(49) **Chatham**, about 11.5 miles southward of Nauset Beach Light, is on fairly high ground on the west side of Chatham Harbor. Two conspicuous standpipes with red and white tops are at 41°41'38"N., 69°58'30"W. **Chatham Light** (41°40'17"N., 69°57'01"W.), 80 feet above the water, is shown from a white conical tower on the west side of the harbor. **Chatham Coast Guard Station** is near Chatham Light.

(50) Chatham Harbor can be entered east of Chatham Light. Shoals are constantly shifting in the entrance, and extreme caution is advised. Currents are extremely strong and dangerous; surf can build up quickly on the outer bar during an ebb tide. Ocean waves carry into the harbor and reportedly break as far north as Tern Island. Anchoring and even mooring in much of Chatham Harbor is not recommended. Mariners unfamiliar with the area are advised to stay east of **Chatham Beach Lighted Whistle Buoy C** (41°39'12"N., 69°55'30"W.).

(51) **Chatham Inlet Bar Guide Light** (41°40'18"N., 69°57'00"W.), 62 feet above the water, is shown from a white skeleton tower near Chatham Light. A white (with orange border) and diamond-shaped dayboard worded **ROUGH BAR** is on the light. The light will be activated when the seas exceed 2 feet in height and are considered hazardous for small boats. Small-boat operators are cautioned, however, that if the light is not flashing it is no guarantee that sea conditions are favorable.

(52) About 2.7 miles south of Chatham Light, at what used to be the end of Nauset Beach, is a large area of shoals that extends southwestward to Monomoy Island. There is no marked channel through the shoals. Small vessels with local knowledge use the area with a smooth sea; strangers should avoid the area. These shoals are dangerous in thick weather and vessels in the vicinity should stay in depths of 8 fathoms or more.

(53) The passage inside the barrier beach from Chatham Light to the head of navigation in **Orleans**, on the west side of Meeting House Pond, is about 7.9 miles long and used by small-craft. The passage, marked by private, seasonal buoys, leads northerly from the break through Chatham Harbor, Pleasant Bay, The Narrows, Little Pleasant Bay and The River to Meeting House Pond. The channel requires local knowledge.

(54) A boat basin is in **Aunt Lydias Cove** between **Tern Island** and Chatham; a fish pier is in the basin. The area is subject to frequent changes. Commercial fishing boats operate from the cove. The **harbormaster** can be contacted on VHF-FM channel 16.

(55) **Bassing Harbor**, at the north end of Chatham Harbor, is the entrance to **Ryder Cove** and **Crows Pond**. A small-craft facility is on the south side of Ryder Cove, about 0.5 mile inside the entrance. A town launching ramp is close westward of the facility. Private seasonal aids mark the channel from Chatham Harbor to the town ramp. A 5 mph **speed limit** is enforced in the cove. A forklift at the facility can haul out craft to 25 feet. Gasoline, water, ice, marine supplies, moorings and storage facilities are available; hull and engine repairs can be made. In 1981, a reported depth of 3 feet could be carried to the small-craft facility.

(56) **Nickersons Neck**, on the north side of Crows Pond and the south side of Pleasant Bay, has a country club on the north side.

(57) **Round Cove**, at the southwest end of Pleasant Bay, has a town wharf and launching ramp. A combination antenna and flagpole on the west bank of the cove is conspicuous.

(58) **The Narrows** is a passage between Sipson Island and the mainland and connects Pleasant Bay with Little Pleasant Bay. The passage is marked by private seasonal buoys.

(59) **Little Pleasant Bay** extends about 1.5 miles northward to Barley Neck. A launching ramp is on the west bank of the entrance to **Paw Wah Pond** on the south side of **Namequoit Point**.

(60) **Namequoit River** leads westward from the head of Little Pleasant Bay to **Areys Pond**. In 1981, depths of 2 feet were reported in Namequoit River, and the channel into the pond had depths of 3 feet. A small-craft facility on the north side of the pond has a 50-foot marine railway, a 2-ton crane, moorings, water, marine supplies, a launching ramp and storage facilities; hull, rigging and sail repairs can be made.

(61) An arm, known as **The River**, extends northward from the entrance to Namequoit River for about 1 mile to **Meeting House Pond**. Private seasonal buoys partially mark the channel from The River to the pond. A town landing and launching ramp are on the north side of the channel leading to the pond. A small-craft facility is on the north side of the pond. In 1981, depths of 4 feet were reported in the channel to the pond. A hydraulic trailer at the facility can handle craft to 50 feet. Berths and moorings in depths of 6 to 10 feet, gasoline, water, ice, a

launching ramp and storage facilities are available; hull and engine repairs can be made. A town ramp is on the east side of the pond southeastward of the small-craft facility.

(62)

Nantucket Sound

(63) **Nantucket Sound** is between the south coast of Cape Cod on the north and Nantucket Island and part of Martha's Vineyard on the south and joins Vineyard Sound on the west to provide an inside passage. Nantucket Sound has a length of about 23 miles in an east-west direction and a width of 6 to 22 miles. At the eastern entrance and within the sound are numerous shoals. Between these shoals are well-marked channels making the navigation of these waters comparatively easy for powered vessels and also sailing vessels with a fair wind. The shoals at the eastern entrance are subject to considerable shifting while those inside are somewhat stable. Boulders are along the shores.

(64) The channel through Nantucket Sound and Vineyard Sound has a controlling depth of about 30 feet and provides an inside passage for vessels of medium draft to avoid Nantucket Shoals. This route is used principally by coastwise vessels and pleasure craft. The navigational aids are colored and numbered for passing through the sound from the eastward.

(65) Monomoy and Nantucket Shoals are eastward and southeastward of the eastern entrance to Nantucket Sound. Owing to the great extent and distance offshore of some parts of these shoals and the strong and baffling tidal currents that set over them, their navigation in thick or foggy weather is hazardous. In clear weather the lights and buoys render navigation of the two principal channels, Pollock Rip and Great Round Shoal, comparatively easy. For the purpose of description Great Round Shoal Channel will be considered as the dividing line between Monomoy and Nantucket Shoals.

(66) Numerous **fishtraps** are located in Nantucket Sound, particularly along the southern shore of Cape Cod. These areas may be marked by private lights.

(67)

Monomoy Shoals to Hankerchief Shoal

(68) **Monomoy Shoals** consist of numerous detached shoals extending about 5.5 miles in an easterly direction and 9.5 miles in a southeasterly direction from **Monomoy Point**, the northeast entrance point of Nantucket Sound. Narrow sloughs separate the many parts of the shoals. It should be remembered that the shoals are shifting in character and are subject to change in location and depth.

(69) A dangerous wreck, reported covered 15 feet, is off Monomoy Island in about 41°35'07"N., 69°57'41"W. Mariners are advised to exercise extreme caution while navigating in the area.

(70) **Bearse Shoal** and **Pollock Rip**, extending about 5 miles eastward of Monomoy Point, are a series of sand shoals and ridges with little water over them in places. Pollock Rip Channel is between the shoals.

(71) **Broken Part of Pollock Rip**, covered 10 to 18 feet, is eastward of Pollock Rip.

(72) **Stone Horse Shoal, Little Round Shoal** and **Great Round Shoal** are portions of a continuous series of sand shoals and ridges covered 4 to 18 feet. These shoals are directly eastward of the entrance to Nantucket Sound and between the two main channels. Southward and eastward of these shoals are numerous shoal spots, including **Orion Shoal**, covered 16 to 19 feet.

(73) **Handkerchief Shoal**, extending for 5 miles southwestward from Monomoy Point, is covered 2 to 18 feet. A spot that uncovers 2 feet is about 2.7 miles southwest of the point. On the northwest side the water shoals gradually and soundings will indicate an approach to danger, but on the southeast side the shoal rises abruptly from the deeper water. Handkerchief Shoal is uneven and shifting in character. Vessels should not attempt to pass northward of the buoys marking the southern end and southeast side of the shoal.

(74) **Nantucket Shoals** is the general name of the numerous broken shoals which extend 23 miles eastward and 39 miles southeastward of Nantucket Island. These extremely dangerous shoals are described in chapter 3; caution must be exercised in this area.

(75) **Halfmoon Shoal**, near the center of Nantucket Sound, is covered 9 feet. Its southern end is marked by a lighted bell buoy. Depths of 17 and 22 feet are 2.5 and 1.5 miles, respectively, southeastward of the shoal. Deep-draft vessels should use care to avoid them. A lighted bell buoy is 1.3 miles east-northeast of the 22-foot spot.

(76) **Cross Rip Shoal**, about 2.5 miles west-southwestward of Halfmoon Shoal, has a least depth of 11 feet. Its northern edge is marked by a lighted gong buoy. A shoal, covered 28 feet, extends 1.2 miles eastward of the buoy. Caution must be exercised in passing between this shoal and the shoal making out southwestward from Halfmoon Shoal.

(77) **Horseshoe Shoal**, about 7.5 miles long, bares in places at extreme low water. Its western side is marked by two buoys and its northern and southeastern sides by lighted buoys. The main channel passes between the southeastern lighted buoy and the lighted gong buoy marking Cross Rip Shoal.

(78) **L'Hommedieu Shoal**, covered 3 feet, and **Hedge Fence**, covered 5 feet, lie in an east-west direction in the western end of Nantucket Sound and the eastern end of Vineyard Sound. The water deepens abruptly at the edge of these shoals, and soundings will give little warning of approaching dangers. The main channel passes southward of Hedge Fence Shoal. L'Hommedieu Shoal is marked by buoys at its north, east and west ends. Hedge Fence is marked by a lighted gong buoy on its southeastern side and a buoy on its western end.

(79) The numerous other shoals in Nantucket Sound are discussed with the land features near them.

(80)

Channels

(81) Two principal channels lead from the eastward into Nantucket Sound. The northerly one is through Pollock Rip Channel and Butler Hole and the southerly one through Great Round Shoal Channel. Between the numerous shoals in Nantucket Sound are two well-marked channels leading to the eastern end of Vineyard Sound. Muskeget Channel, discussed later in this chapter, leads into the sound from the southward, eastward of Chappaquiddick Island.

(82) **Pollock Rip Channel** and **Butler Hole** form the most direct channel leading from points northward of Cape Cod to Nantucket Sound. The channel leads between Bearse Shoal and Pollock Rip, thence eastward of Handkerchief Shoal. Since large-vessel traffic may be encountered in this channel, fishing vessels and small craft should avoid the area during thick or foggy weather. The channel is well marked by navigational aids. Mariners should consult the chart and seek local knowledge before entering Pollock Rip Channel and Butler Hole because numerous shoals exist in this channel. Caution is advised when transiting the area.

(83) Submerged piling, the remains of the former Monomoy Point Light structure, may exist about 0.3 mile southward of Monomoy Point. An abandoned lighthouse about 1.2 miles northward of the point is prominent.

(84) **Great Round Shoal Channel**, about 10 miles southward of Pollock Rip Channel, is used by many large fishing vessels transiting Nantucket Sound from New Bedford to Georges Bank and sometimes by sailboats that are headed by the wind so as to prevent their working through Pollock Rip Channel. The buoyed channel has a controlling depth of about 27 feet between Great Round Shoal and Nantucket Shoals. Great Round Shoal and Great Round Shoal Channel are subject to continual change.

(85) The **Main Channel** of Nantucket Sound leads southward of Halfmoon Shoal, through **Cross Rip Channel**, southward of Horseshoe Shoal, through the fairway between Hedge Fence and Squash Meadow, and thence into the eastern end of Vineyard Sound. The channel is used by most of the vessels bound through Nantucket Sound and is well marked by navigational aids. With care a least depth of 30 feet can be carried through the channel, but the draft of the vessels using it seldom exceeds 24 feet.

(86) **Cross Rip Lighted Gong Buoy 21** (41°26'51"N., 70°17'30"W.), marks the northern edge of Cross Rip Shoal.

(87) **North Channel** leads along the north side of Nantucket Sound, on either side of Bishop and Clerks, northward of Horseshoe Shoal, between Wreck Shoal and Eldridge Shoal, northward of L'Hommedieu Shoal, and through one of the openings in the shoals westward of L'Hommedieu Shoal into Vineyard Sound. This channel is used mostly by craft bound to points on the north shore of Nantucket Sound and by vessels bound through the sound during northerly winds or in winter when the prevailing northerly winds keep the north shore of the sound free from drift ice. The least depth in the channel is about 16 feet. Lighted and unlighted buoys mark the channel.

(88)

Anchorages

(89) Sailing vessels working through the sound against a head wind usually anchor during the night, or if becalmed and drifting toward the shoals it is best to anchor and wait for a favorable current or change of wind. The only anchorages for vessels of over 10-foot draft that afford shelter from all winds are Nantucket Harbor, Hyannis Harbor and Edgartown inner harbor. Vineyard Haven, the anchorage most used by coasters, is exposed to northeasterly winds. In northerly winds the best anchorages are off Dennis Port and Hyannis Port and along the north shore. The anchorage off Falmouth is used in most winds by vessels with good ground tackle. In easterly winds vessels sometimes anchor in smooth water westward of Handkerchief Shoal or inside Great Point. Good shelter from easterly winds can also be found in Chatham Roads and Edgartown outer harbor. In southerly and westerly winds Edgartown Harbor and Vineyard Haven are the best anchorages. With the aid of the chart and the directions given under the discussion of these harbors, strangers can enter the anchorages.

(90) Several **general anchorages** are in Nantucket Sound and its eastern approaches. (See **33 CFR 110.1** and **110.140(c)(3)** through **(c)(7)** and **(d),** chapter 2, for limits and regulations.)

(91)

Routes

(92) Because of the numerous shoals, strong tidal currents, thick fog at certain seasons and vessels that may be encountered in the narrow parts of the channel through Nantucket Sound, the navigator must use more than ordinary care when in these waters.

(93) In clear weather, day or night, the aids are readily distinguished and sufficiently numerous to enable a stranger to follow the channel without difficulty. The strongest currents will be encountered in Pollock Rip Channel, between Pollock Rip Channel Lighted Buoy 8 and Handkerchief Shoal Buoy 14, and off East and West Chop. In some places the current sets directly on the shoals, and in a calm, sailing vessels are sometimes obliged to anchor to prevent going aground. Most of the shoals rise abruptly from deep water and the bottom is very irregular, so soundings alone cannot be depended upon to keep clear of danger. Sailing vessels with a favorable current and with some local knowledge beat through the sound against a head wind in clear weather. If they find they are losing ground, they come to anchor within the prescribed anchorages under the lee of one of the shoals, or in one of the harbors until the wind or current changes.

(94) Vessels off Pollock Rip Channel entrance desiring to anchor, wind and sea permitting, should stand westward and anchor west of a line joining Pollock Rip Channel Buoy 2A and Chatham Beach Lighted Whistle Buoy C. Anchorage may also be had in depths of 5 to 10 fathoms about 1 mile northeastward of Broken Part of Pollock Rip.

(95) In Great Round Shoal Channel, the tidal currents are not as strong as in Pollock Rip Channel. Easterly winds make high tides and strong westerly currents. Westerly winds make low tides and strong easterly currents.

(96) Pollock Rip Channel and Great Round Shoal Channel are subject to change; vessels of deep draft should wait for a favorable tide.

(97) The Main Channel through Nantucket Sound is well marked, and strangers should experience little difficulty in navigating it. Vessels must take care to avoid the 24-foot shoal extending 1.2 miles eastward of the buoy marking Cross Rip Shoal and the 17- to 22-foot shoals 2.5 and 1.5 miles, respectively, southeastward of Halfmoon Shoal.

(98) The North Channel through Nantucket Sound has broken ground with depths of 16 to 17 feet in some places. Strangers should not attempt this channel at night.

(99)

Current

(100) The Tidal Current prediction service at *tidesandcurrents.noaa.gov* contains detailed information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(101) At the eastern entrance to Pollock Rip Channel the flood current sets about 053° and the ebb 212°.

(102) Daily predictions for Butlers Hole at the western end of Pollock Rip Channel can be generated from the Tidal Current prediction service at *tidesandcurrents.noaa.gov*. Links to a user guide for this service can be found in chapter 1 of this book.

(103) Off the southeast end of Great Round Shoal, the tidal current is rotary, turning clockwise. The average velocity at strength is 1.3 knots, and the average minimum velocity is 0.3 knot. Tide rips and water surface agitation caused by upwelling may be observed across the 10-fathom contour east of the entrance to Great Round Shoal Channel. (See the Tidal Current prediction service at *tidesandcurrents.noaa.gov*)

(104) From the eastern entrance of Nantucket Sound to the lighted gong buoy off Hedge Fence, the time of current becomes gradually later; the average velocity at strength varies from about 1 to 2 knots.

(105)

Weather, Nantucket Sound and vicinity

(106) Winter winds and waves along with spring and early summer fogs provide weather hazards in these waters. From October through March gales can be expected about 3 to 6 percent of the time and are frequently out of the west and northwest. Some wind and current interactions can also create problems. The most severe of these is found at the west entrance to the Cape Cod Canal. Rough seas can develop here when the tide ebbing out of the canal opposes a brisk southwest wind. Another well known "rough spot" is the West Chop off the north corner of Martha's Vineyard. At a maximum ebb or flood the current runs 3.5 knots here, and when it is opposed by the wind a nasty chop is set up.

(107) In general, over open waters, waves of 12 feet (3.7 m) or more can be expected 5 to 15 percent of the time from November through February. In the shallow portions of Nantucket Sound these frequencies drop, but waves may break before reaching these heights.

(108) The characteristic advection fog, formed by warm air over cool water, is most frequent from April through August. At this time visibilities drop below 2 miles 10 to 18 percent of the time; May, June and July are the worst and caution is advised, particularly near the numerous shoal areas in these waters. In addition to affecting visibility, fog also distorts sound so the direction of warning bells and horns may be difficult to discern accurately.

(109) Thunderstorms can occur in any season but are most likely in spring and summer. Sometimes they appear as squall lines with strong, gusty winds preceding the rain. Occasionally winds can gust to 60 knots.

(110)

Pilotage, Nantucket Sound

(111) Pilotage is compulsory for foreign vessels of 350 gross tons or more, U.S. vessels under register of 350 gross tons or more and tank barge towing vessels carrying 6,000 barrels or more of petroleum cargoes. Pilotage is available from Northeast Marine Pilots, Inc., Newport, RI, 02840; telephone 401–847–9050 (24 hours), 800–274–1216; FAX 401–847–9052; email: dispatch@nemarinepilots.com.

(112) Vessels requiring a tow usually take a tug from the port of their departure.

(113) **Monomoy Island**, a National Wildlife Refuge on the northeastern side of Nantucket Sound, is a low, narrow spit covered with sand dunes. Vessels sometimes anchor off the east shore of the island in depths of 4 to 6 fathoms to await a favorable current for entering the sound. Off Monomoy Point, the south end of the island, shoals make off up to 5 miles eastward.

(114) **Monomoy National Wildlife Refuge**, a Marine Protected Area (MPA), extends 1 mile offshore from Monomoy and Morris Islands.

(115) **Tidal currents** average about 2 knots at strength in the channel 0.2 mile west of Monomoy Point. The flood current sets 170°, and the ebb 346°.

(116) The large bight formed by Monomoy Island and the north shore of Nantucket Sound, eastward of Point Gammon, has extensive shoals scattered throughout and bordering the shores. Not all of the shoals are marked by buoys.

(119)

(117)

Chatham Roads to Davisville

(118) **Chatham Roads**, at the northeast end of Nantucket Sound, is between the extensive shoals that extend northwestward from the northern end of Monomoy Island and the shoals extending 1.6 miles from the shore of Cape Cod at Harwich Port. The Roads is the approach to **Stage Harbor** and the prominent summer resort of **Chatham** on the hilly ground at the northeast shore of Nantucket Sound.

(120) **Stage Harbor Light** (41°39'30"N., 69°59'04"W.), 28 feet above the water, is shown from a modular tower on the northeast side of Chatham Roads on the north side of the entrance to Stage Harbor.

(121) Among the conspicuous landmarks approaching Chatham Roads and Stage Harbor are the domes of the National Weather Service's installation on the eastern side of Morris Island, a radio tower at South Chatham, church spires, Chatham Light and Stage Harbor Light.

(122) A dredged channel, marked by lighted and unlighted buoys, leads from Chatham Roads into Stage Harbor. The entrance is in an area of shifting sandbars and is subject to shoaling. A narrow, natural channel continues northerly from Stage Harbor through Mitchell River to Mill Pond; the channel is marked by private buoys. A highway bridge crossing Mitchell River between Stage Harbor and Mill Pond has a bascule span with a horizontal clearance of 25 feet and a vertical clearance of 6 feet.

(123)

Anchorages

(124) Good anchorage for vessels up to 18-foot draft can be had in Chatham Roads in depths of 21 to 30 feet, good holding ground. This anchorage is insecure for small craft in heavy southwesterly gales. Small craft can find a well-sheltered anchorage in Stage Harbor.

(125)

Routes

(126) Vessels approaching Chatham Roads from the southward should pass westward of Handkerchief Shoal and the extensive shoals westward of Monomoy Island. Approaching from the westward, pass either side of Bishop and Clerks and thence southward of the seasonal lighted whistle buoy off **Kill Pond Bar**, a shoal covered 4 to 11 feet off the northwest entrance to Chatham Roads. When off the entrance to Chatham Roads, steer **063°** with Stage Harbor and Chatham Lights in range. This course will lead about 500 yards southeastward of Chatham Roads Bell Buoy 3 and north of the buoy marking **Common Flat**, the shoal on the eastern side of the roads, to the Stage Harbor approach buoy about 0.8 mile west-southwestward of Stage Harbor Light. An anchorage may be had northward of the approach buoy in depths of about 28 feet. Boats continuing to Stage Harbor will pick up the

(134)

Saquatucket Harbor, Massachusetts
Image courtesy of Marblehead Power Squadron (2009)

channel entrance buoys about 800 yards southwestward of Stage Harbor Light. The channel is well marked.

(127) A 5 mph **speed limit** is enforced in the harbor.

(128) The harbor is closed by **ice** for short periods each winter. Local fishermen will act as pilots for craft desiring one.

(129) The commercial fish piers in Stage Harbor are on **The Neck** at the head of the dredged channel opposite **Stage Island** and on the west bank of Oyster Pond River just above the first bend about 0.7 mile above the entrance to the river.

(130) There are a marina and boatyard on the north side of Stage Harbor adjacent to the fish piers and a marina on **Mitchell River** just west of the bridge. The marine railway at the boatyard can haul out craft up to 50 feet for hull and engine repairs or dry open or covered storage. Storage is also available at the bridge. Gasoline, diesel fuel, water, ice, marine supplies and moorings are available at Stage Harbor and at the bridge. The marina at the bridge has a 10-ton lift; berthage in 6 feet of water, gasoline, diesel fuel, water, ice, storage facilities, some marine supplies and a launching ramp are also available. A launching ramp is on **Sears Point**.

(131) **Oyster Pond River** extends from Stage Harbor for about 0.7 mile in a northwesterly direction, thence for 0.8 mile in a northeasterly direction into **Oyster Pond**. Private seasonal aids mark the channel at the bend in the river. Shoaling to 2 feet was reported at the bend in 1981. On the west bank at the bend there are a town wharf, a launching ramp and a fish wharf. At 0.3 mile and 0.5 mile above the bend on the west bank are two boatyards and marinas. The largest marine railway at the yards can haul out craft up to 44 feet for hull and engine repairs or dry open or covered storage. Gasoline, diesel fuel, water, ice, marine supplies, launching ramps, berthage and moorings are available.

(132) **Mill Creek**, 1.6 miles northwestward of Stage Harbor Light, is used only at high water by small local craft. The entrance between the jetties was reported to have 1½ feet in 1964. The tall radio tower of Chatham Radio Station WCC is prominent about 0.4 mile west of the jetties. **Cockle Cove** has been entered by small boats through one of the breakthroughs in the sandbar.

(133) **Saquatucket Harbor** is entered about 3.5 miles westward of Stage Harbor Light. A dredged channel leads from Chatham Roads to an anchorage basin at the head of the harbor. The entrance is protected by jetties. Buoys mark the channel, and a light marks the outer end of the east jetty. A marina is on the north side of the anchorage basin. Berths, electricity, gasoline, water, ice, a pump-out station and a launching ramp are available.The marina monitors VHF-FM channel 68. The **harbormaster** here also supervises Wychmere Harbor, Allen Harbor, Herring River and Round Cove in Pleasant Bay. A 5 mph **speed limit** is enforced in these areas. The harbormaster can be contacted at 508–430–7532.

(135) **Wychmere Harbor**, 3.7 miles westward of Stage Harbor Light, is a circular basin with a bulkheaded entrance protected by two jetties. The west jetty is hook shaped and marked on the end by a light. The east jetty is short. The harbor is used by fishing and pleasure craft. The village of **Harwich Port** is west of the harbor. A church spire about 0.5 mile westward of the harbor and a hotel on the west bank of the entrance are conspicuous.

(136) In 1994, the reported controlling depth across the bar was 8 feet. Inside the jetties, the channel has a depth of 6 feet to the harbor. The channel is subject to shoaling. The outer anchorage basin, known locally as Harwich Port Harbor, has a depth of about 8 feet with good holding ground.

(137) There is a large summer club-hotel and wharf on the west side of the jettied entrance. A town wharf is on the east side. A boatyard is at the inner end of the channel. Berthage, gasoline, diesel fuel, water, ice, a pump-out facility, marine supplies and storage facilities are available; hull and engine repairs can be made. A patrol boat enforces a **speed limit** of 5 mph; the patrol boat monitors VHF-FM channels 16, 22 and 68. The **harbormaster** who supervises Wychmere Harbor has his office at Saquatucket Harbor. He can be reached at 508–430–7532 for advice on moorings or local conditions.

(138) **Allen Harbor**, about 4.8 miles west of Stage Harbor Light, has a narrow entrance between two jetties into **Doanes Creek**. Allen Harbor is at the head of the creek. The west jetty is marked by a private seasonal light. An elevated water tank north of the harbor is a good landmark.

(139) In 1979, the controlling depth in the entrance channel was 5 feet. Private seasonal buoys mark the channel. The harbor affords good shelter for small craft.

(140) A town landing and launching ramp at the west end of the bridge were reported to have about 6 feet alongside. A marina and boatyard on the west side at the head of the harbor has a 7-ton forklift and a hydraulic trailer that can handle craft up to 34 feet for hull, engine and electronic repairs. Open and covered storage, electricity, gasoline, diesel fuel, water, ice, a pump-out station, marine supplies and a launching ramp are available. In 2008, the reported alongside depth was 5 feet. The **harbormaster** who supervises Allen Harbor has his office at Saquatucket Harbor. He can be reached at 508–430–7532 for local information.

(141) **Herring River**, 6 miles west of Stage Harbor Light, has a large prominent hotel on the west side and windmill on the east side of the entrance. The entrance, between two small jetties, is subject to shoaling. The approach is marked by private seasonal buoys, and the outer end of the west jetty is marked by a private seasonal light. A basin dredged in the river just below the bridge has moorings for craft drawing up to 3 feet. The fixed bridge, about 0.3 mile above the mouth, has a 14-foot fixed span with a clearance of 10 feet. State Route 28 highway bridge about 0.8 mile above the mouth has a 20-foot fixed span with a clearance of 7 feet. Limited supplies may be obtained at **Dennis Port** about 0.7 mile westward of the river.

(142) **Swan Pond River**, about 1.9 miles west of Herring River, is a narrow shallow creek bordered by marsh, which drains **Swan Pond**. Fishermen and pleasure craft enter at high water. Fish wharves are on the east bank just above the bridge about 0.3 mile above the mouth.

(143) **Bass River**, 9.6 miles westward of Stage Harbor Light, is entered between two jetties. A light is on the west jetty. A seasonal lighted buoy, 1.1 miles southward of the jetty light, marks the approach. A channel marked by private seasonal buoys leads over the bar from about 0.4 mile southward of the jetty light through the jetties to an anchorage basin in the lower part of the river, and thence to the highway bridge at South Dennis, 3.1 miles above the mouth. The bridge has a 25-foot fixed span with a clearance of 10 feet. The entrance to the anchorage basin and the channel over the bar are subject to shoaling. In 2001, the dredged entrance channel had a reported controlling depth of 4.6 feet to the river mouth, just inside the jetties; thence in 1981, the river channel had a reported depth of 5 feet to South Yarmouth, thence 2 feet to the bridge at **South Dennis**. In 1992, severe shoaling was reported across the mouth of the river. Private seasonal buoys mark the channel to Follins Pond, about 6.1 miles above the mouth.

(144) State Route 28 highway bridge crossing the river between **West Dennis** and **South Yarmouth**, about 1.5 miles above the mouth, has a 30-foot fixed span with a clearance of 15 feet.

(145) A 5 mph **speed limit** is enforced on the river.

(146) About 0.8 mile above the mouth, a channel leads eastward to a lagoon; a dredged depth of 10 feet was reported in the channel and lagoon. West Dennis Yacht Club is at the head of the lagoon.

(147) A marina and boatyard, about 0.4 mile below the first highway bridge, has a lift that can haul out craft up to 45 feet for hull and engine repairs or dry open and covered storage. Electricity, gasoline, diesel fuel, water, ice, marine supplies, a pump-out station and a launching ramp are available. In 2008, 6 feet was reported alongside the dock. The marina monitors VHF-FM channel 16. Town landings and launching ramps are on both sides of the river at and below the bridge.

(148) Rental boats, gasoline, bait and tackle can be obtained at a fishing pier just north of the east end of the bridge. A marina, above the pier, has berths, gasoline, diesel fuel, water, ice, a pump-out station, a 35-ton lift and storage facilities; hull and engine repairs can be made.

(149) Two fixed bridges, railroad and highway, cross the river about 0.7 mile above the highway bridge at South Dennis. Least clearances are: 25 feet horizontal and 8 feet vertical.

(150) In 1981, it was reported that about 4 feet could be carried at high water to a marina and boatyard on **Kellys Bay**, about 0.3 mile above the railroad and highway bridges. A mobile lift at the yard can haul out craft to 28 feet for hull and engine repairs or dry open or covered

storage. Gasoline, water, moorings and berths are available. In 1981, depths of 5 to 6 feet were reported at the boatyard.

(151) **Dogfish Bar**, an extensive shoal area off Bass River entrance, is covered 1 to 6 feet. A small breakwater, formerly used as a shelter for small craft, is on the easterly end of the bar, about 1 mile southeastward of Bass River West Jetty Light 11. The area around the breakwater and northeasterly of it has shoaled. Rocks awash at low water are about 0.4 mile northwestward of the breakwater. These rocks are marked by a private seasonal buoy. A fish haven, marked by a private buoy, is about 2 miles south of the breakwater, and a fishtrap marked by a private seasonal light is 1.8 miles southwest of the breakwater.

(152) **Parkers River**, about 1.2 miles west of Bass River, is entered between two jetties and extends 1.3 miles northward to **Seine Pond**. A motel on the east side of the entrance is prominent. Local knowledge should be obtained before entering the river. In 2000, the entrance channel had a reported controlling depth of 4.1 feet. The entrance is reported to shoal quickly after dredging. A fixed highway bridge crosses the river about 1 mile above the entrance.

(153) Local fishermen and pleasure craft enter and moor in dredged slips on the east side of the river. An unnamed creek, about 0.2 mile above the mouth, leads westward 0.5 mile to **Lewis Pond**. Small craft can enter the creek only at high water. Small craft may enter the pond but there are no moorings or services available. A marina is on the east side of the river just below the highway bridge; berths are available.

(154) **Point Gammon**, 12 miles west-northwestward of Monomoy Point, is the eastern entrance point to Hyannis Harbor. The point, prominent and wooded, is marked by an abandoned lighthouse tower. A reef, partly bare at low water, extends about 0.3 mile south of the point. Extensive flats with rocks awash at low water extend 1 mile northwestward of the point. **Gazelle Rock**, covered 5 feet and marked by a seasonal lighted buoy, is about 0.5 mile south-southeastward of the point. **Senator Shoal**, covered 11 feet and unmarked, is about 1 mile southeastward of the point. **Hallets Rock**, covered 17 feet, is about 1 mile south of the point and another rock, covered 13 feet, is about 1.2 miles south-southeast of the point in the vicinity of Hallets Rock. There are several submerged rocks in this area, which extends in a general line running northwest and southeast between Hallets Rock and Gazelle Rock to a private seasonal light 0.5 mile west of the point; the light marks a fishtrap. In 1990, a sunken wreck was reported about 0.2 mile southward of Hallets Rock in about 41°35.3'N., 70°15.7'W.

(155) **Bishop and Clerks**, about 2.2 miles southward of Point Gammon, is an extensive shoal area. The center of the shoal is marked by a light. Several rocks awash at low water are on the arm of the shoal that extends about 0.9 mile south of the light. A rock, covered 5 feet, is 0.7 mile south-southeastward of the light. The rest of the shoal is covered 8 to 18 feet. A lighted gong buoy, about 1.15 miles southward; an unlighted buoy, about 0.75 mile westward; and a lighted bell buoy, about 0.7 mile northeastward of the light, mark the limits of the shoal area. Caution should be exercised when in the vicinity of this shoal.

(156) **Broken Ground**, a shoal area westward of the south end of Bishop and Clerks, has depths of 14 to 18 feet. **West Southwest Ledge**, 1.6 miles southwest of Point Gammon and northwest of Bishop and Clerks, has depths of 15 to 18 feet. A lighted bell buoy is northward of the ledge and marks the approach to Hyannis Harbor. A group of dangerous rocks and obstructions are on the edge of the flat that extends northwestward from Point Gammon and into the approach to Hyannis Harbor. A rock, covered 11 feet in about 41°37'02.9"N., 70°17'21.1"W., is the westernmost and marked close West by Hyannis Harbor Lighted Buoy 4.

(157) **Hyannis Harbor**, protected by a breakwater, is used as a harbor of refuge by coasting vessels and pleasure craft of less than 14-foot draft. A light is on the end of the breakwater. The harbor is the approach to Hyannis Port, on the west side of the harbor, Lewis Bay, and Hyannis at the head of the northwest arm of Lewis Bay.

(158) The most prominent objects when approaching the harbor are the daybeacon on **Great Rock**, two red and white checkered standpipes, a light blue tank, the breakwater light, the abandoned lighthouse tower on Point Gammon and the square gray stone church belfry on the hill overlooking Hyannis Port to the westward.

(159)

Routes

(160) Vessels approaching Hyannis Harbor from the eastward should shape a course to pass about 1,000 yards south of Hallets Rock, exercising caution to avoid the reported wreck mentioned earlier, thence about **317°** to a point about 0.6 mile southwestward of Great Rock Daybeacon 4A, and thence about **012°** to pass about 100 yards or more eastward of the breakwater light. Vessels may anchor inside the breakwater on the east edge of the mooring area in depths of 15 to 20 feet, soft bottom. Small craft can anchor in the northern portion of the harbor in depths of 4 to 8 feet, but care must be taken to keep clear of a charted, submerged wreck, marked by a buoy.

(161) Approaching from the westward, from a position about midway between Horseshoe Shoal Buoy 7 and Wreck Shoal Bell Buoy 8, steer about **054°** to pass about 1400 yards east of Hodges Rock Buoy 2, and thence about **012°** to pass about 100 yards eastward of the breakwater light in entering the harbor.

(162) Ice seldom interferes with the movement of vessels in Hyannis Harbor during normal winters; the prevailing northerly winds keep the harbor clear. However, during severe winters or persistent southwesterly winds, the harbor may be temporarily closed to navigation. During particularly severe winters, the harbor has been closed by ice for up to 3 months.

(165)

(163) **Hyannis Port** is a summer resort with many prominent homes. A privately dredged channel, with reported depths of 6 feet in 2005, leads to the Hyannis Port Yacht Club landing on the west shore of the harbor.

(164) **Lewis Bay**, with depths of 2 to 12 feet, extends northeastward from Hyannis Harbor. In the northwest corner of the bay is the channel to the summer resort of **Hyannis**. The town has a hospital. Hyannis Yacht Club is on the west of the bay. A channel, marked by private seasonal buoys, leads westward to Hyannis Yacht Club. The club can accommodate craft to 140 feet; a reported dockside depth of 8 feet is available.

(166) A dredged channel leads from Hyannis Harbor into Lewis Bay, thence to an anchorage basin north of **Harbor Bluf**, thence to the town wharf at Hyannis, at the westernmost end. The channel is well marked but is subject to shoaling, especially in the vicinity of Lewis Bay Approach Channel Buoy 9. Vessels entering Lewis Bay must be guided by the buoys marking the dredged channel and by the color of the water, deepest where it is darkest. Heavy vessel traffic should be expected during summer months. A riprap jetty extends 1,000 feet southerly from **Dunbar Point**.

(167)

Anchorages

(168) Vessels with drafts up to 13 feet may anchor in the anchorage northeast of Hyannis Breakwater. Small craft can anchor in Lewis Bay west of the channel, off Hyannis Yacht Club and north of Dunbar Point. Limited anchorage is reported available in the basin north of Harbor Bluff.

(169)

Small-craft facilities

(170) Several small-craft facilities and launching ramps are along the northwestern arm of Lewis Bay northward of Harbor Bluff. Limited berths are also available at the town marina on the west side at the head of the arm. A **dockmaster** is usually in attendance at this marina and can be reached at 508–790–6327 or VHF-FM channel 16.

(171) For local information on moorings and berthings, the **harbormaster** can be contacted at 508–790–6273, through the Barnstable police department at 508–775–0387, or VHF-FM channels 16 and 9. A police boat from the town of Barnstable and a Yarmouth Harbor Patrol boat patrol Hyannis Harbor during the summer. A 6 mph **speed limit** is enforced in Lewis Bay north of Harbor Bluff and in Hyannis Harbor in the channel leading to the yacht club.

(172) Ferries to Nantucket and Martha's Vineyard berth in the harbor at Hyannis. Barnstable Municipal Airport is just north of the town.

(173) Westward of Hyannis Harbor breakwater the water is shoal with numerous rocks extending well offshore. **Eddie Woods Rock**, covered 4 feet and unmarked, is 0.6 mile southwestward of the breakwater light. A fishtrap marked by a private light is about 0.4 mile south of the rock.

(174) **Halls Island**, 1 mile westward of Hyannis Harbor Breakwater Light H, is marked by a tower. **Hyannis Point**, the southerly tip of the island, is on the eastern side of Centerville Harbor.

(175) Southward of Hyannis Point and Centerville Harbor are numerous shoals and rocks. **Southwest Ground**, the area about 1.5 miles south of Hyannis Point, has numerous rocks and shoal spots necessitating extreme caution for vessels navigating the area. **Southwest Rock**, about 1.1 miles south of Hyannis Point, is marked by a lighted buoy. Unmarked rocks, some awash at low water and others covered 2 to 6 feet, are between the buoy and Hyannis Point.

(176) **Hodges Rock**, covered 5 feet and marked by a buoy, is 1 mile southward of Southwest Rock lighted buoy. An unmarked rock covered 8 feet is 300 yards east of Hodges Rock. **Bearse Rock**, covered 5 feet and marked by a buoy, is 0.5 mile southwestward of Southwest Rock lighted buoy. **Channel Rock**, covered 5 feet and marked by a buoy, is 0.4 mile west of Bearse Rock. **Gallatin Rock**, covered 4 feet and marked by a buoy, is 0.4 mile southwestward of Bearse Rock. **Collier Ledge**, 1.5 miles west-southwestward of Southwest Rock lighted buoy, is awash at low water. It is marked by a buoy.

(177) **Gannet Ledge**, covered 5 feet and marked by a buoy, is 1.1 miles southwest of Hyannis Point. **Gannet Rocks**, 0.3 mile north of Gannet Ledge, include two unmarked rocks 7 and 3 feet high and a rock covered 4 feet. **Spindle Rock**, awash at low water and marked by a buoy, is near the head of Centerville Harbor. A rock awash at low water and a rock covered 2 feet are 200 yards north of the buoy. Two unmarked rocks covered 6 feet are 1.7 miles southwestward of Hyannis Point.

(178) **Centerville Harbor** is a bight 2 miles wide in the north shore of Nantucket Sound westward of Hyannis Point. A church spire and an elevated tank in **Centerville**, the village inland from the head of the harbor, are used as guides for entering the harbor. **Craigville Beach**, on the north side of the harbor, is a popular bathing beach. The approach to Centerville Harbor is obstructed by the previously mentioned rocks and shoals. The natural channel with depths of 9 to 10 feet leads to the anchorage. Anchorage with good holding ground may be had in depths of 14 to 20 feet; however, vessels seldom anchor here for shelter as the harbor is exposed to southerly winds. The shoals off the entrance somewhat break the force of the seas from southward, but not sufficiently to make it a safe anchorage. Strangers should not enter except in the daytime with clear weather. Ice may close the harbor in the winter.

(179) **East Bay**, on the west side of Centerville Harbor, has depths of 1 to 4 feet. Small pleasure boats enter the bay en route to Centerville River. The entrance to East Bay, protected by a jetty on the southwestern side, had a reported controlling depth of about 5 feet in 1981 but is subject to shoaling. A private light marks the end of the jetty. **Centerville River**, which enters the northeast side of the bay, has been privately dredged for a width of 30 feet to the head of navigation. In 2005, a reported depth of about 3 feet could be taken over the bar into Centerville River. Small boats moor in the river off Centerville or tie up to private piers. A 6 mph **speed limit** is enforced in East Bay and Centerville River.

(180) A conspicuous stone tower with a mushroom-shaped top is on the north side of the river. A town landing is on the north side just above the tower. A launching ramp is on the west shore of East Bay. A conspicuous wooden tower with a balcony on top is 0.3 mile southwestward of the jetty.

(181) **Cotuit Anchorage**, 6.5 miles west of Point Gammon, is an anchorage for small craft between the shoals that make off the shore. The anchorage is exposed to southerly winds and is seldom used except by local craft. The channel to the anchorage is marked by buoys, and vessels of less than 6-foot draft should experience no difficulty in keeping in the best water. **Lone Rock**, covered 4 feet and marked by a buoy, is near the southern side of the anchorage. A long shoal, covered 4 feet and marked by a buoy at its southeast end, is 0.5 mile northeastward of Lone Rock and about 0.7 mile south of the entrance to West Bay.

(182) **West Bay**, on the north side of Cotuit Anchorage and 19 miles west of Stage Harbor, has a jettied entrance about 150 feet wide and is the approach to the village of **Osterville**, on the east side of the bay. A private light marks the end of the east jetty. A seasonal lighted bell buoy about 1 mile southeastward of the entrance marks the approach. Private seasonal buoys mark the channel through West Bay. In 1981–2001, a depth of 2.9 feet was available in the entrance channel, thence 6 feet to the highway bridge. The channel is subject to shoaling, and strangers should obtain local information before entering the bay.

(183) **Ice** closes the bay for about 2 months each year. The wharves at Osterville have reported depths of 6 feet alongside.

(184)

Small-craft facilities

(185) Small-craft facilities are on either side of the channel north of the highway bridge. Berths, moorings, electricity, gasoline, diesel fuel, water, ice, marine supplies, a pump-out station and storage facilities are available. Marine railway to 40 feet, lifts to 70 tons, and hull and motor repairs are available on the east side of the river with an approach depth of 7 feet and alongside depth of 13 feet reported in 2009. Lifts to 75 tons and hull, engine and electronic repairs are available on the west side with an approach depth of 5 feet and alongside depth of 6 feet reported in 2009.

(186) **Little Island**, about 1 mile northward of the entrance to West Bay, separates West Bay from **North Bay** to the northward. In 1981, a reported depth of 5 feet was available in the narrow channel eastward of Little Island into North Bay. Strangers should obtain local information before navigating in North Bay, which has depths of about

6 to 17 feet. The **harbormaster** can be contacted for local information on moorings and berthings, telephone 508–790–6273. A 6 mph **speed limit** is enforced in Cotuit, North and West Bays by the Barnstable harbormaster and police.

(187) The highway bridge across the channel between Osterville and Little Island has a 31-foot bascule span with a clearance of 15 feet. (See **33 CFR 117.1** through **117.49** and **117.622**, chapter 2, for drawbridge regulations.) Advance arrangements for bridge openings can be made through the Department of Public Works.

(188) **Cotuit Bay**, northwestward of Cotuit Anchorage, is separated from West Bay by **Osterville Grand Island**. **Cotuit** is a village on the west side of the bay. A church spire and two elevated water tanks are prominent. A town wharf, with a depth of about 5 feet at its face, and a small-craft launching ramp are at the village. In 1981–2001, a privately dredged channel, with a reported controlling depth of 4 feet, leads from Cotuit Anchorage to off **Cotuit Highlands**; thence in 2001, a winding channel, with a depth of 6 feet, curves between **Sampsons Island** and **Bluf Point** into Cotuit Bay. The channels into Cotuit Bay and North Bay are marked by private seasonal buoys. A reported depth of about 6 feet was in the channel from Cotuit Bay to North Bay in 2005. **Seapuit River**, south of Osterville Grand Island, connects Cotuit Bay and West Bay. The privately dredged channel in the river had a reported controlling depth of 6 feet in 2004. Cotuit Bay is usually closed by **ice** each winter.

(189) **Popponesset Bay**, west of Cotuit Anchorage, is shoal with depths of 1 to 4 feet in the greater part of the bay.

(190) A small marina is on **Daniels Island** near the bridge to **Popponesset Island**. A launching ramp, a pump-out station, gasoline, diesel fuel, water, ice, some marine supplies and some services are available. A privately marked channel with a reported depth of about 3 feet leads to a marina on **Mashpee Neck**. Gasoline, water and engine repairs are available. A flatbed trailer at the marina can haul out craft to 30 feet.

(191) **Wreck Shoal**, about 3 miles south of Cotuit Anchorage, is about 1.4 miles long in an east-west direction and about 0.3 mile wide. Depths on the shoal range from 4 to 13 feet. A lighted bell buoy marks the east end of the shoal and another lighted bell buoy southwestward of the shoal marks the channel between Wreck Shoal and Eldridge Shoal. An unmarked shoal covered 8 to 15 feet is about 1.5 miles northeast of Wreck Shoal and southward of Cotuit Anchorage approach. Broken ground with a least known depth of 13 feet is between this shoal and Wreck Shoal.

(192) **Eldridge Shoal**, about 0.9 mile south of Wreck Shoal, is about 1 mile long in a northeasterly direction and about 0.2 mile wide. Depths on the shoal range from 5 to 14 feet. A buoy marks the northern side. A channel between Eldridge and Wreck Shoals has depths of 23 to 41 feet. A channel between Eldridge and Horseshoe Shoals has depths of 21 feet or more.

(193) **Succonnesset Shoal** extends about 2.4 miles westward from Wreck Shoal to the shoal area off the shore southwestward of **Succonnesset Point**. Depths of 1 to 5 feet are on the shoal. A lighted buoy is off the west end. Between Succonnesset and Wreck Shoals is a narrow unmarked channel. Between Succonnesset and L'Hommedieu Shoals, a shoal area with a least depth of 9 feet is marked by a buoy.

(194) **Waquoit Bay**, 5 miles southwestward of Cotuit Anchorage, has depths of 1 to 8 feet. The entrance, about 250 feet wide, is between two stone jetties. A private seasonal light marks the end of each jetty. In 1971, the controlling depth in the entrance channel was reported to be 4 feet. A seasonal lighted bell buoy, about 0.6 mile west-southwestward of the jetties, marks the approach, and buoys mark a 5-foot channel for about 0.8 mile through the bay. The Waquoit Yacht Club is on the west side at the head of the bay.

(195) The **Waquoit Bay National Estuarine Research Reserve**, a Marine Protected Area (MPA), includes Waquoit Bay and associated waters and protected wetlands.

(196)

Small-craft facility

(197) **Great River** and **Little River** empty into the southeasterly side of Waquoit Bay. A marina is on the west side of Little River, about 0.5 mile above its junction with Great River. Gasoline, water, ice, a launching ramp, limited marine supplies and storage facilities are available. Hull, engine, and electrical repairs can be made; lift to 11 tons. A reported depth of about 3 feet can be carried to the marina.

(198) Between Waquoit Bay and Falmouth Inner Harbor about 3.6 miles to the westward are several ponds formed by the barrier beach, some of which have outlets. Many jetties or groins are built out from the shore for beach erosion control.

(199) **Eel Pond**, about 0.8 mile westward of the entrance to Waquoit Bay, is entered through a narrow jettied entrance. A private seasonal light on the west jetty and a midchannel buoy about 500 yards southeastward of the light mark the approach. The privately marked channel into the pond had a reported controlling depth of 5.1 feet in 2000. The channel is subject to shoaling; extreme caution and local knowledge is advised.

(200) A boatyard is on the west side of the northeasterly arm of Eel Pond at the mouth of **Childs River**. In 2009, a reported approach depth of 4 feet was available. Berths, electricity, gasoline, diesel fuel, water, ice, marine supplies, pump-out station, launching ramp, 40-foot marine railway, 50-ton marine lift and storage facilities are available at the boatyard; hull and engine repairs can be made.

(201) **Seapit River**, about 0.8 mile southward of the mouth of Childs River, connects the northeasterly arm of Eel Pond with the upper part of Waquoit Bay. A reported

(228)

(225) 40°35'56.5"N., 71°38'25.1"W.; thence back to starting point. (See **50 CFR 224.105** in chapter 2 for regulations, limitations and exceptions.)

(226)

Nantucket Harbor

(227) **Nantucket Harbor** is near the middle of the north shore of Nantucket Island. A shallow lagoon about 5 miles long extends northeastward from the harbor. The harbor is the approach to the town of **Nantucket** on the western shore. The principal industry is fishing. Small coastal tankers carry fuel to Nantucket. Year-round passenger, vehicle and cargo ferry service is maintained between Nantucket and the mainland, to either Woods Hole or Hyannis. A passenger ferry also operates from Falmouth and Oak Bluffs during the summer.

(229) Prominent from offshore are a radio tower about 1.2 miles east of Madaket Harbor; a standpipe about 1.5 miles west of Nantucket; a gilded cupola atop a church clock tower, and a church belfry about 500 yards northwestward of it; the spire of a large white church in the town; and the navigational lights at the entrance to Nantucket Harbor.

(230) **Brant Point Light** (41°17'24"N., 70°05'25"W.), 26 feet above the water, is shown from a white cylindrical tower connected to the shore by a footbridge on the west side of the entrance near to the harbor. A sound signal at the light is operated by keying the microphone five times consecutively on VHF-FM channel 83A. **Brant Point Coast Guard Station** is on the point.

(231)

Channels

(232) A dredged channel leads from Nantucket Sound between two submerged breakwaters to deep water in Nantucket Harbor off Brant Point. (See Notice to Mariners and latest editions of charts for controlling depths.) Shallow water extends about 1 mile offshore on both sides of the channel. A lighted bell buoy marks the approach and the channel is marked by lighted and unlighted buoys and a **161.9°** lighted range. A light and sound signal mark the outer end of the east jetty, and a buoy marks the outer end of the west jetty. A 6 mph speed limit is enforced in the harbor.

(233)

Anchorages

(234) Anchorage in Nantucket Harbor may be had in depths of 6 to 17 feet off the south and southwest sides of Brant Point or in depths of 12 to 17 feet in the general anchorage south of Brant Point. (See **33 CFR 110.1** and **110.142**, chapter 2, for limits and regulations.) In general, the bottom is sticky. Although shelter is afforded to vessels it is advisable for small craft to use heavy tackle as the harbor becomes choppy with easterly winds. Caution should be exercised to avoid anchoring in the fairway

and maneuvering area to the ferry wharf or the cable area northeast of Brant Point.

(235) The long sweep of strong northeast winds down the harbor makes anchorage for small craft off the wharves dangerous and uncomfortable. Small craft may find more sheltered anchorage under these conditions in Head of the Harbor or, with local knowledge, in Polpis Harbor.

(236) Launch service is available to craft at moorings or at anchor in the harbor.

(237)

Routes

(238) Vessels approaching Nantucket Harbor from Pollock Rip Channel can set a direct course from Handkerchief Shoal Buoy 14 (41°29'09"N., 70°05'05"W.) to the lighted bell buoy off the entrance. Approaching from the channel northward of Cross Rip Lighted Gong Buoy 21 (41°26'51"N., 70°17'30"W.), pass to the eastward of Tuckernuck Shoal Lighted Bell Buoy 1, and then head for the lighted bell buoy off the entrance to Nantucket Harbor.

(239)

Current

(240) The tidal current off the Nantucket Harbor entrance has a velocity of 0.3 knot; the flood setting eastward and the ebb westward. The tidal current in the entrance channel sets into the harbor at a velocity of 1.2 knots and outward on the ebb at a velocity of 1.5 knots.

(241)

Weather, Nantucket and vicinity

(242) The climate of Nantucket is influenced directly by the proximity of the ocean and is characterized by cool summers and comparatively mild winters. Extremes of either maximum or minimum temperatures are very rare. The mild temperatures of the winter season are neutralized to a degree by sustained periods of high wind. The summers, though cool, are very humid. Heavy fogs are frequent, particularly during the spring and summer. There is a marked lag in the seasons as compared with inland areas.

(243) July and August are relatively cool with average maximum temperatures around 75 °F and average minimums about 61 °F. The average temperature for the island is about 50 °F. January and February are the coldest months, having normal mean temperatures near freezing that is, average maximum of 38 °F and average minimum of 25 °F. The extreme maximum temperature for Nantucket is 100 °F recorded in August 1975 while the extreme minimum is -3 °F recorded in December 1962. An average year sees 100 days with temperatures below 32 °F and only one day with a temperature below 5 °F. Seldom do temperatures exceed 90 °F, a fact that has occurred only during June, July and August.

(244) The average wind velocity is about 11 knots with the highest monthly averages during December through April. Gales have occurred during every month except June and July. Coast storms are frequent during the winter with winds of 40 knots or more. Hurricanes, during the late summer and fall, may cause high winds. Since 1871, forty tropical storms or hurricanes have passed within 50 nautical miles of Nantucket. There have been ten direct hits. The latest was tropical storm Esther in 1961. Esther had been a 125-knot hurricane earlier but was dissipating at the time it crossed the island.

(245) Precipitation is fairly evenly distributed throughout the year averaging about 41 inches (1,041 mm) in any given year. An average of 180 days each year records precipitation with 28 days having greater than 0.50 inches (13 mm). The wettest month is December averaging 4.35 inches (111 mm) and the driest month is June averaging only 2.2 inches (56 mm). Total snowfall for the winter season averages about 30 inches (762 mm); however, melting is usually rapid and snow cover rarely lasts more than a few days. The greatest snowfall in a 24-hour period was 14.9 inches (397) in February 1952. February is the snowiest month averaging nearly nine inches (229 mm). Snow is absent from May through September. An average five days each year record greater than 1.5 inches (38 mm) of snowfall. An average of 18 thunderstorms affect the island in a given year with the greatest frequency occurring during July and August. Fog is present about 200 days each year.

(246) Except in severe winter, the harbor is seldom closed by local formation of ice. However, the harbor is frequently closed by drift ice from the sound that packs and remains across the entrance during northerly winds.

(247) Nantucket Boat Basin, on the west side of Nantucket Harbor, is entered about 0.4 mile south-southwestward of Brant Point Light. The basin is enclosed on the north and south sides by Straight Wharf and Commercial Wharf, respectively, and its entrance is protected by two long bulkheads on the east and southeast sides. Depths in the basin range from 3 to 10 feet. About 180 slips are available in the basin, and yachts 100 feet long and larger can be accommodated. The outer end of the north side of Straight Wharf is used by excursion boats. A private seasonal light is shown off the end of the wharf and is operated only when tour boats are approaching the wharf in fog. Gasoline, diesel fuel and ice can be obtained on the south side of Commercial Wharf. Water and electricity are available at each slip. The basin's dockmaster has his office on the outer end of Commercial Wharf. The dockmaster can be contacted on VHF-FM channel 16.

(248) The Woods Hole-Martha's Vineyard and Nantucket Steamship Authority Wharf is about 0.1 mile northward of the boat basin. A private light is shown from the roof of a shed on the northeast end of the wharf and is operated only when Authority vessels are approaching the wharf in fog. The submerged ruins of a pier that uncover at low water are between the boat basin and the Steamship Authority Wharf. A buoy marks the ruins. Unpainted pile dolphins mark the former pierhead. Mariners are advised to exercise caution in this area.

(249) A boatyard, about 0.2 mile southward of Nantucket Boat Basin, has moorings, gasoline, a 23-ton mobile hoist, storage facilities, water and marine supplies; hull

(266)

and engine repairs can be made. The channel leading to the boatyard, marked by private seasonal buoys, had a reported controlling depth of 6 feet in 1981.

(250) Nantucket maintains ferry service with the mainland and daily airline service with New York and Boston. Seasonal bus and taxi service is also available.

(251)

Head of the Harbor

(252) A narrow unmarked channel leads through the lagoon northeast of Nantucket Harbor to **Head of the Harbor**. In 1981, a reported depth of about 3 feet could be carried with local knowledge as far as the village of **Wauwinet** on the southeast shore of Head of the Harbor. Unmarked shoals and foul areas extend off the several points.

(253) **Polpis Harbor** is at the east end of the harbor just south of Head of the Harbor. The entrance channel, marked by private seasonal buoys, had a reported controlling depth of 6 feet in 1994.

(254)

Muskeget Channel

(255) **Muskeget Channel** is an opening 6 miles wide on the south side of Nantucket Sound between Muskeget and Chappaquiddick Islands. The channel is subject to numerous shifting shoals. Although this channel is partly buoyed, strangers should never attempt it as tidal currents with velocities of 2 to 5 knots make navigation dangerous. The currents through the channel are strong, having a velocity of 3.8 knots on the flood and 3.3 knots on the ebb about 1.5 miles east of Wasque Point. The flood sets north-northeastward and ebbs south-southwestward.

(256) **Wasque Shoal** extends southward of **Wasque Point**, the southeastern extremity of Chappaquiddick Island. The shoal, which dries about 2 miles south of Wasque Point, rises abruptly from the deep water of Muskeget Channel.

(257) **Mutton Shoal**, 0.6 mile east of Wasque Shoal, has a least depth of 5 feet and is marked on its southwestern side by a lighted bell buoy. The best water in Muskeget Channel is between Mutton and Wasque Shoals. Eastward of Mutton Shoal are numerous shoals covered 2 to 6 feet.

(258) Between Muskeget Channel and the main channel north of Cross Rip Shoal are numerous shoals, some of which are separated by unmarked channels. **Tuckernuck Shoal**, northeast of Muskeget Channel, has a least depth of 2 feet; it is marked on the northeastern end by a lighted bell buoy and a buoy on the northern side. **Shovelful Shoal**, westward of Tuckernuck Shoal, is covered 3 to 17 feet. **Long Shoal**, northwestward of Shovelful Shoal, is covered 3 to 16 feet. **Edwards Shoal**, south of Cross Rip Shoal, has a least known depth of 10 feet. **Norton Shoal**, southwestward of Cross Rip Shoal and covered 8 feet, is marked by a buoy on its north side. **Hawes Shoal**,

westward of Norton Shoal, has a least depth of 1 foot; buoys mark its northwestern and southwestern ends.

(259)

Martha's Vineyard and Chappaquiddick Island

(260) **Martha's Vineyard** and **Chappaquiddick Island** have a combined length of 18 miles; the two islands are separated by Edgartown Harbor, Katama Bay and the narrow slough connecting them. The northern extremity of Martha's Vineyard is about 3 miles southeastward of the western end of Cape Cod. Martha's Vineyard is well settled, especially along its northern shore, and is popular as a summer resort. Along the northern shore the island presents a generally rugged appearance. The southern shore is low and fringed with ponds, none of which has navigable outlets to the sea. Approaching from the south, the principal landmarks are a standpipe at Edgartown, an aerolight near the center of the island, a church spire near **Chilmark** in the western part, a tall radar tower north of Chilmark and Gay Head on the west side.

(261) Communication with the mainland is by ferry, airline, cable and telephone. The principal towns are Edgartown, Oak Bluffs and Vineyard Haven.

(262) **Cape Poge**, the northeastern point of Chappaquiddick Island, is a bare, bluff, precipitous head, which may appear from a distance to be a small island. **Cape Poge Light** (41°25'10"N., 70°27'08"W.), 65 feet above the water, is shown from a white conical tower on the cape.

(263) **Cape Poge Flats**, extending about 1.5 miles northeastward from Cape Poge, are marked at the northeast end by a bell buoy. The southerly edge of the white sector of West Chop Light is about 0.9 mile north of the buoy. Shoal water extends about 0.4 mile offshore westward and northwestward of Cape Poge. A buoy, 1 mile west-northwestward of Cape Poge Light, marks the western side of the shoal water.

(264) **Cape Poge Bay**, a lagoon of considerable size in the northern part of Chappaquiddick Island, is entered from Edgartown Harbor. The unmarked entrance is used mostly by local pleasure and fishing craft. In 1981, it was reported that 4 feet could be carried through the entrance channel with local knowledge.

(265) **Edgartown Harbor**, on the eastern side of Martha's Vineyard and westward of Cape Poge, is divided into an outer and an inner harbor. The outer harbor is used principally as a harbor of refuge in southerly and easterly winds and as a night anchorage. At the head of the outer harbor, a narrow arm makes southward into Katama Bay, forming the inner harbor. The inner harbor affords good anchorage and is the approach to **Edgartown**, a fishing and resort town on the western shore. Many yachts and pleasure craft use the harbor during the summer.

(267) **Katama Bay**, used by local fishermen and small pleasure craft, is large and shallow. Extensive shoaling has been reported in the southerly end of the bay. A 4 mph **speed limit** is enforced in the bay.

(268)

Prominent features

(269) **Edgartown Harbor Light** (41°23'27"N., 70°30'11"W.), 45 feet above the water, is shown from a white conical tower on the west side of the head of Edgartown outer harbor. Also prominent are a church belfry in the town, a microwave tower 1 mile to the west-southwestward of the light, a standpipe about 1 mile southwestward of the light and the numerous beach cabanas on Chappaquiddick Point.

(270)

Channels

(271) The buoyed channel through the outer harbor has depths of 20 to 36 feet until nearly to Edgartown Harbor Light. Near the light, the channel narrows and makes a sharp bend westward, leading to the wharves at the town. Abreast the town, the channel narrows and curves southward to Katama Bay, bordered on the eastern side by Middle Ground. Katama Bay is subject to frequent changes, as is the shoreline between the bay and the ocean.

(272)

Anchorages

(273) Anchorage with good shelter from easterly gales is found westward of Cape Poge on the eastern side of the outer harbor. In westerly and southerly gales vessels find shelter in the southern end of the outer harbor about 0.4 mile eastward or east-southeastward from Edgartown Harbor Light. In northerly or northeasterly gales vessels usually go to Woods Hole or Tarpaulin Cove for sheltered anchorage. Vessels should not anchor in the channel abreast the town where the bottom is hard sand, the channel narrow and tidal currents strong. Southeast of the town, anchorage may be found south of Middle Ground, muddy bottom.

(274) Small craft usually anchor in the **special anchorage** in the vicinity of Middle Ground. (See **33 CFR 110.1** and **110.38**, chapter 2, for limits and regulations.)

(275)

Dangers

(276) On the western side of the outer harbor is a shoal area extending 2.8 miles northward of Edgartown Harbor Light. A bell buoy marks the northern edge of the shoal; vessels entering or leaving the harbor pass eastward of this buoy. The depths over the remainder of the shoal are irregular, and there are a rock awash and several rocks covered 3 to 5 feet. Strangers should never attempt to pass across this shoal. The channel into Edgartown Harbor is marked by a lighted buoy and unlighted buoys. An obstruction, covered 19 feet, is at 41°23'32"N., 70°29'28"W.

(277) **Sturgeon Flats**, covered 2 to 18 feet, extend about 600 yards off the southeastern shore of the outer harbor between the narrow entrance to Cape Poge Bay and the entrance to the inner harbor.

(278) A sandbar is making off eastward from Edgartown Harbor Light. A buoy is on the eastern end of the shoal.

(296)

Except for this shoal, the entrance to the inner harbor is not difficult to navigate. **Middle Ground**, in the inner harbor south of the town, has a least depth of 10 feet.

(279)

Routes to Martha's Vineyard

(280) Vessels approaching Edgartown Harbor from the eastward, from a position about 400 yards north of Cross Rip Lighted Gong Buoy 21, can steer **267°**, heading for the standpipe on Martha's Vineyard southward of Oak Bluffs, passing northward of Nantucket Sound Channel Lighted Bell Buoy 21A. When Cape Poge Light bears **155°**, head south-southwestward into the harbor.

(281) Vessels approaching from the westward and passing northward of Squash Meadow can head on a **180°** course from a position about 0.5 mile southward of Hedge Fence Lighted Gong Buoy 22 to enter the harbor. In the daytime, the channel southward of Squash Meadow is sometimes used. Strangers in sailing vessels seldom enter the inner harbor, as a fair wind is necessary to keep in the channel.

(282)

Current

(283) The tidal current in the narrow part of the channel inside Edgartown Harbor Light and off the town has a double flood and a double ebb and in general follows the direction of the channel. Near the middle of each flood or ebb period there is an approximate slack preceded and followed by maximum of velocity. The average velocity is about 1 knot. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book. In 2007, it was reported that a breach had occurred on **South Beach** just southward of Katama Bay. The breach has affected the published tide and tidal current predictions so that mariners are cautioned about the accuracy of the information.

(284) **Fogs** are prevalent during the summer and at times appear without warning. Drift ice from the sound, driven into the entrance by the wind, obstructs the entrance to sailing vessels during a part of the winter. It is reported that the harbor is normally closed by ice during January and February. The Chappaquiddick ferry channel is usually kept open. The tidal currents keep the inner harbor open except for a few days at a time during severe winters.

(285) There are no pilots for Edgartown Harbor. Tugs are seldom used and none are available. Fishing craft or the harbormaster's vessel will act as tugs in an emergency.

(286)

Harbormaster

(287) The harbormaster has control of the anchorage of vessels in the harbor. He will usually be found at the Edgartown Yacht Club and can be contacted on VHF-FM channel 16 or at 508–627–4746. Copies of harbor regulations may be obtained from the harbormaster.

(288) The depth at the Town Wharf is 25 feet. Depths at the other wharves are about 11 feet. A marina, boatyard and a yacht club are at Edgartown. The boatyard has a marine lift that can handle craft to 9 tons for hull and engine repairs and dry open or covered storage. Gasoline, diesel fuel, water, ice, marine supplies and moorings are available. Launch service to moored craft is available.

(289) A small ferry operates between Edgartown and Chappaquiddick Island. No schedule is maintained, but the ferry runs on call. There is seasonal bus service to Oak Bluffs, Vineyard Haven and other island points. Ferries connect Oak Bluffs and Vineyard Haven with Woods Hole, Falmouth, Hyannis and Nantucket.

(290) **Sengekontacket Pond**, about midway between Edgartown and Oak Bluffs, has two entrances that are subject to shoaling. The southerly entrance is the main entrance and had a reported controlling depth of 10 feet in 1981. There are no public landings in the pond and it is used by local and fishing craft only. The south entrance is crossed by a fixed highway bridge with a vertical clearance of 6 feet. The fixed highway bridge over the north entrance has a vertical clearance of 5 feet.

(291) **Squash Meadow** is a shoal south of the main channel through Nantucket Sound and about 4 miles northwest of Cape Poge. The hard sand shoal has depths of 13 to 18 feet and is marked on its southeastern end by a bell buoy and on its western end by a buoy.

(292) **Harthaven** is a small pond northward of Sengekontacket Pond. The entrance is through a privately dredged channel between two short jetties. In 2000, a depth of 4 feet was reported in the entrance. The pond has depths of 2 to 6 feet. There are no services or landings in the pond.

(293) **Lone Rock**, covered 4 feet, is 350 yards offshore about 750 yards southeastward of Oak Bluffs wharf.

(294) **Rhode Island Rock**, covered 14 feet, is about 700 yards northward of the breakwater light.

(295) **Oak Bluffs Harbor**, 4.8 miles northwestward of Edgartown Harbor Light, is a landlocked basin frequented by pleasure craft and some fishing vessels. The entrance is protected by two breakwaters. A light is on the end of the north breakwater. **Oak Bluffs** is a summer resort and fishing village on the harbor. Prominent are a church dome and a cupola in the village and the bluff north of the entrance.

(297) Numerous submerged rocks, covered 10 to 14 feet, are in the harbor approach, in an area within 0.4 mile of shore bounded on the north by a line extending northeasterly from the breakwaters and on the south by Lone Rock. The chart is the best guide for approaching the harbor; however, it is advised that mariners transiting the area exercise extreme caution as other uncharted rocks may exist.

(298) East Chop Yacht Club is on the north side of the harbor, and several private piers are on the west side. The town wharf extends along the bulkhead on the south and east sides of the harbor. The town maintains berths with electricity, a launching ramp, a pump-out station, and guest moorings. Gasoline, diesel fuel, water, ice and some marine supplies are available; hull and engine repairs can be made.

(299) Martha's Vineyard hospital is on the beach road close westward of the town. The **harbormaster** monitors VHF-FM channel 71; telephone 508–693–4355.

(300) A no-wake **speed limit** is enforced in the harbor.

(301) Oak Bluffs Wharf, about 0.2 mile southward of the breakwater light, is reported to have a depth of 13 feet at the head. Several obstructions with lesser depths have been reported about 400 yards northeast of the wharf face. A private seasonal light and sound signal are operated from the seaward end of the wharf when ferry vessels are approaching the wharf in fog. There is seasonal ferry service from the wharf to Woods Hole and Nantucket. Seasonal ferry service is also maintained between Falmouth, Hyannis and New Bedford. Seasonal fast ferry service is maintained from Quonset, RI, and year-round fast ferry passenger service is maintained from New Bedford. Oak Bluffs is a port of call for cruise ships. The Vineyard Transit Authority provides island-wide transit service year-round. There is air service from Martha's Vineyard Airport about 4.5 miles southwestward of the town.

(302) A 530-foot groin, marked at its outer end by a daybeacon and partially submerged at high water, is about 650 yards south of the ferry wharf; caution is advised.

(303) Other ports on the north side of Martha's Vineyard, westward of Oak Bluffs Harbor, are described in chapter 5.

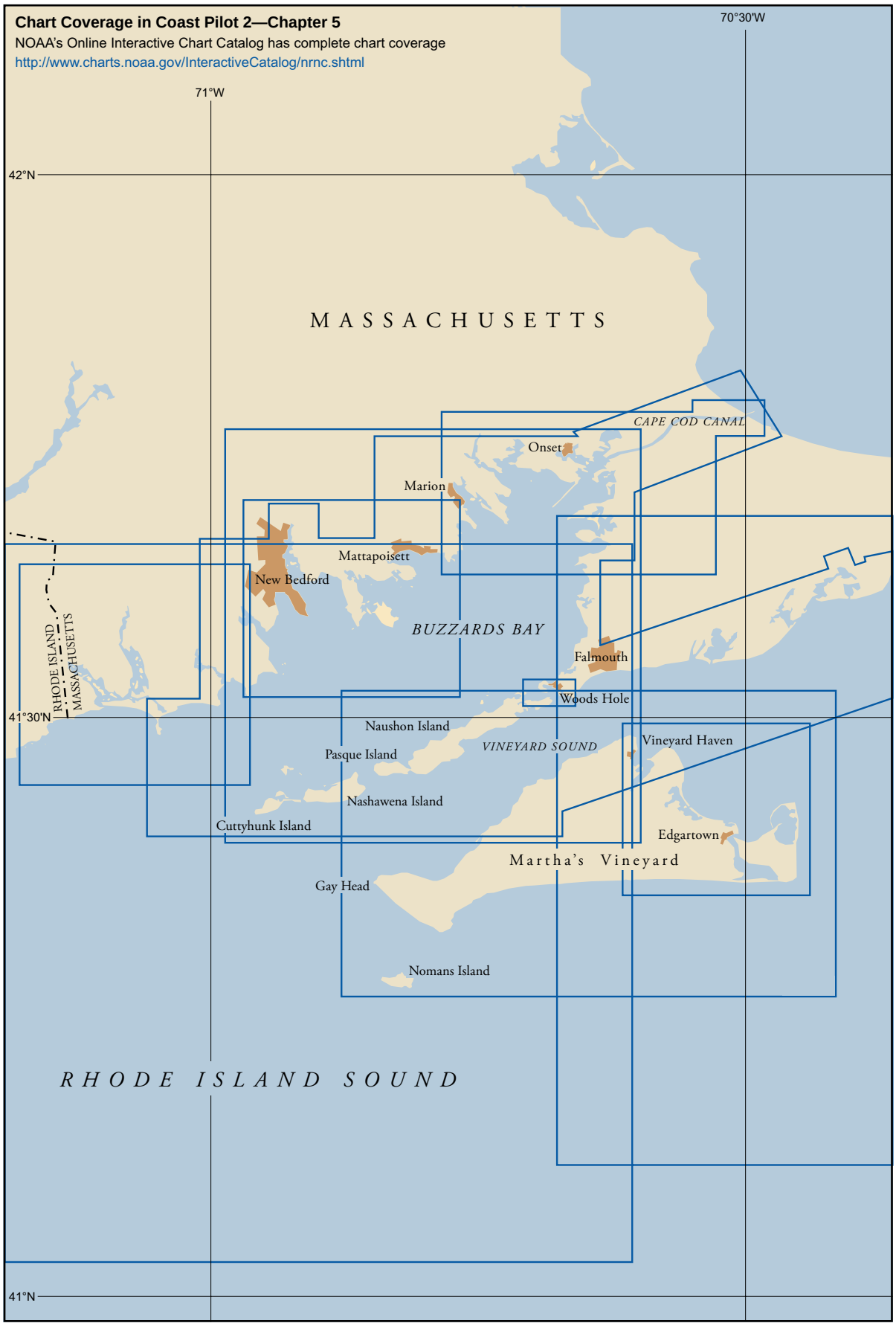


Chart Coverage in Coast Pilot 2—Chapter 5
NOAA's Online Interactive Chart Catalog has complete chart coverage
http://www.charts.noaa.gov/InteractiveCatalog/nrnc.shtml
70°30'W
71°W
42°N
MASSACHUSETTS
CAPE COD CANAL
Onset
Marion
Mattapoisett
New Bedford
BUZZARDS BAY
RHODE ISLAND
MASSACHUSETTS
Falmouth
Woods Hole
41°30'N
Naushon Island
VINEYARD SOUND
Vineyard Haven
Pasque Island
Nashawena Island
Cuttyhunk Island
Edgartown
Martha's Vineyard
Gay Head
Nomans Island
RHODE ISLAND SOUND
41°N

Vineyard Sound and Buzzards Bay

(1) This chapter describes Vineyard Sound and Buzzards Bay following the Massachusetts coast of Vineyard Sound, the northwestern shore of Martha's Vineyard, the eastern shore of Buzzards Bay, the Cape Cod Canal and the western shore of Buzzards Bay. Also described are Woods Hole, Cuttyhunk, Onset, Wareham and the port of New Bedford, as well as the numerous fishing and yachting centers along the sound and bay.

(2)

COLREGS Demarcation Lines

(3) The lines established for this part of the coast are described in **33 CFR 80.145**, chapter 2.

(4)

No-Discharge Zone

(5) The State of Massachusetts, with the approval of the Environmental Protection Agency, has established a No-Discharge Zone (NDZ) in all coastal waters of Massachusetts covered by this chapter except a small area from Woods Hole to Vineyard Haven, extending about 3 miles offshore.

(6) Within the NDZ, discharge of sewage, whether treated or untreated, from all vessels is prohibited. Outside the NDZ, discharge of sewage is regulated by 40 CFR 140 (see chapter 2).

(7)

Vineyard Sound and Buzzards Bay

(8) **Vineyard Sound and Buzzards Bay** are deep and easily navigated day or night. Vineyard Sound, together with Nantucket Sound, provides an inside route from New York to Boston that avoids Nantucket Shoals. Buzzards Bay, together with Cape Cod Canal and Cape Cod Bay, provides the shortest deep-draft route between New York and Boston.

(9) **Vineyard Sound** is bounded on the north by the southwestern part of Cape Cod and the Elizabeth Islands and on the south by part of Martha's Vineyard, which presents a rugged and generally inaccessible shoreline. To the west, it joins Rhode Island Sound on a line between Cuttyhunk Island and Gay Head. To the east, it joins Nantucket Sound on a line between Nobska Point and West Chop and provides an inside passage clear of Nantucket Shoals. The navigational aids are colored and numbered for passing through the sound from the eastward. The channel through the sound is well marked and generally free of dangers. Deep-draft vessels entering or leaving Vineyard Sound should stay at least 3.5 miles southward of the southwest end of Cuttyhunk Island.

(10)

Anchorages

(11) Woods Hole is the only anchorage providing shelter from all winds for vessels drawing more than 10 feet. In northerly and westerly winds, good anchorage may be had in Tarpaulin Cove. In southerly winds, shelter can be had in Menemsha Bight, although Vineyard Haven is generally used. Several general anchorages are in Vineyard Sound. (See **33 CFR 110.1** and **110.140(c)(1)**, **(c)(2)**, and **(d)**, chapter 2, for limits and regulations.)

(12)

Current

(13) The time of current becomes somewhat earlier from Hedge Fence westward through Vineyard Sound. The current velocity increases from 1.2 knots at Hedge Fence Lighted Gong Buoy 22 to about 2.4 knots off Nobska Point and then gradually diminishes to 1.6 knots off Gay Head Light. Tidal current predictions for **Vineyard and Nantucket Sounds** can be generated at *tidesandcurrents.noaa.gov*. Specific information can be obtained about the times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(14) At the western entrance to Vineyard Sound, west-northwestward of Gay Head Light, the tidal current is rotary, turning clockwise. The velocity is only 0.2 to 0.5 knot. Since the tidal current is weak, winds greatly affect it and the current frequently sets approximately with the winds.

(15)

Weather: Vineyard Sound, Buzzards Bay and vicinity

(16) Buzzards Bay is open to winds out of the south and southwest, which are common from spring through fall. Winds increase as they move from the surrounding land out over the Bay. Its northeast-southwest orientation causes southwesterlies to strengthen as they funnel up from the mouth of the Bay to its head. The result is that speeds are often double those at nearby land stations, and southwesterlies may prevail even when land stations are reporting west or northwest winds. However, as a general rule southwesterlies blow harder close to the Elizabeth Islands than in the middle of the Bay. The relatively shallow water of the Bay increases the steepness of waves and their closeness to one another; this can cause a stiff chop. With southerly or westerly gales there is a heavy sea in the westerly entrance to Vineyard Sound, and heavy seas occur at times off the entrance to Quicks Hole.

(25)

(17)

Pilotage, Vineyard Sound and Buzzards Bay

(18) Pilotage is compulsory for foreign vessels of 350 gross tons or more, U.S. vessels under register of 350 gross tons or more and tank barge towing vessels carrying 6,000 barrels or more of petroleum cargoes. Pilotage is available from Northeast Marine Pilots, Inc., Newport, RI, 02840; telephone 401–847–9050 (24 hours), 800–274–1216; FAX 401–847–9052; email: dispatch@nemarinepilots.com.

(19)

Vineyard Haven Harbor

(20) **East Chop** and **West Chop** are prominent points on the north side of Martha's Vineyard and on the east and west side of the entrance to Vineyard Haven. Both points terminate in high wooded bluffs that show prominently from the sounds; each is marked by a light.

(21) **West Chop Light** (41°28'51"N., 70°35'59"W.), 84 feet above the water, is shown from a white conical tower. A mariner radio activated sound signal at the light is initiated by keying the microphone five times consecutively on VHF-FM channel 83A.

(22) A lighted gong buoy, 0.5 mile northeastward of the light, and a buoy, 0.5 mile eastward of the light, mark shoal water and rocks awash to the eastward of West Chop. It has been reported that during strong tidal currents, the buoy may be submerged.

(23) **East Chop Light** (41°28'13"N., 70°34'03"W.), 79 feet above the water, is shown from a white tower on the east side of the entrance to Vineyard Haven. **East Chop Flats**, covered 5 to 18 feet, extend 0.2 mile northward and 0.5 mile eastward of East Chop. A lighted bell buoy, about 0.5 mile east-northeastward, and a buoy, about 350 yards northward of the light, mark the flats.

(24) **Vineyard Haven Harbor** is a funnel-shaped bight in the northern side of Martha's Vineyard between East Chop and West Chop, about 1.4 miles long in a southwest direction and about 1.3 miles wide at the entrance. This haven, easy of access, is the most important harbor of refuge for coasters between Provincetown and Narragansett Bay. The depths range from 46 feet at the entrance to 15 feet near the head of the harbor.

(26) Although Vineyard Haven Harbor is exposed to northeasterly winds, vessels with good ground tackle can ride out most blows. The greatest danger encountered by vessels at anchor in a northeast gale is from vessels with poor ground tackle, which are likely to drift, foul other vessels and then go ashore.

(27) The harbor is the approach to the village of **Vineyard Haven**. A detached breakwater, marked on its southeastern end by a light, is on the flats on the western side of the harbor near the head. The natural channel is

clear; soundings are the best guide for finding anchorage. When well inside the entrance, the water shoals gradually toward the western shore, but the eastern shore is steep and should be given a berth of about 0.2 mile.

(28)

Anchorages

(29) Vessels anchor according to draft, anywhere from the points at the entrance to the head of the harbor. Shallow-draft vessels favor the western shore.

(30) Vessels entering the harbor with a head wind or light breeze, at the end of a favorable current through the sound, should continue on in the channel until the harbor is well opened before standing in for the anchorage. This will help clear the entrance points. Approaching from the eastward, vessels will keep clear of Squash Meadow and East Chop Flats by keeping in the white sector of West Chop Light. The anchorage basin behind the breakwater has depths of 5 to 12 feet and is usually filled to capacity during the summer. When anchoring in the harbor, care must be taken to avoid obstructing the approach to the ferry slip and the approach to the oil wharves on the southerly side of the harbor.

(31)

Dangers

(32) Flats, partly bare at low water and marked by a buoy, make off 300 yards from the eastern shore of the harbor about 1 mile inside East Chop Light. A rock, covered 9 feet, is 0.3 mile northward of the breakwater light.

(33)

Current

(34) The tidal currents have little velocity in the harbor; however, care should be taken on the ebb, which sets westward, not to approach too closely to West Chop as the current in that vicinity sets on the ledges eastward and northward of the point.

(35) The **harbormaster** has control of the anchoring of vessels in the inner harbor; he will usually be found at the town dock on the west side of the harbor and can be contacted on VHF-FM channel 16 or at 508–693–1368. A 4 mph **speed limit** is enforced inside the breakwater and within 150 feet of moored craft.

(36) Pilots are not available at Vineyard Haven. A twin-screw, 500-hp tug, also equipped for salvage work, is based in the harbor.

(37) A ferry terminal, several wharves, two marinas and a boatyard are in the harbor. A yacht club is on the west side of the harbor, about 0.3 mile northward of the breakwater.

(38) Guest moorings, maintained by the town, are available off the municipal wharf, 200 yards northward of the ferry terminal; other moorings can be hired from the boatyard and marinas.

(39) The Martha's Vineyard Hospital is on the beach road near Oak Bluffs.

(40) The Woods Hole-Martha's Vineyard and Nantucket Steamship Authority maintains year-round ferry service from Woods Hole. Air service is available from Martha's Vineyard Airport, about 4 miles south of the town.

(41) **Lagoon Pond**, eastward of the town of Vineyard Haven, has its entrance about 0.5 mile northeastward of Vineyard Haven Breakwater Light 10. A short jetty, marked by a light, extends from the northwestern end of the jutting point on the northerly side of the entrance. In 2024, the controlling depth through the entrance was 7.5 feet (8 feet at midchannel). The channel is marked by private buoys and daybeacons. The pond is primarily used by local fishing craft. **Robbins Rock**, marked by a daybeacon, is off the west shore, about 1 mile above the entrance. The highway bridges across the entrance are divided by two channels. Channel A (west side) is crossed by a bascule bridge and has a horizontal clearance of 30 feet and a (closed) vertical clearance of 15 feet. Channel B (east side) is crossed by a fixed bridge and has a horizontal clearance of 30 feet and a vertical clearance of 19 feet.

(42)

Falmouth Harbor

(43) **Falmouth Harbor**, the open roadstead off the south shore of Cape Cod eastward of Nobska Point Light, affords an anchorage for vessels in 24 to 36 feet about 0.8 mile from shore. Smaller vessels can anchor closer to the shore in 15 to 18 feet. The bottom is generally sticky and good holding ground; the depths shoal gradually toward the shore. The anchorage affords a lee in northerly winds; in southerly winds the sea is somewhat broken by L'Hommedieu Shoal and the shoals westward of it so that a vessel with good ground tackle can ride out a gale in comparative safety. (See **33 CFR 110.1** and **110.140(c)(5)**, and **(d)**, chapter 2, for limits and regulations of the anchorage area.)

(44) Falmouth Harbor is frequently used by vessels with good ground tackle that prefer this anchorage to the anchorage in Vineyard Haven Harbor, which may be crowded in bad weather. Vessels approaching the anchorage are cautioned to stay clear of the two shoal areas with depths of 10 to 16 feet marked by buoys that extend westward of L'Hommedieu Shoal.

(45) Vessels can enter Falmouth Harbor from the southward on a course of **344°** with West Chop Light astern and pass about 0.1 mile westward of Falmouth Harbor East Shoal Buoy 17. Vessels in the vicinity of Nobska Point Light can pass 0.4 mile eastward of the light on a north-northeasterly course, and when Tarpaulin Cove and Nobska Point Lights are nearly in range, stand eastward to an anchorage.

(46) **Falmouth Heights**, about 3 miles east-northeast of Nobska Point Light and east of the town of **Falmouth**, is a prominent yellow bluff on the summit of which are numerous homes and a large hotel.

(47) **Falmouth Inner Harbor**, westward of Falmouth Heights, is a dredged basin about 0.7 mile long and less than 0.1 mile wide, on the north side of Falmouth Harbor. The yacht club dock and flagpole, just inside the entrance on the east side of the harbor, are conspicuous from close

(48)

inshore. The harbor is entered through a dredged channel between two jetties; a light marks the end of each jetty.

(49)

Current

(50) The tidal current in the sound about 1.5 miles south of the harbor sets east-northeastward on the flood at a velocity of 2.3 knots and west-southwestward on the ebb at 1.7 knots.

(51)

Small-craft facilities

(52) There are several small-craft facilities in Falmouth Inner Harbor.

(53) The **harbormaster** is at the town-operated Falmouth Marina, on the west side halfway up the harbor; telephone 508–548–9796. The harbormaster monitors VHF-FM channels 16, 12 and 9. A ferry operates in the summer to Oak Bluffs from the wharf at the head of the harbor.

(54) **Nobska Point**, about 29 miles westward of Monomoy Point, is a bluff with **Nobska Point Light** (41°30'57"N., 70°39'18"W.), 87 feet above the water, shown from a white tower, at the south end. A sound signal at the light is operated by keying the microphone five times consecutively on VHF-FM channel 83A. **Nobska Point Ledges**, partly bare at low water, extends 150 yards eastward and southwestward from the point.

(55)

Middle Ground to Squibnocket Point

(56) **Middle Ground**, covered 6 to 18 feet, is the easterly half of a narrow, somewhat shifting ridge. The ridge extends for about 9 miles westward from West Chop. A buoy is at the northeast end and a lighted bell buoy off the northwest edge. **Lucas Shoal**, with a least depth of 17 feet, is the southwestern end of the ridge. It is separated from the Middle Ground by a natural channel with a depth of 31 feet.

(57) **Lake Tashmoo**, a landlocked pond on the northwest side of Martha's Vineyard, is entered through a narrow jettied entrance that had a reported controlling depth of 2½ feet in 1981. The lake, only used by local craft, has general depths of 3 to 10 feet. A private seasonal light marks the east jetty and private seasonal buoys mark the channel through a shoal area just inside the entrance.

(58) A 4 mph **speed limit** is enforced in the pond. Anchoring is prohibited in the entrance channel.

(59) A small boatyard is on the easterly side of the lake; a flatbed trailer can handle craft up to 30 feet for hull and engine repairs. Guest moorings and limited supplies are available.

(60) **Norton Point** and **Cape Higgon** are prominent bluffs on the northwest side of Martha's Vineyard about 3 and 8 miles, respectively, southwestward of West Chop Light.

(61) **Menemsha Bight**, on the northerly side of the western end of Martha's Vineyard 2.5 miles east of Gay Head, affords shelter from southerly and easterly winds in depths of 25 to 60 feet, sticky bottom. (See **33 CFR 110.1** and **110.140(c)(1)** and **(d)**, chapter 2, for limits and regulations for the anchorage area.) There are no dangers in the bight if the shore is given a berth of 0.3 mile.

(62) **Menemsha Creek**, on the northwestern shore of Martha's Vineyard and about 3 miles eastward of Gay Head Light, is entered from Menemsha Bight through a dredged channel that leads southeastward to **Menemsha Basin**, on the north shore just inside the entrance. From the basin, the dredged channel continues southward through the creek to Menemsha Pond, about 1 mile above the entrance. The entrance to the creek is protected by jetties marked by a lights. A lighted bell buoy, about 300 yards northwestward of the east jetty, marks the channel approach, and buoys and daybeacons mark the channel. The channel south of Menemsha Basin is reported to shoal rapidly after dredging; mariners are advised to seek local knowledge before attempting to go beyond Menemsha Basin.

(63) **Menemsha** is a small fishing village on Menemsha Basin. **Menemsha Pond**, a rectangular basin about 1 mile long and 0.7 mile wide, has general depths of 2 to 18 feet, with the deepest water in the southern half of the pond.

(64)

Current

(65) The tidal currents through the entrance have an estimated velocity of 3 knots or more. Slacks are reported to occur 45 minutes after local high and low waters.

(66) Guest moorings are available in Menemsha Basin, and anchoring is permitted in the pond. Berths are also available at the public facilities at Dutcher Dock, on the northeast side of the basin. Commercial fishing and charter boats berth at the dock or at the private piers on the west side of the basin. **Menemsha Coast Guard Station** is on the south side of the basin. The **harbormaster** controls all berthing and mooring in the basin. The harbormaster has an office at Dutcher Dock and can be contacted by radiotelephone on VHF-FM channel 16 or by telephone at 508–645–2846.

(67) A marina is on the east side of Menemsha Basin. Repairs and hauling of craft to 45 feet can be arranged. Gasoline, diesel fuel, water, ice, a launching ramp and marine supplies are available.

(68) Seasonal bus service is available from Menemsha to Vineyard Haven and other points on the island.

(69) **Gay Head**, the westerly end of Martha's Vineyard, is a prominent high bluff. It is marked by **Gay Head Light** (41°20'54"N., 70°50'04"W.), 175 feet above the water, shown from a 51-foot red brick tower on the head. A lighted gong buoy is 1.6 miles northwestward of the light. Several rocks exist between Gay Head and the lighted gong buoy.

(70) **Devils Bridge** is a reef making off 0.8 mile northwestward of Gay Head. The reef has a depth of 2 feet about 0.4 mile offshore and 17 feet at its end.

(71) **Nomans Land**, about 5.5 miles southward of Gay Head, is a prominent, high, and rocky island. Except for a small section on its northwestern side, the shore consists of clay and gravel cliffs 10 to 18 feet high with boulders lining the shores. In the interior of the island are many hills, the highest over 100 feet high, with considerable marshy area between the hills. A **danger zone** surrounds Nomans Land. (See **33 CFR 334.70**, chapter 2, for limits and regulations.)

(72) Several sunken rocks and ledges are in the passage between Nomans Land and Martha's Vineyard. **Lone Rock**, covered 8 feet, and **Old Man**, a ledge covered 4 feet, are marked by buoys. A buoyed channel about 0.7 mile wide between the islands may be used by small vessels in the daytime. Shoal water extends 0.5 mile southward of **Squibnocket Point**, the southernmost point of Martha's Vineyard.

(73)

Elizabeth Islands

(74) **Elizabeth Islands**, including Nonamesset, Uncatena, Weepecket, Naushon, Pasque, Nashawena, Penikese and Cuttyhunk Islands, extend about 14 miles west-southwest from the southwest end of Cape Cod. The islands, forming part of the northern shore of Vineyard Sound, separate the sound from Buzzards Bay. They are hilly and partly wooded; the shores are, in general, low bluffs. Westward of Woods Hole are several buoyed channels between the islands, but Quicks Hole is the only one recommended for strangers.

(75)

Woods Hole

(76) **Woods Hole** is that water area lying between the southwest tip of Cape Cod and Uncatena and **Nonamesset Island**, the easternmost of the Elizabeth Islands, with Buzzards Bay on the northwest and Vineyard Sound on the southeast; it includes Great and Little Harbors in the eastern part and Hadley Harbor in the western part. Woods Hole is also the approach to the town of **Woods Hole** on the northeastern shore of Great Harbor. The town is a busy commercial center and a transshipping point for passengers and freight to and from Nantucket and Martha's Vineyard. During the summer it is an active resort and frequently a port of call by yachts passing through to Vineyard Sound or Buzzards Bay. There is considerable waterborne commerce in seafood products and general cargo.

(77)

Prominent features

(78) The most prominent landmark approaching Woods Hole is Nobska Point and light. A light marks the south end of **Juniper Point**, the finger of land separating

(84)

Little and Great Harbors. Also prominent is the house high on Juniper Point, a standpipe 0.7 miles north-northwestward of Nobska Point, the dome of the Woods Hole Oceanographic Institution and the buildings of the National Marine Fisheries Service and the Marine Biological Laboratory.

(79)

Channels

(80) **Woods Hole Passage**, a dredged section through the northern part of Woods Hole, connects Vineyard Sound and Great Harbor with Buzzards Bay and consists of **The Strait** and a spur channel known as the **Branch** at the western end of The Strait, and **Broadway**, the southerly entrance to The Strait from Vineyard Sound. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A. The northerly entrance from Great Harbor into The Strait is preferred over Broadway with its sharp turn, which is difficult in strong currents, especially for low-powered vessels and vessels under sail.

(81) Woods Hole Passage is a dangerous waterway surrounded by treacherous shoals, ledges and severe east/west currents. Particular attention should be given to the navigational aids that mark the passage, especially **Woods Hole Passage Junction Buoy SB** at the intersection of The Strait and Broadway. Before attempting to transit the passage, mariners should carefully consult the current edition of nautical charts for the area and the most recent Local Notice to Mariners. Mariners are further cautioned to always use extreme care and prudent speed when transiting the passage. Tidal currents are strong in the passage; buoys in the narrowest part of the channel are sometimes towed under by the current. Strangers to the area should attempt transit only at slack water.

(82) The channels through Woods Hole Passage are marked by buoys and lights, but extreme caution and slack water are required to safely navigate them with drafts greater than 8 feet. Mariners entering from Buzzards Bay should keep in mind that the buoys are colored and marked for passage from Vineyard Sound to Buzzards Bay.

(83) A lighted bell buoy and gong buoy mark the entrance to **Great Harbor** from Vineyard Sound, and a directional light, lighted and unlighted buoys mark the channel. Mariners should guard against the current from Buzzards Bay, which has a tendency to set vessels eastward.

(85)

Anchorages

(86) An anchorage about 0.2 mile square, with poor holding ground and irregular depths ranging from 19 to 62 feet, is at the head of Great Harbor. Shoals covered 5 to 9 feet are northwest of the anchorage. Good anchorage

in depths of 29 to 36 feet is also available about 200 yards northwest of the National Marine Fisheries Service's wharf. Small craft can find good anchorage in Little Harbor and Hadley Harbor. (See **33 CFR 110.1** and **110.140**, chapter 2, for limits and regulations of the deepwater anchorages in the vicinity of Woods Hole.)

(87)

Dangers

(88) Numerous ledges and shoals border the channel through Woods Hole. **Great Ledge**, an extensive rocky shoal awash at low water with a full northwest gale, lies between the entrances to Little and Great Harbors; it is marked by a buoy. **Co f n Rock**, eastward of Great Ledge and covered 5 feet, is marked by a lighted buoy 120 yards eastward of the rock. **Nonamesset Shoal**, covered 10 feet, extends about 0.2 mile eastward from Nonamesset Island, at the entrance to Great Harbor. **Parker Flats** extend as much as 200 yards off the eastern shore of Great Harbor northward of Juniper Point. Most of these dangers are marked by buoys.

(89) Fringing the passage westward of Great Harbor are many other ledges and shoals. **Red Ledge**, grassy, and **Grassy Island**, with its surrounding ledge marked by a light, are on the western side of Great Harbor Channel. **Middle Ledge**, which uncovers 1 foot in places and is marked by buoys, is on the south side of The Strait. A ledge, awash at low water and marked by a light, is about 250 yards westward of Middle Ledge. **Hadley Rock**, covered 5 feet, is some 500 yards west-southwestward of the light west of Middle Ledge. A rocky shoal area extends more than 0.3 mile westward of **Penzance Point**, the southern extremity of **Penzance**, which is the curving peninsula sheltering the west and northwest sides of Great Harbor. Most of the dangers adjoining the passage channel are marked by navigational aids.

(90)

Current

(91) The current velocity at times exceeds 4.5 knots in the narrow part of Woods Hole Passage. Velocities as high as 5.0 knots have been reported by the U.S. Coast Guard. For daily predictions of the current, see the tidal current prediction service at *tidesandcurrents.noaa.gov*. Links to a user guide for this service can be found in chapter 1 of this book.

(92) The velocity of the current is about 3.5 knots in The Strait southward of Penzance Point. (See the tidal current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area.) Both the velocity of the current and time of slack water are affected by strong winds. At the north entrance to Woods Hole in Buzzards Bay, the velocity of the tidal current is 0.8 knot, whereas at the eastern entrance to The Strait in Great Harbor, it is about 1.3 knots. In the upper part of Great Harbor, near the National Marine Fisheries Service's wharf, the currents are barely perceptible, and vessels at anchor lie head to wind.

(93) Drift **ice** is brought through from Buzzards Bay but seldom interferes with navigation except in unusually severe winters, when it may close the entrance from the bay. Small craft may experience difficulty in severe winters, but powered vessels usually proceed through the ice. The strong tidal currents usually keep Great Harbor open.

(94)

Pilotage: Woods Hole

(95) Pilotage service is available for the harbor. (See Pilotage, Vineyard Sound, this chapter.)

(96)

Routes

(97) The following directions are good for medium-draft vessels entering Woods Hole at slack water. Approaching from the eastward, pass about 0.3 mile southward of Nobska Point on a west-southwesterly course until in the white sector of the Great Harbor Directional Light, or from a point close to Nobska Point Lighted Bell Buoy 26, steer 279° until in the white sector. Approaching from the westward in Vineyard Sound, give the south side of the Elizabeth Islands a berth of about 0.5 mile and steer for Nobska Point Light on any bearing between 045° and 051° until in the white sector.

(98)

Towage

(99) Tug service is available at Woods Hole.

(100)

Wharves

(101) The ferry pier of the Woods Hole-Martha's Vineyard and Nantucket Steamship Authority is on the eastern side of Great Harbor. When a ferry is approaching in fog, a private sound signal is sounded, a private quick flashing white light is shown from the southwest corner of the pier, and a private quick flashing yellow light is shown from the southwest corner of the ferry slip. The ferry to Naushon Island lands at the service wharf about 60 yards north of the ferry pier. The buildings and wharf of the Woods Hole Oceanographic Institution are northwestward of the ferry pier. Northwestward of the Oceanographic Institution are the wharves of the Marine Biological Laboratory; the wharf, basin and buildings of the National Marine Fisheries Service; the town pier; and several private buildings.

(102) Depths at the principal piers vary from 11 to 30 feet. A breakwater extends about 90 yards southwestward from the south end of the National Marine Fisheries Service wharf. Foul ground extends about 50 yards northwestward of the outer end of the breakwater.

(103) **Eel Pond**, an extension of Great Harbor to the northeastward, is a basin with depths of 10 to 20 feet. In 2001, the narrow entrance to the pond had a reported controlling depth of 6 feet. A highway bridge over the entrance channel has a 31-foot bascule span with a clearance of 5 feet. (See **33 CFR 117.1** through **117.59** and **117.598**, chapter 2, for drawbridge regulations.) The piers of the Marine Biological Laboratory are along the

southwest side of the pond. A boatyard in Eel Pond has water, some marine supplies and limited berths with electricity and can do engine repairs. The **harbormaster** has an office in Falmouth.

(104) Seasonal ferry service is available from Woods Hole to Nantucket and Oak Bluffs, Martha's Vineyard. Year-round ferry service is available to Vineyard Haven, Martha's Vineyard.

(105) **Woods Hole Coast Guard Station** is on the west side of **Little Harbor** about 450 yards northward of Juniper Point. A federal project provides for a depth of 12 feet from Vineyard Sound through a turning basin off the Coast Guard wharf on the west shore. (See Notice to Mariners and latest editions of charts for controlling depths.) The channel is marked by lighted buoys. The east side of the harbor is used as a mooring area for local craft. Numerous rocks awash are in this part of the harbor; extreme caution is advised. A 6 mph **speed limit** is enforced in the harbor by the Falmouth Harbormaster.

(106) **Hadley Harbor**, in the western portion of Woods Hole at the northwest end of Nonamesset Island, is suitable only for small craft. It is reached by a narrow, crooked channel. The deeper entrance, marked by buoys, is between ledges on both sides. The inner harbor forms a well-sheltered anchorage for small craft.

(107) Two wharves, with depths of about 9 feet at their ends, are on the western side of Hadley Harbor. A private wharf, with a depth of about 7 feet at its end, is at the western end of Nonamesset Island.

(108)

Naushon Island to Cuttyhunk Harbor

(109) **Naushon Island**, the largest of the Elizabeth Islands, extends west-southwestward from Uncatena and Nonamesset Islands.

(110) **Weepecket Islands**, in Buzzards Bay off the northeastern part of Naushon Island, are bare and rocky.

(111) **Weepecket Rock**, on a rocky ledge about 0.7 mile northeastward of the northernmost of the Weepecket Islands, is covered 8 feet and is marked by a lighted gong buoy.

(112) **Lackeys Bay**, between Nonamesset Island and **Jobs Neck**, the southeastern extremity of Naushon Island, is shoal with numerous bare rocks.

(113)

Note

(114) In 1990, unexploded World War II ordnance was reported about 0.5 mile southeastward of Lackeys Bay in about 41°29'34.5"N., 70°41'15.0"W.

(115) **Tarpaulin Cove**, about 5 miles west-southwest of Nobska Point, is a bight about 0.5 mile in diameter, in the south shore of Naushon Island. The cove affords shelter from northerly and westerly winds and is frequently used. A light is on the southwest side of the cove. Anchorage in depths of 14 to 18 feet, good holding ground, is in the cove with the light bearing between 212° and 189°. Deep-draft vessels should anchor farther out in depths of 36 feet or more. The eastern and northern shores should be given a berth of 200 yards. Rocks are near the western shore and should be given a berth of over 300 yards; buoys mark the dangers.

(116) **Robinsons Hole** is a narrow buoyed passage from Vineyard Sound to Buzzards Bay between the western end of Naushon Island and the eastern end of Pasque Island. It has numerous rocks and ledges and strong tidal currents. The buoys often tow under, and the passage should never be attempted by strangers; it is used occasionally by local fishermen. It has been reported that currents sometimes reach a velocity of 5 knots in the passage. The velocity in the narrow part is about 3 knots. The flood sets southeastward and the ebb northwestward into Buzzards Bay. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(117) **Quicks Hole**, between Pasque Island and **Nashawena Island**, is the only passage between Vineyard Sound and Buzzards Bay eastward of Cuttyhunk available for vessels of over 10-foot draft. The clearly defined entrance from Vineyard Sound, about 0.6 mile wide, is about 4 miles southwestward of Tarpaulin Cove and about 5 miles north of Gay Head. The passage is used considerably by tows, especially during westerly or southerly winds, to avoid the very heavy sea in the entrance to Vineyard Sound, and also because a secure anchorage from these winds can be had, if necessary, on the north side of Nashawena Island. The passage is considered unsafe for a long tow at night, but otherwise it may be used by steamers either night or day.

(118) Vessels should follow a midchannel course through the passage. The channel is nearly straight with a width of about 0.2 mile. General depths are 30 feet or more, but there are several shoaler spots. Rocks covered 27 to 37 feet are near the center of the channel. Because of the broken nature of the bottom, the passage is not recommended for a stranger drawing more than 21 feet. Buoys mark the channel.

(119) The aids in Quicks Hole are colored and numbered for passage from Vineyard Sound to Buzzards Bay.

(120) The eastern side of Quicks Hole is foul, and no attempt should be made to pass eastward of the lighted buoy. **Felix Ledge**, 0.2 mile off the eastern shore of Nashawena Island, is covered 15 feet and marked by a buoy.

(121) A sunken wreck, covered 30 feet, is on the west side of the passage in about 41°26.6'N., 70°51.1'W.

(122) **Lone Rock**, covered 8 feet and marked by a lighted buoy, is off the northern entrance, about 0.7 mile northward of **North Point**, the northeastern extremity of Nashawena Island. Tide rips have been observed between North Point and Lone Rock during spring tides.

(123)

Current

(124) The tidal currents have considerable velocity in Quicks Hole, about 2 to 2.5 knots, and a sailing vessel should not attempt to pass through unless with a strong favorable wind on a favorable current. Deep-draft vessels should be careful not to be set off their courses. With a strong westward current through Vineyard Sound, there is a northward current through Quicks Hole; with a strong eastward current in Vineyard Sound, the current sets southward through Quicks Hole. Strong winds affect the regularity of the currents. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(125) **Penikese Island**, grassy and hilly, is about 1.3 miles northwestward of **Knox Point**, the northwestern extremity of Nashawena Island. Shoal water extends from Penikese Island to **Gull Island**, a small islet 0.5 mile southeastward. No attempt should be made to pass between them. Rocky ledges extend southward and westward from Gull Island; buoys are on the southern edge of this area. The channels to Cuttyhunk Harbor from Buzzards Bay are southward of the ledges.

(126) **Cuttyhunk Harbor** is formed by the bight between Nashawena Island and **Cuttyhunk Island**, the westernmost of the Elizabeth Islands. Northward of the harbor are Penikese and Gull Islands and several ledges, which shelter the harbor from winds from that direction. The harbor is exposed to winds from the northeastward. Weather-bound coasting vessels and fishermen sometimes use the anchorage in the harbor. The harbor is the approach to the village of **Cuttyhunk** and to **Cuttyhunk Pond**; the latter is entered through a dredged cut in the eastern end of Cuttyhunk Island. **Copicut Neck** forms the northerly side of Cuttyhunk Pond.

(127) Prominent from offshore is a 50-foot-high monument on an island in **Westend Pond** on the western end of Cuttyhunk Island.

(128) Vessels bound for Cuttyhunk Harbor generally approach from Buzzards Bay. The principal dangers are marked by buoys. Strangers should not enter except in the daytime with clear weather. If entering from the northwestward, a greater draft than 10 feet should not be taken in. The approach from northeastward is deeper.

(129) **Canapitsit Channel**, between the east end of Cuttyhunk Island and Nashawena Island, is used by small boats and is partially marked by buoys. In 2022, the channel had a controlling depth of 4.5 feet. The buoys at this entrance are often dragged off station by strong currents and heavy seas. The channel should never be used during a heavy ground swell. With southerly winds, heavy seas will break across the entrance.

(130)

Channels

(131) A federal project provides a 10-foot channel from Cuttyhunk Harbor into Cuttyhunk Pond to a turning basin at the western terminal in the pond and a 10-foot anchorage basin in the eastern part of the pond north of the channel. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A. The jettied entrance is marked by a light on the north jetty and by a buoy off the end of the submerged south jetty. A bell buoy marks the entrance to the channel.

(132)

Anchorages

(133) Limited anchorage with reported poor holding ground may be found in depths of 10 to 24 feet in Cuttyhunk Harbor. The shores on both sides of the harbor are foul, and the anchorage is in the middle.

(134)

Dangers

(135) Shoals extend 0.6 mile northeastward of Cuttyhunk Island. **Whale Rock** and **Pease Ledge** uncover at low water.**Middle Ground**, covered 11 feet, is 0.5 mile north of **Copicut Neck** in the northwestern approach to the harbor. **Middle Ledge**, covered 14 feet, is about 0.4 mile east of Middle Ground. **Edwards Rock**, covered 5 feet (least depth reported in 2025), is 250 yards northeastward of Whale Rock. These dangers, except for Middle Ledge, are buoyed. An unmarked rocky shoal, covered 10 feet, is in the middle of the northwestern approach about 0.2 mile southeastward of Middle Ledge. Numerous other rocks and ledges covered 4 to 12 feet are between Cuttyhunk Island and the ledges southwestward of Penikese and Gull Islands. The eastern point at the entrance and the eastern shore of the harbor should be given a berth of over 300 yards.

(136)

Routes

(137) In approaching Cuttyhunk Harbor from eastward in Buzzards Bay, take care to avoid Lone Rock, 0.7 mile northward of the northeastern end of Nashawena Island. Thence pass northwestward of Cuttyhunk East Entrance Lighted Bell Buoy CH (41°26.6'N., 70°53.4'W), and thence to anchorage according to draft in the harbor.

(138) In approaching from westward, from a position about midway between Ribbon Reef and the Cuttyhunk Island monument (41°24.8'N., 70°56.8'W.), steer 051° until abeam of Middle Ground Buoy MG, distance 600 yards. Pass midway between Middle Ground and Penikese Island, passing northward of Middle Ledge and the 12-foot spot southeastward of it, and then swinging southeasterly to the harbor anchorage, or southward to the entrance to Cuttyhunk Pond.

(139)

Current

(140) The current velocity in Canapitsit Channel is 2.6 knots on the flood, which sets south-southeastward, and 1.7 knots on the ebb, which sets northwestward. In the channel southward of Penikese Island, the flood sets eastward and the ebb westward at about 0.8 knot. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(141) Drift **ice** is carried into Cuttyhunk Harbor with northerly winds and closes the harbor during severe winters.

(142)

Small-craft facilities

(143) There is a service wharf on the south side of the channel at the entrance to the pond and a marina on the south side of the basin at the head of the channel in the pond. A yacht club and a fishing club are on the island. Gasoline, diesel fuel, water, ice, berths with electricity and some marine supplies are available. Limited lodging in cottages is available on the island. The **harbormaster** can be reached at 508–966–9295. There is daily launch service with New Bedford in the summer and twice weekly in the winter; seaplane service is also available.

(144)

Buzzards Bay

(145) **Buzzards Bay** is the approach to New Bedford, many small towns and villages and the entrance of Cape Cod Canal. The bay indents the south shore of Massachusetts, extending in a northeasterly direction from **Rhode Island Sound**. The bay is enclosed on the south side and separated from Vineyard Sound by the Elizabeth Islands.

(146) The shores are irregular, rocky in character and broken by many bays and rivers. Large boulders are common, in places extending a considerable distance from shore, thus making close approach to the shore dangerous.

(147) The bottom in the main part of the bay and approach is very broken with boulder reefs in places. Vessels should proceed with caution when crossing shoal areas in the tributaries of the bay where the depths are not more than about 6 feet greater than the draft. Caution must also be exercised in the vicinity of the wrecks shown on the chart. Deep water prevails as far as Wings Neck, above which the bay is full of shoals.

(148) **Cape Cod South Closure Area**, a Marine Protected Area (MPA), includes the inshore waters of Buzzards Bay and offshore federal waters of the south coast of Massachusetts.

(149) **Tra f c Separation Scheme (Buzzards Bay)** has been established in the approach to Buzzards Bay through Rhode Island Sound.

(150) The scheme is composed basically of **directed tra f c lanes,** each with one-way inbound and outbound traffic lanes separated by a **def ned tra f c separation zone** and a **precautionary area.** The scheme is recommended for use by vessels approaching or departing from Buzzards Bay but is not necessarily intended for tugs, tows or other small vessels that traditionally operate outside of the usual steamer lanes or close inshore.

(151) **The Tra f c Separation Scheme has been designed to aid in the prevention of collisions at the approaches to the major harbors, but is not intended in any way to supersede or alter the applicable Navigation Rules. Separation zones are intended to separate inbound and outbound tra f c lanes and be free of ship tra f c and should not be used except for crossing purposes. Mariners should use extreme caution when crossing tra f c lanes and separation zones.** (See **33 CFR 167.1** through **167.15** and **167.100** through **167.103**,chapter 2, for limits and regulations and Traffic Separation Schemes, chapter 1, for additional information.)

(152) The **precautionary area** in the southwest part of Rhode Island Sound has a radius of 5.4 miles centered on 41°06'00"N., 71°23'18"W., excluding those areas of the circle bounded by imaginary lines extended between the outer limits of the inbound and outbound traffic lanes. (Note that this precautionary area is common to the Traffic Separation Schemes for the approaches to both Buzzards Bay and Narragansett Bay. The Traffic Separation Scheme for the approach to Narragansett Bay is described in chapter 6.)

(153) The **separation zone** is a 1-mile-wide zone centered in the following positions:

(154) (i) 41°10'12"N., 71°19'06"W.,

(155) (ii) 41°21'48"N., 71°07'06"W.

(156) The **inbound tra f c lane** is a 1-mile-wide lane with a length of about 14.8 miles. Entering the traffic lane at a point in about 41°09'36"N., 71°18'00"W., a course of **038°** follows the centerline of the traffic lane to its end, thence steer usual courses to destination.

(157) The **outbound tra f c lane** is a 1-mile-wide lane with a length of about 14.8 miles. Entering the traffic lane at a point in about 41°22'25"N., 71°08'06"W., a course of **218°** follows the centerline of the traffic lane to a junction with the precautionary area.

(158) The Traffic Separation Scheme is not buoyed.

(159) Buzzards Bay has six entrances, but two of these are so narrow and dangerous as to exclude their use except by small craft with local knowledge. The four major entrances are the main channel, from westward, passing north of Cuttyhunk Island; Cape Cod Canal from northeastward; and Quicks Hole and Woods Hole from the southward. The two hazardous entrances are Canapitsit Channel, between Cuttyhunk and Nashawena Islands, and Robinsons Hole, between Pasque and Naushon Islands.

(160) The western entrance has a clear width of 4.3 miles between Sow and Pigs Reef and Hen and Chickens. The bottom in this entrance is irregular and rocky, and there

are spots with depths of 17 to 34 feet. Because these shoal areas are surrounded by deeper water, vessels of 16-foot draft or more must exercise extra caution when entering the bay. In heavy southwest gales the sea breaks over some of these spots.

(161) The best guides for entering the bay from westward are Buzzards Bay Entrance Light and the lighted buoys in the entrance. Gay Head Light and Buzzards Bay Entrance Light are the guides for vessels approaching from the southward.

(162) **Buzzards Bay Entrance Light** (41°23'49"N., 71°02'05"W.), 67 feet above the water, is shown from a tower on a red square superstructure on red piles about 4 miles 255° from the southwest corner of Cuttyhunk Island. The name BUZZARDS is painted in white on the sides. A racon is at the light and a mariner activated sound signal at the light is initiated by keying the microphone five times on VHF-FM channel 83A.

(163) **Recommended Vessel Route (Buzzards Bay)** has been established in the approach to Buzzards Bay through Rhode Island Sound.

(164) The U.S. Coast Guard Captain of the Port, Providence, in cooperation with the Southeastern Massachusetts and Rhode Island Port Safety and Security Committees, has established a Recommended Vessel Route for deep draft vessels and tugs/barges transiting Rhode Island Sound, Narragansett Bay and Buzzards Bay. Deep draft vessels and tugs/barges are requested to follow the designated routes. These routes were designed to provide safe, established routes for these vessels to reduce the potential for conflict with recreational boaters, fishing gear, and other small craft and to reduce the potential for grounding or collision. Vessels are responsible for their own safety and are not required to remain inside the route nor are fisherman required to keep fishing gear outside the route. Small vessels should exercise caution in and around the Recommended Vessel Routes and monitor VHF channels 16 or 13 for information concerning deep draft vessels and tugs/barges transiting these routes.

(165)

Anchorages

(166) New Bedford Inner Harbor affords anchorage for vessels of 25-foot draft. Cuttyhunk Harbor affords anchorage in depths of 10 to 24 feet; except for the small-craft inner harbor, it is exposed to northerly winds. A good anchorage sheltered from all southerly winds may be had off the north shore of Nashawena Island eastward of Penikese and Gull Islands in depths of 40 to 48 feet. This anchorage, frequently used by tows, is available for vessels of any draft; however, care must be taken to stay clear of the fishtrap area in the vicinity. Two general anchorages are off the western entrance to Cape Cod Canal. (See **33 CFR110.1** and **110.140(b)(1)**, **(b)(2)** and **(d)**, chapter 2, for limits and regulations.)

(167)

Dangers

(168) **Hen and Chickens**, extending 1.4 miles southward of Gooseberry Neck, is a reef consisting of many large boulders, most of them baring a foot or less. The reef is in two large groups; the southerly group is the larger. Numerous covered rocks are well away from the visible part of the danger. A narrow ledge covered 5 to 14 feet extends about 0.4 mile northward from the visible part of Hen and Chickens. A buoy is north of the ledge. **Old Cock**, a rock awash, and **The Wildcat**, covered 5 feet and unmarked, are in the southern shoal area. The south edge of the shoal is marked by a buoy. Strangers are advised to stay outside the 5-fathom curve in this vicinity.

(169) **Sow and Pigs Reef**, much of which is dry or awash, extends about 1.5 miles west-southwestward from Cuttyhunk Island. Its outer end is marked by a lighted bell buoy. An unmarked rock strewn shoal, covered 20 feet, is 0.9 mile westward of Cuttyhunk Island. Numerous obstructions and rocks were reported to extend as much as 3 miles southward of Sow and Pigs Reef.

(170) **Ribbon Reef**, a detached ledge covered 18 feet, is about 1.5 miles northwestward of Cuttyhunk Island. **Coxens Ledge**, covered 28 feet and marked by a lighted bell buoy, is 1.2 miles northward of Ribbon Reef.

(171) **Mishaum Ledge**, a group of several rocky spots with a least depth of 8 feet, extends about 1.7 miles southward of Mishaum Point. It is marked by a lighted gong buoy off its southeast end. A lighted bell buoy marks a rocky shoal covered 22 feet about 1 mile north-northwestward of the north end of Penikese Island. An unmarked rocky shoal covered 18 feet is 0.5 mile north of the island.

(172)

Current

(173) The tidal currents in the passages between Buzzards Bay and Vineyard Sound have considerable velocity and require special attention. At Buzzards Bay Lighted Gong Buoy 3, the tidal current is rotary, turning clockwise. Tide rips occur when a sea is running against the current. Maximum velocities are about 0.5 knot. Minimum velocities average about 0.2 knot. See the Tide and Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(174)

Ice

(175) The head of Buzzards Bay and the harbors in that vicinity are generally closed to navigation during the winter. The approaches to the harbors on the eastern shore are rendered dangerous by drift ice. In severe winters the drift ice extends across the bay and joins the local formations on the western shore, forming an impassable barrier for short periods. Ice forms more rapidly in the bay with winds from north to west as the western shore forms a shelter from such winds. When the field ice extends

sufficiently out toward the channel as to be affected by the winds from north to west, the outer edges are broken up and carried off to the eastern or southern shore where they form drift ice. Under ordinary circumstances a northeast wind, if continued for 48 hours, will clear the bay of ice. Southerly winds, especially southeastern, diminish the extent and weaken the strength of the pack. Some of the lighted buoys are removed from station or replaced by unlighted buoys when endangered by ice.

(176) The southern side of Buzzards Bay from Cuttyhunk to Woods Hole has been discussed previously in this chapter.

(177)

Quissett Harbor to Smith Neck

(178) **Quissett Harbor**, 1.7 miles northeastward of the western entrance to Woods Hole, is used by small pleasure craft. **The Knob**, a small hillock on the north point of the entrance, and the homes on the eastern shore of the harbor are prominent. A standpipe, 1.2 miles northeastward of the entrance, is conspicuous.

(179) A seasonal lighted buoy marks the entrance, and buoys mark the entrance channel, thence private seasonal aids mark the best water to the northeast end of the harbor. In 1981, a depth of about 8 feet was reported available in the channel. Mariners are advised to steer a midchannel course through the entire entrance channel to avoid numerous rocks on both sides of the channel.

(180) Anchorage can be found in the middle of the harbor in depths of 11 to 18 feet, sticky bottom. Local craft generally moor in the northeastern part of the harbor off the boatyard at the town of **Quissett**. The boatyard has an L-shaped pier and a float that was reported to have about 15 feet alongside. Water, ice, moorings, some marine supplies, storage facilities and marine railways up to 40 feet are available; hull and engine repairs can be made. The **harbormaster** can be contacted through the boatyard.

(181) **Hamlin Point**, 2 miles north-northeastward of Quissett Harbor, is marked by a prominent hotel with twin cupolas. A shoal, covered 10 feet near its outer end, extends about 1 mile westward of the point. **Gifford Ledge**, covered 9 feet, is 1.4 miles north-northwestward of Hamlin Point. **Great Sippewisset Rock**, awash and marked by a private seasonal daybeacon, is 0.4 mile offshore about 1.1 miles northward of Hamlin Point. A shoal area, foul with rocks awash and covered, extends 0.3 mile offshore eastward of the daybeacon.

(182) **West Falmouth Harbor**, 5 miles northward of Woods Hole, has depths of 1 to 6 feet and bares in places at low water. The entrance is protected by a breakwater extending about 700 feet southward of **Little Island**, the north point of the entrance, and by a short jetty on the northwest end of **Chappaquoit Point**. A tower and the summer homes on Chappaquoit Point are prominent. The entrance is marked by a seasonal lighted bell buoy and an unlighted buoy on the south side and by an unlighted buoy on the north side; these buoys mark reefs that extend westward from both entrance points. Seasonal private buoys mark the channel in the harbor.

(183) In 1981, it was reported that 4 feet could be taken through the narrow, privately marked channel in the harbor to the anchorage basin off the town wharf at the village of **West Falmouth** on the east shore of the harbor. Depths of 5 feet were reported alongside the wharf in 1981. The **harbormaster** has an office at the town wharf. A 6 mph **speed limit** is enforced in the harbor.

(184) **Wild Harbor** (41°38.3'N., 70°38.9'W.), 7 miles northward of Woods Hole, is a small cove on the south side of **Nyes Neck** affording anchorage in northerly or easterly winds. A tower on Nyes Neck is prominent. The entrance is clear in midchannel, with depths of 13 to 20 feet inside. A seasonal lighted buoy marks the entrance, and buoys mark the shoals extending from the entrance points. The shores are foul, and the easterly part of the harbor is shoal. The reported depth in the privately dredged channel into **Silver Beach Harbor** to a small basin is about 3 feet, but is subject to shoaling. A stone jetty extends off the south side of the entrance to the basin. The basin is a **special anchorage.** (See **33 CFR 110.1** and **110.40**, chapter 2, for limits and regulations.)

(185) A town wharf and surfaced ramp are in the basin. In 1981, depths of 4 feet to bare were reported alongside the wharf. Ice and provisions are available.

(186) **Megansett Harbor**, the approach to the towns of **North Falmouth**, **Megansett** and **Cataumet**, is entered between Nyes Neck on the south and **Scraggy Neck** on the north. The natural channel is buoyed as far as the rock breakwater at Megansett. The breakwater is marked at the end by a light. A yacht club and a town wharf are just inside the breakwater. In 1981, depths of 4 to 5 feet were reported alongside the wharf; water is available. The harbor has extensive shoals and ledges, but by following the buoyed channel a draft of about 8 feet can be carried to an anchorage in the outer harbor in depths of 10 to 22 feet. Inside the breakwater, anchorage is available in 6 to 12 feet, taking care to avoid the shoals on the north side of the harbor and the rock awash near the center in 41°39'27"N., 70°37'31"W. **Cataumet Rock**, covered 6 feet and marked by a buoy, is on the south side of the entrance; Seal Rocks are on the north side and marked by a seasonal lighted buoy.

(187) **Fiddlers Cove** (41°38.9'N., 70°38.2'W.) is a small-craft harbor on the south shore of Megansett Harbor, about 0.5 mile east-southeastward of Cataumet Rock. A channel, privately dredged to a reported depth of 7 feet, leads southward to a marina and boatyard in a dredged basin on the east side of the cove. A seasonal lighted buoy marks the approach, and private buoys mark the channel. Gasoline, diesel fuel, ice, a pump-out station and wet and dry storage are available; lift capacity, 35 tons. Hull, engine and electronic repairs can be made. In April 2002, the reported approach and alongside depth was 7 feet.

(188) **Halftide Rock**, awash at low water, is about 500 yards southwestward of the end of the Megansett breakwater.

Rands Harbor, about 0.3 mile east of Fiddlers Cove, is a private boat basin with little or no water.

(189) **Squeteague Harbor**, northward of Megansett, is entered through a narrow channel from the head of Megansett Harbor. The privately marked channel had a reported depth of about 2 feet in 1981; however, depths of 5 to 7 feet are reported to be available in the channel to the harbor; local knowledge is advised. The village of **Cataumet** is on the northerly shore of the harbor.

(190) **Seal Rocks**, about 0.3 mile southwestward of Scraggy Neck, on the north side of Megansett Harbor entrance, are partly bare at half tide and marked by a buoy about 300 yards southwest of their southern end. Part of an old concrete barge is aground on the rocks. **Southwest Ledge**, extending about 0.7 mile westward of Seal Rocks, consists of two patches of shoals covered by 2 to 18 feet and marked by buoys on its northern, western and southern sides. A rock awash is in the northerly shoal.

(191) **Pocasset Harbor** and Red Brook Harbor share a common entrance between Scraggy Neck and Wings Neck. **Bassetts Island** separates Pocasset Harbor from Red Brook Harbor. Broken ground with depths of 17 to 19 feet in places extends across the entrance. Entering about 250 yards north of buoys marking the north side of Southwest Ledge, vessels of about 14-foot draft can anchor westward of Eustis Rock Buoy in depths of 20 to 30 feet. This anchorage is exposed to westerly winds. **Eustis Rock**, about 0.2 mile north of Scraggy Neck, is covered 5 feet and marked by a buoy. The area eastward of Eustis Rock to Bassetts Island is shoal.

(192) A narrow buoyed channel, with a reported depth of about 8 feet in 1981, leads north of Bassetts Island to Pocasset Harbor. **Barlows Landing**, at the northeast end of the harbor, has a depth of 1½ feet. A small-craft launching ramp is just south of the landing.

(193) **Hospital Cove**, about 0.2 mile southward of the southern end of Bassetts Island, is entered through a natural buoyed channel that leads southeastward from a point about 0.4 mile east-southeastward of Eustis Rock to a small anchorage in the cove. A depth of about 6 feet can be carried in the channel, and there are depths of 8 to 14 feet in the anchorage. In 1981, a shoal at the south tip of Bassetts Island was reported to be shifting westward in the vicinity of Buoy 3; caution is advised. A boulder reef extends northeastward from Scraggy Neck in the approach to the cove. Several private piers with depths of 5 to 8 feet alongside are in the cove.

(194) **Red Brook Harbor**, eastward of Bassetts Island, is approached from the northward through the channel that leads through Pocasset Harbor and from the southward through the channel that leads through Hospital Cove. **Hen Cove** is immediately northward of Red Brook Harbor. The channels are buoyed. In 1981, reported depths of 8 feet and 6 feet could be carried through the north and south channels, respectively. Because of numerous submerged rocks in and near the edges of the channel, local knowledge is advised.

(195) A marina and boatyard are on the east shore of Red Brook Harbor. These facilities can provide berths, electricity, gasoline, diesel fuel, pump-out, water, ice, marine supplies and boat storage and can make hull and engine repairs; electronic repairs can be made at the marina. Marine lifts to 60 tons and a 50-ton marine railway are also available.

(196) **Wings Neck**, 9 miles northward of Woods Hole, extends about 2 miles in a west-southwest direction into Buzzards Bay. The neck is a prominent peninsula, irregular and hilly. The anchorage areas southwestward of the neck are discussed with the Cape Cod Canal. A yacht club and private piers are on the east shore of the neck. Traffic signals for the Cape Cod Canal are displayed from a tower on Wings Neck. (See **33 CFR 207.20**, chapter 2, for details.)

(197) **Pocasset River**, between the northeastern end of Wings Neck and**Bennets Neck**, is entered from the north side of Wings Neck through a privately dredged channel that leads southward between two jetties to a highway bridge about 0.4 mile above the entrance. In 2001, a reported depth of 6 feet could be carried to the bridge. The bridge has a fixed span with a clearance of 7 feet. Only very small boats go above the bridge. A boatyard is on the south side of the river at the bridge. Water, berths with electricity, storage facilities, a 35-foot marine railway and a 5-ton lift are available; hull and engine repairs can be made. A town wharf and floats are on the north side of the river at the bridge.

(198) **Tobys Island**, just northward of the entrance of Pocasset River and on the south side of Phinneys Harbor, is connected to the mainland by a causeway.

(199) **Phinneys Harbor**, between Tobys Island on the east and **Mashnee Island** on the west, is approached from Buzzards Bay through a buoyed channel that leads along the northerly side of Wings Neck to another buoyed channel into the harbor. A light is at the bend of the approach channel. Depths of about 10 feet can be carried in the inner channel, and greater depths are available in the approach channel. The harbor is used as an anchorage by small boats. Mashnee Island, once an island, is now connected with the mainland by a landfill causeway. A group of rocks awash, marked by a buoy, is 0.2 mile off the east shore of the harbor, and another rock, covered 4 feet, and marked by a buoy, is 0.1 mile north of Tobys Island.

(200) The village of **Monument Beach** is on the east shore. A marina is at the long town pier in the cove in the southeastern corner of the harbor. Berthage in 5 feet is available at the pier. Gasoline, water, ice, electricity, a pump-out station and a surfaced ramp are available.

(201) **Back River**, a stream that is nearly bare except near its entrance, empties into the north side of Phinneys Harbor. Small craft sometimes anchor in the entrance. A railroad bridge and a highway bridge crossing the river about 0.2 mile above the mouth have fixed spans with a minimum clearance of 4 feet. A boatyard is on the south bank between the bridges. In 1981, a depth of about 1½

(208)

feet was reported available in the river to the boatyard. A forklift is used to haul out boats up to 26 feet at the yard, while boats up to 37 feet in length can be hauled out on a flatbed trailer at a paved ramp at Barlows Landing and then brought to the yard for hull and engine repairs or dry open or covered storage. A launching ramp and marine supplies are available at the yard.

(202) **Gray Gables** is on the north side of the head of the Back River.

(203) **Cape Cod Canal** is a deep-draft sea-level waterway connecting Buzzards Bay and Cape Cod Bay. The waterway is 15 miles long from Cleveland East Ledge Light to deep water in Cape Cod Bay. The canal shortens the distance between points north and south of Cape Cod by 50 to 150 miles and provides an inside passage to avoid Nantucket Shoals. The canal is maintained by the Federal Government as a free waterway. (See **33 CFR 207.20**, chapter 2, for the regulations governing the use, administration and navigation of the Cape Cod Canal.)

(204) **Tra f c lights** (red, green, and yellow) are located at the easterly canal entrance at Sandwich; at the Canal Electric Terminal basin on the south side of the canal at Sandwich; and at the westerly entrance of Hog Island Channel at Wings Neck. These signals apply to all vessels over 65 feet in length that desire to transit the canal. (See **33 CFR 207.20(h)**, chapter 2, for detailed information on signals.)

(205)

Prominent features

(206) **Cleveland East Ledge Light** (41°37'51"N., 70°41'39"W.), 74 feet above the water, is shown from a white cylindrical tower and dwelling on a red caisson on the east side of the entrance channel approaching Cape Cod Canal from Buzzards Bay. A racon is at the light station. A sound signal at the light is operated by keying the mircrophone five times consecutively on VHF-FM channel 83A. The railroad bridge over the canal at the village of Buzzards Bay and the highway bridge at Bourne are also prominent.

(207) Cape Cod Canal Breakwater Light 6 (41°46'47"N., 70°29'23"W.), 43 feet above the water, is shown from a red cylindrical tower on the end of the north breakwater at the east entrance to Cape Cod Canal from Cape Cod Bay. A mariner radio-activated sound signal is at the light, initiated by keying the microphone five times on VHF-FM channel 83A. The most prominent landmark when approaching from Cape Cod Bay is the tall lighted stack of the powerplant about 1.1 miles west-southwestward of Cape Cod Canal Breakwater Light 6. The high-level highway bridge across the canal at Sagamore, 2.5 miles west of the breakwater light, is also prominent. The breakwaters at the east entrance to the canal should not be confused with the smaller jetties at Sandwich Harbor, 1 mile to the southeastward, nor should the two white church spires back of Sandwich Harbor be mistaken for

(209)

the range structure marking the entrance to the canal.

(210)

Channels

(211) A federal project provides for a channel 32 feet deep through the Cape Cod Canal. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A. Deep-draft vessels should obtain the latest information as to available depths so as to pass through the canal during maximum stages of high water if the draft of the vessel is near the controlling depth. In 1981, it was reported that the east entrance was being privately maintained to a depth of 40 feet to the New England Petroleum Company wharf about 1.2 miles west of Cape Cod Canal Breakwater Light 6.

(212) The approach channels from both the west and the east are marked with lighted ranges and other navigational aids. The canal itself is lighted at night on both banks by mercury vapor lights, generally 500 feet apart.

(213)

Anchorages

(214) General anchorages are on each side of Cleveland Ledge Channel between Cleveland East Ledge Light and Wings Neck. (See **33 CFR 110.1** and **110.140 (b)** and **(d)**, chapter 2, for limits and regulations.) In 1971, a dangerous submerged rock was reported just inside the easterly edge of Anchorage D in about 41°40'05"N., 70°40'17"W. In 1984, an obstruction was reported in Anchorage C in about 41°40'00"N., 70°41'35"W.

(215) Mooring basins, with tieup dolphins, are at both ends of the canal. One is on the east side of Hog Island Channel abreast of Hog Island, where shoaling to bare in about 41°43'49"N., 70°37'53"W. was reported in 1979. The other is just inside the eastern entrance to the canal. A small-boat basin is on the south side of the channel just inside the eastern entrance to the canal; depths of 8 to 13 feet were available in the basin in 1969.

(217)

Current

(218) Daily predictions for the tidal current in Cape Cod Canal at the railroad bridge are available from the Tidal Current prediction service at *tidesandcurrents.noaa.gov*. Links to a user guide for this service can be found in chapter 1 of this book. Under ordinary conditions, the tidal current has a velocity of 4.0 knots on the flood, which sets eastward, and 4.5 knots on the ebb, which sets westward. Large differences in range and timing of the tide between Buzzards Bay and Cape Cod Bay cause strong currents in the canal. Tides may lower the canal level 2 feet below mean low water or even more if attended by heavy offshore winds.

(216)

Structures across Cape Cod Canal

Name•Description•Type	Location	Clear Width of Draw or Span Opening (feet)	Clear Height above Mean High Water (feet)	Information
Railroad Bridge (vertical lift)	41°44'31"N., 70°36'51"W.	500	135 (up) 7 (down)	Notes 1 and 2 Maintained in the raised position
Bourne/Route 25 Bridge (fixed)	41°44'51"N., 70°35'22"W.	500	135	
Overhead power cable	41°46'04"N., 70°33'58"W.		160	
Overhead power cable	41°46'08"N., 70°33'54"W.		165	
Sagamore/Route 3 Bridge (fixed)	41°46'33"N., 70°32'36"W.	500	135	

Note 1 – See **33 CFR 117.1** through **117.59** and **117.589**, chapter 2, for drawbridge regulations.
Note 2 – See **33 CFR 207.20**, chapter 2, for navigation regulations.

(219) Due to the strong tidal currents in the canal, especially during spring tides, low-powered vessels should await slack water or favorable current. Navigators are warned to be on the alert for possible "bank suction" and "bank cushion," the effects of which may cause a vessel to take a sudden and decided sheer.

(220)

Weather: Buzzards Bay and vicinity

(221) Fog is said to be less dense over Cape Cod Canal than outside, but at times a water vapor rises from the canal to such an extent that traffic has to be suspended. The canal proper never has been closed by ice, but occasionally Buzzards Bay and Cape Cod Bay become so congested with ice that navigation through the canal is prevented.

(222)

North Atlantic right whales

(223) Endangered North Atlantic right whales have been reported within the Cape Cod Canal and in the vicinity of the Canal's east entrance. The Cape Cod Marine Traffic Controllers provide information regarding North Atlantic right whale sightings and locations. The Northeast Marine Pilots distribute educational material to mariners in an effort to reduce right whale ship strikes. When right whales are present in Cape Cod Bay (peak season: December through May), vessels transiting Cape Cod Bay are urged to use Recommended Two-Way Whale Avoidance Routes to reduce the likelihood of collisions with right whales. (See North Atlantic right whales, indexed as such, in chapter 3 for more information on right whales and recommended measures to avoid collisions with whales.)

(224) All vessels 65 feet or greater in length overall (LOA) and subject to the jurisdiction of the United States are restricted to speeds of 10 knots or less in the Cape Cod Bay Seasonal Management Area between January 1 and May 15. The area is defined as all waters of Cape Cod Bay with a northern boundary of 42°04'56.5"N., 70°12'W., to 42°12'N., 70°12'W., thence due west back to shore. (See **50 CFR 224.105**, chapter 2, for regulations, limitations and exceptions.)

(225) The waters of Cape Cod Bay lie within the WHALESNORTH Mandatory Ship Reporting Area. Each self-propelled ship of 300 gross tons or greater entering WHALESNORTH must participate in the Mandatory Ship Reporting System (See **33 CFR 169**, chapter 2, for limits and regulations, and chapter 3 for sample reports). Sovereign immune vessels are exempt from the requirement to report but are encouraged to participate.

(226)

Pilotage, Cape Cod Canal and Buzzards Bay

(227) At the canal, Canal Traffic Control "WUA21" or "Cape Cod Canal Control" monitors VHF-FM channels 16, 13 and 14; usually works on 14. The Masters of all vessels required by the Coast Guard to carry a pilot are required to notify Canal Traffic Control prior to entering the waterway with information as specified in **33 CFR 207.20(k)**, Management of Vessels (see chapter 2).

(228) Pilotage is compulsory for Buzzards Bay for foreign vessels of 350 gross tons or more, U.S. vessels under register of 350 gross tons or more and towing vessels in-tow with single hull tank barges carrying 5,000 barrels or more of petroleum and hazardous cargoes. Federal and state pilots are available from Northeast Marine Pilots, Inc. and from Boston Coastwise Pilots. However, all pilotage that is compulsory under M.G.L. C. 103 shall be dispatched through the pilot dispatch office of Northeast Marine Pilots regardless of pilot group affiliation. Northeast Marine Pilots, Inc., Newport, RI 02840; *nemarinepilots.com*; telephone 401–847–9050 (24 hours), 800–274–1216; FAX 401–841–9052; email: dispatch@nemarinepilots.com. Boston Coastwise Pilots is in Winthrop, MA 02152; telephone 617–510–0082, 508–801–4904; email: boscopilots@aol.com. Twenty-four hour advance notice and 3 hour updates are requested.

(229) The Northeast Marine pilot boats that serve the canal and Buzzards Bay are the NORTHEAST I and NORTHEAST V. The NORTHEAST I is a 37-foot boat with a black hull and white superstructure. The NORTHEAST V is a 38-foot boat with a black hull and gray superstructure. Both boats have the word PILOT on both sides.

(230) The pilot boats NORTHEAST II and NORTH-EAST IV serve for boarding eastbound vessels. NORTHEAST II is a 47-foot boat and NORTHEAST IV is a 52 foot boat; both have unpainted aluminum hulls and superstructures

with the word PILOT in red on both sides. All Northeast pilot boats monitor VHF-FM channels 16, 13 and 10 and work on 13 or 10.

(231) Pilots meet westbound vessels off the eastern entrance to the Cape Cod Canal in Cape Cod Bay, ½ mile east of Lighted Bell Buoy CC, in approximate position 41°48.6'N., 70°27.0'W.

(232) Pilots meet eastbound vessels at the Brenton Reef Pilots Station, about 1.5 miles eastward of Narragansett Bay Entrance Lighted Whistle Buoy NB within an area bounded by:

(233) 41°23.6'N., 71°22.4'W.,

(234) 41°22.6'N., 71°22.0'W.,

(235) 41°24.2'N., 71°20.0'W.,

(236) 41°22.6'N., 71°20.6'W. This pilot boarding area is southward of a line extending from Point Judith to Sakonnet Point.

(237) Pilots also meet vessels in the vicinity of Point Judith Lighted Whistle Buoy 2, within a 1-mile radius circle centered in 41°17.2'N., 71°30.4'W.

(238) Eastbound vessels can also meet pilots in the pilot boarding area located about 1 mile northwest of Buzzards Bay Entrance Light in about 41°23'48"N., 71°02'01"W.

(239) Pilot services are generally arranged for in advance by ships' agents or directly by shipping companies.

(240)

Launch service

(241) Launch service is also available; arrange through Boston Coastwise Pilots or Northeast Marine Pilots, Inc.

(242)

Towage

(243) Tugs to 2,200 hp are based at the village of Buzzards Bay; arrangements for their services are usually made through ships' agents. Tugs monitor VHF-FM channel 13 when expecting a vessel and use channel 18A as a working frequency.

(244)

Coast Guard

(245) Cape Cod Canal Coast Guard Station is on the east side of the entrance to the canal about 1 mile northeastward of Sandwich Harbor.

(246)

Wharves

(247) State Pier, site of the **Massachusetts Maritime Academy**, on the north side of Cape Cod Canal, 0.6 mile below the railroad bridge at the village of Buzzards Bay, is 600 feet long with about 25 feet alongside the berthing face. In 1981, shoaling to 10 feet was reported in the basin off the berthing face in about 41°44'15"N., 70°37'35"W. In 1981, the reported controlling depth on the channel side of the pier was 25 feet. Permission to berth at the pier must be obtained from the academy. Vessels should not attempt to go alongside or leave the pier except at periods of slack water. Passing vessels are requested to proceed slowly to avoid damage to lines and other equipment at the pier.

(248) The New England Petroleum Company mooring platform, on the south side of Cape Cod Canal, 1.2 miles westward of Cape Cod Canal Breakwater Light 6, provides berthing for tank vessels up to 750 feet long with drafts of 40 feet. Vessels over 50,000 tons or 32-foot draft moor at high water slack during daylight hours only. Vessels under 50,000 tons moor at high water slack, day or night. Petroleum products are pumped to storage tanks ashore.

(249)

Supplies

(250) Gasoline, diesel fuel, water, ice, marine supplies and a paved ramp are available at the boat basin just inside the Cape Cod Bay entrance to the canal. Temporary berthage and anchorage are also available at the basin.

(251)

Communications

(252) Information on operating conditions, widths, depths or other data on the canal is available at all hours, day or night, by calling the canal office at Buzzards Bay at 508–759–4431.

(253) Vessels that are to transit the Cape Cod Canal will monitor channel 16 continuously to establish contact with traffic controllers. The vessels will be asked to switch to channel 12 or channel 14 as a working channel to pass information between the traffic controllers and the vessel. However, channel 13 may be used only when the above channels are not available.

(254) The radiotelephone at the Cape Cod Canal Office, Buzzards Bay, MA, is in continuous operation. Call letters are WUA-21, and the frequencies are channels 13, 16, 12 and 14. Vessels equipped for communication with the Cape Cod Canal Office are requested to keep their radiotelephone tuned to these frequencies.

(255) **Buttermilk Bay**, at the northeast end of Buzzards Bay, has depths of about 1 to 7 feet. A dredged marked channel and **Cohasset Narrows** connect the bay with Cape Cod Canal. A railroad bridge crossing Cohasset Narrows has a bascule span with a clearance of 6 feet; the bridge is kept in the closed position. A highway bridge, just above the railroad bridge, has a fixed span with a clearance of 8 feet. In 1992, a rock was reported near the railroad bridge in 41°44'46"N., 70°37'22"W. A marina is on the east side of Cohasset Narrows just south of the railroad bridge and can provide transient berths, gasoline, diesel fuel, water, ice, marine supplies, pump-out and a launching ramp. Several small piers for shallow-draft boats are in Buttermilk Bay.

(256) **Onset Bay**, between **Sias Point** on the north and **Hog Neck** on the south, is the approach to the village of **Onset**. A dredged marked channel leads westward from Cape Cod Canal along the southerly side of the bay to a turning basin off the village. Two anchorage areas, one on each side of the channel, are at the head of the channel.

(257) **Wickets Island** is a high and wooded islet in the middle of the bay. The buoys in the entrance channel are frequently towed under because of the strong currents. A

rock, covered 9 feet, is near the channel entrance about 75 yards northeast of Hog Island Channel Light 21. In 1981, two rocks, covered 4 to 5 feet, were reported on the north edge of the channel between Buoys 2 and 4; caution is advised.

(258) A **special anchorage** is in the northern part of Onset Bay. (See **33 CFR 110.1** and **110.45**, chapter 2, for limits and regulations.) Additional anchorages are available at the head of the dredged channel.

(259) A 5 mph **speed limit** is enforced above Wickets Island.

(260) The Onset town wharf, on the north side of the turning basin, has depths of about 14 feet at its face. The **harbormaster** has an office at the wharf. The harbormaster monitors VHF-FM channel 16 and uses channel 9 as a working frequency; call sign KYQ-833. Several small-craft facilities are on the north side of the bay along the southwesterly side of Long Neck.

(261) **East River** empties into Onset Bay southeast of Onset. A draft of 4 feet can be taken to Broad Cove above the highway bridge that connects Onset and Long Neck. The bridge has a fixed span with a clearance of 11 feet.

(262) **Stony Point Dike**, a sandspit breakwater about 5 feet high, extends about 1.8 miles south-southwesterly from **Cedar Island Point** to Abiels Ledge. The breakwater protects Hog Island Channel. Cleveland Ledge Channel Range Front Light is on the outer end of the breakwater. **Abiels Ledge**, between the channel and the south end of the dike, is covered 3 feet. **Dry Ledge**, 1 mile northwestward of Abiels Ledge, bares at half tide; it is marked by a buoy. **Little Bird Island**, 0.8 mile northward, is surrounded by uneven bottom with depths of 2 to 19 feet between it and the northerly shore of Buzzards Bay.

(263) **Wareham River**, which empties into the northern end of Buzzards Bay, is the approach to the town of **Wareham** on the west bank. **Great Hill**, wooded, 124 feet high, and about 1.5 miles southward of **Long Beach Point**, is prominent when approaching the river. A brick stack and a standpipe in Wareham are conspicuous. The buoyed channel to the town is crooked and twisting. A rock covered 6 feet is in the entrance about 350 yards south-southwest of Dry Ledge in about 41°41'59.9"N., 70°41'41.6"W., and several rocks and shoal spots covered 9 to 14 feet are within 0.75 mile west to southwest of the ledge. A shoal makes off southeasterly from **Cromeset Point**, 0.6 mile southward of Long Beach Point. It was reported that Long Beach Point covers at high water; caution is advised. The section near **Quahaug Bar**, north of Long Beach Point, is subject to shoaling. Depths shoal to 2 and 3 feet close to the buoyed channel. Small craft sometimes anchor just north of Long Beach Point. Highway and railroad bridges over the river above the wharves have 22-foot fixed spans with a clearance of 2 feet.

(264) The velocity of the current at the entrance to Wareham River is not great enough to materially interfere with a sailing vessel having a good breeze. During the first half of the ebb the current below the wharves of the town sets across the flats westward of the channel, and during the whole of the ebb it sets across the flats eastward of the channel below Long Beach Point. The river **ices** over for short periods during most winters. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(265) Vessels approaching Wareham River from Buzzards Bay pass 0.8 mile east of Bird Island Light and steer **351°** to the buoyed channel. Strangers should obtain local information regarding channel depths before navigating the river.

(266) The depth at the Wareham town landing was reported to be about 4½ feet in 1981. The Wareham Harbor Patrol patrols the harbor during the summer boating season and enforces a **speed limit** of 5 knots. A boatyard, on the western side of Wareham Neck about 0.3 mile below the bridges, has a marine railway that can handle craft up to 45 feet for hull and engine repairs or storage. Gasoline, diesel fuel, water, marine supplies and a 25-ton mobile hoist are available. In 1981, depths of about 6 feet were reported alongside the boatyard service float. Wareham has bus service.

(267) **Marks Cove**, on the west side of the channel to Wareham River between **Swifts Beach** on the north and **Cromeset Neck** on the south, has depths of 2 to 5 feet. The cove is used by small boats. In 1961, a 6-foot channel was dredged by the state through the cove from the 6-foot contour to **Cedar Island** at the entrance to **Cedar Island Creek**.

(268) **Weweantic River**, entered southward of Cromeset Neck, has a narrow and crooked channel partly obstructed by rocks. A reported depth of about 3 feet can be carried past the rocks and as far as the highway bridge, with local knowledge. The channel is not marked. An overhead power cable crossing the river about 1.4 miles above the mouth has a clearance of 61 feet. The highway bridge, which is 1.7 miles above the mouth, has two fixed spans with a clearance of 5 feet. The channel is through the northeasterly 45-foot span. The river is navigable for 2 miles above the highway bridge by small craft drawing less than 2 feet. A small marina is just above the bridge. Gasoline, water, ice, storage facilities, marine supplies, and hull and engine repairs are available. A flatbed trailer at the marina can haul out boats to 25 feet.

(269) From Wareham River to New Bedford the shore is indented sharply by rocks and ledges extending offshore nearly 2 miles in places.

(270) **Wings Cove**, between **Great Hill Point** southeast of Great Hill and **Piney Point** on the eastern side of **Sippican Neck**, has depths of 8 to 17 feet in its outer section. The cove affords protection from westerly winds; it is used only by small local craft.

(271) **Butler Point** is at the southern tip of Sippican Neck. Shoal water extends about 0.4 mile southward from the

(274)

Sippican Harbor, Massachusetts
Image courtesy of Marblehead Power Squadron (2009)

point to **Bird Island**, a round, low flat island marked by a light.

(272) Southward and westward of Bird Island are several buoyed dangers. **Bird Island Reef**, covered 15 to 18 feet, is about 0.4 mile east-southeastward. About 0.5 mile southwestward is **Centerboard Shoal**, covered 12 feet. **The Bow Bells**, isolated shoals covered 11 to 18 feet, are about 0.6 to 1 mile southwest of Centerboard Shoal. An unlighted gong buoy is about 1.15 miles southward of Bird Island.

(273) **Sippican Harbor**, scene of much pleasure-boat activity, makes into the north shore of Buzzards Bay about 3 miles southward of Wareham River. The harbor is the approach to **Marion**, a small town on the western shore. It is entered between Bird Island on the east and **Converse Point** on the west. Prominent features include the lighthouse on Bird Island and the conspicuous house and flagpole on Converse Point. The standpipe on Sippican Neck can also be seen for a considerable distance.

(275) The town dock at Marion, approached through a channel marked by private seasonal buoys, has reported depths of 4 to 5 feet alongside. Two boatyards at Marion provide limited guest moorings, electricity, gasoline, diesel fuel, water, ice, pump-out, launch ramp, marine lifts to 50 tons and boat storage and hull, engine and electronic repairs can be made. The harbormaster, who controls all mooring and anchoring in the harbor, monitors VHF-FM channel 68.

(276) **Mendells Rock** and **Seal Rocks** are shoal, rocky areas, northward of Converse Point extending up to 0.2 mile off the west shore of the harbor. **Planting Island**, a peninsula extending about 0.6 mile northwesterly from Sippican Neck, is on the eastern side of the harbor. At **Ram Island**, off Marion, the passage between the island and the western shore is less than 275 yards wide. The currents in the narrow portion of the channel have considerable velocity at times. **Little Island** lies on the western side of the channel about 0.2 mile northwestward of Ram Island. The buoyed channel has a reported depth of about 12 feet from the entrance to Marion.

(277) **Blankinship Cove** and **Planting Island Cove**, on the eastern side of Sippican Harbor, have a common entrance northward of Ram Island. They have general depths of 3 to 5 feet. **Meadow Island** separates the two coves. **Gibbs Rock**, marked by a private seasonal daybeacon, is 50 yards off the north point of Ram Island. A rock awash is charted 120 yards north of the daybeacon.

(278) **Hammett Cove** in the northeastern part of Sippican Harbor is shallow and used only by small local craft. The approach to the cove is marked by private seasonal buoys. Charted obstructions are close westward of the buoys.

(279) **Ice** usually closes Sippican Harbor for about a month or more each winter.

(280) **Aucoot Cove**, about 0.8 mile southwestward of Sippican Harbor, has depths of 10 to 19 feet. A 4-foot spot is near the center of the cove in about 41°40'23.2"N., 70°45'23"W., and the head of the cove is foul. The harbor is protected from all winds except southeast. An unmarked channel with depths of less than 1 foot leads to a boatyard west of **Haskell Island**. Local knowledge and a high tide are required to navigate to the boatyard, which can haul out craft up to 40 feet in length. The yard does general repairs and machine work. Gasoline, diesel fuel by truck and marine supplies are available. The other coves between Sippican and Mattapoisett Harbors are foul and seldom entered.

(281) **Mattapoisett Harbor**, about 3.5 miles southwest of Sippican Harbor and 5 miles northeastward of New Bedford Harbor, is the approach to the town of **Mattapoisett**. The harbor is used by numerous yachts during the summer. Although exposed to southeasterly winds, the ledges at the entrance somewhat break the sea from that direction. A light on **Ned Point** marks the approach. A standpipe is in the town. Vessels anchor between Ned Point and the wharves in 13 to 17 feet.

(282) The entrance between **Angelica Point** and **Strawberry Point** on the east and **Mattapoisett Neck** on the west is about 1.5 miles wide. A buoyed natural channel leads through the numerous rocks and ledges in the entrance to the anchorage area off the town. The channel has a depth of about 14 feet but because of the broken bottom, vessels should proceed with caution over areas where the charted depths are not more than 6 feet greater than the draft. Strangers should not attempt to enter at night.

(283) There are many shoals and rocks, most of them buoyed, off the points and in the entrance. Off the west side of the entrance, **Mattapoisett Ledge** extends about a mile southeasterly from Mattapoisett Neck. **Nye Ledge**, covered 7 to 18 feet, about 0.4 mile southeastward of Mattapoisett Ledge, is marked by a seasonal lighted bell buoy. In or near the entrance channel are **Gallatin Rock**, covered 10 feet; **Sunken Ledge**, covered 3 feet; **Snow Rock**, covered 5 feet; and **Barstow Rock**, covered 8 feet. Near the town wharf, a rock, covered 3 feet, is marked by a buoy.

(284) A **special anchorage** is in Mattapoisett Harbor. (See **33 CFR 110.1** and **110.45a**, chapter 2, for limits and regulations.)

(285) The stone wharf at the town has a reported depth of 6 feet alongside. Diesel fuel, gasoline, oil, water, ice, marine supplies and a surfaced ramp are available. A boatyard in town can handle craft to 60 feet long and 7 feet in draft at the town ramp. A 5 mph **speed limit** is enforced in the mooring areas.

(286) A boatyard in the harbor, just northwest of Ned Point, has limited guest berths but can provide gasoline, diesel fuel, electricity, water, ice, pump-out, a 35-ton marine lift and boat storage and hull, engine and electronic repairs. In 2012, a reported depth of 15 feet could be carried to the boatyard with 5 feet alongside.

(287) **Ram Island**, about 1.5 miles southwestward of Mattapoisett Harbor, is a low, grassy island connected to Mattapoisett Neck by a narrow shoal. Rocks and shoal water surround the island.

(288) **Cormorant Rock**, 0.9 mile southeastward of Ram Island, bares at half tide and is marked by a daybeacon. Ledges with very little water over them surround the daybeacon at a distance of 150 yards. A rock, covered 12 feet, is about 0.2 mile northeastward of the daybeacon; depths of 18 to 21 feet extend 0.2 mile southward. The channel between the rock and Ram Island has depths of about 15 feet.

(289) **Nasketucket Bay** is entered between Cormorant Rock on the east and **West Island** on the west. Northward and westward of West Island the bay is greatly obstructed by rocks and small islands. Because of these obstructions, only small craft proceed through the bay to **Little Bay**, at the head, or up **Nasketucket River**. The edges of Little Bay are foul, but excellent anchorage in all but strong southerly winds is available in the center of the bay in 3 to 6 feet, sticky mud. The entrance is made treacherous by obstructions and wind and should not be attempted without local knowledge.

(290) Numerous rocks, including **Whale Rock**, are on the east side of West Island at the entrance. A causeway connects the western side of West Island with **Long Island**. A fixed span in the causeway has a clearance of 5 feet. The depth at the bridge is 6 feet. On the east side of Long Island just north of the bridge is a marina with reported approach and alongside depths of 4 feet in 2011. Berths, moorings, electricity, gasoline, diesel fuel, water, ice, marine supplies, sewage pumpout, a surfaced launching ramp, a 40-ton marine lift and boat storage and hull, engine and electronic repairs are available. The approach from southward to the west side of West Island runs among many sunken rocks and shoals and is very dangerous. The causeway between Long Island and **Sconticut Neck**, the neck of land forming the east side of New Bedford Harbor, completely blocks passage between the two.

(291) Bare rocks and shoaling extend about 1 mile southerly of West Island, and from there to the New Bedford Harbor entrance are numerous isolated rocks and ledges, the most dangerous of which are buoyed. **Mosher Ledge**, about 1.1 miles south of **Wilbur Point**, has a least depth of 6 feet. Strangers should stay south of the buoys marking these dangers.

(292) **New Bedford Harbor**, a tidal estuary at the mouth of **Acushnet River** on the northwestern side of Buzzards Bay, is the approach to the city of New Bedford and the town of Fairhaven. The harbor is about 166 miles from The Battery at New York via Long Island Sound and 83 miles from Boston via Cape Cod Canal. The harbor includes all the tidewater lying northerly of a line from Clarks Point at the southern extremity of New Bedford to Wilbur Point at the southern end of Fairhaven and extends to the head of navigation on Acushnet River at Acushnet. The outer harbor consists of the area south

of the hurricane barrier at Palmer Island, and the inner harbor consists of the area north of the barrier to a short distance above the New Bedford-Fairhaven Bridge.

(293) **New Bedford** is a manufacturing city on the west side of the Acushnet River. **Fairhaven** is on the east side of the river. Principal shipping includes receipt of general cargo and frozen fish; exports are general cargo. Commercial fishing craft operate from the ports. The deepest draft entering is about 30 feet at high water.

(294) The approach from Buzzards Bay and the entrance to New Bedford Harbor are much obstructed by ledges and shoals, between which are several channels leading to the dredged entrance. The bottom is very broken, characterized by large boulders; vessels should proceed with caution when crossing areas off the general track when the charted depths are not more than 6 to 8 feet greater than the draft.

(295)

Prominent features

(296) From the main channel numerous landmarks can be seen on the westerly side. Dumpling Rocks Light 7 off Round Hill Point, about 3 miles west of the channel, is conspicuous. **Clarks Point**, on the west side of the channel, is marked by a granite fort. About 0.7 mile northeast of the point is an abandoned lighthouse. Although there are no landmarks on Sconticut Neck, **Fort Phoenix** is a promontory fairly conspicuous just east of the channel, almost opposite Palmer Island. Several church spires are prominent in Fairhaven. A tall radio tower is on **Popes Island** in the inner harbor. A private light is on the northeast point of **Palmer Island**, about 0.2 mile inside the hurricane barrier. The lights marking the eastern and western sides of the hurricane barrier are also prominent.

(297)

COLREGS Demarcation Lines

(298) The lines established for Buzzards Bay and Vineyard Sound are described in **33 CFR 80.145**, chapter 2.

(299)

Channels

(300) A federal project provides for a 30-foot deep channel from Buzzards Bay to the turning basin just above the New Bedford-Fairhaven3 T0.5 (basin)0.58ier is on

(317)

Structures across Acushnet River

Name•Description•Type	Location	Clear Width of Draw or Span Opening (feet)	Clear Height above Mean High Water (feet)	Information
US 6/New Bedford-Fairhaven Bridge				
New Bedford to Fish Island (fixed span)	41°38'17"N., 70°55'14"W.	52	9	
Fish Island to Popes Island (swing span)	41°38'20"N., 70°55'04"W.	94 (east draw) 95 (west draw)	6	Notes 1 and 2 Call sign WHH-238
Popes Island to Fairhaven (fixed span)	41°38'30"N., 70°54'36"W.	95	6	
Interstate 195 Bridge (fixed)	41°39'17"N., 70°55'00"W.	62	8	
Coggeshall Street Bridge (fixed)	41°39'22"N., 70°55'03"W.	62	8	

Note 1 – See **33 CFR 117.1** through **117.59** and **117.585**, chapter 2, for drawbridge regulations.
Note 2 – Bridgetender monitors VHF-FM channel 16 and works on channel 13.

the entrance passages are marked with buoys. The chart is the best guide.

(313) **Dumpling Rocks**, bare and covered, extend 0.4 mile southeastward from Round Hill Point. A light is on the easterly rock and a gong buoy marks the southeastern portion of the shoal area around the rocks.

(314) **Wilkes Ledge**, 1.8 miles southeastward of Round Hill Point, is the southernmost danger at the entrance to the harbor. It is covered 9 feet with a wreck near the easterly part; a lighted buoy is close south-southwestward of the wreck.

(315)

Regulated navigation area

(316) A **regulated navigation area** has been established south of the western hurricane barrier. (See **33 CFR 165.1** through **165.13** and **165.125**, chapter 2, for limits and regulations.)

(318)

Current

(319) Tidal currents are weak. From a series of current observations conducted by the Corps of Engineers over a 2-day period in 1965 at the center of the navigation opening of the New Bedford Hurricane Barrier, it was revealed that the maximum flooding and ebbing velocities were about 2.4 knots; average flood setting 344° and the ebb 144°. During flooding or ebbing a slight set towards the east barrier abutment may be experienced. During this same period it was further revealed that the time of slack water occurred about 30 minutes before the time of low or high water, that the maximum ebbing velocity occurred about 2 hours after the time of high tide, that the maximum flooding velocity occurred about 4 hours after the time of low tide, and that, generally, the maximum current occurred at about the same time as the most rapid change in the vertical height of the tide was taking place.

(320)

Weather, New Bedford Harbor and vicinity

(321) The prevailing winds during the winter are from north to west, and during the summer from south to southwest. Thick fog is reported to close in quickly with little warning in New Bedford Harbor.

(322)

Ice

(323) The channels and anchorage area usually are navigable throughout the year, although in prolonged periods of extreme cold weather the harbor as well as all of Buzzards Bay may be closed to navigation because of ice. Such conditions are infrequent and of short duration. Steamers generally can make their way through the ice in the harbor.

(324)

Pilotage, New Bedford

(325) Pilotage is compulsory for foreign vessels of 350 gross tons or more and U.S. vessels under register of 350 gross tons or more. Pilotage for New Bedford is available from Northeast Marine Pilots, Inc., Newport, RI 02840; telephone 401–847–9050 (24 hours), 800–274–1216; FAX 401–847–9052; email: dispatch@nemarinepilots.com.

(326) Pilots meet westbound vessels off the eastern entrance to the Cape Cod Canal in Cape Cod Bay, ½ mile east of Lighted Bell Buoy CC, in approximate position 41°48.6'N., 70°27.0'W.

(327) Pilots meet eastbound vessels at the Brenton Reef Pilot Station, about 1.5 miles eastward of Narragansett Bay Entrance Lighted Whistle Buoy NB within an area bounded by

(328) 41°23.6'N., 71°22.4'W.,

(329) 41°22.6'N., 71°22.0'W.,

(330) 41°24.2'N., 71°20.0'W.,

(331) 41°22.6'N., 71°20.6'W. This pilot boarding area is southward of a line extending from Point Judith to Sakonnet Point. Should weather or other conditions prevent pilot boarding in the above location, other arrangements may be made with the pilot office.

(332) The pilot boats NORTHEAST I and NORTHEAST III serve the canal and are owned and operated by Northeast Marine Pilots, Inc. NORTHEAST I is a 38-foot boat with black hull and white superstructure and the word PILOT on the sides; NORTHEAST III is a 33-foot boat with black hull and white superstructure.

(333) The pilot boats NORTHEAST II, NORTHEAST III and NORTHEAST IV serve for boarding eastbound vessels. NORTHEAST II is a 47-foot boat and NORTHEAST IV is a 52 foot boat; both have unpainted

aluminum hulls and superstructures with the word PILOT in red on both sides. All Northeast pilot boats monitor VHF-FM channels 16, 13 and 10 and work on 13 or 10.

(334) Pilot services are generally arranged for in advance by ships' agents.

(335)

Towage

(336) Oceangoing vessels usually require tug assistance when docking and undocking. Tugs up to 2,200 hp are based at New Bedford, and arrangements for their services are usually made through ships' agents. Tugs monitor VHF-FM channel 13 when expecting a vessel and use channel 18A as a working frequency.

(337) New Bedford is a **customs port of entry.**

(338)

Quarantine, customs, immigration and agricultural quarantine

(339) (See chapter 3, Vessel Arrival Inspections, and Appendix A for addresses.)

(340) **Quarantine** is enforced in accordance with regulations of the U.S. Public Health Service. (See Public Health Service, chapter 1.)

(341) New Bedford has several hospitals.

(342) **Coast Guard** vessels moor at the State Pier.

(343)

Harbor regulations

(344) The New Bedford Harbor Development Commission, through the **harbormaster**, enforces the harbor regulations. The state pier traffic manager is the state authority who directs anchoring, berthing and movement of vessels and discharging operations at the state pier. Vessels are expected to proceed slowly in the vicinity of the piers. State laws forbid pollution and dumping of refuse and rocks inside the harbor. The harbormaster has an office just north of the state pier.

(345)

Wharves

(346) The New Bedford waterfront has many piers and wharves. The fishing industry uses most of these facilities. Only the deep-draft facilities are described, and the alongside depths for these facilities are reported; for information on the latest depths contact the operator. All of the facilities described have highway connections, and most have rail connections. Water is available at most piers and wharves. Cargo in the port is usually handled by ship's tackle. A 250-ton floating "A" frame derrick is available for heavy lifts by prior arrangement.

(347) New Bedford South Terminal Wharf: 500 yards westward of Palmer Island, 1,600 feet long, 30 feet alongside, 250,000 cubic feet of refrigerated storage, receipt of seafood products, owned by several seafood companies.

(348) Commonwealth Edison Co. Pier: 300 yards northward of South Terminal Wharf; north side 740 feet long, with dolphins; 30 feet alongside; receipt of petroleum products; vessels usually berth with bow inshore; owned by Commonwealth Electric and operated by New England Petroleum Corp.

(349) State Pier: 500 yards northward of New Bedford Gas and Edison Light Co. Pier; face 450 feet long, north side 600 feet long, south side 775 feet long; 30 feet alongside; 125,000 square feet covered storage; receipt and shipment of general cargo; owned and operated by the Commonwealth of Massachusetts, Division of Waterways.

(350) Maritime Terminal Wharf: westward of Fish Island; 600 feet long; 31 feet alongside; 3 million cubic feet of refrigerated storage; receipt of frozen food, fish, and chilled foodstuffs; shipment of general cargo; owned and operated by Maritime Terminal, Inc.

(351) Bridge Terminal Wharf: northeast side of Fish Island; 450 feet long; 28 feet alongside; 500,000 cubic feet of refrigerated space; receipt of frozen and chilled foodstuffs; owned and operated by Bridge Terminal, Inc.

(352) Frionor Processing and Distribution Center Wharf: 200 yards northwest of Fish Island; 580 feet long; 25 to 28 feet alongside; 63,400 square feet of refrigerated space, 57,500 square feet of freezer space, 34,700 square feet of covered storage space; receipt of frozen fish; owned and operated by Frionor Norwegian Frozen Fish Ltd.

(353) New Bedford North Terminal Wharf: 400 yards northwest of Fish Island, 1,000 feet long; 30 feet alongside, 14 acres of open storage, owned by New Bedford Harbor Development Commission and operated by various tenants.

(354)

Supplies

(355) Gasoline, diesel fuel, water, provisions, marine supplies of all kinds and sewage pumpout are available. Diesel oil and marine bunker fuels are available by truck. The water is excellent for drinking and boiler use; a water boat services craft at anchor.

(356)

Repairs

(357) There are several boatyards at Fairhaven that can make hull, engine and electronic repairs; storage facilities are also available. The largest marine railway in the area can handle vessels up to 210 feet. Lifts to 99 tons are available. Several repair firms in New Bedford are available for above-the-waterline repairs and engine repairs. Derrick lighters, some with air compressors and diving equipment, are also available.

(358)

Communications

(359) There is only rail freight service to Boston and frequent bus service to Providence, Boston and New York. A mail and passenger boat makes trips to Cuttyhunk twice weekly in the winter and daily in the summer. Seasonal passenger ferry service is also available to Martha's Vineyard. Air service is available to Boston, New York, Martha's Vineyard, Nantucket and Cape Cod.

(360) The coast between New Bedford Harbor and the entrance of Sakonnet River is fringed with extensive

shoals, many of them rocky and a considerable number of them well offshore. The entrances to several inlets are shoal and are used only by local fishing and pleasure boats.

(361) **Clarks Cove**, between New Bedford Harbor and Apponagansett Bay, affords anchorage in depths of 12 to 22 feet. It is exposed to southerly weather and is seldom used. Several small piers can accommodate small craft. Several rocks are off **Moshers Point** on the west side of the cove.

(362) **Apponagansett Bay,** about 2 miles southwestward of Clarks Point, shelters numerous pleasure craft and a few fishermen in the summer, but the bay is insecure in southeasterly gales. **Nonquitt** and **Bayview** are villages on the south side of the entrance and **South Dartmouth** is on the northerly shore. **Padanaram Breakwater** is marked on the southern end by a light. The approach to the bay is obstructed by numerous ledges and rocks, and strangers should enter only in the daytime with clear weather. Inside the breakwater, the channel is marked by buoys. **Dartmouth Rock**, covered 4 feet, is on the northeast side of the channel. Private seasonal anchorage buoys mark the area off South Dartmouth, which is usually very crowded in the summer. A highway bridge at the village has a swing span with a channel width of 31 feet and a clearance of 8 feet. (See **33 CFR 117.1** through **117.59** and **117.587**, chapter 2, for drawbridge regulations.) Above the bridge, small craft anchor in a narrow channel near the eastern shore.

(363) The **harbormaster** controls anchoring and berthing in the harbor and can be contacted through the police department or VHF-FM channel 16. The **speed limit** in the harbor is 5 mph.

(364) Southward of the bridge is a landing at a yacht club with reported depth of 10 feet alongside. Diesel fuel, gasoline, water, ice and some marine supplies are available. Two nearby boatyards and a marina can provide limited guest berths, storage complete marine supplies and hull, engine, electronic, rigging and sail repairs. The largest marine railway can handle craft to 55 feet; mobile hoists to 35 tons are also available. In 1981, depths of 3 to 11 feet were reported at the slips.

(365) **Round Hill Point**, about 3.5 miles southwestward of Clarks Point, is marked by a prominent round hill.

(366) Between Round Hill Point and **Salters Point**, 1.1 miles southwestward, **Hunts Rock Breakwater** extends 270 yards in a northeast-southwest direction.

(367) **Mishaum Point**, 1.9 miles southwestward of Round Hill Point, is the southern point of **Smith Neck**. Shoal water extends about 0.2 mile off the point.

(368)

Slocums River to Briggs Point

(369) **Slocums River**, westward of Mishaum Point, has a bar at the entrance nearly bare at low water. The channel inside is narrow, unmarked and little used. **Slocums Ledge**, extending 0.6 mile westward of Mishaum Point, covered 2 to 7 feet, is marked by a buoy. **Pawn Rock** uncovers 3 feet and is 0.2 mile easterly of **Barneys Joy Point**, the point on the west side of the river entrance.

(370) **Gooseberry Neck**, about 4 miles southwestward of Mishaum Point, is marked by several prominent towers. The neck, irregular and elongated, extends about 1 mile southward from **Horseneck Beach** to which it is joined by a narrow roadway over rock fill. The water surrounding the neck is very foul.

(371) Hen and Chickens and the dangers southward of it have been previously discussed under the entrance to Buzzards Bay.

(372) In addition to Hen and Chickens, numerous rocks and reefs surround Gooseberry Neck. Shoal water extends 0.6 mile southwestward of the neck to **Lumber Rock**, covered 4 feet and marked by a buoy, and over 0.5 mile westward to **Browing Ledge**, covered 6 feet. **Little Southwest Rock** is about 0.3 mile northeastward of Lumber Rock.

(373) **Westport River** empties into the large bight between Gooseberry Neck and Sakonnet Point. The mouth of the river is between **Horseneck Point**, 2.7 miles northwest of Gooseberry Neck, and **The Knubble**, a protruding mound of granite marked by a light about 0.2 mile south of Horseneck Point. The river is the approach to **Westport Harbor**, the area just inside the entrance; the village of **Westport Point**, on the north shore of the east branch of the river; and the village of **Acoaxet**, westward of The Knubble. Fishing and pleasure boats use the river as far as Westport Point.

(374) A dredged entrance channel leads northwest and around Horseneck Point into Westport Harbor. The channel is narrow, crooked and marked by buoys. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A. Depths near the entrance are continually changing; mariners are advised to seek local knowledge. Numerous rocks are in the channel below the bridge at Westport; caution is advised.

(375) Boats should not try to enter during strong southerly winds as heavy seas break over the entrance bar. About 1 mile above the entrance the river divides into two branches. The west branch is shallow, with a narrow channel marked by private seasonal buoys, and is used by local craft to opposite **Toms Point**, about 1.6 miles above the entrance. Above the Westport Point bridge, the east branch is marked by private seasonal buoys as far as the Hix Bridge, 4.7 miles above Westport Point. A reported depth of about 4 feet can be carried to this bridge with local knowledge.

(376) A highway bridge with a 49-foot bascule span and a clearance of 21 feet at the center crosses the river at Westport Point, about 2 miles above the mouth. (See **33 CFR 117.1** through **117.59** and **117.620**, chapter 2, for drawbridge regulations.) Hix highway bridge, about 6

miles above the mouth, has a fixed span with a clearance of 7 feet. An overhead power cable on the north side of Hix bridge has a clearance of 37 feet.

(377) Approaching Westport River, boats must take care to pass westward of the dangers off Gooseberry Neck and eastward of the dangers off the river entrance. Numerous rocks and ledges are southward of the entrance to the river. **Twomile Rock**, 1 mile southeastward of Westport Harbor Entrance Light 7 on The Knubble, is marked by a daybeacon. Shoals with depths of 5 to 18 feet are southeasterly of the rock. A buoy is 0.35 mile south-southeast of the rock. **Halfmile Rock**, 3 feet high, is 325 yards southeast of the light on The Knubble. The shoal water surrounding the rock is marked by a buoy. The area south of The Knubble is very foul. Other unmarked dangers include **Twomile Ledge**, extending 1 mile south of The Knubble, and covered 2 to 12 feet; **Joe Burris Ledge** covered 14 feet, midway between Halfmile Rock and Twomile Rock, and **Pinetree Ground**, about 1 mile south of Twomile Rock, covered 25 to 30 feet.

(378) The shore in this vicinity should be given a berth of about 1.3 miles to avoid numerous rocks and ledges extending about 1 mile offshore for 2.5 miles westward of Westport Harbor.

(379)

Current

(380) The tidal current in the entrance has a velocity of 2.5 knots, and caution is recommended when navigating the river. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(381) Two piers, used by fishing and pleasure boats, are at Westport Point. These piers have reported depths of about 10 feet at their faces. Berthing at the piers is under the control of the **harbormaster,** who can be contacted through the town hall or police department.

(382) A 5 mph **speed limit** is enforced in the harbor.

(383) A boatyard and a marina are in the harbor. The boatyard is on Horseneck Point about 0.5 mile west of the highway bridge. The marina is on Westport Point about 0.1 mile west of the highway bridge. Berths, moorings, gasoline, diesel fuel, water, ice, launching ramps and marine supplies are available. The largest marine railway is at the boatyard and can handle craft up to 60 feet for hull and engine repairs and dry open or covered storage.

(384) **Quicksand Point** is about 1.5 miles west of The Knubble. The boundary line between Massachusetts and Rhode Island is near the point. **Cutty Wow Rock**, awash at low water, is 1 mile southwestward of the point.

(385) **Briggs Point**, 2 miles southwestward of Quicksand Point, is surrounded by shoals and rocks. **Halfway Rock**, 2 feet high, is 0.4 mile southeastward of the point.

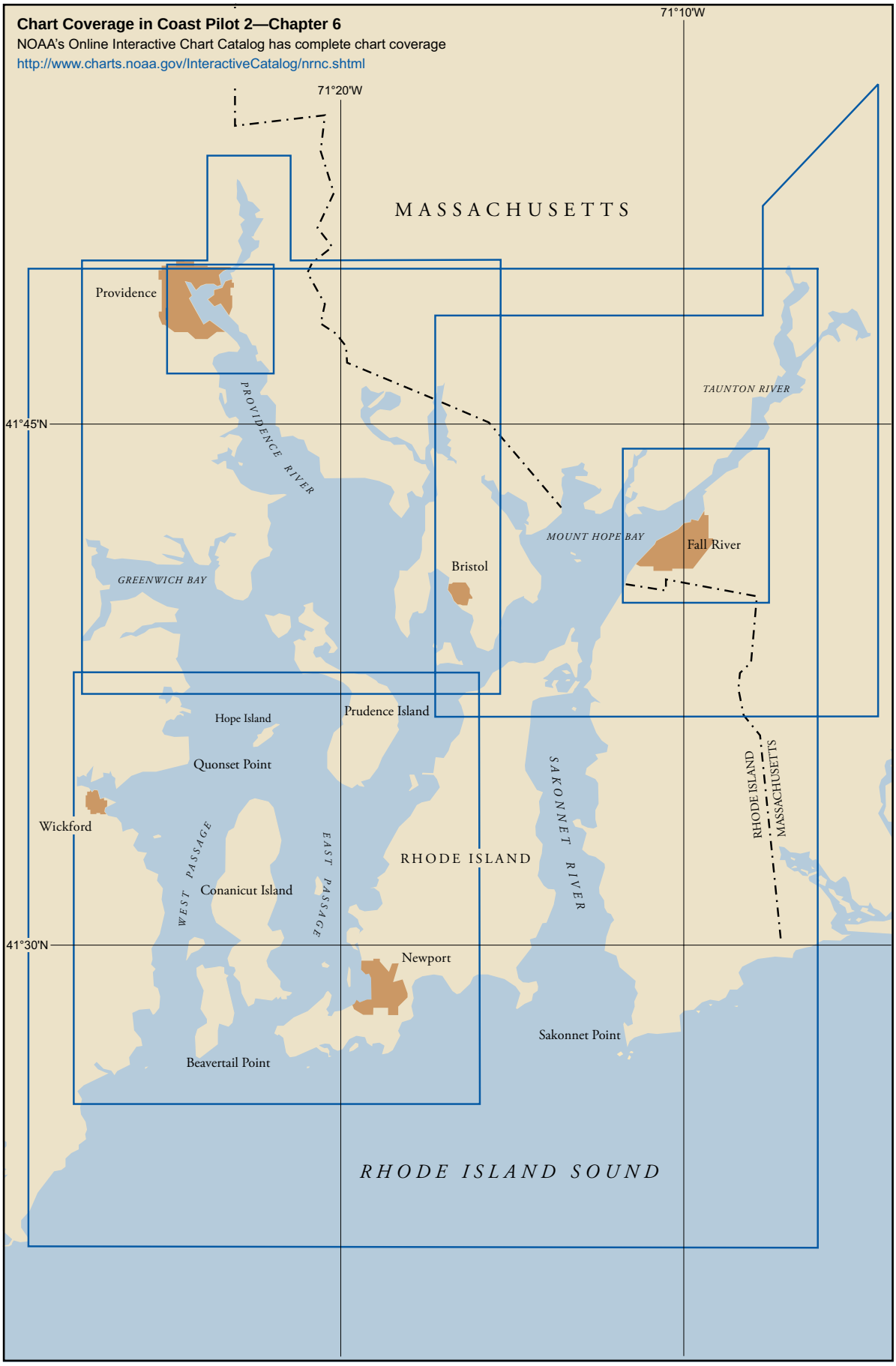


Chart Coverage in Coast Pilot 2—Chapter 6
NOAA's Online Interactive Chart Catalog has complete chart coverage
http://www.charts.noaa.gov/InteractiveCatalog/nrnc.shtml
71°10'W
71°20'W
MASSACHUSETTS
Providence
PROVIDENCE RIVER
TAUNTON RIVER
41°45'N
GREENWICH BAY
MOUNT HOPE BAY
Fall River
Bristol
Hope Island
Prudence Island
Quonset Point
Wickford
WEST PASSAGE
EAST PASSAGE
RHODE ISLAND
SAKONNET RIVER
RHODE ISLAND
MASSACHUSETTS
Conanicut Island
41°30'N
Newport
Sakonnet Point
Beavertail Point
RHODE ISLAND SOUND

Narragansett Bay

(1) This chapter describes the Sakonnet River, Narragansett Bay, Mount Hope Bay and Taunton and Providence Rivers. Also discussed are the ports of Newport, Fall River and Providence, as well as the numerous other yachting and fishing centers in this area.

(2)

COLREGS Demarcation Lines

(3) The lines established for this part of the coast are described in **33 CFR 80.145**, chapter 2.

(4)

No-Discharge Zone

(5) The State of Rhode Island, with the approval of the Environmental Protection Agency, has established a No-Discharge Zone (NDZ) covering all coastal waters of Rhode Island, extending 3 miles offshore.

(6) Within the NDZ, the discharge of sewage, whether treated or untreated, from all vessels is prohibited. Outside the NDZs, discharge of the sewage is regulated by **40 CFR 140** (see chapter 2).

(7)

Narragansett Bay

(8) **Narragansett Bay**, opening into the north side of **Rhode Island Sound** 17 miles westward of Buzzards Bay entrance, is the approach to the cities of Newport, Providence, Fall River and Taunton, as well as numerous towns and villages. **Rhode Island** (Aquidneck Island), the largest island in the bay, forms the eastern shore of the bay proper. The entrance is between Brenton Point, the southwestern part of Rhode Island, on the east, and Point Judith Neck on the west. The bay is about 18 miles long from the entrance to the mouth of Providence River. Navigation of the bay is easy during day or night in clear weather as it is marked by navigational aids. The large **Conanicut Island** and **Prudence Island**, and several smaller islands, divide the bay into two passages. Sakonnet River, although not a part of Narragansett Bay, is included with it in the following discussion.

(9) East Passage is good for a least depth of about 60 feet for about 11 miles up the marked channel to the entrance of the dredged channel to Providence. West Passage is the approach to Dutch Island Harbor, Wickford, Greenwich Bay and Providence River. Vessels of over 16-foot draft rarely go above Dutch Island Harbor without a pilot, but vessels of 16-foot draft or less should have no difficulty in going to the head of the bay and Providence River. Sakonnet River is good for a depth of 16 feet from the mouth to Mount Hope Bay.

(10) **Tra f c Separation Scheme Narragansett Bay** has been established in the approach to Narragansett Bay through Rhode Island Sound.

(11) The scheme is composed basically of **directed tra f c lanes** each with one-way inbound and outbound traffic lanes separated by a **de f ned tra f c separation zone,** and two **precautionary areas,** one at the southern end and the other at the northern end of the directed traffic lanes and separation zones. The scheme is recommended for use by vessels approaching or departing from Narragansett Bay but is not necessarily intended for tugs, tows or other small vessels that traditionally operate outside of the usual steamer lanes or close inshore.

(12) **The Tra f c Separation Scheme has been designed to aid in the prevention of collisions at the approaches to major harbors but is not intended in any way to supersede or alter the applicable Navigation Rules. Separation zones are intended to separate inbound and outbound tra f c lanes and be free of ship tra f c and should not be used except for crossing purposes. Mariners should use extreme caution when crossing tra f c lanes and separation zones.** (See **33 CFR 167.1** through **167.15** and **167.100** through **167.103**, chapter 2, for limits and regulations and Traffic Separation Schemes, chapter 1, for additional information.)

(13) The **southern precautionary area** in the southwest part of Rhode Island Sound has a radius of 5.4 miles centered on 41°06'00"N., 71°23'18"W., excluding those areas of the circle bounded by imaginary lines extending between the outer limits of the inbound and outbound traffic lanes. (Note that the southern precautionary area is common to the Traffic Separation Schemes for the approaches to both Narragansett Bay and Buzzards Bay. The Traffic Separation Scheme for the approach to Buzzards Bay is described in chapter 5.)

(14) The **separation zone** is a 2-mile-wide zone centered upon the following positions:

(15) (i) 41°22'42"N., 71°23'18"W.,

(16) (ii) 41°11'06"N., 71°23'18"W.

(17) The **inbound tra f c lane** is a 1-mile-wide lane with a length of about 11.5 miles. Entering the traffic lane at a point in about 41°11'06"N., 71°21'24"W., a course of **000°** follows the centerline of the traffic lane to a junction with the northern precautionary area.

(18) The **outbound tra f c lane** is a 1-mile-wide lane with a length of about 11.5 miles. Entering the traffic lane at a point in about 41°22'39"N., 71°25'24"W., a course of **180°** follows the centerline of the traffic lane to a junction with the southern precautionary area.

(19) The **northern precautionary area** has a 3.55-mile radius centered on a point in about 41°25'36"N., 71°23'18"W., excluding those areas of the circle bounded by imaginary lines extending between the outer limits of the inbound and outbound traffic lanes.

(20) A 2-mile-wide **restricted area** extends from the northern limits of the Narragansett Bay Approach traffic separation zone to 41°24.7'N. This restricted area within the precautionary area will only be closed to vessel traffic by the Naval Undersea Warfare Center Division, Newport, during periods of daylight and optimum weather conditions for torpedo range use. The closing of the restricted area will be indicated by the activation of red flashing lights on naval vessels supporting the torpedo range activities. There would be no vessel restrictions expected during inclement weather or when the torpedo range is not in use.

(21) The Traffic Separation Scheme is not buoyed.

(22) **Narragansett Bay Entrance Lighted Whistle Buoy NB** (41°23'00"N., 71°23'21"W.) is at the north end of the separation zone and is equipped with a racon.

(23) A **safety zone** has been established about 2 miles northward of Buoy NB for liquefied petroleum gas (LPG) vessels. (See **33 CFR 165.20**, **165.23**, and **165.121,** chapter 2, for limits and regulations.)

(24) **Recommended Vessel Route (Narragansett Bay)** has been established in the approach to Narragansett Bay through Rhode Island Sound.

(25) The U.S. Coast Guard Captain of the Port, Providence, in cooperation with the Southeastern Massachusetts and Rhode Island Port Safety and Security Committees, has established a Recommended Vessel Route for deep draft vessels and tugs/barges transiting Rhode Island Sound, Narragansett Bay and Buzzards Bay. Deep draft vessels and tugs/barges are requested to follow the designated routes. These routes were designed to provide safe, established routes for these vessels, to reduce the potential for conflict with recreational boaters, fishing gear, and other small craft, and to reduce the potential for grounding or collision. Vessels are responsible for their own safety and are not required to remain inside the route nor are fisherman required to keep fishing gear outside the route. Small vessels should exercise caution in and around the Recommended Vessel Routes and monitor VHF channels 16 or 13 for information concerning deep draft vessels and tugs/barges transiting these routes.

(26)

Security Broadcast System, Narragansett Bay

(27) In conjunction with various maritime interests, the Coast Guard has developed a system of recommended radiotelephone procedures for Narragansett and Mount Hope Bays that is designed to supplement the Vessel Bridge-to-Bridge Radiotelephone Regulations (see **33 CFR 26**, chapter 2). These **voluntary** procedures consist of security calls to be made by vessel masters, pilots, or operators on VHF-FM channel 13 (156.65 MHz) at designated points. The procedures are designed to give notice of unseen vessels, give notice of intended movement, clear channel 13 of traffic unrelated to navigation and give each vessel information on all others in the immediate vicinity and to do so at little cost and with as little radiotelephone traffic as possible. These recommendations do not relieve a master, pilot or operator of any requirements of law or regulation. There is no guarantee that every vessel will follow them.

(28) Inbound vessels should make security calls when abeam of Narragansett Bay Entrance Lighted Whistle Buoy NB, when off Castle Hill Light and when at the south end of Prudence Island (state whether bound for Providence or Fall River). The call at Castle Hill Light alerts outbound vessels so that they can pass East Passage Lighted Bell Buoy 11 close aboard, as during ebb current they tend to be set toward the center of the channel. Vessels bound for Providence should make additional security calls when off Popasquash Neck and when approaching Bullock Point Light BP. Vessels bound for Fall River should call Brightman Street Bridge when they enter Mount Hope Bay to allow sufficient time for opening of the bridge.

(29) Vessels outbound from Providence should make security calls when leaving their dock and when off Popasquash Neck. Vessels outbound from Fall River should make calls when leaving their dock, when approaching Mount Hope Bridge, and when off Gould Island.

(30)

Anchorages

(31) The principal anchorages for vessels seeking shelter are Newport Harbor in the East Passage and Dutch Island Harbor in the West Passage. These harbors afford anchorage with good holding ground for deep-draft vessels and are sometimes used by coasting vessels on the passage between Vineyard Sound and Long Island Sound. Good anchorage will be found almost anywhere in the bay under the lee of islands or the shore, where vessels becalmed or at night frequently anchor. Point Judith Harbor of Refuge is just west of Point Judith. General and explosives anchorages are in Narragansett Bay. (See **33 CFR 110.1** and **110.145**, chapter 2, for limits and regulations.)

(32)

Routes

(33) Vessels approaching from eastward should shape their approach to pass well south of Seal Ledge and Brenton Reef. Brenton Reef and other dangers on the easterly side of the entrance will be avoided by keeping Castle Hill Light bearing eastward of 003° and passing westward of the lighted whistle buoy and the gong buoy off Brenton Reef and the bell buoy off Butter Ball Rock. Approaching from westward, from a position with Point Judith Light bearing 344° distant 2 miles, vessels may steer **028°** for about 9 miles to a position 0.5 mile west of Castle Hill Light, thence follow the navigational lights in the bay. The recommended route, however, for

deep-draft vessels is via the Narragansett Bay Approach Traffic Separation Scheme, which is described earlier in this chapter.

(34)

Tides

(35) The tidal movement in Narragansett Bay with its vertical and horizontal constituents–tide and current, respectively–is a continuation of the tide wave of the Atlantic Ocean. This wave sweeps into the three entrances between Sakonnet Point and Point Judith and continues up the bay and into each of its tributaries until stopped by rapids or other obstructions. As is usual when oceanic tidal movements enter inland waterways, the nature of the movement is modified by the hydrographic features encountered. In this area the local features are such that the current movement in particular is subject to considerable distortion. See the tide prediction service at *tidesandcurrents.noaa.gov*. Links to a user guide for this service can be found in chapter 1 of this book.

(36)

Current

(37) The flood current in Narragansett Bay frequently has two maximums of velocity separated by a minimum velocity that at times becomes an ebb flow. Over the greater part of the bay, the usual maximum flood or ebb velocity is from 0.2 knot in the broad portions of the waterways to 1.5 knots in the more constricted sections. Velocities of about 1.4 knots occur at the bridges in Seekonk River, a velocity of about 1.7 knots in the narrows at the mouth of Kickamuit River and a velocity of 2.3 to 2.7 knots at the bridges in Sakonnet River. In Sakonnet River, from the highway bridge to its mouth, current velocities are small, being generally less than 0.5 knot. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(38)

Weather, Narragansett Bay and vicinity

(39) In the entrance to the bay and its approaches, fogs are more prevalent from April to October. The fogs are brought in by winds from east through south to southwest and are cleared off by northerly and westerly winds. The usual duration of the fog is 4 to 12 hours, but periods of 4 to 6 days have been known with only short clear intervals. The head of the bay will sometimes be free from fog while the entrance is completely shut in.

(40) Navigation of the bay and its tributaries is sometimes impeded by floating ice and in severe winters by packs of field ice. The ice that breaks up in Providence River and Mount Hope Bay is set by north and northeast winds down the bay through East Passage. If there is much ice, a gorge is sometimes formed at Fort Adams, but it is of short duration. The passages are rarely closed for any length of time below Gould Island in the East Passage and Dutch Island in the West Passage. During January and February, Mount Hope Bay, Bristol Harbor, Warren, Providence River, Greenwich Bay and Wickford are usually closed to sailing vessels unaided by power. The inner harbor of Newport is also sometimes closed during these months with the exception of a channel kept open by vessels. It can get nasty at the mouth of Narragansett Bay when strong winds oppose the currents. Rounding Point Judith can be rough or interminable, due to the confluence of tidal currents. Also local wind conditions can cause the tide to turn earlier or later than predicted in the tables.

(41)

North Atlantic right whales

(42) Endangered North Atlantic right whales may occur in the Narragansett/Buzzards Bay Traffic Separation Scheme (peak season: November though April). The Northeast Marine Pilots distribute educational material to mariners in an effort to reduce right whale ship strikes. (See North Atlantic Right Whales, indexed as such, in chapter 3 for more information on right whales and recommended measures to avoid collisions.)

(43) All vessels 65 feet or greater in length overall (LOA) and subject to the jurisdiction of the United States are restricted to speeds of 10 knots or less in the Block Island Sound Seasonal Management Area between November 1 and April 30. The area is defined as the waters bounded by:

(44) 40°51'53.7"N., 70°36'44.9"W.;

(45) 41°20'14.1"N., 70°49'44.1"W.;

(46) 41°04'16.7"N., 71°51'21.0"W.;

(47) 40°35'56.5"N., 71°38'25.1"W.; thence back to starting point. (See **50 CFR 224.105**, chapter 2 for regulations, limitations and exceptions.)

(48) The **Narragansett Bay National Estuarine Research Reserve**, a Marine Protected Area (MPA), includes the waters around Prudence, Patience, Dyer and Hope Islands in Narragansett Bay. **Cape Cod South Closure Area** includes inshore waters of Narragansett Bay and offshore federal waters of the south coasts of Massachusetts and Rhode Island.

(49)

Pilotage, Narragansett Bay and other Rhode Island waters

(50) Pilotage is compulsory for foreign vessels and U.S. vessels under register when entering and departing Narragansett Bay and all ports of the waters of the State of Rhode Island.

(51) Federal and state pilots for Narragansett Bay are available from Northeast Marine Pilots, Inc., Newport, RI 02840;*nemarinepilots.com*; telephone 401–847–9050 (24 hours), 800–274–1216; FAX 401–847–9052; email: dispatch@nemarinepilots.com.

(52) Pilots board vessels about 1.5 miles eastward of Narragansett Bay Entrance Lighted Whistle Buoy NB within an area bounded by

(53) 41°23.6'N., 71°22.4'W.,

(54) 41°22.6'N., 71°22.0'W.,

(55) 41°24.2'N., 71°20.0'W.,

(56) 41°22.6'N., 71°20.6'W. This pilot boarding area is southward of a line extending from Point Judith to Sakonnet Point. Vessels arriving from sea should approach this boarding station via the Narragansett Bay Traffic Separation Scheme inbound traffic lane.

(57) Vessels bound for Long Island Sound ports may board pilots at any point south or east of the Point Judith Pilot Station, centered on 41°17.0'N., 071°30.5'W., and outside the waters of the State of Rhode Island.

(58) Pilots board from the Northeast Marine Pilots, Inc. pilot boats NORTHEAST II, NORTHEAST III and NORTHEAST IV. NORTHEAST II is a 47-foot boat and NORTHEAST IV is a 52-foot boat; both have unpainted aluminum hulls and superstructures with the word PILOT in red on both sides. NORTHEAST III is a 33-foot boat with black hull and white superstructure. A vessel should confirm her ETA by VHF-FM radio at least 2 hours before arrival. All Northeast pilot boats monitor VHF-FM channels 16, 13 and 10 and work on 10.

(59) Pilots for Narragansett Bay serving U.S. enrolled vessels in coastwise trade are available from the Northeast Marine Pilots (see above for contact information) and the Connecticut State Pilots (a division of Interport Pilots Agency, Inc.), address: State Pier, New London, CT, telephone 800–346–4877 or 908–787–5554 (24 hours), cable PORTPILOTS Port Monmouth, NJ.

(60) Connecticut State Pilots board vessels from the pilot boat CONNECTICUT PILOT, 65 feet long with blue hull and white superstructure, and from pilot boat CONNECTICUT PILOT II, 47 feet long with blue hull and white superstructure. The boats monitor VHF-FM channels 16 and 13, 2 hours prior to the vessel's scheduled ETA, and work on channel 10. The pilots meet ships bound for Narragansett Bay at any point south or east of the Point Judith Pilot Station, centered on 41°17.0'N., 071°30.5'W., and outside the waters of the State of Rhode Island.

(61) Vessels to be boarded should provide a ladder 3 feet above the water on the lee side.

(62) Pilot services are generally arranged at least 24 hours in advance through ships' agents or directly by shipping companies.

(63)

Sakonnet River to Almy Point

(64) **Sakonnet River**, on the easterly side of Narragansett Bay, is between the mainland and the eastern shore of Rhode Island. The width of the river varies from 0.7 to 2 miles except at its northern end where a least width of 0.3 mile is found. The river is little used except by fishing vessels and small craft. **Sakonnet Light** (41°27'11"N., 71°12'09"W.) is on the eastern side of the southern entrance and marks the western end of the foul ground west-southwest of Sakonnet Point.

(65) The channel of Sakonnet River is good for a depth of 16 feet from the mouth to Mount Hope Bay. There are numerous shoals and outlying rocks, but the dangers are well marked by buoys. Except for the breakwater light off Sakonnet Harbor, no lighted aids are in the river, and strangers should not attempt to navigate it at night.

(66) Good anchorage for vessels drawing up to 17 feet can be had in midriver just below High Hill Point in depths of 21 to 26 feet. Although open to the southward, a heavy sea seldom reaches as far as this anchorage. In southeasterly gales the water is comparatively smooth inside the mouth of the river. Fishermen seeking shelter frequently anchor on the flats in the bight northward of Fogland Point in depths of 10 to 14 feet.

(67) **Sakonnet Point**, at the eastern entrance to Sakonnet River, is surrounded by bare and submerged rocks. Several islets and islands are south of the point. **Schuyler Ledge**, with a least depth of 10 feet, is about 0.8 mile southward of the point, and is marked by a bell buoy. A seasonal fishtrap area marked by private buoys is about 0.7 mile southwest of the point.

(68) **Cormorant Rock**, a bare dark rock off the western side of the entrance to the river, is about 0.8 mile south of **Sachuest Point**, the southeastern extremity of Rhode Island. Vessels should not pass between Cormorant Rock and **Cormorant Reef**, 0.3 mile southward of the rock. The least depth on the reef is 4 feet; it is marked by a bell buoy.

(69) The two bridges and the remains of the abandoned highway bridge at the north end of Sakonnet River act as dams to maintain the water at different levels on either side of them, causing stronger **currents** through the openings. The currents change with great rapidity both in velocity and in direction and are characterized by a double flood. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(70) The river north of Fogland Point is usually closed by **ice** for short periods each winter. Ice packs occur at the railroad bridge.

(71) Vessels proceeding up Sakonnet River should follow a midriver course to the constricted part of the river, thence follow the channel marked by buoys into Mount Hope Bay.

(72) **Sakonnet Harbor**, a small-boat harbor on the northerly side of Sakonnet Point, is protected by an 800-foot breakwater extending in a northerly and easterly direction from **Breakwater Point**. A light marks the outer end. A marina on the southwest side of the harbor provides berths, electricity and a launching ramp. A surfaced launching ramp is also on the south side of the harbor.

(73) The western shore of Sakonnet River from the entrance to Sandy Point should be given a berth of 0.4 mile to avoid shoals with depths of 7 to 17 feet. Rocks extend up to 500 yards offshore between Sachuest Point and **Flint Point**, about 1 mile northward. **Flint Point Ledge**, about 0.5 mile north-northeast of Flint Point, has

a least depth of 10 feet. **Black Point** is a rocky bluff on the western side of the river, 2.6 miles northward of Flint Point. **Sandy Point** and **McCorrie Point**, low and backed by high land, are 3.9 and 5.4 miles, respectively, northward of Flint Point.

(74) The channel passes eastward of **Gould Island**, a high wooded island, 2.5 miles north-northeastward of McCorrie Point. This Gould Island should not be confused with one of the same name in East Passage. A rock with a depth of 1 foot is northwestward of the island and is marked by a buoy.

(75) The eastern side of Sakonnet River is bolder than the western side. The east shore should be given a berth of 0.7 mile from Sakonnet Point to **Church Point**, a flat point with bluffs at the water, about 2.8 miles northward of Sakonnet Point. **Old Bull**, with a depth of 2 feet, is about 0.5 mile southward of Church Point and marked by a buoy. **High Hill Point**, about 3 miles north of Church Point, is a prominent small hill with bluffs at the water. **Fogland Point**, about 0.9 mile northward of High Hill Point, is a projecting prominent point; the westerly and northerly sides should be given a berth of over 200 yards. **Almy Rock**, bare at low water, is 0.3 mile southwest of Fogland Point. The broad bights between Fogland Point and the bridge are shoal.

(76) **Nannaquaket Pond**, on the east side of Sakonnet River eastward of Gould Island, has a narrow entrance 8 feet deep crossed by a fixed bridge with a clearance of 12 feet. The deeper water in the entrance is along the northern shore; several rocks are off the southern shore. The currents have considerable velocity. The northern part of the pond has depths up to 26 feet; the remainder has depths of about 3 feet.

(77) **The Cove**, on the western side of the river south of the bridges, has a depth of about 9 feet in the entrance; the 31-foot-wide fixed highway bridge across the entrance has a clearance of 25 feet. Depths are generally 3 to 4 feet in the cove.

(78) **Tiverton** is a town on the eastern bank of Sakonnet River north and south of the bridge. Oil tankers call at Tiverton. The oil piers northward of the bridge have reported depths of about 32 to 35 feet alongside.

(79) There are small-craft facilities at Tiverton, across the Sakonnet River at **Almy Point** and at Cedar Island Pond approximately 1.1 miles north of Almy Point. The facilities provide berths and mooring, electricity, gasoline, diesel fuel, water, ice, pump-out facilities, a launching ramp, a lift to 60 tons, storage and full repairs. The facility at Tiverton has an approach depth of 35 feet with 12 feet alongside, at Almy Point has an approach depth of 25 feet with 11 feet alongside and at Cedar Island Pond has an approach depth of 12 feet with 8 feet alongside.

(80) The channel at the north end of Sakonnet River, 0.6 mile above Gould Island, is restricted to a width of 100 feet between the abutments of a former highway bridge just south of the entrance to The Cove. A fixed highway bridge, 0.8 mile north, has a clearance of 65 feet; the bridge is under construction (2009). An overhead power, just north of the bridge, has a clearance of 81 feet.

(81)

Newport to Arnold Point

(82) The southern shore of Rhode Island is rocky with numerous offlying rocks and ledges. Numerous prominent residences are on the eastern side of **Newport Neck**, the southwestern part of Rhode Island. A large brick residence with several towers is on the southeastern point of Newport Neck. **Easton Point** is about 1.3 miles eastward of Newport Neck. A stone tower with a short spire at each corner can be seen from offshore, about 0.7 mile northward of Easton Point. Westward of Easton Point is a bathing beach with a prominent pavilion. Several private landings are northward of **Gooseberry Island**, a small islet south of Newport Neck. Local knowledge is required to proceed to the landings.

(83) **East Passage**, the principal passage in Narragansett Bay, extends between Rhode Island on the east and Conanicut and Prudence Islands on the west. It is the most direct route to Newport, Bristol, Providence, Mount Hope Bay and Taunton River.

(84) The Pell Bridge Newport, a fixed highway suspension bridge locally known as the Newport Bridge, crosses East Passage about 3.6 miles above the entrance, between Jamestown and Newport. Clearances through the 1,500-foot center span are 213 feet at the center, 205 for the mid 1,000 feet and 194 feet for the remainder of the center span. A racon and a sound signal are at the bridge.

(85) **Brenton Point** is the southwestern extremity of Rhode Island and the eastern entrance point of East Passage.

(86) **Brenton Reef**, bare in places, extends 0.5 mile south-southwestward of the point and is marked by a gong buoy. Another reef extends 0.5 mile offshore just eastward of the point; **Seal Rock** is at the southeastern end of the reef.

(87) **Seal Ledge**, about 0.5 mile south of Seal Rock, has depths of 15 to 30 feet and is marked by a bell buoy. **Haycock Ledge**, 0.4 mile eastward of Seal Rock, has a least depth of 12 feet.

(88) **Beavertail Point** is the southern extremity of Conanicut Island, on the western side of the entrance to East Passage. **Beavertail Light** (41°26'58"N., 71°23'58"W.), 64 feet above the water, is shown from a square granite tower attached to a white dwelling at Beavertail Point. A mariner radio activated sound signal at the light is initiated by keying the microphone five times consecutively on VHF-FM channel 83A and a radar tower is north of the light. **Newton Rock**, a covered rock, is about 0.1 mile southward of the point; a bell buoy is about 0.2 mile southwestward of the rock.

(89) **Hull Cove**, about 1 mile northeastward of Beavertail Light, is rocky and exposed to southerly winds. **Castle Hill**, the westernmost point of Rhode Island, is marked by Castle Hill Light; a mariner radio activated sound

signal is at the light, initiated by keying the microphone five times on VHF-FM channel 83A. **Castle Hill Coast Guard Station** is close east of the light. **Butter Ball Rock**, about 0.2 mile south of the light and marked by a bell buoy, uncovers 1 foot.

(90) **Mackerel Cove** indents the southern shore of Conanicut Island about 1.6 miles northeastward of Beavertail Light. A house with a cupola is prominent on **Southwest Point**, the eastern entrance point of the cove. The cove is exposed to southerly weather and is seldom used. The Jamestown-Verrazzano Bridge shows prominently over the bar at the head of the cove on entering East Passage. **Kettle Bottom Rock**, about 0.2 mile southeastward of Southwest Point, is bare and marked by a gong buoy.

(91) **Bull Point**, the southeastern point of Conanicut Island, is rugged and rocky, and fringed by shoals that are marked by buoys. **Fort Wetherill** is on the point. **The Dumplings**, northeast of Bull Point, are numerous bare and covered rocks and islets. The most southerly islet has a tall house on it, covering almost the entire islet.

(92) **Fort Adams** is on a peninsula off the north side of Newport Neck. The buildings and masonry of the fort are prominent on the western slope of the peninsula. A dock extending northward from the fort is marked by a light and sound signal.

(93) **Newport Harbor**, on the western shore of Rhode Island and the eastern side of East Passage, 3.5 miles above Beavertail Light, is an important harbor of refuge for coasters, tows and yachts. Its approach is well marked by navigational aids, and the harbor is of easy access day and night. A state regulatory buoy in the entrance to the inner harbor marks a 5 mph **no wake zone**. **Goat Island**, 0.6 mile long in a north-south direction, is a major pleasure boating center and divides Newport Harbor into an outer and inner harbor. The outer harbor, on the western side of Goat Island, is northward of The Dumplings and southward of Gould Island. The inner harbor is on the eastern side of Goat Island and extends along the western front of Newport.

(94) A marina, hotel, recreational and service facilities and marine supplies are available on Goat Island.

(95) **Newport**, a city on the inner harbor, is one of the principal summer resorts on the Atlantic Coast. Some coastwise traffic uses the port, but few foreign vessels enter it. A Naval Education and Training Center is here, from which several Navy ships operate.

(96)

Prominent features

(97) The following objects are prominent when approaching Newport Harbor either from the southward or northward: a hotel on Goat Island, a white building of the yacht club near Ida Lewis Rock in the southerly part of the harbor, church spires in the town and the buildings of the Naval Education and Training Center and Naval War College on Coasters Harbor Island in the north part of the harbor. To the westward on Conanicut Island are several large hotels and a standpipe. Numerous navigational aids mark the passages through the harbor.

(98) The entrance to the outer harbor from the southward is unobstructed; the entrance from northward, passing either side of Gould Island, is clear, but the passage eastward of Rose Island is partly obstructed by the rocks and ledges between Rose and Coasters Harbor Islands.

(99) The inner harbor has two entrances north and south of Goat Island. A fixed highway bridge with a 40-foot span and a clearance of 14 feet connects the Newport mainland to the northern part of Goat Island. This bridge limits the size of vessels that can enter the inner harbor from the northern approach.

(100) **General anchorages** are in the outer and inner harbor and, except in emergencies, vessels must anchor in these areas. (See **33 CFR 110.1** and **110.145**, chapter 2, for limits and regulations.) Vessels of more than 18-foot draft anchor in the outer harbor in depths of 36 to 100 feet with good holding ground.

(101) **Special anchorages** are in Brenton Cove and in the inner harbor east and northeast of Goat Island. (See **33 CFR 110.1** and **110.46**, chapter 2, for limits and regulations.) The anchorage is good in **Brenton Cove** and is used frequently by yachts. When entering the cove, the western shore should be given a berth of 200 yards.

(102) In 1981, the harbormaster requested that transient craft anchor only in the northern part of the Brenton Cove anchorage and in the anchorage northeast of Goat Island.

(103) Shoals with little water over them make out nearly 300 yards from the southern shore of the inner harbor to **Ida Lewis Rock** and **Little Ida Lewis Rock**; the latter is marked by a daybeacon.

(104) Goat Island is marked by a light at its northern end and a lighted bell buoy at its southern end. Buoys mark the shoals along the southeasterly and southwesterly sides of the island.

(105) **Rose Island**, privately owned, is surrounded by a shoal with little water over it. The shoal extends about 0.4 mile northeastward of the island where it rises abruptly from deep water. A rocky area extends southward from Rose Island and is marked by a buoy. A private light marks the southwest point of the island. **Mitchell Rock**, with a depth of 16 feet and marked by a gong buoy, is about 0.1 mile southeast of the dock on the southeast side of the island. **Citing Rock**, 2 feet high, is 350 yards east of the north end of the island and on the edge of the shoal surrounding Rose Island. **Tracey Ledge**, covered 11 feet, is about 0.3 mile eastward of Rose Island and marked by a buoy.

(106) **Gull Rocks** are about midway between Rose Island and Coasters Harbor Island. Buoys mark the ends of the shoals and rocks that extend northward and southward of the rocks. There is deep water between Gull Rocks, Rose Island and Coasters Harbor Island. A rock, covered 19 feet and marked by a buoy, is about 0.3 mile north-northwestward of Gull Rocks.

(107) Off the northern and southern sides of **Coasters Harbor Island** are numerous rocks and ledges. A

private light marks an obstruction in the channel south of the island. **St. Patrick Rock**, covered 5 feet, is about 0.3 mile southeastward of the island. The island has numerous buildings. Three fixed bridges connect the island to Newport. The southern highway bridge and the pedestrian bridge 0.3 mile above it have 31-foot spans with clearances of 3 feet. The northern highway bridge has a span of 85 feet with a clearance of 7 feet. Overhead power cables of unknown clearance cross the harbor below the northern bridge.

(108) The western portion of the outer harbor is generally free of dangers northward of The Dumplings.

(109)

Current

(110) In the entrance off Bull Point the flood current is often irregular. There may be a long period of slack water preceding the flood, or there may be a double flood. The flood reaches a strength of about 1.2 knots; the ebb is regular and averages 1.5 knots at strength.

(111) Northward of Bull Point, tidal current velocities seldom exceed 1 knot. In the inner harbor they are usually less than 0.5 knot. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(112)

Weather, Newport and vicinity

(113) The prevailing winds are southwesterly in the summer and northwesterly in the winter. The heaviest gales are usually from the northwest and northeast.

(114) The harbor and its approaches are navigable throughout the year, although in severe winters **ice** may interfere with navigation in the inner harbor for short periods. Vessels and tugs keep ice well broken up in the main channel through the inner harbor.

(115) July is the warmest month with an average maximum of 79 °F and average minimum of 63 °F. January is the coldest month, having normal mean temperatures near freezing, that is, average maximum of 38 °F and average minimum of 23 °F. The extreme maximum temperature for Newport is 98 °F recorded in August 1975 while the extreme minimum is -9 °F recorded in January 1982.

(116) Precipitation is fairly evenly distributed throughout the year averaging about 45 inches (1,143 mm) in any given year. The wettest month is November averaging 4.57 inches (116 mm) and the driest month is July averaging only 2.94 inches (75 mm). Total snowfall for the winter season averages about 20 inches (508 mm); however, melting is usually rapid and snow cover rarely lasts more than a few days. The greatest snowfall in a 24-hour period was 20.0 inches (508 mm) in February 1978. January is the snowiest month averaging over seven inches (178 mm). Snow is absent from May through September. An average of only five days each year has snowfall amounts greater than 1.5 inches (38 mm).

(117)

Pilotage, Newport

(118) See Pilotage, Narragansett Bay and Other Rhode Island Waters (indexed as such), early this chapter.

(119)

Quarantine, customs, immigration and agricultural quarantine

(120) (See chapter 3, Vessel Arrival Inspections, and Appendix A for addresses.)

(121) **Quarantine** is enforced in accordance with regulations of the U.S. Public Health Service. (See Public Health Service, chapter 1.)

(122) Newport is a **customs port of entry.**

(123) The Coast Guard **vessel documentation** office at Providence serves Newport. (See Appendix A for address.)

(124)

Harbor regulations

(125) The **harbormaster,** under the supervision of the Recreation Department, is charged with the enforcement of harbor regulations, the movement of vessels, and assignment of moorings and anchoring. The harbormaster may be contacted through the Police Department. The **speed limit** inside the harbor is 5 mph.

(126)

Wharves

(127) Facilities include a city wharf and numerous private piers. The depths alongside the principal piers range from about 7 to 18 feet.

(128)

Supplies

(129) Gasoline, diesel oil, diesel fuel, water, provisions and marine supplies may be obtained at Newport.

(130) Launch services are available in the harbor.

(131)

Repairs

(132) Newport has a commercial shipyard specializing in repair, construction and conversion of steel and aluminum vessels. The marine railway is at the shipyard in the inner harbor and can handle vessels to 330 feet long, 63 feet wide and 21.6 feet in draft. Cranes to 60 tons are available.

(133)

Small-craft facilities

(134) There are numerous facilities in Newport harbor. Mobile hoists up to 60 tons are available. Complete small-craft hull and engine repairs can be made.

(135)

Communications

(136) Newport has bus and rail transportation. In the summer the ferry between Block Island and Providence calls at Newport.

(137) **Jamestown** is a town on the east side of Conanicut Island in a bight on the west side of East Passage. A standpipe in the southern part of the town and a hotel near the waterfront are prominent. The bight is a popular

summer anchorage for local craft. A marina basin protected by a detached breakwater is in the center of the bight. Jamestown has two boatyards. The largest marine railway can handle craft to 80 feet, and the largest lift is 50 tons. Berths, electricity, gasoline, diesel fuel, water, ice, some marine supplies and complete hull, engine and electronic repairs are available.

(138) **Potter Cove** is about 1 mile above the marina basin just north of **Taylor Point**. This cove should not be confused with the Potter Cove off Prudence Island. About 1 mile north of Taylor Point are the ruins of a pontoon pier.

(139) **Coddington Point** is about 0.5 mile northward of Coasters Harbor Island on the east shore of East Passage. **Bishop Rock Shoal**, 0.6 mile southwestward of the point, is covered 9 feet and marked by a lighted bell buoy. **The Sisters**, rocks awash westward of the point, are marked by a buoy.

(140) **Coddington Cove**, eastward of Coddington Point, is protected on its north side by a curving breakwater 0.7 mile long, marked at its end by a light and sound signal. Two long finger piers are inside the cove; the north side of the northerly pier is used by the Navy, and the southerly pier is used by a shipyard. Depths of 30 feet are reported alongside both piers. Navy buildings on shore and buildings at the shipyard are conspicuous.

(141)

Restricted Areas

(142) Coddington Cove is within a naval restricted area. (See **33 CFR 334.81**, chapter 2, for limits and regulations).

(143) A naval restricted area covering a large area surrounds Gould Island and extends north to include waters between **Conanicut Island** and Prudence Island. The regulation states in part that no person or vessel shall at any time, under any circumstances, anchor or fish or tow a drag of any kind in the area because of the extensive cable system located therein. (See **33 CFR 334.80**, chapter 2, for limits and regulations.)

(144) **Gould Island**, a military reservation, is about 2 miles north of Rose Island and 0.8 mile east of Conanicut Island. A light is on the south end of the island. The island is sparsely wooded.

(145) **Halfway Rock** and **Fiske Rock** are about 1.8 miles northeastward of Gould Island, on a small ledge bare at its southern end. Halfway Rock is marked by a daybeacon and Fiske Rock by a buoy. Strangers should not pass between these aids.

(146) A mussel farming area is about 1.3 miles northeast of the breakwater at Coddington Cove in the vicinity of the pier at **Lawtons**. Submerged equipment and numerous buoys extend several hundred yards offshore and for about 0.5 mile north of the pier. Caution is advised in the area.

(147) **Dyer Island**, about 0.8 mile eastward of the southern portion of Prudence Island, is low and brush covered. A reef, partly bare, extends 0.4 miles southward and southwestward of the island and is marked by a buoy. Between the shoal area south of Dyer Island and Rhode Island is a bar with depths of 9 to 18 feet over it. North of Dyer Island is a reef with depths of 6 to 18 feet.

(148) A small-craft facility is in a basin at **Coggeshall Point**, just north of Melville. Berths, gasoline, diesel fuel, electricity, water, ice, complete marine supplies and a 50-ton mobile hoist are available; engine, hull and electronic repairs can be made. In 1981, depths of 9 to 15 feet were reported in the basin.

(149) A lighted wind turbine is about 0.9 mile northeast of Coggeshall Point. A boat ramp is about 1.6 miles north-northeast of Coggeshall Point, south of **Arnold Point**.

(150)

Hog Island to Musselbed Shoals

(151) **Hog Island**, about 1 mile north of Arnold Point, lies in the entrance to Bristol Harbor, dividing the waters into two channels. The island has a rolling wooded terrain on which are a few houses and cottages. Shoal water surrounds the island extending as much as 0.4 mile southward and 0.8 mile northward. The shoal area is marked by lights and buoys.

(152) About 0.6 mile east-northeastward of Hog Island Shoal Light is **Musselbed Shoals**, marked on the outer end by a light. From the light structure a directional light is shown to mark the channel to Mount Hope Bay.

(153)

Mount Hope Bay to Somerset

(154) **Mount Hope Bay**, in the northeastern part of Narragansett Bay, is the approach to the city of Fall River and **Taunton River**. There are two approaches to the bay. The approach from the Sakonnet River, previously discussed, is little used. The approach from East Passage is well marked, and with care 34 feet can be carried in the channel into the bay.

(155) **Fall River**, on the eastern shore of the mouth of Taunton River and head of Mount Hope Bay, is an important manufacturing center as well as distribution point of petroleum products. Principal products handled through the port are petroleum products, latex, shellac, cotton and some lumber.

(156) **Somerset**, about 5.3 miles, and **Dighton**, about 7.5 miles above the Fall River, are towns on the west side of Taunton River. **Taunton**, a manufacturing city, is at the head of navigation about 12.5 miles above Fall River.

(157) Mount Hope Bridge, a fixed highway bridge with a clearance of 135 feet, crosses the entrance to Mount Hope Bay between Bristol Point and Rhode Island. The bridge is equipped with a racon and a sound signal. The sound signal is activated by keying the microphone five times consecutively on VHF-FM channel 83A. Two lighted towers at the bridge are visible for many miles in clear weather.

(158) **Mount Hope** is a prominent hill on the western side of the bay 2 miles northeastward of the suspension bridge.

(168)

Structures across Taunton River

Name•Description•Type	Location	Clear Width of Draw or Span Opening (feet)	Clear Height above Mean High Water (feet)	Information
Charles M. Braga Jr. Bridge (fixed)	41°42'23"N., 71°09'57"W.	400	135	A private sound signal is at the bridge
Overhead power cables	41°43'13"N., 71°09'35"W.		145	
Brightman Street Bridge (bascule)	41°43'26"N., 71°09'22"W.	98	27	Notes 1 and 2 Call sign WQA-833
Veterans Memorial Bridge (bascule)	41°43'35"N., 71°09'07"W.	200	60	Note 2
Overhead power cable	41°44'12"N., 71°08'28"W.		150	
Berkley Bridge (swing)	41°50'06"N., 71°06'30"W.	52	7	Bridge is under construction
Overhead power cable	41°50'51"N., 71°06'48"W.		75	
Overhead power cable	41°51'45"N., 71°06'18"W.		65	
Overhead power cable	41°52'29"N., 71°05'41"W.		65	
Weir Bridge (fixed)	41°53'08"N., 71°05'20"W.	35	10	
Railroad Bridge (fixed)	41°53'12"N., 71°05'06"W.	26	9	

Note 1 – Bridgetender monitors VHF-FM channel 16 and works on channel 13.
Note 2 – See **33 CFR 117.1** through **117.59** and **117.619**, chapter 2, for drawbridge regulations.

The eastern and western slopes are wooded. **Spar Island** is a small, low island near the center of Mount Hope Bay.

(159) **Borden Flats**, the shoal area northward of the channel in Fall River Harbor is marked by a light.

(160) Three shallow streams that empty into the northern part of Mount Hope Bay are entered only by local small craft. **Kickamuit River**, the westerly one, has a narrow buoyed entrance through which the currents have considerable velocity. The buoyed channel has a depth of about 6 feet. A ramp is on the western side of the bay, approximately 0.7 mile south of the entrance to Kickamuit River. **Cole River**, the middle of the three, is buoyed on the east side of the entrance. A highway bridge, about 1.5 miles above the entrance, has a 41-foot fixed span with a clearance of 7 feet.

(161) **South Swansea**, on the west shore of **Gardners Neck**, has a boatyard with a 25-ton mobile hoist and a marine railway that can handle craft up to 50 feet for hull, engine and electronic repairs or storage. Berths, electricity, gasoline, diesel fuel, water, ice and marine supplies are available. In 1981, a reported depth of 6 feet could be carried to the boatyard. **Lee River**, the easterly stream, is navigable to a fixed bridge about 1.2 miles above the entrance. A shoal in midchannel just north of the narrow opening through the fill, 0.8 mile above **Brayton Point**, has a depth of 1 foot.

(162)

Channels

(163) A federal project provides for a channel 35 feet deep through Mount Hope Bay to about 0.9 mile above the Brightman Street Bridge across Taunton River at Fall River and a side channel 35 feet deep, about 0.2 mile north of **Common Fence Point** (41°39.3'N., 71°13.3'W.) at the north end of Rhode Island that leads eastward from the main channel into North and South Branch channels. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A.

(164) A privately dredged side channel, about 3.3 miles northeastward of Common Fence Point and marked by buoys and a **325.3°** private lighted range, leads northwestward from the main channel to a powerplant wharf on the east side of Brayton Point.

(165) A dredged channel in Taunton River leads from Somerset to **Peters Point**, 6.7 miles above the Brightman Street Bridge, thence to Taunton, 12.5 miles above Fall River. Local knowledge is required from Dighton to Taunton. Buoys mark the channel to about a mile beyond the Berkley Bridge, about 3.5 miles below Taunton.

(166)

Anchorages

(167) Fall River Harbor has no designated anchorages. Vessels may anchor on either side of the dredged approach channel in the outer harbor or at any locality in Mount Hope Bay where depth and bottom are suitable; the chart is the best guide.

(169)

Current

(170) In Taunton River the currents generally follow the direction of the channel and, except at bridges, do not hinder navigation. The ebb is usually stronger than the flood. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(171)

Weather, Narragansett Bay and vicinity

(172) The prevailing winds are northeasterly for all but the summer months, when the direction is southwesterly. The heaviest gales are usually from the northwest. The

approach channel and harbor are generally free from **ice** and are navigable throughout the year. Taunton River is commonly closed from December to March. During severe winters the harbor and Mount Hope Bay are occasionally frozen over, but the channels to the principal wharves are kept open by vessels and tugs operating in the harbor.

(173)

Pilotage, Fall River

(174) See Pilotage, Narragansett Bay and Other Rhode Island Waters (indexed as such), early this chapter.

(175)

Towage

(176) Tugs to 2,200 hp are available at Fall River and tugs to 3,300 hp are available from Providence for use at Fall River. Vessels are usually met 2 miles below their berth. Large vessels normally require tugs for docking and undocking. Arrangements for tug service should be made at least 6 hours in advance, usually through ships' agents or directly by shipping companies. Tugs are dispatched 24 hours a day. The tugs monitor VHF-FM channels 10, 13, 16 and 18A, and use channel 7A as a working frequency.

(177)

Quarantine, customs, immigration and agricultural quarantine

(178) (See chapter 3, Vessel Arrival Inspections, and Appendix A for addresses.)

(179) Fall River is a **customs port of entry.**

(180) **Quarantine** is enforced in accordance with regulations of the U.S. Public Health Service. (See Public Health Service, chapter 1.) Fall River has several hospitals.

(181) The Coast Guard **vessel documentation** office at New Bedford, MA, serves Fall River. (See Appendix A for address.)

(182) The **harbormaster** can be contacted through Fall River City Hall.

(183)

Wharves

(184) The piers and wharves at Fall River are along the Taunton and Sakonnet Rivers and in Mount Hope Bay. Only the deep-draft facilities are described. The alongside depths are reported; for information on the latest depths contact the operator. All the facilities described have highway connections. Fresh water is available at most of the piers and wharves. Cargo in the port is usually handled by ship's tackle.

(185) The **speed limit** is 5 knots in the channel off the piers and wharves.

(186) **Facilities at Tiverton, RI**

(187) Texaco Inc. Tiverton Terminal: (41°38'50"N., 71°12'40"W.); 50-foot face, 721 feet with dolphins; 35 feet alongside; deck height, 11 feet; receipt of petroleum products; owned by Texaco, Inc.; not being operated in 1983.

(188) Northeast Petroleum Corp., Pier 1: about 250 yards north of Texaco Inc. Terminal; 120-foot face, 700 feet with dolphins; 32 feet alongside; deck height, 9 feet; receipt of petroleum products; owned and operated by Northeast Petroleum Corp.

(189) Northeast Petroleum Corp., Pier 2: about 250 yards north of Northeast Petroleum Corp., Pier 1; 40-foot face, 700 feet with dolphins; 34 feet alongside; deck height, 9 feet; receipt of petroleum products; owned and operated by Northeast Petroleum Corp.

(190) Fuel Storage Co., Tiverton Terminal Pier: east side of Mount Hope Bay, 1.5 miles northward of Northeast Petroleum Corp., Pier 2; 50-foot face, 795 feet with dolphins; 35 feet alongside; deck height, 9 feet; receipt of petroleum products; owned and operated by Fuel Storage Co.

(191) **Facilities at Fall River, east side of Taunton River**: Borden and Remington Corp. Wharf: (41°42'10"N., 71°10'09"W.); 380-foot face, 28 feet alongside; deck height, 10 feet; receipt of latex and caustic soda; owned by Tillotson Co. and operated by Borden and Remington Corp.

(192) State Pier: 0.2 mile northeast of Borden and Remington Corp. Wharf; 398-foot face, 18 to 35 feet alongside; lower side 620 feet long, 35 feet alongside; deck heights, 17 feet; 85,000 square feet covered storage, about 7 acres of open storage; receipt and shipment of general and roll-on/roll-off cargo; owned by the Commonwealth of Massachusetts, operated by Fall River Line Pier, Inc.

(193) The battleship USS MASSACHUSETTS, World War II memorial, and three other U.S. Navy vessels are berthed just northward of the State Pier.

(194) Shell Oil Co. Wharf: about 2 miles above State Pier; 570-foot face, 700 feet with dolphins; 30 feet alongside; deck height, 13 feet; receipt and shipment of petroleum products, receipt of naphtha; owned and operated by Shell Oil Co.

(195) **Facilities on west side of Taunton River**

(196) Brayton Point Station Dock: (41°42'33"N., 71°11'21"W.); 1,017 feet long; 34 feet alongside; deck height, 15 feet; coal unloading tower serves conveyor belt system, unloading rate 1,000 tons per hour; receipt of fuel oil and coal; owned and operated by New England Power Co.

(197) Montaup Electric Co. Wharf: about 2.5 miles above Brayton Station Dock; 645 feet long, 34 feet alongside; deck height, 10 feet; two coal unloading towers serve conveyor belts, combined unloading rate 1,200 tons per hour; receipt of coal and fuel oil; owned and operated by Montaup Electric Co.

(198)

Supplies

(199) Provisions, marine supplies, gasoline and water can be obtained in Fall River. Water is available at most of the berths.

(200)

Repairs

(201) Fall River has no drydocking or major repair facilities for deep-draft vessels; the nearest such facilities are at Boston, MA.

(202) Fall River has two small shipyards, on the west side of the harbor about 0.6 mile above the fixed bridge and on the east side of the harbor about 0.9 mile below the fixed bridge. The northerly shipyard has a marine railway that can handle vessels to 100 feet long with drafts of 7 feet forward and 13 feet aft. The yard can make repairs to wooden, steel and aluminum vessels at their berths. The southerly yard specializes in the construction and conversion of steel vessels. Cranes to 250 tons are available for hauling out vessels. In 1981, depths of 22 to 23 feet were reported alongside.

(203)

Small-craft facilities

(204) Small-craft facilities are at Fall River, **Somerset** opposite Fall River, Taunton and at Dighton. Berths, moorings, electricity, gasoline, diesel fuel, water, ice, pump-out facilities, launching ramps, marine supplies and storage and hull, engine and electronic repairs are available. The largest marine railways, at Dighton, can handle craft to 55 feet. The largest marine lift is at Somerset with a capacity of 80 tons.

(205) A launching ramp is on the west side of Taunton River, about 1.6 miles above Somerset.

(206)

Bristol Harbor to Warren River

(207) **Bristol Harbor**, between **Bristol Neck** on the east end and **Popasquash Neck** on the west, is in a cove about 2 miles long and 1.3 miles wide at its southern end, narrowing to 0.4 mile wide at its northern end. The harbor proper, the northern part of the cove, has depths of 15 to 17 feet.

(208) **Bristol** is a town on the eastern side of the harbor. In approaching the harbor the most prominent mark is Mount Hope Bridge. Also prominent are the navigation lights, a stone tower, a stack and an elevated tank on high ground back of the town. The town has bus service. A ferry operates daily from Bristol to Prudence Island, and summer ferry service is available to Hog Island.

(209) **Hog Island** is in the middle of the entrance to Bristol Harbor. A natural channel with depths of 19 to 28 feet extends on each side of the island. Excellent anchorage may be found in the harbor abreast the town in depths of 15 to 17 feet, soft bottom. A **general anchorage** is in Bristol Harbor. (See **33 CFR 110.1** and **110.145 (c)** and **(d)**, chapter 2, for limits and regulations.)

(210) **Usher Rocks**, about 0.7 mile northeastward of Popasquash Point, are bare at low water. A buoy is eastward of the rocks and a lighted bell buoy is on the western side of the western passage to the harbor.

(211) Depths alongside the piers and wharves range from 9 to 13 feet.

(212) A Coast Guard vessel is moored at Bristol. The Coast Guard pier is marked by a light.

(213) A marina and yacht club are on the west side of the harbor. Guest moorings, electricity, water, ice, marine supplies and hull and engine repairs are available. A mobile hoist at the marina can handle craft to 42 feet long. Launching ramps are available on the east side of the harbor. In 1993, a reported depth of about 8 feet could be carried to the marina.

(214) **Potter Cove**, on the northeast side of **Prudence Island**, is a small nearly landlocked harbor. Buoys mark the entrance channel off **Gull Point**. The north and south ends of Prudence Island are a state park. **Ohio Ledge**, about 2.5 miles northward of Potter Cove, has a least depth of 9 feet and is marked on its east side by a bell buoy.

(215) **Warren River**, emptying into the head of Narragansett Bay westward of Bristol Neck, is the approach to the towns of **Warren** and **Barrington**, and **Barrington River**, which joins Warren River at Warren. A church spire in Warren is prominent.

(216) From the bay, the channel to Warren passes between numerous shoals and rocks and is crooked and winding but well marked.

(217) In 2002, a sunken wreck in about 41°43.65'N., 71°17.25'W. was on the west side of Warren River.

(218) A state regulatory buoy, about 0.9 mile above the mouth of Warren River, marks a **"Slow no wake" zone.**

(219) An excellent anchorage may be found at the mouth of the Warren River about 0.2 mile from the eastern shore in depths of 14 to 15 feet, soft bottom. There is not room for anchorage in the river for any but small craft. Abreast the lower end of Warren the channel is about 0.1 mile wide, with depths of 13 to 17 feet in midchannel, and small vessels can anchor temporarily at this point.

(220) Vessels approaching the river must take care to avoid **Rumstick Shoal**, which extends nearly 0.6 mile south of **Rumstick Point**, the southernmost point of **Rumstick Neck** and the western entrance point of the river. The shoal has depths of 2 to 12 feet and is marked by buoys. **Rumstick Rock**, 6 feet high, and **Rumstick Ledge**, with rocks that uncover 1 to 5 feet, are on the westerly side of the shoal.

(221) The Route 114 Bridge crosses the Barrington River about 0.5 mile above the mouth; it has a fixed span with a clearance of 10 feet. A fixed bridge just north of the highway bridge is under construction (2023). Rocks were reported under the bridge in 1978.

(222) The **tidal current** off the town of Warren has a velocity of about 1 knot. Strong currents may be encountered in Barrington River.

(223) The **harbormaster** controls docking, mooring and anchoring and can be contacted through the Warren Police Department.

(224) A shipyard on the east side of the Warren River at Warren has a hydraulic lift that can handle vessels to 300 tons or 130 feet long. In 1981, a reported depth of 7 feet could be carried to the shipyard.

(225) Berths, electricity, gasoline, diesel fuel, water, ice and marine supplies are available in Warren. Depths reported alongside the major wharves range from 7 to 20 feet. Small craft facilities are also available on the point near the confluence of the two rivers. Several boatyards are in the vicinity; mobile hoists to 25 tons are available. Covered storage and complete engine, electronic and hull repairs are available. Mariners are cautioned of a reported rock, covered 2 feet, just off the gas dock of the Barrington Yacht Club in about 44°44'00"N., 71°17'37"W.

(226)

Providence River

(227) **Providence River**, which empties into the head of Narragansett Bay between **Nayatt Point** and **Conimicut Point**, is the approach to the city of Providence and numerous towns and villages and to Seekonk River.

(228) **Providence** is at the head of navigation on the Providence River, about 7 miles above the entrance, at the junction of the Providence and Seekonk Rivers. The port area includes both sides of the upper navigable channel of the river. The port's chief waterborne commerce is in petroleum products, cement, lumber, steel scrap metal, general cargo and automobiles.

(229) **Occupessatuxet Cove**, on the west side of the river north of Conimicut Point, is a shallow bight south of **Gaspee Point**. The cove is frequented only by small craft with local knowledge.

(230) **Pawtuxet Cove**, used by pleasure and fishing craft, on the west side of Providence River, is entered about 1 mile northward of Gaspee Point through a dredged channel. The entrance channel leads westward to the cove, thence the channel turns northward and leads to a turning basin at the head of the cove. An anchorage basin extends southward from the entrance channel. The entrance channel is marked by buoys. A 12-foot-high protective dike along the east side of the anchorage basin extends southward from **Marsh Island**, on the south side of the entrance channel, to **Rock Island**. **Pawtuxet** is a village on the west side of the cove.

(231) The **harbormaster** in the cove controls anchoring and berthing; he can be contacted through the Warwick City Hall. Berths, electricity, gasoline, diesel fuel, water, ice, marine supplies and a 15-ton mobile hoist are available. A flatbed trailer can haul out craft to 50 feet long for complete hull and engine repairs.

(232) A yacht club is on the west side of Providence River about 1 mile northward of the entrance to Pawtuxet Cove. Gasoline and guest berths are available.

(233) A privately dredged channel leads from the main channel in Providence River, eastward of Pawtuxet, to a small-craft facility about 0.45 mile southwestward of **Fields Point** (41°47.2'N., 71°22.9'W.). The channel is marked by buoys and, in 1998, had a reported controlling depth of 8 feet. The ruins of the piers of a former Naval Reserve facility are on the south side of Fields Point immediately eastward of the yacht club; this area should be avoided.

(234) **Bullock Cove** is on the east side of Providence River, 2 miles north of Conimicut Point. A dredged channel leads from the Providence River to a mooring basin on the east side of Bullock Point, thence northward 0.5 mile to a mooring and turning basin. The entrance channel is marked by buoys and daybeacons. In 1982, a sunken wreck was reported on the west side of the channel at Bullock Point. There are numerous small-craft facilities in Bullock Cove. The largest marine railway, on Bullock Neck about 300 yards northward of Bullock Point, can handle craft up to 60 feet; berths, electricity, gasoline, diesel fuel, water, ice, storage, launching ramps, marine supplies, a pump-out facility, a 45-ton mobile crane and a 99-ton mobile hoist and hull, engine and electronic repairs are available. Sail repairs can be arranged nearby.

(235) **Seekonk River**, which branches off northeasterly from Providence River at Providence, is the approach to **Phillipsdale** and **Pawtucket**. The head of navigation is at Pawtucket, 5 miles above the mouth. Commerce on the river is chiefly in petroleum products. A marina at Pawtucket can provide berths, electricity, water, gasoline, storage facilities, marine supplies and hull and engine repairs. A flatbed trailer at the marina can haul out craft to 40 feet long.

(236)

Prominent features

(237) **Conimicut Light** (41°43'01"N., 71°20'42"W.), 58 feet above the water, is shown from a white conical tower on a brown cylindrical pier on the west side of the entrance to Providence River. A mariner radio-activated sound signal at the light is initiated by keying the microphone five times on VHF-FM channel 83A. An abandoned lighthouse on Nayatt Point is also prominent in the approach to the river. A white masonry conical tower, approximately 20 feet high, is west of the channel, 0.7 mile above Sabin Point. Three wind turbines are prominent on the west side of the river, about 0.6 mile above Fields Point.

(238)

Channels

(239) The federal project for Providence River provides for a channel 40 feet deep from just below Prudence Island Light to **Fox Point** near the junction of Providence and Seekonk Rivers. The channel is well marked with navigational aids. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A.

(240) A **hurricane barrier** crosses the Providence River about 200 yards above Fox Point. The barrier includes three gates that normally will be kept in the opened position until the approach of hurricane weather. The clearances at each of the three river gates are: horizontal,

(247)

Structures across Providence and Seekonk Rivers

Name•Description•Type	Location	Clear Width of Draw or Span Opening (feet)	Clear Height above Mean High Water (feet)	Information
Providence River				
Providence River/I-195 Bridge (fixed)	41°48'55"N., 71°24'06"W.	357	35	
Overhead power cable	41°48'56"N., 71°24'09"W.		145	
Overhead power cable	41°49'00"N., 71°24'12"W.		60	
Point Street Bridge (fixed)	41°49'05"N., 71°24'14"W.	99	7	Note 1
Pedestrian bridge	41°49'13"N., 71°24'20"W.			Bridge under construction
Seekonk River				
Overhead power cable	41°48'58"N., 71°23'22"W.		130	
Washington/I-195/US 6 Bridges (fixed)	41°49'08"N., 71°23'16"W.	100	40	
Tunnel Railroad Bridge (bascule)	41°49'25"N., 71°23'05"W.	92	17	Note 2
New Red Bridge (fixed)	41°49'45"N., 71°22'41"W.	100	42	
Overhead power cables	41°51'56"N., 71°22'48"W.		125	
Overhead power cables	41°52'21"N., 71°26'05"W.		42	
Division Street Bridge (fixed)	41°52'21"N., 71°23'03"W.	50	30	
Pawtucket/I-95 Bridge (fixed)	41°52'24"N., 71°23'04"W.	54	152	

Note 1 – Piers of a former fixed bridge cross the river channel about 300 yards north of Point Street Bridge.
Note 2 – The Tunnel Bridge has been abandoned and the span is locked in the open position.

20 feet; vertical (gate fully opened), 21 feet at mean high water; and depth over the gate sill, 12.9 feet at mean low water. Red lights mark the channel ends of each gate.

(241) **Seekonk River** empties into the easterly side of Providence River at Fox Point. A dredged channel, marked by buoys, leads from **Cold Spring Point**, about 1.3 miles above Fox Point, to a point about 150 yards southward of Division Street Bridge at Pawtucket, about 2.9 miles above Cold Spring Point. The lower section of the river, from Fox Point to Cold Spring Point, is crooked, winding and marked by buoys. Local knowledge and use of the chart are required to carry the best water.

(242) Three areas of submerged boulders with angle iron protrusions, the remains of the approaches and pivot pier of a former swing bridge, are in the river channel at Cold Spring Point in about 41°49'36"N., 71°22'48"W. A 5 mph no-wake zone, marked by state regulatory buoys, is in the entrance channel and above Cold Spring Point.

(243)

Anchorages

(244) Vessels anchor as directed by the **harbormaster** on the edge of the channel between Fields Point and Fox Point. Eastward of Fox Point, a few vessels may anchor in the area where a portion of Green Jacket Shoal was removed. Preferred small-craft anchorages are in Bullock Cove and Pawtuxet Cove.

(245)

Dangers

(246) Numerous rocks and ledges border Providence River Channel on either side. Navigational aids mark the shoal areas off **Bullock Point**, about 1.5 miles above the mouth; off **Sabin Point**, about 3 miles above the Mouth; off **Pomham Rocks**, about 3.5 miles above the mouth; off **Fuller Rock**, about 5 miles above the mouth and **Green Jacket Shoal**, east of Fox Point about 7.4 miles above the mouth. In 2019, a shoal area of 1.5 feet was reported in the Providence River south of the pedestrain bridge at about 41°49’09”N., 71°24’16”W.

(248)

Current

(249) Tidal currents are weak in the approach channel and the harbor, except in the constricted parts of Seekonk River. In Seekonk River the double flood is very pronounced. The velocity near the middle of the flood period is generally less than 0.5 knot and is sometimes in an ebb direction. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(250)

Weather, Providence and vicinity

(251) The proximity of Narragansett Bay and the Atlantic Ocean plays an important part in determining the climate for Providence and vicinity. In winter, the temperatures are modified considerably, and a good many of the major storms drop their precipitation in the form of rain, rather than snow. In summer, many days that would otherwise be uncomfortably warm are cooled by refreshing seabreezes. At other times of the year, sea fog may be advected over land by onshore winds. In fact, most cases of dense fog are produced in this way; but the number of such days is few, averaging 2 or 3 days per month.

(252) The temperature for the entire year averages around 51°F (10.6°C). January is the coldest month averaging

29°F (-1.7°C), and July the hottest month averaging 73°F (22.8°C). Freezing temperatures occur on the average about 117 days per year and the days with minimums below 5°F (-15°C) average six each year. An average nine days each year record maximums in excess of 90°F (32.2°C). The all-time maximum for Providence is 104°F (40°C) recorded in August 1975 and the all-time minimum is -13°F (-25°C) recorded in January 1976.

(253) Measurable precipitation occurs on about 178 days each year. November is the wettest month averaging 4.48 inches (114 mm) and June the driest averaging 2.89 inches (74 mm). Average annual precipitation is 45.12 inches (1,144 mm).

(254) Thunderstorms are responsible for much of the rainfall from May through August. They usually produce heavy, and sometimes even excessive, amounts of rainfall but since the duration is relatively short, damage is ordinarily light. The summer thunderstorms are frequently accompanied by extremely gusty winds, which may cause some damage to property, especially small pleasure and fishing craft.

(255) The first measurable snowfall of winter usually comes in October. The month of greatest snowfall is usually February, which averages about ten inches (254 mm). It is unusual for the ground to remain well covered with snow for any long period of time. The average annual snowfall total is 45 inches (1,143 mm). Snow has fallen in every month October through May. About seven days each year have snowfall greater than 1.5 inches (38 mm).

(256) In early fall, severe coastal storms of tropical origin sometimes bring destructive winds to this area. Even at other times of the year, it is usually coastal storms that produce the most severe weather. Between 1871 and 1996, twelve tropical storms have come within 25 miles of Providence. In August 1991, Hurricane Bob passed within 20 miles east of the city with 85-knot winds. Only twelve hours earlier, Bob was packing winds in excess of 100 knots. In September 1960, Hurricane Donna passed about 20 miles west of the city. At the time, Donna had maximum winds of 90 knots. Coastal areas of Rhode Island, Connecticut and Long Island were raked with winds in excess of 100 knots compliments of Hurricane Donna.

(257)

Ice

(258) The approach channel and the harbor are generally free of ice and navigable throughout the year. During severe winters, the harbor and several miles of Providence River and Upper Narragansett Bay are occasionally broken over, but the ice is usually broken up in the channels to the principal wharves by the traffic in the harbor.

(259) The National Weather Service maintains an office at the T.F. Green State Airport; barometers may be compared here.

(260)

Pilotage, Providence

(261) See Pilotage, Narragansett Bay and Other Rhode Island Waters (indexed as such), early this chapter.

(262)

Towage

(263) Tugs up to 3,300 hp are available at Providence. Large vessels normally require tugs for docking and undocking. Arrangements for tug service should be made 4 hours in advance, and usually through ships' agents or directly by shipping companies. Tugs are dispatched 24 hours a day. Tugs monitor VHF-FM channels 10, 13, 16 and 18 and use channel 7A as a working frequency.

(264) Providence is a **customs port of entry.**

(265)

Quarantine, customs, immigration and agricultural quarantine

(266) (See chapter 3, Vessel Arrival Inspections, and Appendix A for addresses.)

(267) **Quarantine** is enforced in accordance with regulations of the U.S. Public Health Service. (See Public Health Service, chapter 1.)

(268) Providence has several hospitals.

(269)

Coast Guard

(270) A **marine safety o f ce** is in Providence. (See Appendix A for address.)

(271) **Harbor regulations** are enforced by the **harbormaster/port director**, whose headquarters are at the municipal wharf. The harbormaster regulates the movement and anchoring of vessels in the harbor. The **speed** limit in the harbor is 5 knots.

(272)

Wharves

(273) The piers and wharves of the port of Providence are along both sides of the Providence River below Fox Point. Only the deep-draft facilities are described. The alongside depths are reported; for information on the latest depths contact the operator. All the facilities described have highway connections, and most have rail connections. Water is available at most of the piers and wharves. Cargo in the port is usually handled by ship's tackle. Cranes to 200 tons are available.

(274) **Facilities on the east side of Providence River:**

(275) Mobil Oil Corp. Wharf: (41°46'56"N., 71°22'19"W.); 1,225-foot face, 700 feet of berthing space; 20 to 38 feet alongside; deck height, 8 feet; pipelines to storage tanks; receipt and shipment of petroleum products, bunkering vessels; owned and operated by Mobil Oil Corp.

(276) Amoco Oil Co. Wharf: at Kettle Point about 0.9 mile above Mobil Oil Corp. Wharf; 500-foot face, 700 feet with dolphins; 36 feet alongside; deck height, 12 feet; receipt and shipment of petroleum products, bunkering vessels; owned by Amoco Oil Co., operated by Amoco Oil Co. and Atlantic Richfield Co.

(277) Wilkes-Barre Pier: about 2 miles above Mobil Oil Corp. Wharf; 75-foot face, 700 feet of berthing space with

dolphins; 40 feet alongside; deck height, 9 feet; receipt of petroleum products and naphtha; owned by Providence and Worcester Railroad Co., operated by Union Oil Co. of California, Getty Refining and Marketing Co., and Astroline Corp.

(278) **Facilities on the west side of Providence River**

(279) Municipal Wharf, Berths 5 and 6: (41°47'25"N., 71°22'54"W.); 1,283-foot face; 35 to 40 feet alongside; deck height, 10½ feet; two 45-ton container cranes; 60 acres open storage; electrical shore power connections; receipt and shipment of general and containerized cargo and heavy equipment; owned by city of Providence, operated by Cranes Associates.

(280) Municipal Wharf, Berths 1, 2, 3, and 4: immediately northwest of Berths 5 and 6; 2,190-foot face, 35 to 40 feet alongside; deck height, 10½ feet; cranes to 200 tons; 47,000 square feet of covered storage; 12 acres open storage; electrical shore power connections; receipt and shipment of general cargo, lumber, paper products, automobiles, petroleum products, liquefied petroleum gas, scrap metal, pig iron and caustic soda; owned by city of Providence; various operators.

(281) New England Bituminous Wharf: about 500 yards above Municipal Wharf; 384-foot face, 30 feet alongside; deck height, 10½ feet; receipt of asphalt; owned and operated by New England Bituminous, Division of John J. Hudson, Inc.

(282) Lehigh Portland Cement Co. Wharf: about 600 yards above Municipal Wharf; 350-foot face, 20 feet alongside; deck height, 10½ feet; receipt of bulk cement; owned by city of Providence, operated by Lehigh Portland Cement Co.

(283) Lone Star Industries Wharf: about 750 yards above Municipal Wharf; 210-foot face, 28 to 30 feet alongside; deck height, 12 feet; receipt of bulk cement; owned and operated by Lone Star Industries, Inc.

(284) Algonquin LNG Wharf: about 0.4 mile above Municipal Wharf; 450-foot face, 25 feet alongside; deck height, 12 feet; receipt of liquefied natural gas; owned by Providence Gas Co., operated by Algonquin LNG, Inc.

(285) Texaco U.S.A., Harbor Junction Pier: about 0.7 mile above Municipal Wharf; 80-foot face, south side 1,040 feet long, 720 feet usable, 32 feet alongside; north side 1,040 feet long, 600 feet usable, 25 feet alongside; deck height, 9 feet; receipt and shipment of petroleum products, bunkering vessels; owned and operated by Texaco, Inc.

(286) Promet Marine Services Pier: about 1 mile above Municipal Wharf; 120-foot face, 37 to 31 feet alongside; south side 596 feet long, 22 feet alongside; north side 596 feet long, 37 feet alongside; deck height, 12 feet; cranes to 100 tons; 11 acres of open storage; receipt and shipment of general cargo and dry bulk materials; repairs to vessels; owned and operated by Promet Marine Services Corp.

(287) Northeast Petroleum Corp. Pier: about 1.2 miles above Municipal Wharf; south side 620 feet long, 600 feet of berthing space; 30 feet alongside; deck height, 11 feet; receipt of petroleum products; owned and operated by Northeast Petroleum Corp.

(288) C. H. Sprague & Son Co. Pier: about 1.25 miles above Municipal Wharf; north side 500 feet long, 520 feet with platforms, 37 feet alongside; deck height, 12 feet; receipt and shipment of petroleum products; bunkering vessels; owned and operated by C. H. Sprague & Son Co.

(289) The principal wharves at Pawtucket have depths of 9 to 14 feet alongside.

(290)

Supplies

(291) Gasoline, diesel fuel, diesel oil, bunker fuels, provisions and marine supplies of all kinds are available. Oil bunkering facilities, for deep-draft vessels, are available at most of the petroleum companies facilities in Providence. Fuel tank barges are also available for bunkering vessels anywhere in the harbor. Water is available at most of the wharves and piers.

(292)

Repairs

(293) Providence has no facilities for drydocking deep-draft vessels; the nearest such facilities are at Boston, MA. Repairs to boilers, machinery, electrical equipment and hull can be obtained in the port. Several well-equipped machine and welding shops are also available. Some of these concerns also maintain portable equipment for making above-waterline repairs to vessels at their berths.

(294)

Small-craft facilities

(295) Small-craft facilities at Bullock Cove and Pawtuxet have been discussed earlier in this chapter. A marina on the west side of Providence River between Pawtuxet and Fields Point can provide berths, electricity, gasoline, diesel fuel, water, ice, storage, marine supplies and hull and engine repairs; a flatbed trailer can haul out craft to 60 feet long. In 1981, 10 feet was reported in the approach to the marina, with 3 to 6 feet alongside.

(296)

Communications

(297) Providence is served by rail, bus and air. A ferry operates daily in the summer to Newport and Block Island.

(298)

West Passage to Calf Pasture Point

(299) **West Passage**, between Conanicut and Prudence Islands on the east and Boston Neck on the west, is the approach to Dutch Island Harbor, Wickford, Quonset Point and East Greenwich. Vessels may also go to Providence by West Passage, although the route through East Passage is deeper and generally used. Approaching from the eastward, steer for the lighted gong buoy off Whale Rock until southwest of Beavertail Light, and thence lay down a northerly course in midchannel with Dutch Island ahead. On the southward approach from off Point Judith Light, a north-northeasterly course will

bring the vessel to the lighted gong buoy off Whale Rock. At night a careful study of the light characteristics is necessary as the lights marking East Passage will be seen on the starboard bow when approaching from Point Judith.

(300) The course should pass westward of, and 500 yards off, Dutch Island; thence through the Jamestown-Verrazzano Bridge opening; and thence northerly until about 0.8 mile westward of, and abeam, the south tangent of Hope Island. From here a north-northeasterly course will make the buoys marking the entrance of the natural channel westward of Pine Hill Point on Prudence Island. Then a heading toward Warwick Light until abeam the lighted bell buoy off Northwest Point on Patience Island will bring the vessel in position to swing northeasterly and easterly to the dredged channel to Providence.

(301) **Narragansett Pier**, on the west side of West Passage about 3 miles west-southwestward of Beavertail Point, is a summer resort. The large hotels and a square granite tower are prominent. A municipal bathing beach and pavilion at the Upper Pier are prominent from an easterly direction.

(302) **River Ledge**, about 0.9 mile northeastward of Narragansett Pier, has a least depth of 10 feet and is marked by a buoy. **Whale Rock**, on the western side of the passage about 0.8 mile northeast of River Ledge, is marked by a lighted gong buoy to the eastward. **Little Whale**, covered 5 feet, is about 200 yards north of Whale Rock. Strangers should pass eastward of the lighted gong buoy off Whale Rock.

(303) **Bonnet Shores Beach** is on the north shore of the bight formed by **Bonnet Point**, the point about 1.5 miles north of Whale Rock. A bathing pavilion at the beach is prominent from a southeasterly direction.

(304) **The Bonnet**, a prominent hill with the shoreward face bold and rocky, is north of Bonnet Point. The shore between Bonnet Point and **South Ferry**, 1.3 miles northward, should be given a berth of 400 yards. Pilings extend 130 yards eastward just south of the old pier at South Ferry. A 200-foot L-shaped pier of the University of Rhode Island is about 150 yards southeastward of the old pier. In 1981, depths of 20 feet were reported along the outer face with 10 feet reported along the inner face. The buildings of the university, a church spire, and a standpipe are prominent from southward to north of Dutch Island.

(305) **Fox Hill**, on the southern side of the entrance to Dutch Island Harbor, 2.5 miles north of Beavertail Point, is a point which terminates to the northward in Beaverhead, a bluff rocky face.

(306) **Dutch Island Harbor** is in the West Passage of Narragansett Bay about 3 miles north of Beavertail Light. The harbor is a semicircular indentation 0.5 by 1 mile in extent in the west side of Conanicut Island.

(307) A boatyard in the southeastern part of the harbor has a marine railway that can handle craft up to 50 feet for hull and engine repairs. Berths, electricity, gasoline, diesel fuel (by truck), water, ice, storage, a launching ramp, some marine supplies and a 20-ton lift are available.

(308) **Dutch Island**, a state park about 3.2 miles north of Beavertail Point, is surrounded by shoals and foul ground. A bell buoy marks the shoal area off the northern end of the island and a lighted gong buoy is off the southern end. A square white tower on the southern end of the island is prominent marked by a light.

(309) Dutch Island Harbor may be approached from northward or southward. As the harbor is of easy access, it is frequently used as a harbor of refuge. Excellent anchorage may be had in depths of 12 to 46 feet, sticky bottom. A **harbormaster** controls all mooring and berthing. Vessels of over 18-foot draft seeking anchorage should give the eastern shore of the harbor a berth of at least 0.4 mile. The eastern shore of Dutch Island should be given a berth of 100 yards.

(310) **General anchorages** are in West Passage of Narragansett Bay. (See **33 CFR 110.1** and **110.145 (b)** and **(d)**, chapter 2, for limits and regulations.)

(311) A shoal extends about 200 yards westward and 350 yards northward of **Beaverhead**. At its northern extremity this shoal rises abruptly from depths of about 40 feet to a depth of 8 feet; a buoy marks the shoal. The southeastern part of the harbor has a shallow cove.

(312) Tidal currents of 1 to 1.5 knots may be encountered in the vicinity of Dutch Island. Elsewhere in West Passage velocities are usually less than 1 knot. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(313) **Saunderstown** is on the western shore of West Passage abreast Dutch Island. A former ferry dock off the town is in ruins. A yacht club is at Saunderstown.

(314) The **Jamestown-Verrazzano Bridge** crossing the passage from just north of Plum Beach has a fixed span with a clearance of 135 feet. About 300 feet south of the aforementioned highway bridge, the Jamestown-North Kingstown bridge is submerged and in ruins.

(315) From the bridge northward, **Great Ledge** extends along the west shore of Conanicut Island for about 1.5 miles. This ledge is from 175 to 600 yards offshore, culminating in **America Ledge** at its northern end. Numerous rocks are on Great Ledge. Between Plum Beach and **Rome Point**, on the west side of the passage, 1.2 miles north of the bridge, are several rocks, including **Red Rock** and **Old Sergeant**. Bare and covered rocks are northeastward and eastward of Rome Point.

(316) **Fox Island**, 0.4 mile northeast of Rome Point and southward of Wickford Harbor, is small and low. A shoal with numerous submerged rocks including **Seal Rock** extends southward of the island. A narrow channel, suitable only for small craft, is between this shoal and the shoals extending northeastward of Rome Point. **Halfway Ledge** with a depth of 18 feet is about 0.5 mile east of Fox Island.

(317) **Wickford Harbor**, on the western side of Narragansett Bay 8 miles above Beavertail Light,

comprises an outer and an inner harbor. The outer harbor is a broad bight between Quonset Point on the north and **Wild Goose Point**, about 0.6 mile westward of Fox Island, on the south. The entrance is about 2 miles wide. The inner harbor entrance is between **Poplar Point**, 1.3 miles northwest of Fox Island on the south, and **Sauga Point**, about 0.4 mile north of Poplar Point, on the north. The harbor is used chiefly by recreational craft and by oyster and lobster boats. The town of **Wickford** is on the southwestern side of the inner harbor.

(318) Several prominent landmarks are visible when approaching Wickford Harbor. A standpipe and a church spire in Wickford may be seen for many miles. An abandoned lighthouse on Poplar Point and the light off the point are prominent.

(319) The channel to the inner harbor, marked by buoys, is restricted to a width of about 150 yards by the breakwaters and the shoals off Sauga and Poplar Points. The north breakwater is marked by a light. The channel leads to the junction of three coves, **Fishing Cove** to the northward, **Mill Cove** to the northwestward, and Wickford Cove to the southwestward. A state regulatory buoy just inside the jetties marks a **"Slow no wake" zone.**

(320) **Wickford Cove** is the scene of considerable pleasure-boat activity. A dredged channel in the cove extends between flats, many of which are dry at low water, to a highway bridge about 0.9 mile above the breakwaters. Numerous piles, used as moorings, border the channel for about 0.35 mile below the highway bridge.

(321) A marked dredged channel in Mill Cove leads to an anchorage basin about 0.7 mile above the breakwaters.

(322) Good anchorage may be had in the middle and southern parts of outer Wickford Harbor. The northern part of the outer harbor has numerous rocks and ledges. **General Rock**, with a depth of 9 feet over it, is the southerly limit of this shoal, 0.9 mile north-northeastward of Fox Island. A rock with a depth of 10 feet over it is about 500 yards westward of General Rock. **Brig Ledge**, about 0.5 mile north of General Rock, is covered 9 feet. The southern shore of the outer harbor is foul. **Charles Rock**, with a depth of 4 feet, is just inside of the northern breakwater.

(323) Vessels approaching Wickford Harbor from the southward, after passing through the main span of the Jamestown-Verrazzano Bridge, steer 340°. When northeastward of Fox Island, steer for Wickford Harbor Light 1 on any bearing between 313° and 290°, anchoring 0.2 mile or more southeastward of the light in depths of 13 to 15 feet, soft bottom.

(324) In severe winters the inner harbor is closed by ice, but the outer harbor is usually open although drift ice is occasionally encountered.

(325) Wickford has several small-craft facilities and boatyards. The largest marine railway, on the east of Wickford Cove, can handle craft up to 60 feet. Berths, electricity, gasoline, diesel fuel, water, ice, marine supplies, pump-out facilities, a launching ramp, mobile hoists to 30 tons and lifts to 70 tons and hull, engine and electronic repairs are available; wet and dry storage is also available. In 1981, a reported depth of about 7 feet could be carried to the marine railway at Wickford Cove.

(326) **Quonset Point**, on the north side of Wickford Harbor, is marked by elevated tanks. Near the eastern end of the point are the conspicuous buildings of the Quonset Point Industrial Park. The piers at Quonset Point, and at **Davisville**, about 1.5 miles northward, are usually approached from East Passage until north of Conanicut Island, thence through a buoyed dredged channel to a turning basin off the point from which a channel leads to the piers at Davisville. A depth of about 33 feet can be carried in the channel to the turning basin. Depths of 27 to 35 feet are available throughout the basin. An obstruction, covered 26 feet, is at 41°35′09.7"N., 71°24′34.0″W. A depth of about 32 feet can be carried in the channel to Davisville, thence about 26 feet through the turning basin to the piers.

(327) **Hope Island**, about 1.6 miles north of Conanicut Point, has low grassy hills with a few trees. The island is a state estuarine sanctuary and is off limits to visitors during the bird nesting season. Bare and submerged rocks surround the island for about 0.2 mile. **Despair Island** is on the outer end of a rocky ledge extending 0.2 mile northeastward of Hope Island; a buoy is off the northeast side of the ledge. **Scup Rock** and **Round Rock** are off the eastern side of Hope Island, and **Gooseberry Island** and **Seal Rock** are off the western side. A hazardous reef with piles and a boiler awash on it is about 0.4 mile southwestward of the southwest point of Hope Island; a lighted buoy marks the area.

(328) **Allen Harbor** is 2 miles north of Quonset Point. The harbor is entered through a buoyed channel which has a depth of about 8 feet. Depths of 8 to 10 feet are inside. The town docks and a launching ramp are on the southeast side of the harbor.

(329) **Calf Pasture Point** is on the north side of the entrance to Allen Harbor. Abreast the point and for some distance northward of it, a shoal extends 0.5 mile from shore.

(330)

Potowomut River to Providence Point

(331) **Potowomut River**, entering the west side of West Passage, 1.7 miles north of Calf Pasture Point, is separated from Greenwich Bay on the north by **Potowomut Neck**. A dredged channel leads through the entrance of the river. Caution is necessary to avoid rocks with depths of 1 foot over them in the entrance. Strangers should not enter the river.

(332) **Round Rock**, about 0.7 mile eastward of Potowomut River entrance, uncovers 3 feet and is marked by a seasonal lighted buoy close eastward. Several other rocks, awash and submerged, lie between Round Rock and the entrance to the river; caution is advised.

(333) The natural channel of West Passage extends between the shoal area eastward of Round Rock and

the shoal area westward of Patience Island. The channel has depths of 21 to 70 feet. Buoys mark the entrance from the southward, and a lighted bell buoy marks the northwestward edge of the shoal off Patience Island. The channel is the approach from southward to Greenwich Bay, Warwick Point and the channel from West Passage to Providence River.

(334) **Patience Island**, 0.2 mile west of the northern end of Prudence Island, is surrounded by shoals and foul ground. The island is a state park and estuarine sanctuary.

(335) **Warwick Point**, the southernmost point of **Warwick Neck**, 0.7 mile northwest of Patience Island, is marked by a light and sound signal.

(336) **Greenwich Bay**, at the northwestern end of Narragansett Bay, is entered between Warwick Neck and Potowomut Neck. Shoal water borders the shore of the bay, but the general depths are 10 feet or more.

(337) **Warwick Cove**, between Warwick Neck and Horse Neck, is in the northeastern part of Greenwich Bay. A federal project provides for a 6-foot channel from the bay to an anchorage basin at the head of the cove. Other anchorage basins in the cove are on the west side of the channel, 0.5 mile above the channel entrance and on each side of the channel 0.7 mile above the channel entrance. All of the anchorage basins have a project depth of 6 feet. A state regulatory buoy off Horse Neck marks a 5 mph **speed limit**. The cove is the scene of considerable pleasure boat activity.

(338) The **harbormaster** in the cove controls berthing and anchorage; contact can be made through the Warwick City Hall. The cove has several marinas and boatyards. Berths, electricity, gasoline, water, ice, storage, launching ramps, marine supplies and hull and engine repairs are available. The largest lift, on the east side of the cove about 0.3 mile above the mouth, can handle craft up to 70 tons.

(339) **Brush Neck Cove**, about 0.5 mile west of Warwick Cove, is fronted by a flat with a general depth of about 2 feet. This channel is used by small local craft at high water as far as the pier at Oakland Beach. **Oakland Beach**, on **Horse Neck**, between Brush Neck and Warwick Coves, is a summer resort with bus communication.

(340) **Apponaug Cove**, in the northwestern part of Greenwich Bay, is entered through a marked federal channel with a project depth of 6 feet that leads from the bay to an anchorage basin on the southwest side of the channel just below a fixed railroad bridge about 0.7 mile above the channel entrance. A state regulatory buoy at the entrance to the cove marks a 5 mph **speed zone**. (See Notice to Mariners and latest edition of chart for controlling depths.) Small-craft facilities can be found in the cove and at the west end of Greenwich Bay southwest of the entrance channel to the cove.

(341) **Chepiwanoxet Point**, on the western side of the bay northward of the entrance to Greenwich Cove, is a small neck of land with a yellow bluff facing eastward. From this island, shoals with little water over them extend about 500 yards northward and 300 yards eastward and southeastward. Shoals extend about 300 yards northward and westward of **Long Point**, the northwestern extremity of Potowomut Neck.

(342) **Greenwich Cove**, in the southwest end of Greenwich Bay, is about 1.3 miles long and 300 to 600 yards wide. Buoys mark the entrance channel into the cove. State regulatory buoys at the entrance mark a **"slow no wake" zone.** On the western shore is the town of **East Greenwich**. Depths of about 7 to 11 feet are available in the cove to about 0.5 mile from the head. Good anchorage may be had off some of the small-craft facilities on the west side of the cove in depths of 8 to 11 feet.

(343) The cove has several boatyards. Berths, electricity, gasoline, water, diesel fuel, ice, marine supplies, wet and dry storage, launching ramps, lifts to 21 tons and complete engine and hull repairs are available. The **harbormaster** in the cove controls anchoring and berthing; contact can be made through the Warwick City Hall.

(344) From **Sandy Point**, the eastern extremity of Potowomut Neck, shoals with depths of 2 to 9 feet extend northeasterly for about 0.6 mile. Extensive shoals extend off the eastern side of Warwick Neck to Ohio Ledge. **Rocky Point** is on the eastern side of the neck, 1.7 miles north-northeastward of Warwick Point.

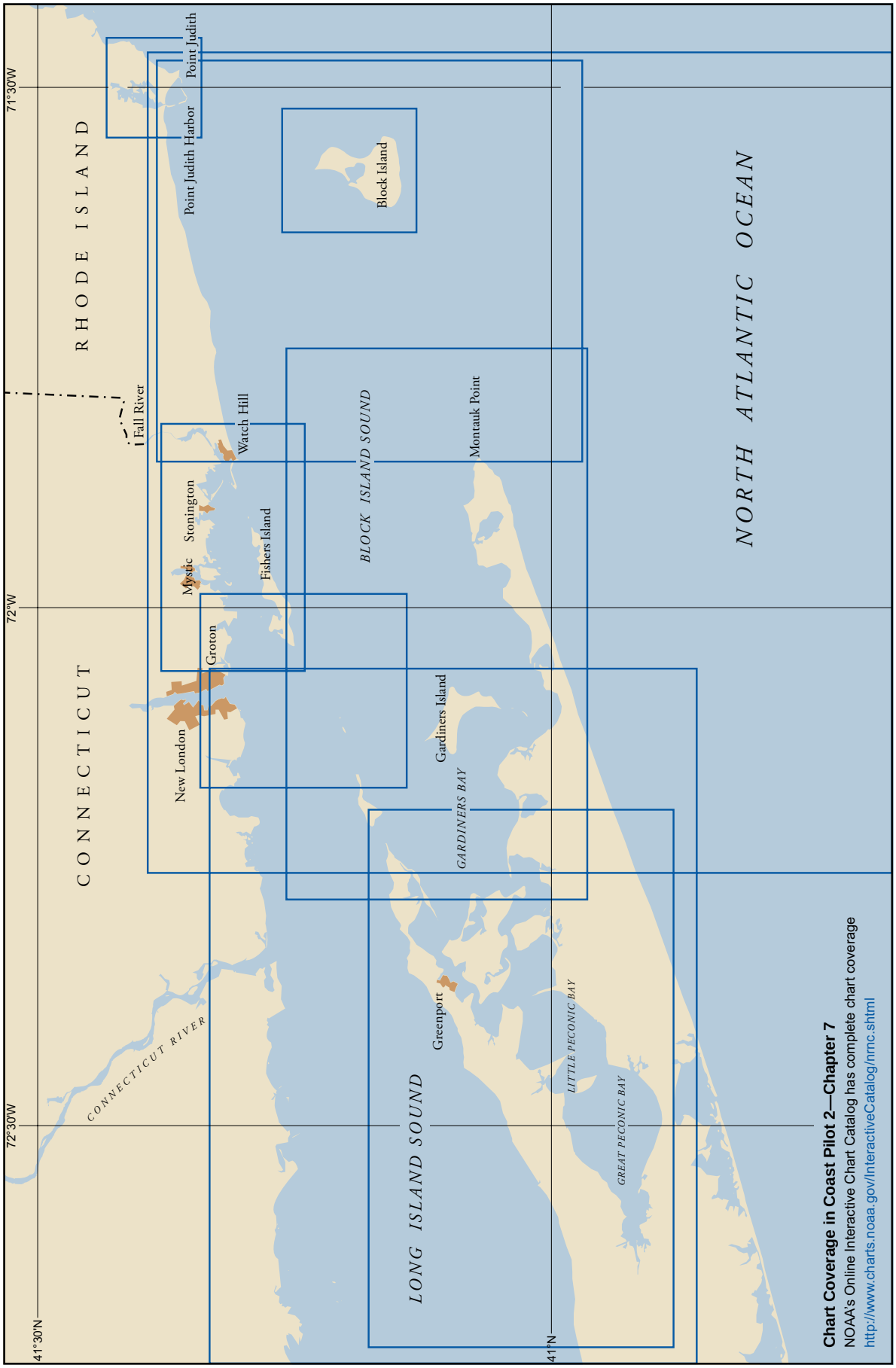
(345) The natural channel between the shoals off Warwick Neck and the shoals northward of Patience and Prudence Islands has depths of 19 to 50 feet. A buoy marks the shoal off **Providence Point**, the northernmost point of Prudence Island.

(346)

Point Judith Neck

(347) The shoreline of **Point Judith Neck** between West Passage and Point Judith should be given a berth of at least 0.6 mile. From Narragansett Pier to **Black Point**, a rocky promontory 1.9 miles southward, the shoreline is a rugged rocky ledge with deep water close inshore. The waters between Black Point and Point Judith are boulder strewn and shoal up gradually.

(348) Two very prominent landmarks are Point Judith Light and the elevated water tank 1.0 mile north of Point Judith. Closer inshore the stone bathing pavilion at the state-operated **Scarborough Beach**, 0.5 mile south of Black Point, and a stone tower on a house 0.4 mile north of Black Point are prominent.


RHODE ISLAND
CONNECTICUT
NORTH ATLANTIC OCEAN
BLOCK ISLAND SOUND
LONG ISLAND SOUND
GARDINERS BAY
LITTLE PECONIC BAY
GREAT PECONIC BAY
CONNECTICUT RIVER
Point Judith
Point Judith Harbor
Block Island
Fall River
Watch Hill
Stonington
Mystic
Fishers Island
Montauk Point
Groton
New London
Gardiners Island
Greenport
71°30'W
72°W
72°30'W
41°30'N
41°N
Chart Coverage in Coast Pilot 2—Chapter 7
NOAA's Online Interactive Chart Catalog has complete chart coverage
http://www.charts.noaa.gov/InteractiveCatalog/nrnc.shtml

Block Island Sound

(1) This chapter describes Block Island Sound, Fishers Island Sound, Gardiners Bay, Little Peconic Bay and Great Peconic Bay and the ports and harbors in the area, the more important of which are Point Judith Harbor, Great Salt Pond, Stonington, Mystic Harbor and Greenport.

(2)

COLREGS Demarcation Lines

(3) The lines established for this part of the coast are described in **33 CFR 80.150** and **80.155**, chapter 2.

(4)

Block Island Sound

(5) **Block Island Sound** is a deep navigable waterway forming the eastern approach to Long Island Sound, Fishers Island Sound and Gardiners Bay from the Atlantic Ocean. The sound is a link for waterborne commerce between Cape Cod and Long Island Sound. It has two entrances from the Atlantic: an eastern entrance from Rhode Island Sound between Block Island and Point Judith and a southern entrance between Block Island and Montauk Point. The sound is connected with Long Island Sound by The Race and other passages to the southwestward and with Fishers Island Sound by several passages between rocky reefs from Watch Hill Point to East Point, Fishers Island.

(6) The north shoreline of Block Island Sound and Fishers Island Sound from Point Judith to New London is generally rocky and broken with short stretches of sandy beach. Many inlets and harbors, especially in the vicinity of Fishers Island, afford harbors of refuge for vessels. Most of the rocks and shoals near the channels are marked with navigational aids.

(7) The southern part of Block Island Sound is bounded by Block Island on the east, the eastern extremity of Long Island and Gardiners Island on the west. Plum Island and Fishers Island are at the western end of the sound.

(8) The deep water in the central part of Block Island Sound will accommodate vessels of the greatest draft.

(9) Westward of Gardiners Island, enclosed between the northeastern and eastern ends of Long Island, are Gardiners Bay, Shelter Island Sound, Little Peconic Bay and Great Peconic Bay. This area is well protected but generally shallow and is not suited for deep-draft vessels. The shoreline is marked by many indentations and shallow harbors. These waters are much used by commercial fishing vessels and small pleasure craft because of the protection afforded and the many anchorages.

(10) **Recommended Vessel Route (Block Island Sound)** has been established for Block Island Sound.

(11) The U.S. Coast Guard Captain of the Port, Providence, in cooperation with the Southeastern Massachusetts and Rhode Island Port Safety and Security Committees, has established a Recommended Vessel Route for deep draft vessels and tugs/barges transiting Rhode Island Sound, Narragansett Bay and Buzzards Bay. Deep draft vessels and tugs/barges are requested to follow the designated routes. These routes were designed to provide safe, established routes for these vessels to reduce the potential for conflict with recreational boaters, fishing gear, and other small craft and to reduce the potential for grounding or collision. Vessels are responsible for their own safety and are not required to remain inside the route nor are fisherman required to keep fishing gear outside the route. Small vessels should exercise caution in and around the Recommended Vessel Routes and monitor VHF channels 16 or 13 for information concerning deep draft vessels and tugs/barges transiting these routes.

(12) **Block Island North Reef** is a sand shoal with a least depth of 11 feet extending 1 mile northward from **Sandy Point** at the north end of Block Island. The shoal should be avoided by all vessels; its depths change frequently, and its position is also subject to a slow change. It is practically steep-to on all sides, so that soundings alone cannot be depended on to clear it. A lighted bell buoy is 1.5 miles northward of the point.

(13) **Southwest Ledge**, 5.5 miles west-southwestward of Block Island Southeast Light, has a least known depth of 21 feet and is marked on its southwest side by Southwest Ledge Lighted Whistle Buoy 2. Rocky patches extend 1.5 miles northeastward from the ledge. The sea breaks on the shoaler places on the ledge in heavy weather.

(14) Several other dangers that must be guarded against are northward and westward of Southwest Ledge Lighted Whistle Buoy 2. These dangers are 37-foot sounding, marked by a lighted buoy, about 2.2 mile 280° from the lighted whistle buoy and numerous rocks up to 1.1 miles north of the lighted whistle buoy.

(15) The deepest passage in the southern entrance to Block Island Sound is just westward of Southwest Ledge and has a width of over 2 miles; this is the best passage for deep draft vessels. The area between Southwest Ledge Lighted Whistle Buoy 2 and Block Island Sound South Entrance Obstruction Lighted Buoy BIS is known locally as Montauk Channel. Mariners should keep in mind that vessels with a draft in excess of 38 feet will not be allowed to transit this area. Further, pilots using Montauk Channel shall consider draft, sea and swell, wind, visibility, current and vessel traffic. When these conditions pose a threat to the safety of any person, vessel, prudent navigation or

safety of the environment, Montauk Channel shall not be used.

(16) Between the inner patch of rocks and the shoals, which extend 0.9 mile from Block Island, is a channel 1.3 miles wide, with a depth of about 31 feet. Vessels using this channel should round the southwest end of Block Island at a distance of 1.5 miles. It is not advisable to use this passage during heavy weather.

(17) The entrance between Point Judith and Block Island is used by vessels coming from the bays and sounds eastward to Long Island Sound. The route generally used is through The Race. Tows of light barges and vessels of 14 feet or less draft sometimes go through Fishers Island Sound, especially during daylight with a smooth sea. This entrance is clear with the exception of Block Island North Reef and the numerous large boulders extending about 4 miles south-southeastward of Point Judith. The coast from Point Judith nearly to Watch Hill should be given a berth of over 1 mile, avoiding the broken ground with depths less than 30 feet.

(18)

Tides and currents

(19) The effect of strong winds, in combination with the regular tidal action, may at times cause the water to fall several feet below or rise the same amount above the plane of reference of the chart.

(20) **Tidal current** data for a number of locations in Block Island Sound are available from the Tidal Current prediction service at *tidesandcurrents.noaa.gov*. Links to a user guide for this service can be found in chapter 1 of this book.

(21) The tidal currents throughout Block Island Sound have considerable velocity; the greatest velocities occur in the vicinity of The Race and in the entrances between Montauk Point, Block Island, and Point Judith. Soundings alone cannot be depended upon to locate the position; the shoaling is generally abrupt in approaching the shores or dangers.

(22) In the middle of the passage between Point Judith and Block Island, the velocity is 0.7 knot. The flood sets westward and the ebb eastward.

(23) In the passage between Block Island and Montauk Point, the flood sets generally northwestward and the ebb southeastward. In the middle of the passage the velocity is 1.5 knots on the flood and 1.9 knots on the ebb. About 1.2 miles eastward of Montauk Point, the flood sets 346°, ebb 162°, with a velocity of 2.8 knots.

(24) In Block Island Sound and in the eastern part of Long Island Sound, **fogs** are generally heaviest with southeast winds. In these waters the usual duration of a fog is from 4 to 12 hours, but periods of from 4 to 6 days have been known with very short clear intervals. In the autumn, **land fogs,** as they are termed locally, sometimes occur with northerly breezes, but are generally burned off before midday.

(25) The Race may be said to be the only locality where tidal currents have any decided influence on the movements of the ice. Large quantities of floe ice usually pass through The Race during the ebb, especially if the wind is westerly, and in severe winters this ice causes some obstruction in Block Island Sound and around Montauk Point. These obstructions are the most extensive around the middle of February.

(26)

Weather, Block Island Sound and vicinity

(27) Land influences the weather only at the northern edge of the Sound, with a northerly wind. Otherwise the waters are open, similar to the nearby ocean. Winds from all other directions have ample time to increase in strength, and the Sound can be as turbulent as any water off the coast. Wind speeds can be double those found on the coast, especially in winter, when average speeds of 16 to 17 knots are common. Gales occur up to 5 percent of the time in winter and are most likely from the west and northwest. Seas built by winds from the southeast through southwest are usually highest since there is no land to interfere with the fetch. Seas of 10 feet (3 m) or more are likely 5 to 7 percent of the time in winter.

(28) Because of relatively cold water, summer fog occurs two to three times more often in these waters than in either Narragansett or Buzzard Bays. For example, in June visibilities drop below ½ mile nearly 9 percent of the time.

(29)

North Atlantic right whales

(30) Endangered North Atlantic right whales may occur in Block Island Sound, in particular in the Narragansett/Buzzards Bay Traffic Separation Scheme. They may also occur 30 miles south of Block Island Sound (peak season: November through April). The Northeast Marine Pilots distribute educational material to mariners in an effort to reduce right whale ship strikes. (See North Atlantic Right Whales, indexed as such, in chapter 3 for more information on right whales and recommended measures to avoid collisions.)

(31) All vessels 65 feet or greater in length overall (LOA) and subject to the jurisdiction of the United States are restricted to speeds of 10 knots or less in the Block Island Sound Seasonal Management Area between November 1 and April 30. The area is defined as the waters bounded by:

(32) 40°51'53.7"N., 70°36'44.9"W.;

(33) 41°20'14.1"N., 70°49'44.1"W.;

(34) 41°04'16.7"N., 71°51'21.0"W.;

(35) 40°35'56.5"N., 71°38'25.1"W.; thence back to starting point. (See **50 CFR 224.105** in chapter 2 for regulations, limitations and exceptions.)

(36)

Pilotage, Block Island Sound and Long Island Sound

(37) Pilotage is compulsory for foreign flag vessels and U.S. vessels that are under register (i.e. engaged in foreign trade) in Block Island Sound and Long Island Sound. Vessels should not enter Block Island Sound or Long Island Sound without a state licensed pilot. See

Pilotage, Long Island Sound (indexed as such), chapter 8. The Point Judith Pilot Station is the primary pilot boarding location for entry into Block Island Sound and Long Island Sound. Vessels bound for Long Island Sound ports may board pilots at the Point Judith Pilot Station, centered on 41°17.0'N., 71°30.5'W. There is a secondary pilot station that may be used with special arrangement at any point south of the Montauk Point Pilot Station, centered on 41°02.0'N., 71°42.0'W.

(38)

Block Island

(39) **Block Island**, 5 miles long, is hilly with elevations up to about 200 feet. The shore of the island is fringed in most places by boulders and should be given a berth of over 0.5 mile even by small craft; the shoaling is generally abrupt in approaching the island.

(40)

Weather, Block Island and vicinity

(41) Block Island, formed by glaciers, consists of nearly 7,000 acres (2,830 hectares) and lies in the Atlantic Ocean about 12 miles east-northeast of Long Island and about the same distance south of Charlestown, RI. Hence, the climate is typically maritime, but under conditions of extreme cold or heat the effect is felt on the island as well as on the mainland. Temperatures of -10°F (-23.3°C, February 1992) and 95°F (35°C, August 1948) have been recorded.

(42) Summers are usually dry. Recorded rainfall for any one month ranges from a trace to 11.51 inches (292 mm). November is the wettest month averaging 4.08 inches (104 mm) and June is the driest averaging 2.46 inches (64 mm).

(43) The warmest month is July with an average high of 76.5°F (24.7°C) and an average low of 63.7°F (17.6°C). The coolest months are January and February. Each average 32°F (0°C). The island is too small to build up cumulonimbus clouds, and local thunderstorms do not occur. Fog occurs on one out of four days in the early summer, when the ocean is relatively cold and foggy days average about 22 each year.

(44) Winters are distinguished for their comparative mildness; maximums average 36°F to 42°F (2.2°C to 5.6°C) and minimums average 26°F (-3.3°C) in January and February. Since the surface winds are usually easterly when snow begins it soon changes to rain or melts rapidly after it piles up. The ocean temperatures are always somewhat above freezing and not far off shore are relatively high.

(45) The ocean has a dampening effect on hot winds in summer and an accelerating effect on cold winds from the mainland in the winter. Katabatic winds from Narrangansett Bay and Long Island reach as high as 35 knots when anticyclonic conditions prevail on the mainland in winter. The wind velocity averages 15 knots for the year, but the mean is 17 knots in the winter, when gales are frequent. In the early fall most of the tropical storms moving up the coast affect the island to some extent. Since 1871 and 1996, 13 storms have come within 25 miles of Block Island. In August 1991, the center of Hurricane Bob passed about ten miles to the west of the island with 85-knot winds.

(46)

Communications

(47) A ferry operates daily from Galilee to Old Harbor, carrying mail, passengers, freight and vehicles. There is summer ferry service from Old Harbor to Galilee, via Newport, Galilee and New London. The island has telephone service to the mainland. Air service is also available.

(48) **Block Island Southeast Light** (41°09'12"N., 71°33'07"W.), 261 feet above the water, is shown from a red-brick octagonal, pyramidal tower attached to a dwelling to **Mohegan Bluf s** on the southeast point of the island and has a sound signal that is operated by keying the microphone five times consecutively on VHF-FM channel 83A. The wreck of the large tanker SS LIGHTBURNE is southeast of the light at 41°08'57"N., 71°32'52"W.

(49) **Block Island North Light** (41°13'39"N., 71°34'33"W.), 58 feet above the water, is shown from a white tower on a house structure on Sandy Point at the north end of the island. At **Clay Head**, on the northeast side of Block Island, is a lone white house on top of the bluff.

(50) **Old Harbor**, frequently used as a harbor of refuge, is an artificial harbor formed by two breakwaters on the east side of Block Island, 1.4 miles northward of Block Island Southeast Light. A federal project provides for a channel 15 feet deep entering the harbor and leading to a basin with a project depth of 15 feet; the inner harbor anchorage area also has a project depth of 15 feet. (See Notice to Mariners and latest editions of nautical charts for controlling depths.) The harbor is the traditional year-round port for lifeline ferry service and is also occupied by pleasure craft during the summer. The eastern part of the inner harbor is left clear for the passage of the ferry to the wharf. The basin in the southeast corner of the inner harbor is usually occupied by fishing boats and local craft that tie up along the sides. Gasoline, diesel fuel and berths are available. The **harbormaster** has an office at the Old Harbor town dock.

(51) The east breakwater extends about 300 yards northward of the entrance of the inner harbor and is marked at its end by a light and sound signal. A bell buoy is 0.55 mile northward of the breakwater. A light marks the end of the breakwater on the west side at the entrance to the inner harbor.

(52) **Great Salt Pond (New Harbor)**, on the west side of Block Island, is the best harbor in Block Island Sound for vessels of 15-foot draft or less. In easterly gales when the sea is too heavy to enter Old Harbor, a landing can be made at Great Salt Pond. The entrance, about 2 miles south-southwestward of Block Island North Light, is a

dredged cut through the narrow beach. The southwestern side of the entrance is protected by a jetty, which is marked by a light and a sound signal at its outer end.

(53) A federal project provides for a channel 18 feet deep entering Great Salt Pond. (See Notice to Mariners and latest editions of the charts for controlling depths.) Local knowledge is advised before entering.

(54) Anchoring is prohibited west of the main channel and in the northern portion of Great Salt Pond. A mooring area is near the southeast end, east of the channel.

(55) Small-craft facilities in Great Salt Pond can provide berths, electricity, gasoline, diesel fuel, water, ice and marine supplies. The marina about 0.3 mile westward of the ferry landing had a reported depth of 16 feet at the face of the dock in 1981. Sail and engine repairs are available nearby.

(56)

Current

(57) **Tidal currents** in the entrance to Great Salt Pond have a velocity of 0.3 knot. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(58)

No-Discharge Zone

(59) The State of Rhode Island, with the approval of the Environmental Protection Agency, has established a **No-Discharge Zone** (NDZ) in Great Salt Pond. The NDZ includes all waters east of a line from the landward end of the jetty located at the northwestern channel entrance to the red marker located approximately 600 feet northwest of the United States Coast Guard facility dock.

(60) Within the NDZ, discharge of sewage, whether treated or untreated, from all vessels is prohibited. Outside the NDZ, discharge of sewage is regulated by **40 CFR 140** (see chapter 2).

(61)

Point Judith Light to Potter Pond

(62) **Point Judith Light** (41°21'40"N., 71°28'53"W.), 65 feet above the water, is shown from an octagonal tower with the lower half white, upper half brown. A sound signal at the light is operated by keying the microphone five times consecutively on VHF-FM channel 83A. About 100 yards north of the light is **Point Judith Coast Guard Station**. A lighted whistle buoy is about 3.2 miles southward of the light.

(63) The area around Point Judith, including the approaches to Point Judith Harbor of Refuge, is irregular with rocky bottom and indications of boulders. Caution is advised to avoid the shoal spots, even with a smooth sea, and to exercise extra care where the depths are not more than 6 feet greater than the draft.

(64) **Point Judith Harbor of Refuge**, on the west side of Point Judith, is formed by a main V-shaped breakwater and two shorearm breakwaters extending to the shore. The harbor is easy of access for most vessels except with a heavy southerly sea. It is little used by tows. The only soft bottom in the harbor is found in the southern part of the deeper water enclosed by the main breakwater. On the north side the shoaling is gradual; the 18-foot curve is about 0.3 to 0.5 mile offshore. A shoal extends from the north to the central part of the harbor with depths of 14 to 18 feet; the shoal is marked by a buoy.

(65) The area within the V-shaped breakwater affords protected anchorage for small craft. The breakwater should be given a berth of 200 yards to avoid broken and hard bottom; a rocky shoal area about 100 yards wide, paralleling the west side of the main breakwater northward from the angle, should be avoided. A good berth for a vessel is on a line between Point Judith Harbor of Refuge East Entrance Light 3 and Point Judith Harbor of Refuge West Entrance Light 2, midway between them in 22 to 30 feet. This position falls on the edge of the east-west thorofare used by pleasure craft and fishing boats.

(66) A wreck, covered 12 feet, is about 570 yards southeast of Point Judith Harbor of Refuge West Entrance Light 2 in about 41°21'33"N., 71°30'28"W. Another wreck, covered 6 feet, is marked by a lighted buoy about 475 yards westward of Point Judith Harbor of Refuge East Entrance Light 3 in 41°21'34"N., 71°30'11"W.

(67) The southern entrance to the Harbor of Refuge, known locally as the East Gap, is 400 yards wide; it has a controlling depth of about 20 feet with deeper water in the western half of the channel.

(68) The western entrance to the Harbor of Refuge, known locally as the West Gap, is 500 yards wide; it has a controlling depth of about 19 feet, with lesser depths on the north side of the entrance. A rock, covered 12 feet, is near the north side of the entrance at 41°21'48.7"N., 71°31'07.4"W. Another rock, covered 16 feet, is 125 yards SE of the end of the north breakwater.

(69)

Current

(70) The tidal currents have a velocity of about 0.7 knot at the south entrance. The currents off the west entrance are rotary, with a velocity at strength of 0.5 knot. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(71) Considerably stronger currents have been reported to develop especially when the tide is ebbing.

(72) **Point Judith Pond** is a saltwater tidal pond entered between two rock jetties at **The Breachway** in the northwestern part of Point Judith Harbor of Refuge. The east jetty is marked near its seaward end by a light. The pond extends 3.3 miles northerly to the town of **Wakefield**. It is used extensively by ferry vessels, small fishing vessels, and pleasure craft. Numerous fish wharves are inside the entrance. The north end of Point

Judith Pond affords good anchorage for boats of 4 feet draft or less during a heavy blow.

(73) The village of **Galilee** is located on the east side of the entrance, **Jerusalem** on the west side at **Succotash Point** , each have state piers and numerous small piers chiefly used by fishermen. A state fisheries laboratory is just above the state pier at Jerusalem. The ferry terminal to Block Island is located at the Galilee State Pier. A state pier superintendent oversees each pier at Galilee and Jerusalem. Their office is located at the head of the Galilee State Pier.

(74) A federal project provides for a depth of 15 feet from Point Judith Harbor of Refuge to the State Pier at Jerusalem along the west side of Point Judith Pond with a branch channel on the east side extending northeasterly from the entrance of the pond to the State Pier at Galilee, and a 10-foot anchorage basin just inside the pond entrance. The western channel extends north the turning basin at Wakefield; portions of the channel are federally maintained to 6 feet. (See Notice to Mariners and latest editions of charts for controlling depths.)

(75)

Tides and Currents

(76) The mean range of tide in the pond is 2.8 feet and occurs later than in the Harbor of Refuge by about 10 minutes just inside the entrance and 30 minutes at the north end. The tidal currents in the entrance have a velocity of 1.8 knots on the flood and 1.5 knots on the ebb and cause slight rips and overfalls at changes of tide. Higher current velocities are reported to occur. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(77) Several boatyards and marinas are at Galilee, Jerusalem and Wakefield and at Snug Harbor, on the west side of the pond about 0.8 mile above the entrance. Berths, electricity, gasoline, diesel fuel, water, ice, marine supplies, storage, launching ramps and hull and engine repairs are available. The largest marine railway in the area, at the southern end of the waterfront at Snug Harbor, can handle craft up to 150 feet long or 400 tons. In 1981, a reported depth of 12 feet could be carried to the railway.

(78) Daily ferry service is available to Block Island from Galilee. Daily bus service is operated to Providence.

(79) **Potter Pond**, shallow and landlocked, is joined with Point Judith by a narrow channel near **Snug Harbor**. Local knowledge should be obtained before using this channel, which has depths of 2 to 4 feet and is crossed by overhead power and telephone cables with a clearance of 30 feet at the channel entrance and by a fixed highway bridge with a clearance of 5 feet about 0.4 mile above the entrance. A current of more than 3 knots develops through the channel on the ebb. The mean range of **tide** in the pond is about 1 foot, and it occurs about 2.5 hours later than in the Harbor of Refuge.

(80)

Matunuck to Old Reef

(81) From Point Judith to Watch Hill the shore is low and for the most part consists of sandy beaches that are broken by several projecting rocky points. Back from the immediate shore are areas of cultivation interspersed with rolling grass-covered or wooded hills. Except for Point Judith Pond, most pond outlets are used only by small local craft. The coast is fringed by broken ground and boulders in places, which should be avoided by deep-draft vessels where the depths are less than 36 to 42 feet.

(82) **Matunuck** is a summer resort about 3 miles west of Point Judith. Southwest of **Matunuck Point** is **Nebraska Shoal**, the shoal is at the south end of broken ground, with depths less than 30 feet. Heading offshore, the water deepens abruptly around the patch.

(83) **Charlestown Breachway**, 4.5 miles westward of Matunuck Point, is a narrow inlet which leads to **Ninigret Pond**, also known as **Charlestown Pond**, to the westward, and the village of **Charlestown** to the northward. In 1994, a reported depth of about 1½ feet could be taken in the inlet, with depths of about 3 to 6 feet inside. The southern part of Ninigret Pond is mostly mud flats. Local knowledge is required in entering and moving about inside. A small-craft facility is at Charlestown, and one is at the western end of Ninigret Pond; berths, gasoline, electricity, marine supplies and launching ramps are at both facilities.

(84) **Quonochontaug**, 10.8 miles westward of Point Judith, is a summer settlement at the outlet of **Quonochontaug Pond**. In 1981, a reported depth of about 3 feet could be carried in **Quonochontaug Breachway**, with depths of 15 to 20 feet reported in the pond. Vessels favor the west side of the entrance to avoid rocks in the easterly half of the entrance.

(85) **Weekapaug Point**, 12.5 miles west of Point Judith, is bold, rocky, and prominent from the southwest and southeast. Two stone jetties, 1,500 feet long, protect the entrance to **Winnapaug Pond** just westward of the point. In 1981, a reported depth of about 5 feet could be carried in **Weekapaug Breachway** to the pond; vessels favor the west side of the breachway above the bridge. Reported depths in Winnapaug Pond vary from bare to 10 feet. There are numerous shoals and sandbars. Southerly winds cause breakers at the ends of the jetties; extreme caution is advised. The fixed bridge over the entrance has a clearance of 6 feet.

(86) **Old Reef**, with a least depth of 7 feet, is about 1.5 miles west of Weekapaug Point and about 0.5 mile offshore.

(87)

Watch Hill to Wicopesset Passage

(88) **Watch Hill**, about 17.5 miles west of Point Judith, is a high bare bluff on its easterly side with several large hotels and summer houses.

(89) **Watch Hill Light** (41°18'14"N., 71°51'30"W.), 61 feet above the water, is shown from a square gray granite tower attached to a white building, on **Watch Hill Point**. A mariner radio activated sound signal at the light is initiated by keying the microphone five times consecutively on VHF-FM channel 83A.

(90) **Gangway Rock**, awash at low water, is part of a boulder reef extending about 0.2 mile southward from Watch Hill Light. A lighted bell buoy marks the south end of the reef. A submerged rock is about 50 yards northward of the buoy.

(91) **Watch Hill Passage** is the principal entrance to Fishers Island Sound from eastward and the only one used by strangers. It has a least depth of about 16 feet. A spot with 13 feet over it in the passage is marked by a buoy; the best channel is northward of this buoy, giving it a berth of about 150 yards.

(92) **Watch Hill Reef**, on the southwest side of Watch Hill Passage, has rocks that bare and is marked by a gong buoy.

(93) **Sugar Reef Passage**, between Watch Hill Reef and Sugar Reef, has a width of 0.3 mile; the least depths are about 22 feet.

(94) **Sugar Reef**, some 500 to 600 yards in extent, is covered 2 to 12 feet and should be avoided; it is marked by a buoy off its north side.

(95) **Catumb Passage**, between Sugar Reef and Catumb Rocks, has a width of 150 yards; its least depth is 13 feet.

(96) **Catumb Rocks**, the highest of which are awash, are marked by buoys on the north, southeast, and southwest sides. Rocks covered 1 to 18 feet extend 0.8 mile westward of Catumb Rocks to the buoy that marks the east side of **Lords Passage**. This passage, about 0.3 mile wide, has a least depth of 16 feet.

(97) **Wicopesset Rock**, on the northwesterly side of Lords Passage, is the easterly part of foul ground extending about 0.3 mile to **Wicopesset Island**, which is low and rocky.

(98) **Wicopesset Passage**, between Wicopesset Island and East Point, is narrow and is obstructed by a rock in the middle marked by a buoy; it is suitable only for small craft and should not be used by strangers. A bell buoy marks the southern entrance. Extreme caution is recommended when using the passage as the ebb current is apt to set boats on the foul ground.

(99) Information about the tidal currents in the passages is given with the discussion of Fishers Island Sound.

(100)

Fishers Island to Constellation Rock

(101) **Fishers Island**, 6 miles long, is hilly and sparsely wooded. **Chocomount**, 136 feet high, is the highest point on the island. **East Point**, at the east end of the island, is marked by several large houses. The former Coast Guard station at East Harbor, about 1 mile from East Point of Fishers Island, is prominent; numerous buildings on the western part of Fishers Island and a large yellow hotel building are conspicuous. The radar antenna on **Mount Prospect**, near the west end of the island, south shore, is the most prominent landmark on Fishers Island from seaward. The south side of the island is fringed with foul ground that rises abruptly from depths of 42 to 48 feet, but by giving the shore a berth of 0.5 mile, all dangers will be avoided.

(102) **Race Point Ledge**, partly bare at low water, extends about 0.2 mile southwestward from **Race Point**, the southwest extremity of Fishers Island, and is marked at its end by a buoy. Inside the buoy are boulders with 2 to 9 feet over them. The passage between the buoy and Race Rock Light has very irregular bottom; the least depth is about 18 feet. It is suitable only for small vessels with a comparatively smooth sea.

(103) **Race Rock**, on the northeast side of The Race, is nearly 200 yards in diameter, with a depth of 8 feet. A ridge with a least depth of 28 feet extends about 120 yards south-southwest of Race Rock. Another ridge, extending in a north-south direction with a least depth of 38 feet is about 320 yards east of Race Rock.

(104) **Race Rock Light** (41°14'37"N., 72°02'50"W.), 67 feet above the water, is shown from a granite tower attached to a dwelling on a granite pier on the rock. A sound signal at the light is operated by keying the microphone five times consecutively on VHF-FM channel 83A. The sound signal is reported at times to be inaudible when a vessel is approaching from eastward and is close southward of Fishers Island.

(105) **The Race**, the main entrance to Long Island Sound from eastward, extends between Fishers Island and Little Gull Island, between which is a width of about 3.5 miles. The only dangers are Valiant Rock, nearly in the middle, and Little Gull Island with its reefs.

(106)

Current

(107) In the middle of The Race, the flood sets 295° and the ebb 100°, with average velocities of 2.9 knots and 3.5 knots, respectively. There are always strong rips and swirls in the wake of all broken ground in The Race, except for about one-half hour at slack water. The rips are exceptionally heavy during heavy weather, and especially when a strong wind opposes the current, or the current sets through against a heavy sea. Predicted times of slack water and times and velocities of strength of current are available from the Tidal Current prediction service at *tidesandcurrents.noaa.gov*. Links to a user guide for this service can be found in chapter 1 of this book.

(108) During the flood stage of the tide, a significant eddy exists on the northwest side of Valiant Rock.

(109) **Little Gull Reef**, with little depth and foul ground, extends 0.3 mile east-northeastward from **Little Gull Island** and is marked at the northeast end by a buoy. Mariners are advised that the buoy is sometimes submerged by the strong current and deep-draft vessels should avoid this locality. **Little Gull Island Light** (41°12'23"N., 72°06'25"W.), 91 feet above the water, is

shown from a gray granite tower, 81 feet high, attached to a red dwelling on a pier. A sound signal is at the light. The light and Race Rock Light are the guides, as soundings cannot be depended upon.

(110) In passing north of Valiant Rock, vessels should keep from 0.5 to 0.8 mile southwestward of Race Rock Light, and craft passing southward of Valiant Rock should hold to a course about 1 mile northeastward of Little Gull Island Light.

(111) **Cerberus Shoal**, 6 miles southeast of Race Rock Light, is about 0.4 mile in diameter, with a least depth of 16 feet on a small rocky patch near its north end. The seas break on this shoal during heavy swells. It is marked by a lighted gong buoy. Near the shoal, tide rips are unusually strong.

(112) **Great Gull Island**, 0.6 mile southwest of Little Gull Island, was formerly a military reservation, but is now privately owned. The pier on the north side is in ruins. A lookout tower on the island is conspicuous.

(113) **Valiant Rock**, with a least depth of 20 feet, is surrounded by shoal area, and the 10-fathom curve surrounding the rock marks the area that should be avoided by deep-draft vessels and preferably all vessels, on account of the heavy swirls and rips. A lighted whistle buoy is northward of the rock.

(114) **The Sluiceway**, the passage between Great Gull Island and Plum Island, has several known dangers and very irregular bottom with boulders and should be avoided. The velocity of the **tidal current** in the passage is 2.6 knots on the flood and 3.2 knots on the ebb; flood sets 299° and ebb 133°. Considerably higher velocities occur at times, and tide rips are very bad in heavy weather. Boulders covered 3 to 10 feet are between **Old Silas Rock** and Plum Island. Old Silas Rock, marked by a buoy, is awash at high water. **Middle Shoal Rock**, 0.3 mile northeastward of Old Silas Rock, has a depth of 8 feet.

(115) **Bedford Reef** is broken ground, on which the least found depths are 14 to 16 feet, extending about 1.5 miles southward from broken ground lying between Great Gull and Plum Islands. It should be avoided. **Constellation Rock**, on the southeasterly extension on this broken ground, has 17 feet over it, is marked by a buoy and lies 1.9 miles southward of Little Gull Island Light.

(116)

Montauk Point to Threemile Harbor

(117) **Montauk Point**, the easterly extremity of Long Island, is a high sandy bluff, on the summit of which is the light. The land is grass covered, with a height of 165 feet at **Prospect Hill**, 2 miles westward of the point. The south side of the point is bold, and the 10-fathom curve is about 0.5 mile from shore; depths of 24 feet and less extend 0.8 mile off the northeast side of the point.

(118) **Montauk Point Light** (41°04'15"N., 71°51'26"W.), 168 feet above the water, is shown from a white conical tower with a red band midway of its height and a covered way to a gray dwelling. A sound signal at the light is operated by keying the microphone five times consecutively on VHF-FM channel 83A.

(119) Surrounding Montauk Point for about 4 miles is a shoal area that has been closely surveyed, the bottom is very broken, and extra caution should be observed where the depths are less than 10 feet greater than the draft. In general, the shoals are a series of long narrow ridges, in places only a few yards wide, and their positions are indicated by the rips over them at the strength of the tidal currents.

(120) **Montauk Shoal**, about 2.5 miles south-southeastward of the light, has least depths of 30 feet. **Great Eastern Rock**, 1.5 miles east-northeast of the light, has a least depth of 25 feet. **Phelps Ledge**, just northerly of Great Eastern Rock, is covered by 24 feet. **Endeavor Shoals**, about 2.3 miles northeast of the light, are covered by 19 to 24 feet on a narrow ridge about 0.4 mile long. A lighted gong buoy is off the eastern end of the ridge.

(121) Vessels drawing up to 20 feet can avoid the dangers eastward and northeastward of Montauk Point in smooth weather by giving the point a berth of over 1 mile and avoiding Great Eastern Rock.

(122) Broken ground with rocky bottom and boulders extends about 2 miles off the north coast west of Montauk Point. **Shagwong Reef**, with a least depth of 6 feet and marked by a lighted bell buoy, is the northern limit of this area. **Shagwong Rock**, with a least depth of 7½ feet and marked by a lighted buoy, and **Washington Shoal**, with a least depth of 12 feet, are between the shore and Shagwong Reef. The principal danger outside Shagwong Reef is a shoal with a depth of 29 feet, 5.3 miles northwestward of Montauk Point.

(123)

Pilotage Pickup Locations Off Montauk Point

(124) Pilots, by special arrangement during favorable weather conditions, may meet a ship with less than 38-foot draft off Montauk Point bound for Long Island Sound. Foreign flag vessels and U.S. vessels that are under register can arrange for a state-licensed pilot by contacting the joint rotation administrator, Block Island Pilots at 243 Spring Street, Newport, RI 02840; telephone 401–847–9050 (24 hours), 800–274–1216; FAX 401–847–9052. U.S. vessels engaged in coastwise trade (enrolled work) in need of pilotage services can contact the various pilot organizations directly. For telephone number, FAX number, cable address, description of the boat, frequencies, etc., consult the name of the association under Pilotage, Narraganset Bay and Other Rhode Island Waters (indexed as such), chapter 6; Pilotage, Long Island Sound (indexed as such), chapter 8; and Pilotage, New York Harbor and Approaches (indexed as such), chapter 11.

(125) Mariners are also reminded that vessels with a draft in excess of 38 feet are advised to not transit the Montauk Channel (as defined by the area between Southwest Ledge Lighted Whistle Buoy 2 and Block Island Sound South Entrance Obstruction Lighted Buoy BIS). Further, pilots

using Montauk Channel shall consider draft, sea and swell, wind, visibility, current and vessel traffic. When these conditions pose a threat to the safety of any person, vessel, prudent navigation or safety of the environment, Montauk Channel shall not be used.

(126) **Montauk Harbor**, in the northern part of **Lake Montauk**, is entered through a dredged channel on the northern shore about 3 miles west of Montauk Point;

(146) A **restricted anchorage** for U.S. Navy submarines is about 3 miles eastward of Gardiners Island. (See **33 CFR 110.1** and **110.150**, chapter 2, for limits and regulations.)

(147) **Gardiners Bay** is at the western end of Block Island Sound from which it is separated by Gardiners Island. The bay is an excellent anchorage easily entered day or night and is the approach to Shelter Island Sound and the Peconic Bays. The principal entrance is northward of Gardiners Point. The entrance from Long Island Sound is through Plum Gut. The entrance southward of Gardiners Island is used by fishing vessels.

(148) The principal guides for the entrance to Gardiners Bay from Block Island Sound are the lighted gong buoy north of Gardiners Point, Little Gull Light and Orient Point Light. When past the lighted gong buoy north of Gardiners Point, vessels can select the anchorage in Gardiners Bay that affords the best lee in the prevailing winds.

(149) The principal dangers in approaching Gardiners Bay from the northward are the broken ground between Constellation Rock and Plum Island and the shoal making out to Gardiners Point. In the bay, Crow Shoal should be avoided. In general, the shoaling is rather abrupt in approaching these dangers and gradual in approaching the shoals on the western side of the bay.

(150)

No-Discharge Zone

(151) The State of New York, with the approval of the Environmental Protection Agency, has established a No-Discharge Zone (NDZ) in the Peconic Estuary. The NDZ includes all open waters, harbors and creeks of the Peconic Estuary west of a line from Orient Point to Montauk Point.

(152) Within the NDZ, discharge of sewage, whether treated or untreated, from all vessels is prohibited. Outside the NDZ, discharge of sewage is regulated by **40 CFR 140** (see chapter 2).

(153) **Plum Island**, about 2 miles westward of Great Gull Island, is 2.5 miles long and hilly and has several large buildings and a prominent tank; it is marked on its western point by Plum Gut Light. The island is a government reservation and closed to the public.

(154) The bight in the southeast side of Plum Island is foul to **Plum Island Rock**, which is 0.5 mile from shore abreast of the middle of the island, has 1 foot over it and is marked by a buoy.

(155) **Plum Gut Harbor**, on the southwest side of Plum Island, has an entrance between jetties with private seasonal lights on dolphins off the outer ends. The lights are shown daily from sundown to 0130. A private sound signal at the west jetty light is sounded occasionally when Department of Agriculture vessels are navigating in the area. A depth of about 14 feet is in the entrance. Small yachts seeking shelter in an emergency lie alongside the wharves. The harbor is under the supervision of the Department of Agriculture and the Coast Guard and may be used only with permission.

(156)

COLREGS Demarcation Lines

(157) The lines established for Plum Gut Harbor are described in **33 CFR 80.155**, chapter 2.

(158) **Plum Gut**, the entrance to Gardiners Bay from Long Island Sound, is nearly 0.6 mile wide and has sufficient water for vessels of the deepest draft; in the passage are several rocks with depths of 16 to 22 feet over them. An obstruction with a least depth of 59 feet at 41°10'06.2"N.,72°12'58.5"W. Tidal currents set through the passage with great velocity. Steamers, or sailing vessels with a strong favorable wind, should have no difficulty in passing through.

(159) Velocities of the current on flood and ebb are 3.5 and 4.3 knots, respectively. The flood sets northwestward and the ebb southeastward. Heavy tide rips occur. In 1983, NOAA Ships RUDE and HECK reported that during the flood a countercurrent normally develops along the north shore of Plum Island. This countercurrent is most prevalent within 0.5 mile of the island. Caution is recommended when using this passage.

(160) **Oyster Pond Reef**, extending about 0.5 mile east-northeastward from **Orient Point**, is marked by a light and sound signal. The sound signal at the light is operated by keying the microphone five times consecutively on VHF-FM channel 83A. Caution is recommended regarding the sound signal, as it may be difficult to hear at times, particularly with an easterly wind. Numerous boulders and little depth are between the light and Orient Point. **Midway Shoal**, about 0.5 mile east of the light, shoals to 16 feet and is marked by a lighted buoy.

(161) When using Plum Gut it is well to give Plum Island and Orient Point Light a berth of 0.2 mile. The best water in the passage will be found on a **295°** course, passing **Pine Point** and the buoy marking Midway Shoal at a distance of 350 yards and passing midway between Orient Point Light and Plum Gut Light on the western end of Plum Island.

(162) A channel, with a reported controlling depth of 7 feet in 1999, leads to a research basin operated by the U.S. Department of Agriculture on the south side of Orient Point, about 1 mile southwest of Orient Point Light. A ferry operates between here, Plum Island and New London.

(163)

Small-craft facility

(164) A small-craft facility is about 0.1 mile westward of the wharf. Berths, electricity, gasoline, diesel fuel, water, ice and a launching ramp are available. In 1981, a reported depth of about 6 feet could be carried to the facility.

(165) **Acabonack Harbor**, at the southeast end of Gardiners Bay, is entered through a privately maintained and marked channel with a reported controlling depth of 8 feet in the entrance in 1996.

(166) **Hog Creek Point**, on the southerly side of Gardiners Bay, is generally flat, with bluffs approximately 25 feet

in height. **Lionhead Rock**, off the point and marked by a buoy, is awash at high water. Fishtraps are westward of the point.

(167) **Threemile Harbor**, on the south side of Gardiners Bay 1.7 miles southwestward of Hog Creek Point, is entered through a channel with two privately dredged sections. In 1980, a portion of the wooden bulkhead on the west side of the entrance collapsed into the channel. In 1981, it was reported that by favoring the east side of the entrance channel a depth of 8 feet could be carried to a point opposite **Maidstone Park**, thence in 1996, a reported depth of 6 feet could be carried to the basin at the head of the harbor. The approach to the harbor is marked by a seasonal lighted bell buoy, and the channel is marked by lighted and unlighted buoys. The jetties at the harbor entrance are marked on the outer ends by private lights. A public commercial landing with reported depths of 8 feet is on the east side of the channel about 0.6 mile above the entrance. A 5 mph **speed limit** is enforced in the harbor.

(168)

Anchorages

(169) Anchorage is available in Threemile Harbor in depths of 9 to 14 feet with soft bottom and good holding ground; this is a good anchorage during strong winds.

(170)

Current

(171) The **tidal current** has a velocity of about 3 knots through the entrance.

(172)

Small-craft facilities

(173) Small-craft facilities on the east and south sides of Threemile Harbor can provide berths, electricity, gasoline, diesel fuel, water, ice, launching ramps, storage and lifts to 40 tons and hull and engine repair. Provisions can be obtained at the town of **East Hampton**, 3.5 miles south of Threemile Harbor.

(174) In 1989, the public pier maintained by the town of East Hampton at the head of the harbor had reported depths of 7 feet at its face and 4 feet on its west side.

(175)

COLREGS Demarcation Lines

(176) The lines established for Threemile Harbor are described in **33 CFR 80.155**, chapter 2.

(177)

Shelter Island Sound to Reeves Bay

(178) **Shelter Island Sound** and Peconic Bays extend westward from Gardiners Bay about 22 miles to Riverhead, the head of navigation on Peconic River. They are much frequented by yachts and other small craft in the summer. Fishtraps and oyster stakes are on many of the shoals.

(179) A depth of about 26 feet can be carried through the channel north of Shelter Island and through Little Peconic Bay as far as Robins Island and about 13 feet through the channel south of Shelter Island. Across the bar between Little and Great Peconic Bays about 13 feet can be carried. With local knowledge greater depths can be carried in the channels and across the bar. A depth of about 6 feet can be taken to South Jamesport and Riverhead.

(180)

Current

(181) The **tidal currents** have considerable velocity wherever the channel is narrowed. The velocity in the narrower places is about 1.8 knots.

(182)

Ice

(183) Ice obstructs navigation in the coves and shallow harbors during January and February. In severe winters, drift ice is reported to interfere with navigation for short periods of time. In the south arm of Shelter Island Sound, the ice is heavy enough at times to destroy structures exposed to it.

(184)

Small-craft facilities

(185) Diesel fuel, gasoline, ice, water, marine supplies and other provisions can best be obtained at Greenport and Sag Harbor. Several boatyards, shipyards, marine railways and enclosed basins with excellent repair facilities are at Greenport.

(186) **Ram Head** is a prominent sandy bluff on the western shore of Gardiners Bay. A lower bluff is nearly 1.5 miles westward of Ram Head with numerous houses along the top. A shoal with 7 to 17 feet over it extends about 2.4 miles southeastward from Ram Head.

(187) A boulder with 1 foot over it is 230 yards from shore about 0.3 mile northeastward of the northern point of the entrance to Coecles Harbor. Other boulders with little depth are between this boulder and Ram Head.

(188) The entrance to **Coecles Harbor** is at the south end of Ram Head; the channel is marked by private seasonal buoys and a private seasonal light. In 1996, the reported controlling depth in the privately maintained entrance channel was 8 feet. The **speed limit** is 5 mph. A marina and boatyard are in the harbor. A mobile hoist at the boatyard can haul out craft up to 35 tons; gasoline, water, ice, diesel fuel, marine supplies, pump-out facilities, berths, guest moorings, storage facilities and complete engine and hull repairs are available. In 1981, a reported depth of 5½ feet could be carried to the marina and boatyard. In 2003, a dangerous rock was reported about 250 yards south of Buoy 10 at 41°04'11.5"N., 72°18'22.5"W. Care should be taken to avoid this hazard.

(189) A **special anchorage** is in Coecles Harbor. (See **33 CFR 110.1** and **110.59**, chapter 2, for limits and regulations.)

(190)

COLREGS Demarcation Lines

(191) The lines established for Coecles Harbor are described in **33 CFR 80.155**, chapter 2.

(192) Extensive flats and an unmarked aquaculture site make off from Ram Head and the shore between it and **Hay Beach Point**, the northernmost point of Shelter

Island, which is a low flat with a clump of scrub at its end and backed by wooded highland.

(193) **Long Beach Point** is a low spit eastward of Hay Beach Point; a light marks the outer end of the point. Shoaling is reported south of the point. Mariners should exercise caution in this area; the shoals extending southward from the point are constantly changing and can be dangerous.

(194)

COLREGS Demarcation Lines

(195) The lines established for the Long Island bays are described in **33 CFR 80.155**, chapter 2.

(196) **Orient Harbor**, about 4 miles northwestward of Ram Head, is an excellent anchorage; the depths range from over 20 feet in its southern part to 16 feet at its northern end. **Orient** is a village at the northeast end of Orient Harbor. At the end of the main wharf the depth is 8½ feet. The eastern part of Orient Harbor has depths of 7 to 9 feet. Fish traps are on the shoals.

(197) About 0.4 mile northeastward of **Cleaves Point**, at the southwest end of Orient Harbor, the shore has been cut through to a small pond which is used as a private basin for small craft. The entrance, between two jetties, has a depth of about 3 feet over the bar, with about 6 feet in the basin. Permission is required before anchoring in the basin. Rocks are 0.2 mile south of the entrance.

(198) **Hallock Bay** makes eastward from Orient Harbor on the north side of Long Beach Point. A channel, marked by uncharted private daybeacons, leads into the bay. The bay is shallow and dangers and shoaling have been reported. Local knowledge is advised prior to entering.

(199) **Gull Pond** is 0.3 mile westward of Cleaves Point at the southwest end of Orient Harbor; a private light marks the entrance. In 1981, a reported depth of 4 feet could be carried through the entrance, with depths of 10 to 15 feet reported in the pond. A state launching ramp is available in the pond.

(200) **Greenport** is an important town and the terminus of a branch of a Class II railroad. The white church spires, near the northern end of town, and a tank and TV radio tower in the center of town are prominent.

(201) **Greenport Harbor** is formed on the northeast by a 5-foot-high breakwater, which extends 0.2 mile southeastward from **Youngs Point**, nearly to the 18-foot curve, and is marked at its outer end by a light. The depths at the wharves range from 5 to 20 feet. The railroad wharf on the south side of the waterfront can accommodate a vessel up to 100 feet.

(202) An entrance channel leads northwest to an anchorage area inside **Stirling Basin**; the entrance channel is marked by private seasonal buoys. Another anchorage area is on the northeast side of the entrance channel.

(203) The **harbormaster** for Greenport Harbor controls mooring and berthing in the basin. The **speed limit** is 5 mph.

(204)

Small-craft facilities

(205) Small-craft facilities at Greenport can provide berths, electricity, gasoline, diesel fuel, water, ice, storage, marine supplies, a pump-out facility and hull and engine repairs. The largest marine railway, at a shipbuilding company at the southeast end of the waterfront, can handle craft up to 500 tons and 15 feet in draft. Mobile hoists to 50 tons are available. A well-equipped machine shop is also in the town.

(206) A ferry operates between Greenport and Shelter Island. During the summer, bus service is available from Greenport to Orient Point where there is ferry service to New London.

(207) **Dering Harbor**, southward of Greenport and at the northwest end of Shelter Island, is a favorite anchorage for yachts and motorboats. The entrance to the harbor, marked by private buoys, is partially constricted by a disposal area in about mid-entrance and shoal area with a reported depth of 4 feet in 1981 that extends from the southwestern entrance point to near the disposal area; caution is advised. In 1989, it was reported that about 10 feet could be carried into the harbor with local knowledge. Depths of 10 to 14 feet are available in the central part of the harbor, with much lesser depths around the edges. Moorings and float landings for small craft are in the bight at the southwest end of the harbor. Vessels too large to enter can anchor outside the harbor in depths of 14 to 30 feet. The **speed limit** is 5 mph.

(208)

Small-craft facilities

(209) Small-craft facilities, on the west side of the harbor, can provide berths, electricity, gasoline, diesel fuel, water, ice, marine supplies, pump-out facilities and hull and outboard engine repairs. A launching ramp is also available.

(210) **Shelter Island Heights** is on the southwestern side of Dering Harbor.

(211) **Fanning Point** is on the north shore at the southwest end of Greenport. A shoal extends 300 yards off the point and is marked by a seasonal lighted buoy. Four dolphins, part of a former oil facility, are northward of the point. Currents of 2 knots, running fair with the channel, have been reported in the vicinity of Fanning Point.

(212) **Conkling Point**, on the north shore 1 mile southwestward of Fanning Point, is low and sandy at the end and has deep water as close as 150 yards. A marina on the southwest side of the point had a reported depth of 6 feet in the approach in 2006. Berths and moorings, electricity, diesel fuel, water, ice, marine supplies, a pump-out, a launching ramp, a 30-ton mobile hoist and winter storage are available. Hull and engine repairs can be made.

(213) **Mill Creek** is the entrance to **Hashamomuck Pond**, about 1.1 miles westward of Conkling Point. In 1981, the privately dredged entrance channel into the creek had a controlling depth of 4 feet, thence 3½ feet was reported in

the channel along the northwest shore of Mill Creek. The entrance channel is marked by private seasonal buoys. About 400 yards eastward of the creek is a small bight entered through a channel with a depth of about 4 feet and marked by private seasonal lights and buoys. In 1992, severe shoaling was reported across the entrance.

(214) **Jennings Point**, the western end of Shelter Island, is high and wooded. Rocks are off the point close-to, and it should be given a berth of over 150 yards. A lighted buoy is off the point. A gazebo on the point is prominent.

(215) The town of **Southold** is at the head of **Southold Bay**, which is the bight at the western end of Shelter Island Sound westward of Jennings Point. For about a mile northeastward of the entrance jetty, shoals with 12 feet or less extend nearly 0.4 mile from shore and are generally steep-to. The southwest part of the bay is shoal for about 0.3 mile from shore. Anchorage can be selected east-southeast of the jetty at a distance of from 0.2 to 0.4 mile, in 12 to 18 feet.

(216) In 1989, the reported controlling depth was about 3 feet in the privately maintained channels in **Town Creek** and **Jockey Creek**. The common entrance to Town Creek and Jockey Creek is marked by private seasonal buoys. The bridge that crosses Jockey Creek has a 45-foot fixed span with a vertical clearance of 6.5 feet. The privately maintained channel in **Goose Creek** had a reported controlling depth of 7 feet in 2008. The fixed highway bridge at the mouth of Goose Creek has a clearance of 9 feet.

(217) On the shore south of Southold entrance jetty is a prominent white tower.

(218)

Small-craft facilities

(219) There are several small-craft facilities on the creeks and along the west shore of Southold Bay from Paradise Point to Conkling Point. Berths, electricity, gasoline, water, ice, marine supplies, launching ramps, storage, lifts and cranes are available. Provisions can be obtained at Southold.

(220) **Paradise Point**, on the west side of Shelter Island Sound, is low and wooded, and from the point a sloping sandspit extends about 0.3 mile eastward and is marked by a lighted buoy. Southward of Paradise Point, shoals with depths of 10 to 15 feet extend from the west shore to midsound; the southeast point of the shoals is marked by a buoy.

(221) The channel south of Shelter Island has numerous shoals but is easily followed by vessels of 13 feet or less draft when the buoys can be seen. The channel is used by vessels going to Sag Harbor. Vessels operating between Greenport and Sag Harbor prefer the inside route around the western end of Shelter Island. The **tidal current** in the channel between Shelter Island and North Haven Peninsula has a velocity of about 2.4 knots. The approach from Gardiners Bay is across a shoal or bar that extends in a southeasterly direction from Ram Head to the south shore, the depths on which vary from 7 to 11 feet about 1.6 miles from Ram Head, and thence 13 to 17 feet to the buoys which mark the entrance.

(222) **Dangerous Rock**, awash at low water in surrounding depths of about 12 feet, is 0.2 mile south of the channel.

(223) A shoal extends 0.3 to 0.4 mile north of the shore of **Cedar Point**, which is marked by a light. The shoal has boulders, and its edge is marked by buoys.

(224) Shoals with boulders and little water over them in places extend nearly 0.5 mile southeastward from **Nicoll Point**. Buoys mark the limit of the channel in this area.

(225) **Northwest Harbor**, between Cedar Island Light and **Barcelona Point**, is strewn with boulders covered by 4 to 6 feet.

(226) **Sand Spit**, an extensive shoal partly bare at half-tide, is between **Mashomack Point**, the southeastern extremity of Shelter Island, and Sag Harbor. The spit is marked by buoys and a light.

(227) A group of rocks locally known as **Gull Island**, showing bare at half-tide, is nearly 0.4 mile northeastward of the breakwater at Sag Harbor.

(228) **Sag Harbor**, about 2.5 miles southwestward of the light on Cedar Point, is protected on the northeast by a breakwater marked at the outer end by a light. A spherical tank, a radio tower and several flagpoles are prominent landmarks.

(229) In entering Sag Harbor, do not round the breakwater too closely, as a depth of about 6 feet is found near its end. Anchor eastward or northeastward of the end of the former ferry wharf, locally known as Long Wharf. A 5 mph **speed limit** is enforced.

(230) The channel to **Sag Harbor Cove** is about 8 feet deep; this channel and the cove are marked by private seasonal lights and buoys. A fixed bridge at the entrance has a clearance of 21 feet. Berths, electricity, gasoline, diesel fuel, storage, marine supplies, water, ice and launching ramps and complete engine, hull, rigging and sail repairs are available at Sag Harbor; a 30-ton mobile hoist, near the inner end of the breakwater, can haul out craft up to about 60 feet.

(231) **Smith Cove**, a small bight on the south side of Shelter Island, is a good anchorage for small craft in northerly weather. Depths range from 11 to 30 feet. A marina on the west side of the cove can provide moorings, limited berths, gasoline, electricity, water and some marine supplies. In 1981, a depth of 6 feet was reported alongside the pier at the marina. A ferry operates between **South Ferry** on the southwest side of the cove to **North Haven Peninsula**.

(232) **West Neck Harbor** and **West Neck Bay** are shallow bodies of water on the southwest side of Shelter Island. In 1989, it was reported that a depth of 2 feet could be carried over the bar and into the harbor from Shelter Island Sound. The entrance is close eastward of the seaward end of a peninsula, marked by a private lighted buoy, that separates the harbor from the sound, and the channel follows along the north side of this peninsula. The channel is marked by private buoys. The harbor has numerous private landings. A boatyard with a marine

railway can handle craft up to 40 feet for hull and engine repairs. Berths, gasoline, water, ice, a launching ramp and some marine supplies are available.

(233)

Anchorages

(234) A special anchorage is in West Neck Harbor. (See **33 CFR 110.1** and **110.59**, chapter 2, for limits and regulations.)

(235) **Noyack (Noyac) Bay** is between North Haven Peninsula and Jessup Neck and southward of the western end of Shelter Island. No dangers will be encountered if the shores are given a berth of 0.4 mile.

(236) **Mill Creek**, in the southern part of Noyack Bay, is entered through a privately dredged channel that leads to a basin. The channel is marked by private seasonal lights and buoys. A clubhouse on the west side of the entrance is prominent.

(237)

Small-craft facilities

(238) Small-craft facilities in the creek can provide berths, electricity, gasoline, water, ice, storage, a launching ramp, marine supplies and hull and engine repairs; a 25-ton mobile hoist is available.

(239) **Jessup Neck** is a long narrow strip, partly high and wooded, separating Noyack Bay from Little Peconic Bay. The north end of the neck is a sandspit from which a shoal with 4 to 12 feet over it extends nearly 0.4 mile north-northwestward. A lighted buoy marks the outer end of the shoal area.

(240) A shoal with depths of 5 to 7 feet extends 1.5 miles southwestward from **Great Hog Neck**, on the northwest side at the entrance to Little Peconic Bay; this shoal is marked by a seasonal lighted buoy.

(241) Heavy tide rips occur southeast of Great Hog Neck during the flood with a southwesterly wind. At such times, small craft can avoid the worst of them by favoring the shore on the northwest side of the passage.

(242) **Richmond Creek** and **Corey Creek** are at the head of **Hog Neck Bay**. A depth of about 7 feet can be taken in the privately dredged channel leading to a basin in Richmond Creek; the channel is marked by private seasonal buoys. The entrance channel is marked by private buoys.

(243) **Little Peconic Bay** is about 5 miles long. The southerly shore of the bay is clear if given a berth of 0.4 mile, but shoals extend 0.6 mile from the south end of the bay.

(244) An aquaculture site, marked by private seasonal buoys, is at the south end of Little Peconic Bay about 1 mile north-northwest of the entrance to North Sea Harbor.

(245) A prominent sandy bluff, known locally as **Holmes Hill**, is just west of the entrance to **North Sea Harbor**. In 2008, the reported controlling depth through the dredged channel and into the harbor was 7 feet. The channel is marked by private seasonal buoys and by a private seasonal light at the entrance. This is an excellent harbor of refuge for small craft with drafts not exceeding 3½ feet. The bottom is soft with good holding ground.

(246) A marina in the harbor has gasoline, ice, water, some marine supplies and a lift that can handle craft to 10 tons; hull and engine repairs can be made.

(247) **Wooley Pond**, 1 mile northeastward of North Sea Harbor, is entered through a dredged channel which, in 2000, had a reported depth of 8 feet. The channel is marked by private seasonal buoys and by a private seasonal light on the north side of the entrance.

(248) A marina in the pond can provide berths, electricity, gasoline, water, ice, storage, marine supplies and hull and engine repairs; a 45-foot marine railway and a 12-ton forklift are available. In 1981, depths of 5 to 6 feet were reported available at the marina.

(249) **Nassau Point**, the long neck on the northwest side of Little Peconic Bay, has high bluffs on the eastern side. A shoal with little depth over it extends 0.5 mile southward from Nassau Point and is marked by a lighted buoy.

(250) **Cutchogue Harbor**, between Nassau Point and New Suffolk, is used by local boats drawing 6 to 10 feet. On the east shore of the harbor, northwestward of Nassau Point, three channels leading into the ponds have been dredged by private interests. At the middle of the three channels, 0.9 mile northwest of the extremity of Nassau Point, are several private wharves. The channel leads between two jetties, and a depth of about 3 feet can be carried into the pond and 1 foot to some of the wharves.

(251) **Haywater Cove,Broadwater Cove**, **Mud Creek**, and **East Creek**, used by local interests and sharing a common entrance, are at the head of Cutchogue Harbor. The entrance channel and the channels through these waterways have been privately dredged. In 1999, a reported depth of 6 feet was available in the entrance channel; thence in 1966, 6 feet in East Creek and 7 feet in Haywater Cove and Broadwater Cove; thence in 1976, 6 feet in Mud Creek. Shoaling is reported to occur in these areas; caution is advised.

(252) A depth of 8 feet can be taken within 100 feet of the wharves at **New Suffolk** by passing eastward and about 200 yards northward of the buoy westward of Nassau Point and steering westward for the wharves. A small basin, with a depth of about 8 feet reported in 1981, is northward of the wharf. In 1981, shoaling to 2 feet was reported in the southern part of Cutchogue Harbor, about 0.4 mile east of New Suffolk.

(253) A larger basin at the north end of New Suffolk, locally known as **School House Creek**, extends to the highway. The entrance channel is protected by a short rock jetty, covered at high water, on the south. The depth to the boatyard at the head of the basin was reported to be 6 feet in 2008. Berths, gasoline, storage, marine supplies, hull and engine repairs and a 30-ton mobile hoist are available at the boatyard.

(254) **Wickham Creek**, locally known as Boatmens Harbor, 0.7 mile north of New Suffolk, is entered through a privately dredged entrance channel with a reported controlling depth of 8 feet in 2002. The channel is marked by private seasonal buoys and bush stakes. Gasoline, water, ice, storage, a launching ramp and some marine

supplies are available in the basin. A flatbed trailer can haul out craft to 32 feet.

(255) In southeast gales, local craft of less than 6-foot draft seek shelter in the small cove, locally known as **Horseshoe Cove**, in the northeast part of Cutchogue Harbor.

(256) The through channel in **North Race**, northward of **Robins Island**, is marked and used only by light-draft boats. **South Race**, the channel southward of Robins Island, has a controlling depth of about 13 feet and is marked by buoys.

(257) An aquaculture site, marked by private buoys, is 0.6 mile southwest of the south end of Robins Island.

(258) Tide rips occur between the mainland and the south end of Robins Island when the tidal current sets against the wind.

(259) **Great Peconic Bay**, about 5 miles in diameter, is used mostly by local motorboats from Shinnecock Canal and by yachts. The bay is generally clear, but extensive shoals make off from the shores, except on its south side. Shinnecock Canal, the entrance from the south, is described in chapter 10.

(260) **Rodgers Rock**, about 1.3 miles west-southwestward of **Cow Neck** and about 1.2 miles south-southwest of Robins Island, has a depth of 6 feet over it and is marked on the northeast side by a buoy. **Robins Island Rock**, 0.8 mile westward of the south end of Robins Island, is awash at low water. It is marked by a buoy. Caution is recommended in this vicinity.

(261) **Sebonac Creek**, on the southeast side of Great Peconic Bay, is used extensively by yachts and serves as a yacht harbor for the town of Southampton. A privately dredged channel, marked by private seasonal lights and buoys, leads into the creek and had a reported controlling depth of 8 feet in 1981. The landings are at **West Neck**, a small settlement northeastward of **Ram Island** in **Bullhead Bay**. An obstruction buoy is locally maintained during the summer to mark a rock, covered 1½ feet, about 100 feet westward of the town landing. In 1981, a reported depth of 5 feet could be carried to the town landing. A 5 mph **speed limit** is enforced.

(262) **Cold Spring Pond**, about 1.6 miles southwestward of Sebonac Creek and 1.1 miles eastward of Shinnecock Canal entrance, is entered through a privately dredged channel that had a reported depth of 6 feet in 2001. In 1992, severe shoaling was reported in the entrance. The entrance channel to the pond is marked by a private seasonal light and buoy. An overhead power cable at the entrance to the pond has a clearance of 34 feet.

(263) **James Creek**, on the north shore of Great Peconic Bay opposite the entrance to Shinnecock Canal, is entered through a privately dredged channel that had a reported controlling depth of 6 feet in 2008. The entrance is marked by private seasonal buoys. Small-craft facilities on the creek can provide berths, electricity, gasoline, diesel fuel, water, ice, some marine supplies, sewage pumpout, launching ramps, lifts to 20 tons and storage and hull, electronic and engine repairs. A flatbed trailer can haul out craft to 30 feet.

(264) **South Jamesport** is a village on **Miamogue Point**, 3.4 miles southwestward of James Creek. Local knowledge is necessary to avoid the shoals in this area, and strangers should take soundings frequently to keep in the best water.

(265)

Small-craft facility

(266) A small-craft facility at South Jamesport can provide berths, electricity, gasoline, water, ice, launching ramps, storage, marine supplies and hull and engine repairs; a 25-ton mobile hoist is available. In 1999, a reported depth of 6 feet could be taken to the facility. The town has railroad passenger and bus service.

(267) **Peconic River** empties into the western end of Flanders Bay, about 1.5 miles westward of South Jamesport. The river is entered through a dredged channel marked by private seasonal lights that leads from Flanders Bay to the head of navigation at **Riverhead**, about 2.4 miles above the channel entrance. The dredged channel is approached from deep water in Great Peconic Bay through a marked channel. In 2001, the controlling depth was 4.3 feet (5.7 feet at midchannel) in the dredged channel. A fixed highway bridge with a clearance of 25 feet crosses the river about 0.9 mile above the mouth.

(268) **Flanders Bay** is the scene of considerable small boat activity. Small-craft facilities are at Riverhead; limited berths, electricity, gasoline, water and a pump-out station are available.

(269) **Meetinghouse Creek,Terrys Creek,** and **Reeves Creek**, which empty into the northwestern part of Flanders Bay, are entered through privately dredged channels. In 1981, the channels had reported controlling depths of 5 feet. The entrance channel leading to, and connecting with, Terrys Creek and Meetinghouse Creek is marked by private seasonal buoys and a private seasonal light. Marinas on Meetinghouse Creek provide berths, electricity, gasoline, diesel fuel, water, ice, marine supplies, pumpout facilities, a 30-ton forklift, 55-ton mobile hoist and storage facilities and hull, motor and electronic repairs. In 2011, a reported depth of 8 feet was available alongside.

(270) **Reeves Bay**, on the southwest side of Flanders Bay, is entered through a privately dredged channel that leads to the town of **Flanders** on the south side of the bay. In 1999, the channel had a reported controlling depth of 2 feet. Other dredged channels lead from the entrance channel into several arms of the bay. A boatyard at Flanders has gasoline, storage facilities, marine supplies and a 10-ton marine railway; hull and engine repairs can be made.

(271)

Fishers Island Sound to Old Mystic

(272) **Fishers Island Sound** extends between the mainland of Connecticut and Fishers Island and forms

one of the entrances into Long Island Sound that is used to some extent by light tows and other vessels up to 14-foot draft. The sound has numerous shoals and lobster trap buoys, and the entire area is exceedingly treacherous, characterized by boulder patches that rise abruptly from deep water. Vessels should follow the deeper channels between the shoals and proceed with caution if obliged to cross shoal areas. In general, all shoal spots or abrupt changes of depth are indications of boulders and should be avoided as anchorages.

(273)

Current

(274) In Watch Hill Passage the tidal currents are strong and necessitate caution in navigating. Buoys may be towed under. The flood current sets nearly in the direction of the channel but has a tendency to northward and the ebb a tendency to southward. The northerly and southerly set is more marked between Napatree Point and Latimer Reef Light.

(275) In Sugar Reef and Catumb Passages the tidal currents set obliquely across the axis of the channel. The flood sets northwestward and the ebb southeastward. The tidal currents in Sugar Reef Passage are about the same velocity as in Watch Hill Passage but are stronger in Catumb Passage.

(276) In Lords Passage the tidal currents set diagonally across the channel and have a velocity of nearly 2 knots, the ebb being greater than the flood.

(277) In the main channel of Fishers Island Sound, the flood sets westward and the ebb eastward. In the main channel between Napatree Point and Wicopesset Island, the velocity of flood is 1.7 knots and ebb 2.2 knots. The flood sets 284° and the ebb 113°.

(278) In the channel south of Ram Island Reef, the velocities of flood and ebb are 1.3 and 1.6 knots, respectively. The flood sets 255° and the ebb 088°. The direction and velocity of the current are affected by strong winds that may change the duration of flood or ebb.

(279) The strong tidal currents prevent the formation of heavy local **ice**, except in shoal tributaries. The only ice to give trouble is that set in from Long Island Sound by wind and current. The ice formations in Little Narragansett Bay are sufficiently heavy to be destructive to structures exposed to them.

(280) On the south side of Fishers Island Sound, off the north side of **East Point** on Fishers Island, are **Seal Rocks**, partly bare at low water and marked by a buoy. A rocky patch covered 11 feet and marked by a buoy is about 500 yards northeastward of Seal Rocks. **Youngs Rock**, about 0.4 mile westward of Seal Rocks, has about 1 foot over it and is marked by a buoy. A rocky patch extends about 400 yards to the east-northeastward.

(281) **East Harbor** and **Chocomount Cove**, in the north shore of Fishers Island, are sometimes used as anchorages by small craft. There is considerable foul ground in East Harbor and in the approach to Chocomount Cove. The harbor and cove are exposed to northerly winds. A former Coast Guard Station with a boathouse and dock is prominent near the south side of East Harbor. Several small private piers with about 6 feet at their ends are in East Harbor.

(282) The north shore of Fishers Island from East Harbor around into West Harbor has several private landings.

(283) **East Clump** is a cluster of rocks partly bare at high water and marked by a buoy about 0.8 mile north of Fishers Island. From East Clump for some 2.8 miles westward to North Dumpling, there are rocky islets and dangers fhat must be avoided. These are 0.5 to 0.8 mile off the Fishers Island shore, and most are buoyed. **North Dumpling**, an islet marked by a light and sound signal, is surrounded by rocks awash and foul ground. The sound signal at the light is operated by keying the microphone five times consecutively on VHF-FM channel 83A. **Seafower Reef**, marked by a light, is near the middle of the western entrance of Fishers Island Sound and 0.8 mile northwestward of North Dumpling Light.

(284) **West Harbor**, on the north side of Fishers Island southeastward of North Dumpling Light, affords shelter from southerly winds. Foul ground extends across the entrance of West Harbor to near the eastern edge of the dredged channel; the northern limits of the foul ground are buoyed.

(285) A yacht club wharf and another small-craft facility are on the southwest side of the harbor. Gasoline, diesel fuel, water, ice, and hull and engine repairs are available. A marine railway can handle craft up to 40 feet. The head of the harbor is used by boats drawing less than 5 feet that enter by the narrow unmarked channel southward of **Goose Island**.

(286) **Hay Harbor**, at the west end of Fishers Island, is used by small craft.

(287) **Silver Eel Cove (Silver Eel Pond)** is on the west side of Fishers Island, 0.6 mile northeastward of Race Point. The entrance, about 75 feet wide and jettied, is marked by a private light and has a depth of about 11 feet, with similar depths inside. Submerged fender pilings are reported on both sides of the entrance. Dolphins are on the northeast side of the cove, and the channel is clear between them and the wharves on the southwest side. Vessels must go to the wharves as there is no room for anchorage. There is very little dockage available. The entrance is difficult with northwesterly or westerly winds. A lighted whistle buoy is about 450 yards off the entrance. A ferry that operates between Fishers Island and New London lands here. During the summer, a Coast Guard unit is stationed inside the entrance to the cove.

(288) On the north side of Fishers Island Sound are Little Narragansett Bay and Pawcatuck River leading to the towns of Westerly and Pawcatuck, Stonington Harbor and the town of Stonington, and Mystic Harbor leading to the towns of Noank and Mystic.

(289) **Napatree Beach**, 1.3 miles long between Watch Hill Point and **Napatree Point**, is bare. **Sandy Point**, about 1.4 miles north-northwestward of Napatree Point, is at the northwestern end of a long and narrow sand island

in Little Narragansett Bay. An extensive sandspit makes off from the northeasterly and southwesterly sides of the island; give these areas a good berth. The island is subject to continual change; caution is advised.

(290) **Napatree Point Ledge**, a boulder reef with little depth, extends nearly 0.4 mile southward of the point. It is marked by a lighted bell buoy. A sunken wreck is about 0.3 mile eastward of the ledge in about 41°18'N., 71°53'W.

(291) The west side of Napatree Point should not be approached closer than 175 yards to avoid a stone jetty that is covered at high water. Between Napatree Point and the Stonington outer breakwater is an extensive flat on which the depths are 2 to 10 feet, rocky bottom. **Middle Ground**, the western part of the flat, is marked by the outer breakwater, which has a light at its western end. A sound signal is at the light.

(292) A depth of 17 feet can be taken to an anchorage inside this breakwater, giving the light on the breakwater a berth of more than 250 yards. In anchoring, give the inside of the breakwater a berth of over 300 yards to avoid shoals and fishweirs. This anchorage provides good shelter except in southwesterly and westerly winds, although it is seldom used.

(293) **Little Narragansett Bay**, at the eastern end of Fishers Island Sound, is entered at its extreme western end southward of Stonington Point. A dredged channel leads around the north side of Sandy Point, thence extends southeast across the bay to the entrance of Pawcatuck River. The channel is marked by lighted and unlighted buoys.

(294) Caution should be exercised in entering Little Narragansett Bay. Shoal water extends for about 200 yards off **Stonington Point**, and the shoal area north of **Sandy Point** is subject to continual change. Strangers are advised to obtain local information before entering because of rocks and shoal water near the edges of the channel.

(295)

Current

(296) In the dredged channel northward of Sandy Point, the currents have a velocity of 1.3 knots. The flood sets eastward and the ebb westward. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(297) **Watch Hill Cove**, in the southeastern part of Little Narragansett Bay, is used by small craft. A dredged channel, marked by lighted and unlighted buoys, leads into the cove. A yacht club and town dock are in Watch Hill Cove; berths, guest moorings, electricity, diesel fuel and water are available.

(298)

Anchorages

(299) A special anchorage is in Watch Hill Cove. (See **33 CFR 110.1** and **110.47**, chapter 2, for limits and regulations.)

(300) **Pawcatuck River**, entered just south of **Pawcatuck Point**, extends about 4 miles to Westerly. A federal project provides for a depth of 10 feet for nearly 4 miles thence 7 feet to the end of the channel. (See Notice to Mariners and latest editions of charts for controlling depths.) The channel is well marked.

(301)

Current

(302) About 1 mile above the entrance to Pawcatuck River the tidal current has a velocity of 0.6 knot on the flood and 0.5 knot on the ebb.

(303)

Ice

(304) The river is generally closed by ice from January to March.

(305) **Colonel Willie Cove**, 0.5 mile above Pawcatuck Point, has a boatyard with a marine railway that can handle craft up to 50 feet for hull and engine repairs. Berths with electricity, gasoline, diesel fuel, water, ice, storage facilities, a pump-out station, marine supplies and a 30-ton lift are also available. Mariners enroute to the boatyard should use the chart as a guide.

(306)

Anchorages

(307) A **special anchorage** is in **Thompson Cove**, 2 miles above Pawcatuck Point. (See **33 CFR 110.1** and **110.48**, chapter 2, for limits and regulations.) A yacht club pier is in the cove. Private seasonal buoys mark the approach to the pier.

(308) **Westerly**, 4 miles above Pawcatuck Point, is an important manufacturing town.

(309)

Small-craft facilities

(310) There are numerous small-craft facilities along both sides of the Pawcatuck River and at the head at Westerly and Pawcatuck, just across the river. The largest marine railway in the area is at Avondale, and it can handle craft to 55 feet. Berths, electricity, gasoline, diesel fuel, water, ice, storage facilities, launching ramps, lifts, some repairs and marine supplies are available. Depths of 7 to 9 feet are reported at the town dock at Pawcatuck.

(311) **Wequetequock Cove** is a shallow cove at the northern end of Little Narrangansett Bay. A narrow unmarked channel leads eastward of **Elihu Island** into the cove. A depth of about 4 feet can be taken as far as **Goat Island**, about a mile above Sandy Point. A fixed railroad bridge with a clearance of 6 feet crosses the cove about 0.2 mile above Goat Island. A small-craft facility is on the west side of the cove near the head. Water, ice, berths, gasoline, storage facilities, launching ramp, 4-ton forklift, marine supplies and hull and engine repairs are

available. In 1981, a reported depth of 2 feet could be carried to the facility.

(312) **Stonington Harbor**, 3 miles northwestward of Watch Hill Point, is protected by breakwaters on each side. Each of the breakwaters is marked at its seaward end by a light. The controlling depth to the inner harbor is about 11 feet. Anchorage can be selected inside the west breakwater in depths of 15 to 18 feet, taking care to keep the south end of Wamphassuc Point bearing northward of 270°. Vessels drawing up to 8 feet can find anchorage in the inner harbor. A rock that bares at low water is about 50 yards southward of the fishing wharf and is marked by a private buoy.

(313)

Anchorages

(314) **Special anchorages** are in Stonington Harbor. (See **33 CFR 110.1** and **110.50**, chapter 2, for limits and regulations.)

(315) Stonington Harbor is approached from southeastward and westward. Vessels with local knowledge sometimes cross Noyes Shoal from southwestward. The southeastern approach is best, with fewer dangers, and the navigational aids serve as excellent guides to avoid them. In daytime with clear weather, no difficulty should be experienced in entering any of the approaches.

(316) From southeastward, the course from south of Napatree Point Ledge should be west-northwestward until off the buoy at the southwest end of Middle Ground, from which a northerly course can be shaped past the breakwater lights and into the harbor.

(317) From southwestward, a northeasterly course can be shaped from the lighted bell buoy south of Ram Island Reef to south of White Rock, and thence eastward past the north side of Noyes Rock to the harbor.

(318) The inner breakwater, about 400 yards northward of Stonington Point on the east side of the entrance, extends westward about 250 yards and is marked by a light.

(319) **Stonington** is on the east side of the harbor. Traffic is mostly fishing and recreational craft. The wharves have depths of 7 to 12 feet alongside. Following southerly weather, a surge is felt by vessels tied to the southern side of the seaward pier.

(320) A boatyard is in the northeast part of the harbor. Berths, electricity, gasoline, diesel fuel, water, ice, storage, 40-ton lift and marine supplies and hull, engine and electronic repairs are available. In 1981, a reported depth of 7 feet could be carried to the yard.

(321) A **harbormaster** is at Stonington.

(322) A railroad causeway with two fixed spans crosses Stonington Harbor 0.4 mile above Stonington; the east span has a clearance of 5 feet and the west span has a clearance of 4 feet. Overhead power cables at the openings have clearances of 41 feet.

(323) **Noyes Rock**, 0.4 mile southward of **Wamphassuc Point**, has a least depth of 7 feet. **Noyes Shoal**, with 10 to 18 feet over it, is nearly 1.5 miles long in a west-northwesterly direction; it is marked by a gong buoy near its eastern end.

(324)

No-Discharge Zone

(325) The State of Connecticut, with the approval of the Environmental Protection Agency, has established a No-Discharge Zone (NDZ) in the Connecticut portion of the Pawcatuck River, Little Narragansett Bay, portions of Fishers Island Sound and Stonington Harbor. The area covered extends from Wamphassuc Point due south past Noyes Shoal to the boundary between Connecticut and New York, easterly following the state boundary to the intersection of the Connecticut, New York and Rhode Island State lines, and following the boundary between Connecticut and Rhode Island to U.S. Route 1 over the Pawcatuck River and including all Connecticut waters seaward of U.S. Route 1.

(326) Within the NDZ, discharge of sewage, whether treated or untreated, from all vessels is prohibited. Outside the NDZ, discharge of sewage is regulated by **40 CFR 140** (see chapter 2).

(327) **Latimer Reef**, about 0.6 mile south of Noyes Shoal, is a very broken and rocky area 0.4 mile long. It is marked by a light at its west end and a buoy at its east end. The eastern end of the reef has a least found depth of 6 feet.

(328) **Latimer Reef Light** (41°18'16"N., 71°56'00"W.), 55 feet above the water, is shown from a white conical tower, brown midway of its height, on a brown cylindrical foundation. A sound signal is at the light.

(329) A detached 11-foot spot, marked by a buoy, is about 0.4 mile northeast of Latimer Reef Light.

(330) **Eel Grass Ground**, about 0.8 mile northwestward of Latimer Reef Light, is a shoal with a least depth of 6 feet, marked by buoys. **White Rock**, about 0.8 mile northeastward of Eel Grass Ground, is bare and prominent. **Red Reef**, covered 2 feet, is 0.2 mile north of White Rock and marked by a buoy. **Ellis Reef**, 0.4 mile northwestward of Eel Grass Ground, is marked on its east side by a daybeacon.

(331) **Mason Island**, 2.5 miles west of Stonington Harbor, is joined to the mainland by a fixed bridge with an 18-foot span and a clearance of 3 feet; the sound end of the island is strewn with boulders. A **special anchorage** is on the east side of Mason Island. (See **33 CFR 110.1** and **110.50a**, chapter 2, for limits and regulations.) An anchorage for small craft is on the west side of the south end of Mason Island where depths range from 8 to 11 feet; caution and local knowledge are required to use this anchorage because of the boulders in the area. A dangerous rock is off the east side of **Mason Point**, the southern extremity of Mason Island, in 41°19'21.6"N., 71°58'05.0"W.

(332) **Enders Island**, 0.3 mile eastward of the southern end of Mason Island, is connected to it by a fixed bridge with a 15-foot span and a clearance of 6 feet.

(333) **Ram Island Reef**, 1.8 miles westward of Latimer Reef Light, has two detached parts: the southerly section

(347)

is covered 8 feet and marked by a lighted bell buoy, and the northerly section, covered by 6 feet, is marked by a daybeacon. Passage between the reef and island is unsafe because of shoals.

(334) **Ram Island**, about 0.4 mile southwest of Mason Island, is wooded and grass fringed. A shoal, on which are two rocky islets, extends about 0.2 mile northeastward from Ram Island. **Ram Island Shoal**, extending nearly 0.5 mile westward from Ram Island, has little water over it, and many rocks bare at low water. **Whaleback Rock** and the islet 300 yards northwestward of it are bare.

(335) The narrow but deep channel along the north side of Ram Island Shoal is the easterly entrance to Mystic Harbor. Between the shoal and Groton Long Point is an area of foul ground and several dangerous rocks, including **Whale Rock**, which bares at low water, at the northwesterly end of Ram Island Shoal. This rock is marked by a seasonal lighted buoy. Leading across the shoal is the buoyed channel, good for about 11 feet, which is used by vessels entering Mystic Harbor from westward.

(336) A rock covered 6 feet is about 0.5 mile southwest of Whale Rock; about 0.65 mile southwest of that rock is **Intrepid Rock**, with 19 feet over it and marked by a buoy, which should be avoided. **Mouse Island**, marked by several dwellings, is 150 yards southwestward of Morgan Point.

(337) In 1983, a rock, covered about 2 feet, was reported 0.2 mile west of Mouse Island in about 41°18'52"N., 71°59'50"W.

(338) **Morgan Point** is located on the west side at the entrance of Mystic Harbor. A privately maintained and marked channel leading to the piers in **West Cove** at Noank westward of the point had a least depth of 4 feet reported in 1981.

(339) **Groton Long Point**, on which is a summer settlement, is about 0.9 mile southwestward of Morgan Point. A reef extends nearly 300 yards southwestward from the point and is marked by a buoy. About 0.3 mile to the west a rock awash at low water is 175 yards off the southwest end of Groton Long Point. It is marked by a buoy.

(340) **Mystic Harbor**, about 6 miles westward of Watch Hill Point, is the approach to the towns of Noank and Mystic. A federal project provides for a 15-foot channel from Morgan Point through Mystic Harbor and into Mystic River to the bascule bridge thence a 12-foot channel to Mystic Seaport Museum Wharf, about 0.6 mile above the bascule bridge. An anchorage basin with a project depth of 9 feet is on the east side of the river opposite Willow Point.

(341)

Anchorages

(342) **Special anchorages** are in Mystic Harbor. (See **33 CFR 110.1**, **110.50b**, and **110.50d**, chapter 2, for limits and regulations.)

(343)

Routes

(344) To enter from eastward, lay a west-northwesterly course from south of the lighted bell buoy marking Napatree Point Ledge for a little over 3 miles to about 400 yards south of the buoy marking the south end of **Cormorant Reef**. From here steer **261°** for 0.8 mile until Mason Point is abeam. Then follow the buoyed channel.

(345) From westward, proceed cautiously from about 100 yards or more southward of the buoy southward of Groton Long Point on an easterly course for about 0.5 mile to Mystic Harbor Channel Buoy 1, then steer a northerly course through the buoyed channel into Mystic Harbor, rounding Noank Light 5 at a distance of about 75 yards.

(346) **Noank** is a town on the west side of the channel through Mystic Harbor. There are several small-craft facilities at Noank and in **West Cove**. Berths, electricity, gasoline, diesel fuel, water, ice, storage facilities, launching ramps, a pump-out station, 30- and 60-ton lift and marine supplies are available; hull, engine, sail and electronic repairs can be made. A **harbormaster** is at Noank.

(348) **Mystic River** flows into Mystic Harbor from northward just below Mystic. The river is used by recreational craft, the local fishing fleet and by transient craft visiting Mystic Seaport. An **anchorage area** with depths of 3½ to 7 feet is in the lower part of the river between Willow Point and Murphy Point. **Ice** usually closes the river during January and February.

(349) **Willow Point**, 0.6 mile below Mystic, has several small-craft facilities that can provide berths, electricity, water, ice, some engine parts and marine supplies. A 12-ton crane and 30-ton mobile hoist are available; hull and engine repairs can be made.

(350) A channel, privately marked by daybeacons, leads from the vicinity of Willow Point for 0.3 mile in an easterly direction, thence about 0.4 mile northeastward to a marina on the west side of the mouth of **Pequotsepos Brook**, just below a railroad bridge. Berths, electricity, water, ice, storage, marine supplies, a 12-ton mobile hoist and hull and engine repairs are available. In 1981, a reported depth of 4 feet could be carried in the channel to the marina.

(351)

Small-craft facilities

(352) Several small-craft facilities are on the northern end of Mason Island. Berths with electricity, gasoline, diesel fuel, water, ice, storage facilities, marine supplies, a pump-out station, 35-ton lift and hull and engine repairs are available. In 1993, a reported depth of 4 feet could be carried to the facilities.

(353) The railroad bridge over Mystic River below Mystic has a swing span with a clearance of 8 feet. The U.S. Route 1 highway bridge at Mystic has a bascule span with a clearance of 4 feet. (See **33 CFR 117.1** through **117.59** and **117.211**, chapter 2, for drawbridge regulations.) The bridgetenders monitor VHF-FM channel 13; call signs KJA-842 and KXR-912, respectively.

(354) **Mystic**, a town about 2 miles above Noank, has several small-craft facilities. Berths, electricity, gasoline, diesel fuel, water, ice, marine supplies, storage facilities, mobile hoists, a 60-ton lift and marine railways up to 110 feet are available; hull and engine repairs can be made.

(355) A **harbormaster** is at Mystic.

(356) The **Mystic Seaport Museum** is about 0.6 mile above the highway bridge at Mystic. The whaler CHARLES W. MORGAN, full-rigged training ship JOSEPH CONRAD and Grand Banks fishing schooner L. A. DUNTON are permanently moored at the museum and open to the public. Along the waterfront of the museum property, a mid-19th Century coastal village has been recreated with shops and lofts of that period. Collections of maritime relics are on exhibit in several formal museum buildings.

(357) Above the Mystic Seaport Museum, the channel is very narrow and is marked by privately maintained seasonal buoys; boats of about 5-foot drafts can be taken to the **Narrows**, and thence depths are 1 and 2 feet to **Old Mystic**. Twin fixed highway bridges crossing the Narrows have clearances of 25 feet. The stream follows the east bank to the next narrows and the west bank to a marina in the bight about 0.3 mile below Old Mystic.

(358)

Mumford Cove to Avery Point

(359) **Mumford Cove** is entered about 2 miles west of Mystic Harbor. A privately dredged channel leads northward from the entrance to the head of the cove; two spur channels lead eastward from the main channel, about 0.3 mile and 0.6 mile, respectively, above the entrance. The channels are marked by private seasonal buoys and daybeacons. In 1981, the channels had a reported controlling depth of 2 feet.

(360) **Special anchorages** are in the cove. (See **33 CFR 110.1** and **110.50c**, chapter 2, for limits and regulations.)

(361) **Venetian Harbor** is a yacht basin on the east side of the entrance to Mumford Cove. A channel 75 feet wide leads through stone breakwaters into a basin with depths of about 3 to 7 feet. A submerged jetty extends along the channel from the outer end of the east breakwater. The entrance to the harbor is marked by a light on the outer end of the west breakwater.

(362) **Horseshoe Reef**, 0.5 mile southward of Mumford Cove entrance, is awash at low water and is marked by a buoy. Broken and rocky grounds extend from the reef to the shore eastward of Mumford Point.

(363) **Vixen Ledge**, with a depth of 10 feet and marked by a buoy, is about 1 mile west of Horseshoe Reef. **Pine Island** is bluff and grassy, about 1.3 miles west

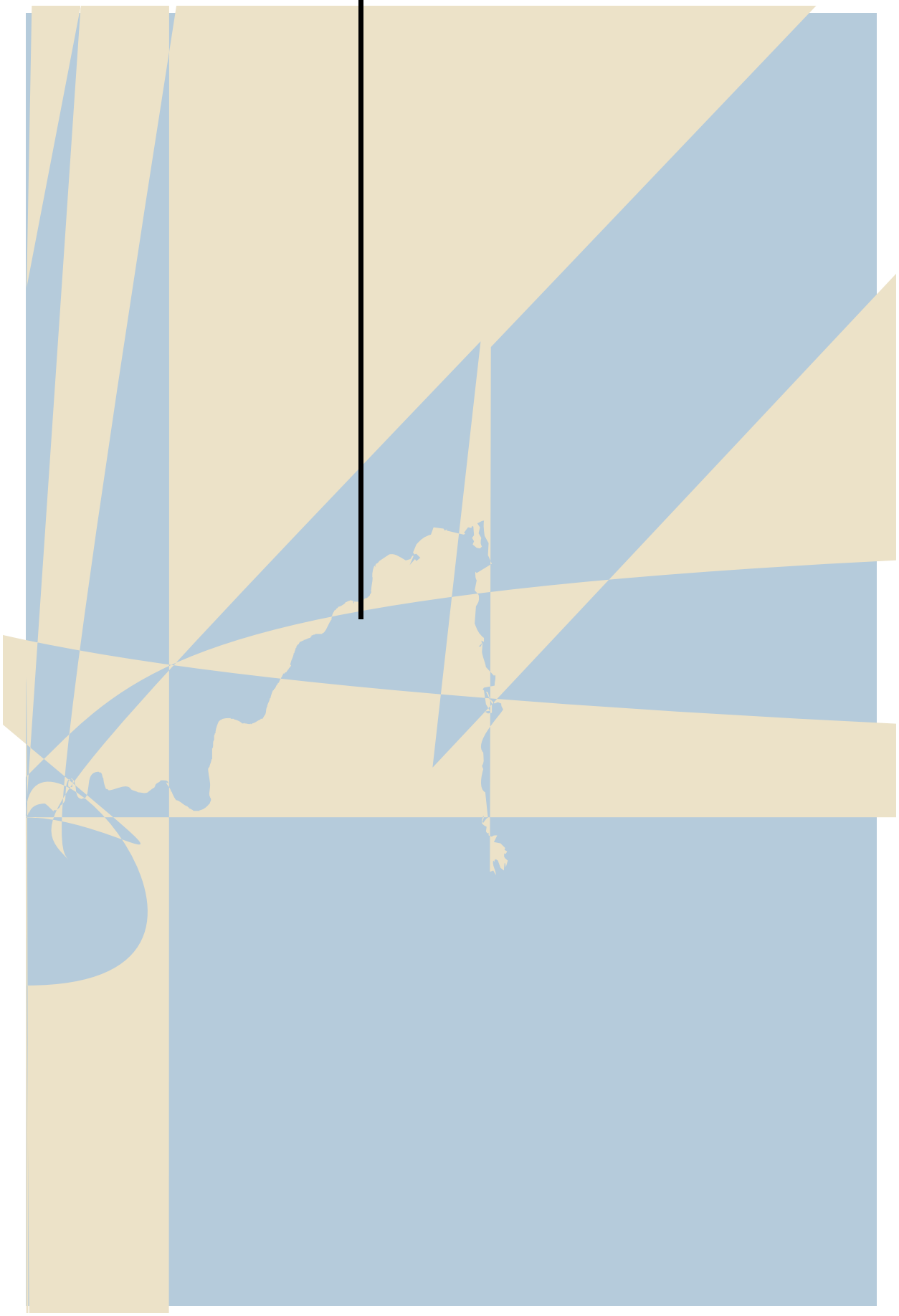
of Mumford Point. It is surrounded by shoal water and rocky bottom and is marked off the southwest side by a lighted bell buoy. A rock, covered 6 feet, in 41°18'35"N., 72°03'16"W., is about 0.3 mile northwestward of Vixen Ledge.

(364) A **special anchorage** is on the north side of Pine Island. (See **33 CFR 110.1** and **110.51**, chapter 2, for limits and regulations.)

(365) **Avery Point Light** (41°18′55″N., 72°03′49″W.) is shown from a white octagonal concrete tower at **Avery Point**. An unmarked rock awash is 0.3 mile south of the light. A cove indents the mainland north of Pine Island and east of Avery Point; the entrance is marked by two buoys eastward of Avery Point. Depths shoal from about 10 feet in the entrance to 1 foot at the head of the cove. A breakwater, marked by a private light, extends southeasterly from the east end of Avery Point. A 5 mph **speed limit** is enforced in the cove.

(366) A yacht club, marina, and launching ramp are in the cove. Berths, guest moorings, gasoline, electricity, water, ice, marine supplies and a 14-ton mobile hoist are available at the marina; hull and engine repairs can be made. In 2000, a reported depth of 7.5 feet could be carried to the marina.

(367) A **special anchorage** is in the cove. (See **33 CFR 110.1** and **110.51**, chapter 2, for limits and regulations.)



Eastern Long Island Sound

(1) This chapter describes the eastern portion of Long Island Sound following the north shore from Thames River to and including the Housatonic River and then the south shore from Orient Point to and including Port Jefferson. Also described are the Connecticut River; the ports of New London, New Haven and Northville; and the more important fishing and yachting centers on Niantic River and Bay, Westbrook Harbor, Guilford Harbor, Branford Harbor and Mattituck Inlet.

(2)

COLREGS Demarcation Lines

(3) The lines established for Long Island Sound are described in **33 CFR 80.155** chapter 2.

(4)

Long Island Sound

(5) **Long Island Sound** is a deep navigable waterway lying between the shores of Connecticut and New York and the northern coast of Long Island.

(6) In this region are boulders and broken ground but little or no natural change in the shoals. The waters are well marked by navigational aids so that strangers should experience no difficulty in navigating them. As all broken ground is liable to be strewn with boulders, vessels should proceed with caution in the broken areas where the charted depths are not more than 6 to 8 feet greater than the draft. All of the more important places are entered by dredged channels; during fog, vessels are advised to anchor until the weather clears before attempting to enter. The numerous oyster grounds in this region are usually marked by stakes and flags. These stakes may become broken off and form obstructions dangerous to small craft. Mariners should proceed with caution especially at night.

(7)

Caution

(8) Submarine operating areas are in the approaches to New London Harbor and Connecticut River and off the northern shore of Long Island. As submarines may be operating submerged in these areas, vessels should proceed with caution.

(9)

Anchorages

(10) New London Harbor is the most important of the anchorages sought for shelter in the eastern part of Long Island Sound. Niantic Bay and the approach between Bartlett Reef and Hatchett Reef are used to some extent by small vessels when meeting unfavorable weather or reaching the eastern part of the sound. Small vessels can select anchorage eastward or westward of Kelsey Point Breakwater, also in Duck Island Roads. Off Madison there is anchorage sheltered from northerly winds. New Haven Harbor is an important harbor of refuge.

(11) Several general anchorages are in Long Island Sound. (See **33 CFR 110.1** and **110.146**, chapter 2, for limits and regulations.)

(12)

No-Discharge Zone

(13) The States of New York and Connecticut, with the approval of the Environmental Protection Agency, have established a No-Discharge Zone (NDZ) in Long Island Sound and a portion of the East River, extending from the Hell Gate Bridge in the west to Block Island Sound in the east.

(14) Within the NDZ, discharge of sewage, whether treated or untreated, from all vessels is prohibited. Outside the NDZ, discharge of sewage is regulated by **40 CFR 140** (see chapter 2).

(15)

Tides

(16) The time of tide is nearly simultaneous throughout Long Island Sound, but the range of tide increases from about 2.5 feet at the east end to about 7.3 feet at the west end. Daily predictions of the times and heights of high and low waters are available from the tide prediction service at *tidesandcurrents.noaa.gov*. Links to a user guide for this service can be found in chapter 1 of this book.

(17) The effect of strong winds, in combination with the regular tidal action, may at times cause the water to fall several feet below the plane of reference of the charts.

(18)

Current

(19) In the eastern portion of Long Island Sound the current turns from ½ to 1½ hours earlier along the north shore than in the middle of the sound.

(20) Proceeding westward from The Race in the middle of the sound, the velocity of current is 1.8 knots off Cornfield Point, about 1 knot off New Haven, 1 knot off Eatons Neck, 0.4 knot between Peningo Neck and Matinecock Point and 0.5 knot eastward of Hart Island.

(21) About 1.5 miles east-southeastward of Bartlett Reef, the velocity of flood is 1.2 knots and ebb 1.6 knots. The flood current sets 285° and the ebb 062°.

(22) At a point about 3 miles southward of Cornfield Point, the flood current sets 256° with a velocity of 2 knots and the ebb sets 094° with a velocity of 1.7 knots.

(23) About 1 mile north of Stratford Shoal (Middle Ground) Light, the velocity is 1 knot, the flood setting

westward and the ebb eastward. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(24)

Weather, Long Island Sound and vicinity

(25) Weather is most favorable from mid-May to mid-October, when the most common hazards are thunderstorms and fog. There is also a rare threat of a tropical cyclone. During June, July and August on the average, there are 20 to 25 days per month with conditions generally considered ideal even for small boaters. Fog is most likely in spring and early summer. Fog, or the lack of it, at inland locations is not a guide to conditions in the Sound or its approaches. Areas along the coast, at the heads of bays and within rivers may be relatively clear, while offshore the fog is thick. For example, on exposed Block Island heavy fog is encountered about 10 to 12 percent of the time from April though August compared to 1 to 3 percent at Westhampton. Thunderstorms on the other hand are more likely over land but can be turbulent in the Sound, especially in a squall line preceding a cold front in spring and early summer. Winter winds are mostly out of the west through north, but gales blow less than 5 percent of the time in these somewhat sheltered waters. Waves are restricted by limited fetch except to the east. However, choppy conditions can create problems.

(26)

Ice

(27) In ordinary winters the floating and pack ice in Long Island Sound, while impeding navigation, does not render it absolutely unsafe, but in exceptionally severe winters the reverse is true; none but powerful steamers can make their way.

(28) Drift ice, which is formed principally along the northern shore of the sound under the influence of the prevailing northerly winds, drifts across to the southern side and accumulates there, massing into large fields, and remains until removed by southerly winds, which drive it back to the northerly shore.

(29) In ordinary winters ice generally forms in the western end of the sound as far as Eatons Neck; in exceptionally severe winters ice may extend to Falkner Island and farther eastward.

(30)

Effects of winds on ice

(31) In Long Island Sound northerly winds drive the ice to the southern shore of the sound and southerly winds carry it back to the northern shore. Northeasterly winds force the ice westward and cause formations heavy enough to prevent the passage of vessels of every description until the ice is removed by westerly winds. These winds carry the ice eastward and, if of long duration, drive it through The Race into Block Island Sound, thence it goes to sea and disappears.

(32) In New Haven Harbor, the influence of the northerly winds clears the harbor and its approaches unless the local formation is too heavy to be moved. Southerly winds force the drift ice in from the sound and prevent the local formations from leaving the harbor. Tides have little effect upon the ice. Additional information concerning ice conditions in the waters adjoining Long Island Sound is given under the local descriptions.

(33) **Vessel Tra f c Service, New York**, operated by the U.S. Coast Guard, serves New York Harbor. (See **33 CFR 161.1** through **161.25**, chapter 2, for regulations.)

(34)

Pilotage, Long Island Sound

(35) Pilotage by a state licensed pilot is compulsory in Long Island Sound for foreign flag vessels and U.S. vessels that are under register (i.e. engaged in foreign trade). Such vessels can arrange for a state licensed pilot by contacting the joint rotation administrator, Block Island Pilots, at 243 Spring Street, Newport, RI 02840; telephone 401–847–9050 (24 hours), 800–274–1216; FAX 401–847–9052. Enrolled vessels (i.e., U.S. vessels engaged in coastwise trade) may be required to have a U.S. Coast Guard Federally licensed pilot unless the master has recency for the intended area.

(36) The pilot boat sets radio guard at least 1 hour before a vessel's ETA.

(37) Vessels to be boarded should provide a ladder 3 feet above the water on the lee side.

(38) Pilot services must be arranged at least 24 hours in advance through ships' agents or directly by shipping companies.

(39) Pilotage, in the waters of Long Island Sound for enrolled vessels (i.e. U.S. vessels engaged in coastwise trade), is available from, but not limited to:

(40) Northeast Marine Pilots, Inc., 243 Spring Street, Newport, RI 02840; *nemarinepilots*.com; telephone 401–847–9050 (24 hours), 800–274–1216; FAX 401–847–9052. Pilot boats are NORTHEAST IV, 52-foot, gray hull and superstructure, word PILOT on both sides and NORTHEAST II, 47-foot, gray hull and superstructure, word PILOT on both sides. The boats monitor VHF-FM channels 16, 10, 13, 14; work on 10.

(41) Interport Pilots Agency, Inc./Connecticut State Pilots, P.O. Box 236, Port Monmouth, NJ 07758; *interportpilots.com*; telephone 732–787–5554 (24 hours); email: interport@verizon.net. Pilot boats are CONNECTICUT PILOT, 65-foot with blue hull and white superstructure and KEN JOHNSON, 48-foot with blue hull and white superstructure. These boats monitor VHF-FM channels 16 and 13, work on 11 and are equipped with AIS.

(42) Constitution State Pilots Association, 9 Nottingham Drive, Old Lyme, CT 06371, telephone 203–627–5057. Pilot boat is CONSTITUTION, 65-foot with black hull and white superstructure displaying the word PILOT on both sides. The boat monitors VHF-FM channels 16, 13 and 9A; works on 13 or 9A.

(47)

(43) Connecticut River Pilots Association (CRPA), P.O. Box 107, Old Saybrook, CT 06475; telephone 860–388–4167. Pilot boat TRUDEE II is 36-foot, with black hull, white superstructure, and with the word PILOT on the house, forward. The boat monitors VHF-FM channels 16 and 13; works on 13.

(44) See Pilotage, New London-Groton (indexed as such), this chapter; Pilotage, New Haven (indexed as such), this chapter; Pilotage, Bridgeport (indexed as such), chapter 9; Pilotage, Offshore Terminal, Northville-Riverhead (indexed as such), this chapter; and Pilotage, Offshore Terminal, Northport (indexed as such), chapter 9.

(45)

Thames River

(46) **New London Harbor**, near the east end of Long Island Sound at the mouth of the **Thames River**, is an important harbor of refuge. Vessels of deep draft can find anchorage here in any weather and at all seasons.

(48) Waterborne commerce in New London Harbor and on the Thames River is chiefly in petroleum products, chemicals, coal, copper, lumber, seafood products and general cargo.

(49) **Security zones** have been established in New London Harbor. (See **33 CFR 165.1** through **165.7**, **165.30**, **165.33**, **165.140** and **165.154** chapter 2, for limits and regulations.)

(50) **New London** is a city on the west bank of Thames River about 2.5 miles above the mouth. The town of **Groton** on the east bank is connected to New London by a highway bridge and a railroad bridge. The main harbor comprises the lower 3 miles of Thames River from Long Island Sound to the bridges and includes Shaw Cove, Greens Harbor and Winthrop Cove. It is approached through the main entrance channel extending from deep water in Long Island Sound to deep water in the upper harbor. The harbor is generally used by vessels drawing 9 to 30 feet; the deepest draft entering is about 36 feet. Petroleum products, seafood products, copper, lumber and other forest products are the principal waterborne commodities handled at the port.

(51) **Greens Harbor**, a small-craft shelter just north of the entrance, has general depths of 6 to 17 feet. **Special anchorages** are in the harbor. (See **33 CFR 110.1** and **110.52**, chapter 2, for limits and regulations.)

(52) **New London Coast Guard Station** and **Fort Trumbull State Park** are on the west side of the main channel northward of Greens Harbor.

(53) **Shaw Cove** is a dredged basin about 0.8 mile northward of Greens Harbor. In 2015, the controlling depth was 14 feet in the entrance channel through the south draw of the bridge, thence 12 feet was available in the basin.

(68)

Structures across Thames River

Name•Description•Type	Location	Clear Width of Draw or Span Opening (feet)	Clear Height above Mean High Water (feet)	Information
Railroad Bridge (swing)	41°20'57"N., 72°05'50"W.	70 (north and south draw)	6	Note 1 Crosses the entrance to Shaw Cove
Railroad Bridge (fixed)	41°21'32"N., 72°05'44"W.	14	4	Crosses the head of Winthrop Cove
Amtrak Thames Railroad Bridge (vertical lift)	41°21'47"N., 72°05'16"W.	150	29 (down), 135 (up)	Notes 2, 3 and 4
Gold Star Memorial Bridges (fixed)	41°21'52"N., 72°05'16"W.	500	135	Vertical clearance is for a middle width of 200 feet. A racon is on the south span.
Overhead power cable	41°26'19"N., 72°05'21"W.		160	
Route 2A Bridge (fixed)	41°28'54"N., 72°04'32"W.	200	75	
Shetucket River				
Route 2/Water Street Bridge (fixed)	41°31'25"N., 72°04'34"W.	123	18	
Railroad Bridge (fixed)	41°31'25"N., 72°04'30"W.	229	13	
Viaduct Road Bridge (fixed)	41°31'24"N., 72°04'30"W.	119	18	
Main Street Bridge (fixed)	41°31'26"N., 72°04'08"W.	94	23	
Yantic River				
Route 32 Bridge (fixed)	41°31'28"N., 72°04'51"W.	62	11	

Note 1 – See **33 CFR 117.1** through **117.59** and **117.223** chapter 2, for drawbridge regulations.
Note 2 – See **33 CFR 117.1** through **117.59** and **117.224** chapter 2, for drawbridge regulations.
Note 3 – Bridgetender monitors VHF-FM channel 13; call sign KT-5473.
Note 4 – In 1998, it was reported that cross currents of 1 to 2 knots can be encountered in the vicinity of this bridge.

(54) **Winthrop Cove**, northward of Shaw Cove, is part of the main waterfront channel.

(55)

Prominent features

(56) **New London Ledge Light** (41°18'21"N., 72°04'39"W.), 58 feet above the water, is shown from a red brick building on a square white pier on the west side of New London Ledge. A sound signal at the light is operated by keying the microphone five times consecutively on VHF-FM channel 83A.

(57) Other prominent features in approaching New London Harbor are New London Harbor Light, on the west side of the entrance channel; the monument at Fort Griswold; the microwave tower atop a building in downtown New London; the large sheds at the shipyard on the east side of the river opposite Fort Trumbull State Park and the highway bridge at New London.

(58)

Channels

(59) A U.S. Navy project for New London Harbor provides for a channel 40 feet deep to Fort Trumbull, thence 38 feet to the Submarine Force Library and Museum, thence 36 feet to the U.S. Navy Submarine Base. A federal project provides for a channel 23 feet deep in the waterfront channels north of Fort Trumbull and in Winthrop Cove. (See Notice to Mariners and latest editions of the charts for controlling depths.) Lighted and unlighted buoys and a **354°** lighted range mark the channel. The range does not mark the center of the lower end of the channel.

(60) **Pine Island Channel**, northeastward of New London Ledge Light, between Pine Island and Black Ledge, has a rocky and very broken bottom on which the least found depth is 9 feet. It is used some by local vessels between New London Harbor and Fishers Island Sound but should be avoided by any vessel drawing more than 10 feet.

(61)

Anchorages

(62) General and naval anchorages are in the approaches to, and in, New London Harbor. (See **33 CFR 110.1** and **110.147**, chapter 2, for limits and regulations.) Special anchorages are in Greens Harbor and in the vicinity of the U.S. Coast Guard Academy. (See **33 CFR 110.1** and **110.52**, chapter 2, for limits and regulations.)

(63)

Dangers

(64) On the west side of the approach to New London Harbor, foul ground extends about 1 mile from shore in the vicinity of **Goshen Point**. The southerly and southeasterly limits of this area are marked by buoys. The area has numerous rocky patches and boulders, some showing above water, and should be avoided by small craft. **Rapid Rock**, marked by a buoy on its southeast side, is about 1.6 miles southwestward of New London Ledge Light; it has a least depth of 10 feet. An unmarked ledge covered 35 feet is about 100 yards south by eastward of Rapid Rock and is the outermost shoal to the southward. **Sarah Ledge**, 0.7 mile northeastward of Rapid Rock and marked by a buoy, has a least depth of 14 feet and is the easternmost shoal on the west side of the main channel approach.

(65) On the east side of the main channel foul ground extends about 1 mile offshore. **New London Ledge**, marked by New London Ledge Light, has a least depth of 7 feet. **Black Ledge**, just to the northeastward of New

London Ledge, has a rocky islet, 2 feet high, on it. Depths are 5 to 18 feet on the ledge. Buoys mark the shoal area.

(66) Broken ground fringes the shore southwestward of New London Harbor Light. A rock covered 6 feet is located about 0.1 mile from shore in the bight just southward of the light.

(67) **White Rock**, an islet in Greens Harbor, is 250 yards from the 18-foot curve on the western edge of the channel. **Hog Back**, a small ledge awash at low water, is 150 yards southwestward of White Rock and about 0.3 mile from the western shore and is marked by a buoy. Rocks, covered 2 to 6 feet, are in the middle of the northern part of Greens Harbor. **Melton Ledge**, northward of White Rock, with one-half foot over it, is 125 yards eastward of **Powder Island** and is marked by a buoy; a rock awash is close westward of Melton Ledge.

(69)

Current

(70) The tidal currents follow the general direction of the channel and usually are not strong. At Winthrop Point, on the west side of the river at New London, the velocity is 0.4 knot, and at Stoddard Hill, about 6.5 miles above New London, 0.7 knot on the flood and 0.4 knot on the ebb. During freshets or when the river is high and the wind is from the north, the current can have considerable southerly set even on the flood.

(71) **Ice** obstructs navigation about 2 months each year above the naval station, which is some 5 miles above New London Ledge Light, but seldom forms below the station. In extremely severe winters, however, heavy ice from the sound, driven in by winds, has been known to extend about 1.8 miles above the entrance. Between New London and the mouth of the river small vessels may navigate with comparative safety in ordinary winters; even in severe weather, it is rare that navigation for small vessels stops for more than a week. Steamers can nearly always enter and leave with safety. Drift ice sometimes forms a decidedly dangerous obstruction in the approaches through Long Island Sound during severe winters, especially during February and March, and small vessels are much hindered in their movements during January, February and March.

(72) **Freshets** usually occur in the river in the spring. It is reported that they seldom exceed 2 feet above high water at Norwich.

(73) New London Harbor and Thames River are easy of access by day or night, but local knowledge is required to take drafts greater than 20 feet above the submarine base.

(74)

Pilotage, New London-Groton

(75) Pilotage by a state-licensed pilot is compulsory in Long Island Sound for foreign flag vessels and U.S. vessels that are under register (i.e., engaged in foreign trade). Such vessels can arrange for a state-licensed pilot by contacting the joint rotation administrator, Block Island Pilots at 243 Spring Street, Newport, RI 02840; telephone 401–847–9050 (24 hours), 800–274–1216; FAX 401–847–9052. Enrolled vessels (i.e., U.S. vessels engaged in coastwise trade) may be required to have a U.S. Coast Guard federally licensed pilot unless the master has recency for the intended area. See Pilotage, Long Island Sound (indexed as such), chapter 8.

(76)

Towage

(77) Tugs to 3,200 hp are available at New London. Vessels usually proceed to the upper harbor without assistance, although a tug may be required when entering with a head wind and contrary current. Large vessels normally require tugs for docking and undocking.

(78) New London is a **customs port of entry.**

(79)

Quarantine, customs, immigration and agricultural quarantine

(80) (See chapter 3, Vessel Arrival Inspections, and Appendix A for addresses.)

(81) **Quarantine** is enforced in accordance with regulations of the U.S. Public Health Service. (See Public Health Service, chapter 1.) New London has several hospitals.

(82) **Harbor regulations** are in force for New London Harbor. The harbormaster has authority to berth vessels, shifting them if necessary, but occasion for doing so seldom arises.

(83)

Wharves

(84) New London Harbor has more than 30 wharves and piers. Most of these facilities are used as repair berths and for mooring recreational craft, fishing vessels, barges, ferries and government vessels. Depths alongside these facilities range from 10 to 40 feet. Only the deep-draft facilities are described. The alongside depths are reported; for information on the latest depths contact the private operator.

(85) Amerada Hess Corp. Wharf (41°20'09"N., 72°04'58"W.): on the east side of the river opposite Greens Harbor; T-head pier with 55-foot face, 960 feet of berthing space with dolphins; 40 feet alongside; deck height, 8 feet; pipelines to storage tanks; fresh water connection; railroad and highway connections; receipt and shipment of petroleum products and receipt of molasses; bunkering vessels; owned and operated by Hess Oil and Chemical Division, Amerada Hess Corp.

(86) Admiral Shear State Pier: the more easterly of the two long piers southwestward of the Thames River bridges, about 1.3 miles northward of Amerada Hess Corp. Wharf; 200-foot face, 26 feet alongside; west side 1,000 feet, 23 to 27 feet alongside; east side 1,020 feet, 34 to 38 feet alongside; deck height, 10 feet; 90,000 square feet of covered storage, 20 acres of open storage; electricity, potable and feed water connections on pier; railroad and highway connections; receipt and shipment of general cargo, copper, zinc, steel and wood products; owned by the State of Connecticut and operated by Logistec U.S.A. Inc., a division of Logistec Stevedoring of Montreal.

(87) Supplies of all kinds are available. Gasoline and diesel oil can be obtained from oil companies on 48 hours' notice by tank truck. Water is available at most of the piers, wharves and marinas.

(88)

Repairs

(89) A shipbuilding company at New London can perform all kinds of repairs on steel-hulled vessels. The company has floating drydocks with lifting capacities from 1,000 to 10,000 tons. The company's largest drydock is on the west side of the river and has a maximum pontoon length of 300 feet and a width between wingwalls of 110 feet and is about 0.9 mile north of the Thames River bridges.

(90) Cranes to 70 tons and floating derricks to 25 tons are available at New London.

(91) Several companies in New London are in the business of wrecking, salvage and marine contracting work. They are equipped with pumps, divers' outfits, floating equipment and other gear.

(92)

Small-craft facilities

(93) There are numerous small-craft facilities in Greens Harbor and Shaw Cove.

(94)

Communications

(95) New London has good railroad and bus communications. Automobile-passenger ferry service is available to Block Island and Fishers Island and to Orient Point, Long Island.

(96) Thames River above New London has a dredged channel to Norwich, the head of navigation. The channel is well marked by navigational aids.

(97)

Caution

(98) The dikes along the Thames River from Easter Point (41°28.2'N., 72°04.5'W.) to Norwich are submerged at half tide.

(99) A wreck debris field with a least depth of 21 feet is charted in about (41°27'41.7"N.,72°04'05.2"W.) Thames River mid-channel in the vicinity of Stoddard Hill.

(100)

Pilotage, Thames River

(101) For Pilotage for the river see Pilotage, New London-Groton (indexed as such) earlier this chapter.

(102) The **U.S. Coast Guard Academy** is on the west side of Thames River about 1 mile north of the center of New London. The administration building, with its white tower and clock, and the lighted chapel spire are very prominent but are not visible until almost abeam of the academy. Depths alongside the 410-foot-long academy pier were reported in 2005 to be 30 feet at the face, 30 feet along the south side and 30 to 34 feet on the north side.

(103) The **U.S. Naval Submarine Base** is on the east side of the Thames River about 2.5 miles above New London. USS Nautilus is permanently moored just south of the base as part of the Submarine Force Library and Museum.

(104) A **restricted area** is off the U.S. Naval Submarine Base. (See **33 CFR 334.75**, chapter 2, for limits and regulations.)

(105) Just below **Gales Ferry**, on the east side about 4 miles above the bridges, are the crew training quarters and boathouses of Harvard and Yale Universities. Opposite Gales Ferry is the town of **Bartlett**, site of a prominent power plant with two tall and conspicuous stacks. A privately dredged channel with depths of about 20 feet leads to the dock and coal tipple.

(106) At **Montville Station**, just above Bartlett, is a dock with a depth of 23 feet at the face. The northeast end of the dock is in ruins. Overhead power cables with a clearance of 160 feet cross the river 0.5 mile above the station near **Kitemaug**.

(107) **Allyn Point**, on the east side about 5 miles above New London, is the site of a large private pier for receiving liquid chemicals, with a reported depth of about 30 feet alongside. It is marked by an elevated water sphere and several small tanks on the pier.

(108) **Fort Point**, on the east side 8 miles above New London, has a long fuel pier marked by privately maintained red lights, and on shore is a building with several stacks. Numerous piles are in the water southward of the pier.

(109) The red brick buildings of the Norwich State Hospital are on a bluff just north of Fort Point and are a conspicuous landmark.

(110) At **Thamesville**, on the west side of the river about 1 mile below Norwich, are two finger piers each with breasting dolphins used to receive petroleum products from barges. Depths of 20 to 25 feet are reported alongside the face of the piers.

(111) **Norwich**, a city at the head of navigation on Thames River at its junction with **Shetucket River** and **Yantic River**, is about 11 miles above New London. Small boats generally anchor in Shetucket River just above the fixed bridges at Norwich.

(112)

Bartlett Reef to Millstone Point

(113) **Bartlett Reef Light** (41°16'28"N., 72°08'14"W.), 35 feet above the water and shown from a skeleton tower with a red and white diamond-shaped dayboard, is about 3.3 miles southwestward of New London Ledge Light and marks the south end of **Bartlett Reef**. A mariner-activated sound signal at the light is initiated by keying the microphone five times on VHF-FM channel 79. The reef, about 1.3 miles long in a general north-south direction and about 0.3 mile wide, is covered 2 to 18 feet and has rocks awash near its northern end. The north end of the reef is marked by a buoy. A lighted bell buoy and an unlighted buoy are about 0.9 mile southward and about 0.3 mile eastward of the light, respectively.

(114) A **general anchorage** is about 0.8 mile northeastward of Bartlett Reef Light. (See **33 CFR 110.1** and **110.147(a)(4)**, and **(b)**, chapter 2, for limits and regulations.)

(115) **Twotree Island**, small and bare, about 1.4 miles northwestward of Bartlett Reef Light, is surrounded by shoals. A buoy marks rocks awash that extend off the northern end of the island.

(116) **Twotree Island Channel** leads northward of Bartlett Reef and Twotree Island. With an adverse current in the sound, this channel is used to some extent by light tows and sailboats with a leading wind in the daytime, as the tidal currents turn about 1 hour earlier along the north shore than in the middle of the sound. About 0.3 mile southwestward of **Seaside**, the tidal currents have a velocity of 1.2 knots and ebb 1.6 knots. Flood sets westerly and the ebb easterly. The channel is buoyed, but strangers are advised to use it with caution and should never attempt to beat through.

(117) From **Goshen Point** (41°18.0'N., 72°06.8'W.) westward, there are scattered boulders that extend offshore as much as 0.2 mile in places. **Jordan Cove**, 1.5 miles west of Goshen Point, is foul in its northerly half, and the southerly part is obstructed by **Flat Rock**, bare at low water and marked by a buoy, and **High Rock**, which shows at high water and is marked by a buoy.

(118) **Millstone Point**, on the east side at the entrance of Niantic Bay, is occupied by the buildings of the Millstone Nuclear Power Station. A 389-foot red and white stack at the station and a radio tower on the point are the most conspicuous landmarks in the area. A cove with depths of 2 to 17 feet is on the west side of the point. A rock with 1 foot over it lies 60 feet off the mouth of the cove. The station maintains channel markers and a range for occasional barge traffic. A dredged area for the power station's water intakes is 0.2 mile northwest of the cove.

(119)

Niantic Bay

(120) **White Rock** is an islet on the east side of the entrance to Niantic Bay 0.5 mile westward of Millstone Point. **Little Rock**, two rocks partly bare at low water, is 150 yards east of White Rock. Rocks with a least depth of 8 feet extend 0.25 mile northwest from Millstone Point. A rock, covered 11 feet, is about 300 yards south-southeast of White Rock and is marked by a lighted bell buoy.

(121) **Niantic Bay**, 4.5 miles westward of New London Harbor, is a good anchorage sheltered from easterly, northerly and westerly winds. It is a harbor of refuge in northerly gales and can be used by small vessels and tows. The general depth of the bay is about 19 feet; the water shoals gradually northward. The entrance is 1.5 miles wide, and the dangers are marked by buoys or show above water.

(122) **Niantic** and **Crescent Beach** are summer resorts with railroad communication at the north end and northwest side of the bay.

(123) The Niantic Bay Yacht Club basin at Crescent Beach is protected on the south, east, and partially on the north side by a U-shaped breakwater; a private seasonal light is near the outer end of the breakwater.

(124) A **special anchorage** is on the west side of Niantic Bay off Crescent Beach. (See **33 CFR 110.1** and **110.53**, chapter 2, for limits and regulations.)

(125) **Niantic River** empties into the northeast end of Niantic Bay and is entered through a dredged channel that leads from the bay, thence through a narrow passage at the entrance, thence to a point about 300 yards northward of the entrance to Smith Cove. The channel is marked by daybeacons and seasonal buoys. Two bridges cross the narrow passage at the entrance. The more southerly is the Amtrak railroad bascule bridge with a clearance of 16 feet. The State Route 156 highway bridge, about 0.1 mile northward, has a bascule span with a clearance of 32 feet. (See **33 CFR 117.1** through **117.59** and **117.215**, chapter 2, for drawbridge regulations.) The bridgetender at each bridge monitors VHF-FM channel 13; call signs KGA-511 and KXR-911, respectively.

(126) Strangers attempting to enter Niantic River are cautioned to pass through the bridges either at slack water or against the current.

(127) Above the head of the dredged channel, small craft can navigate for about another 1.5 miles to **Golden Spur (East Lyme)** with local knowledge. The river from westward of Sandy Point to the stone bulkhead at Golden Spur is deep and clear; vessels generally follow the west bank. **Pine Grove**, **Sandy Point**, and **Saunders Point** are summer resorts on Niantic River.

(128)

Current

(129) The **tidal currents** through the bridges set fair with the channel; the flood velocity is 1.6 knots and the ebb velocity, 0.8 knot. It has been reported that much greater velocities may be expected under storm and freshet conditions. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(130)

Ice

(131) Ice generally closes the river to navigation for about 3 months during the winter.

(132) **Smith Cove** is on the west side of Niantic River about 1.5 miles above the channel entrance. A channel, marked by private daybeacons, leads westward from the river channel into the cove. In 1999, the channel had a reported depth of 5 feet.

(133)

Small-craft facilities

(134) There are several small-craft facilities just above the entrance at Niantic and **Waterford**, on the west side

(155)

Structures across Connecticut River

Name•Description•Type	Location	Clear Width of Draw or Span Opening (feet)	Clear Height above Mean High Water (feet)	Information
Amtrak Old Saybrook-Old Lyme Bridge (bascule)	41°18'39"N., 72°20'54"W.	139	19	Notes 1, 2 and 3
Raymond E. Baldwin/I-95 Bridge (fixed)	41°19'09"N., 72°20'52"W.	258	81	
Overhead power cable	41°19'12"N., 72°20'47"W.		108	
State Route 82 Bridge (swing)	41°27'07"N., 72°27'51"W.	180 (east draw) 200 (west draw)	22	Notes 1 and 4
Overhead power cable	41°27'40"N., 72°27'58"W.		105	
Overhead power cables	41°30'44"N., 72°33'24"W.		101	
Overhead power cable	41°33'30"N., 72°34'38"W.		65	
Overhead power cable	41°33'30"N., 72°35'46"W.		111	
Conrail Middleton-Portland Bridge (swing)	41°34'00"N., 72°38'50"W.	100	25	Note 1
Arrigoni/Sate Route 66 Bridge (fixed)	41°34'09"N., 72°38'55"W.	480	89	
William H. Putman Memorial Bridge (fixed)	41°42'52"N., 72°38'26"W.	300	80	Vertical clearance is over main channel
Overhead power cable	41°45'09"N., 72°39'12"W.		120	
Charter Oak/State Route 15 Bridge (fixed)	41°45'10"N., 72°39'16"W.	215	69	Vertical clearance is over main channel
Overhead power cable	41°45'12"N., 72°39'23"W.		150	
Founders Highway Bridge (fixed)	41°45'57"N., 72°39'55"W.	155	49	Note 5
Bulkeley/I-84 Bridge (fixed)	41°46'10"N., 72°39'55"W.	100	39	
Railroad Bridge (fixed)	41°46'37"N., 72°39'28"W.	125	28	

Note 1 – See **33 CFR 117.1** through **117.59** and **117.205** chapter 2, for drawbridge regulations.
Note 2 – Bridgetender monitors VHF-FM channel 13; call sign KT-5414.
Note 3 – Vessels requesting openings are cautioned to confirm by radiotelephone that the bascule span is safely raised and stabilized before making passage.
Note 4 – Bridgetender monitors VHF-FM channel 13; call sign KXR-913.
Note 5 – When travellers are in use, minimum vertical clearance is 44 feet.

and east side of Niantic River, respectively, and in Smith Cove.

(135) **Harbormasters** are at Niantic and Waterford. A 6 mph **speed limit** is enforced on the river.

(136) **Black Point**, on the west side at the entrance to Niantic Bay, is flat with bluffs at the water and is occupied by many summer cottages. Broken ground extends 0.6 mile south of the southwest side of the point.

(137) Strangers entering the bight between Black Point and Hatchet Point should proceed with caution as there is broken ground with several rocks and ledges. An area with covered rocks and shallow ledges extends about 0.6 mile south of **Griswold Island**. Arock with a least depth of 3 feet is at the outer end of this area; buoys mark the west side of the rocky area and the south side of the outer rock. **North Brother**, in the northwest part of the bight and **South Brother**, in the center, are prominent bare rocks. **Johns Rock**, covered 6 feet, is in 41°17’12”N., 72°14’57”W., about 0.5 mile southwest of South Brother.

(138)

Anchorages

(139) A **special anchorage** is east of **Giants Neck.** (See **33 CFR 110.1** and **110.54**, chapter 2, for limits and regulations.) An unmarked rock is within the anchorage area, about 0.1 mile south of Giants Neck; depth over the rock is not known.

(140) **Hatchett Point** has several large dwellings. A reef extends about 0.2 mile off the southwest side of the point.

(141) **Hatchett Reef**, 0.6 to 1 mile south-southwestward of Hatchett Point, has a least depth of 5 feet and is marked by buoys. Close to the southeast side of the reef the depths are greater than 30 feet. A bar extends westward from Hatchett Reef to Saybrook Bar.

(142)

Connecticut River

(143) **Connecticut River** rises in the extreme northern part of New Hampshire, near the Canadian border, and flows southerly between the States of Vermont and New Hampshire and across Massachusetts and Connecticut to Long Island Sound. It is approximately 375 miles long and is one of the largest and most important rivers in New England. The head of commercial navigation is at Hartford, about 45 miles from the mouth. Waterborne commerce on the river is mostly in petroleum products and chemicals.

(144) The river water is fresh at and above Deep River. Each year after the spring freshets, shoals with least depths of 10 feet are found in places on bars in the upper river; dredging to remove such shoals is begun as soon as the water subsides.

(145) Between the entrance and Middletown the river banks are hard and in some places rocky, but between Middletown and Hartford the river flows through alluvial bottom land, where freshets and ice jams may cause shoaling.

(146)

Channels

(147) A federal project for Connecticut River provides for a 15-foot jettied entrance channel and 15-foot dredged cuts across the bars to Hartford, 45 miles above the entrance. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A.

(148) The channel above the jettied entrance channel usually follows the banks on the outside of the curves of the river, except through the dredged cuts across the bars that are marked by navigational aids.

(149) **Saybrook Breakwater Light** (41°15'48"N., 72°20'34"W.), 58 feet above the water, is shown from a white conical tower on a brown cylindrical pier on the south end of the west jetty at the entrance to Connecticut River. A mariner radio activated sound signal at the light is initiated by keying the microphone five times consecutively on VHF-FM channel 83A.

(150)

Anchorages

(151) Secure anchorage can be had eastward or northeastward of Lynde Point Light. Farther up anchorage can be selected in the wider parts of the channel. Special anchorage areas have been established along the river as far north as Middletown. (See **33 CFR 110.1**, **110.55** and **110.55b**, chapter 2, for limits and regulations.)

(152)

Dangers

(153) **Saybrook Outer Bar**, which obstructs the mouth of the Connecticut River, is shifting, with depths of 2 to 12 feet extending nearly 2 miles off the mouth; it is marked off its southeastern end by a lighted bell buoy.

(154) In 1976, obstructions were reported in the channel at the railroad bascule bridge 3 miles above the mouth of the Connecticut River; a least depth of 13 feet is reported in the channel in area 40 to 50 feet from the east abutment of the bridge. Mariners requiring greater depths are advised to avoid this area of the channel during passages.

(156)

Tides

(157) The time of tide becomes later and the range diminishes in progressing up the river. High water and low water at Hartford occur about 4.5 and 6 hours later, respectively, than at the entrance.

(158)

Current

(159) At the entrance the currents have considerable velocity at times and always require careful attention, as the tidal current of the sound often sets directly across the direction of the current setting out or in between jetties. This condition is reported to be especially dangerous during the first 3 hours of ebb tide. Consult the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for times and velocities of currents at a number of locations in Connecticut River. Links to a user guide for this service can be found in chapter 1 of this book.

(160) During the ebb, a strong current runs from the Lyme Landing toward the center of the railroad bridge. Towboats with vessels in tow should steer for the east pier of the draw and should not swing out for the draw until almost in it, to avoid being set to the west side of the channel. Because of river discharge, the ebb current usually will be considerably stronger than the flood. Ebb current velocities of 1 knot or more have been observed under normal conditions on the bars in Connecticut River between Higganum and Hartford; the velocities of the flood currents are much less.

(161) **Freshets** occur principally in the spring, when the snow is melting, although occasional floods have occurred in every month of the year except July and September. At Hartford the usual rise due to spring freshets is between 16 and 24 feet. The highest freshets are generally of short duration, but the period during which the river at Hartford is at the level of 8 feet or more above mean low water averages nearly 2 months of each year. Below Middletown the height of the crest of a freshet decreases rapidly. At the mouth the variation in water level is due to the tides.

(162) **Ice** closes the river to navigation a part of every winter for wooden hull boats. The duration of closing is about 2 months.

(163)

Weather, Hartford and vicinity

(164) Hartford is well inside the northern temperate climatic zone in a prevailing west to east movement of air carrying the majority of weather systems into Connecticut from the west. The average wintertime position of the "Polar Front" boundary between cold dry polar air and warm moist tropical air is just south of New England, which helps to explain the extensive winter storm activity and the day-to-day variability of local weather. In the summer, the "Polar Front" has an average position along the New England-Canada border and Hartford has a warm and pleasant climate.

(165) The location of Hartford, relative to the continent and ocean, is also significant. Rapid weather changes result when storms move northward along the Mid-Atlantic Coast, frequently producing strong and persistent northeast winds associated with storms known locally as "coastals" or "nor'easter." Seasonally, weather characteristics vary from the cold and dry continental-polar air of winter, to the warm maritime air of summer.

(166) Summer thunderstorms develop in the Berkshire Mountains to the west and northwest, move over the Connecticut Valley and, when accompanied by wind and hail, sometimes cause considerable damage to crops. Thunderstorm days average 20 each year. June, July and August are the most favored months. During the winter, rain often falls through cold air trapped in the valley and creates extremely hazardous ice conditions. On clear nights in the late summer or early autumn, cool

air drainage into the valley and the moisture from the Connecticut River produce steam and/or ground fog that becomes quite dense throughout the valley and temporarily hampers transportation. An average 162 days each year report fog.

(167) The average annual temperature for Hartford is 50°F (10°C). The warmest month is July with an average temperature of 74°F (23.3°C) and the coolest is January with an average temperature of 26°F (-3.3°C). The warmest temperature on record is 102°F (38.9°C) recorded in July 1966 and the coolest temperature on record is -26°F (-3.3°C) recorded in January 1961. Each month except June, July and August has recorded temperatures below freezing. Each month June through September has recorded temperatures in excess of 100°F (37.8°C). An average of 18 days each year record temperatures in excess of 90°F (32.2°C) and an average of 134 days each year have a temperature of 32°F (0°C) or cooler. An average of eleven days each year have temperatures of 5°F (-15°C) or lower.

(168) The average annual precipitation for Hartford is 44.20 inches (1,123 mm). Precipitation is fairly uniform with the difference between the wettest and driest month being less than one inch (25.4 mm). The wettest month is November, averaging 4.07 inches (104 mm), and the driest month is February, averaging 3.13 inches (80 mm). Average snowfall, on an annual basis, totals 44 inches (1,118 mm). February 1961 holds the record of the greatest snowfall in a 24-hour period with 14.3 inches (363 mm).

(169) The National Weather Service office is at Bradley International Airport, northwest of Hartford.

(170)

Routes

(171) To enter Connecticut River from eastward, pass southward of Hatchett Reef and Saybrook Bar, until Saybrook Breakwater Light bears 315°. Steer for Saybrook Breakwater Light on this course through the buoyed opening between the south end of Saybrook Bar and the east end of Long Sand Shoal to the entrance channel between the jetties.

(172) To enter from westward, pass 1 mile southward of Falkner Island Light on course **076°.** This will lead about 0.4 mile northward of the lighted bell buoy on the western end of Long Sand Shoal and about 0.2 mile southward of the lighted bell buoy southward of Cornfield Point. Then steer about **067°,** with Saybrook Breakwater Light a little on the port bow to the entrance channel between the jetties.

(173) Boating regulations for waters within the State of Connecticut can be found at *ct.gov/deep/site/default.asp*.

(174)

Pilotage, Connecticut River

(175) Pilotage by a state-licensed pilot is compulsory in Long Island Sound for foreign flag vessels and U.S. vessels that are under register (i.e., engaged in foreign trade). Such vessels can arrange for a state-licensed pilot by contacting the joint rotation administrator, Block Island Pilots at 243 Spring Street, Newport, RI 02840; telephone 401–847–9050 (24 hours), 800–274–1216; FAX 401–847–9052. Enrolled vessels (i.e., U.S. vessels engaged in coastwise trade) see Pilotage, Long Island Sound (indexed as such), chapter 8.

(176) Pilot services are arranged in advance through ships' agents or directly by shipping companies. A 24-hour advance notice is requested.

(177) Hartford is a **customs port of entry**.

(178)

Wharves

(179) The Connecticut River has more than 20 commercial piers and wharves, most of which handle petroleum products from barges or coastal tankers. Most of the facilities below Rocky Hill, about 34 miles above Saybrook Point, are marginal-type wharves, while those above Rocky Hill are finger-type piers with breasting dolphins. Depths of 11 to 15 feet are reported alongside these facilities.

(180)

Supplies and repairs

(181) Gasoline, diesel fuel, water, ice and marine supplies are available at the principal towns and landings along the Connecticut River. Boatyards along the river can make engine, hull and electronic repairs.

(182)

Old Saybrook to Hamburg

(183) **Old Saybrook** is a village on the west side of Connecticut River, about 1.4 miles northward of Saybrook Breakwater Light. There are several small-craft facilities along the west side of the river from Saybrook Point to **Ferry Point**, about 2 miles to the northward.

(184) A "**Slow no-wake**" speed limit is enforced at Old Saybrook between the railroad bridge and Buoy 20, 0.25 mile above the Raymond E. Baldwin/I-95 Bridge.

(185) **North Cove**, a dredged small-boat basin that affords excellent anchorage, is entered through a dredged channel that leads westward from the main channel about 0.4 mile northward of Saybrook Point. The entrance channel is marked by private buoys. Special anchorage areas are just south of the entrance and in North Cove. (See **33 CFR 110.1** and **110.55b**, chapter 2, for limits and regulations.)

(186) From Saybrook Point to Hartford local knowledge is required to carry the best water. Small craft should have no difficulty in following the channel.

(187) **Lieutenant River**, leading to **Old Lyme**, enters the east side of Connecticut River about 1.4 miles northward of Saybrook Point. Pipe stakes mark the south side of the channel across the bar at the entrance. A midchannel depth of about 3 feet can be carried over the bar to about 0.2 mile above the second bridge. A railroad bridge with a 33-foot fixed span and a clearance of 11 feet crosses the river 0.4 mile above the entrance. An overhead power cable with a reported clearance of about 10 feet is on the north side of the bridge. About 0.3 mile above that

bridge is a highway bridge with a 24-foot fixed span and a clearance of 6 feet. A **harbormaster** is at Old Lyme.

(188) The passage to the east and north of **Calves Island**, about 1 mile above the railroad bridge crossing Connecticut River, is used extensively for mooring small craft in the summer. This passage is subject to shoaling, particularly on the north side of Calves Island; caution is advised. A small-craft facility is on the east side of the passage just above the entrance. Berths, electricity, water, ice, marine supplies, storage facilities, a 25-ton lift and some repairs are available. In 2002, depths of 18 feet were reported at the facility. A special anchorage area is west of Calves Island and the dredged channel across Calves Island Bar. (See **33 CFR 110.1** and **110.55b**, Chapter 2, for limits and regulations.)

(189) **Lord Cove** has its entrance about 300 yards northward of Calves Island. In 1981, a depth of 3½ feet was available through the unmarked entrance. The marshlands surrounding Lord Cove and the other coves between Essex and the river mouth at Saybrook are frequented by duck hunters in October and November. Because of danger of gunfire, mariners are cautioned not to stray too close to the numerous duck blinds that exist in this area.

(190) The dredged section of the main channel in Connecticut River westward of Calves Island has numerous obstructions and sunken rocks close to its edges; mariners are advised to exercise caution and to avoid the edges of the channel.

(191) **Haydens Point**, about 4.6 miles above Saybrook Point, is marked by a light. Foul ground is between the light and the shore.

(192) **Essex**, a town on the west bank about 5 miles above Saybrook Point, is the scene of considerable small-boat activity. Depths alongside the town landing are about 6 feet. **Essex Cove** is the area off the main river channel skirting the waterfront at Essex. A dredged channel, marked by private buoys, leads from the main channel through the cove and thence rejoins the main channel to the northward. A 5 mph **speed limit** is enforced. A privately marked small-boat channel leads westward from the dredged buoyed channel in Essex Cove to a yacht basin in **Middle Cove**, northward of **Thatchbed Island**.

(193)

Small-craft facilities

(194) There are several small-craft facilities at Essex. The facilities provide berths and mooring, electricity, gasoline, diesel fuel, water, ice, marine supplies, pump-out facilities, storage and full repairs. The marina just north of the entrance to Middle Cove has an approach depth of 8 feet and an alongside depth of 11 feet.

(195)

Anchorages

(196) **Special anchorages** are at Essex. (See **33 CFR 110.1** and **110.55**, chapter 2, for limits and regulations.)

(197) **Hamburg Cove** and **Eightmile River**, which empties into the north end of the cove, indent the east side of Connecticut River, 6 miles above Saybrook Point. A dredged channel leads from Connecticut River to a turning basin at **Hamburg**, a village at the head of navigation. There are boulders in places outside the dredged channel and the entrance channel is outlined by grassy flats on each side. Buoys mark the entrance and private aids to navigation mark the channel to Hamburg. The center of the turning basin has piles used for moorings.

(198)

Small-craft facility

(199) A small-craft facility, on the east side of the basin, has sewage pump-out, water, ice and some marine supplies. A 35-ton travel lift is available for hull and engine repairs.

(200)

Eustasia Island to Holyoke

(201) **Eustasia Island**, 8.5 miles above Saybrook Point, divides the Connecticut River into two channels. A light off the southeast end of the island marks the junction of the two channels. The eastern channel crossing **Potash Bar** through a dredged cut is better marked and easier to follow. The western channel leads to **Pratt Creek**, westward of the southerly end of Eustasia Island, and to the landing at **Deep River** and thence crosses **Chester Creek Bar** through a swash channel to **Chester Creek**. A sand shoal and a rocky reef, both bare at low water, are north of Eustasia Island, between the main channel east of the island and Chester Creek.

(202) A rock, covered 3 feet, is on the south side of the entrance to Chester Creek in about 41°24'24.1"N., 72°25'46.6"W.

(203) The Chester-Hadlyme vehicular ferry crosses the river near Fort Hill, 2 miles above Eustasia Island. The ferry operates from April through November.

(204)

Small-craft facilities

(205) There are several small-craft facilities on Pratt Creek and Chester Creek. Berths and moorings, electricity, gasoline, diesel fuel, ice, marine supplies, a pump-out, storage and a launching ramp are available in the area. Lifts to 55 tons are available for complete hull, engine and electronic repairs.

(206)

Anchorages

(207) **Special anchorages** are off Chester Creek and 2 miles up river at Lord Island and Eddy Rock Shoal. (See **33 CFR 110.1** and **110.55**, chapter 2, for limits and regulations.)

(208) On the east side of the river, the turret of the opera house at **East Haddam**, 13.3 miles above Saybrook Point, is prominent. A marina is on the west side of the river just above the swing bridge between East Haddam and **Tylerville**. Limited guest berths, limited marine supplies, electricity, water and ice are available. In 1990,

a reported depth of 5 feet was available in the marina basin.

(209) The shoal off the west side of the river, just north of East Haddam, is reported to be increasing.

(210) **Salmon Cove**, on the east side of the river, 1 mile above East Haddam, is reported to be navigable only by small craft at high tide. The entrance to the cove is subject to shoaling. Considerable grass in the channel and cove makes boat operation difficult.

(211) Overhead power cables with a least clearance of 86 feet cross the cove about 1.2 miles above the mouth.

(212)

Small-craft facility

(213) A small-craft facility is on the west side of the river about 1.1 miles above East Haddam. Berths, electricity, water, ice, a 10-ton mobile hoist and a launching ramp are available; hull and engine repairs can be made. In 1990, a depth of 6 feet was reported at the facility.

(214) **Haddam Island** divides the Connecticut River about 3.2 miles above East Haddam. The main river channel leads eastward of the island through a dredged cut known as Haddam Island Bar Channel. A pinnacle rock, covered 13 feet, is in the approach to Haddam Island Bar Channel in 41°29'31"N., 72°30'49"W.

(215) The passage westward of Haddam Island is closed by a bare sand shoal lying between the island's southerly tip and the westerly shore of the river.

(216) The shoal off the east side of the river opposite **Higganum Creek**, 5.5 miles above East Haddam, is extending westward.

(217) A rock breakwater extends southward from the east side of the river, 1 mile above Higganum Creek. In 1969, the shoal, about 200 yards southward of the breakwater, was found to be extending southward.

(218) A boatyard is on the north side of the river at **Cobalt**, about 3.5 miles above Higganum Creek. Storage facilities and a 15-ton hoist are available. In 1990, a reported depth of 7 feet could be carried to the facility.

(219) After passing through the channel in **Paper Rock Shoal**, 9.7 miles above East Haddam, favor the south side of the river to about 300 yards southeastward of **Bodkin Rock**, then cross to the north side and pass it close-to.

(220) About 0.5 mile westward of Bodkin Rock, a dredged section of the channel leads along the southerly shore of Connecticut River and southward of Mouse Island Bar.

(221)

Anchorages

(222) **Special anchorages** are along the north and east sides of the river, between Bodkin Rock and Portland. (See **33 CFR 110.1** and **110.55(f)** and **(g)**, chapter 2, for limits and regulations.)

(223)

Caution

(224) Caution is recommended when rounding the point on the south side of the river, about 1.5 miles above Bodkin Rock, to avoid a submerged crib that extends northward from the point.

(225) **Portland**, 26.3 miles above Saybrook Point, has several boatyards with marine railways; the largest railway can handle craft to 60 feet for engine and hull repairs. Gasoline, water, berths with electricity, ice, storage facilities, marine supplies, a pump-out station, launching ramps and lifts to 50 tons are available at Portland. In 1990, depths of 7 to 9 feet were reported available.

(226) Berthing and water are available at Harbor Park in **Middletown**, across the river from Portland. Depths of 18 feet are reported to be available along the wharves.

(227) Two small-craft facilities are on the east side of the river at **Gildersleeve**, about 2.5 miles above Portland. Gasoline, diesel fuel, berths with electricity, water, ice, storage facilities, marine supplies, a launching ramp and 15- and 35-ton lifts are available, and hull and engine repairs can be made.

(228) From **Belamose**, 6.5 miles above Portland, northward to Hartford, the land is much lower, and the Connecticut River narrows, its curves become more pronounced, and both of its shores have numerous wood-stake-and-rock groins.

(229) A marina on the east side of the river opposite Belamose has gasoline, berths, electricity, water, ice, marine supplies and a 15-ton lift; engine and hull repairs can be made. In 1983, the privately marked channel into the marina basin had a reported controlling depth of 7 feet.

(230) At **Rocky Hill**, 1 mile above Belamose, a seasonal vehicular ferry crosses the river to South Glastonbury. A small-craft launching ramp is just above the ferry landing.

(231) The cove at **Crow Point**, on the west side of the river about 5.7 miles above Belamose, is used to obtain land fill. Dredging in the cove is uneven, but the bottom is soft ooze. In 1981, it was reported that the entrance had shoaled to bare and could be used only by small outboards.

(232) A rock, covered 5 feet, is on the south side of the dredged channel about 0.8 mile above Crow Point in about 41°42'43.0"N., 72°37'46.5"W.; and a shoal that bares is in 41°43'11"N., 72°38'52"W., on the west side of Connecticut River, about 1.9 miles above Crow Point.

(233) **Wethersfeld Cove**, on the west side of the river 14 miles above Portland, is entered through a narrow dredged channel that leads to a dredged anchorage basin about 0.3 mile above the entrance. (See Notice to Mariners and the latest editions of the charts for controlling depths.) The channel is marked by daybeacons. The Interstate 91 highway bridge over the entrance has a fixed span with a clearance of 38 feet. The speed limit in the channel and cove is 5 knots. Ice, transient berthing and some supplies can be obtained at the yacht club on the south side of the cove. A town marina is on the east side of the cove; a launching ramp is available at the facility. The Wethersfield harbormaster can be contacted through the local police department or town hall.

(234) The only remaining commercial docks at **Hartford** are the bulk fuel handling facility of the Hartford Electric

Light Company's powerplant on the west side of the river, about 0.2 mile below the Charter Oak Bridge, and the Hartford Gas Company's barge unloading facilities on the west side of the river, about 0.5 mile above the Charter Oak Bridge. A public facility with floating docks is on the west side of the river just below the Founders Bridge. A flood control dike is along the west side of the river from just north of the Charter Oak Bridge to the Bulkeley Bridge.

(235) Connecticut River above Hartford is practically unimproved but is navigable about 30 miles to **Holyoke** for boats not exceeding 3-foot draft, when the river is not low. The channel is constantly shifting.

(236)

Long Sand Shoal

(237) **Long Sand Shoal** extends 6 miles westward from off the entrance of Connecticut River and has a greatest width of nearly 0.3 mile; the shoal is constantly shifting. The general depths on the shoal are 4 to 15 feet; bottom is hard and lumpy. Shoaling is abrupt on both sides, but especially on the south side, where the 30-foot curve is only 100 yards from it in places. The shoal is marked at its eastern end by a buoy and on the south side and west end by a lighted buoy and gong buoy, respectively.

(238) At the western end of Long Sand Shoal and 1 mile southward is an area about 0.6 mile long with rocky and broken bottom and with a least found depth of 21 feet.

(239) **Sixmile Reef**, about 3 miles southwestward of Long Sand Shoal, is an area of migrating sandwaves about 2.5 miles long in a west-northwesterly direction with depths of 24 to 31 feet. Shoaling is abrupt in places. A lighted buoy is off the southerly edge of this reef. With extreme low tides, due to northerly and westerly winds, this shoal may be dangerous to vessels with 15-foot draft. Tide rips occur on the reef whenever the direction of the tidal currents is opposed to that of the wind. This is especially true during spring tides and a southwest wind.

(240) A ridge with depths of 27 to 31 feet is near the middle of Long Island Sound southward of Sixmile Reef and 5 miles north-northwestward of Horton Point Light. The ridge is marked by a lighted whistle buoy on the east side.

(241)

Cornfield Point

(242) **Cornfeld Point**, 2 miles westward of Saybrook Breakwater Light, is marked by a large red-roofed stone building. Rocky shoals and foul ground extend 0.5 mile south and 1.9 miles. **Cornfeld Point Shoal**, a small rocky patch covered 3 feet, is about 0.4 mile south of the point. Westward of this shoal are **Hen and Chickens**, bare in spots at low water, and **Crane Reef**, an area of broken ground with a least depth of 3 feet; these dangers are buoyed. About 0.5 mile west of the point is **Halftide Rock**, surrounded by foul ground.

(243)

Westbrook Harbor to Madison

(244) **Westbrook Harbor** is the western part of the open bight between Cornfield Point and Menunketesuck Island. It has many unmarked submerged rocks and is seldom used as an anchorage; the anchorage in Duck Island Roads is better. The bight is characterized by boulders.

(245) **Westbrook**, a town on the north side of Westbrook Harbor, is marked on its east side by a communications tower. A **harbormaster** is at Westbrook and can be contacted through the town hall.

(246) **Menunketesuck Island** is the outermost of several low narrow islands connected to the mainland at low water on the west side of Westbrook Harbor. It has boulders at the south end. A boulder reef extends nearly 0.5 mile south-southeastward from the point to the 18-foot curve. Tide rips frequently occur on this reef. There are two private seasonal buoys off the southern end of Menunketesuck Island. One is about 0.3 mile southeastward and the other is about 0.2 mile westward.

(247) Between Menunketesuck Island and Hammonasset Point, about 4 miles westward, broken ground extends about 1.5 miles offshore. A boulder reef extends 0.5 mile southward from Duck Island to the 18-foot curve and is marked by a buoy. A rock with 1 foot over it is on this reef about 300 yards south of Duck Island. Tide rips have been reported to extend from the vicinity of these rocks to the buoy. During strong flood currents and a southwest wind, tide rips extend from the shoal water southwest of Duck Island to the vicinity of **Southwest Reef** over 1 mile southwestward. Caution is advised when navigating small boats in this vicinity during these conditions.

(248) **Duck Island Roads**, between Menunketesuck Island and **Kelsey Point**, is a harbor of refuge protected by breakwaters 1,100 feet northward and nearly 0.5 mile westward from **Duck Island**, with the added protection of Kelsey Point Breakwater on Stone Island Reef. A prominent landmark on Duck Island is a stone chimney. Both breakwaters extending from Duck Island are marked by lights.

(249) The dredged anchorage enclosed by the breakwaters extending northward and westward from Duck Island is subject to shoaling. General depths of 3 to 8 feet are in the protected area, and 4 to 16 feet in the western end. In addition to the area inside the breakwaters, a small area northward and northeastward of Duck Island North Breakwater Light can be used as an anchorage in southwesterly weather.

(250) The western entrance of Duck Island Roads is easy of access and should be used by vessels with greater draft than 8 feet.

(251)

Routes

(252) Pass southward of Duck Island and keep the light on the end of Kelsey Point Breakwater bearing northward

of 264° until Duck Island West Breakwater Light 2DI bears 010°, then steer northward. Approaching from westward, the main dangers are 15- to 17-foot spots, south-southwestward of Kelsey Point Breakwater Light; the 15-foot spot is marked by a buoy.

(253) The eastern entrance of Duck Island Roads is obstructed by a sand shoal with a least depth of 8 feet about 0.3 mile eastward of Duck Island and by boulder reefs that extend about 0.2 mile off the western side of Menunketesuck Island. This entrance is easy of access for vessels drawing up to 8 feet.

(254) Anchorage, bottom generally sticky, can be had between Duck Island West Breakwater Light 2DI and Kelsey Point. This anchorage is exposed to southerly winds.

(255) **Patchogue River**, used chiefly by fishing and recreational craft, empties into Duck Island Roads just west of Menunketesuck Island. A dredged channel leads north from Duck Island Roads and into the river to just below the first fixed highway bridge, about 0.6 mile above the mouth; an anchorage basin is adjacent to the east channel limit between Daybeacons 8 and 10. The approach channel is marked by lighted and unlighted buoys, and the river channel is marked by private daybeacons. A light is on the outer end of the breakwater on the west side of the river mouth. Several small-craft facilities are on the river.

(256) **Menunketesuck River**, sharing the same entrance channel as Patchogue River, is a shallow stream westward of Patchogue River. A shoal was reported extending south from shore at the junction of Patchogue and Menunketesuck Rivers; caution is advised. The junction is marked by a private seasonal buoy. Small-craft facilities on the river can provide berths, electricity, gasoline, diesel fuel, water, ice, storage, marine supplies, a pump-out station and engine, hull and electronic repairs; a 12-ton mobile hoist and an 80-ton lift are available. The privately maintained channel in the river is reported to be marked by seasonal private aids; local knowledge is advised.

(257) A 6 mph **speed limit** is enforced on both rivers.

(258) **Kelsey Point Breakwater** extends south-southeastward from Stone Island and is marked by a light on the outer end. The rocky, broken ground southwestward of the light has a least depth of 16 feet; the outer shoal is marked by a buoy. Tide rips occur frequently between the end of the breakwater and the buoy. **Stone Island**, at the north end of the breakwater, is mostly covered at high water. There are several rocks of unknown depth between Stone Island and Kelsey Point. Anchoring should be avoided in the area surrounding the breakwater as the bottom is broken and rocky.

(259) The bight at the entrance of Clinton Harbor and westward of Kelsey Point Breakwater affords anchorage but is exposed to southeasterly and southwesterly winds.

(260) **Clinton Harbor**, the bight westward of Kelsey Point Breakwater, is the entrance to **Hammonasset River**, a stream used chiefly by fishing and recreational craft. **Wheeler Rock**, awash at low water, is just outside the bar and is marked by a lighted buoy. A dredged channel leads north, around Cedar Island to the town dock at **Clinton**. The channel is marked by buoys to Cedar Island and thence by seasonal private buoys to the anchorage basin at Clinton. Buoys are shifted often due to changing channel conditions. Local knowledge is advised. From opposite the basin to the upstream limit of the federal project, the southwest and south side of the channel is obstructed by a series of pilings. Boats may be moored between the pilings; caution is advised. Above the dredged channel, the midchannel controlling depth is about 2 feet in the Hammonasset River to the overhead pipeline and bridge crossing about 2 miles above Clinton. Private daybeacons mark this section of the channel.

(261) Several boatyards and marinas are in the harbor. Mooring facilities are available by arrangement with the town **dockmaster**, who can be contacted through the town hall or police department. A 6 mph **speed limit** is enforced in the harbor. The town maintains a fireboat at Clinton Harbor. The vessel can be contacted through the Clinton Police Department or the Coast Guard.

(262) Northeastward of **Cedar Island** in Clinton Harbor are two narrow crooked channels close together, with depths of about 1 foot. The eastern one is usually marked by bush stakes; it leads to a marina and boatyard just inside the mouth of **Hammock River**. The western channel, marked by a private range, leads to a boatyard on **Indian River.**

(263) **Hammonasset Point**, on the southwest side of Clinton Harbor, is a low marshy area with many wooded knolls. The end of the point is a rocky knoll. **Hammonasset State Park** is marked by a conspicuous flagstaff and the buildings at the recreational center. In the summer it is an active resort. Broken ground with rocky irregular bottom and least depths of 10 to 11 feet extends 0.5 mile southward of Hammonasset Point. A reef, with a least depth of 3 feet and a groin on its inner part, extends 0.4 mile southwestward from the point and is marked by a buoy, northeastward of which tide rips frequently occur. When rounding the point, vessels should not pass between the buoy and Hammonasset Point. **West Rock** is the outermost of the bare rocks that extend a short distance off the east end of Hammonasset Point.

(264) **Madison Reef**, over 2 miles westward of Hammonasset Point, extends over a mile east and west. This reef consists of several rocky patches with depths of 4 to 17 feet, with deeper water between them. **Charles Reef**, with a least depth of 7 feet, is about 0.5 mile southwest of Madison Reef and marked by a buoy.

(265) **Kimberly Reef**, about 1.9 miles southward of Charles Reef, is an area of broken ground with a least depth of 12 feet. Rocks with a least depth of 20 feet, marked by a lighted bell buoy, are about 0.2 mile south of the shoal. A bank with depths of 14 to 28 feet extends about 1.5 miles west of Kimberly Reef to Falkner Island.

(266) Vessels of 10-foot draft can anchor northward of Madison Reef but should proceed with caution to avoid the rocky patches at lesser depths.

(267) **Tuxis Island**, northward of Madison Reef and 0.2 mile south of **Middle Beach**, is high and rocky. Between the island and the shore the water is shallow and the ground foul. Rocks awash are 200 to 600 yards eastward of the island, and an islet is 100 yards westward of the island. A steel bulkhead in ruins, the top of which is awash at high water, extends from shore to **Gull Rock**, a high bare ledge about 300 yards east-northeastward of Tuxis Island.

(268) **Madison**, a town on the railroad, has one landing that bares alongside at low water and is in disrepair. A few small craft moor in the cove on its north side. Rocks, bare at low water, are 100 yards eastward of the landing. A beach club building, with a small stone landing, is northward of Tuxis Island. A church with a prominent tower and gilded dome is 0.8 mile northward of Tuxis Island.

(269)

Guilford Harbor to Mansfield Point

(270) **Guilford Harbor**, a bight 5.5 miles westward of Hammonasset Point, is used only by small craft. **East River** and **Sluice Creek** empty into Guilford Harbor from the northward. The approach to the harbor is obstructed by rocks and foul ground. The outermost dangers are **Half Acre Rock** (41°15'17"N., 72°39'10"W.), **Outer White Top** (41°15'03"N., 72°40'00"W.) and **Indian Reef** (41°14'52"N., 72°40'21"W.), extending about 1 mile southwestward of Outer White Top. Indian Reef is marked on its south side by buoys. Stakes and fish traps may exist northward of **Riding Rock** (41°15'32"N., 72°39'52"W.)

(271) The approach channel to Guilford Harbor, marked by buoys, leads along the southeasterly side of Indian Reef, thence westward of Half Acre Rock to a dredged channel about 0.5 mile northwestward of Half Acre Rock. The dredged channel leads northward through the harbor and eastward of **Guilford Point** to a junction with Sluice Creek and East River, about 0.6 mile above the channel entrance. At the junction, the dredged channel leads northwesterly into Sluice Creek for about 0.1 mile and northeasterly into East River for about 0.4 mile to an anchorage basin. A lighted buoy marks the entrance to the channel, and unlighted buoys mark the channel to the junction.

(272) At high water and with local knowledge, small boats can go above the anchorage basin in East River to the fixed railway bridge, about 1.3 miles above the basin. A town marina, just above the entrance to Sluice Creek, has berths with electricity, water, ice and a launching ramp. In 1993, depths of 1½ to 6 feet were reported alongside the marina.

(273) A 5 mph **speed limit** is enforced in the harbor.

(274) **West River** empties into the western side of Guilford Harbor 0.2 mile westward of Guilford Point. The entrance channel is marked by buoys and a **321.3°** lighted range.

(275) There are two boatyards with several marinas and marine railways on West River. The largest marine railway can handle craft up to 40 feet; berths with electricity, water, ice, gasoline, diesel fuel, limited supplies, a 12-ton mobile crane, a 25-ton lift and complete engine and hull repairs are available.

(276) **Falkner Island** and **Goose Islands**, with **Stony Island** to the southward, are about 3 miles south of Guilford Harbor. Each is surrounded by reefs and rocks that bare at low water. A depth of about 16 feet can be carried between Goose Islands and Falkner Island by staying in the middle of the passage and avoiding the 8-foot and 11-foot spots, about 0.35 mile 244° and 0.4 mile 300° from the light on Falkner Island, respectively, and the shoals and reefs extending from the islands. **Falkner Island Light** (41°12'43"N., 72°39'13"W.), 94 feet above the water, is shown from a 46-foot white octagonal tower near the center of Falkner Island. A lighted gong buoy marks the shoal off the northern end of Falkner Island, and a lighted bell buoy is off the southern end of Stony Island.

(277) From Indian Reef westward are rocky shoals and islets extending from 0.2 to 0.7 mile off **Vineyard Point** and **Sachem Head**. **Chimney Corner Reef**, about 0.3 mile south of Sachem Head and marked by a buoy, is a rocky broken area on which the least depth is 9 feet. Westward of it are **Goose Rocks Shoals**, on which are **Goose Rocks**, the northerly of which is bare and the southerly one covered at high water. The outer limit of Goose Rocks Shoals is marked by a lighted bell buoy. To ensure clearing the westerly end of Goose Rocks Shoals, care must be taken not to round the buoy too closely.

(278) **Sachem Head Harbor**, an anchorage for small craft on the southwest side of Sachem Head, is 0.3 mile long and 0.1 mile wide and has depths of 3 to 8 feet at the floats and in the moorings; it is sheltered except from westerly winds. The island forming the south point at the entrance is connected with the shore by a bridge. A yacht clubhouse is on the island. From the north point of the island a breakwater extends 100 yards in a northwesterly direction; a rock awash, marked by a private seasonal light, is off the end of the breakwater. A rock covered at half tide is 50 yards off the southeast side of the harbor, about 350 yards eastward of the end of the breakwater.

(279) The approach to Sachem Head Harbor for small craft from eastward is along the south side of the rocks making off from the south side of Sachem Head. Approaching eastward of Goose Rocks, give the rocks a berth of over 300 yards. The approach from westward is clear between Goose Rocks and Leetes Rocks.

(280) **Uncas Point**, the western extremity of Sachem Head, is marked by a rocky islet on its west side and a privately maintained seasonal light. Just northward of the islet a stone jetty with a bulkhead on its north side extends about 100 yards in a northwesterly direction from

the shore. Vessels can anchor in the angle near the shore where the depth is about 4½ feet.

(281) **Joshua Cove**, northwestward of Sachem Head, is little used, but affords good anchorage in its entrance for small vessels in northerly or easterly winds in 6 to 10 feet, soft bottom. The approach from southwestward is clear between Goose Rocks and Leetes Rocks.

(282) **Leetes Rocks**, midway between Sachem Head and the north end of The Thimbles, are two rocks bare at low water, with an area of broken ground around them. A 9-foot spot is about 200 yards southward of the southerly rock, and a 3-foot spot is 0.3 mile northeast of the southerly rock.

(283) **Leetes Island Quarry** is a prominent feature on the south side of **Hoadley Point**; on the north side of the cove eastward of the point are the ruins of an old dock.

(284) **The Thimbles**, about 1.6 miles west of Sachem Head, comprise many islands, islets and rocks that bare. All of the area, extending over 2 miles from Hoadley Point southwestward to **East Reef**, is foul with rocky bottom and many shoals. To lesser extent, the area from East Reef for 2 miles westward and northwestward to Branford Harbor entrance is dotted with islets and rocks. The whole area is suitable only for small pleasure craft, which are very active here in summer. Many oyster stakes are encountered; these do not mark channels, and caution should be used to avoid fouling them. Caution also is advised to avoid fouling the pipelines and cables in the area.

(285) The outermost of The Thimbles proper is **Outer Island**, marked by a house chimney. A boat landing protected by a stone jetty is on the northeast side of this island, and an unmarked rock, bare at lowest tides, is 200 yards eastward. The reefs southwestward of Outer Island, to and including East Reef and **Browns Reef**, are buoyed.

(286) From eastward a buoyed channel leads through The Thimbles. The channel passes between **Wayland Island** and a buoy marking the foul area southward of **Cat Island**. The channel extends between **Davis Island** and **Dogfish Island**, thence north of **East Crib** and **West Crib** into the more open water westward of The Thimbles; it is good for about 13 feet.

(287) **Stony Creek**, a village on the railroad, extends southward to **Flying Point** (41°15.5'N., 72°45.1'W.). A dredged channel west of Flying Point leads north to a turning basin at Stony Creek. The channel is marked by buoys. Rocks were reported in the northwest corner of the basin. Gasoline, marine supplies, inside storage, and a small-craft launching ramp are available at marinas eastward of the turning basin; small craft can be hauled out on a flatbed trailer for hull and engine repairs. The village dock is on the southeast side of the turning basin.

(288) Between the rocks westward of **Rogers Island** and **Blackstone Rocks**, a privately dredged channel, about 0.9 mile westward of Flying Point, leads northeastward to a quarry wharf on the west side of a dredged basin. In 1995, the reported controlling depths were 14 feet from the channel entrance to the basin, thence a depth of 14 feet was available in the basin except for lesser depths along the north and west edges. The entrance channel is marked by a private **028°** range consisting of a front and middle light and a rear daybeacon.

(289) **Thimble Island Harbor**, in the western part of The Thimbles, affords good shelter for small craft between **Pot Island** and **Money Island** on the east and **High Island** and **West Crib** on the west. Although open southwestward, the sea from that direction loses much of its force before reaching the inner harbor. A rock with 3 feet over it and marked by a buoy is 80 yards off the east side of High Island, just above its south end. Vessels sometimes anchor near midchannel, between this rock and the north end of Pot Island in depths of 13 to 18 feet, soft bottom, but care should be taken to avoid the cables in the area. The harbor is easy of access between Outer Island and Inner Reef.

(290) **Pine Orchard**, about 3 miles westward of Sachem Head, is a summer resort extending northward and westward of **Brown Point**. A breakwater extending about 300 yards southeastward from Brown Point protects a yacht basin entered through a privately dredged channel that leads from southward of **St. Helena Island** north-northwestward to the basin. In 1994, the entrance channel and basin had reported depths of 5 feet. The basin approach northward of St. Helena Island has depths of 3 to 5 feet. Gasoline, diesel fuel, ice and water may be obtained at the yacht club landing.

(291) From Brown Point to Branford Harbor, 2.5 miles westward, bare rocks and shoals extend up to about 2 miles offshore. A seawall extends westward from Brown Point, and the shore is thickly settled. A rock bare at half tide is 600 yards westward of Brown Point and 300 yards from shore.

(292) Rocks bare at low water are eastward of **Haycock Point**, and rocks that bare at half tide are off the southeast side and southwest end of **Green Island**. The foul ground extends about 0.6 mile south-southwestward from Haycock Point, including **Foot Rocks** which are partly above water.

(293) **Branford Reef**, about 1.8 miles southward of Indian Neck and 5 miles eastward of New Haven entrance, is marked by a light. This reef is surrounded by shoal water for a distance of 150 to 450 yards from the light.

(294) Deep water is between Branford Reef and **Totoket Bar**, a reef bare in one place at low water about 0.9 mile northward. Shoreward of Totoket Bar are **Spectacle Island**, **Sumac Island** and **Clam Island**, together with numerous rocks bare and covered.

(295) A private boat landing is on the northwest side of Clam Island. Small craft can enter **Maltby Cove** between the bare rocks off the southwest end of Clam Island and **Jefrey Rock**, favoring the northwest side of Clam Island. Private markers are sometimes at the entrance. The northwest side of the cove is foul, the principal danger being a rock bare at low water near the middle, northwestward of Clam Island; the rock is sometimes marked by a seasonal private spindle.

(296) **Jefrey Point**, the eastern point at the entrance of Branford Harbor, has a bare rock close to its western end.

(297) **Branford Harbor** is a shallow cove between Jeffrey Point and Johnson Point. Vessels up to 10-foot draft can select anchorage in the harbor southward of the Mermaids in 10 to 14 feet, protected against all but southerly and southwesterly winds. Boats up to 5-foot draft can select a well-sheltered anchorage in the upper part of the harbor above the Mermaids. The harbor is used chiefly for recreational boating and by the small local lobster fishing fleet.

(298) The dangers in the approach and entrance to Branford Harbor either show above water or are marked by buoys. **Cow and Calf**, 1.3 miles southwestward of Jeffrey Point, are two boulders close together bare at low water. Boulders, reported covered 10 feet, are about 0.2 mile northward of Cow and Calf. **Five Foot Rock**, 0.5 mile northeastward of Cow and Calf, has 5 feet over it. **Taunton Rock**, 0.9 mile northeastward of Cow and Calf near the middle of the entrance to Branford Harbor, is large but low and bare. **Blyn Rock**, midway between Johnson Point and Taunton Rock, is covered at extreme high tide. **Bird Rock**, 0.2 mile northward of Blyn Rock, has 5 feet over it.

(299) **Little Mermaid**, showing a little above high water, and **Big Mermaid**, a high rock marked by a light, are near the middle of Branford Harbor. Two bare rocks are near the head of the harbor. A rock, bare at low water and usually marked by stakes, is about 100 feet north-northeastward of the north end of **Lovers Island**.

(300)

Routes

(301) To enter Branford Harbor from eastward, pass southward of the lighted buoy marking Totoket Bar, steer about **306°** heading for Taunton Rock, and enter between Taunton and Jeffrey Rocks; or a **333°** course with Branford Reef Light astern will lead into the harbor between Jeffrey and Taunton Rocks. From westward, pass southward and over 100 yards eastward of the lighted bell buoy marking Cow and Calf, thence westward of the buoys marking Blyn Rock and Bird Rock to the buoyed channel in the harbor.

(302) Local craft pass northwestward of Cow and Calf Shoal and midway between Johnson Point and Blyn Rock.

(303) **Branford River**, narrow and crooked, extends northeasterly from Branford Harbor. At low water the channel above **Branford Point** is defined by bare shoals on each side. During the summer numerous stakes used as moorings mark both sides of the channel. A privately dredged channel and basin at a marina 0.5 mile east of Branford Point had reported depths of 9 feet in 1999.

(304) The principal waterborne commerce at Branford is in petroleum products. There are several marinas and boatyards on the river.

(305) A 5 mph **speed limit** is enforced on the river.

(306) The **harbormaster** at Branford controls all moorings and anchoring; he can be contacted through the small-craft facilities.

(307) **Johnson Point** is the western entrance point to Branford Harbor; a rock covered 2 feet is about 100 yards off its south side. A small privately dredged basin on the southwest side of the point is well protected in all but southerly winds. In 1971, it was reported that 4 feet could be carried to and in the basin.

(308) **Gull Rocks**, about 0.3 mile westward of Johnson Point, consist of small islets and submerged rocks that extend about 0.5 mile southwestward from shore on the easterly side of the entrance to a large cove. A rock, bare at half tide, is in the northwestern part of the cove about 350 yards southward of **Short Beach**. The northwest end of the cove has a yacht club landing with a reported depth of 2 feet alongside.

(309) **Farm River Gut**, a small bight on the west side of the cove, is a good anchorage for small craft. Depths range from 4 to 5 feet in the eastern part of the gut with shoaling to bare in the northern and western parts. Two rocks awash are on the north side of the gut about 125 yards inside the entrance. The gut offers good protection from all but easterly winds, mud bottom. A marine railway at a boatyard on the north side of the gut can handle boats to 36 feet for hull repairs; storage facilities are available. The yard can be reached only at high tide. **Old Clump** is a bare rock about 400 yards south of the bight.

(310) **Farm River**, locally known as East Haven River, about 1.5 miles westward of Branford Harbor, is used by local craft. In 1981, it was reported that depths of 3 feet could be carried in the river to the fixed bridge with a clearance of 4 feet about 1 mile above the mouth. Several boatyards on the river provide gasoline, berths, electricity, water, storage and limited marine supplies; diesel fuel can be delivered by truck. A 10-ton mobile hoist and a 12-ton crane can handle vessels for complete engine and hull repairs.

(311) **East Indies Rocks**, about 0.4 mile south of the entrance to Farm River, cover at half tide and are marked by a buoy to the eastward; a rocky shoal with a least depth of 5 feet is 0.2 mile to the eastward. A small ledge, bare at low water, is midway between East Indies Rocks and the south side of Mansfield Point, the western entrance point to Farm River. **Darrow Rocks**, a group of bare rocks, are on the east side of the entrance to the river. The westernmost rocky knoll is marked by a flagstaff. A ledge, bare at low water, with a buoy off its southern end, is 200 yards south of the flagstaff.

(312) **Mansfeld Point** and the shore westward of the entrance to Farm River are thickly settled. Bus communication is available to New Haven.

(313)

New Haven

(314) **New Haven Harbor**, an important harbor of refuge, is about 68 miles from New York, 179 miles from Boston

via Cape Cod Canal, and 171 miles from Nantucket Shoals. It comprises all the tidewater northward of the breakwaters constructed across the mouth of the bay, including the navigable portions of the West, Mill, and Quinnipiac Rivers. It is about 2 miles wide. The inner harbor, northward of Sandy Point and Fort Hale, is shallow for the most part, except where the depths have been increased by dredging. The main entrance channel, between Middle Breakwater and the East Breakwater, leads northward to Tomlinson Bridge at New Haven. Anchorage basins for medium draft vessels are on the west side of the channel north of Sandy Point. Waterborne commerce in the harbor consists of petroleum products, scrap metal, lumber, automobiles, gypsum, paper and pulp products, steel products, chemicals, rock salt and general cargo.

(315) **New Haven**, at the head of the harbor, is an important manufacturing city.

(316)

Prominent features

(317) **New Haven Harbor Lighted Whistle Buoy NH** (41°12'07"N., 72°53'47"W.), is marked with red and white stripes and red spherical topmark.

(318) On the approach from well offshore in clear weather, the prominent landmarks are: on East Rock (41°19.7'N., 72°54.4'W.), the Soldiers and Sailors Monument; in New Haven, the Knights of Columbus Building, a tall rectangular structure with circular pillars at its corners; the lighted stack of the power plant on the east side of the harbor opposite City Point. The lights on the ends of the breakwaters, the aerolight at Tweed-New Haven Airport and the abandoned tower on Lighthouse Point are also prominent.

(319) **Southwest Ledge Light** (41°14'04"N., 72°54'44"W.), 57 feet above the water, is shown from a white octagonal house on a brown cylindrical pier at the westerly end of East Breakwater. A sound signal at the light is operated by keying the microphone five times consecutively on VHF-FM channel 83A.

(320)

Channels

(321) A federal project for New Haven Harbor provides for an entrance channel 35 feet deep to a point just below the junction of Mill River and Quinnipiac River. The channel is well marked. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A.

(322) **West River**, marked by buoys, is located on the west side of the main channel about 3 miles above Southwest Ledge Light. A dredged channel leads west from the main channel into West River; an anchorage area on the south side of the channel is 0.9 mile above the entrance. (See Notice to Mariners and latest editions of charts for controlling depths.) Principal waterfront facilities are at **City Point**.

(323) **Mill River**, on the west side of **Fair Haven** about 4 miles above Southwest Ledge Light, is entered from the main channel through a dredged entrance channel that branches into an east and west fork to the Grand Avenue Bridge, 0.6 mile above the mouth. (See Notice to Mariners and latest edition of the chart for controlling depth information.)

(324) **Quinnipiac River**, on the east side of Fair Haven about 4 miles above Southwest Ledge Light, has a dredged channel to Grand Avenue Bridge, about 1 mile above the mouth. (See Notice to Mariners and latest edition of the chart for controlling depths.)

(325)

Anchorages

(326) Inside West Breakwater and the southwest part of Middle Breakwater, anchorage is available for vessels up to a 19-foot draft. Caution should be exercised to avoid the fish stakes in this area. Vessels anchoring in the area should also be aware that water levels may drop significantly following a long continuous northwesterly wind.

(327) Vessels may anchor northward of Southwest Ledge Light in depths of 18 to 20 feet, soft bottom in places. Care should be taken to avoid the ledges northward of the East Breakwater. Deep-draft vessels awaiting berthing assignments can anchor about 1 mile southward of the sea buoy; holding ground is excellent.

(328) **Morris Cove**, on the east side of the main channel just above Lighthouse Point, affords good anchorage and is used by yachts but is rough in westerly and southerly winds. In 1981, isolated, uncharted 40-foot spots were reported in the cove. Caution is advised when anchoring. **New Haven Coast Guard Station** is on the north side of the jutting point, about 1.5 miles northward of Lighthouse Point.

(329) An anchorage basin on the west side of the main channel southward of New Haven Long Wharf is sometimes used, but considerable shoaling is gradually extending into the anchorage from westward. A sunken barge with 5 feet over it is in this anchorage about 550 yards southward of New Haven Long Wharf. In 1985, depths of 10 to 5 feet were available in the anchorage basin with lesser depths along the edges.

(330) Small craft and scows may anchor northward of the New Haven Long Wharf (Naval Reserve Pier), northwest of the main channel where depths range from about 5 to 6 feet.

(331) No special regulations prescribe the limits within which vessels must anchor, except that the dredged channels must be kept clear.

(332)

Dangers

(333) **Townshend Ledge**, 2.7 miles southeastward of Southwest Ledge Light, has a least depth of 18 feet and is marked by a lighted buoy.

in July 1957. The coldest temperature on record is -7°F (-21.7°C) recorded in January 1961.

(345) Precipitation is quite evenly distributed throughout the year with only a 1.25 inch (32 mm) spread between the wettest and driest months. The annual average precipitation is 42 inches (1067 mm). The wettest month, December, averages 4.24 inches (108 mm) and the driest month, June, averages 2.93 inches (74 mm). The elevation of the land increases northward from the station and results in somewhat higher amounts of precipitation in the northern suburbs as well as a few more thunderstorms each year. During the winter, a variety of precipitation is found in most storms. It is common to have rain along the shore, freezing rain and sleet a short distance inland and snow in the northern parts of the city. Heavy snow is rather uncommon in the immediate coastal area and usually melts in a few days. Farther inland, the snow becomes progressively heavier and a layer of snow covers the ground most of the winter. Annual average snowfall totals 34 inches (864 mm). February is the snowiest month, averaging over nine inches (229 mm). Snow has fallen in each month October through May. The 24-hour record snowfall is 17.1 inches (434 mm), recorded in April 1957.

(346) Prevailing wind direction varies with the seasons. From late spring until fall, winds are predominantly south to southwest due to the effect of the sea breeze. During the winter, the prevailing winds are northerly. Strong southeast winds cause unusually high tides and some local flooding in low-lying coastal areas two or three times a year.

(347) Since 1871, 17 tropical systems have passed within 50 miles of New Haven, Connecticut. The most infamous perhaps was the hurricane of 1938. This storm passed with 15 miles west of the city on September 21, raking the city with 85-knot winds while moving at a forward speed in excess of 40 knots. Most recently, hurricane Gloria passed within 20 miles to the west on September 27, 1985. Highest winds at time of landfall were barely hurricane strength, but two days prior, Gloria had been supporting winds in excess of 125 knots. Due to geographical orientation, all tropical systems approach the coastline from the south or southwest.

(348) The National Weather Service maintains an office at the Tweed-New Haven Airport, about 3 miles southeast of the city.

(349)

Routes

(350) To enter New Haven Harbor from eastward, it is safer for large vessels to pass southward of Branford Reef and Townshend Ledge to the entrance channel. To enter from westward, pass northward of Stratford Shoal Light at a distance of 1.8 miles and head for the entrance channel.

(351) The passage eastward of East Breakwater has boulder patches and is very broken but can be used by small craft drawing less than 6 feet, taking care to avoid the foul ground along the northeast side of the passage. This passage is buoyed, and local vessels of 10- to 12-foot draft use it at high water. Avoid **Quixes Ledge**, which extends about 200 yards southeastward from the eastern end of the breakwater, and pass about 100 yards eastward of the breakwater. The principal danger inside the breakwater is the reef, marked by a buoy, that extends 300 yards southwestward from **Lighthouse Point**. **Adams Fall**, a rock with 5 feet over it and marked by a buoy, is 0.4 mile southwestward of Lighthouse Point.

(352)

Pilotage, New Haven

(353) Pilotage by state-licensed pilot is compulsory in Long Island Sound for foreign flag vessels and U.S. vessels that are under register (i.e., engaged in foreign trade). Such vessels can arrange for a state-licensed pilot by contacting the joint rotation administrator, Block Island Pilots at 243 Spring Street, Newport, RI 02840; telephone 401–487–9050 (24 hours), 800–274–1216; FAX 401–847–9052. Enrolled vessels (i.e., U.S. vessels engaged in coastwise trade) may be required to have a U.S. Coast Guard federally licensed pilot unless the master has recency for the intended area. See Pilotage, Long Island Sound (indexed as such), chapter 8.

(354) Pilot services are arranged in advance through ships' agents or directly by shipping companies.

(355)

Towage

(356) Tugs up to 1,800 hp are available at New Haven, and tugs to 4,000 hp can be obtained by prior arrangement. Vessels usually proceed to the harbor without assistance. Large vessels normally require tugs for docking and undocking. Arrangements for tug service should be made 24 hours in advance, usually through ships' agents or directly by shipping companies. The tugs monitor VHF-FM channels 13 and 16 and use channel 19A as a working frequency; call sign KEE-234.

(357) Launch service to ships at anchor is available. Launches monitor VHF-FM channel 16 and use channel 19A as a working frequency.

(358) New Haven is a **customs port of entry.**

(359)

Quarantine, customs, immigration and agricultural quarantine

(360) (See chapter 3, Vessel Arrival Inspections, and Appendix A for addresses.)

(361) **Quarantine** is enforced in accordance with regulations of the U.S. Public Health Service. (See Public Health Service, chapter 1.)

(362) New Haven has many public and private hospitals.

(363)

Coast Guard

(364) The **Captain of the Port** maintains an office in New Haven. The nearest **vessel documentation** office is in Bridgeport, CT. (See Appendix A for addresses.)

(365) The **harbormaster** at New Haven has charge of the anchoring of vessels; he can be contacted through the local police department.

(366) The city police maintain a harbor patrol during the summer.

(367)

Wharves

(368) The deep-draft facilities at the Port of New Haven are along the north and east sides of the inner portion of New Haven Harbor. Facilities for smaller vessels and barges are along the sides of the harbor and in Mill, Quinnipiac and West Rivers. Depths alongside the facilities in Quinnipiac River range from about 5 to 15 feet; Mill River, 12 to 13 feet; and West River, about 12 to 18 feet. Only the deep-draft facilities are described. The alongside depths for the facilities described are reported; for information on the latest depths contact the private operator. All the facilities have direct highway connections, and most have railroad connections. Water and electrical shore power connections are available at most piers and wharves.

(369) General cargo at the port is usually handled by ship's tackle; special handling equipment, if available, is mentioned in the description of the particular facility. Cranes up to 250 tons and warehouses and cold storage facilities adjacent to the waterfront are available.

(370) Wyatt Light Oil Pier: north end of harbor 0.35 mile northeastward of New Haven Long Wharf; 150-foot face, 715 feet of berthing space with dolphins, 38 feet alongside; deck height, 11 feet; receipt and shipment of petroleum products; owned and operated by Wyatt, Inc.

(371) Wyatt Heavy Oil Wharf: 50 yards east of Wyatt Light Oil Pier; west side 210 feet, 480 feet of berthing space with dolphins; 30 feet alongside; deck height, 11 feet; receipt and shipment of petroleum products, receipt of asphalt; owned and operated by Wyatt, Inc.

(372) Gulf Refining and Marketing Co. Wharf: on each side of harbor, 200 yards south of Tomlinson Bridge; 60-foot face, 735 feet of berthing space with dolphins; 35 feet alongside; deck height, 13 feet; vessels normally moor starboardside-to; receipt and shipment of petroleum products; owned and operated by Gulf Oil Refining and Marketing Co.

(373) Gulf Refining and Marketing Co. Pier: 100 yards southward of Gulf Refining and Marketing Co. Wharf; north side 400 feet, 25 feet alongside; south side 380 feet, 25 feet alongside; deck height, 10 feet; receipt and shipment of petroleum products; owned and operated by Gulf Refining and Marketing Co.

(374) ARCO Petroleum Products Co. Wharf: 300 yards southwestward of Gulf Refining and Marketing Co. Pier; 110-foot face, 760 feet with dolphins; 35 feet alongside; deck height, 15 feet; vessels normally moor starboardside-to; receipt and shipment of petroleum products; owned and operated by ARCO Petroleum Products Co.

(375) New Haven Terminal, Scrap Metal Dock: 275 yards southward of ARCO Petroleum Products Co. Wharf; 640-foot face; 35 feet alongside; deck height, 14 feet; two 30-ton traveling gantry cranes, crawler cranes to 250 tons; receipt and shipment of general and containerized cargo and steel products, shipment of scrap metal, receipt of copper, zinc, and lumber; owned and operated by New Haven Terminal, Inc.

(376) New Haven Terminal Pier: 50 yards southward of Scrap Metal Dock; north and south sides, 650 feet usable, can accommodate tankers up to 700 feet; 35 and 39 feet alongside, north and south sides, respectively; deck height, 13 feet; cranes up to 50 tons; 36,000 square feet covered storage; receipt and shipment of general cargo, receipt of petroleum products, petrochemicals, chemicals, copper, zinc, lumber and steel products; owned and operated by New Haven Terminal, Inc.

(377) Exxon Co. Terminal Wharf: 175 yards southward of New Haven Terminal Pier; 80-foot face, 700 feet with dolphins; 35 feet alongside; deck height, 13 feet; vessels normally moor starboardside-to; receipt and shipment of petroleum products; owned and operated by Exxon Co., U.S.A.

(378)

Supplies

(379) Oil bunkering terminals at New Haven are maintained by the major oil companies. Fuel oil and diesel oil in the usual commercial grades are obtainable. Barges are available for bunkering in the anchorages outside the breakwaters or at the piers; 24-hour advance notice is required, and arrangements should be made through ships' agents. Water, provisions and marine supplies can be procured.

(380)

Repairs

(381) New Haven has no facilities for making major repairs or for drydocking deep-draft vessels; the nearest such facilities are at Boston, MA, and New York. Machine shops in the area can make limited repairs to machinery and boilers and fabricate shafts and other pieces of equipment.

(382)

Small-craft facilities

(383) There are excellent facilities on the east and west sides of the harbor and on West and Quinnipiac Rivers.

(384)

Pond Point to Housatonic River

(385) **Pond Point**, about 5 miles southwestward of the New Haven Harbor entrance, has a rocky shoal with little depth over the greater part of it that extends about 0.3 mile southward. It is marked by a buoy. A prominent white mast is on the point.

(386) **Welches Point**, 0.8 mile westward of Pond Point, forms the east side of the entrance of the Gulf. A reef extends 0.2 mile southward from the point and is marked by a buoy. Several scattered rocks extend a southeasterly direction for about 0.5 mile from the buoy.

(387) **The Gulf**, a bight between Welches Point and Charles Island, about 6.5 miles westward of New Haven Harbor entrance, affords anchorage in 6 to 15 feet and is sheltered in all but southerly and southeasterly winds. The entrance is clear. The shoaling is gradual, and soundings are the best guide on the northwest side of the bight; the western side of Welches Point and the reefs around Charles Island extending to the mainland should be approached with caution, as the shoaling is abrupt.

(388) **Milford Harbor**, comprising the lower portion of the **Wepawaug River**, is entered at the mouth of the river between two jetties at the head of The Gulf. The westerly jetty extends southward from **Burns Point**, and the easterly jetty is marked by Milford Harbor Light 10. The harbor is used chiefly for recreational boating and occasionally for the receipt of shellfish and fish. A dredged channel leads from The Gulf through the jettied entrance to a point about 400 feet above the town wharf, 0.6 mile above Burns Point. The channel is marked by a lighted buoy at the entrance and unlighted buoys in the approach. There are several small-craft facilities, and a 5 mph speed limit is enforced in the harbor. The National Marine Fisheries Service, U.S. Department of Commerce, maintains a laboratory and research vessel base on the west side of the harbor, about 0.2 mile northward of Burns Point.

(389) **Charles Island**, on the southwest side at the entrance to The Gulf, is low and partly covered with trees. The island is connected to the mainland by **The Bar**, anarrow neck about 0.5 mile long and surrounded by rocks awash and shoals. A buoy marks the end of a shoal that extends 250 yards east-northeastward from the island, and a lighted bell buoy marks the end of a rocky area that extends 0.4 mile southward from the island. Northward of Charles Island is a good anchorage in 10 to 16 feet, sheltered from southerly to southwesterly winds.

(390) Between Charles Island and **Stratford Point**, about 3 miles southwestward, several summer resorts are along the shore, and the Housatonic River empties into Long Island Sound just above the point. The shoals that extend southward from Stratford Point toward Stratford Shoal Light consist of narrow ridges of hard sand with deeper water between and have oyster beds marked with stakes. Depths of 12 feet or less extend 1 mile offshore.

(391) **Stratford Point Light** (41°09'07"N., 73°06'12"W.), 52 feet above the water, is shown from a white conical tower, with dark red band midway of its height, from the southerly part of the point.

(392) **Housatonic River** rises in the Berkshire Hills of western Massachusetts and Connecticut and empties into Long Island Sound about 10 miles southwestward of the New Haven Harbor entrance. The river is joined by the **Naugatuck River**, which is not navigable, in the vicinity of Derby, CT. Housatonic River is navigable to a point about 1 mile above Shelton, CT, where it is closed by a power dam. The head of navigation for all practical purposes is at the towns of Derby and Shelton, 11.5 miles above the entrance. Small vessels can anchor in the river abreast of Stratford, where the channel has an available width of about 500 feet. The waterborne commerce on the river is principally in barge shipments of aggregate, fuel oil to the power plant at Devon and seasonal commercial shellfishing. Navigation above Devon is limited to recreational boating.

(393) On the east side of the entrance to Housatonic River, a breakwater extends out from **Milford Point** across the bar and is marked at its south end by Housatonic River Breakwater Light 2A. The inner section of the breakwater is awash at high water.

(394)

Channels

(395) A federal project provides for an 18-foot dredged channel from Long Island Sound between the breakwater on the east and Stratford Point on the west upriver for about 4.3 miles to the lower end of Culver Bar. (See Notice to Mariners and the latest editions of the charts for controlling depths.) Above the lower end of Culver Bar, the river channel extends through several dredged sections across river bars to the towns of Derby and Shelton about 11.5 miles above the river entrance. In 2005, the controlling depths were 2.2 feet in the buoyed channel from the lower end of Culver Bar and across Mill Bar to the naturally deep river channel, thence 5.7 feet in the dredged channels across lower Oronoque Bar and 3.0 feet across upper Oronoque Bar, thence 5.5 feet across Camp Meeting Bar, thence 6.2 feet across Drews Bar except for shoaling to 3.9 feet in the lower part of the dredged channel along the left edge, thence 7 feet across Mouthrops Bar and Hidelom Rock Bar, thence 7 feet in the left outside quarter of the dredged channel across Twomile Island Bar with shoaling to bare in the remainder of the channel, thence 7 feet in the dredged channel near Sow and Pigs Jetty. The channel is marked to a point about 2.5 miles below Derby and Shelton.

(396) **Stratford** is a town on the west side of the river 2.3 miles above the entrance. The principal wharf has a depth of about 9 feet at its end. The **harbormaster** at Stratford controls anchorages and moorings and has jurisdiction from the entrance of the river to the Shelton town line. Harbor regulations may be obtained from the harbormaster, who may be contacted through the Stratford police or at the Town Hall. Stratford has several small-craft facilities.

(397) **Devon** is on the east side about 1 mile above Stratford. Local small craft anchor near the east bank of the river, just north of the highway bridge, in depths up to 10 feet. A 40-foot marine railway at a small-craft facility at Devon can haul out craft for engine and hull repairs; gasoline, water, ice, marine supplies and storage are available. In 1981, depths of 4 feet were reported alongside the facility.

(398) **Shelton**, a town on the west side of the river about 11.5 miles above the entrance is connected to **Derby** by two bridges; the town has several important factories. In

(399)

Structures across Housatonic River

Name•Description•Type	Location	Clear Width of Draw or Span Opening (feet)	Clear Height above Mean High Water (feet)	Information
U.S. Route 1 Bridge (bascule)	41°12'01"N., 73°06'38"W.	125	32	Notes 1 and 2. Call sign KXJ-695
Moses Wheeler/I-95 Bridge (fixed)	41°12'17"N., 73°06'35"W.	99	68	
Metro-North/Devon Railroad Bridge (bascule)	41°12'18"N., 73°06'36"W.	83	19	Notes 1 and 2. Call sign KU-6035
Overhead power cable	41°12'20"N., 73°06'36"W.		91*	*Reported
Overhead power cables	41°13'46"N., 73°06'35"W.		79	
Sikorsky/Route 15 Bridge (fixed)	41°14'47"N., 73°05'27"W.	100	79	
Shelton-Derby Highway Bridge (fixed)	41°18'53"N., 73°05'12"W.	100	34	
Railroad Bridge (fixed)	41°19'03"N., 73°05'21"W.	148	17	
Shelton-Derby Highway Bridge (fixed)	41°19'09"N., 73°05'28"W.	84	30	

Note 1 – Bridgetender monitors VHF-FM channel 13.
Note 2 – See **33 CFR 117.1** through **117.59** and **117.207**, chapter 2, for drawbridge regulations.

1971, the wharves at Derby and Shelton were in ruins and unsuitable for craft of any size.

(400)

Tides

(401) The mean range of tide is 5.5 feet at Stratford and 5 feet at Shelton. The time of the tide becomes later and the range diminishes in progressing up the river. At Stratford the tide is about 0.8 hour later than at the entrance, whereas at Shelton high water is about 1.8 hours later and low water about 2.8 hours later than at the entrance. The river water is fresh about 6 miles above the entrance.

(402)

Current

(403) At the entrance near the end of the breakwater the flood has a strong westerly set. Between Milford Point and **Crimbo Point**, flood and ebb have a velocity of about 1.2 knots. The flood sets about 330° and the ebb 135°. Just north of the draw of the railroad bridge above Stratford, the velocity of flood is 1.1 knots and of ebb, 1.3 knots. In the openings of the bridge the flood current has some easterly set, but the ebb sets fair with the openings. Between that bridge and Shelton the tidal current has a velocity of about 1 knot. Because of the drainage flow of the river, the ebb is usually greater and the flood less than 1 knot. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(404) Spring **freshets** at Shelton rise 10 feet or more above mean high tide.

(405) **Ice** closes the river above Stratford during the winter and sometimes extends to the entrance.

(406)

Routes

(407) The channel in Housatonic River is narrow and crooked, with little depth on either side, and across the bars in the channel are dredged cuts 100 feet wide. The tidal currents are strong, especially in the lower part of the river, and strangers are advised to take a pilot. Small craft, without a pilot, should proceed with caution and preferably on a rising tide.

(408) When entering the river during a flood current, care must be taken to avoid being set on the shoals on the west side by strong westerly currents. In the vicinity of Milford Point care should be exercised to avoid a shoal that reportedly extends from Milford Point to the eastern edge of the channel. Care should also be exercised off the extreme northern end of Nells Island as a shoal is reported to have encroached into the channel. By steering a midchannel course no difficulty should be encountered.

(409) **Pilots** and **tugs** can be obtained at New Haven.

(410) A 5 mph **speed limit** is enforced on the river near anchorage and mooring areas and near boat slips.

(411)

Stratford Shoal Middle Ground to Duck Pond Point

(412) **Stratford Shoal Middle Ground**, 5.4 miles south of Stratford Point and covered 9 to 18 feet, is marked by **Stratford Shoal (Middle Ground) Light** (41°03'35"N., 73°06'05"W.), 60 feet above the water and shown from a gray granite octagonal tower projecting from a house on a pier, and by buoys that mark the outer ends of shoal areas extending 1 mile north, 0.9 mile northeast and 0.5 mile south of the light. A sound signal is at the light.

(413)

North Shore of Long Island

(414) From Orient Point (41°09.6'N., 72°14.0'W.), for about 11 miles to Horton Point, the south shore of Long Island Sound is generally bluff and rocky. The 10-fathom curve is from 0.3 to 0.8 mile from shore, and the shoaling is generally abrupt. The outlying dangers are Orient Shoal and the rocky patch northward of Horton Point.

(415) The prominent features are Browns Hills, a tower at Rocky Point, a tank and television tower at Greenport and Horton Point Light.

(416) Several rocky shoals, including **Orient Shoal** with a least depth of 6 feet, are offshore in the vicinity of **Rocky Point**, about 5 miles westward of Orient Point. The north end of Orient Shoal is marked by a buoy.

(417) Several rocks can be found out to 0.6 mile offshore between Orient Point and Inlet Point. A wreck with a least depth of 29 feet is 0.4 mile north of Inlet Point.

(418) **Horton Point Light** (41°05'06"N., 72°26'44"W.), 103 feet above the water, is shown from a white square tower attached to a dwelling on the northwest part of the point.

(419) A shoal with a least found depth of 29 feet is 1.6 miles northward of Horton Point. The shoal is a ridge having a northeast-southwest direction, with abrupt shoaling on its northwest and southeast sides.

(420) From Horton Point for about 32 miles to Old Field Point, the shore is fringed with shoals that extend off a greatest distance of 1.5 miles and rise abruptly from the deep water of Long Island Sound. Boulders are found near the shore on the shoals that extend off 0.5 mile in places. A sand shoal, about 0.5 mile in extent with a least depth of 26 feet, is about 1.1 miles northwestward of Duck Pond Point.

(421) The bluffs begin about 1 mile westward of Goldsmith Inlet and reach their greatest elevation just eastward of **Duck Pond Point**. A valley, formed by a break in the bluffs, is just westward of the point. Boulders that bare at low water are on the shoals that fringe the shore between Duck Pond Point and Mattituck Inlet.

(422)

Mattituck Inlet

(423) **Mattituck Inlet**, 6.7 miles southwestward of Horton Point Light, is entered between two short jetties. The inlet is marked by a long break in the bluffs. The outer end of the west jetty is marked by a light. A gong buoy about 1 mile north of the jetty light marks the entrance of the inlet. The sides of the channel are sandy, and, although shoaling is liable to occur at the entrance, strangers can enter the inlet without great danger. A federal project provides for depths of 7 feet in the channel from the entrance of Mattituck Creek to the turning basin at Mattituck.The channel is marked by buoys and private markers. The overhead power cable about 1 mile above the entrance has a clearance of 78 feet.

(424)

Current

(425) The tidal currents have an estimated velocity of about 3 knots in the narrow parts of the entrance of Mattituck Inlet. Slack waters occur possibly 1 hour after the time of high and low water. With northerly and westerly winds, the sea is rough in the entrance. The inlet is sometimes closed by **ice** during portions of cold winters.

(426) Several marinas and a boatyard are inside the inlet. A 70-ton mobile hoist at the boatyard can haul out craft for engine, hull and radio repairs. Marine supplies, gasoline, diesel fuel, water and covered and wet storage can be obtained. A transient dock, operated by the Mattituck Park Commission, is at the head of the inlet; depths of about 6 feet are at the dock. A **dockmaster** is at the dock; water is available.

(427) **Mattituck** is a village on the railroad at the head of the inlet. Provisions can be obtained.

(428) **Jacobs Point** is about 11 miles southwestward of Horton Point Light.

(429)

Riverhead Production Platform

(430) An offshore platform for the delivery and receipt of petroleum products is in open roadstead, off Northville, NY (and Riverhead, NY), about 1.2 miles northward of Jacobs Point.

(431) The facility consists of a 45 by 100-foot steel platform structure with breasting dolphins and mooring dolphins providing two berths, one on the northeast side and one on the southwest side. The deck height is 24.5 feet. The northeast berth has depths alongside of 64 feet and can accommodate tankers up to 225,000 DWT and up to 1,150-foot length, of 62-foot maximum draft.

(432) The southwest berth has depths alongside of 50 feet and can accommodate tankers of up to 42,000 DWT and up to 600-foot length, of 42-foot maximum draft. Barges mooring in this berth must be at least 220 feet long.

(433) A private sound signal is on the platform. Private lights are on the northeast and northwest corners, and two lights mark the center of the platform. Lights are also on each of the dolphins.

(434)

Wharf

(435) An 800-foot barge pier is just east of Jacobs Point and southward of the platform. The pier is used for receipt and shipment of petroleum products and has tank storage for 5¼ million barrels. Depth alongside is 13 feet. Lesser depths surround the area, and a shoal with depths of 10 feet is in the recommended southwest approach to the west pier berth. Vessels with draft greater than 12 feet should exercise caution when approaching the pier and should endeavor to arrive or depart at high water.

(436)

Prominent feature

(437) The numerous light green oil storage tanks on Jacobs Point are prominent.

(438)

Communications

(439) Vessels transiting Long Island Sound or approaching the facility may do so through a VHF-FM marine operator. Available marine operator stations' name and channel are:

(440) Riverhead 28

(441) New Bedford 26

(442) New London 26

(443) Bridgeport 24.

(444) Upon the approach of an incoming vessel, the platform, voice call "TOSCO Corporation Offshore Platform," or "Riverhead Platform," or "TOSCO's

Riverhead Terminal," monitors VHF-FM channels 16, 13 and 19A; works channel 19A.

(445) Vessels calling at the platform are moored at any time, weather conditions permitting. The tidal current periods are substantially the same as at The Race. Strong winds from the north and northwest are experienced during the winter and spring. Tidal currents during maximum ebb and flood may reach 3 knots.

(446) Vessels awaiting berth at the platform will normally anchor north of the platform. A vessel drawing more than 50 feet of water may wish to anchor in deeper water northwest of the platform. Pilots are familiar with the best anchorages. Holding ground is good, and a scope of 8 shots (120 feet) is considered adequate.

(447)

Pilotage, Riverhead Production Platform

(448) Pilotage by a state-licensed pilot is compulsory in Long Island Sound for foreign flag vessels and U.S. vessels that are under register (i.e., engaged in foreign trade). Such vessels can arrange for a state-licensed pilot by contacting the joint rotation administrator, Block Island Pilots at 243 Spring Street, Newport, RI 02840; telephone 401–847–9050 (24 hours), 800–274–1216; FAX 401–847–9052. Enrolled vessels (i.e., U.S. vessels engaged in coastwise trade) may be required to have a U.S. Coast Guard federally licensed pilot unless the master has recency for the intended area. See Pilotage, Long Island Sound (indexed as such), Chapter 8 and Pilotage, New York Harbor and Approaches, (indexed as such), chapter 11.

(449) The pilot serves as docking master and remains on board on standby while the vessel is moored at the platform. Pilot services are arranged in advance through ships' agents or directly by shipping companies.

(450)

Tugs

(451) Tug service is available from New Haven, Providence, Brooklyn or Staten Island on advance notice. Normally two or three tugs are used for docking and one or two tugs for undocking.

(452)

Launch service

(453) J & H Launch Service, Port Jefferson (516–331–5336), provides transfer service for vessels at anchor or alongside the platform.

(454)

Supplies

(455) Fueling of a ship alongside the platform is not permitted. A ship may fuel while at anchor from a barge. Water is not available from this facility. Stores may be brought on board via launch while alongside or at anchor.

(456) New York City is the **quarantine, customs, immigration** and **agricultural quarantine** port of entry for Northville. Officials are stationed in New York City. (See Appendix A for addresses.) Arrangements for such inspections must be made by ships' agents in advance, usually not less than 24 hours Monday through Friday and 48 hours on Saturday and Sunday. Officials will board vessels in the anchorage prior to arrival within the vicinity of the offshore mooring facility.

(457)

Horse in Bank to Tuttles White Bank

(458) Between Mattituck Inlet and Port Jefferson the shore is fringed with rock shoals extending in places 1.5 miles offshore. The outer ends of the shoals are marked by buoys.

(459) **Horse in Bank**, 7.3 miles westward of Mattituck Inlet, is an area of white patches in the brush-covered bluff at **Friars Head**. The feature is at the western end of **Roanoke Point Shoal** and 14 miles westward of Horton Point Light.

(460) The valley of **Wading River**, about 20 miles westward of Horton Point Light, forms a broad break in the high bluffs. The entrance to Wading River is protected by a short jetty on the west side. In 1981, a reported depth of about 3 feet could be carried in the river to a town launching ramp 0.1 mile above the entrance. A small canal, about 350 yards westward of the entrance to Wading River, leads southward to the site of a nuclear power station. The canal, closed to general navigation, had a reported depth of about 12 feet in 1989.

(461) **Tuttles White Bank** is a high white bluff 0.6 mile westward of Wading River.

(462)

Mount Sinai Harbor to Setaucket

(463) **Mount Sinai Harbor**, 22.5 miles westward of Mattituck Inlet, is marked by a low break in the beach nearly 1 mile long. The entrance is between two rubble mound jetties; caution should be exercised when near them. The jetties are each marked on the outer end by a private light. In 2015, the west jetty was reported partially submerged at high tide, and the adjacent east beach has receded, creating a breach between the jetty and land. A channel marked by private buoys leads eastward from the entrance to small-craft facilities on the north shore of the harbor.

(464) Small-craft facilities in the harbor can provide transient berths, gasoline, diesel fuel, electricity, water, ice, marine supplies, pump-out facilities and launching ramps. The minimum approach and alongside depths to the facilities is 10 feet.

(465) A **speed limit** of 6 mph is enforced in the harbor by the Suffolk County Police.

(466) **Mount Misery**, 180 feet high, between Mount Sinai Harbor and Port Jefferson, slopes off gradually toward the sound where the bluffs are about 60 feet high and very prominent. Sand banks dug out by sand and gravel companies are very conspicuous.

(467) **Port Jefferson Harbor**, on the south shore of Long Island Sound eastward of Old Field Point, is entered through a dredged channel that leads between two jetties that are in ruins to a docking area near the southwestern

end of the harbor; the jetties are each marked by a light. The approach is marked by a lighted whistle buoy, about 1.1 miles northwest of the entrance. Three stacks on the west side near the head of the harbor are conspicuous landmarks. A 12 mph **speed limit** is enforced in the main entrance channel, and a 5 mph **speed limit** is enforced at the head of the harbor in the vicinity of the mooring areas and wharves.

(468) A **121°–301° measured nautical mile** is westward of the entrance to Port Jefferson Harbor on Old Field Beach. The front markers are orange posts about 8 feet high; the rear markers are rectangles mounted on legs about 12 feet high, painted red with a 6-inch black vertical stripe in the middle.

(469) The approach to Port Jefferson Harbor is clear, taking care to avoid **Mount Misery Shoal** with depths of 7 to 12 feet, about 0.8 mile north-northeast of the east jetty light.

(470) A federal project provides for a channel 26 feet deep from Long Island Sound to the south end of Port Jefferson Harbor. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A. The channel is marked by lighted and unlighted buoys and a directional light with a **145.7°–147.3°** white sector.

(471) Shoals with little depth are on both sides of the channel from the entrance to Port Jefferson to Lighted Bell Buoy 5 inside the entrance. The ground from the east jetty to the lighted bell buoy is broken, with shoals covered 4 to 11 feet. The lighted bell buoy cannot be seen over the breakwater at low tide by small vessels approaching the harbor.

(472)

Current

(473) In the channel between the jetties the velocity of the tidal currents is 2.6 knots on flood and 1.9 on ebb; flood sets 151° and the ebb 323°. It is reported that on the ebb there is a current with a velocity of 1 to 2 knots across the entrance to the harbor.

(474)

Ice

(475) Ice forms over the entire harbor and interrupts navigation in very cold weather but does not endanger shipping in the harbor.

(476)

Pilotage, Port Jefferson

(477) Pilotage by a state-licensed pilot is compulsory in Long Island Sound for foreign flag vessels and U.S. vessels that are under register (i.e., engaged in foreign trade). Such vessels can arrange for a state-licensed pilot by contacting the joint rotation administrator, Block Island Pilots at 243 Spring Street, Newport, RI 02840; telephone 401–847–9050 (24 hours), 800–274–1216; Fax 401–847–9052. Enrolled vessels (i.e., U.S. vessels engaged in coastwise trade) may be required to have a U.S. Coast Guard federally licensed pilot unless the master has recency for the intended area. See Pilotage, Long Island Sound (indexed as such), Chapter 8 and Pilotage, New York Harbor and Approaches, (indexed as such), chapter 11.

(478) Pilot services are arranged in advance through ships' agents or directly by shipping companies.

(479)

Tugs

(480) Tug service is available from New Haven, Providence, Brooklyn or Staten Island on advance notice. Normally, two tugs are used for docking and one for undocking.

(481) **Port Jefferson** is a town at the southern end of the harbor. The principal industries of the port are the shipping of sand and gravel and the distribution of petroleum products. There are small-craft facilities and a launching ramp along the waterfront.

(482)

Wharves

(483) Depths ranging from 2 to 29 feet are reported alongside the commercial wharves and piers at the head of the harbor. The oil wharf on the west side of the harbor, about 400 yards from the head, has depths of 29 feet alongside the face and 20 feet along the north side. The power plant wharf, about 150 yards northwestward, has depths of 29 feet alongside.

(484)

Communications

(485) Port Jefferson is served by railroad and bus. A ferry operates to Bridgeport, CT.

(486) **Conscience Bay** is entered through a long, narrow channel at the northwest end of Port Jefferson Harbor. The bay and entrance have depths of 1 to 2 feet. Strangers should not attempt to enter as there are many rocks at the entrance.

(487) **Setauket Harbor**, on the western side of Port Jefferson Harbor, has a narrow crooked channel. In 1981, a reported depth of about 2½ feet was available in the channel to the boatyard at Setauket. The entrance from Port Jefferson is marked by private seasonal buoys. Gasoline, moorings and limited marine supplies are available at the boatyard; a flatbed trailer can haul out craft to 32 feet long.

(488) **Setauket** is a village on the south shore of Setauket Harbor about 1 mile above the entrance.

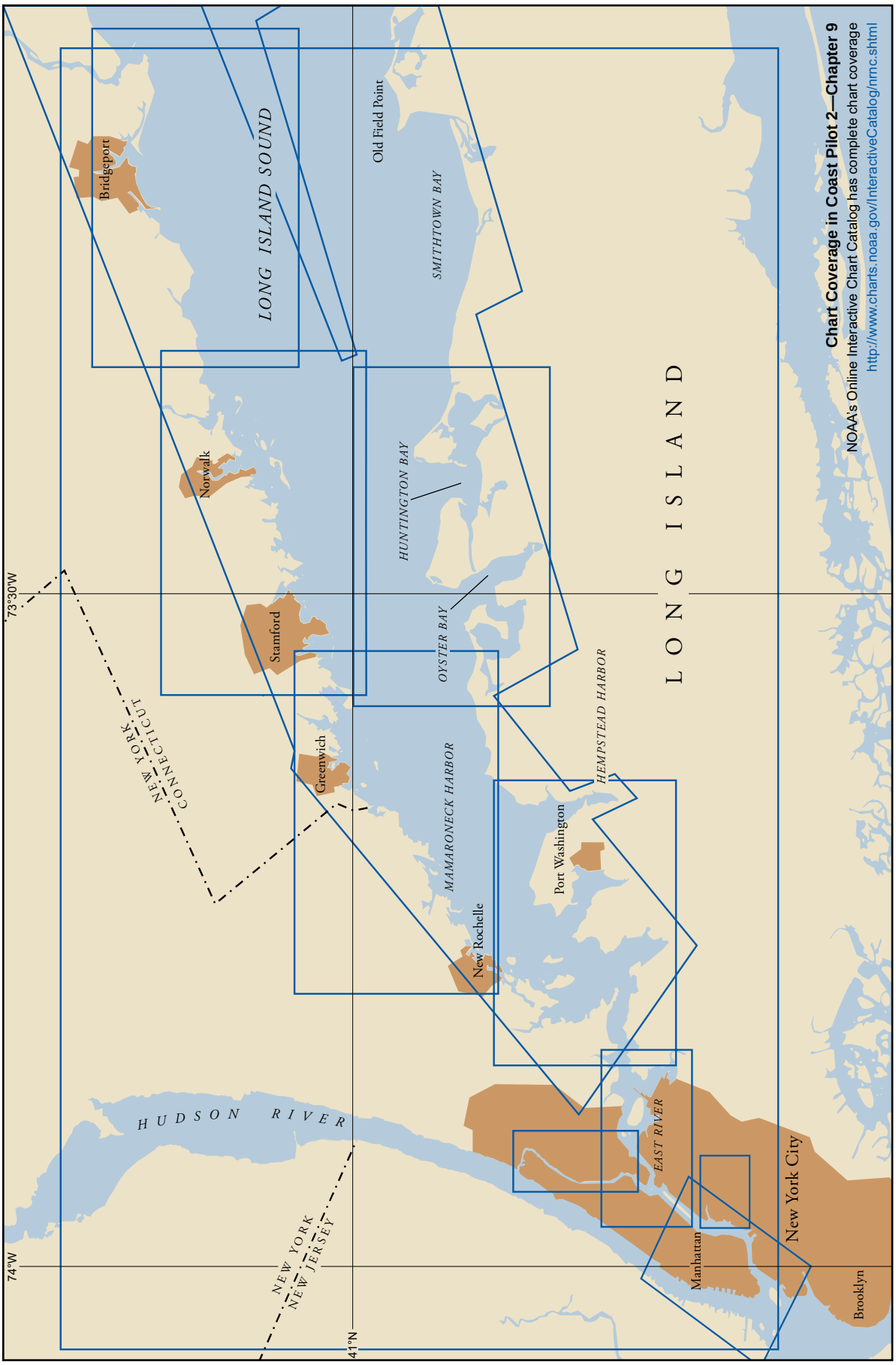


Chart Coverage in Coast Pilot 2—Chapter 9
NOAA's Online Interactive Chart Catalog has complete chart coverage
http://www.charts.noaa.gov/InteractiveCatalog/nrnc.shtml
LONG ISLAND SOUND
Old Field Point
SMITHTOWN BAY
HUNTINGTON BAY
OYSTER BAY
HEMPSTEAD HARBOR
MAMARONECK HARBOR
LONG ISLAND
Bridgeport
Norwalk
Stamford
Greenwich
New Rochelle
Port Washington
HUDSON RIVER
EAST RIVER
New York City
Manhattan
Brooklyn
NEW YORK
CONNECTICUT
NEW YORK
NEW JERSEY
73°30'W
74°W
41°N

Western Long Island Sound

(1) This chapter describes the western part of Long Island Sound along the north shore from Bridgeport to Throgs Neck, the south shore from Old Field Point to Willets Point and the East and Harlem Rivers. Also described are the many bays and their tributaries that make into this part of the sound including Bridgeport Harbor, Stamford Harbor, Captain Harbor, Mamaroneck Harbor, Norwalk Harbor, Eastchester Bay, Huntington Bay, Oyster Bay, Hempstead Harbor, Manhasset Bay, Flushing Bay and New Rochelle Harbor and the commercial and small-craft facilities found in these waters.

(2)

COLREGS Demarcation Lines

(3) The lines established for Long Island Sound are described in **33 CFR 80.155**, chapter 2.

(4)

No-Discharge Zone

(5) The States of New York and Connecticut, with the approval of the Environmental Protection Agency, have established a No-Discharge Zone (NDZ) covering all coastal waters described in this chapter east of the Hell Gate Bridge.

(6) Within the NDZ, discharge of sewage, whether treated or untreated, from all vessels is prohibited. Outside the NDZs, discharge of sewage is regulated by **40 CFR 140** (see chapter 2).

(7)

Western Long Island Sound

(8) **Western Long Island Sound** is that portion of the deep navigable waterway between the shores of Connecticut and New York and the northern coast of Long Island westward of the line between Bridgeport and Old Field Point.

(9) This region has boulders and broken ground, with little or no natural change in the shoals. The waters are well marked by navigational aids so that strangers should experience no difficulty in navigating them. As all broken ground is liable to be strewn with boulders, vessels should proceed with caution when in the vicinity of broken areas where the charted depths are within 8 feet of the draft. All of the more frequented places are entered through dredged channels. During fog, vessels are advised to anchor until the weather clears before attempting to enter. The numerous oyster grounds in this region are usually marked by stakes and flags. These stakes may become broken off and form obstructions dangerous to small craft that, especially at night, should proceed with caution when crossing oyster areas.

(10)

Anchorages

(11) There is anchorage for large vessels in the bight between the entrance channels of Bridgeport Harbor and Black Rock Harbor. Cockenoe Harbor is sometimes used by small vessels, but Sheffield Island Harbor is preferred and is sometimes used by tows. Westward of Norwalk Islands, seagoing vessels can anchor toward the north shore and, with good ground tackle, hold on in northerly winds. Captain Harbor affords good shelter but is rarely used except by local vessels. On the south shore, Huntington Bay and Hempstead Harbor are available for large vessels; Oyster Bay is also used, and Manhasset Bay is available for light-draft vessels. City Island Harbor is a fine resort for coasters.

(12) Several general anchorages are in Long Island Sound. (See **33 CFR 110.1** and **110.146**, chapter 2, for limits and regulations.)

(13)

Tides

(14) The time of tide is nearly simultaneous throughout Long Island Sound, but the range of tide increases from about 2.5 feet at the east end to about 7.3 feet at the west end. Daily predictions of the times and heights of high and low waters are available at the tide prediction service at *tidesandcurrents.noaa.gov*. Links to a user guide for this service can be found in chapter 1 of this book.

(15) The effect of strong winds, in combination with the regular tidal action, may at times cause the water to fall several feet below the plane of reference of the charts.

(16)

Current

(17) About 1.3 miles northward of Eatons Neck Light the ebb runs about 5 hours longer than the flood. The current has a velocity of 1.4 knots; the flood sets 283° and the ebb sets 075°.

(18) The direction and velocity of the currents are affected by strong winds that may increase or diminish the periods of flood or ebb. Currents in East River are described in the latter part of this chapter.

(19)

Weather, Western Long Island Sound and vicinity

(20) These waters are more protected than the eastern Sound resulting in fewer gales. However, winters are colder and summers warmer due to this sheltering effect. Fog is not so frequent either and tends to burn

off quicker than farther east. Winter winds of 16 knots or more are likely about 12 to 15 percent of the time and are predominantly from the west through northwest. Harbors such as Cold Spring, Oyster Bay, Hempstead and Manhasset offer additional shelter. In summer thunderstorms may develop on 4 to 5 days per month. These are most likely during the afternoon or evening.

(21) In Long Island Sound the north and south shores are equally subject to fog, except that on spring and summer mornings, when there is little or no wind, fog will often hang along the Connecticut shore while it is clear offshore and southward.

(22) In the western end of Long Island Sound, although fogs are liable to occur at any time, they are not encountered so often nor do they generally last so long as farther eastward.

(23)

Ice

(24) In ordinary winters the floating and pack ice in Long Island Sound, while impeding navigation, does not render it absolutely unsafe. In exceptionally severe winters, waterways may become impassable for some vessels.

(25) Drift ice, which is formed principally along the northern shore of the sound under the influence of the prevailing northerly winds, drifts across to the southern side and accumulates there, massing into large fields, and remains until removed by southerly winds that drive it back to the northerly shore.

(26) In ordinary winters ice generally forms in the western end of the sound as far as Eatons Neck; in exceptionally severe winters ice may extend to Falkner Island and farther eastward.

(27)

Effects of winds on ice

(28) In Long Island Sound northerly winds drive the ice to the southern shore of the sound and southerly winds carry it back to the northern shore. Northeasterly winds force the ice westward and cause formations heavy enough to prevent the passage of vessels of every description until the ice is removed by westerly winds. These winds carry the ice eastward and, if of long enough duration, drive it through The Race into Block Island Sound, from where it goes to sea and disappears.

(29) In Bridgeport Harbor winds from north to northwest clear the harbor of drift ice, and those from southeast through south to southwest force the ice into the harbor from the sound. The outer buoys may be carried out of position by heavy ice during severe winters.

(30) Additional information concerning ice conditions in the waters adjoining Long Island Sound is given under the local descriptions.

(31) **Vessel Tra f c Service, New York**, operated by the U.S. Coast Guard, serves New York Harbor. (See **33 CFR 161.1** through **161.25**, chapter 2, for regulations).

(32)

Pilotage, Western Long Island Sound

(33) Pilotage by a state-licensed pilot is compulsory in Long Island Sound for foreign flag vessels and U.S. vessels that are under register (i.e., engaged in foreign trade), Enrolled vessels (i.e., U.S. vessels engaged in coastwise trade) may be required to have a U.S. Coast Guard federally licensed pilot unless the master has recency for the intended area. For vessels entering Long Island Sound from the east (from sea via Block Island Sound) see Pilotage, Long Island Sound (indexed as such), chapter 8. For vessels entering Long Island Sound from the west (East River) see Pilotage, New York and Approaches to New York (indexed as such), chapter 11.

(34)

Recommended Vessel Route, Western Long Island Sound

(35) Recommended vessel routes have been established for deep draft vessels (including tugs and barges) transiting Western Long Island Sound and the approaches to the East River. While not mandatory, deep draft commercial vessels (including tugs and barges) are requested to follow the designated routes at the master's discretion. Other vessels, while not excluded from these routes, should exercise caution in and around these areas and monitor VHF-FM channel 16 or 13 for information concerning deep draft vessels (including tugs and barges) transiting these routes. (See NOS charts and Local Notice to Mariners for route limits.)

(36)

Bridgeport Harbor to Sherwood Point

(37) **Bridgeport Harbor**, on the north side of Long Island Sound north-northwestward of Stratford Shoal (Middle Ground) Light and about 52 miles from New York, consists of two widely separated units. The main harbor and its branches serve the east and central portions of the city of **Bridgeport**, and Black Rock Harbor and its tributaries serve the western part. Black Rock Harbor and Cedar Creek are described under separate headings. Waterborne commerce at Bridgeport consists mostly of petroleum products, lumber, sand and gravel, building materials and scrap iron.

(38)

Prominent features

(39) The large red and white horizontally banded stack of a powerplant on Tongue Point is the most prominent landmark in this area. Other prominent landmarks include several church spires, the radio towers at Pleasure Beach and Bridgeport Harbor Light 7. An aerolight about 1.3 miles northwestward of Stratford Point can be seen from offshore.

(40) **Bridgeport Harbor Light 7** (41°09'24"N., 73°10'47"W.), 50 feet above the water, is shown from a black skeleton tower with small white house, on a black base, on the west side of the entrance channel near the end of the west breakwater.

(47)

Structures across Tributaries of Bridgeport Harbor

Name•Description•Type	Location	Clear Width of Draw or Span Opening (feet)	Clear Height above Mean High Water (feet)	Information
Johnsons Creek				
Highway Bridge (swing)	41°09'58"N., 73°10'03"W.	65	7	
Yellow Mill Channel				
Stratford Avenue Bridge (bascule)	41°10'36"N., 73°10'35"W.	82	11	Note 1
Interstate 95 Bridge (fixed)	41°10'42"N., 73°10'33"W.	105	39	
Pequonnock River				
Interstate 95 Bridge (fixed)	41°10'38"N., 73°11'08"W.	134	60	
Overhead power cable	41°10'38"N., 73°11'10"W.		150	
Stratford Avenue Bridge (vertical lift)	41°10'45"N., 73°11'09"W.	103	8 (down), 68 (up)	Notes 2 and 3
Peck Railroad Bridge (bascule)	41°10'58"N., 73°11'09"W.	105, 65 (open)	26	Note 2
Overhead power cables	41°10'58"N., 73°11'11"W.		160	
Congress Street Bridge	41°11'01"N., 73°11'15"W.	–	–	Draw spans removed
East Washington Avenue Bridge (fixed)	41°11'10"N., 73°11'21"W.	69	4	

Note 1 – See **33 CFR 117.1** through **117.59** and **117.225**, chapter 2, for drawbridge regulations.
Note 2 – See **33 CFR 117.1** through **117.59** and **117.219**, chapter 2, for drawbridge regulations.
Note 3 – Bridgetender monitors VHF-FM channel 13; call sign KU-6033.

(41)

Channels

(42) From deep water in Long Island Sound the dredged channel extends north-northeastward between two converging breakwaters into the main harbor, and thence into the three tributaries; Johnsons Creek, Yellow Mill Channel and Pequonnock River. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A.

(43) A power plant is at **Tongue Point**. A privately dredged channel leads from the main channel to the power plant's offshore oil wharf on the south side of the point. In 1980, the channel, except for a 17-foot depth on the southwesterly side of the widener, had a reported controlling depth of about 26 feet; depths of 31 to 37 feet are reported alongside the wharf. Another privately dredged channel, used by barges, leads from the main channel to the powerplant's facilities on the east side of the point. In 2009, the controlling depth in the channel was 13.5 feet.

(44) **Johnsons Creek**, northward of Pleasure Beach, is entered eastward of Tongue Point through a marked dredged channel leading to anchorage basins, two on the west side, and one at the head of the creek. Private yacht clubs and two oil-receiving piers are on the creek.

(45) **Yellow Mill Channel** is entered through a dredged channel that leads for about 0.8 mile north-northeastward from just above the first bend in the main channel to the head of the creek. Flats, largely bare at low water, are on both sides of the channel. Depths at the wharves are 8 to 15 feet.

(46) **Pequonnock River**, the most westerly of the tributaries, is easily followed by small craft, but larger vessels may need the assistance of a tug to get around the sharp bends. The river is entered through a dredged channel that leads northward from the main channel just below Connecticut Turnpike bridge to the head of navigation just below the Berkshire Avenue Dam, about 1.1 miles above the entrance. Depths at some of the wharves are 10 to 15 feet.

(48)

Anchorages

(49) Bridgeport Harbor has three anchorage areas inside the breakwaters. An anchorage is on the east side of the main channel northwestward of Pleasure Beach. A second is on the west side of the channel south of Tongue Point, and a third runs parallel to the west side of the main channel from Tongue Point to Steel Point. The rest of the harbor area consists of broad and shallow sand flats. Vessels seeking shelter from strong northerly winds sometimes anchor off the entrance; the holding ground is good.

(50) A **general anchorage** is in Johnsons Creek. (See **33 CFR 110.1** and **110.148**, chapter 2, for limits and regulations.)

(51)

Dangers

(52) The entrance is clear, and the only dangers are the previously discussed shoals on the east, south of Stratford Point, and on the west, the Penfield Reef shoals.

(53)

Current

(54) The velocity of flood or ebb is about 0.7 knot in the entrance between the breakwaters. Inside the harbor the currents are generally weak. See the Tidal

Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(55)

Ice

(56) Ice does not interfere seriously with navigation in Bridgeport Harbor, although its tributaries are closed at times. The winds from the north and northwest clear the harbor of drift ice, and those from the southeast through the southwest force the ice into the harbor from the sound. The outer buoys may be carried out of position by heavy ice during severe winters.

(57)

Weather, Bridgeport and vicinity

(58) The terrain of the mainland is of glacial origin and rises in a rolling, mostly wooded manner to the foothills of the Berkshires, 30 miles to the north, and the Catskills, about 60 to 70 miles to the northwest. There is some foehn effect (chinook) with north and northwest winds, and the upslope effect with the approach of a coastal low is quite pronounced. The most pronounced topographical effect, however, is that of the land-sea breeze that is most pronounced in the spring, summer, and early autumn. The land-sea breeze effect during this period will inevitably cause a shift in the wind direction, even with a moderately strong isobaric flow.

(59) As a result of the sea breeze, mean monthly temperatures during the summer average 3 to 5 degrees (2 to 3°C) lower than nearby inland stations. Likewise, temperatures during the fall and winter are moderated several degrees owing to the proximity of Long Island Sound. The average annual temperature at Bridgeport is 52°F (11.1°C). The average high is 60°F (15.6°C) and the average low is 44°F (6.7°C). July is the warmest month with average extremes of 82°F (27.8°C) and 66°F (18.9°C). January is the coolest with average extremes of 37°F (2.8°C) and 23°F (-5°C). The record high temperature is 103°F (39.4°C) set in July 1957 while the all-time low temperature is -7°F (-21.7°C) recorded in January 1984.

(60) Precipitation is slightly heavier than at nearby inland stations the year around since coastal low-pressure systems move quite consistently on a track to the south of Bridgeport. One of the greater hazards along the coastal areas in the vicinity of Bridgeport is the accumulation of water (especially during periods of high tide) with the approach of a slowly moving, deepening, low-pressure system from the south. Severe storms occasionally cause inundation of 4 to 5 feet (1.2 to 1.5 m). The average annual precipitation is 41 inches (1041 mm). Precipitation is evenly distributed throughout the year with the difference between the wettest (March) and driest month (February) averaging only 0.89 inches (23 mm). Snowfall averages 26 inches (660 mm) per year and has fallen from October through May. The greatest 24-hour snowfall on record was 16 inches (406 mm) recorded in February 1969.

(61) Bridgeport has been directly affected by many tropical storms since 1871. Tropical storm Belle passed over the site in August 1976. Highest winds were only 60 knots. One day earlier, Belle was packing winds of 105 knots. In September 1985, Hurricane Gloria passed about five miles west of the Bridgeport weather station placing the site in the roughest sector of the storm. Highest gusts approached 75 knots and highest sustained winds were 64 knots. Two days earlier, Gloria had supported winds of 125 knots.

(62) The National Weather Service maintains an office at the Bridgeport Municipal Airport; barometers may be compared here.

(63)

Pilotage, Bridgeport

(64) Pilotage by a state-licensed pilot is compulsory in Long Island Sound for foreign flag vessels and U.S. vessels that are under register (i.e., engaged in foreign trade). Such vessels can arrange for a state licensed pilot by contacting the joint rotation administrator, Block Island Pilots at 243 Spring Street, Newport, RI 02840; telephone 401–847–9050 (24 hours), 800–274–1216; FAX 401–847–9052. Enrolled vessels (i.e., U.S. vessels engaged in coastwise trade) may be required to have a U.S. Coast Guard federally licensed pilot unless the master has recency for the intended area. See Pilotage, Long Island Sound (indexed as such), chapter 8. See also Pilotage, Narragansett Bay and Other Rhode Island Waters (indexed as such), chapter 6, and Pilotage Pickup Locations Off Montauk Point (indexed as such), chapter 7.

(65) Pilot services are generally arranged in advance through ships' agents or directly by shipping companies.

(66)

Towage

(67) Tug service is available from New Haven, Providence, Brooklyn or Staten Island on advance notice. Deep-draft vessels usually require tugs for mooring in Bridgeport Harbor.

(68) Launch service is available to vessels at anchor.

(69) Bridgeport is a **customs port of entry.**

(70)

Quarantine, customs, immigration and agricultural quarantine

(71) (See chapter 3, Vessel Arrival Inspections, and Appendix A for addresses.)

(72) **Quarantine** is enforced in accordance with the regulations of the U.S. Public Health Service. (See Public Health Service, chapter 1.)

(73) Bridgeport has several hospitals.

(74)

Harbormaster

(75) The control of the port is vested in the harbormaster, who maintains an office at the Bridgeport City Hall and

can also be contacted through the Bridgeport Police Department.

(76)

Wharves

(77) Bridgeport has three principal privately owned and operated deep-draft facilities; one is on the south side of Tongue Point and the other two are on the east side of the harbor opposite Tongue Point. Facilities for smaller vessels and barges are along the sides of the harbor and on Johnsons Creek, Yellow Mill Channel and Pequonnock River. Most of the facilities at Bridgeport are of the marginal-type wharf, particularly those in the constricted tributaries. Only the deep-draft facilities are described. The alongside depths for the facilities described are reported; for information on the latest depths contact the private operators. All of these facilities have highway connections, and most have water connections.

(78) Cargo in the port is usually handled by ship's tackle; special handling equipment, if available, is mentioned in the description of the particular facility.

(79) Santa Buckley Energy Terminal: head of the east branch of Cedar Creek, Black Rock Harbor; 150-foot breasting face with two dolphins; 18 feet alongside; 350 feet plus total berthing space; provides marine fuel to large commercial vessels and private yachts up to 350 feet long.

(80) United Illuminating Co. Fuel Oil Dock: on the south side of Tongue Point; an offshore wharf with 345-foot breasting face, 900 feet with dolphins; 31 to 37 feet alongside; deck height, 20 feet; receipt of fuel oil; owned and operated by United Illuminating Co.

(81) Shell Oil Co. Dock: on the east side of the harbor opposite Tongue Point; 190-foot face, 700 feet with shore moorings; 35 feet alongside; deck height, 13 feet; vessels usually moor portside-to; receipt and shipment of petroleum products; owned by Shell Oil Co. and operated by Shell Oil Co. and International Petroleum Terminals Co.

(82) Cilco Terminal Co. Wharf: 0.3 mile northwestward of Shell Oil Co. Dock; 930-foot face; 33 feet alongside; deck height, 13 feet; 90,000 square feet covered storage, 16 acres of open storage; receipt and shipment of general cargo; receipt of lumber, steel products, and pumice, and shipment of scrap metal; owned and operated by Cilco Terminal Co., Inc.

(83) The city-owned recreational pier, seldom used for mooring vessels, is on the northwest end of Pleasure Beach; the end of the pier has depths of about 20 feet.

(84) The municipal dock, a marginal-type wharf, is on the west side of Pequonnock River, just below the Interstate 95 Bridge. A ferry to Port Jefferson ties up at the dock.

(85)

Supplies

(86) Diesel oil, diesel fuel, gasoline, lubricants, water, provisions and marine supplies can be obtained at Bridgeport.

(87)

Repairs

(88) Bridgeport has no facilities for making major repairs or for drydocking deep-draft vessels; the nearest facilities are at the ports of Boston, MA and New York, NY. Bridgeport, however, does have facilities for making above- and below-the-waterline repairs to fishing boats, tugs and recreational craft and hull and engine repair facilities for small craft. The largest marine railway in the area can handle vessels to 120 feet and 400 tons. A 10-ton crane is available.

(89)

Communications

(90) Bridgeport is served by air, rail and bus. Ferry service to Port Jefferson is available year round.

(91) **Black Rock Harbor**, part of Bridgeport Harbor, although not connected with it other than by Long Island Sound, is entered through a dredged channel about 2 miles westward of the main harbor entrance to Bridgeport. The channel leads northward through Black Rock Harbor, and thence to the head of **Cedar Creek** where it divides into **East Branch** and **West Branch**. Black Rock Harbor and Cedar Creek are the approach by water to the large factories of the western part of the city of Bridgeport. The federal project depth in the dredged channel is 18 feet from the entrance to the head of the project. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A. The channel is marked by buoys and lights for about 1.7 miles above the entrance. Caution is advised to avoid an obstruction reported at about 41°08'12"N., 073°13'12"W., on the west side of the entrance channel covered 14 feet.

(92) Anchorage in depths of 18 to 22 feet and exposed to southeasterly and northeasterly winds can be found off the entrance, northeast of the bar that makes out from Shoal Point to Black Rock. Small craft drawing less than 6 feet can select anchorage on either side of the dredged channel as far as the yacht club on the east side of Grover Hill.

(93) Depths of 8 to 18 feet are reported alongside some of the wharves in Black Rock Harbor.

(94) To avoid a shoal off the point separating East Branch and West Branch at the head of Cedar Creek, enter East Branch, pass about 100 feet off the wharf on the southeast side below the entrance and head up the middle. To enter the West Branch, pass 100 feet off the wharves on the southeast side of the branch.

(95) **Fayerweather Island**, on the eastern side of the entrance of Black Rock Harbor, is marked at its south end by the white tower of an abandoned lighthouse. A breakwater and a seawall connect its northern part with the shore eastward.

(96) **Burr Creek**, northward of the town of Black Rock, on the west side of the channel, is the site of a large marina. Berths, gasoline, diesel fuel, electricity, water, ice, a lift and repair facilities are available. In 1986, depths of about 4 to 5 feet were reported at the face of the gasoline dock and alongside the boat slips. Burr Creek has many shoals; mariners are advised to seek local knowledge before entering.

(97) **Ash Creek**, about 0.7 mile westward of Fayerweather Island, is entered through a privately dredged channel protected on its southwest side by a jetty. The entrance channel is marked by private buoys and a private seasonal **314°** lighted range. The channel leads northwestward to a marina. In 2012, depths of 6 feet were reported in the entrance channel, with 5 feet reported in the basin. A 5 mph **speed limit** is enforced in the creek.

(98) **Penfeld Reef**, on which there are rocks bare at low water, is about 1.4 miles south of Black Rock Harbor and 1.3 miles eastward of **Shoal Point**, to which it is joined by a bar that bares at low water. **Black Rock**, marked by a daybeacon and a lighted buoy, is the outermost danger of this reef. A dangerous submerged rock, reported covered 1 foot, is about 40 yards southward of the daybeacon and lighted buoy. **The Little Cows**, about 0.2 mile northward of Black Rock, consist of rocks awash, and is marked by a lighted buoy.

(99) **Penfeld Reef Light** (41°07'02"N., 73°13'20"W.), 51 feet above the water, is shown from a white tower on a granite dwelling on a pier, on the south side of the reef, south of the entrance to Black Rock Harbor. A sound signal at the light is operated by keying the microphone five times consecutively on VHF-FM channel 83A.

(100) A reef, partly bare at low water and with little depth over any part of it, extends over 0.5 mile southward from **Pine Creek Point**, 1.1 miles southwest of Shoal Point. A lighted bell buoy is off the south end of the shoal.

(101) **Southport Harbor**, about 1 mile westward of Pine Creek Point, comprises the lower portion of Mill River and is used primarily for recreational boating. A breakwater, marked at its end by a light, is off the east side of the entrance to the harbor. The harbor is entered through a dredged channel that leads from Long Island Sound to a harbor basin and anchorage, about 1.1 miles above the channel entrance. The channel is marked on its west side by a light and by buoys up to the breakwater. Caution is advised to avoid oyster stakes in the area southeastward of the harbor entrance. A 5 mph **speed limit** is enforced in the harbor.

(102) **Southport** is a village on the west side of the harbor. A yacht club landing and the town dock are on the west side of the harbor; depths of about 6 feet are alongside the town dock and about 6 to 8 feet alongside the yacht club landing. Gasoline, diesel fuel, ice, water and some marine supplies can be obtained. Minor engine repairs can be made. The **harbormaster** can be contacted through the Fairfield Police Department.

(103) **Frost Point**, 1 mile westward of Southport entrance, is marked by many residences and several private piers in disrepair on its southeast side. A reef partly bare at low water extends about 0.4 mile southward from the point.

(104) **Sherwood Point**, a mile westward of Frost Point, is marked by a bare boulder on the reef which extends about 250 yards off the point. A rocky patch, on that the least depth found is 11 feet, is about 0.8 mile southward of the point.

(105)

Saugatuck River to Stamford Harbor

(106) **Saugatuck River**, 6miles westward of Penfield Reef Light and northward of Cockenoe Island, has its entrance between **Cedar Point** on the east and **Bluf Point** on the west. The river is shallow, full of ledges and boulders, and is used chiefly for receipt of petroleum products, sand and gravel and for recreational boating. Freshets do not appreciably affect the height of the water in the navigable part of the river. During the winter, ice usually covers the entire river to its mouth.

(107) Anchorage exposed to southeasterly winds can be had in the entrance to Saugatuck River in 12 to 22 feet, about 0.4 mile southward of **Cedar Point**.

(108) The channel in Saugatuck River is narrow and crooked; vessels should proceed with caution, preferably on a rising tide. In 2001, a reported depth of about 4 feet could be carried in the river from the entrance to about 0.7 mile above the Connecticut Turnpike Bridge at Saugatuck; thence in 2022, the 4-foot channel to Westport had a controlling depth of 1 foot, with shoaling to bare in the east branch. The channel is buoyed to **Stony Point**, about 1.9 miles above the entrance. A 5 mph **speed limit** is enforced on the river.

(109) **Compo Yacht Basin** is in the bight about 0.3 mile northwestward of Cedar Point. In 1995, the privately dredged channel that leads to the basin had a reported depth of 8 feet with 7 feet reported in the basin. The channel is marked by private buoys and a private lighted entrance range. A yacht club with landing and mooring facilities is in the basin. Gasoline, berths, electricity and water are available at the landing.

(110) A yacht club in a privately dredged basin on the west side of Bluff Point has berths with electricity, gasoline and ice.

(111) **Duck Creek**, on the west side of the river about 0.6 mile above Bluff Point, is the site of a private yacht club. The reported controlling depth in the creek was about 7 feet in 1981. The entrance and basin are privately marked.

(112) **Bermuda Lagoon**, southward of Duck Creek, is a large privately owned and maintained basin for the use of the residents in the immediate area.

(113) **Saugatuck**, a village in the town of Westport, is 2.5 miles above the entrance. Commercial traffic consists mostly of barges that call at a sand and gravel company at Saugatuck; depths at the wharf are about 5 feet.

(114) At Saugatuck the river is crossed by a railroad bridge having a bascule span with a clearance of 13 feet. Overhead power cables at the bridge have a clearance

of 192 feet. The Connecticut Turnpike Bridge, 0.1 mile above, has a fixed span with a clearance of 59 feet. About 0.1 mile farther up is a highway swing bridge with a clearance of 7 feet. (See **33 CFR 117.1** through **117.59** and **117.221**, chapter 2, for drawbridge regulations.)

(115) **Westport** is a town at the head of navigation on the Saugatuck River, about 1.4 miles above Saugatuck.

(116) There are several small-craft facilities on the river in the vicinity of the bridges. Gasoline, water, marine supplies and a 3-ton lift are available; hull and engine repairs can be made. Depths of 6 feet are reported alongside the facilities.

(117) **Norwalk Islands**, privately owned with the exception of Shea and Grassy Islands, which are owned by the city of Norwalk, and Cockenoe Island, which is owned by the town of Westport, are 1 to nearly 2 miles off the north shore of Long Island Sound and extend from Georges Rock to Greens Ledge Light, a distance of 6 miles. **Cockenoe Harbor** and **She f eld Island Harbor**, the two approaches to Norwalk River, are good anchorages for drafts of 9 to 12 feet and are easily made. The bottom is very irregular around the islands and rocks in the group; vessels should proceed with caution when crossing shoal areas and avoid all broken ground. In the vicinity are some oyster stakes and spars, which occasionally are towed under or broken off; caution is recommended, especially at night, for small craft.

(118) **Cockenoe Island**, at the eastern end of Norwalk Islands, is marked on its south side by two knolls; the remainder of the island is low and level. A bar, dry in places at low water but with general depths of 1 to 2 feet, connects the island with the mainland at **Seymour Point**.

(119) **Cockenoe Shoal** is an extensive and dangerous area that extends 1.3 miles eastward and east-southeastward from Cockenoe Island. The entire area is exceedingly broken and should be avoided by strangers, even in small craft. **Cockenoe Reef** extends about 0.5 mile eastward from the northern end of Cockenoe Island. **Georges Rock**, with a least depth of 2 feet, is at the eastern end of the shoal; a lighted buoy is off the northeast side of the rock. A lighted bell buoy marks the southeast end of the shoal.

(120) **Channel Rock**, covered 1½ feet, is about 0.2 mile southwestward of Cockenoe Island and is marked by a buoy to the southward. **Peck Ledge**, on the western side of Cockenoe Harbor entrance, is marked by Peck Ledge Light and Norwalk East Approach Buoy 5.

(121) **Cockenoe Harbor**, westward of Cockenoe Island, is marked by Peck Ledge Light. The best anchorage is in depths of 9 to 12 feet, northward and northwestward of the light.

(122)

Routes

(123) To enter Cockenoe Harbor from the eastward, pass southward of Cockenoe Island Shoal Lighted Bell Buoy 24, steer **254°** until Peck Ledge Light bears northward of **285°,** then steer for the light until up with Norwalk East Approach Buoy 4 that marks Channel Rock, and then pass eastward and northward of the light at a distance of 200 to 300 yards.

(124) To enter Cockenoe Harbor from the westward, give the edge of the shoals southward of the Norwalk Islands a good berth until Peck Ledge Light bears westward of **348°**, and then steer **north** and pass 400 yards eastward of the light and midway between Norwalk East Approach Buoy 4 that marks Channel Rock and Norwalk East Approach Buoy 5.

(125) The islands and rocks on the west side of Cockenoe Harbor include **Calf Pasture Island**, with several houses and a few trees; **Sheep Rocks**, which uncover 2 feet; **East White Rock**, high and white; and **Grassy Hammock Rocks**, which uncover and are marked by a light.

(126) The larger islands southwestward are in general hilly and partly settled. **Chimon Island** is marked by several houses; **Copps Island** by large boulders that extend east from it; and **She f eld Island**, the westernmost of the group, by an abandoned lighthouse tower.

(127) Rocks that uncover extend nearly 0.3 mile southwestward of Sheffield Island.

(128) **Greens Ledge** is a rock and sand ridge that extends 1.1 miles southwestward from Sheffield Island. Depths of 10 to 15 feet extend about 400 yards westward and southwestward from Greens Ledge Light. **Greens Ledge Light** (41°02'30"N., 73°26'38"W.), 62 feet above the water, is shown from a conical tower, the upper half white and lower half brown, on a black cylindrical pier on the north side of the west end of the ledge. A mariner radio activated sound signal at the light is initiated by keying the microphone five times consecutively on VHF-FM channel 83A.

(129) **Cable and Anchor Reef** covers an area about 0.4 mile in diameter about 2 miles southeastward of Greens Ledge Light. The least found depth is 25 feet. A lighted bell buoy marks the southern side.

(130) **She f eld Island Harbor**, entered between Greens Ledge and the mainland, is the main approach to Norwalk Harbor and Norwalk River. Anchorage in depths of 12 to 20 feet can be found northwestward of Sheffield Island. The shoal flats on the north side of the harbor have rocks and boulders in places.

(131) **Norwalk River** empties through **Norwalk Harbor** into the north side of Long Island Sound, northward of the Norwalk Islands and about 40 miles east of New York.

(132)

Channels

(133) Norwalk Harbor and River are entered through a dredged channel that extends 3 miles northeasterly from Sheffield Island Harbor between **Manresa Island** on the west and **White Rock** and numerous islets and foul ground on the east, to the first highway bridge at South Norwalk, and thence northerly for another 1.3 miles to the basin at the head of navigation at Norwalk. The tall stack on Manresa Island, marked on top by red lights, is very prominent and can be seen for many miles from sea.

(134) A federal project provides for a depth of 12 feet from Sheffield Island Harbor to the State Route 136 bridge, thence 10 feet to a 10-foot basin at the head of navigation at Norwalk; an anchorage basin opposite Fitch Point has a project depth of 10 feet. The channel is marked by buoys and lights to the South Anchorage Basin. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A.

(135)

Caution

(136) Chemically contaminated material has been buried in the navigation channel off Oyster Shell Point about 140 yards below Interstate Route 95 bridge. The material is covered with a layer of noncontaminated dredged material not less than 3 feet thick.

(137)

Bridges

(138) Three bridges cross Norwalk River between South Norwalk and Norwalk. The first, State Route 136 highway bascule bridge at South Norwalk, has a clearance of 8 feet. The second, a railroad swing bridge just above the highway bridge, has a clearance of 16 feet; an overhead power cable with a clearance of 203 feet crosses the river near the railroad bridge. The third, a turnpike highway fixed bridge, about 0.6 mile above the railroad bridge, has a clearance of 60 feet. (See **33 CFR 117.1** through **117.59** and **117.217**, chapter 2, for drawbridge regulations.) The bridgetenders at the State Route 136 bridge and the railroad bridge monitor VHF-FM channel 13; call signs KXJ-707 and KU-6035, respectively.

(139) **Tavern Island**, with several houses and foul ground on all sides, is just northwestward of the dredged channel entrance to Norwalk Harbor.

(140) **Gregory Point**, marked by a clubhouse and wharf, is on the east side of Norwalk Harbor 1.9 miles above the channel entrance. The boat basin immediately eastward of Gregory Point, locally known as **Norwalk Cove**, is entered through a privately maintained channel. In 1987, the controlling depth was 8 feet in the channel, thence in 1981, 6 feet in the eastern part of the basin. A 220-yard-long detached timber breakwater is on the north side of channel entrance.

(141) **East Norwalk Harbor**, at the town of **East Norwalk**, is on the east side of the river about 2 miles above the main channel entrance. The harbor is entered through a dredged channel that leads westward of **Fitch Point** to the head and to North Anchorage Basin on the westerly side of the harbor. A federal project provides for a depth of 6 feet from Fitch Point Light 1 to and in an anchorage basin at East Norwalk. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A. The channel is marked to near the southern end of the basin.

(142) **South Norwalk** is an important commercial and manufacturing city on the west side of Norwalk River, about 3 miles above the channel entrance. The depths at the wharves below the bridges range from 5 to 10 feet. Commercial traffic is mainly in building materials, petroleum products and shell fishing.

(143) **Norwalk**, 1.3 miles above South Norwalk, is a city on both sides of the river at the head of navigation. The wharves have depths of about 7 feet alongside. The channel from South Norwalk to Norwalk is winding, with extensive flats on both sides, and requires local knowledge to follow it even at high water.

(144) Local regulations provide penalties for exceeding the posted 5 mph **speed limit** or for dumping refuse in the harbor. These regulations are enforced by the Marine Division of the Norwalk Police Department. Police patrol boats operate the year round and are equipped to handle radio traffic on VHF-FM channel 16 (156.80 MHz).

(145) The **harbormaster** at Norwalk can be reached through the police department.

(146)

Current

(147) The tidal currents in Long Island Sound off Norwalk have a velocity of about 1 knot. In Norwalk River, off Gregory Point, the velocity of current is about 0.6 knot. The currents in the harbor follow the direction of the channel, the ebb current being somewhat stronger than the flood. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(148) The channel up to South Norwalk is navigable throughout the year. The harbor and river above South Norwalk are covered with **ice** during a part of the winter. A channel is ordinarily kept open to the highway bridge, but the East Norwalk Channel and the channel in the river are usually closed for about 6 weeks each winter.

(149)

Pilotage, Norwalk

(150) Pilotage by a state-licensed pilot is compulsory in Long Island Sound for foreign flag vessels and U.S. vessels that are under register (i.e., engaged in foreign trade). Such vessels can arrange for a state-licensed pilot by contacting the joint rotation administrator, Block Island Pilots at 243 Spring Street, Newport, RI 02840; telephone 401–847–9050 (24 hours), 800–274–1216; FAX 401–847–9052. Enrolled vessels (i.e., U.S. vessels engaged in coastwise trade) see Pilotage, Long Island Sound (indexed as such), Chapter 8.

(151)

Small-craft facilities

(152) There are excellent small-craft facilities at South Norwalk and East Norwalk and in Norwalk Cove.

(153)

Communications

(154) Rail and bus lines serve the city and area.

(155) **Wilson Cove**, on the north side of Sheffield Island Harbor, is entered about 0.6 mile northwestward of the dredged channel entrance to Norwalk Harbor between **Wilson Point** on the north and Bell Island on the southwest. The ruins of a former oil-receiving pier are on the southwestern extremity of Wilson Point. A yacht club is on the east side of the cove, about 150 yards northward of the wharf ruins, and a marina is at the head of the cove. Gasoline, limited marine supplies, ice, an 18-ton crane, a 20-ton mobile hoist and engine and hull repair facilities are available at the marina.

(156) **Noroton Point**, at the southern end of **Bell Island**, is marked by a flagpole and a prominent house with a cupola. Rocks, bare at low water, are about 300 yards northward of the point. **Pine Point**, just westward of Noroton Point, has a wharf in ruins at its southern end. A shoal with depths of 8 to 12 feet extends about 0.3 mile from the shore westward of Noroton Point. The bottom is broken with boulders in places, and small vessels crossing the shoal should proceed with caution. **Ballast Reef**, about 0.2 mile westward of Pine Point and off the southeast side of the entrance to Fivemile River, is almost bare at low water and extends 300 yards off **Roton Point**; a buoy marks the outer end of the reef.

(157) **Fivemile River** is a narrow inlet about 0.6 mile westward of Noroton Point and about 0.9 mile northward of Greens Ledge Light. A federal project provides for a depth of 8 feet to a point about 1 mile up the river. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A. The river is shallow except in the dredged channel and rocks exposed 2 feet at low water have been reported on the east side of the channel near the channel edge in about 41°03'37"N., 73°26'47"W. The channel is marked by buoys.

(158) In 1981, depths of 2 to 5 feet were reported alongside the small-craft facility wharves on the east side of the river. The river is used chiefly by pleasure craft.

(159) A **special anchorage** is in Fivemile River. (See **33 CFR 110.1** and **110.55a**, chapter 2, for limits and regulations.)

(160) **Rowayton** is a village at the head of Fivemile River. Several **small-craft facilities** are on the east side of the river.

(161) **Scott Cove**, about 0.8 mile westward of Fivemile River and about a mile northwest of Greens Ledge Light, is a rocky shelter with a channel good for about 6 feet to the shallow area northward. There are rocks and broken ground in the entrance. The channel into **Zieglers Cove**, just west of Scott Cove and south of **Great Island**, is good for about 9 feet. A rock, covered 4 feet, lies almost in mid-entrance to this cove. Local knowledge is required to navigate both coves.

(162) **Long Neck Point**, about 2 miles southwestward of Fivemile River, has many summer residences and boat landings on both of its sides. Shoals extend about 0.3 mile off the point. Numerous obstructions exist up to 1.2 miles west-southwest of Long Neck Point.

(163) From Long Neck Point to Shippan Point, about 2.6 miles to the southwestward, there are many reefs and boulders, and the bottom is very broken, necessitating caution. This area is the approach to several shallow coves, none of which is commercially important.

(164) **Goodwives (Darien) River** is a small and shallow stream on the west side of Long Neck Point. Along the western shore and about 0.3 mile above the south end of the point, foul ground extends nearly 200 yards offshore. A private seasonal, **352°** lighted range and buoys mark the best water to a yacht club and basin on the southeast side of **Noroton Neck**. In 2002, a depth of 4.5 feet could be carried to the yacht club landing thence in 1981, 3 feet through **The Gut** to the boat club landing just above **Peartree Point**. Above the boat club landing, the river is practically dry at low water. Goodwives River and its entrance is a **special anchorage**—see **33 CFR 110.1** and **110.56**, chapter 2, for limits and regulations. A 5 mph **speed limit** is enforced on the river.

(165) **Smith Reef**, about 0.9 mile southwestward of Long Neck Point, consists of two rocks that uncover 2 feet. The south end of the reef is marked by a lighted buoy. **Bold Rock**, which uncovers 4 feet, is on the east edge of the rocky ridge extending northward from the reef. Many oyster stakes are on the ridge.

(166) **Cove Harbor**, northward of Smith Reef and about 1 mile westward of Long Neck Point, has depths of about 5 to 10 feet. Local knowledge is necessary to avoid several rocky areas in the approach to the harbor and to the basin at the northwestern end of the harbor at Cove Mills. A depth of about 1 foot can be carried across the bar at the entrance to the basin; private buoys, one of which is a seasonal speed limit buoy, mark the approach. A municipal marina is in the basin.

(167) **Westcott Cove**, just westward of Cove Harbor, has a dredged channel marked by buoys that leads along its westerly side to a basin 0.5 mile above the channel entrance, thence for 0.2 mile through the south arm of the basin. The east side of the entrance to the basin is protected by a jetty. A yacht club is in the northwesterly arm of the basin and a municipal marina is in the southeasterly arm. A marina on the west side of the south arm of the basin can provide gasoline, diesel fuel and water.

(168) **Stamford Harbor**, on the north side of Long Island Sound about 33 miles east of New York, comprises the bay north of a line from Shippan Point on the east through Stamford Harbor Ledge Obstruction Light to the west shore north of Greenwich Point. The harbor is shoal, and the approach is obstructed to a large extent by ledges and rocks. **Shippan Point**, the eastern point at the entrance, is surrounded by rocks which show at low water. **Stamford**

is a manufacturing city on the peninsula at the head of the harbor. Barges and small coastal tankers constitute the main waterborne traffic in the harbor. Petroleum products, scrap metal, sand and gravel and crushed rock are the principal products handled.

(169)

Prominent features

(170) **Stamford Harbor Ledge Obstruction Light** (41°00'49"N., 73°32'34"W.), 80 feet above the water, shown from a white conical tower on a red cylindrical pier, is a private light visible from a considerable distance offshore. Also prominent are a microwave tower westward of the city and the large brown office buildings locally known as Harbor Plaza on Ware Island. **Stamford Harbor West Breakwater Light 3** (41°00'54"N., 73°32'17"W.), 37 feet above the water, is shown from a tower with a square green dayboard at the east end of the west breakwater. **Stamford Harbor East Breakwater Light 4** (41°00'54"N., 73°32'06"W.), 21 feet above the water, is shown from a skeleton tower with a triangular red dayboard at the west end of the east breakwater.

(171)

Channels

(172) Stamford Harbor is entered through a dredged entrance channel that leads northward from Long Island Sound between two detached breakwaters. The breakwater ends nearest the channel are marked by lights. About 1 mile above the entrance, the channel divides into **East Branch** and **West Branch**. The channels are marked by buoys and a **356.8°** lighted range. The 100-foot-wide channel in East Branch is constricted to 90 feet by a hurricane barrier crossing the channel about 300 yards northward of Ware Island. The opening in the barrier will be kept in the open position during fair weather but will be closed on the approach of a storm or unusually high tides. A lighted sign on either side of the barrier indicates whether the barrier is in the open or closed position. A flashing red light is shown from the control tower when the gate is about to be closed.

(173)

Anchorages

(174) A dredged anchorage area with depths of 10 to 18 feet is north of the breakwaters and just westward of the line of the range lights, about 0.1 mile eastward of **Highwater Rock**. Small craft can anchor off the yacht club and southward or southeastward of **Rhode Island Rocks** in depths of 5 to 7 feet. All anchorages in the outer harbor are exposed to southerly and southwesterly winds.

(175)

Dangers

(176) **The Cows** comprise a cluster of rocks, almost bare at low water, about 0.8 mile south-southeast of Shippan Point. Between them and the point is an area of foul ground and rocks bare and awash that extends 0.4 mile southward of Shippan Point. A lighted bell buoy is about 0.2 mile south of The Cows. **Harbor Ledge**, about 200 yards south of the west breakwater, consists of rocks and a ledge marked by a private light.

(177)

Current

(178) The flood current at the entrance to the harbor has a velocity of 0.4 knot and sets 329°; the ebb has a velocity of 0.8 knot and sets 134°. Inside the harbor the currents have little velocity and usually set fair with the channel.

(179)

Ice

(180) The channel in West Branch is usually navigable throughout the year, but in East Branch it is closed by ice for several weeks during severe winters. Ice forms in the harbor during most winters and usually extends to a point just northward of the breakwaters. The channels are kept open as far as practicable by passing traffic.

(181) Prevailing winds are from the south and southwest in the summer and from northeast during the winter season.

(182) No particular directions are required. The range favors the east side of the channel and does not show plainly until eastward of Stamford Harbor West Breakwater Light 3. In East Branch, caution is advised when making the turn abreast Ware Island to avoid a rock nearly awash at high water, eastward of the channel line.

(183) The **harbormaster** at Stamford can be contacted through the Stamford Police Department. A police boat makes routine patrols of the harbor during the boating season. A 6 mph **speed limit** is enforced in the harbor.

(184)

Wharves

(185) The commercial wharves along East Branch and West Branch are of the bulkhead and apron type, all are privately owned, and some are open to the public. Spur tracks from the railroad serve the facilities in East Branch.

(186)

Small-craft facilities

(187) There are excellent facilities for small craft in both East and West Branches.

(188) **Dolphin Cove**, 0.6 mile west of the entrance channel to Stamford Harbor, is a privately owned Lagoon and marine facility. No anchoring is allowed.

(189)

Captain Harbor to Pine Island

(190) **Captain Harbor**, on the north shore of Long Island Sound westward of Greenwich Point and northward of Great and Little Captain Islands, affords shelter from all winds for vessels drawing 12 feet or less. The depths at the anchorage in the deeper part of the harbor, about 0.5 mile northward of Great and Little Captain Islands, are 15 to 30 feet. Vessels of less than 7-foot draft anchor on the flats. The bottom is soft, but the entire harbor and entrances are characterized by boulders. Strangers should proceed with caution, especially on the flats and other shoal areas. The eastern entrance to Captain Harbor, between Flat Neck Point and Little Captain Island, is the

(206)

clearer and better one for strangers. The western entrance, northwestward of Great Captain Island, is easy of access, but the broken ground there requires caution.

(191) **Greenwich Point**, 1.7 miles southwestward of Stamford Harbor West Breakwater Light 3, is characterized by a low grassy hill. Reefs extend 0.3 mile southeastward from Greenwich Point. **Woolsey Rock** near the easterly end of the reefs is covered 2 feet. A buoy marks these dangers.

(192) **Flat Neck Point**, the western end of Greenwich Point, is wooded. A reef with bare and submerged rocks extends nearly 0.3 mile southwestward and westward from Flat Neck Point.

(193) **Greenwich Cove** opens into Captain Harbor from eastward, north of Flat Neck Point. The cove is used for mooring local craft. Depths decrease from 8 feet in the outer cove to less than 3 feet in the eastern part of the cove. **Old Greenwich** is on Greenwich Cove.

(194) **Cos Cob Harbor** is on the northeast side of Captain Harbor. A dredged channel, with its entrance 0.2 mile north of Lowther Point, extends 1.3 miles northward through Mianus River to the head of navigation at Mianus. Shoaling is reported to be abrupt along both edges of the channel. The channel is buoyed to the first bridge; above this point the channel may be followed by steering a midchannel course between the marsh banks.

(195)

Anchorages

(196) **Special anchorages** are in Cos Cob Harbor—see **33 CFR 110.1** and **110.58**, chapter 2, for limits and regulations.

(197) There are several dangers off the entrance of Cos Cob Harbor that must be avoided; most are buoyed. These include **Newfoundland Reef**, covered 4 feet, a mile northeastward of Little Captain Island; **Red Rock**, which uncovers 7 feet, 0.5 mile west of Newfoundland Reef; **Hitchcock Rock**, awash at low water, 0.3 mile northwestward of Newfoundland Reef; and **Pecks Rock**, bare at low water, 0.2 mile north of Hitchcock Rock.

(198) The Riverside Yacht Club, on the east side of Cos Cob Harbor and about 0.5 mile below the first bridge, is prominent.

(199) **Mianus River** is crossed by a railroad bascule bridge with a clearance of 20 feet, and by a highway fixed bridge with a clearance of 45 feet, about 0.4 mile to the northward—see **33 CFR 117.1** through **117.59** and **117.209**, chapter 2, for drawbridge regulations.

(200) Several marinas and boatyards are along the west side of the river from above the railroad bridge to the head of navigation.

(201) **Indian Harbor** is a narrow inlet on the north side of Captain Harbor, about 1 mile west of Cos Cob Harbor. A channel with a depth of about 7 feet passes about 200 feet westward of Tweed Island and follows the west bank to

the bulkhead on the west side of the cove 300 yards above the entrance. Small craft can anchor in the channel just above this point, favoring the bulkhead. A large prominent white residence with red roof and adjacent white clock tower is on the point separating **Smith Cove** and Indian Harbor. A 5 mph **speed limit** is enforced in the harbor.

(202) Depths of 6 feet or less extend 250 yards southward from the point separating Smith Cove and Greenwich Harbor. Bare ledges extend 200 feet southward of the point. The yacht club on the point usually maintains lights on a flagstaff during the summer. The depth is about 7 feet at the landing of the Indian Harbor Yacht Club.

(203) **Greenwich Harbor**, on the north side of Captain Harbor and northeastward of Field Point, is entered through a dredged channel that leads northward 1.2 miles to the head. The channel is buoyed for about 0.8 mile. A 5 mph speed limit is enforced in the harbor.

(204) **Greenwich** is a city on the railroad at the head of the harbor. The wharves are along the point on the east side of Greenwich Harbor. The **harbormaster** at Greenwich can be contacted through the Greenwich Police Department. A police boat patrols the harbor during the summer season.

(205) There are several private yacht and boat clubs within Greenwich Harbor and a pump-out facility at the Grass Island Marina. There are no fuel docks reported to exist within Greenwich Harbor. Fuel is available a few miles to the East on the Mianus River. During the summer, a ferry operates from the town landing at the head of the harbor to Little Captain Island and Great Captain Island.

(207) **Byram Harbor**, a bight used by small craft, is at the northwest end of Captain Harbor, just northward of **Calf Islands**. **Wilson Head**, 2 feet high, on a reef that uncovers, is in the middle of the entrance of the bight and is marked by a buoy off the eastern end. The entrance to Byram Harbor from eastward lies between Otter Rocks and Bowers Island. **Otter Rocks**, which uncover 3 feet, are marked by a lighted buoy about 150 yards to the southward; a submerged rock is close northward of the buoy. **Bowers Island**, just eastward of Calf Islands, is surrounded by a drying reef; a buoy marks the north end of the reef. A rocky ledge makes out from the point 300 yards northwestward of Otter Rocks and is marked by a buoy. A narrow channel also leads to the harbor from southwestward, passing between **Shore Island** and northernmost of the Calf Islands. Private small-craft facilities are on the west side of the harbor.

(208) **Grassy Rocks**, 0.3 mile westward of the southerly tip of Calf Islands, uncover 7 feet. The four large ledges northwestward and westward of Grassy Rocks generally show at low water.

(209) Foul ground extends 0.2 mile east and southeast from the southern tip of Calf Islands. The eastward extent terminates in **Jones Rocks** which are partly bare at high water and marked by a light. The southeastern extent is marked by a buoy at the outer end.

(210) **Cormorant Reef**, northward of Great Captain Island, partly bare at high water, has a rock 4 feet high on the eastern end. A buoy is off the southern end of the reef.

(211) **Great Captain Island**, 2.6 miles southwestward of Greenwich Point, is 0.4 mile long, fringed with reefs and marked near its southeast end by a light. A municipal bathing beach and ferry landing are on the island. The landing has reported depths of about 3 feet. A buoy marks the reef making off 0.3 mile from the southwestern end. The passage between Great and Little Captain Islands is foul and not recommended.

(212) **Great Captain Island Light** (40°58'57"N., 73°37'23"W.), 62 feet above the water, is shown from a skeleton tower with a red and white diamond-shaped dayboard on the southeast part of the island. A sound signal at the light is operated by keying the microphone five times consecutively on VHF-FM channel 83A.

(213) **Little Captain Island**, a summer resort about 0.6 mile northeast of Great Captain Island, has a municipal bathing beach and ferry landing. The landing has reported depths of about 8 feet. A reef extends about 250 yards northeasterly to **Wee Captain Island**. An area of boulders and broken ground extends 0.4 mile eastward and northeastward from the island and is marked by a lighted gong buoy. **Hen and Chickens**, a group of rocks and boulders about 0.4 mile northeastward of Little Captain Island, is marked by a buoy on the north side.

(214)

Current

(215) The tidal current in the entrance between Little Captain Island and Flat Neck Point has a velocity of about 0.7 knot. Between Jones Rocks and Cormorant Reef the estimated velocity is 1 knot.

(216)

Ice

(217) Ice forms in the winter in all the coves and over the greater part of Captain Harbor. It sometimes extends out of the line of Little and Great Captain Islands.

(218)

Routes

(219) From eastward, a course of about **298°** midway between the buoys marking the shoals off Flat Neck Point on the east and Wee Captain Island on the west will bring a vessel to a point 0.2 mile north of Hen and Chickens Buoy 1A. From here a heading of **250°**, with the southerly tip of Calf Islands ahead, will lead to anchorage off the entrance of Greenwich Harbor.

(220) From westward, a course of **014°** for Jones Rocks Light 3 will lead into the Captain Harbor anchorage. Proceed with caution when crossing the broken rocky area on which the least found depth is 12 feet, extending 0.4 mile westward from the western end of Great Captain Island. Vessels should pass 100 yards southeastward of Jones Rocks Light 3 and over 100 yards northward of the buoy northwestward of Cormorant Reef and steer **070°** in the harbor.

(221) **Port Chester Harbor**, about 1.2 miles westward of Great Captain Island, is the entrance to Byram River that leads to the city of **Port Chester** and the town of **Byram (East Port Chester)**. The harbor entrance is between the breakwater that extends southward from **Byram Point** on the north and **North Manursing Island** on the south; a light is on the outer end of the breakwater. The lower section of the river forms the boundary between New York and Connecticut.

(222) The harbor is entered from Long Island Sound through a dredged channel that leads northward for 1.2 miles to a turning basin in **Byram River**, and thence for another 0.15 mile to just below the Mill Street fixed bridge, the head of practical navigation on the river. The channel is marked to a point about 0.3 mile above the entrance. The New England Thruway fixed bridge, with a clearance of 60 feet, crosses the river about 0.8 mile above the channel entrance.

(223)

Routes

(224) The approach to Port Chester is obstructed by rocks but is not difficult with the aid of the chart. From southward it is safer to pass eastward of **Bluefish Shoal**. **Fourfoot Rocks** may be passed on either side, remembering that the buoy is at the south end of the rocks. Entering the harbor, pass westward of Great Captain Rocks, eastward of **Manursing Island Reef** and 150 feet southward of Port Chester Light 4 on the end of the breakwater. The channel in Byram River is fairly well defined at low water but requires local knowledge for the best water; strangers should take it on a rising tide and proceed with caution.

(225) Principal commerce is in building materials, fuel oil and petroleum products, carried in vessels drawing 5 to 14 feet. Barges discharge oil cargoes at a terminal with reported depths of 12 feet alongside.

(226)

Small-craft facilities

(227) There are several small-craft facilities in Port Chester Harbor and on the Byram River at Port Chester and Byram.

(228) The area from Great Captain Island southwestward is fringed with rocks, bare and submerged, and foul ground. **Great Captain Rocks**, part of a reef 0.3 mile southeastward of Port Chester Light 4, uncover 5 to 6 feet; a buoy marks the southern end of the reef. **Transport Rock**, about 0.3 mile south-southwestward of Manursing Island, is part of several ledges generally bare at high water that extend some 0.3 mile offshore. An opening suitable for small craft leads to Rye Beach; it is buoyed.

(229) **Playland**, a recreational center at **Rye Beach**, about 2.4 miles southwest of Great Captain Island, has prominent twin towers at the entrance that are conspicuous from a southeasterly direction. Westward and close to the north breakwater is a former ferry landing in disrepair. A breakwater extends eastward from the south end of Rye Beach. The area between the former ferry landing and the south breakwater is reserved for swimming.

(230) **Forbes Rocks**, about 0.4 mile south of the Rye Beach breakwater, are partly bare at low water, on a reef with depths of 4 to 11 feet that extends 250 yards to the southward and eastward. A buoy marks the east end of the reef.

(231) **Porgy Shoal**, about 0.8 mile south of the Rye Beach breakwater, has a least found depth of 5 feet; it is marked by a lighted buoy.

(232) **Scotch Caps** are three rocky islets 1.4 miles southwestward from Porgy Shoal and on the northwest side of the extensive reefs that make out 0.9 mile southwestward of **Milton Point**. The southerly end of the reefs is marked by a lighted bell buoy about 0.6 mile southward of Scotch Caps. The entire area of the reef northward and northeastward of the lighted bell buoy is very broken and should be avoided even by small craft in the absence of local knowledge.

(233) **West Rock**, just south of the south end of Scotch Caps, is marked by a buoy.

(234) **Milton Harbor**, between **Peningo Neck** and Hen Island, is used as a summer anchorage by small pleasure craft. It is protected from all but southwesterly winds. The harbor depths decrease from 8 feet between Scotch Caps and the southwest end of Hen Island to 6 feet abreast Milton Point.

(235) Foul ground is on the northwest side near **Hen Island**; otherwise the principal danger in the harbor is a rock bare at low water, midway between Milton Point and the northeast end of Hen Island. The best entrance is between the buoys 0.4 mile southwestward of Scotch Caps.

(236) A yacht club and landing are near the southwest end of Milton Point. Near the clubhouse is a prominent white flagstaff from which lights are exhibited from sunset to sunrise during the summer.

(237) A dredged channel, marked by buoys, leads through the harbor from about 400 yards northward of Milton Point to the city boat basin and marina below **Mill Pond**. Two boatyards are in the harbor. The largest marine railway can handle craft up to 40 feet in length; gasoline, water, ice, marine supplies, and complete engine and hull repairs are available. The city **harbormaster** is at the boat basin.

(238) **Mamaroneck Harbor**, an open bight between Hen Island and **Delancey Point**, is exposed to southerly winds but affords shelter against northerly weather. Depths in the outer harbor range from 7 to 12 feet. Important dangers are buoyed; these include **Outer Steamboat Rock**, near the dredged channel entrance, and **Ship Rock**, about 0.5 mile southeastward of Outer Steamboat Rock.

(239) About 1 mile northwest of Outer Steamboat Rock is the incinerator tower, a red brick building with a large glass tower, which is a prominent landmark.

(240)

Channels

(241) A federal project provides for a 10-foot channel entering the harbor and leading about 0.5 mile

(250)

west-northwestward to the intersection with two dredged branch channels leading to basins northward and westward of the junction. The channel leading northward to the east basin has a project depth of 10 feet, thence 6 feet in the main anchorage area; to the west of the junction, the channel and anchorage area have a project depth of 6 feet. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A. The entrance channel and the branch channel to the northern basin are marked by lighted and unlighted buoys. The basins are usually filled with moorings of local craft.

(242)

Caution

(243) A pipeline covered about 6 feet crosses the western branch channel about 50 yards above the junction. Mariners are advised to exercise caution and reduce speed while transiting this area.

(244) The **harbormaster** has an office on the south side of Harbor Island. The harbormaster controls all moorings and can be contacted on VHF-FM channel 16; call sign WZX-8038. A **speed limit** of 5 mph is enforced in the harbor. A village police boat patrols the harbor during the summer season.

(245) The town of **Mamaroneck** extends from both sides of the harbor. Petroleum products, carried by barges, are the main commerce in the harbor.

(246)

Supplies and repairs

(247) There are numerous boatyards and marinas in Mamaroneck Harbor.

(248) Foul ground extends southwesterly from eastward of Delancey Point to the Larchmont Harbor breakwater off **Edgewater Point**, on the east side of the harbor entrance; a light is on the end of the breakwater. **Hen and Chickens**, a reef bare at low water in places, lies off the harbor entrance; surrounding depths are 8 to 17 feet on the outer parts of the reef. About 0.3 mile westward of the breakwater light is **Dauntless Rock**, covered 8 feet and surrounded by depths of 14 to 16 feet. These dangers are buoyed.

(249) **Larchmont Harbor** is between Edgewater Point and **Umbrella Point** and about 2.5 miles northward of Execution Rocks Light. The harbor is the headquarters of the Larchmont Yacht Club. Anchorage depths range from about 12 feet in the entrance to 5 feet near the north central part of the harbor. In summer the harbor is full of mooring buoys for small yachts. The rocks on the west side are marked, whereas unmarked shoals extend 200 yards from the eastern shore. The anchorage for larger vessels is westward of the breakwater.

(251) **Umbrella Rock**, marked by a buoy, is 250 yards eastward of Umbrella Point. A few rocks of a breakwater, which was started on Umbrella Rock, are awash at high water. **North Ledge**, bare at half tide and marked by private daybeacons, is near the western shore southeastward of the yacht club. The principal landing, with a reported depth of about 6 feet alongside, is on the southeast side of the yacht club and is lighted from sunset to sunrise.

(252) Larchmont Harbor may be entered on either side of Hen and Chickens. The easterly entrance, about 100 yards southwestward of the end of the breakwater, is about 300 yards wide and has a depth of about 15 feet.

(253) **Horseshoe Harbor** is a small cove just westward of Larchmont Harbor. A prominent gray building is at the head. The cove is used as a small-boat anchorage.

(254) **Echo Bay**, about 1 mile southwestward of Umbrella Point and 2 miles northwestward of Execution Rocks Light, is the principal approach to New Rochelle. The bay is entered between **Premium Point** on the northeast and **Davenport Neck** on the southwest. **Hicks Ledge**, about 0.5 mile off the entrance, is covered 6 feet and marked on the south side by a buoy.

(255) **Middle Ground**, an extensive shoal with a reef that uncovers 6 feet, lies about 0.5 mile south-southwestward of Hicks Ledge. **Emerald Rock**, covered 9 feet, is off the west side of the shoal and marked by a buoy. A buoy marks the north end of the shoal.

(256) **Bailey Rock** is near the end of a reef that extends about 200 yards off the northeast point of Davenport Neck. The rock uncovers 3 feet and is marked by a lighted buoy.

(257) Echo Bay is an anchorage for small craft and generally is fully occupied during the summer. Depths range from 4 to 15 feet. Small craft can anchor in the shallow cove on the northeast side of the harbor, entering between **Harrison Island** and the rocky, grassy islet off the northwest side of **Echo Island**. Vessels should not anchor near the sewer lines in the middle of the bay. A **special anchorage** is in Echo Bay—see **33 CFR 110.1** and **110.60**, chapter 2, for limits and regulations. A 4 mph **speed limit** is enforced in the bay.

(258) On the northwest side of Echo Bay, a dredged channel leads to a municipal wharf and turning basin at Beaufort Point. The channel is marked by buoys to the turning basin. The area northward of the turning basin, locally known as Ferris Creek, is shoal with extensive mud flats that bare at low water. Southwesterly of the turning basin, the depth varies from 9 feet to bare at the head of the harbor. **New Rochelle** is a city on the western shore of Echo Bay.

(259) The municipal wharf is on the northeast side of **Beaufort Point**. The city police patrol boats usually moor alongside the wharf. A small-craft facility and a municipal marina are in the northern part of Echo Bay. Berths, electricity, gasoline, diesel fuel, water, ice and lifts to 25 tons are available; hull and engine repairs can be made. The municipal marina monitors VHF-FM channel 16.

(260) **Pine Island**, between Davenport Neck and Middle Ground, is rocky and covered with brush. A small private landing is on the west side of the island. A series of bare and submerged rocks and ledges that cover and uncover with foul ground in between are southwestward of the island.

(261)

Davids Island to Locust Point

(262) **Davids Island**, southward of Davenport Neck, is owned by the city of New Rochelle. Reefs, partly bare at low water and marked by a lighted buoy, extend about 0.2 mile northward of the island. Davids Island is surrounded on its east and south sides by a foul area of islands and rocks, the passages between which should not be used by strangers, even in small craft. **Huckleberry Island**, at the eastern end of the group, is wooded. **Pea Island**, about 0.3 mile southeastward of Davids Island, is grass covered, and rocks bare at low water are southeastward of it. **Columbia Island** has been improved by a seawall, making it about 150 feet square, with a pier 150 feet long on the west side.

(263) The wreck of a barge, covered 17 feet, is about 0.3 mile south-southeastward of Pea Island; mariners are advised to exercise caution while navigating in this area.

(264) **Execution Rocks**, about 1.4 miles eastward of Davids Island, consist of many boulders and shoals of considerable extent. **Execution Rocks Light** (40°52'41"N., 73°44'16"W.), 62 feet above the water, is shown from a white stone tower with a brown band midway of its height, attached to a granite dwelling; a racon is at the light. Broken bottom, covered 5 to 18 feet, extends about 0.7 mile northward from the light. Bouys mark the north, east, southeastern and southwestern extent of the broken bottom surrounding the rocks.

(265) **Middle Reef**, 0.5 mile southward of Davids Island, has some boulders that show at high water. **East Nonations** and **South Nonations** are rocks that uncover 4 feet between Middle Reef and Hart Island. South Nonations is marked on its south side by a lighted bell buoy.

(266) **Aunt Phebe Rock**, 300 yards west of Davids Island, is bare at half tide and marked by a light. Mariners are advised to exercise caution while navigating in this area.

(267) **Goose Island**, between Davids Island and Glen Island, is almost completely surrounded by a rock breakwater and has several bare rocks to the westward and southward. A house on pilings is prominent on the island.

(268) **Glen Island**, west of Davids Island, is a public park. Special permits are required prior to using the launching ramp on the island. A light is on the north end of the island. A beach protected by two jetties is on the southeast end of the island. The channel between Glen Island and Davenport Neck is frequently used as an anchorage

by small craft; a **no wake** speed limit is enforced. The channel between Glen Island and Hunter Island is marked by buoys and is entered just north of Hog Island. A special anchorage is on the southwest and west side of Glen Island—see **33 CFR 110.1** and **110.60**, chapter 2, for limits and regulations.

(269) **New Rochelle Harbor** lies between the mainland, and westward of Davenport Neck, and Glen Island; it is off the southerly part of the city of New Rochelle. However, the main access of New Rochelle is through Echo Bay, previously discussed.

(270) New Rochelle Harbor is entered between Glen Island and Davenport Neck. An approach channel, with a depth of about 13 feet, is entered south-southwest of Davids Island and continues north off the east side of Goose Island. The route continues through deeper natural water between **Aunt Phebe Rock**, marked by a light on the east and **Corning Rock**, covered 3 feet and marked by a buoy on the west, thence northward to the harbor entrance. A reef, bare at low water, makes off the west side of Davids Island opposite the buoy marking Corning Rock. Another approach channel, through deeper water, leads from the northeast between Davids Island and Davenport Neck to the entrance—both channels are well marked. In 1990, the narrow dredged channel in the harbor had a controlling depth of 6 feet at midchannel to within 100 yards of the dam at the head.

(271)

Anchorages

(272) Anchorage is not recommended in the harbor because of its congestion. A general anchorage extends to the south from the harbor entrance to City Island and Locust Point—see **33 CFR 110.1, 110.160,** and **110.155**, chapter 2, for limits and regulations.

(273) Several yacht clubs, marinas and boatyards are in New Rochelle Harbor.

(274) A bascule bridge connecting Glen Island with **Neptune Island** has a clearance of 13 feet—see **33 CFR 117.1** through **117.49**, chapter 2, for drawbridge regulations. Just south of the bridge is a yacht club on the east side of Neptune Island.

(275) **Orchard Beach**, about 1 mile southwestward of Davids Island, is a park developed by the State of New York on the filled-in area between Hunter Island, to the north, and Rodman Neck, to the south. The inshore water areas off the crescent beach are a swimming area and are closed to general navigation. The swimming area is marked by private buoys. A bathing pavilion and a flagstaff are prominent. **Chimney Sweeps Islands** are two prominent bare rocks, about 0.4 mile east of the beach.

(276) **Hart Island**, about 1.8 miles southwest of Execution Rocks Light, is the site of a New York Department of Correction facility. Buildings on the island are prominent. A reef extends about 200 yards southeastward from the south end of the island and is marked by a light. There are several wrecks and obstructions off the southwest end of Hart Island which are shown on the chart; caution is advised when navigating the area.

(277) **Rat Island** is a high bare rock about 0.4 mile west of Hart Island. **The Blauzes**, 13 feet high, are a part of the reef that extends 0.3 mile northwestward from the north end of Hart Island.

(278) **City Island**, on the northeast side of Eastchester Bay, is narrow and over 1 mile in length. It is thickly settled and has a commercialized appearance. The west side is residential and the east side is industrialized with several shipyards and other marine-related facilities.

(279)

Pilotage, City Island

(280) A pilot boat of United New York New Jersey Sandy Hook Pilot Association moors at City Island—see Pilotage, New York Harbor from Long Island Sound (indexed as such), chapter 11.

(281) **High Island** is 200 yards northeastward of the north end of City Island to which it is connected by a fixed footbridge with a clearance of 11 feet. The ground under the bridge is reported to bare about 1 foot at low water. A 528-foot-high radio tower, marked on top by red lights, is prominent on High Island.

(282)

Anchorages

(283) The usual anchorage for deep-draft vessels is southeastward of City Island, southward of a line joining the south ends of Hart and City Islands. When anchoring, avoid **Deep Reef**, a small rocky patch covered 29 feet. Other **general** and **special anchorages** are in the vicinity—see **33 CFR 110.1**, **110.60**, and **110.155**, chapter 2, for limits and regulations.

(284) A long pier in ruins and a wide stone pier, the top of which is used as a parking area, are at the south end of City Island at **Belden Point**. The western shore of Hart Island and the wharves on City Island should be given a berth of about 150 yards.

(285) The channel between City Island and Rodman Neck is used extensively as an anchorage by small pleasure craft during the summer. A **no wake** speed limit is enforced. Boat clubs and railways for small craft are on the northwest side of City Island. The shores are generally fringed with boulders and should be approached with caution. The north shores of High Island and City Island northeastward of the bridge are very foul, and boats should avoid the shoals with depths less than 12 feet on that side.

(286) City Island is connected with Rodman Neck by a fixed highway bridge with a clearance of 15 feet. Currents at the bridge are variable and at times exceed 1.5 knots—see the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(301)

Structures across Hutchinson River

Name•Description•Type	Location	Clear Width of Draw or Span Opening (feet)	Clear Height above Mean High Water (feet)	Information
Pelham Parkway Bridge (bascule)	40°51'44"N., 73°48'58"W.	59	13	Notes 1 and 2 Call sign KU-9758 and KU-6095
Amtrak Bridge (rolling lift)	40°51'48"N., 73°49'06"W.	68	8	Notes 1 and 2 Call sign KXS-298
Overhead power cable			130	
Hutchinson River Parkway Bridge (bascule)	40°52'09"N., 73°49'18"W.	130	30	Note 1
New England Thruway/I-95 Bridge (fixed)	40°53'08"N., 73°49'13"W.	100	50	
Boston Post Road Bridge (fixed)	40°53'16"N., 73°49'27"W.	121	50	
Overhead pipeline	40°53'34"N., 73°49'23"W.		130	
South Fulton Avenue Bridge (bascule)	40°53'38"N., 73°49'22"W.	80	6	Note 1

Note 1 – See **33 CFR 117.1** through **117.59** and **117.793**, chapter 2, for drawbridge regulations.
Note 2 – Bridgetender monitors VHF-FM channel 13.

(287) **City Island Harbor**, also called **Hart Island Roads**, is between Hart Island and City Island. It is well sheltered from easterly and westerly winds and is an important anchorage for coasting vessels in the western end of Long Island Sound. Numerous wrecks and obstructions are located throughout the harbor. Besides serving as a harbor of refuge, it is often used by vessels desiring pilots or towboats or awaiting orders. A spire in the center of City Island and a steeple in the northerly part of the island are conspicuous objects.

(288)

Current

(289) The tidal current has a velocity of about 0.3 knot.

(290)

Ice

(291) Ice seldom interferes with navigation of powered vessels.

(292)

Supplies

(293) Gasoline, lubricants and marine supplies of all kinds are available at City Island. Water is piped to some of the wharves; ice, electrical connections, guest moorings and dry and wet storage are readily available.

(294)

Communications

(295) Buses serve the subway system of New York City.

(296) **Eastchester Bay**, between City Island and Throgs Neck, has general depths of 7 to 10 feet in the lower part and 3 to 5 feet in the upper part. The shores of the bay are fringed with boulders, and there are many shoals and several wrecks. Caution is essential, especially where the depths are not more than 3 feet greater than the drafts.

(297) **Hutchinson River** empties into the north end of Eastchester Bay. A dredged channel marked by buoys leads from the river mouth for about 2.5 miles to the head of navigation at the city of Pelham.

(298) **Special anchorages** are in Eastchester Bay—see **33 CFR 110.1** and **110.60**, chapter 2, for limits and regulations.

(299) A **safety and security zone** has been established in Eastchester Bay surrounding much of the shoreline of Rodman Neck.—see **33 CFR 165.169**, chapter 2, for limits and regulations.

(300) The dangers in Eastchester Bay include **Big Tom** on the east side near the entrance, covered 2 feet and marked by buoys on the east and west sides, and **Cuban Ledge**, covered at half tide and marked by a daybeacon and Cuban Ledge Lighted Buoy 2 close southwestward. Numerous rocks and shoals are on both sides of the channel near the entrance to Hutchinson River.

(302) **Eastchester** is a village on the west side of the Hutchinson River about 1.5 miles above the Pelham Parkway Bridge. Commerce on the river to Eastchester is in building materials, fuel oil and petroleum products. **Pelham** is on the east side of the river above Eastchester.

(303) **Weir Creek** is a bight on the west side of the bay near the entrance.

(304) **Locust Point** is about 0.8 mile southeastward of Weir Creek. A cove just southwestward of the point provides small-boat shelter. Rocks, bare at low water, are on the north side of the approach. The entrance has a depth of about 5 feet. Inside the cove, depths range from 20 feet at the south end to about 4 feet at the north end. A yacht club and marina are in the cove. A marina at the head of the cove has a mobile hoist that can handle craft to 30 tons for engine and hull repairs. Gasoline, water, ice and marine supplies are available at the marina; depths of about 7 to 10 feet are reported at the wharf.

(305) The northern approach viaduct of the Throgs Neck Bridge crosses the cove from Locust Point to Throgs Neck. The fixed spans of the viaduct have a minimum clearance of 123 feet.

(306)

Current

(307) Tidal currents have a velocity of 0.4 knot in the vicinity of Big Tom and 0.8 knot at Pelham Bridge.

(308)

Old Field Point to Nissequogue River

(309) **Old Field Point**, about 5 miles southward of Stratford Shoal (Middle Ground) Light, is a low bluff with a light and an abandoned tower on its summit. Boulders extend a short distance off the point, and the light should be given a berth of about 0.3 mile, even by small craft. A gong buoy is 0.6 mile northward of the point. Depths of 14 to 18 feet are found about 0.4 mile northward of the light.

(310) **Crane Neck Point**, 2 miles westward of Old Field Point, is a bare conspicuous bluff about 90 feet high and covered on top with brush.

(311) **Smithtown Bay**, a broad open bight on the south side of the sound, extends 7 miles westward from Crane Neck Point. Rocky shoals extend 1 mile in places from the shore, the water shoaling abruptly from 51 feet in places. A good summer anchorage in 30 to 50 feet sheltered from easterly winds is found about 1 mile southward of Crane Neck Point.

(312) **Stony Brook Harbor** is a narrow shallow bay in the southeastern part of Smithtown Bay. The approach to the harbor from the bay is over a bar that extends 0.8 mile off the entrance; the outer end of the bar is marked by a seasonal lighted buoy and the approach to the harbor is marked by private lighted buoys. In 1981, 3½ feet was reported over the bar. Two branch channels lead from the entrance into the harbor; one leads southwestward to a steel bulkheaded yacht club wharf and pavilion at the village of **Stony Brook**, 0.5 mile inside the entrance, and the other, **Porpoise Channel**, leads westward to a yacht club at the northwestern end of the harbor; gasoline is available at both clubs. In 1994, a depth of 6 feet was reported in both the southwesterly channel and Porpoise Channel. The channels are marked by private seasonal lighted and unlighted buoys. The buoys are periodically moved to mark the best water.

(313) A **speed limit** of 5 mph is enforced in Stony Brook Harbor and Porpoise Channel.

(314)

Small-craft facilities

(315) Small-craft facilities are in the harbor.

(316) The railroad station is about 1 mile from the wharf at Stony Brook.

(317) A high bluff is between Stony Brook Harbor and Nissequogue River, another between Nissequogue River and **Sunken Meadow Creek**, and bluffs in places between Sunken Meadow Creek and Northport Bay.

(318) **Nissequogue River**, a shallow crooked stream about 4 miles westward of the entrance to Stony Brook Harbor, is entered through a privately dredged channel that leads southward from Smithtown Bay for about 1.4 miles into the river. Rocks and shoals, bare at low water, are on the bar outside the entrance. Private seasonal lighted buoys mark the channel. Strong tidal currents are reported in the channel. A **speed limit** of 5 mph is enforced on the river. Guest moorings, gasoline, water and limited supplies are available at a marina on the west side of the river, about 0.9 mile above the channel entrance. In 1995, a depth of 3 feet was reported alongside the marina. A state hospital, a group of buildings with green roofs, and two large red brick chimneys are prominent about 0.5 mile southwestward of the river entrance. Farther westward, a brick building and a stack are also prominent. The railroad station is at **Kings Park**.

(319)

Northport to Oak Neck Creek

(320) **Northport Basin**, about 10.5 miles westward of Old Field Point Light and 2.7 miles southeastward of Eatons Neck Point, is a small privately maintained basin with general depths of 7 to 15 feet and formed by gravel dredges working into the high bank. The channel is marked by a private lighted buoy and unlighted buoys; submerged jetties extend northward from the east and west sides of the entrance. A dangerous rock is close northward of the seaward end of the west jetty. The four stacks of a power and light company on the east side of the basin are prominent. A town launching ramp is in the basin.

(321) An aquaculture site, marked by a private buoy, is about 1.2 miles northwestward of the entrance to Northport Basin.

(322)

Offshore Terminal, Northport

(323) An offshore platform for the receipt of oil is off Northport. The terminal is owned and operated by National Grid Generation LLC. The platform, with offlying mooring buoys, is about 1.6 miles northward of the entrance to Northport Basin and about 2.4 miles eastward of Eatons Neck Light. Submerged pipelines extend from the shore to the platform. The platform is marked at its eastern end by a private light and at the western end by a private light and sound signal.

(324) Upon the scheduled approach of an incoming vessel to the platform, voice call "Northport Power Station." Northport Power Station control room monitors VHF-FM channel 19.

(325)

Pilotage, Offshore Terminal, Northport

(326) Pilotage by a state-licensed pilot is compulsory in Long Island Sound for foreign flag vessels and U.S. vessels that are under register (i.e., engaged in foreign trade). Such vessels can arrange for a state-licensed pilot by contacting the joint rotation administrator, Block Island Pilots at 243 Spring Street, Newport, RI 02840; telephone 401–487–9050 (24 hours), 800–274–1216; FAX 401–847–9052. Enrolled vessels (i.e., U.S. vessels engaged in coastwise trade) may be required to have a U.S. Coast Guard Federally licensed pilot unless the master has recency for the intended area. See Pilotage, Long Island Sound (indexed as such), chapter 8 and Pilotage, New York Harbor and Approaches (indexed as such), chapter 11.

(327)

Tugs

(328) Tug service is available from New Haven, Providence, Brooklyn or Staten Island on advance notice.

(329) **Eatons Neck** is a prominent wooded headland with elevations of 100 feet or more and marked at its north end by a light and tower of **Eatons Neck Coast Guard Station**.

(330) **Eatons Neck Light** (40°57'14"N., 73°23'43"W.), 144 feet above the water, is shown from a white stone tower on the north end of Eatons Neck.

(331) The northwest end of the neck is a spit in the form of a hook that encloses **Eatons Neck Basin**. Eatons Neck Coast Guard Station is at the head of the basin. The basin is entered through a privately dredged cut between two small riprap jetties about 0.5 mile southwestward of the light; the jetties are covered at half tide. The channel between the jetties is buoyed, and there are buoys farther inside the basin. The basin is subject to frequent changes and the buoys in the basin are not charted because they are frequently shifted in position. In 1994, depths of 10 feet could be carried through the entrance. An obstruction is in the entrance channel around 40°56'50"N., 73°24'06"W.

(332)

Caution

(333) Eatons Neck Basin Channel is maintained expressly to enhance the Eatons Neck Coast Guard Station's rescue response. Further, Eatons Neck Basin has become one of the most congested small-boat anchorages in the area in the summer. Mariners are cautioned that heavy wakes from rescue craft departing the station may be experienced by small craft anchoring in this area.

(334) Shoals with depths of 4 to 18 feet extend about 0.9 mile northward of Eatons Neck, and broken ridges extend northward for another 1.8 miles. The northern end of each area is marked by a buoy.

(335) **Huntington Bay**, just westward of Eatons Neck, is the approach to Northport Bay and Harbor, Centerport Harbor, Huntington Harbor and Lloyd Harbor. The bay, protected against all but northerly winds, is an excellent anchorage for large vessels. Depths range from 25 to 36 feet, fairly close to its southern end, and anchorage can be selected according to draft and wind direction.

(336) A **017°56'–197°56' measured half nautical mile** is on the west side of Eatons Neck. Triangular orange shore ranges mark the ends of the course.

(337) Anchorage with shelter from northwesterly winds can be had for small vessels at the southwesterly end of Huntington Bay, 0.4 mile northeastward of Huntington Harbor Light, in 18 to 36 feet. The arms of the bay provide secure harbors; Northport Bay is used generally by the larger vessels.

(338)

Current

(339) In Huntington Bay the velocity of the tidal current is 0.5 knot off East Fort Point and 0.4 knot in the entrance to Northport Bay. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(340) **Duck Island Harbor** is a shallow cove on the north side of Northport Bay westward of **Duck Island Bluf**. Depths range from 6 to 9 feet in the entrance. The south side of Duck Island Bluff and the southeast side of **Winkle Point** should be given berths of 300 and 400 yards, respectively, to avoid shoal water and inshore rocks.

(341) A 5 mph **speed limit** is enforced in Duck Island Harbor.

(342) **Northport Bay**, which opens off the southeast end of Huntington Bay, provides good anchorage in 20 to 50 feet in its western part and in 8 to 11 feet in the eastern half. The entrance to the bay is marked by a lighted buoy, and the entrance channel, privately dredged to about 12 feet, is buoyed.

(343) An amber light, maintained at the public landing by the town of Northport, is a conspicuous mark at night for vessels making the wharves at Northport.

(344) A privately dredged channel at the eastern end of Northport Bay leads to a dredge basin formerly used by a sand and gravel company on the north side of **Bluf Point**. Several private landings and moorings are in the basin.

(345) **Northport Harbor** is at the southeastern end of Northport Bay and is entered by a dredged channel that leads along the waterfront of Northport and an anchorage basin west of the village. The channel is marked by private seasonal buoys. In 2016-2021, the controlling depth was 4.9 feet in the channel; 6 feet was available in the anchorage basin in 2015. A channel leads from the town landing to a boatyard and marina at the southeast end of the harbor and is marked by private seasonal buoys. The boatyard channel is marked by buoys and by a lighted buoy at the entrance; these aids are seasonal and privately maintained. An alternate channel, marked by private buoys, leads from opposite the public landing along the west side of the harbor to the head. A 5 mph **speed limit** marker is in the entrance to the harbor.

(346) **Bird Island**, a bird sanctuary in the southern part of the harbor, is a low, grass-covered man-made island.

(347)

Ice

(348) During severe winters, ice may close the harbor for about 2 months.

(349)

Anchorages

(350) Vessels select anchorage according to draft in the harbor; bottom is soft.

(351) A **special anchorage** is in Northport Harbor. (See **33 CFR 110.1** and **110.60 (a-2)**, chapter 2, for limits and regulations.)

(352) **Northport** is a village with bus communications on the eastern shore of Northport Harbor. Depths at the

principal wharves are about 6 to 8 feet. The greatest depth that can be taken to Northport is about 14 feet at high water.

(353)

Small-craft facilities

(354) Several small-craft facilities are on the east side and the head of the harbor, and a yacht club is on the west side.

(355) **Centerport Harbor** is a shoal bight on the south shore of Northport Bay just eastward of the entrance. The harbor serves the small-boat interests of the village of Centerport. In 1981, a reported depth of about 7 feet could be taken through the privately dredged channel to the spit extending southwesterly from Little Neck, thence about 3 feet to a boatyard on the west side of the harbor just below the bridge. The channel is marked by private seasonal buoys. Berths, moorings, electricity, water, storage, marine supplies and a launching ramp are available. A flatbed trailer can haul out craft to 32 feet; hull and engine repairs can be made.

(356)

Anchorages

(357) A **special anchorage** is in Centerport Harbor. (See **33 CFR 110.1** and **110.60 (a-1)**, chapter 2, for limits and regulations.)

(358) **Huntington Harbor**, at the southwest end of Huntington Bay, is entered through a marked channel that leads to an anchorage off Huntington Town Dock, about 2 miles above the channel entrance. A depth of about 8 feet can be carried in the channel. **Huntington Harbor Light** (40°54'39"N., 73°25'52"W.), 42 feet above the water and shown from a square concrete tower attached to a dwelling on a rectangular pier, is on the west side of the entrance to Huntington Harbor and on the south side of the entrance to Lloyd Harbor. A sound signal at the light is operated by keying the microphone five times consecutively on VHF-FM channel 83A.

(359) The channel is marked by a light and by lighted, unlighted and private unlighted buoys. Some of the private buoys are seasonal.

(360) The wharf just southward of Huntington Town Dock South is used by sand and gravel barges. The **bay constable** has an office at the head of the harbor immediately southward of Huntington Town Dock North.

(361) A boulder reef, on the west side of the entrance, extends out to Huntington Harbor Light. An obstruction, reported covered 4½ feet, is 0.35 mile eastward of the light.

(362) In 1991, a dangerous wreck was reported between Buoys 9 and 11 in about 40°53'54.9"N., 73°25'46.1"W.

(363)

Current

(364) The tidal currents in the entrance channel have an estimated velocity of 2 knots.

(365)

Anchorages

(366) A **special anchorage** is in Huntington Harbor. (See **33 CFR 110.1** and **110.60**, chapter 2, for limits and regulations.)

(367) A 5-mph **speed limit** is enforced in the harbor.

(368) **Huntington** and **Halesite** are villages at the head of the harbor. The yacht club landing on the east side of the harbor has a depth of about 10 feet alongside. Gasoline, diesel fuel, berths, electricity, water and ice can be obtained here. Yachts may anchor off the landing but must keep clear of the channel.

(369) Coindre Hall, a large brick building with a red roof and numerous chimneys at the entrance to the harbor, and Huntington Hospital, well lighted at night, at the head of the harbor are prominent.

(370)

Small-craft facilities

(371) There are several marinas, boatyards and private boat clubs in Huntington Harbor.

(372) **Lloyd Harbor** extends westward from Huntington Bay nearly to Oyster Bay, from which it is separated by a narrow strip of land. Vessels can anchor just inside the entrance, in depths of 7 to 11 feet. The entrance to the harbor is marked by buoys. A **speed limit** of 5 mph is enforced in the harbor.

(373) **Oyster Bay**, on the south side of Long Island Sound about 5 miles westward of Eatons Neck Light, lies between Lloyd Neck and Rocky Point and is the approach to Cold Spring Harbor and Oyster Bay Harbor. The harbor is marked by **Cold Spring Harbor Light** (40°54'51"N., 73°29'35"W.), 37 feet above the water, and shown from a skeleton tower on a caisson with a red and white diamond-shaped dayboard. The entrance and harbor are characterized by extensive shoals, boulder reefs and broken ground making off from the shores. Vessels should proceed with caution if obliged to approach or cross shoal areas. The bay south of Cold Spring Harbor Light is a secure harbor, available for vessels of less than 18-foot draft.

(374) **Lloyd Neck**, between Huntington and Oyster Bays, is high and wooded and has a high, yellow bluff on its north side 0.8 miles eastward of Lloyd Point. Many patches of boulders having least depths of 2 to 8 feet extend 0.2 to 0.5 mile offshore from **East Fort Point** to Lloyd Point. Small craft skirting this shore should keep well outside the line of buoys.

(375) **Lloyd Point**, the north end of Lloyd Neck, is a low spit. A rocky shoal extends 0.5 mile north-northeastward from Lloyd Point. A lighted gong buoy about 1 mile northward of Lloyd Point marks the northern limit of the 30-foot curve in this vicinity. The buoy is removed when endangered by ice.

(376) **Morris Rock**, about 0.5 mile eastward of Lloyd Point, is awash at low water—the rock is marked by a buoy.

(377) The long jetty, about 0.6 mile southwestward of Lloyd Point, forms the southern entrance point to **The Sand Hole**, a pond that has been dredged into the spit by a sand and gravel company. The pond is state controlled and may be entered by steering a midchannel course through the entrance. It is used considerably by local boats as an anchorage and harbor of refuge. The holding ground is good. In 1981, reported depths of about 12 feet were in the entrance channel and about 4 to 22 feet in the basin.

(378) **Rocky Point**, the northern promontory of **Centre Island**, is a small bluff on whose summit is a large prominent house. An extensive foul area with depths of 2 to 17 feet extends about 1 mile northward of Rocky Point. A buoy marks the northern end of this foul area. The area is dangerous and should be avoided.

(379) A shoal area with depths of 4 to 11 feet extends eastward from Rocky Point nearly across Oyster Bay and is marked near its eastern end by Cold Spring Harbor Light. Small craft with local knowledge cross the shoal at a distance of about 0.4 mile westward of the light, but strangers should not attempt it.

(380)

Currents

(381) About 0.4 mile northwest of Cold Spring Harbor Light the velocity is about 0.5 knot; about 0.2 mile north of Cove Point, 1.2 miles southwestward, it is about 0.8 knot. For predictions, see the Tidal Current prediction service at *tidesandcurrents.noaa.gov*. Links to a user guide for this service can be found in chapter 1 of this book.

(382)

Ice

(383) During severe winters ice has been known to extend the full length of the bay during part of January and February.

(384) **Plum Point**, the easternmost point of Centre Island, is marked at its south end by a small stone tower; boat landings are on the southwest side of the point. A yacht club with a prominent flagstaff is about 0.3 mile west of Plum Point. The yacht club landing has reported depths of about 9½ feet.

(385) **Cooper Bluf**, at the northeast end of Cove Neck is prominent. A boulder reef extends nearly 0.3 mile northward from **Cove Point** at the northwest end of **Cove Neck**, and is marked by a lighted buoy that is replaced with an unlighted buoy in the winter.

(386) **Cold Spring Harbor**, the southeasterly end of Oyster Bay, extends about 2.3 miles southward of Cooper Bluff. The tower on top of a dome of a seminary on the hill of **West Neck**, on the east side of the harbor, is prominent. A depth of about 14 feet can be carried to near the head of the harbor by giving the shores a berth of about 0.3 mile.

(387) The village of **Cold Spring Harbor** is on the eastern shore near the head of the harbor. A small-craft facility is on the east side of the cove at the head of Cold Spring Harbor. Gasoline, diesel fuel, water, ice, marine supplies, berthings and dry storage are available. A reported depth of about 3 feet is available alongside the facility. A town launching ramp is available in the harbor. A **speed limit** of 5 mph is enforced in the harbor.

(388)

Anchorages

(389) **Special anchorage areas** are in Cold Spring Harbor and Oyster Bay Harbor—see **33 CFR 110.1** and **110.59**, chapter 2, for limits and regulations.

(390) **Oyster Bay Harbor**, a long, crooked arm in the western side of Oyster Bay, has a channel with a depth over 30 feet leading into the area westward of **Moses Point**. Good anchorage is available southward of Moses Point. West of this point, the channel is narrow and suitable only for vessels drawing less than 10 feet. Vessels of less than 7-foot draft can anchor in the bight between Cove Neck and the wharf at Oyster Bay and also in **West Harbor**, the large bight on the northwest side of Centre Island. A **speed limit** of 5 mph is enforced in the harbor.

(391) The village of **Oyster Bay** is on the shore south of Oyster Bay Harbor. A harbor channel leads westward between the two anchorage areas south of Centre Island. Three separate channels lead south from the harbor channel to an oyster wharf, boat basin and launching ramps—all the channels are marked by private buoys. The oyster wharf has reported depths of about 10 feet along the face and southeast side. Parallel to and about 200 feet off the northwest side of the wharf is a row of sunken barges. An oil receiving wharf is about 125 yards southward of the oyster wharf.

(392) The waters of Oyster Bay Harbor and Mill Neck Creek are part of the **Oyster Bay National Wildlife Refuge**.

(393)

Small-craft facility

(394) Two small-craft facilities are at Oyster Bay, one in the basin and the other just east of the entrance to the basin. Berths and moorings, electricity, gasoline, diesel fuel, water, ice, pump-out facilities, a launching ramp, storage and full repairs are available.

(395) **Brickyard Point**, about 0.5 mile westward of Moses Point, should be given a berth of at least 0.2 mile off its westerly side to avoid several dangerous rocks to the northwestward of the point. None of these rocks is marked. Extensive privately owned oyster beds, marked by stakes, are in this area.

(396) **Mill Neck Creek**, at the northwest end of Oyster Bay Harbor, is crossed by a highway bridge having a bascule span with a clearance of 9 feet—see **33 CFR 117.1** through **117.59** and **117.800**, chapter 2, for drawbridge regulations. A marina, with an approach depth of 6 feet, is on the north side of the river near the bridge. The marina can provide gasoline, diesel fuel, electricity, water, ice, pump-out, marine supplies, launching ramp and full repairs.

(397) **Oak Neck Creek**, northwest of Mill Neck Creek, is entered at high water as the creek is practically bare at low water.

(398)

Oak Neck Point to Little Neck Bay

(399) **Oak Neck Point** (40°54.9'N., 73°34.1'W.), 4 miles west-southwestward of Lloyd Point, is marked by many large residences. Several stone jetties extend a short distance from the shore just westward of the point. A shoal, strewn with boulders and marked by a buoy, extends 0.3 mile from the shore for part of the distance between Oak Neck Point and Matinecock Point to the westward.

(400) **Frost Creek**, locally known as Guthries Creek, 2 miles westward of Oak Neck Point, has a channel at the entrance that is well defined when the water is below half tide. The creek is protected by a stone jetty that extends a short distance from the shore about 50 yards eastward of the channel. The channel has a reported depth of about 1 foot near the entrance. The creek is not recommended without local knowledge.

(401) **Peacock Point** is just west of Frost Creek. A stone jetty to protect a private boat landing extends a short distance from the west side of the point.

(402) **Matinecock Point** is 1.1 miles westward of Frost Creek—a prominent flagpole is on the point. A shoal extends about 600 yards off the point and is marked at its end by a lighted gong buoy that is removed if endangered by ice.

(403) **Hempstead Harbor**, 4 miles wide at the entrance between Matinecock Point and Prospect Point, is free from dangers if the shores, between the entrance and Mosquito Cove, are given a berth of 0.3 mile. It is much used by vessels seeking shelter in any but strong northerly winds and affords excellent anchorage with good holding ground. Vessels can anchor in any part of the harbor according to draft and direction of wind. A good anchorage for vessels drawing less than 20 feet is just inside a line from Mott Point to the breakwater at Glen Cove Landing. Small vessels can anchor behind the breakwater. Vessels should avoid anchoring in the pipeline area between Glenwood Landing and Bar Beach. A 5 mph **speed limit** is enforced in the harbor.

(404) Waterborne commerce in the harbor is in sand, gravel, petroleum products and building material. Vessels engaged in this commerce usually draw from 3 to 12 feet.

(405)

Anchorages

(406) A **special anchorage** is in Hempstead Harbor—see **33 CFR 110.1** and **110.60**, chapter 2, for limits and regulations.

(407) **Weeks Point**, on the eastern side near the entrance, is marked by a breakwater that protects a private boat landing. Nearly 0.5 mile southward of Weeks Point is the entrance to a basin protecting a private wharf that has a reported depth of 8 feet at the end. The basin shoals to the head, and there are rocks bare at low water near the northern end.

(408) **Glen Cove** is a city with rail and bus communication on Glen Cove Creek, about 1 mile back from the eastern shore of the bay. The breakwater extends 500 yards west-southwestward from **Glen Cove Landing** and is marked at its end by a light. The anchorage behind the breakwater has depths ranging from 18 to 22 feet behind its outer half and 7 to 9 feet near shore. A ramp is located north of the Glen Cove Creek entrance.

(409) **Glen Cove Creek**, 0.6 mile southward of the breakwater, is entered through a dredged channel from **Mosquito Cove**. An overhead power cable near the head of the creek has a clearance of 65 feet. The entrance is marked by buoys. There are several small-craft facilities in Glen Cove Creek.

(410) A dredged channel, entered between Bar Beach and Glenwood Landing, leads alongside Glenwood Landing to South Glenwood Landing at Motts Cove. A natural channel continues south through extensive flats for about 0.5 mile with a depth of about 5 feet; local knowledge is advised.

(411) **Sea Cliff** is a village on the steep hill on the south side of Glen Cove Creek. From Sea Cliff southerly to the northerly wharves at Glenwood Landing, a shoal extends 300 yards from the east side of the harbor and is marked by a buoy at the north end and a light at the south end. A dredged entrance channel, marked by two private lights, leads from deep water in the harbor northeastward to a municipal marina just north of Glenwood Landing. In 1999, the reported controlling depths were 8 feet in the entrance channel, thence 7 feet in the marina basin.

(412) **Glenwood Landing** is a village on the eastern shore abreast Bar Beach. An overhead power cable crossing from Bar Beach to Glenwood Landing has a clearance of 90 feet. Depths of about 8 to 10 feet are available at the Glenwood Landing wharves.

(413)

Current

(414) In the channel west of the breakwater the tidal currents are weak and variable. At Bar Beach the tidal currents have a velocity of about 0.8 knot through the narrow channel—see the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(415)

Ice

(416) In severe winters ice has been known to close navigation for about 6 weeks during January and February.

(417) The shore between Prospect Point and **Mott Point** (40°51.4'N., 73°40.6'W.), to the southeastward, is marked by prominent bluffs. A shoal with boulders extends 0.2 mile from shore between the points and for a short distance south of Mott Point. Buoys mark the limits of the

shoal eastward and northeastward of Mott Point. **Picket Rock**, with 2 feet over it, is 350 yards offshore northward of Mott Point.

(418) **Prospect Point**, marked by prominent houses on the bluff, has a rocky shoal making out nearly 0.5 mile northward from it—the shoal rises abruptly from a depth of 60 feet. A lighted gong buoy is off the north end of the shoal.

(419) **Sands Point**, 0.7 mile west of Prospect Point, is marked by a prominent stone tower. A boulder reef extends about 0.3 mile off the point and is marked by a lighted buoy off the northwest end. The boulders show at low water for a distance of about 300 yards from shore. The outer end of these visible boulders is marked by a daybeacon.

(420) **Barker Point**, about 1 mile south-southwest of Sands Point, is a high bluff on the northeast side of the entrance of Manhasset Bay. **Gangway Rock**, marked by a light and gong buoy, is at the northwesterly end of a broken line of rocks and shoal water that extends 0.6 mile northwestward from Barker Point. **Success Rock**, awash at low water and marked by a buoy, is about 0.2 mile southeastward of the light.

(421) **Manhasset Bay**, between Barker Point and Hewlett Point, affords excellent shelter for vessels of about 12 feet or less draft and is much frequented by yachts in the summer. The depths in the outer part of the bay range from 12 to 17 feet and 7 to 12 feet in the inner part inside Plum Point. The extreme south end of the bay is shallow with extensive mudflats. Depths of about 6 to 2 feet can be taken through a natural channel almost to the head of the bay. A 5 mph **speed limit** is enforced.

(422) Waterborne commerce is in petroleum products, carried in vessels drawing 6 to 10 feet.

(423)

Anchorages

(424) Special anchorages are in Manhasset Bay—see **33 CFR 110.1** and **110.60**, chapter 2, for limits and regulations. The bottom is soft and affords good holding ground.

(425) A seaplane **restricted area** is off Manorhaven—see **33 CFR 162.15**, chapter 2, for limits and regulations.

(426) **Plum Point** is a low spit extending southward from the eastern shore about 0.6 mile southward of Barker Point. A lighted entrance buoy is about 150 yards southwest of Plum Point—the buoy is replaced with an unlighted buoy when endangered by ice. The bight eastward of Plum Point is shoal.

(427) **Port Washington** is a village with rail communication on the south side of a shoal bight about 1.2 miles southeastward of Plum Point. An apartment complex on **Toms Point**, 0.9 mile east of Plum Point, is prominent. Depths of about 8 feet can be carried in the buoyed approach from the lighted buoy off Plum Point to the docks at Port Washington, thence through the unmarked channel along the east side of the bight to its north end northeastward of Toms Point. In 1981, depths of 5 feet were reported on the north side of the town dock with 2 and 4 feet on the west and south sides, respectively. Depths at the other wharves are reported to range from 4 to 9 feet. The town's Bay Constable monitors VHF-FM channels 9 and 16 from the town dock.

(428)

Small-craft facilities

(429) There are extensive small-craft facilities at Port Washington and to the eastward and westward of Toms Point at **Manorhaven.**

(430) **Hewlett Point** (40°50.3'N., 73°45.2'W.) is on the west side of the entrance to Manhasset Bay. A boulder reef, mostly bare at low water and marked by a lighted buoy at its northern end, extends about 0.2 mile northward from the point.

(431)

Anchorages

(432) A special anchorage is north of Elm Point—see **33 CFR 110.1** and **110.60**, chapter 2, for limits and regulations.

(433) **Stepping Stones Light** (40°49'28"N., 73°46'29"W.), 46 feet above the water, is shown from a red brick structure on a granite pier, 1.3 miles southwest of Hewlett Point. The **Stepping Stones**, a dangerous boulder reef that dries in places, extends 0.8 mile southeastward from the light to the Long Island shore.

(434) **Kings Point Coast Guard Station** is located at the northern end of the Kings Point boat basin.

(435) **Kings Point**, marked by a private light, is 1.6 miles south-southwestward of Hewlett Point and is the site of the **U.S. Merchant Marine Academy**. The 172-foot unguyed steel flagpole at the academy is said to be the country's tallest; the top of the pole is 216 feet above the water. A boat basin, partially enclosed by an L-shaped pier, is at the point. In 1991, the basin had a reported depth of 11 feet.

(436) **Little Neck Bay** is entered between Kings Point and Willets Point, 1.2 miles to the south-southwestward. Depths are 10 to 12 feet in the entrance, decreasing gradually to the head, about 2 miles inland, where the bay divides into two branches that almost dry; there are boulders in places close to the shores.

(437) The shores of Little Neck Bay are thickly settled, and there are many private boat landings. A much used anchorage, in depths of 2½ to 7 feet, is in the cove midway along the east side of the bay.

(438)

Small-craft facility

(439) A small-craft facility is on the west side of the bay. Water, ice and limited marine supplies are available. In 1981, the facility had a reported depth of 4 feet alongside.

(440)

Anchorages

(441) A **special anchorage** is in Little Neck Bay—see **33 CFR 110.1** and **110.60**, chapter 2, for limits and regulations.

(442)

East River

(443) **East River** is a 14-mile-long tidal strait that connects Long Island Sound with New York Upper Bay and separates the western end of Long Island from the New York mainland. The Sound entrance is between Throgs Neck and Willets Point; the Upper Bay entrance is between The Battery and Governors Island. Hell Gate, about halfway between Throgs Neck and The Battery, is noted for its strong tidal currents. Harlem River extends northward from Hell Gate to the Hudson River. Both sides of the East River, from The Battery to Port Morris, a distance of 9 miles, present an almost continuous line of wharves except where shoals or currents prevent access.

(444)

Channels

(445) A federal project provides for main-channel depths of 35 feet from Throgs Neck to the inactive New York Naval Shipyard, about 2 miles from the western entrance, and thence 40 feet to deep water in New York Upper Bay. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through the USACE hydrographic survey website listed in Appendix A.

(446)

Caution

(447) Mariners transiting East River in the vicinity of Rikers Island and/or South Brother Island Channel are advised of the following:

(448) East River Main Channel Lighted Buoy 5 has been established northeast of Rikers Island in 40°47'47"N., 73°51'57"W. to ensure that no vessel penetration of air space exists over that portion of the East River that coincides with the glide path of the northeast-southwest runway of La Guardia Airport. Vessels with mast heights in excess of 125 feet shall pass 100 yards to the north of this buoy so as to avoid interference with the glide path.

(449) Vessels transiting South Brother Island Channel and using the turning basin at its southern terminus shall ballast prior to entry and are cautioned that mast heights in excess of 125 feet may penetrate the glide path to the northwest-southeast runway to La Guardia Airport. If mast heights cannot be lowered below 125 feet, La Guardia Air Traffic Control Tower shall be notified at 212–779–0242 prior to terminal departure or channel entry.

(450)

Anchorages

(451) Several **general** and **special anchorages** are in East River—see **33 CFR 110.1**, **110.60**, and **110.155**, chapter 2, for limits and regulations.

(452)

Current

(453) In East River the flood current sets eastward and the ebb sets westward. **Note**: this is the direct opposite of conditions in Long Island Sound where the flood is generally westward and the ebb eastward.

(454) The velocity of current is 0.7 knot at Throgs Neck, 1.6 knots at Port Morris, 4 knots in Hell Gate, 3 knots at Brooklyn Bridge, and 1.5 knots north of Governors Island. In Hell Gate (off Mill Rock) the velocity is 3.4 knots for the eastward current and 4.6 knots for the westward current.

(455) The direction and velocity of the currents are affected by strong winds that may increase or diminish the periods of flood or ebb. The currents generally set with the channel, but heavy swirls are found in Hell Gate.

(456)

Tides

(457) See the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for the daily predictions of slack water and times and velocities of strengths of currents in Hell Gate and at other places on the East River. Mariners should exercise caution and discretion in the use of published tidal current predictions.

(458)

Pilotage, East River

(459) See Pilotage, New York Harbor from Long Island Sound (indexed as such), chapter 11.

(460)

Towage

(461) Vessels intending to employ a tug should arrange to do so before proceeding westward of Rikers Island.

(462)

Throgs Neck to Unionport

(463) **Throgs Neck**, on the northwest side of the entrance to East River, is marked by a light. **Throgs Neck Light** (40°48'16"N., 73°47'26"W.), 60 feet above the water, is shown from a skeleton tower with a black and white diamond-shaped dayboard on the outer end of the neck. The shoal ground that extends 0.1 mile southward and eastward from the light is marked by a lighted bell buoy.

(464) **Fort Schuyler**, on the outer end of Throgs Neck, is used as a base for the **New York Maritime College**. The 550-foot-long wharf, on the southwest side of the fort, is used to moor the school's training ship. Depths of about 25 feet are reported alongside the face.

(465) **Throgs Neck Bridge**, a highway suspension bridge with a channel clearance of 138 feet and 152 feet at the center, crosses East River from Throgs Neck to the Long Island Shore.

(466) **Willets Point**, 0.7 mile southeastward across the entrance to East River from Throgs Neck, is marked by **Fort Totten**, the granite walls of which are prominent. **Little Bay**, westward of Willets Point, has general depths of 6 to 10 feet and is used by local small craft. Depths

(469)

of about 9 feet can be taken in the buoyed channel to the piers on the Little Bay side of Willets Point. The southern approach viaduct of the Throgs Neck Bridge crosses the west part of Little Bay. The fixed spans of the viaduct have a minimum clearance of 30 feet.

(467) **Whitestone Point**, 2 miles westward of Willets Point, is a small bluff marked by a light. The town of **Whitestone** is between Little Bay and Whitestone Point. Several private boat clubs are at Whitestone. In 1981, reported depths alongside the boat club docks ranged from ½ to 6 feet.

(468) The **Bronx-Whitestone Bridge** is a suspension structure that crosses East River from Old Ferry Point on the Bronx side to a Long Island landing 0.4 mile southwestward of Whitestone Point. The bridge has a clearance of 130 feet with 135 feet at the center; a traveling maintenance platform reduces vertical clearances by 14 feet when in operation.

(470) **Powell Cove**, between the Long Island end of the Bronx-Whitestone Bridge and Tallman Island, 0.6 mile to the westward, has general depths of 2 to 5 feet. Pier ruins are on the east side of the cove entrance. **Tallman Island**, now joined to the Long Island shore, is marked by the prominent tanks of the NYC DEP Water Pollution Control Plant.

(471) **Old Ferry Point** is on the north side of East River 2 miles westward of Throgs Neck. The bight between Throgs Neck and Old Ferry Point affords anchorage, with good holding ground, in depths of 15 to 35 feet; the water shoals abruptly from 18 feet, 0.3 mile from shore, to depths of 4 to 5 feet. Several private landings are on the north side of this bight. Tug and barge companies maintain unlit commercial mooring buoys in Anchorage Ground 6 for their own vessels.

(472) Numerous obstructions exist in East River between Throgs Neck and the entrance to Westchester Creek. Mariners are advised to use the chart as a guide.

(473) **Westchester Creek**, on the north side of East River, is entered through a dredged channel that leads northward through a shallow bight between Old Ferry Point and Clason Point—0.7 mile westward—to the head of navigation at **Westchester**, about 2.3 miles above the channel entrance. The channel is buoyed to a point about 0.8 mile above the entrance. Waterborne traffic on the creek consists chiefly of petroleum products, sand, gravel and crushed rock.

(474) Several highway bridges, three fixed and one bascule, cross Westchester Creek at **Unionport,** 1.5 miles above the channel entrance. The Bruckner Expressway bascule bridge is under construction (2019), and the fixed bridges have a least clearance of 52 feet—see **33 CFR 117.1** through **117.59** and **117.815**, chapter 2, for drawbridge regulations. The bridgetender at the Bruckner Expressway bridge monitors VHF-FM channel 13; call sign KX-8289.

(475)

Small-craft facilities

(476) There is a small-craft facility on the west side of the creek at Unionport. Water, limited supplies and storage facilities are available.

(477)

Clason Point to Mill Rock

(478) **Clason Point** (40°48.3'N., 73°50.9'W.) is on the north side of East River about 3 miles west of Throgs Neck. **Pugsley Creek**, which empties into Westchester Creek and East River along the east side of Clason Point, is very shallow and should not be entered without local knowledge. Small boats anchor on the flats west of Clason Point.

(479) **College Point** is on the Long Island side of East River opposite Clason Point. **College Point Reef**, covered 6 feet and marked by a light, is 0.2 mile north-northeastward of the point.

(480) The town of **College Point** is south of the point and on the east side of the entrance to Flushing Bay. The wharves on the west side of the town have depths alongside ranging from ½ to 10 feet. The shallow bight north of the town has depths of 2 to 5 feet and is used as a small-boat anchorage.

(481)

Small-craft facilities

(482) Several small-craft facilities are at College Point. Marine railways to 45 feet, mobile cranes to 35 tons, water, ice, marine supplies, storage and hull and engine repairs are available.

(483) **Flushing Bay** extends southeast between the town of College Point and La Guardia Airport, 0.6 mile to the southwest. **Flushing Creek** flows into the east side of the head of the bay. A dredged channel extends from the East River into the creek; the channel is marked by lighted and unlighted buoys. A turning basin is on the west side of the dredged channel west of the entrance to Flushing Creek and a small-craft anchorage area is on the northwest side of the turning basin. Flushing Bay is mostly shallow, with depths of less than 6 feet outside the channel.

(484)

Anchorages

(485) **General** and **special anchorages** are in Flushing Bay—see **33 CFR 110.1**, **110.60** and **110.155**, chapter 2, for limits and regulations. Small-craft anchor south of College Point in depths of 4 to 8 feet.

(486) A **restricted area** is in a portion of the southern part of the channel through Flushing Bay—see **33 CFR 162.20**, chapter 2, for limits and regulations.

(487) A 0.6-mile-long dike, covered at high water, runs close along the west side of the channel to within 0.3 mile of the head of the bay. The dike is marked by lights at the ends and by lighted buoys along its length.

(488) The L-shaped pier at the head of Flushing Bay partially encloses a small-boat basin. Inside the small-boat basin, depths of about 7 feet were reported in 1981. The marina to the westward has a reported depth of about 5 feet inside. Gasoline, diesel fuel, berths, electricity, water, ice, storage and a 30-ton hoist are available; limited electronic and engine repairs can be made.

(489)

Ice

(490) Ice generally obstructs navigation in Flushing Bay and Flushing Creek during a part of January and February.

(491) The fixed spans of the Whitestone Expressway highway bridges over Flushing Creek, 0.2 mile above the mouth, have a clearance of 34 feet. The Van Wyck Expressway fixed highway bridge, also 0.2 mile above the mouth, is under construction (2005). The Northern Boulevard Bridge, 0.4 mile above the mouth, has a fixed span with a clearance of 35 feet.

(492) **Flushing** is on the east side of Flushing Creek. Waterborne traffic consists chiefly of sand, gravel, crushed rock and petroleum products. Drafts of inbound and outbound vessels seldom exceed 12 feet. Vessels must go directly to the marginal wharves because the creek has no room for anchorage.

(493) The east entrance to **Rikers Island Channel**, between Rikers Island and the mainland, is obstructed by a lighted runway approach to **La Guardia Airport**. The approach to Bowery Bay is from westward of Rikers Island.

(494) **Bronx River**, on the north side of East River, has a project depth of 10 feet and is subject to shoaling throughout. It is entered through a dredged channel that leads north-northwestward through a shallow bight between Clason Point and Hunts Point, 1.1 miles to the westward, to the head of river navigation at East 172nd Street, about 2.3 miles above the channel entrance—see Notice to Mariners and latest editions of charts for controlling depths. Waterborne traffic on the Bronx River consists chiefly of sand, gravel and crushed rock.

(495) Bronx River is crossed by four bridges to East 172nd Street. Bruckner Expressway Bridge, 1.7 miles above the entrance, has a bascule span with a clearance of 27 feet—see **33 CFR 117.1** through **117.59** and **117.771**, chapter 2, for drawbridge regulations. Westchester Avenue Bridge, 2 miles above the entrance, has a fixed span with a clearance of 18 feet. The elevated railway structure over Westchester Avenue Bridge has a fixed span with a clearance of 61 feet. The railroad bridge, 2.1 miles above the entrance, has a rolling-lift span with a clearance of 8 feet, but the draw is no longer opened—see **33 CFR 117.771(b)**, chapter 2, for drawbridge regulations. The Bruckner Expressway Bridge is equipped with radiotelephone. The bridgetender can be contacted on VHF-FM channel 13; call sign KX-8189.

(496) **Hunts Point** is on the north side of East River about 4 miles west of Throgs Neck. Small craft anchor in depths of 9 to 17 feet on the flats east of the point.

(497) **Rikers Island**, in the middle of East River between Hunts Point and La Guardia Airport, is partly occupied by buildings of the Department of Correction of New York.

The island is about a mile long, southeast to northwest, and 0.6 mile wide.

(498) East River main channel leads northward of Rikers Island. A much-used general anchorage, with depths of 21 to 30 feet, is between the south side of the channel and the flats off the north side of the island—see **33 CFR 110.1** and **110.155**, chapter 2, for limits and regulations.

(499)

Caution

(500) East River Main Channel Lighted Buoy 5 has been established northeast of Rikers Island in 40°47'47"N., 73°51'59"W. to ensure that no vessel penetration of air space exists over that portion of the East River that coincides with the glide path of the northeast-southwest runway of La Guardia Airport. Vessels with mast heights in excess of 125 feet shall pass 100 yards to the north of this buoy so as to avoid interference with the glide path.

(501) **North Brother Island**, 0.3 mile northwest of Rikers Island, is occupied by the ruins of former municipal buildings. East River main channel leads northward and westward of the island; a light marks the main channel side of the island.

(502) The buoyed channel between North Brother Island and **South Brother Island**, 0.1 mile to the southward, has a controlling depth of about 25 feet. The channel is marked by a light off the north side of South Brother Island. The channel is narrow and subject to strong currents and should not be used by vessels of limited maneuverability.

(503) A ledge, partly bare at low water, extends 0.2 mile southward from South Brother Island; the outer part of the ledge is marked by a light.

(504) **Port Morris**, 0.2 mile westward across East River main channel from North Brother Island, is occupied by several oil terminals.

(505) **South Brother Island Channel**, with a project depth of 35 feet, leads from deep water east of North Brother Island and along the west side of Rikers Island to a turning basin on the west side of Bowery Bay. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through the USACE hydrographic survey website listed in Appendix A. The channel is marked by lighted and unlighted buoys.

(506)

Caution

(507) Vessels transiting South Brother Island Channel and using the turning basin at its southern terminus shall ballast prior to entry and are cautioned that mast heights in excess of 125 feet may penetrate the glide path of the northwest-southeast runway of La Guardia Airport. If mast heights cannot be lowered below 125 feet, La Guardia Air Traffic Control Tower shall be notified at 212–779–0242 prior to terminal departure or channel entry.

(508) **Bowery Bay**, across Rikers Island Channel from Rikers Island, has depths of about 10 feet. A **general** and **special anchorages** are in the west part of the bay—see **33 CFR 110.1**, **110.60** and **110.155**, chapter 2, for limits and regulations. A pipeline area is in the southeast part of the anchorage area. A fixed highway bridge crosses Rikers Island Channel and Bowery Bay and connects Rikers Island with the Borough of Queens, New York; clearance over the channel is 52 feet for a width of 125 feet.

(509) Bowery Bay may be approached from the East River main channel from the northward through South Brother Island Channel and from the northwestward through a 100-yard-wide channel that leads between the ledges that make off from Lawrence Point on the southwest and South Brother Island on the northeast. The controlling depth in the 100-yard-wide channel is about 19 feet. Caution is advised in the northwestern approach as the channel is narrow, the bottom is rocky and uneven and tidal currents are strong.

(510) **Lawrence Point**, on the southeast side of East River 0.7 mile westward of Rikers Island, is occupied by an extensive gas and electric plant. A light marks the outer part of the ledge, partly bare at low water, which extends 0.3 mile northeastward from the point.

(511) **Randalls Island** and Wards Island are on the northwestern side of East River between Port Morris and Hell Gate, separating that river from Harlem River, which is described later. The islands provide recreational facilities for the residents of the city of New York.

(512) **Bronx Kill**, which separates Randalls Island from Port Morris, is a narrow passage that extends westward from the East River to the Harlem River. A fixed railroad bridge with a clearance of 68 feet and a fixed highway bridge with a clearance of 51 feet cross the passage. Bronx Kill is navigable but not recommended as a route of travel. It is shoal and obstructed throughout.

(513) **Sunken Meadow** is the reclaimed area now joined to the northeast end of Wards Island and southeast end of Randalls Island.

(514) **Hell Gate Bridge**, which crosses East River from Wards Island to Long Island 7.1 miles from The Battery, has a fixed railroad span with a clearance of 134 feet.

(515) **Negro Point** is the southernmost point of Wards Island. **Triborough Bridge**, which crosses East River from Negro Point to Long Island 6.8 miles from The Battery, has a highway suspension span with a clearance of 138 feet.

(516) **Holmes Rock** and **Hog Back** are two bare rocks, which are on the eastern and northern parts, respectively, of a reef in the bight on the south side of Wards Island westward of Negro Point. The southwest extremity of this reef is marked by a light.

(517) **Hallets Point**, on the Long Island side of East River about 0.3 mile southwestward of Negro Point, is marked by a light. There are main-channel depths close to the point.

(523)

Structures across Harlem River

Name•Description•Type	Location	Miles*	Clear Width of Draw or Span Opening (feet)	Clear Height above Mean High Water (feet)	Information
103rd Street Bridge (lift)	40°47'10"N., 73°56'14"W.	0.0	300	55 (down) 136 (up)	Notes 1 and 3 Call sign KIL-820
Triborough Bridge (lift)	40°48'02"N., 73°55'42"W.	1.3	204	54 (down) 136 (up)	Notes 1 and 3 Call sign KGW-326
Willis Avenue Bridge (swing)	40°48'14"N., 73°55'45"W.	1.5	109	25	Note 1
Third Avenue Bridge (swing)	40°48'27"N., 73°55'57"W.	1.9	118	27	Note 1
Metro North Railroad Bridge (lift)	40°48'41"N., 73°56'00"W.	2.1	225	25 (down) 135 (up)	Notes 1, 2 and 3 Call sign KAW-326
Madison Avenue/138th Street Bridge (swing)	40°48'51"N., 73°55'59"W.	2.3	104	25	Note 1
149th Street Bridge (swing)	40°49'10"N., 73°55'59"W.	2.8	104	30	Note 1
Macombs Dam Bridge (swing)	40°49'41"N., 73°56'02"W.	3.2	164	27	Note 1
High Bridge (fixed)	40°50'35"N., 73°55'50"W.	4.3	322	77 102 (center)	
Alexander Hamilton Bridge (fixed)	40°50'44"N., 73°55'43"W.	4.5	366	79 103 (center)	
Washington Bridge (fixed)	40°50'48"N., 73°55'40"W.	4.6	354	134	
University Heights Bridge (swing)	40°51'46"N., 73°54'53"W.	6.0	85	25	Note 1
Broadway Bridge (lift)	40°52'25"N., 73°54'40"W.	6.8	288	24 (down) 135 (up)	Note 1
Henry Hudson Bridge (fixed)	40°52'40"N., 73°55'20"W.	7.2	418	53 142 (center)	
Spuyten Duyvil Railroad Bridge (swing)	40°52'42"N., 73°55'32"W.	7.9	100	5	Notes 1, 2 and 3 Call sign KU-9797

* Distance is in nautical miles proceeding from the East River
Note 1 – See **33 CFR 117.1** through **117.59** and **117.789**, chapter 2, for drawbridge regulations
Note 2 – bridge is kept in the open to navigation position except for the passage of trains or maintenance
Note 3 – bridgetenders monitor VHF-FM channel 13

(518) **Hell Gate** is the part of East River between Wards Island and Roosevelt Island, 0.7 mile to the southwest. The crooked channel, the strong tidal currents, and the heavy traffic in Hell Gate require extra caution on the part of the navigator to avoid accident or collision. Vessels navigating Hell Gate on a rising tide sometimes find it necessary to pass starboard-to-starboard because of the strong currents between Negro Point and Hallets Point. This situation may arise when one of the vessels does not maneuver readily or is handling a tow. Northeastward of Negro Point and southwestward of Hallets Point, the customary port passings are made.

(519) **Mill Rock**, on the northwestern side of the main channel through Hell Gate, is 0.2 mile southwest of Wards Island and the same distance northwest of Hallets Point. The islet is marked by lights on its north and south ends.

(520)

Harlem River to Roosevelt Island

(521) **Harlem River**, which joins East River in Hell Gate between Wards Island and Manhattan Island, extends northward about 7 miles and connects with Hudson River through Spuyten Duyvil Creek. The channel through Harlem River is narrow, tortuous, and navigable only for powered vessels. By taking care to avoid several isolated 11- to 13-foot spots, a depth of about 14 feet can be carried to the Hudson River; the chart is the guide.

(522) Traffic is heavy in Harlem River. Vessels with heights too great to pass under the closed drawbridges should make the passage against the current.

(524)

Current

(525) The tidal currents in Harlem River run southward from Hudson River to East River while the east-going current is running in Hell Gate, and the reverse. The south-going current in Harlem River is considered the flood. The times of slack water are subject to variations depending upon freshet conditions in Hudson River. The velocity of the current is 2 knots or more in the narrower parts of the channel—see the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current. Links to a user guide for this service can be found in chapter 1 of this book.

(526) **Roosevelt Island (Welfare Island)**, 1.6 miles long and 0.1 mile wide, is in the middle of East River southwest of Hell Gate. A gray stone tower is on the north end of the island.

(527) The currents on both sides of Roosevelt Island are strong, and caution is advised while navigating in these areas.

(538)

Structures across Newtown Creek and Tributaries

Name•Description•Type	Location	Clear Width of Draw or Span Opening (feet)	Clear Height above Mean High Water (feet)	Information
Newtown Creek				
Pulaski Bridge (bascule)	40°44'21"N., 73°57'09"W.	150	46 (center) 39 (fenders)	Notes 1 and 2 Call sign KX-8178
Greenpoint Avenue Bridge (bascule)	40°44'00"N., 73°56'25"W.	149	30 (center) 24 (fenders)	Notes 1 and 2 Call sign KX-8182
Kosciusko Memorial Bridge (fixed)	40°43'40"N., 73°55'45"W.	130	90	
English Kills				
Metropolitan Avenue Bridge (bascule)	40°42'51"N., 73°55'52"W.	81	10 (center)	Notes 1 and 2 Call sign KX-8179
Montrose Avenue Bridge (fixed)	40°42'33"N., 73°55'50"W.	46	4	
East Branch				
Grand Avenue Bridge (swing)	40°42'59"N., 73°55'22"W.	88 (west draw)	8	Notes 1 and 2 Call sign KX-8187
Dutch Kills				
Long Island Railroad Bridge (swing)	40°44'17"N., 73°56'44"W.	46	2	
Long Island Railroad Bridge (bascule)	40°44'19"N., 73°56'39"W.	50	14	Note 2
Borden Avenue Bridge (retractable span)	40°44'20"N., 73°56'34"W.	49	4	Note 2
Queens Midtown Expressway Bridge (fixed)	40°44'22"N., 73°56'30"W.	90	83	
Hunters Point Avenue Bridge (bascule)	40°44'26"N., 73°56'26"W.	50	5	Note 2

Note 1 – Bridgetender monitors VHF-FM channel 13.
Note 2 – See **33 CFR 117.1** through **117.59**, and **117.801**, chapter 2, for drawbridge regulations.

(528) The 36th Avenue highway bridge that crosses the eastern channel from Roosevelt Island to Long Island 5.6 miles from The Battery has a vertical-lift span with clearances of 40 feet down and 99 feet up—see **33 CFR 117.1** through **117.59** and **117.781**, chapter 2, for drawbridge regulations. The bridgetender monitors VHF-FM channel 13; call sign KX-8184.

(529) **Queensboro Bridge**, which crosses from Manhattan Island to Roosevelt Island and thence to Long Island 5.0 miles from The Battery, has fixed spans with clearances of 131 feet over the main channel and 133 feet over the eastern channel. An overhead cable car with overhead power cables crosses the main channel immediately north of the bridge. The low point of travel of the cabin is not less than 135 feet.

(530)

Roosevelt Island Reef

(531) **Roosevelt Island Reef (Welfare Island Reef)**, with bare islets, rocks awash and submerged rocks, extends 0.3 mile southwestward from the island. **Belmont Island**, near the southwest end of the reef, is marked by a light.

(532)

Newtown Creek

(533) **Newtown Creek** is entered on the eastern side of East River 3.6 miles from The Battery. The creek extends 3.3 miles eastward and southward and has several short tributaries or basins. Traffic is fairly heavy and consists chiefly of petroleum products, sand, gravel and crushed rock; drafts of vessels navigating the creek seldom exceed 15 feet.

(534) Tributary basins are **Dutch Kills**, on the north side of Newtown Creek 0.8 mile from East River; **Whale Creek**, on the south side opposite Dutch Kills; **Maspeth Creek**, on the east side 2.2 miles from East River; **East Branch**, on the east side 2.5 miles from the river; and **English Kills**, which extends westward and southward from the East Branch entrance and forms the last 0.8 mile of Newtown Creek.

(535)

Channels

(536) A federal project provides for a 23-foot channel in Newtown Creek from the East River to and in a turning basin about 240 yards above the Kosciusko Memorial Bridge, thence 20 feet in East Branch and in English Kills to the Metropolitan Avenue bridge, and thence 12 feet in English Kills to the head of the project at Montrose Avenue. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through the USACE hydrographic survey website listed in Appendix A.

(537) The tidal **current** is weak and variable.

(539) From abreast the entrance to Newtown Creek, the 35-foot-project main channel of the East River crosses from the west side of the river to the east side. **Poorhouse Flats Lighted Range** (Front Light; 40°43'28"N., 73°57'46"W.), bearing **160.4°,** is on the Brooklyn side of the river and marks the best water in the crossover.

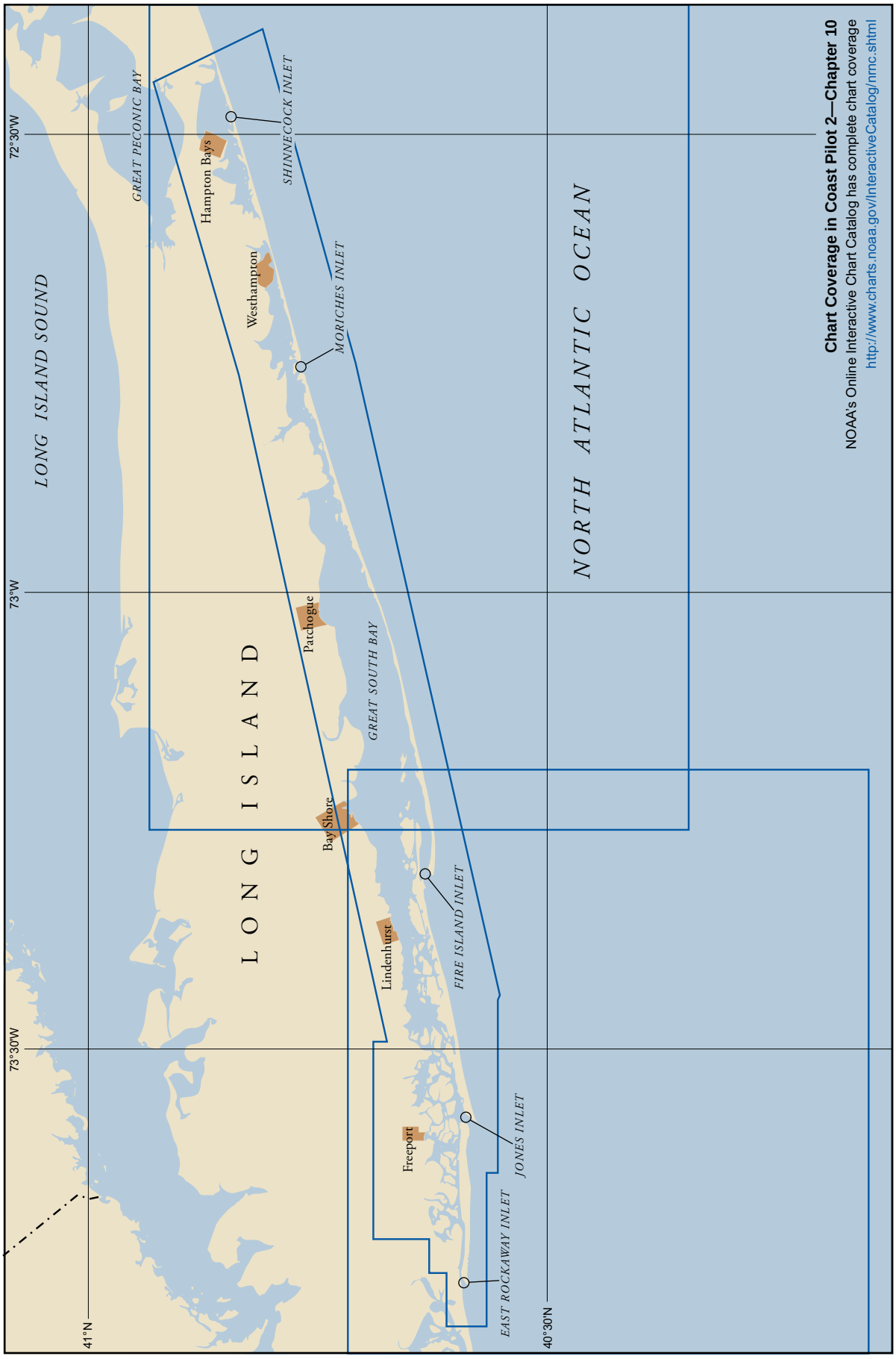
(540) **Williamsburg Bridge**, which crosses East River 2 miles northeast of The Battery, has a suspension span with a clearance of 133 feet.

(541) The site of the inactive **New York Naval Shipyard** is in **Wallabout Bay**, on the Brooklyn side of East River 1.7 miles northeast of The Battery.

(542) **Manhattan Bridge**, which crosses East River 1 mile northeast of The Battery, has a suspension span with a clearance of 134 feet. The clearance under the maintenance platform installed at the west channel edge is 115 feet.

(543) **Brooklyn Bridge**, which crosses East River 0.7 mile northeast of The Battery, has a suspension span with a clearance of 127 feet. The clearance under the maintenance platform is 110 feet.

(544) The channel between The Battery and Governors Island is very congested and subject to strong currents. Caution should be exercised while navigating in the area.


LONG ISLAND SOUND
LONG ISLAND
NORTH ATLANTIC OCEAN
GREAT PECONIC BAY
SHINNECOCK INLET
MORICHES INLET
GREAT SOUTH BAY
FIRE ISLAND INLET
JONES INLET
EAST ROCKAWAY INLET
Hampton Bays
Westhampton
Patchogue
Bay Shore
Lindenhurst
Freeport
72°30'W
73°W
73°30'W
41°N
40°30'N
Chart Coverage in Coast Pilot 2—Chapter 10
NOAA's Online Interactive Chart Catalog has complete chart coverage
http://www.charts.noaa.gov/InteractiveCatalog/nrnc.shtml

South Coast of Long Island

(1) This chapter describes the south coast of Long Island from Shinnecock Inlet to and including East Rockaway Inlet, several other inlets making into the beach along this part of the coast, and the canals, bays, and tributaries inside the beach. Also described are the towns of Patchogue and Oceanside, including Oceanside oil terminals; Bay Shore, a large fishing center; and the many smaller communities that support a large small-craft activity.

(2)

Caution

(3) Eelgrass is found in most of the waters described in this chapter. Eelgrass nets are often placed at the entrances to canals and are sometimes difficult to see.

(4) Fishtrap areas are in Moriches, Shinnecock, Tiana, Quantuck and Great South Bays.

(5) South shore inlets and bays are prone to extreme shoaling and depths as low as 1 to 2 feet at low tide. The location of marked channels is subject to change in order to mark best water.

(6)

COLREGS Demarcation Lines

(7) The lines established for this part of the coast are described in **33 CFR 80.160**, chapter 2.

(8)

Weather, South Coast of Long Island and vicinity

(9) The south coast of Long Island is open to weather from the south and southeast but somewhat sheltered to the west through north. Waves of 8 feet (>2.5 m) or more are most likely in winter when they may be encountered about 6 to 10 percent of the time near the coast. During this period gales are encountered less than 5 percent of the time but are more likely a few hundred miles out to sea. Fogs are more apt to occur in late spring and early summer with a June maximum. Visibilities of less than 2 miles are observed about 5 to 10 percent of the time from May through July. These frequencies are higher at the eastern end in May and June and between Westhampton and Ambrose in July. Locally, Shinnecock Inlet is particularly rough when southerly winds climb to 15 knots or more during ebb tide; breakers fill the entrance.

(10)

North Atlantic right whales

(11) Endangered North Atlantic right whales may occur within 30 miles of the south coast of Long Island, including the approaches to New York Harbor (peak season: November through April). (See North Atlantic Right Whales, indexed as such in chapter 3, for more information on right whales and recommended measures to avoid collisions.)

(12) All vessels 65 feet or greater in length overall (LOA) and subject to the jurisdiction of the United States are restricted to speeds of 10 knots or less in a Seasonal Management Area existing around the Ports of New York/New Jersey between November 1 and April 30. The area is defined as the waters within a 20-nm radius of 40°29'42.2"N., 73°55'57.6"W. (See **50 CFR 224.105** in chapter 2 for regulations, limitations and exceptions.)

(13) The south coast of Long Island has a general trend of 247° for 68 miles from Montauk Point to Fire Island Inlet and thence trends 263° for 36 miles to the western end of Coney Island in the Lower Bay of New York Harbor. It is a clear shore and may be safely approached as close as 1 mile with not less than 30 feet anywhere between Montauk Point and Rockaway Inlet, except off Fire Island Inlet and the inlet's westward side where the shore should be given a berth of at least 1.5 miles. When viewed from seaward it presents only a few prominent features. It is composed of a series of sand dunes backed by low dark woods.

(14) Shinnecock, Moriches, Great South and Hempstead Bays are inside the beach along the south coast of Long Island and form an **inside route** for boats of about 3-foot draft. The three main inlets from the sea to these bays are Fire Island Inlet, Jones Inlet and East Rockaway Inlet. These inlets and all auxiliary channels within the south coast of Long Island have numerous wrecks and obstructions, frequent and extensive changes and, although marked in many areas, should not be used without local knowledge.

(15) Two small inlets, Shinnecock Inlet and Moriches Inlet, which broke through in 1938 and 1931, respectively, are also used by small boats for entrance to these bays, but their use is not advisable without local knowledge.

(16)

No-Discharge Zone

(17) The State of New York, with approval of the Environmental Protection Agency, has established a No-Discharge Zone (NDZ) in the South Shore Estuary Reserve (SSER) and its harbors, bays and creeks. The waterbodies included in the SSER are Shinnecock Bay (East and West), Quantuck Bay, Moriches Bay (East and West), Bellport Bay, Patchogue Bay, Nicoll Bay, Great South Bay (West, East and Great Cove), South Oyster Bay, East Bay Complex, Middle Bay Complex and Western South Shore Bay.

(21)

Structures across Shinnecock Canal

Name•Description•Type	Location	Clear Width of Draw or Span Opening (feet)	Clear Height above Mean High Water (feet)	Information
Overhead power cable	40°53'24"N., 72°30'08"W.		38	
Sunrise Highway/Route 27 Bridge (fixed)	40°53'24"N., 72°30'07"W.	148	23	
Railroad Bridge (fixed)	40°53'14"N., 72°30'06"W.	106	22	
Overhead power cable	40°53'12"N., 72°30'06"W.		48	
Overhead power cable	40°53'10"N., 72°30'06"W.		44	
Route 80 Bridge (fixed)	40°53'06"N., 72°30'03"W.	112	25	
Overhead power and television cable	40°53'05"N., 72°30'03"W.		34	

(18) Within the NDZ, discharge of sewage, whether treated or untreated, from all vessels is prohibited. Outside the NDZ, discharge of sewage is regulated by **40 CFR 140** (see chapter 2).

(19)

Shinnecock Canal to East Rockaway Inlet

(20) **Shinnecock Canal**, 31.5 miles southwestward of Montauk Point, is about 1 mile long and connects Great Peconic Bay with Shinnecock Bay. The canal is owned and maintained by Suffolk County of New York. It is a partly dredged cut and is protected at the north entrance by two jetties marked by lights. In 1985, the east timber jetty was reported to be deteriorating. Protruding timbers and floating debris may be encountered; caution is advised. A lock about midway in the canal is 250 feet long and 41 feet wide, with a depth of 12 feet over the sills. Tide gates are parallel to and westward of the lock. The lock gates and tide gates are constructed so that tidal action opens them to allow the current to set south through the canal and closes them to prevent water from Shinnecock Bay to flow back into Great Peconic Bay. The lock gates are tended 24 hours and are opened mechanically when the tidal current is flowing northward to allow the passage of boats. Red and green traffic lights are at each end of the lock. Vessels are allowed to enter the lock only on the green signal. Mast-stepping cranes are available at both ends of the canal.

(22)

Current

(23) The maximum recorded current is 4.3 knots, southerly, through the lock and tide gates at peak flow when the gates are open. At the railroad bridge, the current has an average speed of 1.5 knots, but it has been reported that greater speeds may be experienced. At times of high southerly current, i.e., when the gates are open, there exists a dangerous eddy system extending from the south end of the lock southerly for approximately 200 yards. Tidal currents throughout the entire canal can be dangerous; caution is advised. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov.*for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(24) A 5 mph **speed limit** is enforced in the canal.

(25) On the east side of Shinnecock Canal just south of the jetties is a boat basin in which the depth ranges from 7 to 10 feet. There are several small-craft facilities on both sides of the canal.

(26) **Hampton Bays**, a station on a Class II railroad just west of Shinnecock Canal, is the nearest post office. **Canoe Place**, the settlement at the canal, has gasoline and some supplies. Small craft and fishing vessels berth in the basins along both sides of the canal.

(27) **Long Island Intracoastal Waterway**.–A federal project provides for a 6-foot channel from Shinnecock Canal to Great South Bay. The cuts provide an inland waterway along the south side of Long Island. This waterway, from the south end of Shinnecock Canal to a point in Great South Bay opposite Patchogue, a distance of about 29.2 miles, is subject to frequent shoaling; mariners are advised to obtain local knowledge.

(28) **Shinnecock Inlet**, 31 miles westward from Montauk Point along the south coast of Long Island, is the easternmost entrance from the Atlantic to **Shinnecock Bay** and the inland water route along the south shore of Long Island. The approach to the inlet is marked by a lighted whistle buoy. The inlet should not be attempted without local knowledge because of the frequent changes in channel depths.

(29)

Current

(30) Tidal currents through the inlet can be dangerous; caution is advised.

(31)

COLREGS Demarcation Lines

(32) The lines established for Shinnecock Inlet are described in **33 CFR 80.160**, chapter 2.

(33) **Shinnecock Light** (40°50'31"N., 72°28'42"W.), 75 feet above the water, is shown from a red skeleton tower on the west side of the inlet. Lights mark the jetties at the entrance to the inlet, and uncharted buoys mark the channel.

(34) The jetties extend about 120 yards beyond the lights marking them. A fish haven is about 2.4 miles south of Shinnecock Inlet entrance.

(35) **Ponquogue Point**, low and sandy, is 1.2 miles northwestward of Shinnecock Light. The west channel from inside Shinnecock Inlet enters the Long Island Intracoastal Waterway southeast of the point. The Ponquogue Bridge, a highway bridge crossing Shinnecock Bay at Ponquogue Point, has a fixed span with a clearance of 55 feet.

(36) **Shinnecock Coast Guard Station** is on Ponquogue Point. An antenna tower, 229 feet above the water and marked by red lights, is also on the point.

(37) **Pine Neck**, 2.3 miles westward of Ponquogue Point and on the west side of Tiana Bay, is low, flat and sandy. A shoal extends southward from **Pine Neck Point** and is marked on the south end by a lighted buoy. About 0.5 mile east of Pine Neck, a privately dredged channel marked by private buoys leads to a basin at **Tiana Beach**, a small summer resort on the south side of Shinnecock Bay.

(38) **Weesuck Creek**, on the north side near the western end of Shinnecock Bay, is entered through a privately dredged channel that leads to the head of the cove at **East Quogue**. The channel is marked by private seasonal buoys. There are two boatyards on the west side near the head of the creek. Berths, electricity, water, marine supplies, storage facilities and lifts to 30 tons are available. The largest marine railway can handle craft up to 65 feet in length; hull and engine repairs can be made.

(39) **Quogue Canal** connects Shinnecock Bay with Quantuck Bay. The canal is crossed by a highway bascule bridge with a clearance of 15 feet and by overhead power and TV cables with clearances of 75 feet. (See **33 CFR 117.1** through **117.59** and **117.799(a)** through**(d)**, chapter 2, for drawbridge regulations.) A 5 mph **speed limit** is enforced in the canal.

(40) **Quantuck Bay** joins Quogue Canal with Quantuck Canal. **Quantuck Creek**, at the head of the bay, is crossed by a fixed bridge, which is the head of navigation except for small pulling boats.

(41) **Quantuck Canal** connects Quantuck Bay and Moriches Bay. The canal is crossed by two highway bascule bridges with a least clearance of 10 feet. (See 33 CFR 117.1 through **117.59** and **117.799(a)** through**(d)**, chapter 2, for drawbridge regulations). The overhead power cable at the westernmost bridge has a clearance of 77 feet. A 5 mph **speed limit** is enforced in the canal.

(42) About 300 yards northeast of the westernmost of the two bridges, a privately dredged channel leads to a yacht basin at **Westhampton Beach**. The channel is along the west bank in the bight and is marked by private seasonal lights and buoys. A dredged cut leads up to Main Street in Westhampton Beach. A reported depth of about 4 feet can be carried to the yacht basin just before reaching the Stevens Lane Bridge, which has a fixed span with a clearance of about 7 feet. Berths, electricity and a launching ramp are at the yacht basin. Gasoline, ice and marine supplies are available at a marina on the east bank of the bight; engine and hull repairs can be made. A forklift can handle craft to 6 tons. In 1981, a reported depth of 2 feet was available at the marina.

(43) **Moriches Bay** extends for about 8 miles from Quantuck Canal to Narrow Bay and provides an inside passage for small boats. The general depths in the bay range from 5 to 7 feet and as low as 1 to 2 feet at low tide, but the southern part is shoal. The marked channel may be shifted significantly to the north in order to mark best water.

(44) When navigating the Long Island Intracoastal Waterway through Moriches Bay northeast of Moriches Inlet, extreme care must be taken as this area is prone to extreme shoaling.

(45) **Speonk Point**, near the eastern end of Moriches Bay on the north shore, is marked by several bulkheaded jetties and a prominent flagstaff.

(46) **Seatuck Cove**, on the north side of Moriches Bay, about 1 mile westward of Speonk Point, is entered through a privately dredged channel that leads northward for about 1.1 miles and then forks into three branch channels: **East Branch**, the easterly branch; **Seatuck Creek**, the northerly branch; and **Little Seatuck Creek**, the westerly branch. Private seasonal buoys mark the entrance channel to the fork and the channel in East Branch to the small-craft facilities just inside the entrance.

(47) A landing at **Eastport** is on the point just above the fork between East Branch and Seatuck Creek. In 1981, depths of 2 to 3 feet were reported at the landing. Berths, moorings, electricity, gasoline, diesel fuel, water, ice, some marine supplies and a pump-out are available at small-craft facilities on East Branch. Lifts to 15 tons can handle craft for engine and hull repairs. Depths of 3 feet are reported alongside.

(48) **Hart Cove**, westward of Seatuck Cove, is entered through a privately dredged channel that leads to the head of the cove.

(49)

Small-craft facilities

(50) Small-craft facilities are near the head on the west side of the cove. Gasoline, water, ice, storage, marine supplies, a launching ramp and a 30-ton mobile hoist are available; hull and engine repairs can be made.

(51) **Tuthill Cove**, locally known as **West Cove**, on the north side of Moriches Bay, 1.5 miles westward of Seatuck Cove, is entered through a privately dredged and partially marked channel that leads to the head of the cove. Several privately dredged channels lead from the main channel to small-craft facilities on the east side of the cove. A Coast Guard Sector Field Office and Station is on the east side of the entrance to Tuthill Cove. **East Moriches** is on the north side of the cove.

(52) **Tuthill Point** is on the west side of the entrance to Tuthill Cove.

(53) **Moriches Inlet**, 44 miles westward of Montauk Point, is a shallow entrance from seaward to the deeper water in Moriches Bay. Weather continuously changes the jettied entrance, which is subject to extreme shoaling. Both east and west jetties are marked by a light, and the approach to the inlet is marked by a lighted whistle buoy.

In 2019, significant erosion observed along the shoreline and exposed riprap mounds in the area of 40°46'11" N., 72°44'33" W. Due to rapidly changing shoaling conditions and existing dangers in Moriches Inlet, it is considered unsafe for mariners to navigate this inlet at any time. Buoys are not maintained in this inlet.

(54) A fish haven, marked by a buoy, is about 2.5 miles south-southwestward of Moriches Inlet East Breakwater Light.

(55)

COLREGS Demarcation Lines

(56) The lines established for Moriches Inlet are described in **33 CFR 80.160**, chapter 2.

(57) **Fire Island** extends west from Moriches Inlet for about 28 miles along the south shore of Long Island to Fire Island Inlet. With the exception of the state park occupying its westernmost 4.6 miles, all of Fire Island is part of the **Fire Island National Seashore**, a Marine Protected Area (MPA).

(58) **Orchard Neck Creek**, 1.7 miles west of Tuthill Point, is extensively used by local small craft as a mooring basin. A reported depth of about 3 feet is available to the head of navigation. A private seasonal lighted buoy marks the entrance.

(59)

Small-craft facility

(60) A small-craft facility is on the west side of the creek near its head. Gasoline, water, marine supplies and a 12-ton lift are available; hull and engine repairs can be made. In 1981, a depth of about 2 feet was reported alongside the facility.

(61) **Areskonk Creek**, immediately westward of Orchard Neck Creek, is used as a harbor by yachtsmen. A privately dredged channel, marked by private seasonal buoys, leads to the head of the creek. In 1981, the channel had a reported controlling depth of 8 feet.

(62) **Senix Creek**, 0.6 mile westward of Orchard Neck Creek, has a narrow entrance. With local knowledge, a reported depth of about 4 feet can be carried in the channel to about 0.5 mile above the entrance.

(63)

Small-craft facilities

(64) Small-craft facilities near the head of the creek have berths, electricity, storage and a 6-ton lift; hull and engine repairs can be made. A marine railway can haul out vessels up to 32 feet in length.

(65) **Mud (West Senix) Creek**, westward of Senix Creek, had a reported controlling depth of about 5 feet in 1981. The creek is used mostly by local residents. A marina on the east side of the creek near the head has berthage, electricity, gasoline, diesel, water, ice, marine supplies, sewage pump-out, surfaced launching ramp and storage; hull and engine repairs can be made. In 2009, approach and alongside depths of 6 feet were reported.

(66) **Forge River**, at the northwest end of Moriches Bay about 0.5 mile westward of the common entrance to Senix and Mud Creeks, is entered through a privately dredged channel that leads from the Intracoastal Waterway to the town dock and turning basin at **Mastic**, about 1.5 miles above the entrance west of **Masury Point**, thence for about 0.2 mile to the head of navigation. In 1981–1999, the reported controlling depth was 6 feet from the Intracoastal Waterway to the head of navigation. Favor the east side of the channel at the entrance. The channel is marked to the turning basin by private seasonal lighted and unlighted buoys. The town dock is available only to the local residents; however, overnight transient berths are available.

(67) **Old Neck Creek** empties into the easterly side of Forge River about 0.5 mile above the entrance. A privately dredged channel leads from the river to the head of the creek. In 1981, the channel had a reported controlling depth of 7 feet. A marina, just inside the easterly entrance to the creek, has berths, electricity, gasoline, diesel, water, ice, marine supplies, lifts to 90 tons and storage; hull, engine and electrical repairs can be made. In 2009, an approach depth of 5 feet was reported with 8 feet alongside.

(68) **Narrow Bay** extends for about 3 miles from Moriches Bay to Bellport Bay and provides a continuation of the inside passage for small boats. The bridge across the bay eastward of **Smith Point** has a bascule span with a clearance of 18 feet. (See **33 CFR 117.1** through **117.59** and **117.799(a)** through **(d)**, chapter 2, for drawbridge regulations.) Caution is recommended when in the vicinity of the bridge because of the piling near the channel. The bridge is an excellent radar target from 5 to 10 miles.

(69) **Bellport Bay** extends for about 3 miles from Narrow Bay to Great South Bay and provides a continuation of the inside passage for small boats. The bay is shoal in its southern part but has depths of 5 to 7 feet in the northern part.

(70) **Carmans River**, on the northeast side of Bellport Bay, has a depth of about 2 feet through the entrance. Sometimes bush stakes are placed on each of the shoals making off from the points at the entrance. Enter in midriver between these stakes, favor the east side for a distance of 0.5 mile and then follow midriver; caution is advised. The river, marked at the entrance by private seasonal lighted buoys, is entered between **Long Point** on the west and Sandy Point on the east. Some of the land areas on both sides of the river just above the entrance are part of the **Werthein National Wildlife Refuge**, a Marine Protected Area (MPA); landing is not permitted. A 5 mph speed limit is enforced on the river.

(71)

Small-craft facility

(72) A small-craft facility is on the west side of the river about 0.6 mile above the entrance. Electricity, water, some marine supplies, a 12-ton lift and storage facilities are available; hull and engine repairs can be made.

(73) **Beaverdam Creek**, on the north side of Bellport Bay about 1.5 miles westward of Carmans River, is

entered through a privately dredged approach channel marked by private seasonal buoys.

(74)

Small-craft facility

(75) A small-craft facility is at the head of the creek and can provide berths, storage, supplies and a 30-ton lift; engine repairs can be made.

(76) The wharf of a yacht club is on the northwest side of Bellport Bay at the town of **Bellport**, about 0.5 mile westward of the entrance to Beaverdam Creek. In 1981, depths of 6 to 8 feet were reported in the basin behind the wharf with about 2½ to 3 feet alongside. Water and a launching ramp are available. A seasonal passenger ferry operates between the yacht club and **Bellport Beach** on Fire Island.

(77) A **dockmaster** manages the village dock adjacent to the yacht club.

(78) **Great South Bay**, on the south shore of Long Island, extends from Bellport Bay on the east to South Oyster Bay on the west. It is about 20 miles long and about 4 miles across its widest part. It can be entered through Fire Island Inlet, from Great Peconic Bay via the inside route, and from westward through Hempstead Bay. The southeast and southwest portions of the bay are shoal. Lights, daybeacons and lighted and unlighted buoys mark the channels.

(79) **Abets Creek** and **Mud Creek**, on the northeast side of Great South Bay, had reported depths of 4 feet in 1981 and 6 feet in 2009, respectively. The entrance to each creek is marked by a private seasonal lighted buoy and stakes. A 5-mph **speed limit** is enforced in Abets Creek.

(80)

Small-craft facilities

(81) Small-craft facilities in both creeks can provide berths, gasoline, diesel fuel, water and storage, and full repairs can be made. A 7-ton marine railway is available in Abets Creek and a 40-ton marine lift is available in Mud Creek.

(82) **Swan River**, about 0.4 mile westward of Mud Creek, is entered through a privately dredged channel that leads to the head of navigation about 1 mile above the mouth. In 1981, the channel had a reported controlling depth of 4 feet. In 1985, a shoal was reported to be encroaching from the west side of the channel at the mouth of the river. A private seasonal lighted buoy marks the entrance, and poles mark the channel above the entrance.

(83) **Patchogue River**, on the north side of Great South Bay, 3.7 miles west of Bellport and 0.9 mile westward of Swan River, is entered through a federal channel that leads from Great South Bay, thence through **Patchogue Bay**, and thence to the head of river navigation about 1 mile above the mouth. The federal channel has a project depth of 8 feet. (See Notice to Mariners and latest edition of chart for controlling depths.) The channel is marked by a lighted and unlighted buoys from the bay to the jettied entrance. The west side of the entrance is protected by a breakwater with a light on the outer end and the east side by a bulkhead and short jetty extending southward from it; a private light is near the end of the jetty.

(84) **Patchogue**, on Patchogue River, is the principal town on Great South Bay. Depths at the wharves and piers at Patchogue range from 3 to 9 feet.

(85) Passenger ferry service, summer only, is maintained from Patchogue to **Davis Park** and Watch Hill on Fire Island.

(86) Marinas and boatyards are on the Patchogue River and can provide gasoline, diesel fuel, electricity, water, marine supplies, pump-out facility, marine lifts to 40 tons and winter storage, and full repairs can be made.

(87) **Corey Creek**, 0.6 mile westward of Patchogue River, is entered between two jetties each marked by a private seasonal light. In 1981, depths of 3 feet were available in the creek. A marina, on the east side of the creek just inside the entrance, has berths, electricity, gasoline, marine supplies, water, ice, storage and a 16-ton lift; hull, engine and electronic repairs can be made. Depths of 5 feet were reported at the marina in 1985.

(88) **Brown Creek**, locally known as Browns River, 3 miles westward of Patchogue, is entered between two short jetties extending out to a depth of about 4 feet. The jetties are marked by lights. A federal project provides for a depth of 6 feet from the entrance to the first bend (at Browns River Road), thence 4 feet to the upstream limit of the project. (See Notice to Mariners and the latest edition of the chart for controlling depths.) Local interests advise that mariners steer a centerline course from a point about 0.75 mile south of the jetty lights through the entrance channel.

(89)

Small-craft facilities

(90) There are several small-craft facilities on the creek. Berths, electricity, gasoline, diesel fuel, water, marine supplies, mobile hoists to 80 tons and hull and engine repairs are available. A marine railway can handle vessels to 15 feet long.

(91) Passenger ferry service, summer only, is available from Sayville to Fire Island Pines, Sailors Haven, Cherry Grove and Barrett Beach on Fire Island.

(92) **Green Creek**, about 1 mile west of Brown Creek, is used by many clam boats. Depths of about 5 feet were reported available in the creek in 1981. Mariners are advised to use care when entering the creek, especially during strong southwest winds. Inside the creek, gasoline, water, marine supplies, storage, a 30-ton lift and complete hull, engine and electronic repairs are available. A 4 mph **speed limit** is enforced in the creek.

(93) **Green Harbor** is a privately maintained two-part harbor just westward of the entrance to Green Creek. The entrance to the outer basin and the connecting channel between the outer and inner basins are very narrow. In 1985, depths were reported to be about 6 feet. Limited berthing is available in the outer basin. A boatyard is in the inner basin. An aquaculture site, marked by a private

buoy, is near the entrance to Green Harbor around 40°43'11"N., 73°05'36"W.

(94) **Connetquot River**, locally known as Great River, is 3 miles westward of Brown Creek. In 1981, a reported depth of 5 feet (with local knowledge) could be carried from **Nicoll Bay** to the boatyards on the east side of the river, thence about 2 feet to the head of navigation at the railroad; favor the east bank of the river above the boatyards. A shoal with depths of 2 feet extends northeast from **Nicoll Island** on the southwest side of the river entrance. A private light marks the entrance to the river. A prominent mansion with a tower, now part of a private school, is on the north shore of the entrance.

(95) **Great River** is a village on the west side of the river.

(96) **Watch Hill**, part of Fire Island National Seashore, is across Great South Bay from Patchogue. A privately dredged channel with a reported depth of 3 feet in 1999 leads from Great South Bay to a seasonally operated marina. The channel is marked by private seasonal lighted buoys and a lighted range. Berths, electricity, water, ice, some supplies and a pump-out facility are available. A passenger ferry operates between Watch Hill and Patchogue.

(97) **Cherry Grove**, a summer resort across Great South Bay from Connetquot River, has a boat landing extending out to a depth of 5 feet. Seasonal ferry service is maintained with Sayville.

(98) **Point o' Woods**, **Ocean Beach**, **Fair Harbor** and **Saltaire** are summer resorts on Great South Bay westward of Cherry Grove. Provisions are available at most of these resorts. Year-round ferry service is maintained between Ocean Beach, Saltaire and Bay Shore, a town northwestward on the north shore of Great South Bay. Seasonal service is available between the surrounding towns and Bay Shore. A privately dredged channel with a reported depth of 9 feet in 1981 leads southward to Fair Harbor. A private light marks the channel.

(99) On the north shore of Great South Bay, in the vicinity of **Nicoll Point**, is **Heckscher State Park**. A boat basin and a small-craft launching ramp are at the park in a cove about 1.6 miles west of Nicoll Point. The park is open during daylight hours only. A 5 mph **speed limit** is enforced.

(100) **Sailors Haven**, across Great South Bay from Nicoll Point, is part of the Fire Island National Seashore. A privately dredged and marked channel with a depth of about 4 feet leads from Great South Bay to a seasonally operated marina at which berthing, water, ice and some supplies are available. A seasonal passenger ferry operates between Sailors Haven and Sayville.

(101)

Weather, Great South Bay and vicinity

(102) **Islip** is on the south side of Long Island just inland from Great South Bay and opposite the Fire Island National Seashore, about halfway between New York city and Montauk Point.

(103) At Islip the average annual temperature is 53°F (11.7°C). The average high is 61°F (16.1°C) and the average low is 44°F (6.7°C). July is the warmest month, with an average high of 82°F (27.8°C) and an average low of 67°F (19.4°C). January is the coolest month, with an average high of 39°F (3.9°C) and an average low of 24°F (-4.4°C). The warmest temperature on record for Islip is 101°F (38.3°C), recorded in July 1991, and the coldest temperature on record is -7°F (-21.7°C), recorded in January 1984. On average, seven days each year record high temperatures in excess of 90°F (32.2°C) and 98 days record minimum temperatures below 32°F (0°C). An average of only two days each year has an extreme minimum below 5°F (-15°C).

(104) Precipitation is both moderate and distributed evenly throughout the year. August is the wettest month. with an average precipitation total of five inches (127 mm), and February the driest, with just over three inches (76 mm). Average annual precipitation is about 45 inches (1,143 mm). Most of the rainfall from June through September comes from thunderstorms; therefore, it is usually of brief duration but relatively intense. Thunderstorm days average 25 each year. From October to April, however, precipitation is generally associated with widespread storm areas, so that day-long rain or snow is common.

(105) Snow falls an average 30 days each year and averages 21 inches (533 mm) in any given year. The snowiest month is February, with an average of six inches (152 mm). Snow has fallen in each month November through April. The greatest 24-hour total snowfall was eight inches (203 mm), which fell in March 1993.

(106) Tropical storms have influenced the area several times since 1871. Hurricane Gloria passed within 10 miles west of Islip in September 1985. Gloria made landfall about halfway between Kennedy and Islip and provided sustained winds of 75 knots at time of landfall for the Islip area. Only two days earlier, Gloria was a 125-knot hurricane.

(107) The Long Island U.S. Courthouse (40°45'35"N., 73°11'25"W.) is a prominent feature in East Islip, across Champlin Creek from Islip. The building is rectangular with a cone-shaped entrance, is constructed of white and gray panels and is 281 feet high; reported to be visible from 20 miles offshore.

(108) **Great Cove**, on the north side of Great South Bay about 4 miles westward of Nicoll Point, has depths of 4 to 8 feet. A line of private orange and white spar buoys across the mouth of Great Cove marks a shellfish closure area.

(109) **Orowoc Creek**, which enters the northeast part of the cove, leads to the boat basin and wharves at the town of **Islip**. The channel in the cove is sometimes marked by stakes and had a reported depth of about 6 feet in 1981. A private light marks the westerly edge of the 3-foot shoal on the east side of the channel near the entrance to the creek.

(110)

Small-craft facilities

(111) Small-craft facilities on the creek can provide gasoline, water, ice, storage, marine supplies and complete engine and hull repairs. A 55-foot marine railway and a 25-ton mobile hoist are available.

(112) Several fish packing plants are on the creek.

(113) **Penataquit Creek** and **Watchogue Creek**, locally known as **West Creek**, about 0.5 mile westward of Orowoc Creek, empty into the northwest end of Great Cove through a common entrance. **Bay Shore** is a large fishing center on the northwest shore of Great Cove at the head of the creeks. The common entrance is protected on its westerly side by a bulkheaded sandspit, which forms a well-protected boat basin. The entrance channel leads between the northeast end of the sandspit and the point to the east. A private light marks the entrance to the creeks. The channel had a reported depth of 6 feet in 1981. A 4 mph **speed limit** is enforced on the creeks.

(114) The ferry landing near the entrance of Penataquit Creek had a reported depth of about 5 feet at its end in 1981. From the landing, ferries connect with Ocean Beach and Saltaire year round and with Point o' Woods, Kismet, Fair Harbor, Dunewood, Atlantique, Sea View and Ocean Bay Park during the summer.

(115) There are several small-craft facilities in Penataquit and Watchogue Creeks.

(116) There are several creeks and a dredged boat basin between Watchogue Creek and Conklin Point to the southwestward. These waterways are for the most part privately maintained and for the exclusive use of the local property owners.

(117) **Fire Island Inlet**, about 28 miles westward along the south coast of Long Island from Moriches Inlet, is the only direct entrance from the Atlantic to Great South Bay. The inlet is subject to extreme shoaling and has been moving westward for many years. Mariners are warned to beware of extreme tidal turbulence especially during times of tidal change and should seek local knowledge of the latest conditions before entering. Navigation of the inlet is difficult even with relatively calm seas, and for small craft it can be extremely dangerous. During heavy weather, the entrance usually is obstructed by breakers.

(118)

COLREGS Demarcation Lines

(119) The lines established for Fire Island Inlet are described in **33 CFR 80.160**, chapter 2.

(120) **Fire Island Light** (40°37'57"N., 73°13'07"W.), 167 feet above the water, is shown from a black and white horizontally banded tower about 4 miles east-northeastward of **Democrat Point**. **Fire Island Coast Guard Station** is about 1.9 miles west-southwestward of the light. A water tower, about 208 feet high, marked by floodlights and visible for 16 miles, is about 0.1 mile southwest of the Fire Island Coast Guard Station.

(121) The **Robert Moses Causeway Bridge** over Fire Island Inlet, 2.1 miles inside the entrance, has a clearance of 65 feet at the 464-foot center span. The bridge is an excellent radar target at a range of more than 12 miles.

(122) Two boat basins at the **Robert Moses (Fire Island) State Park** are entered just westward of the southern end of the bridge. Berths and water are available in the basins between sunrise and sunset. In 1981, depths of 7 feet and 6 feet were reported available in the east and west basins, respectively.

(123)

Currents

(124) The currents in Fire Island Inlet, after crossing the bar, have a velocity of about 2.4 knots at full strength and are influenced greatly by the force and direction of the wind. In the bay, currents have little velocity except in the narrow channels between the shoals and within a radius of 3 miles from Fire Island Coast Guard Station where their estimated velocity is 1 to 1.5 knots. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(125) Fire Island Inlet remains open throughout the year, but ice does become a problem in the inland channels through Great South Bay from early January through about mid-March. Several channels lead from Fire Island Inlet to places in Great South Bay and connecting inside waterways. These channels are marked with buoys that are shifted in position with changing conditions.

(126) The area between Fire Island Inlet and Jones Inlet is characterized by low, sandy beaches and numerous islands fringed by vast stretches of marshy ground. Many shallow areas, irregular in outline, are a serious menace to the navigation of light-draft vessels. An extensive network of bays, creeks, coves, channels and inlets covers the entire area.

(127) The channel connecting Great South Bay with Jones Inlet, East Bay, and South Oyster Bay is narrow and treacherous and has numerous short bends. Caution should be exercised when navigating in these areas in small boats.

(128) From Fire Island Inlet the **State Boat Channel** leads westward through Great South Bay and South Oyster Bay to Zacks Bay at Jones Beach State Park, thence westward in Hempstead Bay through winding channels, well marked by lights, buoys and daybeacons to Reynolds Channel at Point Lookout, just west of Jones Inlet.

(129) The **speed** of vessels is limited to 10.4 knots (12 mph) in the channel and 3.5 knots (4 mph) in the areas designated as basin or anchorage.

(130) A marina on the south side of the channel at the eastern end of **Captree Island** has berthage, gasoline, diesel fuel, water and ice.

(131) The Robert Moses Causeway Bridge over the State Boat Channel, connecting Oak Beach with Captree Island, has twin bascule spans with a clearance of 29 feet at the center. (See **33 CFR 117.1** through **117.59** and

117.799(a) through **(c)** and **(i)**, chapter 2, for drawbridge regulations.) The twin fixed spans of this bridge and causeway over the inside passage in Great South Bay between Captree Island and **Conklin Point** have a clearance of 60 feet for a middle width of 460 feet.

(132) A shellfish closure area, marked by private yellow buoys, extends from the Robert Moses Causeway at Conklin Point westward for about 6.8 miles to Narraskatuck Creek.

(133) **Oak Island Channel**, locally known as Babylon Cut, extends northwestward from the State Boat Channel from a point opposite the northeastern end of **Oak Island** to Great South Bay and **Babylon Cove**. The channel is marked by seasonal buoys to the channel opposite **Grass Island**. From a point about 1.7 miles above the State Boat Channel, Oak Island Channel connects with a privately dredged and marked channel, locally known as **East West Channel**, that leads westward and parallels the northern shore of Great South Bay for about 6 miles to South Oyster Bay. Several channels, some leading northward into the waterways on the north side of Great South Bay and some leading southward to the State Boat Channel, connect with East West Channel. These connecting channels are discussed later in this chapter.

(134) **Babylon** is a town on the north shore of Great South Bay. A flag pole and a church spire are prominent. The public landing, about 0.3 mile northward of Sampawams Point and at the mouth of Sampawams Creek, had a reported depth of 6 feet at the end in 1993. Approaching around **Sampawams Point**, give the point a berth of 0.3 mile when southeastward of it and head northwestward to the wharf.

(135) **Sampawams Creek**, just northward of the wharf, has been dredged to reclaim adjacent lands and is bulkheaded on the west side. The entrance is marked by private seasonal buoys and a private light. It is used as an anchorage by small craft and has a depth of about 5 feet through the entrance and greater depths inside. Boats also anchor between the public landing and Sampawams Point. This anchorage becomes choppy during easterly or southeasterly winds. Several small-craft facilities are on the creek.

(136) **Carlls River**, westward of Sampawams Point, in 1981, had a reported controlling depth of 5 feet in the privately dredged entrance channel leading northward from East West Channel. In 1982, shoaling to an unknown extent was reported in the channel. Small-craft facilities on the river have berthage, gasoline, water, marine supplies, a 9-ton lift and a 40-ton mobile hoist; hull and engine repairs can be made. In 1993, depths of 5 to 6 feet were reported alongside at the facilities.

(137) **West Babylon Creek**, locally known as **Mud Creek**, is about 1 mile westward of Sampawams Point. In 1981, the privately dredged and marked entrance channel leading northward from East West Channel had a reported controlling depth of 4 feet. Creeks to the westward have been improved in a similar manner. Several small-craft facilities are on the creek.

(138) A submerged obstruction was reported at the mouth of the Santapogue River entrance channel in about 40°40'01"N., 73°20'36"W.

(139) **Oak Beach** is primarily a summer resort and fishing village on the north side of Fire Island Inlet. The channel to the village pier, passing eastward of Oak Island, has a depth of about 9 feet. A tall lighted mast on the south side of Oak Beach is prominent.

(140) **Cedar Beach** and **Gilgo Beach**, westward of Oak Beach, are maintained and operated by the County and Township authorities and are not part of the Long Island State Park System. A tower is prominent east of Cedar Beach, and another tower is prominent west of Gilgo Beach.

(141) **Neguntatogue Creek**, on the north side of Great South Bay at the town of **Lindenhurst**, has several small-craft facilities. In 2000, the reported controlling depth in the entrance to the creek was 4 feet. Berths, electricity, gasoline, diesel fuel, water, ice, storage and marine supplies are available; hull and engine repairs can be made.

(142) **Fox Creek Channel**, privately dredged and marked by private seasonal aids, leads from the mouth of Neguntatogue Creek across Great South Bay to a junction with the State Boat Channel just eastward of Cedar Island. In 1982, the reported controlling depth was 3 feet.

(143) **Strongs Creek**, westward of Neguntatogue Creek, in 1981, had a controlling depth of 7 feet in the privately dredged entrance channel leading northward from East West Channel.

(144)

Small-craft facilities

(145) Small-craft facilities in the creek have storage and a 12-ton mobile hoist; hull and engine repairs can be made.

(146) **Great Neck Creek**, westward of **Strongs Point**, has a depth of about 7 feet in the privately dredged entrance channel leading northward from East West Channel; greater depths are inside.

(147) **Woods Creek** is westward of Howell Point and Howell Creek. In 2005, the reported controlling depth in the entrance was 3 feet. Gasoline is available just inside the entrance, and a small-craft facility at the head of the creek can provide limited berths, electricity, gasoline, water, some marine supplies, a pump-out station, winter storage and a 30-ton lift; hull, engine and electrical repairs can be made.

(148) **Amityville Creek**, on the north side of the western extremity of Great South Bay, had a reported controlling depth of about 3 feet in 1981. The entrance to the creek is marked on the west side by a private light. Several boatyards on the creek have marine railways, the largest of which can handle craft up to 50 feet in length; gasoline, water, ice, storage, marine supplies and complete engine and hull repairs are available.

(149) **Amityville** is a small town on the north shore of Great South Bay at its western extremity. The village wharf

bares at low water at its face. **Amityville Cut** extends southward from Amityville Creek and joins the State Boat Channel near Gilgo Beach. The privately maintained and marked channel had a reported controlling depth of 8 feet in 1993.

(150) **Narraskatuck Creek**, 0.5 miles westward of Amityville Creek, had a reported depth of about 3 feet in 1981.

(151)

Small-craft facilities

(152) The small-craft facilities on the creek have gasoline, berths, electricity, water, ice, storage, and marine supplies. Mobile hoists can handle craft up to 20 tons; hull, engine, and electrical repairs can be made.

(153) **Carman Creek**, about 0.8 mile westward of Amityville Creek, is used by boats drawing 4 to 5 feet.

(154) **South Oyster Bay**, lying between Great South Bay and Hempstead Bay, is shoal over its greater part. A channel marked by buoys and daybeacons, good for a draft of 4 feet at high water, extends through the bay. Through traffic uses the State Boat Channel and connecting lanes on the south side of the Bay.

(155) **Gilgo Heading**, a channel and basin between the State Boat Channel and Gilgo Beach, has a depth of about 7 feet.

(156) **Hempstead Bay** is on the south side of Long Island inside the beach extending from the west end of Great South Bay to Far Rockaway. The bay has many sloughs that are subject to change in the vicinity of the inlets and where dredging is done to reclaim land. Navigational aids marking the main channels of the bay are maintained by the town of Hempstead. Many shoal spots, some to a foot or less, have been reported at several areas of the rivers and channels.

(157) **Jones Beach State Park**, on the south coast of Long Island, comprises about 2,500 acres and is under the jurisdiction of the Long Island State Park and Recreation Commission. A prominent red brick water tower, with a pyramid top, 3.5 miles eastward of Jones Inlet, is the center of Central Mall. The tower, flood-lighted at night, is visible for 16 miles. The eastern part of **Zachs Bay**, a dredged basin just east of Jones Beach State Park, is used as an anchorage; a swimming area marked by private buoys is in the western part of the bay.

(158) The Wantagh State Parkway bridge crosses Sloop Channel from Jones Beach State Park to Green Island. The bridge has a bascule span with a horizontal clearance of 76 feet closed and 50 feet open; vertical clearances are 16 feet (20 feet at mid-span.) The middle Wantagh State Parkway fixed bridge over **Goose Creek** between Green Island and Great Island has a clearance of 16 feet. The northern fixed bridge of the Wantagh State Parkway spans Island Creek with a clearance of 12 feet.

(159)

Caution

(160) The current is reported to be swift during periods of maximum flood and ebb at the bridge crossing the Sloop Channel from Green Island to Jones Beach State Park and has a tendency to set boats into the bridge abutments. Mariners are advised to avoid this part of the channel during these periods and to use the secondary route in Goose Creek, north of Green Island.

(161) A privately marked channel, locally known as **Racehorse Channel**, leads northward from Sloop Channel and westward of Green Island to the western entrance of **Island Creek**. **Olivers Channel**, marked by private buoys and daybeacons, leads westward from near the north end of Racehorse Channel to **East Bay**.

(162) A fish haven is near the middle of East Bay.

(163)

Small-craft facilities

(164) On the north side of East Bay, there is a small-craft facility on the west side of **Nicks Point** that had an approach depth of 6 feet in 2010. Another facility is on the north side of Island Creek and can provide gasoline, electricity, water, ice, pump-out, some marine supplies, a 25-ton lift and full repairs; the approach depth is 7 feet.

(165) **Sloop Channel**, the main channel leading east from Jones Inlet, extends along the north side of **Short Beach** and Jones Beach State Park. The channel is marked by buoys and daybeacons; shoaling has been reported in several areas.

(166) A channel joins Haunts Creek east of **East Crow Island** and leads northward through **Broad Creek Channel** to East Bay, thence to **Merrick Creek**. The channel joining Sloop Channel just west of the Meadowbrook State Parkway Bridge leads north through **Swift Creek** and **Neds Creek** to East Bay. The channel between **False Channel Meadow** and Pettit Marsh leads to Freeport Creek.

(167) The Meadowbrook State Parkway Bridge has the following clearances: 21 feet for the bascule span across Sloop Channel between Jones Beach State Park and **Jones Island**, 14 feet for the 29-foot fixed span between **West Crow Island** and **Pettit Marsh**, and 12 feet for the 29-foot fixed span between Pettit Marsh and **Fighting Island**. (See **33 CFR 117.1** through **117.59** and **117.799 (a)**through **(c)** and **(h)**, chapter 2, for drawbridge regulations.)

(168) The Loop Parkway Bridge has the following clearances: 20 feet for the fixed span over Swift Creek between West Crow Island and **Meadow Island**, 21 feet for the bascule span between Meadow Island and **Alder Island**, and 20 feet for the 29-foot fixed span over Reynolds Channel between Alder Island and Point Lookout. (See **33 CFR 117.1** through **117.59** and **117.799(a)**through**(c)** and**(f)**, chapter 2, for drawbridge regulations.)

(169) **Jones Inlet**, about 12 miles westward along the south coast of Long Island from Fire Island Inlet, is the principal entrance from the Atlantic to the inside passages and towns in Hempstead Bay. The inlet, which is used mostly by pleasure craft and fishermen, should not be attempted without local knowledge because the channel and depths are constantly changing.

(170) The approach to Jones Inlet is marked by a lighted whistle buoy, and a light is at the outer end of the jetty on the east side of the entrance. A small-craft basin is inside the inlet on the north side of Jones Beach; berths, electricity, water and a pump-out station are available. **Jones Beach Coast Guard Station** is in the small-craft basin.

(171)

Currents

(172) The **tidal current** in the inlet has a velocity of about 3 knots. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(173)

COLREGS Demarcation Lines

(174) The lines established for Jones Inlet are described in **33 CFR 80.160**, chapter 2.

(175) **Point Lookout** is a village on the east end of the barrier beach on the west side of Jones Inlet. A large lighted tank in the western part of the town is prominent.

(176) **Long Creek**, marked by seasonal lighted and unlighted buoys, leads northward from Jones Inlet between Alder Island and Meadow Island and between **Smith Meadow** and **Pine Marsh** to Freeport. The channel below the Loop Parkway Bridge has been improved by dredging. The channel above the bridge at the intersection with **Sea Dog Creek** is subject to frequent change; local information should be obtained before using these waters.

(177) **Hudson Channel** extends northward to the piers at **Freeport**, a city on the north shore of **Baldwin Bay**. Freeport has rail and bus service to New York City and other points on Long Island.

(178)

Anchorages

(179) A **general anchorage** is in **Randall Bay** at the northeast end of **Baldwin Bay**. (See **33 CFR 110.1** and **110.156**, chapter 2, for limits and regulations.)

(180) **Reynolds Channel** extends westward from Jones Inlet to East Rockaway Inlet and is the main thorofare of the route between the inlets. The channel is crossed by several bridges.

(181)

Currents

(182) Strong currents exist in the western portion of Reynolds Channel, and caution must be exercised when approaching the drawbridges, particularly with a fair current; the signal to open the bridge should be given sufficiently in advance so the bridge can be cleared of traffic and the draw opened before the vessel arrives there. The currents of the two inlets meet at the entrance of the channel leading west from Cinder Creek.

(183) A 5 mph **speed limit** is enforced in the channel between Middle Island and Point Lookout.

(184) A secondary channel extends northwestward through **Cinder Creek** and westward of **Parsonage Island** to Middle Bay; the channel is not marked. **Garrett Lead**, the primary channel extending northeastward from Reynolds Channel to Middle Bay, is marked by buoys and daybeacons.

(185) Barnums Channel, heading west from Garrett Lead, leads to a small-craft facility providing berths, electricity, gasoline, diesel fuel, water, ice, marine supplies, a pump-out facility, a lift to 35 tons, storage and full repairs. The facility has an approach depth of 17 feet and an alongside depth of 14 feet.

(186) **Long Beach** is a seaside resort on the outer beach about 4 miles west of Point Lookout. The waterfront on the bayside is bulkheaded.

(187) The highway bridges crossing Reynolds Channel between Long Beach and **Island Park** have bascule spans with clearances of 20 feet. The railroad bridge about 0.2 mile westward of the highway bridges has a bascule span with a clearance of 14 feet. (See **33 CFR 117.1** through **117.59** and **117.799(a)** through **(c)** and **(g),** chapter 2, for drawbridge regulations.)

(188) A dock of the Long Beach Hospital is on the south side of Reynolds Channel about 0.3 mile eastward of the highway bridge; medical aid to boatmen is available here. The dock is marked by a square white sign with a large red cross.

(189) Just westward of the railroad bridge, a dredged channel passes through Island Park. The fixed footbridge and highway bridge crossing the channel have a least clearance of 7 feet. Several wharves are available at Island Park.

(190) **Hog Island Channel**, the main route to the towns of **Oceanside** and **East Rockaway**, joins Reynolds Channel southwestward of Island Park and leads westward of Island Park, then eastward of West, East and North Meadows. **East Rockaway Channel**, privately marked and an alternate and shallower route to the towns, joins Hog Island Channel about 0.8 mile and 2.4 miles above Reynolds Channel. Oceanside and East Rockaway are along the east and west sides, respectively, of the northern part of East Rockaway Channel.

(191) Mariners of vessels transiting Hog Island Channel in the vicinity of the public beach at the village of Island Park are requested to proceed at a **speed** that will create minimum wave wash and wake and avoid damage to the beach facilities.

(192) There are numerous marginal-type petroleum wharves along the eastern side of Hog Island Channel between 1.75 and 2.25 miles above the junction with Reynolds Channel. In 1981, depths of 6 to 10 feet were reported alongside the wharves; oil barges and coastal tankers berth at or near high tide and ground out at low tide when alongside.

(193) **Woodsburgh Channel** joins Broad Channel about 0.5 mile northward of Hicks Beach and leads northwestward to **Woodsburgh**. The two fixed bridges

over **Woodmere Channel** northwestward of **Brosewere Bay** have a least clearance of 11 feet.

(194) **Atlantic Beach** is an oceanfront and bayside community on the east side of East Rockaway Inlet. Facilities for mooring are eastward and westward of the highway bridge. Gasoline, diesel fuel, water, provisions and other supplies are available.

(195) **Bannister Creek** is just east of the Atlantic Beach Bridge. **Bridge Creek** extends west just above the mouth of Bannister Creek. A small boatyard on the creek can haul out craft up to 6 tons for hull and engine repairs; water, a pump-out and some marine supplies are available.

(196) The highway bridge crossing Reynolds Channel to Atlantic Beach just inside East Rockaway Inlet has a bascule span with a clearance of 25 feet. (See **33 CFR 117.1** through **117.59** and **117.799(a)** through **(c)** and **(e)**, chapter 2, for drawbridge regulations.) The bridgetender monitors VHF-FM channel 13; call sign KFL–348.

(197) **East Rockaway Inlet**, about 8 miles westward along the south coast of Long Island from Jones Inlet, is the westernmost entrance from the Atlantic to Hempstead Bay and the inland water route along the south shore of Long Island. The inlet is subject to frequent changes but is reported to be usually safer to navigate than Jones or Fire Island Inlets. The aids marking the inlet are periodically moved to mark the best water; local knowledge is advised.

(198) Two large identical apartment buildings are prominent about 0.8 mile north-northeastward of the jetty light.

(199)

Currents

(200) The **tidal current** in the inlet has a velocity of about 2.3 knots. Caution should be exercised when passing through the inlet and bridge at times of maximum current. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

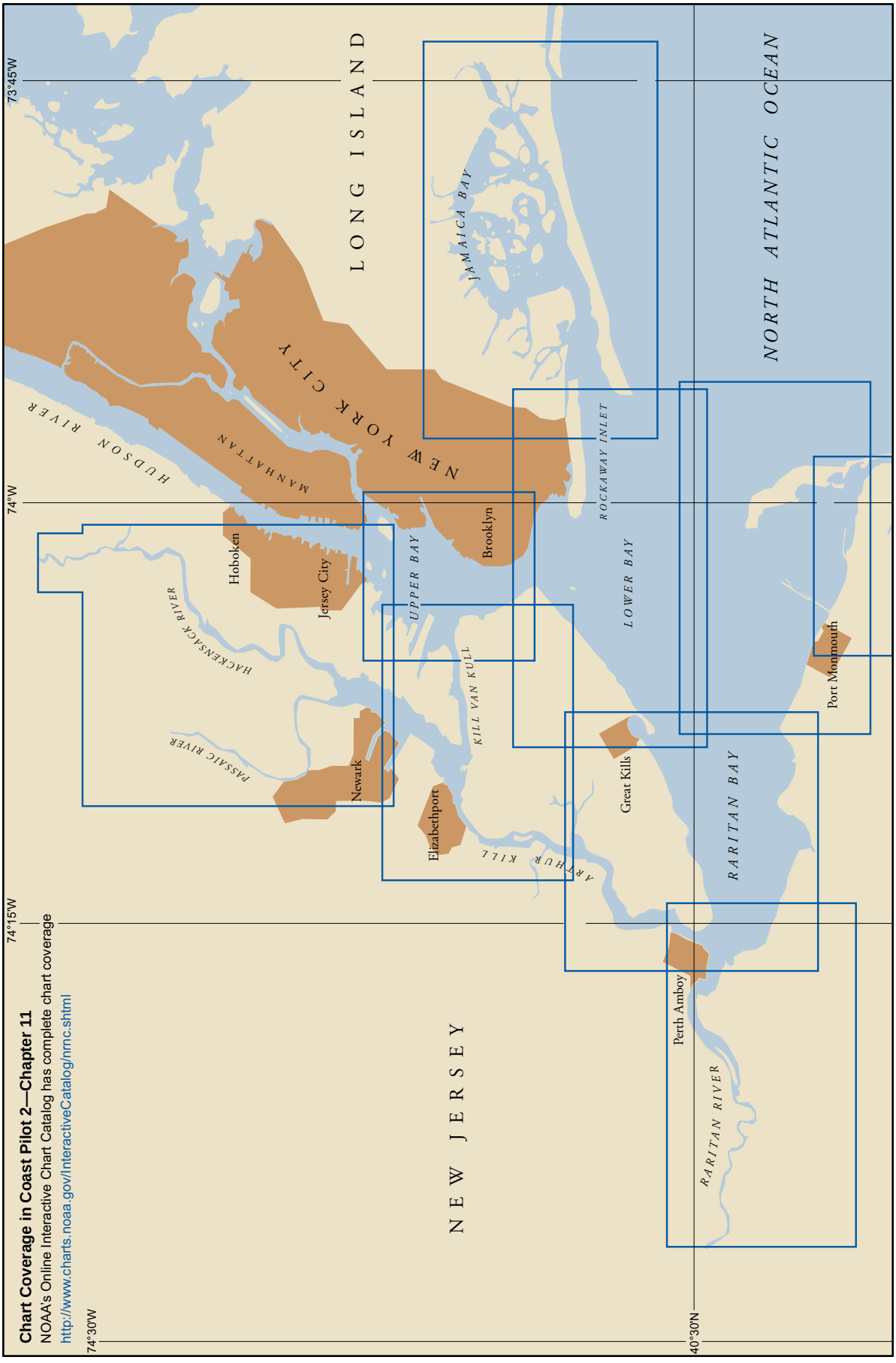


Chart Coverage in Coast Pilot 2—Chapter 11
NOAA's Online Interactive Chart Catalog has complete chart coverage
http://www.charts.noaa.gov/InteractiveCatalog/nrnc.shtml
73°45'W
74°W
74°15'W
74°30'W
40°30'N
LONG ISLAND
NEW YORK CITY
NEW JERSEY
NORTH ATLANTIC OCEAN
HUDSON RIVER
MANHATTAN
JAMAICA BAY
ROCKAWAY INLET
LOWER BAY
UPPER BAY
KILL VAN KULL
ARTHUR KILL
RARITAN BAY
RARITAN RIVER
PASSAIC RIVER
HACKENSACK RIVER
Hoboken
Jersey City
Brooklyn
Newark
Elizabethport
Great Kills
Perth Amboy
Port Monmouth

New York Harbor and Approaches

(1) This chapter describes New York Harbor, its approaches and the areas adjacent to it, bounded by and including Jamaica Bay to the eastward and Sandy Hook Bay to the southward. Included in the text, in addition to the facilities at New York City and Staten Island, are the New Jersey ports of Perth Amboy, Port Elizabeth, Port Newark, Bayonne and others that are accessible through tributaries that empty into New York Harbor, such as Arthur Kill, Kill Van Kull, Passaic River and Hackensack River. The Hudson River above New York City is discussed in chapter 12, and the East River, the approach to New York Harbor from Long Island Sound, is discussed in chapter 9.

(2)

COLREGS Demarcation Lines

(3) The lines established for New York Harbor are described in **33 CFR 80.165**, chapter 2.

(4)

New York Harbor

(5) The approach to **New York Harbor** from seaward is generally along the south coast of Long Island or the east coast of New Jersey, although the harbor is easily approached from any direction between east and south. During the approach, the south shore of Long Island will be seen to northward and the low sandy beaches of the New Jersey shore will be observed to westward. The Long Island shore is readily identified by sand hillocks and densely populated beach communities, whereas the New Jersey shore is characterized by long sandy stretches and many summer resort settlements.

(6)

Prominent features

(7) The four most prominent landmarks, which can be seen for a long distance at sea, are the Fire Island Light, a tower at Jones Beach on the Long Island shore, the Highlands of Navesink and the microwave tower at Atlantic Highlands on the north end of the New Jersey coast. When nearing the Lower Bay of New York Harbor, Ambrose Channel Lighted Whistle Buoy A, will be seen; it marks the entrance to Ambrose Channel, which is the principal deepwater passage through the Lower Bay.

(8) The south coast of Long Island from Fire Island Inlet to Rockaway Inlet has a general 263° trend for 30 miles. It is a clean shore and may be approached as close as 1 mile, with not less than 30 feet except off the inlets where the shore should be given a berth of at least 1.5 miles. This coast is characterized by sandy beaches and summer resorts at the eastern end and amusement parks and densely settled communities at the western end.

(9) The shoreline is broken by three prominent and navigable inlets that lead to the inland waterway along the south shore of Long Island. Fire Island Inlet is at the eastern extremity, and its entrance is marked by lights and buoys. Jones Inlet is about 12 miles to the west of Fire Island Inlet. The entrance is prominently indicated by the 202-foot lighted tower at Jones Beach on the eastern side and by an elevated tank at Point Lookout on the west side of the inlet. Jones Beach State Park is on the east side of the inlet; a lighted tower in the park is a conspicuous landmark.

(10) East Rockaway Inlet, about 8 miles westward of Jones Inlet, is the extreme western entrance to the inland waterway. The inlet entrance is marked by a breakwater with a light on its seaward end. The shoreline between the two inlets is closely built up with large communities. Elevated tanks, towers and other tall structures are prominent in this area.

(11) A **fsh haven** is about 2 miles offshore midway between East Rockaway Inlet and Rockaway Point.

(12) **Rockaway Point**, 17 miles westward of Jones Inlet, is the southwestern extremity of Long Island and the eastern entrance to New York Lower Bay. A breakwater, marked at its seaward end by a light, extends southward from the point. Rockaway Inlet forms a large deep entrance to Jamaica Bay.

(13) **Sandy Hook**, the southern entrance point to New York Harbor, is low and sandy. A Coast Guard station and two radio towers are near the northern extremity of Sandy Hook. The towers and a large green standpipe to the southeast are the most prominent objects on the northern end of Sandy Hook. Southward of the standpipe are several houses and **Sandy Hook Light** (40°27'42"N., 74°00'07"W.), 88 feet above the water and shown from a white stone tower, 85 feet high. This light, established in 1764, is the oldest in continuous use in the United States.

(14) The most prominent landmark southward of the entrance to New York Harbor is the high wooded ridge forming the **Highlands of Navesink**. A tall condominium on the ridge and a microwave tower at Atlantic Highlands to the west are also prominent. The brownstone towers of the abandoned Navesink Lighthouse on the easternmost spur of the highlands are 73 feet above the ground and about 246 feet above the water. The northerly tower is octagonal, and the southerly tower is square. A private seasonal light is shown from the northerly tower.

(15)

COLREGS Demarcation Lines

(16) The lines established for New York Harbor are described in **33 CFR 80.165**, chapter 2.

(17) Soundings will be found most useful to warn vessels of too close an approach to the shore in approaching New York Harbor. Many vessels have been wrecked on the coast of New Jersey and Long Island through failure to take frequent soundings when the position was uncertain. Depth is a better indication of position off this part of the coast than the character of the bottom, as the same characteristics may be found in widely different positions. A frequent use of soundings and close study of the charts will always give sufficient warning of danger. If a vessel is not certain of her position, the depth should not be shoaled to less than 15 fathoms on the south coast of Long Island eastward of Fire Island Light, or 11 fathoms between Fire Island Light and Barnegat Lighted Buoy B (39°45'48"N., 73°46'04"W.) or 9 fathoms southward of Barnegat Lighted Buoy B.

(18) From the position of the two shores relative to each other and to the entrance to New York Harbor it follows that a course of 215° will deepen the water if the vessel is on the Long Island side of the approach and will shoal if she is off the New Jersey coast. A course of 035° will deepen the water if the vessel is off the New Jersey side of the approach and will shoal if she is off the Long Island coast.

(19) Eastward of Fire Island Light the water shoals

(30) The effect of the wind on the current should always be considered. The largest velocities likely to occur during storms are 2.5 knots southward of Nantucket Shoals and 1.5 knots 9 miles south of Fire Island Light. The tidal currents from the latter location have a mean velocity at strength of about 0.2 knot in a westward direction on the flood and an eastward direction on the ebb.

(31) Between Gay Head and Montauk Point the tidal currents set northward on the flood and southward on the ebb. The estimated velocity at strength where the depth is about 25 fathoms is 0.5 knot; closer inshore and near the entrance this velocity increases.

(32)

Weather, New York Harbor and approaches

(33) Winds play an important role by affecting currents in the harbor. During the winter west and northwest winds prevail with northerlies and southwesterlies in secondary roles. The strongest winds are out of the west through northwest at 13 to 15 knots, from January through April. The sheltering effect of the land is apparent when looking at frequencies of winds of 28 knots or more. They blow near the Ambrose Channel entrance about 8 to 9 percent of the time compared to 1 percent at Kennedy Airport and Floyd Bennett Field. Summer winds are often out of the south and southwest with a 10 to 12 knot afternoon peak. Fog in the harbor area is more closely related to land type fogs. In winter it is common on clear, calm mornings and more frequent than in the approaches. Southerlies can also bring winter fogs of the advection type. During the spring and early summer the harbor as well as its approaches are susceptible to advection fog, riding in on east through south winds. A morning peak still exists in the harbor, while the approaches exhibit an afternoon maximum.

(34)

North Atlantic right whales

(35) Endangered North Atlantic right whales may occur within 30 miles of the New York and New Jersey coasts in the approaches to New York Harbor (peak season: November through April). (See North Atlantic Right Whales, indexed as such, in chapter 3 for more information on right whales and recommended measures to avoid collisions.)

(36) All vessels 65 feet or greater in length overall (LOA) and subject to the jurisdiction of the United States are restricted to speeds of 10 knots or less in a Seasonal Management Area existing around the Ports of New York/New Jersey between November 1 and April 30. The area is defined as the waters within a 20-nm radius of 40°29'42.2"N., 73°55'57.6"W. (See **50 CFR 224.105**, chapter 2, for regulations, limitations, and exceptions.)

(37) **Gateway National Recreation Area** and harbor porpoise protection Marine Protected Areas (MPAs) extend from the waters off Long Island along the New Jersey shoreline.

(38) **Information about the coast south of Sandy Hook is contained in United States Coast Pilot 3, Atlantic Coast, Sandy Hook to Cape Henry.**

(39)

New York Harbor

(40) **New York Harbor** is the principal entrance by water to New York City and the surrounding ports. The harbor is divided by The Narrows into Lower Bay and Upper Bay. **The Battery**, the southern tip of Manhattan, is at the junction of East River and Hudson River. The main channel from the sea to the deepwater terminals in Hudson River has a project depth of 45 feet.

(41) **Tra f c Separation Scheme O f New York** has been established in the approaches to New York Harbor from the sea. (See **33 CFR 167.1** through **167.155**, chapter 2, for limits and regulations.)

(42) (See Traffic Separation Schemes, chapter 1, for additional information, and chapter 3 for a discussion of North Atlantic Lane Routes.)

(43)

Pilot boat cruising area, New York Harbor

(44) The pilot boat maintains station approximately 1.5 miles southeast of the Ambrose Channel Lighted Whistle Buoy A. See Pilotage, New York Harbor and Approaches (indexed as such), this chapter.

(45)

Caution

(46) Numerous fishing floats have been reported in the approach to New York Harbor in the Traffic Separation Scheme precautionary area.

(47) **Shipping safety fairways** have been established connecting the eastern approach off Ambrose of Traffic Separation Scheme Off New York and the eastern approach off Nantucket of Traffic Separation Scheme Off New York. (See **33 CFR 166.100** through **166.500**, chapter 2, for limits and regulations.)

(48)

Vessel traffic service, New York

(49) A mandatory vessel traffic service has been established in the navigable waters of Lower New York Harbor**.** (See **33 CFR 161.1** through **161.25**, chapter 2, for limits and regulations.)

(50) In order to prevent groundings and to promote the safety and environmental security of the waterway resources of the Port of New York and New Jersey, the **Harbor Operations Committee of the Port of New York and New Jersey** recommends that all entities responsible for the safe movement of vessels in and through the waters of the Port of New York and New Jersey operate vessels in such a manner as to maintain a minimum clearance of two feet between the deepest draft of their vessel and channel bottom in the following named channels:

(51)

Recommended Minimum Under-keel Clearance for the Ports of New York and New Jersey	
Lower Bay Ambrose Channel* Sandy Hook Chapel Hill Channel	**Hackensack River** Droyers Point to the turning basin at Marion
North River (Hudson River) The Battery to 79th Street	**Passaic River** Kearney Point
Upper Bay Anchorage Channel (The Narrows to The Battery) Bay Ridge Channel Red Hook Channel Buttermilk Channel	**Arthur Kill** Gulfport Reach Pralls Island Reach Tremley Point Reach Fresh Kills Reach Port Reading Reach Port Socony Reach Outer Bridge Reach
Kill Van Kull Constable Hook Reach Bergen Point Reach North of Shooters Island Reach Elizabethport Reach	**Raritan Bay** Raritan Bay West Reach Raritan Bay East Reach Seguine Point Bend Red Bank Reach Ward Point Reach
East River The Battery to Throgs Neck Bridge	**Newark Bay** Newark Bay Reach (Bergan Point to Droyers Point)
*(3 feet minimum under-keel clearance due to wave action)	

(52) A recommended standard of *always afloat* will apply to all other areas, including berths, in the Port District that abut the above listed channels. Ship related factors such as squat, turning heel and other dynamic motions should be considered and, if expected, added to this figure to ensure a minimum clearance of two feet will be maintained throughout a given transit.

(53)

Conformance

(54) The owner, master or person in charge of each vessel has the ultimate responsibility for maintaining this minimum recommended under-keel clearance. Additionally, persons directing the movement of vessels share this responsibility and are expected to advise owners, operators and persons in charge of vessels if, in their judgement, a vessel is not in conformance with these standards.

(55) If at any time a vessel's under-keel clearance is not in conformance with this recommendation and owners, masters or others in charge of the vessel desire to proceed against the pilot's recommendation, pilots are urged to report this to the USCG Captain of the Port (COTP) via VTS New York. Through VTS New York, the COTP will foster communications between the concerned parties in effort to arrive at agreed-upon conditions for safe vessel passage.

(56) It should be recognized that there may be instances when the master, pilot and COTP evaluate a situation and agree that a vessel movement can be made safely even though inconsistent with this recommendation. Such movements may be allowed and should be coordinated through VTS NY so as to ensure the transit of the vessel in question can be assisted as appropriate.

(57) If at any time VTS NY believes a proposed vessel transit may not conform to this recommendation, they will request an assessment be conducted prior to granting a vessel permission to transit within the VTS NY Area. This assessment process will include a review of real-time water level information from the **PORTS®**.

(58) **Physical Oceanographic Real-Time System (PORTS)** is an information acquisition and dissemination technology developed by NOAA. The Port of New York and New Jersey PORTS can be contacted at 866–217–6787 or *co-ops.nos.noaa.gov*.

(59) Also to be considered are the vessel's intended track, including particular areas of inadequate water depth with the pilot's plan for their avoidance, and any other local conditions that might further restrict vessel movement as well as special traffic routing measures that might be required. If VTS NY deems this assessment to be necessary, the VTS Watch Officer will request on VHF-FM that the pilot contact VTS via landline or cellular telephone. Discussion on under-keel clearance plans shall not be conducted on VHF-FM. It is in the best interest of all parties to ensure situations of marginal under-keel clearance are identified and thoroughly discussed well prior to a vessel's underway time. These guidelines became effective in 1996.

(60)

Traffic in New York Harbor

(61) In the East River between the Brooklyn Bridge and Poorhouse Flats Range, shallow-draft vessels customarily keep to the west (Manhattan) side of the channel whether northbound or southbound, thereby reserving the east (Brooklyn) side of the channel for deep-draft vessels. Vessels transiting East River should be aware of this practice and anticipate northbound shallow-draft vessels crossing from east to west in the vicinity of **Corlears Hook**, and from west to east in the vicinity of Newtown Creek.

(62) The New York City Department of Transportation ferries generally follow a prescribed route between The Battery and **St. George** on Staten Island, placing them to the extreme right-hand side of the channel. All mariners are strongly encouraged not to transit close aboard of the ferry slips at The Battery and St. George due to ferries maneuvering.

(63)

Channels

(64) **Ambrose Channel**, the principal entrance, extends from the sea to deep water in Lower Bay. Thence, **Anchorage Channel**, an extension of Ambrose Channel, leads through Upper Bay to The Battery. **Hudson River Channel** continues northward from The Battery for about 5 miles to West 59th Street, Manhattan. Project depth for these channels is 45 feet.

(65) In addition to the usual aids, Ambrose Channel in its outer portion is also marked by **West Bank Light**, shown from a brown conical tower on a black cylindrical pier, in range with **Staten Island Light**, which is shown from a light-colored octagonal brick tower on a gray limestone base on the high ground of Staten Island at Richmond.

(66) **Lower Bay** is that part of New York Harbor extending from Sandy Hook westward to Raritan River and northward to The Narrows.

(67)

Recommended vessel tracks, Ambrose Channel

(68) Recommended vessel tracks for coastwise tug and barge vessels approaching from or leaving toward the south and transiting to New York Harbor via Ambrose Channel, while not mandatory, are recommended by the Harbor Safety, Navigation and Operations Committee of the Port of New York as follows:

(69) Tugs Inbound:

(70) 40°25'20.5"N., 73°52'57.0"W.;

(71) 40°25'48.6"N., 73°52'48.7"W.;

(72) 40°26'31.2"N., 73°52'40.2"W.;

(73) 40°27'09.2"N., 73°52'38.9"W.;

(74) 40°28'05.2"N., 73°52'54.9"W.

(75) Tugs Outbound:

(76) 40°28'26.4"N., 73°53'54.2"W.;

(77) 40°27'52.4"N., 73°53'42.4"W.;

(78) 40°27'32.7"N., 73°53'37.8"W.;

(79) 40°27'05.5"N., 73°53'36.0"W.;

(80) 40°25'42.2"N., 73°53'34.4"W.;

(81) 40°25'22.6"N., 73°53'35.6"W.

(82)

Area to be avoided

(83) To avoid the risk of pollution and damage to the environment, all vessels carrying petroleum or dangerous or toxic cargoes or any other vessel exceeding 1,000 tons should avoid the area enclosed by the following points:

(84) 40°25'44.1"N., 73°52'40.6"W.;

(85) 40°25'51.2"N., 73°50'51.9"W.;

(86) 40°25'28.4"N., 73°50'51.9"W.;

(87) 40°24'43.0"N., 73°51'48.2"W.;

(88) 40°25'13.9"N., 73°52'40.7"W.

(89)

Local magnetic disturbance

(90) Differences of as much as 5° from the normal variation have been reported in Lower Bay in the vicinity of 40°29.6'N., 74°04.2'W.

(91) **Sandy Hook Channel**, project depth 35 feet, provides a secondary route from the sea to deep water in Lower Bay; it connects with **Raritan Bay Channel** to the westward, **Chapel Hill Channel** to the north and **Terminal Channel** to the south. Chapel Hill Channel has a project depth of 30 feet; numerous obstructions with lesser depths are in the channel. The entrance to Sandy Hook Channel is marked by Scotland Lighted Whistle Buoy S, equipped with a racon. The channels are well marked with navigational aids. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A.

(92) **Swash Channel** is a natural buoyed passage between Ambrose Channel and Sandy Hook Channel. Numerous rocks and obstructions are in the entrance to and within the channel; mariners are advised to use the chart as a guide. A lighted range, the rear marker of which is Staten Island Light, leads on a bearing of **305°** to the junction with Chapel Hill Channel.

(93) **False Hook Channel**, along and close to the eastern shore of Sandy Hook, joins Sandy Hook Channel eastward of the north end of Sandy Hook. Strangers should not use the channel.

(94) **Fourteen Foot Channel** enters Lower Bay just north of Ambrose Channel. The channel has a depth of about 16 feet and is unmarked. A shoal area with a least depth of 6 feet is north of the channel in 40°31'55"N., 73°59'00"W.

(95)

Anchorages

(96) General, explosives, naval, and special anchorages have been prescribed for the Port of New York by federal regulations. (See **33 CFR 110.1**, **110.60**, and **110.155**, chapter 2, for limits and regulations.)

(97) Vessels are especially cautioned against anchoring in the vicinity of the pipeline and cable areas as shown on the charts. The pipeline area across The Narrows supplies the water for Staten Island. Extensive cable areas are in the vicinity of Governors Island, The Battery and Ellis Island.

(98) The Harbor Safety, Operations and Navigation Committee of the Port of New York and New Jersey has issued the following recommendations to address the inadequate number of inshore anchorages within the harbor:

(99) Ships awaiting berths will use the offshore anchorages at Ambrose;

(100) All vessels will limit use of Stapleton, Bay Ridge and Gravesend Bay anchorages to the following operations: lightering or loading, bunkering, receiving stores or parts, repairs, Coast Guard inspections, crew changes or emergencies;

(101) Ships will return to the offshore anchorage upon completion of these operations to await berth;

(102) Tugs and barges not engaged in operations described above will refrain from using the deep water anchorages at Stapleton, Bay Ridge (south of Buoy 26) and Gravesend Bay (west of Buoys “A”, “B” and “C”) when there is suitable room east of the Gravesend Bay buoys, north of Buoy 26 in Bay Ridge or in the North River Anchorage or suitable anchorage in Raritan Bay or Perth Amboy.

(103) These recommendations are intended to minimize vessel delays and allow efficient use of current anchorage areas. All vessels are requested to observe these recommendations.

(104)

Dangers

(105) There are five shoal areas in the entrance to New York Harbor that are subject to change in depths and should be avoided by strangers. **False Hook** is off the northeastern side of Sandy Hook. **Flynns Knoll** is between Swash,

Sandy Hook and Chapel Hill Channels. **Romer Shoal**, between Ambrose and Swash Channels, is marked by Romer Shoal Light; a sound signal is sounded from the light station. **East Bank** is northward and eastward of Ambrose Channel. **West Bank** is westward of Ambrose Channel between West Bank (Range Front) Light and Fort Wadsworth. Numerous rocks and obstructions lie between West Bank and the western limit of Ambrose Channel. The chart is the best guide.

(106) The tip of Sandy Hook is changeable, and the area around it is subject to severe shoaling; caution should be exercised in the area.

(107) Mariners are cautioned to maintain a sharp lookout for floating debris in the harbor and channels.

(108)

Current

(109) The flood current entering Lower Bay from the sea attains a velocity of about 2 knots in Ambrose Channel entrance, near the outer extremities of Sandy Hook, Coney Island and The Narrows. It sets generally parallel to the lower straight section of Ambrose Channel and tends to continue to that direction where the channel bends toward The Narrows, setting more or less diagonally across the upper straight section of Ambrose Channel. At the beginning of the flood, the current sets in at the bottom and near the shores while it is still ebbing at the surface in Ambrose Channel.

(110) The ebb in Lower Bay is generally stronger than the flood by 10 percent or more. At its strength it sets from The Narrows approximately parallel to the upper straight end of the lower straight section.

(111) In the channel northward of Governors Island, crosscurrents may be encountered. During the first 2 hours of flood in this channel (eastward), the current in Hudson River is still ebbing (southward). In the first 1.5 hours of ebb (westward) in the channel north of Governors Island, the current in Hudson River is still flooding (northward). At such times large vessels must take special care in navigating the channel. It is reported that the most dangerous time is about 2 hours after high water at The Battery. At this time the current is setting north in the Hudson River and westward from the East River. The effect on a large vessel coming from southward and turning into the East River is to throw her stern to port and her bow to starboard, thus causing a sheer to starboard toward the shoals off the north end of Governors Island. When coming from northward in the Hudson River the same effect tends to prevent the vessel from turning and to cause her to overrun her course. These crosscurrents are known locally as **The Spider**.

(112) At the seaward end of Ambrose Channel the velocity of the flood current is 1.7 knots and of the ebb current 2.3 knots.

(113) When the ebb is strong the currents in both Ambrose and Swash Channels tend to set toward Romer Shoal. Caution should be maintained to prevent being set onto Romer Shoal when using either channel. On the flood and especially with a westerly wind, caution should be exercised to prevent being set onto Romer Shoal when using Swash Channel.

(114) In The Narrows the velocity of the flood current is about 1.7 knots and of the ebb current 2 knots. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for daily predictions of slack water and strength of current. Links to a user guide for this service can be found in chapter 1 of this book.

(115) In the entrance to Hudson River the velocity of the flood and ebb currents is 1.4 knots. Off Grants Tomb, the flood and ebb strengths are 1.6 and 1.9 knots, respectively.

(116) In 1991, tidal currents in The Narrows, Arthur Kill, Kill Van Kull and Hell Gate were reported to deviate significantly from official predictions published by NOAA. Mariners should exercise caution and discretion in the use of published tidal current predictions for these locations.

(117)

Ice

(118) Navigation of the channels in the Port of New York and New Jersey is not restricted by ice. The main channels do not freeze over, and any ice in the smaller waterways is well broken up by tugs and general traffic. Freshwater ice is brought down the Hudson River in large floes during periods of thaws or winter freshets. Occasionally there are large accumulations of ice at Spuyten Duyvil where Harlem River joins the Hudson, and at such times it is difficult for low-powered vessels or tows to make much headway. Under conditions of strong winds the slips on the exposed side of the channel become packed with drift ice, causing difficulty when maneuvering in the slip or when berthing. During extremely severe winters navigation is interfered with seriously for only short periods of time.

(119)

Weather, New York and vicinity

(120) New York City, an area exceeding 300 square statute miles (777 square km), is located on the Atlantic coastal plain at the mouth of the Hudson River. The terrain is flat and diversified by numerous waterways; all but one of the city's five boroughs are situated on islands. Elevations range from less than 50 feet (15.2 m) over most of Manhattan, Brooklyn and Queens to almost 300 feet (91.4 m) in the northern part of Manhattan and the Bronx and over 400 feet (122 m) in Richmond (Staten Island).

(121) Despite its nearness to the ocean and the numerous bays and rivers nearby, New York City has a climate that more closely resembles the continental type of climate than it does the maritime type. Its modified continental climate follows from the fact that weather conditions affecting the city usually approach from a westerly direction and not from the ocean on the east. Some important exceptions to this must be noted, since the oceanic influence is by no means entirely absent. During the summer, local "sea breezes," winds blowing onshore from the cool water surface often moderate the

afternoon heat, and most often in winter, coastal storms, accompanied by easterly winds, produce, on occasion, considerable amounts of precipitation.

(122) From November through April the prevailing winds are from the northwest; for the remainder of the year the prevailing winds are southwesterly. Gales with velocities of 35 knots or more are predominately from the northwest.

(123) At New York/Kennedy the average annual temperature is 54°F (12.2°C). The average high is 61°F (16.1°C) and the average low is 47°F (8.3°C). July is the warmest month, with an average high of 83°F (28°C) and an average low of 69°F (20.6°C). January is the coolest month, with an average high of 39°F (3.9°C) and an average low of 26°F (-3.3°C). The warmest temperature on record for New York/Kennedy is 104°F (40°C), recorded in July 1966, and the coldest temperature on record is -2°F (-18.9°C), recorded in January 1985. On average, ten days each year record high temperatures in excess of 90°F (32.2°C) and 78 days record minimum temperatures below 32°F (0°C). An average of only one day each year has an extreme minimum below 5°F (-15°C).

(124) Precipitation is both moderate and distributed evenly throughout the year with a spread of only 1.06 inches (28 mm) between the wettest and driest months. May is the wettest month with an average precipitation total of 3.92 inches (991 mm) and February the driest with 2.86 inches (74 mm). Average annual precipitation is about 41 inches (1041 mm). Most of the rainfall from June through September comes from thunderstorms; therefore, it is usually of brief duration but relatively intense. Thunderstorm days average 24 each year. From October to April, however, precipitation is generally associated with widespread storm areas, so that day-long rain or snow is common.

(125) Snow falls an average 30 days each year and averages 22 inches (559 mm) in any given year. The snowiest month is February with an average of eight inches (203 mm). Snow has fallen in each month October through April. The greatest 24-hour total snowfall was 20 inches (508 mm), which fell in February 1969.

(126) Many tropical storms have influenced the area. Hurricane Gloria passed within 20 nautical miles east of the Kennedy airport in September 1985. Gloria had winds approaching 75 knots at time of landfall, about halfway between Kennedy airport and Islip. Only two days earlier, Gloria was a more respectable 125-knot hurricane.

(127) The National Weather Service maintains several offices in New York where barometers can be compared.

(128)

Pilotage, New York Harbor and approaches

(129) Foreign vessels and U.S. vessels under register entering or departing from the Port of New York and New Jersey must employ a pilot licensed by the State of New York or New Jersey. Enrolled vessels must have on board or employ a pilot licensed by the federal government.

(130) State and federal pilotage service for vessels entering the Port of New York and New Jersey through Lower Bay and intra-harbor movements is available from the United New York New Jersey Sandy Hook Pilot Association, 201 Edgewater Street, Staten Island, NY 10305, telephone 718–448–3900, FAX 718–876–8055, e-mail: pilotoffice@sandyhookpilots.com.

(131) The Sandy Hook pilot vessel maintains station approximately 1.5 miles southeast of the Ambrose Channel Lighted Whistle Buoy A. All traffic passes through a precautionary area transiting to the pilot station. Most vessels choose to approach the pilot station directly since Ambrose Light was disestablished. Traffic within the precautionary area may consist of vessels making the transition between operating in Ambrose or Sandy Hook Channel and one of the traffic lanes. Mariners are advised to exercise extreme care in navigating within this area. The pilot vessels have a black hull and white superstructure, with the name PILOT NO. 1 or PILOT NO. 2 in yellow on each side and are equipped with AIS and transmit either "PILOT NO.1" or "PILOT NO.2" A pilot vessel is always on station; boarding is made from smaller boats that are also AIS equipped and broadcast "P/B (name)." The pilot vessel monitors VHF-FM channels 16, 13 and 73 and works on 73.

(132) Pilot services are arranged in advance through ships' agents. A 24-hour advance notice of ETA, with a 3-hour update is requested.

(133) Masters of vessels entering the Port of New York and New Jersey are requested, prior to the time of boarding, to contact the pilot boat to ascertain a proper boarding speed, make a lee for the pilot boat and have a pilot ladder over the side about 1 meter above the water.

(134) **Pilot Ladder Requirements:**

(135) 1. Arrangements shall be provided to enable the pilot to embark and disembark safely on either side of the ship.

(136) 2. The rigging of the pilot transfer arrangements and the embarkation of a pilot shall be supervised by a responsible licensed deck officer having means of communication with the navigation bridge. The officer shall also arrange for the escort of the pilot by a safe route to and from the navigation bridge.

(137) 3. The ladder must be made in one length and not consist of two lengths shackled or lashed together and should be equipped with spreaders about 10 feet (3m) apart.

(138) 4. Whenever the distance from the surface of the water to the access point of the ship is more than 9 meters, an accommodation ladder in conjunction with a pilot ladder must be used. The accommodation ladder shall be sited leading aft and shall be secured to the hull. When in use, means shall be provided to secure the lower platform of the accommodation ladder to the ship's side, so as to ensure that the lower end of the accommodation ladder and the lower platform are held firmly against the ship's side and the pilot ladder remains against the ship's side.

(139) 5. In the case of a combination arrangement where the accommodation ladder is fitted with a trap door in

the bottom platform, the pilot ladder shall be rigged up through the trap door and extend above the platform to the height of the handrail.

(140) 6. Adequate lighting shall be provided to illuminate the transfer arrangements alongside and the position on the deck where a pilot embarks and disembarks.

(141) 7. Trailing lines or retrieving lines must not be attached to the lower ends of the ladder.

(142) 8. Ladders must be rigged well clear of the water, clear of any discharge outlets, and at a place near midships.

(143) 9. Means shall be provided to ensure safe and unobstructed passage for any person embarking or disembarking the ship between the head of the pilot ladder and the ship's deck. Adequate handholds or two rigidly secured handhold stanchions shall be fitted.

(144) 10. Vessels must comply with SOLAS Chapter V Regulation 23 – Pilot Transfer Arrangements.

(145) 11. The pilot ladder shall be rigged 1.5 meters above the water's edge. A heaving line must be available.

(146) Pilotage for these waters for U.S. vessels enrolled in coastwise trade is also available from the Interport Pilots Agency, Inc., *interportpilots.com*, 906 Port Monmouth Road, Port Monmouth, NJ 07758-0236, telephone 732–787–5554 (24 hours), email interport@verizon.net. The Interport Pilots office monitors VHF-FM channels 16 and 65A during business hours. Pilot boats are KEN JOHNSON, 47-foot, blue hull and white superstructure with the word PILOT displayed on both sides, and INTERPORT PILOT, 50-foot with the same colors. Boats monitor VHF-FM channels 16 and 13 one and a half hours prior to the vessel's scheduled ETA, work on channel 65A and are equipped with AIS and transmit 'PILOTBOAT (NAME).'

(147) Vessels are generally boarded in the charted, designated pilot boarding area, located southeast of the Ambrose Channel Lighted Whistle Buoy A at 40°26'47"N., 73°48'27"W. Arrangements for pilot services are made in advance through ship's agents or directly to Interport Pilots Agency, Inc.

(148)

Pilotage, New York Harbor from Long Island Sound

(149) Foreign vessels and U.S. vessels under register entering or departing from the Port of New York and New Jersey from Long Island Sound must employ a pilot licensed by the State of New York. Enrolled vessels must have on board or employ a pilot licensed by the Federal Government. Pilotage service for vessels entering the Port of New York and New Jersey from Long Island Sound is available from the United New York New Jersey Sandy Hook Pilot Association (see above). The pilot boat boarding area is off Execution Rocks. The pilot boat ties up at a pier on the east side of City Island about 0.4 mile northward of Belden Point. The pilot station, on the pier, and the pilot boat monitor VHF-FM channel 13 when vessels are scheduled to arrive. The 48-foot pilot boat has a black hull with the word PILOT in red letters on each side of the house. Arrangements for pilots are made in advance either directly by the vessel or through ships' agents. Notification is mandatory 24 hours prior to arrival and ETA updates are required 12 and 6 hours prior to arrival.

(150) Pilotage for U.S. enrolled vessels in the coastwise trade is available from the United New York New Jersey Sandy Hook Pilot Association (see above) and Interport Pilots Agency, Inc. (see above). Pilot boats are KEN JOHNSON, 47-foot with blue hull and white superstructure with the word PILOT displayed on both sides, and INTERPORT PILOT, 50-foot with the same colors. Boats monitor VHF-FM channels 16 and 13 two hours prior to the vessel's scheduled ETA, work on channel 65A, and are equipped with AIS. Interport Pilots board vessels bound from Long Island Sound into New York Harbor via the East River at any Long Island Sound port, Montauk Point or Point Judith Pilot Station or in the vicinity of Execution Rocks.

(151)

Pilotage, Hudson River

(152) See Pilotage, Hudson River (indexed as such), chapter 12.

(153)

Towage

(154) The Port of New York and New Jersey has several towing companies with radio-equipped tugs with over 4,000 hp. Arrangements for tugs are usually made in advance by ships' agents. Fireboats are stationed throughout the harbor.

(155) New York is a **customs port of entry** and the headquarters of the **Regional Commissioner.**

(156)

Quarantine, customs, immigration and agricultural quarantine

(157) (See chapter 3, Vessel Arrival Inspections, and Appendix A for addresses.)

(158) **Quarantine** is enforced in accordance with regulations of the U.S. Public Health Service. (See Public Health Service, chapter 1.)

(159)

Coast Guard

(160) A Coast Guard station is at Rosebank on Staten Island. A Coast Guard Command Center including a **Captain of the Port** office and **Marine Inspection O f ce** are at **Fort Wadsworth**, Staten Island.

(161)

Harbor regulations

(162) The administration of the Port of New York and New Jersey and the enforcement of its laws are vested in no single body but are divided among various departments of the federal, state and municipal Governments.

(163)

Speed

(164) The Coast Guard desires to warn masters and pilots of all types of vessels that possible action may result against their licenses and criminal procedures may be exercised,

when the wash of a vessel proceeding at excessive speed in confined waters endangers life, limb or property. Damage to vessels moored at docks and terminals has been reported. The parting of a mooring line may cause a serious oil fire or damage to pipelines or barges that are being loaded or discharged at chemical and petroleum company terminals. Damage caused by excessive speed may also lead to a possible suit by the injured party against owners, masters or pilots for monetary recovery.

(165) The **New York Economic Development Corporation** administers the piers along the New York City waterfront. The office is at 110 William Street. Additional information can be found at *nycedc.com*.

(166) The **Port Authority of New York and New Jersey** is an executive body appointed by the Governors of New York and New Jersey. The Authority's Port Department serves as a bistate port development, operations, maintenance and promotion organization. The Port Authority administers piers in Manhattan, Brooklyn, Hoboken, Port Newark and Port Elizabeth. The office of the Authority is at the 233 Park Avenue South, New York, NY 10003.

(167)

Wharves

(168) The Port of New York and New Jersey has over 1,100 waterfront facilities. Most of these facilities are privately owned and operated, and the rest are owned or operated by either the railroads serving the port, the Port Authority of New York and New Jersey, the City of New York, the States of New York and New Jersey, the federal government or other municipalities.

(169) The major steamship passenger terminal, the New York City Passenger Ship Terminal, is along the east side of the Hudson River (North River) above The Battery. Containership terminals are throughout the port, but principally at Elizabeth, Newark, Jersey City and Weehawken, NJ. Other containership facilities are at Howland Hook, Staten Island and Brooklyn. Break-bulk general cargo terminals are throughout the port but principally along the east side of Upper New York Bay, on the East River and at Port Newark. Petroleum and other liquid cargo facilities are along Arthur Kill, on the Passaic and Hackensack Rivers and along Newtown Creek, Brooklyn.

(170) General cargo in the port is usually handled to and from vessels by ship's tackle. Heavy lifts up to 500 tons, floating cranes up to 500 tons, and derricks are available in port. Most of the waterfront facilities throughout the port have highway and railroad connections.

(171) The wharves and piers of New York City along the waterfronts of the Hudson and East Rivers are numbered beginning at The Battery and follow in sequence eastward along the East River and northward along the Hudson River.

(172)

Supplies

(173) Provisions and supplies of all kinds are available in the Port of New York and New Jersey. All grades of heavy marine bunker fuel, lubricants and diesel fuel can be obtained. Large vessels are usually bunkered at their berths by tank barges or self-propelled tankers. Water is available at most of the piers and wharves.

(174)

Repairs

(175) The Port of New York and New Jersey has extensive facilities for making all types of repairs to vessels of all sizes. The shipyards at Brooklyn, Hoboken, Staten Island and Queens can drydock some of the largest ocean-going vessels and can make major repairs to hull, electronic equipment, machinery and propulsion plants. Also within the port area, a number of firms without waterfront facilities are engaged in various types of marine repair work. These firms maintain ships and portable equipment for making above-waterline repairs and for installation of equipment, gear and machinery on all types of craft at berth. Several salvage companies also perform all types of salvage work.

(176) The largest floating drydock, east of Red Hook Channel and on the east side of Erie Basin, has a lifting capacity of 16,000 tons, an overall length of 580 feet, a maximum clear inside width of 100 feet, and a depth of 28 feet over the keel blocks. The largest graving dock is on the east side of Wallabout Bay at the site of the former New York Naval Shipyard. The dock has a clear length of 1,092 feet, clear gate width of 143 feet, top and bottom inside widths of 150 feet, and 34 feet over the keel blocks; cranes to 200 tons are available. The largest marine railway, on the east side of East Mill Basin in Jamaica Bay, can handle vessels up to 300 tons of 120 feet long.

(177)

Communications

(178) The Port of New York and New Jersey is served by three trunkline and one short-line railroad, numerous trucking firms engaged in long- and short-haul freight service and several bus companies. Over 100 shipping companies connect the port with the principal U.S. and foreign ports.

(179) Three major airports, John F. Kennedy (New York) International, La Guardia and Newark, provide frequent scheduled service between New York and domestic and overseas points.

(180)

Rockaway Inlet to Bergen Basin

(181) **Rockaway Inlet**, the entrance to Jamaica Bay, is between **Rockaway Point** on the southeast side and **Manhattan Beach** and **Barren Island** on the north side. A breakwater, marked near the outer end by a light, extends south from Rockaway Point. The entrance channel extends westward of the breakwater and is marked by

(185)

Rockaway Inlet and Gil Hodges Memorial Bridge
Image courtesy of Airphoto/Jim Wark (2004)

lighted and unlighted buoys. A shifting sandbar is located about 0.6 mile southeast of the breakwater light. A shoal with depths of less than 1 foot and marked by breakers is west of the entrance channel. Numerous obstructions lie from southeast to southwest of the breakwater light and numerous wrecks are farther inside the inlet; the chart is the best guide.

(182) In 1980, shoaling to about 3 feet was reported in the inlet about 1.75 miles west of the Gil Hodges Memorial Bridge in about 40°34'21"N., 73°55'29.5"W.

(183)

Current

(184) The tidal current in the entrance channel near Rockaway Point has a velocity of about 2.2 knots. In 1975, a strong east-to-west current, believed to have been the result of tidal flow, was observed at the entrance to Rockaway Inlet near the seaward end of the jetty. This current is of sufficient strength to cause a vessel to veer suddenly off course when entering or exiting the channel. East of Barren Island the velocity is is about 1.5 knots. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(186) **Jamaica Bay** is on the south shore of Long Island about 15 miles southeastward of The Battery, New York City. The bay is characterized by numerous meadows, hassocks and marshes. The north and east shores are bordered by marshlands that extend inland for a short distance. Several small tidal creeks enter the bay from the north. Channels and basins have been dredged to project depths of 12 to 20 feet for use of craft operating in the bay. Rockaway Beach forms the south shore. The bay is about 7 miles long and 3.5 miles wide and covers an area of about 22.5 square miles. The greater portion of the bay is in the Boroughs of Brooklyn and Queens, New York City; and a small section of the eastern extremity, consisting of parts of Motts Basin and Head of Bay, is in Nassau County.

(188)

Anchorages

(189) Special anchorages are in Jamaica Bay at Canarsie Beach and adjacent East Broad Channel. (See **33 CFR 110.1** and **110.60**, chapter 2, for limits and regulations.)

(190)

No-Discharge Zone

(191) The State of New York, with the approval of the Environmental Protection Agency, has established a No-Discharge Zone (NDZ) in the open waters and tributaries of Jamaica Bay.

(192) Within the NDZ, discharge of sewage, whether treated or untreated, from all vessels is prohibited. Outside the NDZ, discharge of sewage is regulated by **40 CFR 140**, (see chapter 2).

(187)

Structures across Jamaica Bay and its Tributaries

Name•Description•Type	Location	Clear Width of Draw or Span Opening (feet)	Clear Height above Mean High Water (feet)	Information
Jamaica Bay				
Gil Hodges Memorial Bridge (vertical lift)	40°44'21"N., 73°57'09"W.	475	55 (down), 152 (up)	Notes 1 and 2 Call sign KIL-819
Cross Bay Memorial Bridge (fixed)	40°35'33"N., 73°49'13"W.	200	52	Beach Channel crossing
Railroad Bridge (swing)	40°35'45"N., 73°48'39"W.	101	26	Beach Channel crossing Note 3
Railroad Bridge (fixed)	40°38'40"N., 73°49'31"W.	100	26	North Channel crossing Note 3
Cross Bay Boulevard Bridge (fixed)	40°38'40"N., 73°50'10"W.	102	26	North Channel crossing
Pedestrian bridge (fixed)	40°34'57"N., 73°56'58"W.	39	8	Sheepshead Bay crossing
Shore Parkway Bridge (fixed)	40°35'10"N., 73°54'45"W.	177	35	Gerritsen Inlet crossing
Shore Parkway Bridge (fixed)	40°36'18"N., 73°53'57"W.	131	60	Mill Basin crossing
Shore Parkway Bridge (fixed)	40°37'19"N., 73°53'48"W.	98	29	Paerdegat Basin crossing
Shore Parkway Bridge (fixed)	40°38'18"N., 73°52'43"W.	63	21	Fresh Creek crossing
Shore Parkway Bridge (fixed)	40°38'47"N., 73°52'25"W.	46	20	Hendrix Creek crossing
Pedestrian Bridge (fixed)	40°39'12"N., 73°49'54"W.	63	17	Hawtree Basin crossing
Hook Creek				
Meyer Avenue Bridge (fixed)	40°38'11"N., 73°44'29"W.	24	8	
Rockaway Turnpike Bridge (fixed)	40°38'07"N., 73°44'24"W.	29	4	
Rockaway Turnpike Bridge (fixed)	40°37'55"N., 73°44'22"W.	33	12	
East Avenue Bridge (fixed)	40°38'04"N., 73°44'26"W.	30	3	
Motts Basin				
Overhead power cable	40°37'01"N., 73°45'40"W.		70	Across north arm
Overhead power cable	40°36'42"N., 73°45'39"W.		95	Across south arm
Overhead power cable	40°36'42"N., 73°45'35"W.		92	Across south arm

Note 1 – See **33 CFR 117.1** through 117.59 and **117.795**, chapter 2, for drawbridge regulations.
Note 2 – Bridgetender monitors VHF-FM channel 13.
Note 3 – The trestle crossing East Broad Channel of this railroad is closed to navigation.

(193) The commercial vessel traffic in Jamaica Bay consists of motor tankers, barges, and tugs. The bay is used extensively by pleasure craft.

(194) Jamaica Bay has excellent transportation facilities. Highways connect with all of Long Island and New York City, and a branch of the New York City subway system crosses the central part of the bay and extends eastward and westward along the Rockaway peninsula with stations at Far Rockaway and Inwood serving the Motts Basin area.

(195) **Ice** is a problem in Jamaica Bay, mainly in the tributaries and basins, from early January to about mid-March.

(196) **Sheepshead Bay**, on the northern side of the eastern extremity of Coney Island and northward of **Manhattan Beach**, is well protected and is used by numerous pleasure and party fishing craft. Use caution when navigating and anchoring; approaches are subject to continual change due to shoaling. Buoys may frequently be relocated to mark best water. Numerous wrecks and obstructions are charted from a 2014 hydrographic survey. A private light marks the outer limit of a sewer outfall that extends southward from the bay.

(197)

Anchorages

(198) Special anchorages are in Sheepshead Bay. (See **33 CFR 110.1** and **110.60**, chapter 2, for limits and regulations.)

(199)

Small-craft facility

(200) A small-craft facility in the bay can handle craft to 1½ tons. Mooring, electricity, diesel fuel, water, ice, marine supplies and storage are available.

(201) **Plumb Beach Channel**, northward of Rockaway Inlet, is the common approach to **Gerritsen Inlet**, **Shell Bank Creek**, Gerritsen Creek and Mill Creek. The channel is marked by buoys; mariners are advised to follow the buoys through the inlet closely. Two shoal areas are near Plumb Beach Channel Buoy 7. The first is

north-northeast of the buoy with a least depth of 2 feet; it extends to midchannel where the least depth is 4 feet. The other is about 0.1 mile southeast of the buoy with a least depth of 2 feet.

(202) From the highway bridge over Gerritsen Inlet, Shell Bank Creek leads westerly and **Gerritsen Creek** and **Mill Creek** lead northwesterly. There are dangerous pilings and remains of old barges along the south side of Shell Bank Creek and several submerged wrecks in Gerritsen and Mill Creeks. The fixed highway bridge over Mill Creek is in ruins; mariners are advised to exercise caution in this area as some parts of the bridge structure have fallen into the water and are an obstruction to navigation.

(203)

Small-craft facilities

(204) Small-craft facilities on Shell Bank Creek can provide berths with electricity, gasoline, diesel fuel, water, ice, marine supplies, a pumpout facility, lifts to 60 tons and a 90-foot marine railway; complete hull and engine repairs are available.

(205) **Dead Horse Bay** makes into the southwest side of Barren Island eastward of the highway bridge across Gerritsen Inlet. A marina, on the north side of the bay, has berths and moorings. Numerous wrecks and obstructions are in the entrance to the bay and in the southwest corner of the marina. The chart is the best guide.

(206) **Island Channel** leads northerly from just eastward of Barren Island to Bergen Beach, thence northeasterly in **North Channel** to Howard Beach. In 1998, depths of about 12 feet can be carried to Howard Beach. The channels are marked by lighted and unlighted buoys.

(207) **Big Fishkill Channel** and **Pumpkin Patch Channel** lead in a northeasterly direction from Runway Channel just west of **Ru f e Bar** and join North Channel 0.3 mile west of the North Channel Bridge at Howard Beach.

(208) **Mill Basin** is northward of Barren Island on the west side of Jamaica Bay. Commercial traffic in the basin consists of occasional barge shipments of petroleum.

(209)

Small-craft facilities

(210) Small-craft facilities in the basin can provide berths with electricity, gasoline, water, ice, marine supplies and complete hull and engine repairs; a 50-ton marine railway and lifts to 20 tons are available.

(211) **East Mill Basin** is about 0.4 mile northeastward of Mill Basin. Small-craft facilities in the basin can provide berths with electricity, water, marine supplies, a 15-ton forklift, a 100-ton travel lift and marine railways to 300 tons; complete hull and engine repairs are available.

(212) **Bergen Beach** is a community about 2 miles north of Barren Island. **Paerdegat Basin** is just north of Bergen Beach. A marina at the head of the basin can haul out craft up to 15 tons; gasoline, marine supplies and water are available, however, no repairs are available for the public. In 1981, a reported depth of 8 feet could be taken to the marina. Several yacht clubs are also in the basin.

(213) **Canarsie**, a town on the northwestern shore of Jamaica Bay, is a part of New York City. Canarsie Pier, on the northwest shore of Jamaica Bay between Paerdegat Basin and Fresh Creek, has two prominent flagpoles near its center. The pier is structurally unsafe, and landing is not permitted. The pier is a part of Gateway National Recreation Area.

(214) **Fresh Creek**, 0.6 mile northeastward of the pier at Canarsie, has a midchannel depth of about 8 feet. **Hendrix Creek**, 0.4 mile northeastward of Fresh Creek, is the site of a water pollution control plant. Sludge vessels operate from the pier at the southwestern entrance to the creek. **Old Mill Creek**, 1.1 miles northeastward of Fresh Creek, bares at low water just above the entrance. Fresh, Hendrix and Old Mill Creeks were little used in 1971.

(215) **Howard Beach**, about 2.5 miles eastward of Canarsie, on the north side of Jamaica Bay, has several basins for boats.

(216) **Shellbank Basin**, just west of Howard Beach, extends northward about 1 mile from North Channel. There is shoaling upon entering the channel with a 4-foot obstruction at 40°38'57.5"N., 073°50'10.7"W. The basin has numerous small piers, float landings and other small-craft facilities along the west side. Berths with electricity, water, a 15-ton lift and complete hull and engine repairs are available.

(217) **Hawtree Basin**, about 0.25 mile eastward of Shellbank Basin, has depths ranging from 5 to 24 feet and an obstruction covered 6 feet at 40°39'03.9"N., 073°49’52.0"W.

(218) **Rockaway Beach** is a popular summer resort on the barrier beach forming the southern extremity of Jamaica Bay. Train and bus transportation is available to New York City. Excursion boats operate between New York and Rockaway Beach during the summer only.

(219) **Beach Channel** is on the north side of Rockaway Beach. A federal project provides for a channel 18 feet deep from Rockaway Inlet to about 700 yards above Gil Hodges Memorial Bridge, thence 15 feet deep to the junction with Grass Hassock Channel.

(220) **Barbadoes Basin** is adjacent to the Beach Channel railroad bridge. A facility in the southeast corner of the basin receives broken concrete by barge for recycling.

(221) **Vernam Basin** is northeast of Barbadoes Basin. A facility in the southwest corner of the basin receives petroleum products by barge. In 2007, the controlling depth at the facility was reported to be 15 feet. A facility in the southeast corner of the basin receives sand, gravel and stone by barge. A small-craft facility in the basin has berths, electricity, water, open storage and lifts to 72 tons. Repairs for fishing boats can be made.

(222) **Winhole Channel**, a natural channel marked by buoys, seasonal lights and a daybeacon, extends 1 mile northward to Grassy Bay from the junction of Beach Channel with Grass Hassock Channel. Winhole Channel has a least depth of about 11 feet, except for reported shoaling to 4 feet extending into the channel northeast from Winhole Channel Light 3 in about 40°36.8'N.,

73°48.4'W. Winhole Channel Shoal Daybeacon marks the center of a shoal near the north end of the channel. The daybeacon should not be passed close aboard. A lighted buoy marks the junction of Beach, Grass Hassock and Winhole Channels.

(223) **Grass Hassock Channel** joins Beach Channel off **Brant Point** and continues in a northeasterly direction to Head of Bay. The shallowest water is abeam Brant Point between Buoy 14 and Buoy 16 and at the junction with Negro Bar Channel in the vicinity of Lighted Buoy 23.

(224) **Sommerville Basin**, about 1.2 miles eastward of the railroad bridge at Rockaway Beach, has depths of 27 to 40 feet inside. In 1981, depths of about 15 feet were reported in the approach. Several charted sunken wrecks are in the basin. A boatyard at the head of the basin has berths, electricity, gasoline, water, ice, limited marine supplies, storage facilities, a launching ramp, a 45-foot marine railway and a 7-ton mobile hoist; engine and hull repairs can be made.

(225) **Motts Basin**, a tidal inlet in the eastern part of Jamaica Bay, entered through **Joseph Sanford Jr. Channel**, partially separates the communities of **Inwood** and **Far Rockaway**. Two branch channels lead from inside the entrance to the northeasterly and southeasterly ends of the basin. **Ice** may obstruct vessel movement in the basin during severe winters.

(226) Overhead power cables across Motts Basin have the following clearances: one over the northerly arm, 70 feet; two over the southerly arm, least clearance 92 feet; and one over the cut on the south side of the southerly arm, 60 feet. A retractable boom is on the south shore of the basin about 90 yards northwest of the overhead cable tower. A light is shown from the boom when it is extended into the water.

(227) Depths alongside the wharves in Motts Basin range from 10 to 20 feet. Waterborne commerce in the basin is chiefly in petroleum products.

(228) **Head of Bay** joins Grass Hassock Channel near **Northwest Point** and extends in a northeasterly direction on the south side of **John F. Kennedy (New York) International Airport**. Depths of about 15 feet are in the entrance channel and channel in the bay; aids mark the channels. In 2007, it was reported that a draft of 16 feet could be taken to the facilities at **Uncle Daniels Point**, 14 feet could be taken to facilities on **Motts Creek**, and 14 feet could be taken to facilities at **Norton Point**. Several small marinas in the bay can provide berths, electricity, water, ice, marine supplies, storage facilities and a launching ramp; minor engine and hull repairs can be made.

(229) **Thurston Basin**, at the northeastern extremity of Head of Bay, has reported depths of 10 feet at the entrance decreasing to 2 feet at the head.

(230) **Grassy Bay**, along the southwestern side of John F. Kennedy (New York) International Airport in the northeastern part of Jamaica Bay, is blocked at the southeastern end by an airport runway. The runway continues into the marshlands on the southerly side of the bay.

(231) **Bergen Basin**, at the northern extremity of Grassy Bay, has depths of about 15 feet with lesser depths in the eastern arm of the basin. The entrance is marked by buoys. Conspicuous are a yellow brick circular tank about 40 feet high on the southwestern side of the entrance and the numerous oil storage tanks at the head of the basin on the eastern shore. Coastal tankers and sand-and-gravel barge tows account for most of the commerce in the basin. In 1988, a sunken wreck was reported in the eastern arm of the basin in about 40°39.7'N., 73°49.1'W.

(232) **Safety and security zones** are in the vicinity of John F. Kennedy International Airport, Bergen Basin and Thurston Basin. (See **33 CFR 165.1** through **165.7**, **165.20** through **165.33**, and **165.169**, chapter 2, for limits and regulations.) Within the safety and security zone, a boom 1,500 feet in length is approximately 60 feet offshore on the east side of the Bergen basin.

(233)

Coney Island to Pierhead Channel

(234) **Coney Island**, on the northern side of the entrance to New York Harbor, is a large summer amusement resort. Numerous stacks, towers and amusement rides, including a red steel parachute tower 303 feet high, are prominent on the island. **Coney Island Light** (40°34'36"N., 74°00'42"W.), 75 feet above the water, is shown from a white square skeleton tower on **Norton Point**, the westernmost extremity of the island.

(235) **Coney Island Channel** is a buoyed passage along the south side of Coney Island that leads from deep water in Lower Bay to Rockaway Inlet. It is used principally by vessels going to Jamaica Bay and Coney Island. A shoal area with a least depth of 6 feet is about 0.1 mile west of Buoy 3.

(236) **Gravesend Bay**, northward of Coney Island, affords good anchorage; a **general anchorage** is in the bay. (See **33 CFR 110.1** and **110.155(e)**, chapter 2, for limits and regulations.)

(237) **Coney Island Creek** is at the southeastern end of Gravesend Bay and on the north side of Coney Island. Commercial traffic on the creek consists mainly of occasional barge shipments of sand and gravel. The area northward of the entrance to the creek is being filled, and piling is along the northern side of the creek at the filling site. Numerous obstructions and wrecks are in the creek; mariners are advised to seek local knowledge before entering. The creek is crossed by three fixed bridges having a least clearance of 3 feet. The ruins of a fourth bridge exist about 0.17 mile above the third bridge in about 40°34'49"N., 73°58'42"W. A boatyard about 0.8 mile above the creek entrance provides berths, electricity, gasoline, water, ice, storage, marine supplies, and hull and engine repairs. Lifts to 14 tons are available. In 1981, a reported depth of 4 feet could be carried to the boatyard.

(242)

(238) A buoyed channel with a least depth of 10 feet leads from deep water northward of Coney Island to off the docks in the eastern part of Gravesend Bay.

(239)

Small-craft facility

(240) A small-craft facility on Gravesend Bay can provide berths, electricity, gasoline, diesel fuel, water, ice, storage and marine supplies and hull and engine repairs. Lifts up to 30 tons are available. In 1981, a reported depth of 15 feet could be carried to the marina.

(241) **The Narrows**, connecting Lower Bay and Upper Bay of New York Harbor, has a clear width of over 0.6 mile at its narrowest point between Fort Wadsworth and Fort Hamilton. The**Verrazano Narrows Bridge**, a fixed suspension span, crosses The Narrows at these two points linking Staten Island with Brooklyn. The bridge has a vertical clearance of 215 feet for a midchannel width of 2,000 feet. **Note:** A traveling maintenance platform, when in operation, reduces the vertical clearances by 15 feet. A sound signal is sounded from the eastern end of the bridge. A **safety zone** is near the eastern end of the bridge. (See **33 CFR 165.1** through **165.9**, **165.20** through **165.23**, and **165.172**, chapter 2, for limits and regulations.)

(243)

Coast Guard Station

(244) Coast Guard Station New York is on the east side of Staten Island about 0.6 mile northwest of the Verrazano Narrows Bridge.

(245) **Upper Bay** is that portion of New York Harbor between The Narrows and The Battery. **Anchorage Channel**, marked by lighted buoys, is the main passage through the middle of the bay. **Bay Ridge Flats** is a shoal area with depths of 8 to 20 feet east of Anchorage Channel. **Gowanus Flats** is at the north end of Bay Ridge Flats. **Jersey Flats**, the area on the New Jersey side west of Anchorage Channel, is much shoaler with a least depth of 5 feet. Channels have been dredged through these shoal areas to provide access to the piers on both sides of the bay.

(246)

Channels

(247) **Bay Ridge Channel**, **Red Hook Channel** and **Buttermilk Channel** follow the Brooklyn piers from The Narrows to East River. Midchannel depths in these channels are generally 25 to 40 feet with lesser depths on the sides; the area is subject to shoaling. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A.

(260)

Caution should be exercised when docking and undocking vessels along the southeasterly side of Bay Ridge Channel because the current may flow in a direction opposite to the normal channel flow, especially between the piers. The Brooklyn Cruise Terminal is at Pier 12 on Atlantic Basin. Commuter ferry services operate extensively in Buttermilk Channel.

(248) **Gowanus Bay**, at the junction of Bay Ridge and Red Hook Channels, is a bight in the Brooklyn shore at the mouth of **Gowanus Canal**. A dredged channel leads from Gowanus Bay to the Hamilton Avenue Bridge, about 1 mile above the mouth of the bay.

(249) The improved section of Gowanus Canal above Hamilton Avenue has depths of about 8 to 12 feet. The Third Street, Carroll Street, and Union Street bridges across the canal have the following minimum clearances: drawbridges, 3 feet; fixed bridges, 90 feet. The fixed bridge across that part of the canal that extends southward along Fifth Street has a clearance of 20 feet. (See **33 CFR 117.1** through **117.59** and **117.787**, chapter 2, for drawbridge regulations.)

(250) The Hamilton Avenue and Ninth Street drawbridges, 1 and 1.2 miles above the entrance of Gowanus Bay, respectively, are equipped with radiotelephones. The bridgetenders monitor VHF-FM channel 13; call signs KX-8183 and KX-8186, respectively.

(251) **Erie Basin**, just north of Gowanus Bay, is entered from Red Hook Channel. The entrance is marked by a light and the basin is marked by private lighted and unlighted buoys.

(252) **East River** is a 14-mile-long tidal strait that connects Upper Bay with Long Island Sound. For description of East River and the route to New York Harbor from Long Island Sound, see East River (indexed as such), chapter 9.

(253) **Governors Island** is at the Upper Bay entrance to East River. The hexagonal-shaped **Fort Jay** is prominent on the northeast side of the island, and the circular **Castle William** is on the northwest side. The main channel is westward of the island. Lighted buoys and sound signals are near the southern tip, and a light is on the northwest side of the island, on top of Castle William.

(254) **Liberty Island**, on the eastern part of Jersey Flats across the main channel from Governors Island, is marked by the **Statue of Liberty**, a colossal structure more than 305 feet high; the figure faces southeastward. In 2000, depths of 15 to 21 feet were available in the dredged area near the pier on the west side of the island. The U.S. Park Police marine unit operates from a floating platform on the northwest end of Ellis Island.

(255) **Robbins Reef Light** (40°39'27"N., 74°03'55"W.) 56 feet above the water, is shown from a conical tower, with the lower half brown and the upper half white, on the southeastern part of Jersey Flats.

(256) **Pierhead Channel** leads from the main channel about 0.7 mile southward of Liberty Island, thence

along the New Jersey pierhead line to Kill Van Kull. The channel connects several channels that lead to various facilities along the New Jersey waterfront, including the Army Corps of Engineers Caven Point Terminal, New York Waterway Ferry Landing, Claremont Terminal, New York Cross Island Railroad Terminal, Port Jersey Imported Automobile Terminal and Global Terminal and Container Services. A federal project provides for a depth of 20 feet in the channel. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A. The channels are well marked with lighted and unlighted buoys.

(257)

Sandy Hook Bay

(258) **Sandy Hook Bay** is the southern part of Lower Bay, westward of Sandy Hook and eastward of Point Comfort. The bay is an excellent anchorage, the depths of water ranging from 30 feet just inside Sandy Hook to 15 feet near its southern part; the shoaling is gradual and the bottom is good holding ground. The best anchorage during easterly and southeasterly winds is in the eastern part of the bay. Vessels of more than 24-foot draft will not find good anchorage out of the channel until above Fort Wadsworth. Extensive shoals make off northward and eastward from Point Comfort, but as the depths of water decrease gradually, soundings will give sufficient warning of too close an approach to the shore. Shallow-draft vessels can reportedly find satisfactory anchorage in **Horseshoe Cove**, on the east side of the bay. In 1999, the spit of land that forms Horseshoe Cove was reported visible only at extreme low water. Shoals extend an additional 200 yards southeastward from the end of the charted spit to about 40°26.7'N., 73°59.9'W. Mariners are cautioned not to navigate over this finger of land. Heavy fish traps extend out to a depth of 20 feet in places on the shoals on the southwest side of Sandy Hook Bay between Atlantic Highlands and Point Comfort.

(259) **Sandy Hook**, the southern point at the entrance to New York Harbor and the northern point of the New Jersey coast, is low and sandy. The hook, including Plum Island at the mouth of the Shrewsbury River, is part of Gateway National Recreation Area. Large areas of the park are bird nesting areas, and landing is not permitted. A light, **Sandy Hook Coast Guard Station**, standpipe and a radio tower on the north end of Sandy Hook are prominent. The area around Sandy Hook is changeable and subject to severe shoaling; extreme caution is advised.

(261)

Shrewsbury River and Navesink River

(262) **Shrewsbury River** and **Navesink River** empty through a common entrance into the southern extremity of Sandy Hook Bay eastward of the Highlands of Navesink.

(263) A federal project provides depths of 12 feet from Sandy Hook Bay to a point just above the fixed bridge at Highlands, thence 9 feet in Shrewsbury River to the Branchport Avenue Bridge at Long Branch, about 7.4 miles above the mouth. The Navesink River has a project depth of 6 feet from where it connects with the Shrewsbury River to the head of the project at Red Bank, about 4.9 miles above the mouth. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A.

(264)

Caution

(265) All submarine cables within the area in about 40°24'12″N., 73°59'00″W., in Shrewsbury River have been abandoned. Mariners are cautioned that the cables remain in place.

(266)

No-Discharge Zone

(267) The State of New Jersey, with the approval of the Environmental Protection Agency, has established a No-Discharge Zone (NDZ) in the waters of the Shrewsbury and Navesink Rivers. The NDZ extends south from the Highlands/Route 36 Bridge and covers all waters of the Shrewsbury and Navesink Rivers (see chart for limits).

(268) Within the NDZ, discharge of sewage, whether treated or untreated, from all vessels is prohibited. Outside the NDZ, discharge of sewage is regulated by **40 CFR 140** (see chapter 2).

(269)

Current

(270) At Highlands bridge, the currents have a velocity of about 2.6 knots. At Sea Bright bridge the velocity is about 1.6 knots.

(271)

Ice

(272) Navigation in Shrewsbury and Navesink Rivers is generally suspended because of ice from December to March, inclusive.

(273)

Supplies

(274) Gasoline, lubricants, marine supplies and provisions can be obtained at most of the towns along the shores of the Shrewsbury and Navesink Rivers.

(275)

Communications

(276) Railroad, ferry or bus connects with New York to points on the New Jersey coast.

(277) **Highlands** is a summer resort on the west side of Shrewsbury River 1.5 miles inside the entrance.

(278) The State Route 36 highway bridge (Highlands Bridge) across Shrewsbury River at Highlands has a fixed span with a clearance of 61 feet. The east side of the river northward of the bridge and the west side 0.3 mile southward of the bridge are used as anchorages for small craft.

(279)

Caution

(280) Caution should be exercised at the junction of the Shrewsbury and Navesink Rivers, about 0.6 mile southward of the State Route 36 highway bridge at Highlands, to avoid the submerged stone jetty. Craft entering Navesink River should pass westward of the lighted junction buoy. The submerged jetty is marked by three seasonal buoys.

(281) The State Route 520 highway bridge (Sea Bright Bridge) over Shrewsbury River between **Rumson** and **Sea Bright** has a bascule span (under construction 2022).

(282) **Pleasure Bay**, at the southeast end of Shrewsbury River, is crossed by a fixed highway bridge with a clearance of 25 feet. **Branchport** is a small town on the east side of Pleasure Bay at the head of navigation.

(283) The privately dredged and marked channels in **Little Silver Creek**, **Town Neck Creek**, **Oceanport Creek** and **Blackberry Bay** had controlling depths of about 5 feet in 1965-67. **Parkers Creek** was reported to have a controlling depth of 6 feet in 1999.

(284) A fixed highway bridge with a clearance of 24 feet crosses the westerly part of Shrewsbury River, just eastward of its junction with Parkers and Oceanport Creeks.

(285) The tributaries that empty into the southeasterly and southwesterly sides of Shrewsbury River are crossed by bridges with the following clearances: **Manahassett Creek**, fixed highway, 6 feet; **Troutmans Creek**, fixed highway, 4 feet; Oceanport Creek (Oceanport Bridge) swing railroad, 4 feet; Parkers Creek, fixed railroad, 4 feet.

(286) The channel in Navesink River is crooked but well marked by seasonal buoys. The Oceanic highway bridge across the river between Rumson and **Locust Point** has a bascule span with a clearance of 22 feet. (See **33 CFR 117.1** through **117.59** and **117.734**, chapter 2, for drawbridge regulations.)

(287) **Rumson** is a town on the south side about 1.7 miles above the entrance to Navesink River. Small-craft facilities just west of the bridge at Rumson can provide berths, electricity, gasoline, water, ice and storage. Hull and engine repairs can be made, and a 7-ton mobile hoist is available. In 1981, a reported depth of 5 feet could be carried to the boatyards.

(288) **Fair Haven** is on the south side of Navesink River about 1 mile above the bridge at Rumson. A boatyard and two yacht clubs are at Fair Haven. The boatyard can provide berths, electricity, gasoline, water, ice, storage and marine supplies and hull, engine and radio repairs; lifts to 15 tons are available. In 1987, a reported depth of about 7 feet could be taken to the boatyard.

(289) **Red Bank**, a town near the head of navigation on the Navesink River, has railroad connections with New York.

(290) The dredged channel that extends for 1.5 miles above the landings at Red Bank had a reported midchannel controlling depth of 2 feet to the second highway bridge, and thence less than 1 foot for the rest of the dredged section in 1985. The channel is privately marked by buoys and stakes. Three bridges cross the river near Red Bank: a fixed highway bridge, 4.8 miles above the mouth, with a clearance of 12 feet; a fixed railroad bridge, 450 yards above the fixed bridge, with a clearance of 19 feet; and a fixed highway bridge, connecting Red Bank and River Plaza, with a clearance of 8 feet.

(291)

Atlantic Highlands to Staten Island Flats

(292) **Atlantic Highlands** is a town on the south side of Sandy Hook Bay about 2 miles west of Sandy Hook. A breakwater, marked by a light at its eastern end, forms an anchorage basin.

(293) The basin is used by numerous pleasure and party fishing craft. Numerous piles and ruins of former wharves are westward of the basin.

(294)

Small-craft facilities

(295) Small-craft facilities in the basin can provide berths, electricity, gasoline, diesel fuel, water, ice, storage, marine supplies, launching ramps, pump-out station and hull and engine repair; mobile lifts up to 50 tons are available.

(296) **Terminal Channel**, entered from Sandy Hook Channel about 1 mile west-southwestward of Sandy Hook, leads south-southwestward to a turning basin and to the deepwater ammunition handling piers of the U.S. Naval Ammunition Depot. (See Notice to Mariners and latest editions of the charts for controlling depths.) The channel is marked by lighted and unlighted buoys and by a private **207.5°** lighted range. A side channel leads southward from the southeastern end of the turning basin to an ammunition barge-loading pier. The deepwater piers and barge pier are connected to the shore by a trestle that extends 1.6 miles across the flats from Leonardo.

(297) A **restricted area** surrounds Terminal Channel, turning basin, and piers of the U.S. Naval Ammunition Depot. (See **33 CFR 334.1** through **334.6** and **334.102**, chapter 2, for limits and regulations.)

(298) A dredged channel, about 0.4 mile eastward of the trestle at **Leonardo**, leads southward from Sandy Hook Bay to the entrance and basin of a state marina. The channel is marked by private aids to navigation. Transient berths, electricity and water are available in the basin. A boatyard with a 45-foot marine railway is about 0.5 mile eastward of the boat basin; complete hull and engine repairs can be made.

(321)

Anchorages

(322) A **special anchorage** is in Great Kills Harbor. (See **33 CFR 110.1** and **110.60**, chapter 2, for limits and regulations.)

(323) **Great Kills**, on the west side of Great Kills Harbor, has several small-craft facilities with berths, electricity, gasoline, diesel fuel, water, ice, storage and marine supplies. A public launching ramp is located in the northeast corner of the harbor.

(324) **Lemon Creek**, 0.2 mile westward of Seguine Point, is a narrow shallow stream used only by local boats that enter at high water. The midchannel controlling depth over the bar is about 2 feet with deeper water inside. The abutment of a former bridge is on the south side of the creek just inside the mouth. Overhead power cables crossing the creek at the bridge abutment have a clearance of 47 feet.

(325) A small marina on the creek can haul out craft up to 8 tons for minor engine and hull repairs; berths, electricity, water, ice and outside storage are available.

(326) A prominent tower of a former lighthouse with a statue on top is on the south side of Staten Island, 0.8 mile westward of Seguine Point. Prominent buildings are near the point at **Red Bank**, 0.3 mile southwestward of the tower.

(327) **Keansburg**, on **Point Comfort** on the south side of Raritan Bay, is a summer resort. The wharves on the west side of Point Comfort are in ruins and no longer used.

(328) A **special anchorage** surrounds the shoreline and waters north of Point Comfort. (See **33 CFR 110.1** and **110.60**, chapter 2 for limits and regulations.)

(329) **Waackaack Creek** and **Thorns Creek**, about 0.6 mile southwest of Point Comfort, have a common entrance protected by floodgates. The gates are lowered, thereby closing the harbor, when tides above 4½ feet are sustained for a period of time. An overhead power cable with a clearance of 32 feet crosses the creek entrance at the floodgates. Small-craft facilities on Thorns Creek provide berths, electricity, ice, water, gasoline, marine supplies and a 20-ton forklift and a 12-ton mobile hoist for hull and engine repairs. In 1982, the channels into the creeks were reported dredged to 7 feet.

(330) **Keyport Harbor**, 3 miles westward of Point Comfort, is a shallow harbor on the south side of Raritan Bay between **Conaskonk Point** and **Matawan Point**. A buoyed approach channel leads southward from the bay to a dredged marked channel that leads through the harbor to the mouth of Matawan Creek; the federal project depth is 8 feet. (See Notice to Mariners and latest editions of charts for controlling depths.)

(331) **Matawan Creek**, entered at the head of Keyport Harbor, is used mostly by local craft. In 2023, the controlling depth was 2.5 feet to the first highway bridge, thence 1-foot to the Route 35 highway bridge; thence in 1981, 2 feet to shoaling to bare was reported to the railroad bridge about 1.5 miles above the mouth. Greater depths are available with local knowledge. Three fixed bridges, one railroad and two highway, cross the creek; least clearances are 48 feet horizontal and 8 feet vertical. Least clearance of overhead power cables crossing the creek is 54 feet.

(332) **Keyport** is a town on the east side of the entrance to Matawan Creek.

(333)

Small-craft facilities

(334) There are several small-craft facilities on Matawan Creek and on the southeast side of Keyport Harbor at Keyport. Berths with electricity, gasoline, diesel fuel, water, ice, marine supplies, sewage pump-out, lifts to 30 tons, marine railways to 40 feet and complete hull and engine repairs are available. Vessels proceed to the small-craft facilities at Keyport at high water.

(335) A privately dredged channel, about 25 feet wide in places, leads about 0.3 mile southwesterly from the mouth of Matawan Creek to a marina basin at the entrance to **Luppatatong Creek**. In 1981, a reported depth of 4 feet was available to the marina.

(336) **Cheesequake Creek** and **Stump Creek** share a common entrance on the south side of Raritan Bay, 6 miles westward of Point Comfort. The entrance is between two stone jetties marked by lights on the outer ends. The east jetty is awash at high water. A dredged channel leads between the jetties to the railroad bridge about 0.3 mile above the jetties.

(337) Local boats from Lower Bay usually head for a point about 1.6 miles east-northeastward of the jetties and then shape a course to enter between the jetties at the entrance to Cheesequake and Stump Creeks.

(338)

Caution

(339) Caution should be exercised to avoid the sunken wrecks 0.2 mile eastward of the east jetty and 0.2 mile northeast of the west jetty.

(340) The State Route 35 highway bridge, 0.2 mile inside the jetties, has a bascule span with a clearance of 25 feet. The overhead power cable just north of the bridge has a clearance of 89 feet. The railroad bridge, 0.3 mile inside the jetties, has a bascule span with a clearance of 3 feet. The bridgetender monitors VHF-FM channel 13; call sign KT-3859. (See **33 CFR 117.1** through **117.59** and **117.709**, chapter 2, for drawbridge regulations.) The twin fixed highway bridges over Cheesequake Creek, 1.1 miles inside the jetties, have clearances of 16 feet.

(341) **Laurence Harbor** is a summer resort on the east side of Stump Creek, and **Morgan** is a settlement on the west side of Cheesequake Creek.

(342)

Small-craft facilities

(343) There are small-craft facilities that can provide berths with electricity, gasoline, diesel fuel, water, ice, marine supplies, a pump-out station, storage facilities, lifts to 35 tons and a 50-foot marine railway; complete hull and engine repairs are available.

(346)

(344)

Raritan River and South River

(345) **Raritan River** empties into the western end of Raritan Bay between Perth Amboy and South Amboy. The channel from South Amboy to **New Brunswick** is 11 miles long and very crooked but is well marked with navigational aids. Waterborne commerce on the river is in coal, ore and petroleum products.

(347)

Channels

(348) Vessels enter Raritan River from the east by way of Great Beds Reach and from the north by way of Arthur Kill via Raritan River Cutoff Channel. A federal project provides for a 20-foot channel in Raritan River Cutoff, a 25-foot channel from Great Beds Reach in Raritan Bay to the head of Red Root Reach about 1.9 miles above Garden State Parkway bridge, and thence a 15-foot channel to the junction with Washington Canal. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A. Above Washington Canal, the controlling depth in Raritan River was about 9 feet at midchannel to New Brunswick in 1962.

(349) A dredged channel in Titanium Reach and South Channel branches south from Raritan River about 0.6 mile above Garden State Parkway bridge. The federal project depths are 25 feet in Titanium Reach and 15 to 10 feet in South Channel to Crossman Dock. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A. In 1991, the channels were not being maintained near project depth and the project above Crossman Dock was not being maintained.

(350) A dredged channel in **Washington Canal** branches south from Raritan River about 4.3 miles above Garden State Parkway bridge and connects with **South River**. A dredged channel leads south for about 3.4 miles in South River.

(351) A sunken drydock, marked by a lighted buoy, is on the east side of Raritan River at 40°29'29"N., 74°19'43"W. The ruin extends 60 feet into the channel and is visible at all stages of the tide.

(353)

Current

(354) The tidal current has a velocity of about 1.5 knots at the Victory Highway Bridge at Perth Amboy.

(355) **South Amboy** is a city on the south side of the entrance to Raritan River. Waterborne commerce at the

(352)

Structures across Raritan River and South River

Name•Description•Type	Location	Miles*	Clear Width of Draw or Span Opening (feet)	Clear Height above Mean High Water (feet)	Information
NJTRO bridge (swing)	40°29'46"N., 74°16'51"W.	0.5	124 (north draw) 125 (south draw)	8	Bridge Under Construction (2023) - Note1
Overhead power cable	40°29'46"N., 74°16'51"W.	0.5		140	
Victory Bridge (fixed)	40°30'32"N., 74°17'27"W.	1.6	354	110	
Thomas Edison Memorial Bridges (fixed)	40°30'39"N., 74°18'02"W.	2.1	197	110	
Alfred E. Driscoll Bridge (fixed)	40°30'39"N., 74°18'04"W.	2.2	193	134	
Overhead power cable	40°28'52"N., 74°21'20"W.	5.2		128	
Junction with Washington Canal	40°28'17"N., 74°22'00"W.	6.1			
New Jersey Turnpike bridge (fixed)	40°29'18"N., 74°23'46"W.	10.2	150	45	
Overhead power cables	40°29'24"N., 74°23'55"W.	10.4		114	
US Highway 1 bridges (fixed)	40°29'33"N., 74°24'47"W.	11.1	90	90	
Albany Street bridge (fixed)	40°29'51"N., 74°26'16"W.	12.8	75	16	
South River				103	
Route 535 bridge (fixed)	40°27'19"N., 74°22'17"W.	2.2	80	25	
CONRAIL bridge (swing)	40°26'54"N., 74°22'12"W.	2.8	49	4	Note 2

* Distance is in nautical miles above the mouth
Note 1 – See **33 CFR 117.1** through **117.59** and **117.745**, chapter 2, for drawbridge regulations.
Note 2 – See **33 CFR 117.1** through **117.59** and **117.756**, chapter 2, for drawbridge regulations.

port is in fuel oils, coal, sand and gravel. Depths alongside the wharves and piers range from about 6 to 30 feet. Water, provisions and marine supplies can be obtained here, and berths with electricity, water, ice and winter dry storage are available at a boat club.

(356) **Sayreville** is 6 miles above South Amboy on the south bank of the Raritan River. Most of the wharves are privately owned.

(357) **South River** is a town on the west side of South River 7.5 miles above South Amboy. A marina about 200 yards north of the highway bridge at Old Bridge provides berths, water, marine supplies, a 2-ton lift and engine repairs. In 1981, a reported depth of about 1 foot could be carried to the marina.

(358) The **Delaware and Raritan Canal**, closed to navigation since 1933, had its entrance to the Raritan River at New Brunswick.

(359) **Highland Park** is across Raritan River opposite New Brunswick. In 1981, a reported depth of about 3½ feet was available from the head of the federal project to Highland Park, the practical head of navigation.

(360)

Arthur Kill

(361) **Arthur Kill** is the narrow body of water separating Staten Island from New Jersey. The cities of Perth Amboy, Tottenville and Elizabeth and many large factories, oil refineries and storage facilities are on its shores. Northern Arthur Kill and Kill Van Kull are the major channels for bulk, containerize, and petroleum cargo in New York Harbor.

(362)

Channels

(363) Federal project depth in Arthur Kill is 35 feet. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A.

(364)

Caution

(365) Numerous sunken and visible wrecks are adjacent to both sides of the channel in Arthur Kill; caution is advised.

(366) A liquefied petroleum gas (LPG) facility is on the west side of Arthur Kill immediately south of **Morses Creek**. A moving **safety zone** has been established around loaded LPG vessels transiting between Scotland Lighted Whistle Buoy S at the entrance to Sandy Hook Channel and the LPG facility. (See **33 CFR 165.1** through **165.7**, **165.20** through **165.25**, and **165.160**, chapter 2, for limits and regulations.)

(367)

Anchorages

(368) General anchorages are in Arthur Kill. (See **33 CFR 110.1** and **110.155 (i)**, chapter 2, for limits and regulations.)

(369)

Current

(370) Throughout Arthur Kill the flood sets from Raritan Bay to Newark Bay and the ebb in reverse direction. Velocities of current vary with the location from about 1 to 1.5 knots.

(394)

(371) In 1991, tidal currents in Arthur Kill were reported to deviate significantly from official predictions published by NOAA. Mariners should exercise caution and discretion in the use of published tidal current predictions.

(372) **Perth Amboy** is on the point at the junction of Raritan River and Arthur Kill at the western end of Raritan Bay. The principal wharves are along the west bank of Arthur Kill. The greatest draft entering is about 30 feet. The wharves have depths of 14 to 30 feet alongside. Good anchorage is found abreast some wharves in 30 feet.

(373)

Anchorages

(374) A **special anchorage** is south of Perth Amboy. (See **33 CFR 110.1** and **110.60**, chapter 2, for limits and regulations.)

(375) Perth Amboy is a **customs port of entry.**

(376)

Repairs

(377) Several ship and boat repair yards are in Perth Amboy. Small-craft engine and hull repairs can be made.

(378)

Supplies

(379) Diesel oil, diesel fuel, gasoline, water, lubricants and marine supplies are available at Perth Amboy.

(380) **Outerbridge Crossing Bridge**, 1.7 miles above **Ward Point**, is under construction (2019) across Arthur Kill between Perth Amboy and **Tottenville**. A private sound signal is at the bridge. A marina at Tottenville provides berths, electricity, water, storage facilities and a 15-ton mobile hoist for do-it-yourself repairs.

(381) **Port Socony**, on the east side of Arthur Kill 2.9 miles above Ward Point, is a bulk oil storage terminal. A privately maintained dredged channel leads from the main channel in Arthur Kill to the oil company dock. In 2008, a depth of 22 feet was alongside the south half of the dock with 12 to 28 feet available alongside the northern half in 2007, except for shoaling to 8 feet along the north edge of the channel.

(382) **Smith Creek** enters Arthur Kill from northward about 3.3 miles above Ward Point. The entrance channel is privately marked by buoys. In 1981, a reported depth of 3 feet was available to just above the first bend in the channel. The creek is used principally by small craft.

(383)

Small-craft facilities

(384) Several small-craft facilities are along Smith Creek. Berths with electricity, gasoline, water, ice, marine railways to 40 feet and partial hull and engine repairs are available.

(385) **Port Reading**, 4.5 miles above Ward Point on the north side of Arthur Kill, has several oil storage facilities. Depths of 18 to 36 feet are reported alongside. **Fresh Kills** enters Arthur Kills from eastward about 6 miles above Ward Point.

(386)

Rahway River to Kill Van Kull

(387) **Rahway River** enters Arthur Kill from westward, about 7.2 miles above Ward Point, and extends westward for about 4.5 miles to the town of **Rahway**. It is used only by small craft. In 1981, a reported depth of 5 feet could be taken to Lamberts Wharf about 2.1 miles above the mouth and about 0.5 mile above the New Jersey Turnpike bridge.

(388) An overhead power cable with a clearance of 165 feet crosses Arthur Kill about 1.7 miles north of the Rahway River entrance.

(389) The**Goethals Bridge**, 10 miles above Ward Point, is a fixed highway bridge with a vertical clearance of 140 feet. The railroad bridge above Goethals Bridge has a vertical lift span with a clearance of 31 feet down and 135 feet up. The bridgetender at the railroad bridge monitors VHF-FM channel 13; call sign KXS-237. (See **33 CFR 117.1** through **117.59** and **117.702**, chapter 2, for drawbridge regulations.)

(390) **Elizabethport**, about 11 miles above Ward Point, is the eastern part of the city of **Elizabeth**. It is at the northern end of Arthur Kill at its junction with Newark Bay.

(391) Most of the wharves along the Elizabeth waterfront are of the bulkhead-marginal type. Depths alongside range from 3 to 32 feet. Waterborne commerce at these wharves is in petroleum, sand and gravel, chemicals and petrochemicals and vegetable and animal oils.

(392) **Elizabeth River** enters Arthur Kill from westward at Elizabethport. The overhead power cables just above the entrance have a reported clearance of 23 feet. There are numerous bridges further up the Elizabeth River. (See **33 CFR 117.1** through **117.59** and **117.718**, chapter 2, for drawbridge regulations.)

(393) **Kill Van Kull** separates the southern shore of the city of Bayonne from Staten Island and connects the Upper Bay of New York Harbor with Newark Bay and Arthur Kill. Kill Van Kull is a major channel for petroleum and bulk cargo in New York Harbor, and has extensive through traffic and large factories on its shores.

(395)

Channels

(396) A federal project provides for a 45-foot dredged channel leading through Kill Van Kull and a 35-foot dredged channel north of **Shooters Island** to Arthur Kill. The dredged channel south of Shooters Island has a project depth of 30 feet. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A.

(397) Shoals, obstructions and numerous wrecks are along both sides of the dredged channel in Kill Van Kull. Numerous sunken and visible wrecks are in the channel southward of Shooters Island; caution is advised.

(398) **Bayonne Bridge**, a fixed span with a minimum clearance of 154 feet over the channel (215 feet centerline), crosses Kill Van Kull near the southwestern end of the city of Bayonne and connects **Bergen Point** with Staten Island.

(399)

Current

(400) The flood current sets westward and the ebb eastward. In the bight on the south side of the channel between West New Brighton and Port Richmond there is more or less of an eddy when the current is at strength.

(401) In 1991, tidal currents in Kill Van Kull were reported to deviate significantly from official predictions published by NOAA. Mariners should exercise caution and discretion in the use of published tidal current predictions.

(402) **Constable Hook** and **Port Johnson**, on the north shore of Kill Van Kull, are parts of the city of **Bayonne**. They are commercially important for the shipment of petroleum and other products. A dredged channel 23 feet deep, marked by buoys, leads from the easterly end of Kill Van Kull to the wharf on the north side of Constable Hook.

(403) Several private yacht and boat clubs and a public marina are on the southwestern shore of Bayonne above Bergen Point. A 90-ton crane at the marina can haul out craft for engine and hull repairs; berths, electricity, gasoline, water, ice and marine supplies are available.

(404) **New Brighton**, **Port Richmond** and **Mariners Harbor** are on the south shore of Kill Van Kull. The largest of several shipyards and floating drydocks on the south shore can handle vessels up to 6,400 tons, 400 feet long, 85 feet wide and 26 feet in draft. All kinds of repairs can be made.

(405)

Newark Bay

(406) **Newark Bay** has a length of about 4 miles from Kill Van Kull to the junction of the two channels leading to Passaic and Hackensack Rivers. The greater part of the bay is very shoal, but a dredged channel leads through the bay to the rivers. The channel is well marked by lights and buoys. Strangers in small vessels should have no difficulty when using the chart as a guide. Deep-draft vessels should employ a pilot.

(407)

Channels

(408) Federal project depth in the main channel to about 0.3 mile north of the branch channel to the Port Elizabeth Marine Terminal is 45 feet, thence 40 feet to Port Newark, thence 35 feet to the junction of Passaic and Hackensack Rivers. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available

(427)

Structures across Passaic River

Name•Description•Type	Location	Miles*	Clear Width of Draw or Span Opening (feet)	Clear Height above Mean High Water (feet)	Information
Lincoln Highway bridge (vertical lift)	40°43'57"N., 74°07'05"W.	1.8	300	40 (down) 135 (up)	Note 1
Pulaski Skyway bridge (fixed)	40°44'06"N., 74°07'02"W.	2.0	520	135	
Overhead power cables	40°44'22"N., 74°07'04"W.	2.4		135	
CONRAIL bridge (swing)	40°44'30"N., 74°07'15"W.	2.6			Bridge under construction 2022
Overhead power cable	40°44'30"N., 74°07'17"W.	2.6		135	
New Jersey Turnpike (fixed)	40°44'32"N., 74°07'22"W.	2.7	319	100	
Jackson Street bridge (swing)	40°44'01"N., 74°09'19"W.	4.6	72	18	Note 1
Amtrack bridge (fixed)	40°44'10"N., 74°09'41"W.	5.0			Bridge under construction 2025
Overhead power cables	40°44'10"N., 74°09'41"W.	5.0		170	
Bridge Street bridge (swing)	40°44'42"N., 74°09'57"W.	5.6	80	7	Note 1
NJTRO Newark-Harrison bridge (swing)	40°44'51"N., 74°09'57"W.	5.8	77	15	Notes 1 and 2
Overhead power cables	40°44'51"N., 74°09'57"W.	5.8		135	
US 280 bridge (vertical lift)	40°44'53"N., 74°09'57"W.	5.8	200	35 (down) 135 (up)	Note 1
Clay Street bridge (swing)	40°45'04"N., 74°09'55"W.	6.0	75	8	Note 1
Railroad bridge (bascule)	40°45'16"N., 74°09'52"W.	6.3	126	7	Bridge left open
NJTRO West Arlington bridge (swing and fixed)	40°46'36"N., 74°09'00"W.	8.0	45 (E draw) 48 (W draw) 70 (fixed span)	36 35	Notes 1 and 2
Route 7\Rutgers Street bridge (vertical lift)	40°47'12"N., 74°08'51"W.	8.9	99	7 (down) 51 (up)	
Overhead power cables	40°48'04"N., 74°08'19"W.	9.8		142	
Avondale bridge (swing)	40°48'40"N., 74°08'18"W.	10.7	65	7	Note 1
NJTRO bridge (swing)	40°49'14"N., 74°07'36"W.	11.7	47	26	Notes 1 and 2
Route 3 bridge (fixed)	40°49'22"N., 74°07'26"W.	11.8	125	33	
Union Avenue bridge (fixed)	40°50'30"N., 74°07'22"W.	13.2	60	15	
Gregory Avenue bridge (fixed)	40°51'16"N., 74°07'11"W.	14.0	71 (E draw) 90 (W draw)	12	Note 1
Second Street bridge (fixed)	40°51'36"N., 74°06'57"W.	14.7	100	5	
Eighth Street bridge (fixed)	40°51'19"N., 74°06'34"W.	15.3			Bridge under construction
Overhead power cable	40°51'42"N., 74°06'26"W.	15.7		35	
Wall Street bridge (fixed)	40°51'53"N., 74°06'36"W.	15.9	78	5	

* Distance in nautical miles above the mouth
Note 1 – See **33 CFR 117.1** through **117.59** and **117.739**, chapter 2, for drawbridge regulations.
Note 2 – Bridgetenders monitor VHF-FM channel 13.

through a USACE hydrographic survey website listed in Appendix A.

(409)

Anchorages

(410) A special anchorage is in Newark Bay. (See **33 CFR 110.1** and **110.60**, chapter 2, for limits and regulations.)

(411)

Ice

(412) Ice sometimes closes navigation during a part of January and February.

(413) The **Port Elizabeth Marine Terminal** operated by the Port Authority of New York and New Jersey, is on Newark Bay in Elizabeth, NJ, on the south side of Elizabeth Channel south of Port Newark. The facility is about 8 miles from The Narrows via Kill Van Kull. It is adjacent to the New Jersey Turnpike and Newark Airport in the heart of the New Jersey industrial area, about 25 minutes by highway from Manhattan.

(414) The terminal has 25 deep-draft berths with depths of 32 to 40 feet reported alongside and deck heights of 12 feet. In 1996, a rock with 36 feet of water over it was reported in about 40°40'26.6"N., 74°7'57.1"W., about 200 yards north-northeast of Buoy 14.

(415) A large container-handling complex with extensive lift-on/lift-off and roll-on/roll-off systems is at the terminal. Included in this complex are cranes up to 50 tons, mobile straddle carriers with 32-ton capacities, cargo-handling buildings with more than 1-million square

feet of storage space and a large area for open storage. A Class I railroad provides the terminal with direct rail services. Excellent cargo handling and storage facilities are available.

(416)

Channels

(417) Federal project depth in Elizabeth Channel, leading to the terminal from the main channel in Newark Bay, is 50 feet. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A.

(418) **Port Newark Terminal**, operated by the Port Authority of New York and New Jersey, is on the western side of Newark Bay 2.7 miles above the south entrance, northward of the Port Elizabeth Marine Terminal. It is in the heart of the New Jersey industrial area, adjacent to the New Jersey Turnpike and Newark Airport. There are 37 deep-draft berths; reported depths alongside, 32 to 35 feet; deck heights, 11 to 12 feet; many transit and storage areas and excellent cargo handling facilities, used for the receipt and shipment of general cargo, metals, vegetable oils, petroleum, automobiles and machinery and for the receipt of bananas, rubber products, lumber and pulpwood and chemicals. A Class I railroad provides the terminal with direct rail service.

(419)

Channels

(420) Federal project depth in Port Newark Channel and Port Newark Pierhead Channel, leading to the terminal from the main channel in Newark Bay, is 40 feet. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A.

(421) The New Jersey Turnpike (IS 78) bridge, 0.7 mile above the entrance to Port Newark Terminal, has a fixed span with a clearance of 135 feet. The railroad bridge, 0.2 mile above the New Jersey Turnpike bridge, has a vertical-lift span with a clearance of 35 feet down and 135 feet up. (See **33 CFR 117.1** through **117.59** and **117.735**, chapter 2, for drawbridge regulations.) The bridgetender at the railroad bridge monitors VHF-FM channel 13; call sign KS-9968.

(422) A marina on the east side of Newark Bay about 0.9 mile above the New Jersey Turnpike bridge provides berths, gasoline, diesel fuel, water, electricity, ice, storage, marine supplies and a 25-ton lift; hull and engine repairs can be made.

(423)

Passaic River to Hackensack River

(424) **Passaic River**, which flows into the northwest end of Newark Bay, is used by vessels to **Passaic**, a manufacturing city at the head of navigation 13 miles above the mouth. Above the Wall Street bridge at Passaic the river is obstructed by boulders partly showing above the water for 1.5 miles to the **Dundee Dam**. The city of **Newark** extends along the river for a distance of nearly 5 miles above the mouth. The towns of **Belleville, Arlington, Rutherford** and **Nutley** and several villages are on the river between Newark and Passaic. The channel entrance is well marked. Waterborne commerce on the river consists of barge shipments of sand, gravel and petroleum products.

(425)

Channels

(426) A federal project provides for a 30-foot channel from Newark Bay to a point about 0.5 mile above the Lincoln Highway Bridge; thence 20 feet to the Jackson Street bridge; thence 16 feet to the railroad bridge at Arlington; thence 10 feet to the Eighth Street Bridge at Passaic. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A.

(428) The center pier and approach spans of a former railroad swing bridge remain in Passaic River channel about 1.1 miles above the mouth. An obstruction, covered 15 feet, was reported in the channel east of the center pier. Mariners should use extreme caution when passing between the former bridge remains.

(429) **Freshets** overcome the flood current down as far as Newark and sometimes to the mouth of the river. Ordinary freshets usually of a few hours duration cause a rise of about 2 feet and a current velocity of about 3 knots at Newark. Destructive freshets occasionally occur at intervals of years, generally in the spring and fall.

(430)

Small-craft facilities

(431) There are several boatyards along the Passaic River between the entrance and Passaic. A marine railway at Arlington can handle vessels to 40 feet long for complete engine and hull repairs. Berths, electricity, gasoline, water, ice, storage and marine supplies are available along the river below Kearny.

(432) **Hackensack River** flows into the northeast end of Newark Bay and is navigable for about 17.8 miles to the dams at **New Milford**.

(433)

Channels

(434) A federal project provides for a 30-foot channel from Newark Bay to a 25-foot turning basin about 0.3 mile above the railroad bridge at **Marion**. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through a USACE hydrographic survey website listed in Appendix A. Above this point in 1971, depths of 11 feet were available

vertical clearance of 3 feet. The bridge spans are locked in the closed position. A dam, about 0.8 mile above the mouth, forms the head of navigation on the creek.

Chart Coverage in Coast Pilot 2—Chapter 12
NOAA's Online Interactive Chart Catalog has complete chart coverage
http://www.charts.noaa.gov/InteractiveCatalog/nrnc.shtml
74°30'W
74°W
73°30'W
42°30'N
42°N
41°30'N
41°N
Albany
Troy
Hudson
Saugerties
Kingston
Poughkeepsie
Newburgh
West Point
Ossining
Yonkers
HUDSON RIVER
MASSACHUSETTS
CONNECTICUT
NEW YORK
NEW JERSEY
NEW YORK
CONNECTICUT
LONG ISLAND SOUND

Hudson River

(11)

Structures across Hudson River to Troy, NY

Name•Description•Type	Location	Clearance (feet) Horizontal	Vertical*	Information
George Washington Bridge (fixed)	40°51'06"N., 73°57'11"W.	3060	193 (east end) 205 (center) 206 (west end)	
Governor Mario M. Cuomo Bridge (fixed)	41°04'16"N., 73°53'42"W	1140 (center span) 467 (east and west spans)	139 (center span) 123 (east and west spans)	
Overhead power cables	41°15'47"N., 73°58'16"W.		160	
Bear Mountain Bridge (fixed)	41°19'12"N., 73°59'00"W.	1584	155	
Newburgh-Beacon Bridges (fixed)	41°31'10"N., 73°59'57"W.	960	147 (middle 760 feet) 172 (center)	Channel spans have a private sound signal on the north bridge and a racon on the south bridge.
Mid-Hudson Bridge (fixed)	41°42'12"N., 73°56'44"W.	1080	134	A private sound signal and a racon are at the center of the span.
CSX Railroad (fixed)	41°42'39"N., 73°56'40"W.	490 (center span)	167	
Kingston-Rhinecliff Bridge (fixed)	41°58'45"N., 73°57'03"W.	760	135	A private sound signal and a racon are at the center of the west channel span.
Rip Van Winkle Bridge (fixed)	42°13'26"N., 73°51'07"W.	480	142	A racon is at the center of the main channel span
Overhead power cables	42°14'56"N., 73°48'58"W.		145	
Overhead power cables	42°30'29"N., 73°46'32"W.		185	
Castleton Railroad (fixed)	42°30'33"N., 73°46'30"W.	566 (west span)	139	
Castleton-on-Hudson Bridge (fixed)	42°30'36"N., 73°46'28"W.	552	135	
Overhead power cable	42°35'45"N., 73°45'39"W.		169	
Overhead power cable	42°35'56"N., 73°45'43"W.		194	
Dunn Memorial Bridge (fixed)	42°38'37"N., 73°44'54"W.	300	60	
NYC and Hudson Railroad (swing)	42°39'16"N., 73°44'30"W.	98	25	Kept in the open to navigation position
Overhead power cable	42°39'15"N., 73°44'32"W.		135	
Overhead power cable	42°39'39"N., 73°44'04"W.		88	
Patroon Island Bridge (fixed)	42°39'54"N., 73°43'46"W.	300	60	
Overhead power cable	42°40'52"N., 73°43'03"W.		95	
Overhead power cable	42°41'46"N., 73°42'19"W.		87	
Troy-Menands Bridge (fixed)	42°42'03"N., 73°42'09"W.	317	61	
Congress Street Bridge (fixed)	42°43'44"N., 73°41'48"W.	250	55	
Troy-Green Island Bridge (lift)	42°44'13"N., 73°41'37"W.	184	29 (down), 60 (up)	Note 1
Hoosick Street Bridge (fixed)	42°44'23"N., 73°41'14"W.	201 (east span) 359 (west span)	64 (east span) 61 (west span)	Bridge has two spans on either side of Adams Island

* Vertical clearance measured at Mean High Water

Note – See **33 CFR 165.1** through **165.40** and **165.169**, chapter 2, for safety and security zone regulations pertaining to bridges and overhead cables in the Hudson River.
Note 1 – See **33 CFR 117.1** through **117.59** and **117.791**, chapter 2, for draw bridge regulations.

(1) This chapter describes the Hudson River from New York City to Troy, NY, and includes the principal cities of Yonkers, Newburgh, Poughkeepsie, Kingston and Albany.

(2) **Mileages** shown in this chapter for the Hudson River as Mile 0.9E, Mile 12W, etc., are the nautical miles above The Battery; the letters N, S, E, and W denote by compass points the side of the river where each feature is located. Mile 0.0 is a point at the mouth of the Hudson River in 40°42.1'N., 74°01.5'W. The mileages given are approximations.

(3)

Hudson River

(4) **Hudson River**, sometimes called **North River** in New York City, has its source in the Adirondack Mountains, about 275 miles along its course from a junction with East River at The Battery, NY, and flows in a general southerly direction into New York Upper Bay. Troy Lock and Dam, 134 miles above The Battery, permits vessels to pass from tidewater to the upper river and the New York State Canal System. The river water is usually fresh as far south as Poughkeepsie, halfway from Troy Lock and Dam to The Battery.

(5) New York City extends along the eastern bank of Hudson River for a distance of about 14 miles above The Battery. For about 5 miles northward from The Battery, the New York waterfront is an almost continuous line of wharves and piers, some of which can accommodate the largest transatlantic liners.

(6) On the opposite side of Hudson River from New York City are Jersey City, Hoboken, Weehawken, West New York, Guttenberg, Edgewater, Fort Lee and Englewood Cliffs. The shoreline from Jersey City to Edgewater is lined with ruined piers and piling fields. Mariners must check with local authorities and property owners for approval prior to mooring.

(7)

Channels

(8) The lower Hudson River has depths of 43 feet or more in midchannel from deep water in Upper New York Bay off Ellis Island to the upper limit of New York City's major wharves at 59th Street, about 5.3 miles above the entrance. Above this point, the federal project depth is 32 feet to Albany. For detailed channel information and minimum depths as reported by the U.S. Army Corps of Engineers (USACE), use NOAA Electronic Navigational Charts. Surveys and channel condition reports are available through the USACE hydrographic survey website listed in Appendix A.

(9)

Seasonal buoyage

(10) The lighted buoys marking the Hudson River channel are replaced during the winter by smaller lighted ice buoys or unlighted buoys.

(12)

Anchorages

(13) General anchorages begin 5 miles above The Battery and extend upriver for about 10 miles—see **33 CFR 110.1** and **110.155**, chapter 2, for limits and regulations.

(14) Vessels proceeding from New York to Albany occasionally anchor overnight in the vicinity of Kingston, 79 miles above The Battery and 47 miles below Albany, to await daylight hours for passing through the constricted part of the river.

(15) A buoyed anchorage, 400 feet wide and 2,400 feet long, is on the east side of the channel just above Stuyvesant (42°23'22"N., 73°46'53"W.), about 15 miles below Albany.

(16)

Dangers

(17) Navigation of the river is easy as far north as Kingston, but above Kingston it is more difficult because of the numerous steep-to shoals and middle grounds. In general tows are apt to follow the shoreline that is most favorable as regards wind and current; with a strong northwest wind, tows will follow the west shore regardless of the direction in which they are traveling.

(18)

Regulated navigation area

(19) The Coast Guard established a regulated navigation area on the navigable waters of the Hudson River south of the Troy Locks, effective during certain ice conditions—see **33 CFR 165.165**, chapter 2, for limits and regulations.

(20)

Recreational boaters navigating near commercial shipping channels

(21) Large commercial vessels and tugs with tows are often restricted in their ability to maneuver—as defined in Rule 3 of the Inland Navigation Rules—and therefore have the right of way over all recreational boats including sailboats. In accordance with Rule 9 of the Inland Navigation Rules, vessels less than 20 meters in length shall not cross ahead or otherwise impede the passage of any vessel that can safely navigate only within a narrow channel or fairway. Accordingly, recreational vessels should avoid commercial shipping channels and whenever possible transit them as near to the outer limit of the channel or fairway that lies on the vessel's starboard as is safe and practical. If it becomes necessary to cross a channel, check for other vessels and pass astern of oncoming vessels. Be aware that tugs often tow barges and other objects on long submerged towlines that are difficult to see and should never cross between a tug and its tow. Additional information is available at *www.uscgboating.org*.

(22)

Speed and Wake Damage

(23) Speed and wake damage complaints are an ongoing issue due to the increasing usage by both commercial and recreational users. While there are no federal regulations that address vessel speed limits outside of federal anchorage grounds, all vessel operators are expected to operate at a safe speed and in a manner that does not put others at risk. Licensed commercial mariners are further expected to be familiar with ongoing evolutions within the port and honor the requests of other waterway users as a professional courtesy. This information is published at *https://homeport.uscg.mil* and in the weekly Local Notice to Mariners at *www.navcen.uscg.gov* or by Safety Radio Broadcasts (See chapter 1—Navigation Warnings, Information and Weather). Title 46 Part 185.304 of the Code of Federal Regulations, states: "The operator of a vessel should pay special attention in regards to the

potential caused by their wake." The operation of a vessel in a negligent manner is a violation of federal law that may carry a monetary penalty. In addition, vessel operators may incur civil liability for the damage caused to other persons or property. Parties alleging the creation of an excessive wake may document their concerns via videotape or pictures. This type of documentation could be the basis for opening a civil penalty case.

(24)

No-Discharge Zone

(25) The State of New York, with the approval of the Environmental Protection Agency, has established a No-Discharge Zone in the waters of the Hudson River. The zone extends from the Battery in Manhattan, New York, to the federal dam at Troy, New York. Within the zone, discharge of sewage, whether treated or untreated, from all vessels is prohibited. Outside the zone, discharge of sewage is regulated by **40 CFR 140**—see chapter 2.

(26)

Tides

(27) The tides in Hudson River are affected by freshets, winds and droughts. Because of these variables the predictions given for points above George Washington Bridge are based upon averages for the 6-month period, May to October, when the freshwater discharge is at a minimum. See the tide prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of thetide at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(28)

Current

(29) The currents in Hudson River are influenced by the same variables that affect the tides. The times of slack water and the velocities and durations of flood and ebb are subject to extensive changes; the times of strengths are less likely to be affected. The currents usually set fair with the channels except in the vicinities of bends and wharves.

(30) Velocities of currents are 1.4 knots flood and 1.4 knots ebb northwest of The Battery, 1.6 and 2.2 knots at George Washington Bridge, 0.9 and 1.1 knots at Newburgh, 1.1 and 1.2 knots at Poughkeepsie, 1.3 and 1.6 knots at Kingston and 0.3 knot flood and 0.8 knot ebb at Albany. Near Troy Lock and Dam, the current does not flood and the ebb has a velocity of 0.7 knot. These values are for the summer when the freshwater discharge is at a minimum.

(31) Daily current predictions for The Narrows, New York Harbor, are available from the Tidal Current prediction service at *tidesandcurrents.noaa.gov*. Links to a user guide for this service can be found in chapter 1 of this book. Predictions for places along Hudson River may be obtained by applying the differences and ratios listed for these places in the tidal current predicition tables.

(32) During the summer of 2004, tidal observations were made in the Hudson River near Haverstraw and it was found that there were significant differences in the timing of the tidal current phases as compared with the predicted tidal current phases. The greatest time difference was observed in the slack before ebb, which on average may occur one hour later than the predictions given in the 2005 Tidal Current Tables. NOAA's Center for Operational Oceanographic Products and Services issued special daily tidal current predictions for the Hudson River at eight locations, where data were collected during 2005. Mariners should exercise caution when interpreting tidal predictions for these areas. See the Tidal Current prediction service at *tidesandcurrents.noaa.gov* for specific information about times, directions, and velocities of the current at numerous locations throughout the area. Links to a user guide for this service can be found in chapter 1 of this book.

(33)

Ice

(34) In even extremely severe winters, Coast Guard icebreakers and continuous river traffic maintain an open channel to Albany. The ice season usually starts in early January and ends in mid-March. Normally shipping is affected most seriously in the Hudson River between Tappan Zee and Albany. Modern vessels experience little difficulty maneuvering through the ice but may be slowed by other river traffic. In addition to the problem of getting through the ice, aids to navigation are covered or dragged off station by moving ice.

(35)

Freshets

(36) During March, April and May, freshets have reached heights above normal high water of as much as 18 feet at Albany and 25 feet at Troy Lock and Dam. At the time of the larger freshets the tide may be completely masked, the water continuing to rise and fall for a period of several days without any tidal oscillation. At the time of smaller freshets the range of tide is greatly diminished and the times of high and low waters are somewhat delayed.

(37) During the smaller freshets, the flood current disappears and the ebb current has a velocity of about 1.5 knots. The larger freshets produce an ebb current that varies from 1.5 to nearly 5 knots depending on the size of the freshet and the stage of the tide.

(38)

Pilotage, Hudson River

(39) Pilotage is compulsory on the Hudson River for foreign vessels and U.S. vessels under register. Pilotage north of Yonkers is available from Hudson River Pilots Association, 201 Edgewater Street, Staten Island, NY 10305, telephone 718–815–4316, FAX 718–876–8055. The pilot boat, JOHN E. FLYNN, is 40 feet with a black hull, white superstructure, and the word PILOT in red letters, each side. The boat berths at Yonkers and when underway monitors VHF-FM channel 13 and works channels 13 or 18A. The pilot boat meets vessels in midriver (40°56'21"N., 73°54'41"W.) off Yonkers. Arrangements for pilot services are made in advance

through ships' agents; at least 24-hour advance notice is requested.

(40) Vessels transiting between New York Harbor and Yonkers or between Long Island Sound and Yonkers are serviced by United New York New Jersey Sandy Hook Pilot Association.

(41) U.S. enrolled vessels in the coastwise trade transiting between New York Harbor and Yonkers or between Long Island Sound and Yonkers are also served by Interport Pilots Agency, Inc.

(42) On the Hudson River, pilots maintain bridge-to-bridge communication on channel 13.

(43) Vessels transiting the river to destinations beyond the city of Kingston, NY, will be required to embark another pilot at the Hyde Park Pilot Station (41°49'55"N., 073°56'32"W.) located on the eastern shore of Hudson River in Mills-Norrie State Park. For vessels awaiting daylight transits north of Kingston, a federal anchorage is located just south of the pilot station. The pilot station is manned only while boarding ships in transit and maintains a watch on VHF-FM channel 13 an hour prior to ETA for Norrie Point. The Hudson River Pilot office may be reached at 718–448–3900.

(44)

Towage

(45) Tugs are available in New York Harbor and at Albany—see chapter 11, and Albany later in this chapter.

(46)

Quarantine, customs and immigration

(47) Matters pertaining to these services for places along Hudson River are handled at the Port of New York or at Albany—see chapter 11, New York Harbor, and Albany later in this chapter.

(48) Hudson River averages about 0.6 mile in width along this 5-mile stretch above The Battery. The chart covers most of the principal wharves on the New York City side and those of **Jersey City, Hoboken,** and **Weehawken** on the west, or New Jersey, side. New York Harbor is a commercial/recreational waterway. This section of the Hudson River is used by commercial shipping, tugs and barges, sightseeing vessels, dinner boats, commuter ferries and recreational vessels including hand-powered vessels. Cruise ships operate from the NYC Passenger Ship Terminal Piers 86-92. NYC Department of Sanitation vessels operate from Piers 97 and 99. Con Edison receives fuel shipments at Pier 98.

(49) **Morris Canal Basin** is located across the Hudson River, opposite The Battery, on the New Jersey side. Two marinas, two commuter ferries, one sailing school, one yacht club and various charter boats operate from the basin. Commercial operators occupy the northwest corner of the basin while tour boats operate from the Central Railroad of New Jersey Pier at the southeast entrance to the basin.

(50)

Anchorages

(51) There are no special anchorages or commercial anchorage grounds in this part of the Hudson River. Vessels anchoring inside of the pierhead line shall be lighted in accordance with the Inland Navigation Rules and should check with local authorities for any additional requirements. Hudson River Park extends from Battery Park City to 59th Street. They provide mooring facilities south of Pier 40. The Hudson River Park dockmaster may be contacted at 212–627–2020 for availability.

(52)

Small-craft facilities

(53) Facilities at Manhattan are located at North Cove Yacht Harbor and Pier 59. Facilities in New Jersey are located in Morris Canal Basin, Jersey City, Hoboken and Weehawken. Sailing schools are in Jersey City, Hoboken and Manhattan.

(54)

Caution

(55) Commuter ferries operate between several sites in New Jersey and Manhattan. Extra caution should be used while transiting during the morning and evening rush hours. Hand-powered vessels operate from the New Jersey and New York shores of the Hudson River. Several swimming events are held along the Manhattan shoreline throughout the summer.

(56) On the New Jersey side of the river are **Guttenberg**, Mile 5.5W; **Edgewater**, Mile 7.5W; and **Fort Lee**, Mile 9.5W. Small-craft facilities at Edgewater can provide berths, electricity, gasoline, diesel fuel, water, ice, limited marine supplies, storage and hull and engine repairs. The largest mobile hoist can handle craft up to 25 tons. Commuter ferries operate between Edgewater and Pier 79 in Manhattan.

(57) The New York side of the river is mostly parkway for the length of the chart. The 79th Street Boat Basin, at Mile 5.5E, opposite Guttenberg, can provide berths, electricity, gasoline, diesel fuel, water, ice, marine supplies and minor engine repairs.

(58) **Sailors and Soldiers Monument**, Mile 6.2E, is a prominent landmark at 89th Street and Riverside Drive, Manhattan.

(59) **General Grants Tomb**, Mile 7.7E, is prominent at 123rd Street and Riverside Drive, Manhattan.

(60) **George Washington Bridge**, Mile 10, crosses Hudson River from Fort Lee, NJ, to **Fort Washington Point**, New York City. The suspension span is nearly 0.6 mile long from shore to shore with a least clearance of 195 feet—see *Structures across Hudson River* at the beginning of this chapter. The tops of the towers are about 600 feet above the water. When the traveler platform is in use, the bridge clearance is reduced to 180 feet.

(61)

Anchorages

(62) Three anchorage areas are south of the George Washington Bridge—see **33 CFR 110.1**, **110.155(c)(1)**,

and **110.155(c)(5)**, chapter 2, for limits and regulations. A special anchorage is on the north side of George Washington Bridge at North Manhattan—see **33 CFR 110.1** and **110.60**, chapter 2, for limits and regulations.

(63) From Fort Lee, NJ, the rocky cliffs of **Palisades State Park** and adjoining **Tallman Mountain State Park** extend up the west side of the river for about 12 miles to Piermont, NY. The Palisades are 300 to 500 feet high and in places are thickly wooded with scrub.

(64) **Spuyten Duyvil Creek**, entered at Mile 12E, is marked by the railroad swing bridge over the mouth. The creek is the Hudson River entrance to Harlem River, which is described in chapter 9. Currents are swift and erratic around the mouth of the creek.

(65)

Small-craft facilities

(66) **Englewood Boat Basin**, on the New Jersey side opposite Spuyten Duyvil Creek, can accommodate craft to 50 feet long; berths, gasoline, diesel fuel and water are available. **Alpine Boat Basin** is located at about 40°56'45"N., 73°55'05"W. A boat launching ramp for registered, trailer-towed boats under 24 feet in length, jet skis and car-top boats (canoes and kayaks) is available at Hazard's Launching Ramp south of the George Washington Bridge. On the New York side, Dykman Marina is located at **Tubby Hook**. The Riverdale Yacht Club and the Yonkers Paddling and Rowing Club are about 200 yards north of the **Yonkers Municipal Pier**. Commuter ferries operate between the Yonkers Municipal Pier and Battery Park in Manhattan.

(67) **Yonkers**, Mile 16E, adjoins the north side of New York City. Waterborne commerce is in petroleum products, sugar and syrup products, cement, sand and other building materials.

(68) A sugar refining plant (40°55'41"N., 73°54'21"W.) has a 400-foot marginal wharf with depths of 30 to 32 feet alongside and a deck height of 10 feet. The plant has 20,000 tons of covered storage and is served by a conveyor system with two 20-ton hoppers for the receipt of raw sugar. Vessels berth outboard of two floating cranes moored at the face of the wharf.

(69) Several other private facilities at Yonkers, used mainly by barges, have reported depths of 12 to 30 feet alongside.

(70) **Alpine** is a prominent landing at Mile 16W. A boat basin here, operated by the Palisades Interstate Park Commission, affords shelter for numerous small craft; berths, gasoline, electricity and water are available. In 2013, 4 feet was reported in the basin.

(71)

Anchorages

(72) A **special anchorage** adjoins a yacht club on the Yonkers side of the Hudson River, 17 miles above The Battery; another **special anchorage** is about 0.5 mile to the northward. (See **33 CFR 110.1** and **110.60**, chapter 2, for limits and regulations.)

(73) Several private boat clubs are at **Greystone Station**, just north of Yonkers; guest moorings are available.

(74) **Hastings-on-Hudson**, Mile 19E, has a prominent water tank at its waterfront. A yacht club, north of the waterfront, is adjoined by a **special anchorage**. (See **33 CFR 110.1** and **110.60**, Chapter 2, for limits and regulations.) Limited guest berths are available. In 1981, a reported depth of 4 feet could be carried to the fuel dock.

(75) The **boundary line** between the States of New Jersey and New York extends northwestward from a point on the west side of Hudson River at Mile 19. The river is 0.8 mile wide at this point.

(76) **Dobbs Ferry** is a town at Mile 20.5E. A stack on the waterfront and several cupolas are prominent.

(77) **Irvington**, Mile 22E, has a small private wharf at the northern end of the waterfront; guest moorings are available.

(78) At **Piermont**, Mile 22W, an earthen embankment extends 0.8 mile channelward from the shore to **Piermont Pier**. There is a Class I railroad terminus at the inner end of the embankment; several buildings in Piermont are prominent. A T-head pier, used by Columbia University to moor its geological research vessels, extends from the outer end of Piermont Pier; depths of about 16 feet are reported alongside the face. The ruins of a former ferry slip and other piers, as well as several visible wrecks and an obstruction covered 2 feet, are on the south side of Piermont Pier.

(79) A foul area extends about 300 yards northward from the outer end of Piermont Pier. A sunken wreck is in this area about 200 yards northward from the end of the pier; caution is advised.

(80) In 1981, shoaling to an unknown extent was reported in the area from the outer end of Piermont Pier north to **Lower Nyack Landing**, Mile 24.6W; caution is advised.

(81)

Small-craft facilities

(82) Several small-craft facilities are just northward of Piermont Pier. Berths, electricity, water, ice, storage, marine supplies, mobile hoists up to 10 tons and hull and engine repairs are available. In 1981, reported depths of 4 feet could be carried to the facilities. A scuba diving team of the Piermont Volunteer Fire Department is available for underwater search and rescue work. They can be contacted through the Piermont Police Department; telephone 914–359–0240.

(83) **Tappan Zee** is the 2-mile-wide part of Hudson River between Piermont and Croton Point, 8 miles to the northward.

(84) **Governor Mario M. Cuomo Bridge**, Mile 23.5, crosses Tappan Zee from Nyack to Tarrytown. A racon is atop the center of the main channel span of the southernmost bridge.

(85) **Tarrytown**, Mile 24E, has about 1 mile of developed waterfront, part of which has been improved by dredging. **Tarrytown Light** (41°05'03"N., 73°52'28"W.), 54 feet

above the water, is shown from a white cast iron and concrete tower.

(86) A federal project provides for depths of 12 feet in both the northwest and southwest connecting channels in Tarrytown Harbor and also in the waterfront channel. (See Notice to Mariners and latest edition of the chart for controlling depths.) An obstruction, consisting of rocks, is on the east edge of the waterfront channel in about 41°04'48"N., 73°52'10"W. Both access channels are buoyed.

(87) Tarrytown Harbor usually is open to navigation throughout the year, but in severe winters ice floes from the upper river may temporarily block the channels.

(88)

Anchorages

(89) A **special anchorage** is at Tarrytown. (See **33 CFR 110.1** and **110.60**, chapter 2, for limits and regulations.)

(90) Several waterfront terminals, with depths of 10 feet alongside, are available at Tarrytown, and there are rail connections nearby. The wharves are used mostly for the receipt of petroleum products, sand, gravel and crushed rock.

(91) A marina is southward of the principal wharves; berths, gasoline, diesel fuel, electricity, water, ice, marine supplies and a 15-ton mobile hoist are available. Two private boat clubs are southward of the marina; a launching ramp is available.

(92) **Nyack** is on the west side of Tappan Zee at Mile 25W. Small-craft facilities at Nyack include a boat club on the north side of the waterfront that can provide guest moorings and a municipal marina that does not accommodate transient boaters.

(93) In 1981, shoaling to an unknown extent was reported in the area from Lower Nyack Landing south to the outer end of Piermont Pier, Mile 22W.

(94)

Anchorages

(95) A **special anchorage** is at Nyack. (See **33 CFR 110.1** and **110.60**, chapter 2, for limits and regulations.)

(96) **Upper Nyack** is about 0.6 mile north of Nyack. A boatyard here has a 60-ton mobile boat lift and a 15-ton mobile crane for hauling out small craft. The boatyard has 80 feet of berthing space with a deck height of 8 feet and depths of 6 feet at the face. A basin here has limited space for berthing small craft. Complete engine and hull repairs can be made.

(97) **Hook Mountain**, 730 feet high, is on the west side of Tappan Zee at Mile 27W. The summit is only 0.3 mile inland and is very prominent from the river.

(98) **Ossining** is on the east side of Tappan Zee at Mile 29E. In 1981, depths of 5 to 6 feet were reported on the flats off the oil storage receiving facility piers at Ossining. **Sing Sing Correctional Facility**, a state penitentiary, is on the low flat shore on the south side of Ossining. Two water towers near the prison are prominent. A marina at the north end of town can handle craft to 15 tons for hull and engine repairs; marine supplies are available. There are also two boat clubs and a yacht club at Ossining; gasoline, water, ice and guest berths are available. In 1981, a reported depth of 4 feet could be carried to the yacht club gasoline dock.

(99) From Hook Mountain, Mile 27W, northward to Haverstraw, Mile 33W, the west bank of the Hudson River rises precipitously to heights of more than 800 feet.

(100) **Croton Point**, Mile 30E, is a long peninsula that extends 1.5 miles channelward from the main shore. Croton Point Park is on the southwest part of the peninsula. There are several prominent brick buildings at **Harmon**, near the inner end of Croton Point.

(101) **Haverstraw Bay** is the wide stretch of Hudson River between Croton Point and Stony Point, 5 miles to the northward; the greatest width is about 2.5 miles. The extensive flats in the eastern half of the bay have depths of 5 to 9 feet. The dredged channel through Haverstraw Bay is marked by seasonal lighted buoys and two lighted ranges.

(102) **Croton-on-Hudson**, on the east side of Haverstraw Bay at Mile 31.5E, has a yacht club.

(103) **High Tor**, 820 feet high, is on the west side of Haverstraw Bay at Mile 32W.

(104) **Haverstraw** is on the west side of Haverstraw Bay at Mile 33W. The stacks and large rectangular buildings of a power plant are prominent, back of **Bowline Point**. A T-shaped pier, operated by the power plant and marked by private lights, extends off Bowline Point.

(105) A wharf, used to ship crushed rock by barge, is about 1 mile southward of Bowline Point. The wharf has 580 feet of berthing space with a deck height of 5 feet and 14 feet alongside. A small private boat club is in a cove about 0.5 mile south of Bowline Point.

(106)

Anchorages

(107) A **special anchorage** is at Haverstraw. (See **33 CFR 110.1** and **110.60**, chapter 2, for limits and regulations.)

(108) **Grassy Point** is on the west side of Haverstraw Bay at Mile 34W. A pier, used for receiving gypsum rock, is on the south side of the point. The pier has 500 feet of berthing space with a deck height of 8 feet and 31 feet alongside.

(109) Numerous small-craft facilities are north and south of Grassy Point. Berths, electricity, gasoline, diesel fuel, water, ice, storage, marine supplies, a pump-out facility, lifts to 40 tons and engine and hull repairs are available. In 2001, a reported depth of 17 feet could be carried into the cove south of the point.

(110) **Stony Point**, Mile 35W, is marked at the outer end by a light.

(111) **Verplanck Point**, Mile 35.5E, is marked on its northwestern side by prominent gray eroded banks of tailings from a trap-rock plant. Small-craft facilities on the point can provide berths, electricity, gasoline, diesel fuel, water, ice, storage and limited marine supplies; lifts to 30 tons are available for hull and engine repairs.

(112) **Indian Point**, on the east side of Hudson River, 1.7 miles northward of Verplanck Point, is the site of a nuclear power station. A tall white stack is prominent, lighted on top, and two large domes are conspicuous on the point. A **safety** and **security zone** has been established in the waters of the Hudson River surrounding the station. (See **33 CFR 165.1** through **165.40** and **165.169**, chapter 2, for limits and regulations.)

(113) **Tomkins Cove**, a town at Mile 36W, has a large stone quarry, a rock crusher and a trap-rock plant. The wharf, connected to storage bins by a conveyor system, has 700 feet of berthing space with a deck height of 9 feet and depth of 15 to 20 feet alongside. The wharf is used to ship sand and crushed rock by barge. Numerous beached barges south of the pier are prominent. Northward of the wharf, there are electrical towers, a steel conveyer belt and a large concrete building in ruins, remnants of a power plant.

(114) **Peekskill** is at the head of a shallow bight at Mile 38E. A dredged U-shaped channel extends northeastward from deep water in the Hudson River to the wharf area and thence northwestward back to deep water. The southern channel is marked by buoys and a light. **Caution**—In 1985, it was reported that the channel on the north side of Peekskill Bay was obstructed by a sewer outfall extending across from the entrance to Annsville Creek; caution is advised. A yacht club at Peekskill can provide guest berths, electricity, water, ice and engine repairs.

(115) **Annsville Creek** is a very shallow creek on the north side of Peekskill. The railroad bridge over the entrance has a bascule span with a clearance of 3½ feet. The bridge is maintained in the closed position. The highway bridge about 0.2 mile above the railroad bridge has a fixed span with a clearance of 19 feet.

(116) A wharf, used for receiving petroleum products by barge, is at **Roa Hook**, northwest of Peekskill. The wharf has 150 feet of berthing space with a deck height of 8 feet and 14 feet alongside.

(117) **Dunderberg Mountain**, 1,110 feet high, is a densely wooded mountain at Mile 38W. The mountain slopes eastward to **Jones Point**, which is low and flat.

(118) The river becomes much narrower at Jones Point and has an average width of 0.3 mile for the next 8 miles between the bases of the highlands on both sides. When approaching the sharp turns in this reach, caution should be exercised and a warning signal should be given.

(119) **Iona Island**, formerly a naval depot at Mile 40W, is controlled by the Palisades Interstate Park Commission. A light, shown from a skeleton tower on the north side of the island, is conspicuous.

(120) A rock, with a depth of 10 feet over it and marked by a lighted buoy, is 0.2 mile north-northwestward of the northernmost point of Iona Island. When descending the river, particularly with a strong fair current, a careful watch should be maintained to avoid being set on this rock.

(121) **Bear Mountain**, Mile 40.3W, is 1,305 feet high and has its summit about 1 mile inland. There are wharves at the state park on the riverbank at the foot of the mountain.

(122) **Anthonys Nose**, 900 feet high, is a steep, thickly wooded hill at Mile 40.5E.

(123) **Bear Mountain Bridge**, Mile 40.6 crosses the Hudson River from Bear Mountain to Anthonys Nose. The suspension span has a clearance of 155 feet.

(124) **Con Hook**, a small island at Mile 43W, is marked on its channel side by a light. A rock, with a depth of 7 feet over it and marked by a lighted buoy, is about 0.3 mile southward of Con Hook. When descending the river, particularly with a fair current, there is a tendency to set toward the rock; caution is advised. The area 800 yards north of Con Hook and along the western shoreline is extremely shallow and dangerous and should be avoided due to a large shoal. When southbound on the Hudson River approaching Con Hook, mariners must take care not to confuse the lights on navigation aids with the lights from the railroad track on the west bank, the lights from bridge in the distance, and other background lighting in general to avoid vessel grounding.

(125) A tower at **Highland Falls**, Mile 44W, is prominent. There are a couple of buildings down next to the railroad tracks and there is an old boat ramp at the north end.

(126) A yacht club at **Garrison**, Mile 45E, has depths of about 20 feet alongside its fuel dock. Craft up to 60 feet in length can be accommodated at the slips; gasoline, water, electricity and some marine supplies are available.

(127) **West Point**, Mile 45W, is the site of the **U.S. Military Academy**. The academy is easily recognized from the prominence of the buildings and the road leading up the hillside from the railroad station and wharfs on the riverbank.

(128)

Anchorages

(129) A **special anchorage** is at West Point. (See **33 CFR 110.1** and **110.60**, chapter 2, for limits and regulations.)

(130) The northeastern extremity of West Point descends to **Gees Point**, a rocky feature that is marked by a light. About 0.2 mile south of Gees Point, another light marks the outer edge of a rocky shallow area along the west bank.

(131) **Worlds End**, a sharp bend in the Hudson River at Mile 46, has depths of more than 100 feet. Extreme caution should be exercised when passing through Worlds End; the view is obstructed and vessels should reduce speed and sound a warning signal.

(132) **Constitution Island** is on the upper side of Worlds End at Mile 46.5E. **Magazine Point**, on the channel side of the island, is marked by a light.

(133) **Crows Nest**, Mile 47W, is 1,403 feet high and prominent. A boat club is at **Cold Spring**, Mile 47.3E.

(134) **Little Stony Point**, Mile 48E, is the site of a rock quarry.

(135) **Storm King Mountain**, 1,355 feet high, is prominent at Mile 49W.

(136) **Breakneck Point**, on the opposite side of Hudson River from Storm King Mountain, is marked by one highway tunnel and two railroad tunnels; the lights are prominent at night. Behind Breakneck Point is **Breakneck Ridge**, 1,196 feet high.

(137) **Cornwall-on-Hudson** is at Mile 50W. The wharf at Cornwall is in ruins. A boat club and a yacht club, about 0.6 mile southeastward of the wharf in ruins, can provide gasoline, water and ice; guest moorings and a launching ramp are available. In 1981, the reported depths were 10 feet at the gasoline dock and 3 feet in the basin.

(138) **Pollepel Island**, Mile 50E, is a private estate with buildings that resemble a medieval castle. A light is shown from a skeleton tower 0.1 mile off the west side of the island.

(139) **Newburgh**, Mile 53W, is a major petroleum distribution center. Most of the piers of the major oil companies are at the southern end of the 2-mile waterfront between Newburgh and **Plum Point**. Depths at the piers are reported to range from about 14 feet at the northern end to 35 feet at the southern end of the waterfront.

(140) The yacht club landing near the north end of the Newburgh waterfront has reported depths of about 10 feet alongside. The marine railways here can handle craft up to 46 feet for minor engine and hull repairs; berths, electricity, gasoline, diesel fuel, water, ice, launching ramps and marine supplies are available. A shipbuilding company at Newburgh can make emergency repairs to commercial vessels. A marine railway at the yard can handle vessels to 140 feet, and cranes to 150 tons are available.

(141) **Beacon**, on the east bank of the Hudson River opposite Newburgh, has some manufacturing facilities. A seasonal swimming area in the river at Beacon is marked by private buoys. The **Newburgh-Beacon Bridges**, two spanned fixed highway bridges, with a clearance of 147 feet for a middle 760-foot width and 172 feet at the center, cross the river between Beacon and Newburgh. A private sound signal is at the bridge, and a racon is atop the center of the main channel span of the southernmost bridge.

(142) Two submerged obstructions are reported about 150 yards south of Hudson River Lighted Buoy 52, Mile 55. A submerged obstruction, covered ½ foot, is reported about 700 yards west of Buoy 52.

(143) **Chelsea**, Mile 56.5E, has a boatyard and yacht club; berths, electricity, gasoline, water, ice, marine supplies and complete hull and engine repairs are available. A 12-ton mobile crane is available for do-it-yourself repairs.

(144) **Danskammer Point**, Mile 58W, is marked by a conspicuous power plant with two large buildings, four stacks, a radio tower and an oil receiving pier.

(145) **Wappinger Creek** is entered at Mile 58.5E through a channel that leads to just below **Wappingers Falls**, 1.6 miles above the entrance. In 1977, it was reported that the creek had silted in and was no longer navigable.

(146) **Diamond Reef**, with a depth of 5 feet over it and marked by a seasonal lighted buoy, lies in about the middle of Hudson River 0.2 mile above the entrance to Wappinger Creek. Between Diamond Reef and Poughkeepsie the west side of the river should be favored to avoid two 18-foot spots that are buoyed.

(147) A marina at **New Hamburg**, just north of the entrance to Wappinger Creek, has berths, electricity, gasoline, water, ice, a 12-ton lift and marine supplies; hull and engine repairs can be made. In 2017, depths between 4 and 8 feet were found alongside the edge of the marina.

(148) A boat club at **Marlboro**, Mile 59.7W, can provide gasoline and water. Along the east side of the river, one mile north of Marlboro at **Clinton Point**, is a quarry and wharf used for shipping dolomite by barge. The wharf has 2,025 feet of berthing space with a deck height of 12 feet and a depth of 10 to 11 feet alongside. The wharf is served by a belt conveyor that extends to a processing plant near the quarry.

(149) **Poughkeepsie**, Mile 66E, is an important industrial center specializing in manufactured goods, oil and lumber.

(150) **Mid Hudson Bridge (U.S. 44)**, a fixed span with a clearance of 134 feet, and a fixed railroad bridge with a clearance of 167 feet; 0.5 mile northward, cross the river at Poughkeepsie; both bridges are well lighted at night. The Mid Hudson Bridge is equipped with a private sound signal and a racon in the middle of the span.

(151) A wharf that receives petroleum products by barge is one mile south of the Mid Hudson Highway Bridge. The wharf has 460 feet of berthing space with a deck height of 10 feet and a depth of 13 feet alongside. A town park and a small-craft launching ramp are about 0.2 mile north of the highway bridge.

(152) A marina, on the east side of the river near Mile 68E, has berths, electricity, gasoline, water, ice, a launching ramp, marine supplies and a 20-ton crane; hull, engine and electronic repairs can be made. In 2001, 17 feet was reported alongside the docks.

(153) **Hyde Park**, Mile 71E, is the birthplace of Franklin Delano Roosevelt, the 32nd President of the United States. The residence and library are about 0.4 mile inland.

(154)

Anchorages

(155) A **general anchorage** is just west of Hyde Park. (See **33 CFR 110.1** and **110.155**, chapter 2, for limits and regulations.)

(156) The Hyde Park Pilot Station (41°49'55"N., 073°56'32"W.) is located on the eastern shore of the Hudson River in Mills-Norrie State Park. Vessels transiting the river to destinations above Kingston, NY, will be required to embark another pilot at this point. The pilot station is manned only while boarding ships in transit and maintains a watch on VHF-FM channel 13 an hour prior to ETA for Norrie Point. The Hudson River Pilot office may be reached at 718–448–3900.

(157) The Poughkeepsie Yacht Club, about 0.5 mile north of the anchorage area, has berths, electricity, gasoline, diesel fuel, water, a 15-ton mobile hoist, ice and a sewage

pump-out facility. In 1981, 8 feet was reported available alongside the gasoline dock.

(158) **Esopus Island**, Mile 73, is marked by a light on the south end. A ledge, partly bare at low water and extending about 300 yards from the north end, is marked by a lighted buoy. The better channel is westward of the island. A prominent large gray stone building is on the west side of the river above **Esopus**, about 1 mile north of Esopus Island.

(159) **Indian Kill** flows into the Hudson River at Mile 73.8E. At the entrance to Indian Kill is Mills Norrie State Park Marina. Private seasonal lights mark the entrance to the marina. In 1981, the reported controlling depth was 7½ feet in the entrance channel with 5½ feet available in the basin. The marina has a concrete boat launch ramp, camp sites, showers and a pump out station. Transient boaters are welcome to visit the marina if space allows. 145 slips, both fixed piers and floating docks. Supplies can be obtained nearby.

(160) A shoal about 0.6 mile long and 150 yards wide with a least depth of about 15 feet is just west of the center of the channel, about 1.1 miles above Indian Kill entrance. The shoal is marked by a seasonal lighted buoy about midway along the east edge.

(161) **Esopus Meadows Light**, Mile 75.8, 52 feet above the water, is shown from a white brick lighthouse on the west side of the main channel. Shoals with depths less than 3 feet extend as much as 0.4 mile from either shore from about 1 mile below the light to Rondout Creek at Kingston; the shoals are marked by lighted buoys.

(162) **Rondout Creek** is entered from the Hudson River at Mile 79W through a dredged channel that leads between two long, submerged jetties to **Eddyville**, about 3 miles above the channel entrance. The jetties are marked by lights at the outer ends and by seasonal daybeacons. An obstruction is at 41°55'20.5"N., 73°58'12.4"W. The channel is partially marked by buoys. The head of practical navigation is at the lock of the abandoned Delaware and Hudson Canal, 3.3 miles above the entrance. The lower 2-mile portion of Rondout Creek serves as a harbor for Kingston.

(163) **Kingston** is partly on the lowlands adjacent to the north bank of Rondout Creek and partly on the elevated plateau to the north and westward of it. Waterborne traffic consists chiefly of petroleum products.

(164)

Bridges

(165) Rondout Creek is crossed by a fixed highway bridge with a clearance of 56 feet, about 1 mile above the entrance; a highway suspension bridge with a clearance of 86 feet, about 0.1 mile above the fixed bridge; and the fixed railroad bridge with a clearance of 144 feet, about 2 miles above the entrance. An overhead power cable with a clearance of 75 feet crosses the creek about 0.45 mile above the railroad bridge.

(166)

Small-craft facilities

(167) There are several small-craft facilities on Rondout Creek. Berths, electricity, gasoline, diesel fuel, water, ice, marine supplies, launching ramps, a pump-out facility and wet and dry storage are available as far upstream as Eddyville. Lifts to 35 tons and a 75-foot marine railway can handle craft for hull and engine repairs.

(168) In the Hudson River above Kingston many shoals with depths less than 3 feet are in midriver or extend from the shore on either side. The bottom is rocky at many of the bar crossings. Most of the channels through the critical areas are marked with lights and buoys, but strangers in all except small boats are advised to take a pilot. Pilots are engaged at New York.

(169) An oil terminal is at **Kingston Point**, Mile 80W. The terminal wharf has 250 feet of berthing space with a deck height of 7 feet and a depth of 13 feet alongside. The terminal receives petroleum products by barge.

(170) **Kingston-Rhinecliff Bridge** crosses the Hudson River at Mile 82.7. The fixed channel spans have a clearance of 135 feet. A private sound signal is at the bridge and a racon is in the center of the west channel span.

(171) **Esopus Creek** is entered at Mile 88.5W. The entrance is between two dikes; both are marked by lights. **Saugerties** is on the north bank of the creek about 1 mile above the entrance. A dam crosses the creek about 1.3 miles above the entrance. Just below the dam are many large boulders and several shoals that bare at low water. Small craft with local knowledge sometimes use this area as an anchorage; it should be avoided by strangers.

(172)

Small-craft facilities

(173) Small-craft facilities along the creek can provide berths, electricity, gasoline, diesel fuel, water, ice, outside storage and some marine supplies. A forklift can handle craft to 2 tons for engine and hull repairs; launching ramps are also available.

(174) A rescue vessel of the Ulster County Sheriff's Department is at Saugerties. The Sheriff's office can be contacted through the Coast Guard on VHF-FM channel 16 or directly by telephone at 845–338–3640.

(175) In 2004, shoaling to 9 feet was reported at the southern boundary of **Green Flats**. Vessels are advised to transit along the centerline of the channel in the vicinity of Hudson River Lighted Buoy 94.

(176) **The Maelstrom** is a dangerous whirlpool on the east side of the main channel about 2 miles north of Esopus Creek.

(177) Several large cement manufacturing plants that have prominent buildings and elevators are near **Cementon**, Mile 92.5W. Another cement factory is at **Dewitt Point**, 2 miles above Cementon. A wharf extends out just south of Dewitt Point and has a surveyed depth of 27 feet at the face (2015). However, two charted obstructions exist 30 yards east of the pier face with least depths of 24 feet and

25 feet. The landing for **North Germantown** is across the river opposite this wharf.

(178) **Catskill Creek**, marked at the entrance by buoys, is entered at Mile 97.5W. **Catskill** is about 1 mile above the mouth. A long wharf extends along the north side of Catskill Creek from the entrance to Catskill. A fixed highway bridge, with a clearance of 11 feet, crosses the creek 0.9 mile above the mouth. An overhead power cable with a clearance of 60 feet is about 200 yards above the bridge.

(179)

Small-craft facilities

(180) Several small-craft facilities are on the creek. Berths, electricity, gasoline, diesel fuel, storage facilities, water, ice, marine supplies, a sewage pump-out facility and lifts to 20 tons are available; hull and engine repairs can be made.

(181) **Rip Van Winkle Bridge** crosses the Hudson River at Mile 98.7. The fixed span over the channel has a clearance of 142 feet. A racon is at the center of the main channel span. High-voltage power cables with a clearance of 145 feet cross the river about 2.4 miles above the bridge. Red lights are atop the suspension towers on both sides of the river.

(182) **Hudson**, Mile 102E, is on a slope that rises from the east bank of the Hudson River. Gasoline, berths, electricity, water and a launching ramp are available at a boat club at Hudson.

(183) **Athens** is on the west side of the Hudson River opposite Hudson. Wharves that receive asphalt and calcium chloride are at Athens. Barges call at these facilities, which have reported depths of 7 to 15 feet alongside. In 1981, shoaling to an unknown extent was reported north of **Middle Ground Flats**. Barges approach Athens through the channel south of the flats only.

(184)

Small-craft facility

(185) A small-craft facility at the north end of town has berths, electricity, gasoline, water, ice and limited marine supplies and can make minor engine repairs.

(186) **Coxsackie** is at Mile 108W. Berths, gasoline, electricity, water and ice are available at a yacht club at the north end of town. A state-owned 20-foot concrete launching ramp is also available at Coxsackie.

(187) A boatyard at **New Baltimore**, Mile 113.5W, can provide berths, electricity, gasoline, diesel fuel, water, storage and marine supplies. A launching ramp and a 20-ton mobile hoist are available; hull and engine repairs can be made. In 1981, a reported depth of 20 feet was available at the fuel dock with 6 feet at the berths.

(188) **Coeymans**, Mile 115W, has a boatyard that can provide berths, electricity, gasoline, diesel fuel, water, ice and a 12-ton lift; hull and engine repairs can be made.

(189) Cement and limestone are shipped and gypsum is received at a wharf about 1 mile above Coeymans. The wharf has 666 feet of berthing space with a deck height of 15 feet and 32 feet reported alongside.

(190) A submerged jetty, marked by buoys, is just east of Coeymans.

(191) The fixed railroad bridge with a clearance of 139 feet crosses the Hudson River at Mile 117.8. An overhead power cable just southward of the bridge, has a clearance of 185 feet. The **Castleton-on-Hudson Bridge**, a fixed highway bridge, about 150 yards above the railroad bridge has a clearance of 135 feet.

(192) **Castleton-on-Hudson**, Mile 119E, has a boat club that can provide berths, electricity, gasoline, diesel fuel, water, ice and a launching ramp. Gin poles are available at the boat club for stepping masts. In 1982, depths of 9 feet were reported alongside the docks.

(193) The Castleton Fire Department maintains a rescue vessel at the boat club for emergency medical assistance, firefighting, lifesaving and damage control. The rescue vessel can be contacted through the Coast Guard on VHF-FM channel 16 or by telephone at 518–272–5501.

(194)

Anchorages

(195) A **special anchorage** is at Mile 120W, just below **Cedar Hill**. (See **33 CFR 110.1** and **110.60**, chapter 2, for limits and regulations.)

(196) Overhead power cables crossing the river at Mile 122.9 and Mile 123.1 have clearances of 169 and 194 feet, respectively.

(197) **Albany**, Mile 126W, is the capital of New York State and the principal port on the river above New York City. The port of Albany is the terminus for deep-draft vessels on the Hudson River and serves as a transshipping point for the immediate vicinity, large areas of New England and most of the areas accessible by waterways.

(198) Waterborne commerce at the port is mostly in petroleum products, but grain, automobiles, coal, molasses, scrap iron, aggregates, lumber, wood byproducts, bananas, steel, chemicals and general cargo are also handled.

(199) The Albany Port District includes the lower harbor between points about 0.2 mile below and 1.9 miles above the entrance to Island Creek (42°36'26"N., 73°45'50"W.) and the upper harbor extending northward of this point to the northern limits of Albany on the west side and **Rensselaer** on the east side.

(200)

Channels

(201) The Federal project depth is 32 feet from New York Harbor to Albany. Above the Port of Albany, the project depth is 14 feet to the Troy Lock and Dam. (See Notice to Mariners and latest editions of the charts for controlling depths.)

(202)

Anchorages

(203) The restricted width of the river at Albany is not sufficient to permit vessels to swing at anchor without interfering with passing craft. However, in an emergency, vessels sometimes anchor in midstream to wait for berthing space.

(204)

Bridges

(205) The Dunn Memorial fixed highway bridge with a clearance of 60 feet crosses Hudson River at Albany at Mile 126.4. The railroad bridge has a swing span with a clearance of 25 feet. (See **33 CFR 117.1** through **117.59** and **117.791**, chapter 2, for drawbridge regulations.) An overhead power cable at the railroad bridge has a clearance of 135 feet.

(206)

Weather, Albany and vicinity

(207) The climate at Albany is primarily continental in character but is subject to some modification from the maritime climate that prevails in the extreme southeastern portion of New York State. The moderating effect on temperatures is more pronounced during the warmer months than in the winter season when outbursts of cold air sweep down from Canada with greater vigor than at other times of the year. In the warmer portion of the year temperatures rise rapidly during the daytime to moderate levels. As a rule, temperatures fall rapidly after sunset so that the nights are relatively cool. Very occasionally, the area experiences extended periods of oppressive heat up to a week or more in duration.

(208) The highest temperature of record is 100°F (37.8°C) recorded both in July and September 1953. The extreme minimum temperature at Albany is -28°F (-33.3°C) recorded in January 1971. The average temperature for Albany is 48°F (8.9°C). The average high is 58°F (14.4°C) and the average low is 37°F (2.8°C). July is the warmest month with an average high of 83°F (28.3°C) and an average low of 60°F (15.6°C). January is the coldest month with an average high of 31°F (-0.6°C) and an average low of 13°F (-10.6°C). An average of 11 days each year record maximum temperatures in excess of 90°F (32.2°C) and an average of 147 days record extreme minimums below 32°F (0°C). An average of 22 days each year will have an extreme minimum below 5°F (-15°C).

(209) Precipitation is sufficient to serve the economy of the region in most years, and only occasionally do periods of drought become a threat. A considerable portion of the rainfall in the warmer months is from showers associated with thunderstorms, but hail is not usually of any consequence. Average annual precipitation totals nearly 36 inches (914 mm) and is evenly distributed throughout the year. The difference between the driest month, February, and the wettest month, June, averages exactly one inch (25.4 mm). Precipitation falls an average 205 days each year with the early winter season being the most likely time. Thunderstorms occur on average 24 days each year with June, July and August being the most favored period.

(210) Winters are usually cold and occasionally fairly severe. Maximum temperatures during the colder winter months often are below freezing, and nighttime low temperatures frequently drop to 10°F (-12.2°C) or lower. Sub-zero temperatures (<-17.8°C) occur rather infrequently, about a dozen times a year. Snowfall in the area is quite variable and over some of the higher nearby areas ranges up to 75 inches (1,905 mm) or more for a season. Snow flurries are quite frequent during the cold months. The average annual snowfall is 63 inches (1,600 mm) and snow can be expected each month October through May. January is the snowiest month averaging over 16 inches (406 mm). The 24-hour snowfall record is 22 inches (559 mm), and occurred in March 1993.

(211) On the whole, wind velocities are moderate. The north-south Hudson River Valley has had a marked effect on the lighter winds and the warm months usually average out as a south wind. Destructive winds occur infrequently.

(212) The area enjoys one of the highest percentages of sunshine that can be found in the state. This is true of the Hudson Valley area from Albany southward to the coast with slightly more sunshine progressively southward. Seldom does the area experience extended periods of cloudy days or extended periods of smog. Occasionally during the warm months, there are short periods when high humidity associated with temperatures above 85°F (29.4°C) is rather uncomfortable.

(213)

Pilotage, Albany

(214) See Pilotage, Hudson River (indexed as such), earlier this chapter.

(215)

Towage

(216) Tugs up to 6,800 hp, based at New York City, and tugs up to 1,800 hp, based at Rensselaer, are available at Albany. Arrangements for tugs are usually made in advance by ships' agents.

(217) Albany is a **customs port of entry.**

(218)

Quarantine, customs, immigration and agricultural quarantine

(219) (See chapter 3, Vessel Arrival Inspections, and Appendix A for addresses.)

(220) **Quarantine** is enforced in accordance with the regulations of the U.S. Public Health Service. (See Public Health Service, chapter 1.)

(221) Albany has several hospitals.

(222)

Harbor regulations

(223) Local rules and regulations for the port are handled by the Albany Port District Commission.

(224)

Wharves

(225) There are about 30 waterfront facilities at Albany and Rensselaer; most are located on the west side of the Hudson River at Albany. All have highway connections and, with the exception of the petroleum berths, railroad connections. Cargo is generally handled by ships' tackle. Crawler and truck cranes up to 140 tons can be rented. The alongside depths given for each facility are reported;

(226)

Facilities of Albany

Name	Location	Berthing Spac (feet)	Depths* (feet)	Deck Height (feet)	Storage	Purpose	Owned/Operated by
West side of Hudson River below Island Creek							
Sears Oil Company Tanker Wharf	1.2 miles south of Island Creek	191	31	10	Tank storage (2 million barrels)	Receipt of petroleum products	Sears Oil Company, Inc.
Texaco North Wharf	0.85 miles south of Island Creek	230	32	14	Tank storage (838,000 barrels)	Receipt and shipment of petroleum products	Texaco Inc.
West side of Hudson River below Island Creek							
Agway Petroleum Wharf	0.1 mile north of Island Creek	260	30	11	Tank storage (334,000 barrels)	Receipt and shipment of petroleum products	Agway Petroleum, Inc.
Cibro Petroleum Ship Dock	0.5 mile north of Island Creek	1,000	32	16.5	Tank storage (193,000 barrels asphalt) (955,000 barrels fuel oil) (450,000 barrels crude)	Receipt of crude oil and petroleum products, Receipt and shipment of asphalt	Albany Port District Commission/Cibro Petroleum Products, Inc.
Albany Port District Commission Berths 7, 8 and 9	0.8 mile north of Island Creek	1,270	32	16.5	Grain elevator (10.5 million bushels) Tank storage (4.5 million gallons)	Shipment of grain Receipt and shipment of molasses	Albany Port District Commission/National Molasses Co. and Cargill, Inc.
Albany Port District Commission Berths 5 and 6	Immediately north of Berth 7	750	32	16.5	Covered storage (60,000-square feet) Open storage (35 acres) Tank storage (8.5 million gallons)	Receipt and shipment of general cargo and liquid fertilizer Receipt of molasses	Albany Port District Commission/Pacific Molasses Company and Allied Chemical Corp.
Albany Port District Commission Berth 4	Immediately north of Berth 5	425	32	16.5	Covered storage (26,000-square feet) Open storage (28 acres)	Receipt and shipment of general cargo Receipt of automobiles	Albany Port District Commission
Albany Port District Commission Berth 3	Immediately north of Berth 4	425	32	16.5	Covered storage (72,000-square feet)	Receipt of bananas	Albany Port District Commission/United Brands, Inc.
Albany Port District Commission Berth 2	Immediately north of Berth 3	300	32	16.5	Open storage (28 acres)	Receipt and shipment of general cargo Receipt of automobiles	Albany Port District Commission
Albany Port District Commission Berth 1	Immediately north of Berth 2	600	32	16.5	Covered storage (45,000-square feet)	Receipt and shipment of general cargo Receipt of automobiles	Albany Port District Commission
Mobil Oil Corporation Ship Dock	200 yards north of Berth 1	200	29	16	Tank storage (2.25 million barrels)	Receipt of petroleum products	Mobil Oil Corporation
East side of Hudson River							
Amerada Hess Corporation Wharf	0.3 mile north of Island creek	290	30	8	Tank storage (1 million barrels)	Receipt of petroleum products	Amerada Hess Corporation/Amerada Hess Corp. and Sun Refining and Marketing Co.
Ultramar Petroleum Wharf	0.5 mile north of Island Creek	180	12	11	Tank storage (1 million barrels)	Receipt and shipment of petroleum products	Ultramar Petroleum, Inc.
Atlantic-Richfield Company Rensselaer Wharf	0.65 mile north of Island Creek	295	33	11	Tank storage (1.5 million barrels)	Receipt of petroleum products	Atlantic-Richfield Co./ Atlantic-Richfield Co., Gulf Oil Products Co. and Amoco Oil Co.
Petroleum Fuel and Terminal Company	0.75 mile north of Island Creek	375	32	14	Tank storage (668,000 barrels)	Receipt and shipment of petroleum products	Petroleum Fuel and Terminal Co.
Bray Terminals	0.8 mile north of Island Creek	250	23	12	Tank storage (646,000 barrels)	Receipt and shipment of petroleum products	Bray Terminals, Inc./Bray Terminals, Inc. and Getty Refining and Marketing Co.
Port of Albany Rennsselaer Wharf	1.25 miles north of Island Creek	1,205	32	16.5	Tank storage (500,000 gallons) Open storage (20 acres) Covered storage (43,000 square feet)	Receipt of caustic soda Shipment of scrap metal	Albany Port District Commission/Albany Port District Commission and Ashland Chemical Co.

* The depths given above are reported. For information on the latest depths contact the port authorities or the private operators.

for information on the latest depths, contact the operator. Only the major facilities are listed in the table (Facilities in the Port of Albany.)

(227)

Supplies

(228) Bunkering services for deep-draft vessels are not available at Albany; this service is obtained in New York. Diesel fuel, through metered pumps, is available for small vessels; water, marine supplies and provisions are available.

(229)

Repairs

(230) Repairs to hulls of all types, engine repair and general repairs to small craft are available at a boatyard on the west side of the river just north of Island Creek. A marine railway here will haul/launch vessels of 300 tons, 110 feet (30-foot beam) and a draft of 12 feet.

(231)

Small-craft facilities

(232) A yacht club is on the east side of the Hudson River at Rensselaer at Mile 126.4, about 0.2 mile south of the fixed highway bridge; berths, electricity, gasoline, diesel fuel and water are available. In 1981, reported depths of 15 feet were available on the west side of the yacht club dock with 8 feet on the east side. A municipal launching ramp is at Mile 127.2W.

(233)

Communications

(234) Albany is served by air and rail communications. The Delaware and Hudson Railroad serves facilities on the west side of the river while ConRail serves facilities on both sides of the river. The Albany Port Railroad Corporation, a terminal switching line, serves the waterfront facilities and property owned by the Albany Port District Commission and connects with the main line railroads.

(235) The **Patroon Island Bridge**, a fixed highway bridge, with a clearance of 60 feet crosses the Hudson River just above Albany at Mile 127.8.

(236) The **Troy-Menands Bridge**, a fixed highway bridge, crossing the Hudson River at South Troy, Mile 130.5, has a clearance of 61 feet. The overhead power cables between Albany and Troy have a least clearance of 87 feet. Red lights are shown from the suspension towers on both sides of the river.

(237) **Troy**, Mile 132E, is a manufacturing center. **Watervliet**, on the west side of the river opposite Troy, is the site of the United States Arsenal with a 755-foot stone bulkhead. The harbor extends from the southern limits of the city of Troy to the Troy Lock and Dam. Vessels usually berth on arrival, because the narrow width of the river and character of the bottom are not suitable for anchorage.

(238) The**Congress Street Bridge**connecting Watervliet and Troy, at Mile 132.2, has a fixed span with a clearance of 55 feet. The vertical lift highway bridge 0.5 mile upstream has a clearance of 29 feet down and 60 feet up. (See **33 CFR 117.1** through **117.59** and **117.791**, chapter 2, for drawbridge regulations.) A rock ledge is on the east side of the river at the bridge in about 42°44'07"N., 73°41'22"W. The**Green Island Bridge**, a fixed highway bridge at Mile 132.9, has a clearance of 61 feet.

(239) The **Troy Lock and Dam** is about 8 miles above Albany. The lock dimensions are: length 492.5 feet, width 44.4 feet, depth over upper miter sill 16.3 feet at normal pool level, and depth over lower miter sill 13 feet at lowest low water. The lift at the lowest stages is 17.3 feet. (See **33 CFR 207.50** and **207.60**, chapter 2, for navigation regulations for the lock and operating regulations for the dam.)

(240)

Caution

(241) The area within about 500 feet below the Troy Dam is extremely dangerous because of the turbulence caused by water discharge from the dam. The danger area is marked by buoys.

(242) The Hudson River above the Troy Lock and Dam joins with the New York State Canal System to form a connecting waterway westward to Lake Erie and Lake Ontario and northward to Lake Champlain.

(243) The **New York State Canal System**, comprising Erie Canal, Oswego Canal, Cayuga and Seneca Canal and Champlain Canal, is under the jurisdiction of the State of New York. Navigation on the state canals is free except for mooring, dockage, wharfage, storage or use of canal equipment or facilities for which a permit is required. Detailed data regarding movement through the New York State Canal System may be obtained from the New York State Canal Corporation, Office of Canals, 200 Southern Boulevard, P.O. Box 189, Albany, NY 12201-0189; 1–800–4CANAL4; or *canals.ny.gov*.

(244)

Controlling dimensions of channels, locks and bridges

(245) The **Great Lakes-Hudson River Waterway Improvement** is that part of the barge canal system including the Erie Canal from Waterford west to Three Rivers and thence the Oswego Canal to Lake Ontario. This section of the system, funded by the U.S. Government and maintained by the State of New York, has a project depth of 14 feet at normal pool level between locks and 13 feet at normal pool level through all locks and guard gates. These channels have widths of 104 feet in earth cuts, 120 feet in rock cuts and 200 feet in river and lake sections.

(246) Elsewhere in the New York State Canal System, the project depth is 12 feet in all channels and through all locks and guard gates. These channels have widths of 75 feet in earth cuts, 94 feet in rock cuts and generally 200 feet in canalized rivers.

(247) Usable dimensions of the locks in the New York State Canal System are 300 feet in length and 43½ feet in width. The locks and guard gates have depths of 12 feet over the sills at normal pool level, except 13 feet

over the sills in the Great Lakes-Hudson River Waterway Improvement.

(248) The least clearance of bridges and cables over the Great Lakes-Hudson River Waterway Improvement is 20 feet. The least clearance of bridges and cables over the other waterways of the New York State Canal System is 15 feet.

(249) The navigation season is normally from the first part of May to the latter part of November.

(250) **Erie Canal**, a 294-mile waterway, extends from the pool of the Troy dam in the Hudson River at Waterford westerly through the Mohawk River and landcuts to Oneida Lake; thence through Oneida, Seneca, and Clyde Rivers; landcuts; an artificial channel and Tonawanda Creek to Niagara River at Tonawanda. The Niagara River connects the Erie Canal with Lake Erie at Buffalo.

(251) **Oswego Canal,** a 21-mile waterway, extends northward from the Erie Canal, 141 miles westward of the Troy dam, to Oswego where it joins Lake Ontario. For the most part the canal follows the Oswego River from its confluence with the Oneida and Seneca Rivers.

(252) **Cayuga and Seneca Canal** extends southward from the Erie Canal 177 miles west to the Troy dam. The canal follows the improved Seneca River to Cayuga Lake and extends through the lake to Ithaca at the south end. From the north end of Cayuga Lake, the canal follows Seneca River west to Seneca Lake and extends through the lake to Watkins Glen at the south end. A 2.2-mile canal extends south from Watkins Glen to Montour Falls. These lakes are two of the so-called Finger Lakes of central New York and are each about 30 miles in length.

(253) **Champlain Canal**, a 52-mile waterway, follows the Hudson River northward from Waterford for about 32 miles to Fort Edward, thence through a landcut and Wood Creek to Whitehall at the entrance to Lake Champlain.

(254) **Lake Champlain,** about 97 miles long from Whitehall to the Canadian border and up to 10 miles wide at its widest part, has considerable water commerce between the ports along its shores. The controlling depth is about 12 feet at low lake level through the main channel to the Canadian border and to the principal ports. The least overhead clearance is 92 feet at a fixed bridge at Crown Point, about 32 miles above Whitehall.

(255) An international waterway for commerce is available between the United States and Canada by the use of Champlain Canal, Lake Champlain, and the **Riviere Richelieu** and **Canal de Chambly,** which extend from the northerly end of Lake Champlain for about 70 miles in Canadian waters to the St. Lawrence River, 40 miles below Montreal. The size of vessels that can navigate this route is controlled by the least dimensions of the Canal de Chambly locks which are: usable length, 111 feet, 5 inches; width, 23 feet; depth over sills, 6½ feet. Bridges over the waterway are provided with draws; the least overhead clearance of cables is 120 feet. The least clearance for bridges across Canal de Chambly in the vicinity of the city of St. Jean, Quebec, is 29 feet. The navigation season is from about the middle of April to the middle of November.

(256) Permit requirements and toll charge information for Canal de Chambly and St. Ours Lock may be obtained from the Superintendent, Quebec Canals (see Appendix A for address).

(257)

Charts and Coast Pilot Information

(258) NOAA's nautical chart coverage of the New York State Canal System covers all the canals from the Hudson River at Troy, NY, westward to Lyons, NY, and to Lake Ontario at Oswego; Oneida Lake; and Cayuga and Seneca Lakes. Charts of Lake Champlain are published by NOAA. Coast Pilot information for the above waterways is contained in U.S. Coast Pilot 6.

(259) Coverage of the canal system from Syracuse, west to the Niagara River at Tonawanda, NY, is contained in The Cruising Guide to the New York State Canal System, available from the New York State Canal Corporation at *canals.ny.gov*.

(260) Charts and pilot information for the Riviere Richelieu, Canal de Chambly and other Canadian waters are available from the Canadian Hydrographic Chart Distribution Office (see Appendix A for address).

Navigation Rules

(1) Following is an amalgamation of the **International (72 COLREGS) and Inland Navigation Rules**, their Annexes, and associated Federal rules and regulations.

(2) Text unique to Inland Rules is *italicized* and set apart in a text box or within *‹‹ double angle brackets ››*. International Rules are set apart in a text box or denoted with ‹ single angle brackets ›.

(3) Text within {curly brackets} denotes additions made by the U.S. Coast Guard Office of Navigation Systems.

(4) Disparate paragraph or section numbering are shown side by side separated by a dagger, i.e. (a)ǂ(b).

(5) Instances of "…§§83.xx / in / with / of … this section / subpart / part of this Rule, etc." are redacted, and herein are shown as the enumerated rule(s) they referred to, i.e. 72 COLREGS Rule 18(e) states: "...with the Rules of this Part" and the same Inland Rule states: "...with the Rules of this Subpart (Rules 4-19) (§§83.04 through 83.19)", but, herein it is stated as "...with Rules 4-19.

(6) Instances of paragraph / section (x) are redacted, and herein are shown as §(x).

(7) Rules denoted with an asterisk also have an associated implementing or interpretative rule (i.e. 33 CFR 81-90), which can be found in chapter 2.

(8)

Part A—General

(9)

Rule 1—Application (International)

(a) These Rules shall apply to all vessels upon the high seas and in all waters connected therewith navigable by seagoing vessels.

(b) Nothing in these Rules shall interfere with the operation of special rules made by an appropriate authority for roadsteads, harbors, rivers, lakes, or inland waterways connected with the high seas and navigable by seagoing vessels. Such special rules shall conform as closely as possible to these Rules.

(c) Nothing in these Rules shall interfere with the operation of any special rules made by the Government of any State with respect to additional station or signal lights, shapes or whistle signals for ships of war and vessels proceeding under convoy, or with respect to additional station or signal lights or shapes for fishing vessels engaged in fishing as a fleet. These additional stations or signal lights, shapes or whistle signals shall, so far as possible, be such that they cannot be mistaken for any light, shape, or signal authorized elsewhere under these Rules.

Rule 1—Application (International)

(d) Traffic separation schemes may be adopted by the Organization for the purpose of these Rules.

(e) Whenever the Government concerned shall have determined that a vessel of special construction or purpose cannot comply fully with the provisions of any of these Rules with respect to number, position, range or arc of visibility of lights or shapes, as well as to the disposition and characteristics of sound-signaling appliances, such vessel shall comply with such other provisions in regard to number, position, range or arc of visibility of lights or shapes, as well as to the disposition and characteristics of sound-signaling appliances, as the Government shall have determined to be the closest possible compliance with these Rules in respect to that vessel.

(10)

Rule 1—Application (Inland)

(a) These rules apply to all vessels upon the inland waters of the United States, and to vessels of the United States on the Canadian waters of the Great Lakes to the extent that there is no conflict with Canadian law. These Rules have preemptive effect over State or local regulation within the same field.

(b)(i)These rules constitute special rules made by an appropriate authority within the meaning of Rule 1(b) of the International Regulations for Preventing Collisions at Sea, 1972, including annexes currently in force for the United States ("International Regulations").
(ii) All vessels complying with the construction and equipment requirements of the International Regulations are considered to be in compliance with these Rules.

(c) Nothing in these Rules shall interfere with the operation of any special rules made by the Secretary of the Navy with respect to additional station or signal lights and shapes or whistle signals for ships of war and vessels proceeding under convoy, or by the Secretary with respect to additional station or signal lights and shapes for fishing vessels engaged in fishing as a fleet. These additional station or signal lights and shapes or whistle signals shall, so far as possible, be such that they cannot be mistaken for any light, shape or signal authorized elsewhere under these Rules. Notice of such special rules shall be published in the Federal Register and, after the effective date specified in such notice, they shall have effect as if they were a part of these Rules.

(d) Traffic separation schemes may be established for the purposes of these Rules. Vessel traffic service regulations may be in effect in certain areas.

Rule 1—Application (Inland)

(e) Whenever the Secretary determines that a vessel or class of vessels of special construction or purpose cannot comply fully with the provisions of any of these Rules with respect to the number, position, range, or arc of visibility of lights or shapes, as well as to the disposition and characteristics of sound-signaling appliances, the vessel shall comply with such other provisions in regard to the number, position, range, or arc of visibility of lights or shapes, as well as to the disposition and characteristics of sound-signaling appliances, as the Secretary shall have determined to be the closest possible compliance with these Rules. The Secretary may issue a certificate of alternative compliance for a vessel or class of vessels specifying the closest possible compliance with these Rules. The Secretary of the Navy shall make these determinations and issue certificates of alternative compliance for vessels of the Navy.

(f) The Secretary may accept a certificate of alternative compliance issued by a contracting party to the International Regulations if it determines that the alternative compliance standards of the contracting party are substantially the same as those of the United States.

(g) The operator of each self-propelled vessel 12 meters or more in length shall carry, on board and maintain for ready reference, a copy of these Rules.

(11)

Rule 2—Responsibility

(12) (a) Nothing in these Rules shall exonerate any vessel, or the owner, master, or crew thereof, from the consequences of any neglect to comply with these Rules or of the neglect of any precaution which may be required by the ordinary practice of seamen, or by the special circumstances of the case.

(13) (b) In construing and complying with these Rules due regard shall be had to all dangers of navigation and collision and to any special circumstances, including the limitations of the vessels involved, which may make a departure from these Rules necessary to avoid immediate danger.

(14)

Rule 3—General Definitions

(15) For the purpose of these Rules, except where the context otherwise requires:

(16) (a) The word "vessel" includes every description of watercraft, including non-displacement craft, WIG craft, and seaplanes, used or capable of being used as a means of transportation on water.

(17) (b) The term "power-driven vessel" means any vessel propelled by machinery.

(18) (c) The term "sailing vessel" means any vessel under sail provided that propelling machinery, if fitted, is not being used.

(19) (d) The term "vessel engaged in fishing" means any vessel fishing with nets, lines, trawls, or other fishing apparatus which restrict maneuverability, but does not include a vessel fishing with trolling lines or other fishing apparatus which do not restrict maneuverability.

(20) (e) The term "seaplane" includes any aircraft designed to maneuver on the water.

(21) (f) The term "vessel not under command" means a vessel which through some exceptional circumstance is unable to maneuver as required by these Rules and is therefore unable to keep out of the way of another vessel.

(22) (g) The term "vessel restricted in her ability to maneuver" means a vessel which from the nature of her work is restricted in her ability to maneuver as required by these Rules and is therefore unable to keep out of the way of another vessel. The term "vessels restricted in their ability to maneuver" shall include but not be limited to: (i) A vessel engaged in laying, servicing, or picking up a navigational mark, submarine cable or pipeline; (ii) A vessel engaged in dredging, surveying or underwater operations; (iii) A vessel engaged in replenishment or transferring persons, provisions or cargo while underway; (iv) A vessel engaged in the launching or recovery of aircraft; (v) A vessel engaged in mine clearance operations; (vi) A vessel engaged in a towing operation such as severely restricts the towing vessel and her tow in their ability to deviate from their course.

(23)

Rule 3h (International)

(h) The term "vessel constrained by her draft" means a power-driven vessel which because of her draft in relation to the available depth and width of navigable water is severely restricted in her ability to deviate from the course she is following.

(24) (i) The word "underway" means that a vessel is not at anchor, or made fast to the shore, or aground.

(25) (j) The words "length" and "breadth" of a vessel mean her length overall and greatest breadth.

(26) (k) Vessels shall be deemed to be in sight of one another only when one can be observed visually from the other.

(27) (l) The term "restricted visibility" means any condition in which visibility is restricted by fog, mist, falling snow, heavy rainstorms, sandstorms, or any other similar causes.

(28) (m) The term "Wing-In-Ground (WIG)" craft means a multimodal craft which, in its main operational mode, flies in close proximity to the surface by utilizing surface-effect action.

(29)

Rules 3n–3s (Inland)

(n) "Western Rivers" means the Mississippi River, its tributaries, South Pass, and Southwest Pass, to the navigational demarcation lines {30 CFR 80} dividing the high seas from harbors, rivers and other inland waters of the United States, and the Port Allen-Morgan City Alternate Route, and that part of the Atchafalaya River above its junction with the Port Allen-Morgan City Alternate Route including the Old River and the Red River.

Rules 3n–3s (Inland)
(o) "Great Lakes" means the Great Lakes and their connecting tributary waters including the Calumet River as far as the Thomas J. O'Brien Lock and Controlling Waters (between mile 326 and 327), the Chicago River as far as the east side of the Ashland Avenue Bridge (between mile 321 and 322), and the Saint Lawrence River as far east as the lower exit of Saint Lambert Lock.
(p) "Secretary" means the Secretary of the Department in which the Coast Guard is operating.
(q) "Inland Waters" means the navigable waters of the United States shoreward of the navigational demarcation lines {30 CFR 80} dividing the high seas from harbors, rivers and other inland waters of the United States and the waters of the Great Lakes on the United States side of the International Boundary.
(r) "Inland Rules" or "Rules" means these Inland Navigational Rules and the annexes thereto, which govern the conduct of vessels and specify the lights, shapes, and sound signals that apply on inland waters.
(s) "International Regulations" means the International Regulations for Preventing Collisions at Sea, 1972, including annexes currently in force for the United States.

(30) **Implementing Rule**—See **33 CFR 89.25**, chapter 2, for regulations.

(31)

Part B—Steering and Sailing Rules

(32)

I—Conduct of Vessels in Any Condition of Visibility

(33)

Rule 4—Application

(34) Rules 4 through 10 apply in any condition of visibility.

(35)

Rule 5—Lookout

(36) Every vessel shall at all times maintain a proper look-out by sight and hearing as well as by all available means appropriate in the prevailing circumstances and conditions so as to make a full appraisal of the situation and of the risk of collision.

(37)

Rule 6—Safe Speed

(38) Every vessel shall at all times proceed at a safe speed so that she can take proper and effective action to avoid collision and be stopped within a distance appropriate to the prevailing circumstances and conditions. In determining a safe speed the following factors shall be among those taken into account:

(39) (a) By all vessels:

(40) (i) The state of visibility; (ii) The traffic density including concentrations of fishing vessels or any other vessels; (iii) The maneuverability of the vessel with special reference to stopping distance and turning ability in the prevailing conditions; (iv) At night, the presence of background light such as from shore lights or from back scatter from her own lights; (v) The state of wind, sea and current, and the proximity of navigational hazards; (vi) The draft in relation to the available depth of water.

(41) (b) Additionally, by vessels with operational radar:

(42) (i) The characteristics, efficiency and limitations of the radar equipment; (ii) Any constraints imposed by the radar range scale in use; (iii) The effect on radar detection of the sea state, weather and other sources of interference; (iv) The possibility that small vessels, ice and other floating objects may not be detected by radar at an adequate range; (v) The number, location and movement of vessels detected by radar; (vi) The more exact assessment of the visibility that may be possible when radar is used to determine the range of vessels or other objects in the vicinity.

(43)

Rule 7—Risk of Collision

(44) (a) Every vessel shall use all available means appropriate to the prevailing circumstances and conditions to determine if risk of collision exists. If there is any doubt such risk shall be deemed to exist.

(45) (b) Proper use shall be made of radar equipment if fitted and operational, including long-range scanning to obtain early warning of risk of collision and radar plotting or equivalent systematic observation of detected objects.

(46) (c) Assumptions shall not be made on the basis of scanty information, especially scanty radar information.

(47) (d) In determining if risk of collision exists the following considerations shall be among those taken into account:

(48) (i) Such risk shall be deemed to exist if the compass bearing of an approaching vessel does not appreciably change.

(49) (ii) Such risk may sometimes exist even when an appreciable bearing change is evident, particularly when approaching a very large vessel or a tow or when approaching a vessel at close range.

(50)

Rule 8—Action to Avoid Collision

(51) (a) Any action taken to avoid collision shall be taken in accordance with Rules 4 through 19 and shall if the circumstances of the case admit, be positive, made in ample time and with due regard to the observance of good seamanship.

(52) (b) Any alteration of course and/or speed to avoid collision shall, if the circumstances of the case admit, be large enough to be readily apparent to another vessel observing visually or by radar; a succession of small alterations of course and/or speed should be avoided.

(53) (c) If there is sufficient sea room, alteration of course alone may be the most effective action to avoid a closequarters situation provided that it is made in good time, is substantial and does not result in another close-quarters situation.

(54) (d) Action taken to avoid collision with another vessel shall be such as to result in passing at a safe distance. The

effectiveness of the action shall be carefully checked until the other vessel is finally past and clear.

(55) (e) If necessary to avoid collision or allow more time to assess the situation, a vessel shall slacken her speed or take all way off by stopping or reversing her means of propulsion.

(56) (f)(i) A vessel which, by any of these Rules, is required not to impede the passage or safe passage of another vessel shall, when required by the circumstances of the case, take early action to allow sufficient sea room for the safe passage of the other vessel.

(57) (ii) A vessel required not to impede the passage or safe passage of another vessel is not relieved of this obligation if approaching the other vessel so as to involve risk of collision and shall, when taking action, have full regard to the action which may be required by Rules 4 through 19.

(58) (iii) A vessel, the passage of which is not to be impeded remains fully obliged to comply with Rules 4 through 19 when the two vessels are approaching one another so as to involve risk of collision.

(59)

Rule 9—Narrow Channels

(60) (a) ‹‹*(i)*›› A vessel proceeding along the course of a narrow channel or fairway shall keep as near to the outer limit of the channel or fairway which lies on her starboard side as is safe and practicable.

(61)

Rule 9a (Inland)

(ii) Notwithstanding Rule 9(a)(i) and Rule 14(a), a power-driven vessel operating in narrow channel or fairway on the Great Lakes, Western Rivers, or waters specified by the Secretary, and proceeding downbound with a following current shall have the right-of-way over an upbound vessel, shall propose the manner and place of passage, and shall initiate the maneuvering signals prescribed by Rule 34(a)(i), as appropriate. The vessel proceeding upbound against the current shall hold as necessary to permit safe passing.

(62) (b) A vessel of less than 20 meters in length or a sailing vessel shall not impede the passage of a vessel ‹ which ›‹‹ *that* ›› can safely navigate only within a narrow channel or fairway.

(63) (c) A vessel engaged in fishing shall not impede the passage of any other vessel navigating within a narrow channel or fairway.

(64) (d) A vessel ‹ shall ›‹‹ *must* ›› not cross a narrow channel or fairway if such crossing impedes the passage of a vessel which can safely navigate only within that channel or fairway. The latter vessel ‹ may ›‹‹ *must* ›› use the signal prescribed in Rule 34(d) if in doubt as to the intention of the crossing vessel.

(65)

Rule 9e (International)

(e)(i) In a narrow channel or fairway when overtaking can take place only if the vessel to be overtaken has to take action to permit safe passing, the vessel intending to overtake shall indicate her intention by sounding the appropriate signal prescribed in Rule 34(c)(ii). The vessel to be overtaken shall, if in agreement, sound the appropriate signal prescribed in Rule 34(c)(i) and take steps to permit safe passing. If in doubt she may sound the signals prescribed in Rule 34(d).

Rule 9e (Inland)

(e)(i) In a narrow channel or fairway when overtaking, the power-driven vessel intending to overtake another power-driven vessel shall indicate her intention by sounding the appropriate signal prescribed in Rule 34(c) and take steps to permit safe passing. The power-driven vessel being overtaken, if in agreement, shall sound the same signal and may, if specifically agreed to, take steps to permit safe passing. If in doubt she shall sound the signal prescribed in Rule 34(d).

(66) (e)(ii) This rule does not relieve the overtaking vessel of her obligation under Rule 13.

(67) (f) A vessel nearing a bend or an area of a narrow channel or fairway where other vessels may be obscured by an intervening obstruction shall navigate with particular alertness and caution and shall sound the appropriate signal prescribed in Rule 34(e).

(68) (g) Any vessel shall, if the circumstances of the case admit, avoid anchoring in a narrow channel.

(69)

Rule 10—Traffic Separation Schemes

(70) (a) This Rule applies to traffic separation schemes ‹ adopted by the Organization › and does not relieve any vessel of her obligation under any other rule.

(71) (b) A vessel using a traffic separation scheme shall:

(72) (i) Proceed in the appropriate traffic lane in the general direction of traffic flow for that lane.

(73) (ii) So far as is practicable keep clear of a traffic separation line or separation zone.

(74) (iii) Normally join or leave a traffic lane at the termination of the lane, but when joining or leaving from either side shall do so at as small an angle to the general direction of traffic flow as practicable.

(75) (c) A vessel, shall so far as practicable, avoid crossing traffic lanes but if obliged to do so shall cross on a heading as nearly as practicable at right angles to the general direction of traffic flow.

(76) (d)(i) A vessel shall not use an inshore traffic zone when she can safely use the appropriate traffic lane within the adjacent traffic separation scheme. However, vessels of less than 20 meters in length, sailing vessels and vessels engaged in fishing may use the inshore traffic zone.

(77) (ii) Notwithstanding Rule 10(d)(i), a vessel may use an inshore traffic zone when en route to or from a port, offshore installation or structure, pilot station or any other place situated within the inshore traffic zone, or to avoid immediate danger.

(78) (e) A vessel, other than a crossing vessel or a vessel joining or leaving a lane shall not normally enter a separation zone or cross a separation line except:

(79) (i) in cases of emergency to avoid immediate danger;

(80) (ii) to engage in fishing within a separation zone.

(81) (f) A vessel navigating in areas near the terminations of traffic separation schemes shall do so with particular caution.

(82) (g) A vessel shall so far as practicable avoid anchoring in a traffic separation scheme or in areas near its terminations.

(83) (h) A vessel not using a traffic separating scheme shall avoid it by as wide a margin as is practicable.

(84) (i) A vessel engaged in fishing shall not impede the passage of any vessel following a traffic lane.

(85) (j) A vessel of less than 20 meters in length or a sailing vessel shall not impede the safe passage of a power-driven vessel following a traffic lane.

(86) (k) A vessel restricted in her ability to maneuver when engaged in an operation for the maintenance of safety of navigation in a traffic separation scheme is exempted from complying with this Rule to the extent necessary to carry out the operation.

(87) (l) A vessel restricted in her ability to maneuver when engaged in an operation for the laying, servicing or picking up of a submarine cable, within a traffic separation scheme, is exempted from complying with this Rule to the extent necessary to carry out the operation.

(88)

II—Conduct of Vessels in Sight of One Another

(89)

Rule 11—Application

(90) Rules 11 through 18 apply to vessels in sight of one another.

(91)

Rule 12—Sailing Vessels

(92) (a) When two sailing vessels are approaching one another, so as to involve risk of collision, one of them shall keep out of the way of the other as follows:

(93) (i) when each has the wind on a different side, the vessel which has the wind on the port side shall keep out of the way of the other;

(94) (ii) when both have the wind on the same side, the vessel which is to windward shall keep out of the way of the vessel which is to leeward;

(95) (iii) if a vessel with the wind on the port side sees a vessel to windward and cannot determine with certainty whether the other vessel has the wind on the port or on the starboard side, she shall keep out of the way of the other.

(96) (b) For the purposes of this Rule, the windward side shall be deemed to be the side opposite that on which the mainsail is carried or, in the case of a square-rigged vessel, the side opposite to that on which the largest fore-and-aft sail is carried.

(97)

Rule 13—Overtaking

(98) (a) Notwithstanding anything contained in the Rules 4 through 18, any vessel overtaking any other shall keep out of the way of the vessel being overtaken.

(99) (b) A vessel shall be deemed to be overtaking when coming up with a another vessel from a direction more than 22.5 degrees abaft her beam, that is, in such a position with reference to the vessel she is overtaking, that at night she would be able to see only the sternlight of that vessel but neither of her sidelights.

(100) (c) When a vessel is in any doubt as to whether she is overtaking another, she shall assume that this is the case and act accordingly.

(101) (d) Any subsequent alteration of the bearing between the two vessels shall not make the overtaking vessel a crossing vessel within the meaning of these Rules or relieve her of the duty of keeping clear of the overtaken vessel until she is finally past and clear.

(102)

Rule 14—Head-on Situation

(103) (a) *‹‹ Unless otherwise agreed ››* when two power-driven vessels are meeting on reciprocal or nearly reciprocal courses so as to involve risk of collision each shall alter her course to starboard so that each shall pass on the port side of the other.

(104) (b) Such a situation shall be deemed to exist when a vessel sees the other ahead or nearly ahead and by night she could see the masthead lights of the other in a line or nearly in a line and/or both sidelights and by day she observes the corresponding aspect of the other vessel.

(105) (c) When a vessel is in any doubt as to whether such a situation exists she shall assume that it does exist and act accordingly.

(106)

Rule 14d (Inland)

(d) Notwithstanding Rule 14(a), a power-driven vessel operating on the Great Lakes, Western Rivers, or waters specified by the Secretary, and proceeding downbound with a following current shall have the right-of-way over an upbound vessel, shall propose the manner of passage, and shall initiate the maneuvering signals prescribed by Rule 34(a)(i), as appropriate.

(107)

Rule 15—Crossing Situation

(108) (a) When two power-driven vessels are crossing so as to involve risk of collision, the vessel which has the other on her own starboard side shall keep out of the way and shall, if the circumstances of the case admit, avoid crossing ahead of the other vessel.

(109)

Rule 15b (Inland)

(b) Notwithstanding Rule 15(a), on the Great Lakes, Western Rivers, or water specified by the Secretary, a power-driven vessel crossing a river shall keep out of the way of a power-driven vessel ascending or descending the river.

(110)

Rule 16—Action by Give-way Vessel

(111) Every vessel which is directed to keep out of the way of another vessel shall, so far as possible, take early and substantial action to keep well clear.

(112)

Rule 17—Action by Stand-on Vessel

(113) (a)(i) Where one of two vessels is to keep out of the way, the other shall keep her course and speed.

(114) (ii) The latter vessel may, however, take action to avoid collision by her maneuver alone, as soon as it becomes apparent to her that the vessel required to keep out of the way is not taking appropriate action in compliance with these Rules.

(115) (b) When, from any cause, the vessel required to keep her course and speed finds herself so close that collision cannot be avoided by the action of the giveway vessel alone, she shall take such action as will best aid to avoid collision.

(116) (c) A power-driven vessel which takes action in a crossing situation in accordance with Rule 17(a)(ii) to avoid collision with another power-driven vessel shall, if the circumstances of the case admit, not alter course to port for a vessel on her own port side.

(117) (d) This Rule does not relieve the give-way vessel of her obligation to keep out of the way.

(118)

Rule 18—Responsibilities Between Vessels

(119) Except where Rules 9, 10, and 13 otherwise require:

(120) (a) A power-driven vessel underway shall keep out of the way of: (i) a vessel not under command; (ii) a vessel restricted in her ability to maneuver; (iii) a vessel engaged in fishing; (iv) a sailing vessel.

(121) (b) A sailing vessel underway shall keep out of the way of: (i) a vessel not under command; (ii) a vessel restricted in her ability to maneuver; (iii) a vessel engaged in fishing.

(122) (c) A vessel engaged in fishing when underway shall, so far as possible, keep out of the way of: (i) a vessel not under command; (ii) a vessel restricted in her ability to maneuver.

(123)

Rule 18d (International)

(d)(i) Any vessel other than a vessel not under command or a vessel restricted in her ability to maneuver shall, if the circumstances of the case admit, avoid impeding the safe passage of a vessel constrained by her draft, exhibiting the signals in Rule 28.

(ii) A vessel constrained by her draft shall navigate with particular caution having full regard to her special condition.

(124) (e) A seaplane on the water shall, in general, keep well clear of all vessels and avoid impeding their navigation. In circumstances, however, where risk of collision exists, she shall comply with Rules 4 through 19.

(125) (f)(i) A WIG craft shall, when taking off, landing and in flight near the surface, keep well clear of all other vessels and avoid impeding their navigation;

(126) (ii) a WIG craft operating on the water surface shall comply with Rules 4 through 19 as a power-driven vessel.

(127)

III—Conduct of Vessels in Restricted Visibility

(128)

Rule 19—Conduct of Vessels in Restricted Visibility

(129) (a) This Rule applies to vessels not in sight of one another when navigating in or near an area of restricted visibility.

(130) (b) Every vessel shall proceed at a safe speed adapted to the prevailing circumstances and conditions of restricted visibility. A power-driven vessel shall have her engines ready for immediate maneuver.

(131) (c) Every vessel shall have due regard to the prevailing circumstances and conditions of restricted visibility when complying with Rules 4 through 10.

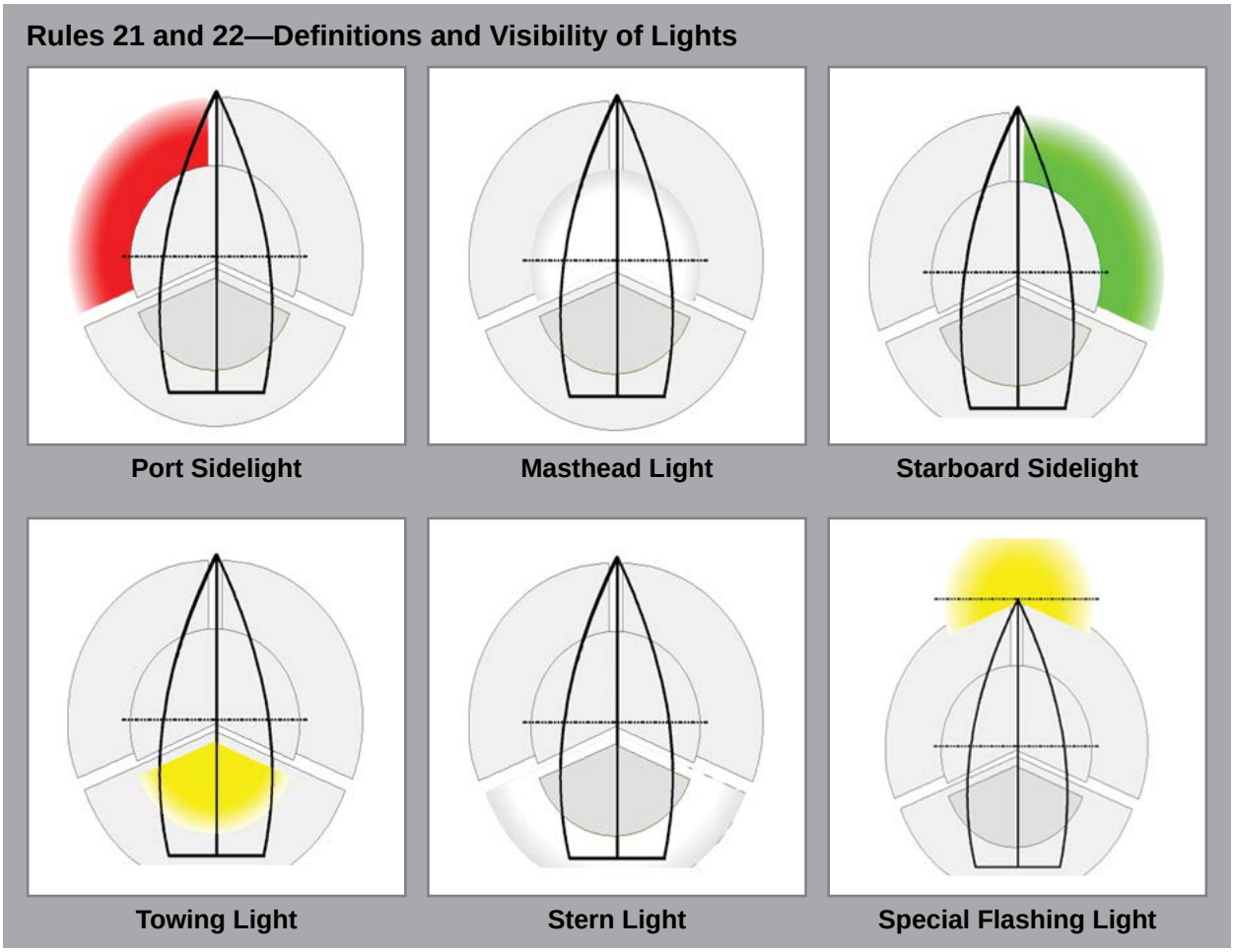
(132) (d) A vessel which detects by radar alone the presence of another vessel shall determine if a closequarters situation is developing and/or risk of collision exists. If so, she shall take avoiding action in ample time, provided that when such action consists of an alteration in course, so far as possible the following shall be avoided:

(133) (i) An alteration of course to port for a vessel forward of the beam, other than for a vessel being overtaken;

(134) (ii) An alteration of course toward a vessel abeam or abaft the beam.

(135) (e) Except where it has been determined that a risk of collision does not exist, every vessel which hears apparently forward of her beam the fog signal of another vessel, or which cannot avoid a close-quarters situation with another vessel forward of her beam, shall reduce her speed to be the minimum at which she can be kept on her course. She shall if necessary take all her way off and in any event navigate with extreme caution until danger of collision is over.

(158)



(136)

Part C—Lights and Shapes

(137)

Rule 20—Application

(138) (a) Rules 20 through 31 shall be complied with in all weathers.

(139) (b) The Rules concerning lights shall be complied with from sunset to sunrise, and during such times no other lights shall be exhibited, except such lights which cannot be mistaken for the lights specified in these Rules or do not impair their visibility or distinctive character, or interfere with the keeping of a proper look-out.

(140) (c) The lights prescribed by these Rules shall, if carried, also be exhibited from sunrise to sunset in restricted visibility and may be exhibited in all other circumstances when it is deemed necessary.

(141) (d) The Rules concerning shapes shall be complied with by day.

(142) (e) The lights and shapes specified in these Rules shall comply with the provisions of Annex I of these Rules.

(143)

Rule 20f (Inland)

(f) A vessel's navigation lights and shapes may be lowered if necessary to pass under a bridge.

(144)

Rule 21—Definitions

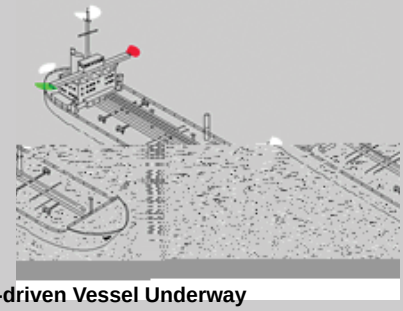
(145) (a) "Masthead light" means a white light placed over the fore and aft centerline of the vessel showing an unbroken light over an arc of the horizon of 225° and so fixed as to show the light from right ahead to 22.5° abaft the beam on either side of the vessel ‹‹ *except that on a vessel of less than 12 meters in length the masthead light shall be placed as nearly as practicable to the fore and aft centerline of the vessel* ››.

(146) (b) "Sidelights" means a green light on the starboard side and a red light on the port side each showing an unbroken light over an arc of the horizon of 112.5° and so fixed as to show the light from right ahead to 22.5° abaft the beam on its respective side. In a vessel of less than 20 meters in length the sidelights may be combined in one lantern carried on the fore and aft centerline of the vessel ‹‹, *except that on a vessel of less than 12 meters in length the sidelights when combined in one lantern*

(166)

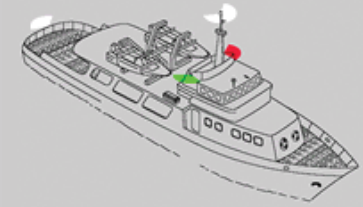
Rule 23—Power-driven Vessels Underway

23(a) **International/Inland**



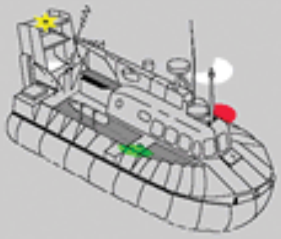
Power-driven Vessel Underway
50 meters or greater in length

23(a) **International/Inland**



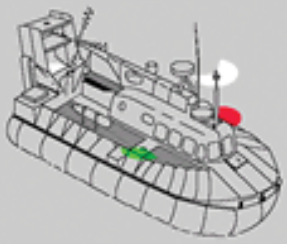
Power-driven Vessel Underway
Less than 50 meters in length

23(b) **International/Inland**



Air-cushion vessel when operating in the *non-displacement* mode—Vessels less than 50 meters in length

23(b) **International/Inland**



Air-cushion vessel when operating in the *displacement* mode—Vessels less than 50 meters in length

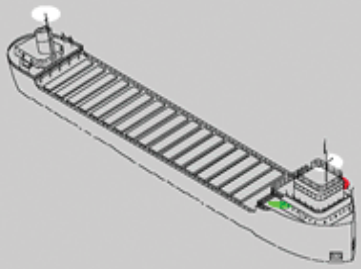
23(d)(i) **International/Inland**

Power-driven Vessel
Less than 12 meters in length

23(d)(ii) **International**

Power-driven Vessel
Less than 7 meters in length

23(e) **Inland**



Power-driven Vessel
on the Great Lakes

shall be placed as nearly as practicable to the fore and aft centerline of the vessel ››.

(147) (c) "Sternlight" means a white light placed as nearly as practicable at the stern showing an unbroken light over an arc of the horizon of 135° and so fixed as to show the light 67.5° from right aft on each side of the vessel.

(148) (d) "Towing light" means a yellow light having the same characteristics as the "sternlight" defined in Rule 21(c).

(149) (e) "All-round light" means a light showing an unbroken light over an arc of the horizon of 360°.

(150) (f) "Flashing light" means a light flashing at regular intervals at a frequency of 120 flashes or more per minute.

(151)

Rule 21g (Inland)

(g) "Special flashing light" means a yellow light flashing at regular intervals at a frequency of 50 to 70 flashes per minute, placed as far forward and as nearly as practicable on the fore and aft centerline of the tow and showing an unbroken light over an arc of the horizon of not less than 180 degrees nor more than 225 degrees and so fixed as to show the light from right ahead to abeam and no more than 22.5 degrees abaft the beam on either side of the vessel.

(152)

Rule 22—Visibility of Lights

(153) The lights prescribed in these Rules (Subpart C) shall have an intensity as specified in Annex I to these

Rules (33 CFR part 84), so as to be visible at the following minimum ranges:

(154) (a) In a vessel of 50 meters or more in length: (i) a masthead light, 6 miles; (ii) a sidelight, 3 miles; (iii) a sternlight, 3 miles; (iv) a towing light, 3 miles; (v) a white, red, green or yellow all-round light, 3 miles; *«and (vi) a special flashing light, 2 miles. »*

(155) (b) In a vessel of 12 meters or more in length but less than 50 meters in length: (i) a masthead light, 5 miles; except that where the length of the vessel is less than 20 meters, 3 miles; (ii) a sidelight, 2 miles; (iii) a sternlight, 2 miles; (iv) a towing light, 2 miles; (v) a white, red, green or yellow all-round light, 2 miles; *«and (vi) a special flashing light, 2 miles. »*

(156) (c) In a vessel of less than 12 meters in length: (i) a masthead light, 2 miles; (ii) a sidelight, 1 mile; (iii) a sternlight, 2 miles; (iv) A towing light, 2 miles; (v) a white, red, green or yellow all-round light, 2 miles; *«and (vi) a special flashing light, 2 miles. »*

(157) (d) In an inconspicuous, partly submerged vessel or objects being towed: (i) A white all-round light, 3 miles. (ii) [Reserved]

(159)

Rule 23—Power-driven Vessels Underway

(160) (a) A power-driven vessel underway shall exhibit: (i) a masthead light forward; (ii) a second masthead light abaft of and higher than the forward one; except that a vessel of less than 50 meters in length shall not be obliged to exhibit such a light but may do so; (iii) sidelights; and (iv) a sternlight.

(161) (b) An air-cushion vessel when operating in nondisplacement mode shall, in addition to the lights prescribed in Rule 23(a) Air Cushion Vessel in Displacement Mode, exhibit an all-round flashing yellow light *‹‹ , where it can best be seen ››*.

(162) (c) A WIG craft only when taking off, landing and in flight near the surface shall, in addition to the lights prescribed in Rule 23(a), exhibit a high intensity allround flashing red light.

(163) (d)(i) A power-driven vessel of less than 12 meters in length may in lieu of the lights prescribed in Rule 23(a) exhibit an all-round white light and sidelights.

(164)

Rule 23d (International)

(ii) a power-driven vessel of less than 7 meters in length whose maximum speed does not exceed 7 knots may in lieu of the lights prescribed in Rule 23(a) exhibit an all-round white light and shall, if practicable, also exhibit sidelights.

(iii) the masthead light or all-round white light on a power-driven vessel of less than 12 metres in length may be displaced from the fore and aft centre line of the vessel if centreline fitting is not practicable, provided that the sidelights are combined in one lantern which shall be carried on the fore and aft centre line of the vessel or located as nearly as practicable in the same fore and aft line as the masthead light or the all-round white light.

Rule 23e (Inland)

(e) A power-driven vessel when operating on the Great Lakes may carry an all-round white light in lieu of the second masthead light and sternlight prescribed in Rule 23(a). The light shall be carried in the position of the second masthead light and be visible at the same minimum range.

(165) Regulations containing specifics on **Law Enforcement** and **Public Safety Vessel** lighting are in **Annex V–Pilot Rules, 33 CFR 88.05** and **33 CFR 88.07**, chapter 2.

(167)

Rule 24—Towing and Pushing

(168) (a) A power-driven vessel when towing astern shall exhibit: (i) instead of the light prescribed in Rule 23(a)(i) or 23(a)(ii), two masthead lights in a vertical line. When the length of the tow, measuring from the stern of the towing vessel to the after end of the tow, exceeds 200 meters, three such lights in a vertical line; (ii) sidelights; (iii) a sternlight; (iv) a towing light in a vertical line above the sternlight; and (v) when the length of the tow exceeds 200 meters, a diamond shape where it can best be seen.

(169) (b) When a pushing vessel and a vessel being pushed ahead are rigidly connected in a composite unit they shall be regarded as a power-driven vessel and exhibit the lights prescribed in Rule 23.

(170) **Interpretive Rule**—See **33 CFR 90.3** and **33 CFR 82.3**, chapter 2, for regulations.

(171) (c) A power-driven vessel when pushing ahead or towing alongside, except ‹ in the case of a composite unit *›‹‹ as required by Rules 24(b) and (i) ››*, shall exhibit: (i) instead of the light prescribed in Rule 23(a)(i) or 23(a)(ii), two masthead lights in a vertical line; (ii) sidelights; and (iii) ‹ a sternlight *›‹‹ two towing lights in a vertical line ››*.

(172) (d) A power-driven vessel to which paragraphs (a) or (c) of this Rule applies shall also comply with Rule 23 *‹‹ (a)(i) and ››* (a)(ii).

(173) (e) A vessel or object being towed, other than those ‹ mentioned *›‹‹ referred ››* in Rule 24(g), shall exhibit: (i) sidelights; (ii) a sternlight; (iii) when the length of the tow exceeds 200 meters, a diamond shape where it can best be seen.

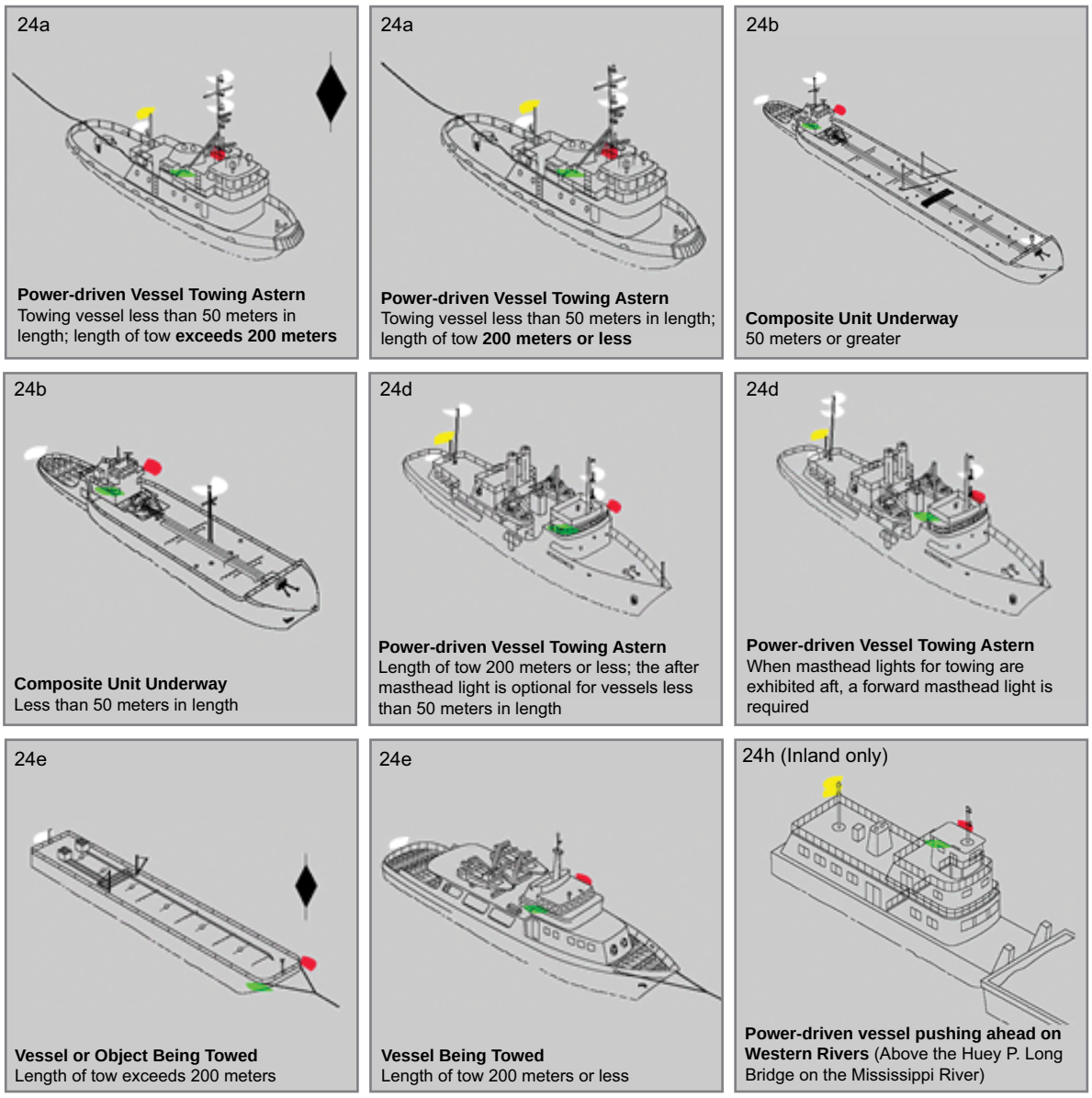
(174) (f) Provided that any number of vessels being towed alongside or pushed in a group shall be lighted as one vessel *‹‹ except as provided in Rule 24(f)(iii) ››*.

(175) (i) a vessel being pushed ahead, not being part of a composite unit, shall exhibit at the forward end, sidelights, and *‹‹ a special flashing light ››*;

(176) (ii) a vessel being towed alongside shall exhibit a sternlight and at the forward end, sidelights, and *‹‹ a special flashing light ››*;

(188)

Rule 24—Towing and Pushing (International/Inland)



(177)

Rule 24f (Inland)

(iii) when vessels are towed alongside on both sides of the towing vessel a sternlight shall be exhibited on the stern of the outboard vessel on each side of the towing vessel, and a single set of sidelights as far forward and as far outboard as is practicable, and a single special flashing light;

(178) (g) An inconspicuous, partly submerged vessel or object, or combination of such vessels or objects being towed, shall exhibit:

(179) (i) if it is less than 25 meters in breadth, one all-round white light at or near the forward end and one at or near the after end except that dracones need not exhibit a light at or near ‹ the forward ›‹‹ *each* ›› end.

(180)

Rule 24g (International)

(ii) if it is 25 meters or more in breadth, two additional all-round white lights at or near the extremities of its breadth;

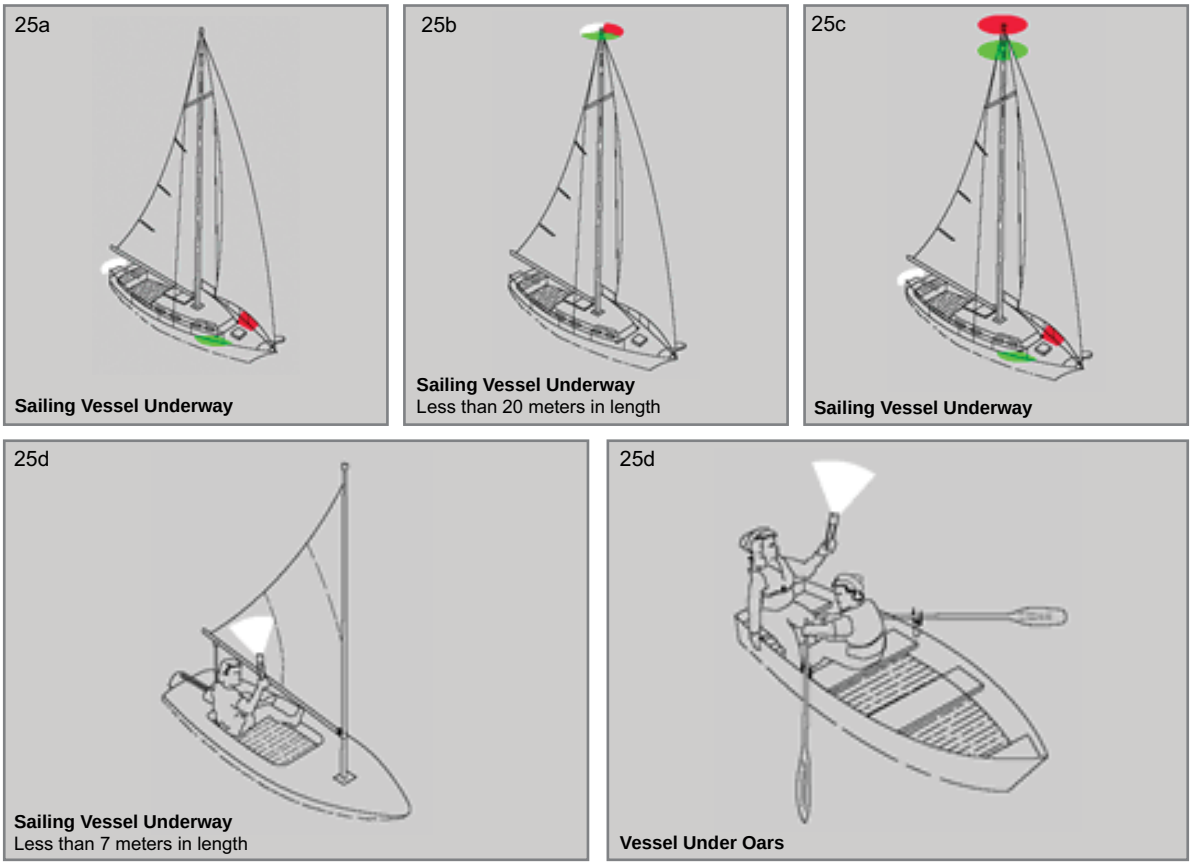
Rule 24g (Inland)

(ii) if it is 25 meters or more in breadth, four all-round white lights to mark its length and breadth;

(181) (iii) if it exceeds 100 meters in length, additional allround white lights between the lights prescribed in Rule 24(g)(i) ‹‹ *and (ii)* ›› and so that the distance between the lights shall not exceed 100 meters. ‹‹ *Provided that*

(196)

Rule 25—Sailing Vessels Underway and Vessels Under Oars (International/Inland)



any vessels or objects being towed alongside each other shall be lighted as one vessel or object ››.

(182) (iv) a diamond shape at or near the aftermost extremity of the last vessel or object being towed; and ‹ if the length of the tow exceeds 200 meters an additional diamond shape where it can best be seen and located as far forward as is practicable. ›

(183)

Rule 24g (Inland)

(v) the towing vessel may direct a searchlight in the direction of the tow to indicate its presence to an approaching vessel.

(184) (h) Where from any sufficient cause it is impracticable for a vessel or object being towed to exhibit the lights or shapes prescribed in Rule 24(e) or (g), all possible measures shall be taken to light the vessel or object towed or at least to indicate the presence of ‹ such *›‹ the unlighted ››* vessel or object.

(185) **Interpretive Rule**—See **33 CFR 90.7** and **33 CFR 82.7**, chapter 2, for regulations.

(186) (i) Where from any sufficient cause it is impracticable for a vessel not normally engaged in towing operations to display the lights prescribed by paragraph (a), (c), *«or (j)»* of this Rule, such vessel shall not be required to exhibit those lights when engaged in towing another vessel in distress or otherwise in need of assistance. All possible measures shall be taken to indicate the nature of the relationship between the towing vessel and the vessel being towed ‹ as authorized by Rule 36, in particular by illuminating the towline *›«and the vessel being assisted. The searchlight authorized by Rule 36 may be used to illuminate the tow».*

(187)

Rule 24j (Inland)

(i) Notwithstanding paragraph (c) of this Rule, on the Western Rivers (except below the Huey P. Long Bridge at mile 106.1 Above Head of Passes on the Mississippi River) and on waters specified by the Secretary, a power-driven vessel when pushing ahead or towing alongside, except as paragraph (b) of this Rule applies, shall exhibit: (i) sidelights; and (ii) two towing lights in a vertical line.

(189)

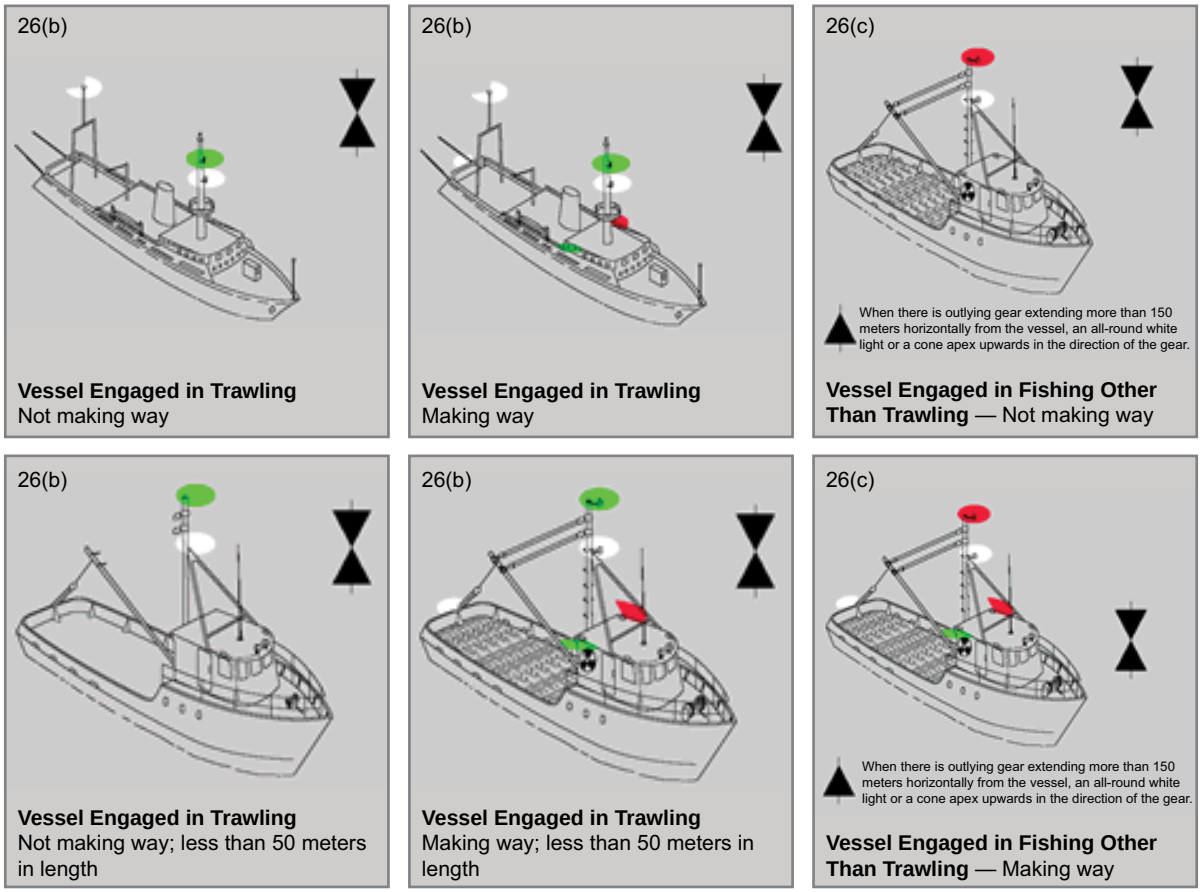
Rule 25—Sailing Vessels Underway and Vessels Under Oars

(190) (a) A sailing vessel underway shall exhibit: (i) sidelights; (ii) a sternlight.

(191) (b) In a sailing vessel of less than 20 meters in length the lights prescribed in Rule 25(a) may be combined in one lantern carried at or near the top of the mast where it can best be seen.

(210)

Rule 26—Fishing Vessels (International/Inland)



(192) (c) A sailing vessel underway may, in addition to the lights prescribed in Rule 25(a), exhibit at or near the top of the mast, where they can best be seen, two allround lights in a vertical line, the upper being red and the lower green, but these lights shall not be exhibited in conjunction with the combined lantern permitted by Rule 25(b).

(193) (d)(i) A sailing vessel of less than 7 meter in length shall, if practicable, exhibit the lights prescribed in Rule 25(a) or (b), but if she does not, she shall *« exhibit an all around white light or »* have ready at hand an electric torch or lighted lantern showing a white light which shall be exhibited in sufficient time to prevent collision.

(194) (ii) A vessel under oars may exhibit the lights prescribed in this rule for sailing vessels, but if she does not, she shall *« exhibit an all around white light or »* have ready at hand an electric torch or lighted lantern showing a white light which shall be exhibited in sufficient time to prevent collision.

(195) (e) A vessel proceeding under sail when also being propelled by machinery shall exhibit forward where it can best be seen a conical shape, apex downwards. *« A vessel of less than 12 meters in length is not required to exhibit this shape, but may do so. »*

(197)

Rule 26—Fishing Vessels

(198) (a) A vessel engaged in fishing, whether underway or at anchor, shall exhibit only the lights and shapes prescribed in this Rule.

(199) (b) A vessel when engaged in trawling, by which is meant the dragging through the water of a dredge net or other apparatus used as a fishing appliance, shall exhibit: (i) two all-round lights in a vertical line, the upper being green and the lower white, or a shape consisting of two cones with their apexes together in a vertical line one above the other; (ii) a masthead light abaft of and higher than the all-round green light; a vessel of less than 50 meters in length shall not be obliged to exhibit such a light but may do so; (iii) when making way through the water, in addition to the lights prescribed in this paragraph, sidelights and a sternlight.

(200) (c) A vessel engaged in fishing, other than trawling, shall exhibit: (i) two all-round lights in a vertical line, the upper being red and the lower white, or a shape consisting of two cones with their apexes together in a vertical line one above the other; (ii) when there is outlying gear extending more than 150 meters horizontally from the vessel, an all-round white light or a cone apex upwards in

the direction of the gear; (iii) when making way through the water, in addition to the lights prescribed in this paragraph, sidelights and a sternlight.

(201)

Rule 26d (International)

(d) The additional signals described in Annex II to these Regulations apply to a vessel engaged in fishing in close proximity to other vessels engaged in fishing.

(202) (e) A vessel ‹ when › not engaged in fishing shall not exhibit the lights or shapes prescribed in this Rule, but only those prescribed for a vessel of her length.

(203) *« (f) Additional signals for fishing vessels in close proximity. »* {Same as International Rules Annex II}

(204) 1ǂ(i) The lights mentioned herein shall ‹,if exhibited in pursuance of Rule 26(d),› be placed where they can best be seen. They shall be at least 0.9 meters apart but at a lower level than lights prescribed in Rule 26. ‹(b)(i) and (c)(i)› The lights shall be visible all round the horizon at a distance of at least 1 mile but at a lesser distance from the lights prescribed by ‹these Rules›*«Rule 26(a)-(c)»* for fishing vessels.

(205) 2ǂ(ii) Signals for trawlers.

(206) (a)ǂ(1) Vessels ‹of 20 meters or more in length› when engaged in trawling, whether using demersal or pelagic gear, ‹shall›*«may»* exhibit: (i)ǂ(A) when shooting their nets—two white lights in a vertical line; (ii)ǂ(B) when hauling their nets—one white light over one red light in a vertical line; (iii)ǂ(C) when the net has come fast upon an obstruction—two red lights in a vertical line.

(207) (b)ǂ(2) ‹A›*«Each»* vessel ‹of 20 meters or more in length› engaged in pair trawling ‹shall›*«may»* exhibit: (i)ǂ(A) by night, a searchlight directed forward and in the direction of the other vessel of the pair; (ii)ǂ(B) when shooting or hauling their nets or when their nets have come fast upon an obstruction, the lights prescribed in Rule 26(f)(2)(a)ǂ(f)(ii)(1).

(208) 3ǂ(iii) Signals for purse seiners.

(209) (a)ǂ(1) Vessels engaged in fishing with purse seine gear may exhibit two yellow lights in a vertical line. These lights shall flash alternately every second and with equal light and occultation duration. These lights may be exhibited only when the vessel is hampered by its fishing gear.

(211)

Rule 27—Vessels Not Under Command or Restricted in Their Ability to Maneuver

(212) (a) A vessel not under command shall exhibit: (i) two all-round red lights in a vertical line where they can best be seen; (ii) two balls or similar shapes in a vertical line where they can best be seen; (iii) when making way through the water, in addition to the lights prescribed in this paragraph, sidelights and a sternlight.

(213) (b) A vessel restricted in her ability to maneuver, except a vessel engaged in mineclearance operations, shall exhibit: (i) three all-round lights in a vertical line where they can best be seen. The highest and lowest of these lights shall be red and the middle light shall be white; (ii) three shapes in a vertical line where they can best be seen. The highest and lowest of these shapes shall be balls and the middle one a diamond; (iii) when makingway through the water, a masthead light(s), sidelights and a sternlight in addition to the lights prescribed in Rule 27(b)(i); (iv) when at anchor, in addition to the lights or shapes prescribed in Rule 27(b)(i) and (ii), the light, lights, or shapes prescribed in Rule 30.

(214) (c) A power-driven vessel engaged in a towing operation such as severely restricts the towing vessel and her tow in their ability to deviate from their course shall, in addition to the lights or shape prescribed in Rule 27(b)(i) and (ii), exhibit the lights or shape prescribed in Rule 24.

(215) (d) A vessel engaged in dredging or underwater operations, when restricted in her ability to maneuver, shall exhibit the lights and shapes prescribed in Rules 27(b)(i), (ii) and (iii) and shall in addition when an obstruction exists, exhibit: (i) two all-round red lights or two balls in a vertical line to indicate the side on which the obstruction exists; (ii) two all-round green lights or two diamonds in a vertical line to indicate the side on which another vessel may pass; and (iii) when at anchor, the lights or shapes prescribed in this paragraph instead of the lights or shapes prescribed in Rule 30.

(216)

Rule 27d (Inland)

(iv) Dredge pipelines that are floating or supported on trestles shall display the following lights at night and in periods of restricted visibility.

(1) One row of yellow lights. The lights must be: (A) flashing 50 to 70 times per minute, (B) visible all round the horizon, (C) visible for at least 2 miles, (D) not less than 1 and not more than 3.5 meters above the water, (E) approximately equally spaced, and (F) not more than 10 meters apart where the pipeline crosses a navigable channel. Where the pipeline does not cross a navigable channel the lights must be sufficient in number to clearly show the pipeline's length and course.

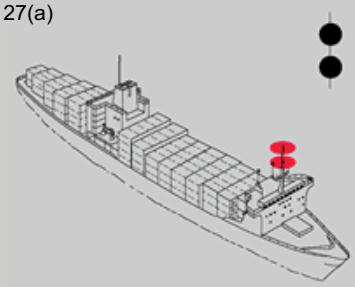
(2) Two red lights at each end of the pipeline, including the ends in a channel where the pipeline is separated to allow vessels to pass (whether open or closed). The lights must be: (A) visible all round the horizon, and (B) visible for at least 2 miles, and (C) one meter apart in a vertical line with the lower light at the same height above the water as the flashing yellow light.

(217) (e) Whenever the size of a vessel engaged in diving operations makes it impracticable to exhibit all lights and shapes prescribed in Rule 27(d), the following shall be exhibited: (i) Three all-round lights in a vertical line where they can best be seen. The highest and lowest of these lights shall be red and the middle light shall be white; (ii) a rigid replica of the International Code flag "A" not less than 1 meter in height. Measures shall be taken to ensure its all-round visibility.

(221)

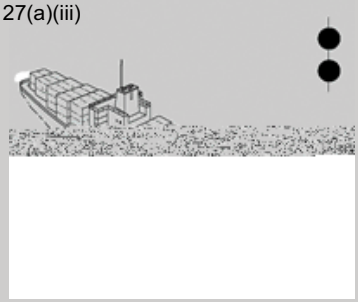
Rule 27—Vessels Not Under Command or Restricted in Their Ability to Maneuver (International/Inland)

27(a)



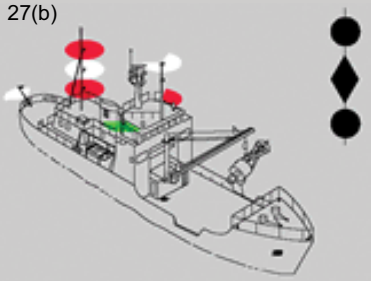
Vessel Not Under Command
Not making way

27(a)(iii)



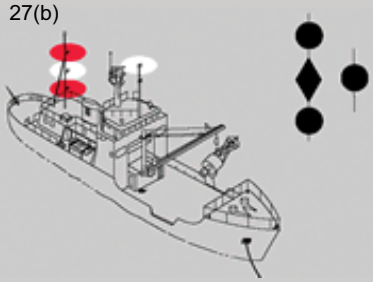
Vessel Not Under Command
Making way

27(b)



Vessel Restricted in Her Ability to Maneuver—Making way; less than 50 meters in length

27(b)

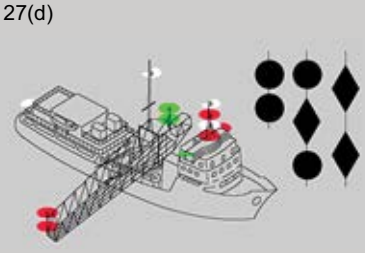


Vessel Restricted in Her Ability to Maneuver—At anchor; less than 50 meters in length

27(c)

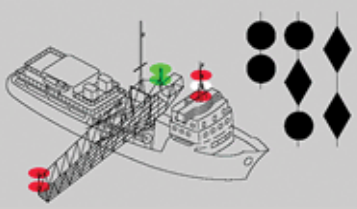
Vessel engaged in towing operation which severely restricts towing vessel and her tow in their ability to deviate from their course—Length of tow does not exceed 200 meters; towing vessel less than 50 meters in length.

27(d)



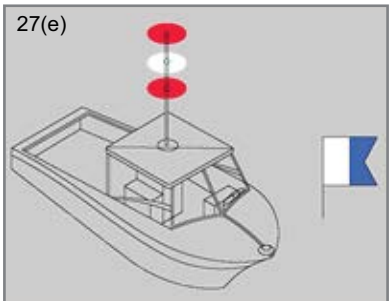
Vessel engaged in dredging or underwater operations when restricted in ability to maneuver—Making way with an obstruction on the starboard side.

27(d)



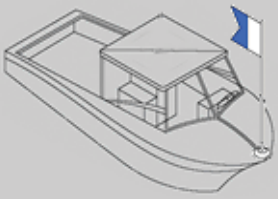
Vessel engaged in dredging or underwater operations when restricted in ability to maneuver—Not making way with an obstruction on the starboard side.

27(e)



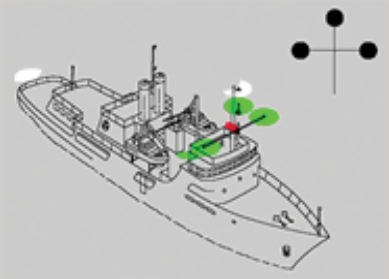
Small vessel engaged in diving operations

27(e)



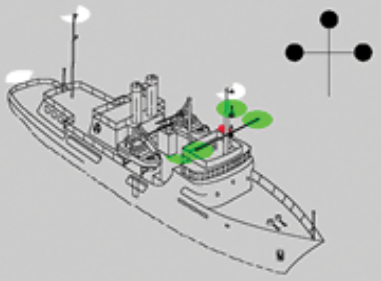
Small vessel engaged in diving operations

27(f)



Vessel engaged in mineclearance operations
Vessel less than 50 meters in length.

27(f)

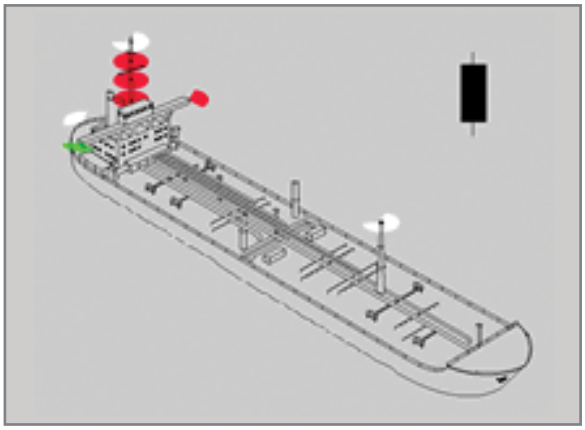


Vessel engaged in mineclearance operations
Vessel 50 meters or greater in length.

(224)

Rule 28—Vessel Constrained by Their Draft (International)

A vessel constrained by her draft may, in addition to the lights prescribed for power-driven vessels in Rule 23, exhibit where they can best be seen three all-round red lights in a vertical line, or a cylinder.



(218) (f) A vessel engaged in mine clearance operations shall, in addition to the lights prescribed for a power-driven vessel in Rule 23 or to the lights or shape prescribed for a vessel at anchor in Rule 30 as appropriate, exhibit three all-round green lights or three balls. One of these lights or shapes shall be exhibited near the foremast head and one at each end of the fore yard. These lights or shapes indicate that it is dangerous for another vessel to approach within 1000 meters of the mineclearance vessel.

(219) (g) Vessels of less than 12 meters in length, except ‹ those ›«*when*» engaged in diving operations, ‹ shall not be ›«*is not*» required to exhibit the lights ‹ and ›«*or*» shapes prescribed in this Rule.

(220) (h) The signals prescribed in this Rule are not signals of vessels in distress and requiring assistance. Such signals are contained in Annex IV to these Rules.

(222)

Rule 28—Vessels Constrained by Their Draft

(223) See graphic, **Rule 28—Vessels Constrained by Their Draft**.

(225)

Rule 29—Pilot Vessels

(226) (a) A vessel engaged on pilotage duty shall exhibit: (i) at or near the masthead, two all-round lights in a vertical line, the upper being white and the lower red; (ii) when underway, in addition, sidelights and a sternlight; (iii) when at anchor, in addition to the lights prescribed in Rule 29(a)(i), the light, lights, or shape prescribed in Rule 30 for vessels at anchor.

(227) (b) A pilot vessel when not engaged on pilotage duty shall exhibit the lights or shapes prescribed for a similar vessel of her length.

(228)

Rule 30—Anchored Vessels and Vessels Aground

(229) (a) A vessel at anchor shall exhibit where it can best be seen: (i) in the fore part, an all-round white light or one ball; (ii) at or near the stern and at a lower level than the light prescribed in Rule 30(a)(i), an all-round white light.

(230) **Interpretive Rule**—See **33 CFR 90.5** and **33 CFR 82.5**, chapter 2, for regulations on vessels at anchor.

(231) (b) A vessel of less than 50 meters in length may exhibit an all-round white light where it can best be seen instead of the lights prescribed in Rule 30(a).

(232) (c) A vessel at anchor may, and a vessel of 100 meters and more in length shall, also use the available working or equivalent lights to illuminate her decks.

(233) (d) A vessel aground shall exhibit the lights prescribed in Rule 30(a) or (b) and in addition, if practicable, where they can best be seen: (i) two all-round red lights in a vertical line; (ii) three balls in a vertical line.

(234) (e) A vessel of less than 7 meters in length, when at anchor not in or near a narrow channel, fairway or where other vessels normally navigate, shall not be required to exhibit the lights or shape prescribed in Rule 30(a) and (b).

(235) (f) A vessel of less than 12 meters in length, when aground, shall not be required to exhibit the lights or shapes prescribed in Rule 30(d)(i) and (ii).

(236)

Rule 30 (Inland)

(g) A vessel of less than 20 meters in length, when at anchor in a special anchorage area designated by the Coast Guard, shall not be required to exhibit the anchor lights and shapes required by this Rule.

(h) The following barges shall display at night and if practicable in periods of restricted visibility the lights described in Rule 30(i):
(i) Every barge projecting into a buoyed or restricted channel.
(ii) Every barge so moored that it reduces the available navigable width of any channel to less than 80 meters.
(iii) Barges moored in groups more than two barges wide or to a maximum width of over 25 meters.
(iv) Every barge not moored parallel to the bank or dock.

(i) Barges described in Rule 30(h) shall carry two unobstructed all-round white lights of an intensity to be visible for at least 1 nautical mile and meeting the technical requirements as prescribed in Annex I.

Rule 30 (Inland)

(j) A barge or a group of barges at anchor or made fast to one or more mooring buoys or other similar device, in lieu of the provisions of Rule 30, may carry unobstructed all-round white lights of an intensity to be visible for at least 1 nautical mile that meet the requirements of Annex I and shall be arranged as follows:
(i) Any barge that projects from a group formation, shall be lighted on its outboard corners.
(ii) On a single barge moored in water where other vessels normally navigate on both sides of the barge, lights shall be placed to mark the corner extremities of the barge.
(iii) On barges moored in group formation, moored in water where other vessels normally navigate on both sides of the group, lights shall be placed to mark the corner extremities of the group.

(k) The following are exempt from the requirements of Rule 30:
(i) A barge or group of barges moored in a slip or slough used primarily for mooring purposes.
(ii) A barge or group of barges moored behind a pierhead.
(iii) A barge less than 20 meters in length when moored in a special anchorage area designated in accordance with 33 CFR 109.10.

(l) Barges moored in well-illuminated areas are exempt from the lighting requirements of Rule 30. These areas are as follows:

CHICAGO SANITARY SHIP CANAL

(1) Mile 293.2 to 293.9	*(15) Mile 314.6*
(2) Mile 295.2 to 296.1	*(16) Mile 314.8 to 315.3*
(3) Mile 297.5 to 297.8	*(17) Mile 315.7 to 316*
(4) Mile 298 to 298.2	*(18) Mile 316.8*
(5) Mile 298.6 to 298.8	*(19) Mile 316.85 to 317.05*
(6) Mile 299.3 to 299.4	*(20) Mile 317.5*
(7) Mile 299.8 to 300.5	*(21) Mile 318.4 to 318.9*
(8) Mile 303 to 303.2	*(22) Mile 318.7 to 318.8*
(9) Mile 303.7 to 303.9	*(23) Mile 320 to 320.3*
(10) Mile 305.7 to 305.8	*(24) Mile 320.6*
(11) Mile 310.7 to 310.9	*(25) Mile 322.3 to 322.4*
(12) Mile 311 to 311.2	*(26) Mile 322.8*
(13) Mile 312.5 to 312.6	*(27) Mile 322.9 to 327.2*
(14) Mile 313.8 to 314.2	

CALUMET SAG CHANNEL

(28) Mile 316.5

LITTLE CALUMET RIVER

(29) Mile 321.2	*(30) Mile 322.3*

CALUMET RIVER

(31) Mile 328.5 to 328.7	*(34) Mile 331.4 to 331.6*
(32) Mile 329.2 to 329.4	*(35) Mile 332.2 to 332.4*
(33) Mile 330 west bank to 330.2	*(36) Mile 332.6 to 332.8*

CUMBERLAND RIVER

(37) Mile 126.8	*(38) Mile 191*

(237)

Rule 31—Seaplanes

(238) Where it is impracticable for a seaplane or a WIG craft to exhibit lights or shapes of the characteristics or in the positions prescribed in Rules 20 through 31 she shall exhibit lights and shapes as closely similar in characteristics and position as is possible.

(239)

Part D—Sound and Light Signals

(240)

Rule 32—Definitions

(241) (a) The word "whistle" means any sound signaling appliance capable of producing the prescribed blasts and which complies with the specifications in Annex III to these Rules.

(242) (b) The term "short blast" means a blast of about one seconds duration.

(243) (c) The term "prolonged blast" means a blast of from four to six seconds duration.

(244)

Rule 33—Equipment for Sound Signals

(245) (a) A vessel of 12 meters or more in length shall be provided with a whistle, a vessel of 20 meters or more in length shall be provided with a bell in addition to a whistle, and a vessel of 100 meters or more in length shall, in addition be provided with a gong, the tone and sound of which cannot be confused with that of the bell. The whistle, bell and gong shall comply with the specifications in Annex III to these Regulations. The bell or gong or both may be replaced by other equipment having the same respective sound characteristics, provided that manual sounding of the prescribed signals shall always be possible.

(246) (b) A vessel of less than 12 meters in length shall not be obliged to carry the sound signaling appliances prescribed in Rule 33(a) but if she does not, she shall be provided with some other means of making an efficient signal.

(247)

Rule 34—Maneuvering and Warning Signs (International)

(a) When vessels are in sight of one, a power-driven vessel underway, when maneuvering as authorized or required by these Rules, shall indicate that manoeuvre by the following signals on her whistle:
–One short blasts to mean "I am altering my course to starboard"
–Two short blasts to mean "I am altering my course to port"
–Three short blasts to mean "I am operating astern propulsion"

(b) Any vessel may supplement the whistle signals prescribed in Rule 34(a) by light signals, repeated as appropriate, while the maneuver is being carried out:
(i) these signals shall have the following significance:
(ii) the duration of each flash shall be about one second, the interval between flashes shall be about one second, and the interval between successive signals shall not be less than ten seconds.
(iii) the light used for this signal shall, if fitted, be an all-round white, visible at a minimum range of 5 miles, and shall comply with the provisions of Annex I to these Regulations.
–One flash to mean "I am altering my course to starboard"
–Two flashes to mean I am altering my course to port"
–Three flashes to mean "I am operating astern propulsion".

Rule 34—Maneuvering and Warning Signs (International)

(c) When in sight of one another in a narrow channel or fairway:
(i) a vessel intending to overtake another shall in compliance with Rule 9(e)(i) indicate her intention by the following signals on her whistle:
–Two prolonged blasts followed by one short blast to mean "I intend to overtake you on your starboard side"
–Two prolonged blasts followed by two short blasts to mean "I intend to overtake you on your port side".
(ii) the vessel about to be overtaken when acting in accordance with Rule 9(e)(i) shall indicate her agreement by the following signal on her whistle:
–one prolonged, one short, one prolonged and one short blast, in that order.

(d) When vessels in sight of one another are approaching each other and from any cause either vessel fails to understand the intentions or actions of the other, or is in doubt whether sufficient action is being taken by the other to avoid collision, the vessel in doubt shall immediately indicate such doubt by giving at least five short and rapid blasts on the whistle. Such signal may be supplemented by at least five short and rapid flashes.

(e) A vessel nearing a bend or an area of a channel or fairway where other vessels may be obscured by an intervening obstruction shall sound one prolonged blast. This signal shall be answered with a prolonged blast by any approaching vessel that may be within hearing around the bend or behind the intervening obstruction.

(f) If whistles are fitted on a vessel at a distance apart of more than 100 meters, one whistle only shall be used for giving maneuvering and warning signals.

(248)

Rule 34—Maneuvering and Warning Signs (Inland)

(a) When power-driven vessels are in sight of one another and meeting or crossing at a distance within half a mile of each other, each vessel underway, when maneuvering as authorized or required by these Rules,
(i) shall indicate that maneuver by the following signals on her whistle:
–One short blasts to mean "I intend to leave you on my port side"
–Two short blasts to mean "I intend to leave you on my starboard side"
–Three short blasts to mean "I am operating astern propulsion"
(ii) upon hearing the one or two blast signal of the other shall, if in agreement, sound the same whistle signal and take the steps necessary to effect a safe passing. If, however, from any cause, the vessel doubts the safety of the proposed maneuver, she shall sound the signal specified in Rule 34(d) and each vessel shall take appropriate precautionary action until a safe passing agreement is made

(b) Any vessel may supplement the whistle signals prescribed in Rule 34(a) by light signals:
(i) these signals shall have the following significance:
(ii) the duration of each flash shall be about one second.
(iii) the light used for this signal shall, if fitted, be an all-round white or yellow, visible at a minimum range of 2 miles, synchronized with the whistle and shall comply with the provisions of Annex I to these Regulations.
–One flash to mean "I intend to leave you on my port side"
–Two flashes to mean "I intend to leave you on my starboard side"
–Three flashes to mean "I am operating astern propulsion"

Rule 34—Maneuvering and Warning Signs (Inland)

(c) When in sight of one another:
(i) a power-driven vessel intending to overtake another power-driven vessel shall indicate her intention by the following signals on her whistle:
–One short blast to mean "I intend to overtake you on your starboard side"
–Two short blasts to mean "I intend to overtake you on your port side"
(ii) the power-driven vessel about to be overtaken shall, if in agreement, sound a similar sound signal. If in doubt she shall sound the signal prescribed in Rule 34(d).

(d) When vessels in sight of one another are approaching each other and from any cause either vessel fails to understand the intentions or actions of the other, or is in doubt whether sufficient action is being taken by the other to avoid collision, the vessel in doubt shall immediately indicate such doubt by giving at least five short and rapid blasts on the whistle. Such signal may be supplemented by at least five short and rapid flashes.

(e) A vessel nearing a bend or an area of a channel or fairway where other vessels may be obscured by an intervening obstruction shall sound one prolonged blast. This signal shall be answered with a prolonged blast by any approaching vessel that may be within hearing around the bend or behind the intervening obstruction.

(f) If whistles are fitted on a vessel at a distance apart of more than 100 meters, one whistle only shall be used for giving maneuvering and warning signals.

(g) When a power-driven vessel is leaving a dock or berth, she shall sound one prolonged blast.

(h) A vessel that reaches agreement with another vessel in a head-on, crossing, or overtaking situation, as for example, by using the radiotelephone as prescribed by the Vessel Bridge-to-Bridge Radiotelephone Act (85 Stat. 164; 33 U.S.C. 1201 et seq.), is not obliged to sound the whistle signals prescribed by this Rule, but may do so. If agreement is not reached, then whistle signals shall be exchanged in a timely manner and shall prevail.

(249)

Rule 35—Sound Signals in Restricted Visibility

(250) In or near an area of restricted visibility, whether by day or night the signals prescribed in this Rule shall be used as follows:

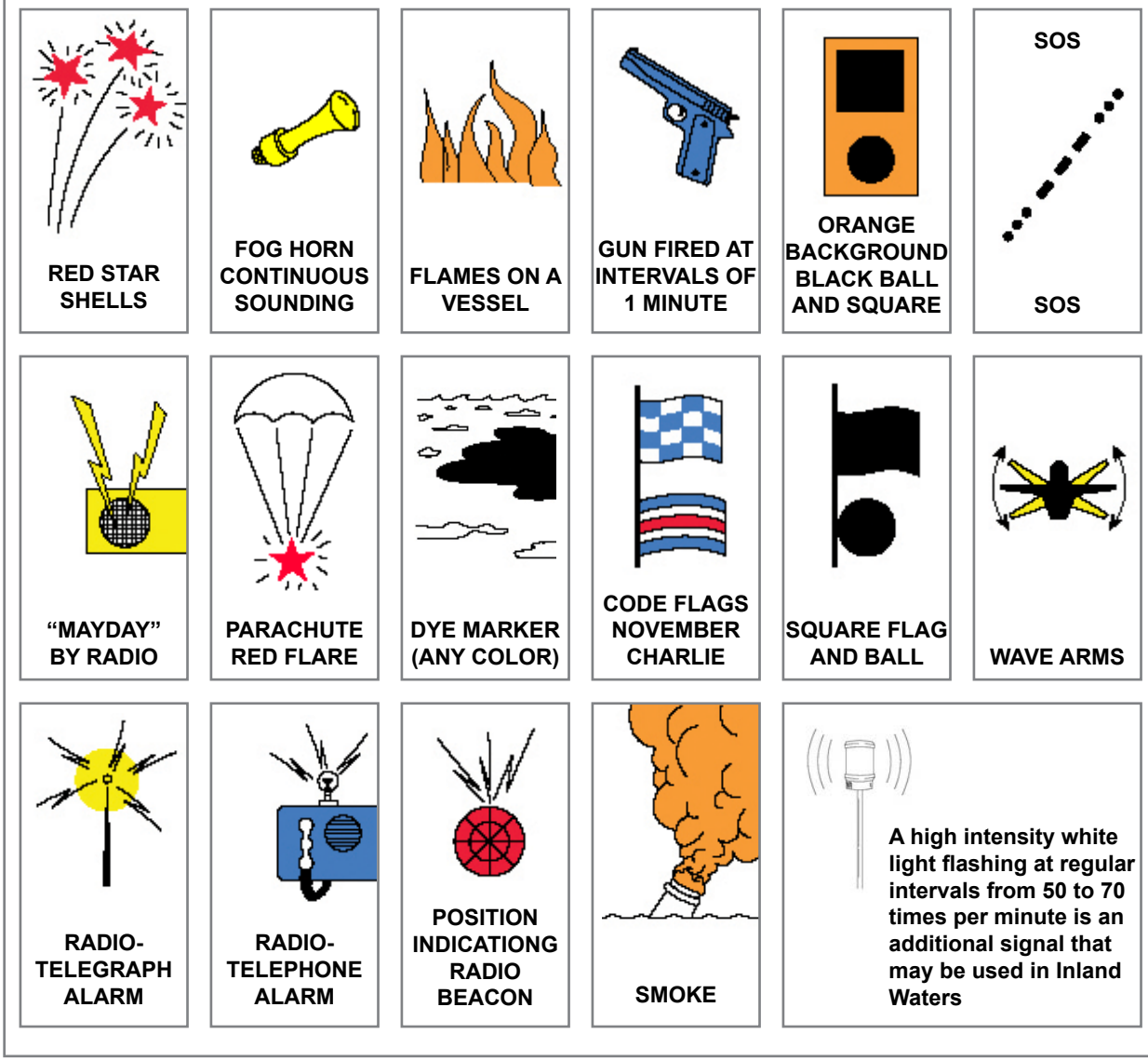
(251) (a) A power-driven vessel making way through the water shall sound at intervals of not more than 2 minutes one prolonged blast.

(252) (b) A power-driven vessel underway but stopped and making no way through the water shall sound at intervals of no more than 2 minutes two prolonged blasts in succession with an interval of about 2 seconds between them.

(253) (c) A vessel not under command, a vessel restricted in her ability to maneuver «*whether underway or at anchor*», ‹ a vessel constrained by her draft › , a sailing vessel, a vessel engaged in fishing and a vessel engaged in towing or pushing another vessel shall, instead of the signals prescribed in Rule 35(a) or (b), sound at intervals of not more than 2 minutes three blasts in succession, namely one prolonged followed by two short blasts.

(268)

Rule 37—Distress Signals (International/Inland)



(254)

Rule 35d (International)

(d) A vessel engaged in fishing, when at anchor, and a vessel restricted in her ability to maneuver when carrying out her work at anchor, shall instead of the signals prescribed in Rule 35(g) sound the signal prescribed in Rule 35(c).

(255) (e) A vessel towed or if more than one vessel is towed the last vessel of the tow, if manned, shall at intervals of not more than 2 minutes sound four blasts in succession, namely one prolonged followed by three short blasts. When practicable, this signal shall be made immediately after the signal made by the towing vessel.

(256) (f) When a pushing vessel and a vessel being pushed ahead are rigidly connected in a composite unit they shall be regarded as a power-driven vessel and shall give the signals prescribed in Rule 35(a) or (b).

(257) (g) A vessel at anchor shall at intervals of not more than 1 minute ring the bell rapidly for about 5 seconds. In a vessel 100 meters or more in length the bell shall be sounded in the forepart of the vessel and immediately after the ringing of the bell the gong shall be sounded rapidly for about 5 seconds in the after part of the vessel. A vessel at anchor may in addition sound three blasts in succession, namely one short, one long and one short blast, to give warning of her position and of the possibility of collision to an approaching vessel.

(258) (h) A vessel aground shall give the bell signal and if required the gong signal prescribed in Rule 35(g) and shall, in addition, give three separate and distinct strokes on the bell immediately before and after the rapid ringing of the bell. A vessel aground may in addition sound an appropriate whistle signal.

(259) (i) A vessel of 12 meters or more but less than 20 meters in length shall not be obliged to give the bell

signals prescribed in Rule 35(g) and (h). However, if she does not, she shall make some other efficient sound signal at intervals of not more than 2 minutes.

(260) (j) A vessel of less than 12 meters in length shall not be obliged to give the above mentioned signals but, if she does not, shall make some other efficient sound signal at intervals of not more than 2 minutes.

(261) (k) A pilot vessel when engaged on pilotage duty may, in addition to the signals prescribed in Rule 35(a), (b) or (g), sound an identity signal consisting of four short blasts.

(262)

Rule 35 (Inland)
(l) The following vessels shall not be required to sound signals as prescribed in Rule 35(g) when anchored in a special anchorage area designated by the Coast Guard:
(i) a vessel of less than 20 meters in length; and
(ii) a barge, canal boat, scow, or other nondescript craft.

(263)

Rule 36—Signals to Attract Attention

(264) If necessary to attract the attention of another vessel, any vessel may make light or sound signals that cannot be mistaken for any signal authorized elsewhere in these Rules, or may direct the beam of her searchlight in the direction of the danger, in such a way as not to embarrass any vessel.

(265)

Rule 36 (International)
Any light to attract the attention of another vessel shall be such that it cannot be mistaken for any aid to navigation. For the purpose of this Rule the use of high intensity intermittent or revolving lights, such as strobe lights, shall be avoided.

(266)

Rule 37—Distress Signals

(267) When a vessel is in distress and requires assistance she shall use or exhibit the signals described in Annex IV to these Rules. (See graphic, **Rule 37—Distress Signals**).

(269)

Part E—Exemptions

(270)

Rule 38—Exemptions (International)
Any vessel (or class of vessel) provided that she complies with the requirements of — the International Regulations for the Preventing of Collisions at Sea, 1960, the keel of which is laid or is at a corresponding stage of construction before the entry into force of these Regulations may be exempted from compliance therewith as follows:
(a) The installation of lights with ranges prescribed in Rule 22, until 4 years after the date of entry into force of these Regulations.
(b) The installation of lights with color specifications as prescribed in §7 of Annex I to these Regulations, until 4 years after the entry into force of these Regulations.
(c) The repositioning of lights as a result of conversion from Imperial to metric units and rounding off measurement figures, permanent exemption.
(d)(i) The repositioning of masthead lights on vessels of less than 150 meters in length, resulting from the prescriptions of §3 (a) of Annex I to these Regulations, permanent exemption. (ii) The repositioning of masthead lights on vessels of 150 meters or more in length, resulting from the prescriptions of §3 (a) of Annex I to these Regulations, until 9 years after the date of entry into force of these Regulations.
(e) The repositioning of masthead lights resulting from the prescriptions of §2(b) of Annex I to these Regulations, until 9 years after the date of entry into force of these Regulations.
(f) The repositioning of sidelights resulting from the prescriptions of §2(g) and 3(b) of Annex I to these Regulations, until 9 years after the date of entry into force of these Regulations.
(g) The requirements for sound signal appliances prescribed in Annex II to these Regulations, until 9 years after the date of entry into force of these Regulations.
(h) The repositioning of all-round lights resulting from the prescription of §9(b) of Annex I to these Regulations, permanent exemption.

(271)

Rule 38—Exemptions (Inland)
Any vessel or class of vessels, the keel of which was laid or which is at a corresponding stage of construction before December 24, 1980, provided that she complies with the requirements of —
(a) The Act of June 7, 1897 (30 Stat. 96), as amended (33 U.S.C. 154-232) for vessels navigating the waters subject to that statute;
(b) §4233 of the Revised Statutes (33 U.S.C. 301-356) for vessels navigating the waters subject to that statute;
(c) The Act of February 8, 1895 (28 Stat. 645), as amended (33 U.S.C. 241-295) for vessels navigating the waters subject to that statute; or

Rule 38—Exemptions (Inland)

(d) §§3, 4, and 5 of the Act of April 25, 1940 (54 Stat. 163), as amended (46 U.S.C. 526 b, c, and d) for motorboats navigating the waters subject to that statute; shall be exempted from compliance with the technical Annexes to these Rules as follows:
(i) The installation of lights with ranges prescribed in Rule 22, vessels of less than 20 meters in length are permanently exempt;
(ii) The installation of lights with color specifications as prescribed in §7 of Annex I to these Rules, until 4 years after the effective date of the Inland Navigational Rules Act of 1980 (Pub. L. 96-591), except that vessels of less than 20 meters in length are permanently exempt;
(iii) The repositioning of lights as a result of a conversion to metric units and rounding off of measurement figures, are permanently exempt.
(iv) The horizontal repositioning of masthead lights prescribed by Annex I to these Rules, vessels of less than 150 meters in length are permanently exempted.
(v) Power-driven vessels of 12 meters or more but less than 20 meters in length are permanently exempt from the provisions of Rule 23(a)(i) and 23(a)(iv) provided that, in place of these lights, the vessel exhibits a white light aft visible all-around the horizon.

(272) **Implementing Rule**—See **33 CFR 81.20**, chapter 2, for regulations.

(273)

Part F—Verification of Compliance with the Provisions of the Convention

(274)

Rule 39—Definitions

(275)

Rule 39 (International)

(a) "Audit" means a systematic, independent and documented process for obtaining audit evidence and evaluating it objectively to determine the extent to which audit criteria are fulfilled.

(b) "Audit Scheme" means the IMO Member State Audit Scheme established by the Organization and taking into account the guidelines developed by the Organization*.

(c) "Code for Implementation" means the IMO Instruments Implementation Code (III Code) adopted by the Organization by resolution A.1070(28).

(d) "Audit Standard" means the Code for Implementation.

(276)

Rule 40—Application

(277)

Rule 40 (International)

Contracting Parties shall use the provisions of the Code for Implementation in the execution of their obligations and responsibilities contained in the present Convention.

(278)

Rule 41—Verification of Compliance

(279)

Rule 41 (International)

(a) Every Contracting Party shall be subject to periodic audits by the Organization in accordance with the audit standard to verify compliance with and implementation of the present Convention.

(b) The Secretary-General of the Organization shall have responsibility for administering the Audit Scheme, based on the guidelines developed by the Organization*.

(c) Every Contracting Party shall have responsibility for facilitating the conduct of the audit and implementation of a programme of actions to address the findings, based on the guidelines developed by the Organization*.

(d) Audit of all Contracting Parties shall be:

(i) based on an overall schedule developed by the Secretary-General of the Organization, taking into account the guidelines developed by the Organization*; and

(ii) conducted at periodic intervals, taking into account the guidelines developed by the Organization*.

* Refer to the Framework and Procedures for the IMO Member State Audit Scheme, adopted by the Organization by resolution A.1067(28).

(280)

Annex I—Positioning and Technical Details of Lights and Shapes

(281)

Definitions

(282) (a) The term "height above the hull" means height above the uppermost continuous deck. This height shall be measured from the position vertically beneath the location of the light.

(283)

Annex I (Inland)

(b) High-speed craft means a craft capable of maximum speed in meters per second (m/s) equal to or exceeding: $3.7\nabla^{0.1667}$; where ∇ = displacement corresponding to the design waterline (cubic meters).

Note: The same formula expressed in pounds and knots is maximum speed in knots (kts) equal to exceeding 1.98(lbs) $3.7\nabla^{0.1667}$; where ∇=displacement corresponding to design waterline in pounds.

(c) The term "practical cut-off" means, for vessels 20 meters or more in length, 12.5 percent of the minimum luminous intensity (Table 14(b)) corresponding to the greatest range of visibility for which the requirements of Annex I are met.

(d) The term "Rule" or "Rules" has the same meaning as in Rule 3(r).

(284)

Vertical Positioning and Spacing of Lights

(285) (a) On a power-driven vessel of 20 meters or more in length the masthead light shall be placed as follows: (i) The forward masthead light, or if only one masthead light is carried, then that light, at a height above the hull of not less than ‹ 6 ›*«5»* meters, and, if the breadth of the vessel exceeds ‹ 6 ›*«5»* meters, then at a height above the hull not less than such breadth, so however that the light need not be placed at a greater height above the hull than ‹ 12 ›*«8»* meters; (ii) when two masthead lights are carried the after one shall be at least ‹ 4.5 ›*«2»* meters vertically higher than the forward one.

(286) (b) The vertical separation of the masthead lights of power-driven vessels shall be such that in all normal conditions of trim the after light will be seen over and separate from the forward light at a distance of 1000 meters from the stem when viewed from ‹ sea ›*«water»* level.

(287) (c) The masthead light of a power-driven vessel of 12 meters but less than 20 meters in length shall be placed at a height above the gunwale of not less than 2.5 meters.

(288)

Annex I (International)

(d) A power-driven vessel of less than 12 meters in length may carry the uppermost light at a height of less than 2.5 meters above the gunwale. When, however, a masthead light is carried in addition to sidelights and a sternlight or the all-round light prescribed in Rule 23(d)(i) is carried in addition to sidelights, then such masthead light or all-round light shall be carried at least 1 meter higher than the sidelights.

Annex I (Inland)

(d) The masthead light, or the all-round light described in Rule 23(d), of a power-driven vessel of less than 12 meters in length shall be carried at least 1 meter higher than the sidelights.

(289) (e) One of the two or three masthead lights prescribed for a power-driven vessel when engaged in towing or pushing another vessel shall be placed in the same position as either the forward masthead light or the after masthead light, provided that ‹ , if carried on the after mast, › the lowest after masthead light shall be at least ‹ 4.5 ›*«2»* meters vertically higher than the *«highest»* forward masthead light.

(290) (f)(i) The masthead lights or lights prescribed in Rule 23(a) shall be so placed as to be above and clear of all other lights and obstructions except as described in §(f)(ii).

(291) (ii) When it is impracticable to carry the all-round lights prescribed by Rule 27(b)(i) ‹ or Rule 28 › below the masthead lights, they may be carried above the after masthead light(s) or vertically in between the forward masthead light(s) and after masthead light(s), provided that in the latter case the requirement of §3(c) shall be complied with.

(292) (g) The sidelights of a power-driven vessel shall be placed at ‹ a height above the hull not greater than three quarters of that ›*«least 1 meter lower»* of ‹ the ›*«than»* forward masthead light. They shall not be so low as to be interfered with by deck lights.

(293)

Annex I (International)

(h) The sidelights, if in a combined lantern and carried on a power-driven vessel of less than 20 meters in length, shall be placed not less than 1 meter below the masthead light.

(294) (i) When the Rules prescribe two or three lights to be carried in a vertical line, they shall be spaced as follows: (i) On a vessel of 20 meters in length or more such lights shall be spaced not less than ‹ 2 ›*«1»* meter apart, and the lowest of these lights shall, except where a towing light is required, be placed at a height of not less than 4 meters above the hull. (ii) On a vessel of less than 20 meters in length such lights shall be spaced not less than 1 meter apart and the lowest of these lights shall, except where a towing light is required, be placed at a height of not less than 2 meters above the gunwale. (iii) When three lights are carried they shall be equally spaced.

(295) (j) The lower of the two all-round lights prescribed for a vessel when engaged in fishing shall be at a height above the sidelights not less than twice the distance between the two vertical lights.

(296) (k) The forward anchor light prescribed in Rule 30(a)(i), when two are carried, shall not be less than 4.5 meters above the after one. On a vessel of 50 meters or more in length this forward anchor light shall be placed at a height or not less than 6 meters above the hull.

(297)

Horizontal Positioning and Spacing of Lights

(298) (a)*«Except as specified in §1(e)*, » when two masthead lights are prescribed for a power-driven vessel, the horizontal distance between them must not be less than one- ‹ quarter ›*«half»* of the length of the vessel but need not be more than ‹ 100 ›*«50»* meters. The forward light must be placed not more than one- ‹ quarter ›*«half»* of the length of the vessel from the stem.

(299) (b) On a power-driven vessel of 20 meters or more in length the sidelights shall not be placed in front of the forward masthead lights. They shall be placed at or near the side of the vessel.

(300) (c) When the lights prescribed in Rule 27(b)(i) ‹or Rule 28 › are placed vertically between the forward masthead light(s) and the after masthead light(s), these all-round lights shall be placed at a horizontal distance of not less than 2 meters from the fore and aft centerline of the vessel in the athwartship direction.

(301) (d) When only one masthead light is prescribed for a power-driven vessel, this light must be exhibited forward of amidships. For a vessel of less than 20 meters in length,

the vessel shall exhibit one masthead light as far forward as is practicable.

(302)

Annex I (Inland)

(e) On power-driven vessels 50 meters but less than 60 meters in length operated on the Western Rivers, and those { waters specified by the Secretary }, the horizontal distance between masthead lights shall not be less than 10 meters.

(303)

Details of Location of Direction-Indicating Lights for Fishing Vessels, Dredgers and Vessels Engaged in Underwater Operations

(304) (a) The light indicating the direction of the outlying gear from a vessel engaged in fishing as prescribed in Rule 26(c)(ii) shall be placed at a horizontal distance of not less than 2 meters and not more than 6 meters away from the two all-round red and white lights. This light shall be placed not higher than the all-round white light prescribed in Rule 26(c)(i) and not lower than the sidelights.

(305) (b) The lights and shapes on a vessel engaged in dredging or underwater operations to indicate the obstructed side and/or the side on which it is safe to pass, as prescribed in Rule 27(d)(i) and (ii), shall be placed at the maximum practical horizontal distance, but in no case less than 2 meters, from the lights or shapes prescribed in Rule 27(b)(i) and (ii). In no case shall the upper of theselights or shapes be at a greater height than the lower of the three lights or shapes prescribed in Rule 27(b)(i) and (ii).

(306)

Screens ‹ For Sidelights ›

(307) (a) The sidelights of vessels of 20 meters or more in length shall be fitted with ‹inboard screens painted› matt black, «*inboard screens*» and meet ‹ing› the requirements of §‹9›*«15»*. On vessels of less than 20 meters in length, the sidelights, if necessary to meet the requirements of §‹9›*«15»*, shall be fitted with ‹ inboard › matt black «*inboard*» screens. With a combined lantern, using a single vertical filament and a very narrow division between the green and red sections, external screens need not be fitted.

(308)

Annex I (Inland)

(b) On power-driven vessels less than 12 meters in length constructed after July 31, 1983, the masthead light, or the all-round light described in Rule 23(d) shall be screened to prevent direct illumination of the vessel forward of the operator's position.

(309)

Shapes

(310) (a) Shapes shall be black and of the following sizes: (i) A ball shall have a diameter of not less than 0.6 meter; (ii) a cone shall have a base diameter of not less than 0.6 meter‹s› and a height equal to its diameter; ‹ (iii) a cylinder shall have a diameter of at least 0.6 meter and a height of twice its diameter; › (iv)ǂ(iii) a diamond shape shall consist of two cones as defined in §(a)(ii) having a common base.

(311) (b) The vertical distance between shapes shall be at least 1.5 meter ‹ s ›.

(312) (c) In a vessel of less than 20 meters in length shapes of lesser dimensions but commensurate with the size of the vessel may be used and the distance apart may be correspondingly reduced.

(313)

Color Specification of Lights

(314) (a) The chromaticity of all navigation lights shall conform to the following standards, which lie within the boundaries of the area of the diagram specified for each color by the International Commission on Illumination (CIE). ‹ , in the "Colors of Light Signals", which is incorporated by reference. It is Publication CIE No. 2.2. (TC-1.6), 1975, and is available from the Illumination Engineering Society, 345 East 47th Street, New York, NY 10017 and is available for inspection at the Coast Guard, Shore Infrastructure Logistics Center, Aids to Navigation and Marine Environmental Response Product Line (CGSILC-ATON/MER), 2703 Martin Luther King, Jr. Ave SE, Mailstop 7714, Washington, DC 20593-7714. It is also available for inspection at the National Archives and Records Administration (NARA). For information on the availability of this material at NARA, call 202–741–6030, or go to: *http://www.archives.gov/federal_register/code_of_federal_regulations/ibr_locations.html*. This incorporation by reference was approved by the Director of the Federal Register. ›

(315) (b) The boundaries of the area for each color are given by indicating the corner coordinates, which are as follows:

(316)

(i) White						
x	0.525	0.525	0.452	0.310	0.310	0.443
y	0.382	0.440	0.440	0.348	0.283	0.382
(ii) Green						
x	0.028	0.009	0.300	0.203		
y	0.385	0.723	0.511	0.356		
(iii) Red						
x	0.680	0.660	0.735	0.721		
y	0.320	0.320	0.265	0.259		
(iv) Yellow						
x	0.612	0.618	0.575	0.575		
y	0.382	0.382	0.425	0.406		

(317)

Intensity of Lights

(318) (a) The minimum luminous intensity of lights shall be calculated by using the formula:

(319)

$$I = 3.43 \times 10^6 \times T \times D^2 \times K^{-D}$$

I is luminous intensity in candelas under service conditions.
T is threshold factor 2 x 10^{-7} lux.
D is range of visibility (luminous range) of the light in nautical miles.
K is atmospheric transmissivity. For prescribed lights the value of K shall be 0.8, corresponding to a meteorological visibility of approximately 13 miles.

(320) (b) A selection of figures derived from the formula is given in the following table:

(321)

Range of visibility (luminous range) of light in nautical miles D	Minimum luminous intensity of light in candelas for K = 0.8 I
1	0.9
2	4.3
3	12
4	27
5	52
6	94

(322) ‹ Note: The maximum luminous intensity of navigation lights should be limited to avoid undue glare. This shall not be achieved by a variable control of the luminous intensity. ›

(323)

Horizontal Sectors

(324) (a)(i) In the forward direction, sidelights as fitted on the vessel shall show the minimum required intensities. The intensities shall decrease to reach practical cut-off between 1 and one degrees outside the prescribed sectors.

(325) (ii) For sternlights and masthead lights and at 22.5 degrees abaft the beam for sidelights, the minimum required intensities shall be maintained over the arc of the horizon up to 5 degrees within the limits of the sectors prescribed in Rule 21. From 5 degrees within the prescribed sectors the intensity may decrease by 50 percent up to the prescribed limits; it shall decrease steadily to reach practical cut-off at not more than 5 degrees outside the prescribed sectors.

(326) (b)(i) All-round lights shall be so located as not to be obscured by masts, topmasts or structures within angular sectors of more than 6 degrees, except anchor lights prescribed in Rule 30, which need not be placed at an impracticable height above the hull *« , and the allround white light described in Rule 23(e), which may not be obscured at all».*

(327) (ii) If it is impracticable to comply with §(b)(i) by exhibiting only one all-round light, two all-round lights shall be used suitably positioned or screened so that they*«to»* appear, as far as practicable, as one light at a *«minimum»* distance of 1*«nautical»* mile.

(328) *«Note: Two unscreened all-round lights that are 1.28 meters apart or less will appear as one light to the naked eye at a distance of 1 nautical mile.»*

(329)

Vertical Sectors

(330) (a) The vertical sectors of electric lights as fitted, with the exception of lights on sailing vessels underway *«and on unmanned barges»*, shall ensure that: (i) At least the required minimum intensity is maintained at all angles from 5 degrees above to 5 degrees below the horizontal; (ii) at least 60 percent of the required minimum intensity is maintained from 7.5 degrees above to 7.5 degrees below the horizontal.

(331) (b) In the case of sailing vessels underway the vertical sectors of electric lights as fitted shall ensure that: (i) At least the required minimum intensity is maintained at all angles from 5 degrees above to 5 degrees below the horizontal; (ii) at least 50 percent of the required minimum intensity is maintained from 25 degrees above to 25 degrees below the horizontal.

(332)

Annex I (Inland)
(c) In the case of unmanned barges the minimum required intensity of electric lights as fitted shall be maintained on the horizontal.

(333) (c)╪(d) In the case of lights other than electric lights these specifications shall be met as closely as possible.

(334)

Intensity of Non-electric Lights

(335) Non-electric lights shall so far as practicable comply with the minimum intensities, as specified in the *« Intensity of Lights »* Table.

(336)

Maneuvering Light

(337) *«Notwithstanding the provisions of §2(f)»*, the maneuvering light described in Rule 34(b) shall be placed *«approximately»* in the same fore and aft vertical plane as the masthead light or lights and, where practicable, at a minimum height of ‹ 2 ›*«1.5»* meter vertically above the forward masthead light, provided that it shall be carried not less than ‹ 2 ›*«1.5»* meter vertically above or below the after masthead light. On a vessel where only one masthead light is carried, the maneuvering light, if fitted, shall be carried where it can best be seen, not less than ‹2 ›*«1.5»* meters vertically apart from the masthead light.

(338)

High-speed Craft

(339) (a) The masthead light of high-speed craft may be placed at a height related to the breadth*« of the craft »* lower than that prescribed in §2(a)(i), provided that the base angle of the isosceles triangle formed by the sidelights and masthead light when seen in end elevation is not less than 27 degrees.

(340) (b) On high-speed craft of 50 meters or more in length, the vertical separation between foremast and mainmast light of 4.5 meters required by §‹ 2(a)(ii) ›*«2(k)»* may be modified provided that such distance shall not be less than the value determined by the following formula:

(341)

$$y = \frac{y = (a+17\Psi)\ C}{1000} + 2$$

y the height of the mainmast light above the foremast light in meters.

a is the height of the foremast light above the water surface in service condition in meters

Y is the trim in service condition in degrees.

C is the horizontal separation of masthead lights in meters.

Note: Refer to the International Code of Safety for High-Speed Craft, 1994 and the International Code of Safety for High-Speed Craft, 2000.

(342)

Approval

(343) The construction of lights and shapes and the installation of lights on board the vessel ‹ shall be to the satisfaction of the appropriate authority of the State whose flag the vessel is entitled to fly ›*«must satisfy the Commandant, U. S. Coast Guard».*

(344)

Annex II—Additional Signals for Fishing Vessels Fishing in Close Proximity

(345) See Rule 26(f).

(346)

Annex III—Technical Details of Sound Signal Appliances

(347) (a) Frequencies and range of audibility. The fundamental frequency of the signal shall lie within the range 70-700 Hz. The range of audibility of the signal from a whistle shall be determined by those frequencies, which may include the fundamental and/or one or more higher frequencies, which lie within the range 180-700 Hz (+/- 1 percent) for a vessel of 20 meters or more in length, or 180-2100 Hz (+/- 1 percent) for a vessel of less than 20 meters in length and which provide the sound pressure levels specified in §1(c).

(348) (b) Limits of fundamental frequencies. To ensure a wide variety of whistle characteristics, the fundamental frequency of a whistle shall be between the following limits: (i) 70-200 Hz, for a vessel 200 meters or more in length; (ii) 130-350 Hz, for a vessel 75 meters but less than 200 meters in length; (iii) 250-700 Hz, for a vessel less than 75 meters in length.

(349) (c) Sound signal intensity and range of audibility. A whistle fitted in a vessel shall provide, in the direction of maximum intensity of the whistle and at a distance of 1 meter from it, a sound pressure level in at least one onethird octave band within the range of frequencies 180-700 Hz (+/- 1 percent) for a vessel of 20 meters ‹ or more in length, or 180-2100 Hz (+/- 1 percent) for a vessel of less than 20 meters in length ›, of not less than the appropriate figure given in the table below.

(350)

Length of vessel in meters	One-third octave band level at 1 meter in dB referred to 2 x 10^{-5}N/m^2	Audible range in nautical miles
200 or more	143	2
75 but less than 200	138	1.5
20 but less than 75	130	1
Less than 20	120* 115** 111***	0.5

* When the measured frequencies lie within the range 180-450 Hz
** When the measured frequencies lie within the range 450-800 Hz
*** When the measured frequencies lie within the range 800-2100 Hz

(351) The range of audibility in the table is for information and is approximately the range at which a whistle may be heard on its forward axis with 90 percent probability in conditions of still air on board a vessel having average background noise level at the listening posts (taken to be 68 dB in the octave band centre120*

(357)

Annex IIIg (International)

(g) Combined whistle systems.

If due to the presence of obstructions the sound field of a single whistle or of one of the whistles referred to in §(f) is likely to have a zone of greatly reduced signal level, it is recommended that a combined whistle system be fitted so as to overcome this reduction. The whistles of a combined system shall be located at a distance apart of not more than 100 meters and arranged to be sounded simultaneously. The frequency of any one whistle shall differ from those of the others by at least 10 Hz.

Annex IIIg (Inland)

(g) Combined whistle systems.

(i) A combined whistle system is a number of whistles (sound emitting sources) operated together. For the purposes of the Rules a combined whistle system is to be regarded as a single whistle.

(ii) The whistles of a combined system shall:

(1) Be located at a distance apart of not more than 100 meters;

(2) Be sounded simultaneously;

(3) Each have a fundamental frequency different from those of the others by at least 10 Hz; and

(4) Have a tonal characteristic appropriate for the length of vessel which shall be evidenced by at least 2-thirds of the whistles in the combined system having fundamental frequencies falling within the limits prescribed in §(b) of this section, or if there are only two whistles in the combined system, by the higher fundamental frequency falling within the limits prescribed in paragraph (b) of this section.

Note: If, due to the presence of obstructions, the sound field of a single whistle or of one of the whistles referred to in §(f) of this section is likely to have a zone of greatly reduced signal level, a combined whistle system should be fitted so as to overcome this reduction.

(358) For the purposes of the Rules a combined whistle system is to be regarded as a single whistle. ‹ (ii) › The whistles of a combined system shall:

(359) (1) Be located at a distance apart of not more than 100 meters;

(360)

Annex III(h) (Inland)

(h) Towing vessel whistles

A power-driven vessel normally engaged in pushing ahead or towing alongside may, at all times, use a whistle whose characteristic falls within the limits prescribed by §1(b) for the longest customary composite length of the vessel and its tow.

(361)

Bell or Gong

(362) (a) Intensity of signal. A bell or gong, or other device having similar sound characteristics shall produce a sound pressure level of not less than 110 dB at ‹ a distance of › 1 meter ‹ from it ›.

(363) (b) Construction. Bells and gongs shall be made of corrosion-resistant material and designed to give clear tone. The diameter of the mouth of the bell shall be not less than 300 mm for vessels of 20 meters or more in length. Where practicable, a power-driven bell striker is recommended to ensure constant force but manual operation shall be possible. The mass of the striker shall be not less than 3 percent of the mass of the bell.

(364)

Approval

(365)

Annex III (International)

The construction of sound signal appliances, their performance and their installation on board the vessel shall be to the satisfaction of the appropriate authority of the State whose flag the vessel is entitled to fly.

(366)

Annex IV—Distress Signals

(367)

«*Need of Assistance*»

(368) The following signals, used or exhibited either together or separately, indicate distress and need of assistance:

(369) (a) a gun or other explosive signal fired at intervals of about a minute;

(370) (b) a continuous sounding with any fog-signaling apparatus;

(371) (c) rockets or shells, throwing red stars fired one at a time at short intervals;

(372) (d) a signal made by any signaling method consisting of the group . . . - - - . . . (SOS) in the Morse Code;

(373) (e) a signal sent by radiotelephony consisting of the spoken word “Mayday”;

(374) (f) the International Code Signal of distress indicated by N.C.;

(375) (g) a signal consisting of a square flag having above or below it a ball or anything resembling a ball;

(376) (h) flames on the vessel (as from a burning tar barrel, oil barrel, etc.);

(377) (i) a rocket parachute flare or a hand flare showing a red light;

(378) (j) a smoke signal giving off orange-colored smoke;

(379) (k) slowly and repeatedly raising and lowering arms outstretched to each side;

(380) (l) a distress alert by means of digital selective calling (DSC) transmitted on: (i) VHF channel 70, or (ii) MF/HF on the frequencies 2187.5 kHz, 8414.5 kHz, 4207.5 kHz, 6312 kHz, 12577 kHz or 16804.5 kHz;

(381) (m) a ship-to-shore distress alert transmitted by the ship’s Inmarsat or other mobile satellite service provider ship earth station;

(382) (n) signals transmitted by emergency position-indicating radio beacons;

(383) (o) approved signals transmitted by radiocommunication systems, including survival craft radar transponders *«meeting the requirements of 47 CFR 80.109»*.

(384) «*(p) A high intensity white light flashing at regular intervals from 50 to 70 times per minute.*»

(385)

«*Exclusive Use*»

(386) The use or exhibition of any of the foregoing signals except for the purpose of indicating distress and need of assistance and the use of other signals which may be confused with any of the above signals is prohibited.

(387)

«*Supplemental Signals*»

(388) Attention is drawn to the relevant sections of the International Code of Signals, the International Aeronautical and Maritime Search and Rescue Manual, Volume III, ‹ the International Telecommunication Union Radio Regulations, › and the following signals:

(389) (a) A piece of orange-colored canvas with either a black square and circle or other appropriate symbol (for identification from the air);

(390) (b) A dye marker.

(391)

Annex V—Pilot Rules

(392)

§88.01 Purpose and applicability.

(393) This part applies to all vessels operating on United States inland waters and to United States vessels operating on the Canadian waters of the Great Lakes to the extent there is no conflict with Canadian law.

(394)

§88.03 Definitions.

(395) The terms used in this part have the same meaning as the terms defined in part 83 of this subchapter.

(396)

§88.05 Law enforcement vessels.

(397) (a) Law enforcement vessels may display a flashing blue light when engaged in direct law enforcement or public safety activities. This light must be located so that it does not interfere with the visibility of the vessel's navigation lights.

(398) (b) The blue light described in this section may be displayed by law enforcement vessels of the United States and the States and their political subdivisions.

(399)

§88.07 Public safety activities.

(400) (a) Vessels engaged in government sanctioned public safety activities, and commercial vessels performing similar functions, may display an alternately flashing red and yellow light signal. This identification light signal must be located so that it does not interfere with the visibility of the vessel's navigation lights. The identification light signal may be used only as an identification signal and conveys no special privilege. Vessels using the identification light signal during public safety activities must abide by the Inland Navigation Rules, and must not presume that the light or the exigency gives them precedence or right of way.

(401) (b) Public safety activities include but are not limited to patrolling marine parades, regattas, or special water celebrations; traffic control; salvage; firefighting; medical assistance; assisting disabled vessels; and search and rescue.

(402)

Implementing Rules

(403) **Alternative Compliance**—see **33 CFR 81** and **33 CFR 89**, chapter 2, for regulations.

(404) **Vessel Bridge-to-Bridge Radiotelephone Regulations**—see **33 CFR 26**, chapter 2, for regulations.

Appendix A

(1)

Sales Information

(2) NOAA publications, nautical charts and unclassified National Geospatial-Intelligence Agency (NGA) nautical charts are sold by authorized sales agents in many U.S. ports and in some foreign ports. Information on obtaining charting products and a listing of authorized agents can be found at *nauticalcharts.noaa.gov*.

(3)

Products and Services–NOAA

(4)

Reporting corrections to Nautical Charts and Coast Pilots

(5) Users are requested to report all significant discrepancies or additions to NOAA charts and Coast Pilots, including depth information in privately maintained channels and basins; obstructions, wrecks and other dangers; new, relocated or demolished landmarks; uncharted fixed private aids to navigation; deletions or additions of small-craft facilities and any other information pertinent to safe navigation. This information may be submitted using the NOAA Office of Coast Survey site at https://www.nauticalcharts.noaa.gov/customer-service/assist/

(6)

Department of Commerce, NOAA
Nautical Data Branch
N/CS26, Station 7505
1315 East-West Highway
Silver Spring, Maryland 20910
ocs.ndb@noaa.gov

(7)

Nautical charts

(8) NOAA maintains the nautical charts and publications for the coast of the United States and the Great Lakes. Over a thousand charts cover 95,000 miles of shoreline and 3.4 million square nautical miles of water. Access to charts, publications and chart catalogs is available through *www.nauticalcharts.noaa.gov*.

(9)

Dates of latest editions

(10) Information concerning the dates of latest editions for the full suite of NOAA's nautical charts and U.S. Coast Pilot volumes can be found at *nauticalcharts.noaa.gov/mcd/dole.htm*.

(11)

Coast Pilots

(12)

U.S. Coast Pilot 1—Atlantic Coast: Eastport to Cape Cod
U.S. Coast Pilot 2—Atlantic Coast: Cape Cod to Sandy Hook
U.S. Coast Pilot 3—Atlantic Coast: Sandy Hook to Cape Henry
U.S. Coast Pilot 4—Atlantic Coast: Cape Henry to Key West
U.S. Coast Pilot 5—Gulf Coast, Puerto Rico and Virgin Islands
U.S. Coast Pilot 6—Great Lakes: Huron, Ontario, Michigan, Erie, Superior, and St. Lawrence River
U.S. Coast Pilot 7—Pacific Coast: California
U.S. Coast Pilot 8—Alaska: Dixon Entrance to Cape Spencer
U.S. Coast Pilot 9—Alaska: Cape Spencer to Beaufort Sea
U.S. Coast Pilot 10—Pacific Coast: Oregon, Washington, Hawaii, and Pacific Islands

(13)

Distance tables

(14) Distances Between United States Ports is available at*https://nauticalcharts.noaa.gov/publications/docs/distances.pdf*

(15)

Center for Operational Oceanographic Products and Services

(16)

1305 East-West Highway
Silver Spring, Maryland 20910
301–713–2815 (phone)
301–713–4500 (fax)
www.tidesandcurrents.noaa.gov

(17)

National Weather Service Offices

(18)

Massachusetts, Rhode Island, Connecticut
NWS Forecast Office Taunton/Boston – *www.weather.gov/box* 445 Myles Standish Boulevard, Taunton, MA 02780 508–822–0634
New York
NWS Forecast Office New York – *www.weather.gov/okx* 175 Brookhaven Avenue, Upton, New York, 11973 631–924–0517

(19)

NOAA Weather Radio

(20) National Weather Service VHF-FM radio stations provide mariners with continuous FM broadcasts of weather warnings, forecasts, radar reports, and selected weather observations. Reception range is typically 20

to 40 nautical miles from the antenna site, but can be as much as 100 nautical miles depending on elevation, terrain, type of receiver, and antenna used. The following VHF-FM radio stations are located in or near the area covered by this Coast Pilot.

(21)

Call Sign	Station	Location	Frequency
KHB-35	Boston, MA	42°12′N., 71°06′W.	162.475
KEC-73	Hyannis, MA	42°41′N., 70°20′W.	162.55
WXJ-39	Providence, RI	41°48′N., 71°28′W.	162.40
WXJ-42	Meriden, CT	41°33′N., 72°50′W.	162.40
KHB-47	New London, CT	41°26′N., 72°08′W.	162.55
WXM-80	Riverhead, NY	40°53′N., 72°43′W.	162.475
KWO-35	New York, NY	40°45′N., 73°58′W.	162.55

(22) The National Weather Service provides **Radio Facsimile Weather Information** for east coast waters through the Coast Guard Communication Station Boston (NMF). Broadcasts are made on the following frequencies: 4235 (02z, 08z), 6340.5, 9110, 12750 (14z) kHz. For carrier frequency, subtract 1.9 kHz. Fax schedules are transmitted at 0243 and 1405 GMT and provide area coverage and descriptions of services. For further information on Marine Radiofax Charts, visit: *https://www.weather.gov/marine/radiofax_charts*

(23)

Marine weather forecasts

(24) Coastal Marine Forecasts are issued four times daily by National Weather Service Offices. For further information on coastal marine forecasts as well as additional types of forecasts, visit: *https://weather.gov/marine/forecast* -and- *https://nowcoast.noaa.gov/*

(25)

Space Weather Prediction Center (SWPC)

(26) The Space Weather Prediction Center provides real-time monitoring and forecasting of solar and geophysical events that impact satellites, power grids, communications, navigation and many other technological systems.

(27)

NOAA, National Weather Service
National Centers for Environmental Predictions
Space Weather Prediction Center, W/NP9
325 Broadway
Boulder, Colorado 80305
www.swpc.noaa.gov

(28)

Products and Services–Other U.S. Government Agencies

(29) A partial list of publications and charts considered of navigational value is included for the ready reference of the mariner. In addition to the agents located in the principal seaports handling publication sales, certain libraries have been designated by the Congress of the United States to receive the publications as issued for public review.

(30)

Government Publishing Office

(31)

U.S. Government Publishing Office
710 North Capitol Street, NW
Washington, DC 20401-0001
202-512-1800
866-512-1800
www.gpo.gov/
ContactCenter@gpo.gov

(32)

Hydrographic Surveys

(33) U.S. Army Corps of Engineers hydrographic survey activity is available at: *https://www.mvr.usace.army.mil/Missions/Navigation/Hydrographic-Surveys/HydrographicSurveysMap/*

(34)

Nautical charts

(35) **Apalachicola, Chattahoochee and Flint Rivers Navigation Charts**, **Alabama River Charts** and **Black Warrior-Tombigbee Rivers River Charts**—available from the U.S. Army Corps of Engineers Mobile District for purchase in bound hard copy or as a PDF at *www.sam.usace.army.mil*.

(36) **Flood Control and Navigation Maps of the Mississippi River, Cairo, IL to the Gulf of America**—available from the U.S. Army Corps of Engineers Memphis District as a PDF at *www.mvm.usace.army.mil*.

(37) **Upper Mississippi River Navigation Charts (Mississippi River, Cairo, Illinois to Minneapolis, Minnesota)** and **Charts of the Illinois Waterway, from Mississippi River at Grafton, Illinois to Lake Michigan at Chicago and Calumet Harbors**—available from the U.S. Army Corps of Engineer Rock Island District for purchase in hard copy or as a PDF at *www.mvr.usace.army.mil*.

(38)

Publications and Services

(39) **Local Notice to Mariners** are posted weekly by the U.S. Coast Guard Navigation Center at *www.navcen.uscg.gov*. The National Geospatial-Intelligence Agency, U.S. Notice to Mariners are available at *msi.nga.mil/NGAPortal/MSI.portal*.

(40) **Special Notices to Mariners** are issued annually in National Geospatial-Intelligence Agency Notice to Mariners 1. These notices contain important information of considerable interest to all mariners. Interested parties are advised to read these notices.

(41) **Light List**—maintained by the United States Coast Guard and available online at *www.navcen.uscg.gov*. Also see Light List, chapter 1, for additional information.

(42) **List of Lights**, **Sailing Directions**, **Radio Navigational Aids (Pub. 117)**, **American Practical Navigator (Pub. 9)** and **International Code of Signals (Pub. 102)**—issued by the National Geospatial-Intelligence Agency and available at *msi.nga.mil/NGAPortal/MSI.portal*.

(43) The **Nautical Almanac**, the **Air Almanac**, and **Astronomical Almanac**—available through the United States Naval Observatory—*https://www.public.navy.mil/fltfor/cnmoc/Pages/usno_test_page.aspx*-and-*https://bookstore.gpo.gov/agency/united-states-naval-observatory-usno*

(44) **Dissemination of Marine Weather Information**, maintained by National Weather Service on the internet at *https://www.weather.gov/marine/nws_dissemination* -and- **NWS Marine Weather Services** at *https://www.weather.gov/marine/*

(45) **Navigation Rules and Regulations Handbook**—publication produced by the United States Coast Guard Navigation Standards Branch, which contains International and Inland Rules of the Road and Navigation Regulations. Available for download or viewing at *www.navcen.uscg.gov*. Navigation Rules are also found near the end of each individual Coast Pilot volume.

(46)

Offices and Services-Other U.S. Government Agencies

(47)

U.S. Army Corps of Engineers (USACE) Offices

(48)

District/Division Office	Contact Information
New England District Office 696 Virginia Road Concord, MA 01742	*www.nae.usace.army.mil* 978–318–8111
New York District Office Jacob K. Javits Federal Bldg. 26 Federal Plaza, Room 2109 New York, NY 10278-0090	*www.nan.usace.army.mil* 917-790-8799

(49)

Environmental Protection Agency (EPA) Offices

(50)

Regional Areas, States and Information
Region 1 New Hampshire, Vermont, Maine, Massachusetts, Connecticut, Rhode Island *www.epa.gov/aboutepa/epa-region-1-new-england*
Region 2 New Jersey, New York, Puerto Rico, Virgin Islands *www.epa.gov/aboutepa/epa-region-2*
Region 3 Delaware, Maryland, Virginia, District of Columbia, Pennsylvania *www.epa.gov/aboutepa/epa-region-3-mid-atlantic*
Region 4 Alabama, Florida, Georgia, Mississippi, South Carolina, North Carolina *https://www.epa.gov/aboutepa/about-epa-region-4-southeast*
Region 5 Illinois, Indiana, Michigan, Minnesota, Ohio, Wisconsin *www.epa.gov/aboutepa/epa-region-5*
Region 6 Louisiana, Texas *www.epa.gov/aboutepa/epa-region-6-south-central*
Region 9 California, Hawaii, Guam *www.epa.gov/aboutepa/epa-region-9-pacific-southwest*
Region 10 Alaska, Oregon, Washington *www.epa.gov/aboutepa/epa-region-10-pacific-northwest*

(51)

U.S. Coast Guard Navigation Center (NAVCEN)

(52) The Coast Guard Navigation Center provides cutting-edge services for safe, secure and efficient maritime transportation. The center operates the Navigation Information Service (NIS), the Maritime Differential GPS (DGPS) and the developing Nationwide Differential Global Positioning System (NDGPS). In addition, NAVCEN serves as the civilian interface for the Global Positioning System and manages other navigation-related projects.

(53) For further information and/or operational questions regarding GPS and DGPS, visit *www.navcen.uscg.gov* or contact:

(54)

Commanding Officer
U.S. Coast Guard Navigation Center
NAVCEN MS 7310
7323 Telegraph Road
Alexandria, VA 20598-7310

(55)

Coast Guard District and Sector Offices

(56)

District/Division Office	Contact Information
New England District Office 696 Virginia Road Concord, MA 01742	*www.nae.usace.army.mil* 978–318–8111
New York District Office Jacob K. Javits Federal Bldg. 26 Federal Plaza, Room 2109 New York, NY 10278-0090	*www.nan.usace.army.mil* 917-790-8799

(57)

Sectors	Contact Information
Sector Southeastern New England	U.S. Coast Guard Sector SE New England 1 Little Harbor Road Woods Hole, MA 02543 508-538-2300
Sector Boston	U.S. Coast Guard Sector Boston 427 Commercial Street Boston, MA 02109 617-223-3123
Sector Long Island Sound	U.S. Coast Guard Sector Long Island Sound 120 Woodward Avenue New Haven, CT 06512 800-774-8724
Sector New York	U.S. Coast Guard Sector New York 212 Coast Guard Drive Staten Island, NY 10305 718-354-4037

(58) .

(59)

Coast Guard Stations

(60) The stations listed are in the area covered by this Coast Pilot. They have search and rescue capabilities and may provide lookout, communication and/or patrol

functions to assist vessels in distress. The National VHF-FM Distress System provides continuous coastal radio coverage outwards to 20 miles on channel 16. After contact on channel 16, communications with the Coast Guard should be on channel 1022 (previously channel 22A). If channel 1022 is not available to the mariner, communications may be made on channel 12. Selected stations guard the International Radiotelephone Distress, Safety and Calling Frequencies.

(61)

U.S. Coast Guard District One Stations	
Massachusetts:	
Cape Cod Canal	East entrance to the canal, near Sandwich, MA. (41°46.4'N., 70°30.0'W.)
Cape Cod Coast Guard Air Station	On Cape Cod at Otis Air Force Base. (41°37.5'N., 70°31.5'W.)
Provincetown	On southwest side of harbor, about 0.4 mile southwest of town pier. (42°02.7'N., 70°11.6'W.)
Chatham	Southeastern Cape Cod, near Chatham Light. (41°40.3'N., 69°57.0'W.)
Woods Hole	On west side of Little Harbor, about 450 yards northward of Juniper Point. (41°31.2'N., 70°40.0'W.)
Brant Point	On west side of entrance to Nantucket Harbor, near Brant Point Light. (41°17.4'N., 70°05.5'W.)
Menemsha	West end of Martha's Vineyard, near Menemsha Light. (41°21.0'N., 70°45.9'W.)
Rhode Island:	
Castle Hill	On west shore of Newport Neck, near Castle Hill Light. (41°27.7'N., 71°21.5'W.)
Point Judith	On Point Judith near Point Judith Light, 0.5 mile east of Point Judith Harbor of Refuge. (41°21.7'N., 71°28.9'W.)
Connecticut:	
New London	At Fort Trumbull, on west side of main channel northward of Greens Harbor. (41°20.7'N., 72°05.7'W.)
New Haven	On the north side of the jutting point, about 1.5 miles northward of Lighthouse Point. (41°16.4'N., 72°54.2'W.)
New York:	
Fishers Island	In Silver Eel Cove, on east end of island (manned during summer months only). (41°15.4'N., 72°01.9'W.)
Eatons Neck	Near Eatons Neck Light, north shore of Long Island, east side of entrance to Huntington Bay. (40°57.3'N., 73°23.9'W.)
Montauk Point	In Montauk Harbor, Long Island. (41°04.3'N., 71°56.1'W.)
Shinnecock	East side of Ponquogue Point, 1.3 miles northwest of Shinnecock Inlet. (40°51.0'N., 72°30.3'W.)
Moriches	On the east side of the entrance to Tuthill Cove. (40°47.3'N., 72°45.0'W.)
Fire Island	Near west end of island, 1.9 miles west-southward of Fire Island Light. (40°37.5'N., 73°15.6'W.)
Jones Beach	Near west end of Jones Beach. (40°35.4'N., 73°33.4'W.)
Kings Point	At the entrance to Little Neck Bay. (40°48.8'N., 73°45.9'W.)
New York	On the west side of the Narrows, about 0.5 mile above Fort Wadsworth at Rosebank, Staten Island. (40°36.7'N., 74°03.7'W.)
New Jersey:	
Sandy Hook	On the Bay side, 0.5 mile south of the northern extremely of Sandy Hook. (40°28.1'N., 74°00.5'W.)

(62) .

(63)

Coast Guard radio broadcasts

(64) Urgent, safety and scheduled marine information broadcasts are made by Coast Guard radio stations. In general, these broadcasts provide information vital to vessels operating in the approaches and coastal waters of the United States including Puerto Rico and U.S. Virgin Islands. Transmissions are as follows:

(65) **By radiotelephone:** (a) upon receipt; (b) repeated 15 minutes later, (for urgent messages only); (c) text only on the first scheduled broadcast unless canceled; (d) additional broadcasts at the discretion of the originator.

(66) **Urgent broadcasts** are preceded by the urgent signal PAN-PAN. Both the urgent signal and message are transmitted on VHF-FM channel 16.

(67) **Safety broadcasts** are preceded by the signal SECURITY. After the preliminary safety signal is broadcast on VHF-FM channel 16, broadcast stations will shift to VHF-FM channel 1022 (previously channel 22A).

(68) Up-to-date U.S. Coast Guard radio broadcast schedules may be found at *nws.noaa.gov/om/marine/home.htm*.

(69)

Radio weather broadcasts

(70) Taped or direct broadcasts of marine weather forecasts and storm warnings are made by commercial and Coast Guard radio stations in the area covered by this Coast Pilot. The Coast Guard broadcasts coastal and offshore marine weather forecasts at the times and frequencies indicated:

(71)

Coast Guard Radio Station	Scheduled Broadcast Times (UTC)
NMN, Portsmouth, VA	
4426.0 kHz	0030, 0500, and 2300 e.s.t.
6501.0 kHz	0030, 0500, 0630, 1100, 1700, 1830, and 2300 e.s.t.
8764.0 kHz	0030, 0500, 0630, 1100, 1230, 1700, 1830, and 2300 e.s.t.
13089.0 kHz	0630, 1100, 1230, 1700, and 1830 e.s.t.
17314.0 kHz	1230 e.s.t.

(72)

U.S. NAVTEX transmitting stations

(73) NAVTEX coverage is reasonably continuous to 200 NM off the U.S. East, Gulf and West Coasts; Puerto Rico;

Southwest Alaska; Hawaii; and 100 NM off Guam. U.S. Coast Guard NAVTEX broadcast stations and message content follow:

(74)

Station	ID	WX Broadcast Schedule (UTC)
Boston (NMF)	F	0050, 0450, 0850, 1250, 1650, 2050
Chesapeake (NMN)	N	0210, 0610, 1010, 1410, 1810, 2210
Charleston (NMN)	E	0040, 0440, 0840, 1240, 1640, 2040
Miami (NMA)	A	0000, 0400, 0800, 1200, 1600, 2000
San Juan (NMR)	R	0250, 0650, 1050, 1450, 1850, 2250
New Orleans (NMG)	G	0100, 0500, 0900, 1300, 1700, 2100

(75) **Boston (NMF)(Station F)**

(76) First Coast Guard District Broadcast Notices to Mariners.

(77) Distress Urgent and Safety messages.

(78) International Ice Patrol Reports (in season).

(79) Gale, storm and hurricane warnings.

(80) Offshore marine weather forecasts for:

(81) New England continental shelf to 1000 fathoms;

(82) Gulf of Maine;

(83) Georges Bank;

(84) South of New England;

(85) South of Nova Scotia.

(86) **Chesapeake (NMN)(Station N)**

(87) Fifth Coast Guard District Broadcast Notices to Mariners.

(88) Distress, Urgent and Safety messages.

(89) Gale, storm and hurricane warnings.

(90) Offshore marine weather forecasts for the west central North Atlantic from 32°N to 40°N and west of 65°W including the continental shelf to 1000 fathoms.

(91) **Charleston (NMN) (Station E)**

(92) Seventh Coast Guard District Broadcast Notices to Mariners.

(93) Distress, Urgent and Safety Messages.

(94) Gale, storm and hurricane warnings.

(95) Offshore Marine Weather Forecast from Murrells Inlet, SC, to Flagler Beach, FL.

(96) **Miami (NMA)(Station A)**

(97) Seventh Coast Guard District Broadcast Notices to Mariners.

(98) Distress, Urgent and Safety messages.

(99) Gale, storm and hurricane warnings.

(100) Offshore marine weather forecasts for the southwest North Atlantic south of 32°N and west of 65°W.

(101) **San Juan (NMR)(Station R)**

(102) Seventh Coast Guard District Broadcast Notices to Mariners.

(103) Distress, Urgent and Safety messages.

(104) Gale, storm and hurricane warnings.

(105) Offshore marine weather forecasts for:

(106) Puerto Rico and Virgin Islands water out 20 NM;

(107) Eastern Caribbean Sea east of 75°W.

(108) **New Orleans (NMG)(Station G)**

(109) Eighth Coast Guard District Broadcast Notices to Mariners.

(110) Distress, Urgent and Safety messages.

(111) Gale, storm and hurricane warnings.

(112) Offshore marine weather forecasts for the Gulf of America.

(113)

Customs ports of entry

(114) Vessels arriving in the United States from a foreign port or place are required to report their arrival to Customs and Border Protection immediately. Field Operations Offices and contact information are listed below and at *cpb.gov*.

(115)

Field Operations Office	Contact Information
Boston	10 Causeway Street Room 801 Boston, MA 02222 617-565-6208
New York	One World Trade Center Suite 50.200 New York, NY 10007 646-733-3100

(116)

Public Health Service Quarantine Stations

(117)

Field Operations Office	Contact Information
CDC Boston	c/o CDC New York Quarantine Station JFK International Airport Terminal 4, Room 219.06 2nd Floor, East Concourse Jamaica, NY 11430-1081 617-561-5701
CDC New York	CDC New York Quarantine Station JFK International Airport Terminal 4, Room 219.06 2nd Floor, East Concourse Jamaica, NY 11430-1081 718-553-1685

(118)

Food and Drug Administration (FDA) Regional Offices

(119)

Northeast Region	158-15 Liberty Avenue Jamaica, New York 11433 718–340–7000
Central Region	20 North Michigan Avenue Suite 510 Chicago, Illinois 60602 215–597–4390
Pacific Region	1301 Clay Street Room 1180N Oakland, California 94612 510–287–2700
Southeast Region	60 Eighth Street NE Atlanta, Georgia 30309 404–253–1171
Southwest Region	4040 North Central Expressway Suite 900 Dallas, Texas 75204 214–253–4901

(120)

Department of Agriculture, Animal and Plant Health Inspection Service (APHIS)

(121) Information on the importation of plants, animals and plant and animal products is available from APHIS,

Department of Agriculture, 4700 River Road, Riverdale, MD 20737. Go to *aphis.usda.gov* for more information.

(122) **Animal Import Centers:**

(123)

USDA Animal and Plant Inspection Service **Animal Import Centers:**
Los Angeles Animal Import Center (LAAIC) 222 Kansas Street El Segundo, CA 90245 310-955-3311
Miami Animal Import Center (MAIC) 6300 NW 36th Street Miami, FL 33122 305-876-2200
New York Animal Import Center (NYAIC) 474 Animal Import Center Newburg, NY 12550 845-838-5500
John F. Kennedy Airport Office 230-59 Rockaway Blvd. Suite 100, Room 101 Jamaica, NY 11413 718-553-3570
Agriculture Select Service Agents 4700 River Road, Unit 2 Riverdale, MD 20737 AgSAS@aphis.usda.gov 301-851-3300 (select option 3)

(124)

U.S. Citizenship and Immigration Services

(125)

Massachusetts	
Boston Field Office	John F. Kennedy Federal Building 15 New Sudbury Street Room E-160 Boston, MA 02203
Lawrence Field Office	2 Mill Street Lawrence, MA 01840
Rhode Island	
Providence Field Office	1543 Atwood Avenue Johnstown, RI 02919
Connecticut	
Hartford Field Office	450 Main Street, 1st Floor Hartford, CT 06103
New York	
Long Island Field Office	30 Barretts Avenue Holtsville, NY 11742
New York City Field Office	Jacob K. Javits Federal Building 26 Federal Plaza 3rd Floor, Room 3-120 New York, NY 10278
Queens Field Office	Jacob K. Javits Federal Building 26 Federal Plaza 8th Floor, Room 8-100 New York, NY 10278

(126)

Federal Communications Commission Offices

(127) **District Field O f ces:**

(128) Boston MA: 1 Batterymarch Park, Quincy, MA 02169-7495.

(129) New York, NY: 201 Varick St., Suite 1151, New York, NY 10014-4870.

(130) Telephone toll-free: 888–225–5322 (888–CALLFCC) to report radio communications interference issues.

(131)

Stations transmitting medical advice

(132) To obtain radio medical advice by reliable voice radio communications urgent calls for assistance may be broadcast using the normal Urgency prowords PAN PAN as follows:

(133)

Broadcast Language	Details
PAN-PAN	(3 times)
All Stations	(3 times or specific station if known)
This is ship name	(3 times)
Call sign	(call sign)
In Position	(give position)
I require medical advice	
Over	

(134)

Measured courses

(135) The positions of measured courses are shown on the chart and their description is included in the Coast Pilots when information is reported to NOAA. Courses are located in the following places covered by this Coast Pilot:

(136) Captain Harbor, on south side of Great Captain Island.

(137) Eatons Neck, on west side of Eatons Neck.

(138) Port Jefferson, off Port Jefferson Harbor.

(139) The pages in the text describing the courses can be obtained by referring to the index for the geographic places; chart numbers follow the names.

(140)

Canadian Government Agencies

(141)

Canadian Government Agencies	
Canadian Hydrographic Service	Director General Canadian Hydrographic Service Department of Fisheries and Oceans 200 Kent Street, Station 15N100 Ottawa, Ontario K1A 0E6 CANADA
Hydrographic Chart Distribution Office	Hydrographic Chart distribution Office Department of Fisheries and Oceans P.O. Box 8080, 1675 Russell Road Ottawa, Ontario K1G 3H6 CANADA
Canadian Coast Guard	Canadian Coast Guard Department of Fisheries and Oceans 340 Slater Street Ottawa, Ontario K1A 0N7 CANADA 613-993-0999
Canadian Communications Group	Canadian Communications Group 45 Sacre-Coeur Boulevard Hull, Quebec K1A0S9 CANADA

Weekly Record of Updates

Week of	Action	Chapter	Paragraph(s)	User notes
18 NOV 2025				U.S. Coast Pilot 2, 55th Edition has been issued.
23 NOV 2025	**No Correction**			
30 NOV 2025	**No Correction**			
07 DEC 2025	**No Correction**			
14 DEC 2025	**No Correction**			
21 DEC 2025	**No Correction**			
28 DEC 2025	**No Correction**			
04 JAN 2026	**No Correction**			
11 JAN 2026	**No Correction**			
18 JAN 2026	**No Correction**			
25 JAN 2026	**Change**	1	450.0020	
	Insert	1	450.0030	
	Insert	1	450.0040	
01 FEB 2026	**No Correction**			
08 FEB 2026	**No Correction**			

This record is intended as a log for critical updates applied to this volume. For online versions or Print on Demand (POD) copies, all weekly critical updates issued and applied to this edition at time of download or purchase are listed.
Affected paragraphs within the chapters are indicated by a gray highlight for ease of identification; e.g. (215)

Updates are available from NOAA at http://nauticalcharts.noaa.gov/nsd/cpdownload.htm.

Week of	Action	Chapter	Paragraph(s)	User notes

Week of	Action	Chapter	Paragraph(s)	User notes

Week of	Action	Chapter	Paragraph(s)	User notes

Index

A

Abets Creek 339
Abiels Ledge 224
Acabonack Harbor 261
Acapesket 198
Acoaxet 230
Acushnet River 226
Adams Fall 294
Agriculture, Department of 26, 419
Aids to navigation 10
Air Almanac 417
Albany 384
Alder Island 343
Allen Harbor 193, 249
Allyn Point 280
Almy Point 237
Almy Rock 237
Alpine 379
Ambrose Channel 350
Amendments 1
America Ledge 248
American Practical Navigator (Pub. 9) 416
Amityville 342
Amityville Creek 342
Amityville Cut 343
Anchorage Channel 350, 360
Anchorage Regulations 45
Anchorages 164
Angelica Point 226
Animal and Plant Health Inspection Service 26, 419
Animal Import Centers 420
Annsville Creek 381
Anthonys Nose 381
Apponagansett Bay 230
Apponaug Cove 250
Area to be avoided 169
Areskonk Creek 338
Areys Pond 187
Arlington 371
Army Corps of Engineers 30
Arnold Point 240
Arthur Kill 367
Articulated Daybeacons 11
Articulated Lights 11
Ash Creek 308
Asia Rip 13204 170
Astronomical Almanac 417
Athens 384
Atlantic Beach 345
Atlantic Highlands 363
Atlantis Canyon 171
Aucoot Cove 226
Aunt Lydias Cove 187
Aunt Phebe Rock 317
auroral borealis 181
Automated Mutual Assistance Vessel Rescue System (AMVER) 14
Automatic Identification System (AIS) Aids to Navigation 12
Avery Point 272
Avery Point Light 272

B

Babylon 342
Babylon Cove 342
Back River 219
Bailey Rock 317
Baldwin Bay 344
Ballast Reef 311
Bannister Creek 345
Barbadoes Basin 358
Barcelona Point 264
Barker Point 325
Barlows Landing 219
Barneys Joy Point 230
Barren Island 355
Barrington 243
Barrington River 243
Barstow Rock 226
Bartlett 280
Bartlett Reef 280
Bassetts Island 219
Bassing Harbor 187
Bass Rip 170
Bass River 193
Bayonne 369
Bayonne Bridge 369
Bay Ridge Channel 360
Bay Ridge Flats 360
Bay Shore 341
Bayview 230
Beach Channel 358
Beacon 382
Bearings 1
Bear Mountain 381
Bear Mountain Bridge 381
Bearse Rock 196
Bearse Shoal 188
Beaufort Point 317
Beaverdam Creek 338
Beaverhead 248
Beavertail Light 237
Beavertail Point 237
Bedford Reef 259
Belamose 286
Belden Point 318
Belford Harbor (Shoal Harbor) . . . 364
Belleville 371
Bell Island 311
Bellport 339
Bellport Bay 338
Bellport Beach 339
Belmont Island 331
Bennets Neck 219
Bergen Basin 359
Bergen Beach 358
Bergen Point 369
Bermuda Lagoon 308
Berrys Creek Canal 372
Big Fishkill Channel 358
Big Mermaid 291
Big Tom 319
Bird Island 225, 321
Bird Island Reef 225
Bird Rock 291
Bishop and Clerks 194
Bishop Rock Shoal 240
Blackberry Bay 363
Black Ledge 278
Black Point 237, 250, 282
Black Rock 293, 308
Black Rock Harbor 307
Blackstone Rocks 290
Blankinship Cove 225
Block Island 255
Block Island North Light 255
Block Island North Reef 253
Block Island Sound 253
Block Island Southeast Light 255
Bluefish Shoal 315
Bluff Point 197, 308, 321
Blyn Rock 291
Bodkin Rock 286
Bold Rock 311
Bonnet Point 248
Bonnet Shores Beach 248
BookletCharts 4
Borden Flats 241
Bostwick Bay 260
Bournes Pond 198
Bowers Island 314
Bowery Bay 329
Bowline Point 380
Branch 212
Branchport 363
Branford Harbor 291
Branford Point 291
Branford Reef 290
Branford River 291
Brant Point 359
Brant Point Light 199
Brayton Point 241
Breakneck Point 382
Breakneck Ridge 382
Breakwater Point 236
Brenton Cove 238
Brenton Point 237
Brenton Reef 237
Brickyard Point 323
Bridge and Cable Clearances 6
Bridge Creek 345
Bridge Lights and Clearance Gages . . 12
Bridgeport 304
Bridgeport Harbor 304
Bridges and Cables 1
Briggs Point 231

Brig Ledge 249
Bristol 243
Bristol Harbor 243
Bristol Neck 243
Broadcast Notices to Mariners 9
Broadcast Notice to Mariners 17
Broad Creek Channel 343
Broadwater Cove 265
Broadway 212
Broken Ground 194
Broken Part of Pollock Rip 188
Bronx Kill 329
Bronx River 328
Bronx-Whitestone Bridge 327
Brooklyn Bridge 332
Brosewere Bay 345
Browing Ledge 230
Brown Creek 339
Brown Point 290
Browns Reef 290
Brush Neck Cove 250
Bullhead Bay 266
Bullock Cove 244
Bullock Point 245
Bull Point 238
Buoys 11
Burns Point 296
Burr Creek 308
Butler Hole 189
Butler Point 224
Butter Ball Rock 238
Buttermilk Bay 223
Buttermilk Channel 360
Buzzards Bay 216
Byram (East Port Chester) 315
Byram Harbor 314
Byram Point 315
Byram River 315

C

Cable and Anchor Reef 309
Cable ferries 2
Calf Islands 314
Calf Pasture Island 309
Calf Pasture Point 249
Calves Island 285
Canadian Government Agencies . . . 415
Canal de Chambly 388
Canapitsit Channel 215
Canarsie 358
Canoe Place 336
Cape Cod 185
Cape Cod Canal 220
Cape Higgon 210
Cape Poge 202
Cape Poge Bay 202
Cape Poge Flats 202
Captain Harbor 312
Captree Island 341
Cargo Care 181
Carlls River 342
Carlstadt 372
Carman Creek 343
Carmans River 338
Cartwright Island 260
Castle Hill 237
Castleton-on-Hudson 384
Castleton-on-Hudson Bridge 384
Castle William 361
Cataumet 218, 219
Cataumet Rock 218
Cat Island 290
Catskill 384
Catskill Creek 384
Catumb Passage 258
Catumb Rocks 258
Cayuga and Seneca Canal 388
Cedar Beach 342
Cedar Creek 307
Cedar Hill 384
Cedar Island 224, 288
Cedar Island Creek 224
Cedar Island Point 224
Cedar Point 264, 308
Cementon 383
Centerboard Shoal 225
Center for Operational Oceanographic Products and Services (CO-OPS) 26
Centerport Harbor 322
Centerville 196
Centerville Harbor 196
Centerville River 196
Centre Island 323
Cerberus Shoal 259
Champlain Canal 388
Channel Markers 12
Channel Rock 196, 309
Chapel Hill Channel 351
Chappaquiddick Island 202
Chappaquoit Point 218
Charles Island 296
Charles Reef 288
Charles Rock 249
Charlestown 257
Charlestown Breachway 257
Charlestown Pond 257
Chart Accuracy 5
Chart Datum, Tidal Waters 5
Chart No. 1 6
Chart Projections 5
Charts and Publications–Other U.S. Government Agencies 416
Chart Scale 4
Charts, Publications and Services–NOAA 415
Chart Symbols, Abbreviations and Terms 6
Chatham 187, 191
Chatham Light 187
Chatham Roads 191
Cheesequake Creek 365
Chelsea 382
Chepiwanoxet Point 250
Cherry Grove 340
Cherry Harbor 260
Cherry Hill Point 260
Chester Creek 285
Chester Creek Bar 285
Childs River 197
Chilmark 202
Chimney Corner Reef 289
Chimney Sweeps Islands 318
Chimon Island 309
Chocomount 258
Chocomount Cove 267
Cholera Bank 348
Church Point 237
Cinder Creek 344
Citing Rock 238
Citizenship and Immigration Services . 31
City Island Harbor 319
City Point 292
Clam Island 290
Clarks Cove 230
Clarks Point 227
Clason Point 328
Clay Head 255
Cleaves Point 263
Cleveland East Ledge Light 220
Clinton 288
Clinton Harbor 288
Clinton Point 382
Coastal Warning Display 28
Coasters Harbor Island 238
Coast Guard District and Sector Offices 417
Coast Guard District Offices 417
Coast Guard Radio Broadcasts . . . 418
Coast Guard Stations 417
Coast Pilot 1
Coast Pilots 415
Cobalt 286
Cockenoe Harbor 309
Cockenoe Island 309
Cockenoe Reef 309
Cockenoe Shoal 309
Cockle Cove 192
Coddington Cove 240
Coddington Point 240
Code of Federal Regulations (CFR) . . . 37
Coecles Harbor 262
Coeymans 384
Coffin Rock 213
Coggeshall Point 240
Cohasset Narrows 223
Cold Fronts 176
Cold Spring 381
Cold Spring Harbor 323
Cold Spring Point 245
Cold Spring Pond 266
Cole River 241
College Point 328
College Point Reef 328
Collier Ledge 196
Colonel Willie Cove 268

COLREGS Demarcation Lines . . 39, 164
Columbia Island 317
Commerce, Department of 26
Commercial Maritime Coast Stations and Weather Nets 17
Common Fence Point 241
Common Flat 191
Compass Roses. 9
Compo Yacht Basin 308
Compton Creek 364
Conanicut Island 233, 240
Conaskonk Point 365
Coney Island 359
Coney Island Channel 359
Coney Island Creek 359
Congress Street Bridge (State Route 2) 387
Con Hook. 381
Conimicut Light 244
Conimicut Point 244
Conkling Point. 263
Conklin Point 342
Connecticut River 282
Connetquot River 340
Conscience Bay 300
Constable Hook 369
Constellation Rock 259
Constitution Island 381
Continental Shelf. 171
Converse Point. 225
Cooper Bluff. 323
Copicut Neck 215
Copps Island. 309
Corey Creek 265, 339
Corlears Hook 350
Cormorant Reef 236, 271, 314
Cormorant Rock 226, 236
Cornfield Point 287
Cornfield Point Shoal 287
Cornwall-on-Hudson 382
Corps of Engineers, U.S. Army . . . 417
Corsair Canyon 171
Cos Cob Harbor 313
COSPAS-SARSAT. 14
Cotuit 197
Cotuit Anchorage. 196
Cotuit Bay 197
Cotuit Highlands 197
Courses 2
Cove Harbor. 311
Cove Neck 323
Cove Point 323
Cow and Calf 291
Cow Neck 266
Coxens Ledge 217
Coxsackie 384
Craigville Beach 196
Crane Neck Point. 320
Crane Reef 287
Crescent Beach 281
Crimbo Point 297
Cromeset Neck. 224
Cromeset Point 224
Cross Rip Channel 189
Cross Rip Shoal 188
Croton-on-Hudson 380
Croton Point 380
Crow Head 260
Crow Point 286
Crow Shoal 260
Crows Nest 381
Crows Pond 187
Cuban Ledge 319
Cultivator Shoal 13204 169
Currents 2, 172
Customs Ports of Entry and Stations . 419
Cutchogue Harbor 265
Cuttyhunk 215
Cuttyhunk Harbor 215
Cuttyhunk Island. 215
Cuttyhunk Pond 215
Cutty Wow Rock 231

D

Dangerous Rock 264
Danger Zones and Restricted Area Regulations 148
Daniels Island 197
Danskammer Point 382
Darrow Rocks 291
Dartmouth Rock 230
Dauntless Rock 316
Davenport Neck 317
Davids Island 317
Davis Bank 13204 170
Davis Island 290
Davis Park 339
Davis South Shoal 13200 170
Davisville 198, 249
Daybeacons 11
Daylight saving time 183
Dead Horse Bay 358
Deep Reef. 318
Deep River 285
Defense, Department of 28
Delancey Point. 315
Delaware and Raritan Canal 367
Democrat Point 341
Dennis Port 193
Department of Agriculture 26, 419
Department of Commerce 26
Department of Defense 28
Department of Health and Human Services 30
Department of Homeland Security. . . 31
Depths 2
Derby 296
Dering Harbor 263
Designated Critical Habitat. 157
Despair Island 249
Destructive Waves 19
Devils Bridge 211
Devon 296
Dewitt Point. 383
Dew Point. 181
Diamond Reef 382
Dighton 240
Digital Selective Calling (DSC) 14
Disposal areas 8
Disposal Sites 8
Distances 2
Distance Tables 415
Distress: Communication Procedures . 13
Doanes Creek 193
Dobbs Ferry 379
Dogfish Bar 194
Dogfish Island 290
Dolphin Cove 312
Drawbridge Operation Regulations . . 65
Drawbridges. 171
Dry Ledge. 224
Duck Creek 308
Duck Island 287
Duck Island Bluff. 321
Duck Island Harbor. 321
Duck Island Roads 287
Duck Pond Point 298
Dumping Grounds 8
Dumping of dredged material. 32
Dumpling Rocks 228
Dunbar Point 13243 195
Dundee Dam 371
Dunderberg Mountain. 381
Dutch Island 248
Dutch Island Harbor 248
Dutch Kills 331
Dyer Island 240

E

East Bank. 352
East Bay 196, 343
East Branch 307, 312, 331, 337
Eastchester 319
Eastchester Bay 319
East Chop. 208
East Chop Flats 208
East Clump 267
East Creek 265
East Crib 290
East Crow Island 343
East Fort Point 322
East Greenwich 250
East Haddam 285
Eastham 186
East Hampton 262
East Harbor 267
East Indies Rocks. 291
East Mill Basin. 358
East Moriches 337
East Nonations. 317
East Norwalk 310
East Norwalk Harbor 310
Easton Point 237

East Passage 237
East Point 258, 267
Eastport 337
East Quogue 337
East Reef 290
East River 224, 289, 326, 361
East Rockaway 344
East Rockaway Channel 344
East Rockaway Inlet 345
East West Channel 342
East White Rock 309
Eatons Neck 321
Eatons Neck Basin 321
Echo Bay 317
Echo Island 317
Echo Soundings 9
Eddie Woods Rock 195
Eddyville 383
Edgartown 202
Edgartown Harbor 202
Edgartown Harbor Light 202
Edgewater 378
Edgewater Point 316
Edwards Rock 215
Edwards Shoal 201
Eel Grass Ground 269
Eel Pond 197, 213
Eightmile River 285
Eldridge Shoal 197
Electronic Navigational Charts (NOAA ENC®) 4
Electronic Positioning Systems 13
Elihu Island 268
Elizabeth 369
Elizabeth Islands 211
Elizabethport 369
Elizabeth River 369
Ellis Reef 269
Emerald Rock 317
Emergency Position Indicating Radiobeacons (EPIRB) 14
Endangered and Threatened Marine Species 155
Endangered Marine and Anadromous Species 155
Endeavor Shoals 259
Enders Island 269
English Kills 331
Environmental Protection Agency (EPA) 32
Environmental Protection Agency (EPA) Offices 417
Erie Basin 361
Erie Canal 388
Esopus 383
Esopus Creek 383
Esopus Island 383
Esopus Meadows Light 383
Essex 285
Essex Cove 285
Eustasia Island 285
Eustis Rock 219
Execution Rocks 317
Extratropical Cyclones 174

F

Fair Harbor 340
Fairhaven 227
Fair Haven 292, 363
Falkner Island 289
Fall River 240
Falmouth 209
Falmouth Harbor 209
Falmouth Heights 209
Falmouth Inner Harbor 209
False Channel Meadow 343
False Hook 351
False Hook Channel 351
Fanning Point 263
Farm River 291
Farm River Gut 291
Far Rockaway 359
Fayerweather Island 307
Federal Communications Commission . 32
Federal Communications Commission Offices 420
Federal Water Pollution Control Act (FWPCA) 21
Felix Ledge 214
Ferry Point 284
Fiddlers Cove 218
Fields Point 244
Fighting Island 343
Fire Island 338
Fire Island Inlet 341
Fire Island Light 341
Fire Island National Seashore 338
Fishers Island 258
Fishers Island Sound 266
Fish havens 8, 30
Fishing Cove 249
Fishing Rip 13204 170
Fishtrap areas 9
fishtraps 30
Fish weirs 171
Fiske Rock 240
Fitch Point 310
Five Foot Rock 291
Fivemile River 311
Flanders 266
Flanders Bay 266
Flat Neck Point 313
Flat Rock 281
Flint Point 236
Flint Point Ledge 236
Float Plan 16
Flushing 328
Flushing Bay 328
Flushing Creek 328
Flying Point 290
Flynns Knoll 351
Fogland Point 237
Food and Drug Administration (FDA) . 30
Food and Drug Administration (FDA) Regional Offices 419
Foot Rocks 290
Forbes Rocks 315
Forge River 338
Fort Adams 238
Fort Jay 361
Fort Lee 378
Fort Phoenix 227
Fort Point 280
Fort Pond Bay 260
Fort Schuyler 326
Fort Totten 326
Fort Trumbull State Park 277
Fort Wadsworth 354
Fort Washington Point 378
Fort Wetherill 238
Fourfoot Rocks 315
Fourteen Foot Channel 351
Fox Creek Channel 342
Fox Hill 248
Fox Island 248
Fox Point 244
Freeport 344
Fresh Creek 358
Fresh Kills 368
Friars Head 299
Frost Creek 324
Frost Point 308
Fuller Rock 245

G

Gales Ferry 280
Galilee 257
Gallatin Rock 196, 226
Gangway Rock 258, 325
Gannet Ledge 196
Gannet Rocks 196
Gardiners Bay 261
Gardiners Island 260
Gardiners Point 260
Gardners Neck 241
Garrett Lead 344
Garrison 381
Gaspee Point 244
Gay Head 211
Gazelle Rock 194
Gees Point 381
General Grants Tomb 378
General Rock 249
Geographic Coordinates 2
Georges Bank 13200, 13204 168
Georges Rock 309
Georges Shoal 13204 169
George Washington Bridge 378
Gerritsen Creek 358
Gerritsen Inlet 357
Giants Neck 282
Gibbs Rock 225
Gifford Ledge 218
Gilbert Canyon 171
Gildersleeve 286
Gilgo Beach 342

Gilgo Heading 343
Glen Cove. 324
Glen Cove Creek 324
Glen Cove Landing 324
Glen Island 317
Glenwood Landing 324
Global Maritime Distress and Safety System (GMDSS). 13
Global Positioning System (GPS) . . . 13
Goat Island 238, 268
Goethals Bridge 369
Golden Spur (East Lyme) 281
Goodwives (Darien) River 311
Gooseberry Island 237, 249
Gooseberry Neck 230
Goose Creek. 264
Goose Island. 267, 317
Goose Islands 289
Goose Rocks. 289
Goose Rocks Shoals. 289
Goshen Point 278, 281
Gould Island. 237, 240
Government Publishing Office . . . 416
Governor Mario M. Cuomo Bridge . . 379
Governors Island 361
Gowanus Bay 361
Gowanus Canal 361
Gowanus Flats 360
Grass Hassock Channel 359
Grass Island 342
Grassy Bay 359
Grassy Hammock Rocks 309
Grassy Island 213
Grassy Point. 380
Grassy Rocks 314
Gravesend Bay 359
Gray Gables 220
Great Captain Island 314
Great Captain Rocks 315
Great Cove 340
Great Eastern Rock 259
Great Gull Island 259
Great Harbor 212
Great Hill 224
Great Hill Point 224
Great Hog Neck 265
Great Island 311
Great Kills 365
Great Kills Harbor 364
Great Lakes-Hudson River Waterway Improvement 387
Great Ledge 213, 248
Great Neck Creek. 342
Great Peconic Bay 266
Great Point 198
Great Rip 170
Great River 197, 340
Great Rock 194
Great Round Shoal 188
Great Round Shoal Channel 189
Great Salt Pond (New Harbor) . . . 255
Great Sippewisset Rock 218
Great South Bay 339
Great South Channel 170
Green Creek. 339
Green Flats 383
Green Harbor 339
Green Island. 290
Green Island Bridge (State Route 7) . 387
Green Jacket Shoal 245
Green Pond 198
Greenport. 263
Greenport Harbor 263
Greens Harbor 277
Greens Ledge 309
Greenwich 314
Greenwich Bay. 250
Greenwich Cove 250, 313
Greenwich Harbor 314
Greenwich Point 313
Gregory Point 310
Greystone Station 379
Griswold Island 282
Groton 277
Groton Long Point 270
Guilford Harbor 289
Guilford Point 289
Gull Island 215, 264
Gull Point. 243
Gull Pond. 263
Gull Rock. 289
Gull Rocks 238, 291
Guttenberg 378

H

Hackensack River 371
Haddam Island. 286
Hadley Harbor 214
Hadley Rock. 213
Halesite 322
Half Acre Rock. 289
Halfmile Rock 231
Halfmoon Shoal 188
Halftide Rock 218, 287
Halfway Ledge 248
Halfway Rock 231, 240
Hallets Point 329
Hallets Rock. 194
Hallock Bay 263
Halls Island 196
Hamburg 285
Hamburg Cove. 285
Hamlin Point 218
Hammett Cove. 225
Hammock River 288
Hammonasset Point 288
Hammonasset River 288
Hammonasset State Park 288
Hampton Bays 336
Handkerchief Shoal. 188
Harbor Bluff. 195
Harbor entrances. 164
Harbor Ledge 312
Harbormasters. 183
Harlem River 330
Harmon 380
Harrison Island 317
Hart Cove. 337
Hartford 286
Harthaven 204
Hart Island 318
Hart Island Roads 319
Harwich Port 193
Hashamomuck Pond 263
Haskell Island 226
Hastings-on-Hudson 379
Hatchett Point. 282
Hatchett Reef 282
Haverstraw 380
Haverstraw Bay 380
Hawes Shoal. 201
Hawtree Basin 358
Hay Beach Point 262
Haycock Ledge. 237
Haycock Point 290
Haydens Point 285
Hay Harbor 267
Haywater Cove. 265
Head of Bay 359
Head of the Harbor 201
Health and Human Services, Department of 30
Heckscher State Park 340
Hedge Fence 188
Heights. 3
Hell Gate 330
Hell Gate Bridge 329
Hempstead Bay 343
Hempstead Harbor 324
Hen and Chickens . . 217, 287, 314, 316
Hen Cove 219
Hendrix Creek 358
Hen Island 315
Herring River 193
Hewlett Point 325
Hicks Island 260
Hicks Ledge 317
Higganum Creek 286
High Hill Point 237
High Island 290, 318
Highland Falls 381
Highland Light 186
Highland Park 367
Highlands. 186, 363
Highlands of Navesink. 347
High Rock 281
High Tor 380
Highwater Rock 312
Hitchcock Rock 313
Hither Creek 198
Hoadley Point 290
Hoboken 378
Hodges Rock 196
Hoffman Island 364

Hog Back 279, 329
Hog Creek Point 261
Hog Island 240, 243
Hog Island Channel 344
Hog Neck 223
Hog Neck Bay 265
Holidays 183
Holmes Hill 265
Holmes Rock 329
Holyoke 287
Homeland Security, Department of . . 31
Hook Mountain 380
Hook of the Cape 185
Hope Island 249
Horizontal Datum 5
Horse in Bank 299
Horse Neck 250
Horseneck Beach 230
Horseneck Point 230
Horseshoe Cove 266, 362
Horseshoe Harbor 317
Horseshoe Reef 271
Horseshoe Shoal 188
Horton Point Light 298
Hospital Cove 219
Housatonic River 296
Howard Beach 358
Huckleberry Island 317
Hudson 384
Hudson Canyon 171, 348
Hudson Channel 344
Hudson River 376
Hudson River Channel 350
Hull Cove 237
Huntington 322
Huntington Bay 321
Huntington Harbor 322
Hunts Point 328
Hunts Rock Breakwater 230
Hurricane 176
Hurricane Havens 177
Hurricane Havens Handbook for the North Atlantic Ocean 177
Hurricanes and Tropical Storms 19
Hurricane Warnings and Forecasts . 177
Hutchinson River 319
Hyannis 195
Hyannis Harbor 194
Hyannis Point 196
Hyannis Port 195
Hyde Park 382
Hydrographer Canyon 171

I

Ice 182
Ida Lewis Rock 238
Immersion Hypothermia 20
Immigration and Naturalization Service Offices 420
Indian Harbor 313
Indian Kill 383
Indian Point 381
Indian Reef 289
Indian River 288
Inland Waterways Navigation Regulations 103
International Code of Signals (Pub. 102) 416
International Ice Patrol (IIP) 182
Intrepid Rock 270
Inwood 359
Iona Island 381
Irvington 379
Island Channel 358
Island Creek 343
Island Park 344
Islip 340

J

Jacobs Point 298
Jamaica Bay 356
James Creek 266
Jamestown 239
Jamestown-Verrazzano Bridge . . . 248
Jeffrey Point 291
Jeffrey Rock 290
Jennings Point 264
Jersey City 378
Jersey Flats 360
Jerusalem 257
Jessup Neck 265
Jobs Neck 214
Jockey Creek 264
Joe Burris Ledge 231
John F. Kennedy (New York) International Airport 359
Johnson Point 291
Johnsons Creek 305
Johns Rock 282
Jones Beach Coast Guard Station . . 344
Jones Beach State Park 343
Jones Inlet 12352 343
Jones Island 343
Jones Point 381
Jones Rocks 314
Jordan Cove 281
Joseph Sanford Jr. Channel 359
Joshua Cove 290
Juniper Point 211

K

Katama Bay 202
Keansburg 365
Kellys Bay 193
Kelsey Point 287
Kelsey Point Breakwater 288
Kettle Bottom Rock 238
Keyport 365
Keyport Harbor 365
Kickamuit River 241
Kill Pond Bar 191
Kill Van Kull 369
Kimberly Reef 288
Kings Park 320
Kings Point 325
Kingston 383
Kingston Point 383
Kingston-Rhinecliff Bridge 383
Kitemaug 280
Knox Point 215

L

Lackeys Bay 214
Lagoon Pond 209
La Guardia Airport 328
Lake Champlain 388
Lake Montauk 260
Lake Tashmoo 210
Larchmont Harbor 316
Latimer Reef 269
Laurence Harbor 365
Law of the Sea Convention 23
Lawrence Point 329
Lawtons 240
Lee River 241
Leetes Island Quarry 290
Leetes Rocks 290
Lemon Creek 365
Leonardo 363
Lewis Bay 195
Lewis Pond 194
L'Hommedieu Shoal 188
Liberty Island 361
Lieutenant River 284
Light and Sound Signal Characteristics . 3
Lighthouse Point 294
Light List 416
Light Lists 12
Light Lists (United States and Possessions) 416
Lights 10
Lindenhurst 342
Lionhead Rock 262
List of Lights 416
List of Lights (Foreign Countries) . . 417
Little Bay 226, 326
Little Bird Island 224
Little Captain Island 314
Little Ferry 372
Little Gull Island 258
Little Gull Reef 258
Little Harbor 214
Little Ida Lewis Rock 238
Little Island 196, 218, 225
Little Mermaid 291
Little Narragansett Bay 268
Little Neck Bay 325
Little Peconic Bay 265
Little Pleasant Bay 187

Little River 197
Little Rock 281
Little Round Shoal 188
Little Seatuck Creek 337
Little Silver Creek 363
Little Southwest Rock 230
Little Stony Point 381
Little Whale 248
Lloyd Harbor 322
Lloyd Neck 322
Lloyd Point 322
Local Magnetic Disturbances 9
Local Notices to Mariners 9
Local Notice to Mariners 416
Locust Point 319, 363
Lone Rock 196, 204, 211, 214
Long Beach 344
Long Beach Point 224, 263
Long Creek 344
Long Island 226
Long Island Intracoastal Waterway . . 336
Long Neck Point 311
Long Point 250, 338
Long Sand Shoal 287
Long Shoal 201
LORAN-C 13
Lord Cove 285
Lords Passage 258
Lovers Island 291
Lower Bay 351
Lower Nyack Landing 379
Lucas Shoal 210
Lumber Rock 230
Luppatatong Creek 365
Lydonia Canyon 171

M

Mackerel Cove 238
Madaket Harbor 198
Madison 289
Madison Reef 288
Magazine Point 381
Maidstone Park 262
Main Channel 189
Maltby Cove 290
Mamaroneck 316
Mamaroneck Harbor 315
Manahassett Creek 363
Manhasset Bay 325
Manhattan Beach 357
Manhattan Bridge 332
Manorhaven 325
Manresa Island 309
Mansfield Point 291
Manursing Island Reef 315
Marine Pollution 21
Marine Product Dissemination Information 28, 417
Marine Protected Area (MPA) 25
Mariners Harbor 369
Marine Sanitation Device Standard . 152
Marine Weather Forecasts 27, 416
Marion 225, 371
Marks Cove 224
Marlboro 382
Marsh Island 244
Martha's Vineyard 202
Mashnee Island 219
Mashomack Point 264
Mashpee Neck 197
Mason Island 269
Mason Point 269
Maspeth Creek 331
Massachusetts Maritime Academy . . 223
Mastic 338
Masury Point 338
Matawan Creek 365
Matawan Point 365
Matinecock Point 324
Mattapoisett 226
Mattapoisett Harbor 226
Mattapoisett Ledge 226
Mattapoisett Neck 226
Mattituck 298
Mattituck Inlet 298
Matunuck 257
Matunuck Point 257
McBlair Shoal 170
McCorrie Point 237
Meadow Island 225, 343
Measured Courses 420
Medical Advice 14
Meetinghouse Creek 266
Meeting House Pond 187
Megansett 218
Megansett Harbor 218
Melton Ledge 279
Menauhant 198
Mendells Rock 225
Menemsha 211
Menemsha Basin 211
Menemsha Bight 211
Menemsha Creek 211
Menemsha Pond 211
Menunketesuck Island 287
Menunketesuck River 288
Merrick Creek 343
Miamogue Point 266
Mianus River 313
Middle Beach 289
Middle Cove 285
Middle Ground 203, 210, 268, 317
Middle Ground Flats 384
Middle Ledge 213, 215
Middle Reef 317
Middle Rip 13204 170
Middle Shoal Rock 259
Middletown 286
Mid Hudson Bridge (U.S. 44) 382
Midland Beach 364
Midway Shoal 261
Milford Harbor 296
Milford Point 296
Mill Basin 358
Mill Cove 249
Mill Creek 192, 263, 265, 358
Mill Neck Creek 323
Mill Pond 315
Mill River 292
Mill Rock 330
Millstone Point 281
Milton Harbor 315
Milton Point 315
Mishaum Ledge 217
Mishaum Point 230
Mitchell River 192
Mitchell Rock 238
Mohegan Bluffs 255
Money Island 290
Monomoy Island 190
Monomoy Point 188
Monomoy Shoals 188
Montauk 260
Montauk Harbor 260
Montauk Point 259
Montauk Shoal 259
Montville Station 280
Monument Beach 219
Morgan 365
Morgan Point 270
Moriches Bay 337
Moriches Inlet 337
Morris Canal Basin 378
Morris Cove 292
Morris Rock 322
Morses Creek 367
Moses Point 323
Mosher Ledge 226
Moshers Point 230
Mosquito Cove 324
Mott Point 324
Motts Basin 359
Motts Creek 359
Mount Hope 240
Mount Hope Bay 240
Mount Misery 299
Mount Misery Shoal 300
Mount Prospect 258
Mount Sinai Harbor 299
Mouse Island 270
Mud Creek 265, 339, 342
Mud Gorge 171, 348
Mud (West Senix) Creek 338
Mumford Cove 271
Muskeget Channel 201
Muskeget Island 198
Musselbed Shoals 240
Mutton Shoal 201
Mystic 271
Mystic Harbor 270
Mystic River 271
Mystic Seaport Museum 271

N

Namequoit Point 187
Namequoit River 187
Nannaquaket Pond 237
Nantucket 199
Nantucket (Great Point) Light . . . 198
Nantucket Harbor 199
Nantucket Island 198
Nantucket Shoals. 188
Nantucket Shoals 13204, 13200 . . . 169
Nantucket Sound. 188
Napatree Beach 267
Napatree Point. 267
Napatree Point Ledge 268
Napeague Bay 260
Napeague Harbor. 260
Narragansett Bay. 233
Narragansett Pier. 248
Narraskatuck Creek. 343
Narrow Bay 338
Narrows 271
Nashawena Island 214
Nasketucket Bay 226
Nasketucket River 226
Nassau Point 265
National Data Buoy Center Meteorological Buoys 27
National Environmental Satellite, Data, and Information Service (NESDIS) 28
National Geospatial-Intelligence Agency (NGA) 28
National Hurricane Center/Tropical Prediction Center. 177
National Institute of Standards and Technology (NIST) 19
National Ocean Service (NOS) 26
National Weather Service Offices . 27, 415
National Weather Service Port Meteorological Officers (PMOs) 28
Naugatuck River 296
Nauset Beach 187
Nauset Beach Light 186
Nauset Harbor 187
Naushon Island 214
Nautical Almanac. 417
Nautical Chart–New Editions and Corrections 4
Nautical Chart Numbering System. . . 4
Naval Observatory 30
Navesink River. 362
Navigational Warnings, Information and Weather 16
Navigation and Navigable Waters . . . 37
Navigation Regulations 140
Navigation Rules and Regulations Handbook 417
NAVTEX 17
Nayatt Point. 244
Nebraska Shoal 257
Ned Point. 226
Neds Creek 343
Negro Point 329
Neguntatogue Creek 342
Neptune Island. 318
Newark. 371
Newark Bay 369
New Baltimore. 384
New Bedford. 227
New Bedford Harbor 226
New Brighton 369
New Brunswick 366
Newburgh 382
Newburgh-Beacon Bridges (IS 84) . . 382
Newfoundland Reef 313
New Hamburg 382
New Haven 292
New Haven Harbor 291
New London. 277
New London Coast Guard Station . . 277
New London Harbor 277
New London Ledge 278
New London Ledge Light 278
New Milford 371
Newport 238
Newport Harbor 238
Newport Neck 237
New Rochelle 317
New Rochelle Harbor 318
New Suffolk 265
Newton Rock 237
Newtown Creek 331
New York Economic Development Corporation 355
New York Harbor. 347, 349
New York Maritime College. 326
New York Naval Shipyard 332
New York State Canal System. . . . 387
Niantic 281
Niantic Bay 281
Niantic River 281
Nickersons Neck 187
Nicks Point 343
Nicoll Bay. 340
Nicoll Island. 340
Nicoll Point 264, 340
Ninigret Pond 257
Nissequogue River 320
NOAA Weather Radio 28, 415
NOAA Weather Radio Broadcasts . . . 17
Noank 271
Nobska Point 210
Nobska Point Ledges 210
No-Discharge Zone 256
No-Discharge Zones 21
Nomans Land 211
Nonamesset Island 211
Nonamesset Shoal 213
Nonquitt 230
Noroton Neck 311
Noroton Point 311
North Atlantic Right Whales 165
North Bay. 196
North Brother 282
North Brother Island 329
North Channel. 189, 358
North Chatham 187
North Cove 284
North Dumpling 267
North Falmouth 218
North Germantown. 384
North Haven Peninsula 264
North Ledge. 317
North Manursing Island 315
North Point 214
Northport. 321
Northport Basin 320
Northport Bay 321
Northport Harbor 321
North Race 266
North River 376
North Sea Harbor 265
Northwest Harbor 264
Northwest Point 359
Norton Point 210, 359
Norton Shoal 201
Norwalk 310
Norwalk Cove 310
Norwalk Harbor 309
Norwalk Islands 309
Norwalk River 309
Norwich 280
Notices to Mariners. 9
Notification of Arrival (NOA) 25
Noyack (Noyac) Bay. 265
Noyes Rock 269
Noyes Shoal 269
Nutley 371
Nyack 380
Nye Ledge. 226
Nyes Neck. 218

O

Oak Beach 342
Oak Bluffs. 204
Oak Bluffs Harbor 204
Oak Island 342
Oak Island Channel. 342
Oakland Beach. 250
Oak Neck Creek 324
Oak Neck Point 324
Obstructions 3
Occupessatuxet Cove 244
Ocean Beach 340
Ocean Dumping 21
Oceanographer Canyon 171
Oceanport Creek 363
Oceanside. 344
Offshore Terminal, Northport. . . . 320
Offshore Traffic Separation Schemes . 134
Ohio Ledge 243
Oil Spill Reporting 21
Old Bull 237
Old Clump 291

Old Cock 217
Old Ferry Point 327
Old Field Point. 320
Old Greenwich. 313
Old Harbor 255
Old Lyme 284
Old Man 211
Old Man Shoal 170
Old Mill Creek 358
Old Mystic 271
Old Neck Creek 12352. 338
Old Orchard Shoal 364
Old Reef 257
Old Saybrook 284
Old Sergeant 248
Old Silas Rock 259
Old South Shoal 13200 170
Olivers Channel 343
Onset 223
Onset Bay. 223
Optical Phenomena. 180
Orchard Beach. 318
Orchard Neck Creek 338
Orient 263
Orient Harbor 263
Orient Point. 261
Orient Shoal. 298
Orion Shoal 188
Orleans. 187
Orowoc Creek 340
Ossining 380
Osterville 196
Osterville Grand Island 197
Oswego Canal 388
Otter Rocks 314
Outer Island. 290
Outer Steamboat Rock 315
Outer White Top 289
Overpeck Creek 372
Oyster Bay 322, 323
Oyster Bay Harbor 323
Oyster Pond 192
Oyster Pond Reef. 261
Oyster Pond River 192

P

Padanaram Breakwater 230
Paerdegat Basin 358
Palisades State Park. 379
Palmer Island 227
Paper Print on Demand Nautical Charts 3
Paper Rock Shoal. 286
Paradise Point 264
Parker Flats 213
Parkers Creek 363
Parkers River 194
Parsonage Island 344
Passaic 371
Passaic River 371
Patchogue 339
Patchogue Bay. 339
Patchogue River 288, 339
Patience Island. 250
Patroon Island Bridge 387
Pawcatuck Point 268
Pawcatuck River 268
Pawn Rock 230
Pawtucket 244
Pawtuxet 244
Pawtuxet Cove 244
Paw Wah Pond. 187
Peacock Point 324
Pea Island. 317
Peaked Hill Bar 186
Peartree Point 311
Pease Ledge 215
Peck Ledge 309
Pecks Rock 313
Peconic River 266
Peekskill 381
Pelham. 319
Penataquit Creek. 341
Penfield Reef 308
Penfield Reef Light 308
Penikese Island 215
Peningo Neck 315
Penzance 213
Penzance Point 213
Pequonnock River 305
Pequotsepos Brook 271
Perth Amboy 368
Peters Point 241
Pettit Marsh. 343
Pews Creek 364
Phelps Bank 13204 170
Phelps Ledge 259
Phillipsdale 244
Phinneys Harbor 219
Picket Rock 325
Pierhead Channel 361
Piermont 379
Piermont Pier 379
Pilotage 183
Pilotage, Albany 385
Pilotage, Block Island Sound and Long Island Sound 254
Pilotage, Bridgeport 306
Pilotage, Cape Cod Canal and Buzzards Bay 222
Pilotage, City Island. 318
Pilotage, Connecticut River 284
Pilotage, East River 326
Pilotage, Fall River 242
Pilotage, Hudson River 354, 377
Pilotage, Long Island Sound 276
Pilotage, Nantucket Sound 190
Pilotage, Narragansett Bay and Other Rhode Island Waters 235
Pilotage, New Bedford 228
Pilotage, New Haven 294
Pilotage, New London-Groton . . . 279
Pilotage, Newport 239
Pilotage, New York Harbor and Approaches 353
Pilotage, New York Harbor from Long Island Sound 354
Pilotage, Norwalk 310
Pilotage, Offshore Terminal, Northport 320
Pilotage Pickup Locations Off Montauk Point 259
Pilotage, Port Jefferson 300
Pilotage, Providence 246
Pilotage, Raritan River and Arthur Kill 364
Pilotage, Riverhead Production Platform 299
Pilotage, Thames River 280
Pilotage, Vineyard Sound and Buzzards Bay 208
Pilotage, Woods Hole 213
Pilot Boat Cruising Area, New York Harbor 349
Pine Creek Point 308
Pine Grove 281
Pine Island 271, 317
Pine Island Channel 278
Pine Marsh 344
Pine Neck. 337
Pine Neck Point 337
Pine Orchard 290
Pine Point 261, 311
Pinetree Ground 231
Piney Point 224
Pipelaying barges. 165
Planting Island. 225
Planting Island Cove 225
Playland 315
Pleasure Bay. 363
Plumb Beach Channel. 357
Plum Gut 261
Plum Gut Harbor. 261
Plum Island 261
Plum Island Rock 261
Plum Point 323, 325, 382
Pocasset Harbor 219
Pocasset River 219
Point Comfort 365
Point Gammon 194
Point Judith Harbor of Refuge . . . 256
Point Judith Light 256
Point Judith Neck 250
Point Judith Pond 256
Point Lookout 344
Point o' Woods. 340
Point Rip 198
Pollepel Island 382
Pollock Rip 188
Pollock Rip Channel 189
Polpis Harbor 201
Pomham Rocks 245
Pond Point 295
Ponquogue Point. 337
Poorhouse Flats Lighted Range . . . 331
Popasquash Neck. 243
Popes Island. 227
Poplar Point. 249

Popponesset Bay 197
Popponesset Island 197
Porgy Shoal 315
Porpoise Channel. 320
Portable Document Format (PDF) Nautical Charts. 3
Port Authority of New York and New Jersey 355
Port Chester. 315
Port Chester Harbor 315
Port Elizabeth Marine Terminal . . . 370
Port Jefferson 300
Port Jefferson Harbor 299
Port Johnson 369
Portland 286
Port Monmouth 364
Port Morris 329
Port Newark Terminal 371
Port Reading 368
Port Richmond 369
PORTS® (Physical Oceanographic Real-Time System). 26, 350
Ports and Waterways Safety-General . . 88
Port Socony 368
Port Washington 325
Potash Bar 285
Pot Island. 290
Potowomut Neck 249
Potowomut River. 249
Potter Cove 240, 243
Potter Pond 257
Poughkeepsie 382
Powder Island 279
Powell Cove 327
Pratt Creek 285
Premium Point 317
Promised Land. 260
Promised Land Channel 260
Prospect Hill 259
Prospect Point 325
Providence 244
Providence Point 250
Providence River 244
Provincetown 186
Provincetown Harbor 186
Prudence Island 233, 243
Public Health Service 30
Public Health Service Quarantine Stations 419
Pugsley Creek 328
Pumpkin Patch Channel 358

Q

Quahaug Bar 224
Quantuck Bay 337
Quantuck Canal 337
Quantuck Creek 337
Quarantine, animal and plant 26
Quarantine Stations 419
Queensboro Bridge 331
Quicksand Point 231
Quicks Hole 214
Quinnipiac River 292
Quissett 218
Quissett Harbor 218
Quixes Ledge 294
Quogue Canal 337
Quonochontaug 257
Quonochontaug Breachway 257
Quonochontaug Pond 257
Quonset Point 249

R

Racehorse Channel 343
Race Point 186, 258
Race Point Ledge. 258
Race Rock. 258
Race Rock Light 258
Radar 164
Radio Navigational Aids 3
Radio Navigational Aids (Pub. 117). . 416
Radio shore stations providing medical advice 420
Radiotelephone Distress Message . . . 13
Radio Weather Broadcasts 418
Rahway. 369
Rahway River 369
Ram Head. 262
Ram Island 225, 226, 266, 270
Ram Island Reef 269
Ram Island Shoal. 270
Randall Bay 344
Randalls Island. 329
Rands Harbor 219
Ranges 3
Rapid Rock 278
Raritan Bay 364
Raritan Bay Channel 351
Raritan River 366
Raster Navigational Charts (NOAA RNC®)4
Rat Island. 318
Recommended Two-Way Whale Avoidance Routes 185
Recommended Two-Way Whale Avoidance Track 185
Recreational Boaters Navigating Near Commercial Shipping Channels 376
Red Bank 363, 365
Red Brook Harbor 219
Red Hook Channel 360
Red Ledge. 213
Red Reef 269
Red Rock 248, 313
Reeves Bay 266
Reeves Creek 266
Regulated Navigation Areas 164
Regulated Navigation Areas and Limited Access Areas 117
Regulated Waters. 22
Rensselaer 384
Repairs-salvage-wrecking 183
Reported information 3
Reynolds Channel 344
Rhode Island 233
Rhode Island Rock 204
Rhode Island Rocks 312
Rhode Island Sound 216, 233
Ribbon Reef 217
Richmond Creek 265
Riding Rock 289
Rikers Island 328
Rikers Island Channel 328
Rip Van Winkle Bridge 384
Riverhead. 266
Riverhead Production Platform . . . 298
River Ledge 248
Riviere Richelieu 388
Roa Hook 381
Roanoke Point Shoal 299
Robbins Reef Light 361
Robbins Rock 209
Robert Moses (Fire Island) State Park 341
Robins Island 266
Robins Island Rock 266
Robinsons Hole 214
Rockaway Beach 358
Rockaway Inlet. 355
Rockaway Point 347, 355
Rock Island 244
Rocky Hill 286
Rocky Point 250, 298, 323
Rodgers Rock 266
Rogers Island 290
Rome Point 248
Romer Shoal 352
Rondout Creek. 383
Roosevelt Island Reef (Welfare Island Reef) 331
Roosevelt Island (Welfare Island) . . 330
Rose and Crown 170
Rose Island 238
Roton Point 311
Round Cove 187
Round Hill Point 230
Round Rock 249
Routes 172
Rowayton 311
Ruffle Bar. 358
Ruins 260
Rumson 363
Rumstick Ledge 243
Rumstick Neck. 243
Rumstick Point 243
Rumstick Rock. 243
Rumstick Shoal 243
Rutherford 371
Ryder Cove 187
Rye Beach. 315

S

Sabin Point 245

Sachem Head 289
Sachem Head Harbor 289
Sachuest Point. 236
Sag Harbor 264
Sag Harbor Cove 264
Sailing Directions 416
Sailing Directions (Foreign Countries) 417
Sailors and Soldiers Monument . . . 378
Sailors Haven 340
Sakonnet Harbor 236
Sakonnet Point 236
Sakonnet River 236
Salmon Cove 286
Saltaire. 340
Salters Point 230
Sampawams Creek 342
Sampawams Point 342
Sampsons Island 197
Sand Spit 264
Sands Point 325
Sandy Hook 347, 362
Sandy Hook Bay 362
Sandy Hook Channel 351
Sandy Point . 237, 250, 253, 267, 268, 281, 293
Sankaty Head Light. 198
Saquatucket Harbor 192
Sarah Ledge. 278
Sauga Point 249
Saugatuck 308
Saugatuck River 308
Saugerties 383
Saunders Point 281
Saunderstown 248
Saybrook Outer Bar. 283
Sayreville 367
Scarborough Beach. 250
School House Creek 265
Schuyler Ledge 236
Sconticut Neck 226
Scotch Caps 315
Scott Cove 311
Scraggy Neck 218
Scup Rock 249
Sea Bright 363
Sea Cliff 324
Sea Dog Creek 344
Seaflower Reef 267
Seal Ledge 237
Seal Rock 237, 248, 249
Seal Rocks 219, 225, 267
Seapit River 197
Seapuit River 197
Search and Rescue 13
SEARCH AND RESCUE 13
Sears Point 192
Seaside 281
Seatuck Cove 337
Seatuck Creek 337
Sebonac Creek. 266
Secaucus 372
Security Broadcast System, Narragansett Bay 234
Seekonk River 244, 245
Seiche 20
Seine Pond 194
Senator Shoal 194
Sengekontacket Pond 204
Senix Creek 338
Sesachacha Pond. 198
Setauket 300
Setauket Harbor 300
Seymour Point. 309
Shag Bank 293
Shagwong Reef 259
Shagwong Rock 259
Shaw Cove 277
Sheep Rocks. 309
Sheepshead Bay 357
Sheffield Island 309
Sheffield Island Harbor 309
Shellbank Basin 358
Shell Bank Creek. 357
Shelter Island Heights. 263
Shelter Island Sound 262
Shelton. 296
Sherwood Point 308
Shetucket River 280
Shinnecock Bay 336
Shinnecock Canal 336
Shinnecock Inlet 336
Shinnecock Light 336
Shippan Point 311
Shipping Safety Fairways 133
Ship Reporting Systems 136
Ship Rock. 315
Shoal Point 308
Shooters Island 369
Shore Island. 314
Short Beach 291, 343
Shovelful Shoal 201
Shrewsbury River 362
Siasconset 198
Sias Point. 223
Silver Beach Harbor 218
Silver Eel Cove (Silver Eel Pond) . . 267
Sing Sing Correctional Facility . . . 380
Sippican Harbor 225
Sippican Neck 224
Sixmile Reef. 287
Slocums Ledge. 230
Slocums River 230
Sloop Channel 343
Sluice Creek. 289
Small-craft facilities 183
Smith Cove 264, 281, 314
Smith Creek. 368
Smith Meadow. 344
Smith Neck 230
Smith Point 338
Smith Reef 311
Smithtown Bay 320
Snow Rock 226
Snug Harbor 257
Somerset 240, 243
Sommerville Basin 359
Sound Signals 12
South Amboy 366
South Beach. 203, 364
South Brother 282
South Brother Island 329
South Brother Island Channel . . . 329
South Dartmouth 230
South Dennis 193
South Ferry 248, 264
South Jamesport 266
South Nonations 317
South Norwalk. 310
Southold 264
Southold Bay 264
South Oyster Bay. 343
Southport. 308
Southport Harbor 308
South Race 266
South River 366, 367
South Swansea. 241
South Wellfleet 186
Southwest Ground 196
Southwest Ledge 219, 253
Southwest Ledge Light 292
Southwest Point 238
Southwest Reef 12374. 287
Southwest Rock 196
South Yarmouth 193
Sow and Pigs Reef 217
Space Weather Prediction Center (SWPC) 28
Spar Island 241
Special Notices to Mariners. 416
Spectacle Island 290
Speed and Wake Damage. 376
Speonk Point 337
Spindle Rock 196
Spoil areas 8
Spuyten Duyvil Creek 379
Squam Head. 198
Squash Meadow 204
Squeteague Harbor 219
Squibnocket Point 211
Stage Harbor 191
Stage Island 192
Stamford 311
Stamford Harbor 311
Standard Abbreviations for Broadcasts . 17
Standard Time 183
Star Island 260
State Boat Channel 341
Staten Island 364
Staten Island Flats 364
Staten Island Light 350
Statue of Liberty 361
St. Elmo's fire 181
Stepping Stones 325
St. George 350
St. Helena Island 290
Stirling Basin 263

Stone Horse Shoal 188
Stone Island 288
Stonington 269
Stonington Harbor 269
Stonington Point 268
Stony Brook 320
Stony Brook Harbor 320
Stony Creek 290
Stony Island 289
Stony Islet 293
Stony Point 308, 380
Stony Point Dike 224
Storm King Mountain 381
Storm Surge 20
St. Patrick Rock 239
Stratford 296
Stratford Point 296
Stratford Shoal Middle Ground . . . 297
Stratford Shoal (Middle Ground) Light 297
Strawberry Point 226
Strongs Creek 342
Strongs Point 342
Stump Creek 365
Sturgeon Flats 202
Submarine Cables and Submerged Pipelines 8
Submarine canyons 171
Success Rock 325
Succonnesset Point 197
Succonnesset Shoal 197
Succotash Point 257
Sugar Reef 258
Sugar Reef Passage 258
Sumac Island 290
Sunken Ledge 226
Sunken Meadow 329
Sunken Meadow Creek 320
Superstructure Icing 180
Supplies 183
Swan Pond 193
Swan Pond River 193
Swan River 339
Swash Channel 351
Swift Creek 343
Swifts Beach 224
Swinburne Island 364

T

Tallman Island 327
Tallman Mountain State Park 379
Tappan Zee 379
Tarpaulin Cove 214
Tarrytown 379
Taunton 240
Taunton River 240
Taunton Rock 291
Tavern Island 310
Taylor Point 240
Terminal Channel 351, 363
Tern Island 187
Territorial Sea 23
Terrys Creek 266
Thames River 277
Thamesville 280
Thatchbed Island 285
The Bar 296
The Battery 349
The Blauzes 318
The Bonnet 248
The Bow Bells 225
The Breachway 256
The Cove 237
The Cows 312
The Dumplings 238
The Gulf 296
The Gut 311
The Knob 218
The Knubble 230
The Little Cows 308
The Maelstrom 383
The Narrows 187, 360
The Neck 192
The Race 258
The River 187
The Sand Hole 323
The Sisters 240
The Sluiceway 259
The Spider 352
The Strait 212
The Thimbles 290
The Wildcat 217
Thimble Island Harbor 290
Thompson Cove 268
Thorns Creek 365
Threemile Harbor 262
Throgs Neck 326
Throgs Neck Bridge 326
Thurston Basin 359
Tiana Beach 337
Tides 3
Time 3
Tiverton 237
Tobys Island 219
Tomkins Cove 381
Toms Point 230, 325
Tongue Point 305
Totoket Bar 290
Tottenville 368
Towage 183
Town Creek 264
Town Neck Creek 363
Townshend Ledge 292
Tracey Ledge 238
Traffic Separation Scheme (Buzzards Bay) 216
Traffic Separation Scheme Narragansett Bay 233
Traffic Separation Scheme Off New York 349
Traffic Separation Schemes (Traffic Lanes) 164
Transport Rock 315
Triborough Bridge 329
Tropical Cyclones 176
Tropical Depression 176
Tropical Storm 176
Tropical Wave 176
Troutmans Creek 363
Troy 387
Troy Lock and Dam 387
Troy-Menands Bridge 387
Truro 186
Tsunamis 19
Tuckernuck Island 198
Tuckernuck Shoal 201
Tuthill Cove 337
Tuthill Point 337
Tuttles White Bank 299
Tuxis Island 289
Twomile Ledge 231
Twomile Rock 231
Twotree Island 281
Twotree Island Channel 281
Tylerville 285

U

Umbrella Point 316
Umbrella Rock 317
Uncas Point 289
Uncle Daniels Point 359
Under-keel clearances 2
Unionport 327
Upper Bay 360
Upper Nyack 380
U.S. Army Corps of Engineers (USACE) Offices 417
U.S. Coast Guard 31
U.S. Coast Guard Navigation Center (NAVCEN) 417
U.S. Customs and Border Protection . . 31
Usher Rocks 243
U.S. Merchant Marine Academy . . . 325
U.S. Merchant Marine Academy 12366 325
U.S. Military Academy 381
U.S. Military Academy 12343 381
U.S. Naval Submarine Base 280

V

Valiant Rock 259
Veatch Canyon 171
Venetian Harbor 271
Vernam Basin 358
Verplanck Point 380
Verrazano Narrows Bridge 360
Vessel Arrival Inspections 183
Vessel Bridge-to-Bridge Radiotelephone Regulations 37
Vessel Identification 14
Vessel Movement Reporting System . 101
Vessel Response Plans 25
Vessel Traffic Management 96

Vessel Traffic Service, New York . 164, 276, 304
Vineyard Haven 208
Vineyard Haven Harbor 208
Vineyard Point 289
Vineyard Sound 207
Visibilities. 179
Vixen Ledge 271
Voluntary Observing Ship Program (VOS) 19

W

Waackaack Creek. 365
Wading River 299
Wakefield 256
Wallabout Bay 332
Wampanoag Ledge 227
Wamphassuc Point 269
Wappinger Creek 382
Wappingers Falls 382
Waquoit Bay. 197
Ward Point 368
Wareham 224
Wareham River 224
Warren 243
Warren River 243
Warwick Cove 250
Warwick Neck 250
Warwick Point 250
Washington Canal 366
Washington Shoal 259
Wasque Point 201
Wasque Shoal 201
Watch Hill 257, 340
Watch Hill Cove 268
Watch Hill Light 258
Watch Hill Passage 258
Watch Hill Point 258
Watch Hill Reef 258
Watchogue Creek. 341
Waterford. 281
Watervliet. 387
Wauwinet. 201
Waves 178
Wayland Island. 290
Wee Captain Island 314
Weehawken 378
Weekapaug Breachway 257
Weekapaug Point. 257
Weeks Point. 324
Weepecket Islands 214
Weepecket Rock 214
Weesuck Creek. 337
Weir Creek 319
Welches Point 295
Welker Canyon. 171
Wepawaug River 296
Wequetequock Cove 268
Werthein National Wildlife Refuge . . 338
West Babylon Creek. 342
West Bank 352, 364
West Bank Light 350
West Bay 196
West Branch. 307, 312
Westbrook 287
Westbrook Harbor 287
Westchester 327
Westchester Creek 327
West Chop 208
Westcott Cove 311
West Cove. 270, 271, 337
West Creek 341
West Crib 290
West Crow Island. 343
West Dennis. 193
Westend Pond 215
Westerly 268
West Falmouth 218
West Falmouth Harbor 218
Westhampton Beach 337
West Harbor. 267
West Harbor 12352 323
West Island 226
West Neck 266, 323
West Neck Bay 264
West Neck Harbor 264
West Passage 247
West Point 381
Westport 309
Westport Harbor 230
Westport Point. 230
Westport River. 230
West River 289, 292
West Rock 288, 315
West Southwest Ledge. 194
Wethersfield Cove 286
Weweantic River 224
Whaleback Rock 270
Whale Creek. 331
Whale Rock 215, 226, 248, 270
Wheeler Rock 288
White Rock 269, 279, 281, 309
Whitestone 327
Whitestone Point. 327
Wickets Island 223
Wickford 249
Wickford Cove 249
Wickford Harbor 248
Wickham Creek 265
Wicopesset Island 258
Wicopesset Passage 258
Wicopesset Rock 258
Wilbur Point 226
Wild Goose Point. 249
Wild Harbor. 218
Wilkes Ledge 228
Willets Point 326
Williamsburg Bridge 332
Willow Point 271
Wilson Cove. 311
Wilson Head. 314
Wilson Point 311
Wind Chill and Frostbite. 20
Wings Cove 224
Wings Neck 219
Winhole Channel. 358
Winkle Point 321
Winnapaug Pond 257
Winthrop Cove. 278
Wood End. 186
Woodmere Channel. 345
Woodmont 293
Woodsburgh. 344
Woodsburgh Channel 344
Woods Creek 342
Woods Hole 211
Woods Hole Passage 212
Wooley Pond 265
Woolsey Rock 313
Worlds End 381
Wreck Shoal. 197
Wychmere Harbor 193

Y

Yantic River 280
Yellow Mill Channel. 305
Yonkers 379
Youngs Point 263
Youngs Rock 267

Z

Zachs Bay. 343
Zieglers Cove 311